

WORK ORDER NO.

At the

PREPARED BY:

Architectural:
Civil:
Structural:

Mechanical:
Electrical:

Submitted By:

Date:

APPROVED BY:

Specifications:

For Commander, NAVFAC:
Date:

HAVING PARTICIPATED IN THE DESIGN OF PROJECT No. 1671552, P-919 - BULK TANK FARM IMPROVEMENTS, ROTA NAVAL STATION, SPAIN, AND HAVING THOROUGHLY REVIEWED THE COMPLETED PROJECT DOCUMENTS, I DECLARE THAT THE FACILITIES DESIGN INCLUDED HEREIN COMPLIES WITH ALL APPLICABLE SPANISH AND EUROPEAN CODES AND LAWS.

HABIENDO PARTICIPADO EN EL DISEÑO DEL PROYECTO N° 1671552, P-919 - BULK TANK FARM IMPROVEMENTS EN LA BASE NAVAL DE ROTA, ESPAÑA, Y HABIENDO REVISADO LOS DOCUMENTOS DEL PROYECTO COMPLETO, DECLARO QUE EL DISEÑO DE LAS INSTALACIONES INCLUIDAS AQUÍ CUMPLE CON TODA LA NORMATIVA Y LEYES APLICABLES ESPAÑOLAS Y EUROPEAS.

Date:
Fecha:

Signature
Firma:



Professional Seal
Sello Profesional:



16 April 2024
16 de Abril de 2024

Juan Ricardo de Andrés
Registered Professional Engineer # 6606
Colegiado nº 6606

PROJECT TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

00 01 15	02/11, CHG 1: 08/14	LIST OF DRAWINGS
00 73 04.00 41	08/08	SUPPLEMENTARY CONDITIONS FOR PROJECTS IN MEDITERRANEAN AREA

DIVISION 01 - GENERAL REQUIREMENTS

01 11 00	08/15, CHG 2: 08/21	SUMMARY OF WORK
01 14 00	11/22, CHG 1: 02/23	WORK RESTRICTIONS
01 20 00	11/20, CHG 3: 02/23	PRICE AND PAYMENT PROCEDURES
01 30 00	11/20, CHG 3: 08/23	ADMINISTRATIVE REQUIREMENTS
01 33 00	08/18, CHG 4: 02/21	SUBMITTAL PROCEDURES
01 35 26	11/20, CHG 4: 08/23	GOVERNMENTAL SAFETY REQUIREMENTS
01 35 29.13	11/15, CHG 1: 08/22	HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES
01 42 00	02/19	SOURCES FOR REFERENCE PUBLICATIONS
01 42 15	02/14	METRIC MEASUREMENTS
01 45 00	08/23	QUALITY CONTROL
01 45 35	11/20	SPECIAL INSPECTIONS
01 50 00	11/20, CHG 2: 08/22	TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS
01 57 19.01 21		ENVIRONMENTAL CONTROLS
01 58 00	08/19, CHG 5: 08/22	PROJECT IDENTIFICATION
01 74 19	02/19, CHG 3: 11/21	CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
01 78 00	05/19, CHG 1: 08/21	CLOSEOUT SUBMITTALS
01 78 23	05/23	OPERATION AND MAINTENANCE DATA
01 78 23.33	08/18, CHG 1: 02/21	OPERATION AND MAINTENANCE MANUALS FOR AVIATION FUEL SYSTEMS
01 78 24.00 20	05/23	FACILITY DATA WORKBOOK (FDW)

DIVISION 02 - EXISTING CONDITIONS

02 41 00	08/22	DEMOLITION
02 61 13	02/10, CHG 1: 02/21	EXCAVATION AND HANDLING OF CONTAMINATED MATERIAL
02 81 00	11/18	TRANSPORTATION AND DISPOSAL OF HAZARDOUS MATERIALS
02 82 00	11/18	ASBESTOS REMEDIATION
02 83 00	11/18	LEAD REMEDIATION

DIVISION 03 - CONCRETE

03 30 00	02/19, CHG 4: 08/22	CAST-IN-PLACE CONCRETE
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DIVISION 05 - METALS

05 50 13	05/17, CHG 1: 08/18	MISCELLANEOUS METAL FABRICATIONS
05 51 00	08/23	METAL STAIRS
05 51 33	02/16, CHG 2: 02/18	METAL LADDERS
05 52 00	08/23	METAL RAILINGS
05 59 10	11/22	ROLLING COVER FOR AVIATION REFUELING VAULTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

07 11 13	08/23	BITUMINOUS DAMPPROOFING
07 12 00	02/16	BUILT-UP BITUMINOUS WATERPROOFING
07 14 00	08/23	FLUID-APPLIED WATERPROOFING
07 21 13	08/23	BOARD AND BLOCK INSULATION
07 21 16	08/23	MINERAL FIBER BLANKET INSULATION
07 60 00	08/23	FLASHING AND SHEET METAL
07 84 00	05/10, CHG 1: 08/13	FIRESTOPPING
07 92 00	08/23	JOINT SEALANTS

DIVISION 08 - OPENINGS

08 51 23	08/20, CHG 1: 02/22	STEEL WINDOWS
08 71 00	02/16, CHG 4: 02/22	DOOR HARDWARE
08 81 00	05/19	GLAZING
08 91 00	08/20	METAL WALL LOUVERS

DIVISION 09 - FINISHES

09 22 00	02/10, CHG 2: 08/18	SUPPORTS FOR PLASTER AND GYPSUM BOARD
09 29 00	08/16, CHG 4: 02/20	GYPSUM BOARD
09 51 00	08/20	ACOUSTICAL CEILINGS
09 69 13	11/15, CHG 1: 08/18	RIGID GRID ACCESS FLOORING
09 90 00	02/21	PAINTS AND COATINGS
09 97 13.17	05/22	THREE COAT EPOXY INTERIOR COATING OF WELDED STEEL PETROLEUM FUEL TANKS
09 97 13.27	02/21, CHG 1: 08/22	HIGH PERFORMANCE COATING FOR STEEL STRUCTURES

DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

23 03 00.00 20	08/10, CHG 3: 08/18	BASIC MECHANICAL MATERIALS AND METHODS
23 05 93	11/15	TESTING, ADJUSTING, AND BALANCING FOR HVAC
23 31 13.00 40	05/16	METAL DUCTS
23 34 23.00 40	02/17	HVAC POWER VENTILATORS
23 81 00	05/18, CHG 1: 02/21	DECENTRALIZED UNITARY HVAC EQUIPMENT

DIVISION 25 - INTEGRATED AUTOMATION

25 05 11	05/21	CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS
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DIVISION 26 - ELECTRICAL

26 08 00	11/22	APPARATUS INSPECTION AND TESTING
26 12 19.10	05/19, CHG 1: 11/19	THREE-PHASE, LIQUID-FILLED PAD-MOUNTED TRANSFORMERS
26 13 00	05/21, CHG 1: 05/22	SF6/HIGH-FIREPOINT FLUIDS INSULATED PAD-MOUNTED SWITCHGEAR
26 20 00	08/23	INTERIOR DISTRIBUTION SYSTEM
26 27 14.00 20	02/21, CHG 1: 05/21	ELECTRICITY METERING
26 32 15.00	05/20	ENGINE-GENERATOR SET STATIONARY 15-2500 KW, WITH AUXILIARIES
26 41 00	08/23	LIGHTNING PROTECTION SYSTEM
26 42 17	05/21	IMPRESSED CURRENT CATHODIC PROTECTION (ICCP) SYSTEM
26 51 00	05/20, CHG 2: 11/21	INTERIOR LIGHTING
26 56 00	08/21	EXTERIOR LIGHTING

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

DIVISION 27 - COMMUNICATIONS

27 10 00	08/11	BUILDING TELECOMMUNICATIONS CABLING SYSTEM
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DIVISION 28 - ELECTRONIC SAFETY AND SECURITY

28 31 70	08/20	INTERIOR FIRE ALARM SYSTEM, ADDRESSABLE
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DIVISION 31 - EARTHWORK

31 00 00	08/23	EARTHWORK
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DIVISION 32 - EXTERIOR IMPROVEMENTS

32 05 33	08/17	LANDSCAPE ESTABLISHMENT
32 11 23	05/22	AGGREGATE BASE COURSE
32 12 13	05/17	BITUMINOUS TACK AND PRIME COATS
32 12 16.16	11/20	ROAD-MIX ASPHALT PAVING
32 15 00	05/17	AGGREGATE SURFACING
32 16 19	05/18	CONCRETE CURBS AND SIDEWALKS
32 17 23	08/16, CHG 5: 11/18	PAVEMENT MARKINGS
32 31 13.53	11/21	HIGH-SECURITY FENCES (CHAIN LINK AND ORNAMENTAL) AND GATES
32 92 19	08/17, CHG 1: 08/21	SEEDING

DIVISION 33 - UTILITIES

33 01 50.55	02/21	CLEANING OF PETROLEUM STORAGE TANKS
33 05 23.13	11/19	UTILITY HORIZONTAL DIRECTIONAL DRILLING
33 08 55.01	08/19	FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT)
33 08 55.02	08/19	FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT)
33 09 52	11/18	FUEL PUMP CONTROL AND ANNUNCIATION SYSTEM (NON-HYDRANT)
33 11 00	02/18, CHG 1: 02/22	WATER UTILITY DISTRIBUTION PIPING
33 40 00	11/21	STORMWATER UTILITIES
33 52 23.15	11/18	POL SERVICE PIPING WELDING
33 52 40	11/18, CHG 2: 11/20	FUEL SYSTEMS PIPING (NON-HYDRANT)
33 52 43.11	08/18	AVIATION FUEL MECHANICAL EQUIPMENT
33 52 43.14	08/18, CHG 1: 02/21	AVIATION FUEL CONTROL VALVES
33 52 43.23	08/18, CHG 1: 02/21	AVIATION FUEL PUMPS
33 52 80	08/22	LIQUID FUELS PIPELINE COATING SYSTEMS
33 56 10	05/19, CHG 1: 11/20	FACTORY-FABRICATED FUEL STORAGE TANKS
33 56 19	11/22	FUEL IMPERMEABLE LINER SYSTEM
33 56 21.17	11/18, CHG 1: 11/20	SINGLE WALL ABOVEGROUND FIXED ROOF STEEL POL STORAGE TANK
33 71 02	08/21	UNDERGROUND ELECTRICAL DISTRIBUTION
33 82 00	04/06	TELECOMMUNICATIONS OUTSIDE PLANT (OSP)

DIVISION 40 - PROCESS INTERCONNECTIONS

40 60 00	05/20	PROCESS CONTROL
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-- End of Project Table of Contents --

SECTION TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

DOCUMENT 00 01 15

LIST OF DRAWINGS

02/11, CHG 1: 08/14

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 CONTRACT DRAWINGS
- 1.3 SUPPLEMENTARY DRAWINGS
 - 1.3.1 Boring Logs
 - 1.3.2 Subsurface Data

-- End of Section Table of Contents --

DOCUMENT 00 01 15

LIST OF DRAWINGS
02/11, CHG 1: 08/14

PART 1 GENERAL

1.1 SUMMARY

This section lists the drawings for the project pursuant to contract clause "DFARS 252.236-7001, Contract Drawings, Maps and Specifications."

1.2 CONTRACT DRAWINGS

NOTE: Tank drawing sheets include a "T" suffix to the Drawing No.
Pipeline drawing sheets include a "P" suffix to the Drawing No.

Contract drawings are as follows:

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DRAWING NO	REVISION NO	NAVFAC Dwg No	TITLE
GENERAL			
G-001	0	14140105	COVER SHEET
G-002	0	14140106	INDEX - SHEET 1
G-003	0	14140107	INDEX - SHEET 2
G-004	0	14140108	INDEX - SHEET 3
G-010T	0	14140109	GENERAL NOTES AND SEQUENCE OF CONSTRUCTION
G-011T	0	14140110	SEQUENCE OF CONSTRUCTION
LIFE SAFETY			
LS001T	0	14140111	BUILDING CODE LIFE SAFETY ANALYSIS - PUMP VAULTS
LS002T	0	14140112	BUILDING CODE/LIFE SAFETY ANALYSIS - INJECTOR BUILDING
LS101T	0	14140113	PUMP VAULT BUILDINGS LIFE SAFETY FLOOR PLANS
LS102T	0	14140114	INJECTOR BUILDING LIFE SAFETY FLOOR PLAN
CIVIL			
C-001T	0	14140115	CIVIL LEGEND AND NOTES
CD100T	0	14140116	DEMOLITION PLAN
C-101T	0	14140117	CIVIL LAYOUT

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

C-102T	0	14140118	GRADING AND DRAINAGE
C-103T	0	14140119	EROSION AND SEDIMENT CONTROL PHASE I
C-104T	0	14140120	EROSION AND SEDIMENT CONTROL PHASE II
C-105T	0	14140121	DRAINAGE PLAN
C-201T	0	14140122	PROFILES
C-501T	0	14140123	CIVIL DETAILS
C-502T	0	14140124	CIVIL DETAILS
C-503T	0	14140125	CIVIL DETAILS
C-504T	0	14140126	CIVIL DETAILS
C-505T	0	14140127	FENCE DETAILS
C-506T	0	14140128	CIVIL NOTES AND DETAILS
STRUCTURAL			
S-001T	0	14140129	STRUCTURAL ABBREVIATIONS, LEGEND AND NOTES
S-002T	0	14140130	STRUCTURAL ABBREVIATIONS, LEGEND AND NOTES
S-003T	0	14140131	STRUCTURAL ABBREVIATIONS, LEGEND AND NOTES
S-101T	0	14140132	STRUCTURAL LAYOUT PLAN
S-401T	0	14140133	BULK TANK FLOOR SLAB REINFORCING PLAN
S-402T	0	14140134	BULK TANK RADIAL GUTTER PLAN
S-403T	0	14140135	BULK TANK COLUMN LAYOUT PLAN
S-404T	0	14140136	BULK TANK ROOF PLAN
S-405T	0	14140137	BULK TANK ROOF SLAB REINFORCING PLAN
S-406T	0	14140138	PUMP VAULT FLOOR PLAN
S-407T	0	14140139	PUMP VAULT ROOF PLAN
S-408T	0	14140140	FOUNDATION PLAN - INJECTOR BUILDING
S-409T	0	14140141	ROOF PLAN - INJECTOR BUILDING
S-410T	0	14140142	ADDITIVE INJECTOR TANKS PAD PLAN
S-411T	0	14140143	TIE-IN VAULT

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

S-412T	0	14140144	TRUCK OFF-LOAD PLAN
S-501T	0	14140145	BULK TANK DETAILS AND SECTIONS
S-502T	0	14140146	BULK TANK DETAILS AND SECTIONS
S-503T	0	14140147	BULK TANK DETAILS AND SECTIONS
S-504T	0	14140148	BULK TANK DETAILS AND SECTIONS
S-505T	0	14140149	BULK TANK DETAILS AND SECTIONS
S-506T	0	14140150	PUMP VAULT REINFORCEMENT ELEVATIONS
S-507T	0	14140151	ADDITIVE INJECTOR BUILDING REINFORCEMENT ELEVATIONS
S-508T	0	14140152	DETAILS AND SECTIONS - INJECTOR BUILDING
S-509T	0	14140153	ADDITIVE INJECTOR TANKS DETAILS AND SECTIONS
S-510T	0	14140154	BERM ACCESS STAIRS DETAILS
S-511T	0	14140155	ROLLING COVER PLAN AND DETAILS
S-512T	0	14140156	ROLLING COVER DETAILS
S-513T	0	14140157	ROLLING COVER PLAN AND DETAILS
S-514T	0	14140158	ROLLING COVER DETAILS
ARCHITECTURAL			
A-001T	0	14140159	ABBREVIATIONS, LEGEND AND NOTES
A-401T	0	14140160	FLOOR PLAN - PUMP VAULTS
A-402T	0	14140161	ROOF PLAN - PUMP VAULTS
A-403T	0	14140162	BUILDING ELEVATIONS - PUMP VAULTS
A-404T	0	14140163	BUILDING SECTIONS - PUMP VAULTS
A-405T	0	14140164	FLOOR PLAN - INJECTOR BUILDING
A-406T	0	14140165	ROOF PLAN - INJECTOR BUILDING
A-407T	0	14140166	REFLECTED CEILING PLAN
A-408T	0	14140167	BUILDING ELEVATIONS - INJECTOR BUILDING
A-409T	0	14140168	BUILDING SECTIONS - INJECTOR BUILDING
A-501T	0	14140169	DETAILS - PUMP VAULTS

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

A-502T	0	14140170	DETAILS - INJECTOR BUILDING
A-503T	0	14140171	DETAILS
A-504T	0	14140172	DETAILS
A-505T	0	14140173	DETAILS
A-601T	0	14140174	SCHEDULES
FIRE PROTECTION			
FA001T	0	14140175	FIRE ALARM LEGEND, NOTES & DETAILS
FA101T	0	14140176	INJECTOR BUILDING FIRE ALARM PLAN
MECHANICAL			
M-001T	0	14140177	MECHANICAL ABBREVIATIONS
M-002T	0	14140178	CUT-AND-COVER FUEL SYSTEM FLOW SCHEMATIC
M-101T	0	14140179	MECHANICAL SITE PLAN
M-201T	0	14140180	FUEL PIPELINE PLAN
M-202T	0	14140181	FUEL PIPELINE PLAN
M-401T	0	14140182	PUMP VAULT PIPING PLAN
M-402T	0	14140183	ADDITIVE INJECTOR BUILDING PIPING PLAN
M-403T	0	14140184	TIE-IN VAULT PIT
M-404T	0	14140185	OFF-LOADING
M-405T	0	14140186	MECHANICAL TANK PLAN
M-501T	0	14140187	DETAILS
M-502T	0	14140188	DETAILS
M-503T	0	14140189	DETAILS
M-504T	0	14140190	SUPPORT DETAILS
M-601T	0	14140191	PIPING & INSTRUMENT DIAGRAM (STORAGE TANKS)
M-602T	0	14140192	PIPING & INSTRUMENT DIAGRAM (ADDITIVE SYSTEM)
M-603T	0	14140193	SCHEDULES
M-604T	0	14140194	SEQUENCE OF OPERATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

M-605T	0	14140195	SEQUENCE OF OPERATION
MECHANICAL HVAC			
MH401T	0	14140196	TANK VAULT HVAC PLAN
MH402T	0	14140197	HVAC PLAN - INJECTOR BUILDING
MH501T	0	14140198	DETAILS
MH502T	0	14140199	DETAILS
MH601T	0	14140200	HVAC SCHEDULES
ELECTRICAL			
E-001T	0	14140201	ELECTRICAL ABBREVIATIONS AND NOTES
E-002T	0	14140202	ELECTRICAL LEGEND
E-101T	0	14140203	ELECTRICAL LAYOUT
E-102T	0	14140204	ELECTRICAL LAYOUT
E-401T	0	14140205	INJECTOR BUILDING - LIGHTNING PROTECTION PLAN
E-402T	0	14140206	INJECTOR BUILDING - POWER AND CONTROL PLAN
E-403T	0	14140207	INJECTOR BUILDING GROUNDING PLAN
E-404T	0	14140208	BULK TANK GROUNDING PLAN
E-405T	0	14140209	PUMP VAULT LIGHTING AND GROUNDING PLANS
E-406T	0	14140210	PUMP VAULT POWER AND CONTROL PLANS
E-407T	0	14140211	ADDITIVE TANK AREA AND LEAK DETECTION PIT ELECTRICAL PLANS
E-408T	0	14140212	FSII OFF-LOADING ELECTRICAL PLANS
E-501T	0	14140213	DETAILS
E-502T	0	14140214	DETAILS
E-503T	0	14140215	DETAILS
E-504T	0	14140216	DETAILS
E-505T	0	14140217	DETAILS
E-601T	0	14140218	ONE-LINE DIAGRAM
E-602T	0	14140219	WIRING DIAGRAMS

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

E-603T	0	14140220	WIRING DIAGRAMS
E-604T	0	14140221	SYSTEM GROUNDING DIAGRAM
E-605T	0	14140222	CABLE, MCC, AND LIGHTING SCHEDULES
E-606T	0	14140223	PANEL SCHEDULES
E-607T	0	14140224	POWER CIRCUIT SCHEDULE
E-608T	0	14140225	CONTROL CIRCUIT SCHEDULE
E-609T	0	14140226	CONTROL AND IS CIRCUIT SCHEDULES
E-701T	0	14140227	INTERIOR LIGHTING DETAILS
E-702T	0	14140228	EXTERIOR LIGHTING DETAILS
E-703T	0	14140229	LIGHTING POLES DETAILS
E-704T	0	14140230	MANHOLE AND HANDHOLE DETAILS
E-705T	0	14140231	HAZARDOUS AREA CLASSIFICATIONS
E-706T	0	14140232	ADDITIVE INJECTOR BUILDING AND DRAIN TANK HAZARDOUS PLAN
E-707T	0	14140233	FSII OFF-LOADING HAZARDOUS PLAN
EC144T	0	14140234	CATHODIC PROTECTION PLAN
ELECTRICAL INSTRUMENTATION			
EI601T	0	14140235	PUMP CONTROL PANEL, SECTION 1 LAYOUT
EI602T	0	14140236	PUMP CONTROL PANEL, SECTION 2 LAYOUT
EI603T	0	14140237	CONTROL SYSTEM BLOCK DIAGRAM
EI604T	0	14140238	TERMINAL BLOCK CONNECTIONS
EI605T	0	14140239	TERMINAL BLOCK CONNECTIONS
CIVIL GENERAL			
C-021P	0	14140240	CIVIL LEGEND - SHEET 1
C-022P	0	14140241	CIVIL LEGEND - SHEET 2
CIVIL DEMOLITION			
CD121P	0	14140242	SITE DEMOLITION KEY PLAN
CD126P	0	14140243	DEMOLITION PLAN

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

CD127P	0	14140244	DEMOLITION PLAN
CD128P	0	14140245	DEMOLITION PLAN
CD129P	0	14140246	DEMOLITION PLAN
CD130P	0	14140247	DEMOLITION PLAN
CD131P	0	14140248	DEMOLITION PLAN
CD132P	0	14140249	DEMOLITION PLAN
CD133P	0	14140250	DEMOLITION PLAN
CD134P	0	14140251	DEMOLITION PLAN
CD135P	0	14140252	DEMOLITION PLAN
CD136P	0	14140253	DEMOLITION PLAN
CD137P	0	14140254	DEMOLITION PLAN
CD138P	0	14140255	DEMOLITION PLAN
CD139P	0	14140256	DEMOLITION PLAN
CD140P	0	14140257	DEMOLITION PLAN
CD141P	0	14140258	DEMOLITION PLAN
CD142P	0	14140259	DEMOLITION PLAN
CD143P	0	14140260	DEMOLITION PLAN
CD144P	0	14140261	DEMOLITION PLAN
CIVIL SITE			
C-121P	0	14140262	OVERALL SITE KEY PLAN
C-126P	0	14140263	SITE PLAN
C-127P	0	14140264	SITE PLAN
C-128P	0	14140265	SITE PLAN
C-129P	0	14140266	SITE PLAN
C-130P	0	14140267	SITE PLAN
C-131P	0	14140268	SITE PLAN
C-132P	0	14140269	SITE PLAN

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

C-133P	0	14140270	SITE PLAN
C-134P	0	14140271	SITE PLAN
C-135P	0	14140272	SITE PLAN
C-136P	0	14140273	SITE PLAN
C-137P	0	14140274	SITE PLAN
C-138P	0	14140275	SITE PLAN
C-139P	0	14140276	SITE PLAN
C-140P	0	14140277	SITE PLAN
C-141P	0	14140278	SITE PLAN
C-142P	0	14140279	SITE PLAN
C-143P	0	14140280	SITE PLAN
C-144P	0	14140281	SITE PLAN
C-427P	0	14140282	ENLARGED SITE PLAN
C-434P	0	14140283	ENLARGED SITE PLAN
C-435P	0	14140284	ENLARGED SITE PLAN
C-437P	0	14140285	ENLARGED SITE PLAN
C-438P	0	14140286	ENLARGED SITE PLAN
C-439P	0	14140287	ENLARGED SITE PLAN
C-440P	0	14140288	ENLARGED SITE PLAN
C-501P	0	14140289	CIVIL DETAILS
C-502P	0	14140290	CIVIL DETAILS
C-503P	0	14140291	CIVIL DETAILS
CIVIL GRADING			
CG-428P	0	14140292	ENLARGED GRADING PLAN
CG-440P	0	14140293	ENLARGED GRADING PLAN
STRUCTURAL			
S-021P	0	14140294	STRUCTURAL NOTES - SHEET 1

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

S-022P	0	14140295	STRUCTURAL NOTES - SHEET 2
S-428P	0	14140296	STRUCTURAL WEST PIT PLAN
S-429P	0	14140297	STRUCTURAL SOUTHEAST PIT PLAN
S-430P	0	14140298	STRUCTURAL NORTHEAST PIT PLAN
S-501P	0	14140299	TYPICAL HDD PIT DETAILS
S-521P	0	14140300	TYPICAL REINFORCING DETAILS
S-522P	0	14140301	TYPICAL FOUNDATION DETAILS
S-524P	0	14140302	TYPICAL LOADING STATION STRUCTURAL DETAILS
S-525P	0	14140303	TYPICAL LOADING STATION STRUCTURAL DETAILS
S-526P	0	14140304	PIG LAUNCH EQUIPMENT PAD PLAN
S-527P	0	14140305	FLOW METER EQUIPMENT PAD PLAN
S-528P	0	14140306	MISCELLANEOUS DETAILS
S-530P	0	14140307	TYPICAL PIPE SUPPORT DETAILS
MECHANICAL GENERAL			
M-021P	0	14140308	MECHANICAL LEGEND
MECHANICAL DEMOLITION			
MD121P	0	14140309	DEMOLITION KEY PLAN
MD122P	0	14140310	MECHANICAL DEMO PLAN
MD123P	0	14140311	MECHANICAL DEMO PLAN
MD124P	0	14140312	MECHANICAL DEMO PLAN
MD125P	0	14140313	MECHANICAL DEMO PLAN
MD126P	0	14140314	MECHANICAL DEMO PLAN
MD127P	0	14140315	MECHANICAL DEMO PLAN
MD128P	0	14140316	MECHANICAL DEMO PLAN
MD129P	0	14140317	MECHANICAL DEMO PLAN
MD130P	0	14140318	MECHANICAL DEMO PLAN
MD141P	0	14140319	DEMOLITION PLAN-WEST SIDE RIO SALADO BRIDGE

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

MD142P	0	14140320	DEMOLITION PLAN-EAST SIDE RIO SALADO BRIDGE
MD421P	0	14140321	DEMOLITION PLAN VP-1 MANHOLE
MD422P	0	14140322	DEMOLITION PLAN BPH-VP-1.5 MANHOLE
MD423P	0	14140323	DEMOLITION PLAN BPH-VP-2 MANHOLE
MD424P	0	14140324	DEMOLITION PLAN BPH-VP-3 & 3.5 MANHOLE
MD425P	0	14140325	DEMOLITION PLAN - SPAGHETTI MANHOLE
MD428P	0	14140326	DEMOLITION PLAN ML-4 MANHOLE
MECHANICAL SITE			
M-121P	0	14140327	OVERALL MECHANICAL SITE & KEY PLAN
M-122P	0	14140328	MECHANICAL SITE PLAN
M-123P	0	14140329	MECHANICAL SITE PLAN
M-124P	0	14140330	MECHANICAL SITE PLAN
M-125P	0	14140331	MECHANICAL SITE PLAN
M-126P	0	14140332	MECHANICAL SITE PLAN
M-127P	0	14140333	MECHANICAL SITE PLAN
M-128P	0	14140334	MECHANICAL SITE PLAN
M-129P	0	14140335	MECHANICAL SITE PLAN
M-130P	0	14140336	MECHANICAL SITE PLAN
M-131P	0	14140337	MECHANICAL SITE PLAN
M-132P	0	14140338	MECHANICAL SITE PLAN
M-133P	0	14140339	MECHANICAL SITE PLAN
M-134P	0	14140340	MECHANICAL SITE PLAN
M-135P	0	14140341	MECHANICAL SITE PLAN
M-136P	0	14140342	MECHANICAL SITE PLAN
M-137P	0	14140343	MECHANICAL SITE PLAN
M-138P	0	14140344	MECHANICAL SITE PLAN
M-139P	0	14140345	MECHANICAL SITE PLAN

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

M-140P	0	14140346	MECHANICAL SITE PLAN
M-141P	0	14140347	MECHANICAL SITE PLAN
M-142P	0	14140348	MECHANICAL SITE PLAN
M-143P	0	14140349	MECHANICAL SITE PLAN
M-421P	0	14140350	SITE PLAN VP-1 MANHOLE
M-422P	0	14140351	SITE PLAN BPH-VP-1.5 MANHOLE
M-423P	0	14140352	SITE PLAN BPH-VP-2 MANHOLE
M-424P	0	14140353	SITE PLAN BPH-VP-3 & 3.5 MANHOLE
M-425P	0	14140354	SITE PLAN - SPAGHETTI MANHOLE
M-428P	0	14140355	WEST HDD PIT
M-429P	0	14140356	SOUTHEAST HDD PIT
M-430P	0	14140357	NORTHEAST HDD PIT
M-439P	0	14140358	SITE PLAN ML-4
M-521P	0	14140359	MECHANICAL DETAILS - SHEET 1
M-522P	0	14140360	MECHANICAL DETAILS - SHEET 2
M-523P	0	14140361	MECHANICAL DETAILS - SHEET 3
M-524P	0	14140362	MECHANICAL DETAILS - SHEET 4
M-525P	0	14140363	TYPICAL PIPING PLAN
M-526P	0	14140364	MECHANICAL SECTION
M-528P	0	14140365	FLOW METER PLAN
M-620P	0	14140366	ALIGNMENT & PIPE SCHEDULES
M-621P	0	14140367	ALIGNMENT & PIPE SCHEDULES
M-622P	0	14140368	ALIGNMENT & PIPE SCHEDULES
M-631P	0	14140369	PROCESS FLOW DIAGRAM - SHEET 1
M-632P	0	14140370	PROCESS FLOW DIAGRAM - SHEET 2
M-633P	0	14140371	PROCESS FLOW DIAGRAM - SHEET 3
M-634P	0	14140372	PROCESS FLOW DIAGRAM - SHEET 4

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

M-635P	0	14140373	PROCESS FLOW DIAGRAM - SHEET 5
M-636P	0	14140374	PROCESS FLOW DIAGRAM - SHEET 6
M-637P	0	14140375	PROCESS FLOW DIAGRAM - SHEET 7
M-638P	0	14140376	PROCESS FLOW DIAGRAM - SHEET 8
ELECTRICAL			
E-021P	0	14140377	ELECTRICAL SYMBOLS & LEGEND SHEET
E-022P	0	14140378	ELECTRICAL SYMBOLS & LEGEND SHEET
E-023P	0	14140379	ELECTRICAL SYMBOLS & LEGEND SHEET
E-024P	0	14140380	ELECTRICAL SYMBOLS & LEGEND SHEET
E-025P	0	14140381	GENERAL NOTES SHEET
E-026P	0	14140382	GENERAL NOTES SHEET
ELECTRIC DEMOLITION			
ED128P	0	14140383	SITE ELECTRICAL DEMOLITION PLAN
ED130P	0	14140384	SITE ELECTRICAL DEMOLITION PLAN
ED134P	0	14140385	SITE ELECTRICAL DEMOLITION PLAN
ED421P	0	14140386	DEMOLITION PLAN PIT VP-1
ED427P	0	14140387	DEMOLITION PLAN - RIO SALADO RIVER CROSSING EAST SIDE
ELECTRIC SITE			
E-121P	0	14140388	ELECTRICAL OVERALL KEY PLAN
E-130P	0	14140389	ELECTRICAL SITE PLAN
E-132P	0	14140390	ELECTRICAL SITE PLAN
E-133P	0	14140391	ELECTRICAL SITE PLAN
E-134P	0	14140392	ELECTRICAL SITE PLAN
E-142P	0	14140393	ELECTRICAL SITE PLAN
E-143P	0	14140394	ELECTRICAL SITE PLAN
E-144P	0	14140395	ELECTRICAL SITE PLAN
E-421P	0	14140396	ELECTRIC VAULT PLAN BPH-VP-1

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

E-427P	0	14140397	RIO SALADO PIPE BRIDGE EAST SIDE
E-428P	0	14140398	METERING STATIONS PLAN
E-429P	0	14140399	SOUTHEAST HDD PIT
E-430P	0	14140400	NORTHEAST HDD PIT
E-501P	0	14140401	ELECTRICAL DETAILS
E-528P	0	14140402	ELECTRICAL METERING STATION PLAN
E-621P	0	14140403	P3-1-VP1 CONTROL PANEL 1-LINE DIAGRAM
E-627P	0	14140404	RIO SALADO EAST 1-LINE DIAGRAM
E-629P	0	14140405	METERING STATION 1-LINE DIAGRAM
ELECTRIC INSTRUMENTATION			
EI121P	0	14140406	FIBER OPTIC OVERALL KEY PLAN
EI122P	0	14140407	METERING STATION FIBER OPTIC CABLE
EI123P	0	14140408	METERING STATION FIBER OPTIC CABLE LAYOUT
EI124P	0	14140409	METERING STATION FIBER OPTIC CABLE
EI125P	0	14140410	BUILDING 603 FIBER OPTIC CABLE
EI133P	0	14140411	FIBER OPTIC SITE PLAN
EI134P	0	14140412	FIBER OPTIC SITE PLAN
EI135P	0	14140413	FIBER OPTIC SITE PLAN
EI136P	0	14140414	FIBER OPTIC SITE PLAN
EI137P	0	14140415	FIBER OPTIC SITE PLAN
EI138P	0	14140416	FIBER OPTIC SITE PLAN
EI143P	0	14140417	FIBER OPTIC SITE PLAN
EI144P	0	14140418	FIBER OPTIC SITE PLAN
EI601P	0	14140419	CONTROL SYSTEM BLOCK DIAGRAM
EI602P	0	14140420	SOUTHEAST PIT CONTROL PANEL
EI603P	0	14140421	SOUTHEAST PIT CONTROL WIRING DIAGRAMS
EI604P	0	14140422	NORTHEAST PIT CONTROL PANEL

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

EI605P	0	14140423	NORTHEAST PIT CONTROL WIRING DIAGRAMS
EI606P	0	14140424	METERING STATION CONTROL PANEL
EI607P	0	14140425	METERING STATION CONTROL WIRING DIAGRAMS
ELECTRIC CATHODIC PROTECTION			
EC121P	0	14140426	CATHODIC PROTECTION KEY PLAN
EC126P	0	14140427	CATHODIC PROTECTION PLAN
EC127P	0	14140428	CATHODIC PROTECTION PLAN
EC128P	0	14140429	CATHODIC PROTECTION PLAN
EC129P	0	14140430	CATHODIC PROTECTION PLAN
EC130P	0	14140431	CATHODIC PROTECTION PLAN
EC131P	0	14140432	CATHODIC PROTECTION PLAN
EC132P	0	14140433	CATHODIC PROTECTION PLAN
EC133P	0	14140434	CATHODIC PROTECTION PLAN
EC134P	0	14140435	CATHODIC PROTECTION PLAN
EC135P	0	14140436	CATHODIC PROTECTION PLAN
EC136P	0	14140437	CATHODIC PROTECTION PLAN
EC137P	0	14140438	CATHODIC PROTECTION PLAN
EC138P	0	14140439	CATHODIC PROTECTION PLAN
EC139P	0	14140440	CATHODIC PROTECTION PLAN
EC140P	0	14140441	CATHODIC PROTECTION PLAN
EC141P	0	14140442	CATHODIC PROTECTION PLAN
EC142P	0	14140443	CATHODIC PROTECTION PLAN
EC143P	0	14140444	CATHODIC PROTECTION PLAN
EC501P	0	14140445	CATHODIC PROTECTION DETAILS
EC502P	0	14140446	CATHODIC PROTECTION DETAILS
REFERENCE DRAWINGS			
P-919 - BULK TANK FARM IMPROVEMENTS - PREFINAL PDFs			

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

SURVEY			
V-001		N/A	NOTES, LEGEND AND SYMBOLS
V-002		N/A	BENCHMARKS
V-003		N/A	BENCHMARKS
V-004		N/A	BENCHMARK SCHEDULES
V-005		N/A	BENCHMARK SCHEDULES
V-006		N/A	BENCHMARK SCHEDULES
V-101		N/A	GENERAL SURVEY - EN -ROUTE SITE
V-102		N/A	GENERAL SURVEY - NEW PIPE
V-401		N/A	ENLARGED SURVEY - AREA I - EN-ROUTE SITE
V-402		N/A	ENLARGED SURVEY - AREA II - EN-ROUTE SITE
V-403		N/A	ENLARGED SURVEY - AREA III - EN- ROUTE SITE
V-404		N/A	ENLARGED SURVEY - AREA IV - EN-ROUTE SITE
V-405		N/A	ENLARGED SURVEY - AREA I - NEW PIPE
V-406		N/A	ENLARGED SURVEY - AREA II - NEW PIPE
V-407		N/A	ENLARGED SURVEY - AREA III - NEW PIPE
V-408		N/A	ENLARGED SURVEY - AREA IV - NEW PIPE
V-409		N/A	ENLARGED SURVEY - AREA V - NEW PIPE
V-410		N/A	ENLARGED SURVEY - AREA VI - NEW PIPE
V-411		N/A	ENLARGED SURVEY - AREA VII - NEW PIPE
V-412		N/A	ENLARGED SURVEY - AREA VIII - NEW PIPE
V-413		N/A	ENLARGED SURVEY - AREA IX - NEW PIPE
V-414		N/A	ENLARGED SURVEY - AREA X - NEW PIPE
V-415		N/A	ENLARGED SURVEY - AREA XI - NEW PIPE
V-416		N/A	ENLARGED SURVEY - AREA XII - NEW PIPE
V-417		N/A	ENLARGED SURVEY - AREA XIII - NEW PIPE
V-418		N/A	ENLARGED SURVEY - AREA XIV - NEW PIPE

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

V-419		N/A	ENLARGED SURVEY - AREA XV - NEW PIPE
V-420		N/A	ENLARGED SURVEY - AREA XVI - NEW PIPE
V-421		N/A	ENLARGED SURVEY - AREA XVII - NEW PIPE
V-422		N/A	ENLARGED SURVEY - AREA XVIII - NEW PIPE
V-423		N/A	ENLARGED SURVEY - AREA XIX - NEW PIPE
V-424		N/A	ENLARGED SURVEY - NEW PIPE
V-425		N/A	ENLARGED SURVEY - DETAIL BPH-VP-1.5 MANHOLE
V-426		N/A	ENLARGED SURVEY - DETAIL BPH-VP-2 MANHOLE
V-427		N/A	ENLARGED SURVEY - DETAIL BPH-VP-3 MANHOLE
V-428		N/A	ENLARGED SURVEY - DETAIL ESPAGUETI MANHOLE
V-429		N/A	ENLARGED SURVEY - PIPING DETAILS SALADO RIVER CROSSING
V-430		N/A	ENLARGED SURVEY - AREA II - NEW PIPE
V-431		N/A	ENLARGED SURVEY - ML-1
V-432		N/A	ENLARGED SURVEY - ML-2
V-433		N/A	ENLARGED SURVEY - ML-3
V-434		N/A	ENLARGED SURVEY - ML-4
GEOTECHNICAL			
B-001		N/A	GENERAL NOTES
BJ101		N/A	BORINGS LOCATION
BJ102		N/A	BORINGS LOCATION - KEY PLAN - NEW PIPE
BJ103		N/A	BORINGS LOCATION - AREA I - NEW PIPE
BJ104		N/A	BORINGS LOCATION - AREA II - NEW PIPE
BJ105		N/A	BORINGS LOCATION - AREA III - NEW PIPE
BJ106		N/A	BORINGS LOCATION - AREA IV - NEW PIPE
BJ107		N/A	BORINGS LOCATION - AREA V - NEW PIPE
BJ108		N/A	BORINGS LOCATION - AREA VI - NEW PIPE
BJ301		N/A	PROFILES

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

BJ302		N/A	PROFILES
BJ601		N/A	BORINGS LOGS - EN ROUTE SITE
BJ602		N/A	BORINGS LOGS - EN ROUTE SITE
BJ603		N/A	BORINGS LOGS - EN ROUTE SITE
BJ604		N/A	BORINGS LOGS - EN ROUTE SITE
BJ605		N/A	BORING LOGS - NEW PIPE SITE
BJ606		N/A	BORING LOGS - NEW PIPE SITE
BJ607		N/A	BORING LOGS - NEW PIPE SITE
BJ608		N/A	BORING LOGS - NEW PIPE SITE
BJ609		N/A	BORING LOGS - NEW PIPE SITE
BJ610		N/A	BORING LOGS - NEW PIPE SITE
BJ611		N/A	BORING LOGS - NEW PIPE SITE
BJ612		N/A	SCHEDULES
BJ613		N/A	SCHEDULES
BJ614		N/A	SCHEDULES
BJ615		N/A	SCHEDULES
BJ001		N/A	GENERAL NOTES
BJ101		N/A	BORINGS LOCATION
BJ301		N/A	SECTIONS
BJ601		N/A	BORING LOGS
BJ602		N/A	BORING LOGS
BJ603		N/A	BORING LOGS
BJ604		N/A	BORING LOGS
BJ605		N/A	BORING LOGS
BJ606		N/A	BORING LOGS
BJ607		N/A	BORING LOGS

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

BJ608		N/A	BORING LOGS
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1.3 SUPPLEMENTARY DRAWINGS

1.3.1 Boring Logs

The Government does not guarantee that borings indicate actual conditions, except for the exact locations and the time that they were made. The data are available for examination by prospective bidders in the office of the Contracting Officer.

1.3.2 Subsurface Data

Subsurface data, not specified or indicated, have been obtained by the Government at the station. The data are available for examination by prospective bidders. The soils report is included as part of the solicitation.

-- End of Document --

SECTION TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

DOCUMENT 00 73 04.00 41

SUPPLEMENTARY CONDITIONS FOR PROJECTS IN MEDITERRANEAN AREA

08/08

PART 1 GENERAL

- 1.1 SUBMITTALS
- 1.2 COOPERATION WITH OTHER CONTRACTORS
- 1.3 REFERENCE TO GENERAL PROVISIONS CLAUSES
- 1.4 PUBLIC RELEASE OF INFORMATION
 - 1.4.1 Prohibition
 - 1.4.2 Subcontracts and Purchase Orders
- 1.5 ENGLISH SPEAKING REPRESENTATIVE
- 1.6 CORRESPONDENCE IN THE ENGLISH LANGUAGE
- 1.7 NORMS AND LAWS
- 1.8 INDUSTRY STANDARDS
- 1.9 PROHIBITED ITEMS
- 1.10 PROPRIETARY NAMES
- 1.11 SUBMITTAL OF PROOF OF QUALIFICATIONS AND EXPERIENCE
- 1.12 ORAL MODIFICATION
- 1.13 NO WAIVER BY GOVERNMENT
- 1.14 PRODUCTS SPECIFIED USING USA STANDARDS
 - 1.14.1 Local Products Specified Using United States of America (USA) Standards
 - 1.14.2 USA Manufactured Products Specified Using USA Standards
 - 1.14.3 Consignment, Marking, and Customs Clearance
 - 1.14.4 Excess
- 1.15 FINAL GOVERNING STANDARDS (FGS)
- 1.16 EXCLUSION OF PRICE REVISION LAWS
- 1.17 OPTIONAL PURCHASE
- 1.18 SECURITY REQUIREMENTS
 - 1.18.1 Contractor Employee Base Access Pass:
 - 1.18.2 Responsibility for Physical Security
 - 1.18.3 Access to Operational Areas
 - 1.18.4 Employment
 - 1.18.5 Vehicle Security Inspections
 - 1.18.6 Vehicle List
 - 1.18.7 Control
 - 1.18.8 Photographs
 - 1.18.9 Compliance

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

DOCUMENT 00 73 04.00 41

SUPPLEMENTARY CONDITIONS FOR PROJECTS IN MEDITERRANEAN AREA
08/08

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor personnel identification; G

1.2 COOPERATION WITH OTHER CONTRACTORS

Other contractors may be engaged in similar and/or supporting work requiring close cooperation. The Contractor shall cooperate and schedule his work to avoid conflict with and interruption of the work of others insofar as practicable. In the case of conflicts with other contractors that cannot be resolved satisfactorily, the matter shall be referred to the Contracting Officer for decision, and such decision shall be final, subject to right of appeal in accordance with terms of the contract.

1.3 REFERENCE TO GENERAL PROVISIONS CLAUSES

Wherever a reference to a clause of the General Provisions occurs in the technical sections of the specifications, it shall be taken to mean reference to a clause of the Contract Clauses having the same title as the referenced General Provisions clause.

1.4 PUBLIC RELEASE OF INFORMATION

1.4.1 Prohibition

There shall be no public release of information or photographs concerning any aspect of the materials or services relating to this bid, contract, or purchase order or other documents resulting therefrom without the prior written approval of the Contracting Officer.

1.4.2 Subcontracts and Purchase Orders

The Contractor agrees to insert the substance of above paragraph "Prohibition" in each subcontract and purchase order generated for this contract.

1.5 ENGLISH SPEAKING REPRESENTATIVE

At all times when any performance of the work at the site is being conducted by any employee of the Contractor or his subcontractors, the Contractor shall have a representative present on the site who is capable of explaining the work operations and receiving instructions in the English language. The Contracting Officer shall have the right to

determine without appeal of such decision, whether the proposed representative has sufficient technical and lingual capabilities and the Contractor shall immediately replace any individual not acceptable to the Contracting Officer. The Construction Contractor's qualified Project Manager, on-site Superintendent, Quality Control Manager, and Quality Control Specialist shall be fluent (can read, speak, and write) in both English and Spanish languages.

1.6 CORRESPONDENCE IN THE ENGLISH LANGUAGE

It shall be the responsibility of the Contractor to prepare all proposals, change orders, shop drawings and submittals, quality control reports, computations, and all correspondence pertaining to this contract, in the English language; but the Contractor may, for his own record purposes, prepare them in the local language. All correspondence to the Contracting Officer shall be in the English language. In case of dispute or claim, the English version will govern.

1.7 NORMS AND LAWS

In case of differences between project specifications or the accompanying drawings and referenced norms and laws, norms and laws shall govern. In case of differences between project specifications and the accompanying drawings, the specifications shall govern.

1.8 INDUSTRY STANDARDS

When both U.S. and European, or Spanish, Industry Standards are applicable to this project and the standards are in conflict with each other, the most stringent of the industry shall govern.

1.9 PROHIBITED ITEMS

Use of the following items in this construction project is prohibited:

- a. Use of aluminum for electrical conductors.
- b. Embedding aluminum conduit in concrete.
- c. Use of fluorescent light ballasts and other products containing PCB's.
- d. Use of urea-formaldehyde foam insulation products.
- e. Use of any paint/coatings having a lead content of over 0.06 percent by weight of non-volatile content. The use of ozone depleting chemicals is prohibited. The use of zinc-chromate is prohibited.
- f. The use of materials containing asbestos is prohibited.

1.10 PROPRIETARY NAMES

Manufacturer's proprietary names indicated for colors, textures and patterns of materials are for the purpose of color, texture and pattern selection only. Other manufacturers materials are acceptable and provided they closely approximate colors, textures and patterns indicated and provided they conform to all other requirements subject to Contracting Officer's approval.

1.11 SUBMITTAL OF PROOF OF QUALIFICATIONS AND EXPERIENCE

Where qualifications or experience requirements are set forth in the specifications with respect to equipment and equipment installers, written proof of such qualifications or experience must be provided within thirty (30) calendar days after Contract award, and before placing any order for such equipment or before dispatching equipment installers to the project site.

1.12 ORAL MODIFICATION

No oral statement of any person other than the Contracting Officer as provided in the clause in this Contract entitled FAR 52.243-4 "Changes" (AUG 1987), shall in any manner or degree modify or otherwise affect the terms of this Contract.

1.13 NO WAIVER BY GOVERNMENT

The failure of the Government, in any one or more instances, to insist upon the strict performance of any of the terms of this Contract or to exercise any option herein conferred shall not be construed as a waiver or relinquishment to any extent of the right to assert or rely upon such terms or option on any future occasion.

1.14 PRODUCTS SPECIFIED USING USA STANDARDS

1.14.1 Local Products Specified Using United States of America (USA) Standards

There are locally available products specified in this contract that meet USA standards. "Locally available products that meet USA standards" are defined as:

- a. "Products that meet the referenced USA standards, and are manufactured and distributed in-country, or manufactured and distributed in the European Union, and are readily available locally from in-country sources or from the European Union sources"; or
- b. "Products manufactured in the USA and distributed in-country or in the European Union or in the USA, and are readily available locally from in-country sources or from the European Union sources, or from USA sources.".

1.14.2 USA Manufactured Products Specified Using USA Standards

There are products to be provided in this contract that are not readily available locally and must be manufactured in the United States of America (USA) and in compliance with USA standards. Products manufactured in another country are considered USA manufactured, if the manufacturers or suppliers are under license by USA manufacturers and the products meet the same applicable specifications and standards.

1.14.3 Consignment, Marking, and Customs Clearance

In order for the Contractor to be exempt from paying duty on material or equipment to be imported for incorporation into the contract work, all shipments shall be consigned to the Contracting Officer and marked for the contract by contract number. Additionally, the Contractor shall supply to the Contracting Officer with certified copies of the confirmed

material/equipment orders and invoices in order that the duty free customs clearance may be accomplished. Failure on the part of the Contractor to comply with the requirements herein will not be grounds for non-enforcement of the provisions of the Clause, "Taxes" of the Contract Clauses.

1.14.4 Excess

The Contractor shall furnish to the Contracting Officer, just prior to completion of this contract, a consolidated inventory of all excess supplies, materials, and equipment imported duty free for use under this contract. The Contract shall either pay required duties on the excesses or re-export the excesses.

1.15 FINAL GOVERNING STANDARDS (FGS)

Final Governing Standards (FGS) for environmental protection at DOD installations and facilities in Spain went into effect June 1994. These standards were developed by comparing and adopting the more protective requirements of the Overseas Environmental Baseline Guidance Document (OEBGD), European Union and national, regional, and local environmental laws and regulations. The FGS are applicable to work under this contract. The FGS (both English and local language versions) may be reviewed at the NAVFAC Rota Office during normal business hours.

1.16 EXCLUSION OF PRICE REVISION LAWS

The Contractor and the United States Government, as the sole parties to this contract, do specifically and irrevocably agree that this contract is a FIRM FIXED-PRICE CONTRACT, and do expressly waive and exclude the application of any and all local Laws, Decrees, Orders, and Regulations, which might mandate, authorize, or permit an increase in the contract price, revision or any other mechanism as a result of increased labor costs, equipment costs, material costs, supplies, subcontractor costs, other overhead costs, direct or indirect, or otherwise.

The risks and consequences of this exclusion of the application of Laws as aforementioned are understood by the parties, and have been taken into consideration in arriving at the contract price set forth herein.

It is the parties' intention that the work contracted for, including any subsequent modifications to this contract, shall be performed on a firm fixed-price basis without price revision of any kind.

Any dispute in this respect arising between the parties will be resolved by strict interpretation of the provision of the contract itself.

1.17 OPTIONAL PURCHASE

All materials and equipment not specifically required to be purchased in the United States of America shall be purchased in Spain and may not be purchased elsewhere without specific written approval of the Contracting Officer. If purchase of material and equipment is not feasible in Spain or it is specified that material and equipment be purchased in the United States, the Clause, "STATUS OF CONTRACTOR AS DIRECT CONTRACTOR" of this Section, will apply.

1.18 SECURITY REQUIREMENTS

1.18.1 Contractor Employee Base Access Pass:

The Contractor shall be responsible for furnishing each employee, and for requiring that each employee engaged in the work display, such identification as may be approved and directed by the Contracting Officer. All prescribed identification shall immediately be delivered to the Contracting Officer for cancellation upon release of any employee.

Contractor personnel identification, including names and photographs, D.N.I. (Documento Nacional de Identidad) numbers, vehicles identification and license plate numbers, of all personnel to be employed in the performance of the work at the site, shall be submitted to the Contracting Officer at least seven (7) days prior to commencement of the work.

Contractor personnel shall be restricted from entering operational buildings or areas without the specific authorization of the Contracting Officer.

1.18.2 Responsibility for Physical Security

The Contractor shall be responsible for physical security of all construction materials, supplies, and equipment of every description (including property which may be Government furnished or owned) provided and/or utilized in the execution of this contract.

1.18.3 Access to Operational Areas

Contractor personnel are expressly prohibited and shall be restricted from entering operational buildings or areas without the specific authorization of the Contracting Officer. To perform work, where "security escorts" are required for access to a facility, the Contractor shall coordinate with the Contracting Officer for access to such facilities.

1.18.4 Employment

Should the continued employment of any person in connection with this contract, or any subcontract hereunder, be deemed by the Contracting Officer to be prejudicial to the interests of the Government, that person shall immediately be removed from the work. In this connection the Contractor agrees that:

a. Cases which may involve disciplinary action against such persons, or the necessity of reassignment or termination of their services, shall be investigated, processed, reported and disposed of, as directed by the Contracting Officer.

b. Employment contracts of all persons employed in connection with this contract, or any subcontract thereunder shall include clauses containing the substance of this clause.

1.18.5 Vehicle Security Inspections

All Contractor and subcontractor vehicles entering the Station including, but not limited to, service vehicles, material delivery vehicles, equipment delivery vehicles, and various other support vehicles, shall first undergo a security inspection.

1.18.6 Vehicle List

Submit an original list of vehicles to be utilized at the work site with the following information for each vehicle:

- a. Make.
- b. Year.
- c. Model.
- d. License number and insurance data's.
- e. Registered owner.

1.18.7 Control

Maintain strict accountability over passes. Immediately report to the source of issue, passes missing or lost and circumstances. If the Contractor has another active contract and one commencing immediately, employee's names may be transferred from one contract to the other. Final payment will not be effected until employees are transferred to another contract or the records are cleared. Furnish a signed letter, countersigned by the source of issue, stating that passes have been turned in.

1.18.8 Photographs

Photographs are prohibited in and around the building unless a written request containing specific justification and details is approved by the security officer via the Contracting Officer. Four days' advance notice is required. Photographs, when approved, shall be taken in the presence of an activity escort.

1.18.9 Compliance

Compliance with the foregoing provisions of this clause by subcontractors shall be the responsibility of the Contractor.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Document --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 11 00

SUMMARY OF WORK

08/15, CHG 2: 08/21

PART 1 GENERAL

- 1.1 SUBMITTALS
- 1.2 WORK COVERED BY CONTRACT DOCUMENTS
 - 1.2.1 Project Description
 - 1.2.2 Location
- 1.3 CONTRACT DRAWINGS
- 1.4 LOCATION OF UNDERGROUND UTILITIES
 - 1.4.1 Notification Prior to Excavation
- 1.5 SALVAGE MATERIAL AND EQUIPMENT

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 11 00

SUMMARY OF WORK
08/15, CHG 2: 08/21

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Salvage Plan; G

1.2 WORK COVERED BY CONTRACT DOCUMENTS

1.2.1 Project Description

The work includes two new "cut-and-cover" 50,000 gallon tanks. Each tank will have a dedicated pump vault, associated pumping equipment and standby generator. Associated Site/Civil work. Remove out-of-service piping and conduct HazMat surveying and testing. Design a new 18 inch diameter fuel pipeline, with associated metering, to transport fuel from Pier 3 to the Bulk Tank Farm. The pipeline will be installed under the Rio Salado River via Horizontal Directional Drilling (HDD) method. Design a cathodic protection system for both the tanks and pipeline. Remove out-of-service piping and conduct HazMat surveying and testing and incidental related work.

1.2.2 Location

The work is located at the NAVSTA Rota, Spain, approximately as indicated. The exact location will be shown by the Contracting Officer.

1.3 CONTRACT DRAWINGS

The following drawings accompany this specification and are a part thereof.

Drawing Nos.:

G-001	COVER SHEET
G-002	INDEX SHEET
G-003	INDEX SHEET
G-004T	GENERAL NOTES AND SEQUENCE OF CONSTRUCTION
G-005T	SEQUENCE OF CONSTRUCTION

LIFE SAFETY

LS001T	BUILDING CODE LIFE SAFETY ANALYSIS - PUMP VAULTS
LS002T	BUILDING CODE LIFE SAFETY ANALYSIS - INJECTOR BUILDING
LS101T	PUMP VAULT BUILDINGS LIFE SAFETY FLOOR PLANS
LS002T	BUILDING CODE LIFE SAFETY ANALYSIS - INJECTOR BUILDING
LS101T	PUMP VAULT BUILDINGS LIFE SAFETY FLOOR PLANS
LS102T	INJECTOR BUILDINGS LIFE SAFETY FLOOR PLAN

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

CIVIL

C-001T	CIVIL LEGEND AND NOTES
CD100T	DEMOLITION PLAN
C-101T	CIVIL LAYOUT
C-102T	GRADING AND DRAINAGE
C-103T	EROSION AND SEDIMENT CONTROL PHASE I
C-104T	EROSION AND SEDIMENT CONTROL PHASE II
C-105T	DRAINAGE PLAN
C-201T	PROFILES
C-501T	CIVIL DETAILS
C-502T	CIVIL DETAILS
C-503T	CIVIL DETAILS
C-504T	CIVIL DETAILS
C-505T	FENCE DETAILS
C-506T	CIVIL NOTES AND DETAILS

STRUCTURAL

S-001T	STRUCTURAL ABBREVIATIONS, LEGEND AND NOTES
S-002T	STRUCTURAL ABBREVIATIONS, LEGEND AND NOTES
S-003T	STRUCTURAL ABBREVIATIONS, LEGEND AND NOTES
S-101T	STRUCTURAL LAYOUT PLAN
S-401T	BULK TANK FLOOR SLAB REINFORCING PLAN
S-402T	BULK TANK RADIAL GUTTER PLAN
S-403T	BULK TANK COLUMN LAYOUT PLAN
S-404T	BULK TANK ROOF PLAN
S-405T	BULK TANK ROOF SLAB REINFORCING PLAN
S-406T	PUMP VAULT FLOOR PLAN
S-407T	PUMP VAULT ROOF PLAN
S-408T	ADDITIVE INJECTOR BUILDING FOUNDATION PLAN - REVIT
S-409T	ADDITIVE INJECTOR BUILDING ROOF PLAN - REVIT
S-410T	ADDITIVE INJECTOR TANKS PAD PLAN
S-411T	TIE-IN VAULT
S-412T	TRUCK OFF-LOAD PLAN
S-501T	BULK TANK DETAILS AND SECTIONS
S-502T	BULK TANK DETAILS AND SECTIONS
S-503T	BULK TANK DETAILS AND SECTIONS
S-504T	BULK TANK DETAILS AND SECTIONS
S-505T	BULK TANK DETAILS AND SECTIONS
S-506T	PUMP VAULT REINFORCEMENT ELEVATIONS
S-507T	ADDITIVE INJECTOR BUILDING REINFORCEMENT ELEVATIONS - REVIT
S-508T	ADDITIVE INJECTOR BUILDING DETAILS AND SECTIONS - REVIT
S-509T	ADDITIVE INJECTOR TANKS DETAILS AND SECTIONS
S-510T	BERM ACCESS STAIRS DETAILS
S-511T	ROLLING COVER PLAN AND DETAILS
S-512T	ROLLING COVER DETAILS
S-513T	ROLLING COVER PLAN AND DETAILS
S-514T	ROLLING COVER DETAILS

ARCHITECTURAL

A-001T	ABBREVIATIONS, LEGEND AND NOTES
A-401T	FLOOR PLAN - PUMP VAULTS
A-402T	ROOF PLAN - PUMP VAULTS
A-403T	BUILDING ELEVATIONS - PUMP VAULTS
A-404T	BUILDING SECTIONS - PUMP VAULTS
A-405T	FLOOR PLAN - INJECTOR BUILDING - REVIT
A-406T	ROOF PLAN - INJECTOR BUILDING - REVIT
A-407T	CEILING PLAN- ADDITIVE INJECTOR BUILDING - REVIT
A-408T	BUILDING ELEVATIONS - INJECTOR BUILDING - REVIT
A-409T	BUILDING SECTIONS - INJECTOR BUILDING - REVIT

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

A-501T	DETAILS - PUMP VAULTS
A-502T	DETAILS - INJECTOR BUILDING - REVIT
A-503T	DETAILS
A-504T	DETAILS
A-505T	DETAILS
A-601T	SCHEDULES

FIRE PROTECTION

FA001T	FIRE ALARM LEGEND, NOTES & DETAILS
FA101T	INJECTOR BUILDING FIRE ALARM PLAN

MECHANICAL

M-001T	MECHANICAL ABBREVIATIONS
M-002T	CUT-AND-COVER FUEL SYSTEM FLOW SCHEMATIC
M-101T	MECHANICAL SITE PLAN
M-201T	FUEL PIPELINE PLAN
M-202T	FUEL PIPELINE PLAN
M-401T	PUMP VAULT PIPING PLAN
M-402T	ADDITIVE INJECTOR BUILDING PIPING PLAN
M-403T	TIE-IN VAULT PIT
M-404T	OFF-LOADING
M-405T	MECHANICAL TANK PLAN
M-501T	DETAILS
M-502T	DETAILS
M-503T	DETAILS
M-504T	SUPPORT DETAILS
M-601T	PIPING & INSTRUMENT DIAGRAM (STORAGE TANKS)
M-602T	PIPING & INSTRUMENT DIAGRAM (ADDITIVE SYSTEM)
M-603T	SCHEDULES
M-604T	SEQUENCE OF OPERATION
M-605T	SEQUENCE OF OPERATION

MECHANICAL HVAC

MH401T	TANK VAULT HVAC PLAN
MH402T	ADDITIVE INJECTOR BUILDING HVAC - REVIT
MH501T	DETAILS
MH502T	DETAILS
MH601T	HVAC SCHEDULES

ELECTRICAL

E-001T	ELECTRICAL ABBREVIATIONS AND NOTES
E-002T	ELECTRICAL LEGEND
E-101T	ELECTRICAL LAYOUT
E-102T	ELECTRICAL LAYOUT
E-401T	INJECTOR BLDG LIGHTING AND LIGHTNING PROTECTION PLAN - REVIT
E-402T	INJECTOR BLDG POWER AND CONTROL PLAN - REVIT
E-403T	INJECTOR BUILDING GROUNDING PLAN
E-404T	BULK TANK GROUNDING PLAN
E-405T	PUMP VAULT LIGHTING AND GROUNDING PLANS
E-406T	PUMP VAULT POWER AND CONTROL PLANS
E-407T	ADDITIVE TANK AREA AND LEAK DETECTION PIT ELECTRICAL PLANS
E-408T	FSII OFF-LOADING ELECTRICAL PLANS
E-501T	DETAILS
E-502T	DETAILS
E-503T	DETAILS
E-504T	DETAILS
E-505T	DETAILS
E-601T	ONE-LINE DIAGRAM

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

E-602T	WIRING DIAGRAMS
E-603T	WIRING DIAGRAMS
E-604T	SYSTEM GROUNDING DIAGRAM
E-605T	CABLE, MCC, AND LIGHTING SCHEDULES
E-606T	PANEL SCHEDULES
E-607T	POWER CIRCUIT SCHEDULE
E-608T	CONTROL CIRCUIT SCHEDULE
E-609T	CONTROL AND IS CIRCUIT SCHEDULES
E-701T	INTERIOR LIGHTING DETAILS
E-702T	EXTERIOR LIGHTING DETAILS
E-703T	LIGHTING POLES DETAILS
E-704T	MANHOLE AND HANDHOLE DETAILS
E-705T	HAZARDOUS AREA CLASSIFICATIONS
E-706T	ADDITIVE INJECTOR BUILDING AND DRAIN TANK HAZARDOUS PLAN
E-707T	FSII OFF-LOADING HAZARDOUS PLAN
EC144T	CATHODIC PROTECTION PLAN

ELECTRICAL INSTRUMENTATION

EI601T	PUMP CONTROL PANEL, SECTION 1 LAYOUT
EI602T	PUMP CONTROL PANEL, SECTION 2 LAYOUT
EI603T	CONTROL SYSTEM BLOCK DIAGRAM
EI604T	TERMINAL BLOCK CONNECTIONS
EI605T	TERMINAL BLOCK CONNECTIONS

CIVIL DEMOLITION

CD121P	SITE DEMOLITION KEY PLAN
CD126P	DEMOLITION PLAN
CD127P	DEMOLITION PLAN
CD128P	DEMOLITION PLAN
CD129P	DEMOLITION PLAN
CD130P	DEMOLITION PLAN
CD131P	DEMOLITION PLAN
CD132P	DEMOLITION PLAN
CD133P	DEMOLITION PLAN
CD134P	DEMOLITION PLAN
CD135P	DEMOLITION PLAN
CD136P	DEMOLITION PLAN
CD137P	DEMOLITION PLAN
CD138P	DEMOLITION PLAN
CD139P	DEMOLITION PLAN
CD140P	DEMOLITION PLAN
CD141P	DEMOLITION PLAN
CD142P	DEMOLITION PLAN
CD143P	DEMOLITION PLAN
CD144P	DEMOLITION PLAN

CIVIL SITE

C-021P	CIVIL LEGEND - SHEET 1
C-022P	CIVIL LEGEND - SHEET 2
C-121P	OVERALL SITE KEY PLAN
C-126P	SITE PLAN
C-127P	SITE PLAN
C-128P	SITE PLAN
C-129P	SITE PLAN
C-130P	SITE PLAN
C-131P	SITE PLAN
C-132P	SITE PLAN
C-133P	SITE PLAN
C-134P	SITE PLAN

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

C-135P	SITE PLAN
C-136P	SITE PLAN
C-137P	SITE PLAN
C-138P	SITE PLAN
C-139P	SITE PLAN
C-140P	SITE PLAN
C-141P	SITE PLAN
C-142P	SITE PLAN
C-143P	SITE PLAN
C-144P	SITE PLAN
C-427P	ENLARGED SITE PLAN
C-434P	ENLARGED SITE PLAN
C-435P	ENLARGED SITE PLAN
C-436P	ENLARGED SITE PLAN
C-437P	ENLARGED SITE PLAN
C-438P	ENLARGED SITE PLAN
C-439P	ENLARGED SITE PLAN
C-440P	ENLARGED SITE PLAN
C-501P	CIVIL DETAILS
C-502P	CIVIL DETAILS
C-503P	CIVIL DETAILS

CIVIL GRADING

CG-428P	ENLARGED GRADING PLAN
CG-440P	ENLARGED GRADING PLAN

STRUCTURAL

S-021P	STRUCTURAL NOTES - SHEET 1
S-022P	STRUCTURAL NOTES - SHEET 2
S-428P	STRUCTURAL WEST PIT PLAN
S-429P	STRUCTURAL SOUTHEAST PIT PLAN
S-430P	STRUCTURAL NORTHEAST PIT PLAN
S-501P	TYPICAL HDD PIPE DETAILS
S-521P	TYPICAL REINFORCING DETAILS
S-522P	TYPICAL FOUNDATION DETAILS
S-523P	TYPICAL SLAB-ON-GRADE DETAILS
S-524P	TYPICAL STRUCTURAL DETAIL
S-525P	TYPICAL LOADING STATION STRUCTURAL DETAILS
S-526P	PIG LAUNCH EQUIPMENT PAD PLAN
S-527P	FLOW METER EQUIPMENT PAD PLAN
S-528P	MISCELLANEOUS DETAILS
S-530P	TYPICAL PIPE SUPPORT DETAILS

MECHANICAL DEMOLITION

MD121P	DEMOLITION KEY PLAN
MD122P	MECHANICAL DEMO PLAN
MD123P	MECHANICAL DEMO PLAN
MD124P	MECHANICAL DEMO PLAN
MD125P	MECHANICAL DEMO PLAN
MD126P	MECHANICAL DEMO PLAN
MD127P	MECHANICAL DEMO PLAN
MD128P	MECHANICAL DEMO PLAN
MD129P	MECHANICAL DEMO PLAN
MD130P	MECHANICAL DEMO PLAN
MD141P	DEMOLITION PLAN-WEST SIDE RIO SALADO BRIDGE
MD142P	DEMOLITION PLAN-EAST SIDE RIO SALADO BRIDGE
MD421P	DEMOLITION PLAN BPH-VP-1 MANHOLE
MD422P	DEMOLITION PLAN BPH-VP-1.5 MANHOLE
MD423P	DEMOLITION PLAN BPH-VP-2 MANHOLE

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

MD424P DEMOLITION PLAN BPH-VP-3 & 3.5 MANHOLE
MD425P DEMOLITION PLAN ML-SPAGHETTI MANHOLE
MD428P DEMOLITION PLAN ML-4 MANHOLE

MECHANICAL SITE

M-021P MECHANICAL LEGEND
M-121P OVERALL MECHANICAL SITE & KEY PLAN
M-122P MECHANICAL SITE PLAN
M-123P MECHANICAL SITE PLAN
M-124P MECHANICAL SITE PLAN
M-125P MECHANICAL SITE PLAN
M-126P MECHANICAL SITE PLAN
M-127P MECHANICAL SITE PLAN
M-128P MECHANICAL SITE PLAN
M-129P MECHANICAL SITE PLAN
M-130P MECHANICAL SITE PLAN
M-131P MECHANICAL SITE PLAN
M-132P MECHANICAL SITE PLAN
M-133P MECHANICAL SITE PLAN
M-134P MECHANICAL SITE PLAN
M-135P MECHANICAL SITE PLAN
M-136P MECHANICAL SITE PLAN
M-137P MECHANICAL SITE PLAN
M-138P MECHANICAL SITE PLAN
M-139P MECHANICAL SITE PLAN
M-140P MECHANICAL SITE PLAN
M-141P MECHANICAL SITE PLAN
M-142P MECHANICAL SITE PLAN
M-143P MECHANICAL SITE PLAN
M-421P SITE PLAN BPH-VP-1 MANHOLE
M-422P SITE PLAN BPH-VP-1.5 MANHOLE
M-423P SITE PLAN BPH-VP-2 MANHOLE
M-424P SITE PLAN BPH-VP- 3 & 3.5 MANHOLE
M-425P SITE PLAN SPAGHETTI MANHOLE
M-428P WEST HDD PIT
M-429P SOUTHEAST HDD PIT
M-430P NORTHEAST HDD PIT
M-439P SITE PLAN ML-4
M-521P MECHANICAL DETAILS - SHEET 1
M-522P MECHANICAL DETAILS - SHEET 2
M523P MECHANICAL DETAILS - SHEET 3
M524P MECHANICAL DETAILS - SHEET 4
M-525P TYPICAL PIPING PLAN
M-526P MECHANICAL SECTION
M-528P FLOW METER PLAN
M-620P ALIGNMENT & PIPE SCHEDULES
M-621P ALIGNMENT & PIPE SCHEDULES
M-622P ALIGNMENT & PIPE SCHEDULES
M-631P PROCESS FLOW DIAGRAM - SHEET 1
M-632P PROCESS FLOW DIAGRAM - SHEET 2
M-633P PROCESS FLOW DIAGRAM - SHEET 3
M-634P PROCESS FLOW DIAGRAM - SHEET 4
M-635P PROCESS FLOW DIAGRAM - SHEET 5
M-636P PROCESS FLOW DIAGRAM - SHEET 6
M-637P PROCESS FLOW DIAGRAM - SHEET 7
M-638P PROCESS FLOW DIAGRAM - SHEET 8

ELECTRICAL

E-021P ELECTRICAL SYMBOLS & LEGEND SHEET

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

E-022P	ELECTRICAL SYMBOLS & LEGEND SHEET
E-023P	ELECTRICAL SYMBOLS & LEGEND SHEET
E-024P	ELECTRICAL SYMBOLS & LEGEND SHEET
E-025P	GENERAL NOTES SHEET
E-026P	GENERAL NOTES SHEET

ELECTRIC DEMOLITION

ED128P	SITE ELECTRICDAL DEMOLITION PLAN
ED130P	SITE ELECTRICDAL DEMOLITION PLAN
ED134P	SITE ELECTRICDAL DEMOLITION PLAN
ED421	DEMOLITION PLAN PIT VP-1
ED427P	DEMOLITION PLAN - RIO SALADO RIVER CROSSING EAST SIDE

ELECTRIC SITE

E-121P	ELECTRICAL OVERALL KEY PLAN
E-130P	ELECTRICAL SITE PLAN
E-132P	ELECTRICAL SITE PLAN
E-133P	ELECTRICAL SITE PLAN
E-134P	ELECTRICAL SITE PLAN
E-142P	ELECTRICAL SITE PLAN
E-143P	ELECTRICAL SITE PLAN
E-144P	ELECTRICAL SITE PLAN
E-421P	ELECTRIC VAULT PLAN BPH-VP-1
E-427P	RIO SALADO PIPE BRIDGE EAST SIDE
E-428P	METERING STATIONS PLAN
E-429P	SOUTHEAST HDD PIT
E-430P	NORTHEAST HDD PIT
E-501P	ELECTRICAL DETAILS
E-528P	ELECTRICAL METERING STATION PLAN
E-621P	P3-1-VP1 CONTROL PANEL 1-LINE DIAGRAM
E-627P	RIO SALADO EAST 1-LINE DIAGRAM
E-629P	METERING STATION 1-LINE DIAGRAM

ELECTRIC INSTRUMENTATION

EI601P	CONTROL SYSTEM BLOCK DIAGRAM
EI602P	SOUTHEAST PIT CONTROL PANEL
EI603P	SOUTHEAST PIT CONTROL PANEL
EI604P	NORTHEAST PIT CONTROL PANEL
EI605P	NORTHEAST PIT CONTROL PANEL
EI606P	METERING STATION CONTROL PANEL
EI607P	METERING STATION CONTROL PANEL

ELECTRIC CATHODIC PROTECTION

EC021P	CATHODIC PROTECTION DETAILS
EC022	CATHODIC PROTECTION DETAILS
EC121P	CATHODIC PROTECTION KEY PLAN
EC126P	CATHODIC PROTECTION PLAN
EC127P	CATHODIC PROTECTION PLAN
EC128P	CATHODIC PROTECTION PLAN
EC129P	CATHODIC PROTECTION PLAN
EC130P	CATHODIC PROTECTION PLAN
EC131P	CATHODIC PROTECTION PLAN
EC132P	CATHODIC PROTECTION PLAN
EC133P	CATHODIC PROTECTION PLAN
EC134P	CATHODIC PROTECTION PLAN
EC135P	CATHODIC PROTECTION PLAN
EC136P	CATHODIC PROTECTION PLAN
EC137P	CATHODIC PROTECTION PLAN
EC138P	CATHODIC PROTECTION PLAN

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

EC139P	CATHODIC PROTECTION PLAN
EC140P	CATHODIC PROTECTION PLAN
EC141P	CATHODIC PROTECTION PLAN
EC142P	CATHODIC PROTECTION PLAN
EC143P	CATHODIC PROTECTION PLAN

REFERENCE DRAWINGS

PREFINAL SUBMITTAL -survey.pdf

V-001	NOTES, LEGEND AND SYMBOLS
V-002	BENCHMARKS
V-101	GENERAL SURVEY - EN -ROUTE SITE
V-401	ENLARGED SURVEY - AREA I
V-402	ENLARGED SURVEY - AREA II
V-403	ENLARGED SURVEY - AREA III
V-404	ENLARGED SURVEY - AREA IV

P-919 - BULK TANK FARM IMPROVEMENTS - SURVEY DRAWINGS - PREFINAL .pdf

V-001	NOTES, LEGEND AND SYMBOLS
V-002	BENCHMARKS
V-003	BENCHMARKS
V-101	GENERAL SURVEY - EN -ROUTE SITE
V-102	GENERAL SURVEY - NEW PIPE
V-401	ENLARGED SURVEY - AREA I - EN-ROUTE SITE
V-402	ENLARGED SURVEY - AREA II - EN-ROUTE SITE
V-403	ENLARGED SURVEY - AREA III - EN-ROUTE SITE
V-404	ENLARGED SURVEY - AREA IV - EN-ROUTE SITE
V-405	ENLARGED SURVEY - AREA I - NEW PIPE
V-406	ENLARGED SURVEY - AREA II - NEW PIPE
V-407	ENLARGED SURVEY - AREA III - NEW PIPE
V-408	ENLARGED SURVEY - AREA IV - NEW PIPE
V-409	ENLARGED SURVEY - AREA V - NEW PIPE
V-410	ENLARGED SURVEY - AREA VI - NEW PIPE
V-411	ENLARGED SURVEY - AREA VII - NEW PIPE
V-412	ENLARGED SURVEY - AREA VIII - NEW PIPE
V-413	ENLARGED SURVEY - AREA IX - NEW PIPE
V-414	ENLARGED SURVEY - AREA X - NEW PIPE
V-415	ENLARGED SURVEY - AREA XI - NEW PIPE
V-416	ENLARGED SURVEY - AREA XII - NEW PIPE
V-417	ENLARGED SURVEY - AREA XIII - NEW PIPE
V-418	ENLARGED SURVEY - AREA XIV - NEW PIPE
V-419	ENLARGED SURVEY - AREA XV - NEW PIPE
V-420	ENLARGED SURVEY - AREA XVI - NEW PIPE
V-421	ENLARGED SURVEY - AREA XVII - NEW PIPE
V-422	ENLARGED SURVEY - AREA XVII - NEW PIPE
V-423	ENLARGED SURVEY - AREA XIX - NEW PIPE
V-424	ENLARGED SURVEY - NEW PIPE
V-425	ENLARGED SURVEY - DETAIL BPH-VP-1.5 MANHOLE
V-426	ENLARGED SURVEY - DETAIL BPH-VP-2 MANHOLE
V-427	ENLARGED SURVEY - DETAIL BPH-VP-3 MANHOLE
V-428	ENLARGED SURVEY - DETAIL ESPAGUETI MANHOLE
V-429	ENLARGED SURVEY - PIPING DETAILS SALADO RIVER CROSSINGS
V-430	ENLARGED SURVEY - AREA II - NEW PIPE
V-431	ENLARGED SURVEY - ML-1
V-432	ENLARGED SURVEY - ML-2
V-433	ENLARGED SURVEY - ML-3
V-434	ENLARGED SURVEY - ML-4

PREFINAL SUBMITTAL -survey.pdf

B-001 GENERAL NOTES

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

BJ101	BORINGS LOCATION
BJ102	BORINGS LOCATION - KEY PLAN - NEW PIPE
BJ103	BORINGS LOCATION - AREA I - NEW PIPE
BJ104	BORINGS LOCATION - AREA II - NEW PIPE
BJ105	BORINGS LOCATION - AREA III - NEW PIPE
BJ106	BORINGS LOCATION - AREA IV - NEW PIPE
BJ107	BORINGS LOCATION - AREA V - NEW PIPE
BJ108	BORINGS LOCATION - AREA VI - NEW PIPE
BJ301	PROFILES
BJ302	PROFILES
BJ601	BORINGS LOGOS - EN ROUTE SITE
BJ602	BORINGS LOGOS - EN ROUTE SITE
BJ603	BORINGS LOGOS - EN ROUTE SITE
BJ604	BORINGS LOGOS - EN ROUTE SITE
BJ605	BORING LOGOS - NEW PIPE SITE
BJ606	BORING LOGOS - NEW PIPE SITE
BJ607	BORING LOGOS - NEW PIPE SITE
BJ608	BORING LOGOS - NEW PIPE SITE
BJ609	BORING LOGOS - NEW PIPE SITE
BJ610	BORING LOGOS - NEW PIPE SITE
BJ611	BORING LOGOS - NEW PIPE SITE
BJ612	SCHEDULES
BJ613	SCHEDULES
BJ614	SCHEDULES
BJ615	SCHEDULES
BJ001	GENERAL NOTES
BJ101	BORINGS LOCATION
BJ301	SECTIONS
BJ601	BORING LOGS
BJ602	BORING LOGS
BJ603	BORING LOGS
BJ604	BORING LOGS
BJ605	BORING LOGS
BJ606	BORING LOGS
BJ607	BORING LOGS
BJ608	BORING LOGS

Sheets 1 through 415

Five sets of full size contract drawings, maps, and specifications will be furnished to the Contractor without charge. Reference publications will not be furnished. Immediately check furnished drawings and notify the Government of any discrepancies.

1.4 LOCATION OF UNDERGROUND UTILITIES

Obtain digging permits prior to start of excavation, and comply with Installation requirements for locating and marking underground utilities. Contact local utility locating service a minimum of 48 hours prior to excavating, to mark utilities, and within sufficient time required if work occurs on a Monday or after a Holiday. Verify existing utility locations indicated on contract drawings, within area of work.

Identify and mark all other utilities not managed and located by the local utility companies. Scan the construction site with Ground Penetrating Radar (GPR), electromagnetic, or sonic equipment, and mark the surface of the ground or paved surface where existing underground utilities or utilities encased in pier structures are discovered. Verify the elevations of existing piping, utilities, and any type of underground or

encased obstruction not indicated, or specified to be removed, that is indicated or discovered during scanning, in locations to be traversed by piping, ducts, and other work to be conducted or installed. Verify elevations before installing new work closer than nearest manhole or other structure at which an adjustment in grade can be made.

1.4.1 Notification Prior to Excavation

Notify the Contracting Officer at least 48 hours prior to starting excavation work.

1.5 SALVAGE MATERIAL AND EQUIPMENT

Items designated by the Contracting Officer to be salvaged remain the property of the Government. Segregate, itemize, deliver and off-load the salvaged property at the Government designated storage area located within 5.0 kilometers of the construction site.

Provide a salvage plan, listing material and equipment to be salvaged, and their storage location. Maintain property control records for material or equipment designated as salvage. Provide a system for property control in the salvage plan. Store and protect salvaged materials and equipment until disposition by the Contracting Officer.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 14 00

WORK RESTRICTIONS

11/22, CHG 1: 02/23

PART 1 GENERAL

- 1.1 SUBMITTALS
- 1.2 SPECIAL SCHEDULING REQUIREMENTS
- 1.3 CONTRACTOR ACCESS AND USE OF PREMISES
 - 1.3.1 Activity Regulations
 - 1.3.1.1 Subcontractors and Personnel Contacts
 - 1.3.1.2 Employee List
 - 1.3.1.3 Personnel Entry Approval
 - 1.3.1.4 No Smoking Policy
 - 1.3.2 Emergency Response Requirement
 - 1.3.3 Working Hours
 - 1.3.4 Work Outside Regular Hours
 - 1.3.5 Exclusionary Period
 - 1.3.6 Occupied and Existing Building
 - 1.3.7 Utility Cutovers and Interruptions
- 1.4 SECURITY REQUIREMENTS
 - 1.4.1 Personnel List
- 1.5 NAVFAC EURAFCENT REQUIREMENTS AND RESTRICTIONS
 - 1.5.1 Building Entrance Restrictions
 - 1.5.2 Public Release of Information
 - 1.5.3 Responsibility for Physical Security
 - 1.5.4 Employment
 - 1.5.5 Compliance
 - 1.5.6 Unexploded Ordnance

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 14 00

WORK RESTRICTIONS
11/22, CHG 1: 02/23

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

List of Contact Personnel

Personnel List

1.2 SPECIAL SCHEDULING REQUIREMENTS

- a. The surrounding defense and airfield facilities will remain in operation during the entire construction period. The Contractor must conduct his operations so as to cause the least possible interference with normal operations of the activity.
- b. Permission to interrupt any Activity roads, railroads, or utility service must be requested in writing a minimum of 15 calendar days prior to the desired date of interruption.
- c. The work under this contract requires special attention to the scheduling and conduct of the work in connection with existing operations. Identify on the construction schedule each factor which constitutes a potential interruption to operations.

The following conditions apply:

- (1) Maintain existing sub-grade piping and surface valves in working condition at all times.
- (2) Work is taking place at an active Naval Station and requires close cooperation with Naval Station ROTA authority. The Contracting Officer will coordinate activities between the Contractor and Naval Station ROTA.

1.3 CONTRACTOR ACCESS AND USE OF PREMISES

1.3.1 Activity Regulations

Ensure that Contractor personnel employed on the Activity become familiar with and obey Activity regulations including safety, fire, traffic and security regulations. Wear appropriate personal protective equipment (PPE) in designated areas. Do not enter any restricted areas unless required to do so and until cleared for such entry. Ensure all Contractor equipment, include delivery vehicles, are clearly identified with their company name.

1.3.1.1 Subcontractors and Personnel Contacts

Provide a [list of contact personnel](#) of the Contractor and subcontractors including addresses and telephone numbers for use in the event of an emergency. As changes occur and additional information becomes available, correct and change the information contained in previous lists.

1.3.1.2 Employee List

The Contractor must provide to the Contracting officer, in writing, the names of two designated representatives authorized to request personnel and vehicle passes for employees and subcontractor's employees prior to commencement of work under this contract.

1.3.1.3 Personnel Entry Approval

Failure to obtain entry approval will not affect the contract price or time of completion.

1.3.1.4 No Smoking Policy

Smoking is prohibited within and outside of all buildings on installation, except in designated smoking areas. This applies to existing buildings, buildings under construction and buildings under renovation. Discarding tobacco materials other than into designated tobacco receptacles is considered littering and is subject to fines. The Contracting Officer will identify designated smoking areas.

1.3.2 Emergency Response Requirement

All emergencies shall be immediately reported at the Emergency Control Center.

1.3.3 Working Hours

Regular working hours will consist of an 9 hour period established by the Contracting Officer, excluding Government holidays.

1.3.4 Work Outside Regular Hours

Work outside regular working hours requires Contracting Officer approval. Make application 15 calendar days prior to such work to allow arrangements to be made by the Government for inspecting the work in progress, giving the specific dates, hours, location, type of work to be performed, contract number and project title. Based on the justification provided, the Contracting Officer may approve work outside regular hours. During periods of darkness, the different parts of the work must be lighted in a manner approved by the Contracting Officer. Make utility cutovers after normal working hours or on Saturdays, Sundays, and Government holidays unless directed otherwise.

1.3.5 Exclusionary Period

No work may be performed during the period 8:00 p.m. to 5:00 a.m., inclusive, without prior written approval of the Contracting Officer. This period has not been considered in computing the time allowed for the performance of this contract.

1.3.6 Occupied and Existing Building

The Contractor shall be working around existing buildings which are occupied. Do not enter the buildings without prior approval of the Contracting Officer.

The existing buildings and their contents must be kept secure at all times. Provide temporary closures as required to maintain security as directed by the Contracting Officer.

Provide dust covers or protective enclosures to protect existing work that remains, during the construction period.

The Government will remove other Government property in the areas of the buildings scheduled to receive work.

1.3.7 Utility Cutovers and Interruptions

- a. Make utility cutovers and interruptions after normal working hours or on Saturdays, Sundays, and Government holidays. Conform to procedures required in paragraph WORK OUTSIDE REGULAR HOURS.
- b. Ensure that new utility lines are complete, except for the connection, before interrupting existing service.
- c. Interruption to water, sanitary sewer, storm sewer, telephone service, and electric service, are considered utility cutovers pursuant to the paragraph WORK OUTSIDE REGULAR HOURS. Interruptions are to be coordinated with the Contracting Officer.
- d. Operation of Station Utilities: The Contractor must not operate nor disturb the setting of control devices in the station utilities system, including water, sewer, electrical, and steam services. The Government will operate the control devices as required for normal conduct of the work. The Contractor must notify the Contracting Officer giving reasonable advance notice when such operation is required.

1.4 SECURITY REQUIREMENTS

1.4.1 Personnel List

Submit for approval, at least 15 days prior to the desired date of entry, an original alphabetical list of personnel who require entry into Government property to perform work on the project. Furnish for each person:

- a. Name
- b. Date and place of birth
- c. Citizenship
- d. Home address
- e. Passport number, place of issue, and expiration date

1.5 NAVFAC EURAFCENT REQUIREMENTS AND RESTRICTIONS

1.5.1 Building Entrance Restrictions

Contractor personnel are restricted from entering operational buildings or areas without the specific authorization of the Contracting Officer. To perform work, where "security escorts" are required for access to a facility, coordinate with the Contracting Officer for access to such facilities.

1.5.2 Public Release of Information

There must be no public release of information or photographs concerning any aspect of the materials or services relating to this contract or other documents resulting therefrom without the prior written approval of the Contracting Officer.

The Contractor agrees to insert the substance of above paragraph in each subcontract and purchase order generated for this contract.

1.5.3 Responsibility for Physical Security

The Contractor is responsible for physical security of construction materials, supplies, and equipment of every description (including property which may be Government furnished or owned) provided or utilized in the execution of this contract.

1.5.4 Employment

Should the continued employment of any person in connection with this contract, or any subcontract thereunder, be deemed by the Contracting Officer to be prejudicial to the interests of the Government, that person must immediately be removed from the work. In this connection the Contractor agrees that:

- a. Cases which may involve disciplinary action against such persons, or the necessity of reassignment or termination of their services, shall be investigated, processed, reported and disposed of, as directed by the Contracting Officer.
- b. Employment contracts of all persons employed in connection with this contract, or any subcontract thereunder shall include clauses containing the substance of this clause.

1.5.5 Compliance

Compliance with the foregoing provisions of this clause by subcontractors is the responsibility of the Contractor.

1.5.6 Unexploded Ordnance

The Contractor is advised that unexploded ordnance may exist in areas where excavation work is required. During the entire life of this Contract, perform both visual and electronic inspection of the work areas to prevent danger to personnel and loss of equipment from inadvertently exploding old ordnance.

During excavation work, use caution when unknown objects are encountered and thoroughly identify the object prior to removal. Upon discovery of

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

suspected unexploded ordnance in the course of contract work, immediately clear the area and contact the Contracting Officer. Resultant exploratory and removal work may be subject to negotiations for a contract change in accordance with the Contract Clauses.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 20 00

PRICE AND PAYMENT PROCEDURES

11/20, CHG 3: 02/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 SCHEDULE OF PRICES
 - 1.3.1 Data Required
 - 1.3.2 Payment Schedule Instructions
 - 1.3.3 Schedule Requirements for HVAC TAB
- 1.4 CONTRACT MODIFICATIONS
- 1.5 CONTRACTOR'S INVOICE AND CONTRACT PERFORMANCE STATEMENT
 - 1.5.1 Content of Invoice
 - 1.5.2 Submission of Invoices
 - 1.5.3 Final Invoice
- 1.6 PAYMENTS TO THE CONTRACTOR
 - 1.6.1 Obligation of Government Payments
 - 1.6.2 Payment for Onsite and Offsite Materials

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 20 00

PRICE AND PAYMENT PROCEDURES
11/20, CHG 3: 02/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EP 1110-1-8

(2021) Engineering and Design --
Construction Equipment Ownership and
Operating Expense Schedule

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Schedule of Prices; G

1.3 SCHEDULE OF PRICES

1.3.1 Data Required

Within 15 calendar days of Contract Award, prepare and deliver to the Contracting Officer a Schedule of Prices as directed by the Contracting Officer. Schedule of Prices must have cost summarized and totals provided for each construction category. Provide a detailed breakdown of the Contract price, giving quantities for each of the various kinds of work, unit prices and extended prices. Contractor overhead and profit including salaries for field office personnel, if applicable, must be proportionately spread over all pay items and not included as individual pay items.

1.3.2 Payment Schedule Instructions

Payments will not be made until the Schedule of Prices has been submitted to and accepted by the Contracting Officer.

1.3.3 Schedule Requirements for HVAC TAB

The field work requirements in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC must be broken down in the Schedule of Prices and in the Construction Progress Documentation by separate line items which reflect measurable deliverables. The value for each pay item listed below will be established on a case by case basis for each Contract. The line items are as follows:

- a. Approval of Design Review Report: The TABS Agency is required to conduct a review of the project plans and specifications to identify any feature, or the lack thereof, that would preclude successful testing and balancing of the project HVAC systems. Submit the resulting findings to the Government to allow correction of the design. The progress payment will not be issued until the report is reviewed and approved.
- b. Approval of the pre-field engineering report: The TABS Agency submits a report which outlines the scope of field work. The report must contain details of what systems will be tested, procedures to be used, sample report forms for reporting test results and a quality control checklist of work items that must be completed before TABS field work commences.
- c. Season I field work: Incremental payments are issued as the TABS field work progresses. The TABS Agency mobilizes to the project site and executes the field work as outlined in the pre-field engineering report. The HVAC water and air systems are balanced and operational data must be collected for one seasonal condition (either summer or winter depending on project timing).
- d. Approval of Season I report: On completion of the Season I field work, the data is compiled into a report and submitted to the Government. The report is reviewed, and approved, after ensuring compliance with the pre-field engineering report scope of work.
- e. Completion of Season I field QA check: Contract QC and Government representatives meet the TABS Agency at the jobsite to retest portions of the systems reported in the Season I report. The purpose of these tests are to validate the accuracy and completeness of the previously submitted Season I report.
- f. Approval of Season II report: The TABS Agency completes all Season II field work, which is normally comprised mainly of taking heat transfer temperature readings, in the season opposite of that under which Season I performance data was compiled. Compile this data into a report and submit to the Government. On completion of submittal review to ensure compliance with the pre-field engineering report scope, progress payment is issued. Progress payment is less than that issued for the Season I report since most of the water and air balancing work effort is completed under Season I.

1.4 CONTRACT MODIFICATIONS

In conjunction with the Contract Clause DFARS 252.236-7000 Modification Proposals-Price Breakdown, and where actual ownership and operating costs of construction equipment cannot be determined from Contractor accounting records, base equipment use rates upon the applicable provisions of the EP 1110-1-8.

1.5 CONTRACTOR'S INVOICE AND CONTRACT PERFORMANCE STATEMENT

1.5.1 Content of Invoice

Requests for payment will be processed in accordance with the Contract Clause FAR 52.232-27 Prompt Payment for Construction Contracts and FAR 52.232-5 Payments Under Fixed-Price Construction Contracts. Invoices not

completed in accordance with contract requirements will be returned to the Contractor for correction of the deficiencies. The requests for payment shall include the documents listed below.

- a. The Contractor's invoice, on NAVFAC Form 7300/30 furnished by the Government, showing in summary form, the basis for arriving at the amount of the invoice. Form 7300/30 must include certification by Quality Control (QC) Manager as required by the Contract.
- b. The Estimate for Voucher/ Contract Performance Statement on NAVFAC Form 4330/54 furnished by the Government. Use NAVFAC Form 4330, unless otherwise directed by the Contracting Officer, on NAVFAC Contracts when a Monthly Estimate for Voucher is required.
- c. Contractor's Monthly Estimate for Voucher and Contractors Certification (NAVFAC Form 4330) with Subcontractor and supplier payment certification. Other documents, including but not limited to, that need to be received prior to processing payment include the following submittals as required. These items are still required monthly even when a pay voucher is not submitted.
- d. Monthly Work-hour report.
- e. Updated Construction Progress Schedule and tabular reports required by the contract.
- f. Contractor Safety Self Evaluation Checklist.
- g. Updated submittal register.
- h. Solid Waste Disposal Report.
- i. Certified payrolls.
- j. Updated testing logs.
- k. Other supporting documents as requested.

1.5.2 Submission of Invoices

If DFARS Clause 252.232-7006 Wide Area WorkFlow Payment Instructions is included in the Contract, provide the documents listed in above paragraph CONTENT OF INVOICE in their entirety as attachments in Wide Area Work Flow (WAWF) for each invoice submitted. The maximum size of each WAWF attachment is two megabytes, but there are no limits on the number of attachments. If a document cannot be attached in WAWF due to system or size restriction, provide it as instructed by the Contracting Officer.

Monthly invoices and supporting forms for work performed through the anniversary award date of the Contract must be submitted to the Contracting Officer within 5 calendar days of the date of invoice. For example, if Contract award date is the 7th of the month, the date of each monthly invoice must be the 7th and the invoice must be submitted by the 12th of the month.

1.5.3 Final Invoice

- a. A final invoice must be accompanied by the certification required by DFARS 252.247.7023 Transportation of Supplies by Sea, and the

Contractor's Final Release. If the Contractor is incorporated, the Final Release must contain the corporate seal. An officer of the corporation must sign and the corporate secretary must certify the Final Release.

- b. For final invoices being submitted via WAWF, the original Contractor's Final Release Form and required certification of Transportation of Supplies by Sea must be provided directly to the respective Contracting Officer prior to submission of the final invoice. Once receipt of the original Final Release Form and required certification of Transportation of Supplies by Sea has been confirmed by the Contracting Officer, the Contractor must then submit final invoice and attach a copy of the Final Release Form and required certification of Transportation of Supplies by Sea in WAWF.
- c. Final invoices not accompanied by the Contractor's Final Release and required certification of Transportation of Supplies by Sea will be considered incomplete and will be returned to the Contractor.

1.6 PAYMENTS TO THE CONTRACTOR

Payments will be made on submission of itemized requests by the Contractor which comply with the requirements of this section, and will be subject to reduction for overpayments or increase for underpayments made on previous payments to the Contractor.

1.6.1 Obligation of Government Payments

The obligation of the Government to make payments required under the provisions of this Contract will, at the discretion of the Contracting Officer, be subject to reductions and suspensions permitted under the FAR and agency regulations including the following in accordance with FAR 32.103 Progress Payments Under Construction Contracts:

- a. Reasonable deductions due to defects in material or workmanship;
- b. Claims which the Government may have against the Contractor under or in connection with this Contract;
- c. Unless otherwise adjusted, repayment to the Government upon demand for overpayments made to the Contractor; and
- d. Failure to maintain accurate "as-built" or record drawings in accordance with FAR 52.236.21.

1.6.2 Payment for Onsite and Offsite Materials

Progress payments may be made to the Contractor for materials delivered on the site, for materials stored off construction sites, or materials that are in transit to the construction sites under the following conditions:

- a. FAR 52.232-5(b) Payments Under Fixed Price Construction Contracts.
- b. Materials delivered on the site but not installed, including completed preparatory work, and off-site materials to be considered for progress payment must be major high cost, long lead, special order, or specialty items, not susceptible to deterioration or physical damage in storage or in transit to the construction site. Examples of materials acceptable for payment consideration include, but are not

limited to, structural steel, non-magnetic steel, non-magnetic aggregate, equipment, machinery, large pipe and fittings, precast/prestressed concrete products, plastic lumber (e.g., fender piles/curbs), and high-voltage electrical cable. Materials not acceptable for payment include consumable materials such as nails, fasteners, conduits, gypsum board, glass, insulation, and wall coverings.

- c. Materials to be considered for progress payment prior to installation must be specifically and separately identified in the Contractor's estimates of work submitted for the Contracting Officer's approval in accordance with Schedule of Prices requirement of this Contract. Requests for progress payment consideration for such items must be supported by documents establishing their value and that the title requirements of the clause at FAR 52.232-5 Payments Under Fixed-Price Construction Contracts have been met.
- d. Materials are adequately insured and protected from theft and exposure.
- e. Provide a written consent from the surety company with each payment request for offsite materials.
- f. Materials to be considered for progress payments prior to installation must be stored either in Hawaii, Guam, Puerto Rico, or the Continental United States. Other locations are subject to written approval by the Contracting Officer.
- g. Materials in transit to the job site or storage site are not acceptable for payment.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS

11/20, CHG 3: 08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 VIEW LOCATION MAP
- 1.4 PROGRESS AND COMPLETION PICTURES
- 1.5 MINIMUM INSURANCE REQUIREMENTS
- 1.6 SUPERVISION
 - 1.6.1 Superintendent Qualifications
 - 1.6.2 Minimum Communication Requirements
 - 1.6.3 Duties
 - 1.6.4 Non-Compliance Actions
- 1.7 PRECONSTRUCTION MEETING
 - 1.7.1 Attendees
- 1.8 FACILITY TURNOVER PLANNING MEETINGS (Red Zone Meetings)
 - 1.8.1 Red Zone Checklist
 - 1.8.2 Meetings
- 1.9 EXPORT LICENSES FOR OVERSEAS PROJECTS
- 1.10 WAIVER FOR WORKER'S COMPENSATION
- 1.11 PARTNERING
 - 1.11.1 Facilitated (Formal) Partnering
- 1.12 MOBILIZATION

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS
11/20, CHG 3: 08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety -- Safety and Health
Requirements Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

15 CFR 772 Definition of Terms

15 CFR 773 Special Licensing Procedures

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

View Location Map

Progress and Completion Pictures

1.3 VIEW LOCATION MAP

Submit, prior to or with the first digital photograph submittals, a sketch or drawing indicating the required photographic locations. Update as required if the locations are moved.

1.4 PROGRESS AND COMPLETION PICTURES

Photographically document site conditions prior to start of construction operations. Provide monthly, and within one month of the completion of work, digital photographs, 1600x1200x24 bit true color minimum resolution in JPEG file format showing the sequence and progress of work. Take a minimum of 20 digital photographs each week throughout the entire project from a minimum of ten different viewpoints selected by the Contractor unless otherwise directed by the Contracting Officer. Submit with the monthly invoice two sets of digital photographs, each set on a separate compact disc (CD) or data versatile disc (DVD), cumulative of all photos to date. Indicate photographs demonstrating environmental procedures. Provide photographs for each month in a separate monthly directory and name each file to indicate its location on the view location sketch. Also

provide the view location sketch on the CD or DVD as a digital file. Include a date designator in file names. Photographs provided are for unrestricted use by the Government.

1.5 MINIMUM INSURANCE REQUIREMENTS

Provide the minimum insurance coverage required by FAR 28.307-2 Liability, during the entire period of performance under this contract. Provide other insurance coverage as required by Spanish law.

1.6 SUPERVISION

1.6.1 Superintendent Qualifications

Provide project superintendent with a minimum of 10 years experience in construction with at least 5 of those years as a superintendent on projects similar in size and complexity. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the alternate superintendent are the same as for the project superintendent. The Contracting Officer may request proof of the superintendent's qualifications at any point in the project if the performance of the superintendent is in question.

For projects where the superintendent is permitted to also serve as the Quality Control (QC) Manager as established in Section 01 45 00 QUALITY CONTROL, the superintendent must have qualifications in accordance with that section.

1.6.2 Minimum Communication Requirements

Have at least one qualified superintendent, or competent alternate, capable of reading, writing, and conversing fluently in the English language, on the job-site at all times during the performance of Contract work. In addition, if a Quality Control (QC) representative is required on the Contract, then that individual must also have fluent English communication skills.

1.6.3 Duties

The project superintendent is primarily responsible for managing subcontractors and coordinating day-to-day production and schedule adherence on the project. The superintendent is required to attend Red Zone meetings, partnering meetings, and quality control meetings. The superintendent or qualified alternative must be on-site at all times during the performance of this contract until the work is completed and accepted.

1.6.4 Non-Compliance Actions

The Project Superintendent is subject to removal by the Contracting Officer for non-compliance with requirements specified in the contract and for failure to manage the project to ensure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

1.7 PRECONSTRUCTION MEETING

Immediately after award, prior to commencing any work at the site, coordinate with the Contracting Officer a time and place to meet for the Preconstruction Meeting. The meeting must take place within 14 calendar days after award of the contract, but prior to commencement of any work at the site. The purpose of this meeting conference is to discuss and develop a mutual understanding of the administrative requirements of the Contract including but not limited to: daily reporting, invoicing, value engineering, safety, base-access, outage requests, hot work permits, schedule requirements, quality control, schedule of prices, shop drawings, submittals, cybersecurity, prosecution of the work, government acceptance, final inspections and contract close-out. Contractor must present and discuss their basic approach to scheduling the construction work and any required phasing.

1.7.1 Attendees

Contractor attendees must include the Project Manager, Superintendent, Site Safety and Health Officer (SSHO), Quality Control Manager and major subcontractors.

1.8 FACILITY TURNOVER PLANNING MEETINGS (Red Zone Meetings)

Meet with the Government to identify strategies to ensure the project is carried to expeditious closure and turnover to the Client. Start planning the turnover process at the Pre-Construction meeting with a discussion of the Red Zone process and convene at regularly scheduled NAVFAC Red Zone Meetings beginning at approximately 75 percent of project completion. Include the following in the facility Turnover effort:

1.8.1 Red Zone Checklist

- a. Contracting Officer's Technical Representative (COTR) will provide the Contractor a copy of the Red Zone Checklist template.
- b. Prior to 75 percent completion, modify the Red Zone Checklist template by adding or deleting critical activities applicable to the project and assign planned completion dates for each activity. Submit the modified Red Zone Checklist to the Contracting Officer. The Contracting Officer may request additional activities be added to the Red Zone Checklist at any time as necessary.

1.8.2 Meetings

- a. Conduct regular Red Zone Meetings beginning at approximately 75 percent project completion, or three to six months prior to Beneficial Occupancy Date (BOD), whichever comes first.
- b. The Contracting Officer will establish the frequency of the meetings, which is expected to increase as the project completion draws nearer. At the beginning, Red Zone meetings may be every two weeks then increase to weekly towards the final month of the project.
- c. Using the Red Zone Checklist as a Plan of Action and Milestones (POAM) and basis for discussion, review upcoming critical activities and strategies to ensure work is completed on time.

- d. During the Red Zone Meetings discuss with the COTR any upcoming activities that require Government involvement.
- e. Maintain the Red Zone Checklist by documenting the actual completion dates as work is completed and update the Red Zone Checklist with revised planned completion dates as necessary to match progress. Distribute copies of the current Red Zone Checklist to attendees at each Red Zone Meeting.

1.9 EXPORT LICENSES FOR OVERSEAS PROJECTS

Obtain individual export licenses and project export licenses required by the Department of Commerce regulations (15 CFR 772 and 15 CFR 773) so that no delays are experienced in shipping from the United States of America to a foreign country. For additional information, the Contractor may contact one of the U.S. and Foreign Commercial Service District Offices of the Department of Commerce which are located in almost every State.

1.10 WAIVER FOR WORKER'S COMPENSATION

In addition to FAR 52.228-4 Workers' Compensation and War-Hazard Insurance Overseas, the Secretary of Labor has granted a waiver. The waiver does not apply to employees who are hired in the United States, or who are residents, or citizens of the United States.

1.11 PARTNERING

Contractor shall host the partnering session within 45 calendar days of contract award. To most effectively accomplish this Contract, the Contractor and Government must form a cohesive partnership with the common goal of drawing on the strength of each organization in an effort to achieve a successful project without safety mishaps, conforming to the Contract, within budget and on schedule. The partnering team must consist of personnel from both the Government and Contractor including project level and corporate level leadership positions. Key Personnel from the supported command, end user, NAVFAC, PWD, Contractor, key subcontractors and the Designer of Record are required to participate in the Partnering process.

1.11.1 Facilitated (Formal) Partnering

- a. Within 35 calendar days after award and prior to the start of work, host a Formal Partnering session with key personnel from the project team including both Contractor and Government personnel. All costs associated with the Partnering session including the third-party independent Facilitator Consultant, meeting room and other incidental items are the responsibility of the Contractor.
- b. Before the Facilitated (Formal) Partnering session, coordinate with the Facilitator all requirements for incidental items (such as audio-visual equipment, easels, flipchart paper, colored markers, note pads, pens/pencils, colored flash cards) and have these items available at the Partnering session. Provide copies of any documents required for distribution to all attendees. Participants will bear their own costs for meals, lodging and transportation associated with Partnering.
- c. The Initial Partnering Session must be a duration of one day and be held at a location off base as agreed to by the Contracting Officer.

Partnering session may take place concurrently with the
Pre-Construction Meeting.

- d. Facilitator must be experienced in conducting corporate Partnering sessions and must be a third-party independent facilitating consultant - not an employee of the Contractor. The Facilitator is responsible for leading all aspects of the Partnering session necessary to achieve the Partnering goal.
- e. An outcome of the Partnering session must be an escalation matrix agreed upon by both the Government and Contractor, which identifies key Government and Contractor decision makers by name and anticipated decision durations.
- f. Host follow-on Partnering Sessions at three- to six-month intervals or more frequently if needed and lasting generally a half day or less. Attendees need only be those required to resolve current issues. The same Facilitator used in the Initial Partnering session must lead the follow-on sessions unless an alternative is permitted by the Contractor Officer. All costs associated with follow-on Partnering sessions are the responsibility of the Contractor.

1.12 MOBILIZATION

Contractor shall mobilize to the jobsite within 45 calendar days after contract award. Mobilize is defined as having equipment AND having a physical presence of at least one person from the contractor's team on the jobsite.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 33 00

SUBMITTAL PROCEDURES

08/18, CHG 4: 02/21

PART 1 GENERAL

- 1.1 DEFINITIONS
 - 1.1.1 Submittal Descriptions (SD)
 - 1.1.2 Approving Authority
 - 1.1.3 Work
- 1.2 SUBMITTALS
- 1.3 SUBMITTAL CLASSIFICATION
 - 1.3.1 Government Approved (G)
 - 1.3.2 For Information Only
- 1.4 FORWARDING SUBMITTALS REQUIRING GOVERNMENT APPROVAL
 - 1.4.1 O&M Data
 - 1.4.2 Submittals Reserved for NAVFAC Approval
 - 1.4.3 Overseas Shop Drawing Submittals
- 1.5 PREPARATION
 - 1.5.1 Transmittal Form
 - 1.5.2 Identifying Submittals
 - 1.5.3 Submittal Format
 - 1.5.3.1 Format of SD-01 Preconstruction Submittals
 - 1.5.3.2 Format for SD-02 Shop Drawings
 - 1.5.3.2.1 Drawing Identification
 - 1.5.3.3 Format of SD-03 Product Data
 - 1.5.3.3.1 Product Information
 - 1.5.3.3.2 Standards
 - 1.5.3.3.3 Data Submission
 - 1.5.3.4 Format of SD-04 Samples
 - 1.5.3.4.1 Sample Characteristics
 - 1.5.3.4.2 Sample Incorporation
 - 1.5.3.4.3 Comparison Sample
 - 1.5.3.5 Format of SD-05 Design Data
 - 1.5.3.6 Format of SD-06 Test Reports
 - 1.5.3.7 Format of SD-07 Certificates
 - 1.5.3.8 Format of SD-08 Manufacturer's Instructions
 - 1.5.3.8.1 Standards
 - 1.5.3.9 Format of SD-09 Manufacturer's Field Reports
 - 1.5.3.10 Format of SD-10 Operation and Maintenance Data (O&M)
 - 1.5.3.11 Format of SD-11 Closeout Submittals
 - 1.5.4 Source Drawings for Shop Drawings
 - 1.5.4.1 Source Drawings
 - 1.5.4.2 Terms and Conditions
 - 1.5.5 Electronic File Format
- 1.6 QUANTITY OF SUBMITTALS
 - 1.6.1 Number of SD-01 Preconstruction Submittal Copies
 - 1.6.2 Number of SD-02 Shop Drawing Copies
 - 1.6.3 Number of SD-03 Product Data Copies
 - 1.6.4 Number of SD-04 Samples

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 1.6.5 Number of SD-05 Design Data Copies
- 1.6.6 Number of SD-06 Test Report Copies
- 1.6.7 Number of SD-07 Certificate Copies
- 1.6.8 Number of SD-08 Manufacturer's Instructions Copies
- 1.6.9 Number of SD-09 Manufacturer's Field Report Copies
- 1.6.10 Number of SD-10 Operation and Maintenance Data Copies
- 1.6.11 Number of SD-11 Closeout Submittals Copies
- 1.7 INFORMATION ONLY SUBMITTALS
- 1.8 PROJECT SUBMITTAL REGISTER AND DATABASE
 - 1.8.1 Submittal Management
 - 1.8.2 Preconstruction Use of Submittal Register
 - 1.8.3 Contractor Use of Submittal Register
 - 1.8.4 Approving Authority Use of Submittal Register
 - 1.8.5 Action Codes
 - 1.8.5.1 Government Review Action Codes
 - 1.8.5.2 Government Review Action Codes
- 1.9 VARIATIONS
 - 1.9.1 Warranting that Variations are Compatible
 - 1.9.2 Review Schedule Extension
- 1.10 SCHEDULING
 - 1.10.1 Reviewing, Certifying, and Approving Authority
 - 1.10.2 Constraints
 - 1.10.3 QC Organization Responsibilities
- 1.11 GOVERNMENT APPROVING AUTHORITY
 - 1.11.1 Review Notations
- 1.12 DISAPPROVED SUBMITTALS
- 1.13 APPROVED SUBMITTALS
- 1.14 APPROVED SAMPLES
- 1.15 WITHHOLDING OF PAYMENT
- 1.16 CERTIFICATION OF SUBMITTAL DATA

PART 2 PRODUCTS

PART 3 EXECUTION

ATTACHMENTS:

ENG Form 4025-R

Appendix A - Submittal Register

-- End of Section Table of Contents --

SECTION 01 33 00

SUBMITTAL PROCEDURES
08/18, CHG 4: 02/21

PART 1 GENERAL

1.1 DEFINITIONS

1.1.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Submittals that are required prior to or commencing with the start of work on site. Submittals that are required prior to or at the start of construction (work) or the next major phase of the construction on a multiphase contract.

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

Certificates Of Insurance

Surety Bonds

List Of Proposed Subcontractors

List Of Proposed Products

Baseline Network Analysis Schedule (NAS)

Submittal Register

Schedule Of Prices

Accident Prevention Plan Health And Safety Plan

Work Plan

Quality Control (QC) plan

Environmental Protection Plan

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-04 Samples

Fabricated or unfabricated physical examples of materials, equipment or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged.

Color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project.

Field samples and mock-ups constructed on the project site establish standards ensuring work can be judged. Includes assemblies or portions of assemblies that are to be incorporated into the project and those that will be removed at conclusion of the work.

SD-05 Design Data

Design calculations, mix designs, analyses or other data pertaining to a part of work.

Design submittals, design substantiation submittals and extensions of design submittals.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. Unless specified in another section, testing must have been within three years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily logs and checklists

Final acceptance test and operational test procedure

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that the product, system, or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits

Text of posted operating instructions

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system or material, including special notices and (SDS)concerning impedances, hazards and safety precautions.

SD-09 Manufacturer's Field Reports

Documentation of the testing and verification actions taken by manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or after installation, to confirm compliance with manufacturer's standards or instructions. The documentation must be signed by an authorized official of a testing laboratory or agency and state the test results; and indicate whether the material, product, or system has passed or failed the test.

Factory test reports.

SD-10 Operation and Maintenance Data

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel.

Data required by operating and maintenance personnel for the safe and efficient operation, maintenance and repair of the item.

Data incorporated in an operations and maintenance manual or control system.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principle Validation (GPV) or Third Party Certification (TPC).

Special requirements necessary to properly close out a construction

contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase contract.

1.1.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.1.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Submittal Register; G

1.3 SUBMITTAL CLASSIFICATION

1.3.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, variations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government.

Government approval is required for any variations from the Solicitation or the Accepted Proposal and for other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

1.3.2 For Information Only

Submittals not requiring Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

1.4 FORWARDING SUBMITTALS REQUIRING GOVERNMENT APPROVAL

As soon as practicable after award of contract, and before procurement or fabrication, forward to the Architect-Engineer, submittals required in the technical sections of this specification, including shop drawings, product data and samples. In addition, forward a copy of the submittals to the Contracting Officer.

1.4.1 O&M Data

Submit data specified for a given item within 30 calendar days after the item is delivered to the contract site.

In the event the Contractor fails to deliver O&M data within the time limits specified, the Contracting Officer may withhold from progress payments 50 percent of the price of the items to which such O&M data apply.

1.4.2 Submittals Reserved for NAVFAC Approval

As an exception to the standard submittal procedure for Government Approval, submit the following to the Commander, NAVFAC:

- a. Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC: All submittals
- b. Section 26 12 19.10 THREE-PHASE, LIQUID-FILLED PAD-MOUNTED TRANSFORMERS: All submittals

1.4.3 Overseas Shop Drawing Submittals

Send submittals via overnight express mail service. All costs associated with the overnight express mail service are borne by the Contractor. Costs associated with the overnight express mail of submittals related to proposed submittal variances of resubmittals necessary as a result of noncompliant or incomplete Contractor submittals are the responsibility of the Contractor.

1.5 PREPARATION

1.5.1 Transmittal Form

Transmit each submittal, except sample installations and sample panels to the office of the approving authority using the transmittal form prescribed by the Contracting Officer. Include all information prescribed by the transmittal form and required in paragraph IDENTIFYING SUBMITTALS. Use the submittal transmittal forms to record actions regarding samples.

Use the ENG Form 4025-R transmittal form for submitting both Government-approved and information-only submittals. Submit in accordance with the instructions on the reverse side of the form. These forms or similar forms will be furnished to the Contractor. Properly complete this form by filling out all the heading blank spaces and identifying each item submitted. Exercise special care to ensure proper listing of the specification paragraph and sheet number of the contract drawings pertinent to the data submitted for each item.

1.5.2 Identifying Submittals

The Contractor's Quality Control Manager must prepare, review and stamp submittals, including those provided by a subcontractor, before submittal to the Government.

Identify submittals, except sample installations and sample panels, with the following information permanently adhered to or noted on each separate component of each submittal and noted on transmittal form. Mark each copy of each submittal identically, with the following:

- a. Project title and location
- b. Construction contract number
- c. Dates of the drawings and revisions
- d. Name, address, and telephone number of Subcontractor, supplier, manufacturer, and any other Subcontractor associated with the submittal.
- e. Section number of the specification by which submittal is required
- f. Submittal description (SD) number of each component of submittal
- g. For a resubmission, add alphabetic suffix on submittal description, for example, submittal 18 would become 18A, to indicate resubmission
- h. Product identification and location in project.

1.5.3 Submittal Format

1.5.3.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.3.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 210 by 297 mm nor more than 1189 by 841 mm, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

- a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.
- b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

Submit an electronic copy of drawings in PDF format.

1.5.3.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than 60 millimeter on the right-hand side of each sheet for the Government disposition stamp.

1.5.3.3 Format of SD-03 Product Data

Present product data submittals for each section as a complete, bound volume. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

1.5.3.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in metric dimensions. Where product data are included in preprinted catalogs with English units only, submit metric dimensions on separate sheet.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

1.5.3.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.3.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will not be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.5.3.4 Format of SD-04 Samples

1.5.3.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

- a. Sample of Equipment or Device: Full size.
- b. Sample of Materials Less Than 50 by 75 mm: Built up to 210 by 297 mm.
- c. Sample of Materials Exceeding 210 by 297 mm: Cut down to 210 by 297 mm and adequate to indicate color, texture, and material variations.
- d. Sample of Linear Devices or Materials: 250 mm length or length to be supplied, if less than 250 mm. Examples of linear devices or materials are conduit and handrails.
- e. Sample Volume of Nonsolid Materials: 750 ml. Examples of nonsolid materials are sand and paint.
- f. Color Selection Samples: 50 by 100 mm. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.
- g. Sample Panel: 1200 by 1200 mm.
- h. Sample Installation: 10 square meters.

1.5.3.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

1.5.3.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.5.3.5 Format of SD-05 Design Data

Provide design data and certificates on 210 by 297 mm paper. Provide a bound volume for submittals containing numerous pages.

1.5.3.6 Format of SD-06 Test Reports

Provide reports on 210 by 297 mm paper in a complete bound volume.

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.3.7 Format of SD-07 Certificates

Provide design data and certificates on 210 by 297 mm paper. Provide a bound volume for submittals containing numerous pages.

1.5.3.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section as a complete, bound volume. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.5.3.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.3.9 Format of SD-09 Manufacturer's Field Reports

Provide reports on 210 by 297 mm paper in a complete bound volume.

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.3.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format.

1.5.3.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

Provide all dimensions in administrative submittals in metric. Where data are included in preprinted material with English units only, submit metric dimensions on separate sheet.

1.5.4 Source Drawings for Shop Drawings

1.5.4.1 Source Drawings

The entire set of source drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

1.5.4.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

1.5.5 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. Compile the submittal file as a single, complete document, to include the Transmittal Form described within. Name the electronic submittal file specifically according to its contents, and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF

files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of a signature.

E-mail electronic submittal documents smaller than 10MB to an e-mail address as directed by the Contracting Officer. Provide electronic documents over 10 MB on an optical disc or through an electronic file sharing system such as the DOD SAFE Web Application located at the following website: <https://safe.apps.mil/>.

1.6 QUANTITY OF SUBMITTALS

1.6.1 Number of SD-01 Preconstruction Submittal Copies

Unless otherwise specified, submit three sets of administrative submittals.

1.6.2 Number of SD-02 Shop Drawing Copies

Submit six copies of submittals of shop drawings requiring review and approval by a QC organization. Submit seven copies of shop drawings requiring review and approval by the Contracting Officer.

1.6.3 Number of SD-03 Product Data Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.4 Number of SD-04 Samples

- a. Submit one sample, or one set of samples showing the range of variation, of each required item.
- b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.
- c. Submit one sample installation, where directed.

1.6.5 Number of SD-05 Design Data Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.6 Number of SD-06 Test Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings, other than field test results that will be submitted with QC reports.

1.6.7 Number of SD-07 Certificate Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.8 Number of SD-08 Manufacturer's Instructions Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.9 Number of SD-09 Manufacturer's Field Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings other than field test results that will be submitted with QC reports.

1.6.10 Number of SD-10 Operation and Maintenance Data Copies

Submit three copies of O&M data to the Contracting Officer for review and approval.

1.6.11 Number of SD-11 Closeout Submittals Copies

Unless otherwise specified, submit three sets of administrative submittals.

1.7 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Provide information-only submittals to the Contracting Officer a minimum of 14 calendar days prior to the Preparatory Meeting for the associated Definable Feature of Work (DFOW). Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.8 PROJECT SUBMITTAL REGISTER AND DATABASE

A sample Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as "Appendix A - Submittal Register."

1.8.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. [Use an electronic submittal register program furnished by the Government.](#) Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required.

Column (c): Lists specification section in which submittal is

required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

Column (f): Lists the approving authority for each submittal.

1.8.2 Preconstruction Use of Submittal Register

Submit the submittal register as an electronic database, using the submittal management program furnished to Contractor. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register database submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.8.3 Contractor Use of Submittal Register

Update the following fields in the Government-furnished submittal register program or equivalent fields in the program used by the Contractor with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.8.4 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.8.5 Action Codes

Entries for columns (j) and (o) are to be used as follows (others may be prescribed by the Transmittal Form):

1.8.5.1 Government Review Action Codes

"A" - "Approved as submitted"; "Completed"

"B" - "Approved, except as noted on drawings"; "Completed"

"C" - "Approved, except as noted on drawings; resubmission required"; "Resubmit"

"D" - "Returned by separate correspondence"; "Completed"

"E" - "Disapproved (See attached)"; "Resubmit"

"F" - "Receipt acknowledged"; "Completed"

"G" - "Other (Specify)"; "Resubmit"

"X" - "Receipt acknowledged, does not comply with contract requirements"; "Resubmit"

1.8.5.2 Government Review Action Codes

"A" - "Approved as submitted"

"AN" - "Approved as noted"

"RR" - "Disapproved as submitted"; "Completed"

"NR" - "Not Reviewed"

"RA" - "Receipt Acknowledged"

1.9 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

1.9.1 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.9.2 Review Schedule Extension

In addition to the normal submittal review period, a period of 14 **calendar working** days will be allowed for the Government to consider submittals

with variations.

1.10 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals.

- a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal.
- b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."
- c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.

Contracting Officer review will be completed within 14 calendar working days after the date of submission.

- d. Except as specified otherwise, allow a review period, beginning with receipt by the approving authority, that includes at least 15 working days for submittals for QC manager approval and 20 working days for submittals where the Contracting Officer is the approving authority. The period of review for submittals with Contracting Officer approval begins when the Government receives the submittal from the QC organization.

1.10.1 Reviewing, Certifying, and Approving Authority

The QC Manager is responsible for reviewing all submittals and certifying that they are in compliance with contract requirements. The approving authority on submittals is the QC Manager unless otherwise specified. At each "Submittal" paragraph in individual specification sections, a notation "G" following a submittal item indicates that the Contracting Officer is the approving authority for that submittal item. Provide an additional copy of the submittal to the Government Approving authority

1.10.2 Constraints

Conform to provisions of this section, unless explicitly stated otherwise for submittals listed or specified in this contract.

Submit complete submittals for each definable feature of the work. At the same time, submit components of definable features that are interrelated as a system.

When acceptability of a submittal is dependent on conditions, items, or materials included in separate subsequent submittals, the submittal will be returned without review.

Approval of a separate material, product, or component does not imply approval of the assembly in which the item functions.

1.10.3 QC Organization Responsibilities

- a. Review submittals for conformance with project design concepts and compliance with contract documents.
- b. Process submittals based on the approving authority indicated in the submittal register.
 - (1) When the QC manager is the approving authority, take appropriate action on the submittal from the possible actions defined in paragraph APPROVED SUBMITTALS.
 - (2) When the Contracting Officer is the approving authority or when variation has been proposed, forward the submittal to the Government, along with a certifying statement, or return the submittal marked "not reviewed" or "revise and resubmit" as appropriate. The QC organization's review of the submittal determines the appropriate action.
- c. Ensure that material is clearly legible.
- d. Stamp each sheet of each submittal with a QC certifying statement or an approving statement, except that data submitted in a bound volume or on one sheet printed on two sides may be stamped on the front of the first sheet only.
 - (1) When the approving authority is the Contracting Officer, the QC organization will certify submittals forwarded to the Contracting Officer with the following certifying statement:

"I hereby certify that the (equipment) (material) (article) shown and marked in this submittal is that proposed to be incorporated with Contract Number (Awarded Contract Number) is in compliance with the contract drawings and specification, can be installed in the allocated spaces, and is submitted for Government approval.

Certified by Submittal Reviewer _____, Date _____
(Signature when applicable)

Certified by QC Manager _____, Date _____"
(Signature)

- (2) When approving authority is the QC manager, the QC manager will use the following approval statement when returning submittals to the Contractor as "Approved" or "Approved as Noted."

"I hereby certify that the (material) (equipment) (article) shown and marked in this submittal and proposed to be incorporated with Contract Number (Awarded Contract Number) is in compliance with the contract drawings and specification, can be installed in the allocated spaces, and is approved for use.

Certified by Submittal Reviewer _____, Date _____
(Signature when applicable)

Approved by QC Manager _____, Date _____"
(Signature)

- e. Sign the certifying statement or approval statement. The QC organization member designated in the approved QC plan is the person signing certifying statements. The use of original ink for signatures is required. Stamped signatures are not acceptable.
- f. Update the submittal register as submittal actions occur, and maintain the submittal register at the project site until final acceptance of all work by the Contracting Officer.
- g. Retain a copy of approved submittals and approved samples at the project site.
- h. For "S" submittals, provide a copy of the approved submittal to the Government Approving authority.

1.11 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

- a. Note the date on which the submittal was received from the QC manager.
- b. Review submittals for approval within the scheduling period specified and only for conformance with project design concepts and compliance with contract documents.
- c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. One copy of the submittal will be retained by the Contracting Officer and one copy of the submittal will be returned to the Contractor. If the Government performs a conformance review of other Designer of Record approved submittals, the submittals will be identified and returned, as described above.

1.11.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

- a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.
- b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.
- c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal

is approved.

- d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.
- e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.12 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

1.13 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only that the general method of construction, materials, detailing, and other information are satisfactory. the design, general method of construction, materials, detailing, and other information appear to meet the Solicitation and Accepted Proposal.

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained within each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.14 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not to be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements.

1.15 WITHHOLDING OF PAYMENT

Payment for materials incorporated in the work will not be made if required approvals have not been obtained. No payment for materials incorporated in the work will be made unless all required DOR approvals or required Government approvals have been obtained. No payment will be made for any materials incorporated into the work for any conformance review submittals or information-only submittals found to contain errors or deviations from the Solicitation or Accepted Proposal.

1.16 CERTIFICATION OF SUBMITTAL DATA

Certify the submittal data as follows on Form ENG 4025: "I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.

____NAME OF CONTRACTOR____ SIGNATURE OF CONTRACTOR

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

INSTRUCTIONS

1. Section I will be initiated by the Contractor in the required number of copies.
2. Each Transmittal shall be numbered consecutively. The Transmittal Number typically includes two parts separated by a dash (-). The first part is the specification section number. The second part is a sequential number for the submittals under that spec section. If the Transmittal is a resubmittal, then add a decimal point to the end of the original Transmittal Number and begin numbering the resubmittal packages sequentially after the decimal.
3. The "Item No." for each entry on this form will be the same "Item No." as indicated on ENG FORM 4288-R.
4. Submittals requiring expeditious handling will be submitted on a separate ENG Form 4025-R.
5. Items transmitted on each transmittal form will be from the same specification section. Do not combine submittal information from different specification sections in a single transmittal.
6. If the data submitted are intentionally in variance with the contract requirements, indicate a variation in column h, and enter a statement in the Remarks block describing the detailed reason for the variation.
7. ENG Form 4025-R is self-transmitting - a letter of transmittal is not required.
8. When submittal items are transmitted, indicate the "Submittal Type" (*SD-01 through SD-11*) in column c of Section I.
Submittal types are the following:

SD-01 - Preconstruction	SD-02 - Shop Drawings	SD-03 - Product Data	SD-04 - Samples	SD-05 - Design Data	SD-06 - Test Reports
SD-07 - Certificates	SD-08 - Manufacturer's Instructions	SD-09 - Manufacturer's Field Reports	SD-10 - O&M Data	SD-11 - Closeout	
9. For each submittal item, the Contractor will assign Submittal Action Codes in column g of Section I. The U.S. Army Corps of Engineers approving authority will assign Submittal Action Codes in column i of Section I. The Submittal Action Codes are:

A -- Approved as submitted. B -- Approved, except as noted on drawings. Resubmission not required. C -- Approved, except as noted on drawings. Refer to attached comments. Resubmission required. D -- Will be returned by separate correspondence. E -- Disapproved. Refer to attached comments.	F -- Receipt acknowledged. X -- Receipt acknowledged, does not comply with contract requirements, as noted. G -- Other action required (<i>Specify</i>) K -- Government concurs with intermediate design. (<i>For D-B contracts</i>) R -- Design submittal is acceptable for release for construction. (<i>For D-B contracts</i>)
--	---
10. Approval of items does not relieve the contractor from complying with all the requirements of the contract.

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		00 73 04.00 41	SD-01 Preconstruction Submittals														
			Contractor personnel	1.18.1	G												
			identification														
		01 11 00	SD-01 Preconstruction Submittals														
			Salvage Plan	1.5	G												
		01 14 00	SD-01 Preconstruction Submittals														
			List of Contact Personnel	1.4.1.1													
			Personnel List	1.5.1													
		01 20 00	SD-01 Preconstruction Submittals														
			Schedule of Prices	1.3	G												
		01 30 00	SD-01 Preconstruction Submittals														
			View Location Map	1.3													
			Progress and Completion	1.4													
			Pictures														
		01 33 00	SD-01 Preconstruction Submittals														
			Submittal Register	1.8	G												
		01 35 26	SD-01 Preconstruction Submittals														
			Accident Prevention Plan (APP)	1.8	G												
			Accident Prevention Plan (APP)	1.8	G												
			APP - Construction	1.8.1	G												
			SD-06 Test Reports														
			Monthly Exposure Reports	1.4													
			Notifications and Reports	1.13													
			Accident Reports	1.13.2	G												
			LHE Inspection Reports	1.13.3													
			SD-07 Certificates														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 26	Contractor Safety Self-Evaluation Checklist	1.5													
			Crane Operators/Riggers	1.7.1.4													
			Standard Lift Plan	1.8.3.2	G												
			Critical Lift Plan	1.8.3.3	G												
			Activity Hazard Analysis (AHA)	1.9													
			Confined Space Entry Permit	1.10.1													
			Hot Work Permit	1.10.1													
			Certificate of Compliance	1.13.4													
			License Certificates	1.15													
			Radiography Operation Planning	1.15.1	G												
			Work Sheet														
			Portable Gauge Operations	1.15.1	G												
			Planning Worksheet														
		01 35 29.13	SD-02 Shop Drawings														
			Work Zones	3.9.1	G												
			Decontamination Facilities	3.10.1	G												
			SD-03 Product Data														
			Amendments to the APP/SSHP	1.4	G												
			Exposure Monitoring/Air	3.5													
			Sampling Program														
			Site Control Log	3.9.2													
			SSHO's Daily Inspection Logs	1.8													
			SD-07 Certificates														
			Certificate Of Worker/Visitor	1.7													
			Acknowledgement														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 29.13	SD-11 Closeout Submittals														
			Safety And Health Phase-Out Report	1.9	G												
		01 45 00	SD-01 Preconstruction Submittals														
			Contractor Quality Control (CQC) Plan	1.5.2	G												
			Additional Requirements For Design Quality Control (DQC) Plan	1.5.2.2	G												
			SD-05 Design Data														
			Discipline-Specific Checklists	1.5.2.2													
			SD-06 Test Reports														
			Verification Statement	1.12.3													
			SD-07 Certificates														
			Certificate Of Readiness	1.6.5.1	G												
		01 45 35	SD-01 Preconstruction Submittals														
			SIOR Letter of Acceptance	3.1.1	G												
			Written NDT Practices	3.1.3													
			SD-06 Test Reports														
			Daily Reports	3.1.3													
			Daily Reports	3.1.3													
			Biweekly Reports	3.1.2													
			SD-07 Certificates														
			AISC Certified Steel Fabricator	2.1													
			Steel Truss Plant Quality Assurance Program	2.1													
			Certificate of Compliance	2.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 45 35	Special Inspector of Record	1.5.5	G												
			Special Inspector	1.5	G												
			Qualification Records	3.1.3													
			SD-11 Closeout Submittals														
			Interim Report	3.1.1	G												
			Comprehensive Final Report	3.1.1	G												
			Comprehensive Final Report	3.1.3	G												
		01 50 00	SD-01 Preconstruction Submittals														
			Construction Site Plan	1.3	G												
			Traffic Control Plan	3.4.1	G												
			Haul Road Plan	2.2.1	G												
			Contractor Computer	1.5.1.4	G												
			Cybersecurity Compliance														
			Statements														
			Contractor Temporary Network	1.5.6	G												
			Cybersecurity Compliance														
			Statements														
		01 57 19.01 21	SD-01 Preconstruction Submittals														
			Preconstruction Environmental,		G												
			Including Common Chameleon														
			and Vegetation, Survey														
			Preconstruction Historical and		G												
			Archaeological Survey														
			Environmental Protection Plan		G												
			(EPP)														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	GOVT CLASS SIF ACTION OR A/E REVIEW WR	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 19.01 21	Spill Prevention, Control and Response Plan (for projects at or near the Bay of Cadiz)		G												
			Solid Waste and Recycling Management Plan		G												
			Training in Environmental Compliance Assessment Training and Tracking System (Ecatts)		G												
			Environmental Manager Qualifications		G												
			Safety Data Sheets (SDS) for Hazardous Materials														
			SD-07 Certificates														
			List of Employee Environmental Training Records		G												
			List of ECATTS Environmental on-line course Certificates of Completion		G												
			SD-11 Closeout Submittals														
			Stormwater Pollution Prevention Plan Compliance Notebook		G												
			Solid Waste and Recycling Report (by weight)		G												
			Disposal Documentation for Hazardous and Regulated Waste		G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 19.01 21	Contractor Hazardous Material Inventory Log		G												
			Regulatory Notifications		G												
			Sales Documentation		G												
			Construction & Demolition (C & D) Debris data and management cost														
		01 58 00	SD-02 Shop Drawings														
			Preliminary Drawing Indicating Layout And Text Content	1.3.1	G												
		01 74 19	SD-01 Preconstruction Submittals														
			Construction Waste Management Plan	1.7	G												
			SD-11 Closeout Submittals														
			Final Construction Waste Diversion Report	1.9	S												
		01 78 00	SD-03 Product Data														
			Warranty Management Plan	1.5.1													
			Warranty Tags	1.5.4													
			Final Cleaning	3.2													
			Spare Parts Data	1.4													
			SD-08 Manufacturer's Instructions														
			Instructions	1.5.1													
			SD-11 Closeout Submittals														
			As-Built Drawings	3.1	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		01 78 00	As-Built Record of Equipment and Materials	1.5.1													
			Certification of EPA Designated Items	2.1	G												
			Interim DD FORM 1354	3.3.1	G												
			Checklist for DD FORM 1354	3.3.2	G												
		01 78 23	SD-10 Operation and Maintenance Data														
			Facility Data Workbook	1.4	G												
			Training Plan	3.1.1	G												
			Training Outline	3.1.3	G												
			Training Content	3.1.2	G												
			Operation And Maintenance Manual, Progress Submittal	3.2.1	G												
			Operation And Maintenance Manual, Prefinal Submittal	3.2.2	G												
			Operation And Maintenance Manual, Final Submittal	3.2.3	G												
			SD-11 Closeout Submittals														
			Training Video Recording	3.1.4	G												
			Validation of Training Completion	3.1.6	G												
			Training Plan	3.1.1	G												
			Record Drawings And Utility Systems	1.6.6.7	G												
		01 78 23.33	SD-10 Operation and Maintenance Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		01 78 23.33	System Instructions	2.1.1	G												
		01 78 24.00 20	SD-11 Closeout Submittals														
			Facility Data Workbook (FDW),	1.3.1	G												
			Progress Submittal														
			Facility Data Workbook (FDW),	1.3.2	G												
			Prefinal Submittal														
			Facility Data Workbook (FDW),	1.3.3	G												
			Final Submittal														
		02 41 00	SD-01 Preconstruction Submittals														
			Demolition Plan	1.2.2	G												
			Existing Conditions	1.10													
		02 61 13	SD-01 Preconstruction Submittals														
			Work Plan	1.3	G												
			SD-02 Shop Drawings														
			Surveys	3.1													
			SD-06 Test Reports														
			Closure Report	3.11	G												
		02 81 00	SD-03 Product Data														
			Packaging Notifications	2.1.1													
			Hazardous Waste Management	3.1	G												
			Plan														
			Onsite Hazardous Waste	3.2	G												
			Management														
			SD-06 Test Reports														
			Recordkeeping	3.7	G												
			Exception Report	3.7	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		02 81 00	Spill Response	3.8													
			SD-07 Certificates														
			Transportation and Disposal	1.4.1	G												
			Coordinator														
			Training	1.4.2	G												
			Certification	1.4.3													
			Shipping Documents and	3.3.3	G												
			Packagings Certification														
			Security Plan	3.3.4													
			Certificates of Disposal	3.3.5													
			Waste Minimization	3.6													
		02 82 00	SD-03 Product Data														
			Amended Water	1.2.2	G												
			Safety Data Sheets (SDS) for All	1.3.9	G												
			Materials														
			Encapsulants	2.1	G												
			Respirators	3.1.2.1	G												
			Local Exhaust Equipment	3.1.7	G												
			Vacuums	3.1.8	G												
			Glovebags	3.1.10	G												
			SD-06 Test Reports														
			Air Sampling Results	1.5.4	G												
			Clearance Sampling	3.2.9.5	G												
			Asbestos Disposal Quantity	3.3.3.2	G												
			Report														
			SD-07 Certificates														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		02 82 00	Employee Training	1.3.4	G												
			Notifications	1.3.5	G												
			Respiratory Protection Program	1.3.7	G												
			Asbestos Hazard Abatement Plan	1.3.10	G												
			Testing Laboratory	1.3.11	G												
			Landfill Approval	1.3.12	G												
			Delivery Tickets	1.3.12	G												
			Waste Shipment Records	1.3.12	G												
			Transporter Certification	1.3.13	G												
			Medical Certification	1.3.14	G												
			Private Qualified Person	1.5.1	G												
			Documentation														
			Designated Competent Person	1.5.2	G												
			Contractor's License	1.5.3	G												
			Federal, State or Local Citations	1.5.6	G												
			on Previous Projects														
			Encapsulants	2.1	G												
			Equipment Used to Contain	3.1	G												
			Airborne Asbestos Fibers														
			Vacuums	3.1.8	G												
			Ventilation Systems	3.1.8	G												
			SD-11 Closeout Submittals														
			Permits and Licenses	1.3.5	G												
			Notifications	1.3.5	G												
			Respirator Program Records	1.3.7.1	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		02 82 00	Protective Clothing	1.5.5	G												
			Decontamination Quality Control														
			Records														
		02 83 00	SD-01 Preconstruction Submittals														
			Competent Person	1.5.1.1	G												
			Training Certification	1.5.1.2	G												
			Medical Examinations	1.5.2.3	G												
			Lead, Cadmium, Chromium	1.5.2.7	G												
			Waste Management Plan														
			Lead, Cadmium, Chromium	1.5.2.2	G												
			Compliance Plan														
			Written Evidence of TSD	3.5.2.1	G												
			Approval														
			SD-03 Product Data														
			Respirators	1.6.1	G												
			Vacuum Filters	1.6.4	G												
			Negative Air Pressure System	1.6.7	G												
			Materials and Equipment	2.1	G												
			Expendable Supplies	2.1.1	G												
			Local Exhaust Equipment	3.1.1.3	G												
			SD-06 Test Reports														
			Sampling and Analysis	1.3.3	G												
			SD-07 Certificates														
			Testing Laboratory	1.5.1.3	G												
			Clearance Certification	3.5.1.1	G												
			SD-11 Closeout Submittals														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		02 83 00	Hazardous Waste Manifest	3.5.2.1	G												
		03 30 00	SD-01 Preconstruction Submittals														
			Concrete Curing Plan	1.6.3.1													
			Quality Control Plan	1.6.6	G												
			Quality Control Personnel	1.6.7	G												
			Certifications														
			Quality Control Organizational Chart	1.6.7													
			Laboratory Accreditation	1.6.9	G												
			Form Removal Schedule	1.6.2.1	G												
			Maturity Method Data	3.3.9													
			SD-02 Shop Drawings														
			Formwork	1.6.2.1													
			Reinforcing Steel	1.6.2.2	G												
			SD-03 Product Data														
			Joint Sealants	2.4.5													
			Joint Filler	2.4.4													
			Formwork Materials	2.1													
			Recycled Aggregate Materials	2.3.3.2													
			Cementitious Materials	2.3.1													
			Vapor Barrier	2.4.6													
			Concrete Curing Materials	2.4.1													
			Reinforcement	2.6													
			Liquid Chemical Floor Hardeners and Sealers	2.4.3.1													
			Admixtures	2.3.4													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		03 30 00	Waterstops	2.2.2													
			Local/Regional Materials	1.8.1													
			Biodegradable Form Release Agent	2.2.3													
			Pumping Concrete	1.6.3.2													
			Nonshrink Grout	2.4.2													
			SD-04 Samples														
			Surface Finish Samples	1.6.5.1													
			SD-05 Design Data														
			Concrete Mix Design	1.6.1.2	G												
			Formwork Calculations	1.6.1.1													
			SD-06 Test Reports														
			Concrete Mix Design	1.6.1.2	G												
			Fly Ash	1.6.4.1													
			Pozzolan	1.6.4.1													
			Slag Cement	1.6.4.2													
			Aggregates	1.6.4.3													
			Compressive Strength Tests	3.13.3.3	G												
			Chloride Ion Concentration	3.13.3.5													
			Air Content	3.13.3.4													
			Slump Tests	3.13.3.1													
			Water	2.3.2													
			SD-07 Certificates														
			Reinforcing Bars	2.6.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		03 30 00	VOC Content for Form Release	1.6.3.3													
			Agents, Curing Compounds, and														
			Concrete Penetrating Sealers														
			Safety Data Sheets	1.6.3.4													
			Forest Stewardship Council	1.8.2													
			(FSC) Certification														
			Field Testing Technician and	1.6.7.2													
			Testing Agency														
			SD-08 Manufacturer's Instructions														
			Liquid Chemical Floor Hardeners	2.4.3.1													
			and Sealers														
			Curing Compound	2.4.1													
		05 50 13	SD-02 Shop Drawings														
			Floor Gratings	2.3	G												
			Bollards	2.4	G												
			Angles and Plates	2.5	G												
			Roof Hatches	2.6	G												
			SD-03 Product Data														
			Floor Gratings	2.3	G												
			Roof Hatches	2.6	G												
			Certified Mill	2.1	G												
		05 51 00	SD-02 Shop Drawings														
			Iron and Steel Hardware	2.1	G												
			Steel Shapes, Plates, Bars, and	2.1	G												
			Strips														
			Metal Stair System	2.2.1	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		05 51 00	SD-03 Product Data														
			Concrete Inserts	2.3.2	G												
			Protective Coating	2.2.3	G												
			Steel Stairs	2.3.1	G												
			SD-06 Test Reports														
			Mill Test Reports	2.5	G												
			SD-07 Certificates														
			Welding Procedures	1.3.1	G												
			Welder Qualification	1.3.1	G												
			SD-08 Manufacturer's Instructions														
			Protective Coating	2.2.3	G												
		05 51 33	SD-02 Shop Drawings														
			Ladders	2.3													
			SD-03 Product Data														
			Ladders	2.3													
			Ladder Safety Devices	2.3.2													
			SD-07 Certificates														
			Fabricator Certification for Ladder	1.3													
			Assembly														
		05 52 00	SD-02 Shop Drawings														
			Fabrication Drawings	1.2.1	G												
			Iron and Steel Hardware	3.2	G												
			Steel Shapes, Plates, Bars and	3.2	G												
			Strips														
			SD-03 Product Data														
			Concrete Inserts	2.2.5	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		05 52 00	Protective Coating	2.1.2	G												
			Steel Railings and Handrails	2.2.7	G												
			Anchorage and Fastening Systems	3.2	G												
			SD-06 Test Reports														
			Welding Inspections	1.4.1	G												
			Mill Test Reports	2.3	G												
			SD-07 Certificates														
			Welder Qualification	1.4.2	G												
			SD-08 Manufacturer's Instructions														
			Installation Instructions	3.2													
		05 59 10	SD-01 Preconstruction Submittals														
			Assembly Tests	2.6.2	G												
			Acceptance Testing	3.3	G												
			SD-02 Shop Drawings														
			Detail Drawings	1.4.3	G												
			SD-03 Product Data														
			Wheel Assemblies	2.1.7	G												
			Materials List	1.4.3	G												
			Welding	2.4	G												
			Welding of Aluminum	2.4.2	G												
			Steel Welding Repairs	2.4.4	G												
			SD-07 Certificates														
			Welder Qualifications	1.4.1													
			Welding of Aluminum	2.4.2													
		07 11 13	SD-07 Certificates														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		07 11 13	Materials	1.3													
		07 12 00	SD-03 Product Data														
			Manufacturer's Standard Details	1.3	G												
			Protection Board	2.6	G												
			Membrane Fabric	2.3	G												
			Reinforcing Fabric	3.3.5.2	G												
			SD-08 Manufacturer's Instructions														
			Installation Instructions	3.3.5.1													
		07 14 00	SD-03 Product Data														
			Fluid-Applied Membrane	2.1													
			Membrane Primer	2.2													
			Elastomeric Sheet	2.7													
			Solvent	3.3													
			Moisture Meter	3.4.1													
			Bond Breaker	2.6													
			SD-11 Closeout Submittals														
			Warranty	1.5													
			Information Card	3.6													
			Maintenance and Repair Training	3.5													
		07 21 13	SD-02 Shop Drawings														
			Manufacturer's Standard Details	1.3	G												
			SD-03 Product Data														
			Block or Board Insulation	2.1	G												
			Accessories	2.3	G												
			SD-08 Manufacturer's Instructions														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		07 21 13	Insulation Installation and Handling	3.3.1													
			Adhesive	2.3.1													
		07 21 16	SD-03 Product Data														
			Blanket Insulation	2.1	G												
			Vapor Retarder	2.3													
			Pressure Sensitive Tape	2.4													
			Accessories	2.5													
			SD-08 Manufacturer's Instructions														
			Insulation	3.3.1													
		07 60 00	SD-02 Shop Drawings														
			Exposed Sheet Metal	2.2.1	G												
			Gutters	3.1.13	G												
			Downspouts	3.1.14	G												
			Gravel Stops and fascia	2.2.1	G												
			Base Flashing	3.1.10	G												
			Counterflashing	3.1.11	G												
			Copings	3.1.15	G												
			SD-04 Samples														
			Finish Samples	1.4.2	G												
			SD-07 Certificates														
			Warranty on Finishes	2.2.5.1	G												
			SD-08 Manufacturer's Instructions														
			Instructions for Installation	1.4.3	G												
			Quality Control Plan	3.5	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		07 60 00	SD-10 Operation and Maintenance Data														
			Cleaning and Maintenance	1.4.3	G												
		07 84 00	SD-02 Shop Drawings														
			Firestopping System	2.1	G												
			SD-03 Product Data														
			Firestopping Materials	2.2	G												
			SD-06 Test Reports														
			Inspection	3.3	G												
			SD-07 Certificates														
			Inspector Qualifications	1.5.2													
			Firestopping Materials	2.2													
		07 92 00	SD-03 Product Data														
			Sealants	2.1	G												
			Primers	2.2	G												
			Bond Breakers	2.3	G												
			Backstops	2.4	G												
			Caulking	2.5	G												
			Cleaning Solvents	2.6	G												
		08 51 23	SD-02 Shop Drawings														
			Windows	2.2													
			SD-03 Product Data														
			Steel Framing Materials	2.1.1													
			Hardware	2.5.3													
			Hardware Materials	2.5.3.1													
			Fasteners	2.5.4													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		08 51 23	Accessories	2.5													
			Operators	2.8.1.1													
			Screens	2.9													
			SD-04 Samples														
			Color Coating	2.7.2	G												
			SD-06 Test Reports														
			Air Infiltration	1.4.1													
			Water Infiltration	1.4.1													
			Minimum Condensation	1.4.2													
			Resistance Factor														
			Resistance to Forced Entry	1.4.2													
			Standard Airblast Test	1.4.2	G												
			SD-10 Operation and Maintenance														
			Data														
			Windows	2.2	G												
		08 71 00	SD-02 Shop Drawings														
			Manufacturer's Detail Drawings	1.3	G												
			Hardware Schedule	1.5	G												
			Keying System	2.2.6	G												
			SD-03 Product Data														
			Hardware Items	2.2	G												
			SD-08 Manufacturer's Instructions														
			Installation	3.1													
			SD-10 Operation and Maintenance														
			Data														
			Hardware Schedule	1.5	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		08 71 00	SD-11 Closeout Submittals														
			Key Bitting	1.6.1													
		08 81 00	SD-03 Product Data														
			Insulating Glass	2.2													
			Glazing Accessories	1.3													
			SD-04 Samples														
			Insulating Glass	2.2													
			SD-07 Certificates														
			Insulating Glass	2.2													
			SD-08 Manufacturer's Instructions														
			Setting and Sealing Materials	2.3													
			Glass Setting	3.2													
			SD-11 Closeout Submittals														
			Insulated Glass Units	1.6.1													
		08 91 00	SD-02 Shop Drawings														
			Wall Louvers	1.4													
			SD-03 Product Data														
			Metal Wall Louvers	2.2													
		09 22 00	SD-03 Product Data														
			Metal Support Systems	2.1													
		09 29 00	SD-03 Product Data														
			Accessories	2.1.5													
			Gypsum Board	2.1.1													
			SD-07 Certificates														
			Asbestos Free Materials	2.1	G												
			SD-08 Manufacturer's Instructions														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		09 29 00	Safety Data Sheets	2.1													
			SD-10 Operation and Maintenance Data														
			Manufacturer Maintenance Instructions	2.1													
		09 51 00	SD-03 Product Data														
			Acoustical Units	2.2													
			SD-04 Samples														
			Acoustical Units	2.2													
		09 69 13	SD-02 Shop Drawings														
			Detailed Installation Drawings	2.2.2	G												
			Fabrication Drawings	2.2.1	G												
			SD-03 Product Data														
			Access Flooring System	2.1	G												
			Indoor Air Quality For Pedestal Adhesive	2.1.5	S												
			Indoor Air Quality For Adhesives	2.2.7	S												
			SD-04 Samples														
			Floor Panels	2.2													
			Floor Covering	2.2.4	G												
			Panel Support System	2.3													
			Accessories	2.2.5	G												
			Cut Outs	2.6	G												
			Seismic Calculations	2.1.7													
			SD-06 Test Reports														
			Factory Tests	2.4													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		09 69 13	Concentrated Load	2.1.1													
			Uniform Live Load	2.1.1													
			Rolling Load	2.1.1													
			Impact Load	2.1.1													
			Ultimate Load	2.1.1													
			Stringer Load	2.1.3													
			Pedestal Axial Load	2.1.4													
			Bonding Strength of Pedestal Adhesive	2.1.5													
			SD-07 Certificates														
			Compliance with ICC-ES AC308	2.1													
			Compliance with ICC IBC	2.1													
			Certificate of Compliance	2.1													
			Qualification of Manufacturer	1.4.1													
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	3.3	G												
			SD-11 Closeout Submittals														
			Lifting Device	2.2.8													
			Warranty	1.6													
		09 90 00	SD-02 Shop Drawings														
			Piping Identification	3.8													
			SD-03 Product Data														
			Coating	2.1	G												
			Product Data Sheets	2.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		09 90 00	SD-04 Samples														
			Color	2.2	G												
			SD-08 Manufacturer's Instructions														
			Mixing	2.1													
			Manufacturer's Safety Data	1.8.1													
			Sheets														
			SD-10 Operation and Maintenance														
			Data														
			Coatings	2.1	G												
		09 97 13.17	SD-01 Preconstruction Submittals														
			Contract Errors, Omissions, and	1.4.1													
			Other Discrepancies														
			Corrective Action Procedures	1.4.2.1													
			Corrective Action Request (CAR)	1.4.2.2													
			Form														
			Coatings Work Plan	1.4.3													
			Inspection Report Form	3.9.1.2													
			SD-05 Design Data														
			Environmental Control System	1.4.4.1													
			SD-06 Test Reports														
			Coatings Qualification Test	1.4.5.1													
			Reports														
			Joint Sealant Qualification Test	1.4.5.2													
			Reports														
			Non-Metallic Abrasive	1.4.5.4	G												
			Qualification Test Reports														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		09 97 13.17	Ferrous Metallic Abrasive	1.4.5.3													
			Qualification Test Reports														
			Coating Field Test Reports	3.1.3													
			Abrasive Field Test Reports	3.1.4													
			Recycled Ferrous Metallic	1.4.5.5													
			Abrasive Field Test Reports (Daily														
			and Weekly)														
			Daily Inspection Reports	3.9.1.3													
			SD-07 Certificates														
			Qualifications of Certified	1.4.6.1													
			Industrial Hygienist (CIH)														
			Qualifications of Certified	1.4.6.2													
			Protective Coatings Specialist														
			(PCS)														
			Qualifications of Coatings	1.4.6.3													
			Inspection Company														
			Qualifications of Quality	1.4.6.4													
			Assurance Coatings Inspector														
			Qualifications of Coatings	1.4.6.5													
			Contractors														
			Qualifications of Individuals	1.4.6.6													
			Performing Abrasive Blasting														
			Qualifications of Individuals	1.4.6.7													
			Applying Coatings														
			Qualifications of Testing	1.4.6.8													
			Laboratory for Coatings														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		09 97 13.17	Qualifications of Testing Laboratory for Abrasive	1.4.6.9													
			Coating Materials Certificate of Conformance	1.4.6.10													
			Joint Sealant Materials Certificate of Conformance	1.4.6.11													
			Joint Sealant Compatibility	1.4.6.12													
			Non-Metallic Abrasive Certificate of Conformance	1.4.6.14													
			Ferrous Metallic Abrasive Certificate of Conformance	1.4.6.13													
			SD-08 Manufacturer's Instructions														
			Joint Sealant Instructions	1.5.1													
			Coating System Instructions	1.5.2													
			SD-11 Closeout Submittals														
			Disposal of Used Abrasive	3.6.6	G												
			Inspection Logbook	3.9.1.4	G												
			Corrective Action Log	1.4.2.3	G												
		09 97 13.27	SD-05 Design Data														
			Containment System	1.4.4.1													
			SD-06 Test Reports														
			Joint Sealant Qualification Test Reports	1.4.5.1													
			Coatings Qualification Test Reports	1.4.5.2													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		09 97 13.27	Metallic Abrasive Qualification Test Reports	1.4.5.3													
			Coating Sample Test Reports	3.1.3													
			Abrasive Sample Test Reports	3.1.4													
			Inspection Report Forms	3.8.2.2													
			Daily Inspection Reports	3.8.2.3													
			Recycled Metallic Abrasive Field Test Reports (Daily and Weekly)	1.4.5.4													
			SD-07 Certificates														
			Contract Errors, Omissions, and Other Discrepancies	1.4.1													
			Corrective Action Procedures	1.4.2.1													
			Coating Work Plan	1.4.3													
			Qualifications of Certified Industrial Hygienist (CIH)	1.4.6.1													
			Qualifications Of Individuals Performing Abrasive Blasting	1.4.6.5													
			Qualifications of Certified Protective Coatings Specialist (PCS)	1.4.6.2													
			Qualifications of Coating Inspection Company	1.4.6.3													
			Qualifications of QC Specialist Coating Inspector	1.4.6.4													
			Qualifications of Testing Laboratory for Coatings	1.4.6.6													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		09 97 13.27	Qualifications of Testing Laboratory for Abrasive	1.4.6.7													
			Qualifications of Coating Contractors	1.4.6.8													
			Joint Sealant Materials	1.4.6.9													
			Coating Materials	1.4.6.10													
			Coating System Component Compatibility	1.4.6.11													
			Non-metallic Abrasive	1.4.6.12													
			Metallic Abrasive	1.4.6.13													
			SD-08 Manufacturer's Instructions														
			Joint Sealant Instructions	1.5.1													
			Coating System Instructions	1.5.2													
			SD-11 Closeout Submittals														
			Disposal of Used Abrasive	3.5.6													
			Inspection Logbook	3.8.2.4	G												
		23 05 93	SD-01 Preconstruction Submittals														
			Independent TAB Agency and Personnel Qualifications	1.5.1	G												
			TAB Design Review Report	1.5.3.2	G												
			SD-02 Shop Drawings														
			TAB Schematic Drawings and Report Forms	1.3.2	G												
			SD-03 Product Data														
			Equipment and Performance Data	1.3	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		23 05 93	TAB Related HVAC Submittals	1.5.1.3	G												
			SD-06 Test Reports														
			Completed Pre-Final DALT Report	3.3.5	G												
			Certified Final DALT Report	3.3.8	G												
			Prerequisite HVAC Work Checkout List	1.5.3.3	G												
			SD-07 Certificates														
			Independent TAB Agency and Personnel Qualifications	1.5.1	G												
			DALT and TAB Submittal and Work Schedule	1.5.3.2	G												
			TAB Pre-Field Engineering Report	1.5.3.3	G												
			Instrument Calibration Certificates	1.5.5	G												
			DALT and TAB Procedures Summary	3.7	G												
			Completed Pre-Final DALT Work Checklist	3.7	G												
			Advance Notice of Pre-Final DALT Field Work	3.3.2	G												
		23 31 13.00 40	SD-01 Preconstruction Submittals														
			Material, Equipment, and Fixture Lists	Part 2	G												
			SD-03 Product Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		23 31 13.00 40	Galvanized Steel Ductwork	2.3.1	G												
			Materials														
			Brazing Materials	2.3.2													
			Mill-Rolled Reinforcing and	2.3.3													
			Supporting Materials														
			Turning Vanes	2.2.3	G												
			Flexible Connectors	2.2.5	G												
			Flexible Duct Materials	2.2.8													
			Power Operated Dampers	2.2.10	G												
			Manual Volume Dampers	2.2.9	G												
			SD-06 Test Reports														
			Ductwork Leakage Tests	3.4.1	G												
			SD-07 Certificates														
			Listing of Product Installations	1.3													
			Brazing Materials	2.3.2													
			Mill-Rolled Reinforcing and	2.3.3													
			Supporting Materials														
			Dampers	2.2.4													
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.5.1	G												
			Manuals														
			Power Operated Dampers	2.2.10	G												
			SD-11 Closeout Submittals														
			Record Drawings	3.5.2	G												
		23 34 23.00 40	SD-02 Shop Drawings														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		23 34 23.00 40	Shop Drawings	2.1.1	G												
			Installation Drawings	3.1	G												
			SD-03 Product Data														
			Housing	2.2.1	G												
			Fan	2.2.2	G												
			Motor	2.2.3	G												
			Bases	2.2.4	G												
			Dampers	2.2.5	G												
			Screens	2.2.6	G												
			SD-06 Test Reports														
			Final Test Reports	3.2.3	G												
			SD-11 Closeout Submittals														
			Record Drawings	3.3	G												
		23 81 00	SD-03 Product Data														
			Spare Parts	3.7.1													
			Posted Instructions	3.4													
			System Performance Tests	3.6													
			Training	3.4	G												
			Inventory	1.5													
			Supplied Products	2.1													
			Manufacturer's Standard Catalog	2.2	G												
			Data														
			SD-06 Test Reports														
			Refrigerant Tests, Charging, and	3.5	G												
			Start-Up														
			System Performance Tests	3.6	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
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		23 81 00	SD-07 Certificates														
			Service Organizations	3.7.2													
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.4	G												
			Manuals														
			SD-11 Closeout Submittals														
			Ozone Depleting Substances	2.2.2.3													
		25 05 11	SD-01 Preconstruction Submittals														
			Device Account Lock Exception	3.3.2	G												
			Request														
			Multiple Ethernet Connection	3.2.2.2	G												
			Device Request														
			Contractor Computer	1.8.1.6	G												
			Cybersecurity Compliance														
			Statements														
			Proposed STIG and SRG	1.6.1	G												
			Applicability Report														
			SD-02 Shop Drawings														
			Network Communication Report	1.6.2	G												
			Cybersecurity Riser Diagram	1.6.5	G												
			SD-03 Product Data														
			Control System Cybersecurity	1.6.7	G												
			Documentation														
			SD-06 Test Reports														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		25 05 11	Control System Cybersecurity Testing Procedures	3.12.1	G												
			Control System Cybersecurity Testing Report	3.12.3	G												
			SD-07 Certificates														
			Software Licenses	1.7	G												
			SD-11 Closeout Submittals														
			Password Change Summary Report	3.4.3.5	G												
			Enclosure Keys	3.3.5	G												
			Software and Configuration Backups	1.6.4	G												
			Auditing Front End Software	3.5.3	G												
			System Maintenance Tool Software	3.9	G												
			Control System Scanning Tools	3.10.2	G												
			STIG, SRG and Vendor Guide	1.6.6	G												
			Compliance Result Report														
			Control System Inventory Report	1.6.3	G												
		26 08 00	SD-06 Test Reports														
			Acceptance Tests and Inspections	3.1	G												
			SD-07 Certificates														
			Qualifications	1.4.1	G												
			Acceptance Test and Inspections Procedure	1.4.3	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		26 12 19.10	SD-02 Shop Drawings														
			Pad-mounted Transformer	1.5.1	G												
			Drawings														
			SD-03 Product Data														
			Pad-mounted Transformers	2.2	G												
			SD-06 Test Reports														
			Acceptance Checks and Tests	3.6.1	G												
			SD-07 Certificates														
			Transformer Efficiencies	2.2.2.1	G												
			SD-09 Manufacturer's Field														
			Reports														
			Transformer Test Schedule	2.7.1	G												
			Design Tests	2.7.2	G												
			Routine and Other Tests	2.7.3	G												
			SD-10 Operation and Maintenance														
			Data														
			Transformer(s)	1.6.1	G												
		26 13 00	SD-02 Shop Drawings														
			Switchgear Drawings	1.5.1	G												
			SD-03 Product Data														
			Electronic Overcurrent Control	1.5.3	G												
			Curves														
			SF6/High-Firepoint Fluid	1.6.1	G												
			Insulated Pad-mounted Switchgear														
			SF6/High-Firepoint Fluid	2.1	G												
			Insulated Pad-mounted Switchgear														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		26 13 00	Insulated High-Voltage Connectors	2.2	G												
			SD-06 Test Reports														
			Acceptance Checks and Tests	3.4.1	G												
			SD-07 Certificates														
			Paint Coating System	1.5.2	G												
			SD-09 Manufacturer's Field Reports														
			Switchgear Design and Production Tests	2.4.1	G												
			SD-10 Operation and Maintenance Data														
			SF6/High-Firepoint Fluid	1.6.1	G												
			Insulated Pad-mounted Switchgear														
			SF6/High-Firepoint Fluid	2.1	G												
			Insulated Pad-mounted Switchgear														
		26 20 00	SD-02 Shop Drawings														
			Panelboards	2.10	G												
			Transformers	2.12	G												
			Motor Control Centers	2.16	G												
			Marking Strips	3.1.8.1	G												
			SD-03 Product Data														
			Receptacles	2.9	G												
			Circuit Breakers	2.10.3	G												
			Switches	2.8	G												
			Transformers	2.12	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		26 20 00	Motor Controllers	2.14	G												
			Combination Motor Controllers	2.16.2	G												
			Manual Motor Starters	2.15	G												
			Metering	2.24	G												
			Primary Bonding Busbar	2.19.3	G												
			Surge Protective Devices	2.25	G												
			Emergency Fuel Shutoff	2.26	G												
			Pushbutton														
			Emergency Fuel Shutoff (EFSO)	2.27	G												
			Panel And Components														
			SD-06 Test Reports														
			600-volt Wiring Test	3.5.2	G												
			Grounding System Test	3.5.5	G												
			Transformer Tests	3.5.3	G												
			Ground-fault Receptacle Test	3.5.4	G												
			SD-09 Manufacturer's Field														
			Reports														
			Transformer Factory Tests	2.29.1													
			SD-10 Operation and Maintenance														
			Data														
			Electrical Systems	1.6.1	G												
			Metering	2.24	G												
		26 27 14.00 20	SD-02 Shop Drawings														
			Installation Drawings	1.3.1	G												
			SD-03 Product Data														
			Electricity Meters	2.1	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		26 27 14.00 20	Current Transformer	2.1.2	G												
			SD-06 Test Reports														
			Acceptance Checks and Tests	3.2.1	G												
			System Functional Verification	3.2.2	G												
			Meter Configuration Report	3.2.1	G												
			SD-10 Operation and Maintenance														
			Data														
			Electricity Meters and	1.4.1	G												
			Accessories														
			SD-11 Closeout Submittals														
			System Functional Verification	3.2.2	G												
		26 32 15.00	SD-02 Shop Drawings														
			Engine-Generator Set and	2.1.1	G												
			Auxiliary Equipment														
			Auxiliary Systems	1.4.6	G												
			Detailed Drawings	1.4.5	G												
			Acceptance	3.14	G												
			SD-03 Product Data														
			Engine-Generator Set	2.1.1	G												
			Efficiencies														
			Emissions	2.12	G												
			filters	2.7.2	G												
			special tools	2.14	G												
			Engine-Generator Parameter	2.1.1													
			Schedule														
			Heat Exchanger	2.8.2													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS SIF CATION OR A/E REVIEW WR	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
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		26 32 15.00	Generator	2.15													
			Spare Parts	1.7.2													
			Onsite Training	3.10													
			Experience	1.4.3													
			Field Engineer	1.4.4													
			General Installation	3.1													
			Exciter	2.16													
			SD-05 Design Data														
			Performance Criteria	2.15													
			Sound Limitations	2.9	G												
			Integral Main Fuel Storage Tank	2.6.4													
			Power Factor	3.4.1.2													
			Heat Exchanger	2.8.2													
			Time-Delay on Alarms	2.19.5													
			Cooling System	2.8													
			Battery Charger	2.13.3.2													
			Capacity Calculations for Engine-Generator Set	2.1.1	G												
			Brake Mean Effective Pressure (BMEP) Calculations	2.1.1	G												
			Torsional Vibration Stress Analysis Computations	2.1.1	G												
			Capacity Calculations for Batteries	2.1.1	G												
			Turbocharger Load Calculations	2.1.1	G												
			SD-06 Test Reports														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		26 32 15.00	Performance Tests	3.4.9													
			Factory Inspection and Tests	2.25													
			Factory Tests	2.25.2													
			Onsite Inspection and Tests	3.4	G												
			Acceptance Checks and Tests	3.7.1	G												
			Functional Acceptance Tests	3.8.2	G												
			Maintenance Procedures	3.10	G												
			Operation and Maintenance Manuals	3.10	G												
			Inspections	3.4.3	G												
			Functional Acceptance Test Procedure	3.7.5	G												
			SD-07 Certificates														
			Cooling System	2.8													
			Reliability and Durability	2.1.5													
			Fuel System Certification	1.4.7	G												
			Start-Up Engineer	3.6	G												
			Instructor's Qualification Resume	3.9.1	G												
			Engine Emission Limits	2.1.2.1	G												
			Sound Limitations	2.9													
			Current Balance	2.15.1													
			Materials and Equipment	2.4													
			Factory Inspection and Tests	2.25													
			Engine Tests	3.4.4	G												
			Generator Tests	3.4.5	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		26 32 15.00	Assembled Engine-Generator Set Tests	3.4.6	G												
			SD-10 Operation and Maintenance Data														
			Preliminary Assembled Operation and Maintenance Manuals	3.7.4	G												
			Training Plan	3.9.2	G												
		26 41 00	SD-06 Test Reports														
			Lightning Protection and Grounding System Test Plan	1.4.2	G												
			Grounding Systems Testing	3.2.2	G												
			SD-07 Certificates														
			Component UL Listed and Labeled	1.4.1	G												
			Lightning Protection System Inspection Certificate	1.4.3	G												
			Roof Manufacturer's Warranty	3.1.1.1	G												
		26 42 17	SD-01 Preconstruction Submittals														
			Preconstruction Survey	1.8.2													
			SD-02 Shop Drawings														
			Drawings	2.1.2.6	G AE												
			Anode junction boxes, bonding boxes, and test stations	2.3.4													
			Joint bonds	3.5.2													
			Contractor's Modifications	2.1.2.8	G AE												
			SD-03 Product Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		26 42 17	Qualifications	1.6.2													
			Equipment	1.5.1	G AE												
			Anodes	1.5.1	G AE												
			Anode junction boxes, bonding boxes, and test stations	2.3.4													
			Isolation flange kits	2.5.9													
			Wires	2.1.2.2													
			Cable and wire	2.4.3													
			Shunts	2.3.2													
			Permanent reference electrodes	2.4.6	G AE												
			Spare Parts	1.5.1													
			SD-06 Test Reports														
			Tests and Measurements	3.6.1	G AE												
			Contractor's Modifications	2.1.2.8	G AE												
			SD-10 Operation and Maintenance Data														
			Cathodic Protection System	3.6	G AE												
			Cathodic Protection System	3.6	G AE												
			Training Course	1.5.1	G AE												
			SD-11 Closeout Submittals														
			Initial Cathodic Protection System Field Testing	3.6.1.12	G AE												
			One Year Warranty Period	3.6.1.14	G AE												
			Cathodic Protection System Field Test Report														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS / FIC / REVIEW / WORK	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		26 42 17	Final Cathodic Protection System	3.6.1.15	G AE												
			Field Test Report														
		26 51 00	SD-02 Shop Drawings														
			Luminaire Drawings	1.5.1	G												
			Lighting Control System One-Line	1.7.2	G												
			Diagram														
			Sequence of Operation for	2.5.1	G												
			Lighting Control System														
			SD-03 Product Data														
			Luminaires	2.2	G												
			Light Sources	2.3	G												
			LED Drivers	2.4	G												
			Luminaire Warranty	1.6.1	G												
			Lighting Controls Warranty	1.6.2	G												
			Lighting Contactor	2.5.2	G												
			Switches	2.5.3.1	G												
			Occupancy/Vacancy Sensors	2.5.3.2	G												
			Photosensors	2.5.3.3	G												
			Power Packs	2.5.3.2.4	G												
			Exit Signs	2.6.1	G												
			Emergency Drivers	2.6.2	G												
			SD-05 Design Data														
			Luminaire Design Data	1.5.2	G												
			SD-06 Test Reports														
			ANSI/IES LM-79 Test Report	1.5.3	G												
			ANSI/IES LM-80 Test Report	1.5.4	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		26 51 00	ANSI/IES TM-21 Test Report	1.5.5	G												
			ANSI/IES TM-30 Test Report	1.5.6	G												
			Occupancy/Vacancy Sensor	3.2.1.1	G												
			Verification Test														
			Photosensor Verification Test	3.2.1.1	G												
			SD-10 Operation and Maintenance														
			Data														
			Lighting System	1.7.1	G												
			Lighting Control System	1.7.2	G												
		26 56 00	SD-02 Shop Drawings														
			Luminaire Drawings	1.5.1.1	G												
			Pole Drawings	1.5.1.2	G												
			Control System One-Line	1.8.1	G												
			Diagram														
			SD-03 Product Data														
			Luminaires	2.2	G												
			Light Sources	2.3	G												
			LED Drivers	2.4	G												
			Luminaire Warranty	1.7.1	G												
			Lighting Controls Warranty	1.7.2	G												
			Pole Warranty	1.7.3	G												
			Photosensors	2.5.2.1	G												
			Lighting Contactor	2.5.2.2	G												
			Poles	2.6	G												
			Brackets	2.6.2													
			SD-05 Design Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
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		26 56 00	Luminaire Design Data	1.5.2	G												
			SD-06 Test Reports														
			ANSI/IES LM-79 Test Report	1.5.3	G												
			ANSI/IES LM-80 Test Report	1.5.4	G												
			ANSI/IES TM-21 Test Report	1.5.5	G												
			SD-08 Manufacturer's Instructions														
			Poles	2.6													
			SD-10 Operation and Maintenance														
			Data														
			Lighting System	1.8.1	G												
		27 10 00	SD-02 Shop Drawings														
			Telecommunications Drawings	1.6.1.1	G												
			Telecommunications Space	1.6.1.2	G												
			Drawings														
			SD-03 Product Data														
			Telecommunications Cabling	2.3	G												
			Patch Panels	2.4.5	G												
			Telecommunications	2.5	G												
			Outlet/Connector Assemblies														
			Equipment Support Frame	2.4.2	G												
			Connector Blocks	2.4.3	G												
			SD-06 Test Reports														
			Telecommunications Cabling	3.5.1	G												
			Testing														
			SD-07 Certificates														
			Telecommunications Contractor	1.6.2.1	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		27 10 00	Key Personnel	1.6.2.2	G												
			Manufacturer Qualifications	1.6.2.3	G												
			Test Plan	1.6.3	G												
			SD-09 Manufacturer's Field Reports														
			Factory Reel Tests	2.9.1	G												
			SD-10 Operation and Maintenance Data														
			Telecommunications Cabling and Pathway System	1.10.1	G												
			SD-11 Closeout Submittals														
			Record Documentation	1.10.2	G												
		28 31 70	SD-01 Preconstruction Submittals														
			Qualified Fire Protection Engineer (QFPE)	1.3.2	G												
			Fire alarm system designer	1.9.2.1	G												
			Supervisor	1.9.2.2	G												
			Technician	1.9.2.3	G												
			Installer	1.9.2.4	G												
			Test Technician	1.9.2.5	G												
			Fire Alarm System Site-Specific Software Acknowledgement	1.7	G												
			SD-02 Shop Drawings														
			Nameplates	1.9.1.3	G												
			Instructions	2.2.4	G												
			Wiring Diagrams	1.9.1.4	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
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		28 31 70	System Layout	1.9.1.5	G												
			Notification Appliances	1.9.1.6	G												
			Initiating devices	1.9.1.7	G												
			Battery Power	1.9.1.8	G												
			Voltage Drop Calculations	1.9.1.9	G												
			SD-03 Product Data														
			Fire Alarm Control Unit (FACU)	2.3	G												
			Manual Stations	2.4	G												
			Smoke Detectors	2.5	G												
			Heat Detectors	2.6	G												
			Notification Appliances	1.9.1.6	G												
			Batteries	2.9.1	G												
			Battery Chargers	2.9.2	G												
			Supplemental Notification	2.9.1.1	G												
			Appliance Circuit Panels														
			Auxiliary Power Supply Panels	2.9.1.1	G												
			Surge Protective Devices	2.10	G												
			Alarm Wiring	2.10	G												
			Back Boxes and Conduit	3.3.4	G												
			Ceiling Bridges	3.2.6	G												
			Terminal Cabinets	3.3.2	G												
			Automatic Fire Alarm	2.12	G												
			Transmitters														
			Radio Transmitter and Interface	2.12.1	G												
			Panels														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		28 31 70	Environmental Enclosures or Guards	2.13	G												
			Document Storage Cabinet	3.11.3	G												
			SD-06 Test Reports														
			Test Procedures	3.7.1	G												
			SD-07 Certificates														
			Verification of Compliant Installation	3.7.2.1	G												
			Request for Government Final Test	3.7.2.2	G												
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance (O&M) Instructions	3.9	G												
			Instruction of Government Employees	3.10	G												
			SD-11 Closeout Submittals														
			As-Built Drawings	1.9.1.12													
			Spare Parts	1.11.1													
		31 00 00	SD-01 Preconstruction Submittals														
			Excavation and Trenching Plan	3.1.3	G												
			Borrow Plan	3.2	G												
			Dewatering Work Plan	3.1.2.2	G												
			Disposition of Surplus Materials	3.9	G												
			Preconstruction Meeting	1.6.4	G												
			SD-03 Product Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		31 00 00	Flowable Fill Mix Design	2.2	G												
			SD-06 Test Reports														
			Dewatering Performance Records	3.1.2.2	G												
			Material Test Report	3.10	G												
			Borrow Site Testing	3.2.2	G												
			Pipe Inspection Report	3.6.1.6	G												
			Geotechnical Evaluation Report	1.6.1	G												
		32 05 33	SD-01 Preconstruction Submittals														
			Integrated Pest Management Plan	2.4	G												
			SD-03 Product Data														
			Fertilizer	2.1	G												
			Mulches Topdressing	2.3													
			Organic Mulch Materials	2.3.2													
			SD-07 Certificates														
			Maintenance Inspection Report	3.4.1													
			Plant Quantities	3.4.2	G												
			SD-10 Operation and Maintenance Data														
			Maintenance	1.6													
			SD-11 Closeout Submittals														
			Tree Staking and Guying Removal	3.4.3													
		32 11 23	SD-03 Product Data														
			Plant, Equipment, and Tools	2.4	G												
			SD-06 Test Reports														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		32 11 23	Initial Tests	2.3.1	G												
			In-Place Tests	3.12.1	G												
		32 12 13	SD-06 Test Reports														
			Sampling and Testing	3.7													
		32 12 16.16	SD-02 Shop Drawings														
			Placement Plan	2.1	G												
			SD-03 Product Data														
			Diamond Grinding Plan	2.1.5	G												
			Mix Design	2.4	G												
			Contractor Quality Control	3.1	G												
			SD-04 Samples														
			Aggregates	2.2													
			Asphalt Cement Binder	2.3													
			SD-06 Test Reports														
			Aggregates	2.2	G												
			QC Monitoring	3.1.3.9													
			SD-07 Certificates														
			Asphalt Cement Binder	2.3	G												
			Laboratory Accreditation and Validation	1.5.11													
		32 15 00	SD-03 Product Data														
			Plant, Equipment, and Tools	1.4	G												
			SD-06 Test Reports														
			Initial Tests	2.3.1	G												
			In-Place Tests	3.9.1	G												
		32 16 19	SD-03 Product Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		32 16 19	Concrete	2.1													
			SD-04 Samples														
			Concrete Curbs	3.4.4													
			SD-06 Test Reports														
			Field Quality Control	3.7													
		32 17 23	SD-03 Product Data														
			Surface Preparation Equipment List	2.1.1.1	G												
			Application Equipment List	2.1.2	G												
			Exterior Surface Preparation	3.2													
			Safety Data Sheets	1.3.1	G												
			Reflective media for roads	2.2.3.1	G												
			Waterborne Paint	2.2.1	G												
			SD-06 Test Reports														
			Reflective Media for Roads	2.2.3.1	G												
			Waterborne Paint	2.2.1	G												
			SD-07 Certificates														
			Qualifications	1.3.2	G												
			Reflective Media for Roads	2.2.3.1													
			Waterborne Paint	2.2.1													
			Volatile Organic Compound	1.3.1	G												
			SD-08 Manufacturer's Instructions														
			Waterborne Paint	2.2.1	G												
		32 31 13.53	SD-02 Shop Drawings														
			Fence Installation Drawings	3.2.1	G												
			SD-03 Product Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		32 31 13.53	Fabric	2.1.1.1													
			Posts	2.1.2													
			Post Caps	2.1.2.2													
			Chain Link Braces	2.1.3													
			Line Posts	2.1.3													
			Rails	2.1.3													
			Tension Wire	2.2.1.3													
			Barbed Wire	2.2.1.2													
			Barbed Wire Supporting Arms	2.1.2.2													
			Latches	2.1.4.3													
			Hinges	2.1.4.3													
			Stops	2.1.4.3													
			Keepers	2.1.4.3													
			Rollers	2.1.4.3													
			Padlocks	2.1.5													
			Wire Ties	2.2.1.1													
			SD-07 Certificates														
			Chain Link Fence	2.1.2.1													
			Fabric	2.1.1.1													
			Barbed Wire	2.2.1.2													
			Gate Hardware and Accessories	3.2.1													
			Concrete	2.2.2													
			SD-10 Operation and Maintenance														
			Data														
			Operating and maintenance	3.2.6													
			instructions														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		32 92 19	SD-03 Product Data														
			Fertilizer	2.4													
			SD-06 Test Reports														
			Topsoil Composition Tests	2.2.3													
			SD-07 Certificates														
			Seed	2.1													
			SD-08 Manufacturer's Instructions														
			Erosion Control Materials	2.6													
		33 01 50.55	SD-03 Product Data														
			Cleaning Agents	2.1.1													
			Abrasive for Blasting	2.1.2.1													
			Gasoline-Oil-Resisting Rubber	2.2													
			Gloves and Boots														
			Cotton Coveralls and Hard Hat	2.2													
			Respiratory Protective Equipment	2.2													
			Disinfectant	2.2													
			SD-06 Test Reports														
			Blasting Abrasive	1.5.12.1													
			Tank Contents	1.5.12.1													
			Cleaning Test Panel Results	1.5.12.1													
			Monitoring Results	1.5.12.2	G												
			SD-07 Certificates														
			Qualifications of Marine Chemist	1.5.11.1													
			Qualifications of Certified	1.5.11.2													
			Industrial Hygienist (CIH)														
			Testing Laboratory	1.5.11.3													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		33 01 50.55	Safety Plan	1.5.11.4													
			Training Certification	1.5.11.4													
			Work Plan	1.5.11.5													
			Hazardous Waste Disposal Plan	1.5.11.6													
			Tank Certification of Safety	1.5.11.7													
			Exhaust	1.7.1													
			Exhaust	1.7.1													
			Respiratory Protective Equipment	2.2													
			Breathing-Air Supply Source	2.2													
			Combustible Gas Indicator	2.2													
			Oxygen Meter	2.2													
			Velometers	2.2													
			Lighting	2.2													
			First Aid Kit	2.2													
			SD-08 Manufacturer's Instructions														
			Cleaning Agents	2.1.1													
			SD-11 Closeout Submittals														
			Safety Permits	1.5.3													
		33 05 23.13	SD-01 Preconstruction Submittals														
			Qualifications	1.3.1	G												
			Horizontal Directional Drilling	1.3.3	G												
			Plan														
			SD-03 Product Data														
			Pipe	2.2.1	G												
			Drilling Fluids	2.2.2	G												
			Additives	2.2.3	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		33 05 23.13	SD-05 Design Data														
			Secondary Containment Plan	3.2.2	G												
			SD-06 Test Reports														
			Soil Test Data	3.1													
			SD-07 Certificates														
			Drill Rod	2.1.1													
			Pipe Welder Qualifications	1.3.5													
			SD-11 Closeout Submittals														
			Record Drawings	3.4													
			Complete Work Logs of Guided	3.4													
			Directional Drill Operations														
		33 08 55.01	SD-01 Preconstruction Submittals														
			Copies of API RP 1595	1.3.2	G												
			Existing System Cleanliness	1.3.3	G												
			Confirmation Plan														
			System Start-Up Plan	1.3.5	G												
			Performance Testing Plan	1.3.1	G												
			Fuel Provisioning Plan	1.3.1	G												
			SD-06 Test Reports														
			Final Reports	1.5.1	G												
			Cathodic Protection Reports	3.2.5	G												
			Equipment Tests	3.9	G												
			Piping Flushing Checklist	3.6.3	G												
			Piping Cleaning Checklist	3.7.4	G												
			SD-07 Certificates														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		33 08 55.01	Fuel Testing Laboratory	1.3.4	G												
			Qualifications														
			Certification (Ready for Start-Up	1.3.8	G												
			and Performance Testing)														
			Certificate of Calibration	2.2.1	G												
			Disposal Of Waste Materials	1.8													
			SD-11 Closeout Submittals														
			Certification of Entire System	1.6.1													
		33 08 55.02	SD-01 Preconstruction Submittals														
			Existing System Cleanliness	1.3.2	G												
			Confirmation Plan														
			System Start-Up Plan	1.3.4	G												
			Performance Testing Plan	1.3.1	G												
			Fuel Provisioning Plan	1.3.1	G												
			SD-06 Test Reports														
			Final Reports	1.5.1	G												
			Cathodic Protection Reports	3.2.3	G												
			Piping Flushing Checklist	3.5.3	G												
			Piping Cleaning Checklist	3.6.3	G												
			SD-07 Certificates														
			Fuel Testing Laboratory	1.3.3	G												
			Qualifications														
			Certification (Ready for Start-Up	1.3.7	G												
			and Performance Testing)														
			Certificate of Calibration	2.2.1	G												
			Disposal Of Waste Materials	1.8													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
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		33 08 55.02	SD-11 Closeout Submittals														
			Certification of Entire System	1.6.1													
		33 09 52	SD-02 Shop Drawings														
			Shop Drawing	3.4.1	G												
			SD-03 Product Data														
			Tools and Spare Parts	1.4													
			Pump Control Panel (PCP) and Components	2.1.1	G												
			Pump Control Panel (PCP) and Components	3.1	G												
			Laptop Computer	2.1.16	G												
			Personal Computer (PC)	2.1.17	G												
			Programmable Logical Controller (PLC) Hardware and Software	2.2	G												
			Programmable Logical Controller (PLC) Hardware and Software	3.2	G												
			Control Wiring Data Lists	3.1.2	G												
			Graphics Display Screen	3.3	G												
			SD-06 Test Reports														
			Certified Pump Control Panel (PCP) Shop Test Report	3.1.3													
			Record of Test	3.4.2													
			SD-07 Certificates														
			Experience and Qualifications	1.5	G												
			Testing Plan	3.4.2	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		33 09 52	Training Plan for Instructing Personnel	3.4.3	G												
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	1.2	G												
		33 11 00	SD-01 Preconstruction Submittals														
			Connections	3.1.1	G												
			SD-02 Shop Drawings														
			Safety Shower Shop Drawing	2.2.9	G												
			SD-03 Product Data														
			Pipe, Fittings, Joints and Couplings	2.1.1	G												
			Valves	2.1.2	G												
			Valve Boxes	2.1.2.4	G												
			Fire Hydrants	2.1.3.1	G												
			Pipe Restraint	2.2.1	G												
			Tapping Sleeves	2.2.3	G												
			Corporation Stops	2.2.8.1	G												
			Backflow Preventer	1.4.2.1.1	G												
			Disinfection Procedures	3.2.3	G												
			SD-06 Test Reports														
			Backflow Preventer Tests	3.3.1.4	G												
			Bacteriological Samples	3.3.1.3	G												
			Leakage Test	3.3.1.2													
			Hydrostatic Test	3.3.1.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		33 11 00	SD-07 Certificates														
			Pipe, Fittings, Joints and Couplings	2.1.1													
			Lining	2.1.1.1.1													
			Lining for Fittings	2.1.1.2.1.2													
			Fire Hydrants	2.1.3.1													
			Backflow Prevention Training Certificate	1.4.2.1.1.1													
			Backflow Certificate	2.1.4													
			SD-08 Manufacturer's Instructions														
			Ductile-Iron Piping	2.1.1.1													
			PVC Piping	2.1.1.2.1.1													
			PVC Piping For Service Lines	2.1.1.2.2													
			Copper Pipe For Service Lines	2.1.1.3													
		33 40 00	SD-03 Product Data														
			Pipe For Culverts And Storm Drains	2.1	G												
			SD-08 Manufacturer's Instructions														
			Placing Pipe and Box Culvert	3.3	G												
			SD-11 Closeout Submittals														
			Post-Installation Inspection Report	3.8.2.1.2	G												
			LID Verification Report	3.8.2.2	G												
		33 52 23.15	SD-01 Preconstruction Submittals														
			Welding Procedure Specifications (WPS)	1.6	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		33 52 23.15	Welder Performance Qualification (WPQ)	1.6	G												
			Procedure Qualification Record (PQR)	1.6	G												
			Welding Operations	3.1	G												
			SD-02 Shop Drawings														
			POL Service Piping	1.6.2	G												
			SD-06 Test Reports														
			Examinations, Inspections and Tests	3.2													
			SD-07 Certificates														
			Qualifications	1.6	G												
		33 52 40	SD-01 Preconstruction Submittals														
			Work Plan	1.5.3	G												
			Pigging Plan	1.5.4	G												
			Hydrotesting Plan	1.5.5	G												
			Quality Assurance Plan	2.2.3	G												
			SD-02 Shop Drawings														
			Grounding and Bonding	2.1.2													
			Pipe Supports	2.5.5	G												
			Pigging System Components	2.6	G												
			SD-03 Product Data														
			Carbon Steel Piping	2.2.2	G												
			Stainless Steel Piping	2.2.3	G												
			External Protective Coatings For Aboveground Piping	2.2.4	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		33 52 40	External Protective Coatings For Buried Steel Piping	2.2.5	G												
			Fittings	2.2.6	G												
			Carbon Steel Fittings	2.2.6.2	G												
			Insulating Flange Kits	2.2.7	G												
			Lightning Surge Arrester	2.2.7	G												
			Bolts, Nuts And Washers	2.2.8	G												
			Ball Valves	2.3.1	G												
			Full Port Ball (DBBV) Valves	2.3.1.2	G												
			Plug (Double Block and Bleed) Valves	2.3.2	G												
			Electric Valve Actuator	2.3.2.5													
			Swing Check Valves	2.3.3													
			Thermal Relief Valve	2.4	G												
			Thermal Relief Valve	2.4.2	G												
			Thermal Relief Valve	2.4.3	G												
			Mechanically Adjustable Segmented Elastomeric Seal	2.5.1	G												
			Pipe Sleeves	2.5.2	G												
			Strainers	2.5.3	G												
			Sample Connections	2.5.6	G												
			Sight Flow Indicators	2.5.7	G												
			Buried Utility Tape	2.7.5	G												
			Pipeline Markers	2.7.6	G												
			Flange Gaskets, Non-Metallic, Non-Electrically Isolating	2.2.9	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		33 52 40	SD-06 Test Reports														
			Exterior Coating Holiday Test	3.13.1.1													
			Pneumatic Test	3.13.2													
			Hydrostatic Test	3.13.3													
			Performance Testing	3.13.5													
			Pipe Pigging Verification	3.15													
			SD-07 Certificates														
			Contractor Qualifications	1.5.1	G												
			System Supplier	1.5.2	G												
			Pipeline Inventory	1.5.6.1	G												
			Qualifications of Owner's Inspector	2.2.3	G												
			Survey Final Elevations	3.5.7	G												
			Demonstrations	3.16													
			SD-10 Operation and Maintenance Data														
			Insulating Flange Kits	2.2.7	G												
			Lightning Surge Arrester	2.2.7	G												
			Ball Valves	2.3.1	G												
			Full Port Ball (DBBV) Valves	2.3.1.2	G												
			Plug (Double Block and Bleed) Valves	2.3.2	G												
			Electric Valve Actuator	2.3.2.5	G												
			Swing Check Valves	2.3.3	G												
			Thermal Relief Valve	2.4	G												
			Thermal Relief Valve	2.4.2	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		33 52 40	Thermal Relief Valve	2.4.3	G												
			Mechanically Adjustable	2.5.1	G												
			Segmented Elastomeric Seal														
			Strainers	2.5.3	G												
			Sample Connections	2.5.6	G												
			Sight Flow Indicators	2.5.7	G												
		33 52 43.11	SD-02 Shop Drawings														
			Meters	2.6	G												
			Bulk Storage Tank Vent	2.14	G												
			High Point Vent and Low Point	2.7	G												
			Drain Pits														
			SD-03 Product Data														
			Pressure Gauges	2.4	G												
			Automatic Pump Controls	2.5	G												
			Meters	2.6	G												
			Bulk Storage Tank Vent	2.14	G												
			High Point Vent and Low Point	2.7	G												
			Drain Pits														
			Bulk Storage Tank Level Indicator	2.11	G												
			Bulk Storage Tank Level	2.12	G												
			Switches														
			Quick Disconnect Couplings	2.10	G												
			Safety Device Against	2.8	G												
			Detonation, Liquid Type														
			Safety Device Against	2.9	G												
			Detonation, Dry Type														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		33 52 43.11	SD-06 Test Reports														
			Coating Testing	3.2.1	G												
			SD-07 Certificates														
			System Supplier	1.4	G												
			Safety Device Against	2.9	G												
			Detonation, Dry Type														
			Safety Device Against	2.8	G												
			Detonation, Liquid Type														
			SD-10 Operation and Maintenance														
			Data														
			Automatic Pump Controls	2.5	G												
			Bulk Storage Tank Vent	2.14	G												
			Bulk Storage Tank Level Indicator	2.11	G												
			Meters	2.6	G												
			Safety Device Against	2.9	G												
			Detonation, Dry Type														
			Safety Device Against	2.8	G												
			Detonation, Liquid Type														
		33 52 43.14	SD-02 Shop Drawings														
			Control Valves	2.2	G												
			SD-03 Product Data														
			Control Valves	2.2	G												
			SD-06 Test Reports														
			Control Valves	2.2													
			SD-07 Certificates														
			Previous Military Projects	1.4.1	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		33 52 43.14	Qualified Engineers	1.4.1	G												
			Field Assistance	1.4.1	G												
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	1.4.2	G												
		33 52 43.23	SD-02 Shop Drawings														
			Withdrawal Pump	2.2	G												
			Offloading Pump	2.3	G												
			Fueling/Water Draw-Off Pump (Vertical Turbine)	2.1	G												
			Leak Detection Pump	2.4	G												
			SD-03 Product Data														
			Offloading Pump	2.3	G												
			Fueling/Water Draw-Off Pump (Vertical Turbine)	2.1	G												
			Leak Detection Pump	2.4	G												
			SD-06 Test Reports														
			Certified Test Curves	1.2													
			SD-07 Certificates														
			Offloading Pump	2.3	G												
			Fueling/Water Draw-Off Pump (Vertical Turbine)	2.1	G												
			SD-10 Operation and Maintenance Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		33 52 43.23	Operation and Maintenance Manuals	1.2													
		33 52 80	SD-01 Preconstruction Submittals														
			Contract Errors, Omissions, and Other Discrepancies	1.5.1													
			Corrective Action Procedures	1.5.2.1													
			Corrective Action Request (CAR) Form	1.5.2.2													
			Coatings Work Plan	1.5.3													
			Inspection Report Forms	3.7.1.2													
			SD-05 Design Data														
			Containment System	1.5.4.1													
			SD-06 Test Reports														
			Daily Inspection Reports	3.7.1.3													
			Ferrous Metallic Abrasive Qualification Test Reports	1.5.5.1													
			Non-Metallic Abrasive Qualification Test Reports	1.5.5.2													
			Coating Field Test Reports	3.1.3													
			Abrasive Field Test Reports	3.1.4													
			Recycled Ferrous Metallic Abrasive Field Test Reports (Daily and Weekly)	1.5.5.3													
			SD-07 Certificates														
			Qualifications of Certified Industrial Hygienist (CIH)	1.5.6.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		33 52 80	Qualifications of Pipe Coating Shop	1.5.6.2													
			Qualifications of Certified Protective Coatings Specialist (PCS)	1.5.6.3													
			Qualifications of Coatings Inspection Company	1.5.6.4													
			Qualifications of Quality Assurance Coatings Inspector	1.5.6.5													
			Qualifications of Coatings Contractors for Field Coating	1.5.6.6													
			Qualifications of Individuals Performing Abrasive Blasting for Field Coating	1.5.6.7													
			Qualifications of Individuals Performing Coating Application for Field Coating	1.5.6.8													
			Qualifications of Testing Laboratory for Coatings	1.5.6.9													
			Qualifications of Testing Laboratory for Abrasive	1.5.6.10													
			Ferrous Metallic Abrasive Certificate of Conformance	1.5.6.11													
			Non-Metallic Abrasive Certificate of Conformance	1.5.6.12													
			SD-08 Manufacturer's Instructions														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		33 52 80	Coating System Instructions	1.6.1													
			SD-11 Closeout Submittals														
			Disposal of Used Abrasive Shop Applied Coating	3.4.1.5													
			Disposal of Used Abrasive Field Applied Coating	3.6.1.6													
			Inspection Logbook	3.7.1.4													
			Corrective Action Log	1.5.2.3	G												
		33 56 10	SD-02 Shop Drawings														
			Grounding and Bonding	2.1.1													
			Underground Drain Tank And Accessories	2.5	G												
			SD-03 Product Data														
			Aboveground Storage Tank (Single Wall Steel)	2.4.1	G												
			Tank Protective Coatings	2.6	G												
			Atmospheric Vent	2.7.6.1	G												
			Pressure/Vacuum Vent	2.7.6.2	G												
			Emergency Vent	2.7.6.4	G												
			Independent Level Alarm System	2.8	G												
			Tank Gauges	2.9	G												
			Automatic Tank Gauging (Atg)	2.9.4													
			Underground Drain Tank And Accessories	2.5	G												
			Desiccant Vent Dryer	2.7.6.3	G												
			SD-06 Test Reports														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

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		33 56 10	Aboveground Storage Tank	3.2.1	G												
			Tightness Tests														
			Underground Storage Tank	3.2.2	G												
			Tightness Tests														
			Tank Manufacturer's Tests	3.2.3													
			Tank Fill Tests	3.4													
			Tank Inspection Reports	3.2.5	G												
			SD-07 Certificates														
			Contractor Qualifications	1.3.1	G												
			Host Nation	1.3.2.2	G												
			Pollution Liability Insurance	1.3.1													
			Permitting	1.3.2.1													
			Registration	1.3.2.2													
			Licensed Personnel	1.3.2.3													
			Demonstrations	3.3													
			STI SP001 Inspector's	3.2.5	G												
			Certification														
			SD-08 Manufacturer's Instructions														
			Aboveground Storage Tank	2.4													
			Independent Level Alarm System	2.8													
			Tank Gauges	2.9													
			Automatic Tank Gauging (Atg)	2.9.4													
			SD-10 Operation and Maintenance														
			Data														
			Aboveground Storage Tank	2.4	G												
			Independent Level Alarm System	2.8	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS SIF CATION OR A/E REVIEW WR	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		33 56 10	Tank Gauges	2.9	G												
			Automatic Tank Gauging (Atg)	2.9.4													
			Underground Drain Tank And Accessories	2.5	G												
		33 56 19	SD-02 Shop Drawings														
			Installation	3.1	G												
			Liner Projections	3.1.4.1	G												
			As-Built Drawings	3.7	G												
			SD-03 Product Data														
			Fabric Reinforced Geomembrane Liner	2.1.1	G												
			Non-Reinforced Smooth Geomembrane Liner	2.1.2	G												
			Nonwoven Geotextile	2.1.3	G												
			Liner Fittings	2.2.1	G												
			Embedment Strips	2.2.2	G												
			Mounting Strip System	2.2.3	G												
			Sealant	2.2.4	G												
			Manufacturer's Warranty	1.7													
			Installer's Warranty	1.7													
			SD-04 Samples														
			Destructive Field Seam Test Sample	3.3.3	G												
			SD-06 Test Reports														
			Trial Seam Logs	3.2.1													
			Non-Destructive Field Seam	3.3.2													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		33 56 19	Destructive Field Seam	3.3.3													
			SD-07 Certificates														
			Field Engineer Qualifications	1.4.2													
			Installer Qualifications	1.4.3													
			Fabricator Certification	1.4.4													
			Liner Manufacturer's Certification	1.4.5													
			Surface Preparation	3.1.2													
			SD-08 Manufacturer's Instructions														
			Liner Manufacturer's Installation	3.1.4													
			Instructions														
			SD-10 Operation and Maintenance														
			Data														
			Geomembrane Liner	3.7	G												
		33 56 21.17	SD-01 Preconstruction Submittals														
			Copies of API Publications	1.3	G												
			Acknowledgement of API Std 650	1.4	G												
			NACE Visual Comparator	3.3.6	G												
			Acknowledgement of Surface	3.3.6	G												
			Finish Requirements														
			SD-02 Shop Drawings														
			Steel Tank	1.8.2	G												
			Water Draw-Off System	2.12.6	G												
			SD-03 Product Data														
			Structural Steel	2.2	G												
			Carbon Steel, Pipe Fittings,	2.3	G												
			Flanges, Gaskets, and Bolting														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		33 56 21.17	Aluminum Piping	2.5	G												
			Aluminum Flanges	2.6	G												
			Gaskets for Manhole Covers, Stilling Well Flanges, and Roof Center Vent	2.9	G												
			Sample Gauge Hatch	2.12.1	G												
			Antiseize Compound	2.12.7	G												
			SD-05 Design Data														
			Steel Tank Design	1.8.3	G												
			SD-06 Test Reports														
			Visual Examination of Vertical Shell-Seam Tack Welds	3.5.2													
			Visual Examination of Initial Pass of Internal Shell-to-Bottom Weld	3.5.2													
			Vacuum Box Testing of Internal Shell-to-Bottom Initial Weld Pass	3.5.2													
			Visual Examination of Completed Internal and External Shell-to-Bottom Welds	3.5.2													
			Radiographic Examination of Shell Butt Weld	3.5.2													
			Visual Examination of Shell Butt Welds	3.5.2													
			Visual Examination of Fillet Welds	3.5.2													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		33 56 21.17	Visual Examination of Tank	3.5.2													
			Bottom Plates														
			Vacuum Box Testing of Tank	3.5.2													
			Bottom Fillet Weld														
			Pneumatic Tests of Reinforcing	3.5.2													
			Plates														
			Hydrostatic Testing	3.5.3	G												
			Shell Settlement Measurements	3.5.3	G												
			Taken Before, During, and After														
			Hydrostatic Testing														
			Internal Bottom Elevation	3.5.3	G												
			Readings Taken Before and After														
			Hydrostatic Testing														
			Shell Plumbness	3.5.3	G												
			Shell Roundness	3.5.3	G												
			Maximum Local Deviations, Shell	3.5.3	G												
			Tightness Test Records	3.5.4	G												
			Tank Bottom Puddle Test	3.5.5	G												
			Roof Puddle Test	3.5.7	G												
			Stilling Well Plumbness Test	3.5.7.1	G												
			SD-07 Certificates														
			Qualifications of Tank Erector	1.6													
			Welding Procedure Specifications	1.6.1	G												
			(WPS)														
			Welding Procedure Qualification	1.6.1	G												
			Records (PQRs)														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		33 56 21.17	Welder Performance Qualification Records (WPQ)	1.6.1	G												
			Tank Calibration Experience	1.7	G												
			Weld Inspector Certification	1.8.4	G												
			NDE Personnel Certification	1.8.6.1	G												
			Qualifications of Testing Agency	1.8.6.2	G												
			Qualifications of API Std 653	1.8.7	G												
			Inspector														
			SD-09 Manufacturer's Field Reports														
			Mill Test Reports	2.2	G												
			Impact Test Data	2.2	G												
			SD-10 Operation and Maintenance Data														
			API Std 653 Inspection Reports	3.2	G												
			Tank Calibration Table	3.5.1	G												
			Electronic Calibration Table	3.5.1	G												
			Maintenance Instructions	3.5.9	G												
			Operator Instructions	3.5.10	G												
		33 71 02	SD-02 Shop Drawings														
			Precast Underground Structures	1.5.1	G												
			SD-03 Product Data														
			Medium Voltage Cable	2.5	G												
			Medium Voltage Cable Joints	2.7	G												
			Medium Voltage Cable Terminations	2.6	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS SIF CATION REVIEWER	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		33 71 02	Precast Concrete Structures	2.13.2.1	G												
			Sealing Material	2.13.2.4													
			Pulling-In Irons	3.3.3													
			Manhole Frames and Covers	2.13.3	G												
			Handhole Frames and Covers	2.13.4	G												
			Composite/Fiberglass Handholes	2.13.5	G												
			Cable Supports	2.14	G												
			SD-06 Test Reports														
			Medium Voltage Cable	2.16.2	G												
			Qualification and Production Tests														
			Field Acceptance Checks and Tests	3.16.1	G												
			Arc-proofing Test	2.16.1	G												
			SD-07 Certificates														
			Cable splicer/terminator	1.5.2	G												
			Cable Installer Qualifications	1.5.3	G												
		33 82 00	SD-02 Shop Drawings														
			Telecommunications Outside Plant	1.6.1.1	G												
			Telecommunications Entrance Facility Drawings	1.6.1.2	G												
			SD-03 Product Data														
			Wire and Cable	2.7	G												
			Cable Splices, and Connectors	2.4	G												
			Closures	2.3	G												
			Building Protector Assemblies	2.2.1	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
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		33 82 00	Protector Modules	2.2.2	G												
			SD-06 Test Reports														
			Pre-installation Tests	3.4.1	G												
			Acceptance Tests	3.4.2	G												
			Outside Plant Test Plan	1.6.3	G												
			SD-07 Certificates														
			Telecommunications Contractor	1.6.2.1	G												
			Key Personnel	1.6.2.2	G												
			Manufacturer's Qualifications	1.6.2.3	G												
			SD-08 Manufacturer's Instructions														
			Building Protector Assembly	2.2.1	G												
			Installation														
			Cable Tensions	3.1.5.1	G												
			Fiber Optic Splices	3.1.6.2	G												
			SD-09 Manufacturer's Field														
			Reports														
			Factory Reel Test Data	2.13.1	G												
			SD-10 Operation and Maintenance														
			Data														
			Telecommunications Outside	1.6.1.1	G												
			Plant (OSP)														
			SD-11 Closeout Submittals														
			Record Documentation	1.8.1	G												
		40 60 00	SD-02 Shop Drawings														
			Contractor Design Drawings	3.6.2	G												
			Draft As-Built Drawings	3.6.2.1	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

CONTRACTOR

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		40 60 00	SD-03 Product Data														
			Control Drawings	3.6													
			Performance Verification Test (PVT)	3.7.2													
			Factory Test Procedure	2.8.2													
			SD-06 Test Reports														
			Factory Test Report	2.8.3													
			Testing, Adjusting and Commissioning	3.7.1													
			Performance Verification Test(PVT)	3.7.2													
			Endurance Test	3.7.3													
			SD-07 Certificates														
			Control and Sensor Wiring	3.4.2.2													
			Ground Rods	3.7.1													
			Wiring	3.2.1													
			Installation	3.2.1													
			SD-10 Operation and Maintenance Data														
			Training Manual	3.8	G												
			Control System	2.1	G												
			SD-11 Closeout Submittals														
			Final As-Built Drawings	3.6.2.2	G												

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS

11/20, CHG 4: 08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Competent Person (CP)
 - 1.2.2 Competent Person, Confined Space
 - 1.2.3 Competent Person, Cranes and Rigging
 - 1.2.4 Competent Person, Excavation/Trenching
 - 1.2.5 Competent Person, Fall Protection
 - 1.2.6 Competent Person, Scaffolding
 - 1.2.7 Competent Person (CP) Trainer
 - 1.2.8 High Risk Activities
 - 1.2.9 High Visibility Accident
 - 1.2.10 Load Handling Equipment (LHE)
 - 1.2.11 Medical Treatment
 - 1.2.12 Near Miss
 - 1.2.13 Operating Envelope
 - 1.2.14 Qualified Person (QP)
 - 1.2.15 Qualified Person, Fall Protection (QP for FP)
 - 1.2.16 Recordable Injuries or Illnesses
 - 1.2.17 Government Property and Equipment
 - 1.2.18 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap
- 1.3 SUBMITTALS
- 1.4 MONTHLY EXPOSURE REPORTS
- 1.5 CONTRACTOR SAFETY SELF-EVALUATION CHECKLIST
- 1.6 REGULATORY REQUIREMENTS
- 1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS
 - 1.7.1 Personnel Qualifications
 - 1.7.1.1 Site Safety and Health Officer (SSHO)
 - 1.7.1.1.1 Additional Site Safety and Health Officer (SSHO) Requirements and Duties
 - 1.7.1.2 Competent Person Qualifications
 - 1.7.1.2.1 Competent Person for Confined Space Entry
 - 1.7.1.2.2 Competent Person for Scaffolding
 - 1.7.1.2.3 Competent Person for Fall Protection
 - 1.7.1.3 Qualified Trainer Requirements
 - 1.7.1.4 Crane Operators/Riggers
 - 1.7.2 Personnel Duties
 - 1.7.2.1 Duties of the Site Safety and Health Officer (SSHO)
 - 1.7.3 Meetings
 - 1.7.3.1 Preconstruction Meeting
 - 1.7.3.2 Safety Meetings
- 1.8 ACCIDENT PREVENTION PLAN (APP)
 - 1.8.1 APP - Construction
 - 1.8.2 Names and Qualifications

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 1.8.3 Plans
 - 1.8.3.1 Confined Space Entry Plan
 - 1.8.3.2 Standard Lift Plan (SLP)
 - 1.8.3.3 Critical Lift Plan - Crane or Load Handling Equipment
 - 1.8.3.3.1 Critical Lift Plan Planning and Schedule
 - 1.8.3.3.2 Lifts of Personnel
 - 1.8.3.4 Multi-Purpose Machines, Material Handling Equipment, and Construction Equipment Lift Plan
 - 1.8.3.5 Fall Protection and Prevention (FP&P) Plan
 - 1.8.3.6 Rescue and Evacuation Plan
 - 1.8.3.7 Hazardous Energy Control Program (HECP)
 - 1.8.3.8 Excavation Plan
 - 1.8.3.9 Lead, Cadmium, and Chromium Compliance Plan
 - 1.8.3.10 Asbestos Hazard Abatement Plan
 - 1.8.3.11 Site Safety and Health Plan
 - 1.8.3.12 Site Demolition Plan
- 1.9 ACTIVITY HAZARD ANALYSIS (AHA)
 - 1.9.1 AHA Management
 - 1.9.2 AHA Signature Log
- 1.10 DISPLAY OF SAFETY INFORMATION
 - 1.10.1 Safety Bulletin Board
 - 1.10.2 Safety and Occupational Health (SOH) Deficiency Tracking System
- 1.11 SITE SAFETY REFERENCE MATERIALS
- 1.12 EMERGENCY MEDICAL TREATMENT
- 1.13 NOTIFICATIONS and REPORTS
 - 1.13.1 Mishap Notification
 - 1.13.2 Accident Reports
 - 1.13.3 LHE Inspection Reports
 - 1.13.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging
- 1.14 HOT WORK
 - 1.14.1 Permit and Personnel Requirements
 - 1.14.2 Work Around Flammable Materials
- 1.15 RADIATION SAFETY REQUIREMENTS
 - 1.15.1 Radiography Operation Planning Work Sheet
 - 1.15.2 Site Access and Security
 - 1.15.3 Loss or Release and Unplanned Personnel Exposure
 - 1.15.4 Site Demarcation and Barricade
 - 1.15.5 Security of Material and Equipment
 - 1.15.6 Transportation of Material
 - 1.15.7 Schedule for Exposure or Unshielding
 - 1.15.8 Transmitter Requirements
- 1.16 CONFINED SPACE ENTRY REQUIREMENTS
 - 1.16.1 Entry Procedures
 - 1.16.2 Forced Air Ventilation
 - 1.16.3 Sewer Wet Wells
 - 1.16.4 Rescue Procedures and Coordination with Local Emergency Responders
- 1.17 SEVERE STORM PLAN

PART 2 PRODUCTS

- 2.1 CONFINED SPACE SIGNAGE

PART 3 EXECUTION

- 3.1 CONSTRUCTION AND OTHER WORK
 - 3.1.1 Worksite Communication

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.1.2 Hazardous Material Use
- 3.1.3 Hazardous Material Exclusions
- 3.1.4 Unforeseen Hazardous Material
- 3.2 UTILITY OUTAGE REQUIREMENTS
- 3.3 OUTAGE COORDINATION MEETING
- 3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)
 - 3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility
 - 3.4.2 Lockout/Tagout Isolation
 - 3.4.3 Lockout/Tagout Removal
- 3.5 FALL PROTECTION PROGRAM
 - 3.5.1 Training
 - 3.5.2 Fall Protection Equipment and Systems
 - 3.5.2.1 Personal Fall Protection Equipment
 - 3.5.3 Fall Protection for Roofing Work
 - 3.5.4 Horizontal Lifelines (HLL)
 - 3.5.5 Guardrails and Safety Nets
 - 3.5.6 Rescue and Evacuation Plan and Procedures
- 3.6 WORK PLATFORMS
 - 3.6.1 Scaffolding
 - 3.6.2 Elevated Aerial Work Platforms (AWPs)
- 3.7 EQUIPMENT
 - 3.7.1 Material Handling Equipment (MHE)
 - 3.7.2 Load Handling Equipment (LHE)
 - 3.7.3 Machinery and Mechanized Equipment
 - 3.7.4 Base Mounted Drum Hoists
- 3.8 EXCAVATIONS
 - 3.8.1 Utility Locations
 - 3.8.2 Utility Location Verification
 - 3.8.3 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces
- 3.9 ELECTRICAL
 - 3.9.1 Conduct of Electrical Work
 - 3.9.2 Qualifications
 - 3.9.3 Arc Flash
 - 3.9.4 Grounding
 - 3.9.5 Testing

-- End of Section Table of Contents --

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS
11/20, CHG 4: 08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B30.3	(2020) Tower Cranes
ASME B30.5	(2021) Mobile and Locomotive Cranes
ASME B30.7	(2021) Winches
ASME B30.8	(2020) Floating Cranes and Floating Derricks
ASME B30.9	(2018) Slings
ASME B30.20	(2018) Below-the-Hook Lifting Devices
ASME B30.22	(2016) Articulating Boom Cranes
ASME B30.23	(2022) Personnel Lifting Systems Safety Standard for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings
ASME B30.26	(2015; R 2020) Rigging Hardware

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.22	(2007; R 2017) Safety Requirements for Rope-Guided and Non-Guided Workers' Hoists
ASSP A10.34	(2021) Protection of the Public on or Adjacent to Construction Sites
ASSP A10.44	(2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations
ASSP Z244.1	(2016) The Control of Hazardous Energy Lockout, Tagout and Alternative Methods
ASSP Z359.0	(2018) Definitions and Nomenclature Used for Fall Protection and Fall Arrest
ASSP Z359.1	(2020) The Fall Protection Code
ASSP Z359.2	(2017) Minimum Requirements for a

Comprehensive Managed Fall Protection
Program

ASSP Z359.3	(2019) Safety Requirements for Lanyards and Positioning Lanyards
ASSP Z359.4	(2013) Safety Requirements for Assisted-Rescue and Self-Rescue Systems, Subsystems and Components
ASSP Z359.6	(2016) Specifications and Design Requirements for Active Fall Protection Systems
ASSP Z359.7	(2019) Qualification and Verification Testing of Fall Protection Products
ASSP Z359.11	(2014) Safety Requirements for Full Body Harnesses
ASSP Z359.12	(2019) Connecting Components for Personal Fall Arrest Systems
ASSP Z359.13	(2013) Personal Energy Absorbers and Energy Absorbing Lanyards
ASSP Z359.14	(2014) Safety Requirements for Self-Retracting Devices for Personal Fall Arrest and Rescue Systems
ASSP Z359.15	(2014) Safety Requirements for Single Anchor Lifelines and Fall Arresters for Personal Fall Arrest Systems
ASSP Z359.16	(2016) Safety Requirements for Climbing Ladder Fall Arrest Systems
ASSP Z359.18	(2017) Safety Requirements for Anchorage Connectors for Active Fall Protection Systems

ASTM INTERNATIONAL (ASTM)

ASTM F855	(2019) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment
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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048	(2016) Guide for Protective Grounding of Power Lines
IEEE C2	(2023) National Electrical Safety Code

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Z535.2	(2011; R 2017) Environmental and Facility Safety Signs
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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10	(2022; ERTA 1 2021) Standard for Portable Fire Extinguishers
NFPA 51B	(2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
NFPA 70	(2023) National Electrical Code
NFPA 70E	(2024) Standard for Electrical Safety in the Workplace
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-222	(2018H; Add 1 2019) Structural Standard for Antenna Supporting Structures and Antennas and Small Wind Turbine Support Structures
TIA-1019	(2012; R 2016) Standard for Installation, Alteration and Maintenance of Antenna Supporting Structures and Antennas

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety -- Safety and Health Requirements Manual
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U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 20	Standards for Protection Against Radiation
29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1910.146	Permit-required Confined Spaces
29 CFR 1910.147	The Control of Hazardous Energy (Lock Out/Tag Out)
29 CFR 1910.333	Selection and Use of Work Practices
29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment
29 CFR 1915.89	Control of Hazardous Energy (Lockout/Tags-Plus)
29 CFR 1926	Safety and Health Regulations for Construction
29 CFR 1926.16	Rules of Construction

29 CFR 1926.450	Scaffolds
29 CFR 1926.500	Fall Protection
29 CFR 1926.552	Material Hoists, Personal Hoists, and Elevators
29 CFR 1926.553	Base-Mounted Drum Hoists
29 CFR 1926.1400	Cranes and Derricks in Construction
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
CPL 02-01-056	(2014) Inspection Procedures for Accessing Communication Towers by Hoist
CPL 2.100	(1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146

1.2 DEFINITIONS

1.2.1 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are dangerous to personnel, and who has authorization to take prompt corrective measures with regards to such hazards.

1.2.2 Competent Person, Confined Space

The CP, Confined Space, is a person meeting the competent person requirements as defined EM 385-1-1 Appendix Q, with thorough knowledge of OSHA's Confined Space Standard, 29 CFR 1910.146, and designated in writing to be responsible for the immediate supervision, implementation and monitoring of the confined space program, who through training, knowledge and experience in confined space entry is capable of identifying, evaluating and addressing existing and potential confined space hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.3 Competent Person, Cranes and Rigging

The CP, Cranes and Rigging, as defined in EM 385-1-1 Appendix Q, is a person meeting the competent person requirements, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the Crane and Rigging Program, who through training, knowledge and experience in crane and rigging is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.4 Competent Person, Excavation/Trenching

A CP, Excavation/Trenching, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and 29 CFR 1926, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the excavation/trenching

program, who through training, knowledge and experience in excavation/trenching is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.5 Competent Person, Fall Protection

The CP, Fall Protection, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and in accordance with ASSP Z359.0, who has been designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the fall protection program, who through training, knowledge and experience in fall protection and rescue systems and equipment, is capable of identifying, evaluating and addressing existing and potential fall hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.6 Competent Person, Scaffolding

The CP, Scaffolding is a person meeting the competent person requirements in EM 385-1-1 Appendix Q, and designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the scaffolding program. The CP for Scaffolding has enough training, knowledge and experience in scaffolding to correctly identify, evaluate and address existing and potential hazards and also has the authority to take prompt corrective measures with regard to these hazards. CP qualifications must be documented including experience on the specific scaffolding systems/types being used, assessment of the base material that the scaffold will be erected upon, load calculations for materials and personnel, and erection and dismantling. The CP for scaffolding must have a documented minimum of 8-hours of scaffold training to include training on the specific type of scaffold being used (e.g. mast-climbing, adjustable, tubular frame), in accordance with EM 385-1-1 Section 22.B.02.

1.2.7 Competent Person (CP) Trainer

A competent person trainer as defined in EM 385-1-1 Appendix Q, who is qualified in the training material presented, and who possesses a working knowledge of applicable technical regulations, standards, equipment and systems related to the subject matter on which they are training Competent Persons. A competent person trainer must be familiar with the typical hazards and the equipment used in the industry they are instructing. The training provided by the competent person trainer must be appropriate to that specific industry. The competent person trainer must evaluate the knowledge and skills of the competent persons as part of the training process.

1.2.8 High Risk Activities

High Risk Activities are activities that involve work at heights, crane and rigging, excavations and trenching, scaffolding, electrical work, and confined space entry.

1.2.9 High Visibility Accident

A High Visibility Accident is any mishap which may generate publicity or high visibility.

1.2.10 Load Handling Equipment (LHE)

LHE is a term used to describe cranes, hoists and all other hoisting equipment (hoisting equipment means equipment, including crane, derricks, hoists and power operated equipment used with rigging to raise, lower or horizontally move a load).

1.2.11 Medical Treatment

Medical Treatment is treatment administered by a physician or by registered professional personnel under the standing orders of a physician. Medical treatment does not include first aid treatment even when provided by a physician or registered personnel.

1.2.12 Near Miss

A Near Miss is a mishap resulting in no personal injury and zero property damage, but given a shift in time or position, damage or injury may have occurred (e.g., a worker falls off a scaffold and is not injured; a crane swings around to move the load and narrowly misses a parked vehicle).

1.2.13 Operating Envelope

The Operating Envelope is the area surrounding any crane or load handling equipment. Inside this "envelope" is the crane, the operator, riggers and crane walkers, other personnel involved in the operation, rigging gear between the hook, the load, the crane's supporting structure (i.e. ground or rail), the load's rigging path, the lift and rigging procedure.

1.2.14 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.2.15 Qualified Person, Fall Protection (QP for FP)

A QP for FP is a person meeting the definition requirements of [EM 385-1-1](#) Appendix Q, and [ASSP Z359.2](#) standard, having a recognized degree or professional certificate and with extensive knowledge, training and experience in the fall protection and rescue field who is capable of designing, analyzing, and evaluating and specifying fall protection and rescue systems.

1.2.16 Recordable Injuries or Illnesses

Recordable Injuries or Illnesses are any work-related injury or illness that results in:

- a. Death, regardless of the time between the injury and death, or the length of the illness;
- b. Days away from work (any time lost after day of injury/illness onset);
- c. Restricted work;
- d. Transfer to another job;

- e. Medical treatment beyond first aid;
- f. Loss of consciousness; or
- g. A significant injury or illness diagnosed by a physician or other licensed health care professional, even if it did not result in (a) through (f) above

1.2.17 Government Property and Equipment

Interpret "USACE" property and equipment specified in USACE EM 385-1-1 as Government property and equipment.

1.2.18 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap

A LHE accident occurs when any one or more of the eight elements in the operating envelope fails to perform correctly during operation, including operation during maintenance or testing resulting in personnel injury or death; material or equipment damage; dropped load; derailment; two-blocking; overload; or collision, including unplanned contact between the load, crane, or other objects. A dropped load, derailment, two-blocking, overload and collision are considered accidents, even though no material damage or injury occurs. A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, or roll over). Document an LHE mishap or accident using the NAVFAC prescribed Navy Crane Center (NCC) accident form.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

Accident Prevention Plan (APP); G

APP - Construction; G

SD-06 Test Reports

Monthly Exposure Reports

Notifications and Reports

Accident Reports; G

LHE Inspection Reports

SD-07 Certificates

Contractor Safety Self-Evaluation Checklist

Crane Operators/Riggers

Standard Lift Plan; G

Critical Lift Plan ; G

Activity Hazard Analysis (AHA)

Confined Space Entry Permit

Hot Work Permit

Certificate of Compliance

License Certificates

Radiography Operation Planning Work Sheet; G

Portable Gauge Operations Planning Worksheet; G

1.4 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report and attach to the monthly billing request. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the voucher.

1.5 CONTRACTOR SAFETY SELF-EVALUATION CHECKLIST

Contracting Officer will provide a "Contractor Safety Self-Evaluation checklist" to the Contractor at the pre-construction meeting. Complete the checklist monthly and submit with each request for payment voucher. An acceptable score of 90 or greater is required. Failure to submit the completed safety self-evaluation checklist or achieve a score of at least 90 may result in retention of up to 10 percent of the voucher. The Contractor Safety Self-Evaluation checklist can be found on the Whole Building Design Guide website at www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-35-26

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, and the following host nation laws, ordinances, criteria, rules and regulations. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Personnel Qualifications

1.7.1.1 Site Safety and Health Officer (SSHO)

Provide an SSHO that meets the requirements of EM 385-1-1 Section 1. The SSHO must ensure that the requirements of 29 CFR 1926.16 are met for the

project. Provide a Safety oversight team that includes a minimum of one person at each project site to function as the Site Safety and Health Officer (SSHO). The SSHO or an equally-qualified Alternate SSHO must be at the work site at all times to implement and administer the Contractor's safety program and Government-accepted Accident Prevention Plan. The SSHO and Alternate SSHO must have the required training, experience, and qualifications in accordance with EM 385-1-1 Section 01.A.17, and all associated sub-paragraphs.

If the SSHO is off-site for a period longer than 24 hours, an equally-qualified alternate SSHO must be provided and must fulfill the same roles and responsibilities as the primary SSHO. When the SSHO is temporarily (up to 24 hours) off-site, a Designated Representative (DR), as identified in the AHA may be used in lieu of an Alternate SSHO, and must be on the project site at all times when work is being performed. Note that the DR is a collateral duty safety position, with safety duties in addition to their full time occupation.

1.7.1.1.1 Additional Site Safety and Health Officer (SSHO) Requirements and Duties

The SSHO may also serve as the Quality Control Manager. The SSHO may also serve as the Superintendent.

1.7.1.2 Competent Person Qualifications

Provide Competent Persons in accordance with EM 385-1-1, Appendix Q and herein. Competent Persons for high risk activities include confined space, cranes and rigging, excavation/trenching, fall protection, and electrical work. The CP for these activities must be designated in writing, and meet the requirements for the specific activity (i.e. competent person, fall protection).

The Competent Person identified in the Contractor's Safety and Health Program and accepted Accident Prevention Plan, must be on-site at all times when the work that presents the hazards associated with their professional expertise is being performed. Provide the credentials of the Competent Persons(s) to the Contracting Officer for information in consultation with the Safety Office.

1.7.1.2.1 Competent Person for Confined Space Entry

Provide a Confined Space (CP) Competent Person who meets the requirements of EM 385-1-1, Appendix Q, and herein. The CP for Confined Space Entry must supervise the entry into each confined space in accordance with EM 385-1-1, Section 34.

1.7.1.2.2 Competent Person for Scaffolding

Provide a Competent Person for Scaffolding who meets the requirements of EM 385-1-1, Section 22.B.02 and herein.

1.7.1.2.3 Competent Person for Fall Protection

Provide a Competent Person for Fall Protection who meets the requirements of EM 385-1-1, Section 21.C.04, 21.B.03, and herein.

1.7.1.3 Qualified Trainer Requirements

Individuals qualified to instruct the 40 hour contract safety awareness course, or portions thereof, must meet the definition of a Competent Person Trainer, and, at a minimum, possess a working knowledge of the following subject areas: EM 385-1-1, Electrical Standards, Lockout/Tagout, Fall Protection, Confined Space Entry for Construction; Excavation, Trenching and Soil Mechanics, and Scaffolds in accordance with 29 CFR 1926.450, Subpart L.

Instructors are required to:

- a. Prepare class presentations that cover construction-related safety requirements.
- b. Ensure that all attendees attend all sessions by using a class roster signed daily by each attendee. Maintain copies of the roster for at least five years. This is a certification class and must be attended 100 percent. In cases of emergency where an attendee cannot make it to a session, the attendee can make it up in another class session for the same subject.
- c. Update training course materials whenever an update of the EM 385-1-1 becomes available.
- d. Provide a written exam of at least 50 questions. Students are required to answer 80 percent correctly to pass.
- e. Request, review and incorporate student feedback into a continuous course improvement program.

1.7.1.4 Crane Operators/Riggers

Provide Operators, Signal Persons, and Riggers meeting the requirements in EM 385-1-1, Section 15.B for Riggers and Section 16.B for Crane Operators and Signal Persons. In addition, for mobile cranes with Original Equipment Manufacturer (OEM) rated capacities of 22,680 kg or greater, designate crane operators qualified by a source that qualifies crane operators (i.e., union, a Government agency, or an organization that tests and qualifies crane operators). Provide proof of current qualification.

1.7.2 Personnel Duties

1.7.2.1 Duties of the Site Safety and Health Officer (SSHO)

The SSHO must:

- a. Conduct daily safety and health inspections and maintain a written log which includes area/operation inspected, date of inspection, identified hazards, recommended corrective actions, estimated and actual dates of corrections. Attach safety inspection logs to the Contractors' daily production report.
- b. Conduct mishap investigations and complete required accident reports. Report mishaps and near misses.
- c. Use and maintain OSHA's Form 300 to log work-related injuries and illnesses occurring on the project site for Prime Contractors and subcontractors, and make available to the Contracting Officer upon

request. Post and maintain the Form 300A on the site Safety Bulletin Board.

- d. Maintain applicable safety reference material on the job site.
- e. Attend the pre-construction meeting, pre-work meetings including preparatory meetings, and periodic in-progress meetings.
- f. Review the APP and AHAs for compliance with EM 385-1-1, and approve, sign, implement and enforce them.
- g. Establish a Safety and Occupational Health (SOH) Deficiency Tracking System that lists and monitors outstanding deficiencies until resolution.
- h. Ensure subcontractor compliance with safety and health requirements.
- i. Maintain a list of hazardous chemicals on site and their material Safety Data Sheets (SDS).
- j. Maintain a weekly list of high hazard activities involving energy, equipment, excavation, entry into confined space, and elevation, and be prepared to discuss details during QC Meetings.
- k. Provide and keep a record of site safety orientation and indoctrination for Contractor employees, subcontractor employees, and site visitors.

Superintendent, QC Manager, and SSHO are subject to dismissal if the above or any other required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.7.3 Meetings

1.7.3.1 Preconstruction Meeting

- a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction meeting. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).
- b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays.
- c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to

begin until an APP is established that is acceptable to the Contracting Officer.

1.7.3.2 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, foremen, or CDSOs must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.8 ACCIDENT PREVENTION PLAN (APP)

1.8.1 APP - Construction

A qualified person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, Appendix A, and as supplemented herein. Cover all paragraph and subparagraph elements in EM 385-1-1, Appendix A. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling authority" for all work site safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed by an officer of the firm (Prime Contractor senior person), the individual preparing the APP, the on-site superintendent, the designated SSHO, the Contractor Quality Control Manager, and any designated Certified Safety Professional (CSP) or Certified Health Physicist (CIH). The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer within 30 calendar days of Contract award and not less than 10 calendar days prior to the date of the preconstruction meeting for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting

Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e. imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ASSP A10.34), and the environment.

1.8.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

- a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.
- b. Qualifications of competent and of qualified persons. As a minimum, designate and submit qualifications of competent persons for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance.

1.8.3 Plans

Provide plans in the APP in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

1.8.3.1 Confined Space Entry Plan

Develop a confined or enclosed space entry plan in accordance with EM 385-1-1, applicable OSHA standards 29 CFR 1910, 29 CFR 1915, and 29 CFR 1926, OSHA Directive CPL 2.100, and any other federal, state and local regulatory requirements identified in this Contract. Identify the qualified person's name and qualifications, training, and experience. Delineate the qualified person's authority to direct work stoppage in the event of hazardous conditions. Include procedure for rescue by Contractor personnel and the coordination with emergency responders. (If there is no confined space work, include a statement that no confined space work exists and none will be created.)

1.8.3.2 Standard Lift Plan (SLP)

Plan lifts to avoid situations where the operator cannot maintain safe control of the lift. Prepare a written SLP in accordance with EM 385-1-1, Section 16.A.03, using Form 16-2 for every lift or series of lifts (if duty cycle or routine lifts are being performed). The SLP must be developed, reviewed and accepted by all personnel involved in the lift in conjunction with the associated AHA. Signature on the AHA constitutes acceptance of the plan. Maintain the SLP on the LHE for the current lift(s) being made. Maintain historical SLPs for a minimum of three months.

1.8.3.3 Critical Lift Plan - Crane or Load Handling Equipment

Provide a Critical Lift Plan as required by EM 385-1-1, Section 16.H.01, using Form 16-3. In addition, Critical Lift Plans are required for the following:

- a. Lifts over 50 percent of the capacity of barge mounted mobile crane's hoist.
- b. When working around energized power lines where the work will get closer than the minimum clearance distance in EM 385-1-1 Table 16-1.
- c. For lifts with anticipated binding conditions.
- d. When erecting cranes.

1.8.3.3.1 Critical Lift Plan Planning and Schedule

Critical lifts require detailed planning and additional or unusual safety precautions. Develop and submit a critical lift plan to the Contracting Officer 30 calendar days prior to critical lift. Comply with load testing requirements in accordance with EM 385-1-1, Section 16.F.03.

1.8.3.3.2 Lifts of Personnel

In addition to the requirements of EM 385-1-1, Section 16.H.02, for lifts of personnel, demonstrate compliance with the requirements of 29 CFR 1926.1400 and EM 385-1-1, Section 16.T.

1.8.3.4 Multi-Purpose Machines, Material Handling Equipment, and Construction Equipment Lift Plan

Multi-purpose machines, material handling equipment, and construction equipment used to lift loads that are suspended by rigging gear, require proof of authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. Written approval from a qualified registered professional engineer, after a safety analysis is performed, is allowed in lieu of the OEM's approval. Demonstrate that the operator is properly trained and that the equipment is properly configured to make such lifts and is equipped with a load chart.

1.8.3.5 Fall Protection and Prevention (FP&P) Plan

The plan must be in accordance with the requirements of EM 385-1-1, Section 21.D and ASSP Z359.2, be site specific, and address all fall hazards in the work place and during different phases of construction. Address how to protect and prevent workers from falling to lower levels when they are exposed to fall hazards above 1.8 m. A competent person or qualified person for fall protection must prepare and sign the plan documentation. Include fall protection and prevention systems, equipment and methods employed for every phase of work, roles and responsibilities, assisted rescue, self-rescue and evacuation procedures, training requirements, and monitoring methods. Review and revise, as necessary, the Fall Protection and Prevention Plan documentation as conditions change, but at a minimum every six months, for lengthy projects, reflecting any changes during the course of construction due to changes in personnel, equipment, systems or work habits. Keep and maintain the accepted Fall Protection and Prevention Plan documentation at the job site

for the duration of the project. Include the Fall Protection and Prevention Plan documentation in the Accident Prevention Plan (APP).

1.8.3.6 Rescue and Evacuation Plan

Provide a Rescue and Evacuation Plan in accordance with EM 385-1-1 Section 21.N and ASSP Z359.2, and include in the FP&P Plan and as part of the APP. Include a detailed discussion of the following: methods of rescue; methods of self-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility.

1.8.3.7 Hazardous Energy Control Program (HECP)

Develop a HECP in accordance with EM 385-1-1 Section 12, 29 CFR 1910.147, 29 CFR 1910.333, 29 CFR 1915.89, ASSP Z244.1, and ASSP A10.44. Submit this HECP as part of the Accident Prevention Plan (APP). Conduct a preparatory meeting and inspection with all effected personnel to coordinate all HECP activities. Document this meeting and inspection in accordance with EM 385-1-1, Section 12.A.02. Ensure that each employee is familiar with and complies with these procedures.

1.8.3.8 Excavation Plan

Identify the safety and health aspects of excavation, and provide and prepare the plan in accordance with EM 385-1-1, Section 25.A and Section 31 00 00 EARTHWORK.

1.8.3.9 Lead, Cadmium, and Chromium Compliance Plan

Identify the safety and health aspects of work involving lead, cadmium and chromium, and prepare in accordance with Section 02 83 00 LEAD REMEDIATION.

1.8.3.10 Asbestos Hazard Abatement Plan

Identify the safety and health aspects of asbestos work, and prepare in accordance with Section 02 82 00 ASBESTOS REMEDIATION.

1.8.3.11 Site Safety and Health Plan

Identify the safety and health aspects, and prepare in accordance with Section 01 35 29.13 HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES.

1.8.3.12 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section 02 41 00 DEMOLITION and referenced sources. Include engineering survey as applicable.

1.9 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting

Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1, Section 1 or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFO. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFO must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.10 DISPLAY OF SAFETY INFORMATION

1.10.1 Safety Bulletin Board

Prior to commencement of work, erect a safety bulletin board at the job site. Where size, duration, or logistics of project do not facilitate a bulletin board, an alternative method, acceptable to the Contracting Officer, that is accessible and includes all mandatory information for employee and visitor review, may be deemed as meeting the requirement for a bulletin board. Include and maintain information on safety bulletin board as required by EM 385-1-1, Section 01.A.07. Additional items required to be posted include:

- a. Confined space entry permit.
- b. Hot work permit.

1.10.2 Safety and Occupational Health (SOH) Deficiency Tracking System

Establish a SOH deficiency tracking system that lists and monitors the status of SOH deficiencies in chronological order. Use the tracking system to evaluate the effectiveness of the APP. A monthly evaluation of the data must be discussed in the QC or SOH meeting with everyone on the project. The list must be posted on the project bulletin board and updated daily, and provide the following information:

- a. Date deficiency identified;

- b. Description of deficiency;
- c. Name of person responsible for correcting deficiency;
- d. Projected resolution date;
- e. Date actually resolved.

1.11 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.12 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with [EM 385-1-1](#). Government has no responsibility to provide emergency medical treatment.

1.13 NOTIFICATIONS and REPORTS

1.13.1 Mishap Notification

Notify the Contracting Officer as soon as practical, but no more than twenty-four hours, after any mishaps, including recordable accidents, incidents, and near misses, as defined in [EM 385-1-1](#) Appendix Q, any report of injury, illness, or any property damage. For LHE or rigging mishaps, notify the Contracting Officer as soon as practical but not more than four hours after mishap. The Contractor is responsible for obtaining appropriate medical and emergency assistance and for notifying fire, law enforcement, and regulatory agencies. Immediate reporting is required for electrical mishaps, to include Arc Flash; shock; uncontrolled release of hazardous energy (includes electrical and non-electrical); load handling equipment or rigging; fall from height (any level other than same surface); and underwater diving. These mishaps must be investigated in depth to identify all causes and to recommend hazard control measures.

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any mishap.

1.13.2 Accident Reports

- a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in [EM 385-1-1](#), to establish the root cause(s) of the accident. [Complete the applicable NAVFAC Contractor Incident Reporting System \(CIRS\), and electronically submit via the NAVFAC Enterprise Safety Applications Management System \(ESAMS\).](#) Complete and submit an accident investigation report in ESAMS within 5 days for mishaps defined in [EM 385-1-1 01.D.03](#) and 10 days for accidents defined by [EM 385-1-1 01.D.05](#). Complete an investigation report within 30 days for those

mishaps defined by EM 385-1-1 01.D.04. Mishaps defined by EM 385-1-1 01.D.04 and 01.D.05 must include a written report submitted as an attachment in ESAMS using the following outline: (1) Mishap summary description to include process, findings and outcomes; (2) Root Cause; (3) Direct Factors; (4) Indirect and Contributing Factors; (5) Corrective Actions; and (6) Recommendations. The Contracting Officer will provide copies of any required or special forms.

- b. Near Misses: For Navy Projects, complete the applicable documentation in NAVFAC Contractor Incident Reporting System (CIRS), and electronically submit via the NAVFAC Enterprise Safety Applications Management System (ESAMS). Near miss reports are considered positive and proactive Contractor safety management actions.
- c. Conduct an accident investigation for any load handling equipment accident (including rigging accidents) to establish the root cause(s) of the accident. Complete the LHE Accident Report (Crane and Rigging Accident Report) form and provide the report to the Contracting Officer within 30 calendar days of the accident. Do not proceed with crane operations until cause is determined and corrective actions have been implemented to the satisfaction of the Contracting Officer. The Contracting Officer will provide a blank copy of the accident report form.

1.13.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.13.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a FORM 16-1 Certificate of Compliance for LHE entering an activity under this Contract and in accordance with EM 385-1-1. Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1, Section 16.H.03 using Form 16-2 Standard Pre-Lift Crane Plan/Checklist for each lift planned. Submit SLP to the Contracting Officer for approval within 15 calendar days in advance of planned lift.

1.14 HOT WORK

1.14.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e. welding or cutting) or operating other flame-producing/spark producing devices, from the Fire Division. A permit is required from the Explosives Safety Office for work in and around where explosives are processed, stored, or handled. CONTRACTORS ARE REQUIRED TO MEET ALL CRITERIA BEFORE A PERMIT IS ISSUED. Provide at least two 9 kg 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of one hour after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize

themselves with the location of the nearest fire alarm boxes and place in memory the emergency Fire Division phone number. REPORT ANY FIRE, NO MATTER HOW SMALL, TO THE RESPONSIBLE FIRE DIVISION IMMEDIATELY.

1.14.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "HOT WORK" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in [EM 385-1-1](#), Section 06.H

1.15 RADIATION SAFETY REQUIREMENTS

Submit [License Certificates](#), employee training records, and Leak Test Reports for radiation materials and equipment to the Contracting Officer and Radiation Safety Office (RSO), [and Contracting Oversight Technician \(COT\)](#) for all specialized and licensed material and equipment proposed for use on the construction project (excludes portable machine sources of ionizing radiation including moisture density and X-Ray Fluorescence (XRF)). Maintain on-site records whenever licensed radiological materials or ionizing equipment are on Government property.

Protect workers from radiation exposure in accordance with [10 CFR 20](#), ensuring any personnel exposures are maintained As Low As Reasonably Achievable.

1.15.1 Radiography Operation Planning Work Sheet

Submit a Gamma and X-Ray [Radiography Operation Planning Work Sheet](#) to Contracting Officer 14 days prior to commencement of operations involving radioactive materials or radiation generating devices. For portable machine sources of ionizing radiation, including moisture density and XRF, use and submit the [Portable Gauge Operations Planning Worksheet](#) instead. The Contracting Officer [and COT](#) will review the submitted worksheet and provide questions and comments.

Contractors must use primary dosimeters process by a National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory.

1.15.2 Site Access and Security

Coordinate site access and security requirements with the Contracting Officer [and COT](#) for all radiological materials and equipment containing ionizing radiation that are proposed for use on a government facility. For gamma radiography materials and equipment, a Government escort is required for any travels on the Installation. The [Navy COT](#) or Government authorized representative will meet the Contractor at a designated location outside the Installation, ensure safety of the materials being transported, and will escort the Contractor for gamma sources onto the Installation, to the job site, and off the Installation. For portable machine sources of ionizing radiation, including moisture density and XRF,

the Navy COT or Government authorized representative will meet the Contractor at the job site.

Provide a copy of all calibration records, and utilization records to the COT for radiological operations performed on the site.

1.15.3 Loss or Release and Unplanned Personnel Exposure

Loss or release of radioactive materials, and unplanned personnel exposures must be reported immediately to the Contracting Officer, RSO, and Base Security Department Emergency Number.

1.15.4 Site Demarcation and Barricade

Properly demark and barricade an area surrounding radiological operations to preclude personnel entrance, in accordance with EM 385-1-1, Nuclear Regulatory Commission, and Applicable State regulations and license requirements, and in accordance with requirements established in the accepted Radiography Operation Planning Work Sheet.

Do not close or obstruct streets, walks, and other facilities occupied and used by the Government without written permission from the Contracting Officer.

1.15.5 Security of Material and Equipment

Properly secure the radiological material and ionizing radiation equipment at all times, including keeping the devices in a properly marked and locked container, and secondarily locking the container to a secure point in the Contractor's vehicle or other approved storage location during transportation and while not in use. While in use, maintain a continuous visual observation on the radiological material and ionizing radiation equipment. In instances where radiography is scheduled near or adjacent to buildings or areas having limited access or one-way doors, make no assumptions as to building occupancy. Where necessary, the Contracting Officer will direct the Contractor to conduct an actual building entry, search, and alert. Where removal of personnel from such a building cannot be accomplished and it is otherwise safe to proceed with the radiography, position a fully instructed employee inside the building or area to prevent exiting while external radiographic operations are in process.

1.15.6 Transportation of Material

Comply with 49 CFR 173 for Transportation of Regulated Amounts of Radioactive Material. Notify Local Fire authorities and the site Radiation Safety Officer (RSO) of any Radioactive Material use.

1.15.7 Schedule for Exposure or Unshielding

Actual exposure of the radiographic film or unshielding the source must not be initiated until after 5 p.m. on weekdays.

1.15.8 Transmitter Requirements

Adhere to the base policy concerning the use of transmitters, such as radios and cell phones. Obey Emissions control (EMCON) restrictions.

1.16 CONFINED SPACE ENTRY REQUIREMENTS

Confined space entry must comply with Section 34 of EM 385-1-1, OSHA 29 CFR 1926, OSHA 29 CFR 1910, OSHA 29 CFR 1910.146, and OSHA Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used.

1.16.1 Entry Procedures

Prohibit entry into a confined space by personnel for any purpose, including hot work, until the qualified person has conducted appropriate tests to ensure the confined or enclosed space is safe for the work intended and that all potential hazards are controlled or eliminated and documented. Comply with EM 385-1-1, Section 34 for entry procedures. Hazards pertaining to the space must be reviewed with each employee during review of the AHA.

1.16.2 Forced Air Ventilation

Forced air ventilation is required for all confined space entry operations and the minimum air exchange requirements must be maintained to ensure exposure to any hazardous atmosphere is kept below its action level.

1.16.3 Sewer Wet Wells

Sewer wet wells require continuous atmosphere monitoring with audible alarm for toxic gas detection.

1.16.4 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.17 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

- a. Secure outside equipment and materials and place materials that could be damaged in protected areas.
- b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.
- c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

2.1 CONFINED SPACE SIGNAGE

Provide permanent signs integral to or securely attached to access covers for new permit-required confined spaces. Signs for confined spaces must comply with NEMA Z535.2. Provide signs with wording: "DANGER--PERMIT-REQUIRED CONFINED SPACE, DO NOT ENTER" in bold letters a minimum of 25 mm in height and constructed to be clearly legible with all paint removed. The signal word "DANGER" must be red and readable from 1520 mm.

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

- a. Hard Hat
- b. Long Pants
- c. Appropriate Safety Shoes
- d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Use

Each hazardous material must receive approval from the Contracting Office or their designated representative prior to being brought onto the job site or prior to any other use in connection with this Contract. Allow a minimum of 10 working days for processing of the request for use of a hazardous material.

3.1.3 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

3.1.4 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e. 29 CFR Part 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during construction operations, stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to FAR 52.243-4 Changes and FAR 52.236-2 Differing Site Conditions.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages at least 10 calendar days in advance. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1, Section 11.A.02 (Isolation). Some examples of energy isolation devices and procedures are highlighted in EM 385-1-1, Section 12.D. In accordance with EM 385-1-1, Section 12.A.01, where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy.

These activities include, but are not limited to, a review of HEC and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1, Section 11.A.02 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1, Section 12.A. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, and the Installation representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HEC training in accordance with EM 385-1-1, Section 12.C. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1 Section 12, 29 CFR 1910.333, 29 CFR 1915.89, ASSP A10.44, NFPA 70E, and paragraph HAZARDOUS ENERGY CONTROL PROGRAM (HECP).

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1, Section 12.A.02. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECF. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1, Sections 05.I and 11.B. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1, Section 12.E.06.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECF. In accordance with EM 385-1-1, Section 12.E.08, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with ASSP Z359.2 and EM 385-1-1, Sections 21.A and 21.D.

3.5.1 Training

Institute a fall protection training program. As part of the Fall Protection Program, provide training for each employee who might be exposed to fall hazards and using personal fall protection equipment. Provide training by a competent person for fall protection in accordance with EM 385-1-1, Section 21.C. Document training and practical application of the competent person in accordance with EM 385-1-1, Section 21.C.04 and ASSP Z359.2 in the AHA.

3.5.2 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1, Section 21.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 Section 21.I, 29 CFR 1926.500 Subpart M, ASSP Z359.0, ASSP Z359.1, ASSP Z359.2, ASSP Z359.3, ASSP Z359.4, ASSP Z359.6, ASSP Z359.7, ASSP Z359.11, ASSP Z359.12, ASSP Z359.13, ASSP Z359.14, ASSP Z359.15, ASSP Z359.16 and ASSP Z359.18.

3.5.2.1 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 1633 kg in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 1.8 m, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1, Section 21.I.06.

3.5.3 Fall Protection for Roofing Work

Implement fall protection controls based on the type of roof being constructed and work being performed. Evaluate the roof area to be accessed for its structural integrity including weight-bearing capabilities for the projected loading.

a. Low Sloped Roofs:

- (1) For work within 1.8 m from unprotected edge of a roof having a slope less than or equal to 4:12 (vertical to horizontal), protect personnel from falling by the use of conventional fall protection systems (personal fall arrest/restraint systems, guardrails, or safety nets) in accordance with EM 385-1-1, Section 21 and 29 CFR 1926.500. A safety monitoring system is not adequate fall protection and is not authorized.
- (2) For work greater than 1.8 m from the unprotected roof edge, addition to the use of conventional fall protection systems the use of a warning line system is also permitted, in accordance with 29 CFR 1926.500 and EM 385-1-1, Section 21.L.

b. Steep-Sloped Roofs: Work on a roof having a slope greater than 4:12 (vertical to horizontal) requires a personal fall arrest system,

guardrails with toe-boards, or safety nets. This requirement also applies to residential or housing type construction.

3.5.4 Horizontal Lifelines (HLL)

Provide HLL in accordance with [EM 385-1-1](#), Section 21.I.08.d.2. Commercially manufactured horizontal lifelines (HLL) must be designed, installed, certified and used, under the supervision of a qualified person, for fall protection as part of a complete fall arrest system which maintains a safety factor of 2 ([29 CFR 1926.500](#)). The competent person for fall protection may (if deemed appropriate by the qualified person) supervise the assembly, disassembly, use and inspection of the HLL system under the direction of the qualified person. Locally manufactured HLLs are not acceptable unless they are custom designed for limited or site specific applications by a Registered Professional Engineer who is qualified in designing HLL systems.

3.5.5 Guardrails and Safety Nets

Design, install and use guardrails and safety nets in accordance with [EM 385-1-1](#), Section 21.F.01 and [29 CFR 1926](#) Subpart M.

3.5.6 Rescue and Evacuation Plan and Procedures

When personal fall arrest systems are used, ensure that the mishap victim can self-rescue or can be rescued promptly should a fall occur. Prepare a Rescue and Evacuation Plan and include a detailed discussion of the following: methods of rescue; methods of self-rescue or assisted-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility. Include the Rescue and Evacuation Plan within the Activity Hazard Analysis (AHA) for the phase of work, in the Fall Protection and Prevention (FP&P) Plan, and the Accident Prevention Plan (APP). The plan must be in accordance with the requirements of [EM 385-1-1](#), [ASSP Z359.2](#), and [ASSP Z359.4](#).

3.6 WORK PLATFORMS

3.6.1 Scaffolding

Provide employees with a safe means of access to the work area on the scaffold. Climbing of any scaffold braces or supports not specifically designed for access is prohibited. Comply with the following requirements:

- a. Scaffold platforms greater than 6 m in height must be accessed by use of a scaffold stair system.
- b. Ladders commonly provided by scaffold system manufacturers are prohibited for accessing scaffold platforms greater than 6 m maximum in height.
- c. An adequate gate is required.
- d. Employees performing scaffold erection and dismantling must be qualified.
- e. Scaffold must be capable of supporting at least four times the maximum intended load, and provide appropriate fall protection as delineated in the accepted fall protection and prevention plan.

- f. Stationary scaffolds must be attached to structural building components to safeguard against tipping forward or backward.
- g. Special care must be given to ensure scaffold systems are not overloaded.
- h. Side brackets used to extend scaffold platforms on self-supported scaffold systems for the storage of material are prohibited. The first tie-in must be at the height equal to 4 times the width of the smallest dimension of the scaffold base.
- i. Scaffolding other than suspended types must bear on base plates upon wood mudsills (51 mm x 254 mm x 203 mm minimum) or other adequate firm foundation.
- j. Scaffold or work platform erectors must have fall protection during the erection and dismantling of scaffolding or work platforms that are more than 1.83 meters.
- k. Delineate fall protection requirements when working above 1.83 meters or above dangerous operations in the Fall Protection and Prevention (FP&P) Plan and Activity Hazard Analysis (AHA) for the phase of work.

3.6.2 Elevated Aerial Work Platforms (AWPs)

Workers must be anchored to the basket or bucket in accordance with manufacturer's specifications and instructions (anchoring to the boom may only be used when allowed by the manufacturer and permitted by the CP). Lanyards used must be sufficiently short to prohibit worker from climbing out of basket. The climbing of rails is prohibited. Lanyards with built-in shock absorbers are acceptable. Self-retracting devices are not acceptable. Tying off to an adjacent pole or structure is not permitted unless a safe device for 100 percent tie-off is used for the transfer.

Use of AWPs must be operated, inspected, and maintained as specified in the operating manual for the equipment and delineated in the AHA. Operators of AWPs must be designated as qualified operators by the Prime Contractor. Maintain proof of qualifications on site for review and include in the AHA.

3.7 EQUIPMENT

3.7.1 Material Handling Equipment (MHE)

- a. Material handling equipment such as forklifts must not be modified with work platform attachments for supporting employees unless specifically delineated in the manufacturer's printed operating instructions. Material handling equipment fitted with personnel work platform attachments are prohibited from traveling or positioning while personnel are working on the platform.
- b. The use of hooks on equipment for lifting of material must be in accordance with manufacturer's printed instructions. Material Handling Equipment Operators must be trained in accordance with OSHA 29 CFR 1910, Subpart N.
- c. Operators of forklifts or power industrial trucks must be licensed in accordance with OSHA.

3.7.2 Load Handling Equipment (LHE)

The following requirements apply. In exception, these requirements do not apply to commercial truck mounted and articulating boom cranes used solely to deliver material and supplies (not prefabricated components, structural steel, or components of a systems-engineered metal building) where the lift consists of moving materials and supplies from a truck or trailer to the ground; to cranes installed on mechanics trucks that are used solely in the repair of shore-based equipment; to crane that enter the activity but are not used for lifting; nor to other machines not used to lift loads suspended by rigging equipment. However, LHE accidents occurring during such operations must be reported.

- a. Equip cranes and derricks as specified in [EM 385-1-1](#), Section 16.
- b. Notify the Contracting Officer 15 working days in advance of any LHE entering the activity, in accordance with [EM 385-1-1](#), Section 16.A.02, so that necessary quality assurance spot checks can be coordinated. [Prior to cranes entering federal activities, a Crane Access Permit must be obtained from the Contracting Officer. A copy of the permitting process will be provided at the Preconstruction Meeting.](#) Contractor's operator must remain with the crane during the spot check. Rigging gear must be in accordance with OSHA, [ASME B30.9](#) Standards and host country safety standards.
- c. Comply with the LHE manufacturer's specifications and limitations for erection and operation of cranes and hoists used in support of the work. Perform erection under the supervision of a designated person (as defined in [ASME B30.5](#)). Perform all testing in accordance with the manufacturer's recommended procedures.
- d. Comply with [ASME B30.5](#) for mobile and locomotive cranes, [ASME B30.22](#) for articulating boom cranes, [ASME B30.3](#) for construction tower cranes, [ASME B30.8](#) for floating cranes and floating derricks, [ASME B30.9](#) for slings, [ASME B30.20](#) for below the hook lifting devices and [ASME B30.26](#) for rigging hardware.
- e. When operating in the vicinity of overhead transmission lines, operators and riggers must be alert to this special hazard and follow the requirements of [EM 385-1-1](#) Section 11, and [ASME B30.5](#) or [ASME B30.22](#) as applicable.
- f. Do not use crane suspended personnel work platforms (baskets) unless the Contractor proves that using any other access to the work location would provide a greater hazard to the workers or is impossible. Do not lift personnel with a line hoist or friction crane. Additionally, submit a specific AHA for this work to the Contracting Officer. Ensure the activity and AHA are thoroughly reviewed by all involved personnel.
- g. Inspect, maintain, and recharge portable fire extinguishers as specified in [NFPA 10](#), Standard for Portable Fire Extinguishers.
- h. All employees must keep clear of loads about to be lifted and of suspended loads, except for employees required to handle the load.
- i. Use cribbing when performing lifts on outriggers.

- j. The crane hook/block must be positioned directly over the load. Side loading of the crane is prohibited.
- k. A physical barricade must be positioned to prevent personnel access where accessible areas of the LHE's rotating superstructure poses a risk of striking, pinching or crushing personnel.
- l. Maintain inspection records in accordance by [EM 385-1-1](#), Section 16.D, including shift, monthly, and annual inspections, the signature of the person performing the inspection, and the serial number or other identifier of the LHE that was inspected. Records must be available for review by the Contracting Officer.
- m. Maintain written reports of operational and load testing in accordance with [EM 385-1-1](#), Section 16.F, listing the load test procedures used along with any repairs or alterations performed on the LHE. Reports must be available for review by the Contracting Officer.
- n. Certify that all LHE operators have been trained in proper use of all safety devices (e.g. anti-two block devices).
- o. Take steps to ensure that wind speed does not contribute to loss of control of the load during lifting operations. At wind speeds greater than 9 m/s, the operator, rigger and lift supervisor must cease all crane operations, evaluate conditions and determine if the lift may proceed. Base the determination to proceed or not on wind calculations per the manufacturer and a reduction in LHE rated capacity if applicable. Include this maximum wind speed determination as part of the activity hazard analysis plan for that operation.
- p. On mobile cranes, lifts where the load weight is greater than 90 percent of the equipment's capacity are prohibited.
- q. Follow FAA guidelines when required based on project location.

3.7.3 Machinery and Mechanized Equipment

- a. Proof of qualifications for operator must be kept on the project site for review.
- b. Manufacture specifications or owner's manual for the equipment must be on-site and reviewed for additional safety precautions or requirements that are sometimes not identified by OSHA or USACE [EM 385-1-1](#). Incorporate such additional safety precautions or requirements into the AHAs.

3.7.4 Base Mounted Drum Hoists

- a. Operation of base mounted drum hoists must be in accordance with [EM 385-1-1](#) and [ASSP A10.22](#).
- b. Rigging gear must be in accordance with applicable ASME/OSHA standards.
- c. When used on telecommunication towers, base mounted drum hoists must be in accordance with [TIA-1019](#), [TIA-222](#), [ASME B30.7](#), [29 CFR 1926.552](#), and [29 CFR 1926.553](#).
- d. When used to hoist personnel, the AHA must include a written standard operating procedure. Operators must have a physical examination in

accordance with EM 385-1-1 Section 16.B.05 and trained, at a minimum, in accordance with EM 385-1-1 Section 16.U and 16.T. The base mounted drum hoist must also comply with OSHA Instruction CPL 02-01-056 and ASME B30.23.

- e. Material and personnel must not be hoisted simultaneously.
- f. Personnel cage must be marked with the capacity (in number of persons) and load limit in kg.
- g. Construction equipment must not be used for hoisting material or personnel or with trolley/tag lines. Construction equipment may be used for towing and assisting with anchoring guy lines.

3.8 EXCAVATIONS

Soil classification must be performed by a competent person in accordance with 29 CFR 1926 and EM 385-1-1.

3.8.1 Utility Locations

Provide a third party, independent, private utility locating company to positively identify underground utilities in the work area in addition to any station locating service and coordinated with the station utility department.

3.8.2 Utility Location Verification

Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within one meter of the underground system.

3.8.3 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. Whenever Contract work involves chipping, saw cutting, or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with station utility departments in addition to location and depth verification by a third party, independent, private locating company. The third party, independent, private locating company must locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement.

3.9 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1, Sections 11 and 12.

3.9.1 Conduct of Electrical Work

As delineated in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the

Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with [ASTM F855](#) and [IEEE 1048](#). Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by [NFPA 70](#), high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety shoes, insulating gloves and electrical arc flash protection for personnel as required by [NFPA 70E](#). Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and [29 CFR 1910.147](#).

3.9.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State, Local and Host Nation requirements applicable to where work is being performed.

3.9.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with [NFPA 70E](#).

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in [NFPA 70E](#) requirements and procedures. Unless permitted by [NFPA 70E](#), no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.9.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with [NFPA 70](#) and [IEEE C2](#) to provide a permanent, continuous and effective path to ground unless otherwise noted by [EM 385-1-1](#).

Check grounding circuits to ensure that the circuit between the ground and a grounded power conductor has a resistance low enough to permit sufficient current flow to allow the fuse or circuit breaker to interrupt the current.

3.9.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 35 29.13

HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES

11/15, CHG 1: 08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 PRECONSTRUCTION SAFETY CONFERENCE
- 1.3 SUBMITTALS
- 1.4 ACCIDENT PREVENTION PLAN/SITE SAFETY AND HEALTH PLAN (APP/SSHP)
 - 1.4.1 Acceptance and Modifications
 - 1.4.2 Availability
- 1.5 STAFF ORGANIZATION, QUALIFICATION AND RESPONSIBILITIES
 - 1.5.1 Safety and Health Manager
 - 1.5.1.1 Additional Qualifications
 - 1.5.1.2 Responsibilities and Duties
 - 1.5.2 Site Safety and Health Officer
 - 1.5.2.1 Qualifications
 - 1.5.2.2 Responsibilities and Duties
 - 1.5.3 Persons Certified in First Aid and CPR
 - 1.5.4 Environmental Observer
 - 1.5.4.1 Qualifications
- 1.6 EMERGENCY RESPONSE AND CONTINGENCY PROCEDURES
- 1.7 CERTIFICATE OF WORKER/VISITOR ACKNOWLEDGEMENT
- 1.8 INSPECTIONS
- 1.9 SAFETY AND HEALTH PHASE-OUT REPORT

PART 2 PRODUCTS

- 2.1 REGULATORY REQUIREMENTS
- 2.2 PERSONAL PROTECTIVE EQUIPMENT
 - 2.2.1 Site Specific PPE Program
 - 2.2.2 Levels of Protection
 - 2.2.2.1 Initial PPE Components
 - 2.2.3 PPE for Government Personnel
- 2.3 EMERGENCY EQUIPMENT AND FIRST AID REQUIREMENTS

PART 3 EXECUTION

- 3.1 SITE DESCRIPTION AND CONTAMINATION CHARACTERIZATION
 - 3.1.1 Project/Site Conditions
 - 3.1.1.1 Storage Tank Documents
 - 3.1.2 Ordnance and Explosives (OE)
- 3.2 TASK SPECIFIC HAZARDS, INITIAL PPE, HAZWOPER MEDICAL SURVEILLANCE AND TRAINING APPLICABILITY
- 3.3 TRAINING
 - 3.3.1 General HTRW Operations Training
 - 3.3.2 Pre-Entry Briefing
 - 3.3.3 Periodic Sessions
- 3.4 MEDICAL SURVEILLANCE PROGRAM

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.5 EXPOSURE MONITORING/AIR SAMPLING PROGRAM
- 3.6 HEAT STRESS MONITORING AND MANAGEMENT
- 3.7 SPILL AND DISCHARGE CONTROL
- 3.8 MATERIALS TRANSFER SAFETY
- 3.9 SITE CONTROL MEASURES
 - 3.9.1 Work Zones
 - 3.9.1.1 Exclusion Zone (EZ)
 - 3.9.1.2 Contamination Reduction Zone (CRZ)
 - 3.9.1.3 Support Zone (SZ)
 - 3.9.2 Site Control Log
 - 3.9.3 Communication
 - 3.9.4 Site Security
- 3.10 PERSONAL HYGIENE AND DECONTAMINATION
 - 3.10.1 Decontamination Facilities
 - 3.10.2 Personnel Decontamination
 - 3.10.3 Equipment Decontamination
 - 3.10.3.1 Facilities for Equipment and Personnel
 - 3.10.3.2 Procedures

ATTACHMENTS:

Example Certificate Of Worker/Visitor Acknowledgement

Task Hazard and Control Sheets

-- End of Section Table of Contents --

SECTION 01 35 29.13

HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES
11/15, CHG 1: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API RP 2219 (2016) Safe Operation of Vacuum Trucks
Handling Flammable and Combustible Liquids
in Petroleum Service

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z358.1 (2014; R 2020) American National Standard
for Emergency Eyewash and Shower Equipment

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

NIOSH 85-115 (1985) Occupational Safety and Health
Guidance Manual for Hazardous Waste Site
Activities

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety -- Safety and Health
Requirements Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1904 Recording and Reporting Occupational
Injuries and Illnesses

29 CFR 1910 Occupational Safety and Health Standards

29 CFR 1910.120 Hazardous Waste Operations and Emergency
Response

29 CFR 1926 Safety and Health Regulations for
Construction

29 CFR 1926.65 Hazardous Waste Operations and Emergency
Response

49 CFR 171 General Information, Regulations, and
Definitions

49 CFR 172 Hazardous Materials Table, Special
Provisions, Hazardous Materials
Communications, Emergency Response

Information, and Training Requirements

1.2 PRECONSTRUCTION SAFETY CONFERENCE

Conduct a preconstruction safety conference prior to the start of site activities and after submission of the Accident Prevention Plan/Site Safety And Health Plan (APP/SSHP). The objective of the meeting is to discuss health and safety concerns related to the impending work, discuss project health and safety organization and expectations, review and answer comments and concerns regarding the APP/SSHP or other health and safety concerns. Ensure that those individuals responsible for health and safety at the project level are available and attend this meeting.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Work Zones; G

Decontamination Facilities; G

SD-03 Product Data

Amendments to the APP/SSHP; G

Exposure Monitoring/Air Sampling Program

Site Control Log

SSH0's Daily Inspection Logs

SD-07 Certificates

Certificate Of Worker/Visitor Acknowledgement

SD-11 Closeout Submittals

Safety And Health Phase-Out Report; G

1.4 ACCIDENT PREVENTION PLAN/SITE SAFETY AND HEALTH PLAN (APP/SSHP)

Develop and implement a Site Safety and Health Plan in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS, and attach to the Accident Prevention Plan (APP) as an appendix (APP/SSHP). Address all occupational safety and health hazards (traditional construction as well as contaminant-related hazards) associated with cleanup operations within the APP/SSHP. Cover each SSHP element in sections 28.A.01 and 33.B of EM 385-1-1 and each APP element in Appendix A of EM 385-1-1. There are overlapping elements in Section 28.A.01 and Appendix A of EM 385-1-1. SSHP appendix elements that overlap with APP elements need not be duplicated in the APP/SSHP provided each safety and occupational health (SOH) issue receives adequate attention and is documented in the APP/SSHP. The APP/SSHP is a dynamic document, subject to change as project operations/execution change. Modify the APP/SSHP to address

changing and previously unidentified health and safety conditions. Ensure that the APP/SSHP is updated accordingly. Submit [amendments to the APP/SSHP](#) to the Contracting Officer as the APP/SSHP is updated. For long duration projects resubmit the APP/SSHP to the Contracting Officer annually for review. The APP/SSHP must contain all updates.

1.4.1 Acceptance and Modifications

Prior to submittal, the APP/SSHP must be signed and dated by the Safety and Health Manager and the Site Superintendent. Submit for review 5 days prior to the Preconstruction Safety Conference. Deficiencies in the APP/SSHP will be discussed at the preconstruction safety conference, and must be revised to correct the deficiencies and resubmitted for acceptance. Onsite work must not begin until the plan has been accepted. Maintain a copy of the written APP/SSHP onsite. Changes and modifications to the APP/SSHP must be made with the knowledge and concurrence of the Safety and Health Manager, the Site Superintendent, and the Contracting Officer. Bring to the attention of the Safety and Health Manager, the Site Superintendent, and the Contracting Officer any unforeseen hazard that becomes evident during the performance of the work, through the Site Safety and Health Officer (SSHO) for resolution as soon as possible. In the interim, take necessary action to re-establish and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public, and the environment. Disregard for the provisions of this specification or the accepted APP/SSHP is cause for stopping work until the matter has been rectified.

1.4.2 Availability

Make available the APP/SSHP in accordance with [29 CFR 1910.120](#), (b)(1)(v) and [29 CFR 1926.65](#), (b)(1)(v).

1.5 STAFF ORGANIZATION, QUALIFICATION AND RESPONSIBILITIES

Provide hazardous waste operations and emergency response organization in accordance with [EM 385-1-1](#), Section 33.

1.5.1 Safety and Health Manager

Safety and Health Manager must be a safety professional certified by the Board of Certified Safety Professionals.

Apply the following in conjunction with the required qualifications and responsibilities stated in [EM 385-1-1](#), Section 33.C.01.

1.5.1.1 Additional Qualifications

The Safety and Health Manager must have the following qualifications:

- a. A minimum of 3 years experience in developing and implementing safety and occupational health programs at underground storage tank removal projects.
- b. Documented experience in supervising professional and technician level personnel.
- c. Documented experience in developing worker exposure assessment programs and air monitoring programs and techniques.

- d. Documented experience in managing personal protective equipment (PPE) programs and conducting PPE hazard evaluations for the types of activities and hazards likely to be encountered on the project.
- e. Working knowledge of Host Nation and Federal occupational safety and health regulations.

1.5.1.2 Responsibilities and Duties

- a. Development, implementation, oversight, and enforcement of the APP/SSHP.
- b. Provide onsite consultation as needed to ensure the APP/SSHP is fully implemented.
- c. Conduct initial site-specific training.
- d. Be available for consultation during the first 3 days of remedial activities and at the startup of each new major phase of work.
- e. Visit the site as needed and at least once per week for the duration of activities, to audit the effectiveness of the APP/SSHP.
- f. Be available for emergencies.
- g. Coordinate any modifications to the APP/SSHP with the Site Superintendent, the SSHO, and the Contracting Officer.
- h. Be responsible for evaluating air monitoring data and recommending changes to engineering controls, work practices, and PPE.
- i. Provide continued support for upgrading/downgrading of the level of personal protection.
- j. Serve as a member of the quality control staff.
- k. Review accident reports and results of daily inspections.
- l. Sign and date the APP/SSHP prior to submittal.

1.5.2 Site Safety and Health Officer

Designate an individual and one alternate as the Site Safety and Health Officer (SSHO). Include the name, qualifications (education and training summary and documentation), and work experience of the Site Safety and Health Officer and alternate in the APP/SSHP.

The Apply the following in conjunction with the required qualifications and responsibilities stated in EM 385-1-1, Section 33.C.02.

1.5.2.1 Qualifications

The following requirements are in addition to those in Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

- a. A minimum of 1 year experience in implementing SOH programs at underground storage tank removal projects where personal protective equipment was required.

- b. Meet 29 CFR 1910.120/29 CFR 1926.65 requirements for 40-hour initial and 8-hour supervisor training and, maintain 8-hour refresher training requirements.
- c. Specific training in personal and respiratory protective equipment, confined space entry and in the proper use of air monitoring instruments and air sampling methods including monitoring for ionizing radiation.
- d. Documented experience in construction techniques and construction safety procedures.
- e. Working knowledge of Federal and Host Nation occupational SOH regulations.

1.5.2.2 Responsibilities and Duties

The following requirements are in addition to those in Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

- a. Assist and represent the Safety and Health Manager in onsite training and the day to day onsite implementation and enforcement of the accepted APP/SSHP.
- b. Be assigned to the site on a full time basis for the duration of field activities. The SSHO can have collateral duties in addition to SOH related duties. If operations are performed during more than 1 work shift per day, a site Safety and Health Officer must be present for each shift and when applicable, act as the radiation safety officer (RSO) as defined in paragraph 06.F.02 of EM 385-1-1 on radioactive waste cleanup projects.
- c. Have authority to stop work if unacceptable health or safety conditions exist, and take necessary action to re-establish and maintain safe working conditions.
- d. Have authority to ensure site compliance with specified SOH requirements, Federal and Host Nation regulations and all aspects of the APP/SSHP including, but not limited to, activity hazard analyses, air monitoring, monitoring for ionizing radiation, use of PPE, decontamination, site control, standard operating procedures used to minimize hazards, safe use of engineering controls, the emergency response plan, confined space entry procedures, spill containment program, and preparation of records by performing a daily SOH inspection and documenting results on the Daily Safety Inspection Log in accordance with 29 CFR 1904.
- e. In coordination with site management and the Safety and Health Manager, recommend corrective actions for identified deficiencies and oversee the corrective actions.
- f. Consult with and coordinate any modifications to the APP/SSHP with the Safety and Health Manager, the Site Superintendent, and the Contracting Officer.
- g. Conduct daily safety inspection and document SOH findings into the Daily Safety Inspection Log. Track noted SOH deficiencies to ensure that they are corrected.

- h. Conduct accident investigations and prepare accident reports.
- i. Serve as a member of the quality control staff on matters relating to SOH.

1.5.3 Persons Certified in First Aid and CPR

At least two persons who are currently certified in first aid and CPR by the American Red Cross or other approved agency must be onsite at all times during site operations. These persons may perform other duties but must be immediately available to render first aid when needed.

1.5.4 Environmental Observer

Provide an Environmental Observer to observe and document the dewatering process if the dewatering effluent is found to be contaminated, has a petroleum odor, or leaves a sheen. If the effluent is contaminated, the Environmental Observer must approve the plan for capturing the water, monitoring the dewatering process, and documenting the process for inclusion in the Accident Prevention Plan/Site Safety and Health Plan. The Environmental Observer must be present when the dewatering begins and must visit the site at least weekly during the process to document the flow/quantity of water, monitor the treatment process, testing, take photos and observe/audit the effectiveness of the of the process. The observer must be available for consultation. When the process is completed, summarize the results in a report.

1.5.4.1 Qualifications

The Environmental Observer must be approved by the Navy and the Spanish ECCA.

1.6 EMERGENCY RESPONSE AND CONTINGENCY PROCEDURES

Develop and implement an Emergency Response Plan, that meets the requirements of EM 385-1-1 Section 33.G, 29 CFR 1910.120 (l) and 29 CFR 1926.65 (l), as a section of the APP/SSHP. In the event of any emergency associated with remedial action, without delay, alert all onsite employees and as necessary offsite emergency responders that there is an emergency situation; take action to remove or otherwise minimize the cause of the emergency; alert the Contracting Officer; and institute measures necessary to prevent repetition of the conditions or actions leading to, or resulting in, the emergency. Train employees that are required to respond to hazardous emergency situations to their level of responsibility according to 29 CFR 1910.120 (q) and 29 CFR 1926.65 (q) requirements. Rehearse the plan regularly as part of the overall training program for site operations. Review the plan periodically and revised as necessary to reflect new or changing site conditions or information. Provide copies of the Emergency Response Portion of the accepted APP/SSHP to the affected local emergency response agencies. Address, as a minimum, the following elements in the plan:

- a. Pre-emergency planning. Coordinate with local emergency response providers during preparation of the Emergency Response Plan. At a minimum, coordinate with local fire, rescue, hazardous materials response teams, police and emergency medical providers to assure all organizations are capable and willing to respond to and provide services for on-site emergencies. Ensure the Emergency Response Plan for the site is compatible and integrated with the local fire, rescue,

medical and police security services available from local emergency response planning agencies.

- b. Personnel roles, lines of authority, communications for emergencies.
- c. Emergency recognition and prevention.
- d. Site topography, layout, and prevailing weather conditions.
- e. Criteria and procedures for site evacuation (emergency alerting procedures, employee alarm system, emergency PPE and equipment, safe distances, places of refuge, evacuation routes, site security and control).
- f. Route maps to nearest prenotified medical facility. Site-support vehicles must be equipped with maps. At the beginning of project operations, drivers of the support vehicles must become familiar with the emergency route and the travel time required.
- g. Specific procedures for decontamination and medical treatment of injured personnel.
- h. Emergency alerting and response procedures including posted instructions and a list of names and telephone numbers of emergency contacts (physician, nearby medical facility, fire and police departments, ambulance service, Federal, and local environmental agencies; as well as Safety and Health Manager, the Site Superintendent, the Contracting Officer and their alternates).
- i. Criteria for initiating community alert program, contacts, and responsibilities.
- j. Procedures for reporting incidents to appropriate government agencies. In the event that an incident such as an explosion or fire, or a spill or release of toxic materials occurs during the course of the project, the appropriate government agencies must be immediately notified. In addition, verbally notify the Contracting Officer and the local district safety office immediately and submit a written notification within 24 hours. Include within the report the following items:
 - (1) Name, organization, telephone number, and location of the Contractor.
 - (2) Name and title of the person(s) reporting.
 - (3) Date and time of the incident.
 - (4) Location of the incident, i.e., site location, facility name.
 - (5) Brief summary of the incident giving pertinent details including type of operation ongoing at the time of the incident.
 - (6) Cause of the incident, if known.
 - (7) Casualties (fatalities, disabling injuries).
 - (8) Details of any existing chemical hazard or contamination.

- (9) Estimated property damage, if applicable.
- (10) Nature of damage, effect on contract schedule.
- (11) Action taken to ensure safety and security.
- (12) Other damage or injuries sustained, public or private.

k. Procedures for critique of emergency responses and follow-up.

1.7 CERTIFICATE OF WORKER/VISITOR ACKNOWLEDGEMENT

A copy of a [certificate of worker/visitor acknowledgement](#) must be completed and submitted for each visitor allowed to enter contamination reduction or exclusion zones, and for each employee, following the Example Certificate Of Worker/Visitor Acknowledgement at the end of this section.

1.8 INSPECTIONS

Attach to and submit with the Daily Quality Control reports the [SSHO's Daily Inspection Logs](#). Include with each entry the following: date, work area checked, employees present in work area, PPE and work equipment being used in each area, special SOH issues and notes, and signature of preparer.

1.9 SAFETY AND HEALTH PHASE-OUT REPORT

Submit a [Safety and Health Phase-Out Report](#) in conjunction with the project close out report, prior to final acceptance of the work. Include the following minimum information :

- a. Summary of the overall performance of SOH (e.g., accidents or incidents including near misses, unusual events, lessons learned).
- b. Final decontamination documentation including procedures and techniques used to decontaminate equipment, vehicles, and on site facilities.
- c. Summary of exposure monitoring and air sampling accomplished during the project.
- d. Signatures of Safety and Health Manager and SSHO.

PART 2 PRODUCTS

2.1 REGULATORY REQUIREMENTS

Comply with [EM 385-1-1](#), [29 CFR 1926.65](#), [29 CFR 1910.120](#), OSHA requirements in [29 CFR 1910](#) and [29 CFR 1926](#) with work performed under this contract, and Host Nation specific requirements where applicable. Submit to the Contracting Officer for resolution matters of interpretation of standards before starting work. The most stringent requirements apply where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary.

2.2 PERSONAL PROTECTIVE EQUIPMENT

2.2.1 Site Specific PPE Program

Provide onsite personnel exposed to contaminants with appropriate personal

protective equipment. Components of levels of protection (B, C, D and modifications) must be relevant to site-specific conditions, including heat and cold stress potential and safety hazards. Use only respirators approved by NIOSH.

Keep protective equipment and clothing clean and well maintained. Include site-specific procedures to determine PPE program effectiveness and for onsite fit-testing of respirators, cleaning, maintenance, inspection, cartridge change out, and storage of PPE within the PPE section of the APP/SSHP.

2.2.2 Levels of Protection

The Safety and Health Manager must establish and evaluate as the work progresses the levels of protection for each work activity. Also establish action levels for upgrade or downgrade in levels of PPE. Describe in the SSHP the protocols and the communication network for changing the level of protection. Address air monitoring results, potential for exposure, changes in site conditions, work phases, job tasks, weather, temperature extremes, and individual medical considerations within the PPE evaluation protocol.

2.2.2.1 Initial PPE Components

The following items constitute initial minimum protective clothing and equipment ensembles.

Level D	Disposable coveralls, gloves, and boot covers, hard hat, safety glasses, and hearing protection
Level C	Level D equipment plus air purifying respirator
Level B	Level D equipment plus Self Contained Breathing Apparatus (SCBA)

2.2.3 PPE for Government Personnel

Three clean sets of personal protective equipment (excluding air-purifying negative-pressure respirators and safety shoes, which will be provided by individual visitors), as required for entry into the Exclusion Zone and Contamination Reduction Zone, must be available for use by the Contracting Officer or official visitors. The items must be cleaned, maintained, and clearly marked: "FOR USE BY GOVERNMENT ONLY." Provide basic training in the use and limitations of the PPE provided.

2.3 EMERGENCY EQUIPMENT AND FIRST AID REQUIREMENTS

Maintain, as a minimum, the following items onsite and available for immediate use:

- First aid equipment and supplies approved by the consulting physician.
- Emergency eyewashes and showers that comply with [ANSI/ISEA Z358.1](#).
- Provide fire extinguishers of sufficient size and type at site facilities and in all vehicles and at any other site locations where flammable or combustible materials present a fire risk.

PART 3 EXECUTION

3.1 SITE DESCRIPTION AND CONTAMINATION CHARACTERIZATION

3.1.1 Project/Site Conditions

Refer to the following reports and information for the site description and contamination characterization.

3.1.1.1 Storage Tank Documents

Subsurface Soil and Groundwater Investigation Report – September 2023
Environmental Assessment for project: P-919 - Bulk Tank Farm Improvements, Transfer of hydrocarbons across the Salado River at the Naval Base of Rota (Spain).

3.1.2 Ordnance and Explosives (OE)

Stop work and contact the Contracting Officer if ordnance and explosives (OE), explosive media or chemical agent contaminated media (CACM) are discovered during HTRW site cleanup activities.

3.2 TASK SPECIFIC HAZARDS, INITIAL PPE, HAZWOPER MEDICAL SURVEILLANCE AND TRAINING APPLICABILITY

Task specific occupational hazards, task specific HAZWOPER medical surveillance and training applicability and task specific initial PPE requirements for the project are listed on the Task Hazard and Control Sheets at the end of this section. Reevaluate occupational safety and health hazards as the work progresses and to adjust the PPE and onsite operations, if necessary, so that the work is performed safely and in compliance with occupational safety and health regulations.

3.3 TRAINING

In conjunction with EM 385-1-1, Section 33D, meet the training program requirements for workers performing cleanup operations and who will be exposed to contaminants.

3.3.1 General HTRW Operations Training

All Personnel performing duties with potential for exposure to onsite contaminants must meet and maintain the following 29 CFR 1910.120/29 CFR 1926.65 (e) training requirements:

- a. 40 hours of off site HTRW instruction.
- b. 3 days actual on-the-job field experience under the direct supervision of a trained, experienced supervisor.
- c. 8 hours refresher training annually.

Onsite supervisors must have an additional 8 hours management and supervisor training specified in 29 CFR 1910.120/29 CFR 1926.65 (e) (4).

3.3.2 Pre-Entry Briefing

Prior to commencement of onsite field activities, all site employees,

including those assigned only to the Support Zone, must attend a site-specific SOH training session. This session will be conducted by the Safety and Health Manager and the Site Safety and Health Officer to ensure that all personnel are familiar with requirements and responsibilities for maintaining a safe and healthful work environment. Thoroughly discuss procedures and contents of the accepted APP/SSHP and Sections 01.B.02 and 28.D.03 of EM 385-1-1. Each employee must sign a training log to acknowledge attendance and understanding of the training. Notify the Contracting Officer at least 5 days prior to the initial site-specific training session so government personnel involved in the project may attend.

3.3.3 Periodic Sessions

Conduct periodic onsite training by the SSHO at least weekly for personnel assigned to work at the site during the following week. Address SOH procedures, work practices, any changes in the APP/SSHP, activity hazard analyses, work tasks, or schedule; results of previous week's air monitoring, review of safety discrepancies and accidents. Convene a meeting prior to implementation of the change should an operational change affecting onsite field work be made, to explain SOH procedures. Conduct a site-specific training sessions for new personnel, visitors, and suppliers by the SSHO using the training curriculum outlines developed by the Safety and Health Manager. Each employee must sign a training log to acknowledge attendance and understanding of the training.

3.4 MEDICAL SURVEILLANCE PROGRAM

Meet all requirements of 29 CFR 1910.120/29 CFR 1926.65 medical surveillance program and EM 385-1-1, Section 33.G for workers performing cleanup operations and who will be exposed to contaminants. Ensure the Occupational Physician or the physician's designee performs the physical examinations and reviews examination results. Participation in the medical surveillance program is without cost to the employee, without loss of pay and at a reasonable time and place.

3.5 EXPOSURE MONITORING/AIR SAMPLING PROGRAM

Prepare and implement by the Safety and Health Manager an exposure monitoring/air sampling program to identify and quantify SOH hazards and airborne levels of hazardous substances in order to assure proper selection of engineering controls, work practices and personal protective equipment for affected site personnel. Include action levels for upgrading/downgrading PPE in the program.

3.6 HEAT STRESS MONITORING AND MANAGEMENT

Document in the APP/SSHP and implement the procedures and practices in section 06.J. in EM 385-1-1 to monitor and manage heat stress.

3.7 SPILL AND DISCHARGE CONTROL

Develop and implement written spill and discharge containment/control procedures. Address radioactive wastes, shock sensitive wastes, laboratory waste packs, material handling equipment, as well as drum and container handling, opening, sampling, shipping and transport. Describe prevention measures, such as building berms or dikes; spill control measures and material to be used (e.g. booms, vermiculite); location of the spill control material; personal protective equipment required to

cleanup spills; disposal of contaminated material; and who is responsible to report the spill. Storage of contaminated material or hazardous materials must be appropriately bermed, diked and contained to prevent any spillage of material on uncontaminated soil. If the spill or discharge is reportable, or human health or the environment are threatened, notify the Contracting Officer as soon as possible. Provide control as required by Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS.

3.8 MATERIALS TRANSFER SAFETY

Remove liquids and residues from the tanks using explosion-proof or air-driven pumps. In accordance with EM 385-1-1, Section 9, electrically bond the tank and ground pump motors and suction hoses to prevent electrostatic ignition hazards. Use of a hand pump will be permitted to remove the last of the liquid from the bottom of the tanks. If a vacuum truck is used for removal of liquids or residues, the area of operation for the vacuum truck must be vapor free. Locate the truck upwind from the tank and outside the path of probable vapor travel. Discharge the vacuum pump exhaust gases through a hose of adequate size and length downwind of the truck and tank area. Vacuum truck operating and safety practices must conform to API RP 2219. Collect tank residues in drums, tanks, or tank trucks labeled according to 49 CFR 171 and 49 CFR 172 and disposed of as specified. Disconnect and drain fittings and lines of their contents after the materials have been transferred and the tanks have been exposed. Do not spill contents into the environment during cutting or disconnecting of tank fittings. Transfer materials drained into DOT-approved drums for storage and transportation. Use only non-sparking or non-heat producing tools to disconnect and drain or to cut through tank fittings. Electrical equipment (e.g., pumps, portable hand tools) used for tank preparation must be explosion-proof. Following cutting or disconnecting of the fittings, plug openings leading to the tanks.

3.9 SITE CONTROL MEASURES

Coordinate site control measures with Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS.

3.9.1 Work Zones

Initial anticipated work zone boundaries (exclusion zone, contamination reduction zone, support zone, all access points and decontamination areas) are to be clearly delineated on the site drawings. Base delineation of work zone boundaries on the contamination characterization data and the hazard/risk analysis to be performed as described in EM 385-1-1 06.A.02. As work progresses and field conditions are monitored, work zone boundaries may be modified (and site drawings modified) with approval of the Contracting Officer. Clearly identify work zones and mark in the field (using fences, tape, or signs). Submit and post a site map, showing work zone boundaries and locations of decontamination facilities in the onsite office. Work zones must consist of the following:

3.9.1.1 Exclusion Zone (EZ)

The exclusion zone is the area where hazardous contamination is either known or expected to occur and the greatest potential for exposure exists. Control entry into this area and exit may only be made through the Contamination Reduction Zone (CRZ).

3.9.1.2 Contamination Reduction Zone (CRZ)

The CRZ is the transition area between the Exclusion Zone and the Support Zone. The personnel and equipment decontamination areas must be separate and unique areas located in the CRZ.

3.9.1.3 Support Zone (SZ)

The Support Zone is defined as areas of the site, other than exclusion zones and contamination reduction zones, where workers do not have the potential to be exposed to hazardous substances or dangerous conditions resulting from HTRW operations. Secure the Support Zone against active or passive contamination. Site offices, parking areas, and other support facilities must be located in the Support Zone.

3.9.2 Site Control Log

A log of personnel visiting, entering, or working on the site must be maintained. Include the following: date, name, agency or company, time entering and exiting site, time entering and exiting the exclusion zone (if applicable). Before visitors are allowed to enter the Contamination Reduction Zone or Exclusion Zone, they must show proof of current training, medical surveillance and respirator fit testing (if respirators are required for the tasks to be performed) and fill out a Certificate of Worker or Visitor Acknowledgment. Record this visitor information, including date, in the log.

3.9.3 Communication

Provide and install an employee alarm system that has adequate means of on and off site communication in accordance with 29 CFR 1910 Section .165. The means of communication must be able to be perceived above ambient noise or light levels by employees in the affected portions of the workplace. The signals must be distinctive and recognizable as messages to evacuate or to perform critical operations.

3.9.4 Site Security

Provide the following site security: warning signs. Print signs in bold large letters on contrasting backgrounds. Signs must be visible from all points where entry might occur and at such distances from the restricted area that employees may read the signs and take necessary protective steps before entering.

3.10 PERSONAL HYGIENE AND DECONTAMINATION

Personnel entering the Exclusion or Contamination Reduction Zones or otherwise exposed to hazardous chemical vapors, gases, liquids, or contaminated solids must decontaminate themselves and their equipment prior to exiting the contamination reduction zone (CRZ) and entering the support zone. Consult Chapter 10.0 of NIOSH 85-115 when preparing decontamination procedures. Submit a detailed discussion of personal hygiene and decontamination facilities and procedures to be followed by site workers as part of the APP/SSHP. Train employees in the procedures and enforce the procedures throughout site operations.

3.10.1 Decontamination Facilities

Submit drawings showing the layout of the personnel and equipment

decontamination areas.

3.10.2 Personnel Decontamination

Initially set up a decontamination line in the CRZ. Employees must exit the exclusion zone through the CRZ and implement the following decontamination procedures and techniques: remove disposable garments. Showers, if needed, must comply with 29 CFR 1910, Section.141 and EM 385-1-1, 02 F, Washing Facilities. It is the Site Safety and Health Officer's responsibility to recommend techniques to improve personnel decontamination procedures, if necessary.

3.10.3 Equipment Decontamination

Decontaminate the vehicles and equipment used in the EZ in the CRZ prior to leaving the EZ.

3.10.3.1 Facilities for Equipment and Personnel

Provide a decontamination station within the CRZ for decontaminating vehicles and equipment leaving the EZ. Provide a designated "clean area" in the CRZ for performing equipment maintenance. Use this area when personnel are required by normal practices to come in contact with the ground, i.e., crawling under a vehicle to change engine oil. Equipment within the EZ or CRZ must be decontaminated before maintenance is performed.

3.10.3.2 Procedures

Procedures for equipment decontamination must be developed and utilized to prevent the spread of contamination into the SZ and offsite areas. These procedures must address disposal of contaminated products and spent materials used on the site, including, as a minimum, containers, fluids, and oils. Assume any item taken into the EZ to be contaminated and perform an inspection and decontaminate. Vehicles, equipment, and materials must be cleaned and decontaminated prior to leaving the site. Handle construction material in such a way as to minimize the potential for contaminants being spread or carried offsite. Prior to exiting the site, vehicles and equipment must be monitored to ensure the adequacy of decontamination.

-- End of Section --

Visitor Acknowledgment of Risk

In consideration of the services of _____ their officers, agents, employees, and stockholders, and all other persons or entities associated with those businesses (hereafter collectively referred to as “_____”) I agree as follows:

Although _____ has taken reasonable steps to provide me with appropriate equipment and skilled guides so I can enjoy an activity for which I may not be skilled, _____ has informed me this activity is not without risk. Certain risks are inherent in each activity and cannot be eliminated without destroying the unique character of the activity. These inherent risks are some of the same elements that contribute to the unique character of this activity and can be the cause of loss or damage to my equipment, or accidental injury, illness, or in extreme cases, permanent trauma or death. _____ does not want to frighten me or reduce my enthusiasm for this activity, but believes it is important for me to know in advance what to expect and to be informed of the inherent risks. The following describes some, but not all, of those risks.

Description of Risks:

I am aware that _____ entails risks of injury or death to any participant. I understand the description of these inherent risks is not complete and that other unknown or unanticipated inherent risks may result in injury or death. I agree to assume and accept full responsibility for the inherent risks identified herein and those inherent risks not specifically identified. My participation in this activity is purely voluntary; no one is forcing me to participate, and I elect to participate in spite of and with full knowledge of the inherent risks.

I acknowledge that engaging in this activity may require a degree of skill and knowledge different from other activities and that I have responsibilities as a participant. I acknowledge that the staff of _____ has been available to more fully explain to me the nature and physical demands of this activity and the inherent risks, hazards, and dangers associated with this activity.

I certify that I am fully capable of participating in this activity. Therefore, I assume and accept full responsibility for myself, including all minor children in my care, custody, and control, for bodily injury, death, or loss of personal property and expenses as a result of those inherent risks and dangers identified herein and those inherent risks and dangers not specifically identified, and as a result of my negligence in participating in this activity.

I have carefully read, clearly understood, and accepted the terms and conditions stated herein and acknowledge that this agreement shall be effective and binding upon me, my heirs, assigns, personal representative, and estate and for all members of my family, including minor children.

Signature

Date

Signature of Parent of Guardian, if participant is under 18 years of age

Signature

Date

Task Hazard and Control Requirements Sheet	
Task	
Initial Anticipated Hazards	
Initial PPE	
Initial Controls	
Initial Exposure Monitoring	
[Yes][No]	HAZWOPER Medical Surveillance Required
[Yes][No]	HAZWOPER Training Required

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS

02/19

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 ORDERING INFORMATION

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS
02/19

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)
30 West University Drive
Arlington Heights, IL 60004-1893
Ph: 847-394-0150
Fax: 847-253-0088
E-mail: communications@amca.org
Internet: <http://www.amca.org>

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)
2111 Wilson Blvd, Suite 400
Arlington, VA 22201
Ph: 703-524-8800
Internet: <http://www.ahrinet.org>

ALUMINUM ASSOCIATION (AA)
1400 Crystal Drive
Suite 430
Arlington, VA 22202
Ph: 703-358-2960
E-Mail: info@aluminum.org
Internet: <https://www.aluminum.org/>

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)
1900 E Golf Rd, Suite 1250
Schaumburg, IL 60173
Ph: 847-303-5664
E-mail: customerservice@aamanet.org
Internet: <https://aamanet.org/>

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)
444 North Capital Street, NW, Suite 249

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Washington, DC 20001
Ph: 202-624-5800
Fax: 202-624-5806
E-Mail: info@ashto.org
Internet: <https://www.transportation.org/>

AMERICAN CONCRETE INSTITUTE (ACI)
38800 Country Club Drive
Farmington Hills, MI 48331-3439
Ph: 248-848-3700
Fax: 248-848-3701
Internet: <https://www.concrete.org/>

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)
1330 Kemper Meadow Drive
Cincinnati, OH 45240
Ph: 513-742-2020
Fax: 513-742-3355
Internet: <https://www.acgih.org/>

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)
130 East Randolph, Suite 2000
Chicago, IL 60601
Ph: 312-670-5444
Fax: 312-670-5403
Steel Solutions Center: 866-275-2472
E-mail: solutions@aisc.org
Internet: <https://www.aisc.org/>

AMERICAN LADDER INSTITUTE (ALI)
330 N. Wabash, Suite 2000
Chicago, IL 60611
Ph: 312-321-6806
Fax: 312-673-6929
E-mail: info@americanladderinstitute.org
Internet: <https://www.americanladderinstitute.org>

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
1899 L Street, NW, 11th Floor
Washington, DC 20036
Ph: 202-293-8020
Fax: 202-293-9287
E-mail: storemanager@ansi.org
Internet: <https://www.ansi.org/>

AMERICAN PETROLEUM INSTITUTE (API)
1220 L Street, NW
Washington, DC 20005-4070
Ph: 202-682-8000
Internet: <https://www.api.org/>

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)
P.O. Box 28518
1711 Arlingate Lane
Columbus, OH 43228-0518
Ph: 800-222-2768 or 614-274-6003
Fax: 614-274-6899
E-mail: tjones@asnt.org
Internet: <https://www.asnt.org/>

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)
1801 Alexander Bell Drive
Reston, VA 20191
Ph: 800-548-2723; 703-295-6300
Internet: <https://www.asce.org/>

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)
1791 Tullie Circle, NE
Atlanta, GA 30329
Ph: 404-636-8400 or 800-527-4723
Fax: 404-321-5478
E-mail: ashrae@ashrae.org
Internet: <https://www.ashrae.org/>

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Two Park Avenue
New York, NY 10016-5990
Ph: 800-843-2763
Fax: 973-882-1717
E-mail: customercare@asme.org
Internet: <https://www.asme.org/>

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)
520 N. Northwest Highway
Park Ridge, IL 60068
Ph: 847-699-2929
E-mail: customerservice@assp.org
Internet: <https://www.assp.org/>

AMERICAN WATER WORKS ASSOCIATION (AWWA)
6666 W. Quincy Avenue
Denver, CO 80235 USA
Ph: 303-794-7711 or 800-926-7337
Fax: 303-347-0804
Internet: <https://www.awwa.org/>

AMERICAN WELDING SOCIETY (AWS)
8669 NW 36 Street, #130
Miami, FL 33166-6672
Ph: 800-443-9353
Internet: <https://www.aws.org/>

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)
P.O. Box 361784
Birmingham, AL 35236-1784
Ph: 205-733-4077
Fax: 205-733-4075
Internet: <http://www.awpa.com>

ASPHALT INSTITUTE (AI)
2696 Research Park Drive
Lexington, KY 40511-8480
Ph: 859-288-4960
Fax: 859-288-4999
E-mail: info@asphaltinstitute.org
Internet: <http://www.asphaltinstitute.org>

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

ASSOCIATED AIR BALANCE COUNCIL (AABC)
1220 19th St NW, Suite 410
Washington, DC 20036
Ph: 202-737-0202
Fax: 202-315-0285
E-mail: info@aabc.com
Internet: <https://www.aabc.com/>

ASSOCIATION OF EDISON ILLUMINATING COMPANIES (AEIC)
600 North 18th Street
P.O. Box 2641
Birmingham, AL 35291
Ph: 205-257-3839
Fax: 205-257-2540
Internet: <https://aeic.org/>

ASTM INTERNATIONAL (ASTM)
100 Barr Harbor Drive, P.O. Box C700
West Conshohocken, PA 19428-2959
Ph: 610-832-9500
Fax: 610-832-9555
E-mail: service@astm.org
Internet: <https://www.astm.org/>

BRITISH STANDARDS INSTITUTE (BSI)
Ph: +44 345-086-9001
E-mail: cservices@bsigroup.com
Internet: <https://www.bsigroup.com/>

BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)
355 Lexington Avenue, 15th Floor
New York, NY 10017
Ph: 212-297-2122
Fax: 212-370-9047
Internet: <https://www.buildershardware.com/>

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)
PO Box 997377, MS 0500
Sacramento, CA 95899-7377
Ph: 916-558-1784
Internet: <https://www.cdph.ca.gov/>

CEILINGS AND INTERIOR SYSTEMS CONSTRUCTION ASSOCIATION (CISCA)
1010 Jorie Blvd, Suite 30
Oak Brook, IL 60523
Ph: 630-584-1919
Fax: 866-560-8537
E-mail: cisca@cisca.org
Internet: <https://www.cisca.org>

CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC)
1600 Clifton Road
Atlanta, GA 30329-4027
Ph: 800-232-4636
TTY: 888-232-6348
Internet: <https://www.cdc.gov>

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

COMPRESSED GAS ASSOCIATION (CGA)
14501 George Carter Way, Suite 103
Chantilly, VA 20151-1788
Ph: 703-788-2700
Fax: 703-961-1831
E-mail: cga@cganet.com
Internet: <https://www.cganet.com/>

ELECTRICAL GENERATING SYSTEMS ASSOCIATION (EGSA)
1650 South Dixie Highway, Suite 400
Boca Raton, FL 33432-7462
Ph: 561-750-5575
Fax: 561-395-8557
Internet: <http://www.egsa.org>

ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)
310 Maxwell Road, Suite 200
Alpharetta, GA 30009
Ph: 678-393-9990
Fax: 678-393-9998
E-mail: emikoski@ecianow.org
Internet: <https://www.ecianow.org>

EUROPEAN UNION (EU)
European Commission
Rue de la Loi 200
1000 Bruxelles
Belgium
Ph: +32 2 299 96 96
Internet: https://ec.europa.eu/info/index_en

FM GLOBAL (FM)
270 Central Avenue
Johnston, RI 02919-4949
Ph: 401-275-3000
Fax: 401-275-3029
Internet: <https://www.fmglobal.com/>

FOREST STEWARDSHIP COUNCIL (FSC)
708 First Street North, Suite 235
Minneapolis, MN 55401
Ph: 612-353-4511
E-mail: info@us.fcs.org
Internet: <https://us.fsc.org/>

FOUNDATION FOR CROSS-CONNECTION CONTROL AND HYDRAULIC RESEARCH
(FCCCHR)
USC Foundation Office
Research Annex 219
Los Angeles, CA 90089-7700
Ph: 866-545-6340
Fax: 213-740-8399
E-mail: fccchr@usc.edu
Internet: <https://fccchr.usc.edu/>

GLASS ASSOCIATION OF NORTH AMERICA (GANA)
National Glass Association
1945 Old Gallows Rd., Suite 750
Vienna, VA 22182

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Ph: 866-342-5642
Ph: 703-442-4890
Fax: 703-442-0630
Internet: <http://www.glasswebsite.com>

GREEN SEAL (GS)
1001 Connecticut Avenue, NW
Suite 827
Washington, DC 20036-5525
Ph: 202-872-6400
Fax: 202-872-4324
E-mail: green seal@green seal.org
Internet: <https://www.green seal.org/>

GYP SUM ASSOCIATION (GA)
962 Wayne Ave., Suite 620
Silver Spring, MD 20910
Ph: 301-277-8686
Fax: 301-277-8747
E-mail: info@gyp sum.org
Internet: <https://www.gyp sum.org/>

ICC EVALUATION SERVICE, INC. (ICC-ES)
3060 Saturn Street, Suite 100
Brea, CA 92821
Ph: 800-423-6587
Fax: 562-695-4694
E-mail: es@icc-es.org
Internet: <https://icc-es.org/>

ILLUMINATING ENGINEERING SOCIETY (IES)
120 Wall Street, Floor 17
New York, NY 10005-4001
Ph: 212-248-5000
Fax: 212-248-5018
E-mail: membership@ies.org
Internet: <https://www.ies.org/>

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
445 and 501 Hoes Lane
Piscataway, NJ 08854-4141
Ph: 732-981-0060 or 800-701-4333
Fax: 732-981-9667
E-mail: onlinesupport@ieee.org
Internet: <https://www.ieee.org/>

INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)
P.O. Box 493
Miamitown, OH 45041-9998
E-mail: info@icea.net
Internet: <https://www.icea.net/>

INSULATING GLASS MANUFACTURERS ALLIANCE (IGMA)
27 N. Wacker Dr. Suite 365
Chicago, IL 60606-2800
Ph: 613-233-1510
Fax: 613-482-9436
E-mail: enquiries@igmaonline.org
Internet: <https://www.igmaonline.org/>

INTERNATIONAL CODE COUNCIL (ICC)
500 New Jersey Avenue, NW
6th Floor, Washington, DC 20001
Ph: 800-786-4452 or 888-422-7233
Fax: 202-783-2348
E-mail: order@iccsafe.org
Internet: <https://www.iccsafe.org/>

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)
3050 Old Centre Ave. Suite 101
Portage, MI 49024
Ph: 269-488-6382
Fax: 269-488-6383
Internet: <https://www.netaworld.org/>

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)
3, rue de Varembe, 1st floor
P.O. Box 131
CH-1211 Geneva 20, Switzerland
Ph: 41-22-919-02-11
Fax: 41-22-919-03-00
E-mail: info@iec.ch
Internet: <https://www.iec.ch/>

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)
ISO Central Secretariat
BIBC II
Chemin de Blandonnet 8
CP 401 - 1214 Vernier, Geneva
Switzerland
Ph: 41-22-749-01-11
E-mail: central@iso.ch
Internet: <https://www.iso.org>

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)
1901 North Moore Street
Arlington, VA 22209-1762
Ph: 703-525-1695
Fax: 703-528-2148
Internet: <https://safetyequipment.org/>

INTERNATIONAL SOCIETY OF AUTOMATION (ISA)
67 T.W. Alexander Drive
PO Box 12277
Research Triangle Park, NC 27709
Ph: 919-549-8411
Fax: 919-549-8288
E-mail: info@isa.org
Internet: <https://www.isa.org/>

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS
INDUSTRY (MSS)
127 Park Street, NE
Vienna, VA 22180-4602
Ph: 703-281-6613
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Internet: <https://www.sae.org/>

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Diamond Bar, CA 91765
Ph: 909-396-2000
E-mail: webinquiry@aqmd.gov
Internet: <http://www.aqmd.gov>

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Fax: 703-907-7727
E-mail: marketing@tiaonline.org
Internet: <https://www.tiaonline.org/>

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Londonderry, NH 03053
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PART 2 PRODUCTS

Not used

PART 3 EXECUTION

Not used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 42 15

METRIC MEASUREMENTS

02/14

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 GENERAL
- 1.3 USE OF MEASUREMENTS IN SPECIFICATIONS
 - 1.3.1 Hard Metric
 - 1.3.2 Soft Metric
 - 1.3.3 Neutral
- 1.4 COORDINATION
- 1.5 RELATIONSHIP TO SUBMITTALS

-- End of Section Table of Contents --

SECTION 01 42 15

METRIC MEASUREMENTS
02/14

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM SI10

(2016) American National Standard for Use
of the International System of Units (SI):
The Modern Metric System

1.2 GENERAL

This project includes metric units of measurements. The metric units used are the International System of Units (SI) developed and maintained by the General Conference on Weights and Measures (CGPM); the name International System of Units and the international abbreviation SI were adopted by the 11th CGPM in 1960. When both metric and English inch-pound (I-P) measurements are included the specification may contain measurements for products that are manufactured to an industry recognized rounded metric (hard metric) dimensions but are allowed to be substituted by I-P products, to indicate industry and/or Government standards, test values or other controlling factors, such as the code requirements where I-P values are needed for clarity, or to trace back to the referenced standards, test values or codes.

1.3 USE OF MEASUREMENTS IN SPECIFICATIONS

Measurements in specifications are either in SI or I-P units as indicated, except as otherwise authorized. When only SI or I-P measurements are specified for a product, procure the product in the specified units (SI or I-P) unless otherwise authorized by the Contracting Officer. The Contractor is responsible for all associated labor and materials when authorized to substitute one system of units for another and for the final assembly and performance of the specified work and/or products.

1.3.1 Hard Metric

Hard metric measurements are often used for field data such as distance from one point to another or distance above the floor. Products are considered to be hard metric when they are manufactured to metric dimensions or have an industry recognized metric designation.

1.3.2 Soft Metric

- a. A soft metric measurement is a non-mathematical, industry related conversion. Soft metric measurements are used for measurements pertaining to products, test values, and other situations where the I-P units are the standard for manufacture, verification, or other

controlling factor.

- b. A soft metric measurement is also indicated for products that are manufactured in industry designated metric dimensions but are required by law to allow substitute I-P products.

1.3.3 Neutral

A neutral measurement is indicated by an identifier which has no expressed relation to either an SI or an I-P value (e.g., American Wire Gage (AWG) which indicates thickness but in itself is neither SI nor I-P).

1.4 COORDINATION

Bring discrepancies, such as mismatches or product unavailability, arising from use of both metric and non-metric measurements and discrepancies between the measurements in the specifications and the measurements in the drawings to the attention of the Contracting Officer for resolution.

1.5 RELATIONSHIP TO SUBMITTALS

Submittals for Government approval or for information only covers the SI or I-P products actually being furnished for the project. Submit the required drawings and calculations in the same units used in the contract documents describing the product or requirement unless otherwise instructed or approved. Use [ASTM SI10](#) as the basis for establishing metric measurements required to be used in submittals.

-- End of Document --

SECTION TABLE OF CONTENTS
DIVISION 01 - GENERAL REQUIREMENTS
SECTION 01 45 00
QUALITY CONTROL
08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 PAYMENT
- 1.3 SUBMITTALS
- 1.4 GENERAL REQUIREMENTS
- 1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS
 - 1.5.1 Meetings
 - 1.5.1.1 Quality Control Plan Meeting
 - 1.5.1.2 Coordination and Mutual Understanding Meeting
 - 1.5.1.2.1 Purpose
 - 1.5.1.2.2 Coordination of Activities
 - 1.5.1.2.3 Attendees
 - 1.5.1.3 Quality Control (QC) Meetings
 - 1.5.2 Contractor Quality Control (CQC) Plan
 - 1.5.2.1 Content of Contractor Quality Control (CQC) Plan
 - 1.5.2.2 Additional Requirements for Design Quality Control (DQC) Plan
 - 1.5.3 Acceptance of the Quality Control (QC) and Design Quality Control (DQC) Plan
 - 1.5.4 Preliminary Construction Work Authorized Prior to Acceptance
 - 1.5.5 Notification of Changes
 - 1.5.6 Special Inspections
- 1.6 QUALITY CONTROL (QC) ORGANIZATION
 - 1.6.1 Quality Control (QC) Manager
 - 1.6.1.1 Duties
 - 1.6.1.2 Qualifications
 - 1.6.1.3 Construction Quality Management Training
 - 1.6.1.4 Quality Control Requirements for Multiple Work Sites
 - 1.6.2 Organizational Changes
 - 1.6.3 Alternate Quality Control (QC) Manager Duties and Qualifications
 - 1.6.4 Assistant Quality Control (QC) Manager Duties and Qualifications
 - 1.6.5 Commissioning
 - 1.6.5.1 Certificate of Readiness
 - 1.6.6 Quality Control (QC) Specialists
 - 1.6.7 Submittal Reviewers Duties and Qualifications
- 1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL
- 1.8 THREE PHASES OF CONTROL
 - 1.8.1 Preparatory Phase
 - 1.8.2 Initial Phase
 - 1.8.3 Follow-Up Phase
 - 1.8.4 Additional Preparatory and Initial Phases
 - 1.8.5 Notification of Three Phases of Control for Off-Site Work
 - 1.8.6 Deficiency/Rework Items List

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NAVSTA ROTA, SPAIN

1.9 TESTING

- 1.9.1 Accreditation Requirements
- 1.9.2 Laboratory Accreditation Authorities
- 1.9.3 Capability Check
- 1.9.4 Test Results
- 1.9.5 Test Reports and Monthly Summary Report of Tests

1.10 COMPLETION INSPECTIONS

- 1.10.1 Punch-Out Inspection
- 1.10.2 Pre-Final Inspection
- 1.10.3 Final Acceptance Inspection

1.11 QUALITY CONTROL (QC) CERTIFICATIONS

- 1.11.1 Contractor Quality Control (CQC) Report Certification
- 1.11.2 Completion Certification
- 1.11.3 Invoice Certification

1.12 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER

- 1.12.1 Construction Documentation
- 1.12.2 Quality Control Activities
- 1.12.3 Verification Statement
- 1.12.4 Reports from the Quality Control (QC) Specialist(s)
- 1.12.5 Quality Control Validation
- 1.12.6 Testing Plan and Log
- 1.12.7 As-Built Drawings

1.13 NOTIFICATION ON NON-COMPLIANCE

1.14 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 45 00

QUALITY CONTROL
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C1077	(2017) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation
ASTM D3666	(2016) Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials
ASTM D3740	(2019) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
ASTM E329	(2021) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection
ASTM E543	(2021) Standard Specification for Agencies Performing Non-Destructive Testing

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety -- Safety and Health Requirements Manual
ER 1110-3-12	(2021) Military Engineering and Design Quality Management

1.2 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program. Include all associated costs in the applicable Pricing Schedule item.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G

Additional Requirements For Design Quality Control (DQC) Plan; G

SD-05 Design Data

Discipline-Specific Checklists

SD-06 Test Reports

Verification Statement

SD-07 Certificates

Certificate Of Readiness; G

1.4 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system that complies with FAR 52.246-12 Inspection of Construction. QC is comprised of plans, procedures, and organization necessary to produce an end product that complies with the Contract requirements. The QC system covers all design and construction operations, both onsite and offsite, and must be keyed to the proposed design and construction sequence. The Quality Control Manager, Superintendent, Site Safety and Health Officer (SSHO), and all on-site supervisors are responsible for the quality of work and are subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the Contract. The Quality Control Manager must maintain a physical presence at the work site at all times and is the primary individual responsible for all quality control.

1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this section. This QC program is a key element in meeting the objectives of the Commissioning Process (Cx). The QC program consists of a QC Organization, QC Plan, QC Plan Meeting(s), a Coordination and Mutual Understanding Meeting, QC meetings, three phases of control, submittal review and approval, testing, completion inspections, QC certifications, independent Special Inspections in accordance with Section 01 45 35 SPECIAL INSPECTIONS, and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations that comply with the requirements of this Contract. The QC program must cover on-site and off-site work and be keyed to the work sequence. No construction work or testing may be performed unless the QC Manager is on the work site. The QC Manager must report to an officer of the firm and not be subordinate to the Project Superintendent or the Project Manager. The QC Manager, Project Superintendent and Project Manager must work together effectively. Although the QC Manager is the primary individual responsible for quality control, all individuals will be held responsible for the quality of work on the job.

1.5.1 Meetings

1.5.1.1 Quality Control Plan Meeting

Prior to submission of the QC Plan, the Contractor may request a meeting

with the Contracting Officer to discuss the QC Plan requirements of this Contract.

The purpose of this meeting is to develop a mutual understanding of the QC Plan requirements prior to plan development and submission and to agree on the Contractor's list of Definable Feature of Work (DFOW).

1.5.1.2 Coordination and Mutual Understanding Meeting

After the **Post Award Conference**, before start of **design or** construction, and prior to acceptance by the Government of the CQC Plan, meet with the Contracting Officer and discuss the Contractor's quality control system. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations, **design activities**, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the QC Manager and signed by the Contractor, **the Architect/Engineer (A/E)**, and the Government. Provide a copy of the signed minutes to all attendees and include in the QC Plan. At a minimum the Coordination and Mutual Understanding Meeting must be repeated when a new QC Manager is appointed. There can be other occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

1.5.1.2.1 Purpose

The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site work, design intent, environmental requirements and procedures, coordination of activities to be performed, and the coordination of the Contractor's management, production, and QC personnel. At the meeting, the Contractor must explain in detail how three phases of control will be implemented for each DFOW, as well as how each DFOW will be affected by each management plan or requirement as listed below:

- a. Waste Management Plan.
- b. Environmental Protection Plan.
- c. Environmental regulatory requirements.

1.5.1.2.2 Coordination of Activities

Coordinate activities included in various sections to assure efficient and orderly installation of each component. Coordinate operations included under different sections that are dependent on each other for proper installation and operation.

1.5.1.2.3 Attendees

As a minimum, the Contractor's personnel required to attend include an officer of the firm, the Project Manager, Project Superintendent, QC Manager, Alternate QC Manager, Environmental Manager, and subcontractor representatives. Each subcontractor who will be assigned QC responsibilities must have a principal of the firm at the meeting.

1.5.1.3 Quality Control (QC) Meetings

After the start of construction, conduct weekly QC meetings led by the QC Manager at the work site with the Project Superintendent, and the other personnel as necessary. The QC Manager is to prepare the minutes of the meeting and provide a copy to the Contracting Officer within two working days after the meeting. The Contracting Officer may attend these meetings. As a minimum, accomplish the following at each meeting:

- a. Review the minutes of the previous meeting.
- b. Review the schedule and the status of work and deficiencies/rework. Review the most current approved schedule (in accordance with schedule specification) and the status of work and deficiencies/rework.
- c. Review the status of submittals and Request For Information (RFIs).
- d. Review the work to be accomplished in the next three weeks as defined by the schedule section paragraph and all documentation required for that work.
- e. Review Testing Plan and Log including status of tests performed since last QC Meeting.
- f. Resolve QC and production problems. Discuss status of pending change orders.
- g. Address items that may require revising the QC Plan.
- h. Review Accident Prevention Plan (APP) and effectiveness of the safety program.
- i. Review environmental requirements and procedures.
- j. Review Environmental Management Plan.
- k. Review Waste Management Plan.
- l. Review the status of training completion.

1.5.2 Contractor Quality Control (CQC) Plan

Submit no later than 30 days **after Contract Award**, the CQC Plan proposed to implement the requirements FAR 52.246-12 Inspection of Construction. **Design and Construction** will be permitted to begin only after acceptance of the CQC Plan and other Contract requirements **Design work is not permitted to start prior to approval of a Design Quality Control Plan.**

1.5.2.1 Content of Contractor Quality Control (CQC) Plan

Provide a CQC Plan, prior to start of construction that includes a table of contents, with major sections identified, pages numbered sequentially, and that documents the proposed methods and responsibilities for accomplishing quality control during the construction of the project. The CQC Plan must at a minimum include the following sections:

- a. A description of the quality control organization and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified.

- b. An organizational chart showing the quality control organization with individual names and job titles and lines of authority up to an executive of the company at the home office.
- c. NAMES AND QUALIFICATIONS: Names and qualifications, in resume format, (including position titles and durations for qualifying experiences) for each person in the QC organization. Include the Construction Quality Management (CQM) for Contractors course certifications for the QC personnel as required by the paragraph CONSTRUCTION QUALITY MANAGEMENT TRAINING.
- d. DUTIES, RESPONSIBILITY AND AUTHORITY OF QC PERSONNEL: Duties, responsibilities, and authorities of each person in the QC organization.
- e. OUTSIDE ORGANIZATIONS: A listing of outside organizations, such as architectural and consulting engineering firms, that will be employed by the Contractor and a description of the services these firms will provide.
- f. APPOINTMENT LETTERS: Letters signed by an officer of the firm appointing the QC Manager and Alternate QC Manager, [Design Quality Control Manager](#), and stating that they are responsible for implementing and managing the QC program as described in this Contract. Include in this letter the responsibility of the QC Manager and Alternate QC Manager to implement and manage the three phases of control, and their authority to stop work that is not in compliance with the Contract. Letters of direction are to be issued by the QC Manager to the Assistant QC Manager and all other QC Specialists or quality control representatives outlining their duties, authorities, and responsibilities. Include copies of the letters in the QC Plan.
- g. TESTING LABORATORY INFORMATION: Testing laboratory information required by the paragraph ACCREDITATION REQUIREMENTS, as applicable.
- h. TESTING PLAN AND LOG: A Testing Plan and Log that includes the tests required, associated feature of work required, referenced by the specification paragraph number requiring the test, the frequency, and the person responsible for each test.
- i. Procedures to complete [design and](#) construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected. This phase is performed prior to beginning work on each definable feature of work, after all required plans, documents, materials are approved, and after copies are at the work site.
- j. Reporting procedures, including proposed reporting formats.
- k. Procedures for submitting and reviewing design changes/variations prior to submission to the Contracting Officer.
- l. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task that is separate and distinct from other tasks and has control requirements and work crews unique to that task. A DFOW is identified by different trades or disciplines, or it is work by the same trade in a different environment. A DFOW is by definition any item or activity on the construction schedule, and the schedule specification provides

direction regarding how the DFOWs are to be structured. Include in the list of DFOWs for all activities on the Construction Schedule. Provide separate DFOWs in the Network Analysis Schedule for each design development stage and submittal package. Although each section of the specifications can generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section. Identify the specification section number and schedule activity ID for each DFOW listed. The DFOW list will be reviewed in coordination with the construction schedule and agreed upon during the Coordination of Mutual Understanding Meeting.

- m. PROCEDURES FOR PERFORMING AND TRACKING THE THREE PHASES OF CONTROL: Identify procedures used to ensure the three phases of control to manage the quality on this project. For each Definable Feature of Work (DFOW), a Preparatory and Initial phase checklist will be filled out during the Preparatory and Initial phase meetings. Conduct the Preparatory and Initial Phases and meetings with a view towards obtaining quality construction by planning ahead and identifying potential problems for each DFOW.
- o. Coordinate scheduled work with Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Special Inspections Project Manual. Where the applicable code issued by the International Code Council (ICC) calls for inspections by the Building Official, the Contractor must include the inspections in the Quality Control Plan and must perform the inspections required by the applicable ICC. The Contractor must perform these inspections using independent qualified inspectors. Include the Special Inspections Project Manual requirements in the QC Plan.
- p. PROCEDURES FOR COMPLETION INSPECTION: Procedures for identifying and documenting the completion inspection process. Include in these procedures the responsible party for punch out inspection, pre-final inspection, and final acceptance inspection.
- q. TRAINING PROCEDURES AND TRAINING LOG: Procedures for coordinating and documenting the training of personnel required by the Contract.
- r. ORGANIZATION AND PERSONNEL CERTIFICATIONS LOG: Procedures for coordinating, tracking and documenting all certifications required for entities such as subcontractors, testing laboratories, suppliers, and personnel. The QC Manager will ensure that certifications are current, appropriate for the work being performed, and will not lapse during any period of the Contract that the work is being performed.

1.5.2.2 Additional Requirements for Design Quality Control (DQC) Plan

The following additional requirements apply to the DQC Plan:

- a. Submit and maintain a DQC Plan as an effective quality control program which assures that all services required by this Contract are performed and provided in a manner that meets professional architectural and engineering quality standards. As a minimum, all documents must be technically reviewed by competent, independent reviewers identified in the DQC Plan. The same element that produced the product may not perform the Design Quality Control Review. Correct errors and deficiencies in the design documents prior to submitting them to the Government.

- b. Include the design schedule in the master project schedule, showing the sequence of events involved in carrying out the project design tasks within the specific Contract period. Schedule must include sufficient detail to identify all major design tasks, including those that control the flow of work. Include review and correction periods associated with each item. Schedule must be a forward planning as well as a project monitoring tool. The schedule reflects calendar days and not dates for each activity. If the schedule is changed, submit a revised schedule reflecting the change within 7 calendar days.
- c. Include in the DQC Plan the **discipline-specific checklists** to be used during the design and quality control of each submittal. Submit at each design phase as part of the project documentation these completed discipline-specific checklists. **ER 1110-3-12** provides some useful information in developing checklists.
- d. Implement the DQC Plan by a Design Quality Control Manager who has the responsibility of being cognizant of and assuring that all documents on the project have been coordinated and who coordinates work with and reports to the Contracting Officer via the QC Manager. Notify the Contracting Officer, in writing, of the name of the individual and the name of an alternate person assigned to the position.
- e. Provide Quality Control Documentation procedures such as QC review sets and QC comments to demonstrate that cross checking of all engineering discipline's design drawings and specifications has taken place. The QC review documentation must exhibit a checking process of the design documents for completeness, accuracy, and constructability.
- f. Names of the individuals, including their respective firm names, who will be serving as the Designer of Record (DOR) in their respective design discipline.
- g. Submit a formal Communication Plan that indicates the frequency of design meetings and what information is covered in those meetings, key design decision points tied to the Network Analysis Schedule and how the DOR plans to include the Government in those decisions, peer review procedures, interdisciplinary coordination, design review procedures, and comment resolution.

The Communication Plan must emphasize key decisions and possible problems the Contractor and Government may encounter during the design phase of the project. Provide a plan to discuss design alternatives and design coordination with the stakeholders at the key decision points as they arise on the project. Identify individual stakeholders and suggested communication methods that will be employed to expedite and facilitate each anticipated critical decision. Communication methods may include: Concept Design Workshop, over-the-shoulder review meetings, presentation at client's office, lifecycle cost analysis presentation, technical phone conversation, and formal review meeting. The design portion of the Communication Plan must be written by the DQC Manager and confirmed during the Post Award Kick-off Partnering. Update the Communication Plan at every Partnering meeting.

- h. A statement of Life Safety and Fire Protection Features Inspections and Testing must be prepared by the Fire Protection DOR. Examples of life safety and fire protection features include, but are not limited to, water distribution systems including fire pumps and fire hydrants,

fire resistive assemblies such as fire rated walls/partitions, through-penetration firestop systems, spray-applied fireproofing of structural components, fire alarm and detection systems, fire suppression and standpipe systems, means of egress components, emergency and exit lighting fixtures. The plan must include a listing of the individuals, approved agencies or firms that will be retained for conducting the required inspections and tests accompanied by a description of individual inspector's experience, and a copy of all required certifications. Additional copies of this plan must be submitted to the NAVFAC Fire Protection Engineer and the Installation Fire Chief. This plan must include the following:

- (1) Comprehensive list of systems, components or features to be inspected and tested.
 - (2) Description of performance verification testing activities for each system or component.
 - (3) Procedures and schedules for functional performance tests of all systems requiring functional testing.
- i. Procedures for ensuring the design documents are submitted in accordance with FC 1-300-09N, Navy and Marine Corps Design Procedures and other procedures to ensure disciplines have been properly coordinated to eliminate conflicts.

1.5.3 Acceptance of the Quality Control (QC) and Design Quality Control (DQC) Plan

The Contracting Officer's acceptance of the Contractor QC Plan and Design Quality Control Plan is required prior to the start of design and construction. The Contracting Officer reserves the right to require changes in the QC and DQC Plan and operations as necessary, including removal or addition of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC and DQC organization at any time to verify the submitted qualifications. All QC and DQC organization personnel are subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the Contract.

1.5.4 Preliminary Construction Work Authorized Prior to Acceptance

The only construction work that is authorized to proceed prior to the acceptance of the QC Plan is mobilization of storage and office trailers, temporary utilities, and surveying with specific prior approval of the Contracting Officer.

1.5.5 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed changes in the QC Plan or changes to the QC organization personnel. Proposed changes are subject to acceptance by the Contracting Officer.

1.5.6 Special Inspections

Perform all required Special Inspections per Section 01 45 35 SPECIAL INSPECTIONS, the statement of Special Inspections and the Schedule of Special Inspections.

1.6 QUALITY CONTROL (QC) ORGANIZATION

1.6.1 Quality Control (QC) Manager

1.6.1.1 Duties

Provide a QC Manager at the work site to implement and manage the QC program. The only duties and responsibilities of the QC Manager are to manage and implement the QC program on this Contract. The QC Manager must attend the [partnering meetings, QC Plan Meetings](#), Coordination and Mutual Understanding Meeting, [conduct the QC meetings](#), perform the three phases of control, perform submittal review and approval, ensure testing is performed and provide QC certifications and documentation required in this Contract. The QC Manager is responsible for managing and coordinating the three phases of control and documentation performed by the QC Specialists, testing laboratory personnel and any other inspection and testing personnel required by this Contract. The QC Manager is the manager of all QC activities. [The QC manager is responsible for notifying the Special Inspector of activities which require their review. The QC manager is responsible for coordinating the Special Inspection activities, see paragraph CONTRACTOR'S QUALITY CONTROL \(QC\) MANAGER, in Section 01 45 35 SPECIAL INSPECTIONS.](#)

1.6.1.2 Qualifications

The QC Manager must be an individual with a minimum of 10 years combined experience in the following positions: Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction Contracts which included the major trades that are part of this Contract. The individual must have at least 2 years experience as a QC Manager. The individual must be familiar with the requirements of [EM 385-1-1](#) and have experience in the areas of hazard identification, safety compliance, and sustainability.

The QC Manager and all members of the QC organization must be capable of reading, writing, and conversing fluently in the English language.

1.6.1.3 Construction Quality Management Training

In addition to the above experience and education requirements, the QC Manager and all members of the QC team must have completed the CQM for Contractors course. If the QC Manager does not have a current certification, obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Systems Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

The Construction Quality Management Training certificate expires after 5 years. If the QC Manager's certificate has expired, retake the course to remain current.

1.6.1.4 Quality Control Requirements for Multiple Work Sites

Provide separate full-time QC Managers at the work sites listed below. Provide a separate full-time QC Manager at the following work sites:

- a. JP5 PIPELINE

b. JP8 BULK TANK FARM

The QC Managers for the work sites listed above are each responsible for implementing and managing the QC Program at the designated work site. They must perform the three phases of control, perform submittal review, ensure testing is performed, and prepare QC certifications and documentation required by this Contract. The qualification requirements for the QC Managers at each site are listed in the paragraph QUALIFICATIONS.

1.6.2 Organizational Changes

Maintain the QC staff with personnel as required by the specification section at all times. When it is necessary to make changes to the QC staff, revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

1.6.3 Alternate Quality Control (QC) Manager Duties and Qualifications

Designate an alternate for the QC Manager at the work site to serve in the event of the designated QC Manager's absence. [The period of absence may not exceed two weeks at one time, and not more than 30 workdays during a calendar year.](#) The qualification requirements for the Alternate QC Manager must be the same as for the QC Manager.

1.6.4 Assistant Quality Control (QC) Manager Duties and Qualifications

Provide a full-time assistant to the QC Manager at the work site to perform the three phases of control, perform submittal review, ensure testing is performed, and prepare QC certifications and documentation required by this Contract. The qualification requirements for the Assistant QC Manager must be equal or exceed the experience and qualifications of the QC Manager. The individual must be familiar with the requirements of [EM 385-1-1](#) and have experience in the areas of hazard identification and safety compliance.

The Assistant QC Manager must be capable of reading, writing, and conversing fluently in the English language.

1.6.5 Commissioning

[Commissioning \(Cx\) is a systematic, quality-focused process for delivery of a project focusing on verifying and documenting all commissioned systems and assemblies are installed, tested, and operating as they were planned and designed to meet the project requirements. The Quality Control requirements outlined in this specification section are key in supporting the objectives of the Cx process, specifically coordinating testing, documenting, and verifying proper system operation. Properly executed the Quality Control support of Cx ensures timely execution of necessary tasks to deliver the fully commissioned and operating systems in coordination with the overall construction and project schedule.](#)

[Provide Cx in addition to the quality control requirements of this section and not as a substitute for quality control requirements. The QC Manager is responsible for carrying out the three phases of control while ensuring the functional performance and integrated systems tests are coordinated with the Cx provider as required for each system to be commissioned.](#)

1.6.5.1 Certificate of Readiness

The QC Manager must issue a Certificate of Readiness for Government approval for each system to be commissioned. Schedule Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. The Certificate of Readiness certifies that all required inspections have been completed and deficiencies that were identified through any prior review, inspection, or test activity have been corrected before the start of Functional Performance Tests.

1.6.6 Quality Control (QC) Specialists

Provide a separate QC Specialist at the work site for each of the areas as listed in the Matrix listed below, who must assist and report to the QC Manager and who may perform production related duties but must be allowed sufficient time to perform their assigned quality control duties. These individuals or specialized technical companies are employees of the Prime or subcontractor [including Designer of Record \(DOR\)](#). QC Specialists must be physically present at the work site with frequency as indicated in the Experience Matrix below, to participate in the QC Meetings, perform the three phases of control, including participation in Preparatory and Initial Phase meetings, and to perform and document Follow-up inspections as an extension of the QC Manager for each definable feature of work in their area of responsibility. QC Specialist must assist and be present for training events, and Critical System Acceptance inspections by the Government. Qualification, experience, Area of Responsibility, and frequency of QC surveillance are provided in Matrix listed herein.

Experience Matrix		
1. Area	2-1. Qualification 2-2. Experience	3-1. Area of Responsibility 3-2. Frequency
Civil	2-1. Graduate Civil Engineer or Construction Manager with a professional engineer registration, and 2-2. 2 years experience in the type of work being performed on this project or technician with 5 years related experience	3-1. Area of Responsibility: POL Fuels Heavy Civil Work; Horizontal Directional Drilling 3-2. Frequency of QC surveillance and inspection will be Bi-monthly during installation and including final inspection.

Mechanical	2-1. Graduate Mechanical Engineer with a professional engineer registration and 2-2. 2 years experience or person with 5 years of experience supervising mechanical features of work in the field with a construction company	3-1. Area of Responsibility: POL Fuels Distribution & Storage Systems 3-2. Frequency of QC surveillance and inspection will be Bi-monthly during installation and including final inspection.
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1.6.7 Submittal Reviewers Duties and Qualifications

Each submittal must be reviewed by a registered architect or professional engineer.

1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL

Procedures for submission, review and approval of submittals are described in Section 01 33 00 SUBMITTAL PROCEDURES. Procedures must include field verification of relevant dimensions and component characteristics by the QC organization prior to submittal being sent to the Contracting Officer. The CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the Contract.

1.8 THREE PHASES OF CONTROL

CQC enables the Contractor to ensure that the construction, including that of subcontractors and suppliers, complies with the requirements of the Contract. At least three phases of control must be conducted by the QC Manager to adequately cover both on-site and off-site work for each definable feature of the construction work as follows:

1.8.1 Preparatory Phase

Document the results of the preparatory phase actions by separate minutes prepared by the QC Manager and attach to the daily CQC report. Instruct applicable workers as to the acceptable level of workmanship required to meet Contract specifications.

Notify the Contracting Officer at least two business days in advance of each preparatory phase meeting. The meeting will be conducted by the QC Manager and attended by the QC Specialists, the Project Superintendent, the CxC, and the foreman responsible for the DFO. When the DFO will be accomplished by a subcontractor, that subcontractor's foreman must attend the preparatory phase meeting. This phase is performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. Perform the following prior to beginning work on each DFO:

- a. Review each paragraph of the applicable specification sections, reference codes, and standards. Make available during the preparatory inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.

- b. Review the Contract drawings.
- c. Verify that field measurements are as indicated on construction or shop drawings or both before confirming product orders, to minimize waste due to excessive materials.
- d. Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.
- e. Review the testing plan and ensure that provisions have been made to provide the required QC testing.
- f. Examine the work area to ensure that the required preliminary work has been completed and complies with the Contract and ensure any deficiencies/rework items in the preliminary work have been corrected and confirmed by the Contracting Officer.
- g. Review coordination of product/material delivery to designated prepared areas to execute the work.
- h. Examine the required materials, equipment and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data and are properly stored.
- i. Check to assure that all materials and equipment have been tested, submitted, and approved.
- j. Discuss specific controls to be used, construction methods, construction tolerances, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each DFO. Ensure any portion of the plan requiring separate Contracting Officer acceptance has been approved.
- k. Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.
- l. Review Special Inspections required by Section 01 45 35 SPECIAL INSPECTION, the Statement of Special Inspections and the Schedule of Special Inspections.

1.8.2 Initial Phase

Notify the Contracting Officer at least two business days in advance of each initial phase. When construction crews are ready to start work on a DFO, conduct the initial phase with the QC Specialists, the Project Superintendent, the Special Inspector, and the foreman responsible for that DFO. Observe the initial segment of the DFO to ensure that the work complies with Contract requirements. Document the results of the initial phase in the daily CQC Report and in the Initial Phase Checklist. Repeat the initial phase for each new crew to work on-site when acceptable levels of specified quality are not being met. Indicate the exact location of initial phase for definable feature of work for future reference and comparison with follow-up phases. Perform the following for each DFO:

- a. Check work to ensure that it is in full compliance with Contract requirements. Review minutes of the preparatory meeting.
- b. Verify adequacy of controls to ensure full Contract compliance. Verify required control inspection and testing comply with the Contract.
- c. Establish level of workmanship and verify that it meets the minimum acceptable workmanship standards. Compare with required sample panels as appropriate.
- d. Resolve any workmanship issues.
- e. Ensure that testing is performed by the approved laboratory.
- f. Check work procedures for compliance with the APP and the appropriate AHA to ensure that applicable safety requirements are met.
- g. Review project specific work plans (i.e., Cx, HAZMAT Abatement, Stormwater Management) to ensure all preparatory work items have been completed and documented.
- h. Coordinate scheduled work with Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Schedule of Special Inspections.

1.8.3 Follow-Up Phase

Perform the following for on-going DFOW daily, or more frequently as necessary, until the completion of each DFOW. The Final Follow-Up for any DFOW will clearly note in the daily report the DFOW is completed, and all deficiencies/rework items have been completed in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Each DFOW that has completed the Initial Phase and has not completed the Final Follow-up must be included on each daily report. If no work was performed on that DFOW for the period of that daily report, it must be so noted. Document all Follow-Up activities for DFOWs in the daily CQC Report:

- a. Ensure the work including control testing complies with Contract requirements until completion of that particular work feature. Record checks in the CQC documentation.
- b. Maintain the quality of workmanship required.
- c. Ensure that testing is performed by the approved laboratory.
- d. Ensure that deficiencies/rework items are being corrected. Conduct final follow-up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work.
- e. Do not build upon nor conceal non-conforming work.
- f. Assure manufacturers' representatives have performed necessary inspections if required and perform safety inspections.
- g. Coordinate scheduled work with Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Schedule of Special Inspections.

1.8.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same DFOW if the quality of on-going work is unacceptable, if there are changes in the applicable QC organization, if there are changes in the on-site production supervision or work crew, if work on a DFOW has not started within 45 days of the initial preparatory meeting or has resumed after 45 days of inactivity, or if other problems develop.

1.8.5 Notification of Three Phases of Control for Off-Site Work

Notify the Contracting Officer at least two weeks prior to the start of the preparatory and initial phases.

1.8.6 Deficiency/Rework Items List

The QC Manager must maintain a list of work that does not comply with the Contract, identifying what items need to be corrected, the activity ID number associated with the item, the date the item was originally discovered, the date the item will be corrected by, and the date the item was corrected.

The list shall be reviewed at each weekly QC Meeting:

- a. There is no requirement to report a deficiency/rework item that is corrected the same day it is discovered.
- b. No successor task may be advanced beyond the preparatory phase meeting until all deficiencies/rework items have been cleared by the QC Manager and concurred with by the Contracting Officer. This must be confirmed as part of the Preparatory Phase activities.
- c. Attach a copy of the "Deficiency/Rework Items List" to the last daily CQC Report of each month.
- d. The Contractor is responsible for including those items identified by the Contracting Officer.
- e. All deficiencies/rework items must be confirmed as corrected by the QC Manager, and concurred by the Contracting Officer, prior to commencement of any completion inspections per paragraph COMPLETION INSPECTIONS unless specifically exempted by the Contracting Officer.
- f. Non-Compliance with these requirements shall be grounds for removal in accordance with paragraph ACCEPTANCE OF THE QUALITY CONTROL (QC) PLAN.
- g. All delays, concurrent or related to failure to manage, monitor, control, and correct deficiencies/rework items are entirely the responsibility of the Contractor and shall not be made the subject, or any component of any request for additional time or compensation.

1.9 TESTING

Perform specified or required tests to verify that control measures are adequate to provide a product which conforms to Contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance tests when specified. Procure the services of an

U.S. Army Corps of Engineers approved testing laboratory or establish an approved testing laboratory at the project site or within 100 miles. Perform the following activities and record and provide the following data:

- a. Verify that testing procedures comply with Contract requirements.
- b. Verify that facilities and testing equipment are available and comply with testing standards.
- c. Check test instrument calibration data against certified standards.
- d. Verify that recording forms and test identification control number system, including all test documentation requirements, have been prepared.
- e. Record results of all tests taken, both passing and failing on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. If approved by the Contracting Officer, actual test reports are submitted later with a reference to the test number and date taken. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this Contract.

1.9.1 Accreditation Requirements

Construction materials testing laboratories must be accredited by a laboratory accreditation authority and must submit a copy of the Certificate of Accreditation and Scope of Accreditation. The laboratory's scope of accreditation must include the appropriate ASTM standards (ASTM E329, ASTM C1077, ASTM D3666, ASTM D3740, ASTM E543) listed in the technical sections of the specifications. Laboratories engaged in Hazardous Materials Testing must meet the requirements of OSHA and EPA. The policy applies to the specific laboratory performing the actual testing, not just the Corporate Office.

1.9.2 Laboratory Accreditation Authorities

Laboratory Accreditation Authorities include the National Voluntary Laboratory Accreditation Program (NVLAP) administered by the National Institute of Standards and Technology at <https://www.nist.gov/nvlap>, the American Association of State Highway and Transportation Officials (AASHTO) Accreditation Program at <http://www.aashtoresource.org/aap/overview>, International Accreditation Services, Inc. (IAS) at <https://www.iasonline.org/>, U.S. Army Corps of Engineers Materials Testing Center (MTC) at <https://www.erdc.usace.army.mil/Media/Fact-Sheets/Fact-Sheet-Article-View/Article/476661/materials-testing-center/>, the American Association for Laboratory Accreditation (A2LA) program at <https://a2la.org/>, the Washington Association of Building Officials (WABO) at <https://www.wabo.org/> (Approval authority for WABO is limited to projects within Washington State), and the Washington Area Council of Engineering Laboratories (WACEL) at <https://www.wacel.org/lab-accreditation-and-inspection-agency-audit-programs/laboratory-accreditation-program/> (Approval authority by WACEL is limited to projects within Facilities Engineering Command (FEC) Washington geographical area).

1.9.3 Capability Check

The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract. Laboratories utilized for testing soils, concrete, asphalt, and steel must meet criteria detailed in [ASTM D3740](#) and [ASTM E329](#).

1.9.4 Test Results

Cite applicable Contract requirements, tests or analytical procedures used. Provide actual results and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify the Contracting Officer immediately. Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results must be signed by a testing laboratory representative authorized to sign certified test reports. Furnish the signed reports, certifications, and other documentation to the Contracting Officer via the QC Manager. [Furnish a summary report of field tests at the end of each month, in accordance with paragraph DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER.](#)

1.9.5 Test Reports and Monthly Summary Report of Tests

[Furnish the signed reports, certifications, and a summary report of field tests at the end of each month to the Contracting Officer. Attach a copy of the summary report to the last daily CQC Report of each month. Provide a copy of the signed test reports and certifications to the Operation and Maintenance Support Information \(OMSI\) preparer for inclusion into the OMSI documentation, in accordance with Sections 01 78 23 OPERATION AND MAINTENANCE DATA and 01 78 24.00 20 FACILITY DATA WORKBOOK \(FDW\).](#)

1.10 COMPLETION INSPECTIONS

1.10.1 Punch-Out Inspection

Near the completion of all work or any increment thereof, established by a completion time stated in the Contract Clause entitled "Commencement, Prosecution, and Completion of Work," or stated elsewhere in the specifications, the QC Manager must conduct an inspection of the work and develop a "punch list" of items which do not conform to the approved drawings, specifications, and Contract. Include in the punch list any remaining items on the "Deficiency/Rework Items List", that were not corrected prior to the Punch-Out Inspection as approved by the Contracting Officer in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Include within the punch list the estimated date by which the deficiencies will be corrected. Provide a copy of the punch list to the Contracting Officer.

The QC Manager, or staff, must make follow-on inspections to ascertain that all deficiencies have been corrected. All punch list items must be confirmed as corrected by the QC Manager and concurred by the Contracting Officer. Once this is accomplished, notify the Government that the facility is ready for the Government "Pre-Final Inspection".

1.10.2 Pre-Final Inspection

The Government and QC Manager will perform this inspection to verify that the facility is complete and ready to be occupied. A Government "Pre-Final Punch List" will be documented by the QC Manager as a result of this inspection. The QC Manager will ensure that all items on this list are corrected and concurred by the Contracting Officer prior to notifying the Government that a "Final" inspection with the Client can be scheduled. All items noted on the "Pre-Final" inspection must be corrected and concurred by the Contracting Officer in a timely manner and be accomplished before the Contract completion date for the work, or any increment thereof, if the project is divided into increments by separate completion dates unless exceptions are directed by the Contracting Officer.

1.10.3 Final Acceptance Inspection

Notify the Contracting Officer at least 14 calendar days prior to the date a final acceptance inspection can be held. State within the notice that all items previously identified on the pre-final punch list will be corrected and acceptable, along with any other unfinished Contract work, by the date of the final acceptance inspection. The Contractor must be represented by the QC Manager, the Project Superintendent, and others deemed necessary. Attendees for the Government will include the Contracting Officer, other Government QA personnel, and personnel representing the Client. Failure of the Contractor to have all Contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with the Contract Clause entitled "Inspection of Construction."

1.11 QUALITY CONTROL (QC) CERTIFICATIONS

1.11.1 Contractor Quality Control (CQC) Report Certification

Contain the following statement within the CQC Report: "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used, and work performed during this reporting period is in compliance with the Contract drawings and specifications to the best of my knowledge, except as noted in this report."

1.11.2 Completion Certification

Upon completion of work under this Contract, the QC Manager must furnish a certificate to the Contracting Officer attesting that "the work has been completed, inspected, tested and is in compliance with the Contract." Provide a copy of this final QC Certification for completion to the preparer of the Operation & Maintenance (O&M) documentation.

1.11.3 Invoice Certification

Furnish a certificate to the Contracting Officer with each payment request, signed by the QC Manager, attesting that as-built drawings are current, coordinated and attesting that the work for which payment is requested, including stored material, complies with Contract requirements.

1.12 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER

1.12.1 Construction Documentation

Reports are required for each day that work is performed and must be attached to the Contractor Quality Control Report prepared for the same day. Maintain current and complete records of on-site and off-site QC program operations and activities. Reports are required for each day work is performed. Account for each calendar day throughout the life of the Contract.

The Project Superintendent and the QC Manager must prepare and sign the Contractor Production and CQC Reports, respectively. Every space on the forms must be filled in. Use N/A if nothing can be reported in one of the spaces. The reporting of work must be identified by terminology consistent with the construction schedule. In the "Remarks" sections of the reports, enter pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, delays encountered, a record of visitors to the work site, quality control problem areas, deviations from the QC Plan, construction deficiencies encountered, and meetings held. For each entry in the report(s), identify the Schedule Activity No. that is associated with the entered remark.

1.12.2 Quality Control Activities

CQC and Contractor Production reports will be prepared daily to maintain current records providing factual evidence that required quality control activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

- a. The name and area of responsibility of the Contractors and any subcontractors.
- b. Operating plant/equipment with hours worked, idle, or down for repair.
- c. Work performed each day, giving location, description, and by whom. When a Network Analysis Schedule (NAS) is used, identify each item of work performed each day by NAS activity number.
- d. Control phase activities performed. Preparatory, and Initial phase Checklists associated with the DFOW referenced to the construction schedule. Follow-up phase activities identified to the DFOW. If testing or specific QC Specialist activities are associated with the Follow-up phase activities for a specific DFOW note this and include those reports.
- e. Test and control activities performed with results and references to specifications and drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST.
- f. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications and drawings requirements.

- g. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.
- h. Offsite surveillance activities, including actions taken.
- i. Job safety evaluations stating what was checked, results, and instructions or corrective actions.
- j. Instructions given/received and conflicts in plans and specifications.

1.12.3 Verification Statement

Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract.

Furnish the original and one copy of these records in report form to the Government by 10:00 AM the next working day after the date covered by the report. As a minimum, prepare and submit one report for every seven days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the Contract. The first report following a day of no work will be for that day only. Reports need to be signed and dated by the QC Manager. Include copies of test reports and copies of reports prepared by all subordinate quality control personnel within the QC Manager Report.

1.12.4 Reports from the Quality Control (QC) Specialist(s)

Document inspection results on a QC specialist report prepared each day work is performed in their area of responsibility. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract documents, and signature of the QC Specialist. In person inspections must be documented with Video/photographs. Video/photographic documentation of deficiencies must include before and after conditions and physical measurements, as necessary. Forward the QC daily report to the QC Manager who must include the report with the submission of their daily QC Report to the Government each day. Every site visit by the QC Specialist must be documented on a QC Specialist daily report.

1.12.5 Quality Control Validation

Establish and maintain the following in an electronic folder. Divide folder into a series of tabbed sections as shown below. Ensure folder is updated at each required progress meeting.

- a. CQC Meeting minutes in accordance with paragraph QUALITY CONTROL (QC) MEETINGS.
- b. All completed Preparatory and Initial Phase Checklists, arranged by specification section, further sorted by DFOV referenced to the construction schedule. Submit each individual Phase Checklist the day the phase event occurs as part of the CQC daily report.
- c. All milestone inspections, arranged by Activity Number referenced to the construction schedule.

- d. An up-to-date copy of the Testing Plan and Log with supporting field test reports, arranged by specification section referenced to the DFW to which individual reports results are associated. Individual field test reports will be submitted within two working days after the test is performed in accordance with the paragraph QUALITY CONTROL ACTIVITIES. **Monthly Summary Report of Tests: Submit the report as an electronic attachment to the CQC Report at the end of each month.**
- e. Copies of all Contract modifications, arranged in numerical order. Also include documentation that modified work was accomplished.
- f. An up-to-date copy of the paragraph DEFICIENCY/REWORK ITEMS LIST.
- g. Special Inspection reports.**
- h. Upon commencement of Completion Inspections of the entire project or any defined portion, maintain up-to-date copies of all punch lists issued by the QC staff to the Contractor and subcontractors and all punch lists issued by the Government in accordance with the paragraph COMPLETION INSPECTIONS.

1.12.6 Testing Plan and Log

As tests are performed, **the CxC and** the QC Manager will record on the "Testing Plan and Log" the date the test was performed and the date the test results were forwarded to the Contracting Officer. Attach a copy of the updated "Testing Plan and Log" to the last daily CQC Report of each month. Provide a copy of the final "Testing Plan and Log" to the preparer of the Operation & Maintenance (O&M) documentation.

1.12.7 As-Built Drawings

The QC Manager must ensure the as-built drawings, required by Section **01 78 00** CLOSEOUT SUBMITTALS are kept current on a daily basis and marked to show deviations which have been made from the Contract drawings. The as-built drawings document shall commence with the QC Manager ensuring all amendments, or changes to the Contract prior to Contract award are accurately noted in the initial document set creating the accurate baseline of the Contract prior to any work starting. Ensure each deviation has been identified with the appropriate modifying documentation (e.g., PC No., Modification No., Request for Information No.). The QC Manager or QC Specialist assigned to an area of responsibility must initial each revision. Upon completion of work, the QC Manager will furnish a certificate attesting to the accuracy of the as-built drawings prior to submission to the Contracting Officer.

1.13 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected non-compliance with the Contract. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, is deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of a claim for extension of time for excess costs or damages by the Contractor.

1.14 DELIVERY, STORAGE, AND HANDLING

Designate receiving/storage areas for incoming material to be delivered according to installation schedule and to be placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. Store and handle materials in a manner as to prevent loss from weather and other damage. Keep materials, products, and accessories covered and off the ground, and store in a dry, secure area. Prevent contact with material that may cause corrosion, discoloration, or staining. Protect all materials and installations from damage by the activities of other trades.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 35

SPECIAL INSPECTIONS

11/20

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 GENERAL REQUIREMENTS
- 1.3 DEFINITIONS
 - 1.3.1 Continuous Special Inspections
 - 1.3.2 Perform
 - 1.3.3 Observe
 - 1.3.4 Special Inspector (SI)
 - 1.3.5 Third Party
 - 1.3.6 Special Inspector of Record (SIOR)
 - 1.3.7 Contracting Officer
 - 1.3.8 Contractor's Quality Control (QC) Manager
 - 1.3.9 Structural Engineer of Record (SER)
 - 1.3.10 Statement of Special Inspections (SSI)
 - 1.3.11 Schedule of Special Inspections (SSI)
 - 1.3.12 Designated Seismic Systems (DSS)
 - 1.3.13 Definable Feature of Work (DFOW)
- 1.4 SUBMITTALS
- 1.5 SPECIAL INSPECTOR QUALIFICATIONS
 - 1.5.1 Welding Structural Steel
 - 1.5.1.1 Special Inspector
 - 1.5.2 Nondestructive Testing of Welds
 - 1.5.2.1 Special Inspector
 - 1.5.3 Concrete Construction
 - 1.5.3.1 Special Inspector
 - 1.5.4 Verification of Site Soil Condition, Fill Placement and Load-Bearing Requirements
 - 1.5.4.1 Special Inspector
 - 1.5.5 Special Inspector of Record (SIOR)

PART 2 PRODUCTS

- 2.1 FABRICATOR SPECIAL INSPECTIONS

PART 3 EXECUTION

- 3.1 RESPONSIBILITIES
 - 3.1.1 Special Inspector of Record
 - 3.1.2 Quality Control Manager
 - 3.1.3 Special Inspectors
- 3.2 DEFECTIVE WORK

-- End of Section Table of Contents --

SECTION 01 45 35

SPECIAL INSPECTIONS
11/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2021) International Building Code

1.2 GENERAL REQUIREMENTS

Perform Special Inspections in accordance with the Statement of Special Inspections, Schedule of Special Inspections and Chapter 17 of ICC IBC. The Statement of Special Inspections and Schedule of Special Inspections are included as an attachment to this specification. Special Inspections are to be performed by an independent third party and are intended to ensure that the work of the Prime Contractor is in accordance with the Contract Documents and applicable building codes. Special inspections do not take the place of the three phases of control inspections performed by the Contractor's QC Manager or any testing and inspections required by other sections of the specifications.

Structural observations will be performed separately by the Government. The Contractor must provide notification to the Contracting Officer 14 days prior to the following points of construction to allow for structural observation:

- a. Footing placement for tanks
- b. Slab base preparation for tanks
- c. Concrete slab placement for tanks
- d. Concrete wall placement for tanks
- e. Verification of site soils condition

1.3 DEFINITIONS

1.3.1 Continuous Special Inspections

Continuous Special Inspections is the constant monitoring of specific tasks by a special inspector. These inspections must be carried out

continuously over the duration of the particular tasks.

1.3.2 Perform

Perform these Special Inspections tasks for each welded joint or member.

1.3.3 Observe

Observe these Special Inspections items on a periodic daily basis. Operations need not be delayed pending these inspections.

1.3.4 Special Inspector (SI)

A qualified person retained by the Contractor and approved by the Contracting Officer as having the competence necessary to inspect a particular type of construction requiring Special Inspections. The SI must be an independent third party hired directly by the Prime Contractor.

1.3.5 Third Party

A Special inspector must not be an employee of the Contractor or of any Sub-Contractor performing the work to be inspected.

1.3.6 Special Inspector of Record (SIOR)

A licensed engineer in responsible charge of supervision of all special inspectors for the project and approved by the Contracting Officer. The SIOR must be an independent third party entity hired directly by the Prime Contractor.

1.3.7 Contracting Officer

The Government official having overall authority for administrative contracting actions. Certain contracting actions may be delegated to the Contracting Officer's Representative (COR).

1.3.8 Contractor's Quality Control (QC) Manager

An individual retained by the Prime Contractor and qualified in accordance with the Section 01 45 00 QUALITY CONTROL having the overall responsibility for the Contractor's QC organization.

1.3.9 Structural Engineer of Record (SER)

A registered design professional employed by the Government responsible for the overall design and review of submittal documents prepared by others. The SER is registered or licensed to practice their respective design profession as defined by the statutory requirements of the professional registration laws in the state in which the design professional works. The SER is also referred to as the Engineer of Record (EOR) in design code documents.

1.3.10 Statement of Special Inspections (SSI)

A document developed by the SER identifying the material, systems, components and work required to have Special Inspections. This statement is included at the end of this specification.

1.3.11 Schedule of Special Inspections (SSI)

A schedule which lists each of the required Special Inspections, the extent to which each Special Inspection is to be performed, and the required frequency for each in accordance with [ICC IBC](#) Chapter 17. This schedule is included at the end of this specification.

1.3.12 Designated Seismic Systems (DSS)

Those nonstructural components that require design in accordance with [ASCE 7-16](#) Chapter 13 and for which the component importance factor, I_p , is greater than 1.0. This designation applies to systems that are required to be operational following the Design Earthquake for RC I - IV structures and following the MCER for RC V structures. All systems in RC V facilities designated as MC-1 in accordance with UFC 3-301-02 are considered part of the Designated Seismic Systems. Designated Seismic Systems will have an Importance Factor $I_p = 1.5$.

1.3.13 Definable Feature of Work (DFOW)

An inspection group that is separate and distinct from other inspection groups, having inspection requirements or inspectors that are unique.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section [01 33 00](#) SUBMITTAL PROCEDURES:

[SD-01 Preconstruction Submittals](#)

[SIOR Letter of Acceptance; G](#)

Special Inspections Agency's [Written NDT Practices](#) with method and evidence of regular equipment calibration where applicable

[SD-06 Test Reports](#)

[Special Inspections Daily Reports](#)

[Special Inspections Biweekly Reports](#)

[SD-07 Certificates](#)

[AISC Certified Steel Fabricator](#)

[Steel Truss Plant Quality Assurance Program](#)

[Certificate of Compliance](#)

[Special Inspector of Record Qualifications; G](#)

[Special Inspector Qualifications; G](#)

[Qualification Records](#) for NDT technicians

[SD-11 Closeout Submittals](#)

Interim Report of Special Inspections for Each DFOW; G

Comprehensive Final Report of Special Inspections; G

1.5 SPECIAL INSPECTOR QUALIFICATIONS

Submit qualifications for each special inspector.

1.5.1 Welding Structural Steel

1.5.1.1 Special Inspector

- a. ICC Structural Welding Special Inspector certificate with one year of related experience, or
- b. AWS Certified Welding Inspector

1.5.2 Nondestructive Testing of Welds

1.5.2.1 Special Inspector

NDT Level III Certificate

1.5.3 Concrete Construction

1.5.3.1 Special Inspector

- a. ICC Reinforced Concrete Special Inspector Certificate with one year of related experience, or
- b. ACI Concrete Construction Special Inspector, or
- c. Registered Professional Engineer with three years of related experience

1.5.4 Verification of Site Soil Condition, Fill Placement and Load-Bearing Requirements

1.5.4.1 Special Inspector

- a. ICC Soils Special Inspector Certificate with one year of related experience, or
- b. NICET Soils Technician Level II Certificate in Construction Material Testing, or
- c. Geologist-In-Training with three years of related experience, or
- d. Registered Professional Engineer with three years of related experience

1.5.5 Special Inspector of Record (SIOR)

Registered Professional Engineer with five years of related experience.

PART 2 PRODUCTS

2.1 FABRICATOR SPECIAL INSPECTIONS

Special Inspections of fabricator's work performed in the fabricator's shop is required to be inspected in accordance with the Statement of

Special Inspections and the Schedule of Special Inspections unless the fabricator is certified by the approved agency to perform such work without Special Inspections. Submit the following certification to the Contracting Officer for information to allow work performed in the fabricator's shop to not be subjected to Special Inspections.

[AISC Certified Steel Fabricator.](#)

Truss Plate Institute (TPI) [steel truss plant quality assurance program](#) certification.

At the completion of fabrication, submit a [certificate of compliance](#), to be included with the comprehensive final report of Special Inspections, stating that the materials supplied and work performed by the fabricator are in accordance with the construction documents.

PART 3 EXECUTION

3.1 RESPONSIBILITIES

3.1.1 Special Inspector of Record

- a. Supervise all Special Inspectors required by the Contract Documents and the IBC.
- b. Submit a [SIOR Letter of Acceptance](#) to the Contracting Officer attesting to acceptance of the duties of SIOR, signed and sealed by the SIOR.
- c. Verify the qualifications of all of the Special Inspectors.
- d. Verify the qualifications of fabricators.
- e. Discrepancies that are observed during Special Inspections must be reported to the QC Manager for correction. If discrepancies are not corrected before the special inspector leaves the site the observed discrepancies must be documented in the daily report.
- f. Submit a biweekly Special Inspections report until all work requiring Special Inspections is complete. A report is required for each biweekly period in which Special Inspections activity occurs, and must include the following:
 - (1) A brief summary of the work performed during the reporting time frame.
 - (2) Changes and discrepancies with the drawings, specifications and mechanical component certification, that were observed during the reporting period.
 - (3) Discrepancies which were resolved or corrected.
 - (4) A list of nonconforming items requiring resolution.
 - (5) All applicable test results including nondestructive testing reports.
- g. At the completion of each Definable Feature of Work (DFOW) requiring Special Inspections, submit an [interim report](#) that documents the

Special Inspections completed for that DFOW including corrections of all discrepancies noted in the daily reports. Interim reports of Special Inspections must be signed and dated by the SIOR.

- h. At the completion of the project submit a [comprehensive final report](#) of Special Inspections that documents the Special Inspections completed for the project including corrections of all discrepancies noted in the daily reports. The comprehensive final report of Special Inspections must be signed, dated and bear the seal of the SIOR.

3.1.2 Quality Control Manager

- a. Supervise all Special Inspectors required by the Contract Documents and the IBC.
- b. Verify the qualifications of all of the Special Inspectors.
- c. Verify the qualifications of fabricators.
- d. Maintain a 3-ring binder for the Special Inspector's daily and [biweekly reports](#). This file must be located in a conspicuous place in the project trailer/office to allow review by the Contracting Officer and the SER.
- e. Maintain a rework items list that includes discrepancies noted on the Special Inspectors daily report.

3.1.3 Special Inspectors

- a. Inspect all elements of the project for which the special inspector is qualified to inspect and are identified in the Schedule of Special Inspections.
- b. Attend preparatory phase meetings related to the Definable Feature of Work (DFOW) for which the special inspector is qualified to inspect.
- c. Submit Special Inspections agency's [written NDT practices](#) for the monitoring and control of the agency's operations to include the following:
 - (1) The agency's procedures for the selection and administration of inspection personnel, describing the training, experience and examination requirements for qualifications and certification of inspection personnel.
 - (2) The agency's inspection procedures, including general inspection, material controls, and visual welding inspection.
- d. Submit [qualification records](#) for nondestructive testing (NDT) technicians designated for the project.
- e. Submit NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.
- f. Submit a copy of the [daily reports](#) to the QC Manager.
- g. Report discrepancies that are observed during Special Inspections to the QC Manager for correction. If discrepancies are not corrected before the special inspector leaves the site the observed

discrepancies must be documented in the daily report.

- h. Submit a biweekly Special Inspection Report until all inspections are complete. A report is required for each biweekly period in which Special Inspections activity occurs, and must include the following:
 - (1) A brief summary of the work performed during the reporting time frame.
 - (2) Changes and discrepancies with the drawings, specifications that were observed during the reporting period.
 - (3) Discrepancies which were resolved or corrected.
 - (4) A list of nonconforming items requiring resolution.
 - (5) All applicable test result including nondestructive testing reports.
- i. At the completion of the project submit a [comprehensive final report](#) of Special Inspections that documents the Special Inspections completed for the project and corrections of all discrepancies noted in the daily reports. The comprehensive final report of Special Inspections must be signed, dated and indicate the certification of the special inspector qualifying them to conduct the inspection.
- k. Submit [daily reports](#) to the SIOR.

3.2 DEFECTIVE WORK

Check work as it progresses, but failure to detect any defective work or materials must in no way prevent later rejection if defective work or materials are discovered, nor obligate the Contracting Officer to accept such work.

-- End of Section --

Project: MILCON P-919 Bulk Tank Farm Improvements
 Location: NAS Rota, Spain
 Project #:
 Date:



STATEMENT OF SPECIAL INSPECTIONS

Project Seismic Design Category: D
 Project Risk Category: III
 Project Design Wind Speed (m/s): 177
 Number of Stories: 1
 Structure Height Above Grade (m): 6.1
 Hazardous Occupancy or attached to such? No Group H Occupancies

Special Inspector of Record (SIOR)

A Special Inspector of Record (SIOR) IS required (per UFGS 01 45 35, Section 1.3.8)

SIOR Name (Registered Professional): Contractor to provide prior to construction start.
 Professional Registration Number:
 Consulting Firm Name (if any):
 SIOR Office AND Mobile Phone Number:

Lateral Force Resisting System (LFRS)

2018 IBC 1704.3.2 and 1704.3.3

Following is a listing of critical main wind/seismic force resisting systems for this structure. Carefully inspect these elements as part of the roles and responsibilities of the Special Inspector (reference the Schedule of Special Inspections for inspection checklists).

Vertical LFRS Elements	Notes
Ordinary Reinforced Concrete Shear Walls	Pump Vaults and Additive Injector Building
Horizontal LFRS Elements	Notes
Cast in Place Concrete Roof Beams and Roof Slabs	Pump Vaults and Additive Injector Building
Ordinary Reinforced Concrete Shear Walls	Pump Vaults and Additive Injector Building

Project: MILCON P-919 Bulk Tank Farm Improvements
 Location: NAS Rota, Spain
 Project #: 0
 Date: 1/0/1900

Designated Seismic Systems (DSS)

(2018 IBC 1705.13.3) (ASCE 7-16, 13.2.2, C13.2.2) (UFC 3-301-1, 2-5.3)

Non-structural 'Designated Seismic Systems' (DSS) must remain operable and contain hazardous substances following a design earthquake. Accordingly, all Designated Seismic Systems must be listed below and must be certified by the manufacturer to remain both operable and/or to contain hazardous substances after a design earthquake per UFC 3-301-01, Section 3-6.2. Submit said Certificates of Compliance to the Contracting Officer for each DSS after they have been reviewed and accepted by the EOR/DOR.

Additionally, the below listed Designated Seismic Systems must be carefully inspected by the Special Inspector according to the requirements noted in the Schedule of Special Inspections, Section AA.

ELECTRICAL Designated Seismic Systems (DSS) Requiring a Certificate of Compliance	
1.	N/A
2.	
3.	
4.	
5.	

If additional space is required, append an additional sheet listing the remaining DSS

MECHANICAL/PLUMBING Designated Seismic Systems (DSS) Requiring a Certificate of Compliance	
1.	N/A
2.	
3.	
4.	
5.	
6.	

If additional space is required, append an additional sheet listing the remaining DSS

OTHER Designated Seismic Systems (DSS) Requiring a Certificate of Compliance	
1.	N/A
2.	
3.	
4.	
5.	
6.	

Final Walk Down Inspection and Report

(UFC 3 301 01 SECTION 2-5.4)

Designated Seismic Systems shall receive a final walk-down inspection by the Registered Design Professional in Responsible Charge

Final Walk Down Report, Prepared by the Registered Design Professional in Responsible Charge, Must Include:

1. Record observations of Final Walk Down Inspection
2. Document that Inspections were performed in accordance with the Schedule of Special Inspections
3. Document that all Designated Seismic Systems are installed according to construction/manufacture document requirements, and that Compliance Certificates have been collected (UFC 03 301 01, 3-6.2, 13.2.2.2).

SCHEDULE OF SPECIAL INSPECTIONS

Reference UFGS 01 45 35 for all requirements not noted as part of this schedule.

INSPECTION DEFINITIONS:

- PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and noted verification.
- OBSERVE:** Observe these items randomly during the course of each work day to insure that applicable requirements are being met. Operations need not be delayed pending these inspections at contractor's risk.
- DOCUMENT:** Document, with a report, that the work has been performed in accordance with the contract documents. This is in addition to any other reports required in the Special Inspections guide specification.
- CONTINUOUS:** Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

The Seismic Design Category for this project (according to ASCE 7-16) is: ☐ A, ☐ B, ☐ C, ☒ D, ☐ E, ☐ F

STRUCTURAL - STEEL – WELDING SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒**

STEEL INSPECTION <u>PRIOR TO WELDING</u> – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.2.1, AISC 360-16: Table C-N5.4-1		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Verify that the welding procedures specification (WPS) is available	PERFORM	
2. Verify manufacturer certifications for welding consumables are available	PERFORM	
3. Verify material identification	PERFORM	Type and grade.
4. Welder Identification System	PERFORM	The fabricator or erector, as applicable, shall maintain a system by which a welder who has welded a joint or member can be identified. Stamps, if used, shall be the low-stress type.
5. Fit-up of groove welds (including joint geometry)	OBSERVE	✓ Joint preparation ✓ Dimensions (alignment, root opening, root face, bevel) ✓ Cleanliness (condition of steel surfaces) ✓ Tacking (tack weld quality and location) ✓ Backing type and fit (if applicable)
6. Configuration and finish of access holes	OBSERVE	
7. Fit-up of fillet welds	OBSERVE	✓ Dimensions (alignment, gaps at root) ✓ Cleanliness (condition of steel surfaces) ✓ Tacking (tack weld quality and location)
STEEL INSPECTION <u>DURING WELDING</u> – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.2.1, AISC 360-16: Table C-N5.4-2		
TASK	INSPECTION TYPE	DESCRIPTION
8. Use of qualified welders	PERFORM	Welding by welders, welding operators, and tack welders who are qualified in conformance with requirements.
9. Control and handling of welding consumables	OBSERVE	✓ Packaging ✓ Electrode atmospheric exposure control
10. No welding over cracked tack welds	OBSERVE	
11. Environmental conditions	OBSERVE	✓ Wind speed within limits ✓ Precipitation and temperature
12. Welding Procedures Specification followed	OBSERVE	✓ Settings on welding equipment ✓ Travel speed ✓ Selected welding materials ✓ Shielding gas type/flow rate ✓ Preheat applied ✓ Interpass temperature maintained (min./max.) ✓ Proper position (F, V, H, OH) ✓ Intermix of filler metals avoided
13. Welding techniques	OBSERVE	✓ Interpass and final cleaning ✓ Each pass within profile limitations ✓ Each pass meets quality requirements

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.

OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

STRUCTURAL - STEEL – WELDING SECTION (CONTINUED)

STEEL INSPECTION AFTER WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.2.1, AISC 360-16: Table C-N5.4-3		
TASK	INSPECTION TYPE ¹	DESCRIPTION
14. Welds cleaned	OBSERVE	
15. Size, length, and location of all welds	PERFORM	Size, length, and location of all welds conform to the requirements of the detail drawings.
16. Welds meet visual acceptance criteria	PERFORM AND DOCUMENT	✓ Crack prohibition ✓ Weld/base-metal fusion ✓ Crater cross section ✓ Weld profiles ✓ Weld size ✓ Undercut ✓ Porosity
17. Arc strikes	PERFORM	
18. k-area	PERFORM	When welding of doubler plates, continuity plates or stiffeners has been performed in the k-area, visually inspect the web k-area for cracks.
19. Backing removed, weld tabs removed and finished, and fillet welds added where required	PERFORM	
20. Repair activities	PERFORM AND DOCUMENT	
21. Document acceptance or rejection of welded joint or member	PERFORM	

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

STRUCTURAL - STEEL – BOLTING SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒**

STEEL INSPECTION TASKS PRIOR TO BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.2.1, AISC 360-16: Table C-N5.6-1		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Manufacture's certifications available for fastener materials	PERFORM	
2. Fasteners marked in accordance with ASTM requirements	OBSERVE	
3. Proper fasteners selected for joint detail (grade, type, bolt length if threads are to be excluded from shear plane)	OBSERVE	
4. Proper bolting procedure selected for joint detail	OBSERVE	
5. Connecting elements, including appropriate faying surface condition and hole preparation, if specified, meet applicable requirements	OBSERVE	
6. Proper storage provided for bolts, nuts, washers, and other fastener components	OBSERVE	
STEEL INSPECTION TASKS DURING BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.2.1, AISC 360-16: Table C-N5.6-2		
TASK	INSPECTION TYPE ¹	DESCRIPTION
7. Fastener assemblies of suitable condition, placed in all holes and washers (if required) are positioned as required	OBSERVE	
8. Joint brought to the snug-tight condition prior to pretensioning operation	OBSERVE	
9. Fastener component not turned by the wrench prevented from rotating	OBSERVE	
10. Bolts are pretensioned in accordance with RCSC Specification, progressing systematically from the most rigid point toward the free edges	OBSERVE	
STEEL INSPECTION TASKS AFTER BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.1, AISC 360-10: Table C-N5.6-3		
TASK	INSPECTION TYPE ¹	DESCRIPTION
11. Document acceptance or rejection of all bolted connections	DOCUMENT	

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

STRUCTURAL - STEEL - NON DESTRUCTIVE TESTING SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒**

NONDESTRUCTIVE TESTING OF WELDED JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.2.1, AISC 360-16: Section N5.5		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Use of qualified nondestructive testing personnel	PERFORM	Visual weld inspection and nondestructive testing (NDT) shall be conducted by personnel qualified in accordance with AWS D1.8 clause 7.2
2. CJP groove welds	OBSERVE	Dye penetrant testing (DT) and ultrasonic testing (UT) shall be performed on 20% of CJP groove welds for materials greater than 5/16" (8mm) thick. Testing rate must be increased to 100% if greater than 5% of welds tested have unacceptable defects.
3. Welded joints subject to fatigue	OBSERVE	Dye penetrant testing (DT) and Ultrasonic testing (UT) shall be performed on 100% of welded joints identified on contract drawings as being subject to fatigue.
4. Weld tab removal sites	OBSERVE	At the end of welds where weld tabs have been removed, magnetic particle testing shall be performed on the same beam-to-column joints receiving UT

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.

OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

STRUCTURAL - STEEL – AISC 341 REQUIREMENTS (SEISMIC PROVISIONS) SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐**

NONDESTRUCTIVE TESTING OF WELDED JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.2.1, AISC 341-16: Section J6.2		
TASK	INSPECTION TYPE ¹	DESCRIPTION
[NOTE: DOR may uncheck this section for projects NOT designed in accordance with AISC 341 (Seismic Provisions) or for projects designed according to AISC 341, but using an R value equal to 3]		
5. CJP groove welds	OBSERVE	Dye penetrant testing (DT) and ultrasonic testing (UT) shall be performed on 100% of CJP groove welds for materials greater than 5/16" thick (8mm).
6. Beam cope and access hole.	OBSERVE	At welded splices and connections, thermally cut surfaces of beam copes and access holes shall be tested using magnetic particle testing (MT) or dye penetrant testing (DT), when the flange thickness exceeds 1 1/2 in. for rolled shapes, or when the web thickness exceeds 1 1/2 in. for built-up shapes.
7. K-area NDT (AISC 341)	PERFORM	Where welding of doubler plates, continuity plates or stiffeners has been performed in the k-area, the web shall be tested for cracks using magnetic particle testing (MT). The MT inspection area shall include the k-area base metal within 3-inches of the weld. The MT shall be performed no sooner than 48 hours following completion of the welding.
8. Placement of reinforcing or contouring fillet welds	DOCUMENT	

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

STRUCTURAL - STEEL - COMPOSITE CONSTRUCTION ¹**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒**

COMPOSITE CONSTRUCTION PRIOR TO PLACING CONCRETE – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.2.1, AISC 360-16: Table N6.1, AISC 341-16: Table J9.1		
TASK	INSPECTION TYPE ²	DESCRIPTION
1. Placement and installation of steel headed stud anchors	PERFORM	
2. Material identification of reinforcing steel (Type/Grade)	OBSERVE	
3. Determination of carbon equivalent for reinforcing steel other than ASTM A706	OBSERVE	
4. Proper reinforcing steel size, spacing, clearances, support, and orientation	OBSERVE	
5. Reinforcing clearances have been provided	OBSERVE	
6. Reinforcing steel has been tied and supported as required	OBSERVE	

END SECTION**STRUCTURAL - STEEL - OTHER INSPECTIONS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒**

OTHER STEEL INSPECTIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.2.1, AISC 341-16: Tables J8.1 & J10.1		
TASK	INSPECTION TYPE ²	DESCRIPTION
1. Anchor rods and other embedments supporting structural steel	PERFORM	Verify the diameter, grade, type, and length of the anchor rod or embedded item, and the extent or depth of embedment prior to placement of concrete.
2. Fabricated steel or erected steel frame	OBSERVE	Verify compliance with the details shown on the construction documents, such as braces, stiffeners, member locations and proper application of joint details at each connection.

END SECTION

¹ See Concrete Construction Section for all concrete related inspection of composite steel construction.

² **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

STRUCTURAL - COLD-FORMED METAL DECK - PLACEMENT SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒**

METAL DECK INSPECTION PRIOR TO DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.1		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Verify compliance of materials (deck and all deck accessories) with construction documents, including profiles, material properties, and base metal thickness	PERFORM	
2. Document acceptance or rejection of deck and deck accessories	DOCUMENT	
METAL DECK INSPECTION DURING DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.2		
TASK	INSPECTION TYPE ¹	DESCRIPTION
3. Verify compliance of deck and all deck accessories installation with construction documents	PERFORM	
4. Verify deck materials are represented by the mill certifications that comply with the construction documents	PERFORM	
5. Document acceptance or rejection of installation of deck and deck accessories	DOCUMENT	
METAL DECK INSPECTION AFTER DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.3		
TASK	INSPECTION TYPE ¹	DESCRIPTION
6. Welding procedure specification (WPS) available	PERFORM	
7. Manufactures certifications for welding consumables available	OBSERVE	
8. Material identification (type/grade)	OBSERVE	
9. Check welding equipment	OBSERVE	

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

STRUCTURAL - COLD-FORMED METAL DECK – WELDING SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒**

METAL DECK INSPECTION <u>DURING</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.4		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Use of qualified welders	OBSERVE	
2. Control and handling of welding consumables	OBSERVE	
3. Environmental conditions (wind speed, moisture, temperature)	OBSERVE	
4. WPS followed	OBSERVE	
METAL DECK INSPECTION <u>AFTER</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.5		
TASK	INSPECTION TYPE ¹	DESCRIPTION
5. Verify size and location of welds, including support, sidelap, and perimeter welds.	PERFORM	
6. Welds meet visual acceptance criteria	PERFORM	
7. Verify repair activities	PERFORM	
8. Document acceptance or rejection of welds	DOCUMENT	

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

STRUCTURAL - COLD-FORMED METAL DECK – FASTENING SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒**

METAL DECK INSPECTION <u>BEFORE</u> MECHANICAL FASTENING – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.6		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Manufacturer installation instructions available for mechanical fasteners	OBSERVE	
2. Proper tools available for fastener installation	OBSERVE	
METAL DECK INSPECTION <u>DURING</u> MECHANICAL FASTENING – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.7		
TASK	INSPECTION TYPE ¹	DESCRIPTION
3. Fasteners are positioned as required	OBSERVE	
4. Fasteners are installed in accordance with manufacturer's instructions	OBSERVE	
METAL DECK INSPECTION <u>AFTER</u> MECHANICAL FASTENING – VERIFY THE FOLLOWING ARE IN COMPLIANCE SDI QA/QC-2011, Appendix 1, Table 1.8		
TASK	INSPECTION TYPE ¹	DESCRIPTION
5. Check spacing, type, and installation of support fasteners	PERFORM	
6. Check spacing, type, and installation of sidelap fasteners	PERFORM	
7. Check spacing, type, and installation of perimeter fasteners	PERFORM	
8. Verify repair activities	PERFORM	
9. Document acceptance or rejection of mechanical fasteners	DOCUMENT	

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

STRUCTURAL - LIGHT GAUGE STEEL FRAMING AND/OR LIGHT GAUGE TRUSSES SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒**

LIGHT GAUGE STEEL CONSTRUCTION AND CONNECTIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.2.2, 1705.11.2, 1705.11.3, UFC 4 023 03		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Trusses spanning 60-feet or greater where/if applies	PERFORM	Verify that temporary and permanent truss restraint/bracing is installed in accordance with approved truss submittal package.
2. Welded connections (seismic and/or wind resisting system)	OBSERVE	Visually inspect all welds composing part of the main wind or seismic force resisting system, including shearwalls, braces, collectors (drag struts), and hold-downs.
3. Connections (seismic and/or wind resisting system)	OBSERVE	Visually inspect all screw attachment, bolting, anchoring and other fastening of components within the main wind or seismic force resisting system, including roof deck, roof framing, exterior wall covering, wall to roof/floor connections, braces, collectors (drag struts) and hold-downs.
4. Cold-formed steel (progressive collapse resisting system where/if applies)	OBSERVE	Verify proper welding operations, screw attachment, bolting, anchoring and other fastening of components within the progressive collapse resisting system, including horizontal tie force elements, vertical tie force elements and bridging elements (UFC 4 023 03).

END SECTION**STRUCTURAL - OPEN-WEB STEEL JOISTS SECTION****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐**

OPEN-WEB STEEL JOISTS AND JOIST GIRDERS – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC TABLE 1705.2.3		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Installation of open-web steel joists and joist girders	OBSERVE	✓ End connections – welded or bolted ✓ Bridging – horizontal and diagonal

END SECTION

¹ **PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.

OBSERVE: Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

STRUCTURAL - CONCRETE CONSTRUCTION SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒**

CONCRETE CONSTRUCTION, INCLUDING COMPOSITE DECK – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC TABLE 1705.3 (ACI 318 REFERENCES NOTED IN IBC TABLE)		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Inspect reinforcement, including prestressing tendons, and verify placement.	OBSERVE	Verify prior to placing concrete that reinforcing is of specified type, grade and size; that it is free of oil, dirt and unacceptable rust; that it is located and spaced properly; that hooks, bends, ties, stirrups and supplemental reinforcement are placed correctly; that lap lengths, stagger and offsets are provided; and that all mechanical connections are installed per the manufacturer's instructions and/or evaluation report.
2. Reinforcing bar welding	OBSERVE	✓ Verify weldability of reinforcing bars other than ASTM A 706 ✓ Inspect single-pass fillet welds, maximum 5/16" in accordance with AWS D1.4
3. All other welding	CONTINUOUS	Visually inspect all welds in accordance with AWS D1.4
4. Cast in place anchors and post installed drilled anchors (downward inclined)	OBSERVE	Verify prior to placing concrete that cast in place anchors and post installed drilled anchors have proper embedment, spacing and edge distance.
5. Post-installed adhesive anchors in horizontal or upward inclined orientations	CONTINUOUS AND DOCUMENT	✓ Inspect as required per approved ICC-ES report ✓ Verify that installer is certified for installation of horizontal and overhead installation applications ✓ Inspect proof loading as required by the contract documents
6. Verify use of required mix design	OBSERVE	Verify that all mixes used comply with the approved construction documents
7. Prior to concrete placement, fabricate specimens for strength tests, perform slump and air content tests, and determine the temperature of the concrete	CONTINUOUS	At the time fresh concrete is sampled to fabricate specimens for strength test verify these tests are performed by qualified technicians.
8. Inspect concrete and/or shotcrete placement for proper application techniques	CONTINUOUS	Verify proper application techniques are used during concrete conveyance and depositing avoids segregation or contamination. Verify that concrete is properly consolidated.
9. Verify maintenance of specified curing temperature and technique	OBSERVE	Inspect curing, cold weather protection, and hot weather protection procedures.

CONTINUED ON FOLLOWING PAGE

¹ **OBSERVE:** Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

STRUCTURAL - CONCRETE CONSTRUCTION (CONTINUED)

CONCRETE CONSTRUCTION, INCLUDING COMPOSITE DECK – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC TABLE 1705.3 (ACI 318 REFERENCES NOTED IN IBC TABLE)		
TASK	INSPECTION TYPE ¹	DESCRIPTION
10. Inspect erection of precast concrete members	OBSERVE	
11. Verify in-situ concrete strength, prior to stressing of tendons in post-tensioned concrete and prior to removal of shores and forms from beams and structural slabs.	OBSERVE	
12. Inspect formwork for shape, location and dimensions of the concrete member being formed.	OBSERVE	

END SECTION

¹ **OBSERVE:** Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

STRUCTURAL - MASONRY CONSTRUCTION SECTION (ALL RISK CATEGORIES)**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐**

MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE <u>AT START</u> OF CONSTRUCTION IBC 1705.4 (ACI 530-13 TABLE 3.1.2 & 3.1.3)		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Compliance with approved submittals prior to start	OBSERVE	
2. Proportions of site-mixed mortar.	OBSERVE	
3. Grade and type of reinforcement, anchor bolts, and prestressing tendons and anchorages	OBSERVE	
4. Prestressing technique	OBSERVE	
5. Properties of thin bed mortar for AAC masonry	OBSERVE	
MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE <u>PRIOR TO</u> GROUTING IBC 1705.4 (ACI 530-13 TABLE 3.1.2 & 3.1.3)		
TASK	INSPECTION TYPE ¹	DESCRIPTION
6. Grout space	OBSERVE CONTINUOUS	[NOTE: DOR must either delete 'OBSERVE' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/ III]
7. Proportions of site-prepared grout and prestressing grout for bonded tendons	OBSERVE	
8. Proportions of site-mixed grout and prestressing grout for bonded tendons	OBSERVE	
9. Placement of masonry units and mortar joints	OBSERVE	
10. Welding of reinforcement	CONTINUOUS	
MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE <u>DURING</u> CONSTRUCTION IBC 1705.4 (ACI 530-13 TABLE 3.1.2 & 3.1.3)		
TASK	INSPECTION TYPE ¹	DESCRIPTION
11. Size and location of structural elements is in compliance	OBSERVE	
12. Preparation, construction, and protection of masonry during cold weather (temperature below 40°F (4.4°C) or hot weather (temp above 90°F (32.2°C))	OBSERVE	
13. Application and measurement of prestressing force	CONTINUOUS	
14. Placement of grout and prestressing grout for bonded tendons	CONTINUOUS	
15. Placement of AAC masonry units and construction of thin bed mortar joints	CONTINUOUS	Continuous for first 5000 square feet only (465 square meters).
16. Observe preparation of grout specimens, mortar specimens, and/or prisms	OBSERVE	
17. Type, size and placement of reinforcement, connectors, anchor bolts and prestressing tendons and anchorages, including details of anchorage of masonry to structural members, frames, or other construction	OBSERVE CONTINUOUS	[NOTE: DOR must either delete 'OBSERVE' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/III]

END SECTION

¹ **OBSERVE:** Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

STRUCTURAL - WOOD CONSTRUCTION – SPECIALTY ITEMS SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

WOOD CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.5		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. High-load diaphragms where applicable	OBSERVE	Verify thickness and grade of sheathing, size of framing members at panel edges, nail diameters and length, and the number of fastener lines and that fastener spacing is per approved contract documents.
2. Metal-plate connected wood trusses spanning 60 feet or greater	OBSERVE	Verify that the temporary installation restraint/bracing and the permanent individual truss member restraint/bracing are installed in accordance with the approved truss submittal package

END SECTION**STRUCTURAL - WOOD CONSTRUCTION - SEISMIC & WIND SECTION****THIS SECTION IS APPLICABLE IF BOX IS CHECKED:** ☐

WOOD CONSTRUCTION SEISMIC AND WIND – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.11 & 1705.12.2		
TASK	INSPECTION TYPE ¹	DESCRIPTION
[NOTE: DOR may uncheck this section where sheathing nailing/fasteners (both shearwall and roof) are consistently greater than 4" on center, or if the design wind speed (ASD) is less than 110 mph (49 meters/sec) AND the seismic design category is A or B]		
1. Nailing, bolting, anchoring and other fastening of elements of the main wind/seismic force-resisting system	OBSERVE (CONTINUOUS FOR GLUING)	Includes connectors for: shearwall sheathing, roof/floor sheathing, drag struts/collectors (double top plates), braces, hold downs, roof connections to exterior walls.

END SECTION**STRUCTURAL – ISOLATION AND ENERGY DISSIPATION SYSTEMS SECTION****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

ISOLATION AND ENERGY DISSIPATION SYSTEMS – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC TABLE 1705.12.8		
[NOTE: This section is <u>not</u> applicable to Seismic Design Category A. Uncheck this section if this category applies]		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Fabrication and installation	OBSERVE	Verify that fabrication and installation of isolator units and energy dissipation devices conform to manufacturer's recommendations and approved construction documents
2. Testing of seismic isolation Systems in seismically isolated structures		Seismic Isolation Systems in seismically isolated structures shall be tested accordance with ASCE 7, Section 17.8

END SECTION

¹ **OBSERVE:** Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

GEOTECHNICAL - SOILS INSPECTION SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒**

SOILS INSPECTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.6		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Materials below shallow foundations are adequate to achieve the design bearing capacity.	OBSERVE	
2. Excavations are extended to proper depth and have reached proper material	OBSERVE	
3. Perform classification and testing of compacted fill materials	OBSERVE	
4. Verify use of proper materials, densities and lift thicknesses during placement and compaction of compacted fill	CONTINUOUS	
5. Prior to placement of compacted fill, inspect subgrade and verify that site has been prepared properly.	OBSERVE	During fill placement, the special inspector shall verify that proper materials and procedures are used in accordance with the provisions of the approved geotechnical report

END SECTION**GEOTECHNICAL - DRIVEN DEEP FOUNDATION ELEMENTS SECTION****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐**

DEEP DRIVEN FOUNDATION CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE IBC 1705.7		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Verify element materials, sizes and lengths comply with requirements	CONTINUOUS	
2. Inspect driving operations and maintain complete and accurate records for each element	CONTINUOUS	
3. Verify placement locations and plumbness, confirm type and size of hammer, record number of blows per foot of penetration, determine required penetrations to achieve design capacity, record tip and butt elevations and document any damage to foundation element	CONTINUOUS	
4. Determine capacities of test elements and conduct additional load tests if required.	CONTINUOUS	
5. For steel or concrete elements, perform additional special inspections in accordance with the Steel and Concrete sections in this schedule		

END SECTION

¹ **OBSERVE:** Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

GEOTECHNICAL - HELICAL PILE FOUNDATIONS SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

HELICAL PILE FOUNDATIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE
2018 IBC 1705.9

TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Record installation equipment used, pile dimensions, tip elevations, final depth, final installation torque and other pertinent installation data as required. The approved geotechnical report and the contract documents shall be used to determine compliance	CONTINUOUS	

END SECTION**GEOTECHNICAL - CAST IN PLACE DEEP FOUNDATION ELEMENTS SECTION****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

CAST IN PLACE DEEP FOUNDATION ELEMENTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE
2018 IBC 1705.8

TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Inspect drilling operations and maintain complete and accurate records for each element.	CONTINUOUS	
2. Verify placement locations and plumbness, confirm element diameters, bell diameters (if applicable), lengths, embedment into bedrock (if applicable) and adequate end-bearing strata capacity. Record concrete or grout volumes	CONTINUOUS	For concrete elements, perform additional special inspections in accordance with the Concrete section in this schedule

END SECTION

¹ **CONTINUOUS:** Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

FIRE PROTECTION - SPRAYED FIRE-RESISTANT MATERIALS SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

SPRAYED FIRE RESISTANT MATERIALS (SFRM) – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.14		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Substrate condition	OBSERVE	Prior to application, confirm that surfaces have been prepared according to the approved fire-resistance design and manufacturer's instructions.
2. Material thickness	OBSERVE	Verify SFRM thickness according to 2018 IBC 1705.14.4
3. Material density	OBSERVE	Verify SFRM density according to 2018 IBC 1705.14.5
4. Bond strength	OBSERVE	Verify bond strength of cured SFRM according to IBC 1705.14.6

END SECTION**FIRE PROTECTION - MASTIC AND INTUMESCENT COATINGS SECTION****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

MASTIC AND INTUMESCENT FIRE-RESISTANT COATINGS – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.15		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Inspect according to AWCI 12-B and the contract documents	OBSERVE	Inspections shall be performed in accordance with AWCI 12-B, Standard Practice for the Testing and Inspection of Field Applied Thin Film Intumescent Fire-Resistive Materials.

END SECTION**FIRE PROTECTION – FIRE RESISTANT PENETRATIONS AND JOINTS SECTION****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☒

FIRE RESISTANT PENETRATIONS AND JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.17		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Inspections of penetration firestop systems conducted in accordance with ASTM E 2174.	OBSERVE	
2. Inspections of fire-resistant joint systems conducted in accordance with ASTM E 2393	OBSERVE	

END SECTION

¹ **OBSERVE:** Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

FIRE PROTECTION – SMOKE CONTROL SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐**SMOKE CONTROL – VERIFY THE FOLLOWING ARE IN COMPLIANCE**
2018 IBC 1705.18

TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Verify device locations and perform leakage testing	OBSERVE	Perform during erection of ductwork and prior to concealment
2. Pressure difference testing, flow measurements and detection and control verification	OBSERVE	Perform prior to occupancy and after sufficient completion

END SECTION

¹ **OBSERVE:** Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

ARCHITECTURAL - EXTERIOR INSULATION AND FINISH SYSTEMS SECTION**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

EXTERIOR INSULATION AND FINISH SYSTEMS (EIFS) – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.16		
TASK	INSPECTION TYPE ¹	DESCRIPTION
1. Water resistive barrier coating applied over a sheathing substrate.	OBSERVE	Verify that water resistive barrier coating complies with ASTM E 2570. [NOTE: not applicable to masonry or concrete wall applications. Uncheck this section in those cases]

END SECTION**ARCHITECTURAL – ARCHITECTURAL COMPONENTS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

ARCHITECTURAL COMPONENTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE 2018 IBC 1705.12.5, 1705.12.7		
TASK	INSPECTION TYPE ¹	DESCRIPTION
[NOTE: This section is not applicable to Seismic Design Categories A, B, & C. Uncheck this section if one of those categories applies. Confirm Seismic Design Category with the structural engineer]		
1. Erection and fastening of exterior cladding and interior and exterior veneer.	OBSERVE	Verify appropriate materials, fasteners and attachment at commencement of work and at completion. Inspector Note: Inspection not required if height is less than 30 feet or weight is less than 5psf
2. Interior and exterior non-load bearing walls	OBSERVE	Verify appropriate materials, fasteners and attachment at commencement of work and at completion. Inspector Note: Inspection not required if interior non-load bearing walls weigh less than 15psf
3. Access floors	OBSERVE	Verify that anchorage complies with approved construction documents.
4. Storage racks	OBSERVE	Verify that anchorage complies with approved construction documents. Inspection of post-installed anchors shall comply with approved ICC-ES report. Inspector Note: Not required for racks less than 8 feet in height

END SECTION

¹ **OBSERVE:** Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

PLUMBING/MECHANICAL/ELECTRICAL DESIGNATED SEISMIC SYSTEMS SECTION

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

PLUMBING, MECHANICAL AND ELECTRICAL IBC 1705.12.6		
TASK	INSPECTION TYPE ¹	DESCRIPTION
[NOTE: This section is not applicable to Seismic Design Categories A or B. Uncheck this section if one of those categories applies. Confirm Seismic Design Category with structural engineer]		
1. Anchorage of electrical equipment for emergency and standby power systems	OBSERVE	✓ Check for general conformance
2. Anchorage of all other electrical equipment in Seismic Design Categories E and F only (See first page of this schedule for Seismic Design Category)	OBSERVE	✓ Check for general conformance
3. Installation and anchorage of piping designed to carry hazardous materials and their associated mechanical units.	OBSERVE	✓ Check for general conformance
4. Installation and anchorage of vibration isolation systems where the construction documents require a nominal clearance of ¼" or less between support framing and restraint.	OBSERVE	✓ Check for general conformance
5. Verification of clearance between fire sprinkler piping and surrounding mechanical and electrical equipment, including ductwork, piping and their structural supports.	OBSERVE	✓ Check for minimum clearances noted in ASCE7 13.2.3 or a nominal clearance of not less than 3 inches

END SECTION

¹ **OBSERVE:** Observe these items on a random sampling basis daily to insure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

11/20, CHG 2: 08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 CONSTRUCTION SITE PLAN
- 1.4 DOD CONDITION OF READINESS (COR)
- 1.5 CYBERSECURITY DURING CONSTRUCTION
 - 1.5.1 Contractor Computer Equipment
 - 1.5.1.1 Operating System
 - 1.5.1.2 Anti-Malware Software
 - 1.5.1.3 Passwords and Passphrases
 - 1.5.1.4 Contractor Computer Cybersecurity Compliance Statements
 - 1.5.2 Temporary IP Networks
 - 1.5.2.1 Network Boundaries and Connections
 - 1.5.3 Government Access to Network
 - 1.5.4 Temporary Wireless IP Networks
 - 1.5.5 Passwords and Passphrases
 - 1.5.6 Contractor Temporary Network Cybersecurity Compliance Statements

PART 2 PRODUCTS

- 2.1 TEMPORARY SIGNAGE
 - 2.1.1 Bulletin Board
 - 2.1.2 Project Identification Signs
 - 2.1.3 Warning Signs
- 2.2 TEMPORARY TRAFFIC CONTROL
 - 2.2.1 Haul Roads
 - 2.2.2 Barricades
- 2.3 FENCING
 - 2.3.1 Polyethylene Mesh Safety Fencing
 - 2.3.2 Chain Link Panel Fencing
 - 2.3.3 Post-Driven Chain Link Fencing
- 2.4 TEMPORARY WIRING

PART 3 EXECUTION

- 3.1 EMPLOYEE PARKING
- 3.2 AVAILABILITY AND USE OF UTILITY SERVICES
 - 3.2.1 Temporary Utilities
 - 3.2.2 Payment for Utility Services
 - 3.2.3 Meters and Temporary Connections
 - 3.2.4 Advance Deposit
 - 3.2.5 Final Meter Reading
 - 3.2.6 Utilities at Special Locations
 - 3.2.7 Sanitation

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.2.8 Telephone
- 3.2.9 Fire Protection
- 3.3 STATION OPERATION AFFECT ON CONTRACTOR OPERATIONS
 - 3.3.1 Restricted Access Areas
- 3.4 TRAFFIC PROVISIONS
 - 3.4.1 Maintenance of Traffic
 - 3.4.2 Protection of Traffic
 - 3.4.3 Rush Hour Restrictions
 - 3.4.4 Dust Control
- 3.5 CONTRACTOR'S TEMPORARY FACILITIES
 - 3.5.1 Administrative Field Offices
 - 3.5.2 Quality Control Manager Records and Field Office
 - 3.5.3 Storage Area
 - 3.5.4 Supplemental Storage Area
 - 3.5.5 Appearance of Trailers
 - 3.5.6 Trailers or Storage Buildings
 - 3.5.7 Safety Systems
 - 3.5.8 Special Storage Requirements
 - 3.5.8.1 Storage Size and Location
 - 3.5.9 Weather Protection of Temporary Facilities and Stored Materials
 - 3.5.9.1 Building and Site Storm Protection
- 3.6 PLANT COMMUNICATIONS
- 3.7 TEMPORARY PROJECT SAFETY FENCING
- 3.8 CLEANUP
- 3.9 RESTORATION OF STORAGE AREA

-- End of Section Table of Contents --

SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS
11/20, CHG 2: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

NFPA 241 (2022) Standard for Safeguarding
Construction, Alteration, and Demolition
Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety -- Safety and Health
Requirements Manual

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

MUTCD (2009; Rev 2012) Manual on Uniform Traffic
Control Devices

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Site Plan; G

Traffic Control Plan; G

Haul Road Plan; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

1.3 CONSTRUCTION SITE PLAN

Prior to the start of work, submit for Government approval a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage area (onsite and offsite), and access and haul routes, avenues of ingress/egress to the

fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.4 DOD CONDITION OF READINESS (COR)

DOD will set the Condition of Readiness (COR) based on the weather forecast for sustained winds 50 knots (93 km/hr) or greater. Contact the Contracting Officer for the current COR setting.

Monitor weather conditions a minimum of twice a day and take appropriate actions according to the approved Emergency Plan in the accepted Accident Prevention Plan, EM 385-1-1 Section 01 Emergency Planning and the instructions below.

Unless otherwise directed by the Contracting Officer, comply with:

- a. Condition FOUR (Sustained winds of 93 km/hr or greater expected within 72 hours): Normal daily jobsite cleanup and good housekeeping practices. Collect and store in piles or containers scrap lumber, waste material, and rubbish for removal and disposal at the close of each work day. Maintain the construction site including storage areas, free of accumulation of debris. Stack form lumber in neat piles less than one m high. Remove all debris, trash, or objects that could become missile hazards. Review requirements pertaining to "Condition THREE" and continue action as necessary to attain "Condition FOUR" readiness. Contact Contracting Officer for weather and COR updates and completion of required actions.
- b. Condition THREE (Sustained winds of 93 km/hr or greater expected within 48 hours): Maintain "Condition FOUR" requirements and commence securing operations necessary for "Condition ONE" which cannot be completed within 18 hours. Cease all routine activities which might interfere with securing operations. Commence securing and stow all gear and portable equipment. Make preparations for securing buildings. Reinforce or remove formwork and scaffolding. Secure machinery, tools, equipment, materials, or remove from the jobsite. Expend every effort to clear all missile hazards and loose equipment from general base areas. Contact Contracting Officer for weather and COR updates and completion of required actions. Review requirements pertaining to "Condition TWO" and continue action as necessary to attain "Condition THREE" readiness.
- c. Condition TWO (Sustained winds of 93 km/hr or greater expected within 24 hours): Secure the jobsite, and leave Government premises.
- d. Condition ONE. (Sustained winds of 93 km/hr or greater expected within 12 hours): Contractor access to the jobsite and Government premises is prohibited.

1.5 CYBERSECURITY DURING CONSTRUCTION

{For Reference Only: This subpart (and its subparts) relates to AC-18, SA-3, CCI-00258.} Meet the following requirements throughout the construction process.

1.5.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. When used, contractor computers must meet the following requirements:

1.5.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.5.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. All computers used on this project must be scanned using the installed software at least once per day.

1.5.1.3 Passwords and Passphrases

The passwords and passphrases for all computers must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.5.1.4 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Each Statement must be signed by a cybersecurity representative for the relevant company.

1.5.2 Temporary IP Networks

Temporary contractor-installed IP networks may be used during construction. When used, temporary contractor-installed IP networks must meet the following requirements:

1.5.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than IP networks provided under this project or Government furnished IP networks provided for this purpose. Any and all network access from outside the project site is prohibited.

1.5.3 Government Access to Network

Government personnel, as defined, prescribed, and identified by the Contracting Officer, must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification. Or if there is a Government agency that's responsible, identify that agency.

1.5.4 Temporary Wireless IP Networks

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks must not interfere with existing wireless network and must use WPA2 security. Network names (SSID) for wireless networks must be changed from their default values.

1.5.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.5.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Prior to the commencement of work activities, provide a clear weatherproof covered bulletin board not less than 915 by 1220 mm in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, Safety and Health Information as required by EM 385-1-1 Section 01 and other information approved by the Contracting Officer. Coordinate requirements herein with 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, and in location as approved by the Contracting Officer.

2.1.2 Project Identification Signs

The requirements for the signs, their content, and location are as specified in Section 01 58 00 PROJECT IDENTIFICATION. Erect signs within 15 days after receipt of the notice to proceed. Correct the data required by the safety sign daily, with light colored metallic or non-metallic numerals.

2.1.3 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to the EM 385-1-1 Section 04. Attach signs to the perimeter fencing every 150 feet warning the public of the presence of construction hazards. Signs must require unauthorized persons to keep out of the construction site. Correct the data required by safety signs daily. Post signs at all points of entry designating the construction site as a hard hat area.

2.2 TEMPORARY TRAFFIC CONTROL

2.2.1 Haul Roads

Construct access and haul roads necessary for proper prosecution of the work under this Contract in accordance with EM 385-1-1 Section 04. Construct with suitable grades and widths; avoid sharp curves, blind corners, and dangerous cross traffic. Submit haul road plan for approval. Provide necessary lighting, signs, barricades, and distinctive markings for the safe movement of traffic. The method of dust control, although optional, must be adequate to ensure safe operation at all times. Location, grade, width, and alignment of construction and haul roads are subject to approval. Lighting must be adequate to assure full and clear visibility for full width of haul road and work areas during any night work operations.

2.2.2 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Barricades are required whenever safe public access to paved areas such as roads, parking areas or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.3 FENCING

Provide fencing along the construction site and at all open excavations and tunnels to control access by unauthorized personnel. Safety fencing must be highly visible to be seen by pedestrians and vehicular traffic. All fencing must meet the requirements of EM 385-1-1. Remove the fence upon completion and acceptance of the work.

2.3.1 Polyethylene Mesh Safety Fencing

Temporary safety fencing must be a high visibility orange colored, high density polyethylene grid, a minimum of 1.2 m high and maximum mesh size of 50 mm. Fencing must extend from the grade to a minimum of 1.2 m above the grade and be tightly secured to T-posts spaced as necessary to maintain a rigid and taut fence. Fencing must remain rigid and taut with a minimum of 90.7 kg of force exerted on it from any direction with less than 100 mm of deflection.

2.3.2 Chain Link Panel Fencing

Temporary panel fencing must be galvanized steel chain link panels 1.8 m high. Multiple fencing panels may be linked together at the bases to form long spans as needed. Each panel base must be weighted down using sand bags or other suitable materials in order for the fencing to withstand anticipated winds while remaining upright. Fencing must remain rigid and taut with a minimum of 90.7 kg of force exerted on it from any direction with less than 100 mm of deflection.

2.3.3 Post-Driven Chain Link Fencing

Temporary post-driven fencing must be galvanized chain link fencing 1.8 m high supported by an tightly secured to galvanized steel posts driven below grade. Fence posts must be located on minimum 3 meter centers.

Posts may be set in various surfaces such as sand, soil, asphalt or concrete as necessary. Chain link fencing must remain rigid and taut with a minimum of 90.7 kg of force exerted on it from any direction with less than 100 mm of deflection. Completely remove fencing and posts at the completion of construction and restore surfaces disturbed or damaged to its original condition. Locate and identify underground utilities prior to setting fence posts. Equip fence with a lockable gate. Gate must remain locked when construction personnel are not present.

2.4 TEMPORARY WIRING

Provide temporary wiring in accordance with EM 385-1-1 Section 11, NFPA 241 and NFPA 70. Include monthly inspection and testing of all equipment and apparatus.

PART 3 EXECUTION

3.1 EMPLOYEE PARKING

Construction Contract employees must park privately owned vehicles in an area designated by the Contracting Officer. Employee parking must not interfere with existing and established parking requirements of the Government installation.

3.2 AVAILABILITY AND USE OF UTILITY SERVICES

3.2.1 Temporary Utilities

Provide temporary utilities required for construction. Materials may be new or used, must be adequate for the required usage, not create unsafe conditions, and not violate applicable codes and standards.

3.2.2 Payment for Utility Services

- a. The Government will make all reasonably required utilities available from existing outlets and supplies, as specified in the Contract. Unless otherwise provided in the Contract, the amount of each utility service consumed will be charged to or paid at prevailing rates charged to the Government or, where the utility is produced by the Government, at reasonable rates determined by the Contracting Officer. Carefully conserve utilities furnished without charge.
- b. Reasonable amounts of the following utilities will be made available at the prevailing rates.
- c. The point at which the Government will deliver such utilities or services and the quantity available is must be coordinated with the Contracting Officer. Pay all costs incurred in connecting, converting, and transferring the utilities to the work. Make connections, including providing meters; and providing transformers; and make disconnections. Under no circumstances will taps to base fire hydrants be allowed for obtaining domestic water.

3.2.3 Meters and Temporary Connections

Provide and maintain necessary temporary connections, distribution lines, and meter bases (Government will provide meters) required to measure the amount of each utility used for the purpose of determining charges. Notify the Contracting Officer, in writing, 5 working days before final

electrical connection is desired so that a utilities contract can be established. The Government will provide a meter and make the final hot connection after inspection and approval of the Contractor's temporary wiring installation. Do not make the final electrical connection.

3.2.4 Advance Deposit

An advance deposit for utilities consisting of an estimated month's usage or a minimum of \$50.00 will be required. The last monthly bills for the fiscal year will normally be offset by the deposit and adjustments will be billed or returned as appropriate. Services to be rendered for the next fiscal year, beginning 1 October, will require a new deposit. Notification of the due date for this deposit will be mailed prior to the end of the current fiscal year.

3.2.5 Final Meter Reading

Before completion of the work and final acceptance of the work by the Government, notify the Contracting Officer, in writing, 5 working days before termination is desired. The Government will take a final meter reading, disconnect service, and remove the meters. Then remove all the temporary distribution lines, meter bases, and associated appurtenances. Pay all outstanding utility bills before final acceptance of the work by the Government.

3.2.6 Utilities at Special Locations

- a. Reasonable amounts of utilities will be made available at the prevailing Government rates. These rates may be obtained upon application to the Commanding Officer, Naval Station ROTA, by way of the Contracting Officer. Make connections, provide transformers and meters, and make disconnections; and provide backflow preventer devices on connections to domestic water lines.
- b. Reasonable amounts of utilities will be made available without charge. Make connections, provide transformers and meters, and make disconnections; and provide backflow preventer devices on connections to domestic water lines. Under no circumstances will taps to base fire hydrants be allowed for obtaining domestic water.

3.2.7 Sanitation

Provide and maintain within the construction area minimum field-type sanitary facilities in accordance with EM 385-1-1 Section 02. Locate the facilities behind the construction fence or out of the public view. Clean units and empty wastes at least once a week or more frequently into a municipal, district, or station sanitary sewage system, or remove waste to a commercial facility. Obtain approval from the system owner prior to discharge into a municipal, district, or commercial sanitary sewer system. Penalties or fines associated with improper discharge will be the responsibility of the Contractor. Coordinate with the Contracting Officer and follow station regulations and procedures when discharging into the station sanitary sewer system. Maintain these conveniences at all times. Include provisions for pest control and elimination of odors. Government toilet facilities will not be available to Contractor's personnel.

3.2.8 Telephone

Make arrangements and pay all costs for telephone facilities desired.

3.2.9 Fire Protection

Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials daily to minimize potential hazards.

3.3 STATION OPERATION AFFECT ON CONTRACTOR OPERATIONS

3.3.1 Restricted Access Areas

The Government will monitor work in areas listed below. Notify Contracting Officer at least 14 calendar days prior to starting work in these areas.

- a. Pier 3 and related access bridge to Pier 3.

3.4 TRAFFIC PROVISIONS

3.4.1 Maintenance of Traffic

- a. Conduct operations in a manner that will not close a thoroughfare or interfere with traffic on railways or highways except with written permission of the Contracting Officer at least 15 calendar days prior to the proposed modification date, and provide a [Traffic Control Plan](#) for Government approval detailing the proposed controls to traffic movement for approval. The plan must be in accordance with local regulations and the [MUTCD](#), Part VI. Make all notifications and obtain all permits required for modification to traffic movements outside Station's jurisdiction. Contractor may move oversized and slow-moving vehicles to the worksite provided requirements of the highway authority have been met.
- b. Conduct work so as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times. Obtain approval from the Contracting Officer prior to starting any activity that will obstruct traffic.
- c. Provide, erect, and maintain, at Contractor's expense, lights, barriers, signals, passageways, detours, and other items, that may be required by the Life Safety Signage, overhead protection authority having jurisdiction.
- d. Provide cones, signs, barricades, lights, or other traffic control devices and personnel required to control traffic. Do not use foil-backed material for temporary pavement marking because of its potential to conduct electricity during accidents involving downed power lines.

3.4.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction period except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment the work, and the erection and maintenance of adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Provide self-illuminated (lighted) barricades during hours of darkness.

Brightly-colored (orange) vests are required for all personnel working in roadways. Protect the traveling public from damage to person and property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. Contractor is responsible for the repair of damage to roads caused by construction operations.

3.4.3 Rush Hour Restrictions

Do not interfere with the peak traffic flows preceding and during normal operations without notification to and approval by the Contracting Officer.

3.4.4 Dust Control

Dust control methods and procedures must be approved by the Contracting Officer. Coordinate dust control methods with 01 57 19.01 21 TEMPORARY ENVIRONMENTAL CONTROLS.

3.5 CONTRACTOR'S TEMPORARY FACILITIES

Contractor is responsible for security of their property. Provide adequate outside security lighting at the temporary facilities. Trailers must be anchored to resist high winds and meet applicable state or local standards for anchoring mobile trailers. Coordinate anchoring with EM 385-1-1 Section 04. The Contract Clause entitled "FAR 52.236-10, Operations and Storage Areas" and the following apply:

3.5.1 Administrative Field Offices

Provide and maintain administrative field office facilities within the construction area at the designated site. Government office and warehouse facilities will not be available to the Contractor's personnel.

3.5.2 Quality Control Manager Records and Field Office

Provide on the jobsite an office with approximately 9 square meter of useful floor area for the exclusive use of the QC Manager. Provide a weathertight structure with adequate heating and cooling, toilet facilities, lighting, ventilation, a 1200 by 2400 mm plan table, a standard size office desk and chair, computer station, and working communications facilities. Provide either a 1,500 watt radiant heater and a window-mounted air conditioner rated at 2.6 kW minimum or a window-mounted heat pump of the same minimum heating and cooling ratings. Provide a door with a cylinder lock and windows with locking hardware. Make utility connections. Locate as directed. File quality control records in the office and make available at all times to the Government. After completion of the work, remove the entire structure from the site.

3.5.3 Storage Area

Construct a temporary 1.8 m high chain link fence around trailers and materials. Include plastic strip inserts, colored brown, so that visibility through the fence is obstructed. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the installation boundaries. Trailers, equipment, or materials must not be open to public view with the exception

of those items which are in support of ongoing work on the current day. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

Keep fencing in a state of good repair and proper alignment. If the Contractor elects to traverse grassed or unpaved areas which are not established roadways with construction equipment or other vehicles, cover the grassed or unpaved areas with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways; gravel gradation must be at the Contractor's discretion. Mow and maintain grass located within the boundaries of the construction site for the duration of the project. Grass and vegetation along fences, structures, under trailers, and in areas not accessible to mowers must be edged or trimmed neatly.

3.5.4 Supplemental Storage Area

Upon request, and pending availability, the Contracting Officer will designate another or supplemental area for the use and storage of trailers, equipment, and materials. This area may not be in close proximity of the construction site but will be within the installation boundaries. Maintain the area in a clean and orderly fashion and secured if needed to protect supplies and equipment. Utilities will not be provided to this area by the Government.

3.5.5 Appearance of Trailers

- a. Trailers must be roadworthy and comply with all appropriate state and local vehicle requirements. Trailers which are rusted, have peeling paint or are otherwise in need of repair will not be allowed on Installation property. Trailers must present a clean and neat exterior appearance and be in a state of good repair.
- b. Maintain the temporary facilities. Failure to do so will be sufficient reason to require their removal at the Contractor's expense.

3.5.6 Trailers or Storage Buildings

- a. Trailers or storage buildings will be permitted, where space is available, subject to the approval of the Contracting Officer.
- b. Mount a sign not smaller than 600 by 600 mm on the trailer or building that shows the company name, business phone number, emergency phone number and conforms to the following requirements and sketch:

Graphic panel	Aluminum, painted blue
Copy	Screen painted or vinyl die-cut, white
Typeface	Univers 65 u/lc
See Sketch No. 01500 (graphic).	

3.5.7 Safety Systems

Protect the integrity of all installed safety systems or personnel safety

devices. Obtain prior approval from the Contracting Officer if entrance into systems serving safety devices is required. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish Contract requirements, provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and obtain approval from the Contracting Officer.

3.5.8 Special Storage Requirements

The following special storage requirements apply:

3.5.8.1 Storage Size and Location

The open site available for storage must be confined to the indicated operations area.

3.5.9 Weather Protection of Temporary Facilities and Stored Materials

Take necessary precautions to ensure that roof openings and other critical openings in the building are monitored carefully. Take immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the building from damage.

3.5.9.1 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions must include, but are not limited to, closing openings; removing loose materials, tools and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

3.6 PLANT COMMUNICATIONS

Whenever the individual elements of the plant are located so that operation by normal voice between these elements is not satisfactory, install a satisfactory means of communication, such as telephone or other suitable devices and make available for use by Government personnel.

3.7 TEMPORARY PROJECT SAFETY FENCING

As soon as practicable, but not later than 15 days after the date established for commencement of work, furnish and erect temporary project safety fencing at the work site. Maintain the safety fencing during the life of the Contract and, upon completion and acceptance of the work, remove from the work site.

3.8 CLEANUP

Remove construction debris, waste materials, packaging material and the like from the work site daily. Any dirt or mud which is tracked onto paved or surfaced roadways must be cleaned away. Store all salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

3.9 RESTORATION OF STORAGE AREA

Upon completion of the project remove the bulletin board, signs, barricades, haul roads, and all other temporary products from the site. After removal of trailers, materials, and equipment from within the fenced area, remove the fence. Restore areas used during the performance of the Contract to the original or better condition. Remove gravel used to traverse grassed areas and restore the area to its original condition, including top soil and seeding as necessary.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 57 19.01 20

ENVIRONMENTAL CONTROLS

PART 1 GENERAL

1.1 REFERENCES

1.2 DEFINITIONS

- 1.2.1 Class I and II Ozone Depleting Substance (ODS)
 - 1.2.2 Contractor Generated Hazardous Waste
 - 1.2.3 Cultural Resources
 - 1.2.4 Electronics Waste
 - 1.2.5 Environmental Pollution and Damage
 - 1.2.6 Environmental Protection
 - 1.2.7 Hazardous Debris
 - 1.2.8 Hazardous Materials
 - 1.2.9 Hazardous Waste
 - 1.2.10 Installation Pest Management Coordinator
 - 1.2.11 Land Application
 - 1.2.12 Material not regulated as solid waste
 - 1.2.13 Non-Hazardous Waste
 - 1.2.14 Oily Waste
 - 1.2.15 Pesticide
 - 1.2.16 Pesticide Treatment Plan
 - 1.2.17 Pests
 - 1.2.18 Pest Management Coordinator
 - 1.2.19 Recyclables
 - 1.2.20 Recycling
 - 1.2.21 Regulated Waste
 - 1.2.22 Reportable Release
 - 1.2.23 Scrap Metal
 - 1.2.24 Sediment
 - 1.2.25 Solid Waste
 - 1.2.25.1 Debris
 - 1.2.25.2 Green Waste
 - 1.2.25.3 Bulky Waste
 - 1.2.26 Spill Event
 - 1.2.27 Stormwater
 - 1.2.28 Surface Discharge
 - 1.2.29 Universal Waste
 - 1.2.30 Wastewater
 - 1.2.31 Waters of Spain
 - 1.2.31.1 Continental Waters
 - 1.2.31.2 Coastal Waters
 - 1.2.32 Wetlands
 - 1.2.33 Wood
- 1.3 SUBMITTALS
- 1.4 ENVIRONMENTAL PROTECTION REQUIREMENTS
- 1.4.1 Training in Environmental Compliance Assessment Training and Tracking System (ECATTS)
 - 1.4.1.1 Personnel Requirements
 - 1.4.1.2 Certification

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 1.4.1.3 Refresher Training
- 1.4.2 Conformance with the Environmental Management System
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Preconstruction Survey and Protection of Features
 - 1.5.2 Regulatory Notifications
 - 1.5.3 Environmental Protection Plan (EPP)
 - 1.5.4 Preconstruction Environmental Brief/Discussion
 - 1.5.5 Environmental Coordinator (EC)
 - 1.5.6 Employee Training Records
 - 1.5.6.1 Pest Control Training
 - 1.5.7 Pesticide Delivery, Storage, and Handling
 - 1.5.7.1 Delivery and Storage
 - 1.5.7.2 Handling Requirements
 - 1.5.8 Non-Compliance Notifications
- 1.6 LICENSES AND PERMITS
- 1.7 SOLID WASTE AND RECYCLING MANAGEMENT PLAN
 - 1.7.1 Solid Waste Management Report

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 PROTECTION AND MANAGEMENT OF RESOURCES
 - 3.1.1 Natural Resources
 - 3.1.2 Protection of Endangered Species
 - 3.1.3 Vegetation
 - 3.1.4 Surface Waters
 - 3.1.4.1 Flow Ways
 - 3.1.4.2 Rivers and Streams
- 3.2 STORMWATER
 - 3.2.1 Stormwater Pollution Prevention Plan (SWPPP)
 - 3.2.2 Erosion and Sediment Control Measures
- 3.3 SURFACE AND GROUNDWATER
 - 3.3.1 Cofferdams, Diversions, and Dewatering
- 3.4 PROTECTION OF HISTORICAL AND ARCHAEOLOGICAL RESOURCES
- 3.5 WASTE MANAGEMENT AND DISPOSAL AND RECYCLING
 - 3.5.1 Solid Waste
 - 3.5.2 Hazardous Waste
 - 3.5.2.1 Waste Storage/Satellite Accumulation/90 Day Storage Areas
 - 3.5.3 Universal Waste
 - 3.5.4 Electronics End-of-Life Management
- 3.6 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING
 - 3.6.1 Fuel Tanks
 - 3.6.2 Used Oil Management
 - 3.6.3 Oil Storage Including Fuel Tanks
- 3.7 RELEASES/SPILLS OF OIL, HAZARDOUS SUBSTANCES AND AQUEOUS FILM FORMING FOAM (AFFF)
- 3.8 WATER AND AIR QUALITY PROTECTION MEASURES
- 3.9 AIR RESOURCES
 - 3.9.1 Oil or Dual-fuel Boilers and Furnaces
 - 3.9.2 Burning
 - 3.9.3 Class I and II ODS Prohibition
 - 3.9.4 Accidental Venting of Refrigerant
 - 3.9.5 Dust Control
 - 3.9.5.1 Particulates
 - 3.9.5.2 Abrasive Blasting
 - 3.9.6 Odors
- 3.10 WASTEWATER

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.10.1 Treatment
- 3.10.2 Surface Discharge
- 3.10.3 Land Application
- 3.11 PREVIOUSLY USED EQUIPMENT
- 3.12 CONTROL AND MANAGEMENT OF ASBESTOS-CONTAINING MATERIAL (ACM)
- 3.13 CONTROL AND MANAGEMENT OF LEAD-BASED PAINT (LBP)
- 3.14 CONTROL AND MANAGEMENT OF POLYCHLORINATED BIPHENYLS (PCBS)
- 3.15 CONTROL AND MANAGEMENT OF LIGHTING BALLAST AND LAMPS CONTAINING
- 3.16 MILITARY MUNITIONS
- 3.17 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS
WASTES
- 3.18 CHLORDANE
- 3.19 SOUND INTRUSION
- 3.20 POST CONSTRUCTION CLEANUP
- 3.21 LAYDOWN AREA
- 3.22 DRINKING WATER

-- End of Section Table of Contents --

SECTION 01 57 19.01 20

ENVIRONMENTAL CONTROLS

(Naval Station Rota, Spain, March 2021)

ANTIGUA - SE ENVIO UNA CON SPILLS 7 JUNIO 2024

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NAVSTA ROTA ENVIRONMENTAL REFERENCE DOCUMENTS (NAVSTA)

LG (2016) NAVSTA Rota Landscaping Guidelines

FGS-SPAIN (2014) Final Governing Standards for Spain

MINISTERIO DE LA PRESIDENCIA (MP)

RD 9/2005 (2005) Real Decreto 9/2005, de 14 de enero, por el que se establece la relación de actividades potencialmente contaminantes del suelo y los criterios y estándares para la declaración de suelos contaminados.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Health Requirements Manual

1.2 DEFINITIONS

1.2.1 Class I and II Ozone Depleting Substance (ODS)

Class I and II ODS include the chemicals listed in FGS-SPAIN Chapter 2, Table 2.8.

1.2.2 Contractor Generated Hazardous Waste

Hazardous waste as defined in FGS-SPAIN or as defined by applicable Spanish regulations. Any waste which displays hazardous properties and is denoted as an asterisk (*) next to the six digit code in the European List of Wastes.

Excluded from this definition are domestic sewage sludge, and household and medical wastes not possessing the properties in Chapter 6 and Appendix A of FGS-SPAIN. It includes the following:

- a. Discarded materials that contain one or more of the characteristics or properties described in Appendix A of FGS-SPAIN.
- b. All hazardous waste described Chapter 6 of FGS-SPAIN.

- c. Any other form of waste defined as hazardous by the Competent Authority.

These waste streams typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e., methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment waste water.

1.2.3 Cultural Resources

Physical remains of any prehistoric or historic district, site, building, structure, or object significant in world, national or local history, architecture, archeology, engineering, or culture. The term includes artifacts, archeological resources, records, and material remains that are related to such a district, site, building, structure, or object, and also includes natural resources (plants, animals, landscape features, etc.) that may be considered important as a part of a country's traditional culture and history. The term also includes any property listed on the World Heritage List or the Spanish equivalent of the National Register of Historic Places. Spanish lists of properties should be evaluated to determine if they are equivalent with the National Register of Historic Places prior to application.

1.2.4 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, or disposal.

1.2.5 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

1.2.6 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.7 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with FGS-SPAIN. Hazardous debris also includes debris that exhibits a characteristic of hazardous waste in accordance with FGS-SPAIN. And as defined in paragraph entitled "Debris" of this section, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) per FGS-SPAIN and applicable national, regional, and local regulations.

1.2.8 Hazardous Materials

Hazardous materials as defined in FGS-SPAIN or the Country of Spain. Hazardous material is any substance, compound, mixture, or material that is capable of posing an unreasonable risk to health, safety, or the environment if improperly handled, stored, issued, transported, labeled, or disposed because it displays a characteristic listed in Table 5.1 of FGS-SPAIN "Typical Hazardous Materials Characteristics," or the material is listed in Appendix A of FGS-SPAIN, "List of Hazardous Substances and Materials." Or,

- a. Is regulated as a hazardous material or
- b. Requires a Safety Data Sheet (SDS) per EM 385-1-1, or
- c. During end use, treatment, handling, packaging, storage, transportation, or disposal meets or has components that meet or have potential to meet the definition of a hazardous waste.

Designation of a material by this definition, when separately regulated or controlled by other instructions or directives, does not eliminate the need for adherence to that hazard-specific guidance which takes precedence over this instruction for "control" purposes. Such material include ammunition, weapons, explosive actuated devices, propellants, pyrotechnics, chemical and biological warfare materials, medical and pharmaceutical supplies, medical waste and infectious materials, bulk fuels, radioactive materials, and other materials such as asbestos, mercury, and polychlorinated biphenyls (PCBs). Nonetheless, the exposure may occur incident to manufacture, storage, use and demilitarization of these items.

1.2.9 Hazardous Waste

Hazardous waste as defined in FGS-SPAIN or as defined by applicable national, regional and local regulations. Any discarded material, liquid, solid, or gaseous, which meets the definition of hazardous material or is designated hazardous waste by FGS-SPAIN or the Country of Spain. Comply with FGS-SPAIN and the installation Hazardous Waste Management Plan.

1.2.10 Installation Pest Management Coordinator

Installation Pest Management Coordinator (IPMC) is the individual officially designated by the Installation Commander to oversee the Installation Pest Management Program and the Installation Pest Management Plan. All Pesticides to be used at the Installation must be approved by the NAVFAC Atlantic Entomologist/Professional Pest Management Consultant (PPMC).

1.2.11 Land Application

Land Application means spreading or spraying discharge water at a rate that allows the water to percolate into the soil. No sheeting action, soil erosion, discharge into storm sewers, discharge into defined drainage areas, must occur. Comply with FGS-SPAIN and applicable Spanish laws.

1.2.12 Material not regulated as solid waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as

amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.13 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, hazardous waste criteria in accordance with Chapter 6 of FGS-SPAIN.

1.2.14 Oily Waste

Oily waste are those materials that are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from that POLs. Oily wastes also means materials, including wastewaters, centrifuge solids, filter residues or sludges, bottom sediments, tank bottoms, and absorbents which have come into contact with and have been contaminated by, POLs and may be appropriately tested and discarded in a manner which is in compliance with Chapter 6 of FGS-SPAIN. A Spanish waste manifest (Hoja de Control y Seguimiento) shall be prepared and approved by authorized firm to properly dispose of hazardous waste in accordance with Spanish Regulations. This definition includes materials such as oily rags, "kitty litter" sorbent clay and organic sorbent material. Centrifuge solids, filter residues or sludge, bottom sediments, tank bottoms, oily rags, kitty litter, sorbents, etc. are to be containerized and turned in for proper disposal to an authorized firm for proper treatment and/or disposal. Chapter 9 of FGS-SPAIN describes the POL related requirements on NAVSTA Rota, Spain.

1.2.15 Pesticide

Pesticide is any substance or mixture of substances intended for preventing, destroying, repelling, or mitigating any pest, or intended for use as a plant regulator, defoliant or desiccant. All Pesticides to be used at the Installation must be approved by the NAVFAC Atlantic Entomologist/Professional Pest Management Consultant (PPMC).

1.2.16 Pesticide Treatment Plan

A plan for the prevention, monitoring, and control to eliminate pest infestation.

1.2.17 Pests

Pests are arthropods, birds, rodents, nematodes, fungi, bacteria, viruses, algae, snails, marine borers, snakes, weeds and other organisms (except for human or animal disease-causing organisms) that adversely affect readiness, military operations, or the well-being of personnel and animals; attack or damage real property, supplies, equipment, or vegetation; or are otherwise undesirable.

1.2.18 Pest Management Coordinator

The Pest Management Coordinator is an individual who works at NAVSTA Rota Public Works Department and who is responsible for implementing the Pest Management Plan.

1.2.19 Recyclables

Recyclables are materials that can be recycled instead of disposed of in landfills, including but not limited to; pipes, doors, windows, door and window frames, plumbing fixtures, glazing and mirrors, wiring, insulated/non-insulated copper wire cable, wire rope, and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Metal meeting the definition of lead contaminated or lead based paint contaminated may not be included as recyclable. Paint cans may not be included as recyclable.

1.2.20 Recycling

Recycling is the process of recovering material from waste and turning it into new products. The original product is destroyed in the process and is used to form new products. Examples are scrap metal, glass, cardboard, paper, plastic and many more items. On NAVSTA Rota all scrap metal, including wiring, remains the property of the U. S. Government and will be brought by the Contractor to the NAVSTA Rota Recycling Center near Tow Way. Other recyclable materials will be taken off-base to a local recycling center.

1.2.21 Regulated Waste

Regulated waste are solid wastes that have specific additional FGS-SPAIN, regional, or local controls for handling, storage, or disposal.

1.2.22 Reportable Release

A reportable release means any spilling, leaking, pumping, pouring, emitting, emptying, discharging, injecting, escaping, leaching, dumping, or disposing into the environment of a known or unknown material or hazardous substance that poses an immediate threat to human health or the environment to the air, soil, or water. In the event of a spill on base call the fire department at 9-1-1.

Report all spills to the NAVSTA Rota Environmental Division.

1.2.23 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included.

1.2.24 Sediment

Sediment is soil and other debris that has eroded and has been transported by runoff water or wind.

1.2.25 Solid Waste

Solid waste is garbage, refuse, sludge, and other discarded materials, including solid, semi-solid, liquid, and contained gaseous materials resulting from industrial and commercial operations and from community

activities. It does not include solids or dissolved material in domestic sewage or other significant pollutants in water resources, such as silt, dissolved or suspended solids in industrial wastewater effluent, dissolved materials in irrigation return flows, or other common water pollutants.

Types of solid waste typically generated at construction sites may include:

1.2.25.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure that exceeds 60 mm (2.5-inch) particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials may not be reinforced with or contain ferrous wire, rods, accessories and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.25.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps and plant roots. Marketable trees, grasses and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.25.3 Bulky Waste

Large items of solid waste such as household appliances, furniture, large auto parts, trees, branches, stumps, and other oversize wastes whose large size precludes or complicates their handling by normal solid waste collection, processing, or disposal methods.

1.2.26 Spill Event

A spill is any release of oil or hazardous substances to the water or ground that is not controlled or permitted. This includes any spilling, leaking, pumping, emitting, discharging, injecting, escaping, leaching, disposing, or dumping of liquid or solid material that is not authorized in writing by the Contracting Officer.

1.2.27 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.28 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, creeks or "waters of Spain". Surface discharges are discrete, identifiable sources and require a permit from the governing agency. Comply with FGS-SPAIN and national, regional, and local laws and regulations.

1.2.29 Universal Waste

Universal wastes are certain hazardous wastes in the following categories: batteries, pesticides, mercury-containing equipment (for example, thermostats), and lamps (for example, fluorescent bulbs). These wastes are collected and sent for recycling or proper disposal.

1.2.30 Wastewater

Wastewater is the used water and solids from a community that flow to a treatment plant. Wastewater is also any water which during manufacturing or processing, comes into direct contact with, or results from the production or use of, any raw material, intermediate product, finished product, by-product, or waste product.

1.2.31 Waters of Spain

Waters included in the following categories:

1.2.31.1 Continental Waters

Any water body within the Spanish territory other than coastal waters; this includes groundwater and surface waters (e.g., underground aquifers, rivers, streams, lakes, reservoirs, springs, irrigation channels).

1.2.31.2 Coastal Waters

The zone between the coastline of low waters and a line parallel to the coastline that is approximately 12 miles (20 km) from the shore.

Exclusions to Waters of Spain are domestic or industrial waste treatment systems, including treatment ponds or lagoons designed to meet the requirements of FGS-SPAIN, are not waters of Spain. This exclusion applies only to manmade bodies of water that were neither originally waters of Spain nor resulted from impoundment of waters of Spain.

1.2.32 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

1.2.33 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, hardboard. Treated or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.

Submit the following in accordance with Section 01 33 00 SUBMITTAL

PROCEDURES:

SD-01 Preconstruction Submittals

Preconstruction Environmental, Including Common Chameleon and Vegetation, Survey; G (NAVSTA Rota Environmental Division Review).

Preconstruction Historical and Archaeological Survey; G (NAVSTA Rota Environmental Division Review).

Environmental Protection Plan (EPP); G (NAVSTA Rota Environmental Division Review). Write the EPP following the template in section 1.5.3.

Spill Prevention, Control and Response Plan (for projects at or near the Bay of Cadiz); G (NAVSTA Rota Environmental Division Review).

Solid Waste and Recycling Management Plan

Stormwater Pollution Prevention Plan; G (NAVSTA Rota Environmental Division Review).

Training in Environmental Compliance Assessment Training and Tracking System (Ecatts); G

Environmental Manager Qualifications; G

Safety Data Sheets (SDS) for Hazardous Materials

SD-07 Certificates

List of Employee Environmental Training Records; G

List of ECATTS Environmental on-line course Certificates of Completion; G

SD-11 Closeout Submittals

Stormwater Pollution Prevention Plan Compliance Notebook; G

Solid Waste and Recycling Report (by weight); G (NAVSTA Rota Environmental Division Review).

Disposal Documentation for Hazardous and Regulated Waste; G

Contractor Hazardous Material Inventory Log; G

Regulatory Notifications; G

Sales Documentation; G

Construction & Demolition (C & D) Debris data and management cost

1.4 ENVIRONMENTAL PROTECTION REQUIREMENTS

All work must be performed in accordance with the requirements of FGS-SPAIN. All work must also adhere to the Spanish Navy and the Junta de Andalucia (Regional Spanish Government) natural and cultural resource requirements.

The Contractor must identify, obtain, and comply with all applicable (Spanish and US) environmental requirements, authorizations, and permits. Examples of work elements that typically have environmental requirements, authorizations, and permits are removal and disposal of contaminated soils, storm water, biological and archaeological resources, hazardous materials, hazardous waste, pesticides, and in-water work requirements. A portion of this project includes the installation of underground fuel pipes through Rio Salado river, using HDD technology. This portion of the works requires authorizations and permits from the Spanish Regulator. Contractor must coordinate all Spanish Navy and Junta de Andalucia authorization and permitting actions with Installation Environmental Division and NAVFAC Host Nation Liaison.

The Contractor may obtain from the Contracting Officer the two following documents with relevant information of contamination and Spanish code regulations:

- "Subsurface Soil and Groundwater Investigation Report" prepared for NAVFAC Atlantic by CH2M HILL, Inc.
- Environmental Assessment for project: P-919 - Bulk Tank Farm Improvements, Transfer of hydrocarbons across the Salado River at the Naval Base of Rota (Spain) prepared by Tecnoambiente.

These documents, combined with actual findings during excavation shall be used in the design of the documents needed to request from the Spanish Regulator the authorizations for removal and disposal of the contaminated soils and for crossing Rio Salado with underground piping. The documents related with contaminated soils will include a risk analysis in conformance with RD 9/2005. The quantities of contaminated soil for such risk analysis must be estimated using the above referenced documents. Refer to drawing G-004T for supplementary instructions. Transportation and disposal of the contaminated soils will be performed by a separate contract.

Provide and maintain, during the life of the contract, environmental protection as defined and described in the EPP. Plan for and provide environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract.

Tests and procedures assessing whether construction operations comply with applicable environmental laws may be required. Analytical work must be performed by qualified laboratories; and where required by law, the laboratories must be accredited/certified.

1.4.1 Training in Environmental Compliance Assessment Training and Tracking System (ECATTS)

Coordinate with the NAVSTA Environmental Division to add additional staff that require this on-line training.

1.4.1.1 Personnel Requirements

Appoint in writing to the Environmental Division the Environmental Coordinators (ECs) (Primary and an Alternate) for the project. The ECs and construction workers must complete applicable ECATTS training modules (installation specific or general) prior to starting respective portions of on-site work under this Contract. If personnel changes occur for any of these positions after starting work, replacement personnel must complete applicable ECATTS training within 14 days of assignment to the project.

All personnel assigned to handling Hazardous Waste and Hazardous Materials are properly trained to meet, at a minimum, OSHA standards, 29 CFR 1910.120 Contractor shall assign properly trained Environmental coordinators to ensure proper management of hazardous materials and hazardous waste, and to properly respond to oil and hazardous substance (OHS) spills. Contractor assigned ECs must have at least successfully completed the 24-hr Hazardous Waste Management and Emergency response (HAZWOPER) or its equivalent. Prepare and maintain employee training records throughout the term of the contract.

Refrigerant technicians shall be trained in proper recovery/recycling procedures, leak detection, safety, shipping, and disposal in accordance with recognized industry standards or Spanish equivalent.

1.4.1.2 Certification

Maintain ECATTS certificates of completion for personnel who have completed the required ECATTS training. This training is web-based and can be accessed from any computer with Internet access using the following instructions.

Register for NAVFAC Environmental Compliance Training and Tracking System, by logging on to <https://environmentaltraining.ecatts.com/>. Obtain the password for registration from the Contracting Officer.

1.4.1.3 Refresher Training

This training has been structured to allow Contractor personnel to receive credit under this contract and to carry forward credit to future contracts. Ensure the Environmental Manager review their training plans for new modules or updated training requirements prior to beginning work.

1.4.2 Conformance with the Environmental Management System

Perform work under this contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities

affect work performed under the contract.

Coordinate with the installation's EMS manager to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Division will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

1.5 QUALITY ASSURANCE

1.5.1 Preconstruction Survey and Protection of Features

Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer, and take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record. Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the Contractor's assigned storage area and access route(s), as applicable. The Contractor and the Contracting Officer will sign this survey report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract.

If the project work area include areas of suitable chameleon habitat where indicated on contract drawings, chameleons found during night surveys in these areas must be relocated by a Spanish approved chameleon specialist prior to construction activities. Follow the Protected Species Protocol (PSP) for the common chameleon as applicable.

If the project work area include areas of archaeological concern, the site must be evaluated by an archaeological specialist prior to construction activities. If required by the Junta de Andalucia, an archaeological monitor may be required to be present during earth moving activities.

If the project work area include areas suitable for bird nests, a bird nest survey will be conducted and if any nests are found to contain eggs or chicks, the nest shall not be disturbed until the chicks have fledged and left the nest.

1.5.2 Regulatory Notifications

Provide regulatory notification requirements in accordance with Spanish regulations. Typically, regulatory notifications must be provided for the following (this listing is not all-inclusive): demolition, renovation, construction, removal or use of an air emissions sources, and remediation of controlled substances (asbestos, hazardous waste, lead paint).

1.5.3 Environmental Protection Plan (EPP)

The purpose of the EPP is to present an overview of known or potential environmental issues that must be considered and addressed during

construction or demolition activities.

The EPP (including drawings and figures) addresses environmental categories, as identified below. An EPP must be developed by the Contractor and submitted to the Contracting Officer for review by the NAVSTA Rota Environmental Division. The EPP must be reviewed by the NAVSTA Rota Environmental Division prior to the start of construction. Submit the EPP using the following format:

Environmental Protection Plan (EPP) Template
Naval Station Rota Spain

1. Cover Page - Name of project; Navy contract and task order number; name of Navy Base and specific locations; Contractor's name, address, e-mail and phone number. Company's environmental point of contact name and phone number.
2. Table of contents.
3. Project description and general site information. Show work area plan of the proposed activity in each portion of the area and identify the areas of limited use or nonuse, including laydown areas. Include measures for marking the limits of use areas, including methods for protection of features to be preserved within authorized work areas and methods to control runoff and to contain materials on site.
4. Existing environmental conditions of the project site. For example: known archaeological sites; protected species (i.e. common chameleon potential habitat); erosion issues; Andalusian forest trees present such as Stone Pine; nesting birds, and other natural resources present.
5. Conclusions of environmental or archaeological surveys and studies done for the project, if any.
6. Spanish environmental permits needed, including those needed for the removal and disposal of soils contaminated with hydrocarbons or heavy metals and those needed for the crossing of Rio Salado river with underground piping. Environmental and Historical/Archaeological Resource Permits are also, including Regulatory Notification. Certification of Environmental and Cultural Permits Obtained by Contractor or HN Installation Commander.
7. Compliance with NAVSTA Rota's Protected Species Protocol for the Common Chameleon (2016), and NAVSTA Rota's Landscaping Guidelines (2016).
8. Measures/Steps to address resources and materials (as are applicable to the project):

Archaeological and historical site protection measures: Include the statement, "If during digging an archaeological artifact is uncovered, work will be stopped and the NAVSTA Rota Environmental Division will be contacted. Work will continue when approved by the Contracting Officer Representative";

Hazardous materials and hazardous waste containment/storage and disposal;

Recycling and Solid waste management/disposal: Include the statement,

"Scrap metal and wiring shall remain the property of the U.S. Government and will be taken to the NAVSTA Rota Recycling Center";

Vegetation and wildlife protection and revegetation: including Spanish protected species, Andalusian forest trees, nesting birds, and other sensitive species;

Spill prevention plan and planned actions in the event of a spill: include requirement to notify EV of all spills, regardless of quantity;

Storm water management and soil erosion control: storm water pollution prevention plan, if required;

Wastewater management;

Water pollution protection (i.e. ocean/bay, groundwater, streams, rivers, and other water bodies);

Air Quality Protection (e.g., dirt and dust) Submit truck and material haul routes along with a Dirt and Dust Control Plan for controlling dirt, debris, and dust on Installation roadways. At a minimum, identify in the plan the subcontractor and equipment for cleaning along the haul route and measures to reduce dirt, dust, and debris from roadways;

Planned air pollution-generating equipment and processes and management control measures (including, but not limited to, spray painting, abrasive blasting, demolition, material handling, fugitive dust, and fugitive emissions). Log hours of operations and track quantities of materials used;

Identify portable and stationary internal combustion engines that will be supplied, used or serviced. At minimum, include the make, model, serial number, manufacture date, size (engine brake horsepower), and emission certification status of each engine. Maintain applicable records and log hours of operation and fuel use. Logs must include reasons for operation and delineate between emergency and non-emergency operation;

Noise control;

Dirt and Dust Control: include measures to reduce dirt, dust, and debris from roadways, adjacent areas, and building interiors;

Other environmental concerns specific to the project: i.e., in-water dredging and/or pile installments, traffic and transportation, etc.;

Hazardous materials and hazardous waste containment/storage and disposal;

Water pollution protection (i.e., groundwater, streams, rivers, lakes and other water bodies);

9. Monitoring and Quality Control Procedures;

10. Employee Environmental Trainings List: Do not submit training certificates, only list the environmental trainings completed by the employees.

1.5.4 Preconstruction Environmental Brief/Discussion

The Construction Manager/Contracting Officer representative, Contractor, and NAVSTA Rota Environmental Division staff must attend an environmental brief/discussion that is to be included in the preconstruction or another meeting before the beginning of construction or demolition activities. Provide the following: types, quantities, use of hazardous materials and copy of Safety Data Sheet (SDS) that will be brought onto the installation for Environmental review and approval; and types and quantities of solid wastes/Hazardous waste/wastewater that may be generated during the Contract. All pesticides must be approved by the NAVFAC Atlantic Entomologist/Professional Pest Management Consultant (PPMC). Discuss the environmental protection plan, environmental protection measures to be taken, and the results of the Preconstruction Survey at this time.

1.5.5 Environmental Coordinator (EC)

The designated Contractor EC will be directly responsible for coordinating Contractor compliance with DoD, Navy and Spanish requirements. This can be a collateral position; however the person in this position must be trained to adequately accomplish the duties and responsibilities. ECs are required to appear for a face-to-face interaction with Environmental Division staff to understand the base environmental policies, regulations and compliance requirements. The ECs are directly responsible for coordinating Contractor compliance with Spanish and NAVSTA Rota requirements. The EC must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, manifesting, and disposal); implement the EPP; ensure environmental permits are obtained (as required), maintained, and closed out; ensure compliance with Stormwater Program requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; coordinate any remediation of regulated substances (lead, asbestos, PCB transformers); ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure Contractor personnel are trained in accordance with their position requirements; coordinate removal of waste containers; and maintain the Environmental Records binder and required documentation, including environmental permits compliance, Spanish Hazardous Waste Manifests, and close-out. Submit EC Qualifications to the Contracting Officer.

1.5.6 Employee Training Records

Prepare and maintain Employee Training Records throughout the term of the contract and maintain in the Environmental records binder. Provide Employee Training Records in the Environmental Records Binder. Ensure every employee completes a program of classroom instruction or on-the-job training that teaches them to perform their duties in a way that ensures compliance with FGS-SPAIN. Provide a Position Description for each employee, evaluating the employee's association with hazardous and regulated wastes. This Position Description will include training requirements for a Hazardous Waste Generator facility. Submit these Assembled Employee Training Records to the Contracting Officer at the conclusion of the project, unless otherwise directed.

Train personnel to meet FGS-SPAIN requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting

agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of Spain, and endangered species and their habitat that are known to be in the area. Provide copy of the Erosion and Sediment Control Inspector qualifications.

1.5.6.1 Pest Control Training

Train personnel in pest control. Conduct a pest control meeting for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting a pest infestation; familiarization with statutory and contractual pest control standards; installation and care of devices and instruments, if required, for monitoring purposes to ensure adequate and continuous pest control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of waters of the Host Nation, and endangered species and their habitat that are known to be in the area. Provide a Certificate of Competency for the personnel who will be conducting the pesticide application and management of pest control. All Pesticides to be used at the Installation must be approved by the NAVFAC Atlantic Entomologist/Professional Pest Management Consultant (PPMC).

1.5.7 Pesticide Delivery, Storage, and Handling

1.5.7.1 Delivery and Storage

Deliver pesticides to the site in the original, unopened containers bearing legible labels. Store pesticides according to manufacturer's instructions and under lock and key when unattended. Contractors must remove all pesticides from the installation at the end of the working day. Contractors are not permitted to store pesticides at the Installation after working hours.

1.5.7.2 Handling Requirements

Formulate, treat with, and dispose of pesticides and associated containers in accordance with label directions and use the clothing and personal protective equipment specified on the labeling for use during each phase of the application. Furnish SDSs for pesticide products.

1.5.8 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with FGS-SPAIN and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions will be granted or equitable adjustments allowed for any such suspensions. This is in addition to any other actions the Contracting Officer may take under the contract, or in accordance with the FAR.

1.6 LICENSES AND PERMITS

Obtain licenses and permits required for the construction of the project and in accordance with FAR 52.236-7. Notify the Government of all general use permitted equipment the Contractor plans to use on site. This paragraph supplements the Contractor's responsibility under FAR 52.236-7.

1.7 SOLID WASTE AND RECYCLING MANAGEMENT PLAN

Provide the Contracting Officer and NAVSTA Rota Environmental Division with written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Also, the anticipated quantity (weight) of materials to be recycled. Include in the report the locations where various types of waste will be disposed of or recycled. For instance, all scrap metal will be recycled at the NAVSTA Rota Recycling Center. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location license showing such entities approval of the disposal plan before transporting wastes off Government property.

1.7.1 Solid Waste Management Report

Monthly, submit a solid waste disposal and recycled materials report to the Contracting Officer and the NAVSTA Rota Environmental Division. For each waste, the report will state the classification (using the definitions provided in this section), amount, location, and name of the business receiving the solid waste.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 PROTECTION AND MANAGEMENT OF RESOURCES

3.1.1 Natural Resources

Minimize interference with, disturbance to, and damage to fish, wildlife, and plants, including their habitats. Prior to the commencement of activities, consult with the Installation Environmental Division regarding rare species or sensitive habitats that need to be protected. The protection of rare, threatened, and endangered animal and plant species identified, including their habitats, is the Contractor's responsibility.

Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work that is consistent with the requirements of the Installation Environmental Division or as otherwise specified. Confine construction activities to within the limits of the work indicated or specified.

3.1.2 Protection of Endangered Species

Known endangered or threatened species and Spanish protected species and their habitats must be protected.

The Protected Species Protocol for the Common Chameleon (March 2016) will be followed, as applicable. This includes submitting a plan for

anticipated vegetation removal and site restoration and a survey of potentially suitable chameleon habitat in the project area. If chameleons are present in the project area they must be relocated by a specialist in advance of construction activities.

Preserve natural resources within the work area limits to the maximum extent possible. Restore natural resources to an equivalent or improved condition upon completion of work.

If a natural resource of interest is encountered during construction (e.g. chameleon or other sensitive species), the Contractor must stop all work until the resource is properly assessed by the Environmental Division and the Contracting Officer Representative has given the notice to resume work.

3.1.3 Vegetation

All planted vegetation must follow the NAVSTA Rota Landscaping Guidelines (LG). This guideline contains an approved list of vegetation and trees that may be planted on NAVSTA Rota.

Areas of Stone Pine and other Andalusian forest tree species must be avoided for new construction to the maximum extent possible. If trees must be removed, approval must be granted by the Contracting Officer. Trees that are approved for removal must be replaced at a 3:1 ratio at an area designated by the Contracting Officer. The replacement trees must be saplings (not seedlings) and must be irrigated for one year to ensure successful growth.

Do not damage trees. Do not fasten or attach ropes, cables, or guys to existing nearby trees or shrubs without the Contracting Officer's permission. With Contracting Officer approval, remove trees with 30 percent or more of their root systems destroyed as well as trees and other landscape features scarred or damaged by equipment operations. With Contracting Officer approval, replace damaged natural resources with equivalent trees and landscape features.

3.1.4 Surface Waters

Do not enter, disturb, destroy, or allow discharge of contaminants into surface waters except as authorized herein. The protection of surface waters is the Contractor's responsibility. Authorization to enter specific waters identified does not relieve the Contractor from any obligation to protect other waters within, adjacent to, or in the vicinity of the construction site and associated boundaries.

3.1.4.1 Flow Ways

Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as specified.

3.1.4.2 Rivers and Streams

River and stream crossings must allow movement of materials or equipment without violating water pollution control standards of FGS-SPAIN. Construction of stream crossing structures must be in compliance with any required Spanish government permits.

The Contracting Officer's approval and appropriate permits are required

before any equipment will be permitted to cross rivers and streams. In areas where frequent crossings are required, install temporary culverts or bridges. Obtain Contracting Officer's approval prior to installation. Remove temporary culverts or bridges upon completion of work, and repair the area to its original condition unless otherwise required by the Contracting Officer.

3.2 STORMWATER

3.2.1 Stormwater Pollution Prevention Plan (SWPPP)

Develop and implement a storm water pollution prevention plan (SWPPP) for activities where there is a potential for storm water pollution or erosion. Activities include: construction activities; hazardous material storage, outdoor material storage areas; outdoor painting/depainting operations; pesticide operations; power production; vehicle storage yards; aircraft ground support equipment maintenance; aircraft/runway deicing; Aircraft/vehicle fueling operations; aircraft vehicle maintenance and repair; aircraft/vehicle washing; bulk fuel storage areas; and corrosion control activities. Include the SWPPP as part of the Environmental Protection Plan (EPP) submittal.

3.2.2 Erosion and Sediment Control Measures

Implement measures to control effluent in stormwater discharges.

Control stormwater volume and velocity to minimize soil erosion in order to minimize pollutant discharges. Control stormwater discharges, including both peak flowrates and total stormwater volume, to minimize channel and streambank erosion and scour in the immediate vicinity of discharge points.

Provide and maintain natural buffers around waters of the HN. Direct stormwater to vegetated areas and maximize stormwater infiltration through use of low-impact development practices (identified by UFC 3-210-10) to reduce pollutant discharges, unless infeasible.

Minimize soil compaction. Minimizing soil compaction is not required where the intended function of a specific area of the site dictates that it be compacted. Preserve topsoil, unless impractical. Preserving topsoil is not required where the intended function of a specific area of the site dictates that the topsoil be disturbed or removed.

Initialize soil stabilization whenever any clearing, grading, excavating, or other earth disturbing activities have permanently ceased on any portion of the site, or temporarily ceased on any portion of the site and will not resume for a period exceeding 14 calendar days. In arid, semi-arid, and drought-stricken areas where initiating vegetative stabilization measures immediately is infeasible, alternative stabilization measures must be employed as appropriate.

Minimize the exposure of building materials, building products, construction wastes, trash, landscape materials, fertilizers, pesticides, herbicides, detergents, sanitary waste, and other materials present on the site to precipitation and to stormwater. Minimization of exposure is not required in cases where the exposure to precipitation and to stormwater will not result in a discharge of pollutants, or where exposure of a specific material or product poses little risk of stormwater contamination (such as final products and materials intended for outdoor use).

Personnel who handle hazardous substances or perform activities that could contribute pollution in wet weather events must be trained in appropriate BMPs.

3.3 SURFACE AND GROUNDWATER

3.3.1 Cofferdams, Diversions, and Dewatering

Construction operations for dewatering, removal of cofferdams, tailrace excavation, and tunnel closure must be constantly controlled to maintain compliance with existing Spanish standards and designated uses of the surface water body. Do not discharge excavation ground water to the sanitary sewer, storm drains, or to surface waters without prior specific authorization in writing from the Installation Environmental Division. Discharge of hazardous substances will not be permitted under any circumstances. Use sediment control BMPs to prevent construction site runoff from directly entering any storm drain or surface waters.

If the construction dewatering is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is treated and authorized by the Installation Environmental Division. Discharge of hazardous substances will not be permitted under any circumstances.

3.4 PROTECTION OF HISTORICAL AND ARCHAEOLOGICAL RESOURCES

Comply with Junta de Andalucia requirements for archaeological site protection. This may include having an archaeological monitor on site during earth moving activities. Contractor must coordinate all Spanish Navy and Junta de Andalucia authorization and permitting actions with Installation Environmental Division and NAVFAC Host Nation Liaison.

Construction of any type, that may alter the character of a historic and cultural resource or disturb its view, is prohibited without proper monitoring and documentation.

If an archeological item is uncovered during excavation, the Contractor must stop all work and contact the Contracting Officer immediately. Work can continue once the archaeological resource has been evaluated and the Contracting Officer gives the direction to proceed.

3.5 WASTE MANAGEMENT AND DISPOSAL AND RECYCLING

Scrap metal, including wiring, shall remain the property of the U.S. Government and the Contractor shall transport all recovered scrap metal from the project site to the NAVSTA Rota Recycling Facility (near Tow Way). Other recyclable materials become the property of the Contractor upon demolition and must be taken by the Contractor off of the Navy Base to a local recycling center.

3.5.1 Solid Waste

All solid waste must be regularly removed from the project site to maintain a clean work area. Recycle solid waste at a recycling center to the greatest extent possible. Solid waste that cannot be recycled must be treated, stored, and disposed of properly at an authorized land-fill facility.

Report the weight (tonnage) of all materials disposed of as solid waste and of all materials recycled on a monthly basis. This report must be provided to the NAVSTA Rota's Environmental Division's Solid Waste/Recycling Manager. Document the weight of disposed materials in daily reports.

Pick up solid wastes, and place in covered containers that are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Employ segregation measures so that no hazardous or toxic waste will become co-mingled with non-hazardous solid waste. Transport solid waste off Government property and dispose of it in compliance with Chapter 7 of FGS-SPAIN for solid waste disposal. Verify that the selected transporters and disposal facilities have the necessary permits and licenses to operate. Segregate and separate treated wood components disposed at a lined landfill approved to accept this waste in accordance with Spanish regulations. Disposal must comply with Spanish regulations.

3.5.2 Hazardous Waste

No hazardous material or hazardous waste shall be brought onto the NAVSTA Rota property that does not directly relate to requirements for the performance of this contract. Manage hazardous material used in construction, including but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, in accordance with Chapters 5 and 6 of FGS-SPAIN. Do not dispose of hazardous waste on Government property. Do not discharge any waste to a sanitary sewer, storm drain, or to surface waters or conduct waste treatment or disposal on Government property without written approval of the Contracting Officer.

Comply with the Safety Data Sheets (SDSs) requirements per FGS-SPAIN.

All toxic or hazardous materials and hazardous wastes will be transported, stored, treated, and properly disposed of per FGS requirements and the Naval Station Rota Waste Analysis Plan. All hazardous materials and waste generated from the project will be characterized and properly stored. Hazardous waste must be stored outside the installation while awaiting disposition. Hazardous Material storage must be coordinated with the NAVSTA Rota Safety Office and Environmental Division.

Storage and handling of hazardous materials will adhere to the Department of Defense Component policies, including Joint Service Publication on Storage and Handling of Hazardous Materials. Defense Logistics Agency Instruction 4145.11, Army Technical Manual 38-410, Naval Supply Publication 573, Air Force Joint Manual (AFJMAN) 23-209, and Marine Corps Order 4450.12A "Storage and Handling of Hazardous Materials" provide additional guidance on the storage and handling of hazardous materials.

Hazardous material dispensing areas must be properly maintained. Drums/containers must not be leaking. Drip pans/absorbent materials must be placed under containers, as necessary, to collect drips or spills. Container contents will be clearly marked. The dispensing area shall be designed and operated to ensure appropriate segregation of incompatible hazardous materials.

All containers of hazardous materials shall have manufacturer-generated labels or a Hazardous Chemical Warning Label in accordance with DoD Instruction 600.05 and have SDSs information available on site. These

requirements apply throughout the life-cycle of these materials.

The use of materials listed in Table 5.3 of FGS-SPAIN is either prohibited or restricted in Spain, under specified conditions, with the exception of mission critical materials. The use of perfluorooctane sulfonates (PFOS)-based Fire-fighting foams are prohibited.

Completed Spanish Disposal Manifests of outgoing hazardous waste are required. They must specify the treatment and/or disposal method and location and must be provided to Contracting Officer.

3.5.2.1 Waste Storage/Satellite Accumulation/90 Day Storage Areas

Accumulate hazardous waste at satellite accumulation points and in compliance with Chapter 6 of FGS-SPAIN. Individual waste streams will be limited to 208 liters of accumulation (or 0.95 liter for acutely hazardous wastes). If the Contractor expects to generate hazardous waste at a rate and quantity that makes satellite accumulation impractical, the Contractor may request a temporary 90 day accumulation point be established. Submit a request in writing to the Contracting Officer and provide the following information (Attach Site Plan to the Request):

Contract Number	_____
Contractor	_____
Haz/Waste or Regulated Waste POC	_____
Phone Number	_____
Type of Waste	_____
Source of Waste	_____
Emergency POC	_____
Phone Number	_____
Location of the Site	_____

Attach a Waste Determination form for the expected waste streams. Allow 10 working days for processing this request. Additional compliance requirements (e.g. training and contingency planning) that may be required are the responsibility of the Contractor. Barricade the designated area where waste is being stored and post a sign identifying as follows:

"DANGER - UNAUTHORIZED PERSONNEL KEEP OUT"

3.5.3 Universal Waste

Manage the following categories of universal waste in accordance with FGS-SPAIN and installation instructions:

- a. Batteries
- b. Lamps
- c. Mercury-containing equipment
- d. Pesticides

Mercury is prohibited in the construction of this facility, unless specified otherwise, and with the exception of mercury vapor lamps and fluorescent lamps. Dumping of mercury-containing materials and devices such as mercury vapor lamps, fluorescent lamps, and mercury switches, in rubbish containers is prohibited. Remove without breaking, pack to prevent breakage, and transport out of the activity in an unbroken condition for disposal or recycling as directed.

3.5.4 Electronics End-of-Life Management

Recycle or dispose of electronics waste, including, but not limited to, used electronic devices such computers, monitors, hard-copy devices, televisions, mobile devices, in accordance with FGS-SPAIN and installation instructions.

3.6 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING

POL products include flammable or combustible liquids, such as gasoline, diesel, lubricating oil, used engine oil, hydraulic oil, mineral oil, and cooking oil. Store POL products and fuel equipment and motor vehicles in a manner that affords the maximum protection against spills into the environment. Manage and store POL products in accordance with FGS-SPAIN. Use secondary containments, dikes, curbs, and other barriers, to prevent POL products from spilling and entering the ground, storm or sewer drains, stormwater ditches or canals, or Spanish waters. Describe in the EPP how POL tanks and containers will be stored, managed, and inspected and what protections will be provided. Storage of oil, including fuel, on the project site is not allowed. Fuel must be brought to the project site each day that work is performed.

3.6.1 Fuel Tanks

Petroleum products and lubricants required to sustain up to 30 days of construction activity may be kept on site. Storage and refilling practices shall comply with FGS-SPAIN and Spanish Government requirements. Secondary containment shall be provided and be no less than 110 percent of the tank volume plus five inches of free-board. If a secondary berm is used for containment then the berm shall be impervious to oil for 72 hours and be constructed so that any discharge will not permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Drips pans are required and the tanks must be covered during inclement weather. Rain water collected in secondary containment that has been contaminated with POL must be treated and disposed of as hazardous waste.

3.6.2 Used Oil Management

Manage used oil generated on site in accordance with FGS-SPAIN. Determine if any used oil generated while onsite exhibits a characteristic of hazardous waste. Used oil containing 1,000 parts per million of solvents is considered a hazardous waste and disposed of at the Contractor's expense. Used oil mixed with a hazardous waste is also considered a hazardous waste. Dispose in accordance with paragraph HAZARDOUS WASTE DISPOSAL. Contractors can turn-in used oil that meets the recycling criteria to the Environmental Division for recycling.

3.6.3 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection and flow shut off

valve for oil storage tanks. A berm used to provide secondary containment must be of sufficient size and strength to contain the contents of the tanks plus 12 centimeters (5 inches) freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks. If tanks and containers with an aggregate aboveground capacity greater than 5,000 liter (1,320 gallons) will be used onsite (only containers with a capacity of 208 liter (55 gallons) or greater are counted), provide and implement a SPCC plan meeting the requirements of FGS-SPAIN and installation requirements. Do not bring underground storage tanks to the installation for Contractor use during a project. Submit the SPCC plan to the Contracting Officer for approval.

Proper labels are required on POL storage containers (Flammable, No Smoking, Capacity of tank, Type of POL etc.).

Generator (if inbuilt tank is not double walled and not equipped with overfill protection and shut off valve) and other fuel assemblies will be placed on a secondary containment and drainage of storm water from containment areas will be controlled by a valve that is locked closed when not in active use. Stormwater will be inspected for petroleum sheen before being drained from containment areas. If petroleum sheen is present it must be collected with sorbent materials prior to drainage, or treated using an oil-water separator. Sorbent material shall be disposed of in accordance with NAVSTA Rota Environmental Waste Management Procedures.

All flammable tanks shall be grounded properly to withstand the static current that may be generated during fuel transfer (FGS-SPAIN and UFC 3-460-03 address grounding and bonding of fuel oil storage tanks and fuel transfers).

Conduct the fueling and lubricating of equipment and motor vehicles in a manner that protects against spills and evaporation. Prevent oil or hazardous substances from entering the ground, drainage areas, or navigable waters. All fueling operations including fuel delivery on base shall follow NAVSTA Rota fuel management procedures.

3.7 RELEASES/SPILLS OF OIL, HAZARDOUS SUBSTANCES AND AQUEOUS FILM FORMING FOAM (AFFF)

Exercise due diligence to prevent, contain, and respond to spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated by environmental law. Maintain spill cleanup equipment and materials at the work site. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of any releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the NAVSTA Rota Fire Department, phone # 9-1-1 or Dispatch phone # 956-82-2911. If the Contractor's response is inadequate, the U.S. Navy may respond. If this should occur, the Contractor will be required to reimburse the government for spill response assistance and analysis. If the spill can be contained and cleaned up using available spill kit/supplies and trained manpower capabilities, complete response operations/clean up the spill, if it is safe to do so. Then report

incidents to base Environmental Division.

Should the potential exist for fuel and other spills, a spill prevention, control, and response plan is required specific to this project.

For all in-water or above water projects, a thorough spill prevention, control, and response plan is required.

Conduct the fueling and lubricating of equipment and motor vehicles in a manner that protects against spills and evaporation. Manage all used oil generated on site in accordance with FGS-SPAIN and Spanish requirements. Determine if any used oil generated while on-site exhibits a characteristic of hazardous waste. Used oil containing 1000 parts per million of solvents will be considered a hazardous waste and disposed of at Contractor's expense. Used oil mixed with a hazardous waste will also be considered a hazardous waste.

Submit verbal and written notifications as required by Installation instructions and Spill Prevention Plan immediately. Spill response must be in accordance with Chapter 18 of FGS-SPAIN and the Installation Spill Prevention Plan. Contain and clean up these spills without cost to the Government.

The person discovering/causing the spill shall conduct the following immediate first responder actions as appropriate; but only if properly trained and they can be done safely:

- Shutdown all appropriate manifold storage tank and truck valves and/or pumps.
- Restrict all ignition sources and utilize spill control devices and equipment.
- Notify supervisor and people working nearby; and evacuate the immediate area to a safe distance upwind.
- Aid exposed personnel by assisting in safely removing personnel from immediate spill area, and requesting medical assistance for an ambulance (as needed)
- Secure storm drains, if any with storm drain covers or diversion barriers.

While reporting provide as much of the following information as possible

- Name, telephone number, and identification of caller;
- Time and location of spill;
- Identity and quantity of spill materials;
- Origin and cause of spill;
- Description of spill (behavior of spill, affected areas);
- Anticipated movement of spill;
- Planned or initiated actions;
- Type of assistance required;
- Stay in the area, if it is safe to do so, until FRT arrives and direct them to the spill;
- Provide SDSs as needed.

Report any Aqueous Film Forming Foam (AFFF) intentional or unintentional releases at any location (regardless of any amount) to the Environmental Division.

3.8 WATER AND AIR QUALITY PROTECTION MEASURES

All precautionary measures practicable shall be taken to prevent and minimize any leakage and emissions of ozone-depleting substances.

To reduce air pollution emissions, minimize the number of vehicle trips and vehicle idling.

Protect all water supply aquifers (groundwater) and surface water sources from contamination.

Use only lead-free and aluminum free pipe, solder, flux, and fittings in the installation or repair of potable water distribution and plumbing. Use lead check swab test kit on pipes, solder, flux, and fittings to verify compliance.

Use Best Management Practices (BMPs) that will minimize or eliminate the possibility of pollution being introduced into the continental and coastal waters of Spain. Should a discharge occur, immediately contact Naval Station Rota Fire Department (phone # 956-82-2911) or Dispatch (phone # 9-1-1).

3.9 AIR RESOURCES

3.9.1 Oil or Dual-fuel Boilers and Furnaces

Provide product data and details for new, replacement, or relocated fuel fired boilers, heaters, or furnaces to the Installation Environmental Division through the Contracting Officer. Data to be reported include: equipment purpose (water heater, building heat, process), manufacturer, model number, serial number, fuel type (oil type, gas type) size (MMBTU heat input).

3.9.2 Burning

Burning is prohibited on the Government premises.

3.9.3 Class I and II ODS Prohibition

Certain reclaimed Class I and II ODS must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Division to determine the appropriate management of Class I and II ODS.

3.9.4 Accidental Venting of Refrigerant

Accidental venting of a refrigerant is a release and must be reported immediately to the Contracting Officer. Any class I or class II ODS, HFC, and PFC refrigerant listed in FGS Tables 2.2 and 2.3 shall not be intentionally released in the course of maintaining, servicing, repairing, or disposing of appliances, industrial process refrigeration units, air conditioning units, or motor vehicle air conditioners.

All repairs, including leak repairs or services to appliances, industrial process refrigeration units, air conditioning units, or motor vehicle air conditioners, must be performed using commercially available refrigerant recovery/recycling equipment operated by trained personnel.

3.9.5 Dust Control

If Contractor vehicles transfer mud or dirt onto roads, they must remove the materials from the roads. Keep dust down at all times, including during nonworking periods. Sprinkle or treat, with dust suppressants, the soil at the site, haul roads, and other areas disturbed by operations. Avoid use of potable water for these purposes if practicable. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Do not unnecessarily shake bags of cement, concrete mortar, or plaster.

3.9.5.1 Particulates

Dust particles, aerosols and gaseous by-products from construction activities, and processing and preparation of materials (such as from asphaltic batch plants) must be controlled at all times, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed FGS standards or that would cause a hazard or a nuisance. Sprinkling, chemical treatment of an approved type, baghouse, scrubbers, electrostatic precipitators, or other methods will be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp. Avoid use of potable water for these purposes if practicable. Provide sufficient, competent equipment available to accomplish these tasks. Perform particulate control as the work proceeds and whenever a particulate nuisance or hazard occurs.

3.9.5.2 Abrasive Blasting

Blasting operations cannot be performed without prior approval of the Installation Environmental Division. The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive agent, paint chips, and other debris. Perform work involving removal of hazardous material in accordance with Chapter 5 of FGS-SPAIN. Blasting residues will be treated and disposed of as hazardous waste in accordance with Chapter 6 of FGS-SPAIN.

3.9.6 Odors

Control odors from construction activities. The odors may not constitute a health hazard.

3.10 WASTEWATER

3.10.1 Treatment

Do not allow wastewater from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, and forms to enter water ways or to be discharged prior to being treated to remove pollutants. Dispose of the construction-related waste water off-Government property by collecting and

placing it in a retention pond where suspended material can be settled out or the water can evaporate to separate pollutants from the water or other means as agreed to by the Contracting Officer. The site for the retention pond must be coordinated and approved with the Environmental Division and Contracting Officer. The residue left in the pond prior to completion of the project must be removed, tested, and disposed of off-Government property in accordance with Chapter 6 of FGS-SPAIN. Backfill the area to the original grade, top-soiled, and seeded or sodded. Coordinate with the Environmental Division on test requirements for the water in the retention pond for and have the results reviewed and approved by the Contracting Officer prior to being discharged or disposed of off-Government property.

3.10.2 Surface Discharge

For discharge of ground water obtain authorization from the Environmental Division prior to surface discharging.

3.10.3 Land Application

Water generated from the flushing of lines after disinfection or disinfection in conjunction with hydrostatic testing may be land-applied after de-chlorination or discharged into the sanitary sewer with prior approval by the Environmental Division.

3.11 PREVIOUSLY USED EQUIPMENT

Clean previously used construction equipment prior to bringing it onto the project site. Equipment must be free from soil residuals, egg deposits from plant pests, noxious weeds, and plant seeds.

3.12 CONTROL AND MANAGEMENT OF ASBESTOS-CONTAINING MATERIAL (ACM)

Manage and dispose of asbestos-containing waste in accordance with FGS-SPAIN. Manifest asbestos-containing waste and provide the Final Spanish Manifest to the Contracting Officer. Notifications to the Installation Air Program Manager are required before starting any asbestos work.

The Contractor needs to contact the Public Works Asbestos Program Manager and Environmental Division in identifying Asbestos Containing Material (ACM) locations and procedures to follow during any construction/maintenance to ensure ACM detected areas are managed properly. If asbestos is found at any stage of the project in confined areas or left unnoticed during the environmental survey - DO NOT disturb the area, stop work in the area of the questionable material or condition and notify both the Contracting Officer and Environmental Division immediately for further guidance. Sealant and adhesives used in the project must not contain asbestos. Submit a certificate proving the asbestos-free nature of products for materials that suspect asbestos-containing materials (Sealant, Adhesives, gasket etc.).

3.13 CONTROL AND MANAGEMENT OF LEAD-BASED PAINT (LBP)

Manage and dispose of lead-contaminated waste in accordance with FGS-SPAIN. Manifest any lead-contaminated waste and provide the manifest to the Contracting Officer.

3.14 CONTROL AND MANAGEMENT OF POLYCHLORINATED BIPHENYLS (PCBS)

Manage and dispose of PCB-contaminated waste in accordance with FGS-SPAIN.

3.15 CONTROL AND MANAGEMENT OF LIGHTING BALLAST AND LAMPS CONTAINING PCBS

Manage and dispose of contaminated waste in accordance with Chapter 6 of FGS-SPAIN.

3.16 MILITARY MUNITIONS

In the event military munitions, as defined in 40 CFR 260, are discovered or uncovered, immediately stop work in that area and immediately inform the Contracting Officer.

3.17 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS WASTES

If petroleum-contaminated soil, groundwater or suspected hazardous waste is found during construction that was not identified in the Contract documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.18 CHLORDANE

Evaluate excess soils and concrete foundation debris generated during the demolition of housing units or other wooden structures for the presence of chlordane or other pesticides prior to reuse or final disposal.

3.19 SOUND INTRUSION

Make the maximum use of low-noise emission products. Blasting or use of explosives are not permitted without written permission from the Contracting Officer, and then only during the designated times. Confine pie driving from 8 a.m. and 4 p.m., Monday through Friday, exclusive of holidays, unless otherwise specified. Keep construction activities under surveillance and control to minimize environment damage by noise.

3.20 POST CONSTRUCTION CLEANUP

Clean up areas used for construction in accordance with Contract Clause: "Cleaning Up". Unless otherwise instructed in writing by the Contracting Officer, remove traces of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. Grade parking area and similar temporarily used areas to conform to surrounding contours.

3.21 LAYDOWN AREA

The Contractor laydown area will be subjected to periodic environmental inspections to ensure compliance with applicable environmental regulations. During demobilization and prior to returning the laydown area back to Government, a walk through site inspection shall be coordinated in advance with the Environmental Division to ensure the area is free from contamination/spills, etc. If required, remediation processes will be conducted at Contractor's expense.

3.22 DRINKING WATER

Identify and protect all above or below-ground drinking water distribution lines and wellhead equipment during all phases of construction. All aspects of drinking water systems in the installation, whether for repair and/or new systems, shall use certified lead-free and aluminum free materials, including pipe, solder, flux, appurtenances and all fixtures in accordance with 40 CFR 143. Use lead check swab test kit on pipes, solder, flux, and fittings to verify compliance.

- a. Submit results of lead check swap test kit to the Environmental Division for approval.
- b. Testing is required if material is made in another country to ensure it meets FGS-SPAIN Criteria.

Pipes, blow-offs, valves, fittings, pipe materials and products including, but not limited to, paints, linings, coatings, adhesives, and lubricants in direct contact with potable water should be certified to NSF/ANSI Standard 61. Prior to purchase, materials and products will be submitted to the NAVSTA Rota Public Works Department for approval.

Any connection to the existing water distribution system shall be coordinated with the NAVSTA Rota Public Works Department. Contractor shall ensure Cross Connection (CC) control inspections are done and Back Flow Prevention (BFP) devices are properly installed and certified prior to looping or installation of any new plumbing devices to the distribution line. All new pipelines shall be flushed in accordance with an approved flushing plan that follows the Requirements of American Water Works Association (AWWA) Standard 651 for Disinfecting Water Mains latest edition. Flushing plan shall be submitted to the installation water management board for review/approval.

Upon completion of project, Contractor shall coordinate with the NAVSTA ROTA Environmental Division and Public Works Department to arrange for residual chlorine and lead swab (on fixtures, solder etc.) testing a minimum of 24 hours in advance.

Use energy/water conservation products (such as energy star, high efficiency - low flow flushing system etc.). Coordinate with the NAVSTA Rota Public Works Department, UEM Energy Manager. Water distributed onboard shall not be used for any construction purpose.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 58 00

PROJECT IDENTIFICATION

08/19, CHG 5: 08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 PROJECT IDENTIFICATION SIGN
 - 1.3.1 Project Identification Signboard

PART 2 PRODUCTS

PART 3 EXECUTION

ATTACHMENTS:

project sign detail

-- End of Section Table of Contents --

SECTION 01 58 00

PROJECT IDENTIFICATION
08/19, CHG 5: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)

AWPA C1 (2003) All Timber Products - Preservative Treatment by Pressure Processes

AWPA C2 (2003) Lumber, Timber, Bridge Ties and Mine Ties - Preservative Treatment by Pressure Processes

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Preliminary Drawing Indicating Layout And Text Content; G

1.3 PROJECT IDENTIFICATION SIGN

Prior to initiating any work on site, provide one project identification sign at the location designated. Construct the sign in accordance with project sign detail, which can be downloaded at: <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Maintain sign throughout the life of the project. Upon completion of the project, remove the sign from the site.

1.3.1 Project Identification Signboard

Provide a project identification signboard in accordance with attached Plates 1, 3, and 4. Provide a preliminary drawing indicating layout and text content. Erect a signboard at a conspicuous location on the job site where directed by the Contracting Officer.

- a. The field of the sign consists of a 1200 by 2400 mm sheet of grade B-B medium density overlaid exterior plywood.
- b. Lumber is B or better Southern pine, pressure-preservative treated in accordance with AWPA C1 and AWPA C2. Nails are aluminum or galvanized steel.
- c. Give one coat of exterior alkyd primer and two coats of exterior alkyd

enamel paint to the entire signboard and supports. Perform the lettering and sign work by a skilled sign painter using paint known in the trade as bulletin colors. The colors, lettering sizes, and lettering styles are as indicated. Where preservative-treated lumber is required, utilize only cured pressure-treated wood which has had the chemicals leached from the surface of the wood prior to painting.

- d. Use spray applied automotive quality high gloss acrylic white enamel paint as background for the NAVFAC logo. NAVFAC logo is an applied 0.0508 mm film sticker/decal with either transparent or white background or paint the logo by stencil onto the sign. The weather resistant sticker/decal film is rated for a minimum of 2-year exterior vertical exposure. Mount the self-adhering sticker to the sign with pressure sensitive, permanent acrylic adhesive. Shop cut sticker/decal to rectangular shape and provide pull-off backing sheet on adhesive side of design sticker for shipping.
- e. Sign paint colors (manufacturer's numbers/types listed below for color identification only)
 - (1) Blue = To match dark blue color in the NAVFAC logo.
 - (2) White = To match Brilliant White color in the NAVFAC logo.
- f. NAVFAC logo must retain proportions and design integrity. NAVFAC logos in electronic format may be obtained from the WBDG at the following link:
<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>.
Use the following to choose color values for the paint to be used:
 - (1) Dark Blue = equivalent to CMYK values 100, 72, 0, 8.
 - (2) Light Blue = equivalent to CMYK values 69, 34, 0, 0.
 - (3) Cyan = equivalent to CMYK values 100, 9, 0, 6.
 - (4) Yellow = equivalent to CMYK values 0.9, 94, 0.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

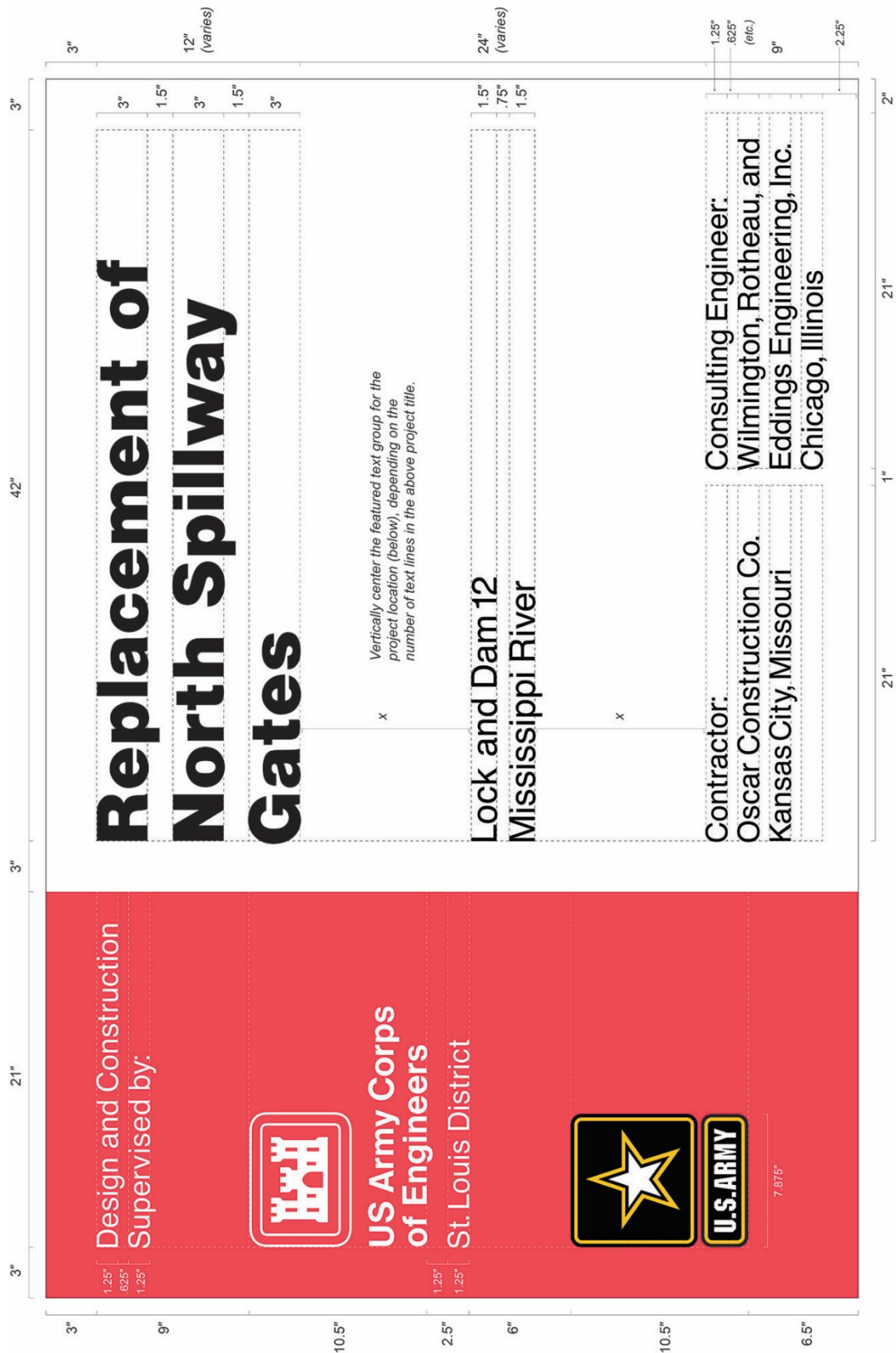
Not Used

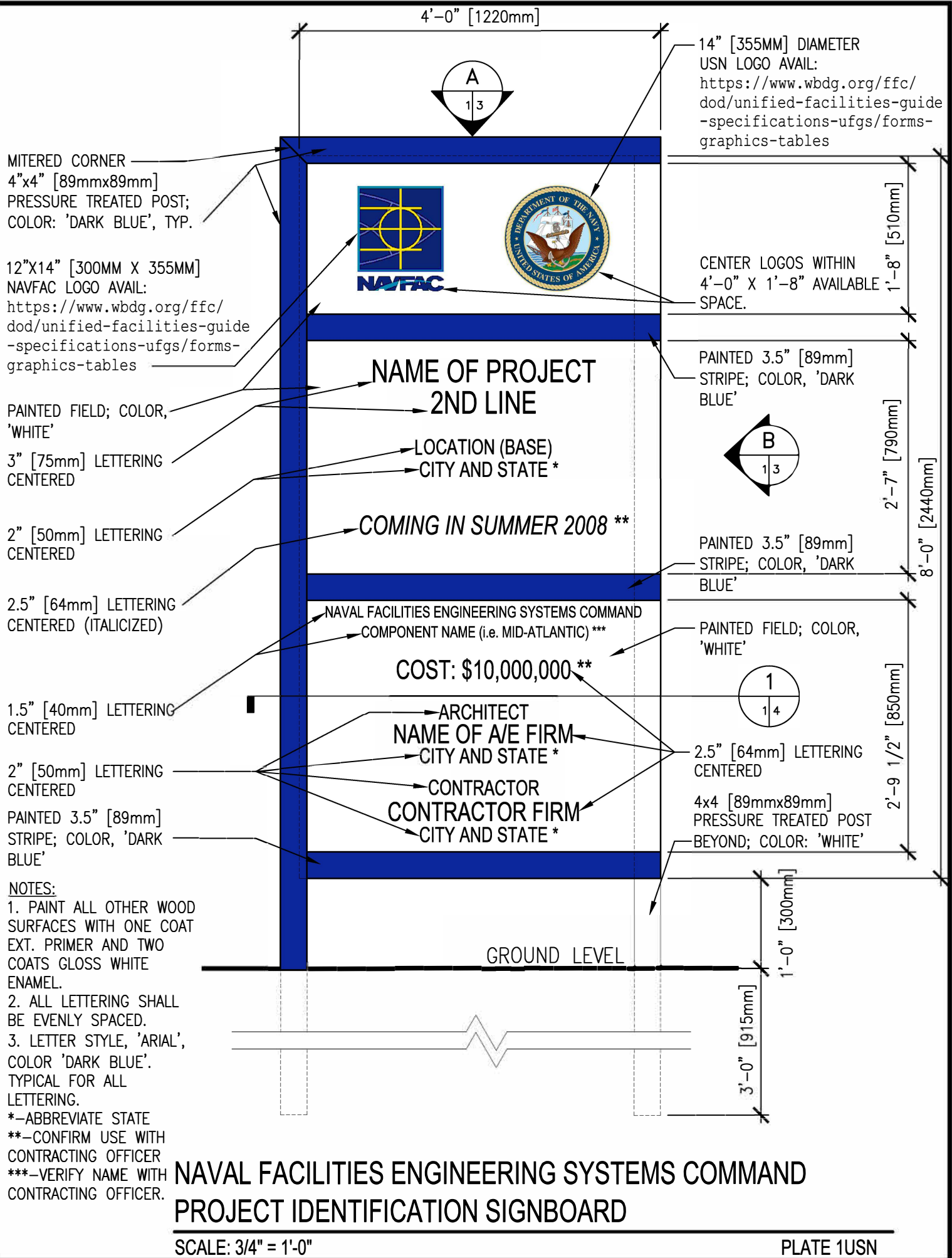
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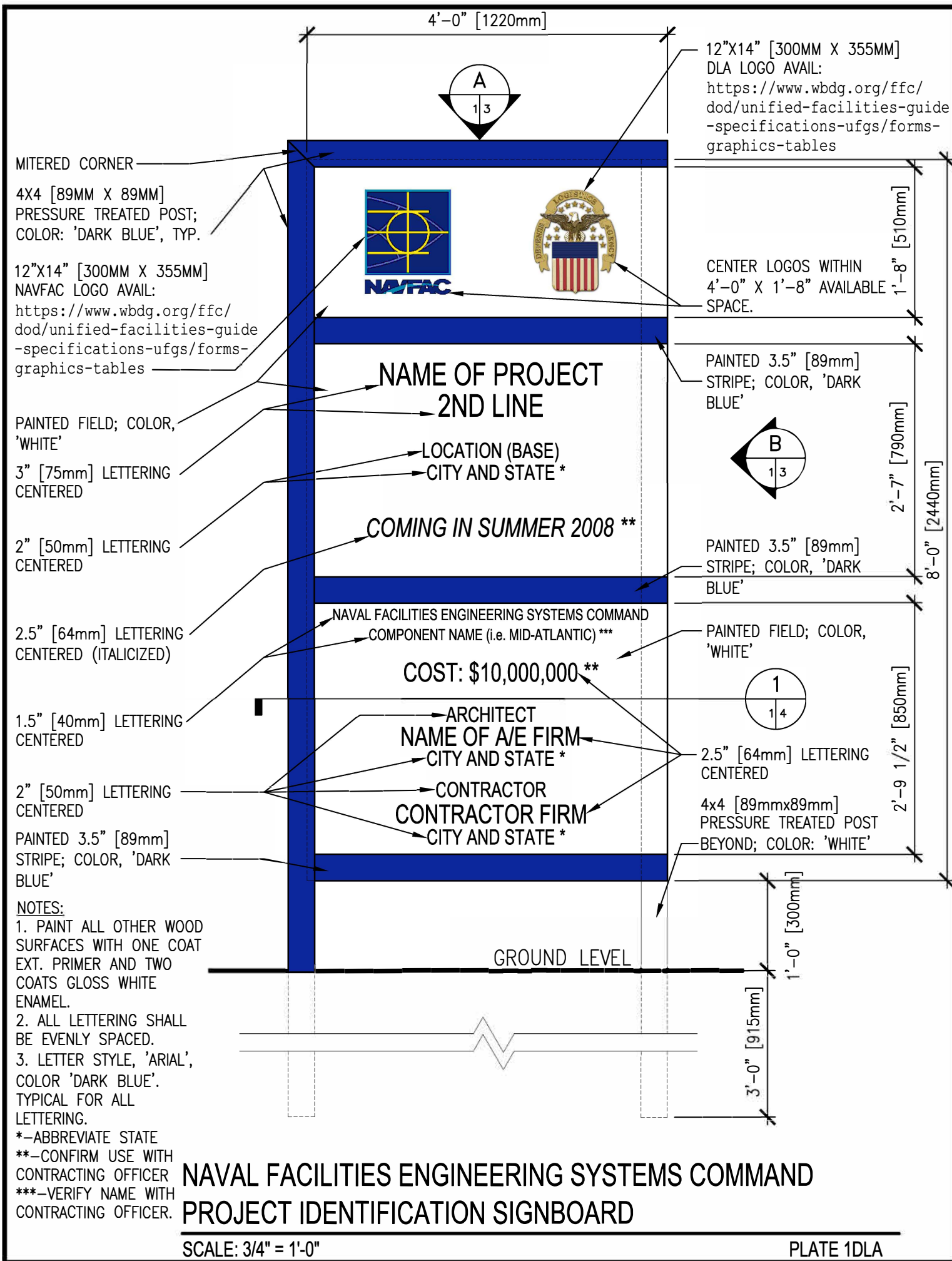
ATTACHMENT A: Example Graphic of Signage with Dimensions for MILCON Project

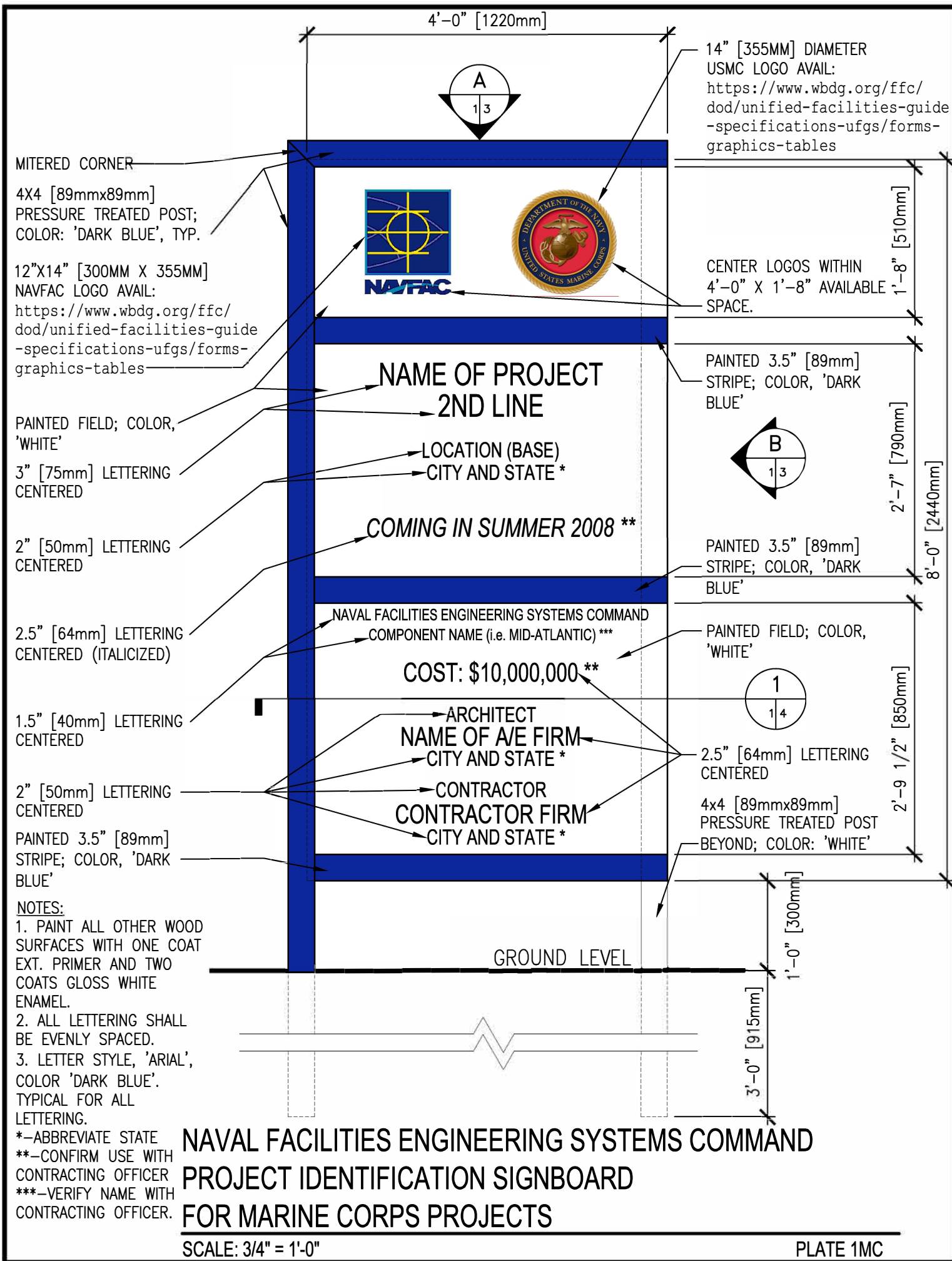


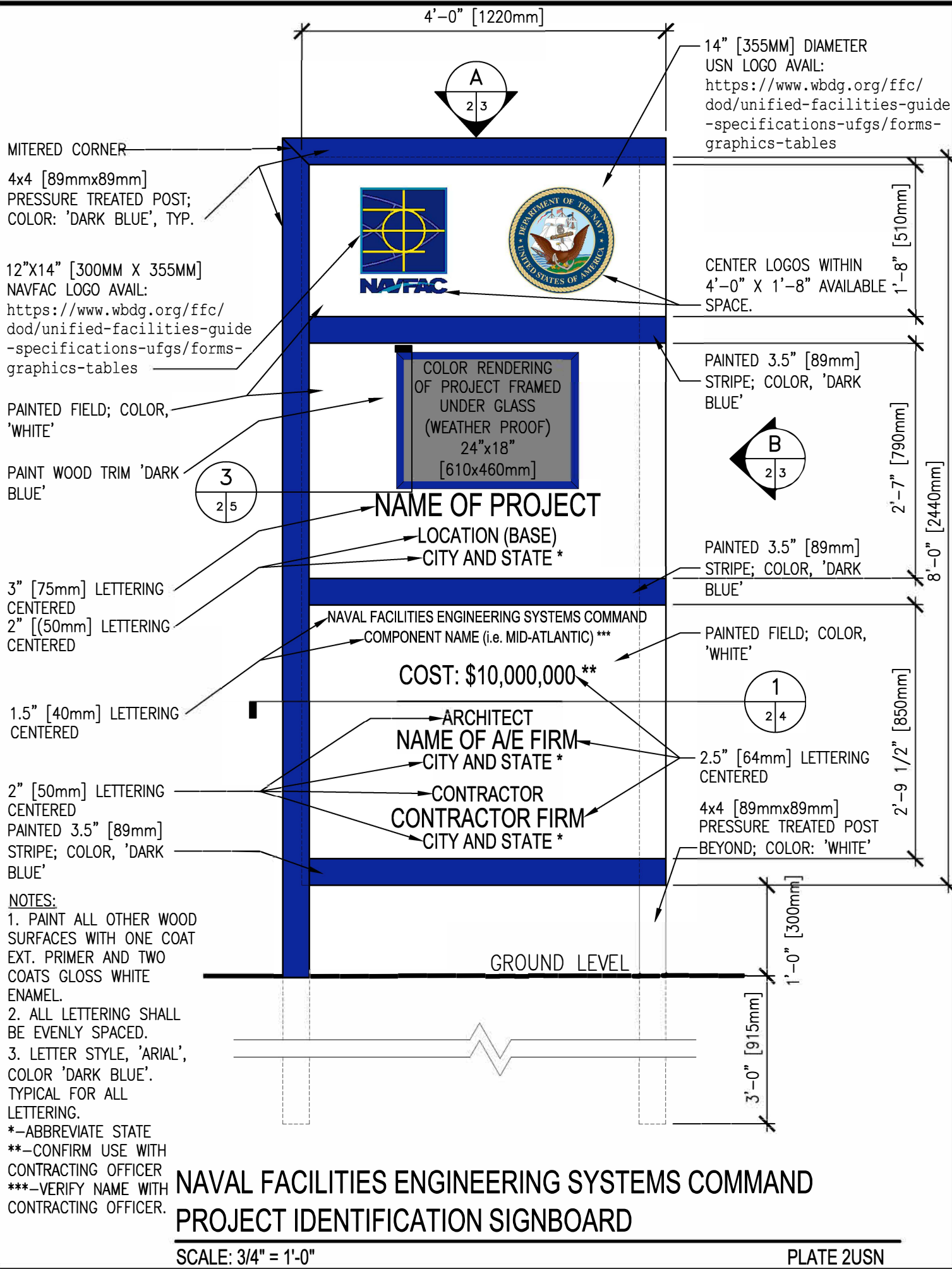
ATTACHMENT B: Example Graphic of Signage with Dimensions for Civil Works Project

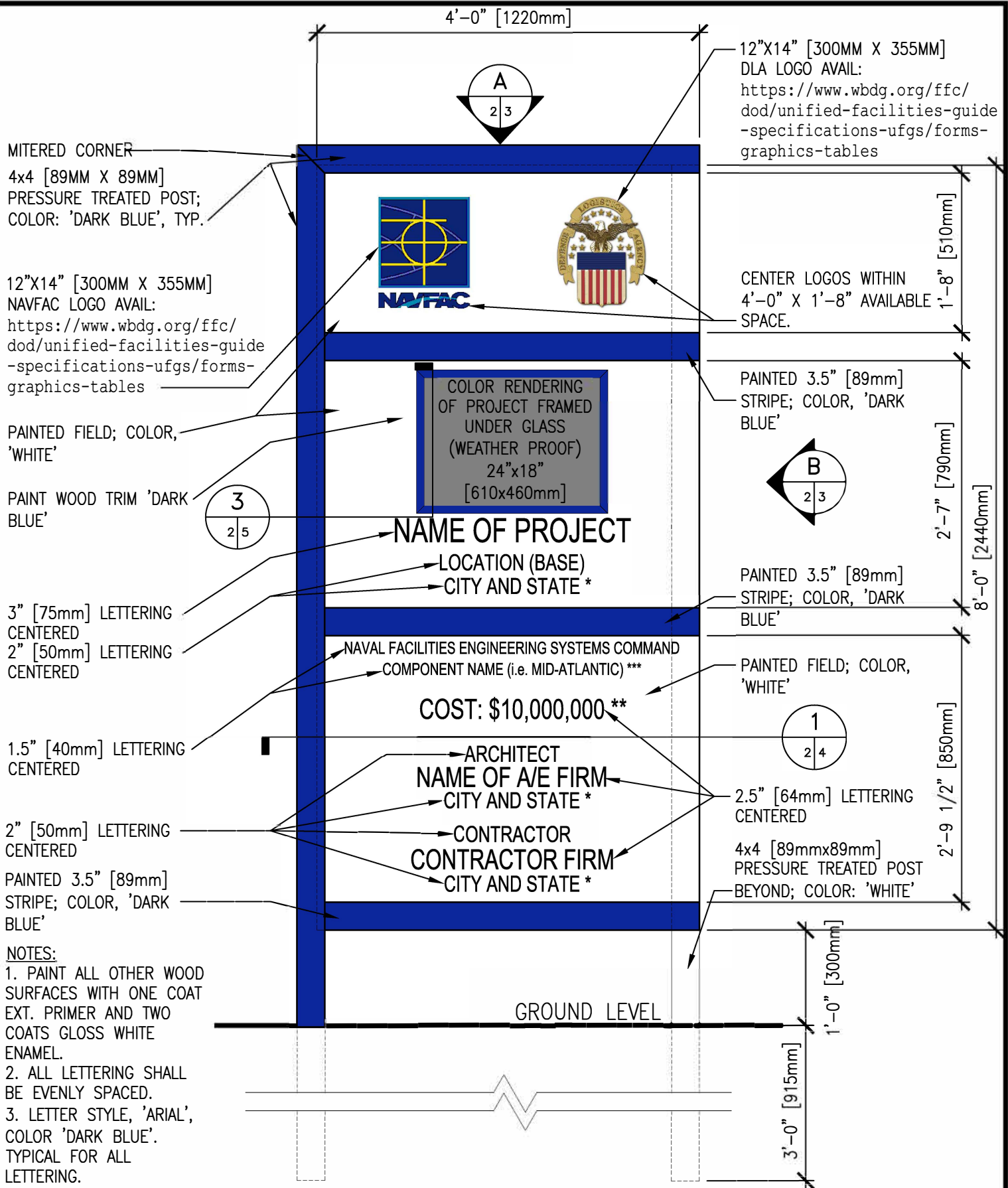








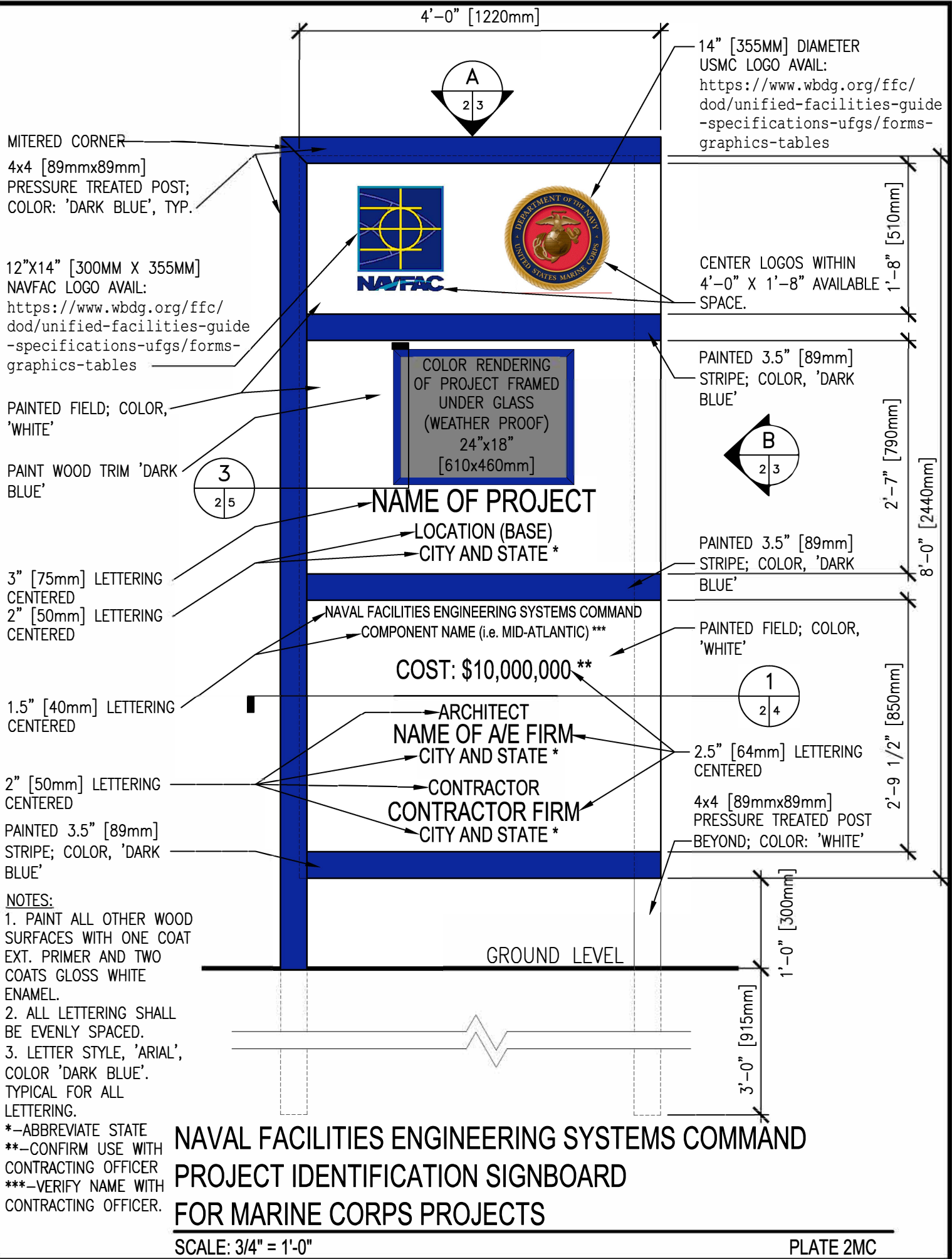


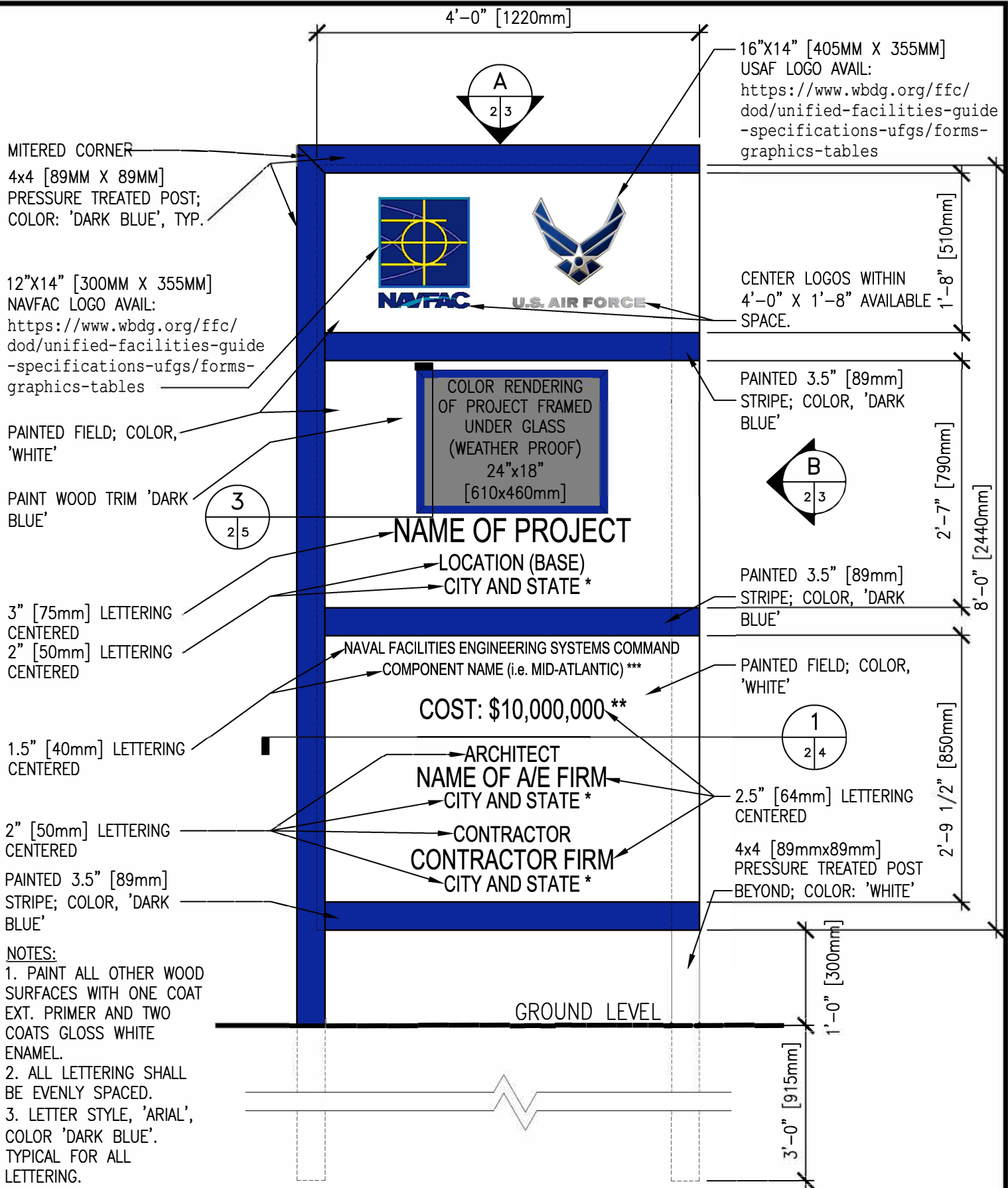


NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND PROJECT IDENTIFICATION SIGNBOARD

SCALE: 3/4" = 1'-0"

PLATE 2DLA

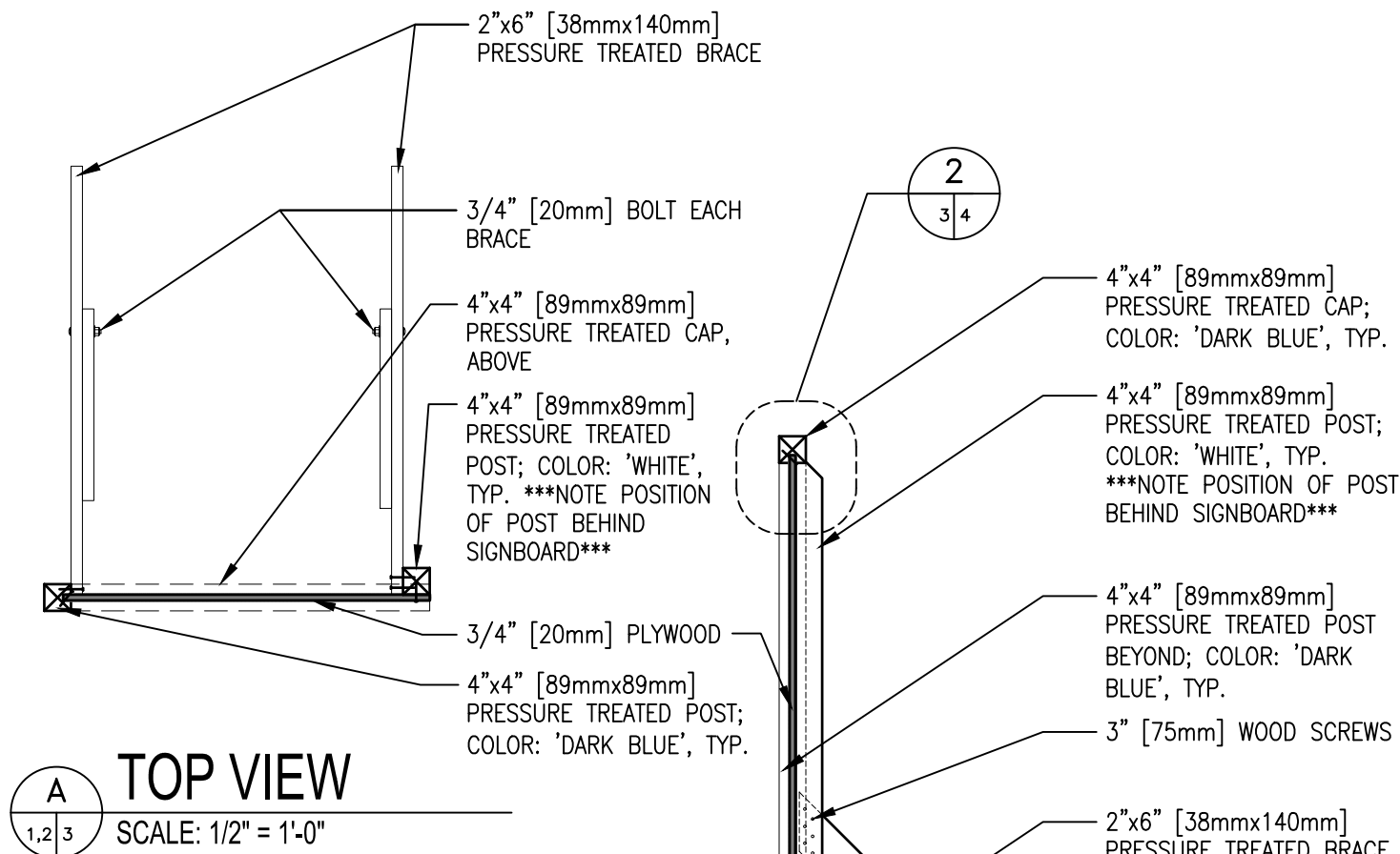




NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND PROJECT IDENTIFICATION SIGNBOARD

SCALE: 3/4" = 1'-0"

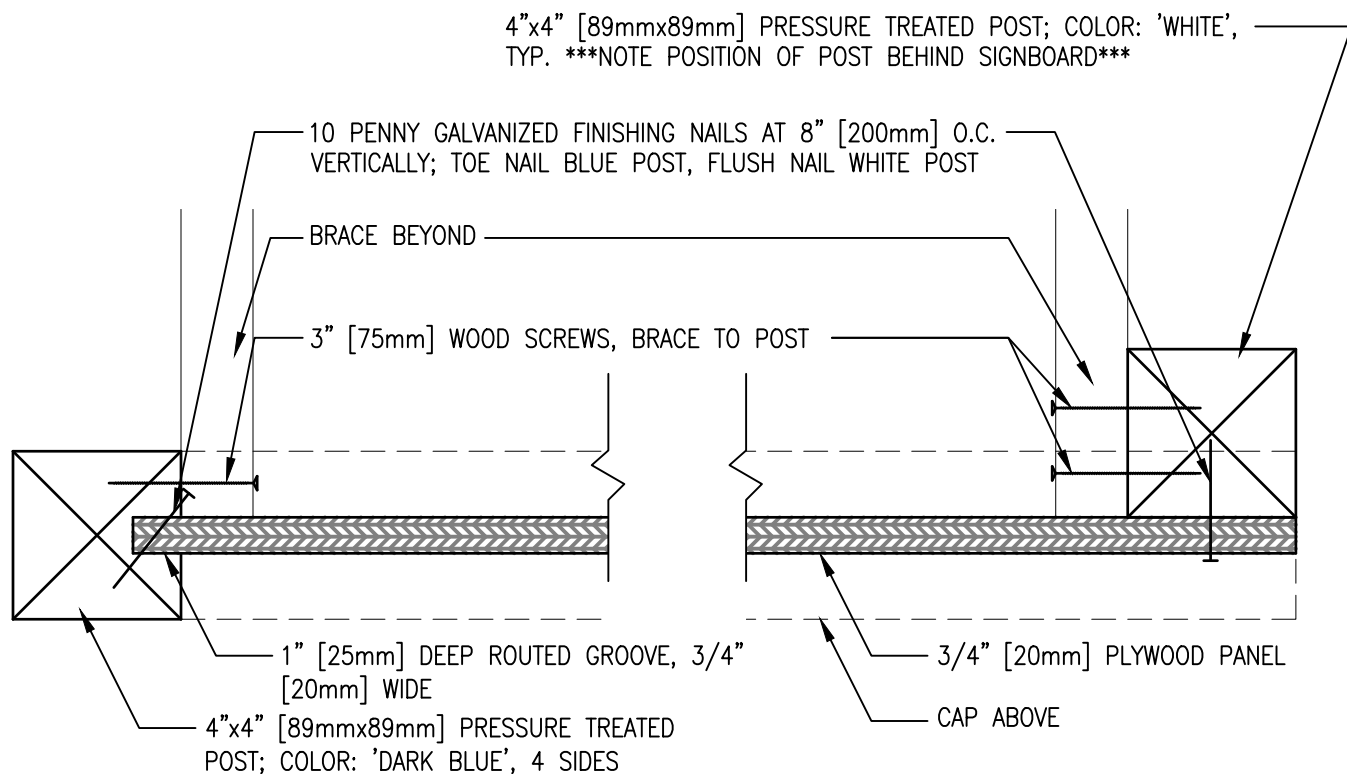
PLATE 2USAF



NOTES:

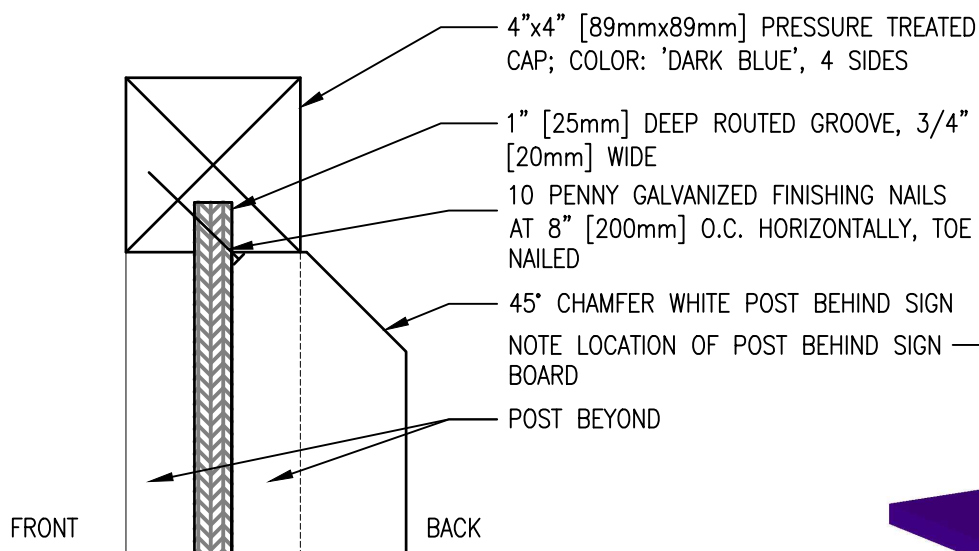
1. POSTS AND BRACES SHALL BE PRESSURE TREATED.
2. ALL FASTENERS SHALL BE ZINC COATED.
3. BRACING IS REQUIRED IN ALL SOIL CONDITIONS AND HIGH WIND ENVIRONMENTS.

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD
SUPPORT DETAILS



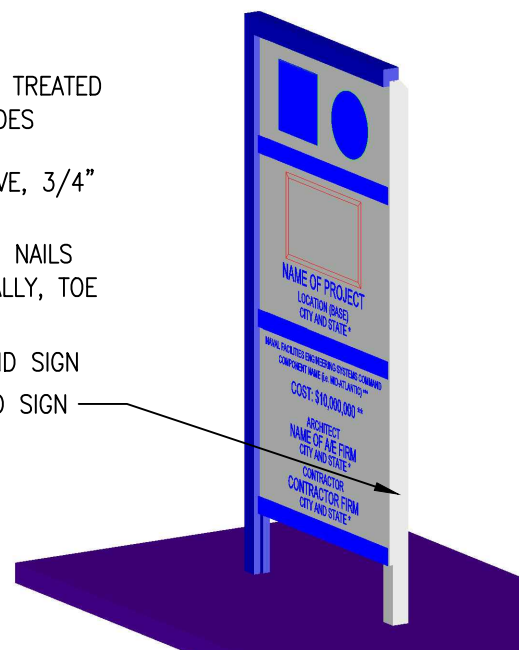
1 PLAN SECTION

SCALE: 3" = 1'-0"



2 SECTION AT TOP

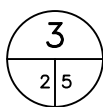
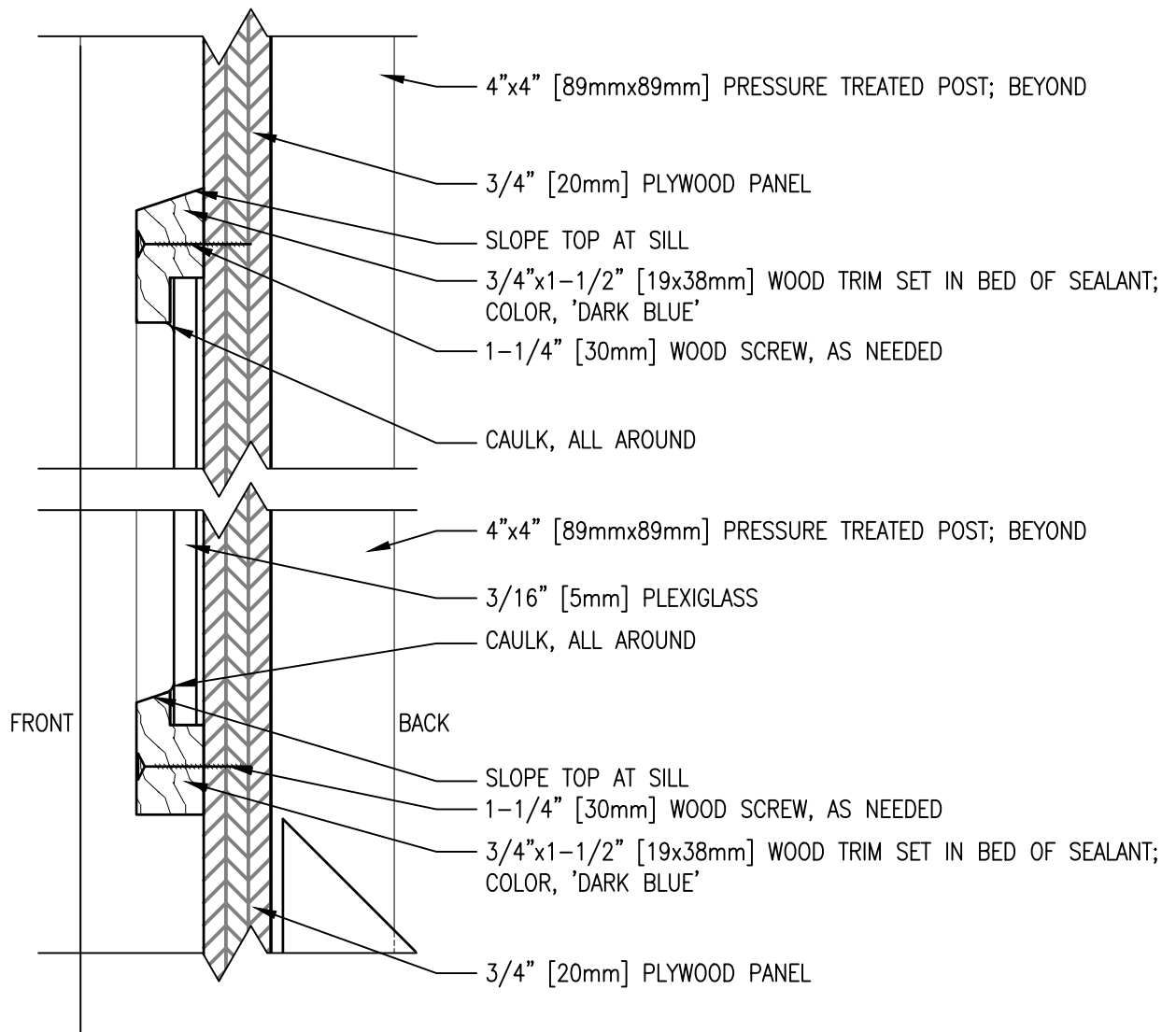
SCALE: 3" = 1'-0"



3 ISO VIEW

SCALE: NONE

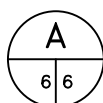
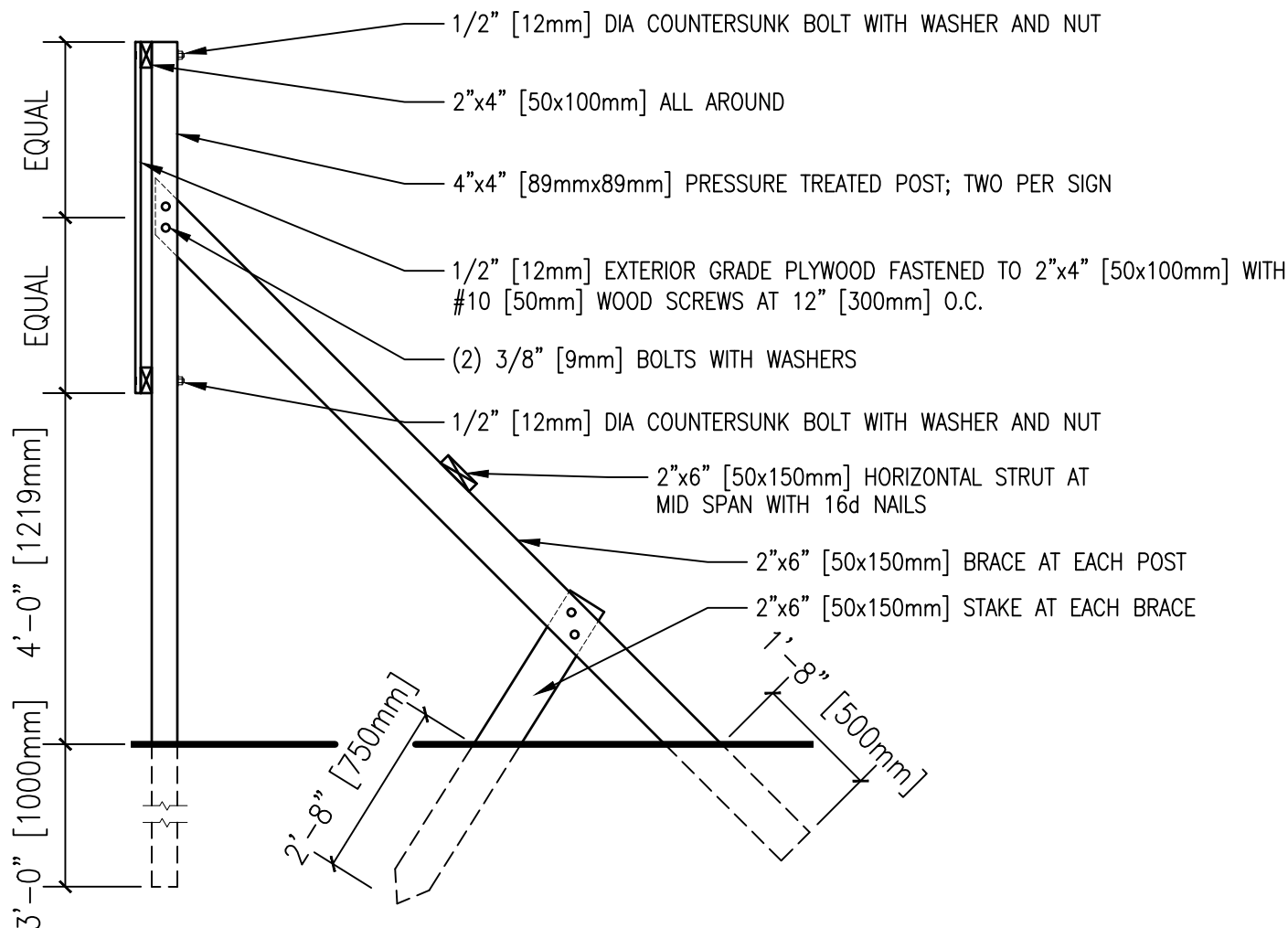
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD SECTIONS



SECTION AT RENDERING FRAME

SCALE: 6" = 1'-0"

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD SECTION



SECTION

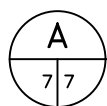
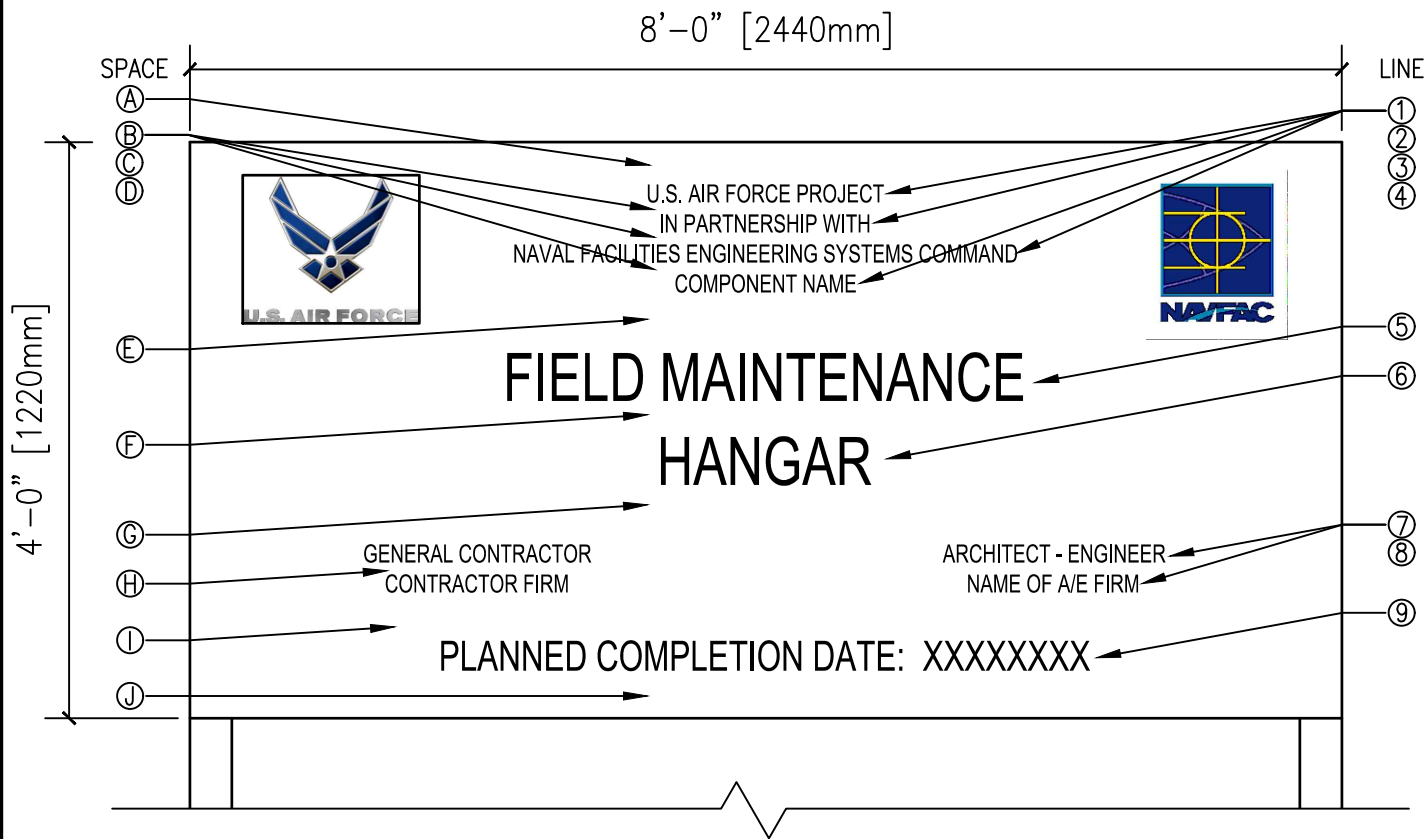
SCALE: 1/2" = 1'-0"

NOTES:

1. THE SIGN MUST BE CONSTRUCTED OF 4'x8' [1200mmx2400mm] 1/2" [12mm] GRADE A-C, EXTERIOR TYPE PLYWOOD MOUNTED ON A FRAME OF TREATED LUMBER.
2. THE HOST MAJCOM WILL PROVIDE SIGN COLORS AND DETAILS TO THE DESIGN MANAGER IN THE REQUIREMENTS AND MANAGEMENT PLAN (RAMP).
3. ALL LETTERING SHALL BE UPPERCASE HELVETICA MEDIUM.
4. PRESERVATIVE TREAT ALL WOOD SURFACES BURIED OR IN CONTACT WITH THE EARTH.
5. ALL FASTENERS SHALL BE GALVANIZED COATED.
6. LUMBER SHALL BE NON STRESS GRADE: $F_t = 8.5^5 \text{ KGS/M}^2 \text{ (1200PSI)}$
 $F_t = 10.5^5 \text{ KGS/M}^2 \text{ (1500PSI)}$

7. THE ENTIRE SIGNBOARD AND SUPPORTS SHALL BE GIVEN ONE COAT OF EXTERIOR ALKYD PRIMER AND TWO COATS OF EXTERIOR ALKYD ENAMEL PAINT. THE LETTERING AND SIGN WORK SHALL BE PERFORMED BY A SKILLED SIGN PAINTER USING PAINT KNOWN IN THE TRADE AS BULLETIN COLORS. WHERE PRESERVATIVE-TREATED LUMBER IS REQUIRED, UTILIZE ONLY CURED PRESSURE-TREATED WOOD WHICH HAS HAD THE CHEMICALS LEACHED FROM THE SURFACE OF THE WOOD PRIOR TO PAINTING.

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD SECTION
U.S. AIR FORCE PROJECTS



SIGNBOARD

SCALE: 3/4" = 1'-0"

SPACE	HEIGHT	LINE	DESCRIPTION	LETTER HEIGHT	STROKE
A	3.5"	①	U.S. AIR FORCE PROJECT	1.5"	3/16"
B	1"	②	IN PARTNERSHIP WITH	1.5"	3/16"
C	1"	③	NAVAL FACILITIES ENGINEERING COMMAND	1.5"	3/16"
D	1"	④	COMPONENT NAME	1.5"	3/16"
E	5"	⑤	PROJECT NAME	4"	1/2"
F	3"	⑥	PROJECT NAME CONT'D (IF REQUIRED)	4"	1/2"
G	5"	⑦	GENERAL CONTRACTOR/ A-E	1.5"	3/16"
H	1"	⑧	GENERAL CONTRACTOR/ A-E	1.5"	3/16"
I	4"	⑨	PLANNED COMPLETION DATE:	2.5"	1/4"
J	4"				

NOTES:

1. ALL LETTERING SHALL BE UPPERCASE HELVETICA MEDIUM.

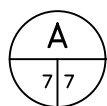
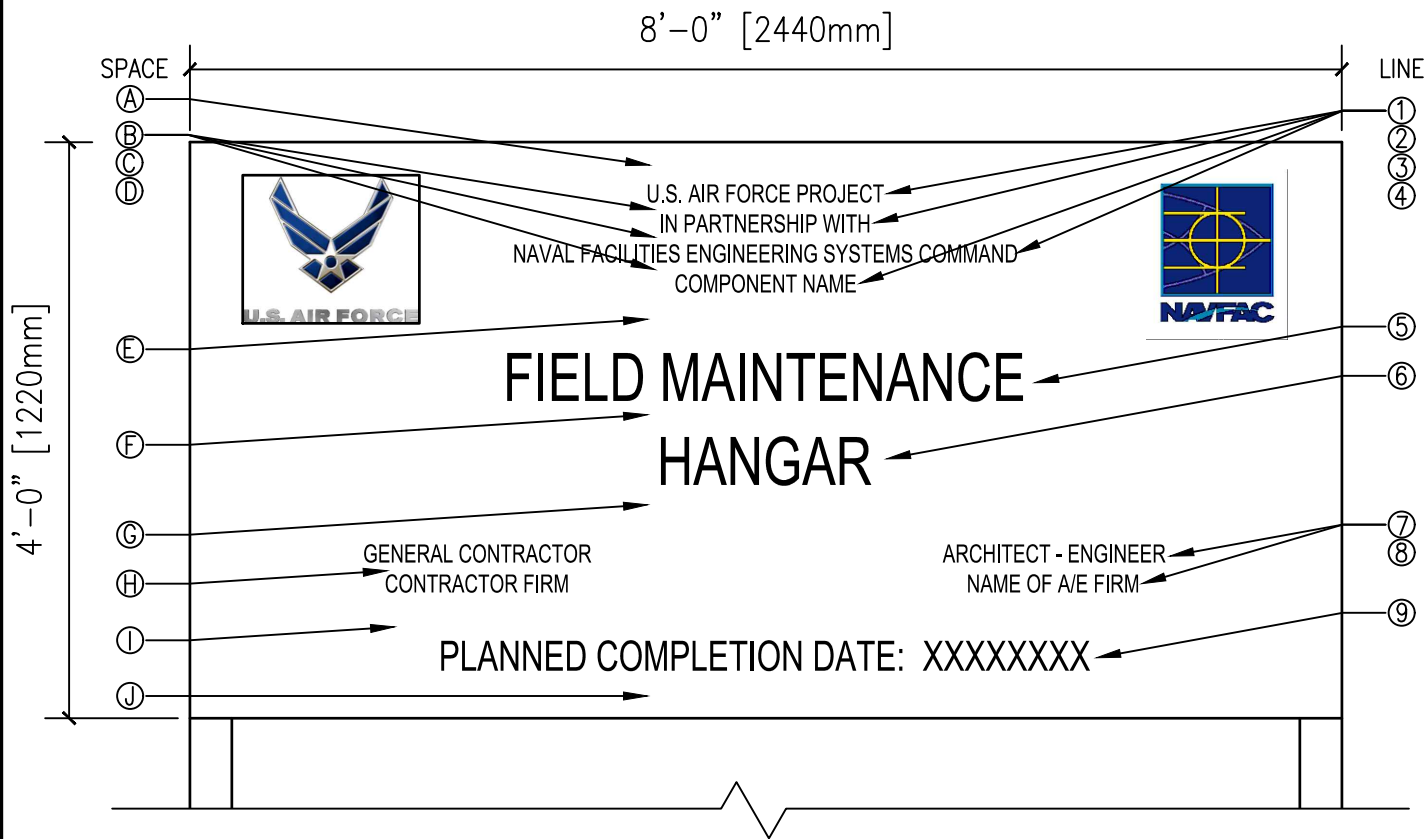
2. NAVFAC LOGO AVAILABLE AT:

<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>

3. AIR FORCE LOGO AVAILABLE AT:

<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD
U.S. AIR FORCE PROJECTS



SIGNBOARD

SCALE: 3/4" = 1'-0"

SPACE	HEIGHT	LINE	DESCRIPTION	LETTER HEIGHT	STROKE
①	3.5"	①	U.S. AIR FORCE PROJECT	1.5"	3/16"
②	1"	②	IN PARTNERSHIP WITH	1.5"	3/16"
③	1"	③	NAVAL FACILITIES ENGINEERING COMMAND	1.5"	3/16"
④	1"	④	COMPONENT NAME	1.5"	3/16"
⑤	5"	⑤	PROJECT NAME	4"	1/2"
⑥	3"	⑥	PROJECT NAME CONT'D (IF REQUIRED)	4"	1/2"
⑦	5"	⑦	GENERAL CONTRACTOR/ A-E	1.5"	3/16"
⑧	1"	⑧	GENERAL CONTRACTOR/ A-E	1.5"	3/16"
⑨	4"	⑨	PLANNED COMPLETION DATE:	2.5"	1/4"
⑩	4"				

NOTES:

1. ALL LETTERING SHALL BE UPPERCASE HELVETICA MEDIUM.

2. NAVFAC LOGO AVAILABLE AT:

<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>

3. AIR FORCE LOGO AVAILABLE AT:

<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD
U.S. AIR FORCE PROJECTS

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

02/19, CHG 3: 11/21

PART 1 GENERAL

- 1.1 DEFINITIONS
 - 1.1.1 Co-mingle
 - 1.1.2 Construction Waste
 - 1.1.3 Demolition Debris/Waste
 - 1.1.4 Disposal
 - 1.1.5 Diversion
 - 1.1.6 Final Construction Waste Diversion Report
 - 1.1.7 Recycling
 - 1.1.8 Reuse
 - 1.1.9 Salvage
 - 1.1.10 Source Separation
- 1.2 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)
- 1.3 CONSTRUCTION WASTE MANAGEMENT
 - 1.3.1 Implementation of Construction Waste Management Program
 - 1.3.2 Oversight
 - 1.3.3 Special Programs
 - 1.3.4 Special Instructions
 - 1.3.5 Waste Streams
- 1.4 SUBMITTALS
- 1.5 MEETINGS
- 1.6 CONSTRUCTION WASTE MANAGEMENT PLAN
- 1.7 RECORDS (DOCUMENTATION)
 - 1.7.1 General
 - 1.7.2 Accumulated
- 1.8 FINAL CONSTRUCTION WASTE DIVERSION REPORT
- 1.9 COLLECTION
 - 1.9.1 Source Separation Method
 - 1.9.2 Co-Mingled Method
 - 1.9.3 Other Methods
- 1.10 DISPOSAL
 - 1.10.1 Reuse
 - 1.10.2 Recycle
 - 1.10.3 Waste

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
02/19, CHG 3: 11/21

PART 1 GENERAL

1.1 DEFINITIONS

1.1.1 Co-mingle

The practice of placing unrelated materials together in a single container, usually for benefits of convenience and speed.

1.1.2 Construction Waste

Waste generated by construction activities, such as scrap materials, damaged or spoiled materials, temporary and expendable construction materials, and other waste generated by the workforce during construction activities.

1.1.3 Demolition Debris/Waste

Waste generated from demolition activities, including minor incidental demolition waste materials generated as a result of Intentional dismantling of all or portions of a building, to include clearing of building contents that have been destroyed or damaged.

1.1.4 Disposal

Depositing waste in a solid waste disposal facility, usually a managed landfill or incinerator, regulated in the US under the Resource Conservation and Recovery Act (RCRA).

1.1.5 Diversion

The practice of diverting waste from disposal in a landfill or incinerator, by means of eliminating or minimizing waste, or reuse of materials.

1.1.6 Final Construction Waste Diversion Report

A written assertion by a material recovery facility operator identifying constituent materials diverted from disposal, usually including summary tabulations of materials, weight in short-ton.

1.1.7 Recycling

The series of activities, including collection, separation, and processing, by which products or other materials are diverted from the solid waste stream for use in the form of raw materials in the manufacture of new products sold or distributed in commerce, or the reuse of such materials as substitutes for goods made of virgin materials, other than fuel.

1.1.8 Reuse

The use of a product or materials again for the same purpose, in its original form or with little enhancement or change.

1.1.9 Salvage

Usable, salable items derived from buildings undergoing demolition or deconstruction, parts from vehicles, machinery, other equipment, or other components.

1.1.10 Source Separation

The practice of administering and implementing a management strategy to identify and segregate unrelated waste at the first opportunity.

1.2 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)

Divert a minimum of 60 percent by weight of the project construction waste and demolition debris/waste from the landfill or incinerator. Follow applicable industry standards in the management of waste. Apply sound environmental principles in the management of waste. (1) Practice efficient waste management when sizing, cutting, and installing products and materials and (2) use all reasonable means to divert construction waste and demolition debris/waste from landfills and incinerators and to facilitate the recycling or reuse of excess construction materials.

1.3 CONSTRUCTION WASTE MANAGEMENT

Implement a Construction Waste Management Program for the project. Take a pro-active, responsible role in the management of construction construction waste, recycling process, disposal of demolition debris/waste, and require all subcontractors, vendors, and suppliers to participate in the Construction Waste Management Program. Establish a process for clear tracking, and documentation of construction waste and demolition debris/waste.

1.3.1 Implementation of Construction Waste Management Program

Develop and document how the Construction Waste Management Program will be implemented in a Construction Waste Management Plan. Submit a Construction Waste Management Plan to the Contracting Officer for approval. Construction waste and demolition debris/waste materials include un-used construction materials not incorporated in the final work, as well as demolition debris/waste materials from demolition activities or deconstruction activities. In the management of waste, consider the availability of viable markets, the condition of materials, the ability to provide material in suitable condition and in a quantity acceptable to available markets, and time constraints imposed by internal project completion mandates.

1.3.2 Oversight

The Environmental Manager, as specified in Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS, is responsible for overseeing and documenting results from executing the Construction Waste Management Plan for the project.

1.3.3 Special Programs

Implement special programs involving rebates or similar incentives related to recycling of construction waste and demolition debris/waste materials. Retain revenue or savings from salvaged or recycling, unless otherwise directed. Ensure firms and facilities used for recycling, reuse, and disposal are permitted for the intended use to the extent required by federal, state, and local regulations.

1.3.4 Special Instructions

Provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the projects. Designation of single source separating or commingling will be clearly marked on the containers.

1.3.5 Waste Streams

Delineate waste streams and characterization, including estimated material types and quantities of waste, in the Construction Waste Management Plan. Manage all waste streams associated with the project. Typical waste streams are listed below. Include additional waste streams not listed:

- a. Land Clearing Debris
- b. Asphalt
- c. Concrete
- d. Metals (Includes, but is not limited to, banding, stud trim, ductwork, piping, rebar, roofing, other trim, steel, iron, galvanized, stainless steel, aluminum, copper, zinc, bronze.)
- e. Paper
- f. Plastics (PET, HDPE, PVC, LDPE, PP, PS, Other)
- g. Ceiling Tiles
- h. Insulation
- i. Beverage Containers

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Waste Management Plan; G

SD-11 Closeout Submittals

Final Construction Waste Diversion Report; S

1.5 MEETINGS

Conduct Construction Waste Management meetings. After award of the Contract and prior to commencement of work, schedule and conduct a meeting with the Contracting Officer to discuss the proposed Construction Waste Management Plan and to develop a mutual understanding relative to the management of the Construction Waste Management Program and how waste diversion requirements will be met.

The requirements of this meeting may be fulfilled during the coordination and mutual Understanding meeting outlined in Section 01 45 00 QUALITY CONTROL. At a minimum, discuss and document waste management goals at following meetings:

- a. Preconstruction meeting.
- b. Regular Quality Control meetings.
- c. Work safety meeting (if applicable).

1.6 CONSTRUCTION WASTE MANAGEMENT PLAN

Submit Construction Waste Management Plan within 45 calendar days after contract award. Revise and resubmit Construction Waste Management Plan as necessary, in order for construction to begin. Execute demolition or deconstruction activities in accordance with Section 02 41 00 DEMOLITION AND DECONSTRUCTION. Manage demolition debris/waste or deconstruction materials in accordance with the approved construction waste management plan.

An approved Construction Waste Management Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations or meeting project cumulative waste diversion requirement. Ensure all subcontractors receive a copy of the approved Construction Waste Management Plan. The plan demonstrates how to meet the project waste diversion requirement. Also, include the following in the plan:

- a. Identify the names of individuals responsible for waste management and waste management tracking, along with roles and responsibilities on the project.
- b. Actions that will be taken to reduce solid waste generation, including coordination with subcontractors to ensure awareness and participation.
- c. Description of the regular meetings to be held to address waste management.
- d. Description of the specific approaches to be used in recycling/reuse of the various materials generated, including the areas on site and equipment to be used for processing, sorting, and temporary storage of materials.
- e. Name of landfill and incinerator to be used.
- f. Identification of local and regional re-use programs, including non-profit organizations such as schools, local housing agencies, and organization that accept used materials such as material exchange networks and resale stores. Include the name, location, phone number for each re-use facility identified, and provide a copy of the permit or license for each facility.

- g. List of specific materials, by type and quantity, that will be salvaged for resale, salvaged and reused on the current project, salvaged and stored for reuse on a future project, or recycled. Identify the recycling facilities by name, address, and phone number.
- h. Identification of materials that cannot be recycled or reused with an explanation or justification, to be approved by the Contracting Officer.
- i. Description of the means by which materials identified in item (g) above will be protected from contamination.
- j. Description of the means of transportation of the recyclable materials (whether materials will be site-separated and self-hauled to designated centers, or whether mixed materials will be collected by a waste hauler and removed from the site).
- k. Copy of training plan for subcontractors and other services to prevent contamination by co-mingling materials identified for diversion and waste materials.
- n. Identify local jurisdiction requirements for waste management. Include local requirements and points of contact.

Distribute copies of the waste management plan to each subcontractor, [Quality Control Manager](#) [Environmental Manager](#), and the Contracting Officer.

1.7 RECORDS (DOCUMENTATION)

1.7.1 General

Maintain records to document the types and quantities of waste generated and diverted through re-use, recycling and sale to third parties; through disposal to a landfill or incinerator facility. Provide explanations for materials not recycled, reused or sold. Collect and retain manifests, weight tickets, sales receipts, and invoices specifically identifying diverted project waste materials or disposed materials.

1.7.2 Accumulated

Maintain a running record of materials generated and diverted from landfill disposal, including accumulated diversion rates for the project. Make records available to the Contracting Officer during construction or incidental demolition activities. Provide a copy of the diversion records to the Contracting Officer upon completion of the construction, incidental demolitions or minor deconstruction activities.

1.8 FINAL CONSTRUCTION WASTE DIVERSION REPORT

A Final Construction Waste Diversion Report is required at the end of the project. Provide [Final Construction Waste Diversion Report](#) 60 days prior to the Beneficial Occupancy Date (BOD).

1.9 COLLECTION

Collect, store, protect, and handle reusable and recyclable materials at the site in a manner which prevents contamination, and provides protection from the elements to preserve their usefulness and monetary value. Provide

receptacles and storage areas designated specifically for recyclable and reusable materials and label them clearly and appropriately to prevent contamination from other waste materials. Keep receptacles or storage areas neat and clean.

Train subcontractors and other service providers to either separate waste streams or use the co-mingling method as described in the Construction Waste Management Plan. Handle hazardous waste and hazardous materials in accordance with applicable regulations and coordinate with Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS and Section 02 81 00 TRANSPORTATION AND DISPOSAL OF HAZARDOUS MATERIALS. Separate materials by one of the following methods described herein:

1.9.1 Source Separation Method

Separate waste products and materials that are recyclable from trash and sort as described below into appropriately marked separate containers and then transport to the respective recycling facility for further processing. Deliver materials in accordance with recycling or reuse facility requirements (e.g., free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process). Separate materials into the category types as defined in the Construction Waste Management Plan.

1.9.2 Co-Mingled Method

Place waste products and recyclable materials into a single container and then transport to an authorized recycling facility, which meets all applicable requirements to accept and dispose of recyclable materials in accordance with all applicable local, state and federal regulations. The Co-mingled materials must be sorted and processed in accordance with the approved Construction Waste Management Plan.

1.9.3 Other Methods

Other methods proposed by the Contractor may be used when approved by the Contracting Officer.

1.10 DISPOSAL

Control accumulation of waste materials and trash. Recycle or dispose of collected materials off-site at intervals approved by the Contracting Officer and in compliance with waste management procedures as described in the waste management plan. Except as otherwise specified in other sections of the specifications, dispose of in accordance with the following:

1.10.1 Reuse

Give first consideration to reusing construction and demolition materials as a disposition strategy. Recover for reuse materials, products, and components as described in the approved Construction Waste Management Plan. Coordinate with the Contracting Officer to identify onsite reuse opportunities or material sales or donation available through Government resale or donation programs. Sale of recovered materials is not allowed on the Installation. Consider the use of surplus industrial supply broker services, who match entities with reusable or repurpose industrial materials with entities with need of such materials.

1.10.2 Recycle

Recycle non-hazardous construction and demolition/debris materials that are not suitable for reuse. Track rejection of contaminated recyclable materials by the recycling facility. Rejected recyclables materials will not be counted as a percentage of diversion calculation. Recycle all fluorescent lamps, HID lamps, mercury (Hg) -containing thermostats and ampoules, and PCBs-containing ballasts and electrical components as directed by the Contracting Officer. Do not crush lamps on site as this creates a hazardous waste stream with additional handling requirements.

1.10.3 Waste

Dispose by landfill or incineration only those waste materials with no practical use, economic benefit, or recycling opportunity.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 78 00

CLOSEOUT SUBMITTALS

05/19, CHG 1: 08/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 As-Built Drawings
- 1.3 SUBMITTALS
- 1.4 SPARE PARTS DATA
- 1.5 WARRANTY MANAGEMENT
 - 1.5.1 Warranty Management Plan
 - 1.5.2 Performance Bond
 - 1.5.3 Pre-Warranty Conference
 - 1.5.4 Warranty Tags

PART 2 PRODUCTS

- 2.1 CERTIFICATION OF EPA DESIGNATED ITEMS

PART 3 EXECUTION

- 3.1 AS-BUILT DRAWINGS
 - 3.1.1 Markup Guidelines
 - 3.1.2 As-Built Drawings Content
- 3.2 CLEANUP
- 3.3 REAL PROPERTY RECORD
 - 3.3.1 Interim DD FORM 1354
 - 3.3.2 Completed DD FORM 1354

-- End of Section Table of Contents --

SECTION 01 78 00

CLOSEOUT SUBMITTALS
05/19, CHG 1: 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1971 (2005; R 2011) Standard Guide for Stewardship for the Cleaning of Commercial and Institutional Buildings

GREEN SEAL (GS)

GS-37 (2017) Cleaning Products for Industrial and Institutional Use

U.S. DEPARTMENT OF DEFENSE (DOD)

FC 1-300-09N (2014; with Change 6, 2021) Navy and Marine Corps Design

UFC 1-300-08 (2023) Criteria for Transfer and Acceptance of DoD Real Property

1.2 DEFINITIONS

1.2.1 As-Built Drawings

As-built drawings are the marked-up drawings, maintained by the Contractor on-site, that depict actual conditions and deviations from the Contract Documents. These deviations and additions may result from coordination required by, but not limited to: contract modifications; official responses to submitted Requests for Information (RFI's); direction from the Contracting Officer; design that is the responsibility of the Contractor, and differing site conditions. Maintain the as-builts throughout construction as red-lined hard copies on site. These files serve as the basis for the creation of the record drawings.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Warranty Management Plan

Warranty Tags

Final Cleaning

Spare Parts Data

SD-08 Manufacturer's Instructions

Posted Instructions

SD-11 Closeout Submittals

As-Built Drawings; G

As-Built Record of Equipment and Materials

Certification of EPA Designated Items; G

Interim DD FORM 1354; G

Checklist for DD FORM 1354; G

1.4 SPARE PARTS DATA

Submit two copies of the Spare Parts Data list.

- a. Indicate manufacturer's name, part number, and stock level required for test and balance, pre-commissioning, maintenance and repair activities. List those items that may be standard to the normal maintenance of the system.

1.5 WARRANTY MANAGEMENT

1.5.1 Warranty Management Plan

Develop a warranty management plan which contains information relevant to FAR 52.246-21 Warranty of Construction. At least 30 days before the planned pre-warranty conference, submit one set of the warranty management plan. Include within the warranty management plan all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan narrative must contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesmen, or of engineering background, not necessarily familiar with this contract. The term "status" as indicated below must include due date and whether item has been submitted or was accomplished. Submit warranty information, made available during the construction phase, to the Contracting Officer for approval prior to each monthly pay estimate. Assemble approved information in a binder and turn over to the Government upon acceptance of the work. The construction warranty period must begin on the date of project acceptance and continue for the full product warranty period. Conduct a joint 4 month and 9 month warranty inspection, measured from time of acceptance; with the Contractor, Contracting Officer and the Customer Representative. The warranty management plan must include, but is not limited to, the following:

- a. Roles and responsibilities of personnel associated with the warranty process, including points of contact and telephone numbers within the organizations of the Contractors, subcontractors, manufacturers or suppliers involved.

- b. For each warranty, the name, address, telephone number, and e-mail of each of the guarantor's representatives nearest to the project location.
- c. A list and status of delivery of Certificates of Warranty for extended warranty items, including roofs, HVAC balancing, pumps, motors, transformers, and for commissioned systems, such as fire protection and alarm systems, sprinkler systems, and lightning protection systems.
- d. [As-Built Record of Equipment and Materials](#) list for each warranted equipment, item, feature of construction or system indicating:
 - (1) Name of item.
 - (2) Model and serial numbers.
 - (3) Location where installed.
 - (4) Name and phone numbers of manufacturers or suppliers.
 - (5) Names, addresses and telephone numbers of sources of spare parts.
 - (6) Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have warranties longer than one year must be indicated with separate warranty expiration dates.
 - (7) Cross-reference to warranty certificates as applicable.
 - (8) Starting point and duration of warranty period.
 - (9) Summary of maintenance procedures required to continue the warranty in force.
 - (10) Cross-reference to specific pertinent Operation and Maintenance manuals.
 - (11) Organization, names and phone numbers of persons to call for warranty service.
 - (12) Typical response time and repair time expected for various warranted equipment.
- e. The plans for attendance at the 4 and 9 month post-construction warranty inspections conducted by the Government.
- f. Procedure and status of tagging of equipment covered by warranties longer than one year.
- g. Copies of [instructions](#) to be posted near selected pieces of equipment where operation is critical for warranty or safety reasons.

1.5.2 Performance Bond

The Performance Bond must remain effective throughout the construction and warranty period.

- a. In the event the Contractor fails to commence and diligently pursue any construction warranty work required, the Contracting Officer will have the work performed by others, and after completion of the work, will charge the remaining construction warranty funds of expenses incurred by the Government while performing the work, including, but not limited to administrative expenses.
- b. In the event sufficient funds are not available to cover the construction warranty work performed by the Government at the Contractor's expense, the Contracting Officer will have the right to recoup expenses from the bonding company.

- c. Following oral or written notification of required construction warranty repair work, respond in a timely manner. Written verification will follow oral instructions. Failure to respond will be cause for the Contracting Officer to proceed against the Contractor.

1.5.3 Pre-Warranty Conference

Prior to contract completion, and at a time designated by the Contracting Officer, meet with the Contracting Officer to develop a mutual understanding with respect to the requirements of this section. At this meeting, establish and review communication procedures for Contractor notification of construction warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the Contracting Officer for the execution of the construction warranty. In connection with these requirements and at the time of the Contractor's quality control completion inspection, furnish the name, telephone number and address of a licensed and bonded company which is authorized to initiate and pursue construction warranty work action on behalf of the Contractor. This point of contact must be located within the local service area of the warranted construction, be continuously available, and be responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of its responsibilities in connection with other portions of this provision.

1.5.4 Warranty Tags

At the time of installation, tag each warranted item with a durable, oil and water resistant tag approved by the Contracting Officer. Attach each tag with a copper wire and spray with a silicone waterproof coating. Also, submit two record copies of the warranty tags showing the layout and design. The date of acceptance and the QC signature must remain blank until the project is accepted for beneficial occupancy. Show the following information on the tag.

Type of product/material	
Model number	
Serial number	
Contract number	
Warranty period from/to	
Inspector's signature	
Construction Contractor	
Address	
Telephone number	

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Warranty contact	
Address	
Telephone number	
Warranty response time priority code	
WARNING - PROJECT PERSONNEL TO PERFORM ONLY OPERATIONAL MAINTENANCE DURING THE WARRANTY PERIOD.	

PART 2 PRODUCTS

2.1 CERTIFICATION OF EPA DESIGNATED ITEMS

Submit the [Certification of EPA Designated Items](#) as required by FAR 52.223-9 Estimate of Percentage of Recovered Material Content for EPA Designated Items and FAR 52-223-17 Affirmative Procurement of EPA designated items in Service and Construction Contracts. Include on the certification form the following information: project name, project number, Contractor name, license number, Contractor address, and certification. The certification will read as follows and be signed and dated by the Contractor. "I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current EPA standards for recycled/recovered materials content. The following exemptions may apply to the non-procurement of recycled/recovered content materials:

- a. The product does not meet appropriate performance standards;
- b. The product is not available within a reasonable time frame;
- c. The product is not available competitively (from two or more sources);
- d. The product is only available at an unreasonable price (compared with a comparable non-recycled content product)."

PART 3 EXECUTION

3.1 AS-BUILT DRAWINGS

Provide and maintain two black line print copies of the PDF contract drawings for As-Built Drawings. Maintain the as-builts throughout construction as red-lined hard copies on site or red-lined PDF files. Submit As-Built Drawings 30 days prior to Beneficial Occupancy Date (BOD).

3.1.1 Markup Guidelines

Make comments and markup the drawings complete without reference to letters, memos, or materials that are not part of the As-Built drawing. Show what was changed, how it was changed, where item(s) were relocated and change related details. These working as-built markup prints must be neat, legible and accurate as follows:

- a. Use base colors of red, green, and blue. Color code for changes as

follows:

- (1) Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes.
 - (2) Deletions (Red) - Over-strike deleted graphic items (lines), lettering in notes and leaders.
 - (3) Additions (Green) - Added items, lettering in notes and leaders.
- b. Provide a legend if colors other than the "base" colors of red, green, and blue are used.
 - c. Add and denote any additional equipment or material facilities, service lines, incorporated under As-Built Revisions if not already shown in legend.
 - d. Use frequent written explanations on markup drawings to describe changes. Do not totally rely on graphic means to convey the revision.
 - e. Use legible lettering and precise and clear digital values when marking prints. Clarify ambiguities concerning the nature and application of change involved.
 - f. Wherever a revision is made, also make changes to related section views, details, legend, profiles, plans and elevation views, schedules, notes and call out designations, and mark accordingly to avoid conflicting data on all other sheets.
 - g. For deletions, cross out all features, data and captions that relate to that revision.
 - h. For changes on small-scale drawings and in restricted areas, provide large-scale inserts, with leaders to the applicable location.
 - i. Indicate one of the following when attaching a print or sketch to a markup print:
 - 1) Add an entire drawing to contract drawings
 - 2) Change the contract drawing to show changes on the drawing.
 - 3) Provided for reference only to further detail the initial design.
 - j. Incorporate all shop and fabrication drawings into the markup drawings.

3.1.2 As-Built Drawings Content

Show on the as-built drawings, but not limited to, the following information:

- a. The actual location, kinds and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become covered over or obscured, show by offset dimensions to two permanently fixed surface features the end of each run including each change in direction on the record drawings. Locate valves, splice boxes and similar appurtenances by dimensioning along the utility run from a reference point. Also record the average depth below the

surface of each run.

- b. The location and dimensions of any changes within the building structure.
- c. Layout and schematic drawings of electrical circuits and piping.
- d. Correct grade, elevations, cross section, or alignment of roads, earthwork, structures or utilities if any changes were made from contract plans.
- e. Changes in details of design or additional information obtained from working drawings specified to be prepared or furnished by the Contractor; including but not limited to shop drawings, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment, and foundations.
- f. The topography, invert elevations and grades of drainage installed or affected as part of the project construction.
- g. Changes or Revisions which result from the final inspection.
- h. Where contract drawings or specifications present options, show only the option selected for construction on the working as-built markup drawings.
- i. If borrow material for this project is from sources on Government property, or if Government property is used as a spoil area, furnish a contour map of the final borrow pit/spoil area elevations.
- j. Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarm, fire sprinkler, and irrigation systems.
- k. Changes in location of equipment and architectural features.
- l. Modifications and compliance with FC 1-300-09N procedures.
- m. Actual location of anchors, construction and control joints, etc., in concrete.
- n. Unusual or uncharted obstructions that are encountered in the contract work area during construction.
- o. Location, extent, thickness, and size of stone protection particularly where it will be normally submerged by water.

3.2 CLEANUP

Provide final cleaning in accordance with ASTM E1971 and submit two copies of the listing of completed final clean-up items. Leave premises "broom clean." Comply with GS-37 for general purpose cleaning and bathroom cleaning. Use only nonhazardous cleaning materials, including natural cleaning materials, in the final cleanup. Clean interior and exterior glass surfaces exposed to view; remove temporary labels, stains and foreign substances; polish transparent and glossy surfaces; vacuum carpeted and soft surfaces. Clean equipment and fixtures to a sanitary condition. Replace filters of operating equipment and comply with the Indoor Air Quality (IAQ) Management Plan. Clean debris from roofs, gutters, downspouts and drainage systems. Sweep paved areas and rake

clean landscaped areas. Remove waste and surplus materials, rubbish and construction facilities from the site. Recycle, salvage, and return construction and demolition waste from project in accordance with Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS, and 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.

3.3 REAL PROPERTY RECORD

Refer to UFC 1-300-08 for instruction on completing the DD FORM 1354. Contact the Contracting Officer for any project specific information necessary to complete the DD FORM 1354.

3.3.1 Interim DD FORM 1354

Near the completion of Project, but a minimum of 60 days prior to final acceptance of the work, complete and submit an accounting of all installed property with Interim DD FORM 1354. Include any additional assets, improvements, and alterations from the Draft DD FORM 1354.

3.3.2 Completed DD FORM 1354

Submit the completed Checklist for DD FORM 1354 of Installed Building Equipment items. Attach this list to the updated DD FORM 1354.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA

05/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 MEETINGS
 - 1.3.1 Pre-Construction Meeting
 - 1.3.2 Operation and Maintenance Manual and Facility Data Workbook Coordination Meeting
 - 1.3.3 Submittal Coordination Meeting
 - 1.3.4 Facility Turnover Meeting
- 1.4 FACILITY DATA WORKBOOK
- 1.5 OPERATION AND MAINTENANCE MANUAL MEDIA
 - 1.5.1 CD or DVD Label and Disk Holder or Case
- 1.6 O&M MANUAL CONTENT
 - 1.6.1 Part 1: Executive Summary
 - 1.6.2 Part 2: Facility Design and Construction
 - 1.6.2.1 General Facility and Systems Description
 - 1.6.2.2 Basis of Design
 - 1.6.2.3 Contract Documents, RFP, Amendments, and Modifications
 - 1.6.3 Part 3: Facilities, Systems, and Assemblies Information
 - 1.6.3.1 Organization
 - 1.6.3.2 Related Specifications
 - 1.6.3.3 Manufacturer's Operations and Maintenance Data
 - 1.6.3.4 Approved Submittals and Certificates
 - 1.6.3.5 Approved Coordination/Shop Drawings
 - 1.6.4 Sequence of Operation for Operating Equipment
 - 1.6.4.1 Safety Precautions and Hazards
 - 1.6.4.2 Operator Prestart
 - 1.6.4.3 Startup, Shutdown, and Post-Shutdown Procedures
 - 1.6.4.4 Normal Operations
 - 1.6.4.5 Emergency Operations
 - 1.6.4.6 Operator Service Requirements
 - 1.6.4.7 Environmental Conditions
 - 1.6.4.8 Operating Log
 - 1.6.4.9 Additional Requirements for Equipment Control Systems
 - 1.6.4.10 Testing Equipment Information and Performance Data
 - 1.6.5 Routine Maintenance Requirements
 - 1.6.5.1 Preventive Maintenance Plan, Schedule, and Procedures
 - 1.6.5.2 Lubrication Data
 - 1.6.6 Repair Procedures
 - 1.6.6.1 Troubleshooting Guides and Diagnostic Techniques
 - 1.6.6.2 Wiring Diagrams and Control Diagrams
 - 1.6.6.3 Removal and Replacement Instructions
 - 1.6.6.4 Repair Work-Hours
 - 1.6.6.5 Warranty Information
 - 1.6.6.6 Extended Warranty Information

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 1.6.6.7 Record Drawings and Utility Systems
- 1.6.6.8 Personnel Training Requirements
- 1.6.6.9 Contractor / Supplier Listing and Contact Information
- 1.6.7 Part 4: Facility Operations
 - 1.6.7.1 Completed Facility Operating Plan
 - 1.6.7.2 Testing Equipment and Special Tool Information
 - 1.6.7.3 Testing and Performance Data
 - 1.6.7.4 Approved Field Test Reports and Manufacturer's Field Reports
 - 1.6.7.5 Maintenance Plans, Procedures, Checklists, Records, and Spare Parts Inventory
 - 1.6.7.5.1 Maintenance Schedules
 - 1.6.7.5.2 Ongoing Commissioning Operational and Maintenance Record Keeping
 - 1.6.7.5.3 Janitorial and Cleaning Plans and Procedures
 - 1.6.7.6 Utility Record Drawings
 - 1.6.7.6.1 Utility Schematic Diagrams
 - 1.6.7.6.2 Enlarged Connection and Cutoff Plans
 - 1.6.7.6.2.1 Description of Utility Metering and Monitoring Systems
 - 1.6.7.6.2.2 Procedures for Tracking Utility Use and Reporting
 - 1.6.7.6.2.3 One-Line Diagrams and Meter Location of System
 - 1.6.7.6.3 Spare Parts and Supply Lists
- 1.6.8 Part 5: Training
- 1.6.9 Part 6: Cx Project Report and TAB Report
- 1.6.10 Part 7: Regulatory Requirements
- 1.6.11 Part 8: Permits
- 1.6.12 Part 9: Operations and Maintenance Manual Approval
- 1.7 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES
 - 1.7.1 Package Quality
 - 1.7.2 Data Package 1
 - 1.7.3 Data Package 2
 - 1.7.4 Data Package 3
 - 1.7.5 Changes to Submittals

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 TRAINING
 - 3.1.1 Training Plan
 - 3.1.2 Training Content
 - 3.1.3 Training Outline
 - 3.1.4 Training Video Recording
 - 3.1.5 Unresolved Questions from Attendees
 - 3.1.6 Validation of Training Completion
 - 3.1.7 Quality Control Coordination
- 3.2 SUBMITTAL SCHEDULING
 - 3.2.1 Operation and Maintenance Manual, Progress Submittal
 - 3.2.2 Operation and Maintenance Manual, Prefinal Submittal
 - 3.2.3 Operation and Maintenance Manual, Final Submittal
 - 3.2.4 Final Operation and Maintenance Manual Submittal Translation

-- End of Section Table of Contents --

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA
05/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE GUIDELINE 1.4 (2019) Preparing Systems Manuals for
Facilities

ASTM INTERNATIONAL (ASTM)

ASTM E1971 (2005; R 2011) Standard Guide for
Stewardship for the Cleaning of Commercial
and Institutional Buildings

ASTM E2166 (2016) Standard Practice for Organizing
and Managing Building Data

U.S. DEPARTMENT OF DEFENSE (DOD)

FC 1-300-09N (2014; with Change 6, 2021) Navy and
Marine Corps Design

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-10 Operation and Maintenance Data

Facility Data Workbook; G

Training Plan; G

Training Outline; G

Training Content; G

Operation And Maintenance Manual, Progress Submittal; G

Operation And Maintenance Manual, Prefinal Submittal; G

Operation And Maintenance Manual, Final Submittal; G

SD-11 Closeout Submittals

Training Video Recording; G

Validation of Training Completion; G

Training Plan; G

Record Drawings And Utility Systems; G

1.3 MEETINGS

To assure that Operation and Maintenance Manual and Facility Data Workbook requirements are being met through the duration of the project, organize the following meetings and discuss the subsequent topics:

1.3.1 Pre-Construction Meeting

At a minimum, discuss the following:

- a. The requirement for Operation and Maintenance Manuals and Facility Data deliverables under this contract including coordination meetings
- b. Processes and method of gathering Facility Data information during construction
- c. Primary roles and responsibilities associated with the development and delivery of the Operation and Maintenance Manuals and Facility Data deliverables, and
- d. Identify and agree upon a date and attendance list for the meetings described below:

1.3.2 Operation and Maintenance Manual and Facility Data Workbook Coordination Meeting

Facilitate a meeting after the Pre-Construction Meeting prior to the submission of the Operation and Maintenance Manual Progress Submittal. Meeting attendance must include the Contractor's Operation and Maintenance Manual and Facility Data Workbook Preparer, Designer of Record (DOR), Quality Control Manager, the Government's Design Manager (DM), Contracting Officer's Representative, and Government's facility data reviewer. Include any Mechanical, Electrical, and Fire Protection Sub-Contractors.

The purpose of this meeting is to reach a mutual understanding of the scope of work concerning the contract requirements for Operation and Maintenance Manual and coordinate the efforts necessary by both the Government and Contractor to ensure an accurate collection, preparation and timely Government review of Operation and Maintenance Manual.

1.3.3 Submittal Coordination Meeting

Facilitate a meeting following submission and Government review of each design or progress submittal of the Operation and Maintenance Manual and Facility Data Workbook.

- a. Include personnel from the Coordination meeting and any additional personnel identified.
- b. The purpose of this meeting is to demonstrate ongoing compliance with

the requirements identified in this specification. Discuss Government review comments and unresolved items preventing completion and Government approval of the Operation and Maintenance Manuals and Facility Data Workbook (FDW).

- c. The applicable deliverables, along with Government remarks associated with review of these submittals serve as the primary guide and agenda for this meeting.

1.3.4 Facility Turnover Meeting

Include Operation and Maintenance Manual in NAVFAC Red Zone (NRZ) facility turnover meetings as specified in Section 01 30 00, ADMINISTRATIVE REQUIREMENTS.

1.4 FACILITY DATA WORKBOOK

Develop an editable, electronic spreadsheet based on the equipment in the Operation and Maintenance Manuals that contains the information required to start a preventive maintenance program as outlined in 01 78 24.00 20 FACILITY DATA WORKBOOK (FDW). As a minimum, provide Facility Data Workbook as a list of system equipment, location installed, warranty expiration date, manufacturer, model, and serial number.

1.5 OPERATION AND MAINTENANCE MANUAL MEDIA

Assemble Operation and Maintenance Manual into an electronically bookmarked file using the most current version of Adobe Acrobat or similar software capable of producing PDF file format. Provide compact disks (CD) or data digital versatile disk (DVD) as appropriate, so that each one contains operation, maintenance and record files, project record documents, and training videos. Include a complete bookmarked operation and maintenance directory.

1.5.1 CD or DVD Label and Disk Holder or Case

Provide the following information on the disk label and disk holder or case:

- a. Building Number
- b. Project Title
- c. Activity and Location
- d. Construction Contract Number
- e. Prepared For: (Contracting Agency)
- f. Prepared By: (Name, title, phone number and email address)
- g. Include the disk content on the disk label
- h. Date
- i. Virus scanning program used

1.6 O&M MANUAL CONTENT

Organize the bookmarked Operation and Maintenance Manual into the following Parts in accordance with **ASHRAE GUIDELINE 1.4**, and as modified and detailed below. Word template for O&M Manual is available at: <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-78-23>.

1.6.1 Part 1: Executive Summary

Provide a summary of the information found in the O&M manual including the purpose of the manual and a description of the manual's organization.

1.6.2 Part 2: Facility Design and Construction

1.6.2.1 General Facility and Systems Description

Provide an overview of the intent for design and use of the facility. Provide a PDF of the Record Drawings prepared in accordance with **FC 1-300-09N** and **01 78 00** CLOSEOUT SUBMITTALS and bookmarked using the sheet title and sheet number. Include uncluttered **28 by 43 mm** floor plans with room numbers, type or function of space, and overall facility dimensions on the floor plans. Do not include items such as construction instructions, references, or frame numbers.

Detail the overall dimensions of the facility, number of floors, foundation type, expected number of occupants, and facility Category Code list and generally describe all the facility systems and any special building features (for example, HVAC Controls, Sprinkler Systems, Cranes, Elevators, and Generators). Include photographs marked up and labeled to show key operating components and the overall facility appearance.

1.6.2.2 Basis of Design

Provide a copy of the contract Basis of Design.

1.6.2.3 Contract Documents, RFP, Amendments, and Modifications

Provide the contract construction documents complete, to include specifications, drawings, Request for Proposal, amendments, and modifications.

1.6.3 Part 3: Facilities, Systems, and Assemblies Information

1.6.3.1 Organization

Bookmark information in this section using the current version of **ASTM E2166** Unifomat II, UFGS numbers, and document type as outlined in the example below. Bookmark/tab each item to the third level for easy navigation of the manual.

Example as shown in Table below:

PARTS AND SUBPART NUMBERING
3.1 B20 EXTERIOR CLOSURE (System)

PARTS AND SUBPART NUMBERING	
3.1.1 B2030 EXTERIOR DOORS (Subsystem)	
3.1.1.1 B2030110 GLAZED DOORS (Component)	
3.1.1.1.1 Applicable specifications List in UFGS Format	
3.1.1.1.2 Manufacturer's Operations and Maintenance Data	
3.1.1.1.3 Approved Submittal	
3.1.1.1.4 Coordination/Shop Drawings	
3.1.1.1.5 Sequence of Operation for Operating Equipment	
3.1.1.1.6 Testing Equipment Information and Performance Data	
3.1.1.1.7 Routine Maintenance Requirements	
3.1.1.1.8 Repair Procedures	
3.1.1.1.9 Emergency Procedures & Locations of Applicable Controls	
3.1.1.1.10 Warranties	
3.1.1.1.11 Record Drawings and Utility Systems	
3.1.1.1.12 Contractor / Supplies Listing and Contact Information	

1.6.3.2 Related Specifications

Reference each specification related to the subsystem in this section, and locate the actual specification section in Part 2 of the O&M Manual. List specifications in table format as shown in the below example.

UFGS Number	Specification Title	Page Spec Begins in Part 2

1.6.3.3 Manufacturer's Operations and Maintenance Data

Provide a copy of all manufacturer specifications and cutsheets. Provide text-searchable, high-quality document files from the manufacturer's online or electronic documentation. Color documents are preferred. Provide documents specific to the product(s) installed under this Contract. Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Provide Uniformat II Level 3 identification for D20, D30, D40 installed equipment. When possible, do not submit document files containing multiple product catalogs from the same manufacturer, or product data from multiple manufacturers in the same files. Provide documents directly from the manufacturer whenever possible. Do not provide scanned copies of hardcopy documents. Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware

requirements, such as requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number that will cross-reference the illustrated part to the listed part. Group the parts shown in the listings by components, assemblies, and subassemblies in accordance with the manufacturer's standard practice. Parts data may cover more than one model or series of equipment, components, assemblies, subassemblies, attachments, or accessories, such as typically shown in a master part catalog.

1.6.3.4 Approved Submittals and Certificates

Provide a copy of all submittals documented with the required approval as applicable for each UFGS specification listed in the table outlined in applicable specifications. Include copies of SD-07 Certificates submittals documented with the required approval, SD-08 Manufacturer's Instructions submittals documented with the required approval, and SD-10 Operation and Maintenance Data submittals documents with the required approval.

1.6.3.5 Approved Coordination/Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work. Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project. Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

1.6.4 Sequence of Operation for Operating Equipment

Provide record one-line diagrams for each floor, delineating mechanical equipment location within the building. Provide specific instructions, procedures, and illustrations for the following phases of operation for the installed model and features of each system:

1.6.4.1 Safety Precautions and Hazards

List personnel hazards and equipment or product safety precautions for operating conditions. List all residual hazards identified in the Activity Hazard Analysis provided under Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Provide recommended safeguards for each identified hazard. Specify if any certifications or licenses are required to operate the equipment.

1.6.4.2 Operator Prestart

Provide procedures required to install, set up, and prepare each system for use.

1.6.4.3 Startup, Shutdown, and Post-Shutdown Procedures

Provide narrative description for Startup, Shutdown and Post-shutdown operating procedures including the control sequence for each procedure.

1.6.4.4 Normal Operations

Provide Control Diagrams with data to explain operation and control of systems and specific equipment. Provide narrative description of Normal Operating Procedures.

1.6.4.5 Emergency Operations

Provide Emergency Procedures for equipment malfunctions to permit a short period of continued operation or to shut down the equipment to prevent further damage to systems and equipment. Provide Emergency Shutdown Instructions for fire, explosion, spills, or other foreseeable contingencies. Provide guidance and procedures for emergency operation of utility systems including required valve positions, valve locations and zones or portions of systems controlled.

1.6.4.6 Operator Service Requirements

Provide instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and recording gauge readings.

1.6.4.7 Environmental Conditions

Provide a list of Environmental Conditions (temperature, humidity, and other relevant data) that are best suited for the operation of each product, component or system. Describe conditions under which the item equipment should not be allowed to run.

1.6.4.8 Operating Log

Provide forms, sample logs, and instructions for maintaining necessary operating records.

1.6.4.9 Additional Requirements for Equipment Control Systems

Provide Data Package 5 and the following for all control systems:

- a. Provide a narrative description on how to perform and apply functions, features, modes, and other operations, including unoccupied operation, seasonal changeover, manual operation, and alarms. Include detailed technical manual for programming and customizing control loops and algorithms.
- b. Submit complete controls equipment schedules, full as-built sequence of operations, wiring and logic diagrams, Input/Output Tables, equipment schedules, copies of checkout tests and calibrations performed by the Contractor (not Cx tests), and all associates information.
- c. Full print out of all schedules and set points after testing and acceptance of the system.
- d. Full as-built print out of software program.
- e. Marking of system sensors and thermostats on the as-built floor plan and mechanical drawings with their control system designations.

1.6.4.10 Testing Equipment Information and Performance Data

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.5 Routine Maintenance Requirements

1.6.5.1 Preventive Maintenance Plan, Schedule, and Procedures

Provide manufacturer's schedule for routine preventive maintenance, inspections, condition monitoring (predictive tests) and adjustments required to ensure proper and economical operation and to minimize repairs. Provide instructions stating when the systems should be retested. Provide manufacturer's projection of preventive maintenance work-hours on a daily, weekly, monthly, and annual basis including requirements by type of activity. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

- a. Define the anticipated time required to perform each of each test (work-hours), test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventive maintenance, inspection, adjustment, lubrication and cleaning necessary to minimize repairs.
- b. Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the system after acceptance.

1.6.5.2 Lubrication Data

Include the following preventive maintenance lubrication data, in addition to instructions for lubrication required under paragraph OPERATOR SERVICE REQUIREMENTS:

- a. A table showing recommended lubricants for specific temperature ranges and applications.
- b. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities. Provide procedural instructions for Oil Sampling for all equipment.
- c. A Lubrication Schedule showing service interval frequency.

1.6.6 Repair Procedures

Provide instructions and a list of tools required to repair or restore the

product or equipment to proper condition or operating standards. Provide manufacturer's recommended procedures and instructions for correcting problems and making repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials. Specify if any certifications or licenses are required to repair the equipment.

1.6.6.1 Troubleshooting Guides and Diagnostic Techniques

Provide step-by-step procedures to promptly isolate the cause of typical malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

1.6.6.2 Wiring Diagrams and Control Diagrams

Provide point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type, identically to actual installation configuration and numbering.

1.6.6.3 Removal and Replacement Instructions

Provide step-by-step procedures and a list of required specialty tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments. Provide tolerances, dimensions, settings and adjustments required. Use a combination of text and illustrations.

1.6.6.4 Repair Work-Hours

Provide manufacturer's projection of repair work-hours including requirements by type of craft. Identify, and tabulate separately, repair that requires the equipment manufacturer to complete or to participate.

1.6.6.5 Warranty Information

List and explain the various warranties and clearly identify the servicing and technical precautions prescribed by the manufacturers or contract documents in order to keep warranties in force. Identify if replacement of a subassembly, attachment or accessory requires the entire assembly to be replaced. Include warranty information for primary components of the system. Provide copies of warranties required by Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.6.6 Extended Warranty Information

List all warranties for products, equipment, components, and sub-components whose duration exceeds one year. For each warranty listed, indicate the applicable specification section, duration, start date, end date, and the point of contact for warranty fulfillment. Also, list or reference the specific operation and maintenance procedures that must be performed to keep the warranty valid. Provide copies of warranties required by Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.6.7 Record Drawings and Utility Systems

The record drawings are the final compilation of actual conditions reflected in the as-built drawings. Provide record drawings as outlined in 01 78 00 CLOSEOUT SUBMITTALS.

Using Record Source Drawings, show and document details of the actual installation of the utility systems, annotate and highlight the Operation and Maintenance information. Provide the following drawings at a large enough scale to differentiate designated isolation units from surrounding valves and switches.

1.6.6.8 Personnel Training Requirements

Provide information available from the manufacturers that is needed for use in training designated personnel to properly operate and maintain the equipment and systems.

1.6.6.9 Contractor / Supplier Listing and Contact Information

Provide a list that includes the name, address, telephone number, email and website of the General Contractor and each Subcontractor who installed the product or equipment, or system. For each item, also provide the name address and telephone number of the manufacturer's representative and service organization that can provide replacements most convenient to the project site. Provide the name, address, and telephone number of the product, equipment, and system manufacturers.

1.6.7 Part 4: Facility Operations

1.6.7.1 Completed Facility Operating Plan

Provide a plan that documents that procedures for the operation of systems and assemblies in the facility. The systems that should be included in the Operating Plan include, but are not limited to:

- a. Electrical systems and equipment
- b. Mechanical systems and equipment
- c. Fire Protection systems and equipment
- d. Control Systems and equipment
- e. Structural systems, fixtures, structures, and equipment
- f. Vertical transportation such as elevators and escalators

1.6.7.2 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

1.6.7.3 Testing and Performance Data

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.7.4 Approved Field Test Reports and Manufacturer's Field Reports

Compile and provide approved Field Test Reports (SD-06) and Manufacturer's Field Reports (SD-09) submittals.

1.6.7.5 Maintenance Plans, Procedures, Checklists, Records, and Spare Parts Inventory

1.6.7.5.1 Maintenance Schedules

Include recommended maintenance schedules for systems and equipment.

1.6.7.5.2 Ongoing Commissioning Operational and Maintenance Record Keeping

Include ongoing commissioning and optimization procedures and documentation to monitor and improve the performance of facility systems.

1.6.7.5.3 Janitorial and Cleaning Plans and Procedures

Include a copy of facility cleaning and janitorial plan with procedures and intended chemicals and equipment.

Provide environmentally friendly cleaning recommendations in accordance with [ASTM E1971](#).

1.6.7.6 Utility Record Drawings

1.6.7.6.1 Utility Schematic Diagrams

Provide a one-line schematic diagram for each utility system such as power, water, wastewater, and gas/fuel. Schematic diagram must show from the point where the utility line is connected to the mainline up to the **1.5 meter** connection point to the facility. Indicate location or area designation for route of transmission or distribution lines; locations of duct banks, manholes/handholes or poles; isolation units such as valves and switches; and utility facilities such as pump stations, lift stations, and substations.

1.6.7.6.2 Enlarged Connection and Cutoff Plans

Provide enlarged floor plans and provide information between the **1.5 meter** utilities connection point and where utilities connect to facility distribution. Enlarge floor plans/elevations of the rooms where the utility enters the building and indicate on these plans the locations of the main interiors and exterior connection and cutoff points for the utilities. Also enlarge floor plans/elevations of the rooms where equipment is located. Include enough information to enable someone unfamiliar with the facility to locate the connection and cutoff points. Indicate designations such as room number, panel number, circuit breaker, or valve number of each utility and equipment connection and cutoff point, and what that connection and cutoff point controls.

1.6.7.6.2.1 Description of Utility Metering and Monitoring Systems

Provide in narrative format a description of the utility metering and monitoring systems. Include locations, function, and related systems.

1.6.7.6.2.2 Procedures for Tracking Utility Use and Reporting

Procedures for usage reporting and tracking in support of establishing and monitoring utility budgets and costs, and in developing annual energy reports.

1.6.7.6.2.3 One-Line Diagrams and Meter Location of System

Provide one-line diagrams and design drawings that highlight meter locations on the site.

1.6.7.6.3 Spare Parts and Supply Lists

Provide lists of spare parts and supplies required for repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead-time to obtain.

1.6.8 Part 5: Training

Provide a copy of training plans used for each type of equipment along with training materials used, arranged in specification sequence. Provide a copy of training records, sign-in sheets, and agendas. Include training and documentation on the updating and continued use of the O&M Manual.

1.6.9 Part 6: Cx Project Report and TAB Report

Provide the final Cx Plan and complete Cx reports with evaluation and testing forms and records for each building system. Include relevant commissioned system assemblies test reports including installers checklists of assemblies. Provide all Cx Progress Reports, issues and resolutions logs with resolution or status of each item, and a list of any open items and seasonal or additional testing required.

1.6.10 Part 7: Regulatory Requirements

Provide information describing regulatory and policies compliance requirements or provide a reference to where it is stored.

1.6.11 Part 8: Permits

Provide information requiring frequently asked questions and associated answers or provide a reference to where it is stored.

1.6.12 Part 9: Operations and Maintenance Manual Approval

Provide a signed document stating that the project O&M Manual has been reviewed and confirming agreement with the approach it presents. Include contact information for the signer for coordination of any future changes.

1.7 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Provide the O&M data packages specified in individual technical sections. O&M Data Packages are one of the components of the O&M Manual. The information required in each type of data package follows:

1.7.1 Package Quality

Documents must be fully legible. Operation and Maintenance data must be

consistent with the manufacturer's standard brochures, schematics, printed instructions, general operating procedures, and safety precautions.

1.7.2 Data Package 1

- a. Safety precautions and hazards
- b. Cleaning recommendations
- c. Maintenance and repair procedures
- d. Warranty information
- e. Extended warranty information
- f. Contractor information
- g. Spare parts and supply list

1.7.3 Data Package 2

- a. Safety precautions and hazards
- b. Normal operations
- c. Environmental conditions
- d. Lubrication data
- e. Preventive maintenance plan, schedule, and procedures
- f. Cleaning recommendations
- g. Maintenance and repair procedures
- h. Removal and replacement instructions
- i. Spare parts and supply list
- j. Parts identification
- k. Warranty information
- l. Extended warranty information
- m. Contractor information

1.7.4 Data Package 3

- a. Safety precautions and hazards
- b. Operator prestart
- c. Startup, shutdown, and post-shutdown procedures
- d. Normal operations
- e. Emergency operations

- f. Environmental conditions
- g. Operating log
- h. Lubrication data
- i. Preventive maintenance plan, schedule, and procedures
- j. Cleaning recommendations
- k. Troubleshooting guides and diagnostic techniques
- l. Wiring diagrams and control diagrams
- m. Maintenance and repair procedures
- n. Removal and replacement instructions
- o. Spare parts and supply list
- p. Product submittal data
- q. O&M submittal data
- r. Parts identification
- s. Warranty information
- t. Extended warranty information
- u. Testing equipment and special tool information
- v. Testing and performance data
- w. Contractor information
- x. Field test reports

1.7.5 Changes to Submittals

Provide manufacturer-originated changes or revisions to submitted data if a component of an item is so affected subsequent to acceptance of the O&M Data. Submit changes, additions, or revisions required by the Contracting Officer for final acceptance of submitted data within 30 calendar days of the notification of this change requirement.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 TRAINING

Prior to acceptance of the facility by the Contracting Officer for Beneficial Occupancy, provide comprehensive training for the systems and equipment specified in the technical specifications. The training must be targeted for the building maintenance personnel, and applicable building occupants. Instructors must be well-versed in the particular systems that

they are presenting. Address aspects of the Operation and Maintenance Manual submitted in accordance with Section 01 78 00 CLOSEOUT SUBMITTALS. Training must include classroom or field lectures based on the system operating requirements. The location of classroom training requires approval by the Contracting Officer.

3.1.1 Training Plan

Submit a written training plan to the Contracting Officer for approval at least 60 calendar days prior to the scheduled training. Training plan must be approved by the Quality Control Manager (QC) prior to forwarding to the Contracting Officer. Also, coordinate the training schedule with the Contracting Officer and QC. Include within the plan the following elements:

- a. Equipment included in training
- b. Intended audience
- c. Location of training
- d. Dates of training
- e. Objectives
- f. Outline of the information to be presented and subjects covered including description
- g. Start and finish times and duration of training on each subject
- h. Methods (e.g. classroom lecture, video, site walk-through, actual operational demonstrations, written handouts)
- i. Instructor names and instructor qualifications for each subject
- j. List of texts and other materials to be furnished by the Contractor that are required to support training
- k. Description of proposed software to be used for video recording of training sessions.

3.1.2 Training Content

The core of this training must be based on manufacturer's recommendations and the operation and maintenance information. The QC is responsible for overseeing and approving the content and adequacy of the training. Spend 95 percent of the instruction time during the presentation on the OPERATION AND MAINTENANCE DATA. Include the following for each system training presentation:

- a. Start-up, normal operation, shutdown, unoccupied operation, seasonal changeover, manual operation, controls set-up and programming, troubleshooting, and alarms.
- b. Relevant health and safety issues.
- c. Discussion of how the feature or system is environmentally responsive. Advise adjustments and optimizing methods for energy conservation.

- d. Design intent.
- e. Use of O&M Manual Files.
- f. Review of control drawings and schematics.
- g. Interactions with other systems.
- h. Special maintenance and replacement sources.
- i. Tenant interaction issues.

3.1.3 Training Outline

Provide the Operation and Maintenance Manual Files (Bookmarked PDF) and a written course outline listing the major and minor topics to be discussed by the instructor on each day of the course to each trainee in the course. Provide the course outline 14 calendar days prior to the training.

3.1.4 Training Video Recording

Record classroom training session(s) on video. Provide to the Contracting Officer two copies of the training session(s) in DVD video recording format. Capture within the recording, in video and audio, the instructors' training presentations including question and answer periods with the attendees. The recording camera(s) must be attended by a person during the recording sessions to assure proper size of exhibits and projections during the recording are visible and readable when viewed as training.

3.1.5 Unresolved Questions from Attendees

If, at the end of the training course, there are questions from attendees that remain unresolved, the instructor must send the answers, in writing, to the Contracting Officer for transmittal to the attendees, and the training video must be modified to include the appropriate clarifications.

3.1.6 Validation of Training Completion

Ensure that each attendee at each training session signs a class roster daily to confirm Government participation in the training. At the completion of training, submit a signed validation letter that includes a sample record of training for reporting what systems were included in the training, who provided the training, when and where the training was performed, and copies of the signed class rosters. Provide two copies of the validation to the Contracting Officer, and one copy to the Operation and Maintenance Manual Preparer for inclusion into the Manual's documentation.

3.1.7 Quality Control Coordination

Coordinate this training with the QC in accordance with [Section 01 45 00 QUALITY CONTROL](#).

3.2 SUBMITTAL SCHEDULING

3.2.1 Operation and Maintenance Manual, Progress Submittal

Submit the Progress submittal when construction is approximately 50 percent complete, to the Contracting Officer for approval. Provide Operation and Maintenance Manual Files (Bookmarked PDF). Include the elements and portions of system construction completed up to this point. The purpose of this submittal is to verify progress is in accordance with contract requirements as discussed during the Operation and Maintenance Manual Coordination Meeting.

3.2.2 Operation and Maintenance Manual, Prefinal Submittal

Submit the 100 percent submittal of the Operation and Maintenance Prefinal Submittal to the Contracting Officer for approval within 90 calendar days of the Beneficial Occupancy Date (BOD). This submittal must provide a complete, working document that can be used to operate and maintain the facility. Any portion of the submittal that is incomplete or inaccurate requires the entire submittal to be returned for correction. Any discrepancies discovered during the Government's review of Operation and Maintenance Progress submittal must be corrected prior to the Prefinal submission. The Prefinal Submittal must include Operation and Maintenance Manual Files (Bookmarked PDF).

3.2.3 Operation and Maintenance Manual, Final Submittal

Submit completed Operation and Maintenance Manual Files (Bookmarked PDF). The Final submittal is due at BOD. Any discrepancies discovered during the Government's review of the Prefinal submittal, including the Field Verification, must be corrected prior to the Final submission.

3.2.4 Final Operation and Maintenance Manual Submittal Translation

Provide a translation in Spanish of the Operation and Maintenance Manual in electronic format. Provide drawings, charts and tables in both English and the foreign language. If required by Contracting Officer, provide a split format showing the foreign language on the left and English translation on the right.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 78 23.33

OPERATION AND MAINTENANCE MANUALS FOR AVIATION FUEL SYSTEMS

08/18, CHG 1: 02/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 CONTRACTOR RESPONSIBILITY
- 1.3 SUBMITTALS

PART 2 PRODUCTS

- 2.1 DEVELOPMENT OF SUBMITTALS
 - 2.1.1 Operation and Maintenance System Instructions (OMSI) Submittal Requirements
 - 2.1.2 Assembly
- 2.2 IDENTIFICATION

PART 3 EXECUTION

- 3.1 OPERATING INSTRUCTIONS
 - 3.1.1 Safety
 - 3.1.2 Operator Prestart
 - 3.1.3 Starting and Shutdown Procedures and Controls
 - 3.1.4 Normal Operating Instructions
 - 3.1.5 Emergency Operating Procedures
 - 3.1.6 Operator Service Requirements
- 3.2 OPERATION INSTRUCTION TO GOVERNMENT PERSONNEL
- 3.3 MAINTENANCE, SERVICE AND REPAIR INSTRUCTIONS
 - 3.3.1 Lubrication Instructions
 - 3.3.2 Table of Preventative Maintenance Instructions
 - 3.3.3 Preventative Maintenance Inspection
 - 3.3.4 Troubleshooting Guides and Diagnostic Techniques
 - 3.3.5 Removal and Replacement Instructions
 - 3.3.6 Maintenance and Repair Procedure
- 3.4 PARTS MANUAL
 - 3.4.1 Contents
 - 3.4.2 Illustrations, Drawings, and/or Exploded Views
 - 3.4.3 End Item Manufacturer's Part Numbers
 - 3.4.4 Components Assemblies/Parts
 - 3.4.5 Appendices
- 3.5 VALIDATION
- 3.6 SPECIFIC EQUIPMENT SUBMITTALS
 - 3.6.1 Pressure Gages
 - 3.6.2 Automatic Pump Controls
 - 3.6.3 Meters
 - 3.6.4 Additives, Storage Tank, Water Drain Tank, and Accessories
 - 3.6.5 Truck Offload System
 - 3.6.6 Operating Tank Level Indicator
 - 3.6.7 Piping and Fittings

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.6.8 Manual Valves
- 3.6.9 Gaskets and Isolating Gasket Kits
- 3.6.10 Strainers
- 3.6.11 Protective Coatings
- 3.6.12 Sample Connections
- 3.6.13 Pumps, Offload, Fuel Transfer, Water Draw-off Pump
- 3.6.14 Flexible Hoses
- 3.6.15 Control Valves
- 3.6.16 Engine-Generator
- 3.6.17 Fire Alarm and Fire Detecting System
- 3.6.18 Motor Control Center
- 3.6.19 Non-Automatic Transfer Switch
- 3.6.20 Pump Control Panel (PCP)

-- End of Section Table of Contents --

SECTION 01 78 23.33

OPERATION AND MAINTENANCE MANUALS FOR AVIATION FUEL SYSTEMS
08/18, CHG 1: 02/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

U.S. DEFENSE LOGISTICS AGENCY (DLA)

DLA J4 Handbook

(Part IV) Federal Supply Class Assignments
(Numeric and Alphabetic Listing)
(<http://www.dla.mil/>)

1.2 CONTRACTOR RESPONSIBILITY

The Contractor is responsible for providing the technical publications specified herein for all of the components, assemblies, sub-assemblies, attachments, and accessories, required to be supplied in accordance with submittal requirements of each specification section, regardless of whether the item was manufactured and assembled in-house or obtained from other sources. The System Supplier is responsible to the Contractor for providing the technical publications specified herein for all of the components, assemblies, sub-assemblies, attachments, and accessories that the System Supplier provided.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-10 Operation and Maintenance Data

System Instructions; G

PART 2 PRODUCTS

2.1 DEVELOPMENT OF SUBMITTALS

2.1.1 Operation and Maintenance System Instructions (OMSI) Submittal Requirements

OMSI submittals are required in order that complete documentation can be assembled to provide the Government "Activity" with the necessary information and orientation to adequately operate and maintain the new structures/facilities of this project. Submit the OEMSI documents and information specified for the equipment listed under the OEMSI submittal paragraphs in each technical section. Provide copies of each Electronic Operation and Maintenance Support Information (e-OESI) submittal to the Contracting Officer no later than 120-days prior to contract completion.

In addition to requirements of this section, see Section 01 78 23 OPERATION AND MAINTENANCE DATA for additional requirements on assembly of operation and maintenance data. OMSI submittals are to be submitted separate from and in addition to Contractor's product approval submittals.

2.1.2 Assembly

Provide submittals in separate folders consistent with the Contractor's standard practice. For hard copy manufacturer's manuals or data for the components, assemblies, subassemblies, and other operating parts which are provided must be assembled into a loose-leaf notebook-type folder, indexed by major assembly and component in sequential order. Manuals must be complete in all respects for all equipment, controls, and accessories provided. In addition, provide an electronic copy of the manuals in Adobe Acrobat 11.0 or later (CD-ROM or DVD-ROM). Utilize Bookmarks to display indexing, and assembly and component requirements.

2.2 IDENTIFICATION

On each folder identify and mark as follows:

- a. Inscribe on the cover, the words, "FUEL SYSTEM OPERATION AND MAINTENANCE MANUAL", the name and location of the building, and the contract number.
- b. Equipment manufacturer and/or Contractor's address and telephone number; names, address and telephone numbers of each subcontractor installing equipment; and local representative for each item of equipment.
- c. Volume number and title of the folder.
- d. The manual must have a table of contents and be assembled to conform to the table of contents with the tab sheets placed before instructions covering the subject. Sub-divide manuals or provide separate manuals for each of the following categories.
 - (1) Operating Instructions
 - (2) Maintenance, Service, and Repair Instructions
 - (3) Parts Manual

PART 3 EXECUTION

3.1 OPERATING INSTRUCTIONS

The operator's instructions must include specific instructions and illustrations of the equipment operation required or recommended by the manufacturer as follows:

3.1.1 Safety

Include manufacturer's safety precautions to be observed while operating under all conditions for which the equipment was designed. Clearly list all major hazards to personnel and equipment safety that are peculiar to systems and equipment described in the manual.

3.1.2 Operator Prestart

Include instructions for prestart checks, lubrication, and service

requirements necessary for setting up or preparing each system for use, warm up procedures, and verification of normal operation. Include control diagrams with data to explain detailed operation and control of each item of the equipment.

3.1.3 Starting and Shutdown Procedures and Controls

Include a control sequence describing start up operation and provide shutdown procedures and post-shutdown requirements.

3.1.4 Normal Operating Instructions

Instructions must be sufficient to enable the mechanic to adjust, stop and start, and operate the equipment properly. Special startup precautions must be noted, as well as other items requiring action before the equipment may be put into service. Include detailed drawings indicating procedure and valve numbers and status as to normally open/closed.

3.1.5 Emergency Operating Procedures

Include action to be taken in the event of a malfunction of the unit, either to permit a short period of continued operation or to prevent further damage to the unit and to the system in which it is installed.

3.1.6 Operator Service Requirements

Include instructions for operator service requirements during operation of the equipment.

3.2 OPERATION INSTRUCTION TO GOVERNMENT PERSONNEL

Furnish the services of competent instructors who will give full instruction to the designated personnel in the adjustment, operation and maintenance, including pertinent safety requirements, of the equipment or system specified. Each instructor must be thoroughly familiar with all parts of the installation and must be trained in operating theory as well as practical operation and maintenance work. Instruction must be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. A minimum of 1 man-day (8-hours) of instruction must be furnished for each system specified in other sections. When more than 4 man-days of instruction are specified, approximately half of the time must be used for classroom instruction. All other time must be used for instruction with the equipment or system. When significant changes or modifications in the equipment or system are made under the terms of the contract, additional instruction must be provided to acquaint the operating personnel with the changes or modifications. Government representatives must be allowed to video tape all classroom and field instructions.

3.3 MAINTENANCE, SERVICE AND REPAIR INSTRUCTIONS

The shop or maintenance manual must include manufacturer's instructions to maintain the equipment in a safe and serviceable condition. The maintenance or shop manual must contain all necessary instructions, illustrations, charges and diagrams covering, as a minimum, the items listed below.

3.3.1 Lubrication Instructions

- a. Include a table showing recommended lubricants for specific temperature ranges and applications.
- b. Include chart(s) with schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities. Provide a lubrication schedule showing service interval frequency.

3.3.2 Table of Preventative Maintenance Instructions

Include frequency in time, miles or hours covering routine servicing, lubrication, and adjustments.

3.3.3 Preventative Maintenance Inspection

Points and checklist should be clearly spelled out as part of operator-type inspection in this section. Include chart with schematic diagram and/or a separate inspection checklist indicating what should be examined for wear or possible malfunction and what should be reported for repair.

3.3.4 Troubleshooting Guides and Diagnostic Techniques

Provide step-by-step procedures to enable prompt isolation of the cause of a malfunction with corrective maintenance instructions. Instructions must clearly indicate why the check out is performed and what conditions are to be sought.

3.3.5 Removal and Replacement Instructions

Provide step-by-step procedures for removal, replacement, disassembly and assembly of all components, assemblies, sub-assemblies, accessories, and attachments normally subjected to wear, damage, malfunction, and frequent replacement. These instructions should provide for a judicious combination of text and illustrations.

3.3.6 Maintenance and Repair Procedure

Provide instructions for tolerances, dimensions, settings, and adjustments normally required for performing routine maintenance servicing. Instructions must provide the necessary information to bring equipment up to the required serviceable standard when it becomes unserviceable. Include instructions for examining equipment for needed repairs and adjustment, and any tests or inspections required to determine whether or not parts must be replaced.

3.4 PARTS MANUAL

3.4.1 Contents

The parts manual must provide positive identification and coverage for all of the parts of components, assemblies, sub-assemblies, and accessories of the end item normally subject to wear, malfunctioning, damage, or loss. Include any special hardware requirements (e.g., high-strength bolts and nuts). The parts manual may cover more than one model or series or equipments, components, assemblies, subassemblies, attachments, or accessories, such as a master parts catalog, in accordance with the

manufacturer's standard commercial practice. Identification of the parts must be such that all parts may be ordered and centrally stocked by the government without further identification to the make, model, and serial number of the equipment being provided.

3.4.2 Illustrations, Drawings, and/or Exploded Views

Provide clear and legible illustrations, drawings, and/or views to enable easy identification of all individual parts, components, assemblies, sub-assemblies, and accessories of the end item. Show part numbers and description on illustrations or list separately. When the illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number which will cross-reference the illustrated part to listed part. Parts shown in the listings must be grouped by components, assemblies, and sub-assemblies with individual parts identified to the assembly.

3.4.3 End Item Manufacturer's Part Numbers

Include parts for which the end item manufacturer has proprietary rights or has exercised design control, and for which the end item manufacturer is the logical supplier. The end item manufacturer must also assign numbers to purchased production parts, if such parts are altered to meet the prime manufacturer's design configuration. (Repainting, marking, or other insignificant operations are not adequate cause for use of exclusively assigned numbers).

3.4.4 Components Assemblies/Parts

Include those components assemblies/parts purchased by the end item manufacturer for which the end item manufacturer does not have control, and must be identified by the actual manufacturer's name and part numbers. Detail parts in the manufacturer's assembly, as well as attaching parts, for which the manufacturer does not have design control must also be identified by the applicable actual manufacturer's parts numbers. This paragraph does not restrict the end item manufacturer from assigning part numbers as long as the actual manufacturer's part number and the Federal Supply Code for Manufacturer (DLA J4 Handbook) or manufacturer is shown.

3.4.5 Appendices

End item manufacturer may add an appendix for cross-reference to implement components assemblies/parts requirements when implementation in manual form varies drastically with the style, format, and method of Contractor's standard commercial practice. Subject cross-referenced in an appendix will appear in the following format:

End Item Manufacturer's Alpha Numeric Seq.	Actual Manufacturer's Name and/or FCSM* from DLA J4 Handbook	Actual Manufacturer Part No.
100001	John Doe & Co. 000000	2000002
*Federal Supply Code for Manufacturers Cataloging Handbook, Name to Code		

3.5 VALIDATION

Each submittal must be validated by the Contractor or Manufacturer as being accurate and applicable to the systems and equipment provided.

3.6 SPECIFIC EQUIPMENT SUBMITTALS

The technical sections of this specification identify the specific equipment or systems for which OMSI submittals are required. This paragraph and its subparagraph contain a general list of various types of equipment and systems together with the OMSI information required for each type. The applicable OMSI information contained in this paragraph must be submitted for each specific piece of equipment or system listed under the "OMSI Submittals" paragraph in the technical sections. Operating instructions; maintenance, service, and repair instructions; and parts manuals must conform to the requirements of their respective paragraph herein. Provide validation in accordance with paragraph VALIDATION for all submittals.

3.6.1 Pressure Gages

- a. Manufacturer's descriptive literature, general.
- b. Parts manuals and recommended spare parts list.
- c. Maintenance, service and repair instructions.
- d. Manufacturer's name, model number, serial number, Federal Stock Number (if any).

3.6.2 Automatic Pump Controls

Includes Pressure Indicating Transmitters, Flow Switches, Differential Pressure Transmitters.

- a. Manufacturer's descriptive literature, general.
- b. Parts manual.
- c. Maintenance, service and repair instructions.
- d. Operating Instructions.
- e. Manufacturer's name, model number, serial number, Federal Stock Number (if any).
- f. Performance data at specified conditions.
- g. Control wiring diagrams showing all terminations of conductors (and all control devices) labeled to permit identification in the field; part numbers of all control devices; normally open or normally closed; voltage of all control components.
- h. Name, address and telephone number of the nearest manufacturer's representative.

3.6.3 Meters

- a. Manufacturer's descriptive literature, general.

- b. Parts manual and recommended spare parts list.
- c. Maintenance, service, calibration instructions, and repair instructions.
- d. Operating Instructions.
- e. Manufacturer's name, model number, serial number, Federal Stock Number (if any).
- f. Performance data at specified conditions.
- g. Name, address and telephone number of the nearest manufacturer's representative.

3.6.4 Additives, Storage Tank, Water Drain Tank, and Accessories

- a. Manufacturer's descriptive literature, general.
- b. Parts manual and recommended spare parts list.
- c. Maintenance, service and repair instructions.
- d. Operating Instructions.
- e. Manufacturer's name, model number, serial number, Federal Stock Number (if any).
- f. Performance data at specified conditions.
- g. Name, address and telephone number of the nearest manufacturer's representative.

3.6.5 Truck Offload System

- a. General description and specifications.
- b. Comprehensive discussion of operating program.
- c. Installation and initial checkout procedures.
- d. Detailed electrical description.
- e. Complete troubleshooting procedures, diagrams, and guidelines.
- f. Complete alignment and calibration procedures for components.
- g. Preventive maintenance requirements.
- h. Detailed system schematics, system field assembly drawings, and system component specifications and dimensions.
- i. Complete spare parts lists.
- j. Complete as-built bill of materials, control drawings, schedules, and sequence of operations.
- k. Safety precautions.

- l. Control sequence describing start-up, operation, and shutdown. Control sequence must be integrated with startup and operation of the motor control center.
- m. Provide part list that clearly indicates sources of supply, recommended spare parts, and name of servicing organization.
- n. Manufacturer's name, address, and telephone number.

3.6.6 Operating Tank Level Indicator

- a. Manufacturer's descriptive literature, general.
- b. Parts manual and recommended spare parts list.
- c. Maintenance, service and repair instructions.
- d. Operating Instructions.
- e. Manufacturer's name, model number, serial number, Federal Stock Number (if any).
- f. Performance data at specified conditions.
- g. Control wiring diagrams showing all terminations of conductors (and all control devices) labeled to permit identification in the field; part numbers of all control devices; normalcy open or normally closed; voltage of all control components.
- h. Name, address and telephone number of the nearest manufacturer's representative.

3.6.7 Piping and Fittings

- a. Certificates of Compliance.
- b. Batch run numbers.
- c. Manufacturer's descriptive literature, general.
- d. Name address and telephone number of manufacturer.

3.6.8 Manual Valves

- a. Manufacturer's descriptive literature, general.
- b. Parts manual and recommended spare parts list.
- c. Maintenance, service and repair instructions.
- d. Operating Instructions.
- e. Manufacturer's name, model number, serial number, Federal Stock Number (if any).
- f. Performance data at specified conditions.
- g. Where specified to have limit switches, control wiring diagrams

showing all terminations of conductors (and all control devices) labeled to permit identification in the field; part numbers of all control devices; normally open or normally closed; voltage of all control components.

- h. Name, address and telephone number of the nearest manufacturer's representative.

3.6.9 Gaskets and Isolating Gasket Kits

- a. Manufacturer's descriptive literature, general.
- b. Manufacturer's name, model number, serial number, Federal Stock Number (if any).
- c. Name, address and telephone number of the nearest manufacturer's representative.

3.6.10 Strainers

- a. Manufacturer's descriptive literature, general.
- b. Parts manual and recommended spare parts list.
- c. Maintenance, service and repair instructions.
- d. Manufacturer's name, model number, serial number, Federal Stock Number (if any).
- e. Name, address and telephone number of the nearest manufacturer's representative.

3.6.11 Protective Coatings

- a. Manufacturer's descriptive literature, general.
- b. Maintenance, service and repair instructions.
- c. Manufacturer's name, model number, serial number, Federal Stock Number (if any).
- d. Name, address and telephone number of the nearest manufacturer's representative.
- e. Product standards compliance and the materials system data sheet.

3.6.12 Sample Connections

- a. Manufacturer's descriptive literature, general.
- b. Parts manual and recommended spare parts list.
- c. Maintenance, service and repair instructions.
- d. Operating Instructions.
- e. Manufacturer's name, model number, serial number, Federal Stock Number (if any).

- f. Name, address and telephone number of the nearest manufacturer's representative.

3.6.13 Pumps, Offload, Fuel Transfer, Water Draw-off Pump

- a. Manufacturer's descriptive literature, general.
- b. Parts manual and recommended spare parts list.
- c. Maintenance, service and repair instructions.
- d. Manufacturer's name, model number, serial number, Federal Stock Number (if any).
- e. Performance data at specified flow rates. Performance must include:
 - (1) Head developed, horsepower required and efficiency.
 - (2) Pump curves, flow and power requirements, efficiency, head and operating speed. Curves to show operating points at full range of operating conditions.
- f. Control wiring diagrams showing all terminations of conductors (and all control devices) labeled to permit identification in the field; part numbers of all control devices; normally open or normally closed; voltage of all control components and operational description.
- g. Plan and elevation views of equipment showing clearance required for maintenance and/or replacement.
- h. Name, address and telephone number of the nearest manufacturer's representative.
- i. Shipping and operating weights.
- j. Operating instructions.
- k. Factory run test curves indicating flow, head rpm, vibration amplitude and BHP.

3.6.14 Flexible Hoses

- a. Manufacturer's descriptive literature, general.
- b. Maintenance service and repair instructions.
- c. Manufacturer's name, model number, serial number.
- d. Name, address and telephone number of the nearest manufacturer's representative.

3.6.15 Control Valves

Submit for each type control valve specified

- a. Manufacturer's descriptive literature, general.
- b. Operational description of valve and control pilots.

- c. Description of valve assembly complete with parts list.
- d. Recommended spare parts list for main valve and pilot control systems.
- e. Instructions for trouble shooting.
- f. Maintenance, service and repair instructions.
- g. Manufacturer's name, model number and stock number.
- h. Operational Test Data.

3.6.16 Engine-Generator

- a. Internal and interconnecting wiring and control diagrams with data to explain detailed operation and control of the system or equipment.
- b. A control sequence describing startup', operation, and shutdown.
- c. Description of the function of each principal item of equipment.
- d. Installation and maintenance instructions.
- e. Safety precautions.
- f. Diagrams and illustrations.
- g. Testing methods.
- h. Performance data.
- i. Lubrication schedule including type, grade, temperature range, and frequency.
- j. Parts list: Provide list that clearly indicates sources of supply, recommended spare parts, and name of servicing organization.
- k. List qualified permanent servicing organizations for support of the equipment, including addresses and certified qualifications.

3.6.17 Fire Alarm and Fire Detecting System

- a. Manufacturer's descriptive literature, general.
- b. Parts manual.
- c. Maintenance, service and repair instructions.
- d. Operating Instructions.
- e. Drawing of component arrangement, schedule of components with sizes, types, and ratings, and wiring diagrams.
- f. Manufacturer's name, model number, serial number, Federal Stock Number (if any).
- g. Name, address and telephone number of the nearest manufacturer's representative.

3.6.18 Motor Control Center

- a. Internal and interconnecting wiring and control diagrams with data to explain detailed operation and control of the system or equipment.
- b. A control sequence describing startup, operation, and shutdown.
- c. Description of the function of each principal item of equipment.
- d. Installation and maintenance instructions.
- e. Safety precautions.
- f. Diagrams and illustrations.
- g. Parts list.
- h. Drawing of component arrangement, schedule of components with sizes, types, and ratings.
- i. Manufacturer's name, model number, serial number.
- j. Name, address and telephone number of the nearest manufacturer's representative.

3.6.19 Non-Automatic Transfer Switch

- a. Manufacturer's descriptive literature, general.
- b. Parts list.
- c. Maintenance, service and repair instructions.
- d. Operating Instructions.
- e. Drawing of component arrangement, schedule of components with sizes, types, and ratings, and wiring diagrams.
- f. Manufacturer's name, model number, serial number, Federal Stock Number (if any).
- g. Name, address and telephone number of the nearest manufacturer's representative.

3.6.20 Pump Control Panel (PCP)

- a. General description and specifications.
- b. Comprehensive discussion of both hardware and operating program.
- c. Installation and initial checkout procedures.
- d. Detailed electrical and logical description.
- e. Complete troubleshooting procedures, diagrams, and guidelines.
- f. Complete alignment and calibration procedures for components.
- g. Preventive maintenance requirements.

- h. Detailed system schematics, system field assembly drawings, and system component specifications and dimensions.
- i. Complete spare parts lists.
- j. Interface requirements and capabilities.
- k. Signal identification and timing diagrams.
- l. Complete as-built bill of materials, control drawings, schedules, and sequence of operations.
- m. Safety precautions.
- n. Control sequence describing start-up, operation, and shutdown. Control sequence must be integrated with startup and operation of the motor control center.
- o. Provide part list that clearly indicates sources of supply, recommended spare parts, and name of servicing organization.
- p. Manufacturer's name, address, and telephone number.
- q. Supplier name, manufacturer and version of all software including: PLC, desktop computer, laptop computer, and "alternate" desktop computer.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 78 24.00 20

FACILITY DATA WORKBOOK (FDW)

05/23

PART 1 GENERAL

- 1.1 DEFINITIONS AND ABBREVIATIONS
 - 1.1.1 Facility Data Workbook (FDW)
 - 1.1.2 Systems
 - 1.1.3 KTR
- 1.2 OPERATION AND MAINTENANCE MANUAL AND FACILITY DATA WORKBOOK MEETINGS
 - 1.2.1 Pre-Construction Meeting
- 1.3 SUBMITTAL SCHEDULING
 - 1.3.1 Facility Data Workbook (FDW), Progress Submittal
 - 1.3.2 Facility Data Workbook (FDW), Prefinal Submittal
 - 1.3.3 Facility Data Workbook (FDW), Final Submittal
- 1.4 UNITS OF MEASURE
- 1.5 SUBMITTALS

PART 2 PRODUCTS

- 2.1 eOMSI FACILITY DATA WORKBOOK

PART 3 EXECUTION

- 3.1 FIELD VERIFICATION

ATTACHMENTS:

Facility Data Workbook

-- End of Section Table of Contents --

SECTION 01 78 24.00 20

FACILITY DATA WORKBOOK (FDW)
05/23

PART 1 GENERAL

1.1 DEFINITIONS AND ABBREVIATIONS

1.1.1 Facility Data Workbook (FDW)

A Microsoft Excel file containing required facility information populated by the Contractor.

1.1.2 Systems

The words "system", "systems", and "equipment", when used in this document refer to as-built systems and equipment.

1.1.3 KTR

An abbreviation for "Contractor."

1.2 OPERATION AND MAINTENANCE MANUAL AND FACILITY DATA WORKBOOK MEETINGS

1.2.1 Pre-Construction Meeting

Meeting requirements are identified in Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.3 SUBMITTAL SCHEDULING

1.3.1 Facility Data Workbook (FDW), Progress Submittal

Submit the Progress submittal when construction is approximately 50 percent complete, to the Contracting Officer for approval. Provide Facility Data Workbook (Excel). Include the elements and portions of system construction completed up to this point.

The purpose of this submittal is to verify progress is in accordance with contract requirements as discussed during the Operation and Maintenance Manual and Facility Data Workbook Meetings. Field verify a portion of the eOMSI information in accordance with paragraph FIELD VERIFICATION.

1.3.2 Facility Data Workbook (FDW), Prefinal Submittal

Submit the 100 percent submittal of the FDW Prefinal Submittal to the Contracting Officer for approval within 90 calendar days of the Beneficial Occupancy Date (BOD). This submittal must provide a complete, working document that can be used to operate and maintain the facility. Any portion of the submittal that is incomplete or inaccurate requires the entire submittal to be returned for correction. Any discrepancies discovered during the Government's review of FDW Progress submittal must be corrected prior to the Prefinal submission.

1.3.3 Facility Data Workbook (FDW), Final Submittal

Submit completed Facility Data Workbook (Excel). The Final submittal is due at BOD. Any discrepancies discovered during the Government's review of the Prefinal eOMSI submittal, including the Field Verification, must be corrected prior to the Final eOMSI submission.

1.4 UNITS OF MEASURE

Provide FDW utilizing the units of measure used in the Government generated contract documents. Metric eOMSI must be in SI (System International) metric units exclusively.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-11 Closeout Submittals

Facility Data Workbook (FDW), Progress Submittal; G

Facility Data Workbook (FDW), Prefinal Submittal; G

Facility Data Workbook (FDW), Final Submittal; G

PART 2 PRODUCTS

2.1 eOMSI FACILITY DATA WORKBOOK

An initial, pre-edited draft of the Model & Facility Data Matrix tab within the Facility Data Workbook is attached to this Section. The Government will provide this Facility Data Workbook electronically to the Contractor upon award. Add, delete, and update Mastersystems, Systems, and Subsystems that may have changed during construction, or any items that may have been omitted or missed during design, at no additional cost to the Government. The following tabs are included in the eOMSI Facility Data File Workbook and serve the purpose stated:

- a. Instructions Tab: Instructions for completing Model & Facility Data Matrix Tab and KTR Facility Data File Tab. If a discrepancy exists between what is required in this Section and the Workbook, the instructions within the workbook take precedence.
- b. Model & Facility Data Matrix Tab: - The Matrix lists Required Facility Asset Fields for each SYSTEM and SUBSYSTEM. The Designer of Record selects SYSTEMS and SUBSYSTEMS that are within the project scope, which the Contractor needs to include and populate in KTR Facility Data File tab. The "Required Facility Asset Field Position Numbers," one through thirty-five, are pre-populated, and are not editable.
- c. Required Facility Asset Fields Tab: Defines the 35 Required Facility Asset Field Position Numbers used in Model and Facility Data Matrix and KTR Facility Data File tabs.
- d. KTR Sample Facility Data File Tab: Sample KTR eOMSI facility data file. This tab provides an example of the mandatory fields of

equipment installed by the Contractor, and populated in the KTR eOMSI Facility Data File Tab, along with their descriptions.

- e. KTR Facility Data File Tab: Required eOMSI facility data file deliverable provided to the Government. Provide a separate and unique new row for each facility component or piece of equipment installed. Coordinate with the Government's Contracting Officer's Representative and NAVFAC PW FMD for specific facility component naming convention.

PART 3 EXECUTION

3.1 FIELD VERIFICATION

Field verify Facility Data Workbook information with Contractor and Government personnel. Include the following personnel in this meeting: Contractor's Facility Data Workbook Preparer and Quality Control Manager, [Commissioning Authority](#), and the Government's Contracting Officer's Representative and NAVFAC PW FMD. Request, and provide, an eOMSI Field Verification Meeting no sooner than 14 calendar days after submission of the Progress FDW submittal, and another, no sooner than 14 calendar days after submission of the Prefinal FDW submittal. During this meeting, the Government and Contractor will verify that the Facility Data Workbook is complete and accurate.

Field verify that at least 5 Subsystems under each of the Mastersystems are accurate, for a total of 25 Subsystems. For each of these items, verify that the required facility asset field, as defined in the "Model & Facility Data Matrix" tab, contains the specified data and it is accurate (i.e., item description, manufacturer, model no., serial no.). 100 percent accuracy of eOMSI information is required for successful field verification. If data discrepancies are discovered amongst the 25 Subsystems verified, resubmit an updated eOMSI FDW, and request a make-up field verification meeting. At the make-up field verification meeting 25 new Subsystems and their associated required facility asset fields will be field verified; the 25 new Subsystems must be 100 percent accurate. Any discrepancies discovered must be corrected prior to next eOMSI Facility Data Workbook Submittal.

- (1) D20 - PLUMBING
- (2) D30 - HVAC
- (3) D50 - ELECTRICAL

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 41 00

DEMOLITION

08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 PROJECT DESCRIPTION
 - 1.2.1 Definitions
 - 1.2.1.1 Demolition
 - 1.2.1.2 Demolition Plan
 - 1.2.2 Demolition Plan
 - 1.2.3 General Requirements
- 1.3 ITEMS TO REMAIN IN PLACE
 - 1.3.1 Existing Construction Limits and Protection
 - 1.3.2 Trees
 - 1.3.3 Utility Service
 - 1.3.4 Facilities
- 1.4 BURNING
- 1.5 AVAILABILITY OF WORK AREAS
- 1.6 SUBMITTALS
- 1.7 QUALITY ASSURANCE
 - 1.7.1 Dust and Debris Control
- 1.8 PROTECTION
 - 1.8.1 Traffic Control Signs
 - 1.8.2 Protection of Personnel
- 1.9 RELOCATIONS
- 1.10 EXISTING CONDITIONS

PART 2 PRODUCTS

- 2.1 FILL MATERIAL

PART 3 EXECUTION

- 3.1 EXISTING FACILITIES TO BE REMOVED
 - 3.1.1 Structures
 - 3.1.2 Utilities and Related Equipment
 - 3.1.2.1 General Requirements
 - 3.1.2.2 Disconnecting Existing Utilities
 - 3.1.3 Chain Link Fencing
 - 3.1.4 Paving and Slabs
 - 3.1.5 Concrete
 - 3.1.6 Miscellaneous Metal
 - 3.1.7 Patching
 - 3.1.8 Mechanical Equipment and Fixtures
 - 3.1.8.1 Piping
 - 3.1.9 Electrical Equipment and Fixtures
 - 3.1.9.1 Fixtures
 - 3.1.9.2 Electrical Devices

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.1.9.3 Wiring Ducts or Troughs
- 3.1.9.4 Conduit and Miscellaneous Items
- 3.1.10 Items With Unique/Regulated Disposal Requirements
- 3.2 CONCURRENT EARTH-MOVING OPERATIONS
- 3.3 DISPOSITION OF MATERIAL
 - 3.3.1 Title to Materials
 - 3.3.2 Reuse of Materials and Equipment
 - 3.3.3 Salvaged Materials and Equipment
 - 3.3.4 Unsalvageable and Non-Recyclable Material
- 3.4 CLEANUP
- 3.5 DISPOSAL OF REMOVED MATERIALS
 - 3.5.1 Regulation of Removed Materials
 - 3.5.2 Burning on Government Property
 - 3.5.3 Removal to Spoil Areas on Government Property
 - 3.5.4 Removal from Government Property
- 3.6 REUSE OF SALVAGED ITEMS

-- End of Section Table of Contents --

SECTION 02 41 00

DEMOLITION
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO M 145 (1991; R 2012) Standard Specification for Classification of Soils and Soil-Aggregate Mixtures for Highway Construction Purposes

AASHTO T 180 (2017) Standard Method of Test for Moisture-Density Relations of Soils Using a 4.54-kg (10-lb) Rammer and a 457-mm (18-in.) Drop

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.6 (2006) Safety & Health Program Requirements for Demolition Operations - American National Standard for Construction and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety -- Safety and Health Requirements Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61 National Emission Standards for Hazardous Air Pollutants

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Demolition

Demolition is the process of tearing apart and removing any feature of a facility together with any related handling and disposal operations.

1.2.1.2 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

1.2.2 Demolition Plan

Prepare a [Demolition Plan](#) and submit proposed salvage, demolition, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials specified to be salvaged, coordination with other work in progress, a disconnection schedule of utility services, a detailed description of methods and equipment to be used for each operation and of the sequence of operations. [Identify components and materials to be salvaged for reuse or recycling with reference to paragraph Existing Facilities to be Removed.](#) [Append tracking forms for all removed materials indicating type, quantities, condition, destination, and end use.](#) Coordinate with Waste Management Plan in accordance with Section [01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL](#). Provide procedures for safe conduct of the work in accordance with [EM 385-1-1](#). Plan must be approved by Contracting Officer prior to work beginning.

1.2.3 General Requirements

Do not begin demolition until authorization is received from the Contracting Officer. [The work of this section is to be performed in a manner that maximizes the value derived from the salvage and recycling of materials.](#) The work includes demolition, salvage of identified items and materials, and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with [EM 385-1-1](#), Section 23, Demolition, and other applicable Sections.

1.3 ITEMS TO REMAIN IN PLACE

Comply with FAR 52.236-9 to protect existing vegetation, structures, equipment, utilities, and improvements. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, or demolition work performed under this contract. Do not overload structural elements. Provide new supports and reinforcement for existing construction weakened by demolition or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove dust, dirt, and debris from work areas daily.

1.3.2 Trees

Protect trees within the project site which might be damaged during demolition, and which are indicated to be left in place, by a [1.8 m](#) high fence. Erect and secure fence a minimum of [1.5 m](#) from the trunk of individual trees or follow the outer perimeter of branches or clumps of

trees. Replace any tree designated to remain that is damaged during the work under this contract with like-kind or as approved by the Contracting Officer.

1.3.3 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition operations. Prior to start of work, utilities serving each area of alteration or removal will be shut off by the Government and disconnected and sealed by the Contractor.

1.3.4 Facilities

Protect electrical and mechanical services and utilities. Where removal of existing utilities and pavement is specified or indicated, provide approved barricades, temporary covering of exposed areas, and temporary services or connections for electrical and mechanical utilities. Floors, roofs, walls, columns, pilasters, and other structural components that are designed and constructed to stand without lateral support or shoring, and are determined to be in stable condition, must remain standing without additional bracing, shoring, or lateral support until demolished, unless directed otherwise by the Contracting Officer. Ensure that no elements determined to be unstable are left unsupported and place and secure bracing, shoring, or lateral supports as may be required as a result of any cutting, removal, or demolition work performed under this contract.

1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted.

1.5 AVAILABILITY OF WORK AREAS

Areas in which the work is to be accomplished will be available at contract award.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Demolition Plan; G

Existing Conditions

1.7 QUALITY ASSURANCE

Notify the Contracting Officer in writing 10 working days prior to the commencement of work in accordance with 40 CFR 61, Subpart M. Comply with local hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in ASSP A10.6. Comply with the Environmental Protection Agency requirements specified. Use of explosives will not be permitted.

1.7.1 Dust and Debris Control

Prevent the spread of dust and debris and avoid the creation of a nuisance in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, flooding, or pollution.

1.8 PROTECTION

1.8.1 Traffic Control Signs

Where pedestrian and driver safety is endangered in the area of removal work, use traffic barricades with flashing lights. Anchor barricades in a manner to prevent displacement by wind. Notify the Contracting Officer prior to beginning such work.

1.8.2 Protection of Personnel

Before, during and after the demolition work continuously evaluate the condition of the site specific features or structure being demolished and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

1.9 RELOCATIONS

Perform the removal and reinstallation of relocated items as indicated with workmen skilled in the trades involved. Repair or replace items to be relocated which are damaged by the Contractor with new undamaged items as approved by the Contracting Officer.

1.10 EXISTING CONDITIONS

Before beginning any demolition work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer showing the condition of structures and other facilities adjacent to areas of alteration or removal. Photographs or electronic images with a minimum resolution of 3072 x 2304 pixels, capable of a print resolution of 300 dpi, will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundation walls, finish floor elevations, possible conflicting electrical conduits, plumbing lines, alarms systems, the location and extent of existing cracks and other damage and description of surface conditions that exist prior to starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results to the Contracting Officer.

PART 2 PRODUCTS

2.1 FILL MATERIAL

- a. Comply with excavating, backfilling, and compacting procedures for soils used as backfill material to fill voids, depressions or excavations resulting from demolition of structures. Provide fill

material consisting of waste products from demolition until all waste appropriate for this purpose is consumed.

- b. Proposed fill material must be sampled and tested by an approved soil testing laboratory, as follows:

Soil classification	AASHTO M 145
Moisture-density relations	AASHTO T 180, Method B or D

PART 3 EXECUTION

3.1 EXISTING FACILITIES TO BE REMOVED

Disassemble existing construction scheduled to be removed for reuse. Dismantled and removed materials are to be separated, set aside, and prepared as specified, and stored or delivered to a collection point for reuse, remanufacture, recycling, or other disposal, as specified. Designate materials for reuse onsite whenever possible.

3.1.1 Structures

- a. Locate demolition equipment throughout the structure and remove materials so as to not impose excessive loads to supporting walls, floors, or framing.

3.1.2 Utilities and Related Equipment

3.1.2.1 General Requirements

Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by the Contracting Officer. Do not interrupt existing utilities serving facilities occupied and used by the Government except when approved in writing and then only after temporary utility services have been approved and provided. Do not begin demolition work until all utility disconnections have been made. Shut off and cap utilities for future use, as indicated.

3.1.2.2 Disconnecting Existing Utilities

Remove existing utilities, as indicated and terminate in a manner conforming to the nationally recognized code covering the specific utility and approved by the Contracting Officer. When utility lines are encountered but are not indicated on the drawings, notify the Contracting Officer prior to further work in that area. Remove meters and related equipment and deliver to a location in accordance with instructions of the Contracting Officer.

3.1.3 Chain Link Fencing

Remove chain link fencing, gates and other related salvaged items scheduled for removal and transport to designated areas. Remove gates as whole units. Cut chain link fabric to 7 m lengths and store in rolls off the ground.

3.1.4 Paving and Slabs

Sawcut and remove concrete and asphaltic concrete paving and slabs including aggregate base as indicated. Move, grind and store pavement and slabs designated to be recycled and utilized in this project as directed by the Contracting Officer. Remove pavement and slabs not to be used in this project from the project site at Contractor's expense. If pavement remnant is 1 m or less, then remove and restore pavement.

3.1.5 Concrete

Saw concrete along straight lines to a depth of a minimum 50 mm. Make each cut in walls perpendicular to the face and in alignment with the cut in the opposite face. Break out the remainder of the concrete provided that the broken area is concealed in the finished work, and the remaining concrete is sound. At locations where the broken face cannot be concealed, grind smooth or saw cut entirely through the concrete. Sawcuts in existing concrete sidewalk, roads, concrete pads, and parking areas must be at the nearest existing expansion joint or weakened plane joint and at full depth.

3.1.6 Miscellaneous Metal

Salvage shop-fabricated items such as access doors and frames, steel gratings, metal ladders, wire mesh partitions, metal railings, metal windows and similar items as whole units. Salvage light-gage and cold-formed metal framing, such as steel studs, steel trusses, metal gutters, roofing and siding, metal toilet partitions, toilet accessories and similar items. Recycle scrap metal as part of demolition operations. Provide separate containers to collect scrap metal and transport to a scrap metal collection or recycling facility, in accordance with the Waste Management Plan.

3.1.7 Patching

Where removals leave holes and damaged surfaces exposed in the finished work, patch and repair these holes and damaged surfaces to match adjacent finished surfaces, using on-site materials when available. Where new work is to be applied to existing surfaces, perform removals and patching in a manner to produce surfaces suitable for receiving new work. Make finished surfaces of patched area flush with the adjacent existing surface and match the existing adjacent surface as closely as possible to texture and finish. Provide patching as specified and indicated, and include the following:

- a. Concrete: Completely fill holes and depressions, caused by previous physical damage or left as a result of removals in existing concrete to remain, with an approved concrete material, applied in accordance with specifications.

3.1.8 Mechanical Equipment and Fixtures

3.1.8.1 Piping

Disconnect piping at unions, flanges and valves, and fittings as required to reduce the pipe into straight lengths for practical storage. Store salvaged piping according to size and type. If the piping that remains can become pressurized due to upstream valve failure, attach end caps, blind flanges, or other types of plugs or fittings with a pressure gage

and bleed valve to the open end of the pipe to ensure positive leak control. Carefully dismantle piping that previously contained gas, gasoline, oil, or other dangerous fluids, with precautions taken to prevent injury to persons and property. Store piping outdoors until all fumes and residues are removed. Box prefabricated supports, hangers, plates, valves, and specialty items according to size and type. Wrap sprinkler heads individually in plastic bags before boxing. Classify piping not designated for salvage, or not reusable, as scrap metal.

3.1.9 Electrical Equipment and Fixtures

Salvage motors, motor controllers, and operating and control equipment that are attached to the driven equipment. Salvage wiring systems and components. Box loose items and tag for identification. Disconnect primary, secondary, control, communication, and signal circuits at the point of attachment to their distribution system.

3.1.9.1 Fixtures

Remove and salvage electrical fixtures. Salvage unprotected glassware from the fixture and salvage separately. Salvage incandescent, mercury-vapor, and fluorescent lamps and fluorescent ballasts manufactured prior to 1978, boxed and tagged for identification, and protected from breakage.

3.1.9.2 Electrical Devices

Remove and salvage switches, switchgear, transformers, conductors including wire and nonmetallic sheathed and flexible armored cable, regulators, meters, instruments, plates, circuit breakers, panelboards, outlet boxes, and similar items. Box and tag these items for identification according to type and size.

3.1.9.3 Wiring Ducts or Troughs

Remove and salvage wiring ducts or troughs. Dismantle plug-in ducts and wiring troughs into unit lengths. Remove plug-in or disconnecting devices from the busway and store separately.

3.1.9.4 Conduit and Miscellaneous Items

Salvage conduit except where embedded in concrete or masonry. Consider corroded, bent, or damaged conduit as scrap metal. Sort straight and undamaged lengths of conduit according to size and type. Classify supports, knobs, tubes, cleats, and straps as debris to be removed and disposed.

3.1.10 Items With Unique/Regulated Disposal Requirements

Remove and dispose of items with unique or regulated disposal requirements in the manner dictated by law or in the most environmentally responsible manner.

3.2 CONCURRENT EARTH-MOVING OPERATIONS

Do not begin excavation, filling, and other earth-moving operations that are sequential to demolition work in areas occupied by structures to be demolished until all demolition in the area has been completed and debris removed. Fill holes and other hazardous openings.

3.3 DISPOSITION OF MATERIAL

3.3.1 Title to Materials

Except for salvaged items, all materials and equipment removed and not reused or salvaged, become the property of the Contractor and must be removed from Government property. Materials approved for storage by the Contracting Officer must be removed before completion of the contract. Title to materials resulting from demolition, and materials and equipment to be removed, is vested in the Contractor upon approval by the Contracting Officer. The Government will not be responsible for the condition or loss of, or damage to, such property after contract award. Showing for sale or selling materials and equipment on site is prohibited.

3.3.2 Reuse of Materials and Equipment

Remove and store materials and equipment to be reused or relocated to prevent damage, and reinstall as the work progresses. Coordinate the re-use of materials and equipment with the re-use requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture re-use of materials in the diversion calculations for the project.

3.3.3 Salvaged Materials and Equipment

Remove materials and equipment that are indicated to be salvaged or are to be removed by the Contractor and that are to remain the property of the Government, and deliver to a storage site.

- a. Salvage items and material to the maximum extent possible.
- b. Store all materials salvaged for the Contractor as approved by the Contracting Officer and remove from Government property before completion of the contract. Coordinate the salvaged materials with tracking requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture salvaged materials in the diversion calculations for the project.
- c. Remove salvaged items to remain the property of the Government in a manner to prevent damage, and packed or crated to protect the items from damage while in storage or during shipment. Items damaged during removal or storage must be repaired or replaced to match existing items. Properly identify the contents of containers.
- d. Remove the following items reserved as property of the using service prior to commencement of work under this contract: signs, concrete, barriers, vinyl fencing, bus stop shelter, and guardrail.

3.3.4 Unsalvageable and Non-Recyclable Material

Dispose of unsalvageable and non-recyclable noncombustible and noncombustible material per Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS.

3.4 CLEANUP

Remove debris and rubbish from project site and similar excavations. Remove and transport the debris in a manner that prevents spillage on

streets or adjacent areas. Apply local regulations regarding hauling and disposal.

3.5 DISPOSAL OF REMOVED MATERIALS

3.5.1 Regulation of Removed Materials

Dispose of debris, rubbish, scrap, and other nonsalvageable materials resulting from removal operations with all applicable local regulations as contractually specified.

3.5.2 Burning on Government Property

Burning of materials removed from demolished structures will not be permitted on Government property.

3.5.3 Removal to Spoil Areas on Government Property

Transport noncombustible materials removed from demolished structures to designated spoil areas on Government property.

3.5.4 Removal from Government Property

Transport waste materials removed from demolished structures, except waste soil, from Government property for legal disposal. Dispose of waste soil per Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS.

3.6 REUSE OF SALVAGED ITEMS

Recondition salvaged materials and equipment designated for reuse before installation. Replace items damaged during removal and salvage operations or restore them as necessary to usable condition.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 61 13

EXCAVATION AND HANDLING OF CONTAMINATED MATERIAL

02/10, CHG 1: 02/21

PART 1 GENERAL

- 1.1 MEASUREMENT AND PAYMENT
 - 1.1.1 Measurement
 - 1.1.2 Payment
 - 1.1.2.1 Excavation and Transportation
 - 1.1.2.2 Backfilling
 - 1.1.2.3 Stockpiling
- 1.2 REFERENCES
- 1.3 DESCRIPTION OF WORK
 - 1.3.1 Scheduling
 - 1.3.2 Work Plan
 - 1.3.3 Other Submittal Requirements
- 1.4 SUBMITTALS
- 1.5 REGULATORY REQUIREMENTS
 - 1.5.1 Permits and Licenses
 - 1.5.2 Air Emissions

PART 2 PRODUCTS

- 2.1 SPILL RESPONSE MATERIALS
- 2.2 BACKFILL

PART 3 EXECUTION

- 3.1 SURVEYS
- 3.2 EXISTING STRUCTURES AND UTILITIES
- 3.3 CLEARING
- 3.4 CONTAMINATED MATERIAL REMOVAL
 - 3.4.1 Excavation
 - 3.4.2 Shoring
 - 3.4.3 Dewatering
- 3.5 CONFIRMATION SAMPLING AND ANALYSIS
- 3.6 CONTAMINATED MATERIAL STORAGE
 - 3.6.1 Stockpiles
 - 3.6.2 Roll-Off Units
 - 3.6.3 Liquid Storage
- 3.7 SAMPLING
 - 3.7.1 Sampling of Stored Material
- 3.8 SPILLS
- 3.9 BACKFILLING
 - 3.9.1 Confirmation Test Results
- 3.10 DISPOSAL REQUIREMENTS
- 3.11 CLOSURE REPORT

-- End of Section Table of Contents --

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

SECTION 02 61 13

EXCAVATION AND HANDLING OF CONTAMINATED MATERIAL
02/10, CHG 1: 02/21

PART 1 GENERAL

1.1 MEASUREMENT AND PAYMENT

1.1.1 Measurement

Measurement for excavation and onsite transportation shall be based on the actual number of cubic **meters** of contaminated material in-place prior to excavation. Determination of the volume of contaminated material excavated shall be based on cross-sectional volume determination reflecting the differential between the original elevations of the top of the contaminated material and the final elevations after removal of the contaminated material. Measurement for backfilling of excavated areas shall be based on in-place cubic **meters** of compacted fill. Measurement for construction of stockpile areas shall be based on the number of square **meters** of stockpile liner constructed.

1.1.2 Payment

1.1.2.1 Excavation and Transportation

Compensation for excavation and onsite transportation of contaminated material will be paid as a unit cost. This unit cost shall include any other items incidental to excavation and handling not defined as having a specific unit cost.

1.1.2.2 Backfilling

Compensation for backfill soil, transportation of backfill, backfill soil conditioning, backfilling, compaction, and geotechnical testing will be paid as a single unit cost.

1.1.2.3 Stockpiling

Compensation for construction of stockpile areas will be paid for as a unit cost. This unit cost shall include all aspects of grading, preparation, handling, placement, maintenance, removal, treatment, and disposal of stockpile cover materials and liner materials and all other items incidental to construction of stockpiles.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D2487

(2017; E 2020) Standard Practice for
Classification of Soils for Engineering
Purposes (Unified Soil Classification
System)

ASTM D5434

(2012) Field Logging of Subsurface
Explorations of Soil and Rock

MINISTERIO DE LA PRESIDENCIA (MP)

RD 9/2005

(2005) Real Decreto 9/2005, de 14 de
enero, por el que se establece la relación
de actividades potencialmente
contaminantes del suelo y los criterios y
estándares para la declaración de suelos
contaminados.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1

(2014) Safety -- Safety and Health
Requirements Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926

Safety and Health Regulations for
Construction

1.3 DESCRIPTION OF WORK

The work consists of excavation, temporary storage and disposal of contaminated material. Approximate locations of contaminated material are indicated in the report of the assessment completed by the Government. This report may be available to the contractor during the bidding period if requested to the Contracting Officer. Characterization data on the nature and extent of the contaminated material is shown in Appendix A. Subsurface conditions are shown on the drawings. Submit a [Work Plan](#) as specified below. Notify the Contracting Officer within 24 hours, and before excavation, if contaminated material is discovered that has not been previously identified or if other discrepancies between data provided and actual field conditions are discovered. Backfill material is not available onsite. Ground water is approximately 2 [meters](#) below pre-excavation ground surface, but it may be found at shallower elevation during construction. Required sampling and chemical analysis shall be conducted in accordance with applicable facility requirements to comply with [RD 9/2005](#).

1.3.1 Scheduling

Notify the Contracting Officer 10 calendar days prior to the start of excavation of contaminated material. The Contractor shall be responsible for contacting regulatory agencies in accordance with the applicable reporting requirements.

1.3.2 Work Plan

Submit a Work Plan within 30 calendar days after notice to proceed. No work at the site, with the exception of site inspections and surveys, shall be performed until the Work Plan is approved. Allow 30 calendar days in the schedule for the Government's review. No adjustment for time or money will be made if resubmittals of the Work Plan are required due to deficiencies in the plan. At a minimum, the Work Plan shall include:

- a. Schedule of activities.

- b. Method of excavation and equipment to be used.
- c. Shoring or side-wall slopes proposed.
- d. Dewatering plan.
- e. Storage methods and locations for liquid and solid contaminated material.
- f. Borrow sources and haul routes.
- g. Decontamination procedures per RD 9/2005.
- h. Spill contingency plan.

1.3.3 Other Submittal Requirements

Submit separate cross-sections of each area before and after excavation and after backfilling, test results, and three copies of the Closure Report within 14 calendar days of work completion at the site.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Work Plan; G

SD-02 Shop Drawings

Surveys

SD-06 Test Reports

Closure Report; G

1.5 REGULATORY REQUIREMENTS

1.5.1 Permits and Licenses

Obtain required federal, state, and local permits for excavation and storage of contaminated material. Permits shall be obtained at no additional cost to the Government.

1.5.2 Air Emissions

Air emissions shall be monitored and controlled in accordance with local requirements.

PART 2 PRODUCTS

2.1 SPILL RESPONSE MATERIALS

Provide appropriate spill response materials including, but not limited to the following: containers, adsorbents, shovels, and personal protective equipment. Spill response materials shall be available at all times when contaminated materials/wastes are being handled or transported. Spill response materials shall be compatible with the type of materials and contaminants being handled.

2.2 BACKFILL

Backfill material shall be obtained from offsite sources approved by the Contracting Officer. Backfill shall be classified in accordance with [ASTM D2487](#) as GW, GP, GM, SW, SP, [or](#) SM and shall be free from roots and other organic matter, trash, debris, snow, ice or frozen materials.

PART 3 EXECUTION

3.1 SURVEYS

Perform surveys immediately prior to and after excavation of contaminated material to determine the volume of contaminated material removed.

3.2 EXISTING STRUCTURES AND UTILITIES

No excavation shall be performed until site utilities have been field located. Take the necessary precautions to ensure no damage occurs to existing structures and utilities. Damage to existing structures and utilities resulting from the Contractor's operations shall be repaired at no additional cost to the Government. Utilities encountered that were not previously shown or otherwise located shall not be disturbed without approval from the Contracting Officer.

3.3 CLEARING

Clearing shall be performed to the limits shown on the drawings in accordance with Section [02 41 00](#) DEMOLITION.

3.4 CONTAMINATED MATERIAL REMOVAL

3.4.1 Excavation

Areas of contamination shall be excavated to the depth and extent shown on the drawings and not more than [60 mm](#) beyond the depth and extent shown on the drawings unless directed by the Contracting Officer. Excavation shall be performed in a manner that will limit spills and the potential for contaminated material to be mixed with uncontaminated material. An excavation log describing visible signs of contamination encountered shall be maintained for each area of excavation. Excavation logs shall be prepared in accordance with [ASTM D5434](#).

3.4.2 Shoring

If workers must enter the excavation, it shall be evaluated, shored, sloped or braced as required by [EM 385-1-1](#) and [29 CFR 1926](#) section 650.

3.4.3 Dewatering

Surface water shall be diverted to prevent entry into the excavation. Dewatering shall be limited to that necessary to assure adequate access, a safe excavation, prevent the spread of contamination, and to ensure that compaction requirements can be met. No dewatering shall be performed without prior approval of the Contracting Officer.

3.5 CONFIRMATION SAMPLING AND ANALYSIS

The Contracting Officer shall be present to inspect the removal of contaminated material from each site. After all material suspected of

being contaminated has been removed, the excavation shall be examined for evidence of contamination. If the excavation appears to be free of contamination, field analysis shall be used to determine the presence of contamination using a real time vapor monitoring instrument. Excavation of additional material shall be as directed by the Contracting Officer.

3.6 CONTAMINATED MATERIAL STORAGE

Material shall be placed in temporary storage immediately after excavation. The following paragraphs describe acceptable methods of material storage. Storage units shall be in good condition and constructed of materials that are compatible with the material or liquid to be stored. If multiple storage units are required, each unit shall be clearly labeled with an identification number and a written log shall be kept to track the source of contaminated material in each temporary storage unit.

3.6.1 Stockpiles

Stockpiles shall be constructed to isolate stored contaminated material from the environment. Stockpiles shall be constructed to include:

- a. A chemically resistant geomembrane liner free of holes and other damage. Non-reinforced geomembrane liners shall have a minimum thickness of 0.5 mm. Scrim reinforced geomembrane liners shall have a minimum weight of 20 kg/100 square m. The ground surface on which the geomembrane is to be placed shall be free of rocks greater than 13 mm in diameter and any other object which could damage the membrane.
- b. Geomembrane cover free of holes or other damage to prevent precipitation from entering the stockpile. Non-reinforced geomembrane covers shall have a minimum thickness of 0.25 mm. Scrim reinforced geomembrane covers shall have a minimum weight of 13 kg/100 square m. The cover material shall be extended over the berms and anchored or ballasted to prevent it from being removed or damaged by wind.
- c. Berms surrounding the stockpile, a minimum of 300 mm in height. Vehicle access points shall also be bermed.
- d. The liner system shall be sloped to allow collection of leachate. Storage and removal of liquid which collects in the stockpile, in accordance with paragraph Liquid Storage.

3.6.2 Roll-Off Units

Roll-off units used to temporarily store contaminated material shall be water tight. A cover shall be placed over the units to prevent precipitation from contacting the stored material. Liquid which collects inside the units shall be removed and stored in accordance with paragraph Liquid Storage.

3.6.3 Liquid Storage

Liquid collected from excavations and stockpiles shall be temporarily stored in 220 L barrels. Liquid storage containers shall be water-tight.

3.7 SAMPLING

3.7.1 Sampling of Stored Material

Samples of stored material shall be collected at a frequency based on applicable local requirements. Samples shall be tested for parameters in which contamination is suspected.

Stored material with contaminant levels that exceed the action levels shall be managed offsite. Analyses for contaminated material to be taken to an offsite facility shall conform to local criteria as well as to the requirements of the facility. Documentation of all analyses performed shall be furnished to the Contracting Officer. Additional sampling and analyses to the extent required by the approved offsite treatment, storage or disposal (TSD) facility shall be the responsibility of the Contractor and shall be subject to approval by the Contracting Officer.

3.8 SPILLS

In the event of a spill or release of a hazardous substance. Immediate containment actions shall be taken to minimize the effect of any spill or leak. Cleanup shall be in accordance with applicable federal, state, and local regulations. As directed by the Contracting Officer, additional sampling and testing shall be performed to verify spills have been cleaned up. Spill cleanup and testing shall be done at no additional cost to the Government.

3.9 BACKFILLING

3.9.1 Confirmation Test Results

Excavations shall be backfilled immediately after all contaminated materials have been removed and confirmation test results have been approved. Backfill shall be placed and compacted to the lines and grades shown on the drawings.

3.10 DISPOSAL REQUIREMENTS

Offsite disposal of contaminated material shall be in accordance with applicable Spanish national and regional (Junta de Andalucia) requirements.

3.11 CLOSURE REPORT

Submit three copies of a Closure Report within 14 calendar days of completing work at the site. The report shall be labeled with the contract number, project name, location, date, name of general Contractor, and the Corps of Engineers District contracting for the work. The Closure Report shall include the following information as a minimum:

- a. A cover letter signed by a responsible company official certifying that all services involved have been performed in accordance with the terms and conditions of the contract documents and regulatory requirements.
- b. A narrative report including, but not limited to, the following:
 - (1) site conditions, ground water elevation, and cleanup criteria;
 - (2) excavation logs;

- (3) field screening readings;
 - (4) quantity of materials removed from each area of contamination;
 - (5) quantity of water/product removed during dewatering;
 - (6) sampling locations and sampling methods;
 - (7) sample collection data such as time of collection and method of preservation;
 - (8) sample chain-of-custody forms; and
 - (9) source of backfill.
- c. Copies of all chemical and physical test results.
- d. Copies of all manifests and land disposal restriction notifications.
- e. Copies of all certifications of final disposal signed by the responsible disposal facility official.
- f. Waste profile sheets.
- g. Scale drawings showing limits of each excavation, limits of contamination, known underground utilities within **15 m** of excavation, sample locations, and sample identification numbers. On-site stockpile, storage, treatment, loading, and disposal areas shall also be shown on the drawings.
- h. Progress Photographs. Color photographs shall be used to document progress of the work. A minimum of four views of the site showing the location of the area of contamination, entrance/exit road, and any other notable site conditions shall be taken before work begins. After work has been started, activities at each work location shall be photographically recorded daily. Photographs shall be a minimum of **76.2 by 127.0 mm** and shall include:
- (1) Soil removal and sampling.
 - (2) Dewatering operations.
 - (3) Unanticipated events such as spills and the discovery of additional contaminated material.
 - (4) Contaminated material/water storage, handling, treatment, and transport.
 - (5) Site or task-specific employee respiratory and personal protection.
 - (6) Fill placement and grading.
 - (7) Post-construction photographs. After completion of work at each site, take a minimum of four views of each excavation site.

A digital version of all photos shown in the report shall be included with the Closure Report. Photographs shall be a minimum of **76 by 127 mm** and shall be mounted back-to-back in double face plastic sleeves punched to fit standard three ring binders. Each print shall have an information box attached. The box shall be typewritten and arranged as follows:

Project Name:	Direction of View:
Location:	Date/Time:
Photograph No.:	Description of View:

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 81 00

TRANSPORTATION AND DISPOSAL OF HAZARDOUS MATERIALS

11/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Hazardous Material
 - 1.2.2 Hazardous Waste
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Transportation and Disposal Coordinator
 - 1.4.2 Training
 - 1.4.3 Certification
 - 1.4.4 Laws and Regulations Requirements

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Packagings
 - 2.1.2 Markings
 - 2.1.3 Labeling
 - 2.1.4 Placards
 - 2.1.5 Spill Response Materials
- 2.2 EQUIPMENT AND TOOLS

PART 3 EXECUTION

- 3.1 HAZARDOUS WASTE MANAGEMENT PLAN
- 3.2 ONSITE HAZARDOUS WASTE MANAGEMENT
 - 3.2.1 Hazardous Waste Classification
- 3.3 OFFSITE HAZARDOUS WASTE MANAGEMENT
 - 3.3.1 Treatment, Storage, and Disposal Facility and Transporter
 - 3.3.2 Facility Status Information
 - 3.3.3 Shipping Documents and Packagings Certification
 - 3.3.4 Transportation
 - 3.3.5 Treatment and Disposal of Hazardous Wastes
- 3.4 RADIOACTIVE MATERIALS MANAGEMENT
 - 3.4.1 Identification of Proper Shipping Names
 - 3.4.2 Packaging, Labeling, and Marking
 - 3.4.3 Shipping Documents
 - 3.4.3.1 Asbestos Waste Shipment Documents
 - 3.4.3.2 Other Hazardous Material Shipment Documents
- 3.5 SPECIAL REQUIREMENTS FOR ASBESTOS WASTES
- 3.6 WASTE MINIMIZATION
- 3.7 RECORDKEEPING
- 3.8 SPILL RESPONSE
- 3.9 EMERGENCY CONTACTS

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

-- End of Section Table of Contents --

SECTION 02 81 00

TRANSPORTATION AND DISPOSAL OF HAZARDOUS MATERIALS
11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61	National Emission Standards for Hazardous Air Pollutants
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 266	Standards for the Management of Specific Hazardous Wastes and Specific Types of Hazardous Waste Management Facilities
40 CFR 279	Standards for the Management of Used Oil
40 CFR 300	National Oil and Hazardous Substances Pollution Contingency Plan
40 CFR 302	Designation, Reportable Quantities, and Notification
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
49 CFR 178	Specifications for Packagings

1.2 DEFINITIONS

1.2.1 Hazardous Material

A substance or material which has been determined by the Secretary of Transportation to be capable of posing an unreasonable risk to health, safety, and property when transported in commerce, and which has been so designated pursuant to the Hazardous Materials Transportation Act, 49 U.S.C. Appendix Section 1801 et seq. The term includes materials designated as hazardous materials under the provisions of 49 CFR 172, Sections .101 and .102 and materials which meet the defining criteria for hazard classes and divisions in 49 CFR 173. EPA designated hazardous wastes are also hazardous materials.

1.2.2 Hazardous Waste

A waste which meets criteria established in RCRA or specified by the EPA in 40 CFR 261 or which has been designated as hazardous by a RCRA authorized state program.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Packaging Notifications
Hazardous Waste Management Plan; G
Onsite Hazardous Waste Management; G

SD-06 Test Reports

Recordkeeping; G
Exception Report; G
Spill Response

SD-07 Certificates

Transportation and Disposal Coordinator; G
Training; G
Certification
Shipping Documents and Packagings Certification; G
Security Plan
Certificates of Disposal
Waste Minimization

1.4 QUALITY ASSURANCE

1.4.1 Transportation and Disposal Coordinator

Designate, by position and title, one person to act as the Transportation and Disposal Coordinator (TDC) for this contract. The TDC must serve as the single point of contact for all environmental regulatory matters and have overall responsibility for total environmental compliance at the site including, but not limited to, accurate identification and classification of hazardous waste and hazardous materials; determination of proper

shipping names; identification of marking, labeling, packaging and placarding requirements; completion of waste profiles, hazardous waste manifests, asbestos waste shipment records, PCB manifests, bill of lading, exception and discrepancy reports; and all other environmental documentation. The TDC must have, at a minimum, five years of specialized experience in the management and transportation of hazardous waste and have been certified by the Junta de Andalucia Regulator.

1.4.2 Training

Hazardous materials employees must be trained, tested, and certified to safely and effectively carry out their assigned duties in accordance with host nation requirements. Employees transporting hazardous materials or preparing hazardous materials for transportation, including samples, must be trained, tested, and certified in accordance with Junta de Andalucia, including security awareness and any applicable security plans.

1.4.3 Certification

The hazardous materials transporter must possess a current certificate of registration issued by Junta de Andalucia. Submit copies of the certificates or written statements certifying exemption from these requirements.

1.4.4 Laws and Regulations Requirements

Comply with Spanish national and regional (Junta de Andalucia) laws and regulations which are applicable. These requirements are amended frequently and compliance with amendments is required as they become effective. Notify the Contracting Officer immediately if compliance exceeds the scope of work or conflicts with specific requirements of the contract.

PART 2 PRODUCTS

2.1 MATERIALS

Provide all the materials required for the packaging, labeling, marking, placarding and transportation of hazardous wastes and hazardous materials in conformance with host nation requirements. Details in this specification must not be construed as establishing the limits of the Contractor's responsibility.

2.1.1 Packagings

Provide bulk and non-bulk containers for packaging hazardous materials/wastes consistent with the authorizations referenced in the Hazardous Materials Table in 49 CFR 172, Section .101, Column 8. Bulk and non-bulk packaging must meet the corresponding specifications in 49 CFR 173 referenced in the Hazardous Materials Table, 49 CFR 172, Section .101. Packaging must conform to the general packaging requirements of Subpart B of 49 CFR 173, to the requirements of 49 CFR 178 at the specified packing group performance level, to the requirements of special provisions of column 7 of the Hazardous Materials Table in 49 CFR 172, Section .101, and be compatible with the material to be packaged as required by 40 CFR 262. Also provide other packaging related materials such as materials used to cushion or fill voids in overpacked containers. The hazardous materials being packaged must not react dangerously with, decompose or ignite the sorbent packaging materials. Additionally, sorbents used to treat free

liquids to be disposed of in landfills must be non-biodegradable as specified in 40 CFR 264, Section .314. In addition, packaging notifications will be provided to the Government in accordance with 49 CFR 172, Section .178.2(c) regarding type and dimensions of closures, including gaskets, needed to satisfy performance test requirements.

2.1.2 Markings

Provide markings for each hazardous material/waste package, freight container, and transport vehicle consistent with host nation requirements and the requirements of 49 CFR 172, Subpart D and 40 CFR 262, Section .32 (for hazardous waste), 40 CFR 61, Section .149(d) (for asbestos). Markings must withstand a 180 day exposure to conditions reasonably expected to be encountered during container storage and transportation, without deterioration or substantial color change.

2.1.3 Labeling

Provide primary and subsidiary labels for hazardous materials/wastes consistent with the requirements in the Hazardous Materials Table in 49 CFR 172, Section .101, Column 6. Labels must meet design specifications required by 49 CFR 172, Subpart E including size, shape, color, printing, and symbol requirements. Labels must be durable weather resistant and withstanding a 180 day exposure to conditions reasonably expected to be encountered during container storage and transportation, without deterioration or substantial color change.

2.1.4 Placards

For each offsite shipment of hazardous material/waste, provide primary and subsidiary placards consistent with the requirements of 49 CFR 172, Subpart F. Provide placards for each side and each end of bulk packaging, freight containers, transport vehicles, and rail cars requiring such placarding. Placards may be plastic, metal, or other material capable of withstanding, without deterioration, a 30 day exposure to open weather conditions and must meet design requirements specified in 49 CFR 172, Subpart F.

2.1.5 Spill Response Materials

Provide spill response materials including, but not limited to, containers, adsorbent, shovels, and personal protective equipment. Spill response materials must be available at all times when hazardous materials/wastes are being handled or transported. Spill response materials must be compatible with the type of material being handled.

2.2 EQUIPMENT AND TOOLS

Provide miscellaneous equipment and tools necessary to handle hazardous materials and hazardous wastes in a safe and environmentally sound manner.

PART 3 EXECUTION

3.1 HAZARDOUS WASTE MANAGEMENT PLAN

Prepare a Hazardous Waste Management Plan detailing the manner in which hazardous wastes will be managed and describing the types and volumes of hazardous wastes anticipated to be managed. The plan must address both onsite and offsite hazardous waste management. Describe the methods to be

used to ensure accurate piece counts or weights of shipments; describe waste minimization methods; identify and describe facilities to be used for treatment, storage, and disposal (TSD); identify areas onsite where hazardous wastes are to be handled; and identify whether transfer facilities are to be used; and if so, how the wastes will be tracked to ultimate disposal. Submit the plan to the Contracting Officer for approval prior to start of work. Submit written documentation of weekly hazardous waste inspections on a monthly basis.

3.2 ONSITE HAZARDOUS WASTE MANAGEMENT

Coordinate the onsite management of all hazardous materials and waste with the installation environmental function and the Contracting Officer. The Contractor is responsible for ensuring compliance with Federal, state, and local hazardous waste laws and regulations and verifying those requirements when preparing reports, waste shipment records, hazardous waste manifests, or other documents. Identify hazardous wastes using criteria set forth in 40 CFR 261 or applicable state and local laws, regulations, and ordinances. Comply with generator requirements in 40 CFR 262 and applicable state or local law or regulations when accumulating hazardous waste onsite. Onsite accumulation times must be restricted to applicable time frames referenced in 40 CFR 262, Section .34 and applicable state or local law or regulation. Only use containers in good condition and compatible with the waste to be stored. Ensure containers are closed except when adding or removing waste, and immediately mark all hazardous waste containers with the words "hazardous waste" and other information required by 40 CFR 262, Section .32 and applicable state or local law or regulation as soon as the waste is containerized. Inspect containers for signs of deterioration and for responding to any spills or leaks. Inspect all hazardous waste areas weekly and provide written documentation of the inspection. Include date and time of inspection, name of individual conducting the inspection, problems noted, and corrective actions taken on the inspection logs.

3.2.1 Hazardous Waste Classification

Identify, in consultation with the Contracting Officer, all waste codes applicable to each hazardous waste stream based on requirements in 40 CFR 261 or applicable state or local law or regulation. Submit waste profiles, analyses, classification and treatment standards information to Contracting Officer for review and approval.

3.3 OFFSITE HAZARDOUS WASTE MANAGEMENT

Coordinate the off site transfer of all hazardous materials and waste with the installation environmental function and the Contracting Officer.

3.3.1 Treatment, Storage, and Disposal Facility and Transporter

Provide the Contracting Officer with names, locations, and telephone numbers of TSD facilities and transporters. This information must be contained in the Hazardous Waste Management Plan and be approved by the Contracting Officer prior to waste disposal.

3.3.2 Facility Status Information

Facilities receiving hazardous waste must be permitted in accordance with host nation requirements. Submit this information to the Contracting Officer in the Hazardous Waste Management Plan.

3.3.3 Shipping Documents and Packagings Certification

Prior to shipment of any hazardous material offsite and a minimum of 14 days prior to anticipated pickup, provide for review written certification to the Contracting Officer that hazardous materials have been properly packaged, labeled, and marked in accordance with Department of Transportation and EPA requirements. Furnish designated disposal facility packaging assurances not later than 35 days after acceptance of the shipment. The Contractor's TDC must also provide written certification regarding waste minimization efforts documenting that efforts have been taken to reduce the volume and toxicity of waste to the degree economically practicable and that the method of treatment, storage, or disposal selected minimizes threats to human health and the environment.

3.3.4 Transportation

Prior to conducting hazardous materials activities, the Contractor responsible for pre-transportation activities must either certify to the Government that a [Security Plan](#) is in place which meets the requirements of [49 CFR 172](#), Subpart I or in the event that the types or amounts of hazardous materials are excluded from the security planning requirements, a written statement to that effect detailing the basis for the exception. Use manifests for transporting hazardous wastes as required by [40 CFR 263](#) or applicable state or local law or regulation. Transportation must comply with all host nation requirements and the requirements in the Department of Transportation referenced regulations in the 49 CFR series. Prepare hazardous waste manifests for each shipment of hazardous waste shipped offsite. Complete manifests using instructions in [40 CFR 262](#), Subpart B and applicable state or local law or regulation. Submit manifests and waste profiles to Contracting Officer for review and approval.

3.3.5 Treatment and Disposal of Hazardous Wastes

Coordinate any off site shipments of hazardous materials or hazardous wastes with the installation environmental function. Initial, or satellite hazardous waste accumulation is limited to 55 gallons (or 1 quart of acutely hazardous waste). Once a waste stream exceeds 55 gallons, it must be transferred to an on-site 90 day (180 day small quantity generator) accumulation area, or a permitted hazardous waste treatment, storage or disposal facility within three days. Ship hazardous wastes only to facilities which are properly permitted to accept the hazardous waste or operating under interim status. Propose TSD facilities via submission of the Hazardous Waste Management Plan, subject to the approval of the Contracting Officer. Submit [Certificates of Disposal](#) documenting the ultimate disposal, destruction or placement of hazardous wastes and asbestos within 180 days of initial shipment. Receipt of these certificates will be required for final payment.

3.4 RADIOACTIVE MATERIALS MANAGEMENT

3.4.1 Identification of Proper Shipping Names

Use [49 CFR 172](#), Section .101 to identify proper shipping names for each hazardous material (including hazardous wastes) to be shipped offsite. Submit proper shipping names to the Contracting Officer in the form of draft shipping documents for review and approval.

3.4.2 Packaging, Labeling, and Marking

Package, label, and mark hazardous materials/wastes using the specified materials and in accordance with the referenced authorizations. Mark each container of hazardous waste of 416 L or less with the following:

"HAZARDOUS WASTE - Federal Law Prohibits Improper Disposal.

If found, contact the nearest police or public safety authority.

Generator's name _____

Manifest Document Number _____".

3.4.3 Shipping Documents

Ensure that each shipment of hazardous material sent offsite is accompanied by properly completed shipping documents. This includes shipments of samples that may potentially meet the definition of a Department of Transportation regulated hazardous material.

3.4.3.1 Asbestos Waste Shipment Documents

Prepare waste shipment records, as required by 40 CFR 61, for shipments of asbestos. Submit waste shipment records to the Contracting Officer for review and approval. Waste shipment records must be signed by the Contractor.

3.4.3.2 Other Hazardous Material Shipment Documents

Prepare a bill of lading for each shipment of hazardous material which is not accompanied by a hazardous waste manifest or asbestos waste shipment record which fulfills the shipping paper requirements. The bill of lading must satisfy the requirements of 49 CFR 172, Subpart C, and 40 CFR 279 if shipping used oil and applicable state or local law or regulation, and must be submitted to the Contracting Officer for review and approval. Bill of ladings requiring shipper's certifications must be signed by the Contractor.

3.5 SPECIAL REQUIREMENTS FOR ASBESTOS WASTES

If work involves asbestos containing wastes, manage these wastes in accordance with specification Section 02 82 00 ASBESTOS REMEDIATION.

3.6 WASTE MINIMIZATION

Minimize the generation of hazardous waste to the maximum extent practicable and take all necessary precautions to avoid mixing clean and contaminated wastes. Identify and evaluate recycling and reclamation options as alternatives to land disposal. Requirements of 40 CFR 266 apply to: hazardous wastes recycled in a manner constituting disposal; hazardous waste burned for energy recovery; lead-acid battery recycling; and hazardous wastes with economically recoverable precious metals. Submit written certification that waste minimization efforts have been undertaken to reduce the volume and toxicity of waste to the degree economically practicable and that the method of treatment, storage, or disposal selected minimizes threats to human health and the environment.

3.7 RECORDKEEPING

Maintain adequate records to support information provided to the Contracting Officer regarding exception reports, annual reports, and

biennial reports; maintain asbestos waste shipment records for a minimum of 3 years from the date of shipment or any longer period required by applicable law or regulation or other provision of this contract; and maintain bill of lading for a minimum of 375 days from the date of shipment or longer period required by applicable law or regulation or other provision of this contract. Submit information necessary to file state annual or EPA biennial reports for hazardous waste transported, treated, stored, or disposed of under this contract. Do not forward these data directly to the regulatory agency but to the Contracting Officer at the specified time. Submit the information necessary for filing of the formal reports in the form and format required by the governing Federal or state regulatory agency. A cover letter must accompany the data to include the contract number, Contractor name, and project location. In the event that a manifest copy documenting receipt of hazardous waste at the treatment storage and disposal facility is not received within 35 days of shipment initiation, or that a manifest copy documenting receipt of PCB waste at the designated facility is not received within 35 days of shipment initiation, prepare and submit an [exception report](#) to the Contracting Officer within 37 days of shipment initiation.

3.8 SPILL RESPONSE

In the event of a spill or release of a hazardous substance (as designated in [40 CFR 302](#)), or pollutant or contaminant, or oil (as governed by the Oil Pollution Act (OPA), 33 U.S.C. 2701 et seq.), notify the Contracting Officer immediately. Direction from the Contracting Officer concerning a spill or release is not considered a change under the contract. If the spill exceeds a reporting threshold, follow the pre-established procedures for immediate reporting to the Contracting Officer. Comply with applicable requirements of Federal, state, or local laws or regulations regarding any spill incident.

3.9 EMERGENCY CONTACTS

Comply with the emergency contact provisions in [49 CFR 172](#), Section .604. Whenever the Contractor ships hazardous materials, provide a 24 hr emergency response contact and phone number of a person knowledgeable about the hazardous materials being shipped and who has comprehensive emergency response and incident mitigation information for that material, or has immediate access to a person who possesses such knowledge and information. Monitor the phone on a 24 hour basis at all times when the hazardous materials are in transportation, including during storage incidental to transportation. Ensure that information regarding this emergency contact and phone number are placed on all hazardous material shipping documents. Designate an emergency coordinator and post the following information at areas in which hazardous wastes are managed:

- a. The name of the emergency coordinator.
- b. Phone number through which the emergency coordinator can be contacted on a 24 hour basis.
- c. The telephone number of the local fire department.
- d. The location of fire extinguishers and spill control materials.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Attachment A SAMPLE OFF-SITE POLICY CERTIFICATION MEMO	
Project/Contract #:	
Waste Stream:	
Primary TSD Facility, EPA ID # and Location:	
Alter. TSD Facility, EPA ID # and Location:	
EPA Region	Contact
I	888-372-7341
II	212-673-4040
III	800-438-2474 or 215-814-5000
IV	800-241-1754 or 404-562-9900
V	312-353-2000
VI	800-887-6063 or 214-665-2210
VII	800-223-0425
VIII	800-424-8802
IX	415-947-8713
X	800-424-4372 or 206-553-4973
EPA representative contacted:	
EPA representative phone number:	
Date contacted:	
Comment:	
The above EPA representative was contacted on _____. As of that date the above sites were considered acceptable in accordance with the Off-Site Policy in 40 CFR 300.440.	
Date:	Signature:
Phone number:	

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 82 00

ASBESTOS REMEDIATION

11/18

PART 1 GENERAL

1.1 REFERENCES

1.2 DEFINITIONS

- 1.2.1 ACM
- 1.2.2 Amended Water
- 1.2.3 Area Sampling
- 1.2.4 Asbestos
- 1.2.5 Asbestos Control Area
- 1.2.6 Asbestos Fibers
- 1.2.7 Asbestos Permissible Exposure Limit
- 1.2.8 Authorized Person
- 1.2.9 Background
- 1.2.10 Competent Person (CP)
- 1.2.11 Contractor
- 1.2.12 Disposal Bag
- 1.2.13 Disturbance
- 1.2.14 Encapsulation
- 1.2.15 Encapsulants
- 1.2.16 Friable Asbestos Material
- 1.2.17 Glovebag Technique
- 1.2.18 Government Consultant (GC)
- 1.2.19 HEPA Filter Equipment
- 1.2.20 Model Accreditation Plan (MAP)
- 1.2.21 Negative Pressure Enclosure (NPE)
- 1.2.22 NESHAP
- 1.2.23 Nonfriable Asbestos Material
- 1.2.24 Permissible Exposure Limits (PELs)
 - 1.2.24.1 PEL-Time Weighted Average(TWA)
 - 1.2.24.2 PEL-Excursion Limit
- 1.2.25 Personal Sampling
- 1.2.26 Private Qualified Person (PQP)
- 1.2.27 Qualified Person (QP)
- 1.2.28 TEM
- 1.2.29 Time Weighted Average (TWA)
- 1.2.30 Transite
- 1.2.31 Wetting Agent
- 1.2.32 Worker

1.3 REQUIREMENTS

- 1.3.1 Description of Work
- 1.3.2 Unexpected Discovery of Asbestos
- 1.3.3 Medical Requirements
 - 1.3.3.1 Medical Examinations
 - 1.3.3.2 Medical Records
- 1.3.4 Employee Training
- 1.3.5 Permits, Licenses and Notifications

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 1.3.6 Environment, Safety and Health Compliance
- 1.3.7 Respiratory Protection Program
 - 1.3.7.1 Respirator Program Records
 - 1.3.7.2 Respirator Fit Testing
 - 1.3.7.3 Respirator Selection and Use Requirements
- 1.3.8 Asbestos Hazard Control Supervisor
- 1.3.9 Hazard Communication
- 1.3.10 Asbestos Hazard Abatement Plan
- 1.3.11 Testing Laboratory
- 1.3.12 Landfill Approval
- 1.3.13 Transporter Certification
- 1.3.14 Medical Certification
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Private Qualified Person Documentation
 - 1.5.2 Designated Competent Person Documentation
 - 1.5.3 Contractor's License
 - 1.5.4 Air Sampling Results
 - 1.5.5 Protective Clothing Decontamination Quality Control Records
 - 1.5.6 Federal, State or Local Citations on Previous Projects
 - 1.5.7 Preconstruction Conference
- 1.6 SECURITY
- 1.7 EQUIPMENT
 - 1.7.1 Rental Equipment

PART 2 PRODUCTS

- 2.1 ENCAPSULANTS
 - 2.1.1 Removal Encapsulants
 - 2.1.2 Bridging Encapsulant
- 2.2 ENCASEMENT PRODUCTS
- 2.3 DUCT TAPE
- 2.4 DISPOSAL CONTAINERS
- 2.5 SHEET PLASTIC
 - 2.5.1 Flame Resistant
 - 2.5.2 Reinforced
- 2.6 MASTIC REMOVING SOLVENT
- 2.7 LEAK-TIGHT WRAPPING
- 2.8 VIEWING INSPECTION WINDOW
- 2.9 WETTING AGENTS

PART 3 EXECUTION

- 3.1 EQUIPMENT
 - 3.1.1 Air Monitoring Equipment
 - 3.1.2 Respirators
 - 3.1.2.1 Respirators for Handling Asbestos
 - 3.1.3 Exterior Whole Body Protection
 - 3.1.3.1 Outer Protective Clothing
 - 3.1.3.2 Work Clothing
 - 3.1.3.3 Eye Protection
 - 3.1.4 Regulated Areas
 - 3.1.5 Load-out Unit
 - 3.1.6 Warning Signs and Labels
 - 3.1.6.1 Warning Sign
 - 3.1.6.2 Warning Labels
 - 3.1.7 Local Exhaust System
 - 3.1.8 Tools
 - 3.1.9 Rental Equipment

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.1.10 Glovebags
- 3.2 WORK PROCEDURE
 - 3.2.1 Protection of Existing Work to Remain
 - 3.2.2 Precleaning
 - 3.2.3 Asbestos Control Area Requirements
 - 3.2.3.1 Negative Pressure Enclosure
 - 3.2.3.2 Glovebag
 - 3.2.3.3 Regulated Area for Class II Removal
 - 3.2.4 Removal Procedures
 - 3.2.4.1 Sealing Contaminated Items Designated for Disposal
 - 3.2.4.2 Exposed Pipe Insulation Edges
 - 3.2.5 Methods of Compliance
 - 3.2.5.1 Mandated Practices
 - 3.2.5.2 Control Methods
 - 3.2.5.3 Unacceptable Practices
 - 3.2.6 Class I Work Procedures
 - 3.2.7 Class II Work Procedures
 - 3.2.8 Specific Control Methods for Class II Work
 - 3.2.8.1 Sealants and Mastic
 - 3.2.8.2 Gaskets
 - 3.2.9 Air Sampling
 - 3.2.9.1 Sampling Prior to Asbestos Work
 - 3.2.9.2 Sampling During Asbestos Work
 - 3.2.9.3 Final Clearance Requirements, NIOSH PCM Method
 - 3.2.9.4 Final Clearance Requirements, EPA TEM Method
 - 3.2.9.5 Sampling After Final Clean-Up (Clearance Sampling)
 - 3.2.9.6 Air Clearance Failure
 - 3.2.10 Lock-Down
 - 3.2.11 Site Inspection
- 3.3 CLEAN-UP AND DISPOSAL
 - 3.3.1 Housekeeping
 - 3.3.2 Title to Materials
 - 3.3.3 Disposal of Asbestos
 - 3.3.3.1 Procedure for Disposal
 - 3.3.3.2 Asbestos Disposal Quantity Report

-- End of Section Table of Contents --

SECTION 02 82 00
ASBESTOS REMEDIATION
11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

- | | |
|--------------------|---|
| UNE-EN 166 | (2002) Personal eye-protection - Specifications. |
| UNE-EN 13053 | (2007+A1:2012) Ventilation for buildings - Air handling units - Rating and performance for units, components and sections |
| UNE-EN ISO 1519 | (2011) Paints and varnishes - Bend test (cylindrical mandrel) (ISO 1519:2011) |
| UNE-EN ISO 4892-2 | (2014) Plastics - Methods of exposure to laboratory light sources - Part 2: Xenon-arc lamps (ISO 4892-2:2013) |
| UNE-EN ISO 16890-1 | (2017) Air filters for general ventilation - Part 1: Technical specifications, requirements and classification system based upon particulate matter efficiency (ePM) (ISO 16890-1:2016) |

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

- | | |
|-----------|---|
| ASSP Z9.2 | (2018) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems |
|-----------|---|

ASTM INTERNATIONAL (ASTM)

- | | |
|---------------|--|
| ASTM E96/E96M | (2022) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials |
| ASTM E1368 | (2014) Visual Inspection of Asbestos Abatement Projects |

COMPRESSED GAS ASSOCIATION (CGA)

- | | |
|---------|--|
| CGA G-7 | (2014) Compressed Air for Human Respiration; 6th Edition |
|---------|--|

EUROPEAN PARLIAMENT (EP)

DIRECTIVE 2003/18/EC (2003) Directive 2003/18/EC of the European Parliament And Of The Council of 27 March 2003 amending Council Directive 83/477/EEC on the protection of workers from the risks related to exposure to asbestos at work

MINISTERIO DE LA PRESIDENCIA (MP)

RD 396/2006 (2006) REAL DECRETO 396/2006, de 31 de marzo, por el que se establecen las disposiciones mínimas de seguridad y salud aplicables a los trabajos con riesgo de exposición al amianto

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

NIOSH NMAM (2016; 5th Ed) NIOSH Manual of Analytical Methods

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety and Health Requirements Manual

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 340/1-90/018 (1990) Asbestos/NESHAP Regulated Asbestos Containing Materials Guidance

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.147 The Control of Hazardous Energy (Lock Out/Tag Out)

29 CFR 1926.59 Hazard Communication

29 CFR 1926.103 Respiratory Protection

29 CFR 1926.200 Accident Prevention Signs and Tags

29 CFR 1926.1101 Asbestos

40 CFR 61-SUBPART A General Provisions

40 CFR 61-SUBPART M National Emission Standard for Asbestos

40 CFR 763 Asbestos

49 CFR 107 Hazardous Materials Program Procedures

49 CFR 171 General Information, Regulations, and Definitions

49 CFR 172 Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response

Information, and Training Requirements

49 CFR 173

Shippers - General Requirements for
Shipments and Packagings

U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

ND OPNAVINST 5100.23

(2005; Rev G) Navy Occupational Safety and
Health (NAVOSH) Program Manual

UNDERWRITERS LABORATORIES (UL)

UL 586

(2009; Reprint Dec 2017) UL Standard for
Safety High-Efficiency Particulate, Air
Filter Units

1.2 DEFINITIONS

1.2.1 ACM

Asbestos Containing Materials.

1.2.2 Amended Water

Water containing a wetting agent or surfactant with a maximum surface
tension of 2.9 Pa.

1.2.3 Area Sampling

Sampling of asbestos fiber concentrations which approximates the
concentrations of asbestos in the theoretical breathing zone but is not
actually collected in the breathing zone of an employee.

1.2.4 Asbestos

The term asbestos includes chrysotile, amosite, crocidolite, tremolite
asbestos, anthophyllite asbestos, and actinolite asbestos and any of these
minerals that has been chemically treated or altered. Materials are
considered to contain asbestos if the asbestos content of the material is
determined to be at least one percent.

1.2.5 Asbestos Control Area

That area where asbestos removal operations are performed which is
isolated by physical boundaries which assist in the prevention of the
uncontrolled release of asbestos dust, fibers, or debris.

1.2.6 Asbestos Fibers

Those fibers having an aspect ratio of at least 3:1 and longer than 5
micrometers.

1.2.7 Asbestos Permissible Exposure Limit

0.1 fibers per cubic centimeter of air as an 8-hour time weighted average
measured in the breathing zone as defined by DIRECTIVE 2003/18/EC and
RD 396/2006, Article 4.

1.2.8 Authorized Person

Any person authorized by the Contractor and required by work duties to be present in the regulated areas.

1.2.9 Background

The ambient airborne asbestos concentration in an uncontaminated area as measured prior to any asbestos hazard abatement efforts. Background concentrations for other (contaminated) areas are measured in similar but asbestos free locations.

1.2.10 Competent Person (CP)

A person capable of identifying existing asbestos hazards in the workplace and selecting the appropriate control strategy for asbestos exposure, who has the authority to take prompt corrective measures to eliminate them, and is specifically trained in a training course which meet the criteria of EPA's Model Accreditation Plan (40 CFR 763) for project designer or supervisor, or permanent worker of a firm registered in RERA (Registro de Empresas con Riesgo de Asfalto).

1.2.11 Contractor

The Contractor is that individual, or entity under contract to perform the herein listed work, and must hold an active valid registration in RERA.

1.2.12 Disposal Bag

A 0.15 mm thick, leak-tight plastic bag, pre-labeled in accordance with RD 396/2006, used for transporting asbestos waste from containment to disposal site.

1.2.13 Disturbance

Activities that disrupt the matrix of ACM, crumble or pulverize ACM, or generate visible debris from ACM. Disturbance includes cutting away small amounts of ACM, no greater than the amount which can be contained in one standard sized glovebag or waste bag, not larger than 1.5 m in length and width in order to access a building component.

1.2.14 Encapsulation

The abatement of an asbestos hazard through the appropriate use of chemical encapsulants.

1.2.15 Encapsulants

Specific materials in various forms used to chemically or physically entrap asbestos fibers in various configurations to prevent these fibers from becoming airborne. There are four types of encapsulants as follows which must comply with performance requirements as specified herein.

- a. Removal Encapsulant (can be used as a wetting agent)
- b. Bridging Encapsulant (used to provide a tough, durable surface coating to asbestos containing material)
- c. Penetrating Encapsulant (used to penetrate the asbestos containing

material encapsulating all asbestos fibers and preventing fiber release due to routine mechanical damage)

- d. Lock-Down Encapsulant (used to seal off or "lock-down" minute asbestos fibers left on surfaces from which asbestos containing material has been removed).

1.2.16 Friable Asbestos Material

A term defined in 40 CFR 61-SUBPART M and EPA 340/1-90/018 meaning any material which contains more than 1 percent asbestos, as determined using the method specified in 40 CFR 763, Polarized Light Microscopy (PLM), that when dry, can be crumbled, pulverized, or reduced to powder by hand pressure.

1.2.17 Glovebag Technique

Those asbestos removal and control techniques put forth in 29 CFR 1926.1101.

1.2.18 Government Consultant (GC)

That qualified person employed directly by the Government to monitor, sample, inspect the work or in some other way advise the Contracting Officer. The GC is normally a private consultant, but can be an employee of the Government.

1.2.19 HEPA Filter Equipment

High efficiency particulate air (HEPA) filtered vacuum and exhaust ventilation equipment with a filter system capable of collecting and retaining asbestos fibers. Filters must retain 99.97 percent of particles 0.3 microns or larger as indicated in UL 586.

1.2.20 Model Accreditation Plan (MAP)

USEPA training accreditation requirements for persons who work with asbestos as specified in 40 CFR 763.

1.2.21 Negative Pressure Enclosure (NPE)

That engineering control technique described as a negative pressure enclosure in 29 CFR 1926.1101.

1.2.22 NESHAP

National Emission Standards for Hazardous Air Pollutants. The USEPA NESHAP regulation for asbestos is at 40 CFR 61-SUBPART M.

1.2.23 Nonfriable Asbestos Material

Material that contains asbestos in which the fibers have been immobilized by a bonding agent, coating, binder, or other material so that the asbestos is well bound and will not normally release asbestos fibers during any appropriate use, handling, storage or transportation. It is understood that asbestos fibers may be released under other conditions such as demolition, removal, or mishap.

1.2.24 Permissible Exposure Limits (PELs)

1.2.24.1 PEL-Time Weighted Average(TWA)

Concentration of asbestos not in excess of 0.1 fibers per cubic centimeter of air (f/cc) as an 8-hour time weighted average (TWA).

1.2.24.2 PEL-Excursion Limit

An airborne concentration of asbestos not in excess of 1.0 f/cc of air as averaged over a sampling period of 30 minutes.

1.2.25 Personal Sampling

Air sampling which is performed to determine asbestos fiber concentrations within the breathing zone of a specific employee, as performed in accordance with 29 CFR 1926.1101.

1.2.26 Private Qualified Person (PQP)

That qualified person hired by the Contractor to perform the herein listed tasks.

1.2.27 Qualified Person (QP)

A Registered Architect, Professional Engineer, Certified Industrial Hygienist, consultant or other qualified person who has successfully completed training and is therefore accredited under a legitimate State Model Accreditation Plan as described in 40 CFR 763 as a Building Inspector, Contractor/Supervisor Abatement Worker, and Asbestos Project Designer; and has successfully completed the National Institute of Occupational Safety and Health (NIOSH) 582 course "Sampling and Evaluating Airborne Asbestos Dust" or equivalent. The QP must be qualified to perform visual inspections as indicated in ASTM E1368.

1.2.28 TEM

Refers to Transmission Electron Microscopy.

1.2.29 Time Weighted Average (TWA)

The TWA is an 8-hour time weighted average airborne concentration of asbestos fibers.

1.2.30 Transite

A generic name for asbestos cement wallboard and pipe.

1.2.31 Wetting Agent

A chemical added to water to reduce the water's surface tension thereby increasing the water's ability to soak into the material to which it is applied. An equivalent wetting agent must have a surface tension of at most 2.9 Pa.

1.2.32 Worker

Individual (not designated as the Competent Person or a supervisor) who performs asbestos work and has completed asbestos worker training required

by 29 CFR 1926.1101, to include EPA Model Accreditation Plan (MAP) "Worker" training; accreditation, if required by the OSHA Class of work to be performed or by the state where the work is to be performed.

1.3 REQUIREMENTS

1.3.1 Description of Work

The work covered by this section includes the handling and control of asbestos containing materials and describes some of the resultant procedures and equipment required to protect workers, the environment and occupants of the building or area, or both, from contact with airborne asbestos fibers. The work also includes the disposal of any asbestos containing materials generated by the work. More specific operational procedures must be outlined in the Asbestos Hazard Abatement Plan called for elsewhere in this specification. The work plan must be submitted to the labor authority of Junta de Andalucía. The Contractor shall not start any work related with asbestos until the work plan is approved by the Junta de Andalucía. The asbestos work includes the removal of ACM located in the gaskets of the flanged fittings in the fuel piping. Provide abatement techniques as outlined in this specification. The work area will be evacuated during the asbestos abatement work. A competent person must supervise asbestos removal work as specified herein.

1.3.2 Unexpected Discovery of Asbestos

Notify the Contracting Officer if any previously untested building components suspected to contain asbestos are impacted by the work.

1.3.3 Medical Requirements

Provide medical requirements including but not limited to medical surveillance and medical record keeping as listed in 29 CFR 1926.1101 and Article 16 of RD 396/2006.

1.3.3.1 Medical Examinations

Before exposure to airborne asbestos fibers, provide workers with a comprehensive medical examination as required by 29 CFR 1926.1101 or RD 396/2006. This requirement must have been satisfied within the 12 months prior to the start of work on this contract. The same medical examination must be given on an annual basis to employees engaged in an occupation involving asbestos and within 30 calendar days before or after the termination of employment in such occupation. Specifically identify x-ray films of asbestos workers to the consulting radiologist and mark medical record jackets with the word "ASBESTOS."

1.3.3.2 Medical Records

Maintain complete and accurate records of employees' medical examinations, medical records, and exposure data for a period of 50 years after termination of employment and make records of the required medical examinations and exposure data available for inspection and copying to: The Labor for Occupational Safety and Health (INSHT), or authorized representatives of them.

1.3.4 Employee Training

Submit certificates, prior to the start of work but after the main

abatement submittal, signed by each employee indicating that the employee has received training in the proper handling of materials and wastes that contain asbestos in accordance with 40 CFR 763 and Articles 13 & 14 of RD 396/2006; understands the health implications and risks involved, including the illnesses possible from exposure to airborne asbestos fibers; understands the use and limits of the respiratory equipment to be used; and understands the results of monitoring of airborne quantities of asbestos as related to health and respiratory equipment as indicated in 29 CFR 1926.1101 on an initial and annual basis. Organize certificates by individual worker, not grouped by type of certification. Train personnel involved in the asbestos control work in accordance with RD 396/2006. Document the training by providing: dates of training, training entity, course outline, names of instructors, and qualifications of instructors upon request by the Contracting Officer. Furnish each employee with respirator training and fit testing administered by the PQP as required by 29 CFR 1926.1101 and 29 CFR 1926.103. Fully cover engineering and other hazard control techniques and procedures.

1.3.5 Permits, Licenses and Notifications

Prior to the start of work, obtain necessary permits and licenses in conjunction with asbestos removal, encapsulation, hauling, and disposition, and furnish notification of such actions required by RD 396/2006. Notify the Contracting Officer and other appropriate Government agencies in writing 20 working days prior to the start of asbestos work as indicated in applicable laws, ordinances, criteria, rules, and regulations. Submit copies of all Notifications to the Contracting Officer.

1.3.6 Environment, Safety and Health Compliance

In addition to detailed requirements of this specification, comply with those applicable laws, ordinances, criteria, rules, and regulations of Federal, State, regional, and local authorities regarding handling, storing, transporting, and disposing of asbestos waste materials. Comply with the applicable requirements of the current issue of EM 385-1-1, 29 CFR 1926.1101, 40 CFR 61-SUBPART A, 40 CFR 61-SUBPART M, 40 CFR 763 and ND OPNAVINST 5100.23. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting the work. Where the requirements of this specification, applicable laws, rules, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirement as defined by the Government apply.

1.3.7 Respiratory Protection Program

Establish and implement a respirator program as required by 29 CFR 1926.1101, and 29 CFR 1926.103. Submit a written description of the program to the Contracting Officer. Submit a written program manual or operating procedure including methods of compliance with regulatory statutes.

1.3.7.1 Respirator Program Records

Submit records of the respirator program as required by 29 CFR 1926.103, and 29 CFR 1926.1101.

1.3.7.2 Respirator Fit Testing

The Contractor's PQP must conduct a qualitative or quantitative fit test

conforming to 29 CFR 1926.103 for each worker required to wear a respirator, and any authorized visitors who enter a regulated area where respirators are required to be worn. A respirator fit test must be performed prior to initially wearing a respirator and every 12 months thereafter. If physical changes develop that will affect the fit, a new fit test must be performed. Functional fit checks must be performed each time a respirator is put on and in accordance with the manufacturer's recommendation.

1.3.7.3 Respirator Selection and Use Requirements

Provide respirators, and ensure that they are used as required by 29 CFR 1926.1101 and in accordance with CGA G-7 and the manufacturer's recommendations. Respirators must be approved by the National Institute for Occupational Safety and Health (NIOSH), for use in environments containing airborne asbestos fibers. For air-purifying respirators, the particulate filter must be high-efficiency particulate air (HEPA)/(N-,R-,P-100). The initial respirator selection and the decisions regarding the upgrading or downgrading of respirator type must be made by the Contractor's Designated IH based on the measured or anticipated airborne asbestos fiber concentrations to be encountered.

1.3.8 Asbestos Hazard Control Supervisor

The Contractor must be represented on site by a supervisor, trained for all portions of the herein listed work.

1.3.9 Hazard Communication

Adhere to all parts of 29 CFR 1926.59 and provide the Contracting Officer with a copy of the Safety Data Sheets (SDS) for all materials brought to the site.

1.3.10 Asbestos Hazard Abatement Plan

Submit a detailed plan of the safety precautions such as lockout, tagout, tryout, fall protection, and confined space entry procedures and equipment and work procedures to be used in the removal of materials containing asbestos. The plan, not to be combined with other hazard abatement plans, must be prepared, signed, and sealed by the PQP. Provide a Table of Contents for each abatement submittal, which follows the sequence of requirements in the contract. The plan must include but not be limited to the precise personal protective equipment to be used including, but not limited to, respiratory protection, type of whole-body protection, the location of asbestos control areas including clean and dirty areas, buffer zones, showers, storage areas, change rooms, removal method, interface of trades involved in the construction, sequencing of asbestos related work, disposal plan, type of wetting agent and asbestos sealer to be used, locations of local exhaust equipment, planned air monitoring strategies, and a detailed description of the method to be employed in order to control environmental pollution. The plan must also include (both fire and medical emergency) response plans and an Activity Hazard Analyses (AHAs) in accordance with EM 385-1-1. The Asbestos Hazard Abatement Plan must be approved in writing by the labor authority of Junta de Andalucía prior to starting any asbestos work. The Contractor, Asbestos Hazard Control Supervisor,, CP and PQP must meet with the Contracting Officer prior to beginning work, to discuss in detail the Asbestos Hazard Abatement Plan, including work procedures and safety precautions. Once approved by the Contracting Officer, the plan will be enforced as if an

addition to the specification. Any changes required in the specification as a result of the plan must be identified specifically in the plan to allow for free discussion and approval by the Contracting Officer prior to starting work.

1.3.11 Testing Laboratory

Submit the name, address, and telephone number of each testing laboratory selected for the analysis and reporting of airborne concentrations of asbestos fibers along with evidence that each laboratory selected holds the appropriate regional license and permits and certification that each laboratory is accredited by INSHT and complies with the requirements of Annex II of [RD 396/2006](#). Where analysis to determine asbestos content in bulk materials or transmission electron microscopy is required, submit evidence that the laboratory is accredited for asbestos analysis. The testing laboratory firm must be independent of the asbestos contractor and must have no employee or employer relationship which could constitute a conflict of interest.

1.3.12 Landfill Approval

Submit written evidence that the landfill is approved for asbestos disposal by the regional Environmental Protection Agency Junta de Andalucía. Within three working days after delivery, submit detailed [delivery tickets](#), prepared, signed, and dated by an agent of the landfill, certifying the amount of asbestos materials delivered to the landfill. Submit a copy of the [waste shipment records](#) within one day of the shipment leaving the project site.

1.3.13 Transporter Certification

Submit written evidence that the transporter is approved to transport asbestos waste in accordance with the DOT requirements of [49 CFR 171](#), [49 CFR 172](#) and [49 CFR 173](#) as well as registration requirements of [49 CFR 107](#) and all other regional and local regulatory agency requirements.

1.3.14 Medical Certification

Provide a written certification for each worker and supervisor, signed by a licensed physician indicating that the worker and supervisor has met or exceeded all of the medical prerequisites listed herein and in [29 CFR 1926.1101](#) and [29 CFR 1926.103](#) as prescribed by law. Submit certificates prior to the start of work but after the main abatement submittal.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section [01 33 00](#) SUBMITTAL PROCEDURES:

[SD-03 Product Data](#)

[Amended Water; G](#)

[Safety Data Sheets \(SDS\) for All Materials; G](#)

[Encapsulants; G](#)

Respirators; G

Local Exhaust Equipment; G

Vacuums; G

Glovebags; G

SD-06 Test Reports

Air Sampling Results; G

Clearance Sampling; G

Asbestos Disposal Quantity Report; G

SD-07 Certificates

Employee Training; G

Notifications; G

Respiratory Protection Program; G

Asbestos Hazard Abatement Plan; G

Testing Laboratory; G

Landfill Approval; G

Delivery Tickets; G

Waste Shipment Records; G

Transporter Certification; G

Medical Certification; G

Private Qualified Person Documentation; G

Designated Competent Person; G

Contractor's License; G

Federal, State or Local Citations on Previous Projects; G

Encapsulants; G

Equipment Used to Contain Airborne Asbestos Fibers; G

Vacuums; G

Ventilation Systems; G

SD-11 Closeout Submittals

Permits and Licenses; G

Notifications; G

Respirator Program Records; G

Protective Clothing Decontamination Quality Control Records; G

1.5 QUALITY ASSURANCE

1.5.1 Private Qualified Person Documentation

Submit the name, address, and telephone number of the Private Qualified Person (PQP) selected to prepare the Asbestos Hazard Abatement Plan, direct monitoring and training, and documented evidence that the PQP has successfully completed training in and is accredited and where required is certified as, a Building Inspector, Contractor/Supervisor Abatement Worker, and Asbestos Project Designer as described by 40 CFR 763 and has successfully completed the National Institute of Occupational Safety and Health (INSHT) or equivalent.

1.5.2 Designated Competent Person Documentation

The Designated Competent Person must be experienced in the administration and supervision of asbestos abatement projects including exposure assessment and monitoring, work practices, abatement methods, protective measures for personnel, setting up and inspecting asbestos abatement work areas, evaluating the integrity of containment barriers, placement and operation of local exhaust systems, ACM generated waste containment and disposal procedures, decontamination units installation and maintenance requirements, site safety and health requirements, notification of other employees onsite. The Designated Competent Person must be on-site at all times when asbestos abatement activities are underway. Submit training certification and a current Contractor's License. Submit evidence that the Designated Competent Person has a minimum of 2 years of on-the-job asbestos abatement experience.

1.5.3 Contractor's License

Submit a copy of the asbestos contractor's license issued by RERA. Submit the following certification along with the license: "I certify that the personnel I am responsible for during the course of this project fully understand the contents of 29 CFR 1926.1101, 40 CFR 61-SUBPART MEM 385-1-1, and the Federal, State and local requirements for those asbestos abatement activities that they will be involved in." This certification statement must be signed by the Company's President or Chief Executive.

1.5.4 Air Sampling Results

Complete fiber counting and provide results to the PQP and GC for review within 16 hours of the "time off" of the sample pump. Notify the Contracting Officer immediately of any airborne levels of asbestos fibers in excess of the acceptable limits. Submit sampling results to the Contracting Officer and the affected Contractor employees where required by law within three working days, signed by the testing laboratory employee performing air sampling, the employee that analyzed the sample, and the PQP and GC. Notify the Contractor and the Contracting Officer immediately of any variance in the pressure differential which could cause adjacent unsealed areas to have asbestos fiber concentrations in excess of 0.01 fibers per cubic centimeter or background whichever is higher. In no circumstance must levels exceed 0.1 fibers per cubic centimeter.

1.5.5 Protective Clothing Decontamination Quality Control Records

Provide all records that document quality control for the decontamination of reusable outer protective clothing.

1.5.6 Federal, State or Local Citations on Previous Projects

Submit a statement, signed by an officer of the company, containing a record of any citations issued by Federal, State or local regulatory agencies relating to asbestos activities within the last 5 years (including projects, dates, and resolutions); a list of penalties incurred through non-compliance with asbestos project specifications, including liquidated damages, overruns in scheduled time limitations and resolutions; and situations in which an asbestos-related contract has been terminated (including projects, dates, and reasons for terminations). If there are none, a negative declaration signed by an officer of the company must be provided.

1.5.7 Preconstruction Conference

Conduct a safety preconstruction conference to discuss the details of the Asbestos Hazard Abatement Plan, Accident Prevention Plan (APP) including the AHAs required in specification Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. The safety preconstruction conference must include the Contractor and their Designated Competent Person, Designated IH and Project Supervisor and the Contracting Officer. Deficiencies in the APP will be discussed. Onsite work must not begin until the APP has been accepted.

1.6 SECURITY

A log book must be kept documenting entry into and out of the regulated area. Entry into regulated areas must only be by personnel authorized by the Contractor and the Contracting Officer. Personnel authorized to enter regulated areas must be trained, medically evaluated, and wear the required personal protective equipment.

1.7 EQUIPMENT

1.7.1 Rental Equipment

Keep a copy of the written notification to the rental company concerning the intended use of the equipment and the possibility of asbestos contamination of the equipment. Provide the copy if required by the Contracting Officer.

PART 2 PRODUCTS

2.1 ENCAPSULANTS

Encapsulants must conform to current USEPA requirements, contain no toxic or hazardous substances as defined in 29 CFR 1926.59, and conform to the following performance requirements.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

2.1.1 Removal Encapsulants

<u>Requirement</u>	<u>Test Standard</u>
Fire Class	A1 or A2 or B
Life Expectancy - 20 years	UNE-EN ISO 4892-2
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Impact Resistance - Minimum 245.5 mm/N	
Flexibility - no rupture or cracking	UNE-EN ISO 1519

2.1.2 Bridging Encapsulant

<u>Requirement</u>	<u>Test Standard</u>
Fire Class	A1 or A2 or B
Life Expectancy - 20 years	UNE-EN ISO 4892-2
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Impact Resistance - Minimum 245.5 mm/N	
Flexibility - no rupture or cracking	UNE-EN ISO 1519

2.2 ENCASEMENT PRODUCTS

Encasement must consist of primary cellular polymer coat, polymer finish coat, and any other finish coat as approved by the Contracting Officer.

2.3 DUCT TAPE

Industrial grade duct tape of appropriate widths suitable for bonding sheet plastic and disposal container.

2.4 DISPOSAL CONTAINERS

Leak-tight (defined as solids, liquids, or dust that cannot escape or spill out) disposal containers must be provided for ACM wastes as required by 29 CFR 1926.1101. Disposal containers can be in the form of:

- a. Disposal Bags
- b. Fiberboard Drums
- c. Cardboard Boxes

2.5 SHEET PLASTIC

Sheet plastic must be polyethylene of 0.15 mm minimum thickness and must be provided in the largest sheet size necessary to minimize seams. Film must be clear.

2.5.1 Flame Resistant

Where a potential for fire exists, flame-resistant sheets must be provided.

2.5.2 Reinforced

Reinforced sheets must be provided where high skin strength is required, such as where it constitutes the only barrier between the regulated area and the outdoor environment. The sheet stock must consist of translucent, nylon-reinforced or woven-polyethylene thread laminated between 2 layers of polyethylene film.

2.6 MASTIC REMOVING SOLVENT

Mastic removing solvent must be nonflammable and must not contain methylene chloride, glycol ether, or halogenated hydrocarbons. Solvents used onsite must have a flash point greater than 60 degrees C.

2.7 LEAK-TIGHT WRAPPING

Two layers of 0.15 mm minimum thick polyethylene sheet stock must be used for the containment of removed asbestos-containing components or materials such as large tanks, boilers, insulated pipe segments and other materials. Upon placement of the ACM component or material, each layer must be individually leak-tight sealed with duct tape.

2.8 VIEWING INSPECTION WINDOW

Where feasible, a minimum of one clear, 3 mm thick, acrylic sheet, 450 by 610 mm, must be installed as a viewing inspection window at eye level on a wall in each containment enclosure. The windows must be sealed leak-tight with industrial grade duct tape.

2.9 WETTING AGENTS

Removal encapsulant must be provided when conducting removal abatement activities that require a longer removal time or are subject to rapid evaporation of amended water. The removal encapsulant must be capable of wetting the ACM and retarding fiber release during disturbance of the ACM greater than or equal to that provided by amended water. Performance requirements for penetrating encapsulants are specified in paragraph ENCAPSULANTS above.

PART 3 EXECUTION

3.1 EQUIPMENT

Provide the Contracting Officer or the Contracting Officer's Representative, with at least two complete sets of personal protective equipment as required for entry to and inspection of the asbestos control area. Provide equivalent training to the Contracting Officer or a

designated representative as provided to Contractor employees in the use of the required personal protective equipment. Provide manufacturer's certificate of compliance for all [equipment used to contain airborne asbestos fibers](#).

3.1.1.1 Air Monitoring Equipment

The Contractor's PQP must approve air monitoring equipment. The equipment must include, but must not be limited to:

- a. High-volume sampling pumps that can be calibrated and operated at a constant airflow up to 16 liters per minute.
- b. Low-volume, battery powered, body-attachable, portable personal pumps that can be calibrated to a constant airflow up to approximately 3.5 liters per minute, and a self-contained rechargeable power pack capable of sustaining the calibrated flow rate for a minimum of 10 hours. The pumps must also be equipped with an automatic flow control unit which must maintain a constant flow, even as filter resistance increases due to accumulation of fiber and debris on the filter surface.
- c. Single use standard 25 mm diameter cassette, open face, 0.8 micron pore size, mixed cellulose ester membrane filters and cassettes with 50 mm electrically conductive extension cowl, and shrink bands for personal air sampling.
- d. A flow calibrator capable of calibration to within plus or minus 2 percent of reading over a temperature range of [minus 20 to plus 60 degrees C](#) and traceable to a NIST primary standard.

3.1.2 Respirators

Select respirators from those approved by the National Institute for Occupational Safety and Health of Spain (INSHT).

3.1.2.1 [Respirators](#) for Handling Asbestos

Provide personnel engaged in pre-cleaning, cleanup, handling, encapsulation, removal or demolition of asbestos materials with respiratory protection as indicated in [29 CFR 1926.1101](#) and [29 CFR 1926.103](#). Breathing air must comply with [CGA G-7](#).

3.1.3 Exterior Whole Body Protection

3.1.3.1 Outer Protective Clothing

Provide personnel exposed to asbestos with disposable "non-breathable" whole body outer protective clothing, head coverings, gloves, and foot coverings. Provide disposable plastic or rubber gloves to protect hands. Cloth gloves may be worn inside the plastic or rubber gloves for comfort, but must not be used alone. Make sleeves secure at the wrists, make foot coverings secure at the ankles, and make clothing secure at the neck by the use of tape.

3.1.3.2 Work Clothing

Provide cloth work clothes for wear under the outer protective clothing and foot coverings and either dispose of or properly decontaminate them as

recommended by the GC after each use.

3.1.3.3 Eye Protection

Provide eye protection that complies with **UNE-EN 166** when operations present a potential eye injury hazard. Provide goggles to personnel engaged in asbestos abatement operations when the use of a full face respirator is not required.

3.1.4 Regulated Areas

All Class I, II, and III asbestos work must be conducted within regulated areas. The regulated area must be demarcated to minimize the number of persons within the area and to protect persons outside the area from exposure to airborne asbestos. Control access to regulated areas, ensure that only authorized personnel enter, and verify that Contractor required medical surveillance, training and respiratory protection program requirements are met prior to allowing entrance.

3.1.5 Load-out Unit

Provide a temporary load-out unit that is adjacent and connected to the regulated area. Attach the load-out unit in a leak-tight manner to each regulated area.

3.1.6 Warning Signs and Labels

Provide bilingual warning signs printed in English and Spanish at all approaches to asbestos control areas. Locate signs at such a distance that personnel may read the sign and take the necessary protective steps required before entering the area. Provide labels and affix to all asbestos materials, scrap, waste, debris, and other products contaminated with asbestos. Containers with preprinted warning labels conforming to the requirements are acceptable

3.1.6.1 Warning Sign

Provide vertical format conforming to **29 CFR 1926.200**, and **29 CFR 1926.1101** minimum **500 by 355 mm** displaying the following legend in the lower panel:

Legend	Notation
DANGER	25 mm Sans Serif Gothic or Block
ASBESTOS	25 mm Sans Serif Gothic or Block
MAY CAUSE CANCER	25 mm Sans Serif Gothic or Block
CAUSES DAMAGE TO LUNGS	6 mm Sans Serif Gothic or Block
AUTHORIZED PERSONNEL ONLY	6 mm Sans Serif Gothic or Block

Legend	Notation
WEAR RESPIRATORY PROTECTION AND PROTECTIVE CLOTHING IN THIS AREA	6 mm Sans Serif Gothic or Block

Spacing between lines must be at least equal to the height of the upper of any two lines.

3.1.6.2 Warning Labels

Provide labels conforming to 29 CFR 1926.1101 of sufficient size to be clearly legible, displaying the following legend:

DANGER
CONTAINS ASBESTOS FIBERS
MAY CAUSE CANCER
CAUSES DAMAGE TO LUNGS
DO NOT BREATHE DUST AVOID CREATING DUST

3.1.7 Local Exhaust System

Provide a local exhaust system in the asbestos control area in accordance with ASSP Z9.2 and 29 CFR 1926.1101 or alternatively UNE-EN ISO 16890-1 and UNE-EN 13053 that will provide at least four air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until the asbestos control area is removed and must be leak proof to the filter and equipped with HEPA filters. Maintain a minimum pressure differential in the control area of minus 0.51 mm of water column relative to adjacent, unsealed areas. Filters on exhaust equipment must conform to UNE-EN ISO 16890-1. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.8 Tools

Vacuums must be leak proof to the filter and equipped with HEPA filters. Filters on vacuums must conform to ASSP Z9.2 and UL 586 or UNE-EN ISO 16890-1. Do not use power tools to remove asbestos containing materials unless the tool is equipped with effective, integral HEPA filtered exhaust ventilation systems. Remove all residual asbestos from reusable tools prior to storage or reuse. Reusable tools must be thoroughly decontaminated prior to being removed from the regulated areas.

3.1.9 Rental Equipment

If rental equipment is to be used, furnish written notification to the rental agency concerning the intended use of the equipment and the possibility of asbestos contamination of the equipment.

3.1.10 Glovebags

Submit written manufacturers proof that glovebags will not break down under expected temperatures and conditions.

3.2 WORK PROCEDURE

Perform asbestos related work in accordance with 29 CFR 1926.1101, 40 CFR 61-SUBPART M and as specified herein. Use wet removal procedures and appropriate encapsulation procedures as listed in the asbestos hazard abatement plan. Wear and utilize protective clothing and equipment as specified herein. No eating, smoking, drinking, chewing gum, tobacco, or applying cosmetics is permitted in the asbestos work or control areas. Personnel of other trades not engaged in the removal and demolition of asbestos containing material must not be exposed at any time to airborne concentrations of asbestos unless all the personnel protection and training provisions of this specification are complied with by the trade personnel. Power to the regulated area must be locked-out and tagged in accordance with 29 CFR 1910.147. Stop abatement work in the regulated area immediately when the airborne total fiber concentration: (1) equals or exceeds 0.01 f/cc, or the pre-abatement concentration, whichever is greater, outside the regulated area; or (2) equals or exceeds 1.0 f/cc inside the regulated area. Correct the condition to the satisfaction of the Contracting Officer, including visual inspection and air sampling. Work must resume only upon notification by the Contracting Officer. Corrective actions must be documented. If an asbestos fiber release or spill occurs outside of the asbestos control area, stop work immediately, correct the condition to the satisfaction of the Contracting Officer including clearance sampling, prior to resumption of work.

3.2.1 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent work. Where such work is damaged or contaminated as verified by the Contracting Officer using visual inspection or sample analysis, it must be restored to its original condition or decontaminated by the Contractor at no expense to the Government as deemed appropriate by the Contracting Officer. This includes inadvertent spill of dirt, dust, or debris in which it is reasonable to conclude that asbestos may exist. When these spills occur, stop work immediately. Then clean up the spill. When satisfactory visual inspection and air sampling results are obtained from the PQP work may proceed at the discretion of the Contracting Officer.

3.2.2 Precleaning

Wet wipe and HEPA vacuum all surfaces potentially contaminated with asbestos prior to establishment of an enclosure.

3.2.3 Asbestos Control Area Requirements

3.2.3.1 Negative Pressure Enclosure

Provide local exhaust system in the asbestos control area. Openings will be allowed in enclosures of asbestos control areas for personnel and equipment entry and exit, the supply and exhaust of air for the local exhaust system and the removal of properly containerized asbestos containing materials. Replace local exhaust system filters as required to maintain the efficiency of the system.

3.2.3.2 Glovebag

If the construction of a negative pressure enclosure is infeasible, use alternate techniques as indicated in 29 CFR 1926.1101. Establish designated limits for the asbestos regulated area with the use of rope or other continuous barriers, and maintain all other requirements for asbestos control areas. The PQP must conduct personal samples of each worker engaged in asbestos handling (removal, disposal, transport and other associated work) throughout the duration of the project. If the quantity of airborne asbestos fibers monitored at the breathing zone of the workers at any time exceeds background or 0.01 fibers per cubic centimeter whichever is greater, stop work, evacuate personnel in adjacent areas or provide personnel with approved protective equipment at the discretion of the Contracting Officer. This sampling may be duplicated by the Government at the discretion of the Contracting Officer. If the air sampling results obtained by the Government differ from those obtained by the Contractor, the Government will determine which results predominate. If adjacent areas are contaminated as determined by the Contracting Officer, clean the contaminated areas, monitor, and visually inspect the area as specified herein.

3.2.3.3 Regulated Area for Class II Removal

Removal of asbestos containing sealants are Class II removal activities. Establish designated limits for the asbestos regulated work area with the use of red barrier tape; install critical barriers, splash guards and signs, and maintain all other requirements for asbestos control area except local exhaust. Place impermeable dropcloths on surfaces beneath removal activity extending out 3 feet in all directions. A detached decontamination system may be used. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If workers the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.4 Removal Procedures

Wet asbestos material with a fine spray of amended water during removal, cutting, or other handling so as to reduce the emission of airborne fibers. Remove material and immediately place in 0.15 mm plastic disposal bags. Remove asbestos containing material in a gradual manner, with continuous application of the amended water or wetting agent in such a manner that no asbestos material is disturbed prior to being adequately wetted. Where unusual circumstances prohibit the use of 0.15 mm plastic bags, submit an alternate proposal for containment of asbestos fibers to the Contracting Officer for approval. For example, in the case where both piping and insulation are to be removed, the Contractor may elect to wet the insulation, wrap the pipes and insulation in plastic and remove the pipe by sections. Containerize asbestos containing material while wet. Do not allow asbestos material to accumulate or become dry. Lower and otherwise handle asbestos containing material as indicated in 40 CFR 61-SUBPART M.

3.2.4.1 Sealing Contaminated Items Designated for Disposal

Remove contaminated architectural, mechanical, and electrical appurtenances such as venetian blinds, full-height partitions, carpeting, duct work, pipes and fittings, radiators, light fixtures, conduit, panels,

and other contaminated items designated for removal by completely coating the items with an asbestos lock-down encapsulant at the demolition site before removing the items from the asbestos control area. These items need not be vacuumed. The asbestos encapsulant must be tinted a contrasting color and spray-applied by airless method. Thoroughness of sealing operation must be visually gauged by the extent of colored coating on exposed surfaces. Encapsulants must comply with the performance requirements specified herein.

3.2.4.2 Exposed Pipe Insulation Edges

Contain edges of asbestos insulation to remain that are exposed by a removal operation. Wet and cut the rough ends true and square with sharp tools and then encapsulate the edges with a 6 mm thick layer of non-asbestos containing insulating cement troweled to a smooth hard finish. When cement is dry, lag the end with a layer of non-asbestos lagging cloth, overlapping the existing ends by at least 100 mm. When insulating cement and cloth is an impractical method of sealing a raw edge of asbestos, take appropriate steps to seal the raw edges as approved by the Contracting Officer.

3.2.5 Methods of Compliance

3.2.5.1 Mandated Practices

The specific abatement techniques and items identified must be detailed in the Contractor's AHAP. Use the following engineering controls and work practices in all operations, regardless of the levels of exposure:

- a. Vacuum cleaners equipped with HEPA filters.
- b. Wet methods or wetting agents except where it can be demonstrated that the use of wet methods is unfeasible due to the creation of electrical hazards, equipment malfunction, and in roofing.
- c. Prompt clean-up and disposal.
- d. Inspection and repair of polyethylene.
- e. Cleaning of equipment and surfaces of containers prior to removing them from the equipment room or area.

3.2.5.2 Control Methods

Use the following control methods:

- a. Local exhaust ventilation equipped with HEPA filter;
- b. Enclosure or isolation of processes producing asbestos dust;
- c. Where the feasible engineering and work practice controls are not sufficient to reduce employee exposure to or below the PELs, use them to reduce employee exposure to the lowest levels attainable and must supplement them by the use of respiratory protection.

3.2.5.3 Unacceptable Practices

The following work practices must not be used:

- a. High-speed abrasive disc saws that are not equipped with point of cut ventilator or enclosures with HEPA filtered exhaust air.
- b. Compressed air used to remove asbestos containing materials, unless the compressed air is used in conjunction with an enclosed ventilation system designed to capture the dust cloud created by the compressed air.
- c. Dry sweeping, shoveling, or other dry clean up.
- d. Employee rotation as a means of reducing employee exposure to asbestos.

3.2.6 Class I Work Procedures

In addition to requirements of paragraphs MANDATED PRACTICES and CONTROL METHODS, the following engineering controls and work practices must be used:

- a. A Competent Person must supervise the installation and operation of the control methods.
- b. For jobs involving the removal of more than 7.5 m or 0.9 square m of TSI or surfacing material, place critical barriers over all openings to the regulated area.
- c. HVAC systems must be isolated in the regulated area by sealing with a double layer of plastic or air-tight rigid covers.
- d. Impermeable dropcloths (0.15 mm or greater thickness) must be placed on surfaces beneath all removal activity.
- e. Where a negative exposure assessment has not been provided or where exposure monitoring shows the PEL was exceeded, the regulated area must be ventilated with a HEPA unit and employees must use PPE.

3.2.7 Class II Work Procedures

In addition to the requirements of paragraphs MANDATED PRACTICES and CONTROL METHODS, the following engineering controls and work practices must be used:

- a. A Competent Person must supervise the work.
- b. For indoor work, critical barriers must be placed over all openings to the regulated area.
- c. Impermeable dropcloths must be placed on surfaces beneath all removal activity.

3.2.8 Specific Control Methods for Class II Work

3.2.8.1 Sealants and Mastic

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers and signs, and maintain all other requirements for asbestos control area except local exhaust. Spread 0.15 mm plastic sheeting on the ground around the perimeter of the work area extending out in all directions. Using adequately wet methods, carefully remove the ACM sealants and mastics using a scraper or knife

blade. As it is removed place the material into a disposal bag. Make every effort to keep the asbestos material from falling to the ground or work area floor below. Dry sweeping is prohibited. Use vacuums equipped with HEPA filter and disposable dust bag. Place debris into a 0.15 mm minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Lower and otherwise handle asbestos containing materials as indicated in 40 CFR 61-SUBPART M. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or at designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.8.2 Gaskets

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. Gaskets must be thoroughly wetted with amended water prior to removal and immediately placed in a disposal container. If a gasket is visibly deteriorated and unlikely to be removed intact, removal must be undertaken within a glovebag. Any scraping to remove residue must be performed wet. Place debris into a 0.15 mm minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.9 Air Sampling

Perform sampling of airborne concentrations of asbestos fibers in accordance with 29 CFR 1926.1101, the Contractor's air monitoring plan and as specified herein. Sampling performed in accordance with 29 CFR 1926.1101 must be performed by the PQP. Unless otherwise specified, use NIOSH Method 7400 or alternatively Method MTA/MA-051 from the INSHT "Método del Filtro de Membrana/Microscopia Optica de Contraste de Fases" for sampling and analysis. Monitoring may be duplicated by the Government at the discretion of the Contracting Officer. If the air sampling results obtained by the Government differ from those results obtained by the Contractor, the Government will determine which results predominate. Results of breathing zone samples must be posted at the job site and made available to the Contracting Officer. Submit all documentation regarding initial exposure assessments, negative exposure assessments, and air-monitoring results.

3.2.9.1 Sampling Prior to Asbestos Work

Provide area air sampling and establish the baseline one day prior to the masking and sealing operations for each removal site. Establish the background by performing area sampling in similar but uncontaminated sites in the building.

3.2.9.2 Sampling During Asbestos Work

The PQP must provide personal and area sampling as indicated in [29 CFR 1926.1101](#) and governing environmental regulations. Breathing zone samples must be taken for at least 25 percent of the workers in each shift, or a minimum of two, whichever is greater. Air sample fiber counting must be completed and results provided within 24-hours. In addition, provided the same type of work is being performed, provide area sampling at least once every work shift close to the work inside the enclosure, outside the clean room entrance to the enclosure, and at the exhaust opening of the local exhaust system. If sampling outside the enclosure shows airborne levels have exceeded background or 0.01 fibers per cubic centimeter, whichever is greater, stop all work, correct the condition(s) causing the increase, and notify the Contracting Officer immediately.

3.2.9.3 Final Clearance Requirements, NIOSH PCM Method

For PCM sampling and analysis using [NIOSH NMAM](#) Method 7400, the fiber concentration inside the abated regulated area, for each airborne sample, must be less than 0.01 f/cc. The abatement inside the regulated area is considered complete when every PCM final clearance sample is below the clearance limit. If any sample result is greater than 0.01 total f/cc, the asbestos fiber concentration (asbestos f/cc) must be confirmed from that same filter using [NIOSH NMAM](#) Method 7402 (TEM) at Contractor's expense. If any confirmation sample result is greater than 0.01 asbestos f/cc, abatement is incomplete and cleaning must be repeated at the Contractor's expense. Upon completion of any required recleaning, resampling with results to meet the above clearance criteria must be done at the Contractor's expense.

3.2.9.4 Final Clearance Requirements, EPA TEM Method

For EPA TEM sampling and analysis, using the EPA Method specified in [40 CFR 763](#), abatement inside the regulated area is considered complete when the arithmetic mean asbestos concentration of the five inside samples is less than or equal to 70 structures per square millimeter (70 S/mm). When the arithmetic mean is greater than 70 S/mm, the three blank samples must be analyzed. If the three blank samples are greater than 70 S/mm, resampling must be done. If less than 70 S/mm, the five outside samples must be analyzed and a Z-test analysis performed. When the Z-test results are less than 1.65, the decontamination must be considered complete. If the Z-test results are more than 1.65, the abatement is incomplete and cleaning must be repeated. Upon completion of any required recleaning, resampling with results to meet the above clearance criteria must be done at the Contractor's expense.

3.2.9.5 Sampling After Final Clean-Up ([Clearance Sampling](#))

Provide area sampling of asbestos fibers and establish an airborne asbestos concentration of less than 0.01 fibers per cubic centimeter after final clean-up but before removal of the enclosure or the asbestos work

control area. After final cleanup and the asbestos control area is dry but prior to clearance sampling, the PQP must perform a visual inspection in accordance with [ASTM E1368](#) to ensure that the asbestos control and work area is free of any accumulations of dirt, dust, or debris. Prepare a written report signed and dated by the PQP documenting that the asbestos control area is free of dust, dirt, and debris and all waste has been removed. The asbestos fiber counts from these samples must be less than 0.01 fibers per cubic centimeter or be not greater than the background, whichever is greater. Should any of the final samples indicate a higher value take appropriate actions to re-clean the area and repeat the sampling analysis at the Contractor's expense.

3.2.9.6 Air Clearance Failure

If clearance sampling results fail to meet the final clearance requirements, pay all costs associated with the required recleaning, resampling, and analysis, until final clearance requirements are met.

3.2.10 Lock-Down

Prior to removal of plastic barriers and after pre-clearance clean up of gross contamination, the PQP must conduct a visual inspection of all areas affected by the removal in accordance with [ASTM E1368](#). Inspect for any visible fibers.

3.2.11 Site Inspection

While performing asbestos engineering control work, the Contractor must be subject to on-site inspection by the Contracting Officer who may be assisted by or represented by safety or industrial hygiene personnel. If the work is found to be in violation of this specification, the Contracting Officer or his representative will issue a stop work order to be in effect immediately and until the violation is resolved. All related costs including standby time required to resolve the violation must be at the Contractor's expense.

3.3 CLEAN-UP AND DISPOSAL

3.3.1 Housekeeping

Essential parts of asbestos dust control are housekeeping and clean-up procedures. Maintain surfaces of the asbestos control area free of accumulations of asbestos fibers. Give meticulous attention to restricting the spread of dust and debris; keep waste from being distributed over the general area. Use HEPA filtered vacuum cleaners. DO NOT BLOW DOWN THE SPACE WITH COMPRESSED AIR. When asbestos removal is complete, all asbestos waste is removed from the work-site, and final clean-up is completed, the Contracting Officer will attest that the area is safe before the signs can be removed. After final clean-up and acceptable airborne concentrations are attained but before the HEPA unit is turned off and the enclosure removed, remove all pre-filters on the building HVAC system and provide new pre-filters. Dispose of filters as asbestos contaminated materials. Reestablish HVAC mechanical, and electrical systems in proper working order. The Contracting Officer will visually inspect all surfaces within the enclosure for residual material or accumulated dust or debris. The Contractor must re-clean all areas showing dust or residual materials. If re-cleaning is required, air sample and establish an acceptable asbestos airborne concentration after re-cleaning. The Contracting Officer must agree that the area is safe in

writing before unrestricted entry will be permitted. The Government must have the option to perform monitoring to determine if the areas are safe before entry is permitted.

3.3.2 Title to Materials

All waste materials, except as specified otherwise, become the property of the Contractor and must be disposed of as specified in applicable local, State, and Federal regulations and herein.

3.3.3 Disposal of Asbestos

3.3.3.1 Procedure for Disposal

Coordinate all waste disposal manifests with the Contracting Officer and NAVFAC EV. Collect asbestos waste, contaminated waste water filters, asbestos contaminated water, scrap, debris, bags, containers, equipment, and asbestos contaminated clothing which may produce airborne concentrations of asbestos fibers and place in sealed fiber-proof, waterproof, non-returnable containers (e.g. double plastic bags 0.15 mm thick, cartons, drums or cans). Wastes within the containers must be adequately wet in accordance with 40 CFR 61-SUBPART M. Affix a warning and Department of Transportation (DOT) label to each container including the bags or use at least 0.15 mm thick bags with the approved warnings and DOT labeling preprinted on the bag. Clearly indicate on the outside of each container the name of the waste generator and the location at which the waste was generated. Prevent contamination of the transport vehicle (especially if the transport vehicle is a rented truck likely to be used in the future for non-asbestos purposes). These precautions include lining the vehicle cargo area with plastic sheeting (similar to work area enclosure) and thorough cleaning of the cargo area after transport and unloading of asbestos debris is complete. Dispose of waste asbestos material at an Environmental Protection Agency (EPA) or State-approved asbestos landfill off Government property. For temporary storage, store sealed impermeable bags in asbestos waste drums or skids. An area for interim storage of asbestos waste-containing drums or skids will be assigned by the Contracting Officer or his authorized representative. Comply with 40 CFR 61-SUBPART M, State, regional, and local standards for hauling and disposal. Sealed plastic bags may be dumped from drums into the burial site unless the bags have been broken or damaged. Damaged bags must remain in the drum and the entire contaminated drum must be buried. Uncontaminated drums may be recycled. Workers unloading the sealed drums must wear appropriate respirators and personal protective equipment when handling asbestos materials at the disposal site.

3.3.3.2 Asbestos Disposal Quantity Report

Direct the PQP to record and report, to the Contracting Officer, the amount of asbestos containing material removed and released for disposal. Deliver the report for the previous day at the beginning of each day shift with amounts of material removed during the previous day reported in linear meters or square meters as described initially in this specification and in cubic meters for the amount of asbestos containing

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

material released for disposal.

Allow the GC to inspect, record and report the amount of asbestos
containing material removed and released for disposal on a daily basis.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 83 00

LEAD REMEDIATION

11/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Abatement
 - 1.2.2 Action Level
 - 1.2.3 Area Sampling
 - 1.2.4 Cadmium Permissible Exposure Limit (PEL)
 - 1.2.5 Certified Industrial Hygienist (CIH)
 - 1.2.6 Chromium Permissible Exposure Limit (PEL)
 - 1.2.7 Competent Person (CP)
 - 1.2.8 Contaminated Room
 - 1.2.9 Decontamination Shower Facility
 - 1.2.10 Deleading
 - 1.2.11 Eight-Hour Time Weighted Average (TWA)
 - 1.2.12 High Efficiency Particulate Air (HEPA) Filter Equipment
 - 1.2.13 Lead
 - 1.2.14 Lead-Based Paint (LBP)
 - 1.2.15 Lead-Based Paint Activities
 - 1.2.16 Lead-Based Paint Hazards
 - 1.2.17 Lead, Cadmium, Chromium Control Area
 - 1.2.18 Lead Permissible Exposure Limit (PEL)
 - 1.2.19 Material Containing Lead/Paint with Lead (MCL/PWL)
 - 1.2.20 Personal Sampling
 - 1.2.21 Physical Boundary
- 1.3 DESCRIPTION
 - 1.3.1 Protection of Existing Areas To Remain
 - 1.3.2 Coordination with Other Work
 - 1.3.3 Sampling and Analysis
 - 1.3.3.1 Dust Wipe Materials, Sampling and Analysis
 - 1.3.3.2 Clearance Monitoring
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Qualifications
 - 1.5.1.1 Competent Person (CP)
 - 1.5.1.2 Training Certification
 - 1.5.1.3 Testing Laboratory
 - 1.5.1.4 Certified Risk Assessor
 - 1.5.2 Requirements
 - 1.5.2.1 Competent Person (CP) Responsibilities
 - 1.5.2.2 Lead, Cadmium, Chromium Compliance Plan
 - 1.5.2.3 Medical Examinations
 - 1.5.2.4 Training
 - 1.5.2.5 Respiratory Protection Program
 - 1.5.2.6 Hazard Communication Program
 - 1.5.2.7 Lead, Cadmium, Chromium Waste Management

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 1.5.2.8 Environmental, Safety and Health Compliance
- 1.5.3 Pre-Construction Conference
- 1.6 EQUIPMENT
 - 1.6.1 Respirators
 - 1.6.2 Special Protective Clothing
 - 1.6.3 Rental Equipment Notification
 - 1.6.4 Vacuum Filters
 - 1.6.5 Equipment for Government Personnel
 - 1.6.6 Abrasive Removal Equipment
 - 1.6.7 Negative Air Pressure System
 - 1.6.7.1 Minimum Requirements
 - 1.6.8 Vacuum Systems
- 1.7 PROJECT/SITE CONDITIONS
 - 1.7.1 Protection of Existing Work to Remain

PART 2 PRODUCTS

- 2.1 MATERIALS AND EQUIPMENT
 - 2.1.1 Expendable Supplies
 - 2.1.1.1 Polyethylene Bags
 - 2.1.1.2 Polyethylene Leak-tight Wrapping
 - 2.1.1.3 Polyethylene Sheeting
 - 2.1.1.4 Tape and Adhesive Spray
 - 2.1.1.5 Containers
 - 2.1.1.6 Chemical Paint Strippers
 - 2.1.1.7 Chemical Paint Stripper Neutralizer
 - 2.1.1.8 Detergents and Cleaners

PART 3 EXECUTION

- 3.1 PREPARATION
 - 3.1.1 Protection
 - 3.1.1.1 Notification
 - 3.1.1.2 Lead, Cadmium, Chromium Control Area
 - 3.1.1.3 Local Exhaust System
 - 3.1.1.4 Negative Air Pressure System Containment
 - 3.1.1.5 Eye Wash Station
 - 3.1.1.6 Mechanical Ventilation System
 - 3.1.1.7 Personnel Protection
- 3.2 ERECTION
 - 3.2.1 Lead, Cadmium, Chromium Control Area Requirements
- 3.3 APPLICATION
 - 3.3.1 Lead, Cadmium, Chromium Work
 - 3.3.2 Paint with Lead, Cadmium, Chromium or Material Containing Lead, Cadmium, Chromium Removal
 - 3.3.2.1 Paint with Lead, Cadmium, Chromium or Material Containing Lead, Cadmium, Chromium - Indoor Removal
 - 3.3.2.2 Paint with Lead, Cadmium, Chromium or Material Containing Lead, Cadmium, Chromium - Outdoor Removal
 - 3.3.3 Personnel Exiting Procedures
- 3.4 FIELD QUALITY CONTROL
 - 3.4.1 Tests
 - 3.4.1.1 Air and Wipe Sampling
- 3.5 CLEANING AND DISPOSAL
 - 3.5.1 Cleanup
 - 3.5.1.1 Clearance Certification
 - 3.5.2 Disposal
 - 3.5.2.1 Disposal Documentation

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

-- End of Section Table of Contents --

SECTION 02 83 00

LEAD REMEDIATION
11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP Z9.2 (2018) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems

ASTM INTERNATIONAL (ASTM)

ASTM E1613 (2012) Standard Test Method for Determination of Lead by Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-AES), Flame Atomic Absorption Spectrometry (FAAS), or Graphite Furnace Atomic Absorption Spectrometry (GFAAS) Techniques

ASTM E1644 (2021) Standard Practice for Hot Plate Digestion of Dust Wipe Samples for the Determination of Lead

ASTM E1728/E1728M (2020) Standard Practice for Collection of Settled Dust Samples Using Wipe Sampling Methods for Subsequent Lead Determination

ASTM E1792 (2020) Standard Specification for Wipe Sampling Materials for Lead in Surface Dust

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety -- Safety and Health Requirements Manual

U.S. DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT (HUD)

HUD 6780 (1995; Errata Aug 1996; Rev Ch. 7 - 1997) Guidelines for the Evaluation and Control of Lead-Based Paint Hazards in Housing

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926.21 Safety Training and Education

29 CFR 1926.33 Access to Employee Exposure and Medical Records

29 CFR 1926.55	Gases, Vapors, Fumes, Dusts, and Mists
29 CFR 1926.59	Hazard Communication
29 CFR 1926.62	Lead
29 CFR 1926.65	Hazardous Waste Operations and Emergency Response
29 CFR 1926.103	Respiratory Protection
29 CFR 1926.1126	Chromium
29 CFR 1926.1127	Cadmium
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 268	Land Disposal Restrictions
40 CFR 745	Lead-Based Paint Poisoning Prevention in Certain Residential Structures

U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

ND OPNAVINST 5100.23	(2005; Rev G) Navy Occupational Safety and Health (NAVOSH) Program Manual
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UNDERWRITERS LABORATORIES (UL)

UL 586	(2009; Reprint Dec 2017) UL Standard for Safety High-Efficiency Particulate, Air Filter Units
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1.2 DEFINITIONS

1.2.1 Abatement

Measures defined in 40 CFR 745, Section 223, designed to permanently eliminate lead-based paint hazards.

1.2.2 Action Level

Employee exposure, without regard to use of respirators, to an airborne concentration of lead of 30 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of cadmium of 2.5 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of chromium (VI) of 2.5 micrograms per cubic meter of air averaged over an 8-hour period.

1.2.3 Area Sampling

Sampling of lead, cadmium, chromium concentrations within the lead, cadmium, chromium control area and inside the physical boundaries which is representative of the airborne lead, cadmium, chromium concentrations but is not collected in the breathing zone of personnel (approximately 1.5 to 1.8 meters above the floor).

1.2.4 Cadmium Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1127. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 40/\text{No. hrs worked per day}$$

1.2.5 Certified Industrial Hygienist (CIH)

As used in this section refers to a person retained by the Contractor who is certified as an industrial hygienist and who is trained in the recognition and control of lead, cadmium and chromium hazards in accordance with current federal, State, and local regulations. CIH must be certified for comprehensive practice by the American Board of Industrial Hygiene. The Certified Industrial Hygienist must be independent of the Contractor and must have no employee or employer relationship which could constitute a conflict of interest.

1.2.6 Chromium Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1126. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 40/\text{No. hrs worked per day}$$

1.2.7 Competent Person (CP)

As used in this section, refers to a person employed by the Contractor who is trained in the recognition and control of lead, cadmium and chromium hazards in accordance with current federal, State, and local regulations and has the authority to take prompt corrective actions to control the lead, cadmium and chromium hazard. The Contractor may provide more than one CP as required to supervise and monitor the work.

1.2.8 Contaminated Room

Refers to a room for removal of contaminated personal protective equipment (PPE).

1.2.9 Decontamination Shower Facility

That facility that encompasses a clean clothing storage room, and a contaminated clothing storage and disposal rooms, with a shower facility in between.

1.2.10 Deleading

Activities conducted by a person who offers to eliminate lead-based paint or lead-based paint hazards or paints containing cadmium/chromium or to plan such activities in commercial buildings, bridges or other structures.

1.2.11 Eight-Hour Time Weighted Average (TWA)

Airborne concentration of lead, cadmium, chromium to which an employee is exposed, averaged over an 8-hour workday as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.

1.2.12 High Efficiency Particulate Air (HEPA) Filter Equipment

HEPA filtered vacuuming equipment with a UL 586 filter system capable of collecting and retaining lead, cadmium, chromiumcontaminated particulate. A high efficiency particulate filter demonstrates at least 99.97 percent efficiency against 0.3 micron or larger size particles.

1.2.13 Lead

Metallic lead, inorganic lead compounds, and organic lead soaps. Excludes other forms of organic lead compounds. The use of the term Lead in this section also refers to paints which contain detectable concentrations of Cadmium and Chromium. For the purposes of the section lead-based paint (LBP) and paint with lead (PWL) also contains cadmium and chromium.

1.2.14 Lead-Based Paint (LBP)

Paint or other surface coating that contains lead in excess of 1.0 milligrams per centimeter squared or 0.5 percent by weight.

1.2.15 Lead-Based Paint Activities

In the case of target housing or child occupied facilities, lead-based paint activities include; a lead-based paint inspection, a risk assessment, or abatement of lead-based paint hazards.

1.2.16 Lead-Based Paint Hazards

Paint-lead hazard, dust-lead hazard or soil-lead hazard as identified in 40 CFR 745, Section 65. Any condition that causes exposure to lead from lead-contaminated dust, lead-contaminated soil, lead-based paint that is deteriorated or present in accessible surfaces, friction surfaces, or impact surfaces that would result in adverse human health effects.

1.2.17 Lead, Cadmium, Chromium Control Area

A system of control methods to prevent the spread of lead, cadmium, chromium dust, paint chips or debris to adjacent areas that may include temporary containment, floor or ground cover protection, physical boundaries, and warning signs to prevent unauthorized entry of personnel. HEPA filtered local exhaust equipment may be used as engineering controls

to further reduce personnel exposures or building/outdoor environmental contamination.

1.2.18 Lead Permissible Exposure Limit (PEL)

Fifty micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.62. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 400/\text{No. hrs worked per day}$$

1.2.19 Material Containing Lead/Paint with Lead (MCL/PWL)

Any material, including paint, which contains lead as determined by the testing laboratory using a valid test method. The requirements of this section does not apply if no detectable levels of lead are found using a quantitative method for analyzing paint or MCL using laboratory instruments with specified limits of detection (usually 0.01 percent). An X-Ray Fluorescence (XRF) instrument is not considered a valid test method.

1.2.20 Personal Sampling

Sampling of airborne lead, cadmium, chromium concentrations within the breathing zone of an employee to determine the 8-hour time weighted average concentration in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. Samples must be representative of the employees' work tasks. Breathing zone must be considered an area within a hemisphere, forward of the shoulders, with a radius of 150 to 225 mm and centered at the nose or mouth of an employee.

1.2.21 Physical Boundary

Area physically roped or partitioned off around lead, cadmium, chromium control area to limit unauthorized entry of personnel.

1.3 DESCRIPTION

Construction activities impacting PWL or material containing lead, cadmium, chromium which are covered by this specification include the demolition or removal of material containing lead, cadmium, chromium in dry condition, located in the coatings and as indicated on the drawings. The work covered by this section includes work tasks and the precautions specified in this section for the protection of building occupants and the environment during and after the performance of the hazard abatement activities.

1.3.1 Protection of Existing Areas To Remain

Project work including, but not limited to, lead, cadmium, chromium hazard abatement work, storage, transportation, and disposal must be performed without damaging or contaminating adjacent work and areas. Where such work or areas are damaged or contaminated, restore work and areas to the original condition.

1.3.2 Coordination with Other Work

Coordinate with work being performed in adjacent areas to ensure there are no exposure issues. Explain coordination procedures in the Lead, Cadmium, Chromium Compliance Plan and describe how the Contractor will prevent

lead, cadmium and chromium exposure to other contractors and Government personnel performing work unrelated to lead, cadmium and chromium activities.

1.3.3 Sampling and Analysis

Submit a log of the analytical results from sampling conducted during the abatement. Keep the log of results current with project activities and brief the results to the Contracting Officer as analytical results are reported.

1.3.3.1 Dust Wipe Materials, Sampling and Analysis

Sampling must conform to ASTM E1728/E1728M or ASTM E1792. Analysis must conform to ASTM E1613 and ASTM E1644.

1.3.3.2 Clearance Monitoring

- a. Collect dust wipe samples inside the lead, cadmium and chromium hazard control area after the final visual inspection in the quantities and at the locations specified.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Competent Person Qualifications; G

Training Certification; G

Medical Examinations; G

Lead, Cadmium, Chromium Waste Management Plan; G

Lead, Cadmium, Chromium Compliance Plan; G

Written Evidence of TSD Approval; G

SD-03 Product Data

Respirators; G

Vacuum Filters; G

Negative Air Pressure System; G

Materials and Equipment; G

Expendable Supplies; G

Local Exhaust Equipment; G

SD-06 Test Reports

Sampling and Analysis; G

SD-07 Certificates

Testing Laboratory; G

Clearance Certification; G

SD-11 Closeout Submittals

Hazardous Waste Manifest; G

1.5 QUALITY ASSURANCE

1.5.1 Qualifications

1.5.1.1 Competent Person (CP)

Submit name, address, and telephone number of the CP selected to perform responsibilities specified in paragraph COMPETENT PERSON (CP) RESPONSIBILITIES. Provide documented construction project-related experience with implementation of OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium standard (29 CFR 1926.1126), Cadmium standard (29 CFR 1926.1127) which shows ability to assess occupational and environmental exposure to lead, cadmium, chromium; experience with the use of respirators, personal protective equipment and other exposure reduction methods to protect employee health. Demonstrate a minimum of 3 years experience implementing OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127). Submit proper documentation that the CP is trained in accordance with federal, State and local laws.

1.5.1.2 Training Certification

Submit a certificate for each worker and supervisor, signed and dated by the training provider, stating that the employee has received the required lead, cadmium and chromium training specified in 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.

1.5.1.3 Testing Laboratory

Submit the name, address, and telephone number of the testing laboratory selected to perform the air analysis, testing, and reporting of airborne concentrations of lead, cadmium and chromium.

1.5.1.4 Certified Risk Assessor

The Certified Risk Assessor must be certified pursuant to 40 CFR 745, Section 226 and be responsible to perform the clearance sampling, clearance sample data evaluation and summarize clearance sampling results in a section of the abatement report. The risk assessor must sign the abatement report to indicate clearance requirements for the contract have been met.

1.5.2 Requirements

1.5.2.1 Competent Person (CP) Responsibilities

- a. Verify training meets all federal, State, and local requirements.

- b. Review and approve Lead, Cadmium, Chromium Compliance Plan for conformance to the applicable referenced standards.
- c. Continuously inspect LBP/PWL or MCL work for conformance with the approved plan.
- d. Perform (or oversee performance of) air sampling. Recommend upgrades or downgrades (whichever is appropriate based on exposure) on the use of PPE (respirators included) and engineering controls.
- e. Ensure work is performed in strict accordance with specifications at all times.
- f. Control work to prevent hazardous exposure to human beings and to the environment at all times.
- g. Supervise final cleaning of the lead, cadmium, chromium control area, take clearance wipe samples if necessary; review clearance sample results and make recommendations for further cleaning.
- h. Certify the conditions of the work as called for elsewhere in this specification.
- i. The CP must be certified pursuant to 40 CFR 745, Section 226 and is responsible for development and implementation of the abatement report and supervise lead, cadmium and chromium hazard abatement work activities.

1.5.2.2 Lead, Cadmium, Chromium Compliance Plan

Submit a detailed job-specific plan of the work procedures to be used in the disturbance of lead, cadmium and chromium, LBP/PWL or MCL. Include in the plan a sketch showing the location, size, and details of lead, cadmium, chromium control areas, critical barriers, physical boundaries, location and details of decontamination facilities, viewing ports, and mechanical ventilation system. Include a description of equipment and materials, work practices, controls and job responsibilities for each activity from which lead, cadmium, chromium is emitted. Include in the plan, eating, drinking, smoking, hygiene facilities and sanitary procedures, interface of trades, sequencing of lead, cadmium, chromium related work, collected waste water and dust containing lead, cadmium, chromium and debris, air sampling, respirators, personal protective equipment, and a detailed description of the method of containment of the operation to ensure that lead, cadmium, chromium is not released outside of the lead, cadmium, chromium control area. Include site preparation, cleanup and clearance procedures. Include occupational and environmental sampling, training and strategy, sampling and analysis strategy and methodology, frequency of sampling, duration of sampling, and qualifications of sampling personnel in the air sampling portion of the plan. Include a description of arrangements made among contractors on multicontractor worksites to inform affected employees and to clarify responsibilities to control exposures.

1.5.2.3 Medical Examinations

Submit pre-work blood lead levels and post-work blood lead levels for all workers performing lead, cadmium, chromium activities during the execution of the work. Initial medical surveillance as required by 29 CFR 1926.62,

29 CFR 1926.1126, 29 CFR 1926.1127 must be made available to all employees exposed to lead, cadmium, chromium at any time (one day) above the action level. Full medical surveillance must be made available to all employees on an annual basis who are or may be exposed to lead, cadmium and chromium in excess of the action level for more than 30 days a year or as required by 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. Adequate records must show that employees meet the medical surveillance requirements of 29 CFR 1926.33, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 29 CFR 1926.103. Provide medical surveillance to all personnel exposed to lead, cadmium, chromium as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. Maintain complete and accurate medical records of employees for the duration of employment plus 30 years.

1.5.2.4 Training

Train each employee performing work that disturbs lead, cadmium, chromium, who performs LBP/MCL/PWL disposal, and air sampling operations prior to the time of initial job assignment and annually thereafter, in accordance with 29 CFR 1926.21, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, 40 CFR 745 and State, regional and local regulations where appropriate.

1.5.2.5 Respiratory Protection Program

- a. Provide each employee required to wear a respirator a respirator fit test at the time of initial fitting and at least annually thereafter as required by 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.
- b. Establish and implement a respiratory protection program as required by 29 CFR 1926.103, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 29 CFR 1926.55.

1.5.2.6 Hazard Communication Program

Establish and implement a Hazard Communication Program as required by 29 CFR 1926.59.

1.5.2.7 Lead, Cadmium, Chromium Waste Management

The [Lead, Cadmium, Chromium Waste Management Plan](#) must comply with applicable requirements of federal, State, and local hazardous waste regulations and address:

- a. Identification and classification of wastes associated with the work.
- b. Estimated quantities of wastes to be generated and disposed of.
- c. Names and qualifications of each contractor that will be transporting, storing, treating, and disposing of the wastes. Include the facility location and a 24-hour point of contact.
- d. Names and qualifications (experience and training) of personnel who will be working on-site with hazardous wastes.
- e. List of waste handling equipment to be used in performing the work, to include cleaning, volume reduction, and transport equipment.
- f. Spill prevention, containment, and cleanup contingency measures including a health and safety plan to be implemented in accordance with 29 CFR 1926.65.

- g. Work plan and schedule for waste containment, removal and disposal. Proper containment of the waste includes using acceptable waste containers (e.g., 55-gallon drums) as well as proper marking/labeling of the containers. Clean up and containerize wastes daily.
- h. Include any process that may alter or treat waste rendering a hazardous waste non hazardous.
- i. Unit cost for hazardous waste disposal according to this plan.

1.5.2.8 Environmental, Safety and Health Compliance

In addition to the detailed requirements of this specification, comply with laws, ordinances, rules, and regulations of federal, State, and local authorities regarding lead, cadmium and chromium. Comply with the applicable requirements of the current issue of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, EM 385-1-1, ND OPNAVINST 5100.23. Submit matters regarding interpretation of standards to the Contracting Officer for resolution before starting work. Where specification requirements and the referenced documents vary, the most stringent requirements apply.

1.5.3 Pre-Construction Conference

Along with the CP, meet with the Contracting Officer to discuss in detail the Lead, Cadmium, Chromium Waste Management Plan and the Lead, Cadmium, Chromium Compliance Plan, including procedures and precautions for the work.

1.6 EQUIPMENT

1.6.1 Respirators

Furnish appropriate respirators approved by the National Institute for Occupational Safety and Health (NIOSH), Department of Health and Human Services, for use in atmospheres containing lead, cadmium and chromium dust, fume and mist. Respirators must comply with the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.

1.6.2 Special Protective Clothing

Personnel exposed to lead, cadmium, chromiumcontaminated dust must wear proper uncontaminated, reusable protective whole body clothing, head covering, gloves, eye, and foot coverings as required by 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. Furnish proper disposable plastic or rubber gloves to protect hands. Reduce the level of protection only after obtaining approval from the CP.

1.6.3 Rental Equipment Notification

If rental equipment is to be used during PWL or MCL handling and disposal, notify the rental agency in writing concerning the intended use of the equipment.

1.6.4 Vacuum Filters

UL 586 labeled HEPA filters.

1.6.5 Equipment for Government Personnel

Furnish the Contracting Officer with two complete sets of personal protective equipment (PPE) daily, as required herein, for entry into and inspection of the lead, cadmium and chromium removal work within the lead, cadmium and chromium controlled area. Personal protective equipment must include disposable whole body covering, including appropriate foot, head, eye, and hand protection. PPE remains the property of the Contractor. The Government will provide respiratory protection for the Contracting Officer.

1.6.6 Abrasive Removal Equipment

The use of powered machine for vibrating, sanding, grinding, or abrasive blasting is prohibited unless equipped with local exhaust ventilation systems equipped with high efficiency particulate air (HEPA) filters.

1.6.7 Negative Air Pressure System

1.6.7.1 Minimum Requirements

Do not proceed with work in the area until containment is set up and HEPA filtration systems are in place. The negative air pressure system must meet the requirements of [ASSP Z9.2](#) including approved HEPA filters in accordance with [UL 586](#). Negative air pressure equipment must be equipped with new HEPA filters, and be sufficient to maintain a minimum pressure differential of minus [0.005 kPa](#) relative to adjacent, unsealed areas. Negative air pressure system minimum requirements are listed as follows:

- a. The unit must be capable of delivering its rated volume of air with a clean first stage filter, an intermediate filter and a primary HEPA filter in place.
- b. The HEPA filter must be certified as being capable of trapping and retaining mono-disperse particles as small as 0.3 micrometers at a minimum efficiency of 99.97 percent.
- c. The unit must be capable of continuing to deliver no less than 70 percent of rated capacity when the HEPA filter is 70 percent full or measures [0.625 kPa](#) static pressure differential on a magnehelic gauge.
- d. Equip the unit with a manometer-type negative pressure differential monitor with minor scale division of [0.005 kPa](#) and accuracy within plus or minus 1.0 percent. The manometer must be calibrated daily as recommended by the manufacturer.
- e. Equip the unit with a means for the operator to easily interpret the readings in terms of the volumetric flow rate of air per minute moving through the machine at any given moment.
- f. Equip the unit with an electronic mechanism that automatically shuts the machine off in the event of a filter breach or absence of a filter.
- g. Equip the unit with an audible horn that sounds an alarm when the machine has shut itself off.
- h. Equip the unit with an automatic safety mechanism that prevents a worker from improperly inserting the main HEPA filter.

1.6.8 Vacuum Systems

Vacuum systems must be suitably sized for the project, and filters must be capable of trapping and retaining all mono-disperse particles as small as 0.3 micrometers (mean aerodynamic diameter) at a minimum efficiency of 99.97 percent. Properly dispose of used filters that are being replaced.

1.7 PROJECT/SITE CONDITIONS

1.7.1 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent areas. Where existing work is damaged or contaminated, restore work to its original condition or better as determined by the Contracting Officer.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

Keep materials and equipment needed to complete the project available and on the site. Submit a description of the materials and equipment required; including Safety Data Sheets (SDSs) for material brought onsite to perform the work.

2.1.1 Expendable Supplies

Submit a description of the expendable supplies required.

2.1.1.1 Polyethylene Bags

Disposable bags must be polyethylene plastic and be a minimum of 0.15 mm thick (0.1 mm thick if double bags are used) or any other thick plastic material shown to demonstrate at least equivalent performance; and capable of being made leak-tight. Leak-tight means that solids, liquids or dust cannot escape or spill out.

2.1.1.2 Polyethylene Leak-tight Wrapping

Wrapping used to wrap lead, cadmium, chromium contaminated debris must be polyethylene plastic that is a minimum of 0.15 mm thick or any other thick plastic material shown to demonstrate at least equivalent performance.

2.1.1.3 Polyethylene Sheeting

Sheeting must be polyethylene plastic with a minimum thickness of 0.15 mm, or any other thick plastic material shown to demonstrate at least equivalent performance; and be provided in the largest sheet size reasonably accommodated by the project to minimize the number of seams. Where the project location constitutes an out of the ordinary potential for fire, or where unusual fire hazards cannot be eliminated, provide flame-resistant polyethylene sheets.

2.1.1.4 Tape and Adhesive Spray

Tape and adhesive must be capable of sealing joints between polyethylene sheets and for attachment of polyethylene sheets to adjacent surfaces. After dry application, tape or adhesive must retain adhesion when exposed to wet conditions, including amended water. Tape must be minimum 50 mm wide, industrial strength.

2.1.1.5 Containers

When used, containers must be leak-tight.

2.1.1.6 Chemical Paint Strippers

Chemical paint strippers must not contain methylene chloride and be formulated to prevent stain, discoloration, or raising of the substrate materials.

2.1.1.7 Chemical Paint Stripper Neutralizer

Neutralizers for paint strippers must be compatible with the substrate and suitable for use with the chemical stripper that has been applied to the surface.

2.1.1.8 Detergents and Cleaners

Detergents or cleaning agents must not contain trisodium phosphate and have demonstrated effectiveness in lead, cadmium and chromium control work using cleaning techniques specified by HUD 6780 guidelines.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Protection

3.1.1.1 Notification

- a. Notify the Contracting Officer 20 days prior to the start of any lead, cadmium and chromium work.

3.1.1.2 Lead, Cadmium, Chromium Control Area

- a. Physical Boundary - Provide physical boundaries around the lead, cadmium, chromium control area by roping off the area designated in the work plan or providing curtains, portable partitions or other enclosures to ensure that lead, cadmium and chromium will not escape outside of the lead, cadmium and chromium control area. Prohibit the general public from accessing the lead, cadmium, chromium control areas.
- b. Warning Signs - Provide warning signs at approaches to lead, cadmium, chromium control areas. Locate signs at such a distance that personnel may read the sign and take the necessary precautions before entering the area. Signs must comply with the requirements of 29 CFR 1926.62.

3.1.1.3 Local Exhaust System

Provide a local exhaust system in the lead, cadmium, chromium control area in accordance with ASSP Z9.2, 29 CFR 1926.62, 29 CFR 1926.1126 and 29 CFR 1926.1127 that will provide at least 4 air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until the lead, cadmium, chromium control area is removed and must be leak proof to the filter and equipped with HEPA filters. Maintain a minimum pressure differential in the lead, cadmium,

chromium control area of minus 0.51 mm of water column relative to adjacent, unsealed areas. Filters on exhaust equipment must conform to ASSP Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.1.4 Negative Air Pressure System Containment

- a. Operate the negative air pressure systems to provide at least 4 air changes per hour inside the containment. Operate the local exhaust unit equipment continuously until the containment is removed. Notify the Contracting Officer immediately if the pressure differential falls below the prescribed minimum. Terminate the local exhaust system out of doors unless the Contracting Officer allows an alternate arrangement. All filters must be new at the beginning of the project and be periodically changed as necessary to maintain specified pressure differential and disposed of as lead, cadmium and chromium contaminated waste.
- b. Discontinuing Negative Air Pressure System. Operate the negative air pressure system continuously during abatement activities unless otherwise authorized by the Contracting Officer. At the completion of the project, units must be run until full cleanup has been completed and final clearance testing requirements have been met. Dismantling of the negative air pressure systems must conform to written decontamination procedures. Seal the HEPA filter machine intakes with polyethylene to prevent environmental contamination.

3.1.1.5 Eye Wash Station

Provide suitable facilities within the work area for quick drenching or flushing of the eyes where eyes may be exposed to injurious corrosive materials.

3.1.1.6 Mechanical Ventilation System

- a. Use adequate ventilation to control personnel exposure to lead, cadmium and chromium in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. To the extent feasible, use local exhaust ventilation or other collection systems, approved by the CP. Evaluate and maintain local exhaust ventilation systems in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.
- b. Vent local exhaust outside the building and away from building ventilation intakes or ensure system is connected to HEPA filters.
- c. Use locally exhausted, power actuated tools or manual hand tools.

3.1.1.7 Personnel Protection

Personnel must wear and use protective clothing and equipment as specified herein. Eating, smoking, or drinking or application of cosmetics is not permitted in the lead, cadmium, chromium control area. No one will be permitted in the lead, cadmium, chromium control area unless they have been appropriately trained and provided with protective equipment.

3.2 ERECTION

3.2.1 Lead, Cadmium, Chromium Control Area Requirements

Establish a lead, cadmium, chromium control area by completely establishing barriers and physical boundaries around the area or structure where PWL or MCL removal operations will be performed.

3.3 APPLICATION

3.3.1 Lead, Cadmium, Chromium Work

Perform lead, cadmium, chromium work in accordance with approved Lead, Cadmium, Chromium Compliance Plan. Use procedures and equipment required to limit occupational exposure and environmental contamination with lead, cadmium, chromium when the work is performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and as specified herein. Dispose of all PWL or MCL and associated waste in compliance with federal, State, and local requirements.

3.3.2 Paint with Lead, Cadmium, Chromium or Material Containing Lead, Cadmium, Chromium Removal

Manual or power sanding or grinding of lead, cadmium, chromium surfaces or materials is not permitted unless tools are equipped with HEPA attachments or wet methods. The dry sanding or grinding of surfaces that contain lead, cadmium, chromium is prohibited. Provide methodology for removing lead, cadmium, chromium in the Lead, Cadmium, Chromium Compliance Plan. Select lead, cadmium, chromium removal processes to minimize contamination of work areas outside the control area with lead, cadmium, chromium contaminated dust or other lead, cadmium, chromium contaminated debris or waste and to ensure that unprotected personnel are not exposed to hazardous concentrations of lead, cadmium, chromium. Describe this removal process in the Lead, Cadmium, Chromium Compliance Plan.

3.3.2.1 Paint with Lead, Cadmium, Chromium or Material Containing Lead, Cadmium, Chromium - Indoor Removal

Perform manual removal and thermal cutting in the lead, cadmium, chromium control areas using enclosures, barriers or containments. Collect residue and debris for disposal in accordance with federal, State, and local requirements.

3.3.2.2 Paint with Lead, Cadmium, Chromium or Material Containing Lead, Cadmium, Chromium - Outdoor Removal

Perform outdoor removal as indicated in federal, State, and local regulations and in the Lead, Cadmium, Chromium Compliance Plan. The worksite preparation (barriers or containments) must be job dependent and presented in the Lead, Cadmium, Chromium Compliance Plan.

3.3.3 Personnel Exiting Procedures

Whenever personnel exit the lead, cadmium, chromium controlled area, they must perform the following procedures and must not leave the work place wearing any clothing or equipment worn in the control area:

- a. Vacuum all clothing before entering the contaminated change room.

- b. Remove protective clothing in the contaminated change room, and place them in an approved impermeable disposal bag.
- c. Change to clean clothes prior to leaving the clean clothes storage area.

3.4 FIELD QUALITY CONTROL

3.4.1 Tests

3.4.1.1 Air and Wipe Sampling

Conduct sampling for lead, cadmium, chromium in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and as specified herein. Air and wipe sampling must be directed or performed by the CP.

- a. The CP must be on the job site directing the air and wipe sampling and inspecting the PWL or MCL removal work to ensure that the requirements of the contract have been satisfied during the entire PWL or MCL operation.
- b. Collect personal air samples on employees who are anticipated to have the greatest risk of exposure as determined by the CP. In addition, collect air samples on at least twenty-five percent of the work crew or a minimum of two employees, whichever is greater, during each work shift.
- c. Submit results of air samples, signed by the CP, within 72-hours after the air samples are taken.
- d. Conduct area air sampling daily, on each shift in which lead, cadmium and chromium and lead-based paint removal operations are performed, in areas immediately adjacent to the lead, cadmium and chromium control area. Conduct sufficient area monitoring to ensure unprotected personnel are not exposed at or above 30 micrograms of lead per cubic meter of air or 2.5 micrograms of cadmium/chromium per cubic meter of air. If 30 micrograms of lead per cubic meter of air or 2.5 micrograms of cadmium/chromium per cubic meter of air is reached or exceeded, stop work, correct the condition(s) causing the increased levels. Notify the Contracting Officer immediately. Determine if condition(s) require any further change in work methods. Resume removal work only after the CP and the Contracting Officer give approval.

3.5 CLEANING AND DISPOSAL

3.5.1 Cleanup

Maintain surfaces of the lead, cadmium, chromium control area free of accumulations of dust and debris. Restrict the spread of dust and debris; keep waste from being distributed over the work area. Do not dry sweep or use pressurized air to clean up the area. At the end of each shift and when the lead, cadmium, chromium operation has been completed, clean the controlled area of all visible contamination by vacuuming with a HEPA filtered vacuum cleaner, wet mopping the area and wet wiping the area as indicated by the Lead, Cadmium, Chromium Compliance Plan. Reclean areas showing dust or debris. After visible dust and debris is removed, wet wipe and HEPA vacuum all surfaces in the controlled area. If adjacent areas become contaminated at any time during the work, clean, visually inspect, and then wipe sample all contaminated areas. The CP must then certify in

writing that the area has been cleaned of lead, cadmium and chromium contamination before clearance testing.

3.5.1.1 Clearance Certification

The CP must certify in writing that air samples collected outside the lead, cadmium, chromium control area during paint removal operations are less than 30 micrograms of lead per cubic meter of air and less than 2.5 micrograms of cadmium/chromium per cubic meter of air; the respiratory protection used for the employees was adequate; the work procedures were performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127; and that there were no visible accumulations of material and dust containing lead, cadmium, chromium left in the work site. Do not remove the lead, cadmium, chromium control area or roped off boundary and warning signs prior to the Contracting Officer's acknowledgement of receipt of the CP certification.

3.5.2 Disposal

- a. Dispose of material, whether hazardous or non-hazardous in accordance with all laws and provisions and all federal, State or local regulations. Ensure all waste is properly characterized. The result of each waste characterization (TCLP for RCRA materials) will dictate disposal requirements.
- b. Contractor is responsible for segregation of waste. Collect lead, cadmium, chromium contaminated waste, scrap, debris, bags, containers, equipment, and lead, cadmium, chromium contaminated clothing that may produce airborne concentrations of lead, cadmium, chromium particles. Label the containers in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 40 CFR 261, 40 CFR 262 and corresponding state regulations.
- c. Dispose of lead, cadmium, chromium contaminated material classified as hazardous waste at an approved hazardous waste treatment, storage, or disposal facility off Government property.
- d. Handle, store, transport, and dispose lead, cadmium, chromium or lead, cadmium, chromium contaminated waste in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, and 40 CFR 265. Comply with land disposal restriction notification requirements as required by 40 CFR 268.
- e. All lead, cadmium, and chromium waste generation, management, and disposal will be coordinated with the host installation environmental function.

3.5.2.1 Disposal Documentation

Coordinate all disposal or off-site shipments of lead, cadmium, and chromium waste with the host installation environmental function. Submit written evidence of TSD approval to demonstrate the hazardous waste treatment, storage, or disposal facility (TSD) is approved for lead, cadmium, chromium disposal by the EPA, State or local regulatory agencies. Submit one copy of the completed hazardous waste manifest, signed and dated by the initial transporter in accordance with 40 CFR 262. Provide a certificate that the waste was accepted by the disposal facility.

-- End of Section --

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

SECTION TABLE OF CONTENTS

DIVISION 03 - CONCRETE

SECTION 03 30 00

CAST-IN-PLACE CONCRETE

02/19, CHG 4: 08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
- 1.3 SUBMITTALS
- 1.4 MODIFICATION OF REFERENCES
- 1.5 DELIVERY, STORAGE, AND HANDLING
 - 1.5.1 Reinforcement
- 1.6 QUALITY ASSURANCE
 - 1.6.1 Design Data
 - 1.6.1.1 Formwork Calculations
 - 1.6.1.2 Concrete Mix Design
 - 1.6.2 Shop Drawings
 - 1.6.2.1 Formwork
 - 1.6.2.2 Reinforcing Steel
 - 1.6.3 Control Submittals
 - 1.6.3.1 Concrete Curing Plan
 - 1.6.3.2 Pumping Concrete
 - 1.6.3.3 VOC Content for form release agents, curing compounds, and concrete penetrating sealers
 - 1.6.3.4 Safety Data Sheets
 - 1.6.4 Test Reports
 - 1.6.4.1 Fly Ash and Pozzolan
 - 1.6.4.2 Slag Cement
 - 1.6.4.3 Aggregates
 - 1.6.5 Field Samples
 - 1.6.5.1 Surface Finish Samples
 - 1.6.6 Quality Control Plan
 - 1.6.7 Quality Control Personnel Certifications
 - 1.6.7.1 Quality Manager Qualifications
 - 1.6.7.2 Field Testing Technician and Testing Agency
 - 1.6.8 Laboratory Qualifications for Concrete Qualification Testing
 - 1.6.9 Laboratory Accreditation
- 1.7 ENVIRONMENTAL REQUIREMENTS
 - 1.7.1 Submittals for Environmental Performance
- 1.8 SUSTAINABLE DESIGN REQUIREMENTS
 - 1.8.1 Local/Regional Materials
 - 1.8.2 Forest Stewardship Council (FSC) Certification

PART 2 PRODUCTS

- 2.1 FORMWORK MATERIALS
 - 2.1.1 Wood Forms
 - 2.1.1.1 Concrete Form Plywood (Standard Rough)
 - 2.1.2 Steel Forms
- 2.2 FORMWORK ACCESSORIES

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.2.1 Form Ties
- 2.2.2 Waterstops
 - 2.2.2.1 Thermoplastic Elastomeric Rubber Waterstop
- 2.2.3 Biodegradable Form Release Agent
- 2.2.4 Chamfer Materials
- 2.2.5 Construction and movement joints
- 2.2.6 Other Embedded items
- 2.3 CONCRETE MATERIALS
 - 2.3.1 Cementitious Materials
 - 2.3.1.1 Portland Cement
 - 2.3.1.2 Fly Ash
 - 2.3.1.3 Slag Cement
 - 2.3.1.4 Other Supplementary Cementitious Materials
 - 2.3.2 Water
 - 2.3.3 Aggregate
 - 2.3.3.1 Normal-Weight Aggregate
 - 2.3.3.2 Recycled Aggregate Materials
 - 2.3.4 Admixtures
- 2.4 MISCELLANEOUS MATERIALS
 - 2.4.1 Concrete Curing Materials
 - 2.4.2 Nonshrink Grout
 - 2.4.3 Floor Finish Materials
 - 2.4.3.1 Liquid Chemical Floor Hardeners and Sealers
 - 2.4.4 Expansion/Contraction Joint Filler
 - 2.4.5 Joint Sealants
 - 2.4.5.1 Horizontal Surfaces, 3 Percent Slope, Maximum
 - 2.4.5.2 Vertical Surfaces Greater Than 3 Percent Slope
 - 2.4.6 Vapor Barrier
- 2.5 CONCRETE MIX DESIGN
 - 2.5.1 Properties and Requirements
 - 2.5.2 Durability
 - 2.5.2.1 Alkali-Aggregate Reaction
 - 2.5.2.2 Freezing and Thawing Resistance
 - 2.5.2.3 Corrosion and Chloride Content
 - 2.5.2.4 Sulfate Resistance
 - 2.5.2.5 Concrete Temperature
 - 2.5.2.6 Concrete permeability
 - 2.5.3 Ready-Mix Concrete
- 2.6 REINFORCEMENT
 - 2.6.1 Reinforcing Bars
 - 2.6.1.1 Headed Reinforcing Bars
 - 2.6.2 Welded Wire Reinforcement
 - 2.6.3 Reinforcing Bar Supports
 - 2.6.4 Dowels for Load Transfer in Floors

PART 3 EXECUTION

- 3.1 EXAMINATION
- 3.2 PREPARATION
 - 3.2.1 General
 - 3.2.2 Subgrade Under Foundations and Footings
 - 3.2.3 Subgrade Under Slabs on Ground
 - 3.2.4 Edge Forms and Screed Strips for Slabs
 - 3.2.5 Reinforcement and Other Embedded Items
- 3.3 FORMS
 - 3.3.1 Coating
 - 3.3.2 Reshoring
 - 3.3.3 Reuse
 - 3.3.4 Forms for Standard Rough Form Finish

- 3.3.5 Forms for Standard Smooth Form Finish
- 3.3.6 Form Ties
- 3.3.7 Tolerances for Form Construction
- 3.3.8 Removal of Forms and Supports
- 3.3.9 Strength of Concrete Required for Removal of Formwork
- 3.4 WATERSTOP INSTALLATION AND SPLICES
 - 3.4.1 Thermoplastic Elastomeric Rubber Waterstop
 - 3.4.2 Hydrophilic Waterstop
- 3.5 PLACING REINFORCEMENT AND MISCELLANEOUS MATERIALS
 - 3.5.1 Reinforcement Supports
 - 3.5.2 Splicing
 - 3.5.3 Setting Miscellaneous Material
 - 3.5.4 Fabrication
 - 3.5.5 Placing Reinforcement
 - 3.5.6 Spacing of Reinforcing Bars
 - 3.5.7 Concrete Protection for Reinforcement
- 3.6 BATCHING, MEASURING, MIXING, AND TRANSPORTING CONCRETE
 - 3.6.1 Measuring
 - 3.6.2 Mixing
 - 3.6.3 Transporting
- 3.7 PLACING CONCRETE
 - 3.7.1 Pumping
 - 3.7.2 Cold Weather
 - 3.7.3 Hot Weather
 - 3.7.4 Bonding
- 3.8 WASTE MANAGEMENT
 - 3.8.1 Mixing Equipment
 - 3.8.2 Reinforcing Steel
 - 3.8.3 Other Waste
- 3.9 SURFACE FINISHES EXCEPT FLOOR, SLAB, AND PAVEMENT FINISHES
 - 3.9.1 Defects
 - 3.9.2 Not Against Forms (Top of Walls)
 - 3.9.3 Formed Surfaces
 - 3.9.3.1 Tolerances
 - 3.9.3.2 As-Cast Rough Form
 - 3.9.3.3 Standard Smooth Finish
 - 3.9.4 Smooth-Rubbed Finish
- 3.10 FLOOR, SLAB, AND PAVEMENT FINISHES AND MISCELLANEOUS CONSTRUCTION
 - 3.10.1 Finish
 - 3.10.1.1 Scratched
 - 3.10.1.2 Floated
 - 3.10.1.3 Steel Troweled
 - 3.10.1.4 Broomed
 - 3.10.1.5 Pavement
 - 3.10.2 Concrete Walks
 - 3.10.3 Pits and Trenches
 - 3.10.4 Curbs and Gutters
 - 3.10.5 Splash Blocks
- 3.11 JOINTS
 - 3.11.1 Construction Joints
 - 3.11.1.1 Construction Joints for Constructability Purposes
 - 3.11.2 Isolation Joints in Slabs on Ground
 - 3.11.3 Contraction Joints in Slabs on Ground
 - 3.11.4 Sealing Joints in Slabs on Ground
- 3.12 CURING AND PROTECTION
 - 3.12.1 Curing Periods
 - 3.12.2 Curing Formed Surfaces
 - 3.12.3 Curing Unformed Surfaces
 - 3.12.4 Temperature of Concrete During Curing

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.12.5 Protection from Mechanical Injury
- 3.12.6 Protection After Curing
- 3.13 FIELD QUALITY CONTROL
 - 3.13.1 Aggregate Testing
 - 3.13.1.1 Fine Aggregate
 - 3.13.1.2 Coarse Aggregate
 - 3.13.2 Concrete Sampling
 - 3.13.3 Concrete Testing
 - 3.13.3.1 Slump Tests
 - 3.13.3.2 Temperature Tests
 - 3.13.3.3 Compressive Strength Tests
 - 3.13.3.4 Air Content
 - 3.13.3.5 Chloride Ion Concentration
 - 3.13.3.6 Strength of Concrete Structure
 - 3.13.3.7 Non-Conforming Materials
 - 3.13.3.8 Testing Concrete Structure for Strength
- 3.14 REPAIR, REHABILITATION AND REMOVAL
 - 3.14.1 Crack Repair
 - 3.14.2 Repair of Weak Surfaces
 - 3.14.3 Failure of Quality Assurance Test Results

-- End of Section Table of Contents --

SECTION 03 30 00

CAST-IN-PLACE CONCRETE
02/19, CHG 4: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 36061	(2014) Steel welded fabric for structural use in concrete reinforcement. Steel welded fabric made out of steel bars B 500 S
UNE 36068	(2011) Ribbed bars of weldable steel for the reinforcement of concrete.
UNE 36092	(2014) Steel welded fabric for structural use in concrete reinforcement. Steel welded fabric made out of steel wires B 500 T
UNE 36097-1	(1981) Hot Rolled Plain Bars for the Reinforcement of Concrete. Characteristics.
UNE 36731	(1996) Plains Steel Wire For Welded Fabric And Lattice Girders.
UNE 41107	(1961) Prefabricated Products. Elastics and of Low Transversal Dilation for the Filling of Expansion Joints in Concrete Paving
UNE 146508	(2018) Test For Aggregates. Determination Of The Alkali-Silica And Alkali-Silicate Potential Reactivity Of Aggregates. Accelerated Mortar Bar Test.
UNE 146509	(2018) Determination Of The Potential Reactivity Of Aggregates. Concrete Prisms Method
UNE-EN 196-5	(2011) Methods of testing cement - Part 5: Pozzolanicity test for pozzolanic cement
UNE-EN 197-1	(2011) Cement - Part 1: Composition, specifications and conformity criteria for common cements.
UNE-EN 450-1	(2013) Fly ash for concrete - Part 1: Definition, specifications and conformity

criteria.

UNE-EN 932-1	(1997) Test For Geometrical Properties Of Aggregates. Part 1: Methods for Sampling.
UNE-EN 932-3	(A1:2004) Tests For General Properties Of Aggregates. Part 3: Procedure And Terminology For Simplified Petrographic Description.
UNE-EN 933-1	(2012) Tests for geometrical properties of aggregates - Part 1: Determination of particle size distribution - Sieving method
UNE-EN 933-3	(2012) Tests for geometrical properties of aggregates - Part 3: Determination of particle shape - Flakiness index
UNE-EN 933-4	(2008) Tests for geometrical properties of aggregates - Part 4: Determination of particle shape - Shape index
UNE-EN 934-2	(2010+A1:2012) Admixtures for concrete, mortar and grout. Part 2: Concrete admixtures. Definitions, requirements, conformity, marking and labeling.
UNE-EN 1367-2	(2010) Tests for thermal and weathering properties of aggregates - Part 2: Magnesium sulfate test
UNE-EN 10080	(2006) Steel for the reinforcement of concrete - Weldable reinforcing steel - General.
UNE-EN 12350-1	(2020) Testing fresh concrete - Part 1: Sampling and common apparatus
UNE-EN 12350-2	(2020) Testing fresh concrete - Part 2: Slump-test
UNE-EN 12390-2	(2020) Testing hardened concrete. Part 2: Making and curing specimens for strength tests.
UNE-EN 12390-3	(2020) Testing hardened concrete - Part 3: Compressive strength of test specimens.
UNE-EN 12504-1	(2020) Testing concrete in structures. Part 1: Cored specimens. Taking, examining and testing in compression.
UNE-EN 12620	(2003+A1:2009) Aggregates for concrete
UNE-EN 15167-1	(2008) Ground granulated blast furnace slag for use in concrete, mortar and grout - Part 1: Definitions, specifications and conformity criteria

UNE-EN ISO/IEC 17025 (2017) General requirements for the competence of testing and calibration laboratories (ISO/IEC 17025:2017)

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 117 (2010; Errata 2011) Specifications for Tolerances for Concrete Construction and Materials and Commentary

ACI 121R (2008) Guide for Concrete Construction Quality Systems in Conformance with ISO 9001

ACI 301 (2016) Specifications for Structural Concrete

ACI 302.1R (2015) Guide for Concrete Floor and Slab Construction

ACI 304R (2000; R 2009) Guide for Measuring, Mixing, Transporting, and Placing Concrete

ACI 305R (2020) Guide to Hot Weather Concreting

ASTM INTERNATIONAL (ASTM)

ASTM A934/A934M (2016) Standard Specification for Epoxy-Coated Prefabricated Steel Reinforcing Bars

ASTM C150/C150M (2021) Standard Specification for Portland Cement

ASTM C173/C173M (2016) Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method

ASTM C231/C231M (2017a) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method

ASTM C595/C595M (2021) Standard Specification for Blended Hydraulic Cements

ASTM C618 (2023; E 2023) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete

ASTM C920 (2018) Standard Specification for Elastomeric Joint Sealants

ASTM C989/C989M (2022) Standard Specification for Slag Cement for Use in Concrete and Mortars

ASTM C1074 (2019) Standard Practice for Estimating Concrete Strength by the Maturity Method

ASTM C1107/C1107M (2020) Standard Specification for Packaged

	Dry, Hydraulic-Cement Grout (Nonshrink)
ASTM C1157/C1157M	(2020a) Standard Performance Specification for Hydraulic Cement
ASTM C1218/C1218M	(2020c) Standard Test Method for Water-Soluble Chloride in Mortar and Concrete
ASTM C1602/C1602M	(2018) Standard Specification for Mixing Water Used in Production of Hydraulic Cement Concrete
ASTM D6690	(2015) Standard Specification for Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavements

FOREST STEWARDSHIP COUNCIL (FSC)

FSC STD 01 001	(2015) Principles and Criteria for Forest Stewardship
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MINISTERIO DE LA PRESIDENCIA (MP)

RD 470/2021	(2021) Real Decreto 470/2021, de 29 de junio, por el que se aprueba el Código Estructural.
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MINISTERIO DE VIVIENDA (MV)

RD 410/2010	(2010) Requirements to Building Control Entities and to Testing Laboratories for building quality control.
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U.S. GREEN BUILDING COUNCIL (USGBC)

LEED NC	(2013) Leadership in Energy and Environmental Design(tm) New Construction Rating System
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1.2 DEFINITIONS

- a. "Cementitious material" as used herein must include all portland cement, pozzolan, fly ash, slag cement.
- b. "Exposed to public view" means situated so that it can be seen from eye level from a public location after completion of the building. A public location is accessible to persons not responsible for operation or maintenance of the building.
- c. "Chemical admixtures" are materials in the form of powder or fluids that are added to the concrete to give it certain characteristics not obtainable with plain concrete mixes.
- d. "Supplementary cementing materials" (SCM) include coal fly ash, slag cement, natural or calcined pozzolans, and ultra-fine coal ash when used in such proportions to replace the portland cement that result in improvement to sustainability and durability and reduced cost.

- e. "Design strength" (f'_c) is the specified compressive strength of concrete at time(s) specified in this section to meet structural design criteria.
- f. "Mass Concrete" is any concrete system that approaches a maximum temperature of 70 degrees C within the first 72 hours of placement. In addition, it includes all concrete elements with a section thickness of 1 meter or more regardless of temperature.
- g. "Mixture proportioning" is the process of designing concrete mixture proportions to enable it to meet the strength, service life and constructability requirements of the project while minimizing the initial and life-cycle cost.
- h. "Mixture proportions" are the masses or volumes of individual ingredients used to make a unit measure (cubic meter or cubic yard) of concrete.
- i. "Pozzolan" is a siliceous or siliceous and aluminous material, which in itself possesses little or no cementitious value but will, in finely divided form and in the presence of moisture, chemically react with calcium hydroxide at ordinary temperatures to form compounds possessing cementitious properties.
- j. "Workability (or consistence)" is the ability of a fresh (plastic) concrete mix to fill the form/mould properly with the desired work (vibration) and without reducing the concrete's quality. Workability depends on water content, chemical admixtures, aggregate (shape and size distribution), cementitious content and age (level of hydration).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Concrete Curing Plan

Quality Control Plan; G

Quality Control Personnel Certifications; G

Quality Control Organizational Chart

Laboratory Accreditation; G

Form Removal Schedule; G

Maturity Method Data

SD-02 Shop Drawings

Formwork

Reinforcing Steel; G

SD-03 Product Data

Joint Sealants; (LEED NC)
Joint Filler; (LEED NC)
Formwork Materials
Recycled Aggregate Materials; (LEED NC)
Cementitious Materials; (LEED NC)
Vapor Barrier
Concrete Curing Materials
Reinforcement; (LEED NC)
Liquid Chemical Floor Hardeners and Sealers
Admixtures
Waterstops
Local/Regional Materials; (LEED NC)
Biodegradable Form Release Agent
Pumping Concrete
Nonshrink Grout

SD-04 Samples

Surface Finish Samples

SD-05 Design Data

Concrete Mix Design; G
Formwork Calculations

SD-06 Test Reports

Concrete Mix Design; G
Fly Ash
Pozzolan
Slag Cement
Aggregates
Compressive Strength Tests; G
Chloride Ion Concentration
Air Content

Slump Tests

Water

SD-07 Certificates

Reinforcing Bars

VOC Content for Form Release Agents, Curing Compounds, and
Concrete Penetrating Sealers

Safety Data Sheets

Forest Stewardship Council (FSC) Certification

Field Testing Technician and Testing Agency

SD-08 Manufacturer's Instructions

Liquid Chemical Floor Hardeners and Sealers

Curing Compound

1.4 MODIFICATION OF REFERENCES

Accomplish work in accordance with ACI publications and [RD 470/2021](#), except as modified herein. Consider the advisory or recommended provisions to be mandatory. Interpret reference to the "Building Official," the "Structural Engineer," and the "Architect/Engineer" to mean the Contracting Officer.

1.5 DELIVERY, STORAGE, AND HANDLING

Follow [ACI 301](#), [ACI 304R](#) and [ASTM A934/A934M](#) and [RD 470/2021](#), Art. 14, 15, 49.2.3 and Anejo 4 requirements and recommendations. Do not deliver concrete until forms, reinforcement, embedded items, and chamfer strips are in place and ready for concrete placement. Do not store concrete curing compounds or sealers with materials that have a high capacity to adsorb volatile organic compound (VOC) emissions. Do not store concrete curing compounds or sealers in occupied spaces.

1.5.1 Reinforcement

[RD 470/2021](#), Art. 49.7 and the following. Store reinforcement of different sizes and shapes in separate piles or racks raised above the ground to avoid excessive rusting. Protect from contaminants such as grease, oil, and dirt. Ensure bar sizes can be accurately identified after bundles are broken and tags removed.

1.6 QUALITY ASSURANCE

1.6.1 Design Data

1.6.1.1 Formwork Calculations

Water shoring for concrete placement will not be permitted at any location on this project. Include design calculations indicating arrangement of forms, sizes and grades of supports (lumber), panels, and related

components. Furnish drawings and calculations of shoring and re-shoring methods proposed for retaining walls and horizontal slabs. Calculations must indicate concrete pressure with both live and dead loads, along with material types.

1.6.1.2 Concrete Mix Design

RD 470/2021, Art. 33.6 and the following. Sixty days minimum prior to concrete placement, submit a mix design for each strength and type of concrete. Submit a complete list of materials including type; brand; source and amount of cement, supplementary cementitious materials, and admixtures; and applicable reference specifications. Submit mill test and all other test for cement, supplementary cementitious materials, aggregates, and admixtures. Provide documentation of maximum nominal aggregate size, gradation analysis, percentage retained and passing sieve, and a graph of percentage retained verses sieve size. Provide mix proportion data using at least three different water-cementitious material ratios for each type of mixture, which produce a range of strength encompassing those required for each type of concrete required. If source material changes, resubmit mix proportion data using revised source material. Provide only materials that have been proven by trial mix studies to meet the requirements of this specification, unless otherwise approved in writing by the Contracting Officer. Indicate clearly in the submittal where each mix design is used when more than one mix design is submitted. Resubmit data on concrete components if the qualities or source of components changes. Required average strength can be documented by field experience if field strength test data are available and represent a single group of at least 10 consecutive strength tests for one mixture, using materials and conditions similar to those expected for work, and encompassing a period of not less than 45 days. The average of field strength tests shall equal or exceed f_{cr} . Changes in materials, conditions, and proportions within the test record shall not have been more closely restricted than those for the proposed work. Test records shall not be more than 24 months old. Obtain mix design approval from the contracting officer prior to concrete placement.

1.6.2 Shop Drawings

1.6.2.1 Formwork

Drawings showing details of formwork including, but not limited to; joints, supports, studding and shoring, and sequence of form and shoring removal. Indicate placement schedule, construction, location and method of forming control joints. Include locations of inserts, conduit, sleeves and other embedded items. Reproductions of contract drawings are unacceptable. Submit [form removal schedule](#) indicating element and minimum length of time for form removal.

Design, fabricate, erect, support, brace, and maintain formwork so that it is able to support, without failure, all vertical and lateral loads that may reasonably be anticipated to be applied to the formwork.

1.6.2.2 Reinforcing Steel

RD 470/2021, Figure 49.5.1.1 and Art. 49.5.1. Indicate bending diagrams, assembly diagrams, splicing and laps of bars, shapes, dimensions, and details of bar reinforcing, accessories, and concrete cover. Do not scale dimensions from structural drawings to determine lengths of reinforcing bars. Reproductions of contract drawings are unacceptable.

1.6.3 Control Submittals

1.6.3.1 Concrete Curing Plan

Submit proposed materials, methods and duration for curing concrete elements in accordance with [RD 470/2021](#).

1.6.3.2 Pumping Concrete

Submit proposed materials and methods for pumping concrete. Submittal must include mix designs, pumping equipment including type of pump and size and material for pipe, and maximum length and height concrete is to be pumped.

1.6.3.3 VOC Content for form release agents, curing compounds, and concrete penetrating sealers

Submit certification for the form release agent, curing compounds, and concrete penetrating sealers that indicate the VOC content of each product.

1.6.3.4 Safety Data Sheets

Submit Safety Data Sheets (SDS) for all materials that are regulated for hazardous health effects. SDS must be readily accessible during each work shift to employees when they are at the construction site.

1.6.4 Test Reports

1.6.4.1 Fly Ash and Pozzolan

Submit test results in accordance with [ASTM C618](#) or alternatively [UNE-EN 196-5](#) and [UNE-EN 450-1](#) for fly ash and pozzolan. Submit test results performed within 6 months of submittal date.

1.6.4.2 Slag Cement

Submit test results in accordance with [ASTM C989/C989M](#) or [UNE-EN 15167-1](#) for slag cement. Submit test results performed within 6 months of submittal date.

1.6.4.3 Aggregates

Submit test results in accordance with [UNE-EN 1367-2](#), [UNE-EN 12620](#), [UNE 146508](#) and [UNE 146509](#), as required in the paragraph titled ALKALI-AGGREGATE REACTION.

1.6.5 Field Samples

1.6.5.1 Surface Finish Samples

Provide a sample concrete panel for each finish. Once approved, each panel must be representative of each of the finishes specified and of the workmanship and finish(es) required. Do not remove or destroy samples until directed by the Contracting Officer.

1.6.6 Quality Control Plan

Develop and submit for approval a concrete quality control program in

accordance with the guidelines of [ACI 121R](#) or alternatively [RD 470/2021](#), Art. 17 through 23 and Art. 63 and as specified herein. The plan must include approved laboratories. Provide direct oversight for the concrete qualification program inclusive of associated sampling and testing. All quality control reports must be provided to the Contracting Officer, Quality Manager and Concrete Supplier. Maintain a copy of the reports at project site.

1.6.7 Quality Control Personnel Certifications

The Contractor must submit for approval the responsibilities of the various quality control personnel, including the names and qualifications of the individuals in those positions and a [quality control organizational chart](#) defining the quality control hierarchy and the responsibility of the various positions. Quality control personnel must be employed by the Contractor.

Submit certification for the following:

- a. Personnel responsible for inspection of concrete operations.
- b. Lead Foreman or Journeyman of the Concrete Placing, Finishing, and Curing Crews.
- c. Field Testing Technicians

1.6.7.1 Quality Manager Qualifications

The quality manager must hold a current license as a professional engineer in Spain, with experience on at least five similar projects. Evidence of extraordinary proven experience may be considered by the Contracting Officer as sufficient to act as the Quality Manager.

1.6.7.2 Field Testing Technician and Testing Agency

Submit data on qualifications of proposed testing agency and technicians for approval by the Contracting Officer prior to performing testing on concrete.

- a. Work on concrete under this contract must be performed by a professional Engineer, Ingeniero de Caminos, Canales y Puertos, Ingeniero Industrial or Ingeniero de la Edificación.
- b. Testing agencies that perform testing services on reinforcing steel must meet the requirements of [UNE-EN ISO/IEC 17025](#).
- c. Testing agencies that perform testing services on concrete materials must meet the requirements of [UNE-EN ISO/IEC 17025](#).

1.6.8 Laboratory Qualifications for Concrete Qualification Testing

The concrete testing laboratory must have the necessary equipment and experience to accomplish required testing. The laboratory must meet the requirements of [UNE-EN ISO/IEC 17025](#) and be accredited by ENAC and comply with [RD 410/2010](#).

1.6.9 Laboratory Accreditation

Laboratory and testing facilities must be provided by and at the expense

of the Contractor. The laboratories performing the tests must be accredited by ENAC and comply with RD 410/2010, including UNE-EN 12390-2 and UNE 146508. The accreditation must be current and must include the required test methods, as specified. Furthermore, the testing must comply with the following requirements:

- a. Aggregate Testing and Mix Proportioning: Aggregate testing and mixture proportioning studies must be performed by an accredited laboratory and under the direction of a licensed professional engineer in Spain who is competent in concrete materials and must sign all reports and designs.
- b. Acceptance Testing: Furnish all materials, labor, and facilities required for molding, curing, testing, and protecting test specimens at the site and in the laboratory. Furnish and maintain boxes or other facilities suitable for storing and curing the specimens at the site while in the mold within the temperature range stipulated by UNE-EN 12390-2.
- c. Contractor Quality Control: All sampling and testing must be performed by an approved, onsite, independent, accredited laboratory.

1.7 ENVIRONMENTAL REQUIREMENTS

RD 470/2021, Art. 6 and Anejo 2 and the following. Provide space ventilation according to material manufacturer recommendations, at a minimum, during and following installation of concrete curing compound and sealer.

1.7.1 Submittals for Environmental Performance

- a. Provide data indication the percentage of post-industrial pozzolan (fly ash, slag cement) cement substitution as a percentage of the full product composite by weight.
- b. Provide data indicating the percentage of post-industrial and post-consumer recycled content aggregate.
- c. Provide product data indicating the percentage of post-consumer recycled steel content in each type of steel reinforcement as a percentage of the full product composite by weight.
- d. Provide product data stating the location where all products were manufactured
- e. For projects using reusable formwork, provide data showing how formwork is reused.
- f. Provide SDS product information data showing that form release agents meet any environmental performance goals such as using vegetable and soy based products.
- g. Provide SDS product information data showing that concrete adhesives meet any environmental performance goals including low emitting, low volatile organic compound products.

1.8 SUSTAINABLE DESIGN REQUIREMENTS

1.8.1 Local/Regional Materials

Use materials or products extracted, harvested, or recovered, as well as manufactured, within a 805 kilometer radius from the project site, if available from a minimum of three sources.

1.8.2 Forest Stewardship Council (FSC) Certification

Use FSC-certified wood where specified. Provide letter of certification signed by lumber supplier. Indicate compliance with FSC STD 01 001 and identify certifying organization. Submit FSC certification numbers; identify each certified product on a line-item basis. Submit copies of invoices bearing the FSC certification numbers.

PART 2 PRODUCTS

2.1 FORMWORK MATERIALS

- a. Form-facing material in contact with concrete must be lumber or plywood. Submit product information on proposed form-facing materials if different from that specified herein.
- b. Design formwork, shores, reshores, and backshores to support loads transmitted to them and to comply with applicable building code requirements.
- c. Design formwork to withstand pressure resulting from placement and vibration of concrete and to maintain specified tolerances.
- d. Design formwork to accommodate waterstop materials in joints at locations indicated in Contract Documents.
- e. Provide temporary openings in formwork if needed to facilitate cleaning and inspection.
- f. Design formwork joints to inhibit leakage of mortar.
- g. Limit deflection of facing materials for concrete surfaces exposed to view to 1/400 of center-to-center spacing of facing supports.
- h. Submit product information on proposed form-facing materials if different from that specified herein.
- i. Submit shop drawings for formwork, shoring, reshoring, and backshoring. Shop drawings must be signed and sealed by a licensed design engineer.
- j. Submit design calculations for formwork, shoring, reshoring, and backshoring. Design calculations must be signed and sealed by a licensed design engineer.
- k. Submit procedure for reshoring and backshoring, including drawings signed and sealed by a licensed design engineer. Include on shop drawings the formwork removal procedure and magnitude of construction loads used for design of reshoring or backshoring system. Indicate in procedure the magnitude of live and dead loads assumed for required capacity of the structure at time of reshoring or backshoring.

1. Submit manufacturer's product data on form liner proposed for use with each formed surface.

2.1.1.1 Wood Forms

Provide lumber that is square edged or tongue-and-groove boards, free of raised grain, knotholes, or other surface defects. Provide plywood panels or hardboard for smooth form lining.

2.1.1.1.1 Concrete Form Plywood (Standard Rough)

Provide plywood concrete form not less than 16 mm thick.

2.1.2 Steel Forms

Provide steel form surfaces that do not contain irregularities, dents, or sags.

2.2 FORMWORK ACCESSORIES

- a. Use commercially manufactured formwork accessories, including ties and hangers.
- b. Form ties and accessories must not reduce the effective cover of the reinforcement.

2.2.1 Form Ties

- a. Use form ties with ends or end fasteners that can be removed without damage to concrete.
- b. Use form ties with integral water barrier plates or other acceptable positive water barriers in below-grade walls.
- c. The breakback distance for ferrous ties must be at least 50 mm.
- d. Submit manufacturer's data sheet on form ties.

2.2.2 Waterstops

Submit manufacturer's data sheet on waterstop materials and splices.

2.2.2.1 Thermoplastic Elastomeric Rubber Waterstop

Thermoplastic elastomeric rubber waterstops must be jet fuel resistant.

2.2.3 Biodegradable Form Release Agent

- a. Provide form release agent that is colorless, biodegradable, and water-based.
- b. Provide product that does not bond with, stain, or adversely affect concrete surfaces and does not impair subsequent treatments of concrete surfaces.
- c. Provide form release agent that reduces formwork moisture absorption, and does not contain diesel fuel, petroleum-based lubricating oils, waxes, or kerosene. Submit documentation indicating type of biobased

material in product and biobased content. Indicate relative dollar value of biobased content products to total dollar value of products included in project.

- d. Submit manufacturer's product data on formwork release agent for use on each form-facing material.

2.2.4 Chamfer Materials

Use lumber materials with dimensions of 19 x 19 mm.

2.2.5 Construction and movement joints

- a. Submit details and locations of construction joints in accordance with the requirements herein.
- b. Locate construction joints within middle one-third of spans of slabs, beams, and girders.
- c. Locate construction joints in walls and columns at underside of slabs, beams, or girders and at tops of footings or slabs.
- d. Make construction joints perpendicular to main reinforcement.
- e. Provide movement joints where indicated in Contract Documents or in accepted alternate locations.
- f. Submit location and detail of movement joints if different from those indicated in Contract Documents.
- g. Submit manufacturer's data sheet on expansion joint materials.
- h. Provide keyways where indicated in Contract Documents.

2.2.6 Other Embedded items

Use sleeves, inserts, anchors, and other embedded items of material and design indicated in Contract Documents.

2.3 CONCRETE MATERIALS

2.3.1 Cementitious Materials

2.3.1.1 Portland Cement

- a. Unless otherwise specified, provide cement that conforms to RD 470/2021, Art. 28 and Table 28, and to UNE-EN 197-1, CEM I or CEM II, 42.5, Types BS, BP or BV.
- b. Use one brand and type of cement for formed concrete having exposed-to-view finished surfaces.
- c. Supplier must certify that no hazardous waste is used in the fuel mix or raw materials.
- d. Submit information along with evidence demonstrating compliance with referenced standards. Submittals must include types of cementitious materials, manufacturing locations, shipping locations, and certificates showing compliance.

- e. Cementitious materials must be stored and kept dry and free from contaminants.

2.3.1.2 Fly Ash

- a. [RD 470/2021](#), Art. 32.1 and [UNE-EN 450-1](#), Class F, except that the maximum allowable loss on ignition must not exceed 6 percent.
- b. If fly ash is used it shall range from 15 to 35 percent by weight of cementitious material, provided the fly ash does not reduce the amount of cement in the concrete mix below the minimum requirements of local building codes. Where the use of fly ash cannot meet the minimum level, it shall not be used. Report the chemical analysis of the fly ash in accordance with [UNE-EN 197-1](#).

2.3.1.3 Slag Cement

[UNE-EN 15167-1](#), Grade 100.

2.3.1.4 Other Supplementary Cementitious Materials

Natural pozzolan must be raw or calcined and conform to [UNE-EN 450-1](#) and [UNE-EN 196-5](#), Class N, including the optional requirement for uniformity.

Ultra Fine Fly Ash (UFFA) and Ultra Fine Pozzolan (UFP) must conform to [UNE-EN 196-5](#) Class F or N, and the following additional requirements:

- a. The strength activity index at 28 days of age must be at least 95 percent of the control specimens.
- b. The average particle size must not exceed 6 microns.
- c. The sum of $\text{SiO}_2 + \text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3$ must be greater than 77 percent.

2.3.2 Water

- a. Water must comply with the requirements of [RD 470/2021](#), Art. 29.
- b. Minimize the amount of water in the mix. Improve workability by adjusting the grading of the aggregate and using admixture rather than by adding water.
- c. Water must be potable; free from injurious amounts of oils, acids, alkalis, salts, organic materials, or other substances deleterious to concrete.
- d. Protect mixing water and ice from contamination during storage and delivery.
- e. Submit test report showing water complies with [ASTM C1602/C1602M](#).

2.3.3 Aggregate

2.3.3.1 Normal-Weight Aggregate

- a. Aggregates must conform to [RD 470/2021](#), Art. 30 and to [UNE-EN 12620](#). Comply with the following articles from [RD 470/2021](#): for max. size, Art. 30.3; for grading, Art. 30.4; for fines content, Art. 30.4.1; for

coarse aggregate shape, Art. 30.5; for physical requirements, Art. 30.6; and for chemical requirements, Art. 30.7.

- b. Aggregates used in concrete must be obtained from the same sources and have the same size range as aggregates used in concrete represented by submitted field test records or used in trial mixtures.
- c. Store and handle aggregate in a manner that will avoid segregation and prevents contamination by other materials or other sizes of aggregates. Store aggregates in locations that will permit them to drain freely. Do not use aggregates that contain frozen lumps.
- d. Submit types, pit or quarry locations, producers' names, aggregate supplier statement of compliance with [UNE-EN 932-1](#), [UNE-EN 932-3](#), [UNE-EN 933-3](#) and [UNE-EN 933-4](#).

2.3.3.2 Recycled Aggregate Materials

Use recycled aggregate, depending on local availability and conforming to requirements of the mix design. Recycled aggregate to include: recovered concrete or recovered stone that meets the aggregate requirements specified. Submit recycled material request with the aggregate certification submittals and do not use until approved by the Contracting Officer.

2.3.4 Admixtures

[RD 470/2021](#), Art. 31 and the following.

- a. Chemical admixtures must conform to [UNE-EN 934-2](#).
- b. Air-entraining admixtures must conform to [UNE-EN 934-2](#).
- c. Chemical admixtures for use in producing flowing concrete must conform to [UNE-EN 934-2](#).
- d. Do not use calcium chloride admixtures.
- e. Use an ASR-inhibiting admixture for concrete containing aggregate susceptible to ASR.
- f. Admixtures used in concrete must be the same as those used in the concrete represented by submitted field test records or used in trial mixtures.
- g. Protect stored admixtures against contamination, evaporation, or damage.
- h. To ensure uniform distribution of constituents, provide agitating equipment for admixtures used in the form of suspensions or unstable solutions. Protect liquid admixtures from temperature changes that would adversely affect their characteristics.
- i. Submit types, brand names, producers' names, manufacturer's technical data sheets, and certificates showing compliance with standards required herein.

2.4 MISCELLANEOUS MATERIALS

2.4.1 Concrete Curing Materials

Provide concrete curing material in accordance with [RD 470/2021](#), Art. 52.5. Submit product data for concrete [curing compounds](#). Submit manufactures instructions for placement of curing compound.

2.4.2 Nonshrink Grout

Nonshrink grout in accordance with [ASTM C1107/C1107M](#).

2.4.3 Floor Finish Materials

2.4.3.1 Liquid Chemical Floor Hardeners and Sealers

- a. Hardener must be a colorless aqueous solution containing a blend of inorganic silicate or siliconate material and proprietary components combined with a wetting agent; that penetrates, hardens, and densifies concrete surfaces. Submit manufactures instructions for placement of liquid chemical floor hardener.
- b. Use concrete penetrating sealers with a low (maximum 100 grams/liter, less water and less exempt compounds) VOC content. Submit manufactures instructions for placement of sealers.

2.4.4 Expansion/Contraction Joint Filler

[UNE 41107](#). Material must be [13 mm](#) thick, unless otherwise indicated.

2.4.5 Joint Sealants

Submit manufacturer's product data, indicating VOC content.

2.4.5.1 Horizontal Surfaces, 3 Percent Slope, Maximum

[ASTM D6690](#) or [ASTM C920](#), Type M, Class 25, Use T, fuel resistant.

2.4.5.2 Vertical Surfaces Greater Than 3 Percent Slope

[ASTM C920](#), Type M, Grade NS, Class 25, Use NT. No sag, fuel resistant.

2.4.6 Vapor Barrier

Polyethylene sheeting, minimum [0.25 mm](#) thickness or other equivalent material with a maximum permeance rating of 0.04 perms.

2.5 CONCRETE MIX DESIGN

2.5.1 Properties and Requirements

- a. Use materials and material combinations listed in this section and the contract documents.
- b. Cementitious material content must be adequate for concrete to satisfy the specified requirements for strength, w/cm, durability, and finishability described in this section and the contract documents.

The minimum cementitious material content for concrete used in floors

must meet the following requirements:

Nominal maximum size of aggregate, mm	Minimum cementitious material content, kg per cubic meter
25	310

- c. Selected target slump must meet the requirements this section, the contract documents, and must not exceed 230 mm. Concrete must not show visible signs of segregation.
- d. The target slump must be enforced for the duration of the project. Determine the slump by UNE-EN 12350-2. Slump tolerances must meet the requirements of RD 470/2021, Art. 33.5 and Tables 51.2.4 and 57.5.2.2..
- e. The nominal maximum size of coarse aggregate for a mixture must not exceed three-fourths of the minimum clear spacing between reinforcement, one-fifth of the narrowest dimension between sides of forms, or one-third of the thickness of slabs or toppings.
- f. Concrete for slabs to receive a hard-troweled finish must not contain an air-entraining admixture or have a total air content greater than 3 percent.
- g. Concrete properties and requirements for each portion of the structure are specified in the table below. Refer to the paragraph titled DURABILITY for more details on exposure categories and their requirements.

	Minimum $f'c$ MPa	Miscellaneous Requirements
All concrete related to the pipeline	30 at 28 days	Max. slump: 15 cm for footings Nominal maximum aggregate size must be 25 mm
All concrete related to the tanks	34.5	Max. slump: 15 cm for footings Nominal maximum aggregate size must be 25 mm

2.5.2 Durability

2.5.2.1 Alkali-Aggregate Reaction

RD 470/2021, Art. 30.7.5 and the following. Do not use any aggregate susceptible to alkali-carbonate reaction (ACR). Use one of the three options below for qualifying concrete mixtures to reduce the potential of alkali-silica reaction (ASR):

- a. For each aggregate used in concrete, the expansion result determined

in accordance with **UNE 146509** must not exceed 0.04 percent at one year.

- b. For each aggregate used in concrete, the expansion result of the aggregate and cementitious materials combination determined in accordance with **UNE 146509** must not exceed 0.10 percent at an age of 16 days.
- c. Alkali content in concrete (LBA) must not exceed **2.4 kg per cubic meter** for moderately reactive aggregate or **1.8 kg per cubic meter** for highly reactive aggregate. Reactivity must be determined by testing in accordance with **UNE 146509**. Alkali content is calculated as follows:

$$\text{LBA} = (\text{cement content, kg per cubic meter}) \times (\text{equivalent alkali content of portland cement in percent}/100 \text{ percent})$$

2.5.2.2 Freezing and Thawing Resistance

- a. Provide concrete meeting the following requirements based on exposure class assigned to members for freezing-and-thawing exposure in Contract Documents:

Exposure class	Maximum w/cm^*	Minimum $f'c$, MPa	Air content	Additional Requirements
F2	0.45	31	Depends on aggregate size	See limits on maximum cementitious material by mass

- b. Concrete must be air entrained for members assigned to Exposure Class F2. The total air content must meet the requirements of the following table:

Nominal maximum aggregate size, mm	Total air content, percent [^]	
	Exposure Class F2 and F3	Exposure Class F1
25.0	6.0	4.5
37.5	5.5	4.5

*Tolerance on air content as delivered must be plus/minus 1.5 percent.

[^]For $f'c$ greater than 5000 psi, reducing air content by 1.0 percentage point is acceptable.

- c. Submit documentation verifying compliance with specified requirements.

2.5.2.3 Corrosion and Chloride Content

- a. Provide concrete meeting the requirements of the following table based on the exposure class assigned to members requiring protection against reinforcement corrosion in Contract Documents.
- b. Submit documentation verifying compliance with specified requirements.

- c. Water-soluble chloride ion content contributed from constituents including water, aggregates, cementitious materials, and admixtures must be determined for the concrete mixture at age 28 days.
- d. The maximum water-soluble chloride ion (Cl-) content in concrete, percent by mass of cement is as follows:

Exposure class	Maximum w/cm*	Minimum f'c, MPa	Maximum water-soluble chloride ion (Cl-) content in
Reinforced concrete			
C2	0.4	35	0.15

*The maximum w/cm limits do not apply to lightweight concrete.

2.5.2.4 Sulfate Resistance

- a. Provide concrete meeting the requirements of the following table based on the exposure class assigned to members for sulfate exposure.

Exposure class	Maximum w/cm	Minimum f'c, MPa	Required cementitious materials-types			Calcium chloride admixture
			ASTM C150/C150M	ASTM C595/C595M	ASTM C1157/C1157M	
S0	N/A	17	N/A	N/A	N/A	No restrictions

2.5.2.5 Concrete Temperature

The temperature of concrete as delivered must not exceed 35°C.

2.5.2.6 Concrete permeability

- a. Provide concrete meeting the requirements of the following table based on exposure class assigned to members requiring low permeability in the Contract Documents.

Exposure class	Maximum w/cm*	Minimum f'c, MPa	Additional minimum requirements
W0 & W1	0.5	28	None

- b. Submit documentation verifying compliance with specified requirements.

2.5.3 Ready-Mix Concrete

Provide concrete that meets the requirements of RD 470/2021 Art. 33, 43.2 and Anejo 4.

Ready-mixed concrete manufacturer must provide duplicate delivery tickets with each load of concrete delivered. Provide delivery tickets with the following information in addition to that required by **RD 470/2021** Anejo 4:

- a. Type and brand cement
- b. Cement and supplementary cementitious materials content in **kilogram** per cubic **meter** of concrete
- c. Maximum size of aggregate
- d. Amount and brand name of admixtures
- e. Total water content expressed by water cementitious material ratio

2.6 REINFORCEMENT

- a. Bend reinforcement cold. Fabricate reinforcement in accordance with fabricating tolerances of **RD 470/2021**, Art. 34, 49 and 59.2, and Anejo 14 Art. 5.
- b. Submit manufacturer's certified test report for reinforcement.
- c. Submit placing drawings showing fabrication dimensions and placement locations of reinforcement and reinforcement supports. Placing drawings must indicate locations of splices, lengths of lap splices, and details of mechanical and welded splices.
- d. Submit request with locations and details of splices not indicated in Contract Documents.
- e. Submit request to place column dowels without using templates.
- f. Submit request and procedure to field-bend or straighten reinforcing bars partially embedded in concrete.
- g. Submit request for field cutting, including location and type of bar to be cut and reason field cutting is required.

2.6.1 Reinforcing Bars

- a. Reinforcing bars must be deformed, except load-transfer dowels, and welded wire reinforcement, which may be plain.
- b. **UNE-EN 10080** or **UNE 36068**, B-500-T or B-500-S.
- c. Reinforcing bars may contain post-consumer or post-industrial recycled content.
- d. Submit mill certificates for reinforcing bars.

2.6.1.1 Headed Reinforcing Bars

Headed reinforcing bars must conform to **UNE-EN 10080** including Annex A1, and other specified requirements.

2.6.2 Welded Wire Reinforcement

- a. Use welded wire reinforcement specified in Contract Documents and conforming to one or more of the specifications given herein.
- b. Plain welded wire reinforcement must conform to **RD 470/2021**, Art. 35.2.1, to **UNE 36731** and to **UNE 36092**, with welded intersections spaced no greater than **300 mm** apart in direction of principal reinforcement.
- c. Deformed welded wire reinforcement must conform to **UNE 36061**, with welded intersections spaced no greater than **400 mm** apart in direction of principal reinforcement.

2.6.3 Reinforcing Bar Supports

- a. Provide reinforcement support types within structure as required by Contract Documents.
- b. Legs of supports in contact with formwork must be hot-dip galvanized, or plastic coated after fabrication, or stainless-steel bar supports.

2.6.4 Dowels for Load Transfer in Floors

Provide greased dowels for load transfer in floors of the type, design, weight, and dimensions indicated. Provide dowel bars that are plain-billet steel conforming to **UNE 36097-1**, 235 Mpa minimum yield stress.

PART 3 EXECUTION

3.1 EXAMINATION

- a. Do not begin installation until substrates have been properly constructed; verify that substrates are level. Comply with **RD 470/2021**, Art. 48.1.
- b. If substrate preparation is the responsibility of another installer, notify Contracting Officer of unsatisfactory preparation before processing.
- c. Check field dimensions before beginning installation. If dimensions vary too much from design dimensions for proper installation, notify Contracting Officer and wait for instructions before beginning installation.

3.2 PREPARATION

Determine quantity of concrete needed and minimize the production of excess concrete. Designate locations or uses for potential excess concrete before the concrete is poured.

3.2.1 General

Comply with **RD 470/2021**, Art. 48.2 and the following.

- a. Surfaces against which concrete is to be placed must be free of debris, loose material, standing water, and other deleterious substances before start of concrete placing.

- b. Remove standing water without washing over freshly deposited concrete. Divert flow of water through side drains provided for such purpose.

3.2.2 Subgrade Under Foundations and Footings

- a. When subgrade material is semi-porous and dry, sprinkle subgrade surface with water as required to eliminate suction at the time concrete is deposited, or seal subgrade surface by covering surface with specified vapor barrier.
- b. When subgrade material is porous, seal subgrade surface by covering surface with specified vapor barrier.

3.2.3 Subgrade Under Slabs on Ground

- a. Before construction of slabs on ground, have underground work on pipes and conduits completed and approved.
- b. Previously constructed subgrade or fill must be cleaned of foreign materials
- c. Finish surface of capillary water barrier under interior slabs on ground must not show deviation in excess of 6.4 mm when tested with a 3000 mm straightedge parallel with and at right angles to building lines.
- d. Finished surface of subgrade or fill under exterior slabs on ground must not be more than 6.10 mm above or 30.50 mm below elevation indicated.

3.2.4 Edge Forms and Screed Strips for Slabs

- a. Set edge forms or bulkheads and intermediate screed strips for slabs to obtain indicated elevations and contours in finished slab surface and must be strong enough to support vibrating bridge screeds or roller pipe screeds if nature of specified slab finish requires use of such equipment.
- b. Align concrete surface to elevation of screed strips by use of strike-off templates or approved compacting-type screeds.

3.2.5 Reinforcement and Other Embedded Items

- a. Secure reinforcement, joint materials, and other embedded materials in position, inspected, and approved before start of concrete placing.
- b. When concrete is placed, reinforcement must be free of materials deleterious to bond. Reinforcement with rust, mill scale, or a combination of both will be considered satisfactory, provided minimum nominal dimensions, nominal weight, and minimum average height of deformations of a hand-wire-brushed test specimen are not less than applicable ASTM specification requirements.

3.3 FORMS

- a. Provide forms, shoring, and scaffolding for concrete placement. Set forms mortar-tight and true to line and grade.
- b. Chamfer above grade exposed joints, edges, and external corners of

concrete 19 mm unless indicated otherwise. Place chamfer strips in corners of formwork to produce beveled edges on permanently exposed surfaces.

- c. Provide formwork with clean-out openings to permit inspection and removal of debris.
- d. Inspect formwork and remove foreign material before concrete is placed.
- e. At construction joints, lap form-facing materials over the concrete of previous placement. Ensure formwork is placed against hardened concrete so offsets at construction joints conform to specified tolerances.
- f. Provide positive means of adjustment (such as wedges or jacks) of shores and struts. Do not make adjustments in formwork after concrete has reached initial setting. Brace formwork to resist lateral deflection and lateral instability.
- g. Fasten form wedges in place after final adjustment of forms and before concrete placement.
- h. Provide anchoring and bracing to control upward and lateral movement of formwork system.
- i. Construct formwork for openings to facilitate removal and to produce opening dimensions as specified and within tolerances.
- j. Provide runways for moving equipment. Support runways directly on formwork or structural members. Do not support runways on reinforcement. Loading applied by runways must not exceed capacity of formwork or structural members.
- k. Position and support expansion joint materials, waterstops, and other embedded items to prevent displacement. Fill voids in sleeves, inserts, and anchor slots temporarily with removable material to prevent concrete entry into voids.
- l. Clean surfaces of formwork and embedded materials of mortar, grout, and foreign materials before concrete placement.

3.3.1 Coating

- a. Cover formwork surfaces with an acceptable material that inhibits bond with concrete.
- b. If formwork release agent is used, apply to formwork surfaces in accordance with manufacturer's recommendations before placing reinforcement. Remove excess release agent on formwork prior to concrete placement.
- c. Do not allow formwork release agent to contact reinforcement or hardened concrete against which fresh concrete is to be placed.

3.3.2 Reshoring

- a. Do not allow structural members to be loaded with combined dead and construction loads in excess of loads indicated in the accepted procedure.

- b. Install and remove reshores or backshores in accordance with accepted procedure.

3.3.3 Reuse

- a. Reuse forms providing the structural integrity of concrete and the aesthetics of exposed concrete are not compromised.
- b. Wood forms must not be clogged with paste and must be capable of absorbing high water-cementitious material ratio paste.
- c. Remove leaked mortar from formwork joints before reuse.

3.3.4 Forms for Standard Rough Form Finish

Provide formwork in accordance with [ACI 301](#) Section 5 with a surface finish, SF-1.0, for formed surfaces that are to be concealed by other construction.

3.3.5 Forms for Standard Smooth Form Finish

Provide formwork in accordance with [ACI 301](#) Section 5 with a surface finish, SF-3.0, for formed surfaces that are exposed to view.

3.3.6 Form Ties

- a. After ends or end fasteners of form ties have been removed, repair tie holes in accordance with [ACI 301](#) Section 5 requirements.

3.3.7 Tolerances for Form Construction

- a. Construct formwork so concrete surfaces conform to tolerances in [ACI 117](#).
- b. Position and secure sleeves, inserts, anchors, and other embedded items such that embedded items are positioned within tolerances.
- c. To maintain specified elevation and thickness within tolerances, install formwork to compensate for deflection and anticipated settlement in formwork during concrete placement. Set formwork and intermediate screed strips for slabs to produce designated elevation, camber, and contour of finished surface before formwork removal. If specified finish requires use of vibrating screeds or roller pipe screeds, ensure that edge forms and screed strips are strong enough to support such equipment.

3.3.8 Removal of Forms and Supports

- a. If vertical formed surfaces require finishing, remove forms as soon as removal operations will not damage concrete.
- b. Remove top forms on sloping surfaces of concrete as soon as removal will not allow concrete to sag. Perform repairs and finishing operations required. If forms are removed before end of specified curing period, provide curing and protection.
- c. Do not damage concrete during removal of vertical formwork for columns, walls, and sides of beams. Perform needed repair and

finishing operations required on vertical surfaces. If forms are removed before end of specified curing period, provide curing and protection.

- d. Leave formwork and shoring in place to support construction loads and weight of concrete in beams, slabs, and other structural members until in-place required strength of concrete is reached.
- e. Form-facing material and horizontal facing support members may be removed before in-place concrete reaches specified compressive strength if shores and other supports are designed to allow facing removal without deflection of supported slab or member.

3.3.9 Strength of Concrete Required for Removal of Formwork

If removal of formwork, reshoring, or backshoring is based on concrete reaching a specified in-place strength, mold and field-cure cylinders in accordance with [UNE-EN 12390-2](#). Test cylinders in accordance with [UNE-EN 12390-3](#).

- a. Tests of cast-in-place cylinders. This option is limited to slabs with concrete depths from [12.5 to 30 cm](#).
- b. Maturity method in accordance with [ASTM C1074](#). Submit [maturity method data](#) using project materials and concrete mix proportions used on the project to demonstrate the correlation between maturity and compressive strength of laboratory cured test specimens to the Contracting Officer.

3.4 WATERSTOP INSTALLATION AND SPLICES

- a. Provide waterstops in construction joints as indicated.
- b. Install formwork to accommodate waterstop materials. Locate waterstops in joints where indicated in Contract Documents. Minimize number of splices in waterstop. Splice waterstops in accordance with manufacturer's written instructions. Install factory-manufactured premolded mitered corners.
- c. Install waterstops to form a continuous diaphragm in each joint. Make adequate provisions to support and protect waterstops during progress of work. Protect waterstops protruding from joints from damage.

3.4.1 Thermoplastic Elastomeric Rubber Waterstop

Fittings must be shop made using a machine specifically designed to mechanically weld the waterstop. A portable power saw must be used to miter or straight cut the ends to be joined to ensure good alignment and contact between joined surfaces. Maintain continuity of the characteristic features of the cross section of the waterstop (for example ribs, tabular center axis, and protrusions) across the splice.

3.4.2 Hydrophilic Waterstop

Miter cut ends to be joined with sharp knife or shears. The ends must be adhered with adhesive.

3.5 PLACING REINFORCEMENT AND MISCELLANEOUS MATERIALS

- a. Unless otherwise specified, placing reinforcement and miscellaneous materials must be in accordance to and [RD 470/2021](#) Art. 49. Comply with Art. 49.4.1 for spacing and Art. 49.4.3 for general requirements of reinforcement and Art. 49.8 for placing. Provide bars, welded wire reinforcement, wire ties, supports, and other devices necessary to install and secure reinforcement.
- b. Reinforcement must not have rust, scale, oil, grease, clay, or foreign substances that would reduce the bond. Rusting of reinforcement is a basis of rejection if the effective cross-sectional area or the nominal weight per unit length has been reduced. Remove loose rust prior to placing steel. Tack welding is prohibited.
- c. Nonprestressed cast-in-place concrete members must have concrete cover for reinforcement as indicated on the contract documents.

3.5.1 Reinforcement Supports

Provide reinforcement support in accordance with [RD 470/2021](#), Art. 49.4.3.1 and 49.8.2.

3.5.2 Splicing

As indicated in the Contract Documents. For splices not indicated follow [RD 470/2021](#) Art. 49.5.1 and 49.5.2.2. Do not splice at points of maximum stress. Overlap welded wire reinforcement the spacing of the cross wires, plus [50 mm](#).

3.5.3 Setting Miscellaneous Material

Place and secure anchors and bolts, pipe sleeves, conduits, and other such items in position before concrete placement and support against displacement. Plumb anchor bolts and check location and elevation. Temporarily fill voids in sleeves with readily removable material to prevent the entry of concrete.

3.5.4 Fabrication

Shop fabricate reinforcing bars to conform to shapes and dimensions indicated for reinforcement, and as follows:

- a. Provide fabrication tolerances that are in accordance with [ACI 117](#) and [RD 470/2021](#) Art. 34.2 and 49.2.4.
- b. Provide hooks and bends that are in accordance with the Contract Documents and [RD 470/2021](#), Art. 49.3.4 and Figure 49.5.1.1.

Reinforcement must be bent cold to shapes as indicated. Bending must be done in the shop. Rebending of a reinforcing bar that has been bent incorrectly is not be permitted. Bending must be in accordance with standard approved practice and by approved machine methods.

Deliver reinforcing bars bundled, tagged, and marked. Tags must be metal with bar size, length, mark, and other information pressed in by machine. Marks must correspond with those used on the placing drawings.

Do not use reinforcement that has any of the following defects:

- a. Bar lengths, depths, and bends beyond specified fabrication tolerances
- b. Bends or kinks not indicated on drawings or approved shop drawings
- c. Bars with reduced cross-section due to rusting or other cause

Replace defective reinforcement with new reinforcement having required shape, form, and cross-section area.

3.5.5 Placing Reinforcement

Place reinforcement in accordance with [ACI 301](#) and [RD 470/2021](#) Art. 49.8.

For slabs on grade (over earth or over capillary water barrier) and for footing reinforcement, support bars or welded wire reinforcement on precast concrete blocks, spaced at intervals required by size of reinforcement, to keep reinforcement the minimum height specified above the underside of slab or footing.

For slabs other than on grade, supports for which any portion is less than [25 mm](#) from concrete surfaces that are exposed to view or to be painted must be of precast concrete units, plastic-coated steel, or stainless steel protected bar supports. Precast concrete units must be wedge shaped, not larger than [90 by 90 mm](#), and of thickness equal to that indicated for concrete protection of reinforcement. Provide precast units that have cast-in galvanized tie wire hooked for anchorage and blend with concrete surfaces after finishing is completed.

Provide reinforcement that is supported and secured together to prevent displacement by construction loads or by placing of wet concrete, and as follows:

- a. Provide supports for reinforcing bars that are sufficient in number and have sufficient strength to carry the reinforcement they support. Do not use supports to support runways for concrete conveying equipment and similar construction loads.
- b. Equip supports on ground and similar surfaces with sand-plates.
- c. Support welded wire reinforcement as required for reinforcing bars.
- d. Secure reinforcements to supports by means of tie wire. Wire must be black, soft iron wire, not less than [1.6 mm](#).
- e. Reinforcement must be accurately placed, securely tied at intersections, and held in position during placing of concrete by spacers, chairs, or other approved supports. Point wire-tie ends away from the form. Unless otherwise indicated, numbers, type, and spacing of supports must conform to the Contract Documents.
- f. Bending of reinforcing bars partially embedded in concrete is permitted only as specified in the Contract Documents.

3.5.6 Spacing of Reinforcing Bars

- a. Spacing must be as indicated in the Contract Documents.
- b. Reinforcing bars may be relocated to avoid interference with other

reinforcement, or with conduit, pipe, or other embedded items. If any reinforcing bar is moved a distance exceeding one bar diameter or specified placing tolerance, resulting rearrangement of reinforcement is subject to preapproval by the Contracting Officer.

3.5.7 Concrete Protection for Reinforcement

Additional concrete protection must be in accordance with the Contract Documents.

3.6 BATCHING, MEASURING, MIXING, AND TRANSPORTING CONCRETE

In accordance with **RD 470/2021** Art. 51.2 and 51.3, except as modified herein. Batching equipment must be such that the concrete ingredients are consistently measured within the following tolerances: 1 percent for cement and water, 2 percent for aggregate, and 3 percent for admixtures. Furnish mandatory batch ticket information for each load of ready mix concrete.

3.6.1 Measuring

Make measurements at intervals as specified in paragraphs SAMPLING and TESTING.

3.6.2 Mixing

- a. Mix concrete in accordance with **RD 470/2021** Art. 51.2.3 and 51.2.4.
- b. Machine mix concrete. Begin mixing within 30 minutes after the cement has been added to the aggregates. Place concrete within 90 minutes of either addition of mixing water to cement and aggregates or addition of cement to aggregates if the concrete temperature is less than **29 degrees C**.
- c. Place concrete within 60 minutes if the concrete temperature is greater than **29 degrees C** except as follows: if set retarding admixture is used and slump requirements can be met, limit for placing concrete may remain at 90 minutes. Additional water may be added, provided that both the specified maximum slump and submitted water-cementitious material ratio are not exceeded and the required concrete strength is still met. When additional water is added, an additional 30 revolutions of the mixer at mixing speed is required.
- d. Dissolve admixtures in the mixing water and mix in the drum to uniformly distribute the admixture throughout the batch. Do not reconstitute concrete that has begun to solidify.
- e. When fibers are used, add fibers together with the aggregates and never as the first component in the mixer. Fibers must be dispensed into the mixing system using appropriate dispensing equipment and procedure as recommended by the manufacturer.

3.6.3 Transporting

Transport concrete from the mixer to the forms as rapidly as practicable. Prevent segregation or loss of ingredients. Clean transporting equipment thoroughly before each batch. Do not use aluminum pipe or chutes. Remove concrete which has segregated in transporting and dispose of as directed.

3.7 PLACING CONCRETE

Place concrete in accordance with RD 470/2021 Art. 52.1 and 52.2. Concrete shall be placed within 15 minutes of discharge into non-agitating equipment.

3.7.1 Pumping

Pumping must not result in separation or loss of materials nor cause interruptions sufficient to permit loss of plasticity between successive increments. Loss of slump in pumping equipment must not exceed 50 mm at discharge/placement. Do not convey concrete through pipe made of aluminum or aluminum alloy. Avoid rapid changes in pipe sizes. Limit maximum size of coarse aggregate to 33 percent of the diameter of the pipe. Limit maximum size of well-rounded aggregate to 40 percent of the pipe diameter. Take samples for testing at both the point of delivery to the pump and at the discharge end.

3.7.2 Cold Weather

Cold weather concrete must meet the requirements of RD 470/2021 Art. 52.3.1 unless otherwise specified. Do not allow concrete temperature to decrease below 10 degrees C. Obtain approval prior to placing concrete when the ambient temperature is below 4 degrees C or when concrete is likely to be subjected to freezing temperatures within 24 hours. Cover concrete and provide sufficient heat to maintain 10 degrees C minimum adjacent to both the formwork and the structure while curing. Limit the rate of cooling to 3 degrees C in any one hour and 10 degrees C per 24 hours after heat application.

3.7.3 Hot Weather

Hot weather concrete must meet the requirements of RD 470/2021 Art. 52.3.2 unless otherwise specified. Maintain required concrete temperature using Figure 4.2 in ACI 305R to prevent the evaporation rate from exceeding 1 kg per square meter of exposed concrete per hour. Cool ingredients before mixing or use other suitable means to control concrete temperature and prevent rapid drying of newly placed concrete. Shade the fresh concrete as soon as possible after placing. Start curing when the surface of the fresh concrete is sufficiently hard to permit curing without damage. Provide water hoses, pipes, spraying equipment, and water hauling equipment, where job site is remote to water source, to maintain a moist concrete surface throughout the curing period. Provide burlap cover or other suitable, permeable material with fog spray or continuous wetting of the concrete when weather conditions prevent the use of either liquid membrane curing compound or impervious sheets. For vertical surfaces, protect forms from direct sunlight and add water to top of structure once concrete is set.

3.7.4 Bonding

Surfaces of set concrete at joints, must be roughened and cleaned of laitance, coatings, loose particles, and foreign matter. Roughen surfaces in a manner that exposes the aggregate uniformly and does not leave laitance, loosened particles of aggregate, nor damaged concrete at the surface.

Obtain bonding of fresh concrete that has set as follows:

- a. At joints between footings and walls or columns, between walls or columns and the beams or slabs they support, and elsewhere unless otherwise specified; roughened and cleaned surface of set concrete must be dampened, but not saturated, immediately prior to placing of fresh concrete.
- b. At joints in exposed-to-view work; at vertical joints in walls; at joints near midpoint of span in girders, beams, supported slabs, other structural members; in work designed to contain liquids; the roughened and cleaned surface of set concrete must be dampened but not saturated and covered with a cement grout coating.
- c. Provide cement grout that consists of equal parts of portland cement and fine aggregate by weight with not more than 22.5 liters of water per sack of cement. Apply cement grout with a stiff broom or brush to a minimum thickness of 1.6 mm. Deposit fresh concrete before cement grout has attained its initial set.

3.8 WASTE MANAGEMENT

Provide as specified in the Waste Management Plan and as follows.

3.8.1 Mixing Equipment

Before concrete pours, designate Contractor-owned site meeting environmental standards for cleaning out concrete mixing trucks. Minimize water used to wash equipment.

3.8.2 Reinforcing Steel

Collect reinforcing steel and place in designated area for recycling.

3.8.3 Other Waste

Identify concrete manufacturer's or supplier's policy for collection or return of construction waste, unused material, deconstruction waste, and/or packaging material.

3.9 SURFACE FINISHES EXCEPT FLOOR, SLAB, AND PAVEMENT FINISHES

3.9.1 Defects

Repair surface defects in accordance with ACI 301 Section 5 and RD 470/2021 Art. 53.3.

3.9.2 Not Against Forms (Top of Walls)

Surfaces not otherwise specified must be finished with wood floats to even surfaces. Finish must match adjacent finishes.

3.9.3 Formed Surfaces

3.9.3.1 Tolerances

Tolerances in accordance with ACI 117 and RD 470/2021 Art. 53.3 and Anejo 14, and as indicated.

3.9.3.2 As-Cast Rough Form

Provide for surfaces not exposed to public view a surface finish SF-1.0. Patch holes and defects.

3.9.3.3 Standard Smooth Finish

Provide for surfaces exposed to public view a surface finish SF-3.0. Patch holes and defects.

3.9.4 Smooth-Rubbed Finish

Provide a smooth-rubbed finish per [ACI 301](#) Section 5 in the locations exposed to view.

3.10 FLOOR, SLAB, AND PAVEMENT FINISHES AND MISCELLANEOUS CONSTRUCTION

In accordance with [ACI 301](#) and [ACI 302.1R](#) and [RD 470/2021](#) Art. 53.3, unless otherwise specified. Slope floors uniformly to drains where drains are provided. Where straightedge measurements are specified, Contractor must provide straightedge.

3.10.1 Finish

Place, consolidate, and immediately strike off concrete to obtain proper contour, grade, and elevation before bleedwater appears. Permit concrete to attain a set sufficient for floating and supporting the weight of the finisher and equipment. If bleedwater is present prior to floating the surface, drag the excess water off or remove by absorption with porous materials. Do not use dry cement to absorb bleedwater. Grate tampers ("jitterbugs") shall not be used.

3.10.1.1 Scratched

Use for surfaces intended to receive bonded applied cementitious applications. Finish concrete in accordance with [ACI 301](#) Section 5 for a scratched finish.

3.10.1.2 Floated

Use for exterior slabs where not otherwise specified. Finish concrete in accordance with [ACI 301](#) Section 5 for a floated finish.

3.10.1.3 Steel Troweled

Use for floors intended as walking surfaces. Finish concrete in accordance with [ACI 301](#) Section 5 for a steel troweled finish.

3.10.1.4 Broomed

Use on surfaces of exterior walks, platforms, patios, and ramps, unless otherwise indicated. Finish concrete in accordance with [ACI 301](#) Section 5 for a broomed finish.

3.10.1.5 Pavement

Screed the concrete with a template advanced with a combined longitudinal and crosswise motion. Maintain a slight surplus of concrete ahead of the template. After screeding, float the concrete longitudinally. Use a

straightedge to check slope and flatness; correct and refloat as necessary. Obtain final finish by a burlap drag. Drag a strip of clean, wet burlap from 900 to 3000 mm wide and 600 mm longer than the pavement width across the slab. Produce a fine, granular, sandy textured surface without disfiguring marks. Round edges and joints with an edger having a radius of 3 mm.

3.10.2 Concrete Walks

Provide 100 mm thick minimum. Provide contraction joints spaced every 1500 lineal mm unless otherwise indicated. Cut contraction joints 25 mm deep, or one fourth the slab thickness whichever is deeper, with a jointing tool after the surface has been finished. Provide 13 mm thick transverse expansion joints at changes in direction where sidewalk abuts curb, steps, rigid pavement, or other similar structures; space expansion joints every 15 m maximum. Give walks a broomed finish. Unless indicated otherwise, provide a transverse slope of 1/48. Limit variation in cross section to 6 mm in 1500 mm.

3.10.3 Pits and Trenches

Place bottoms and walls monolithically or provide waterstops and keys.

3.10.4 Curbs and Gutters

Provide contraction joints spaced every 3 m maximum unless otherwise indicated. Cut contraction joints 20 mm deep with a jointing tool after the surface has been finished. Provide expansion joints 13 mm thick and spaced every 30 m maximum unless otherwise indicated. Perform pavement finish.

3.10.5 Splash Blocks

Provide at outlets of downspouts emptying at grade. Splash blocks may be precast concrete, and must be 600 mm long, 300 mm wide and 100 mm thick, unless otherwise indicated, with smooth-finished countersunk dishes sloped to drain away from the building.

3.11 JOINTS

RD 470/2021, Art. 52.4 and the following:

3.11.1 Construction Joints

Make and locate joints not indicated so as not to impair strength and appearance of the structure, as approved. Joints must be perpendicular to main reinforcement. Reinforcement must be continued and developed across construction joints. The maximum allowable construction joint spacing shall be 18.4 m in any horizontal direction in walls and not in excess of 111.5 m in slabs on ground. Locate construction joints as follows:

3.11.1.1 Construction Joints for Constructability Purposes

- a. In walls, at top of footing; at top of slabs on ground; at top and bottom of door and window openings or where required to conform to architectural details; and at underside of deepest beam or girder framing into wall.
- b. In columns or piers, at top of footing; at top of slabs on ground; and

at underside of deepest beam or girder framing into column or pier.

- c. Near midpoint of spans for supported slabs, beams, and girders unless a beam intersects a girder at the center, in which case construction joints in girder must offset a distance equal to twice the width of the beam. Make transfer of shear through construction joint by use of inclined reinforcement.

Provide keyways at least 40 mm deep in construction joints in walls and slabs and between walls and footings; approved bulkheads may be used for slabs.

3.11.2 Isolation Joints in Slabs on Ground

- a. Provide joints at points of contact between slabs on ground and vertical surfaces, such as column pedestals, foundation walls, grade beams, and elsewhere as indicated.
- b. Fill joints with premolded joint filler strips 13 mm thick, extending full slab depth. Install filler strips at proper level below finish floor elevation with a slightly tapered, dress-and-oiled wood strip temporarily secured to top of filler strip to form a groove not less than 19 mm in depth where joint is sealed with sealing compound and not less than 6 mm in depth where joint sealing is not required. Remove wood strip after concrete has set. Contractor must clean groove of foreign matter and loose particles after surface has dried.

3.11.3 Contraction Joints in Slabs on Ground

- a. Provide joints to form panels as indicated.
- b. Under and on exact line of each control joint, cut 50 percent of welded wire reinforcement before placing concrete.
- c. Sawcut contraction joints into slab on ground.

3.11.4 Sealing Joints in Slabs on Ground

- a. Contraction and control joints which are to receive finish flooring material must be sealed with joint sealing compound after concrete curing period. Slightly underfill groove with joint sealing compound to prevent extrusion of compound. Remove excess material as soon after sealing as possible.
- b. Sealed groove must be left ready to receive filling material that is provided as part of finish floor covering work.

3.12 CURING AND PROTECTION

Curing and protection in accordance with RD 470/2021, unless otherwise specified. Begin curing immediately following form removal. Avoid damage to concrete from vibration created by blasting, pile driving, movement of equipment in the vicinity, disturbance of formwork or protruding reinforcement, and any other activity resulting in ground vibrations. Protect concrete from injurious action by sun, rain, flowing water, frost, mechanical injury, tire marks, and oil stains. Do not allow concrete to dry out from time of placement until the expiration of the specified curing period. Do not use membrane-forming compound on surfaces where appearance would be objectionable, on any surface to be painted, where

coverings are to be bonded to the concrete, or on concrete to which other concrete is to be bonded. If forms are removed prior to the expiration of the curing period, provide another curing procedure specified herein for the remaining portion of the curing period. Provide moist curing for those areas receiving liquid chemical sealer, hardener, or epoxy coating. Allow curing compound/sealer installations to cure prior to the installation of materials that adsorb VOCs.

3.12.1 Curing Periods

ACI 301 Section 5 and RD 470/2021 Art. 52.5, except 10 days for retaining walls, pavement or chimneys. Begin curing immediately after placement. Protect concrete from premature drying, excessively hot temperatures, and mechanical injury; and maintain minimal moisture loss at a relatively constant temperature for the period necessary for hydration of the cement and hardening of the concrete. The materials and methods of curing are subject to approval by the Contracting Officer.

3.12.2 Curing Formed Surfaces

Accomplish curing of formed surfaces, including undersurfaces of girders, beams, supported slabs, and other similar surfaces by moist curing with forms in place for full curing period or until forms are removed. If forms are removed before end of curing period, accomplish final curing of formed surfaces by any of the curing methods specified above, as applicable.

3.12.3 Curing Unformed Surfaces

- a. Accomplish initial curing of unformed surfaces, such as monolithic slabs, floor topping, and other flat surfaces, by membrane curing.
- b. Accomplish final curing of unformed surfaces by any of curing methods specified, as applicable.
- c. Accomplish final curing of concrete surfaces to receive liquid floor hardener or finish flooring by moisture-retaining cover curing.

3.12.4 Temperature of Concrete During Curing

When temperature of atmosphere is 5 degrees C and below, maintain temperature of concrete at not less than 13 degrees C throughout concrete curing period or 7 degrees C when the curing period is measured by maturity. When necessary, make arrangements before start of concrete placing for heating, covering, insulation, or housing as required to maintain specified temperature and moisture conditions for concrete during curing period.

When the temperature of atmosphere is 27 degrees C and above or during other climatic conditions which cause too rapid drying of concrete, make arrangements before start of concrete placing for installation of wind breaks, of shading, and for fog spraying, wet sprinkling, or moisture-retaining covering of light color as required to protect concrete during curing period.

Changes in temperature of concrete must be uniform and not exceed 3 degrees C in any one hour nor 27 degrees C in any 24-hour period.

3.12.5 Protection from Mechanical Injury

During curing period, protect concrete from damaging mechanical disturbances, particularly load stresses, heavy shock, and excessive vibration and from damage caused by rain or running water.

3.12.6 Protection After Curing

Protect finished concrete surfaces from damage by construction operations.

3.13 FIELD QUALITY CONTROL

3.13.1 Aggregate Testing

3.13.1.1 Fine Aggregate

At least once during each shift when the concrete plant is operating, there shall be one sieve analysis and fineness modulus determination in accordance with [UNE-EN 933-1](#) for the fine aggregate or for each fine aggregate if it is batched in more than one size or classification. The location at which samples are taken may be selected by the Contractor as the most advantageous for control. However, the Contractor is responsible for delivering fine aggregate to the mixer within specification limits. When the amount passing on any sieve is outside the specification limits, the fine aggregate shall be immediately resampled and retested. If there is another failure on any sieve, the fact shall be immediately reported to the Contracting Officer, concreting shall be stopped, and immediate steps taken to correct the grading.

3.13.1.2 Coarse Aggregate

At least once during each shift in which the concrete plant is operating, there shall be a sieve analysis in accordance with [UNE-EN 933-1](#) for each size of coarse aggregate. The location at which samples are taken may be selected by the Contractor as the most advantageous for production control. However, the Contractor shall be responsible for delivering the aggregate to the mixer within specification limits. A test record of samples of aggregate taken at the same locations shall show the results of the current test as well as the average results of the five most recent tests including the current test. The Contractor may adopt limits for control coarser than the specification limits for samples taken other than as delivered to the mixer to allow for degradation during handling. When the amount passing any sieve is outside the specification limits, the coarse aggregate shall be immediately resampled and retested. If the second sample fails on any sieve, that fact shall be reported to the Contracting Officer. Where two consecutive averages of 5 tests are outside specification limits, the operation shall be considered out of control and reported to the Contracting Officer. Concreting shall be stopped and immediate steps shall be taken to correct the grading.

3.13.2 Concrete Sampling

[UNE-EN 12350-1](#). Collect samples of fresh concrete to perform tests specified. [UNE-EN 12390-2](#) for making test specimens.

3.13.3 Concrete Testing

3.13.3.1 Slump Tests

UNE-EN 12350-2. Take concrete samples during concrete placement/discharge.

The maximum slump may be increased as specified with the addition of an approved admixture provided that the water-cementitious material ratio is not exceeded. Perform tests at commencement of concrete placement, when test cylinders are made, and for each batch (minimum) or every 16 cubic meters (maximum) of concrete.

3.13.3.2 Temperature Tests

Test the concrete delivered and the concrete in the forms. Perform tests in hot or cold weather conditions (below 10 degrees C and above 27 degrees C) for each batch (minimum) or every 16 cubic meters (maximum) of concrete, until the specified temperature is obtained, and whenever test cylinders and slump tests are made.

3.13.3.3 Compressive Strength Tests

UNE-EN 12390-3. Make six 150 mm by 300 mm test cylinders for each set of tests in accordance with UNE-EN 12390-2 applicable requirements of RD 470/2021, Art. 57.5.3, 57.5.4 and 57.5.5. Take precautions to prevent evaporation and loss of water from the specimen. Test two cylinders at 7 days, two cylinders at 28 days, and hold two cylinder in reserve. Take samples for strength tests of concrete placed each day not less than once a day, nor less than once for each 75 cubic meters of concrete for the first 380 cubic meters, then every 380 cubic meters thereafter, nor less than once for each 500 square meters of surface area for slabs or walls. For the entire project, take no less than five sets of samples and perform strength tests for each mix design of concrete placed. Each strength test result must be the average of two cylinders from the same concrete sample tested at 28 days. Concrete compressive tests must meet the requirements of this section and the Contract Document. Retest locations represented by erratic core strengths. Where retest does not meet concrete compressive strength requirements submit a mitigation or remediation plan for review and approval by the contracting officer. Repair core holes with nonshrink grout. Match color and finish of adjacent concrete.

3.13.3.4 Air Content

ASTM C173/C173M or ASTM C231/C231M for normal weight concrete. Test air-entrained concrete for air content at the same frequency as specified for slump tests.

3.13.3.5 Chloride Ion Concentration

Chloride ion concentration must meet the requirements of the paragraph titled CORROSION AND CHLORIDE CONTENT. Determine water soluble ion concentration in accordance with ASTM C1218/C1218M. Perform test once for each mix design.

3.13.3.6 Strength of Concrete Structure

The strength of the concrete structure will be considered to be deficient if any of the following conditions are identified:

- a. Failure to meet compressive strength tests as evaluated.

- b. Reinforcement not conforming to requirements specified.
- c. Concrete which differs from required dimensions or location in such a manner as to reduce strength.
- d. Concrete curing and protection of concrete against extremes of temperature during curing, not conforming to requirements specified.
- e. Concrete subjected to damaging mechanical disturbances, particularly load stresses, heavy shock, and excessive vibration.
- f. Poor workmanship likely to result in deficient strength.

Where the strength of the concrete structure is considered deficient submit a mitigation or remediation plan for review and approval by the contracting officer.

3.13.3.7 Non-Conforming Materials

Factors that indicate that there are non-conforming materials include (but not limited to) excessive compressive strength, inadequate compressive strength, excessive slump, excessive voids and honeycombing, concrete delivery records that indicate excessive time between mixing and placement, or excessive water was added to the mixture during delivery and placement. Any of these indicators alone are sufficient reason for the Contracting Officer to request additional sampling and testing.

Investigations into non-conforming materials must be conducted at the Contractor's expense. The Contractor must be responsible for the investigation and must make written recommendations to adequately mitigate or remediate the non-conforming material. The Contracting Officer may accept, accept with reduced payment, require mitigation, or require removal and replacement of non-conforming material at no additional cost to the Government.

3.13.3.8 Testing Concrete Structure for Strength

When there is evidence that strength of concrete structure in place does not meet specification requirements or there are non-conforming materials, make cores drilled from hardened concrete for compressive strength determination in accordance with [UNE-EN 12504-1](#), and as follows:

- a. Take at least three representative cores from each member or area of concrete-in-place that is considered potentially deficient. Location of cores will be determined by the Contracting Officer.
- b. Test cores after moisture conditioning in accordance with [UNE-EN 12504-1](#) if concrete they represent is more than superficially wet under service.
- c. Air dry cores, (16 to 27 degrees C with relative humidity less than 60 percent) for 7 days before test and test dry if concrete they represent is dry under service conditions.
- d. Strength of cores from each member or area are considered satisfactory if their average is equal to or greater than 85 percent of the 28-day design compressive strength of the class of concrete.

Fill core holes solid with patching mortar and finished to match adjacent concrete surfaces.

Correct concrete work that is found inadequate by core tests in a manner approved by the Contracting Officer.

3.14 REPAIR, REHABILITATION AND REMOVAL

Before the Contracting Officer accepts the structure the Contractor must inspect the structure for cracks, damage and substandard concrete placements that may adversely affect the service life of the structure. A report documenting these defects must be prepared which includes recommendations for repair, removal or remediation must be submitted to the Contracting Officer for approval before any corrective work is accomplished.

3.14.1 Crack Repair

Prior to final acceptance, all cracks in excess of 0.50 mm wide must be documented and repaired. The proposed method and materials to repair the cracks must be submitted to the Contracting Officer for approval. The proposal must address the amount of movement expected in the crack due to temperature changes and loading.

3.14.2 Repair of Weak Surfaces

Weak surfaces are defined as mortar-rich, rain-damaged, uncured, or containing exposed voids or deleterious materials. Concrete surfaces with weak surfaces less than 6 mm thick must be diamond ground to remove the weak surface. Surfaces containing weak surfaces greater than 6 mm thick must be removed and replaced or mitigated in a manner acceptable to the Contracting Officer.

3.14.3 Failure of Quality Assurance Test Results

Proposed mitigation efforts by the Contractor must be approved by the Contracting Officer prior to proceeding.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 50 13

MISCELLANEOUS METAL FABRICATIONS

05/17, CHG 1: 08/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALIFICATION OF WELDERS
- 1.4 DELIVERY, STORAGE, AND PROTECTION
- 1.5 MISCELLANEOUS REQUIREMENTS
 - 1.5.1 Fabrication Drawings
 - 1.5.2 Installation Drawings

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Structural Carbon Steel
 - 2.1.2 Structural Tubing
 - 2.1.3 Steel Pipe
 - 2.1.4 Fittings for Steel Pipe
 - 2.1.5 Gratings
 - 2.1.6 Floor Plates, Patterned
 - 2.1.7 Anchor Bolts
 - 2.1.7.1 Expansion Anchors and Adhesive Anchors
 - 2.1.7.2 Lag Screws and Bolts
 - 2.1.7.3 Toggle Bolts
 - 2.1.7.4 Bolts, Nuts, Studs and Rivets
 - 2.1.7.5 Powder Actuated Fasteners
 - 2.1.7.6 Screws
 - 2.1.7.7 Washers
 - 2.1.7.8 Welded Headed Shear Studs
- 2.2 FABRICATION FINISHES
 - 2.2.1 Galvanizing
 - 2.2.2 Galvanize
 - 2.2.3 Repair of Zinc-Coated Surfaces
 - 2.2.4 Shop Cleaning and Painting
 - 2.2.4.1 Surface Preparation
 - 2.2.4.2 Pretreatment, Priming and Painting
 - 2.2.5 Nonferrous Metal Surfaces
- 2.3 FLOOR GRATINGS
- 2.4 BOLLARDS
- 2.5 MISCELLANEOUS PLATES AND SHAPES
- 2.6 ROOF HATCHES (SCUTTLES)

PART 3 EXECUTION

- 3.1 GENERAL INSTALLATION REQUIREMENTS
- 3.2 WORKMANSHIP
- 3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.4 BUILT-IN WORK
- 3.5 WELDING
- 3.6 DISSIMILAR METALS
- 3.7 PREPARATION
 - 3.7.1 Material Coatings and Surfaces
 - 3.7.2 Environmental Conditions
- 3.8 ROOF HATCH (SCUTTLES)
- 3.9 INSTALLATION OF BOLLARDS

-- End of Section Table of Contents --

SECTION 05 50 13

MISCELLANEOUS METAL FABRICATIONS
05/17, CHG 1: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 104231	(1999) Waterproofing. Bituminous And Bituminous Modified Materials. Asphalt Emulsions
UNE-EN 1562	(2020) Founding - Malleable cast irons
UNE-EN 1993-1-1	(2013) Eurocode 3: Design of steel structures - Part 1-1: General rules and rules for buildings
UNE-EN 10025-2	(2020) Hot rolled products of structural steels - Part 2: Technical delivery conditions for non-alloy structural steels
UNE-EN 10025-3	(2006) Hot rolled products of structural steels - Part 3: Technical delivery conditions for normalized/normalized rolled weldable fine grain structural steels
UNE-EN 10083-1	(2008) Steels for quenching and tempering - Part 1: General technical delivery conditions.
UNE-EN 10162	(2005) Cold formed steel sections - Technical Supply Conditions. Dimensional, tolerances and on Cross-section.
UNE-EN 10216-1	(2014) Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties
UNE-EN 10217-1	(2019) Welded steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties
UNE-EN 10219-1	(2007) Cold formed welded structural hollow sections of non-alloy and fine grain steels - Part 1: Technical delivery conditions

UNE-EN 10363	(2016) Continuously hot-rolled patterned steel strip and plate/sheet cut from wide strip - Tolerances on dimensions and shape
UNE-EN ISO 898-1	(2015) Mechanical properties of fasteners made of carbon steel and alloy steel - Part 1: Bolts, screws and studs with specified property classes - Coarse thread and fine pitch thread (ISO 898-1:2013)
UNE-EN ISO 898-2	(2013) Mechanical properties of fasteners made of carbon steel and alloy steel - Part 2: Nuts with specified property classes - Coarse thread and fine pitch thread (ISO 898-2:2012)
UNE-EN ISO 1461	(2010) Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods (ISO1461:1999)
UNE-EN ISO 1482	(2011) Slotted countersunk (flat) head tapping screws (ISO 1482:2011)
UNE-EN ISO 9606-1	(2017) Qualification testing of welders - Fusion welding - Part 1: Steels (ISO 9606-1:2012 including Cor 1:2012)
UNE-EN ISO 9606-2	(2005) Qualification test of welders-Fusion welding-Part 2: Aluminium and aluminium alloys (ISO 9606-2:2004)
UNE-EN ISO 10673	(2010) Plain washers for screw and washer assemblies. Small, normal and large series. Product grade A (ISO 10673:2009)

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318	(2014; Errata 1-2 2014; Errata 3-5 2015; Errata 6 2016; Errata 7-9 2017) Building Code Requirements for Structural Concrete (ACI 318-14) and Commentary (ACI 318R-14)
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AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 303	(2016) Code of Standard Practice for Steel Buildings and Bridges
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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1	(2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)
ASME B18.6.2	(2020) Square Head Set Screws and Slotted Headless Set Screws (Inch Series)
ASME B18.6.3	(2013; R 2017) Machine Screws, Tapping Screws, and Machine Drive Screws (Inch

Series)

- ASME B18.21.1 (2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)
- ASME B18.21.2M (1999; R 2014) Lock Washers (Metric Series)
- ASME B18.22M (1981; R 2017) Metric Plain Washers

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

- ASSP A10.3 (2020) Safety Requirements for Powder-Actuated Fastening Systems American National Standard for Construction and Demolition Operations

AMERICAN WELDING SOCIETY (AWS)

- AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

- ASTM A53/A53M (2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
- ASTM A108 (2013) Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
- ASTM A780/A780M (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
- ASTM C1513 (2018) Standard Specification for Steel Tapping Screws for Cold-Formed Steel Framing Connections

MASTER PAINTERS INSTITUTE (MPI)

- MPI 79 (2016) Primer, Alkyd, Anti-Corrosive for Metal

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

- NAAMM MBG 531 (2017) Metal Bar Grating Manual

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

- SSPC SP 3 (2018) Power Tool Cleaning
- SSPC SP 6/NACE No.3 (2007) Commercial Blast Cleaning

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance

with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Floor Gratings, Installation Drawings; G

Bollards; G

Embedded Angles and Plates, Installation Drawings; G

Roof Hatches, Installation Drawings; G

SD-03 Product Data

Floor Gratings; G

Roof Hatches; G

SD-07 Certificates

Certified Mill Test Reports for Chemistry and Mechanical Properties; G

1.3 QUALIFICATION OF WELDERS

Qualify welders in accordance with AWS D1.1/D1.1M or in accordance with UNE-EN ISO 9606-1 for steel and UNE-EN ISO 9606-2 for aluminum. Use procedures, materials, and equipment of the type required for the work.

1.4 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

1.5 MISCELLANEOUS REQUIREMENTS

1.5.1 Fabrication Drawings

Submit fabrication drawings showing layout(s), connections to structural system, and anchoring details as specified in AISC 303 and in accordance with UNE-EN 1993-1-1.

1.5.2 Installation Drawings

Submit templates, erection, and installation drawings indicating thickness, type, grade, class of metal, and dimensions. Show construction details, reinforcement, anchorage, and installation in relation to the building construction.

PART 2 PRODUCTS

2.1 MATERIALS

Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals). Coordinate color and finish with the material to which fastenings are applied. Submit the manufacturer's certified mill reports which clearly show the applicable ASTM mechanical and chemical requirements together with the actual test results for the supplied

materials.

2.1.1 Structural Carbon Steel

Provide in accordance with [UNE-EN 10025-2](#) and [UNE-EN 10025-3](#).

2.1.2 Structural Tubing

Provide in accordance with [UNE-EN 10219-1](#).

2.1.3 Steel Pipe

Provide in accordance with [UNE-EN 10216-1](#) or [UNE-EN 10217-1](#). Minimum tensile strength 330 Mpa.

2.1.4 Fittings for Steel Pipe

Provide standard malleable iron fittings in accordance with [UNE-EN 1562](#).

2.1.5 Gratings

a. Provide metal bar type grating in accordance with [NAAMM MBG 531](#).

2.1.6 Floor Plates, Patterned

Provide floor plate in accordance with [UNE-EN 10363](#). Provide steel plate not less than [1.9 mm](#).

2.1.7 Anchor Bolts

Provide in accordance with [UNE-EN ISO 898-1](#). Where exposed, provide anchor bolts of the same material, color, and finish as the metal to which they are applied.

2.1.7.1 Expansion Anchors and Adhesive Anchors

Provide expansion anchors and adhesive anchors as indicated in the Contract Documents to the minimum embedment indicated.

a. Provide minimum allowable pullout value as required by design. Calculate pullout capacity according to [ACI 318](#).

b. Provide minimum allowable shear value as required by design. Calculate shear capacity according to [ACI 318](#).

2.1.7.2 Lag Screws and Bolts

Provide in accordance with [ASME B18.2.1](#), type and grade best suited for the purpose.

2.1.7.3 Toggle Bolts

Provide in accordance with [ASME B18.2.1](#).

2.1.7.4 Bolts, Nuts, Studs and Rivets

Provide in accordance with [UNE-EN ISO 898-1](#) and [UNE-EN ISO 898-2](#).

2.1.7.5 Powder Actuated Fasteners

Follow safety provisions in accordance with [ASSP A10.3](#).

2.1.7.6 Screws

Provide in accordance with [ASME B18.2.1](#), [ASME B18.6.2](#), [ASME B18.6.3](#) and [ASTM C1513](#) or [UNE-EN ISO 1482](#).

2.1.7.7 Washers

Provide plain washers in accordance with [ASME B18.22M](#), [ASME B18.21.1](#) or [UNE-EN ISO 10673](#). Provide beveled washers for American Standard beams and channels, square or rectangular, tapered in thickness, and smooth. Provide lock washers in accordance with [ASME B18.21.2M](#), [ASME B18.21.1](#).

2.1.7.8 Welded Headed Shear Studs

Provide in accordance with [ASTM A108](#) or [UNE-EN 10162](#) or [UNE-EN 10083-1](#).

2.2 FABRICATION FINISHES

2.2.1 Galvanizing

Hot-dip galvanize items specified to be zinc-coated, after fabrication where practicable. Provide galvanizing in accordance with [UNE-EN ISO 1461](#), Z275 G90.

2.2.2 Galvanize

Anchor bolts, grating fasteners, washers, and parts or devices necessary for proper installation, unless indicated otherwise.

2.2.3 Repair of Zinc-Coated Surfaces

Repair damaged surfaces with galvanizing repair method and paint in accordance with [ASTM A780/A780M](#) or [UNE-EN ISO 1461](#), or by application of stick or thick paste material specifically designed for repair of galvanizing, as approved by Contracting Officer. Clean areas to be repaired and remove slag from welds. Heat, with a torch, surfaces to which stick or paste material will be applied. Heat to a temperature sufficient to melt the metals in the stick or paste. Spread molten material uniformly over surfaces to be coated and wipe off excess material.

2.2.4 Shop Cleaning and Painting

2.2.4.1 Surface Preparation

Blast clean surfaces in accordance with [SSPC SP 6/NACE No.3](#). Surfaces that will be exposed in spaces above ceiling or in attic spaces, crawl spaces, furred spaces, and chases may be cleaned in accordance with [SSPC SP 3](#) in lieu of being blast cleaned. Wash cleaned surfaces which become contaminated with rust, dirt, oil, grease, or other contaminants with solvents until thoroughly clean. Steel to be embedded in concrete must be free of dirt and grease prior to embed. Do not paint or galvanize bearing surfaces, including contact surfaces within slip critical joints. Shop coat these surfaces with rust prevention.

2.2.4.2 Pretreatment, Priming and Painting

Apply pre-treatment, primer, and paint in accordance with manufacturer's printed instructions. On surfaces concealed in the finished construction or not accessible for finish painting, apply an additional prime coat to a minimum dry film thickness of 0.03 mm. Tint additional prime coat with a small amount of tinting pigment.

2.2.5 Nonferrous Metal Surfaces

Protect by plating, anodic, or organic coatings.

2.3 FLOOR GRATINGS

- a. NAAMM MBG 531, band ends of gratings with bars of the same or greater thickness than the metal used for grating. Weld banding bars to bearing bars or channels at least every fourth bar or channel and in every corner. Tack weld intervening bars or channels. Band diagonal or round cuts by welding bars of the same or greater thickness as the grating and in accordance with the manufacturer's standard for trim unless otherwise indicated.
- b. Anchor gratings to structural members with bolts, toggle bolts, or expansion shields and bolts.
- c. Provide slip resistant surface finishes.

2.4 BOLLARDS

Provide 200 mm prime coated standard weight steel pipe in accordance with ASTM A53/A53M or UNE-EN 10216-1. Anchor posts in concrete as indicated and fill solidly with concrete with minimum compressive strength of 20 MPa.

2.5 MISCELLANEOUS PLATES AND SHAPES

Provide items that do not form a part of the structural framework, such as miscellaneous mountings and frames. Provide with connections and fasteners or welds.

Provide angles and plates in accordance with UNE-EN 10025-2 and UNE-EN 10025-3, for embedment as indicated. Galvanize embedded items exposed to the elements in accordance with UNE-EN ISO 1461.

2.6 ROOF HATCHES (SCUTTLES)

Provide aluminum sheets not less than 1.9 mm with 75 mm beaded flange, welded and ground at corners. Provide a minimum clear opening of 760 by 900 mm. Insulate cover and curb with 25 mm thick rigid fiberboard insulation, covered and protected by aluminum sheet of not less than 0.45 mm. Provide with 300 mm high curb, formed with 75 mm mounting flanges with holes for securing to the roof deck.

PART 3 EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated in accordance with manufacturer's instructions. Verify all field dimensions prior to fabrication. Include materials and parts necessary to complete each assembly, whether indicated

or not. Miss-alignment and miss-sizing of holes for fasteners is cause for rejection. Conceal fastenings where practicable. Joints exposed to weather must be watertight.

3.2 WORKMANSHIP

Provide miscellaneous metalwork that is true and accurate in shape, size, and profile. Make angles and lines continuous and straight. Make curves consistent, smooth and unfaceted. Provide continuous welding along the entire area of contact except where tack welding is permitted. Do not tack weld exposed connections. Unless otherwise indicated and approved, provide a smooth finish on exposed surfaces. Provide countersunk rivets where exposed. Provide coped and mitered corner joints aligned flush and without gaps.

3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS

Provide anchorage as necessary, whether indicated or not, for fastening miscellaneous metal items securely in place. Include slotted inserts, expansion shields, powder-driven fasteners, toggle bolts (when approved for concrete), headed shear studs, machine and carriage bolts for steel. Provide non-ferrous attachments for non-ferrous metal. Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals), that generally match in color and finish the surfaces to which they are applied. Conceal fastenings where practicable. Provide all fasteners flush with the surfaces they fasten, unless indicated otherwise.

3.4 BUILT-IN WORK

Where necessary and not otherwise indicated, form built-in metal work for anchorage with concrete. Provide built-in metal work in ample time for securing in place as the work progresses.

3.5 WELDING

Perform welding, welding inspection, and corrective welding in accordance with [AWS D1.1/D1.1M](#) or [UNE-EN ISO 9606-1](#). Use continuous welds on all exposed connections. Grind visible welds smooth in the finished installation.

3.6 DISSIMILAR METALS

Where dissimilar metals are in contact, protect surfaces with a coating in accordance with [MPI 79](#) to prevent galvanic or corrosive action. Where aluminum is in contact with concrete, plaster, mortar, masonry, wood, or absorptive materials subject to wetting, protect in accordance with [UNE 104231](#), asphalt-base emulsion. Clean surfaces with metal shavings from installation at the end of each work day.

3.7 PREPARATION

3.7.1 Material Coatings and Surfaces

Remove rust preventive coating just prior to field erection, using a remover approved by the metal manufacturer. Surfaces, when assembled, must be free of rust, grease, dirt and other foreign matter.

3.7.2 Environmental Conditions

Do not clean or paint surfaces when damp or exposed to foggy or rainy weather, when metallic surface temperature is less than minus 15 degrees C above the dew point of the surrounding air, or when surface temperature is below 7 degrees C or over 35 degrees C, unless approved by the Contracting Officer. Metal surfaces to be painted must be dry for a minimum of 48 hours prior to the application of primer or paint.

3.8 ROOF HATCH (SCUTTLES)

Construction and accessories as follows:

- a. Provide insulated cover and curb with mounting flanges for securing to roof deck. Provide curbs with integral metal cap flashing of the same gage and metal as the curb, fully welded and ground at corners for weather tightness.
- b. Provide hatches completely assembled, with pintle hinges, compression spring operators enclosed in telescopic tubes, positive snap latches with turn handles on inside and outside, and neoprene draft seals. Provide fasteners for padlocking from the inside. Provide covers with automatic hold-open arms complete with grip handle to permit one hand release. Cover action must be smooth through its entire range of motion with an operating pressure of approximately 130 N.

3.9 INSTALLATION OF BOLLARDS

Set bollards vertically in concrete piers. Fill hollow cores with concrete having a compressive strength of 30 MPa.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 51 00

METAL STAIRS

08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY CONTROL
 - 1.3.1 Qualifications for Welding Work

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
- 2.2 FABRICATION
 - 2.2.1 General Fabrication
 - 2.2.2 Floor Grating Treads and Platforms
 - 2.2.3 Protective Coating
- 2.3 COMPONENTS
 - 2.3.1 Steel Stairs
 - 2.3.1.1 Materials
 - 2.3.2 Concrete Inserts
 - 2.3.3 Fasteners
- 2.4 MATERIALS
 - 2.4.1 Structural Steel Plates, Shapes and Bars
 - 2.4.2 Structural Steel Tubing
 - 2.4.3 Hot-Rolled Carbon Steel Bars
 - 2.4.4 Hot-Rolled Carbon Steel Sheets and Strips
 - 2.4.5 Galvanized Carbon Steel Sheets
 - 2.4.6 Steel Pipe
- 2.5 Mill Test Reports

PART 3 EXECUTION

- 3.1 PREPARATION
- 3.2 INSTALLATION
 - 3.2.1 Field Preparation
 - 3.2.2 Field Welding
 - 3.2.3 Safety Nosings
 - 3.2.4 Touchup Painting

-- End of Section Table of Contents --

SECTION 05 51 00

METAL STAIRS
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

- | | |
|-------------------|--|
| UNE-EN ISO 3834-3 | (2006) Quality requirements for fusion welding of metallic materials - Part 3: Standard quality requirements (ISO 3834-3:2005) |
| UNE-EN ISO 9606-1 | (2017) Qualification testing of welders - Fusion welding - Part 1: Steels (ISO 9606-1:2012 including Cor 1:2012) |
| UNE-EN ISO 15607 | (2020) Specification and qualification of welding procedures for metallic materials - General rules (ISO 15607:2019) |

AMERICAN WELDING SOCIETY (AWS)

- | | |
|----------------|---|
| AWS D1.1/D1.1M | (2020; Errata 1 2021) Structural Welding Code - Steel |
|----------------|---|

ASTM INTERNATIONAL (ASTM)

- | | |
|-----------------|---|
| ASTM A6/A6M | (2022) Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling |
| ASTM A29/A29M | (2020) Standard Specification for General Requirements for Steel Bars, Carbon and Alloy, Hot-Wrought |
| ASTM A36/A36M | (2019) Standard Specification for Carbon Structural Steel |
| ASTM A53/A53M | (2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless |
| ASTM A123/A123M | (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware |

ASTM A283/A283M	(2013) Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates
ASTM A500/A500M	(2021a) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM A568/A568M	(2019a) Standard Specification for Steel, Sheet, Carbon, Structural, and High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, General Requirements for
ASTM A575	(2020) Standard Specification for Steel Bars, Carbon, Merchant Quality, M-Grades
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A924/A924M	(2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process
ASTM A1011/A1011M	(2023) Standard Specification for Steel Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength
ASTM F568M	(2007) Standard Specification for Carbon and Alloy Steel Externally Threaded Metric Fasteners

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 898-1	(2013) Mechanical Properties of Fasteners Made of Carbon Steel and Alloy Steel – Part 1: Bolts, Screws and Studs with Specified Property Classes – Coarse Thread and Fine Pitch Thread
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NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM MBG 531	(2017) Metal Bar Grating Manual
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Iron and Steel Hardware; G

Steel Shapes, Plates, Bars, and Strips; G

Metal Stair System; G

SD-03 Product Data

Concrete Inserts; G

Protective Coating; G

Steel Stairs; G

SD-06 Test Reports

Mill Test Reports; G

SD-07 Certificates

Welding Procedures; G

Welder Qualification; G

SD-08 Manufacturer's Instructions

Protective Coating; G

1.3 QUALITY CONTROL

1.3.1 Qualifications for Welding Work

Submit [welding procedures](#) in accordance with [AWS D1.1/D1.1M](#) or [UNE-EN ISO 15607](#) and [UNE-EN ISO 3834-3](#). Make test specimens in the presence of the Contracting Officer, and have the specimens tested by an approved testing laboratory at the Contractor's expense.

Certify [welder qualification](#) by tests in accordance with [AWS D1.1/D1.1M](#) or in accordance with [***UNE-EN ISO 9606-1](#) for steel, or under an equivalent approved qualification test. In addition, perform tests on test pieces in positions and with clearances equivalent to those actually encountered. If a test weld fails to meet requirements, ensure that two test welds are retested immediately and that each test weld is made and passes. Failure in the immediate retest requires that the welder be retested after further practice or training and a complete set of test welds be made.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Submit complete and detailed fabrication drawings for all [iron and steel hardware](#), and for all [steel shapes, plates, bars, and strips](#) used in accordance with the design specifications referenced in this section.

2.2 FABRICATION

Preassemble items in the shop to the greatest extent possible. Disassemble units only to the extent necessary for shipping and handling. Clearly mark units for reassembly and coordinated installation.

For the fabrication of work exposed to view, use only materials that are smooth and free of surface blemishes, including pitting, seam marks, roller marks, rolled trade names, and roughness. Remove blemishes by grinding, or by welding and grinding, before cleaning and treating surfaces and applying surface finishes, including zinc coatings.

2.2.1 General Fabrication

Prepare and submit **metal stair system** shop drawings with detailed plans and elevations at scales not less than **1 to 12 scale** and with details of sections and connections at scales not less than **1 to 4 scale**. Also detail the placement drawings, diagrams, and templates for installation of anchorages, including concrete inserts, anchor bolts, and miscellaneous metal items having integral anchorage devices.

Use materials of size and thicknesses indicated or, if not indicated, of the size and thickness necessary to produce a finished product that is strong enough and durable enough for its intended use. Work the materials to the dimensions indicated on approved detail drawings, using proven methods of fabrication and support. Use the type of materials indicated or specified for the various components of work.

Form exposed work true to line and level, with accurate angles and surfaces and with straight sharp edges. Ease exposed edges to a radius of approximately **0.8 millimeter**, and bend metal corners to the smallest radius possible without causing grain separation or otherwise impairing the work.

Continuously weld corners and seams in accordance with the recommendations of **AWS D1.1/D1.1M**. Grind exposed welds smooth and flush to match and blend with adjoining surfaces.

Form exposed connections with hairline joints that are flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of the type indicated or, if not indicated, use Phillips flat-head (countersunk) screws or bolts.

Provide and coordinate anchorage of the type indicated for the supporting structure. Fabricate anchoring devices, and space them as indicated and as necessary to provide adequate support for the intended use of the work.

Use hot-rolled steel bars for work fabricated from bar stock unless work is indicated or specified as fabricated from cold-finished or cold-rolled stock.

2.2.2 Floor Grating Treads and Platforms

Provide floor grating treads and platforms conforming to **ASTM A6/A6M**, **ASTM A29/A29M** and **NAAMM MBG 531**, "Metal Bar Grating Manual." Provide the pattern, spacing, and bar sizes as indicated:

- a. Galvanized finish, conforming to **ASTM A123/A123M**.

Fabricate grating treads with steel plate nosings on one edge and with steel angle or steel plate carriers at each end for string connections. Secure treads to strings with bolts.

Match the nosings of grating platforms with the nosing of grating treads at landings. Provide toeplates where the open-sided edges of floor

grating meet platform framing members.

2.2.3 Protective Coating

Hot-dip galvanize steelwork as indicated in accordance with [ASTM A123/A123M](#). Touch up abraded surfaces and cut ends of galvanized members with zinc-dust, zinc-oxide primer, or an approved galvanizing repair compound.

2.3 COMPONENTS

2.3.1 Steel Stairs

Provide steel stairs complete with stringers, grating treads, landings, columns, handrails, and necessary bolts and other fastenings. Hot-dip-galvanize steel stairs and accessories.

2.3.1.1 Materials

Provide steel stairs of welded construction except that bolts may be used where welding is not practicable. Do not use screw or screw-type connections.

- a. Structural Steel: [ASTM A36/A36M](#).
- b. Gratings for Treads and Landings: [NAAMM MBG 531](#). Provide gratings with nonslip nosings.
- c. Support steel grating on angle cleats welded to stringers or treads with integral cleats, welded or bolted to the stringer. Close exposed ends. For exterior stairs, form all exposed joints to exclude water.
- d. Before fabrication, obtain necessary field measurements and verify drawing dimensions.
- e. Clean metal surfaces free of mill scale, flake rust, and rust pitting before shop finishing. Weld permanent connections. Finish welds flush and smooth on surfaces that will be exposed after installation.

2.3.2 Concrete Inserts

Carbon steel bolts having special wedge-shaped heads, nuts, washers, and shims and galvanized in accordance with [ASTM A153/A153M](#). Provide slotted-type concrete inserts consisting of galvanized [3 millimeter](#) thick pressed steel plate conforming to [ASTM A283/A283M](#); of box-type welded construction with slot designed to receive [M20](#) diameter square-head bolt with knockout cover; and be hot-dip-galvanized in accordance with [ASTM A123/A123M](#).

2.3.3 Fasteners

Select galvanized zinc-coated fasteners conforming to [ASTM A153/A153M](#) for exterior applications or where the fasteners are built into exterior walls or floor systems. Select the fasteners for the type, grade, and class required for the installation of steel stair items:

- a. Standard/regular hexagon-head bolts and nuts, conforming to [ASTM F568M](#),.
- b. Square-head lag bolts conforming to [ISO 898-1](#),.

2.4 MATERIALS

2.4.1 Structural Steel Plates, Shapes and Bars

Structural size shapes and plates, conforming to [ASTM A36/A36M](#), unless otherwise noted, except bent or cold-formed plates.

Steel plates - bent or cold-formed, conforming to [ASTM A283/A283M](#), Grade C.

Steel bars and bar-size shapes, conforming to [ASTM A36/A36M](#), unless otherwise noted for steel bars and bar-size shapes.

2.4.2 Structural Steel Tubing

Provide the following:

- a. Structural steel tubing, hot-formed, welded or seamless, conforming to [ASTM A500/A500M](#), Grade B, unless otherwise noted.

2.4.3 Hot-Rolled Carbon Steel Bars

Provide the following:

- a. Hot-rolled carbon steel bars and bar-size shapes, conforming to [ASTM A575](#), grade as selected by the fabricator.

2.4.4 Hot-Rolled Carbon Steel Sheets and Strips

Provide the following:

- a. Hot-rolled carbon sheets and strips conforming to [ASTM A568/A568M](#) and [ASTM A1011/A1011M](#), pickled and oiled.

2.4.5 Galvanized Carbon Steel Sheets

Provide the following:

- a. Galvanized carbon steel sheets conforming to [ASTM A653/A653M](#), with galvanizing conforming to [ASTM A653/A653M](#) and [ASTM A924/A924M](#).

2.4.6 Steel Pipe

Provide the following:

- a. Steel pipe conforming to [ASTM A53/A53M](#), type as selected, Grade B; primed finish, unless galvanizing is required; standard weight (Schedule 40).

2.5 Mill Test Reports

Provide mill test reports for structural steel plates, shapes, and bars; structural steel tubing; hot-rolled carbon steel bars; hot-rolled carbon steel sheets and strips; galvanized carbon steel sheets. Mill test report will include but is not limited to material heat number, material grade, edition year and type of specification met, material dimension, mechanical properties, chemical analysis, heat treatment (if applicable), and certified inspector signature.

PART 3 EXECUTION

3.1 PREPARATION

Clean surfaces thoroughly before installation. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions. Examine materials upon arrival at site. Notify the carrier and manufacturer of any damage.

Protect installed products until completion of project. Touch up, repair or replace, damaged products before substantial completion

3.2 INSTALLATION

Install in accordance with the manufacturer's instructions and approved submittals. Install in proper relationship with adjacent construction.

Install items at locations indicated, according to the manufacturer's instructions. Verify all measurements and take all field measurements necessary before fabrication. Ensure that exposed fastenings are compatible with generally match the color and finish of, and harmonize with the material to which they are applied. Include materials and parts necessary to complete each item, even though such work is not definitely shown or specified. Poor matching of holes for fasteners is cause for rejection. Conceal fastenings where practicable. Select thickness of metal and details of assembly and supports that adequately strengthen and stiffen the construction. Form joints exposed to the weather to exclude water.

3.2.1 Field Preparation

Remove rust-preventive coating just before field erection, using a remover approved by the coating manufacturer. Provide surfaces, when assembled, free of rust, grease, dirt and other foreign matter.

3.2.2 Field Welding

Comply with AWS D1.1/D1.1M in executing manual shielded-metal arc welding, (for appearance and quality of new welds) and in correcting existing welding.

3.2.3 Safety Nosings

Completely embed nosing in concrete before the initial set of the concrete occurs and finish flush with the top of the concrete surface.

3.2.4 Touchup Painting

Immediately after installation, clean all field welds, bolted connections, and abraded areas of the shop-painted material, and repaint exposed areas with the same paint used for shop painting. Apply paint by brush or spray to provide a minimum dry-film thickness of 0.051 millimeter.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 51 33

METAL LADDERS

02/16, CHG 2: 02/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 CERTIFICATES
- 1.4 QUALIFICATION OF WELDERS
- 1.5 DELIVERY, STORAGE, AND PROTECTION

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Structural Carbon Steel
 - 2.1.2 Structural Tubing
 - 2.1.3 Steel Pipe
 - 2.1.4 Fittings for Steel Pipe
 - 2.1.5 Aluminum Alloy Products
- 2.2 FABRICATION FINISHES
 - 2.2.1 Galvanizing
 - 2.2.2 Galvanize
 - 2.2.3 Repair of Zinc-Coated Surfaces
 - 2.2.4 Nonferrous Metal Surfaces
 - 2.2.5 Aluminum Surfaces
 - 2.2.5.1 Surface Condition
 - 2.2.5.2 Aluminum Finishes
- 2.3 LADDERS
 - 2.3.1 Phasing out of Ladder Cages and Wells (29 CFR 1910.28, Nov 2016)
 - 2.3.2 Ladder Safety Devices (Climbing Ladder Fall Arrest Systems)

PART 3 EXECUTION

- 3.1 GENERAL INSTALLATION REQUIREMENTS
- 3.2 WORKMANSHIP
- 3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS
- 3.4 WELDING
- 3.5 FINISHES
 - 3.5.1 Dissimilar Materials
- 3.6 LADDERS

-- End of Section Table of Contents --

SECTION 05 51 33

METAL LADDERS

02/16, CHG 2: 02/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 104231	(1999) Waterproofing. Bituminous And Bituminous Modified Materials. Asphalt Emulsions
UNE-EN 485-1	(2017) Aluminium and aluminium alloys - Sheet, strip and plate - Part 1: Technical conditions for inspection and delivery
UNE-EN 573-3	(2020) Aluminium and aluminium alloys - Chemical composition and form of wrought products - Part 3: Chemical composition and form of products
UNE-EN 755-1	(2016) Aluminium and aluminium alloys - Extruded rod/bar, tube and profiles - Part 1: Technical conditions for inspection and delivery
UNE-EN 10025-1	(2006) Hot Rolled Products of Structural Steels -Part 1: General Technical Delivery Conditions
UNE-EN 10083-1	(2008) Steels for quenching and tempering - Part 1: General technical delivery conditions.
UNE-EN 10216-1	(2014) Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties
UNE-EN 10217-1	(2019) Welded steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties
UNE-EN ISO 1461	(2010) Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods (ISO1461:1999)
UNE-EN ISO 3834-2	(2006) Quality requirements for fusion

welding of metallic materials - Part 2:
Comprehensive quality requirements (ISO
3834-2:2005)

UNE-EN ISO 9606-1 (2017) Qualification testing of welders -
Fusion welding - Part 1: Steels (ISO
9606-1:2012 including Cor 1:2012)

UNE-EN ISO 9606-2 (2005) Qualification test of
welders-Fusion welding-Part 2: Aluminium
and aluminium alloys (ISO 9606-2:2004)

ALUMINUM ASSOCIATION (AA)

AA DAF45 (2003; Reaffirmed 2009) Designation System
for Aluminum Finishes

AMERICAN LADDER INSTITUTE (ALI)

ALI A14.3 (2008; R 2018) Ladders - Fixed - Safety
Requirements

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP Z359.16 (2016) Safety Requirements for Climbing
Ladder Fall Arrest Systems

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding
Code - Steel

MASTER PAINTERS INSTITUTE (MPI)

MPI 79 (2016) Primer, Alkyd, Anti-Corrosive for
Metal

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.23 (Nov 2016) Ladders

29 CFR 1910.28 (Nov 2016) Duty to Have Fall Protection and
Falling Object Protection

29 CFR 1910.29 (Nov 2016) Fall Protection System and
Falling Object Protection - Criteria and
Practices

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S"
classification. Submittals not having a "G" or "S" classification are for
Contractor Quality Control approval. Submit the following in accordance
with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Ladders, Installation Drawings

SD-03 Product Data

Ladders

Ladder Safety Devices (Climbing Ladder Fall Arrest Systems)

SD-07 Certificates

Fabricator Certification for Ladder Assembly

1.3 CERTIFICATES

Provide fabricator certification for ladder assembly stating that the ladder and associated components have been fabricated according to the requirements of 29 CFR 1910.23.

1.4 QUALIFICATION OF WELDERS

Qualify welders in accordance with AWS D1.1/D1.1M or UNE-EN ISO 9606-1 for steel and or UNE-EN ISO 9606-2 for aluminum. Use procedures, materials, and equipment of the type required for the work.

1.5 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Structural Carbon Steel

UNE-EN 10083-1 or UNE-EN 10025-1.

2.1.2 Structural Tubing

UNE-EN 10216-1 or UNE-EN 10217-1.

2.1.3 Steel Pipe

UNE-EN 10216-1 or UNE-EN 10217-1.

2.1.4 Fittings for Steel Pipe

Standard malleable iron fittings.

2.1.5 Aluminum Alloy Products

Conform to UNE-EN 485-1 for sheet plate, UNE-EN 755-1 for extrusions and UNE-EN 573-3 for castings, as applicable. Provide aluminum extrusions at least 3 mm thick and aluminum plate or sheet at least 1.3 mm thick.

2.2 FABRICATION FINISHES

2.2.1 Galvanizing

Hot-dip galvanize items specified to be zinc-coated, after fabrication

where practicable. Galvanizing: **UNE-EN ISO 1461, Z275**, as applicable.

2.2.2 Galvanize

Anchor bolts, washers, and parts or devices necessary for proper installation, unless indicated otherwise.

2.2.3 Repair of Zinc-Coated Surfaces

Repair damaged surfaces with galvanizing repair method and paint conforming to **UNE-EN ISO 1461** or by application of stick or thick paste material specifically designed for repair of galvanizing, as approved by Contracting Officer. Clean areas to be repaired and remove slag from welds. Heat surfaces to which stick or paste material is applied, with a torch to a temperature sufficient to melt the metallics in stick or paste; spread molten material uniformly over surfaces to be coated and wipe off excess material.

2.2.4 Nonferrous Metal Surfaces

Protect by plating, anodic, or organic coatings.

2.2.5 Aluminum Surfaces

2.2.5.1 Surface Condition

Before finishes are applied, remove roll marks, scratches, rolled-in scratches, kinks, stains, pits, orange peel, die marks, structural streaks, and other defects which will affect uniform appearance of finished surfaces.

2.2.5.2 Aluminum Finishes

Unexposed plate and extrusions may have mill finish as fabricated. Sandblast castings' finish, medium, **AA DAF45**. Unless otherwise specified, provide all other aluminum items with standard mill finish. Provide a coating thickness not less than that specified for protective and decorative type finishes for items used in interior locations or architectural Class I type finish for items used in exterior locations in **AA DAF45**.

2.3 LADDERS

Fabricate vertical ladders conforming to **29 CFR 1910.23** and Section 5 of **ALI A14.3**. Ladders shall be capable of supporting their maximum intended load. Use **65 by 13 mm** steel flats for stringers and **25 mm** diameter steel rods for rungs. Ladder rungs, step and cleats must be spaced not less than **25 cm** and not more than **300 mm** wide (measured before installation of ladder safety system), spaced no more than **30 cm** apart, plug welded or shouldered and headed into stringers. Install ladders so that the maximum perpendicular distance from the centerline of the steps or rungs, or grab bars, or both, to the nearest permanent object in the back of the ladder or to the finished wall surface will not be less than **175 mm**. Provide heavy clip angles as indicated. Provide intermediate clip angles not over **1200 mm** on centers. The top rung of the ladder must be level with the top of the access level, parapet or landing served by the ladder except for hatches or wells. Extend the side rails of through or side step ladders **105 centimeters** above the access level. Provide ladder access protective swing gates at the top of access/egress level. The drawings must indicate

ladder locations and details of critical dimensions and materials.

2.3.1 Phasing out of Ladder Cages and Wells (29 CFR 1910.28, Nov 2016)

Conform to 29 CFR 1910.28 (Nov 2016).

Each newly installed ladder over 6,000 mm in length shall only be equipped with a personal fall arrest system or climbing ladder fall arrest system (ladder safety device), cages and wells are prohibited. When a fixed ladder, cage, or well, or any portion of a section thereof, is replaced, a personal fall arrest system or climbing ladder fall arrest system (ladder safety device) is installed in at least that section of the fixed ladder, cage, or well where the replacement is located. On and after November 18, 2036, all fixed ladders shall only be equipped with a personal fall arrest system or a ladder safety device (climbing ladder Fall Arrest System).

2.3.2 Ladder Safety Devices (Climbing Ladder Fall Arrest Systems)

Conform to 29 CFR 1910.29, Section 7 of ALI A14.3 and ASSP Z359.16. Install ladder safety devices on ladders over 6000 mm long or more. The ladder safety systems must meet the design requirement of the ladders which they serve. The ladder safety system must be capable of sustaining a minimum static load of 4.44kN. The applied loads transferred to the climbing ladder mounting locations as a result of a fall shall be specified by the manufacturer of the climbing ladder fall arrest system. Each ladder safety system must allow the worker to climb up and down using both hands and does not require the employee continuously, hold, push, or pull any part of the system while climbing. The connection between the carrier or lifeline and the point of attachment to the body harness does not exceed 23 cm. The ladder safety system consists of a rigid or flexible carrier. Mountings for the rigid carries are attached at each end of the carrier, with intermediate mountings spaced as necessary, along the entire length of the carrier. Mountings for flexible carrier are attached at each end of the carrier and cable guides for flexible carriers are installed at least 7.6 m apart but not more than 12.2 m apart along the entire length of the carrier. The design and installation of mountings and cable guides does not reduce the design strength of the ladder.

PART 3 EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated, according to manufacturer's instructions. Verify all measurements and take all field measurements necessary before fabrication. Provide Exposed fastenings of compatible materials, generally matching in color and finish, and harmonize with the material to which fastenings are applied. Include materials and parts necessary to complete each item, even though such work is not definitely shown or specified. Poor matching of holes for fasteners will be cause for rejection. Conceal fastenings where practicable. Thickness of metal and details of assembly and supports must provide strength and stiffness. Formed joints exposed to the weather to exclude water. Items listed below require additional procedures.

3.2 WORKMANSHIP

Metalwork must be well formed to shape and size, with sharp lines and angles and true curves. Drilling and punching must produce clean true

lines and surfaces. Continuously weld along the entire area of contact. Do not tack weld exposed connections of work in place. Grind smooth exposed welds. Provide smooth finish on exposed surfaces of work in place, unless otherwise approved. Where tight fits are required, mill joints. Cope or miter corner joints, well formed, and in true alignment. Install in accordance with manufacturer's installation instructions and approved drawings, cuts, and details.

3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS

Provide anchorage where necessary for fastening metal items securely in place. Include for anchorage not otherwise specified or indicated slotted inserts, expansion anchors, and powder-actuated fasteners, when approved for concrete; toggle bolts and through bolts for masonry; machine bolts, carriage bolts and powder-actuated threaded studs for steel; through bolts, lag bolts, and screws for wood. Do not use wood plugs in any material. Provide non-ferrous attachments for non-ferrous metal. Make exposed fastenings of compatible materials, generally matching in color and finish, to which fastenings are applied. Conceal fastenings where practicable.

3.4 WELDING

Perform welding, welding inspection, and corrective welding, in accordance with or [UNE-EN ISO 3834-2](#). Use continuous welds on all exposed connections. Grind visible welds smooth in the finished installation.

3.5 FINISHES

3.5.1 Dissimilar Materials

Where dissimilar metals are in contact, protect surfaces with a coat conforming to [MPI 79](#) to prevent galvanic or corrosive action. Where aluminum is in contact with concrete, plaster, mortar, masonry, wood, or absorptive materials subject to wetting, protect with [UNE 104231](#), asphalt-base emulsion.

3.6 LADDERS

Secure to the adjacent construction with the clip angles attached to the stringer. Install intermediate clip angles not over **1200 mm** on center. Install brackets as required for securing of ladders welded or bolted to structural steel or built into the concrete. Ends of ladders must not rest upon floor.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 52 00

METAL RAILINGS

08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 ADMINISTRATIVE REQUIREMENTS
 - 1.2.1 Preinstallation Meetings
- 1.3 SUBMITTALS
- 1.4 QUALITY CONTROL
 - 1.4.1 Welding Procedures
 - 1.4.2 Welder Qualification

PART 2 PRODUCTS

- 2.1 FABRICATION
 - 2.1.1 Steel Handrails
 - 2.1.2 Protective Coating
- 2.2 COMPONENTS
 - 2.2.1 Structural Steel Plates, Shapes and Bars
 - 2.2.2 Structural-Steel Tubing
 - 2.2.3 Hot-Rolled Carbon Steel Bars
 - 2.2.4 Steel Pipe
 - 2.2.5 Concrete Inserts
 - 2.2.6 Fasteners
 - 2.2.7 Steel Railings And Handrails
 - 2.2.7.1 Steel Handrails
 - 2.2.8 Swing Gate
- 2.3 Mill Test Reports

PART 3 EXECUTION

- 3.1 PREPARATION
- 3.2 INSTALLATION
 - 3.2.1 Steel Handrail
 - 3.2.2 Touchup Painting
- 3.3 FIELD QUALITY CONTROL
 - 3.3.1 Field Welding

-- End of Section Table of Contents --

SECTION 05 52 00

METAL RAILINGS
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE-EN 10025-1	(2006) Hot Rolled Products of Structural Steels -Part 1: General Technical Delivery Conditions
UNE-EN 10210-1	(2007) Hot finished structural hollow sections of non-alloy and fine grain steels - Part 1: Technical delivery conditions
UNE-EN 10216-1	(2014) Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties
UNE-EN 10217-1	(2019) Welded steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties
UNE-EN 10346	(2015) Continuously hot-dip coated steel flat products for cold forming - Technical delivery conditions
UNE-EN ISO 898-1	(2015) Mechanical properties of fasteners made of carbon steel and alloy steel - Part 1: Bolts, screws and studs with specified property classes - Coarse thread and fine pitch thread (ISO 898-1:2013)
UNE-EN ISO 1461	(2010) Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods (ISO1461:1999)
UNE-EN ISO 3834-2	(2006) Quality requirements for fusion welding of metallic materials - Part 2: Comprehensive quality requirements (ISO 3834-2:2005)
UNE-EN ISO 3834-3	(2006) Quality requirements for fusion welding of metallic materials - Part 3: Standard quality requirements (ISO

3834-3:2005)

UNE-EN ISO 9606-1 (2017) Qualification testing of welders -
Fusion welding - Part 1: Steels (ISO
9606-1:2012 including Cor 1:2012)

UNE-EN ISO 10673 (2010) Plain washers for screw and washer
assemblies. Small, normal and large
series. Product grade A (ISO 10673:2009)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding
Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A153/A153M (2023) Standard Specification for Zinc
Coating (Hot-Dip) on Iron and Steel
Hardware

ASTM A283/A283M (2013) Standard Specification for Low and
Intermediate Tensile Strength Carbon Steel
Plates

1.2 ADMINISTRATIVE REQUIREMENTS

1.2.1 Preinstallation Meetings

Within 30 days of contract award, submit [fabrication drawings](#) for the
following items:

- a. Iron and steel hardware
- b. Steel shapes, plates, bars and strips
- c. Steel railings and handrails
- d. Anchorage and fastening systems

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S"
classification. Submittals not having a "G" or "S" classification are for
Contractor Quality Control approval. Submit the following in accordance
with Section [01 33 00](#) SUBMITTAL PROCEDURES:

[SD-02 Shop Drawings](#)

[Fabrication Drawings; G](#)

[Iron and Steel Hardware; G](#)

[Steel Shapes, Plates, Bars and Strips; G](#)

[SD-03 Product Data](#)

[Concrete Inserts; G](#)

Protective Coating; G

Steel Railings and Handrails; G

Anchorage and Fastening Systems; G

SD-06 Test Reports

Welding Inspections; G

Mill Test Reports; G

SD-07 Certificates

Welder Qualification; G

SD-08 Manufacturer's Instructions

Installation Instructions

1.4 QUALITY CONTROL

1.4.1 Welding Procedures

Submit results of [welding inspections](#) testing in accordance with [AWS D1.1/D1.1M](#) or [UNE-EN ISO 3834-2](#) by an approved testing laboratory at the Contractor's expense. Provide laboratory welding inspection results to Contracting Officer.

1.4.2 Welder Qualification

Submit certified [welder qualification](#) by tests in accordance with [AWS D1.1/D1.1M](#) or [UNE-EN ISO 9606-1](#), or under an equivalent approved qualification test. In addition, perform tests on test pieces in positions and with clearances equivalent to those actually encountered. If a test weld fails to meet requirements, conduct an immediate retest of two test welds and ensure that each test weld passes. Failure in the immediate retest will require that the welder be retested after further practice or training and make a complete set of test welds.

PART 2 PRODUCTS

2.1 FABRICATION

Preassemble items in the shop to the greatest extent possible. Disassemble units only to the extent necessary for shipping and handling. Clearly mark units for reassembly and coordinated installation.

For the fabrication of work exposed to view, use only materials that are smooth and free of surface blemishes, including pitting, seam marks, roller marks, rolled trade names, and roughness. Remove blemishes by grinding, or by welding and grinding, before cleaning, treating, and applying surface finishes, including zinc coatings.

Provide railing and handrail detail plans and elevations at not less than **1 to 12 scale**. Provide details of sections and connections at not less than **1 to 4 scale**. Also detail setting drawings, diagrams, templates for installation of anchorages, including concrete inserts, anchor bolts, and miscellaneous metal items having integral anchors.

Use materials of size and thicknesses indicated or, if not indicated, of the size and thickness necessary to produce adequate strength and durability in the finished product for its intended use. Work the materials to the dimensions indicated on approved detail drawings, using proven details of fabrication and support. Use the type of materials indicated or specified for the various components of work.

Form exposed work true to line and level, with accurate angles and surfaces and straight sharp edges. Ensure that all exposed edges are eased to a radius of approximately **0.8 millimeter**. Bend metal corners to the smallest radius possible without causing grain separation or otherwise impairing the work.

Weld corners and seams continuously and in accordance with the recommendations of or **UNE-EN ISO 3834-2**. Grind exposed welds smooth and flush to match and blend with adjoining surfaces.

Form the exposed connections with hairline joints that are flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of the type indicated or, if not indicated, use countersunk Phillips flathead screws or bolts.

Provide anchorage of the type indicated and coordinated with the supporting structure. Fabricate anchoring devices and space as indicated and as required to provide adequate support for the intended use of the work.

Use hot-rolled steel bars for work fabricated from bar stock unless work is indicated or specified to be fabricated from cold-finished or cold-rolled stock.

2.1.1 Steel Handrails

Fabricate joint posts, rail, and corners by one of the following methods:

- a. Flush-type rail fittings of commercial standard, welded and ground smooth, with railing splice locks secured with **10 mm** hexagonal-recessed-head setscrews.
- b. Mitered and welded joints made by fitting post to top rail and intermediate rail to post, mitering corners, groove-welding joints, and grinding smooth. Butt railing splices and reinforce them by a tight-fitting interior sleeve not less than **150 mm** long.
- c. Railings may be bent at corners in lieu of jointing, provided that bends are made in suitable jigs and the pipe is not crushed.

2.1.2 Protective Coating

Provide hot-dipped galvanized steelwork as indicated in accordance with or **UNE-EN ISO 1461**, Z275. Touch up abraded surfaces and cut ends of galvanized members with zinc-dust, zinc-oxide primer, or an approved galvanizing repair compound.

2.2 COMPONENTS

2.2.1 Structural Steel Plates, Shapes and Bars

Provide structural-size shapes and plates, except plates to be bent or cold-formed, conforming to [UNE-EN 10025-1](#), unless otherwise noted.

Provide steel plates, to be bent or cold-formed, conforming to [ASTM A283/A283M](#), Grade C, or [UNE-EN 10346](#).

Provide steel bars and bar-size shapes conforming to [UNE-EN 10025-1](#), unless otherwise noted.

2.2.2 Structural-Steel Tubing

Provide structural-steel tubing, hot-formed, welded or seamless, conforming to [UNE-EN 10216-1](#) or [UNE-EN 10217-1](#), unless otherwise noted.

2.2.3 Hot-Rolled Carbon Steel Bars

Provide bars and bar-size shapes conforming to [UNE-EN 10025-1](#), grade as selected by the fabricator.

2.2.4 Steel Pipe

Provide pipe conforming to [UNE-EN 10216-1](#) or [UNE-EN 10217-1](#) type as selected; primed finish, unless galvanizing is required; standard weight (Schedule 40).

2.2.5 Concrete Inserts

Provide carbon steel bolts having special wedge-shaped heads, nuts, washers, and shims, galvanized in accordance with [UNE-EN ISO 1461](#). Provide slotted-type concrete inserts consisting of a galvanized [3 millimeter](#) thick pressed-steel plate, made of box-type welded construction with a slot designed to receive [M20](#) diameter square-head bolt with knockout cover; and hot-dip galvanized in accordance with [UNE-EN ISO 1461](#).

2.2.6 Fasteners

Provide galvanized zinc-coated fasteners in accordance with [ASTM A153/A153M](#) used for exterior applications or where built into exterior walls or floor systems. Select fasteners for the type, grade, and class required for the installation of steel stair items.

Provide standard hexagon-head bolts, conforming to [UNE-EN ISO 898-1](#).

Provide plain round, general-assembly-grade, carbon steel washers conforming to [UNE-EN ISO 10673](#).

2.2.7 Steel Railings And Handrails

2.2.7.1 Steel Handrails

Provide steel handrails, including inserts in concrete, steel pipe conforming to [UNE-EN 10216-1](#) or [UNE-EN 10217-1](#), or structural tubing conforming to [UNE-EN 10210-1](#), Grade A or B of equivalent strength. Provide steel railings of [40 mm](#) nominal size, hot-dip galvanized.

Provide kickplates between railing posts where indicated, and consisting of 4 millimeter steel flat bars not less than 150 millimeter high. Secure kickplates as indicated.

Galvanize exterior railings, including pipe, fittings, brackets, fasteners, and other ferrous metal components. Provide black steel pipe for interior railings.

Provide galvanized railings, including pipe, fittings, brackets, fasteners, and other ferrous metal components.

2.2.8 Swing Gate

Provide swing gate in accordance with Section 05 51 33 METAL LADDERS.

2.3 Mill Test Reports

Provide mill test reports for structural steel plates, shapes, and bars; structural steel tubing; hot-rolled carbon steel bars. Mill test report will include but is not limited to material heat number, material grade, edition year and type of specification met, material dimension, mechanical properties, chemical analysis, heat treatment (if applicable), and certified inspector signature.

PART 3 EXECUTION

3.1 PREPARATION

Adjust stair railings and handrails before securing in place in order to ensure proper matching at butting joints and correct alignment throughout their length. Space posts not more than distance indicated. Plumb posts in each direction. Secure posts and rail ends to building construction as indicated.

3.2 INSTALLATION

Submit manufacturer's installation instructions for the following products to be used in the fabrication of steel, stair railing, and hand rail work:

- a. Structural-steel plates, shapes, and bars
- b. Structural-steel tubing
- c. Hot-rolled carbon steel bars
- d. Protective coating
- e. Steel railings and handrails
- f. Anchorage and fastening systems

Provide complete, detailed fabrication and installation drawings for all iron and steel hardware, and for all steel shapes, plates, bars, and strips used in accordance with the design specifications cited in this section.

3.2.1 Steel Handrail

Install handrail as indicated.

3.2.2 Touchup Painting

Immediately after installation, clean field welds, bolted connections, abraded areas of the shop paint, and exposed areas painted with the paint used for shop painting. Apply paint by brush or spray to provide a minimum dry-film thickness of 0.051 millimeter.

3.3 FIELD QUALITY CONTROL

3.3.1 Field Welding

Ensure that procedures of manual shielded metal arc welding, appearance and quality of welds made, and methods used in correcting welding work comply with UNE-EN ISO 3834-3.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 59 10

ROLLING COVER FOR AVIATION REFUELING VAULTS

11/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 ADMINISTRATIVE REQUIREMENTS
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Welder Qualifications
 - 1.4.2 Workmanship
 - 1.4.3 Detail Drawings

PART 2 PRODUCTS

- 2.1 MATERIALS AND EQUIPMENT
 - 2.1.1 Bolts and Cap Screws
 - 2.1.2 Nuts
 - 2.1.3 Washers
 - 2.1.4 Tube Steel
 - 2.1.5 Structural Steel
 - 2.1.6 Rolling Cover Shell
 - 2.1.6.1 Aluminum
 - 2.1.7 Wheel Assemblies
 - 2.1.7.1 Wheels
 - 2.1.7.2 Axles
- 2.2 FABRICATION
 - 2.2.1 General
 - 2.2.2 Dimensional Tolerances
 - 2.2.3 Steel
 - 2.2.4 Aluminum
 - 2.2.5 Bolted Connections
 - 2.2.5.1 Bolted Steel Connections
 - 2.2.5.2 Bolted Aluminum Connections
- 2.3 MACHINE WORK
 - 2.3.1 Finished Surfaces
 - 2.3.2 Unfinished Surfaces
- 2.4 WELDING
 - 2.4.1 Welding of Structural Steel
 - 2.4.2 Welding of Aluminum
 - 2.4.3 Welding Inspection
 - 2.4.3.1 Visual Examination
 - 2.4.4 Steel Welding Repairs
- 2.5 MISCELLANEOUS PROVISIONS
 - 2.5.1 Galvanizing
 - 2.5.2 Cleaning of Stainless Steel
- 2.6 SHOP TESTING
 - 2.6.1 Wheel Assembly Testing
 - 2.6.2 Assembly Tests

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

2.7 PREPARATION FOR SHIPPING

PART 3 EXECUTION

3.1 ASSEMBLY

3.2 PROTECTION OF FINISHED WORK

3.2.1 Lubrication After Assembly

3.2.2 Aluminum

3.3 ACCEPTANCE TESTING

-- End of Section Table of Contents --

SECTION 05 59 10

ROLLING COVER FOR AVIATION REFUELING VAULTS
11/22

PART 1 GENERAL

This specification covers the factory fabrication, assembly, testing, and shipping requirements for custom fabricated rolling covers having steel or aluminum shells as indicated in the vault schedule on the vault drawings. Covers are to be field installed on variously sized new and/or existing hydrant fueling system vaults and tanks.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE-EN 485-1	(2017) Aluminium and aluminium alloys - Sheet, strip and plate - Part 1: Technical conditions for inspection and delivery
UNE-EN 10025-1	(2006) Hot Rolled Products of Structural Steels -Part 1: General Technical Delivery Conditions
UNE-EN 10088-2	(2015) Stainless steels - Part 2: Technical delivery conditions for sheet/plate and strip of corrosion resisting steels for general purposes
UNE-EN 10210-1	(2007) Hot finished structural hollow sections of non-alloy and fine grain steels - Part 1: Technical delivery conditions
UNE-EN ISO 898-1	(2015) Mechanical properties of fasteners made of carbon steel and alloy steel - Part 1: Bolts, screws and studs with specified property classes - Coarse thread and fine pitch thread (ISO 898-1:2013)
UNE-EN ISO 898-2	(2013) Mechanical properties of fasteners made of carbon steel and alloy steel - Part 2: Nuts with specified property classes - Coarse thread and fine pitch thread (ISO 898-2:2012)
UNE-EN ISO 1461	(2010) Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods (ISO1461:1999)
UNE-EN ISO 3834-3	(2006) Quality requirements for fusion

welding of metallic materials - Part 3:
Standard quality requirements (ISO
3834-3:2005)

UNE-EN ISO 9606-1

(2017) Qualification testing of welders -
Fusion welding - Part 1: Steels (ISO
9606-1:2012 including Cor 1:2012)

UNE-EN ISO 9606-2

(2005) Qualification test of
welders-Fusion welding-Part 2: Aluminium
and aluminium alloys (ISO 9606-2:2004)

ALUMINUM ASSOCIATION (AA)

AA ADM

(2020) Aluminum Design Manual

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B4.1

(1967; R 1994; R 2004; R 2009; R 2020)
Preferred Limits and Fits for Cylindrical
Parts

ASME B46.1

(2020) Surface Texture, Surface Roughness,
Waviness and Lay

ASME BPVC SEC IX

(2017; Errata 2018) BPVC Section
IX-Welding, Brazing and Fusing
Qualifications

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M

(2020; Errata 1 2021) Structural Welding
Code - Steel

AWS D1.2/D1.2M

(2014; Errata 1 2014; Errata 2 2020)
Structural Welding Code - Aluminum

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M

(2019) Standard Specification for Carbon
Structural Steel

ASTM A123/A123M

(2017) Standard Specification for Zinc
(Hot-Dip Galvanized) Coatings on Iron and
Steel Products

ASTM A307

(2021) Standard Specification for Carbon
Steel Bolts, Studs, and Threaded Rod 60
000 PSI Tensile Strength

ASTM A563

(2021; E 2022a) Standard Specification for
Carbon and Alloy Steel Nuts

ASTM A1085/A1085M

(2015) Standard Specification for
Cold-Formed Welded Carbon Steel Hollow
Structural Sections (HSS)

ASTM B695

(2021) Standard Specification for Coatings
of Zinc Mechanically Deposited on Iron and

Steel

ASTM F1136/F1136M

(2011) Standard Specification for
Zinc/Aluminum Corrosion Protective
Coatings for Fasteners

ASTM F2329/F2329M

(2015) Standard Specification for Zinc
Coating, Hot-Dip, Requirements for
Application to Carbon and Alloy Steel
Bolts, Screws, Washers, Nuts, and Special
Threaded Fasteners

ASTM F2833

(2011; R 2017) Standard Specification for
Corrosion Protective Fastener Coatings
with Zinc Rich Base Coat and Aluminum
Organic/Inorganic Type

1.2 ADMINISTRATIVE REQUIREMENTS

Submit manufacturer's catalogue cuts and dimensional sheets. Include a description of the item, materials of construction, and dimensions. Provide data sufficient to indicate compliance with specifications. Mark items pertaining to specifications with a heavy black arrow.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Assembly Tests; G

Acceptance Testing; G

SD-02 Shop Drawings

Detail Drawings; G

SD-03 Product Data

Wheel Assemblies; G

Materials List; G

Welding; G

Welding of Aluminum; G

Steel Welding Repairs; G

SD-07 Certificates

Welder Qualifications

Welding of Aluminum

1.4 QUALITY ASSURANCE

1.4.1 Welder Qualifications

Submit certification stating that the welders, welding operators, and tack welders, who will perform structural steel welding, have been qualified for the particular type of work to be done in accordance with the requirements of AWS D1.1/D1.1M, Section 4, or UNE-EN ISO 9606-1 or UNE-EN ISO 9606-2, prior to commencing fabrication. The certificate must list the qualified welders by name, specify the code and procedures for which they are qualified, and provide the date of qualification. Prior qualification will be accepted if welders have performed satisfactory work under the code for which they have been qualified within the preceding three months. Welders will be required to repeat the qualifying tests when their work indicates a reasonable doubt as to proficiency. Those welders passing the requalification tests will be recertified. Those welders not passing will be disqualified until they have passed the requalification tests. All expenses in connection with qualification and requalification must be borne by the Contractor.

1.4.2 Workmanship

Workmanship must be of the highest grade and in accordance with the best modern practices to conform with the specifications for the item of work being furnished. Welding must be continuous along the entire area of contact, except where tack welding is permitted. Exposed connections of work in place must not be tack welded. Exposed welds must be ground smooth. Exposed surfaces of work in place must have a smooth finish.

1.4.3 Detail Drawings

Submit detail drawings for metalwork and machine work prior to fabrication. Submit a materials list for fabricated items with the detail drawings. Detail drawings for metalwork and machine work must include catalog cuts, templates, fabrication and assembly details, and type, grade, and class of material, as appropriate. Also include a sketch showing final wheel to axle mounting (i.e., washers, nuts, and spacers). Elements of fabricated items inadvertently omitted on contract drawings must be detailed by the fabricator and indicated on the detail drawings. Drawings must include all dimensional and tolerance data for each size of vault being fabricated.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

2.1.1 Bolts and Cap Screws

All bolts must be steel and must conform to ASTM A307, Grade A, Hex, or equivalent per UNE-EN ISO 898-1.

2.1.2 Nuts

Must conform to ASTM A563, Grade A, Hex, or equivalent per UNE-EN ISO 898-2 and must be of the same finish as the fasteners they are used with.

2.1.3 Washers

Flat washers must conform to the requirements of UNE-EN 10088-2 (stainless

steel).

2.1.4 Tube Steel

Structural tubing must conform to [ASTM A1085/A1085M](#) or [UNE-EN 10210-1](#).

2.1.5 Structural Steel

Carbon steel must conform to [ASTM A36/A36M](#) or [UNE-EN 10025-1](#).

2.1.6 Rolling Cover Shell

Cover shell material must be as indicated on the drawings and specified as follows:

2.1.6.1 Aluminum

Aluminum sheets and strips must comply with or [UNE-EN 485-1](#), alloy and temper best suited for the purpose.

2.1.7 Wheel Assemblies

2.1.7.1 Wheels

Provide heavy duty industrial type wheels that are the product of a company regularly engaged in the production of wheels. The wheels must have solid rubber tires that are molded onto spoked or solid centers that are either cast, forged, or machined. The rubber must have a hardness rating of 80-95 Shore A durometer. Tires must not stretch or work loose from the metal center. The wheels must have roller bearings and must be pressure lubricated from a grease fitting when available. The wheels must work in a temperature range of [-40 to +82 degrees C](#). The entire wheel assembly must be symmetrical and must spin concentrically around the bearing. Finish metal centers with either an epoxy paint, a powder coating, or manufacture galvanized. The diameter of the carrier wheels must be [150 mm](#) and the diameter of the side wheels must be [80 mm](#). The fabricator must use appropriate washers and spacers to lock the inner bearing bushing to the axle. Wheels must be similar or equal to the following:

- a. [150 mm Dia. x 50 mm wide](#)
- b. [80 mm Dia. x 40 mm wide](#)

2.1.7.2 Axles

The axle assembly must be stainless steel and must be eccentrically machined. A slotted adjustment cam plate must be attached to the axle by welding, as indicated on the drawings. Eccentric offset must be a minimum of [6 mm](#). Diameter, tolerance, and finish of the mating axle shaft must be coordinated with the wheel manufacturer's diameters and tolerances for a close fit. Submit shop drawings showing all fits and tolerances. Material, weld, and nut must all be a 300 series stainless steel.

2.2 FABRICATION

2.2.1 General

Material must be straight before being laid off or worked. If

straightening is necessary, it must be done by methods that will not impair the metal. Sharp kinks or bends must be cause for rejection of the material. Material with welds will not be accepted, except where welding is definitely specified, indicated, or otherwise approved. Bends must be made by approved dies, press brakes, or bending rolls. Where heating is required, precautions must be taken to avoid overheating or warping the metal, and it must be allowed to cool in a manner that will not impair the original properties of the metal. Proposed flame cutting of material other than structural steel must be subject to approval and must be indicated on detail drawings. Shearing must be accurate, and all portions of the work must be neatly finished. Corners must be square and true unless otherwise shown on the drawings. Re-entrant cuts must be filleted to a minimum radius of 20 mm unless otherwise approved. Finished members must be free of twists, bends, and open joints. Bolts, nuts, and screws must be tight.

2.2.2 Dimensional Tolerances

Dimensions must be measured by a calibrated steel tape of approximately the same temperature as the material being measured. The overall dimensions of an assembled structural unit must be within the tolerances indicated on the drawings or as specified in the particular section of these specifications for the item of work. Where tolerances are not specified in other sections of these specifications or shown on the drawings, an allowable variation of 1 mm is permissible in the overall length of component members with both ends milled. Component members without milled ends must not deviate from the dimensions shown by not more than 1.5 mm for members 30 feet or less in length and by more than 3 mm for members over 9 m in length.

2.2.3 Steel

Structural steel may be cut, when approved, by mechanically guided or hand-guided torches as long as an accurate profile with a surface that is smooth and free from cracks and notches is obtained. Surfaces and edges to be welded must be prepared in accordance with AWS D1.1/D1.1M. Where structural steel is not to be welded, chipping or grinding will not be required, except as necessary to remove slag and sharp edges of mechanically guided or hand-guided cuts not exposed to view. Hand-guided cuts, which are to be exposed or visible, must be chipped, ground, or machined to sound metal.

2.2.4 Aluminum

Laying out and cutting of aluminum must be in accordance with AA ADM.

2.2.5 Bolted Connections

2.2.5.1 Bolted Steel Connections

Bolts, nuts, and washers must be of the type specified or indicated. Beveled washers must be used where bearing faces have a slope of more than 1:20 with respect to a plane normal to the bolt axis. Bolt holes must be accurately located, smooth, perpendicular to the member, and cylindrical. Holes for bolts must be drilled or subdrilled and reamed in the shop and must not be more than 1.5 mm larger than the diameter of the bolt, unless otherwise approved, indicated on the drawings, or specified below. Poor matching of holes will be cause for rejection. Drifting occurring during assembly must not distort the metal or enlarge the holes. Reaming to a

larger diameter of the next standard size bolt will be allowed for slight mismatching.

2.2.5.2 Bolted Aluminum Connections

Punching, drilling, reaming, and bolting for bolted aluminum connections must conform to the requirements of AA ADM.

2.3 MACHINE WORK

Tolerances, allowances, and gauges for metal fits between plain, non-threaded, cylindrical parts must conform to ASME B4.1 for the class of fit shown or required, unless otherwise shown on approved detail drawings. Where fits are not shown, they must be suitable as approved. Tolerances for machine-finished surfaces designated by non-decimal dimensions must be within 0.4 mm, unless otherwise indicated on the drawings. Sufficient machining stock must be allowed to ensure true surfaces of solid material. Assembled parts must be accurately machined, and all like parts must be interchangeable. All drilled holes must be accurately located.

2.3.1 Finished Surfaces

Surface finishes indicated or specified herein must be in accordance with ASME B46.1. Values of required roughness heights are arithmetical average deviations expressed in microinches. These values are maximum. Lesser degrees will be satisfactory unless otherwise indicated. Compliance with surface requirements must be determined by sense of feel and visual inspection of the work compared to Roughness Comparison Specimens in accordance with the provisions of ASME B46.1. Values of roughness width and waviness height must be consistent with the general type of finish specified by roughness height. Where the finish is not indicated or specified, the finish selected must be the most suitable for that particular surface. Provide the class of fit required and indicate it on the detail drawings with a symbol, which conforms to ASME B46.1 when machine finishing is provided. Flaws such as scratches, ridges, holes, peaks, cracks, or checks, which will make the part unsuitable for the intended use, will be cause for rejection.

2.3.2 Unfinished Surfaces

All work must be laid out to secure proper matching of adjoining unfinished surfaces, unless otherwise directed. Where there is a large discrepancy between adjoining unfinished surfaces, it must be chipped and ground smooth or machined to secure proper alignment. Unfinished surfaces must be true to the lines and dimensions shown and must be chipped or ground free of all projections and rough spots. Depressions or holes not affecting the strength or usefulness of the parts must be filled in an approved manner.

2.4 WELDING

Submit the Welding Procedure Specification (WPS).

2.4.1 Welding of Structural Steel

Welding must be in accordance with AWS D1.1/D1.1M or UNE-EN ISO 3834-3. Welding procedures which are considered prequalified as specified in AWS D1.1/D1.1M will be accepted without further qualification. Submit for

approval a listing or an annotated drawing to indicate the joints not prequalified. Procedure qualification must be required for these joints.

2.4.2 Welding of Aluminum

Welding of aluminum must conform to AA ADM or AWS D1.2/D1.2M, Sections 1 through 5. The welding process and welding operators must be prequalified as required by AWS D1.2/D1.2M, Section 3 or AA ADM or UNE-EN ISO 9606-2, in accordance with the methods described in ASME BPVC SEC IX. Submit a certified report giving the results of the aluminum welding qualification tests. Also, submit a complete schedule of the welding process for each aluminum fabrication to be welded prior to commencing fabrication.

2.4.3 Welding Inspection

Maintain an approved inspection system and perform required inspections in accordance with Section 01 45 35 SPECIAL INSPECTIONS. Welding must be subject to inspection to determine conformance with the requirements of AWS D1.1/D1.1M, the approved welding procedures, and provisions stated in other sections of these specifications.

2.4.3.1 Visual Examination

All completed welds must be cleaned and carefully examined for insufficient throat or leg sizes, cracks, undercutting, overlap, excessive convexity or reinforcement, and other surface defects to ensure compliance with the requirements of AWS D1.1/D1.1M, Section 5 and Section 10, Part D.

2.4.4 Steel Welding Repairs

Defective welds must be repaired in accordance with AWS D1.1/D1.1M, Section 7. Defective weld metal must be removed to sound metal by use of air carbon arc or oxygen gouging. The surfaces must be thoroughly cleaned before welding. Welds that have been repaired must be retested by the same methods used in the original inspection. Costs for repairs and retesting must be borne by the Contractor. Submit repair procedure prior to doing repair.

2.5 MISCELLANEOUS PROVISIONS

2.5.1 Galvanizing

ASTM F2329/F2329M, ASTM F1136/F1136M, ASTM F2833 or ASTM B695 for threaded parts.

ASTM A123/A123M or UNE-EN ISO 1461 for structural steel members, as applicable. Unless specified otherwise, provide hot-dipped galvanization after fabrication where practical. Hot-dipped galvanizing must be applied in a manner and of a thickness and quality conforming to ASTM A123/A123M or UNE-EN ISO 1461.

Where zinc coatings are destroyed by cutting, welding, or other causes, the affected areas must be regalvanized. Coatings 2 ounces or heavier must be regalvanized with a suitable low-melting zinc base alloy similar to the recommendations of the American Hot-Dip Galvanizers Association to the thickness and quality specified for the original zinc coating.

2.5.2 Cleaning of Stainless Steel

Oil, paint, and other foreign substances must be removed from stainless steel surfaces after fabrication. Cleaning must be done by vapor degreasing or by the use of cleaners of the alkaline, emulsion, or solvent type.

2.6 SHOP TESTING

2.6.1 Wheel Assembly Testing

The first wheel assembly must be tested for correct fit and operation in the presence of the Contracting Officer's Technical Representative (COTR), unless otherwise waived in writing. The wheel must rotate concentrically and smoothly on the bearings. The cam adjuster must provide at least 3 mm of adjustment in each vertical direction. Waiving of tests will not relieve the Contractor of responsibility for any fault in operation, workmanship, or material that occurs before the completion of the contract or guarantee.

2.6.2 Assembly Tests

Each rolling cover, including the shell, carrier, frame, and temporary brackets, must be assembled in the shop to determine the correctness of the fabrication and matching of the component parts. Tolerances must not exceed those shown on the drawings. Each cover assembly must be closely checked to ensure that all necessary clearances have been provided and that binding does not occur in any moving part. Assembly in the shop must be done on a straight and level floor or platform; the frame must be mounted on temporary supports in a level position. The carrier must move smoothly and with minimal effort. Misalignment, poor operation, or disclosed defects must be immediately remedied by the Contractor without cost to the Government. Assembly, testing, and disassembly work must be performed in the presence of the Contracting Officer, unless waived in writing. Provide ten working days notice, in writing, of the first and each proceeding rolling cover assembly to the COTR.

2.7 PREPARATION FOR SHIPPING

Before disassembly for shipment, each rolling cover subassembly must be match-mark stamped (or as otherwise approved) to facilitate correct reassembly in the field. The location of stampings must be indicated by circling with a ring of white chalk after the shop finish has been applied or as otherwise directed. Each subassembly must be wood crated, slatted, skid mounted, or otherwise packaged such that abrasion does not occur during shipment.

PART 3 EXECUTION

3.1 ASSEMBLY

All parts to be assembled must be thoroughly cleaned. Packing compounds, rust, dirt, grit, and other foreign matter must be removed. Holes and grooves for lubrication must be cleaned. Enclosed chambers or passages must be examined to make sure that they are free from damaging materials. Where units or items are shipped as assemblies, they will be inspected prior to installation. Pipe wrenches, cold chisels, or other tools likely to cause damage to the surfaces of rods, nuts, or other parts must not be used for assembling and tightening parts. Bolts and screws must be

tightened firmly and uniformly but care must be taken not to overstress the threads. When a half nut is used for locking a full nut, the half nut must be placed first and followed by the full nut. Threads of all bolts, nuts, and screws must be lubricated with a lubricant before assembly. Threads of corrosion-resisting steel bolts and nuts must be coated with an approved antigalling compound. Driving and drifting bolts or keys will not be permitted.

3.2 PROTECTION OF FINISHED WORK

3.2.1 Lubrication After Assembly

After assembly, all wheels must be pressure lubricated or oiled.

3.2.2 Aluminum

Aluminum in contact with structural steel in the area of the cover shell fastener angle clips must be protected against galvanic or corrosive action by being given a coat of zinc-chromate primer and a coat of aluminum paint.

3.3 ACCEPTANCE TESTING

The rolling cover must be field tested to ensure proper wheel adjustments to eliminate binding and track misalignment. In addition, demonstrate to the Contracting Officer that the cover and cover tracks are level. The rolling cover must be rolled the full distance of the tracks. The test must be repeated a sufficient number of times (minimum of three) to demonstrate proper operation. Misalignment, poor operation, or disclosed defects must be immediately remedied without cost to the Government. Provide all personnel necessary to conduct the tests. Testing must be performed in the presence of the Contracting Officer. Notify the Contracting Officer, in writing, at least 7 days prior to testing operations.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 11 13

BITUMINOUS DAMPPROOFING

08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY AND STORAGE

PART 2 PRODUCTS

- 2.1 EMULSION-BASED ASPHALT DAMPPROOFING
 - 2.1.1 Fibrated Emulsion-Based Asphalt

PART 3 EXECUTION

- 3.1 SURFACE PREPARATION
- 3.2 APPLICATION
 - 3.2.1 Surface Priming
 - 3.2.2 Cold-Application Method
 - 3.2.2.1 Emulsion-Based Asphalt

-- End of Section Table of Contents --

SECTION 07 11 13

BITUMINOUS DAMPPROOFING
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 104231 (1999) Waterproofing. Bituminous And Bituminous Modified Materials. Asphalt Emulsions

ASTM INTERNATIONAL (ASTM)

ASTM D1227/D1227M (2013; R 2019; e1) Standard Specification for Emulsified Asphalt Used as a Protective Coating for Roofing

ASTM D4263 (1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-07 Certificates

Materials

1.3 DELIVERY AND STORAGE

Deliver materials in sealed containers bearing manufacturer's original labels. Labels must include date of manufacture, contents of each container, performance standards that apply to the contents and recommended shelf life. While in storage, do not allow water based bituminous dampproofing to freeze.

PART 2 PRODUCTS

2.1 EMULSION-BASED ASPHALT DAMPPROOFING

2.1.1 Fibrated Emulsion-Based Asphalt

Provide cold-applied fibrated emulsion-based asphalt dampproofing conforming to ASTM D1227/D1227M Type II, Class 1 or UNE 104231, asbestos-free, manufactured of refined asphalt, emulsifiers and selected

clay, fibrated with mineral fibers. For spray or brush application, emulsion must contain a minimum of 59 percent solids by weight, 56 percent solids by volume. For trowel application, emulsion must contain a minimum of 58 percent solids by weight, 55 percent solids by volume.

PART 3 EXECUTION

3.1 SURFACE PREPARATION

Remove or cut form ties and repair surface defects as required in Section 03 30 00 CAST-IN-PLACE CONCRETE. Clean concrete and masonry surfaces to receive dampproofing of foreign matter and loose particles. Apply dampproofing to clean dry surfaces. Moisture test in accordance with ASTM D4263. If test indicates moisture, allow a minimum of seven additional days after test completion for curing. If moisture still exists, redo test until substrate is dry.

3.2 APPLICATION

Prime surfaces to receive fibrous asphaltic dampproofing unless recommended otherwise by dampproofing materials manufacturer. Apply dampproofing after priming coat is dry, but prior to any deterioration of primed surface, and when ambient temperature is above 4 degrees C.

3.2.1 Surface Priming

Prime surfaces to receive fibrous asphalt dampproofing with asphalt primer. Apply primer when ambient temperature is above 4 degrees C and at rate of approximately 4 liters per 10 square meters, fully covering entire surface to be dampproofed.

3.2.2 Cold-Application Method

3.2.2.1 Emulsion-Based Asphalt

Do not perform emulsion-based asphalt dampproofing work in temperatures below 4 degrees C. Emulsions must have a smooth and uniform consistency at time of application. Apply dampproofing materials in accordance with manufacturer's published instructions to produce a smooth uniform dry film of not less than 0.3 mm thick without voids or defects. Recoat any dull or porous spots. Seal dampproofing materials tightly around pipes and other items projecting through dampproofing. Apply materials at rates as follows:

- a. Primer: 0.2 liters per square meter, cold-applied.
- b. Fibrated Dampproofing: 0.8 liters per square meter, cold-applied with spray, brush or trowel.
- c. Non-fibrated Dampproofing: 0.8 liters per square meter, cold-applied with spray, brush or trowel.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 12 00

BUILT-UP BITUMINOUS WATERPROOFING

02/16

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 MANUFACTURER'S DETAIL
- 1.4 ENVIRONMENTAL CONDITIONS
- 1.5 DELIVERY AND STORAGE
 - 1.5.1 Materials Packaging
 - 1.5.2 Materials Storage
 - 1.5.2.1 Asphalt
 - 1.5.2.2 Reinforcement Fabrics
- 1.6 Flame Heated Equipment
 - 1.6.1 Fire Protection
 - 1.6.2 Operational Requirements
 - 1.6.3 Dripping of Bitumen

PART 2 PRODUCTS

- 2.1 BITUMEN
- 2.2 BITUMINOUS PLASTIC CEMENT
- 2.3 MEMBRANE FABRIC
 - 2.3.1 Cotton Fabrics
 - 2.3.2 Woven Burlap Fabrics
- 2.4 NAILS
- 2.5 PRIMER
- 2.6 PROTECTION BOARD

PART 3 EXECUTION

- 3.1 INSPECTION OF SURFACES
- 3.2 PREPARATION OF SURFACES
- 3.3 APPLICATION
 - 3.3.1 Building Envelope Requirements
 - 3.3.2 General Installation Requirements
 - 3.3.3 Protection of Surrounding Areas
 - 3.3.4 Heating and Application of Bitumen Coatings
 - 3.3.5 Membrane Waterproofing
 - 3.3.5.1 Below Grade Wall Waterproofing
 - 3.3.5.2 Floor Waterproofing
 - 3.3.6 Fabric Membrane Reinforcement
 - 3.3.7 Keyed Joint Footings
 - 3.3.8 Flashing Flanges
 - 3.3.9 Clamping Devices
 - 3.3.10 Reglets
- 3.4 PROTECTIVE COVERING
 - 3.4.1 Vertical Surfaces

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

3.4.2 Horizontal Surfaces
3.5 CLEAN UP
3.6 SCHEDULE OF MATERIALS

-- End of Section Table of Contents --

SECTION 07 12 00

BUILT-UP BITUMINOUS WATERPROOFING
02/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE-EN ISO 14118 (2018) Safety of machinery - Prevention of unexpected start-up (ISO 14118:2017)

ASTM INTERNATIONAL (ASTM)

ASTM C208 (2012; R 2017; E 2017; E 2019) Standard Specification for Cellulosic Fiber Insulating Board

ASTM C726 (2017) Standard Specification for Mineral Wool Roof Insulation Board

ASTM D41/D41M (2011; R 2016) Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing

ASTM D173/D173M (2003; R 2011; E 2012) Bitumen-Saturated Cotton Fabrics Used in Roofing and Waterproofing

ASTM D226/D226M (2017) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing

ASTM D449/D449M (2003; R 2014; E 2014) Asphalt Used in Dampproofing and Waterproofing

ASTM D517 (1998; R 2008) Asphalt Plank

ASTM D1327/D1327M (2004; R 2020) Standard Specification for Bitumen-Saturated Woven Burlap Fabrics Used in Roofing and Waterproofing

ASTM D1668/D1668M (1997a; R 2014; E 2014) Glass Fabrics (Woven and Treated) for Roofing and Waterproofing

ASTM D2178/D2178M (2015a) Asphalt Glass Felt Used in Roofing and Waterproofing

ASTM D4586/D4586M (2007; R 2018) Asphalt Roof Cement, Asbestos-Free

MINISTERIO DE FOMENTO (MF)

FOM/2523/2014

(2014) Orden de 12 de diciembre, por la que se actualizan determinados artículos del pliego de prescripciones técnicas generales para obras de carreteras y puentes, relativos a materiales básicos, a firmes y pavimentos, y a señalización, balizamiento y sistemas de contención de vehículos.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Standard Details, G

Protection Board; G

Membrane Fabric; G

Reinforcing Fabric; G

SD-08 Manufacturer's Instructions

Installation Instructions

1.3 MANUFACTURER'S DETAIL

Submit manufacturer's standard details indicating methods of attachment and spacing, transition and termination conditions, installation details.

1.4 ENVIRONMENTAL CONDITIONS

Apply the primers and waterproofing specified herein when the ambient temperature is above 4 degrees C.

1.5 DELIVERY AND STORAGE

1.5.1 Materials Packaging

Deliver materials in bundles, rolls, and sealed containers in accordance with manufacturer's printed handling instructions and bearing manufacturer's original labels. Material labels indicate dates for use or shelf life; remove outdated material, damaged, and deteriorated material from the jobsite. Keep materials wrapped and separated from off-gassing materials (such as drying paints and adhesives). Do not use materials that have visible moisture or biological growth.

1.5.2 Materials Storage

1.5.2.1 Asphalt

Protect asphalt from freezing. Store asphalt in a weathertight enclosure, free from contact with soil. Store and maintain at not less than **10 degrees C** for at least 24 hours before use.

1.5.2.2 Reinforcement Fabrics

Handle and store reinforcement fabrics in accordance with manufacturer's printed instructions. Protect fabrics from moisture damage and absorption in a weathertight enclosure or off the ground on pallets, and covered on top and all sides with breathable-type canvas tarpaulins. Plastic sheets cause condensation buildup therefore do not use them to cover waterproofing materials.

1.6 Flame Heated Equipment

1.6.1 Fire Protection

Locate melt kettles no closer than **8 meters** from buildings or combustible materials. Provide and maintain two approved 4-A:40-B:C fire extinguishers within **8 meters** of each operating kettle. Fire extinguishers, operations and locations must comply with NFPA 1 Section Tar Kettles. Equip asphalt (tar) kettles with tight fitting lids.

1.6.2 Operational Requirements

Equip kettles with automatic thermostatic control capable of maintaining asphalt temperature. Calibrate and maintain controls in working order for the duration of the work. Equip kettles with means of agitation and ensure they are operating as necessary to produce a controlled uniform temperature throughout kettle contents to prevent spot heating. Do not heat contents above flash point. Do not place flame heated equipment on the roof.

1.6.3 Drillage of Bitumen

Seal joints in and at edges as necessary to prevent drillage of asphalt into the building or onto adjacent surfaces.

PART 2 PRODUCTS

2.1 BITUMEN

Asphalt; **ASTM D449/D449M**, Type I.

2.2 BITUMINOUS PLASTIC CEMENT

ASTM D4586/D4586M, Type I or **FOM/2523/2014** for asphalt.

2.3 MEMBRANE FABRIC

The following requirements apply:

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

<u>Felt or Fabric Material</u>	<u>Saturant or Impregnant</u>	<u>Specification</u>
Glass (felt) mat	Asphalt	ASTM D2178/D2178M, Type III or UNE-EN ISO 14118
Cellulose fiber mat-based (organic) felt	Asphalt	ASTM D226/D226M
Reinforcing glass fabric	Asphalt	ASTM D1668/D1668M REV A, Type I or UNE-EN ISO 14118
Reinforcing cotton fabric	Asphalt	ASTM D173/D173M
Reinforcing woven burlap fabric	Asphalt	ASTM D1327/D1327M

2.3.1 Cotton Fabrics

Provide cotton fabrics woven entirely of cotton in accordance with ASTM D173/D173M and thoroughly and uniformly saturated with asphalt.

2.3.2 Woven Burlap Fabrics

Provide woven burlap fabrics in accordance with ASTM D1327/D1327M composed of 100 percent jute fiber and two cotton threads at each selvage, and thoroughly and uniformly saturated with asphalt. Fabric cannot be completely closed or sealed by the process of saturation and is to have sufficient porosity to allow successive moppings of plying asphalt to seep through. Fabric surface cannot be coated or covered with talc or any other substance that interferes with the adhesion between fabric and plying asphalt. Provide fabric surface uniformly smooth and free of irregularities, folds, knots, ragged or untrue edges, breaks, cracks, and other visible defects.

2.4 NAILS

Provide galvanized roofing nails or nails in accordance with fabric and protection board manufacturer's written recommendations. If fabric contains metal, provide nails as necessary to avoid electrolytic action due to proximity of dissimilar metals.

2.5 PRIMER

ASTM D41/D41M for asphalt.

2.6 PROTECTION BOARD

ASTM D517, plain, asphalt plank; ASTM C208, construction grade building board, 12.7 mm thick, asphalt saturated or coated; ASTM C726, 11 mm thick, covered on one side with waterproof paper or asphalt-saturated felt.

PART 3 EXECUTION

3.1 INSPECTION OF SURFACES

Before starting the work, inspect all surfaces that must be waterproofed to determine if in satisfactory condition. Check the location and setting of all embedded items. Place backing and blocking and perimeter framing

for recessed items as required by the various trades on the project. Complete conduit, piping, and other required rough-in. Notify the Contracting Officer of serious defects or conditions that prevent satisfactory application. Start application after such defects and conditions have been corrected.

3.2 PREPARATION OF SURFACES

Ensure surfaces to receive treatment are clean and dry, smooth and free from deleterious and excess materials and projections. Ensure concrete surfaces are properly cured, free of release agents, oil, grease, dirt, laitance, loose material, frost, debris and other contaminants. Thoroughly wet holes, joints, cracks, and voids in concrete with water, then fill with Portland cement mortar, strike flush, and permit to dry. Cut off or grind high spots smooth. Coat surfaces to receive asphalt membrane waterproofing with a priming coat of asphalt primer. Apply priming coat at a rate of not less than 4 liters per 10 square meters, covering the entire waterproofed surface. Allow primer to dry per manufacturer's printed instructions before applying waterproofing.

3.3 APPLICATION

3.3.1 Building Envelope Requirements

Provide a continuous waterproofing system at all material and building transitions. Lap, wrap, fasten and seal products in accordance with manufacturer's printed instructions. Locate waterproofing components within envelope assemblies in locations indicated on the Drawings. Envelope assembly variations are not permitted without written approval from the Contracting Officer's Representative.

3.3.2 General Installation Requirements

Provide waterproofing where indicated. Provide ventilation for enclosed spaces when using bituminous membrane waterproofing.

3.3.3 Protection of Surrounding Areas

Before starting waterproofing work, protect surrounding areas and surfaces from spillage and migration of asphalt onto other work. Provide non-combustible protective coverings at surfaces adjacent to hoists and kettles. Lap protective coverings at least 150 mm, secure against wind, and vent to prevent collection of moisture on covered surfaces. Keep protective coverings in place for the duration of asphalt work. Protect drains and conductors from clogging with asphalt.

3.3.4 Heating and Application of Bitumen Coatings

Heat solid bitumen in kettle equipped with an automatic heating device or control unit for positive control of the specified temperature. Provide an accurate and clearly readable thermometer on all kettles. Heat bitumen to flow freely but not above 190 degrees C. Apply bitumen over the primer, between each ply and as a top coating at the rate of not less than 10 kilograms of asphalt per 10 square meters of surface.

3.3.5 Membrane Waterproofing

3.3.5.1 Below Grade Wall Waterproofing

Provide 3-ply hot-applied asphalt membrane system for foundation walls. Install fabrics in accordance with manufacturer's printed [installation instructions](#). Caulk joints before applying primer. Apply primer at a rate of **0.2 liters per square meter**. Overlap fabrics at ends and stagger a minimum of **610 mm** for 3-ply system. End-to-end taping is not acceptable. Firmly embed each fabric in a solid uniform coating of hot asphalt at a rate of **0.98 kg per square meter kg per square meter**. Allow asphalt to penetrate each fabric and to provide required adhesion. Avoid excessive applications of asphalt between fabrics in order to prevent slippage. Provide waterproofing system consisting of two or more fabrics with fabric reinforcement at corners, angles, over construction joints, and in locations where subject to increased stress.

3.3.5.2 Floor Waterproofing

Apply primer at a rate of **0.2 liters per square meter**. Do not allow primer to puddle. Confirm primer is dry to the touch before application of asphalt. Where slab abuts walls, extend first [reinforcing fabric](#) a minimum of **150 mm** on slabs and **200 mm** on walls. At vertical corners, extend first fabric a minimum of **125 mm** from corners on each side. Lap second fabric with the first fabric a minimum of **50 mm**. At floor drains, and elsewhere as indicated, extend fabric into a clamping device, set in a heavy coating of flashing cement, and securely clamp.

3.3.6 Fabric Membrane Reinforcement

Provide fabric membranes to reinforce felts at intersections. Provide reinforcement consisting of two plies of fabric membrane cemented in place and to each other with bituminous plastic cement not less than **2 mm** thick for each coating. At the intersection of slabs and vertical surfaces, extend the first ply at least **150 mm** on the slab and **100 mm** up the vertical surface. At intersections of two vertical surfaces, extend the first ply at least **250 mm** on each side of the intersection. Place second ply to lap the first by not less than **50 mm**.

3.3.7 Keyed Joint Footings

Provide membrane flashing, neatly formed, to the contours of keyed joints in foundation wall footings. Extend flashing to the outside edge of the footing, and turn the flashing down **100 mm**. Continue the flashing through the joint to the inside of the walls and lap the flashing into the waterproofing membrane under the slab. Protect the flashing until it is lapped by the waterproofing membranes for the subsurface floor slabs and foundation walls. Provide flashing membrane made up of the same number and type materials as the waterproofing membrane or a thermoplastic material compatible with the waterproofing materials, as recommended by the manufacturer in writing.

3.3.8 Flashing Flanges

Prime flashing flanges of pipe sleeves and ducts penetrating the waterproofing membrane. Allow primer to dry. Provide flanges with two fabric membrane collars cemented in place and to each other with bituminous plastic cement. Extend collars **100 and 150 mm**, respectively, beyond the edge of the flanges, cover the flanges, and fit the flanges

tight against the sleeve. Extend waterproofing connected to work exposed to weather to the back of the adjoining work, or counter flash to form a watertight connection.

3.3.9 Clamping Devices

At floor drains and elsewhere, as indicated, extend membrane into clamping device set in heavy coating of bituminous plastic cement, and clamp securely.

3.3.10 Reglets

Install continuous reglets as specified in Section 07 60 00 FLASHING AND SHEET METAL to receive the exposed edges of membrane waterproofing. After placement of waterproofing, completely fill reglets with bitumen.

3.4 PROTECTIVE COVERING

3.4.1 Vertical Surfaces

Protect membrane waterproofing against which backfill must be placed by providing protective covering pressed into the final mopping while the mopping of bitumen is still hot. Butt edges of protection board against adjacent edges of protection boards. Cover exposed surfaces with a coating of bitumen. Where surfaced fiberboard or mineral fiberboard is used, place surface side facing outward. Fit board around pipes and projections so as to cover the entire surface of the membrane waterproofing.

3.4.2 Horizontal Surfaces

Place protective covering over membrane immediately after application has thoroughly dried. Remove protective covering immediately before proceeding with work that will conceal the membrane waterproofing.

3.5 CLEAN UP

Use a cleaner recommended by the waterproofing manufacturer to clean other work surfaces that are stained with waterproofing material.

3.6 SCHEDULE OF MATERIALS

Some metric measurements in this section are based on mathematical conversion of inch-pound measurement, and not on metric measurement commonly agreed to by the manufacturers or other parties. The inch-pound and metric measurements shown are as follows:

<u>Products</u>	<u>Inch-Pound</u>	<u>Metric</u>
Protection Board		12.7 mm
		11 mm
Polyethylene Sheet		13.6 kg
Laminated Sheet		1.95 kg per sq. m

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

<u>Products</u>	<u>Inch-Pound</u>	<u>Metric</u>
Copper Sheet		0.92 kg/sq m
		1.52 kg/sq m
		2.14 kg/sq m

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 14 00

FLUID-APPLIED WATERPROOFING

08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY, STORAGE, AND HANDLING
- 1.4 ENVIRONMENTAL CONDITIONS
- 1.5 WARRANTY
 - 1.5.1 Membrane Manufacturer Warranty
 - 1.5.2 Roofing System Installer Warranty
 - 1.5.3 Continuance of Warranty

PART 2 PRODUCTS

- 2.1 FLUID-APPLIED MEMBRANE
- 2.2 MEMBRANE PRIMER
- 2.3 SEALANT
- 2.4 SEALANT PRIMER
- 2.5 BACKING MATERIAL
- 2.6 BOND BREAKER
- 2.7 ELASTOMERIC SHEET
- 2.8 ELASTOMERIC SHEET ADHESIVE
- 2.9 MEMBRANE REINFORCEMENT

PART 3 EXECUTION

- 3.1 PREPARATION
 - 3.1.1 Flashings
 - 3.1.1.1 Drains
 - 3.1.1.2 Penetrations and Projections
 - 3.1.1.3 Walls and Vertical Surfaces
 - 3.1.2 Cracks and Joints
 - 3.1.3 Priming
- 3.2 SPECIAL PRECAUTIONS
- 3.3 APPLICATION
- 3.4 FIELD QUALITY CONTROL
 - 3.4.1 Moisture Test
 - 3.4.2 Film Thickness
- 3.5 MAINTENANCE AND REPAIR TRAINING
- 3.6 INFORMATION CARD

-- End of Section Table of Contents --

SECTION 07 14 00

FLUID-APPLIED WATERPROOFING
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 104231 (1999) Waterproofing. Bituminous And Bituminous Modified Materials. Asphalt Emulsions

ASTM INTERNATIONAL (ASTM)

ASTM C836/C836M (2018; R 2022) Standard Specification for High Solids Content, Cold Liquid-Applied Elastomeric Waterproofing Membrane for Use With Separate Wearing Course

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Fluid-Applied Membrane

Membrane Primer

Elastomeric Sheet

Solvent

Moisture Meter

Bond Breaker

SD-11 Closeout Submittals

Warranty

Information Card

Maintenance and Repair Training

Include copies of Safety Data Sheets for maintenance/repair materials.

1.3 DELIVERY, STORAGE, AND HANDLING

Deliver waterproofing materials in manufacturer's original, unopened containers, with labels intact and legible. Containers of materials covered by a referenced specification number must bear the specification number, type, and class of the contents. Deliver materials in sufficient quantity to continue work without interruption. Store and protect materials in accordance with manufacturer's instructions, and use within their indicated shelf life. When hazardous materials are involved, adhere to special precautions of the manufacturer, unless precautions conflict with local, state, and federal regulations. Promptly remove from the site materials or incomplete work adversely affected by exposure to moisture or freezing. Store materials on pallets and cover from top to bottom with canvas tarpaulins.

1.4 ENVIRONMENTAL CONDITIONS

Apply materials when ambient temperature is 4 degrees C or above for a period of 24 hours prior to the application and when there is no ice, frost, surface moisture, or visible dampness on the substrate surface. Apply materials when air temperature is expected to remain above 4 degrees C during the cure period recommended by the manufacturer. Moisture test for substrate is specified under paragraph MOISTURE TEST. Work may be performed within heated enclosures, provided the surface temperature of the substrate is maintained at a minimum of 4 degrees C for 24 hours prior to the application of the waterproofing, and remains above that temperature during the cure period recommended by the manufacturer.

1.5 WARRANTY

Provide system material and workmanship warranties meeting specified requirements. Provide revisions or amendment to standard membrane manufacturer warranty to comply with the specified requirements. Provide a manufacturer's warranty that has no dollar limit, covers full system water-tightness, and has a minimum duration of 20 years.

1.5.1 Membrane Manufacturer Warranty

Provide the membrane manufacturer's 20-year no dollar limit system materials and installation workmanship warranty, including flashing, insulation, and accessories necessary for a watertight system construction. Write the warranty directly to the Government commencing at time of Government's acceptance of the work. Provide the following statements for such warranty:

- a. If within the warranty period the system, as installed for its intended use in the normal climatic and environmental conditions of the facility, becomes non-watertight, shows evidence of moisture intrusion within the assembly, blisters, splits, tears, cracks, delaminates, separates at the seams, or shows evidence of excessive weathering due to defective materials or installation workmanship, the repair or replacement of the defective and damaged materials of the system assembly and correction of defective workmanship are the responsibility of the membrane manufacturer. All costs associated with the repair or replacement work are the responsibility of the membrane manufacturer.
- b. The warranty must remain in full force and effect, including emergency

temporary repairs performed by others, when the manufacturer or his approved applicator fail to perform the repairs within 72 hours of notification.

1.5.2 Roofing System Installer Warranty

The system installer must warrant for a minimum period of two years that the system, as installed, is free from defects in installation workmanship, to include the membrane, flashing, insulation, accessories, attachments, and sheet metal installation integral to a complete watertight system assembly. Write the warranty directly to the Government. The system installer is responsible for correction of defective workmanship and replacement of damaged or affected materials. The system installer is responsible for all costs associated with the repair or replacement work.

1.5.3 Continuance of Warranty

Repair or replacement work that becomes necessary within the warranty period must be approved and accomplished in a manner so as to restore the integrity of the system assembly and validity of the membrane manufacturer warranty for the remainder of the manufacturer warranty period.

PART 2 PRODUCTS

2.1 FLUID-APPLIED MEMBRANE

ASTM C836/C836M or UNE 104231. Fluid-applied single component fully reinforced system with fiberglass or polyester reinforcement.

2.2 MEMBRANE PRIMER

As recommended by the fluid-applied membrane manufacturer unless specifically prohibited by the manufacturer of the fluid-applied membrane.

2.3 SEALANT

As specified in Section 07 92 00 JOINT SEALANTS.

2.4 SEALANT PRIMER

As specified in Section 07 92 00 JOINT SEALANTS.

2.5 BACKING MATERIAL

Premolded, closed-cell, polyethylene, or polyurethane foam rod having a diameter 25 percent larger than joint width before being compressed into joint. Provide bond breaker of polyethylene film or other suitable material between backing material and sealant.

2.6 BOND BREAKER

As recommended by the fluid-applied membrane manufacturer. Bond breaker must not interfere with the curing process or other performance properties of the fluid-applied membrane. Submit material description and physical properties, application details, and recommendations regarding shelf life, application procedures, and precautions on flammability and toxicity.

2.7 ELASTOMERIC SHEET

Preformed; as recommended by the fluid-applied membrane manufacturer. Bond strength between the fluid-applied membrane and the preformed elastomeric sheet must be a minimum of 7 kPa when tested in accordance with ASTM C836/C836M.

2.8 ELASTOMERIC SHEET ADHESIVE

As recommended by the elastomeric sheet manufacturer.

2.9 MEMBRANE REINFORCEMENT

As recommended by the fluid-applied membrane manufacturer, fiberglass or polyester.

PART 3 EXECUTION

3.1 PREPARATION

Coordinate work with that of other trades to ensure that components to be incorporated into the waterproofing system are available when needed. Inspect and approve surfaces immediately before application of waterproofing materials. Remove laitance, loose aggregate, sharp projections, grease, oil, dirt, curing compounds, and other contaminants which could adversely affect the complete bonding of the fluid-applied membrane to the concrete surface.

3.1.1 Flashings

Make penetrations through sleeves in structural slab watertight before application of waterproofing. After flashing is completed, cover elastomeric sheet with fluid-applied waterproofing during waterproofing application.

3.1.1.1 Drains

Make drain flanges flush with surface of structural slab. Apply a full elastomeric sheet around the drain, with edges fully adhered to drain flange and to structural slab. Do not adhere elastomeric sheet over joint between drain and concrete slab. Do not plug drainage or weep holes. Cover elastomeric sheet with fluid-applied waterproofing during waterproofing application. Lap elastomeric sheet a minimum of 100 mm onto concrete slab.

3.1.1.2 Penetrations and Projections

Flash penetrations and projections through structural slab with an elastomeric sheet adhered to the concrete slab and the penetration. Leave elastomeric sheet unadhered for 25 mm over joint between penetration and concrete slab. Adhere elastomeric sheet a minimum of 150 mm onto horizontal deck.

3.1.1.3 Walls and Vertical Surfaces

Flash wall intersections which are not of monolithic pour or constructed with reinforced concrete joints with an elastomeric sheet adhered to both vertical wall surfaces and concrete slab. Flash intersections which are monolithically poured or constructed with reinforced concrete joints with

either an elastomeric sheet or a vertical grade of fluid-applied waterproofing adhered to vertical wall surfaces and concrete slab. Leave sheet unadhered for a distance of 25 mm from the corner on both vertical and horizontal surfaces.

3.1.2 Cracks and Joints

Prepare visible cracks and joints in substrate to receive fluid-applied waterproofing membrane by placing a bond breaker and an elastomeric slip sheet between membrane and substrate. Apply a 50 mm bond breaker to cracks that show movement followed by an elastomeric sheet adhered to the deck. Nonmoving cracks must be double coated with fluid-applied waterproofing.

3.1.3 Priming

Prime surfaces to receive fluid-applied waterproofing membrane. Apply primer as required by membrane manufacturer's printed instructions.

3.2 SPECIAL PRECAUTIONS

Protect waterproofing materials during transport and application. Do not dilute primers and other materials, unless specifically recommended by materials manufacturer. Keep containers closed except when removing contents. Do not mix remains of unlike materials. Thoroughly remove residual materials before using application equipment for mixing and transporting materials. Do not permit equipment on the project site that has residue of materials used on previous projects. Use cleaners only for cleaning, not for thinning primers or membrane materials. Ensure that workers and others who walk on cured membrane wear clean, soft-soled shoes to avoid damaging the waterproofing materials.

3.3 APPLICATION

Over primed surfaces, provide a uniform, wet, monolithic coating of fluid-applied membrane, both base layer and top layer by following manufacturer's printed instructions. Apply material by trowel, squeegee, roller, brush, spray apparatus, or other method recommended by membrane manufacturer. Check wet film thickness as specified in paragraph FILM THICKNESS and adjust application rate as necessary to provide a uniform coating of the thickness specified. Where possible, mark off surface to be coated in equal units to facilitate proper coverage. At expansion joints, control joints, prepared cracks, flashing, and terminations, carry membrane over preformed elastomeric sheet in a uniform 1.5 mm thick, plus or minus 0.125 mm, wet thickness to provide a monolithic coating. If membrane cures before next application, wipe previously applied membrane with a solvent to remove dirt and dust that could inhibit adhesion of overlapping membrane coat. Use solvent recommended by the membrane manufacturer, as approved.

3.4 FIELD QUALITY CONTROL

3.4.1 Moisture Test

Prior to application of fluid-applied waterproofing, measure moisture content of substrate with a moisture meter in the presence of the Contracting Officer. Do not begin application until meter reading indicates "dry" range as per manufacturer's installation instructions.

3.4.2 Film Thickness

Measure wet film thickness every **10 square meters** during application by placing flat metal plates on the substrate or using a mil-thickness gage especially manufactured for the purpose.

3.5 MAINTENANCE AND REPAIR TRAINING

Provide written and verbal instructions on proper maintenance procedures to designated Government personnel. Verbal instructions must be provided by a competent representative of the membrane manufacturer and include a minimum of 4 hours on maintenance and emergency repair of the membrane. Include a demonstration of membrane repair, and give sources of required special tools. Provide information on safety requirements during maintenance and emergency repair operations.

3.6 INFORMATION CARD

For each system application, provide a minimum **215 mm** information card for facility records and a card laminated in plastic and framed for interior display at access point, or a photoengraved **one mm** thick aluminum card for exterior display. Identify facility name and number; location; Contract number; approximate area; detailed system description, including deck type, membrane, number of plies, method of application, manufacturer, insulation and cover board system and thickness; presence of tapered insulation for primary drainage, presence of vapor retarder; date of completion; installing Contractor identification and Contract information; membrane manufacturer warranty expiration, warranty reference number, and contact information. Install card at system access location as directed by the Contracting Officer and provide a paper copy to the Contracting Officer.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

FORM 1	
FLUID-APPLIED WATERPROOFING SYSTEM COMPONENTS	
1. Contract Number	
2. Date Work Completed	
3. Project Specification Designation	
4. Substrate Material	
5. Slope of Substrate	
6. Drains Type/Manufacturer	
7. Waterproofing	
a. Membrane	
b. Sealant	
c. Elastomeric Sheet	
d. Materials Manufacturer(s)	
8. Warranty	
a. Manufacturer warranty expiration	
b. Warranty reference number	
9. Statement of Compliance or Exception	
Contractor's Signature	Date Signed
Inspector's Signature	Date Signed

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 21 13

BOARD AND BLOCK INSULATION

08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 MANUFACTURER'S DETAILS
- 1.4 PRODUCT DATA
- 1.5 DELIVERY, STORAGE, AND HANDLING
 - 1.5.1 Delivery
 - 1.5.2 Storage
- 1.6 SAFETY PRECAUTIONS
 - 1.6.1 Respirators
 - 1.6.2 Other Safety Considerations

PART 2 PRODUCTS

- 2.1 BLOCK OR BOARD INSULATION
 - 2.1.1 Thermal Resistance
 - 2.1.2 Fire Protection Requirements
 - 2.1.3 Prohibited Materials
- 2.2 PROTECTION BOARD OR COATING
- 2.3 ACCESSORIES
 - 2.3.1 Adhesive
 - 2.3.2 Mechanical Fasteners

PART 3 EXECUTION

- 3.1 EXISTING CONDITIONS
- 3.2 PREPARATION
 - 3.2.1 Blocking Around Heat Producing Devices
- 3.3 INSTALLATION
 - 3.3.1 Insulation Installation and Handling
 - 3.3.2 Electrical Wiring
 - 3.3.3 Continuity of Insulation
 - 3.3.4 Coordination
- 3.4 INSTALLATION ON UNDERSIDE OF CONCRETE SLAB
 - 3.4.1 Mechanically Fastened Systems
 - 3.4.2 Adhesively Bonded Systems

-- End of Section Table of Contents --

SECTION 07 21 13

BOARD AND BLOCK INSULATION
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE-EN 13162 (2013) Thermal insulation products for buildings - Factory made mineral wool (MW) products - Specification

UNE-EN 13164 (2013) Thermal insulation products for buildings - Factory made extruded polystyrene foam (XPS) products - Specification

UNE-EN 13501-1 (2019) Fire classification of construction products and building elements - Part 1: Classification using data from reaction to fire tests

ASTM INTERNATIONAL (ASTM)

ASTM C578 (2022) Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation

ASTM C612 (2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation

ASTM C930 (2019) Standard Classification of Potential Health and Safety Concerns Associated with Thermal Insulation Materials and Accessories

ASTM C1902 (2022) Standard Specification for Cellular Glass Insulation Used in Building and Roof Applications

ASTM E84 (2023) Standard Test Method for Surface Burning Characteristics of Building Materials

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2021) International Building Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.134 Respiratory Protection

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Manufacturer's Standard Details; G

SD-03 Product Data

Block or Board Insulation; G

Accessories including sealants; G

SD-08 Manufacturer's Instructions

Insulation Installation and Handling
Adhesive

1.3 MANUFACTURER'S DETAILS

Submit manufacturer's standard details indicating methods of attachment and spacing, transition and termination details, and installation details. Include verification of existing conditions.

1.4 PRODUCT DATA

Include environmental data for materials descriptions, recommendations for product shelf life, requirements for protection board or coatings, and precautions for flammability and toxicity. Include data to verify compatibility of sealants with insulation.

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 Delivery

Deliver materials to the site in original sealed wrapping bearing manufacturer's name and brand designation, specification number, type, grade, R-value, and class. Store and handle to protect from damage. Do not allow insulation materials to become wet, soiled, crushed, or covered with ice or snow. Comply with manufacturer's recommendations for handling, storing, and protecting of materials before and during installation.

1.5.2 Storage

Inspect materials delivered to the site for damage and store out of weather in manufacturer's original packaging. Store only in dry

locations, not subject to open flames or sparks, and easily accessible for inspection and handling. Keep materials wrapped and separated from off-gassing materials (such as drying paints and adhesives). Do not use materials that have visible moisture or biological growth. Comply with manufacturer's recommendations for handling, storage, and protection of materials before and during installation.

1.6 SAFETY PRECAUTIONS

1.6.1 Respirators

Provide installers with dust/mist respirators, training in their use, and protective clothing, all approved by the National Institute for Occupational Safety and Health (NIOSH)/Mine Safety and Health Administration (MSHA) and in accordance with 29 CFR 1910.134.

1.6.2 Other Safety Considerations

Comply with the safety requirements of ASTM C930.

PART 2 PRODUCTS

2.1 BLOCK OR BOARD INSULATION

Provide thermal insulating materials as recommended by manufacturer for each type of application indicated. Provide insulation with the following physical properties and in accordance with the following standards:

- a. Extruded Preformed Cellular Polystyrene (XPS): ASTM C578 REV A, Type IV or UNE-EN 13164.
- b. Mineral Fiber Block and Board: ASTM C612, Type IVB or UNE-EN 13162.
- c. Cellular Glass: ASTM C1902 or UNE-EN 13162.

2.1.1 Thermal Resistance

Provide R-values as indicated on the drawings.

2.1.2 Fire Protection Requirements

- a. Flame spread index of 75 or less when tested in accordance with ASTM E84 or Class A1 or A2 per UNE-EN 13501-1.
- b. Smoke developed index of 450 or less when tested in accordance with ASTM E84 or Class S1 per UNE-EN 13501-1.
- c. Provide insulated assemblies in accordance ICC IBC Chapter Fire and Smoke Protection Features.

2.1.3 Prohibited Materials

Do not provide materials containing asbestos.

2.2 PROTECTION BOARD OR COATING

As recommended by insulation manufacturer.

2.3 ACCESSORIES

2.3.1 Adhesive

As recommended by insulation manufacturer.

2.3.2 Mechanical Fasteners

Corrosion resistant fasteners as recommended by the insulation manufacturer.

PART 3 EXECUTION

3.1 EXISTING CONDITIONS

Prior to installation, ensure all areas that will be in contact with the insulation are dry and free of projections that could cause voids, compressed insulation, or punctured vapor retarders. For foundation perimeter or under slab applications, check that subsurface fill is flat, smooth, dry, and well tamped. Do not proceed with installation if moisture or other conditions are present and notify the Contracting Officer of such conditions. Do not proceed with the work until conditions have been corrected and verified to be dry.

3.2 PREPARATION

3.2.1 Blocking Around Heat Producing Devices

Provide noncombustible blocking at all spaces between heat producing devices and the floors, ceilings and roofs through which they pass. Provide in accordance with ICC IBC Section 2111.13 Fireplace Fireblocking and with the following clearances:

- a. Recessed lighting fixtures, including wiring compartments, ballasts, and other heat producing devices, unless certified for installation surrounded by insulation: 75 mm from outside face of fixtures and devices or as required by NFPA 70 and, if insulation is placed above fixture or device, 600 mm above fixture.

3.3 INSTALLATION

3.3.1 Insulation Installation and Handling

Provide insulation in accordance with the manufacturer's printed installation instructions. Keep material dry and free of extraneous materials.

3.3.2 Electrical Wiring

Do not install insulation in a manner that would enclose electrical conductors, that are not installed in conduit, between two layers of insulation.

3.3.3 Continuity of Insulation

Butt tightly against adjoining boards, studs, rafters, joists, sill plates, headers and obstructions. Provide continuity and integrity of insulation at corners, wall to ceiling joint, roof, and floor. Avoid creating thermal bridges and voids. Provide and verify continuity of

insulative barrier throughout the building enclosure.

3.3.4 Coordination

Verify final installed insulation thicknesses comply with thicknesses indicated and with the approved insulation submittal(s).

3.4 INSTALLATION ON UNDERSIDE OF CONCRETE SLAB

3.4.1 Mechanically Fastened Systems

Size insulation to cover underside of slab. Apply adhesive to slab and set fasteners in adhesive as recommended by manufacturer. After curing of adhesive, install insulation over fasteners and bend split prongs to provide a flush condition with the insulation. Butt all edges of insulation and seal with tape.

3.4.2 Adhesively Bonded Systems

Apply adhesive to underside of slab and completely cover wall with insulation.

- a. As recommended by insulation manufacturer.
- b. Butt all edges of insulation and seal with tape.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 21 16

MINERAL FIBER BLANKET INSULATION

08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY, STORAGE, AND HANDLING
 - 1.3.1 Delivery
 - 1.3.2 Storage
- 1.4 SAFETY PRECAUTIONS
 - 1.4.1 Respirators
 - 1.4.2 Other Safety Concerns

PART 2 PRODUCTS

- 2.1 BLANKET INSULATION
 - 2.1.1 Prohibited Materials
- 2.2 BLOCKING
- 2.3 VAPOR RETARDER
- 2.4 PRESSURE SENSITIVE TAPE
- 2.5 ACCESSORIES
 - 2.5.1 Adhesive
 - 2.5.2 Mechanical Fasteners
 - 2.5.3 Wire Mesh

PART 3 EXECUTION

- 3.1 EXISTING CONDITIONS
- 3.2 PREPARATION
 - 3.2.1 Blocking at Attic Vents and Access Doors
 - 3.2.2 Blocking Around Heat Producing Devices
- 3.3 INSTALLATION
 - 3.3.1 Insulation
 - 3.3.1.1 Electrical wiring
 - 3.3.1.2 Continuity of Insulation
 - 3.3.1.3 Installation at Bridging and Cross Bracing
 - 3.3.1.4 Insulation Blanket with Affixed Vapor Retarder
 - 3.3.1.5 Insulation without Affixed Vapor Retarder
 - 3.3.1.6 Sizing of Blankets
 - 3.3.2 Installation of Separate Vapor Retarder

-- End of Section Table of Contents --

SECTION 07 21 16

MINERAL FIBER BLANKET INSULATION
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE-EN 13162 (2013) Thermal insulation products for buildings - Factory made mineral wool (MW) products - Specification

UNE-EN 13501-1 (2019) Fire classification of construction products and building elements - Part 1: Classification using data from reaction to fire tests

ASTM INTERNATIONAL (ASTM)

ASTM C665 (2023) Standard Specification for Mineral-Fiber Blanket Thermal Insulation for Light Frame Construction and Manufactured Housing

ASTM C930 (2019) Standard Classification of Potential Health and Safety Concerns Associated with Thermal Insulation Materials and Accessories

ASTM E84 (2023) Standard Test Method for Surface Burning Characteristics of Building Materials

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350 (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers

GREEN SEAL (GS)

GS-36 (2013) Adhesives for Commercial Use

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168

(2017) Adhesive and Sealant Applications

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.134

Respiratory Protection

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Blanket Insulation; G

Vapor Retarder

Pressure Sensitive Tape

Accessories

SD-08 Manufacturer's Instructions

Insulation

1.3 DELIVERY, STORAGE, AND HANDLING

1.3.1 Delivery

Deliver materials to site in original sealed wrapping bearing manufacturer's name and brand designation, specification number, type, grade, R-value, and class. Store and handle to protect from damage. Do not allow insulation materials to become wet, soiled, crushed, or covered with ice or snow. Comply with manufacturer's recommendations for handling, storing, and protecting of materials before and during installation.

1.3.2 Storage

Inspect materials delivered to the site for damage; unload and store out of weather in manufacturer's original packaging. Store only in dry locations, not subject to open flames or sparks, and easily accessible for inspection and handling.

1.4 SAFETY PRECAUTIONS

1.4.1 Respirators

Provide installers with dust/mist respirators, training in their use, and protective clothing, all approved by National Institute for Occupational Safety and Health (NIOSH)/Mine Safety and Health Administration (MSHA) in accordance with 29 CFR 1910.134.

1.4.2 Other Safety Concerns

Consider other safety concerns and measures as outlined in [ASTM C930](#).

PART 2 PRODUCTS

2.1 BLANKET INSULATION

[ASTM C665](#), Type I or [UNE-EN 13162](#), blankets without membrane coverings and II, blankets with non-reflecting coverings; Class A, membrane-faced surface with a flame spread of 25 or less, except a flame spread rating of 25 or less and a smoke developed rating of 150 or less when tested in accordance with [ASTM E84](#) or Class A1 or A2 per [UNE-EN 13501-1](#).

Provide insulation with R-Value as indicated on drawings

2.1.1 Prohibited Materials

Do not provide asbestos-containing materials.

2.2 BLOCKING

Wood, metal, unfaced mineral fiber blankets in accordance with [ASTM C665](#), Type I, or [UNE-EN 13162](#) or other approved materials.

2.3 VAPOR RETARDER

- a. [0.15 mm](#) thick polyethylene sheeting and having a water vapor permeance of [57.2 ng/\(Pa * s * m2\)](#) or less.
- b. Membrane with the following properties:

Water Vapor Permeance: [57.2 ng/\(Pa * s * m2\)](#)
Maximum Flame Spread: [ASTM E84](#): 25

2.4 PRESSURE SENSITIVE TAPE

As recommended by the vapor retarder manufacturer and having a water vapor permeance rating of [57.2 ng/\(Pa * s * m2\)](#) or less.

2.5 ACCESSORIES

2.5.1 Adhesive

As recommended by the insulation manufacturer. Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of [CDPH SECTION 01350](#) (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of [SCAQMD Rule 1168](#). Provide aerosol adhesives used on the interior of the building that meet either emissions requirements of [CDPH SECTION 01350](#) (use the office or classroom requirements, regardless of space type) or VOC content requirements of [GS-36](#). Provide certification or validation of indoor air quality for adhesives.

2.5.2 Mechanical Fasteners

Corrosion resistant fasteners as recommended by the insulation manufacturer.

2.5.3 Wire Mesh

Corrosion resistant and as recommended by the insulation manufacturer.

PART 3 EXECUTION

3.1 EXISTING CONDITIONS

Before installing insulation, ensure that surfaces that will be in contact with the insulation are dry and free of projections which could cause voids, compressed insulation, or punctured vapor retarders. If moisture or other conditions are found that do not allow the workmanlike installation of the insulation, do not proceed but notify Contracting Officer of such conditions.

3.2 PREPARATION

3.2.1 Blocking at Attic Vents and Access Doors

Prior to installation of insulation, install permanent blocking to prevent insulation from slipping over, clogging, or restricting air flow through soffit vents at eaves.

3.2.2 Blocking Around Heat Producing Devices

Install non-combustible blocking around heat producing devices to provide the following clearances:

- a. Recessed lighting fixtures, including wiring compartments, ballasts, and other heat producing devices, unless these are certified by the manufacturer for installation surrounded by insulation: 75 mm from outside face of fixtures and devices or as required by NFPA 70 and, if insulation is to be placed above fixture or device, 600 mm above fixture.

3.3 INSTALLATION

3.3.1 Insulation

Install and handle insulation in accordance with manufacturer's instructions. Keep material dry and free of extraneous materials. Any materials that show visual evidence of biological growth due to presence of moisture must not be installed on the building project. Ensure personal protective clothing and respiratory equipment is used as required. Observe safe work practices.

3.3.1.1 Electrical wiring

Do not install insulation in a manner that would sandwich electrical conductors, that are not installed in conduit, between two layers of insulation.

3.3.1.2 Continuity of Insulation

Install blanket insulation to butt tightly against adjoining blankets and to studs, rafters, joists, sill plates, headers and any obstructions. Provide continuity and integrity of insulation at corners, wall to ceiling joints, roof, and floor. Avoid creating thermal bridges.

3.3.1.3 Installation at Bridging and Cross Bracing

Insulate at bridging and cross bracing by splitting blanket vertically at center and packing one half into each opening. Butt insulation at bridging and cross bracing; fill in bridged area with loose or scrap insulation.

3.3.1.4 Insulation Blanket with Affixed Vapor Retarder

Locate vapor retarder as indicated. Do not install blankets with affixed vapor retarders unless so specified. Unless the insulation manufacturer's instructions specifically recommend not to staple the flanges of the vapor retarder facing, staple flanges of vapor retarder at 150 mm intervals flush with face or set in the side of truss, joist, or stud. Avoid gaps and bulges in insulation and "fishmouth" in vapor retarders. Overlap both flanges when using face method. Seal joints and edges of vapor retarder with pressure sensitive tape. Stuff pieces of insulation into small cracks between trusses, joists, studs and other framing, such as at attic access doors, door and window heads, jambs, and sills, band joists, and headers. Cover these insulated cracks with vapor retarder material and tape joints with pressure sensitive tape to provide air and vapor tightness.

3.3.1.5 Insulation without Affixed Vapor Retarder

Provide snug friction fit to hold insulation in place. Stuff pieces of insulation into cracks between trusses, joists, studs and other framing, such as at attic access doors, door and window heads, jambs, and sills, band joists, and headers.

3.3.1.6 Sizing of Blankets

Provide only full width blankets when insulating between trusses, joists, or studs. Size width of blankets for a snug fit where trusses, joists or studs are irregularly spaced.

3.3.2 Installation of Separate Vapor Retarder

Apply continuous vapor retarder as indicated. Overlap joints at least 150 mm and seal with pressure sensitive tape. Seal at sill, header, windows, doors and utility penetrations. Repair punctures or tears with pressure sensitive tape.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 60 00

FLASHING AND SHEET METAL

08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 GENERAL REQUIREMENTS
 - 1.2.1 General Material Requirements
- 1.3 SUBMITTALS
- 1.4 MISCELLANEOUS REQUIREMENTS
 - 1.4.1 Product Data
 - 1.4.2 Finish Samples
 - 1.4.3 Operation and Maintenance Data
- 1.5 DELIVERY, HANDLING, AND STORAGE

PART 2 PRODUCTS

- 2.1 RECYCLED CONTENT
- 2.2 MATERIALS
 - 2.2.1 Exposed Sheet Metal Items
 - 2.2.2 Drainage
 - 2.2.3 Stainless Steel
 - 2.2.4 Aluminum Alloy Sheet and Plate
 - 2.2.5 Finishes
 - 2.2.5.1 Warranty on Finishes
 - 2.2.6 Aluminum Alloy, Extruded Bars, Rods, Shapes, and Tubes
 - 2.2.7 Solder
 - 2.2.8 Bituminous Plastic Cement
 - 2.2.9 Asphalt Primer
 - 2.2.10 Fasteners

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Workmanship
 - 3.1.2 Nailing
 - 3.1.3 Cleats
 - 3.1.4 Bolts, Rivets, and Screws
 - 3.1.5 Seams
 - 3.1.5.1 Flat-lock Seams
 - 3.1.5.2 Lap Seams
 - 3.1.5.3 Loose-Lock Expansion Seams
 - 3.1.5.4 Standing Seams
 - 3.1.5.5 Flat Seams
 - 3.1.6 Soldering
 - 3.1.6.1 Edges
 - 3.1.7 Welding and Mechanical Fastening
 - 3.1.7.1 Welding of Aluminum
 - 3.1.7.2 Mechanical Fastening of Aluminum

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.1.8 Protection from Contact with Dissimilar Materials
 - 3.1.8.1 Copper or Copper-bearing Alloys
 - 3.1.8.2 Aluminum
 - 3.1.8.3 Metal Surfaces
 - 3.1.8.4 Wood or Other Absorptive Materials
- 3.1.9 Expansion and Contraction
- 3.1.10 Base Flashing
- 3.1.11 Counterflashing
- 3.1.12 Gravel Stops and fascia
 - 3.1.12.1 Edge Strip
 - 3.1.12.2 Joints
- 3.1.13 Gutters
- 3.1.14 Downspouts
 - 3.1.14.1 Terminations
- 3.1.15 Copings
- 3.2 PAINTING
 - 3.2.1 Aluminum Surfaces
- 3.3 CLEANING
- 3.4 REPAIRS TO FINISH
- 3.5 FIELD QUALITY CONTROL
 - 3.5.1 Procedure

ATTACHMENTS:

Table II

-- End of Section Table of Contents --

SECTION 07 60 00
FLASHING AND SHEET METAL
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

- | | |
|-------------------|--|
| UNE 104231 | (1999) Waterproofing. Bituminous And Bituminous Modified Materials. Asphalt Emulsions |
| UNE-EN 485-1 | (2017) Aluminium and aluminium alloys - Sheet, strip and plate - Part 1: Technical conditions for inspection and delivery |
| UNE-EN 755-1 | (2016) Aluminium and aluminium alloys - Extruded rod/bar, tube and profiles - Part 1: Technical conditions for inspection and delivery |
| UNE-EN ISO 3834-2 | (2006) Quality requirements for fusion welding of metallic materials - Part 2: Comprehensive quality requirements (ISO 3834-2:2005) |

AMERICAN WELDING SOCIETY (AWS)

- | | |
|----------------|---|
| AWS D1.2/D1.2M | (2014; Errata 1 2014; Errata 2 2020) Structural Welding Code - Aluminum |
|----------------|---|

ASTM INTERNATIONAL (ASTM)

- | | |
|---------------|---|
| ASTM B32 | (2020) Standard Specification for Solder Metal |
| ASTM B221M | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric) |
| ASTM D41/D41M | (2011; R 2016) Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing |
| ASTM D2244 | (2021) Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates |
| ASTM D4214 | (2007; R 2015) Standard Test Method for |

Evaluating the Degree of Chalking of
Exterior Paint Films

ASTM D4586/D4586M

(2007; R 2018) Asphalt Roof Cement,
Asbestos-Free

NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)

NRCA 0429

(2022) The NRCA Roofing Manual:
Architectural Metal Flashing, Condensation
and Air Leakage Control and Reroofing

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION
(SMACNA)

SMACNA 1793

(2012) Architectural Sheet Metal Manual,
7th Edition

1.2 GENERAL REQUIREMENTS

Finished sheet metal assemblies must form a weathertight enclosure without waves, warps, buckles, fastening stresses or distortion, while allowing for expansion and contraction without damage to the system. The sheet metal installer is responsible for cutting, fitting, drilling, and other operations in connection with sheet metal modifications required to accommodate the work of other trades. Coordinate installation of sheet metal items used in conjunction with roofing with roofing work to permit continuous, uninterrupted roofing operations.

1.2.1 General Material Requirements

All materials specified in this Section installed in conjunction with the roofing system must be provided by the roofing system manufacturer, or by a manufacturer approved by the roofing system manufacturer for use in the roofing system, and must form a part of the Warranty as required by the applicable roofing system Section.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Exposed Sheet Metal Coverings; G

Gutters; G

Downspouts; G

Gravel Stops and fascia; G

Base Flashing; G

Counterflashing; G

Copings; G

SD-04 Samples

Finish Samples; G

SD-07 Certificates

Warranty on Finishes; G

SD-08 Manufacturer's Instructions

Instructions for Installation; G

Quality Control Plan; G

SD-10 Operation and Maintenance Data

Cleaning and Maintenance; G

1.4 MISCELLANEOUS REQUIREMENTS

1.4.1 Product Data

Indicate thicknesses, dimensions, fastenings, anchoring methods, expansion joints, and other provisions necessary for thermal expansion and contraction. Scaled manufacturer's catalog data may be submitted for factory fabricated items.

1.4.2 Finish Samples

Submit two color charts and two finish sample chips from manufacturer's standard color and finish options for each type of finish indicated.

1.4.3 Operation and Maintenance Data

Submit detailed [instructions for installation](#) and quality control during installation, [cleaning and maintenance](#), for each type of assembly indicated.

1.5 DELIVERY, HANDLING, AND STORAGE

Package and protect materials during shipment. Uncrate and inspect materials for damage, dampness, and wet-storage stains upon delivery to the job site. Remove from the site and replace damaged materials that cannot be restored to like-new condition. Handle sheet metal items to avoid damage to surfaces, edges, and ends. Store materials in dry, weather-tight, ventilated areas until installation.

PART 2 PRODUCTS

2.1 RECYCLED CONTENT

Provide products with recycled content. Provide data for each product with recycled content, identifying percentage of recycled content.

2.2 MATERIALS

Use any metal listed by [NRCA 0429](#) or [SMACNA 1793](#) for a particular item, unless otherwise indicated. Provide materials and configurations in

accordance with [NRCA 0429](#) or [SMACNA 1793](#) for each material, while also meeting the minimum thickness requirements specified in this Section. Different items need not be of the same metal, except that contact between dissimilar metals must be avoided.

Provide sheet metal items in [2400 to 3000 mm](#) lengths. Single pieces less than [2400 mm](#) long may be used to connect to factory-fabricated inside and outside corners, and at ends of runs. Factory fabricate corner pieces with minimum [300 mm](#) legs. Provide accessories and other items essential to complete the sheet metal installation. Provide accessories made of the same or compatible materials as the items to which they are applied. Fabricate sheet metal items of the materials specified below and to the gage, thickness, or weight shown in Table I at the end of this section. Provide sheet metal items with mill finish unless specified otherwise. Where more than one material is listed for a particular item in Table I, each is acceptable and may be used, except as follows:

2.2.1 [Exposed Sheet Metal](#) Items

Provide exposed sheet metal items of the same material. Consider the following as exposed sheet metal: gutters, including hangers; downspouts; [gravel stops and fascia](#); cap, valley, steeped, base, and eave flashings and related accessories.

2.2.2 Drainage

Do not use copper for an exposed item if drainage from that item will pass over exposed masonry, stonework or other metal surfaces. In addition to the metals listed in Table I, lead-coated copper may be used for such items.

2.2.3 Stainless Steel

Provide Type 302 or 304, 2D Finish, fully annealed, dead-soft temper, a minimum of [0.64 mm](#).

2.2.4 Aluminum Alloy Sheet and Plate

Provide in accordance with [UNE-EN 485-1](#) anodized color form alloy, and temper appropriate for use. Provide material not less than [0.813 mm](#) in thickness.

2.2.5 Finishes

Provide exposed exterior sheet metal and aluminum with a baked on, factory applied coil fluoropolymer color coating or approved equal siliconized polyester coating. Dry film thickness of coatings must be [0.020 to 0.033 mm](#). Color to be selected from manufacturer's full range of color choices. Field applications of color coatings are prohibited and will be rejected.

2.2.5.1 [Warranty on Finishes](#)

Provide a manufacturer's warranty to repair, or replace, sheet metal flashing and trim that shows evidence of deterioration of factory-applied finishes within 20 years from date of project substantial completion. Deterioration includes the following:

- a. Color fading more than 5 Delta E units when tested in accordance with

ASTM D2244.

- b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
- c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.

2.2.6 Aluminum Alloy, Extruded Bars, Rods, Shapes, and Tubes

ASTM B221M or UNE-EN 755-1.

2.2.7 Solder

Provide in accordance with ASTM B32, 95-5 tin-antimony.

2.2.8 Bituminous Plastic Cement

Provide in accordance with ASTM D4586/D4586M, Type I.

2.2.9 Asphalt Primer

Provide in accordance with ASTM D41/D41M or UNE 104231.

2.2.10 Fasteners

Use the same metal as, or a metal compatible with the item fastened. Use stainless steel fasteners to fasten. Confirm compatibility of fasteners and items to be fastened to avoid galvanic corrosion due to dissimilar materials.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Workmanship

Make lines and angles sharp and true. Free exposed surfaces from visible wave, warp, buckle, and tool marks. Fold back exposed edges neatly to form a 13 mm hem on the concealed side. Make sheet metal exposed to the weather watertight with provisions for expansion and contraction.

Make surfaces to receive sheet metal plumb and true, clean, even, smooth, dry, and free of defects and projections. For installation of items not shown in detail or not covered by specifications conform to the applicable requirements of SMACNA 1793, Architectural Sheet Metal Manual. Provide sheet metal flashing in the angles formed where roof decks abut walls, curbs, ventilators, pipes, or other vertical surfaces and wherever indicated and necessary to make the work watertight. Join sheet metal items together as shown in Table II.

3.1.2 Nailing

Confine nailing of sheet metal generally to sheet metal having a maximum width of 450 mm. Confine nailing of flashing to one edge only. Space nails evenly not over 75 mm on center and approximately 13 mm from edge unless otherwise specified or indicated. Face nailing is not permitted. Where sheet metal is applied to other than wood surfaces, include in shop drawings, the locations for sleepers and nailing strips required to secure

the work. Secure flashing at one-half the normal interval to ensure a wind-resistant installation.

3.1.3 Cleats

Provide cleats for sheet metal 450 mm and over in width. Space cleats evenly not over 300 mm on center unless otherwise specified or indicated. Unless otherwise specified, provide cleats of 50 mm wide by 75 mm long and of the same material and thickness as the sheet metal being installed. Secure one end of the cleat with two nails and the cleat folded back over the nailheads. Lock the other end into the seam. Where the fastening is to be made to concrete or masonry, use screws and drive in expansion shields set in concrete or masonry. Pre-tin cleats for soldered seams.

3.1.4 Bolts, Rivets, and Screws

Install bolts, rivets, and screws where indicated or required. Provide compatible washers where required to protect surface of sheet metal and to provide a watertight connection. Provide mechanically formed joints in aluminum sheets 1.0 mm or less in thickness.

3.1.5 Seams

Straight and uniform in width and height with no solder showing on the face.

3.1.5.1 Flat-lock Seams

Finish not less than 20 mm wide.

3.1.5.2 Lap Seams

Finish soldered seams not less than 25 mm wide. Overlap seams not soldered, not less than 75 mm.

3.1.5.3 Loose-Lock Expansion Seams

Not less than 75 mm wide; provide minimum 25 mm movement within the joint. Completely fill the joints with the specified sealant, applied at not less than 3 mm thick bed.

3.1.5.4 Standing Seams

Not less than 25 mm high, double locked without solder.

3.1.5.5 Flat Seams

Make seams in the direction of the flow.

3.1.6 Soldering

Where soldering is specified, apply to copper, terne-coated stainless steel, zinc-coated steel, and stainless steel items. Pre-tin edges of sheet metal before soldering is begun. Seal the joints in aluminum sheets of 1.0 mm or less in thickness with specified sealants. Do not solder aluminum.

3.1.6.1 Edges

Scrape or wire-brush the edges of lead-coated material to be soldered to produce a bright surface. Flux brush the seams in before soldering. Treat with soldering acid flux the edges of stainless steel to be pre-tinned. Seal the joints in aluminum sheets of 1.0 mm or less in thickness with specified sealants. Do not solder aluminum.

3.1.7 Welding and Mechanical Fastening

Use welding for aluminum of thickness greater than 1.0 mm. Aluminum 1.0 mm or less in thickness must be butted and the space backed with formed flashing plate; or lock joined, mechanically fastened, and filled with sealant as recommended by the aluminum manufacturer.

3.1.7.1 Welding of Aluminum

Use welding of the inert gas, shield-arc type. For procedures, appearance and quality of welds, and the methods used in correcting welding work, conform to AWS D1.2/D1.2M or UNE-EN ISO 3834-2.

3.1.7.2 Mechanical Fastening of Aluminum

Use No. 12, aluminum alloy, sheet metal screws or other suitable aluminum alloy or stainless steel fasteners. Drive fasteners in holes made with a No. 26 drill in securing side laps, end laps, and flashings. Space fasteners 300 mm maximum on center. Where end lap fasteners are required to improve closure, locate the end lap fasteners not more than 50 mm from the end of the overlapping sheet.

3.1.8 Protection from Contact with Dissimilar Materials

3.1.8.1 Copper or Copper-bearing Alloys

Paint with heavy-bodied bituminous paint surfaces in contact with dissimilar metal, or separate the surfaces by means of moistureproof building felts.

3.1.8.2 Aluminum

Do not allow aluminum surfaces in direct contact with other metals except stainless steel, zinc, or zinc coating. Where aluminum contacts another metal, paint the dissimilar metal with a primer followed by two coats of aluminum paint. Where drainage from a dissimilar metal passes over aluminum, paint the dissimilar metal with a non-lead pigmented paint. Aluminum may be used over concrete construction, provided that required reglets are of stainless steel and aluminum surface in contact with concrete or masonry is coated with bituminous paint or zinc chromate primer.

3.1.8.3 Metal Surfaces

Paint surfaces in contact with mortar, concrete, or other masonry materials with alkali-resistant coatings such as heavy-bodied bituminous paint.

3.1.8.4 Wood or Other Absorptive Materials

Paint surfaces that may become repeatedly wet and in contact with metal

with two coats of aluminum paint or a coat of heavy-bodied bituminous paint.

3.1.9 Expansion and Contraction

Provide expansion and contraction joints at not more than 9750 mm intervals for aluminum and at not more than 12 meter intervals for other metals. Provide an additional joint where the distance between the last expansion joint and the end of the continuous run is more than half the required interval. Space joints evenly. Join extruded aluminum gravel stops and fascia by expansion and contraction joints spaced not more than 3600 mm apart.

3.1.10 Base Flashing

Extend up vertical surfaces of the flashing not less than 200 mm and not less than 100 mm under the roof covering. Where finish wall coverings form a counterflashing, extend the vertical leg of the flashing up behind the applied wall covering not less than 150 mm. Overlap the flashing strips with the previously laid flashing not less than 75 mm. Fasten the strips at their upper edge to the deck. Solder end laps and provide for expansion and contraction. Extend the metal flashing over crickets at the up-slope side of vertical surfaces extending through sloping roofs, the metal flashings. Extend the metal flashings onto the roof covering not less than 115 mm at the lower side of vertical surfaces extending through the roof decks. Install and fit the flashings so as to be completely weathertight. Provide factory-fabricated base flashing for interior and exterior corners. Do not use metal base flashing on built-up roofing.

3.1.11 Counterflashing

Except where indicated or specified otherwise, insert counterflashing in reglets located from 230 to 250 mm above roof decks, extend down vertical surfaces over upturned vertical leg of base flashings not less than 75 mm. Fold the exposed edges of counterflashings 13 mm. Where stepped counterflashings are required, they may be installed in short lengths a minimum 200 mm by 200 mm or may be of the preformed single piece type. Provide end laps in counterflashings not less than 75 mm and make it weathertight with plastic cement. Do not make lengths of metal counterflashings exceed 3000 mm. Form flashings to the required shapes before installation. Factory form corners not less than 300 mm from the angle. Secure the flashings in the reglets with lead wedges and space not more than 450 mm apart; on short runs, place wedges closer together. Fill caulked-type reglets or raked joints which receive counterflashing with caulking compound. Turn up the concealed edge of counterflashings built into masonry or concrete walls not less than 6 mm and extend not less than 50 mm into the walls. Install counterflashing to provide a spring action against base flashing.

3.1.12 Gravel Stops and fascia

Prefabricate in the shapes and sizes indicated and in lengths not less than 2400 mm. Extend flange at least 100 mm onto roofing. Provide prefabricated, mitered corners internal and external corners. Install gravel stops and fascia after all plies of the roofing membrane have been applied, but before the flood coat of bitumen is applied. Prime roof flange of gravel stops and fascia on both sides with an asphalt primer. After primer has dried, set flange on roofing membrane and strip-in. Nail flange securely to wood nailer with large-head, barbed-shank roofing nails

38 mm long spaced not more than 75 mm on center, in two staggered rows.

3.1.12.1 Edge Strip

Hook the lower edge of fascia at least 20 mm over a continuous strip of the same material bent outward at an angle not more than 45 degrees to form a drip. Nail hook strip to a wood nailer at 150 mm maximum on center. Where fastening is made to concrete or masonry, use screws spaced 300 mm on center driven in expansion shields set in the concrete or masonry. Where horizontal wood nailers are slotted to provide for insulation venting, install strips to prevent obstruction of vent slots. Where necessary, install strips over 2 mm thick compatible spacer or washers.

3.1.12.2 Joints

Leave open the section ends of gravel stops and fascia 6 mm and backed with a formed flashing plate, mechanically fastened in place and lapping each section end a minimum of 100 mm set laps in plastic cement. Face nailing is not permitted. Install prefabricated aluminum gravel stops and fascia in accordance with the manufacturer's printed instructions and details.

3.1.13 Gutters

The hung type of shape indicated and supported on underside by brackets that permit free thermal movement of the gutter. Provide gutters in sizes indicated complete with mitered corners, end caps, outlets, brackets, and other accessories necessary for installation. Bead with hemmed edge or reinforce the outer edge of gutter with a stiffening bar not less than 20 by 5 mm of material compatible with gutter. Fabricate gutters in sections not less than 2400 mm. Lap the sections a minimum of 25 mm in the direction of flow or provide with concealed splice plate 150 mm minimum. Join the gutters, other than aluminum, by riveted and soldered joints. Join aluminum gutters with riveted sealed joints. Provide expansion-type slip joints midway between outlets. Support gutters on by continuous cleats or by cleats spaced not less than 900 mm apart. Adjust gutters to slope uniformly to outlets, with high points occurring midway between outlets. Fabricate hangers and fastenings from compatible metals.

3.1.14 Downspouts

Space supports for downspouts according to the manufacturer's recommendation for the masonry substrate. Types, shapes and sizes are indicated. Provide complete including elbows and offsets. Provide downspouts in approximately 3000 mm lengths. Provide end joints to telescope not less than 13 mm and lock longitudinal joints. Provide gutter outlets with wire ball strainers for each outlet. Provide strainers to fit tightly into outlets and be of the same material used for gutters. Keep downspouts not less than 25 mm away from walls. Fasten to the walls at top, bottom, and at an intermediate point not to exceed 1500 mm on center with leader straps or concealed rack-and-pin type fasteners. Form straps and fasteners of metal compatible with the downspouts.

3.1.14.1 Terminations

Neatly fit into the drainage connection the downspouts terminating in drainage lines and fill the joints with a portland cement mortar cap sloped away from the downspout. Provide downspouts terminating in splash

blocks with elbow-type fittings. Provide splash pans as specified.

3.1.15 Copings

Provide coping with locked and soldered seam. Terminate outer edges in edge strips. Install with sealed cover plate joints as indicated.

3.2 PAINTING

Touch ups in the field may be applied only after metal substrates have been cleaned and pretreated in accordance with manufacturer's written instructions and products.

Field-paint dissimilar sheet metals in contact to separate and deter galvanic interactions.

3.2.1 Aluminum Surfaces

Clean with solvent and apply one coat of zinc-molybdate primer and one coat of aluminum paint.

3.3 CLEANING

Clean exposed sheet metal work at completion of installation. Remove grease and oil films, handling marks, contamination from steel wool, fittings and drilling debris, and scrub-clean. Free the exposed metal surfaces of dents, creases, waves, scratch marks, and solder or weld marks.

3.4 REPAIRS TO FINISH

Scratches, abrasions, and minor surface defects of finish may be repaired in accordance with the manufacturer's printed instructions and as approved. Repair damaged surfaces caused by scratches, blemishes, and variations of color and surface texture. Replace items which cannot be repaired.

3.5 FIELD QUALITY CONTROL

Establish and maintain a [Quality Control Plan](#) for sheet metal used in conjunction with roofing to assure compliance of the installed sheet metalwork with the Contract requirements. Remove work that is not in compliance with the Contract and replace or correct. Include quality control, but not be limited to, the following:

- a. Observation of environmental conditions; number and skill level of sheet metal workers; condition of substrate.
- b. Verification that specified material is provided and installed.
- c. Inspection of sheet metalwork, for proper size(s) and thickness(es), fastening and joining, and proper installation.

3.5.1 Procedure

Submit for approval prior to start of roofing work. Include a checklist of points to be observed. Document the actual quality control observations and inspections. Provide a copy of the documentation to the Contracting Officer at the end of each day.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES					
Sheet Metal Items	Copper kilograms per square meter	Aluminum, mm	Stainless Steel, mm	Terne-Coated Stainless Steel, mm	Zinc-Coated Steel, mm
Building Expansion Joints					
Cover	4.9	1.60	0.635	0.635	0.6
Waterstop-bellows or flanged, U-type.	4.9	-	0.635	0.635	-
Covering on minor flat, pitched or curved surfaces	6.125	1.60	0.635	0.635	-
Downspouts and leaders	4.9	1.60	0.635	0.635	0.6
Downspout clips and anchors	-	1.02 clip 3.175 anchor	-	-	-
Downspout straps, 50 mm	14.7 (a)	1.52	1.27	-	-
Conductor heads	4.9	1.60	0.635	0.635	-
Scupper lining	6.125	1.60	0.635	0.635	-
Strainers, wire diameter or gage	4.0 gage	3.66 diameter	2.77 diameter	-	
Flashings:					
Base	6.125	1.60	0.635	0.635	0.6
Cap (Counter-flashing)	4.9	1.60	0.635	0.635	0.5
Eave	4.9	-	0.635	0.635	0.6
Spandrel beam	3.1	-	0.635	0.635	-
Bond barrier	4.9	-	0.635	0.635	-
Stepped	4.9	1.60	0.635	0.635	-
Valley	4.9	1.60	0.635	0.635	-
Roof drain	4.9 (b)				

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES					
Sheet Metal Items	Copper kilograms per square meter	Aluminum, mm	Stainless Steel, mm	Terne-Coated Stainless Steel, mm	Zinc-Coated Steel, mm
Pipe vent sleeve (d)					
Coping	4.9	-	-	-	-
Gravel stops and fascia:					
Extrusions	-	1.91	-	-	-
Sheets, corrugated	4.9	1.60	0.635	0.635	-
Sheets, smooth	6.125	1.60	0.635	0.635	0.6
Edge strip	7.35	1.60	0.635	-	-
Gutters:					
Gutter section	4.9	0.81	0.38	0.38	0.6
Continuous cleat	4.9	0.81	0.38	0.38	0.6
Hangers, dimensions	25 mm by 3 mm (a)	25 mm by 2 mm (c)	25 mm by 1 mm	-	-
Joint Cover plates (See Table II)	4.9	1.60	0.635	0.635	0.6
Reglets (c)	3.1	-	0.635	0.635	-
Splash pans	4.9	1.60	0.635	0.635	-
(a) Brass.					
(b) May be lead weighing 19.6 kilograms per square meter.					
(c) May be polyvinyl chloride.					
(d) 12.25 kilogram minimum lead sleeve with 100 mm flange. Where lead sleeve is impractical, refer to paragraph SINGLE PIPE VENTS for optional material.					

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

TABLE II. SHEET METAL JOINTS			
TYPE OF JOINT			
Item Designation	Copper, Terne-Coated Stainless Steel, Zinc-Coated Steel and Stainless Steel	Aluminum	Remarks
Joint cap for building expansion seam, cleated joint at roof	30 mm single lock, standing seam, cleated	30 mm single lock, standing	--
Flashings			
Base	25 mm 75 mm lap for expansion joint	25 mm flat locked, soldered; sealed; 75 mm lap for expansion joint	Aluminum manufacturer's recommended hard setting sealant for locked aluminum joints. Fill each metal expansion joint with a joint sealing compound.
Cap-in reglet	75 mm lap	75 mm lap	Seal groove with joint sealing compound.
Reglets	Butt joint	--	Seal reglet groove with joint sealing compound.
Eave	25 mm flat locked, cleated. 25 mm loose locked, sealed expansion joint, cleated.	25 mm flat locked, locked, cleated 25 mm loose locked, sealed expansion joints, cleated	Same as base flashing.
Stepped	75 mm lap	75 mm lap	--
Valley	150 mm lap cleated	150 mm lap cleated	--
Edge strip	Butt	Butt	--
Gravel stops:			

TABLE II. SHEET METAL JOINTS			
TYPE OF JOINT			
Item Designation	Copper, Terne-Coated Stainless Steel, Zinc-Coated Steel and Stainless Steel	Aluminum	Remarks
Extrusions	--	Butt with 13 mm space	Use sheet flashing beneath and a cover plate
Sheet, smooth	Butt with 6 mm space	Butt with 6 mm space	Use sheet flashing backup plate.
Sheet, corrugated	Butt with 6 mm space	Butt with 6 mm space	Use sheet flashing beneath and a cover plate or a combination unit
Gutters	40 mm lap, riveted and soldered	25 mm flat locked riveted and sealed	Aluminum producers recommended hard setting sealant for locked aluminum joints.
(a) Provide a 75 mm lap elastomeric flashing with manufacturer's recommended sealant.			
(b) Seal Polyvinyl chloride reglet with manufacturer's recommended sealant.			

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 84 00

FIRESTOPPING

05/10, CHG 1: 08/13

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 REFERENCES
- 1.3 SEQUENCING
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Installer
 - 1.5.2 Inspector Qualifications
- 1.6 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

- 2.1 FIRESTOPPING SYSTEM
- 2.2 FIRESTOPPING MATERIALS
 - 2.2.1 Fire Hazard Classification
 - 2.2.2 Toxicity
 - 2.2.3 Fire Resistance Rating
 - 2.2.3.1 Through-Penetrations
 - 2.2.3.1.1 Penetrations of Fire Resistance Rated Walls and Partitions
 - 2.2.3.2 Construction Joints and Gaps
 - 2.2.4 Material Certification

PART 3 EXECUTION

- 3.1 PREPARATION
- 3.2 INSTALLATION
 - 3.2.1 Insulated Pipes and Ducts
 - 3.2.2 Data and Communication Cabling
 - 3.2.2.1 Re-Enterable Devices
 - 3.2.2.2 Re-Sealable Products
- 3.3 INSPECTION
 - 3.3.1 Inspection Standards
 - 3.3.2 Inspection Reports

-- End of Section Table of Contents --

SECTION 07 84 00

FIRESTOPPING

05/10, CHG 1: 08/13

PART 1 GENERAL

1.1 SUMMARY

Furnish and install tested and listed firestopping systems, combination of materials, or devices to form an effective barrier against the spread of flame, smoke and gases, and maintain the integrity of fire resistance rated walls, partitions, floors, and ceiling-floor assemblies, including through-penetrations and construction joints and gaps.

- a. Through-penetrations include the annular space around pipes, tubes, conduit, wires, cables and vents.
- b. Construction joints include those used to accommodate expansion, contraction, wind, or seismic movement; firestopping material shall not interfere with the required movement of the joint.

Gaps requiring firestopping include gaps between the curtain wall and the floor slab and between the top of the fire-rated walls and the roof or floor deck above and at the intersection of shaft assemblies and adjoining fire resistance rated assemblies.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E84	(2023) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E119	(2020) Standard Test Methods for Fire Tests of Building Construction and Materials
ASTM E699	(2009) Standard Practice for Evaluation of Agencies Involved in Testing, Quality Assurance, and Evaluating of Building Components
ASTM E814	(2013a; R 2017) Standard Test Method for Fire Tests of Penetration Firestop Systems
ASTM E1399/E1399M	(1997; R 2017) Standard Test Method for Cyclic Movement and Measuring the Minimum and Maximum Joint Widths of Architectural Joint Systems
ASTM E1966	(2015; R 2019) Standard Test Method for

Fire-Resistive Joint Systems

- ASTM E2174 (2020a) Standard Practice for On-Site Inspection of Installed Firestop Systems
- ASTM E2307 (2020) Standard Test Method for Determining Fire Resistance of Perimeter Fire Barrier Systems Using Intermediate-Scale, Multi-story Test Apparatus
- ASTM E2393 (2020a) Standard Practice for On-Site Inspection of Installed Fire Resistive Joint Systems and Perimeter Fire Barriers

FM GLOBAL (FM)

- FM 4991 (2013) Approval of Firestop Contractors
- FM APP GUIDE (updated on-line) Approval Guide
<http://www.approvalguide.com/>

UNDERWRITERS LABORATORIES (UL)

- UL 723 (2018) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials
- UL 1479 (2015; Reprint May 2021) Fire Tests of Through-Penetration Firestops
- UL 2079 (2015; Reprint Jul 2020) Tests for Fire Resistance of Building Joint Systems
- UL Fire Resistance (2014) Fire Resistance Directory

1.3 SEQUENCING

Coordinate the specified work with other trades. Apply firestopping materials, at penetrations of pipes and ducts, prior to insulating, unless insulation meets requirements specified for firestopping. Apply firestopping materials, at building joints and construction gaps, prior to completion of enclosing walls or assemblies. Cast-in-place firestop devices shall be located and installed in place before concrete placement. Pipe, conduit or cable bundles shall be installed through cast-in-place device after concrete placement but before area is concealed or made inaccessible. Firestop material shall be inspected and approved prior to final completion and enclosing of any assemblies that may conceal installed firestop.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Firestopping System; G

SD-03 Product Data

Firestopping Materials; G

SD-06 Test Reports

Inspection; G

SD-07 Certificates

Inspector Qualifications

Firestopping Materials

1.5 QUALITY ASSURANCE

1.5.1 Installer

Engage an experienced Installer who is:

- a. FM Research approved in accordance with FM 4991, operating as a UL Certified Firestop Contractor, or

1.5.2 Inspector Qualifications

The inspector shall meet the criteria contained in ASTM E699 for agencies involved in quality assurance and shall have a minimum of two years experience in construction field inspections of firestopping systems, products, and assemblies. The inspector shall be completely independent of, and divested from, the installer, the manufacturer, and the supplier of any material or item being inspected. The inspector shall not be a competitor of the installer, the contractor, the manufacturer, or supplier of any material or item being inspected. Include in the qualifications submittal a notarized statement assuring compliance with the requirements stated herein.

1.6 DELIVERY, STORAGE, AND HANDLING

Deliver materials in the original unopened packages or containers showing name of the manufacturer and the brand name. Store materials off the ground, protected from damage and exposure to elements and temperatures in accordance with manufacturer requirements. Remove damaged or deteriorated materials from the site. Use materials within their indicated shelf life.

PART 2 PRODUCTS

2.1 FIRESTOPPING SYSTEM

Submit detail drawings including manufacturer's descriptive data, typical details conforming to UL Fire Resistance or other details certified by another nationally recognized testing laboratory, installation instructions or UL listing details for a firestopping assembly in lieu of fire-test data or report. For those firestop applications for which no UL tested system is available through a manufacturer, a manufacturer's engineering judgment, derived from similar UL system designs or other tests, shall be submitted for review and approval prior to installation. Submittal must indicate the firestopping material to be provided for each

type of application. When more than a total of 5 penetrations and/or construction joints are to receive firestopping, provide drawings that indicate location, "F" "T" and "L" ratings, and type of application.

Also, submit a written report indicating locations of and types of penetrations and types of firestopping used at each location; record type by UL list printed numbers.

2.2 FIRESTOPPING MATERIALS

Provide firestopping materials, supplied from a single domestic manufacturer, consisting of commercially manufactured, asbestos-free, nontoxic products FM APP GUIDE approved, or UL listed, for use with applicable construction and penetrating items, complying with the following minimum requirements:

2.2.1 Fire Hazard Classification

Material shall have a flame spread of 25 or less, and a smoke developed rating of 50 or less, when tested in accordance with ASTM E84 or UL 723. Material shall be an approved firestopping material as listed in UL Fire Resistance or by a nationally recognized testing laboratory.

2.2.2 Toxicity

Material shall be nontoxic and carcinogen free to humans at all stages of application or during fire conditions and shall not contain hazardous chemicals or require harmful chemicals to clean material or equipment.

2.2.3 Fire Resistance Rating

Firestop systems shall be UL Fire Resistance listed or FM APP GUIDE approved with "F" rating at least equal to fire-rating of fire wall or floor in which penetrated openings are to be protected. Where required, firestop systems shall also have "T" rating at least equal to the fire-rated floor in which the openings are to be protected.

2.2.3.1 Through-Penetrations

Firestopping materials for through-penetrations, as described in paragraph SUMMARY, shall provide "F", "T" and "L" fire resistance ratings in accordance with ASTM E814 or UL 1479. Fire resistance ratings shall be as follows:

2.2.3.1.1 Penetrations of Fire Resistance Rated Walls and Partitions

F Rating = 3 hour Rating of wall or partition being penetrated.

2.2.3.2 Construction Joints and Gaps

Fire resistance ratings of construction joints, as described in paragraph SUMMARY, and gaps such as those between floor slabs and curtain walls shall be the same as the construction in which they occur. Construction joints and gaps shall be provided with firestopping materials and systems that have been tested in accordance with ASTM E119, ASTM E1966 or UL 2079 to meet the required fire resistance rating. Curtain wall joints shall be provided with firestopping materials and systems that have been tested in accordance with ASTM E2307 to meet the required fire resistance rating. Systems installed at construction joints shall meet the cycling

requirements of [ASTM E1399/E1399M](#) or [UL 2079](#). All joints at the intersection of the top of a fire resistance rated wall and the underside of a fire-rated floor, floor ceiling, or roof ceiling assembly shall provide a minimum class II movement capability.

2.2.4 Material Certification

Submit certificates attesting that firestopping material complies with the specified requirements. For all intumescent firestop materials used in through penetration systems, manufacturer shall provide certification of compliance with [UL 1479](#).

PART 3 EXECUTION

3.1 PREPARATION

Areas to receive firestopping must be free of dirt, grease, oil, or loose materials which may affect the fitting or fire resistance of the firestopping system. For cast-in-place firestop devices, formwork or metal deck to receive device prior to concrete placement must be sound and capable of supporting device. Prepare surfaces as recommended by the manufacturer.

3.2 INSTALLATION

Completely fill void spaces with firestopping material regardless of geometric configuration, subject to tolerance established by the manufacturer. Firestopping systems for filling floor voids [100 mm](#) or more in any direction must be capable of supporting the same load as the floor is designed to support or be protected by a permanent barrier to prevent loading or traffic in the firestopped area. Install firestopping in accordance with manufacturer's written instructions. Provide tested and listed firestop systems in the following locations, except in floor slabs on grade:

- a. Penetrations of duct, conduit, tubing, cable and pipe through floors and through fire-resistance rated walls, partitions, and ceiling-floor assemblies.
- b. Penetrations of vertical shafts such as pipe chases, elevator shafts, and utility chutes.
- c. Gaps at the intersection of floor slabs and curtain walls, including inside of hollow curtain walls at the floor slab.
- d. Gaps at perimeter of fire-resistance rated walls and partitions, such as between the top of the walls and the bottom of roof decks.
- e. Construction joints in floors and fire rated walls and partitions.
- f. Other locations where required to maintain fire resistance rating of the construction.

3.2.1 Insulated Pipes and Ducts

Thermal insulation shall be cut and removed where pipes or ducts pass through firestopping, unless insulation meets requirements specified for firestopping. Replace thermal insulation with a material having equal thermal insulating and firestopping characteristics.

3.2.2 Data and Communication Cabling

Cabling for data and communication applications shall be sealed with re-enterable firestopping products and devices as indicated.

3.2.2.1 Re-Enterable Devices

Firestopping devices shall be pre-manufactured modular devices, containing built-in self-sealing intumescent inserts. Firestopping devices shall allow for cable moves, additions or changes without the need to remove or replace any firestop materials. Devices must be capable of maintaining the fire resistance rating of the penetrated membrane at 0 percent to 100 percent visual fill of penetrants; while maintaining "L" rating of <10 cfm/sf measured at ambient temperature and 205 degrees C at 0 percent to 100 percent visual fill.

3.2.2.2 Re-Sealable Products

Provide firestopping pre-manufactured modular products, containing self-sealing intumescent inserts. Firestopping products shall allow for cable moves, additions or changes. Devices shall be capable of maintaining the fire resistance rating of the penetrated membrane at 0 percent to 100 percent visual fill of penetrants.

3.3 INSPECTION

For Navy projects, install one of each type of penetration and have it inspected and accepted by the Atlantic Division, Naval Facilities Engineering Command, Fire Protection Engineer prior to the installation of the remainder of the penetrations. At this inspection, the manufacturer's technical representative of the firestopping material shall be present.

3.3.1 Inspection Standards

Inspect all firestopping in accordance with ASTM E2393 and ASTM E2174 for firestop inspection, and document inspection results to be submitted.

3.3.2 Inspection Reports

Submit inspection report stating that firestopping work has been inspected and found to be applied according to the manufacturer's recommendations and the specified requirements.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 92 00

JOINT SEALANTS

08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 PRODUCT DATA
- 1.4 ENVIRONMENTAL CONDITIONS
- 1.5 DELIVERY AND STORAGE
- 1.6 QUALITY ASSURANCE
 - 1.6.1 Compatibility with Substrate
 - 1.6.2 Joint Tolerance
 - 1.6.3 Mock-Up
 - 1.6.4 Adhesion

PART 2 PRODUCTS

- 2.1 SEALANTS
 - 2.1.1 Interior Sealants
 - 2.1.2 Exterior Sealants
 - 2.1.3 Floor Joint Sealants
- 2.2 PRIMERS
- 2.3 BOND BREAKERS
- 2.4 BACKSTOPS
 - 2.4.1 Rubber
 - 2.4.2 Butyl Rubber Based
- 2.5 CAULKING
- 2.6 CLEANING SOLVENTS

PART 3 EXECUTION

- 3.1 FIELD QUALITY CONTROL
- 3.2 SURFACE PREPARATION
 - 3.2.1 Steel Surfaces
 - 3.2.2 Aluminum or Bronze Surfaces
 - 3.2.3 Concrete and Masonry Surfaces
 - 3.2.4 Wood Surfaces
- 3.3 SEALANT PREPARATION
- 3.4 APPLICATION
 - 3.4.1 Joint Width-To-Depth Ratios
 - 3.4.2 Unacceptable Sealant Use
 - 3.4.3 Masking Tape
 - 3.4.4 Backstops
 - 3.4.5 Primer
 - 3.4.6 Bond Breaker
 - 3.4.7 Sealants
- 3.5 PROTECTION AND CLEANING
 - 3.5.1 Protection

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

3.5.2 Final Cleaning

-- End of Section Table of Contents --

SECTION 07 92 00

JOINT SEALANTS
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM C1193	(2016) Standard Guide for Use of Joint Sealants
ASTM C1311	(2022) Standard Specification for Solvent Release Sealants
ASTM C1521	(2019; R 2020) Standard Practice for Evaluating Adhesion of Installed Weatherproofing Sealant Joints
ASTM D1056	(2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber
ASTM D2452	(2015; R 2019) Standard Test Method for Extrudability of Oil- and Resin-Base Caulking Compounds
ASTM D2453	(2015; R 2020; E 2020) Standard Test Method for Shrinkage and Tenacity of Oil- and Resin-Base Caulking Compounds

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350	(2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers
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SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168	(2017) Adhesive and Sealant Applications
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Sealants; G

Primers; G

Bond Breakers; G

Backstops; G

Caulking; G

Cleaning Solvents; G

1.3 PRODUCT DATA

Include storage requirements, shelf life, curing time, instructions for mixing and application, and accessories. Provide manufacturer's Safety Data Sheets (SDS) for each solvent, primer and sealant material proposed.

1.4 ENVIRONMENTAL CONDITIONS

Apply sealant when the ambient temperature is between 4 and 32 degrees C.

1.5 DELIVERY AND STORAGE

Deliver materials to the jobsite in unopened manufacturers' sealed shipping containers, with brand name, date of manufacture, and material designation clearly marked thereon. Label elastomeric sealant containers to identify type, class, grade, and use. Handle and store materials in accordance with manufacturer's printed instructions. Prevent exposure to foreign materials or subjection to sustained temperatures exceeding 32 degrees C or lower than 4 degrees C. Keep materials and containers closed and separated from absorptive materials such as wood and insulation.

1.6 QUALITY ASSURANCE

1.6.1 Compatibility with Substrate

Verify that each sealant is compatible for use with each joint substrate in accordance with sealant manufacturer's printed recommendations for each application.

1.6.2 Joint Tolerance

Provide joint tolerances in accordance with manufacturer's printed instructions.

1.6.3 Mock-Up

Provide a mock-up of each type of sealant using materials, colors, and techniques approved for use on the project. Approved mock-ups may be incorporated into the work.

1.6.4 Adhesion

Provide in accordance with ASTM C1193 or ASTM C1521.

PART 2 PRODUCTS

2.1 SEALANTS

Provide sealant products that have been tested, found suitable, and documented as such by the manufacturer for the particular substrates to which they will be applied.

In areas with ambient temperatures that exceed **43.33 degrees C**, do not use polybutene, bituminous, acrylic-latex, polyvinyl acetate latex sealants, polychloroprene (neoprene), polyvinyl chloride (PVC), and polyurethane foams, and neoprene, PVC, and styrene butadiene rubber extruded seals and closure strips due to these materials having maximum recommended surface temperature ranges from **54.44 degrees C to 82.22 degrees C**.

2.1.1 Interior Sealants

Provide fuel-resistant **ASTM C920**, Type S or M, Grade NS, Class 12.5, Use NT. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of **CDPH SECTION 01350** (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of **SCAQMD Rule 1168**. Provide certification or validation of indoor air quality for interior sealants. Location(s) and color(s) of sealant for the following (Note, color "as selected" refers to manufacturer's full range of color options):

LOCATION	COLOR
a. Small voids between walls or partitions and adjacent lockers, casework, shelving, door frames, built-in or surface mounted equipment and fixtures, and similar items.	White
b. Perimeter of frames at doors, windows, and access panels which adjoin exposed interior concrete and masonry surfaces.	White
c. Joints of interior masonry walls and partitions which adjoin columns, pilasters, concrete walls, and exterior walls unless otherwise detailed.	White
d. Joints between edge members for acoustical tile and adjoining vertical surfaces.	White
e. Interior locations, not otherwise indicated or specified, where small voids exist between materials specified to be painted.	White
f. Joints between bathtubs and ceramic tile; joints between shower receptors and ceramic tile; joints formed where non-planar tile surfaces meet.	
g. Joints formed between tile floors and tile base cove; joints between tile and dissimilar materials; joints occurring where substrates change.	

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

LOCATION	COLOR
h. Behind escutcheon plates at valve pipe penetrations and showerheads in showers.	

2.1.2 Exterior Sealants

For joints in vertical surfaces, provide fuel-resistant **ASTM C920**, Type S or M, Grade NS, Class 25, Use NT. For joints in horizontal surfaces, provide fuel-resistant **ASTM C920**, Type S or M, Grade P, Class 25, Use T. Provide location(s) and color(s) of sealant as follows (Note, color "as selected" refers to manufacturer's full range of color options):

LOCATION	COLOR
a. Joints and recesses formed where frames and subsills of windows, doors, louvers, and vents adjoin masonry, concrete, or metal frames. Use sealant at both exterior and interior surfaces of exterior wall penetrations.	Match adjacent surface color
b. Joints between new and existing exterior masonry walls.	
c. Masonry joints where shelf angles occur.	
d. Joints in wash surfaces of stonework.	
e. Expansion and control joints.	Match adjacent surface color
f. Interior face of expansion joints in exterior concrete or masonry walls where metal expansion joint covers are not required.	Match adjacent surface color
g. Voids where items pass through exterior walls.	Match adjacent surface color
h. Metal reglets, where flashing is inserted into masonry joints, and where flashing is penetrated by coping dowels.	
i. Metal-to-metal joints where sealant is indicated or specified.	Match adjacent surface color
j. Joints between ends of gravel stops, fascia, copings, and adjacent walls.	Match adjacent surface color

2.1.3 Floor Joint Sealants

Fuel-resistant [ASTM C920](#), Type S or M, Grade P, Class 25, Use T. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of [CDPH SECTION 01350](#) (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of [SCAQMD Rule 1168](#). Provide certification or validation of indoor air quality for interior floor joint sealants. Provide location(s) and color(s) of sealant as follows (Note, color "as selected" refers to manufacturer's full range of color options):

LOCATION	COLOR
a. Seats of metal thresholds for exterior doors.	Gray
b. Control and expansion joints in floors, slabs, ceramic tile, and walkways.	Gray

2.2 PRIMERS

Non-staining, quick drying type and consistency as recommended by the sealant manufacturer for the particular application. Provide primers for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

2.3 BOND BREAKERS

Type and consistency as recommended by the sealant manufacturer to prevent adhesion of the sealant to the backing or to the bottom of the joint. Provide bond breakers for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

2.4 BACKSTOPS

Provide glass fiber roving, neoprene, butyl, polyurethane, or polyethylene foams free from oil or other staining elements as recommended by sealant manufacturer. Provide 25 to 33 percent oversized backing for closed cell and 40 to 50 percent oversized backing for open cell material, unless otherwise indicated. Provide backstop material that is compatible with sealant. Do not use oakum or other types of absorptive materials as backstops.

2.4.1 Rubber

Provide in accordance with [ASTM D1056](#), Type 2, closed cell, Class B, round cross section for cellular rubber sponge backing.

2.4.2 Butyl Rubber Based

Provide in accordance with [ASTM C1311](#), from a single component, with solvent release. color as selected from manufacturer's full range of color choices.

2.5 CAULKING

For interior use and only where there is little or no anticipated joint movement. Provide in accordance with [ASTM D2452](#) and [ASTM D2453](#). Provide products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of [CDPH SECTION 01350](#) (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of [SCAQMD Rule 1168](#). Provide certification or validation of indoor air quality for interior caulking.

2.6 CLEANING SOLVENTS

Provide type(s) recommended by the sealant manufacturer and in accordance with environmental requirements herein. Protect adjacent aluminum and bronze surfaces from solvents. Provide solvents for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

PART 3 EXECUTION

3.1 FIELD QUALITY CONTROL

Perform a field adhesion test in accordance with manufacturer's instructions and [ASTM C1193](#), Method A or ASTM C1521, Method A, Tail Procedure. Remove sealants that fail adhesion testing; clean substrates, reapply sealants, and re-test. Test sealants adjacent to failed sealants. Submit field adhesion test report indicating tests, locations, dates, results, and remedial actions taken.

3.2 SURFACE PREPARATION

Prepare surfaces according to manufacturer's printed installation instructions. Clean surfaces from dirt, frost, moisture, grease, oil, wax, lacquer, paint, or other foreign matter that would destroy or impair adhesion. Remove oil and grease with solvent; thoroughly remove solvents prior to sealant installation. Wipe surfaces dry with clean cloths. When resealing an existing joint, remove existing caulk or sealant prior to applying new sealant. For surface types not listed below, provide in accordance with sealant manufacturer's printed instructions for each specific surface.

3.2.1 Steel Surfaces

Remove loose mill scale by sandblasting or, if sandblasting is impractical or would damage finished work, scraping and wire brushing. Remove protective coatings by sandblasting or using a residue free solvent. Remove resulting debris and solvent residue prior to sealant installation.

3.2.2 Aluminum or Bronze Surfaces

Remove temporary protective coatings from surfaces that will be in contact with sealant. When masking tape is used as a protective coating, remove tape and residual adhesive prior to sealant application. For removing protective coatings and final cleaning, use non-staining solvents recommended by the manufacturer of the item(s) containing aluminum or bronze surfaces.

3.2.3 Concrete and Masonry Surfaces

Where surfaces have been treated with curing compounds, oil, or other such materials, remove materials by sandblasting or wire brushing. Remove laitance, efflorescence and loose mortar from the joint cavity. Remove resulting debris prior to sealant installation.

3.2.4 Wood Surfaces

Ensure wood surfaces that will be in contact with sealants are free of splinters, sawdust and other loose particles.

3.3 SEALANT PREPARATION

Do not add liquids, solvents, or powders to sealants. Mix multicomponent elastomeric sealants in accordance with manufacturer's printed instructions.

3.4 APPLICATION

3.4.1 Joint Width-To-Depth Ratios

Acceptable Ratios:

<u>JOINT WIDTH</u>	<u>JOINT DEPTH</u>	
	Minimum	Maximum
For metal, glass, or other nonporous surfaces:		
6 mm (minimum)	6 mm	6 mm
over 6 mm	1/2 of width	Equal to width
For wood, concrete, masonry, stone, or :		
6 mm (minimum)	6 mm	6 mm
over 6 mm to 13 mm	6 mm	Equal to width
over 13 mm to 25 mm	50 mm	16 mm
Over 25 mm	prohibited	

Unacceptable Ratios: Where joints of acceptable width-to-depth ratios have not been provided, clean out joints to acceptable depths and grind or cut to acceptable widths without damage to the adjoining work. Grinding is prohibited at metal surfaces.

3.4.2 Unacceptable Sealant Use

Do not install sealants in lieu of other required building enclosure weatherproofing components such as flashing, drainage components, and joint closure accessories, or to close gaps between walls, floors, roofs, windows, and doors, that exceed acceptable installation tolerances. Remove sealants that have been used in an unacceptable manner and correct

building enclosure deficiencies to comply with contract documents requirements.

3.4.3 Masking Tape

Place masking tape on the finished surface on one or both sides of joint cavities to protect adjacent finished surfaces from primer or sealant smears. Remove masking tape within 10 minutes of joint filling and tooling.

3.4.4 Backstops

Provide backstops dry and free of tears or holes. Tightly pack the back or bottom of joint cavities with backstop material to provide joints in specified depths. Provide backstops where indicated and where backstops are not indicated but joint cavities exceed the acceptable maximum depths specified in JOINT WIDTH-TO-DEPTH RATIOS Table.

3.4.5 Primer

Clean out loose particles from joints immediately prior to application of. Apply primer to joints in concrete masonry units, wood, and other porous surfaces in accordance with sealant manufacturer's printed instructions. Do not apply primer to exposed finished surfaces.

3.4.6 Bond Breaker

Provide bond breakers to surfaces not intended to bond in accordance with, sealant manufacturer's printed instructions for each type of surface and sealant combination specified.

3.4.7 Sealants

Provide sealants compatible with the material(s) to which they are applied. Do not use a sealant that has exceeded its shelf life or has jelled and cannot be discharged in a continuous flow from the sealant gun. Apply sealants in accordance with the manufacturer's printed instructions with a gun having a nozzle that fits the joint width. Work sealant into joints so as to fill the joints solidly without air pockets. Tool sealant after application to ensure adhesion. Apply sealant uniformly smooth and free of wrinkles. Upon completion of sealant application, roughen partially filled or unfilled joints, apply additional sealant, and tool smooth as specified. Apply sealer over sealants in accordance with the sealant manufacturer's printed instructions.

3.5 PROTECTION AND CLEANING

3.5.1 Protection

Protect areas adjacent to joints from sealant smears. Masking tape may be used for this purpose if removed 5 to 10 minutes after the joint is filled and no residual tape marks remain.

3.5.2 Final Cleaning

Upon completion of sealant application, remove remaining smears and stains and leave the work in a clean and neat condition.

- a. Masonry and Other Porous Surfaces: Immediately remove fresh sealant

that has been smeared on adjacent masonry, rub clean with a solvent, and remove solvent residue, in accordance with sealant manufacturer's printed instructions. Allow excess sealant to cure for 24 hours then remove by wire brushing or sanding. Remove resulting debris.

- b. Metal and Other Non-Porous Surfaces: Remove excess sealant with a solvent moistened cloth. Remove solvent residue in accordance with solvent manufacturer's printed instructions.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 08 - OPENINGS

SECTION 08 51 23

STEEL WINDOWS

08/20, CHG 1: 02/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY ASSURANCE
 - 1.3.1 Shop Drawing Information
 - 1.3.2 Color Coating Samples Information
- 1.4 TEST REPORT REQUIREMENTS
 - 1.4.1 Air and Water Infiltration
 - 1.4.2 Blast Testing
- 1.5 WINDOW PERFORMANCE
 - 1.5.1 Structural Performance
 - 1.5.2 Thermal Performance
- 1.6 WARRANTY
- 1.7 DELIVERY AND STORAGE

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 General System Requirements
 - 2.1.2 Steel Bars
 - 2.1.3 Sheet Steel
 - 2.1.4 Zinc-Coated Sheet Steel
 - 2.1.5 Zinc Coating
 - 2.1.6 Screws and Bolts
- 2.2 FABRICATION OF WINDOWS
- 2.3 PROVISIONS FOR GLAZING
- 2.4 METAL-TO-METAL JOINTS
- 2.5 ACCESSORIES
 - 2.5.1 Anchors
 - 2.5.2 Weatherstripping
 - 2.5.3 Hardware
 - 2.5.3.1 Hardware Materials and Finish
 - 2.5.4 Fasteners
 - 2.5.5 Metal Sub-frames and Stools
- 2.6 GLASS AND GLAZING
- 2.7 WINDOW FINISH
 - 2.7.1 Shop Primed Finish
 - 2.7.2 Factory Finish
- 2.8 WINDOW TYPES
 - 2.8.1 Awning Windows
 - 2.8.1.1 Operators
 - 2.8.1.2 Ventilators
- 2.9 SCREENS
 - 2.9.1 Construction
 - 2.9.2 Insect Screening

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

PART 3 EXECUTION

- 3.1 INSTALLATION
- 3.2 ANCHORS AND FASTENINGS
- 3.3 OPERATORS
- 3.4 WEATHERSTRIPPING
- 3.5 ADJUSTMENTS AFTER INSTALLATION
- 3.6 CLEANING

-- End of Section Table of Contents --

SECTION 08 51 23

STEEL WINDOWS
08/20, CHG 1: 02/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

- | | |
|------------------|---|
| UNE-EN 10051 | (2012) Continuously hot-rolled strip and plate/sheet cut from wide strip of non-alloy and alloy steels - Tolerances on dimensions and shape |
| UNE-EN 10143 | (2007) Continuously Hot Dip Coated Steel Sheet and Strip - Tolerances on Dimensions and Shape |
| UNE-EN 12207 | (2017) Windows And Doors - Air Permeability - Classification. |
| UNE-EN ISO 898-1 | (2015) Mechanical properties of fasteners made of carbon steel and alloy steel - Part 1: Bolts, screws and studs with specified property classes - Coarse thread and fine pitch thread (ISO 898-1:2013) |
| UNE-EN ISO 1461 | (2010) Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods (ISO1461:1999) |

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

- | | |
|------------------------------|---|
| AAMA/WDMA/CSA 101/I.S.2/A440 | (2017) North American Fenestration Standard/Specification for Windows, Doors, and Skylights |
|------------------------------|---|

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

- | | |
|--------------|---|
| ASME B18.6.3 | (2013; R 2017) Machine Screws, Tapping Screws, and Machine Drive Screws (Inch Series) |
|--------------|---|

ASTM INTERNATIONAL (ASTM)

- | | |
|-----------------|---|
| ASTM A123/A123M | (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products |
| ASTM A653/A653M | (2023) Standard Specification for Steel |

Sheet, Zinc-Coated (Galvanized) or
Zinc-Iron Alloy-Coated (Galvannealed) by
the Hot-Dip Process

ASTM A1011/A1011M

(2023) Standard Specification for Steel
Sheet and Strip, Hot-Rolled, Carbon,
Structural, High-Strength Low-Alloy,
High-Strength Low-Alloy with Improved
Formability, and Ultra-High Strength

ASTM D3656/D3656M

(2013) Standard Specification for Insect
Screening and Louver Cloth Woven from
Vinyl-Coated Glass Yarns

ASTM E283

(2019) Standard Test Method for
Determining the Rate of Air Leakage
Through Exterior Windows, Curtain Walls,
and Doors Under Specified Pressure
Differences Across the Specimen

ASTM E331

(2000; R 2016) Standard Test Method for
Water Penetration of Exterior Windows,
Skylights, Doors, and Curtain Walls by
Uniform Static Air Pressure Difference

ASTM F1642/F1642M

(2017) Standard Test Method for Glazing
and Glazing Systems Subject to Airblast
Loadings

NATIONAL FENESTRATION RATING COUNCIL (NFRC)

NFRC 100

(2020) Procedure for Determining
Fenestration Product U-Factors

NFRC 200

(2020) Procedure for Determining
Fenestration Product Solar Heat Gain
Coefficient and Visible Transmittance at
Normal Incidence

STEEL WINDOW INSTITUTE (SWI)

SWI SWS

(2017; R 2018) Steel Window Specifications

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Windows

SD-03 Product Data

Steel Framing Materials

Hardware

Hardware Materials

Fasteners

Accessories

Operators

Screens

SD-04 Samples

Color Coating; G

SD-06 Test Reports

Air Infiltration

Water Infiltration

Minimum Condensation Resistance Factor

Resistance to Forced Entry

Standard Airblast Test; G

SD-10 Operation and Maintenance Data

Windows, Data Package 1; G

1.3 QUALITY ASSURANCE

1.3.1 Shop Drawing Information

Indicate elevations of windows, full-size sections, thicknesses and gages of metal, fastenings, proposed method of anchoring, size and spacing of anchors, details of construction, method of glazing, details of operating hardware, other related items, and installation details.

1.3.2 Color Coating Samples Information

Submit chart of manufacturer's color coatings if factory finish is to be provided in lieu of field painting.

1.4 TEST REPORT REQUIREMENTS

1.4.1 Air and Water Infiltration

ASTM E283 and ASTM E331 or UNE-EN 12207. Do not exceed maximum air infiltration of 0.05 cubic meter per minute per meter of crack length when subjected to a static pressure of 75 Pa (equivalent to a wind velocity of 40 kilometers per hour (kph)). Water infiltration must be "zero."

1.4.2 Blast Testing

Submit test reports for each type of window attesting that identical windows have been tested and meet the requirements specified herein for conformance to AAMA/WDMA/CSA 101/I.S.2/A440 including test size, minimum

condensation resistance factor (CRF), and resistance to forced entry, and, for Antiterrorism windows, in lieu of a Design Analysis, results of a Standard Airblast Test. For Antiterrorism windows, in lieu of a Design Analysis, results of airblast testing, whether by arena test or shock tube, must be included in a test report, providing information in accordance with ASTM F1642/F1642M, as prepared by the independent testing agency performing the test. The test results must demonstrate the ability of each window proposed for use to withstand the airblast loading parameters and achieve the hazard level rating specified in paragraph STANDARD AIRBLAST TEST METHOD.

1.5 WINDOW PERFORMANCE

Steel windows must meet the following performance requirements. Perform testing requirements by an independent testing laboratory or agency.

1.5.1 Structural Performance

Structural test pressures on window units must be for positive load (inward) and negative load (outward). After testing, there will be no glass breakage, permanent damage to fasteners, hardware parts, support arms or actuating mechanisms or any other damage which could cause window to be inoperable. There must be no permanent deformation of any main frame, sash or ventilator member in excess of the requirements established by AAMA/WDMA/CSA 101/I.S.2/A440 for the window types and classification specified in this section.

1.5.2 Thermal Performance

Non-residential glazed systems (including frames and glass) must be certified by the National Fenestration Rating Council with a whole-window Solar Heat Gain Coefficient (SHGC) maximum of 0.30 determined according to NFRC 200 procedures and a U-factor maximum of 3.4 W per square m by K in accordance with NFRC 100.

1.6 WARRANTY

Provide Manufacturer's standard performance guarantees or warranties that extend beyond a one year period.

1.7 DELIVERY AND STORAGE

Deliver to project site in undamaged condition. Store windows and components on edge, out of contact with the ground, under weathertight covering, and arranged to avoid bending, warping, or other damage.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 General System Requirements

Steel framing materials must contain a minimum of 40 percent total recycled content. Provide data identifying percentage of recycled content for steel framing materials.

2.1.2 Steel Bars

SWI SWS.

2.1.3 Sheet Steel

ASTM A1011/A1011M or UNE-EN 10051.

2.1.4 Zinc-Coated Sheet Steel

ASTM A653/A653M or UNE-EN 10143.

2.1.5 Zinc Coating

ASTM A123/A123M or UNE-EN ISO 1461.

2.1.6 Screws and Bolts

ASME B18.6.3 or UNE-EN ISO 898-1 as applicable.

2.2 FABRICATION OF WINDOWS

Form permanent joints by welding or mechanically fastening as specified for each type window. Use joints of strength to maintain structural value of members connected. Weld joints solid, remove excess metal, and dress smooth on exposed and contact surfaces. Closely fit joints formed with mechanical fastenings and make permanently watertight. Assemble frames and sash, including ventilators and thermal breaks, at the plant and ship as a unit with hardware unattached. Provide the following construction:

- a. Where fixed window sections adjoin ventilator sections, provide fixed sash, fabricated from similar frame members, and of manufacturer's standard type suitable for the purpose.
- b. Roll weathering surfaces integrally to provide two-point parallel-surface contact with overlap at both inside and outside points of closure.
- c. Provide drips and weep holes as required to return water to outside.
- d. Design glazed windows and rabbets suitable for glass thickness shown on drawings or specified.
- e. Use flathead, cross recessed type, exposed head screws and bolts with standard threads on windows, trim and accessories. Screw heads must finish flush with adjoining surfaces. Self tapping sheet-metal screws are not acceptable.
- f. For hot-dipped galvanized windows, use stainless steel or hot-spun galvanized steel fasteners. For windows with painted finish use electro-galvanized fasteners. Finish exposed heads to match finish of windows.

2.3 PROVISIONS FOR GLAZING

Design sash for inside glazing and for securing glass with metal beads and glazing compound. Where insulating glass is indicated, use rabbets of adequate weight and depth to receive and properly support glass and glazing accessories.

2.4 METAL-TO-METAL JOINTS

Set in mastic, using type recommended by window manufacturer to provide weathertight joints. Remove excess mastic before it hardens.

2.5 ACCESSORIES

Provide windows with hardware, clips, fins, anchors, glazing beads, and fastenings, necessary for complete installation and operation of ventilators.

2.5.1 Anchors

Use hot-dip galvanized steel anchors. Secure anchors and fastenings to heads, jambs, and sills of openings, and fasten securely to windows or frames. Use anchors recommended by window manufacturer for specific type of construction and conceal. Anchor each frame at jambs with minimum of three adjustable steel anchors.

2.5.2 Weatherstripping

Provide on all operable windows so that, when tested before leaving factory, in accordance with [ASTM E283](#), do not exceed a maximum air infiltration of [0.05 cubic meter per minute per meter](#) of crack length when subjected to static pressure of [75 Pa](#) equivalent to wind velocity of [40 kmh](#).

2.5.3 Hardware

Equip all operable sash with latching device which can be secured from inside. The item, type, and function of hardware required are specified under individual window type. Attach hardware securely to windows with corrosion resisting bolts or machine screws; do not use sheet metal screws. At fixed screens, adapt hardware to permit operation of ventilators. Fit and test hardware for each window at factory to ensure satisfactory operation and security.

2.5.3.1 Hardware Materials and Finish

Provide non-magnetic type stainless steel exposed hardware with satin finish. Use galvanized steel or malleable iron hinges, with nonferrous pins, or with steel pins and non-ferrous bushings or washers.

2.5.4 Fasteners

Stainless steel or aluminum materials Prime exposed heads of coated or plated fasteners and finish to match adjacent material.

2.5.5 Metal Sub-frames and Stools

Manufacturer's standard type designed to suit the particular window. Match exposed surfaces to windows.

2.6 GLASS AND GLAZING

Provide materials in accordance with Section [08 81 00](#) GLAZING.

2.7 WINDOW FINISH

2.7.1 Shop Primed Finish

After fabrication, clean all surfaces of windows, fins, mullions, cover plates, and screen frames and provide a hot-dip galvanized, phosphate-treated and shop primed finish. Conform to **SWI SWS** for the methods of cleaning, chemical treatment, galvanizing, and painting.

2.7.2 Factory Finish

In lieu of shop primed finish, factory finish may be provided using the following method, in which case finish field painting will not be required:

- a. Chemically clean and bonderize windows. Apply dip coat of epoxy primer baked on for not less than 15 minutes at not less than **149 degrees C**, followed by finish coat of alkyd-amine enamel of not less than **0.025 mm** thickness, baked on for 15 minutes at not less than **149 degrees C**.
- b. Finish **color coating** to be selected from manufacturer's standard color chart.
- c. Touch up abraded surfaces with enamel as specified for factory finish.

2.8 WINDOW TYPES

Conform to **SWI SWS**. Provide combinations, types and sizes indicated. Each window must consist of a unit including frame, sash, hardware, trim, insect screen, and anchors. Design windows indicated to have screen to accommodate items to be furnished.

2.8.1 Awning Windows

Provide compression-type weatherstripping. Heavy Intermediate materials in group of top-hinged or projected out-swinging ventilators:

2.8.1.1 Operators

Control must be simultaneous by means of cam-type lever handle fastener for hand push-pull operation. For windows with screens, provide with underscreen push bar operators. For operators more than **2 meters** above floor, provide with hardware designed for pole operation.

2.8.1.2 Ventilators

Support on two hinges and two arms, or on two steel slide arms pivoted to vent and to principal frame member. Provide bronze-brushed pivots and hinges with bronze pins. Design ventilators to close and weather on each other, or on independent meeting rails assembled as part of window frame. Provide for positive adjustment of individual vents to ensure positive contact between sash and frame when closed.

2.9 SCREENS

Provide one insect screen for each operable exterior sash or ventilator. Locate screen units either inside or outside, depending upon window type and method of operation. Provide full-length top-hung type screens. Design screens to fit closely around entire perimeter of ventilator or

opening, to be rewirable, easily removable from inside building, and interchangeable for same size ventilators of similar type windows, with minimum of exposed fasteners and latches. Provide all guides, stops, clips, bolts, and screws, as necessary, for a secure and insect-tight attachment to window. Where wickets are necessary, use sliding or hinged type, with friction catches, framed and trimmed for durability and tight fit. Provide wicket opening frames of similar material and cross-section as screen frames. Provide continuous framing bar between the two sides of screen frames.

2.9.1 Construction

Provide screen frames of steel with finish matching that of windows. Equip frames with removable splines of steel or vinyl. Form groove in frame for holding screen cloth in place with noncylindrical splines. Make spline and groove assembly so that cloth cannot be removed from groove by pressure on cloth. Make splines of such size and shape that rotation of spline in groove will be prevented and spline will tightly hold cloth in place.

2.9.2 Insect Screening

ASTM D3656/D3656M, Class 2, 18 by 14 mesh, color charcoal . Install with weave parallel to frames. Stretch tight for smooth appearance. Conceal edges in spline channels.

PART 3 EXECUTION

3.1 INSTALLATION

Install in accordance with window manufacturer's printed instructions and details. Coordinate installation with commissioning as specified in Section . Build in windows as work progresses or install without forcing into prepared window openings. Set at proper elevation, location, and reveal; plumb, square, level, and in alignment. Brace and stay to prevent distortion and misalignment. Protect ventilators and operating parts against dirt and building materials by keeping closed and locked to frame. Bed screws or bolts in sill members, joints at mullions, contacts of windows with sills, built-in fins, and subframes in mastic sealant recommended by window manufacturer. Install and seal windows in a manner that will prevent entrance of water and wind. Fasten insect screens securely in place.

Any materials that show visual evidence of biological growth due to the presence of moisture must not be installed on the building project.

3.2 ANCHORS AND FASTENINGS

Make provision for securing units to each other and to adjoining construction. Design head and jamb members to enter into masonry not less than 11 mm where windows are installed in direct contact with masonry. Where windows are set in prepared masonry openings, build in anchors and fastenings to jambs of openings and fasten securely to windows or frames and to adjoining construction. Space anchors not more than 450 mm apart on jambs and sills, and install a minimum of three anchors on each side of each opening. Anchors and fastenings must have sufficient strength to hold member firmly in position. Where type, size, or spacing of anchors is not shown or specified, use expansion or toggle bolts or screws as best suited to construction material. Provide expansion shield and bolt

assemblies of type designed to give holding power beyond tensile and shearing strength of bolt. Minimum fastener penetration must be not less than that recommended by manufacturer for type fastener and wall material involved.

3.3 OPERATORS

Install operators before glazing. Plumb and level shaft risers and runs. Adjust ventilators for free opening and tight closing. Secure housings and adjustable supports to wall. Anchor operator parts to steel window mullions with 13 mm bolts. Couple individual lengths of shafting with steel rivets or bolts. Leave mechanical equipment and ventilators in proper operating condition.

3.4 WEATHERSTRIPPING

Use bronze, spring-brass, or stainless steel and secure with non-ferrous screws. Secure weatherstripping or rubbing-blocks to parting-strip and each end of meeting-rails. For solid bar stock windows, use manufacturer's standard weatherstripping inserted into groove.

3.5 ADJUSTMENTS AFTER INSTALLATION

After installation of windows and completion of glazing and field painting, adjust all ventilators and hardware to operate smoothly and to provide weathertight sealing when ventilators are closed and locked. Lubricate hardware and operating parts. Adjust weatherstripping to assure weathertight contact with frames when ventilators are closed and locked. Weatherstripping must not cause binding of sash, or prevent closing and locking of ventilator. Verify products are properly installed, connected, and adjusted.

3.6 CLEANING

Clean interior and exterior surfaces of window units of mortar, plaster, paint spattering spots, and other foreign matter to present a neat appearance and to prevent fouling of weathering surfaces and weatherstripping, or interference with operation of hardware. Clean and touch up abraded surfaces. Replace with new windows any stained, discolored, or abraded windows that cannot be restored to original condition.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 08 - OPENINGS

SECTION 08 71 00

DOOR HARDWARE

02/16, CHG 4: 02/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 SHOP DRAWINGS
- 1.4 PRODUCT DATA
- 1.5 HARDWARE SCHEDULE
 - 1.5.1 Hardware Item List:
 - 1.5.2 Hardware Schedule
- 1.6 KEY BITTING CHART REQUIREMENTS
 - 1.6.1 Requirements
- 1.7 QUALITY ASSURANCE
 - 1.7.1 Hardware Manufacturers and Modifications
 - 1.7.2 Key Shop Drawings Coordination Meeting
- 1.8 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

- 2.1 TEMPLATE HARDWARE
- 2.2 HARDWARE ITEMS
 - 2.2.1 Hinges
 - 2.2.2 Continuous Hinges
 - 2.2.3 Pivots
 - 2.2.4 Locks and Latches
 - 2.2.4.1 Mortise Locks and Latches
 - 2.2.4.2 Auxiliary Locks
 - 2.2.5 Exit Devices
 - 2.2.6 Keying System
 - 2.2.7 Keys
 - 2.2.8 Door Bolts
 - 2.2.9 Closers
 - 2.2.9.1 Identification Marking
 - 2.2.10 Overhead Holders
 - 2.2.11 Door Stops and Silencers
 - 2.2.12 Weatherstripping Gasketing
 - 2.2.12.1 Extruded Aluminum Retainers
- 2.3 FASTENERS
- 2.4 FINISHES

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Weatherstripping Installation
 - 3.1.2 Threshold Installation
- 3.2 FIRE DOORS AND EXIT DOORS
- 3.3 HARDWARE LOCATIONS

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

3.4 FIELD QUALITY CONTROL

-- End of Section Table of Contents --

SECTION 08 71 00

DOOR HARDWARE
02/16, CHG 4: 02/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E283	(2019) Standard Test Method for Determining the Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen
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BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)

ANSI/BHMA A156.1	(2021) Butts and Hinges
ANSI/BHMA A156.3	(2020) Exit Devices
ANSI/BHMA A156.4	(2013) Door Controls - Closers
ANSI/BHMA A156.7	(2016) Template Hinge Dimensions
ANSI/BHMA A156.8	(2021) Door Controls - Overhead Stops and Holders
ANSI/BHMA A156.13	(2017) Mortise Locks & Latches Series 1000
ANSI/BHMA A156.16	(2018) Auxiliary Hardware
ANSI/BHMA A156.17	(2019) Self Closing Hinges & Pivots
ANSI/BHMA A156.18	(2020) Materials and Finishes
ANSI/BHMA A156.22	(2021) Gasketing
ANSI/BHMA A156.26	(2012) Continuous Hinges
ANSI/BHMA A156.36	(2010) Auxiliary Locks

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 72	(2022) National Fire Alarm and Signaling Code
NFPA 80	(2022) Standard for Fire Doors and Other Opening Protectives
NFPA 101	(2021) Life Safety Code

NFPA 252 (2022) Standard Methods of Fire Tests of
Door Assemblies

STEEL DOOR INSTITUTE (SDI/DOOR)

SDI/DOOR A250.8 (2017) Specifications for Standard Steel
Doors and Frames

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191 Americans with Disabilities Act (ADA)
Accessibility Guidelines for Buildings and
Facilities; Architectural Barriers Act
(ABA) Accessibility Guidelines

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Manufacturer's Detail Drawings; G

Hardware Schedule; G

Keying System; G

SD-03 Product Data

Hardware Items; G

SD-08 Manufacturer's Instructions

Installation

SD-10 Operation and Maintenance Data

Hardware Schedule Items, Data Package 1; G

SD-11 Closeout Submittals

Key Bitting

1.3 SHOP DRAWINGS

Submit manufacturer's detail drawings indicating all hardware assembly components and interface with adjacent construction.

1.4 PRODUCT DATA

Indicate fire-ratings at applicable components. Provide documentation of ABA/ADA accessibility compliance of applicable components, as required by 36 CFR 1191 Appendix D - Technical.

1.5 **HARDWARE SCHEDULE**

Provide Hardware Item List and Hardware Schedule containing the following information, and additional information as needed to identify the complete make up of each hardware set and its application to each opening:

1.5.1 Hardware Item List:

- a. Hardware Type
- b. Item Number
- c. Quantity
- d. Size(s)
- e. Reference Publication / Type Number
- f. Manufacturer's Name / Catalog Number
- g. Key Control Symbols
- h. UL Mark (If fire rated and listed)
- i. BHMA Finish(es)
- j. Remarks

1.5.2 Hardware Schedule

- a. Hardware Set Number
- b. Opening Number(s)
- c. Opening Description (single/double leaf, hand, size, door/frame material)
- d. Fire Rating
- e. Sound Rating
- f. Hardware Items
- g. Quantity
- h. Size
- i. BHMA Finish
- j. Remarks

In addition, submit hardware schedule data package 1 in accordance with Section **01 78 23** OPERATION AND MAINTENANCE DATA.

1.6 KEY BITTING CHART REQUIREMENTS

1.6.1 Requirements

Submit [key bitting](#) charts to the Contracting Officer prior to completion of the work. Include:

- a. Complete listing of all keys (e.g. AA1 and AA2).
- b. Complete listing of all key cuts (AA1-123456, AA2-123458).
- c. Tabulation showing which key fits which door.
- d. Copy of floor plan showing doors and door numbers.
- e. Listing of 20 percent more key cuts than are presently required in each master system.

1.7 QUALITY ASSURANCE

1.7.1 Hardware Manufacturers and Modifications

Provide, as far as feasible, locks, hinges, pivots, and closers of one lock, hinge, pivot, or closer manufacturer's make. Modify hardware as necessary to provide features indicated or specified.

1.7.2 Key Shop Drawings Coordination Meeting

Prior to the submission of the key shop drawing, the Contracting Officer, Contractor, Door Hardware Subcontractor, using Activity and Base Locksmith must meet to discuss and coordinate key requirements for the facility.

1.8 DELIVERY, STORAGE, AND HANDLING

Deliver hardware in original individual containers, complete with necessary appurtenances including fasteners and instructions. Mark each individual container with item number as shown on hardware schedule. Deliver permanent keys and removable cores to the Contracting Officer, either directly or by certified mail. Deliver construction master keys with the locks.

PART 2 PRODUCTS

2.1 TEMPLATE HARDWARE

Hardware applied to metal doors must be manufactured using a template. Provide templates to door and frame manufacturers in accordance with [ANSI/BHMA A156.7](#) for template hinges. Coordinate hardware items to prevent interference with other hardware.

2.2 [HARDWARE ITEMS](#)

Clearly and permanently mark with the manufacturer's name or trademark, hinges, pivots, locks, latches, exit devices, bolts and closers where the identifying mark is visible after the item is installed. For closers with covers, the name or trademark may be beneath the cover.

2.2.1 Hinges

Provide in accordance with ANSI/BHMA A156.1. Provide hinges that are 114 by 114 mm unless otherwise indicated. Construct loose pin hinges for interior doors and reverse-bevel exterior doors so that pins are non-removable when door is closed. Other anti-friction bearing hinges may be provided in lieu of ball bearing hinges.

2.2.2 Continuous Hinges

Where continuous hinges are required, provide in accordance with ANSI/BHMA A156.26.

2.2.3 Pivots

Provide in accordance with ANSI/BHMA A156.17.

2.2.4 Locks and Latches

2.2.4.1 Mortise Locks and Latches

Provide in accordance with ANSI/BHMA A156.13, Series 1000, Operational Grade 1, Security Grade 2. Provide knobs and roses of mortise locks with screwless shanks and no exposed screws.

2.2.4.2 Auxiliary Locks

Provide in accordance with ANSI/BHMA A156.36, Grade 1.

2.2.5 Exit Devices

Provide in accordance with ANSI/BHMA A156.3, Grade 1. Provide adjustable strikes for rim type and vertical rod devices. Provide open back strikes for pairs of doors with mortise and vertical rod devices.

2.2.6 Keying System

Provide cylinders of Grade 1 products from one manufacturer. Notify the Contracting Officer 90 days prior to the required delivery of the cylinders. Provide temporary cores and keys for the Contractor's use during construction, and for testing of locksets.

2.2.7 Keys

Furnish one file key, one duplicate key, and one working key for each key change and for each master keying system. Furnish one additional working key for each lock of each keyed-alike group. Stamp each key with appropriate key control symbol and "U.S. property - do not duplicate." Do not place room number on keys.

2.2.8 Door Bolts

Provide in accordance with ANSI/BHMA A156.16. Provide dustproof strikes for bottom bolts, except at doors having metal thresholds. Provide automatic latching flush bolts in accordance with ANSI/BHMA A156.3, Type 25.

2.2.9 Closers

Provide in accordance with ANSI/BHMA A156.4, Series C02000, Grade 1, with PT 4C. Provide with brackets, arms, mounting devices, fasteners, full size covers, except at storefront mounting, and other features necessary for the particular application. Size closers in accordance with manufacturer's printed recommendations, or provide multi-size closers, Sizes 1 through 6, and list sizes in the Hardware Schedule. Provide manufacturer's 10 year warranty.

Use stainless steel inside bracketed or door mounted closers on exterior doors. Non-ferrous closers, such as aluminum or cast bronze, are permissible where door utilization is minimal. On interior doors use closers of 302 or 304 stainless steel or non-ferrous materials. On surface-mounted closers use or apply rust inhibiting finish on all ferrous parts. Also apply this finish on concealed closers.

2.2.9.1 Identification Marking

Engrave each closer with manufacturer's name or trademark, date of manufacture, and manufacturer's size designation in locations that will be visible after installation.

2.2.10 Overhead Holders

Provide in accordance with ANSI/BHMA A156.8.

2.2.11 Door Stops and Silencers

Provide in accordance with ANSI/BHMA A156.16. Silencers Type L03011. Provide three silencers for each single door, two for each pair.

2.2.12 Weatherstripping Gasketing

Provide in accordance with ANSI/BHMA A156.22. Provide the type and function designation where specified in paragraph HARDWARE SCHEDULE. Provide a set to include head and jamb seals and, for pairs of doors, astragals. Air leakage of weatherstripped doors not to exceed 2.19 by 10-5 cms per minute of air per square meter of door area when tested in accordance with ASTM E283. Provide weatherstripping with one of the following:

2.2.12.1 Extruded Aluminum Retainers

Extruded aluminum retainers not less than 1.25 mm wall thickness with vinyl, neoprene, silicone rubber, or polyurethane inserts. Provide clear (natural) anodized aluminum.

2.3 FASTENERS

Provide fasteners of type, quality, size, and quantity appropriate to the specific application. Fastener finish to match hardware. Provide stainless steel or nonferrous metal fasteners in locations exposed to weather. Verify metals in contact with one another are compatible and will avoid galvanic corrosion when exposed to weather.

2.4 FINISHES

Provide in accordance with ANSI/BHMA A156.18. Provide hardware in BHMA

630 finish (satin stainless steel), unless specified otherwise. Provide items not manufactured in stainless steel in BHMA 626 finish (satin chromium plated) over brass or bronze, except finish for surface door closers, for steel hinges. Provide hinges for exterior doors in stainless steel with BHMA 630 finish. Furnish exit devices in BHMA 626 finish in lieu of BHMA 630 finish. Match exposed parts of concealed closers to lock and door trim. Match hardware finish for aluminum doors to the doors.

PART 3 EXECUTION

3.1 INSTALLATION

Provide hardware in accordance with manufacturers' printed installation instructions. Fasten hardware to wood surfaces with full-threaded wood screws or sheet metal screws. Provide machine screws set in expansion shields for fastening hardware to solid concrete and masonry surfaces. Provide toggle bolts where required for fastening to hollow core construction. Provide through bolts where necessary for satisfactory installation.

3.1.1 Weatherstripping Installation

Provide full contact, weathertight seals that allow operation of doors without binding the weatherstripping.

3.1.2 Threshold Installation

Extend thresholds the full width of the opening and notch end for jamb stops. Set thresholds in a full bed of sealant and anchor to floor with cadmium-plated, countersunk, steel screws. For aluminum thresholds placed on top of concrete surfaces, coat the underside surfaces that are in contact with the concrete with fluid applied waterproofing as a separation measure prior to placement.

3.2 FIRE DOORS AND EXIT DOORS

Provide hardware in accordance with NFPA 72 for door alarms, NFPA 80 for fire doors, NFPA 101 for exit doors, and NFPA 252 for fire tests of door assemblies.

3.3 HARDWARE LOCATIONS

Provide in accordance with SDI/DOOR A250.8, unless indicated or specified otherwise.

- a. Kick and Armor Plates: Push side of single-acting doors. Both sides of double-acting doors.
- b. Mop Plates: Bottom flush with bottom of door.

3.4 FIELD QUALITY CONTROL

After installation, protect hardware from paint, stains, blemishes, and other damage until acceptance of work. Submit notice of testing 15 days before scheduled, so that testing can be witnessed by the Contracting Officer. Adjust hinges, locks, latches, bolts, holders, closers, and other items to operate properly. Demonstrate that permanent keys operate respective locks, and give keys to the Contracting Officer. Correct, repair, and finish, errors in cutting and fitting and damage to adjoining

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

work.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 08 - OPENINGS

SECTION 08 81 00

GLAZING

05/19

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 SYSTEM DESCRIPTION
- 1.4 DELIVERY, STORAGE, AND HANDLING
- 1.5 ENVIRONMENTAL REQUIREMENTS
- 1.6 WARRANTY
 - 1.6.1 Warranty for Insulated Glass Units

PART 2 PRODUCTS

- 2.1 GLASS
 - 2.1.1 Clear Glass
 - 2.1.2 Tempered Glass
- 2.2 INSULATING GLASS UNITS
- 2.3 SETTING AND SEALING MATERIALS
 - 2.3.1 Putty and Glazing Compound
 - 2.3.2 Glazing Compound
 - 2.3.3 Sealants
 - 2.3.3.1 Elastomeric Sealant
 - 2.3.4 Joint Backer
 - 2.3.5 Glazing Tapes
 - 2.3.5.1 Back-Bedding Mastic Glazing Tapes
 - 2.3.5.2 Expanded Cellular Glazing Tapes
 - 2.3.6 Sealing Tapes
 - 2.3.7 Setting Blocks and Edge Blocks
 - 2.3.8 Glazing Gaskets
 - 2.3.8.1 Fixed Glazing Gaskets
 - 2.3.8.2 Wedge Glazing Gaskets
 - 2.3.8.3 Aluminum Framing Glazing Gaskets
 - 2.3.9 Accessories

PART 3 EXECUTION

- 3.1 PREPARATION
- 3.2 GLASS SETTING
 - 3.2.1 Sheet Glass
 - 3.2.2 Insulating Glass Units
- 3.3 CLEANING
- 3.4 PROTECTION
- 3.5 SCHEDULE

-- End of Section Table of Contents --

SECTION 08 81 00

GLAZING
05/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

- | | |
|----------------------|---|
| UNE-EN 572-1 | (2012) Glass in building - Basic soda lime silicate glass products - Part 1: Definitions and general physical and mechanical properties |
| UNE-EN 1279-1 | (2019) Glass in Building - Insulating glass units - Part 1: Generalities, system description, rules for substitution, tolerances and visual quality |
| UNE-EN 12150-1 | (2016) Glass in building - Thermally toughened soda lime silicate safety glass - Part 1: Definition and description. |
| UNE-EN 14188-3 | (2007) Joint fillers and sealants - Part 3: Specifications for preformed joint seals |
| UNE-EN ISO/IEC 17025 | (2017) General requirements for the competence of testing and calibration laboratories (ISO/IEC 17025:2017) |

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

- | | |
|----------|---|
| AAMA 800 | (2016) Voluntary Specifications and Test Methods for Sealants |
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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

- | | |
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| ANSI Z97.1 | (2015) Safety Glazing Materials Used in Buildings - Safety Performance Specifications and Methods of Test |
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ASTM INTERNATIONAL (ASTM)

- | | |
|------------|--|
| ASTM C509 | (2006; R 2021) Standard Specification for Elastomeric Cellular Preformed Gasket and Sealing Material |
| ASTM C920 | (2018) Standard Specification for Elastomeric Joint Sealants |
| ASTM C1021 | (2008; R 2014) Standard Practice for |

Laboratories Engaged in Testing of
Building Sealants

ASTM C1036	(2021) Standard Specification for Flat Glass
ASTM C1048	(2018) Standard Specification for Heat-Strengthened and Fully Tempered Flat Glass
ASTM C1087	(2016) Standard Test Method for Determining Compatibility of Liquid-Applied Sealants with Accessories Used in Structural Glazing Systems
ASTM C1281	(2016) Standard Specification for Preformed Tape Sealants for Glazing Applications
ASTM E1300	(2016) Standard Practice for Determining Load Resistance of Glass in Buildings
ASTM E2190	(2010) Standard Specification for Insulating Glass Unit Performance and Evaluation

GLASS ASSOCIATION OF NORTH AMERICA (GANA)

GANA Glazing Manual	(2008) Glazing Manual
GANA Sealant Manual	(2008) Sealant Manual
GANA Standards Manual	(2008) Engineering Standards Manual

INSULATING GLASS MANUFACTURERS ALLIANCE (IGMA)

IGMA TB-1200	(1983; R 2016) Guidelines for Insulating Glass Dimensional Tolerances
IGMA TB-3001	(2001) Guidelines for Sloped Glazing
IGMA TM-3000	(1990; R 2016) North American Glazing Guidelines for Sealed Insulating Glass Units for Commercial & Residential Use

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

16 CFR 1201	Safety Standard for Architectural Glazing Materials
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Insulating Glass

Glazing Accessories

SD-04 Samples

Insulating Glass

SD-07 Certificates

Insulating Glass

SD-08 Manufacturer's Instructions

Setting and Sealing Materials

Glass Setting

SD-11 Closeout Submittals

Warranty for Insulated Glass Units

1.3 SYSTEM DESCRIPTION

Fabricate and install watertight and airtight glazing systems to withstand thermal movement and wind loading without glass breakage, gasket failure, deterioration of [glazing accessories](#), or defects in the work. Glazed panels must comply with the safety standards, in accordance with [ANSI Z97.1](#), and comply with indicated wind loading in accordance with [ASTM E1300](#).

1.4 DELIVERY, STORAGE, AND HANDLING

Deliver products to the site in unopened containers, labeled plainly with manufacturers' names and brands. Store glass and setting materials in safe, enclosed dry locations and do not unpack until needed for installation. Handle and install materials in a manner that will protect them from damage.

1.5 ENVIRONMENTAL REQUIREMENTS

Do not start glazing work until the outdoor temperature is above [4 degrees C](#) and rising, unless procedures recommended by the glass manufacturer and approved by the Contracting Officer are made to warm the glass and rabbet surfaces. Provide ventilation to prevent condensation of moisture on glazing work during installation. Do not perform glazing work during damp or rainy weather.

1.6 WARRANTY

1.6.1 Warranty for Insulated Glass Units

Warranty insulating glass units against development of material obstruction to vision (such as dust, fogging, or film formation on the inner glass surfaces) caused by failure of the hermetic seal, other than through glass breakage, for a 10-year period following acceptance of the work. Provide new units for any units failing to comply with terms of this warranty within 45 working days after receipt of notice from the Government.

PART 2 PRODUCTS

2.1 GLASS

ASTM C1036 or UNE-EN 572-1 unless specified otherwise. In doors and sidelights, provide safety glazing material conforming to 16 CFR 1201.

2.1.1 Clear Glass

Type I, Class 1 (clear), Quality q4 (A) . Provide for glazing openings not indicated or specified otherwise. Use double-strength sheet glass or 3 mm float glass for openings up to and including 1.39 square meters, 4.5 mm for glazing openings over 1.39 square meters but not over 2.79 square meters, and 6 mm for glazing openings over 2.79 square meters but not over 4.18 square meters.

2.1.2 Tempered Glass

ASTM C1048 or UNE-EN 12150-1, Kind FT (fully tempered), Condition A (uncoated), Type I, Class 1 (transparent), Quality q3, 6 mm thick conforming to ASTM C1048 or UNE-EN 12150-1 and GANA Standards Manual. Color must be clear.

2.2 INSULATING GLASS UNITS

Two panes of glass separated by a dehydrated airspace, filled with argon gas and hermetically sealed, conforming to ASTM E2190 or UNE-EN 1279-1. Submit performance and compliance documentation for each type of insulating glass.

Dimensional tolerances must be as specified in IGMA TB-1200. Spacer must be black, roll-formed, , with bent or tightly welded or keyed and sealed joints to completely seal the spacer periphery and eliminate moisture and hydrocarbon vapor transmission into airspace through the corners. Primary seal must be compressed polyisobutylene and the secondary seal must be a specially formulated silicone.

The inner light must be ASTM C1048 or UNE-EN 12150-1, Grade B (fully tempered), Style I (uncoated), Type I, Class 1 (transparent), Quality q4, 6 mm thick. The outer light must be ASTM C1048 or UNE-EN 12150-1, Grade B (fully tempered), Style I (uncoated), Type I, Class 1 (clear) , Quality q4, 6 mm thick.

2.3 SETTING AND SEALING MATERIALS

Provide as specified in the GANA Glazing Manual, IGMA TM-3000, IGMA TB-3001, and manufacturer's recommendations, unless specified otherwise herein. Do not use metal sash putty, nonskinning compounds, nonresilient preformed sealers, or impregnated preformed gaskets. Materials exposed to view and unpainted must be gray or neutral color. Sealant testing must be performed by a testing agency qualified according to ASTM C1021 or UNE-EN ISO/IEC 17025.

Submit glass manufacturer's recommendations for setting and sealing materials and for installation of each type of glazing material specified.

2.3.1 Putty and Glazing Compound

Provide glazing compound as recommended by manufacturer for face-glazing

metal sash. Putty must be linseed oil type. Do not use putty and glazing compounds with insulating glass or laminated glass.

2.3.2 Glazing Compound

Use for face glazing metal sash. Do not use with insulating glass units or laminated glass.

2.3.3 Sealants

Provide elastomeric sealants.

2.3.3.1 Elastomeric Sealant

ASTM C920, Type S, Grade NS, Class 12.5, Use G. Use for channel or stop glazing metal sash. Sealants must be chemically compatible with setting blocks, edge blocks, and sealing tapes. Color of sealant must be white.

2.3.4 Joint Backer

Joint backer must have a diameter size at least 25 percent larger than joint width; type and material as recommended in writing by glass and sealant manufacturer.

2.3.5 Glazing Tapes

2.3.5.1 Back-Bedding Mastic Glazing Tapes

Preformed, butyl-based, 100 percent solids elastomeric tape; nonstaining and nonmigrating in contact with nonporous surfaces; with or without spacer rod as recommended in writing by tape and glass manufacturers for application indicated; and complying with ASTM C1281 and AAMA 800 for products indicated below:

- a. AAMA 804.3 tape, where indicated.
- b. AAMA 806.3 tape, for glazing applications in which tape is subject to continuous pressure.
- c. AAMA 807.3 tape, for glazing applications in which tape is not subject to continuous pressure.

2.3.5.2 Expanded Cellular Glazing Tapes

Closed-cell, PVC foam tapes; factory coated with adhesive on both surfaces; and complying with AAMA 800 for the following types:

- a. AAMA 810.1, Type 1, for glazing applications in which tape acts as the primary sealant.
- b. AAMA 810.1, Type 2, for glazing applications in which tape is used in combination with a full bead of liquid sealant.

2.3.6 Sealing Tapes

Preformed, semisolid, PVC-based material of proper size and compressibility for the particular condition. Use only where glazing rabbet is designed for tape and tape is recommended by the glass or sealant manufacturer. Provide spacer shims for use with compressible

tapes. Tapes must be chemically compatible with the product being set.

2.3.7 Setting Blocks and Edge Blocks

Closed-cell neoprene setting blocks must be dense extruded type conforming to [ASTM C509](#) or [UNE-EN 14188-3](#), Shore A durometer between 70 and 90. Edge blocking must be Shore A durometer of 50 (plus or minus 5). Provide silicone setting blocks when blocks are in contact with silicone sealant. Profiles, lengths and locations must be as required and recommended in writing by glass manufacturer. Block color must be black.

2.3.8 Glazing Gaskets

Glazing gaskets must be extruded with continuous integral locking projection designed to engage into metal glass holding members to provide a watertight seal during dynamic loading, building movements and thermal movements. Glazing gaskets for a single glazed opening must be continuous one-piece units with factory-fabricated injection-molded corners free of flashing and burrs. Glazing gaskets must be in lengths or units recommended by manufacturer to ensure against pull-back at corners. Provide glazing gasket profiles as recommended by the manufacturer for the intended application.

2.3.8.1 Fixed Glazing Gaskets

Fixed glazing gaskets must be closed-cell (sponge) smooth extruded compression gaskets of cured elastomeric virgin neoprene compounds conforming to [ASTM C509](#) or [UNE-EN 14188-3](#), Type 2, Option 1.

2.3.8.2 Wedge Glazing Gaskets

Wedge glazing gaskets must be high-quality extrusions of cured elastomeric virgin neoprene compounds, ozone resistant, Shore A durometer between 65 and 75.

2.3.8.3 Aluminum Framing Glazing Gaskets

Glazing gaskets for aluminum framing must be permanent, elastic, non-shrinking, non-migrating, watertight and weathertight.

2.3.9 Accessories

Provide as required for a complete installation, including glazing points, clips, shims, angles, beads, and spacer strips. Provide noncorroding metal accessories. Provide primer-sealers and cleaners as recommended by the glass and sealant manufacturers. Use [ASTM C1087](#) to determine whether priming and other specific joint preparation techniques are required to obtain rapid, optimum adhesion of glazing sealants to surface.

PART 3 EXECUTION

Any materials that show visual evidence of biological growth due to the presence of moisture must not be installed on the building project.

3.1 PREPARATION

Preparation, unless otherwise specified or approved, must conform to applicable recommendations in the [GANA Glazing Manual](#), [GANA Sealant Manual](#), [IGMA TB-3001](#), [IGMA TM-3000](#), and manufacturer's recommendations. Determine

the sizes to provide the required edge clearances by measuring the actual opening to receive the glass. Grind smooth in the shop glass edges that will be exposed in finish work. Leave labels in place until the installation is approved, except remove applied labels on heat-absorbing glass and on insulating glass units as soon as glass is installed. Securely fix movable items or keep in a closed and locked position until glazing compound has thoroughly set.

3.2 GLASS SETTING

Shop glaze or field glaze items to be glazed using glass of the quality and thickness specified or indicated. Glazing, unless otherwise specified or approved, must conform to applicable recommendations in the [GANA Glazing Manual](#), [GANA Sealant Manual](#), [IGMA TB-3001](#), [IGMA TM-3000](#), and manufacturer's recommendations. Aluminum windows, wood doors, and wood windows may be glazed in conformance with one of the glazing methods described in the standards under which they are produced, except that face puttying with no bedding will not be permitted. Handle and install glazing materials in accordance with manufacturer's instructions. Use beads or stops which are furnished with items to be glazed to secure the glass in place. Verify products are properly installed, connected, and adjusted.

3.2.1 Sheet Glass

Cut and set with the visible lines or waves horizontal.

3.2.2 Insulating Glass Units

Do not grind, nip, or cut edges or corners of units after the units have left the factory. Springing, forcing, or twisting of units during setting will not be permitted. Handle units so as not to strike frames or other objects. Installation must conform to applicable recommendations of [IGMA TB-3001](#) and [IGMA TM-3000](#).

3.3 CLEANING

Clean glass surfaces and remove labels, paint spots, putty, and other defacement as required to prevent staining. Glass must be clean at the time the work is accepted.

3.4 PROTECTION

Protect glass work immediately after installation. Identify glazed openings with suitable warning tapes, cloth or paper flags, attached with non-staining adhesives. Protect reflective glass with a protective material to eliminate any contamination of the reflective coating. Place protective material far enough away from the coated glass to allow air to circulate to reduce heat buildup and moisture accumulation on the glass. Upon removal, separate protective materials for reuse or recycling. Remove and replace glass units which are broken, chipped, cracked, abraded, or otherwise damaged during construction activities with new units.

3.5 SCHEDULE

Some metric measurements in this section are based on mathematical conversion of inch-pound measurements, and not on metric measurement commonly agreed to by the manufacturers or other parties. The inch-pound

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

and metric measurements are as follows:

<u>PRODUCTS</u>	<u>INCH-POUND</u>	<u>METRIC</u>
Glass	1/8 inch	3 mm
	3/16 inch	4.5 mm
	7/32 inch	6 mm
	1/4 inch	6 mm
	3/8 inch	10 mm
Interlayer	0.015 inch	0.38 mm
Glazing Channels	1/4 inch	6 mm

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 08 - OPENINGS

SECTION 08 91 00

METAL WALL LOUVERS

08/20

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY, STORAGE, AND PROTECTION
- 1.4 DETAIL DRAWINGS
- 1.5 COLOR SAMPLES

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Galvanized Steel Sheet
 - 2.1.2 Aluminum Sheet
 - 2.1.3 Extruded Aluminum
 - 2.1.4 Cold Rolled Steel Sheet
- 2.2 METAL WALL LOUVERS
 - 2.2.1 Extruded Aluminum Louvers
 - 2.2.2 Mullions and Mullion Covers
 - 2.2.3 Screens and Frames
- 2.3 FASTENERS AND ACCESSORIES
- 2.4 FINISHES
 - 2.4.1 Aluminum
 - 2.4.1.1 Anodic Coating
 - 2.4.1.2 Organic Coating

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Wall Louvers
 - 3.1.2 Screens and Frames
- 3.2 PROTECTION FROM CONTACT OF DISSIMILAR MATERIALS
 - 3.2.1 Copper or Copper-Bearing Alloys
 - 3.2.2 Aluminum
 - 3.2.3 Metal
 - 3.2.4 Wood

-- End of Section Table of Contents --

SECTION 08 91 00

METAL WALL LOUVERS
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE-EN 485-1 (2017) Aluminium and aluminium alloys - Sheet, strip and plate - Part 1: Technical conditions for inspection and delivery

UNE-EN 755-1 (2016) Aluminium and aluminium alloys - Extruded rod/bar, tube and profiles - Part 1: Technical conditions for inspection and delivery

UNE-EN 10131 (2007) Cold rolled uncoated and zinc or zinc-nickel electrolytically coated low carbon and high yield strength steel flat products for cold forming - Tolerances on dimensions and shape

UNE-EN 10346 (2015) Continuously hot-dip coated steel flat products for cold forming - Technical delivery conditions

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 500-L (2015) Laboratory Methods of Testing Louvers for Rating

AMCA 511 (2010; R 2016) Certified Ratings Program for Air Control Devices

ALUMINUM ASSOCIATION (AA)

AA DAF45 (2003; Reaffirmed 2009) Designation System for Aluminum Finishes

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 611 (2014) Voluntary Specification for Anodized Architectural Aluminum

AAMA 2605 (2020) Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels

ASTM INTERNATIONAL (ASTM)

ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A1008/A1008M	(2021a) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable
ASTM B209M	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
ASTM B221M	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Wall Louvers

SD-03 Product Data

Metal Wall Louvers

1.3 DELIVERY, STORAGE, AND PROTECTION

Deliver materials to the site in an undamaged condition. Carefully store materials off the ground to provide proper ventilation, drainage, and protection against dampness. Louvers must be free from nicks, scratches, and blemishes. Replace defective or damaged materials with new.

1.4 DETAIL DRAWINGS

Show all information necessary for fabrication and installation of wall louvers. Indicate materials, sizes, thicknesses, fastenings, and profiles.

1.5 COLOR SAMPLES

Colors of finishes for wall louver samples and door louver samples must closely approximate colors indicated. Where color is not indicated, submit the manufacturer's standard colors to the Contracting Officer for selection.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Galvanized Steel Sheet

ASTM A653/A653M or UNE-EN 10346, coating designation Z275.

2.1.2 Aluminum Sheet

ASTM B209M or UNE-EN 485-1, alloy 3003 or 5005 with temper as required for forming.

2.1.3 Extruded Aluminum

ASTM B221M or UNE-EN 755-1, alloy 6063-T5 or -T52.

2.1.4 Cold Rolled Steel Sheet

ASTM A1008/A1008M or UNE-EN 10131, Class 1, with matte finish. Use for interior louvers only.

2.2 METAL WALL LOUVERS

Wind driven rain resistant type, with bird screens and made to withstand a wind load of not less than 1.44 kilopascals. Wall louvers must bear the AMCA certified ratings program seal for air performance and water penetration in accordance with AMCA 500-L and AMCA 511. The rating must show a water penetration of 0.06 kilograms or less per square meter of free area at a free velocity of 244 meters per minute.

2.2.1 Extruded Aluminum Louvers

Fabricated of extruded 6063-T5 or -T52 aluminum with a wall thickness of not less than 2 mm.

2.2.2 Mullions and Mullion Covers

Same material and finish as louvers. Provide mullions for all louvers more than 1500 mm in width at not more than 1500 mm on centers. Provide mullion covers on both faces of joints between louvers.

2.2.3 Screens and Frames

For aluminum louvers, provide 12.5 mm square mesh, 1.8 or 1.5 mm aluminum or 6 mm square mesh, 1.5 mm aluminum bird screening. For steel louvers, provide 12.5 mm square mesh, 2.5 or 1.5 mm zinc-coated steel; 12.5 mm square mesh, 1.5 mm copper; or 6 mm square mesh, 1.5 mm thick zinc-coated steel or copper bird screening. Mount screens in removable, rewirable frames of same material and finish as the louvers.

2.3 FASTENERS AND ACCESSORIES

Provide stainless steel screws and fasteners for aluminum louvers and zinc-coated or stainless steel screws and fasteners for steel louvers. Provide other accessories as required for complete and proper installation.

2.4 FINISHES

2.4.1 Aluminum

Exposed aluminum surfaces must be factory finished with an anodic coating or organic coating. Color must be as indicated. Louvers must have the same finish.

2.4.1.1 Anodic Coating

Clean exposed aluminum surfaces and provide an anodized finish conforming to AA DAF45 and AAMA 611. Finish must be:

- a. Architectural Class II (0.01 to 0.0175 mm), designation AA-M10-C22-A31, clear (natural) anodized.

2.4.1.2 Organic Coating

Clean and prime exposed aluminum surfaces. Provide a superior performance finish in accordance with AAMA 2605 with total dry film thickness of not less than 0.03 mm, color as indicated.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Wall Louvers

Install using stops or moldings, flanges, strap anchors, or jamb fasteners as appropriate for the wall construction and in accordance with manufacturer's recommendations.

3.1.2 Screens and Frames

Attach frames to louvers with screws or bolts.

3.2 PROTECTION FROM CONTACT OF DISSIMILAR MATERIALS

3.2.1 Copper or Copper-Bearing Alloys

Paint copper or copper-bearing alloys in contact with dissimilar metal with heavy-bodied bituminous paint or separate with inert membrane.

3.2.2 Aluminum

Where aluminum contacts metal other than zinc, paint the dissimilar metal with a primer and two coats of aluminum paint.

3.2.3 Metal

Paint metal in contact with mortar, concrete, or other masonry materials with alkali-resistant coatings such as heavy-bodied bituminous paint.

3.2.4 Wood

Paint wood or other absorptive materials that may become repeatedly wet and in contact with metal with two coats of aluminum paint or a coat of heavy-bodied bituminous paint.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 22 00

SUPPORTS FOR PLASTER AND GYPSUM BOARD

02/10, CHG 2: 08/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Materials for Attachment of Gypsum Wallboard
 - 2.1.1.1 Suspended and Furred Ceiling Systems
 - 2.1.1.2 Non-load Bearing Wall Framing and Furring

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Systems for Attachment of Gypsum Wallboard
 - 3.1.1.1 Suspended and Furred Ceiling Systems
 - 3.1.1.2 Non-load Bearing Wall Framing and Furring
- 3.2 ERECTION TOLERANCES

-- End of Section Table of Contents --

SECTION 09 22 00

SUPPORTS FOR PLASTER AND GYPSUM BOARD
02/10, CHG 2: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

- | | |
|-----------------|---|
| UNE 102043 | (2013) Installation of construction systems with gypsum plasterboards. Partitions, wall linings and ceilings. Definitions, applications and recommendations |
| UNE-EN 10162 | (2005) Cold formed steel sections - Technical Supply Conditions. Dimensional, tolerances and on Cross-section. |
| UNE-EN ISO 1461 | (2010) Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods (ISO1461:1999) |
| UNE-EN ISO 2063 | (2005) Thermal spraying - Metallic and other inorganic coatings - Zinc, aluminium and their alloys (ISO 2063:2005) |

ASTM INTERNATIONAL (ASTM)

- | | |
|-----------------|---|
| ASTM A463/A463M | (2015; R 2020; E 2020) Standard Specification for Steel Sheet, Aluminum-Coated, by the Hot-Dip Process |
| ASTM A653/A653M | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process |
| ASTM C645 | (2014; E 2015) Nonstructural Steel Framing Members |
| ASTM C754 | (2020) Standard Specification for Installation of Steel Framing Members to Receive Screw-Attached Gypsum Panel Products |

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for

Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Metal Support Systems

1.3 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the job site and store in ventilated dry locations permitting easy access for inspection and handling. If materials are stored outdoors, stack materials off the ground, supported on a level platform, and fully protected from the weather. Handle materials carefully to prevent damage. Remove damaged items and provide new items.

PART 2 PRODUCTS

2.1 MATERIALS

Provide steel materials for metal support systems with galvanized coating ASTM A653/A653M or UNE-EN ISO 1461, Z180; aluminum coating ASTM A463/A463M, T1-75 or UNE-EN ISO 2063; or a 55-percent aluminum-zinc coating.

2.1.1 Materials for Attachment of Gypsum Wallboard

2.1.1.1 Suspended and Furred Ceiling Systems

ASTM C645 or UNE-EN 10162.

2.1.1.2 Non-load Bearing Wall Framing and Furring

ASTM C645 or UNE-EN 10162, but not thinner than 0.45 mm thickness, with 0.85 mm minimum thickness supporting wall hung items such as cabinetwork, equipment and fixtures.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Systems for Attachment of Gypsum Wallboard

3.1.1.1 Suspended and Furred Ceiling Systems

ASTM C754 or UNE 102043, except provide framing members 400 mm o.c. unless indicated otherwise.

3.1.1.2 Non-load Bearing Wall Framing and Furring

ASTM C754 or UNE 102043, except as indicated otherwise.

3.2 ERECTION TOLERANCES

Provide framing members which will be covered by finish materials such as wallboard, plaster, or ceramic tile set in a mortar setting bed, within the following limits:

- a. Layout of walls and partitions: 6 mm from intended position;
- b. Plates and runners: 5 mm in 1.9 meters from a straight line;
- c. Studs: 5 mm in 1.9 meters out of plumb, not cumulative; and

- d. Face of framing members: 5 mm in 1.9 meters from a true plane.

Provide framing members which will be covered by ceramic tile set in dry-set mortar, latex-portland cement mortar, or organic adhesive within the following limits:

- a. Layout of walls and partitions: 6 mm from intended position;
- b. Plates and runners: 5 mm in 3.8 meters from a straight line;
- c. Studs: 5 mm in 3.8 meters out of plumb, not cumulative; and
- d. Face of framing members: 5 mm in 3.8 meters from a true plane.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 29 00

GYPSUM BOARD

08/16, CHG 4: 02/20

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY, STORAGE, AND HANDLING
 - 1.3.1 Delivery
 - 1.3.2 Storage
 - 1.3.3 Handling
- 1.4 QUALIFICATIONS
- 1.5 ENVIRONMENTAL REQUIREMENTS

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Gypsum Board
 - 2.1.1.1 Regular
 - 2.1.2 Joint Treatment Materials
 - 2.1.2.1 All-Purpose Compound
 - 2.1.2.2 Joint Tape
 - 2.1.3 Fasteners
 - 2.1.3.1 Screws
 - 2.1.4 Adhesives
 - 2.1.4.1 Adhesive for Fastening Gypsum Board to Metal Framing
 - 2.1.5 Accessories

PART 3 EXECUTION

- 3.1 EXAMINATION
 - 3.1.1 Framing and Furring
 - 3.1.2 Building Construction Materials
- 3.2 APPLICATION OF GYPSUM BOARD
- 3.3 FINISHING OF GYPSUM BOARD
 - 3.3.1 Uniform Surface
- 3.4 SEALING
- 3.5 PATCHING

-- End of Section Table of Contents --

SECTION 09 29 00

GYPSUM BOARD
08/16, CHG 4: 02/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

- | | |
|-----------------|---|
| UNE 102043 | (2013) Installation of construction systems with gypsum plasterboards. Partitions, wall linings and ceilings. Definitions, applications and recommendations |
| UNE-EN 520 | (2005 & AI: 2010) Gypsum Plasterboards -Definitions, Requirements and Test Methods |
| UNE-EN 13963 | (2014) Jointing materials for gypsum boards - Definitions, requirements and test methods |
| UNE-EN 14566 | (A1:2009) Mechanical fasteners for gypsum plasterboard systems - Definitions, requirements and test methods |
| UNE-EN ISO 7049 | (2011) Cross-recessed pan head tapping screws (ISO 7049:2011) |

ASTM INTERNATIONAL (ASTM)

- | | |
|-------------------|---|
| ASTM C475/C475M | (2017) Standard Specification for Joint Compound and Joint Tape for Finishing Gypsum Board |
| ASTM C840 | (2020) Standard Specification for Application and Finishing of Gypsum Board |
| ASTM C1002 | (2020) Standard Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs |
| ASTM C1047 | (2019) Standard Specification for Accessories for Gypsum Wallboard and Gypsum Veneer Base |
| ASTM C1396/C1396M | (2017) Standard Specification for Gypsum Board |

GYPSUM ASSOCIATION (GA)

GA 214 (2010) Recommended Levels of Gypsum Board Finish

GA 216 (2010) Application and Finishing of Gypsum Panel Products

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Accessories

Submit for each type of gypsum board and for cementitious backer units.

Gypsum Board

SD-07 Certificates

Asbestos Free Materials; G

Certify that gypsum board types, gypsum backing board types, cementitious backer units, and joint treating materials do not contain asbestos.

SD-08 Manufacturer's Instructions

Safety Data Sheets

SD-10 Operation and Maintenance Data

Manufacturer Maintenance Instructions

1.3 DELIVERY, STORAGE, AND HANDLING

1.3.1 Delivery

Deliver materials in the original packages, containers, or bundles with each bearing the brand name, applicable standard designation, and name of manufacturer, or supplier.

1.3.2 Storage

Keep materials dry by storing inside a sheltered building. Where necessary to store gypsum board and cementitious backer units outside, store off the ground, properly supported on a level platform, and protected from direct exposure to rain, snow, sunlight, and other extreme weather conditions. Provide adequate ventilation to prevent condensation. Store per manufacturer's recommendations for allowable temperature and humidity range. Do not store panels near materials that may offgas or emit harmful fumes, such as kerosene heaters, fresh paint, or adhesives. Do not use materials that have visible moisture or biological growth.

1.3.3 Handling

Neatly stack gypsum board and cementitious backer units flat to prevent sagging or damage to the edges, ends, and surfaces.

1.4 QUALIFICATIONS

Furnish type of gypsum board work specialized by the installer with a minimum of 3 years of documented successful experience.

1.5 ENVIRONMENTAL REQUIREMENTS

Do not expose the gypsum board to excessive sunlight prior to gypsum board application. Maintain a continuous uniform temperature of not less than 10 degrees C and not more than 27 degrees C for at least one week prior to the application of gypsum board work, while the gypsum board application is being done, and for at least one week after the gypsum board is set. Shield air supply and distribution devices to prevent any uneven flow of air across the plastered surfaces. Provide ventilation to exhaust moist air to the outside during gypsum board application, set, and until gypsum board jointing is dry. In glazed areas, keep windows open top and bottom or side to side 75 to 100 mm. Reduce openings in cold weather to prevent freezing of joint compound when applied. For enclosed areas lacking natural ventilation, provide temporary mechanical means for ventilation. In unglazed areas subjected to hot, dry winds or temperature differentials from day to night of 10 degrees C or more, screen openings with cheesecloth or similar materials. Avoid rapid drying. During periods of low indoor humidity, provide minimum air circulation following gypsum boarding and until gypsum board jointing complete and is dry.

PART 2 PRODUCTS

2.1 MATERIALS

Conform to specifications, standards and requirements specified. Provide gypsum board types, gypsum backing board types, cementitious backing units, and joint treating materials manufactured from asbestos free materials only. Submit Safety Data Sheets and manufacturer maintenance instructions for gypsum materials including adhesives.

2.1.1 Gypsum Board

ASTM C1396/C1396M or UNE-EN 520.

2.1.1.1 Regular

1200 mm wide, 15.9 mm thick, tapered.

2.1.2 Joint Treatment Materials

ASTM C475/C475M or UNE-EN 13963. Product must be low emitting VOC types with VOC limits not exceeding 50 g/L. Provide data identifying VOC content of joint compound.

2.1.2.1 All-Purpose Compound

Specifically formulated and manufactured to serve as both a taping and a finishing compound and compatible with tape, substrate and fasteners.

2.1.2.2 Joint Tape

Use cross-laminated, tapered edge, reinforced paper, or fiber glass mesh tape recommended by the manufacturer.

2.1.3 Fasteners

2.1.3.1 Screws

ASTM C1002 or UNE-EN 14566, Type "G", Type "S" or Type "W" steel drill screws for fastening gypsum board to gypsum board, wood framing members and steel framing members less than 0.84 mm thick. UNE-EN 14566 or UNE-EN ISO 7049 steel drill screws for fastening gypsum board to steel framing members 0.84 to 2.84 mm thick. Provide cementitious backer unit screws with a polymer coating.

2.1.4 Adhesives

2.1.4.1 Adhesive for Fastening Gypsum Board to Metal Framing

Not permitted.

2.1.5 Accessories

ASTM C1047. Fabricate from plastic designed for intended use. Accessories manufactured with paper flanges are not acceptable. Flanges must be free of dirt, grease, and other materials that may adversely affect bond of joint treatment. Provide prefinished or job decorated materials.

PART 3 EXECUTION

3.1 EXAMINATION

3.1.1 Framing and Furring

Verify that framing and furring are securely attached and of sizes and spacing to provide a suitable substrate to receive gypsum board and cementitious backer units. Verify that all blocking, headers and supports are in place to support plumbing fixtures and to receive soap dishes, grab bars, towel racks, and similar items. Do not proceed with work until framing and furring are acceptable for application of gypsum board and cementitious backer units.

3.1.2 Building Construction Materials

Do not install building construction materials that show visual evidence of biological growth.

3.2 APPLICATION OF GYPSUM BOARD

Apply gypsum board to framing and furring members in accordance with ASTM C840 or GA 216 or UNE 102043 and the requirements specified. Apply gypsum board with separate panels in moderate contact; do not force in place. Stagger end joints of adjoining panels. Neatly fit abutting end and edge joints. Use gypsum board of maximum practical length; select panel sizes to minimize waste. Cut out gypsum board to make neat, close, and tight joints around openings. In vertical application of gypsum board, provide panels in lengths required to reach full height of vertical

surfaces in one continuous piece. Lay out panels to minimize waste; reuse cutoffs whenever feasible. Surfaces of gypsum board and substrate members may not be bonded together with an adhesive. Treat edges of cutouts for plumbing pipes, screwheads, and joints with water-resistant compound as recommended by the gypsum board manufacturer. Provide type of gypsum board for use in each system specified herein as indicated.

3.3 FINISHING OF GYPSUM BOARD

Tape and finish gypsum board in accordance with [ASTM C840](#) and [UNE 102043](#), [GA 214](#) and [GA 216](#). Finish plenum areas above ceilings to Level 1 in accordance with [GA 214](#). Finish water resistant gypsum backing board, [ASTM C1396/C1396M](#), to receive ceramic tile to Level 2 in accordance with [GA 214](#). Finish walls and ceilings to receive a heavy-grade wall covering or heave textured finish before painting to Level 3 in accordance with [GA 214](#). Finish walls and ceilings without critical lighting to receive flat paints, light textures, or wall coverings to Level 4 in accordance with [GA 214](#). Unless otherwise specified, finish all gypsum board walls, partitions and ceilings to Level 5 in accordance with [GA 214](#). Provide joint, fastener depression, and corner treatment. Tool joints as smoothly as possible to minimize sanding and dust. Do not use self-adhering fiber glass mesh tape with conventional drying type joint compounds; use setting or hardening type compounds only. Provide treatment for water-resistant gypsum board as recommended by the gypsum board manufacturer. Protect workers, building occupants, and HVAC systems from gypsum dust.

3.3.1 Uniform Surface

Wherever gypsum board is to receive eggshell, semigloss or gloss paint finish, or where severe, up or down lighting conditions occur, finish gypsum wall surface in accordance to [GA 214](#) Level 5. In accordance with [GA 214](#) Level 5, apply a thin skim coat of joint compound to the entire gypsum board surface, after the two-coat joint and fastener treatment is complete and dry.

3.4 SEALING

Seal openings around pipes, fixtures, and other items projecting through gypsum board and cementitious backer units as specified in Section [07 92 00](#) JOINT SEALANTS. Apply material with exposed surface flush with gypsum board or cementitious backer units.

3.5 PATCHING

Patch surface defects in gypsum board to a smooth, uniform appearance, ready to receive finishes.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 51 00

ACOUSTICAL CEILINGS

08/20

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY, STORAGE. AND HANDLING
- 1.4 ENVIRONMENTAL REQUIREMENTS
- 1.5 SCHEDULING
- 1.6 WARRANTY
- 1.7 EXTRA MATERIALS

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Light Reflectance
- 2.2 ACOUSTICAL UNITS
 - 2.2.1 Units for Exposed-Grid System
 - 2.2.1.1 Type
 - 2.2.1.2 Flame Spread
 - 2.2.1.3 Minimum Light Reflectance Coefficient
 - 2.2.1.4 Nominal Size
 - 2.2.1.5 Edge Detail
 - 2.2.1.6 Finish
- 2.3 SUSPENSION SYSTEM
- 2.4 HANGERS
 - 2.4.1 Wires
- 2.5 FINISHES
- 2.6 COLORS AND PATTERNS

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Suspension System
 - 3.1.1.1 Plumb Hangers
 - 3.1.2 Wall Molding
- 3.2 CLEANING

-- End of Section Table of Contents --

SECTION 09 51 00

ACOUSTICAL CEILINGS
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

- | | |
|----------------|---|
| UNE 102043 | (2013) Installation of construction systems with gypsum plasterboards. Partitions, wall linings and ceilings. Definitions, applications and recommendations |
| UNE-EN 10244-2 | (2010) Steel wire and wire products - Non-ferrous metallic coatings on steel wire - Part 2: Zinc or zinc alloy coatings |
| UNE-EN 13964 | (2016) Suspended ceilings - Requirements and test methods |

ASTM INTERNATIONAL (ASTM)

- | | |
|-----------------|---|
| ASTM A641/A641M | (2019) Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire |
| ASTM C635/C635M | (2017) Standard Specification for Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-In Panel Ceilings |
| ASTM C636/C636M | (2013) Standard Practice for Installation of Metal Ceiling Suspension Systems for Acoustical Tile and Lay-In Panels |
| ASTM E1264 | (2019) Acoustical Ceiling Products |
| ASTM E1477 | (1998a; R 2017; E 2018) Standard Test Method for Luminous Reflectance Factor of Acoustical Materials by Use of Integrating-Sphere Reflectometers |

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Acoustical Units

SD-04 Samples

Acoustical Units

1.3 DELIVERY, STORAGE. AND HANDLING

Deliver materials to the site in the manufacturer's original unopened containers with brand name and type clearly marked. Carefully handle and store materials in dry, watertight enclosures. Immediately before installation, store acoustical units for not less than 24 hours at the same temperature and relative humidity as the space where they will be installed in order to assure proper temperature and moisture acclimation.

1.4 ENVIRONMENTAL REQUIREMENTS

Maintain a uniform temperature of not less than 16 degrees C nor more than 29 degrees C and a relative humidity of not more than 70 percent for 24 hours before, during, and 24 hours after installation of acoustical units.

1.5 SCHEDULING

Complete and dry interior finish work such as plastering, concrete and terrazzo work before ceiling installation. Complete mechanical, electrical, and other work above the ceiling line; install and start operating heating, ventilating, and air conditioning systems in order to maintain temperature and humidity requirements.

1.6 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials and workmanship including but not limited to, sagging and warping of panels and rusting and of grid systems, for a period of ten years from date of final acceptance of the work.

1.7 EXTRA MATERIALS

Furnish spare tiles, from the same lot as those installed, of each color at the rate of 10 percent of units installed.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide sound controlling units mechanically mounted on a ceiling suspension system for acoustical treatment. Provide the unit size, texture, finish, and color as specified. Coordinate the entire ceiling system with other details, like the location of access panels and ceiling penetrations, for instance, shown on the drawings. The Contractor is responsible for the final assembly and performance of the specified work. Prepare drawings showing suspension system, method of anchoring and fastening, details, and reflected ceiling plan.

2.1.1 Light Reflectance

Determine light reflectance factor in accordance with ASTM E1477 test method.

2.2 ACOUSTICAL UNITS

Submit samples of each type of acoustical unit and each type of suspension grid tee section showing texture, finish, and color. Conform acoustical units to [ASTM E1264](#) or [UNE-EN 13964](#), Class A, and the following requirements:

2.2.1 Units for Exposed-Grid System

2.2.1.1 Type

III (non-asbestos mineral fiber with painted finish).

2.2.1.2 Flame Spread

Class A, 25 or less

2.2.1.3 Minimum Light Reflectance Coefficient LR-1, 0.75 or greater

2.2.1.4 Nominal Size

600 by 600 mm

2.2.1.5 Edge Detail

Square

2.2.1.6 Finish

Factory-applied standard finish. See paragraph COLORS AND STANDARDS.

2.3 SUSPENSION SYSTEM

Provide standard [exposed-grid](#) standard width flange suspension system conforming to [ASTM C635/C635M](#) and [UNE-EN 13964](#). Provide surfaces exposed to view of aluminum or steel with a factory-applied white color baked-enamel finish. Provide wall molding having a flange of not less than 24 mm. Provide standard corners. Provide a suspension system with a maximum deflection of 1/360 of the span length capable of supporting the finished ceiling, light fixtures, air diffusers, and accessories, as shown.

2.4 HANGERS

Provide hangers and attachment capable of supporting a minimum 1334 N ultimate vertical load without failure of supporting material or attachment.

2.4.1 Wires

Conform wires to [ASTM A641/A641M](#) or [UNE-EN 10244-2](#), Class 1, 2.0 mm in diameter.

2.5 FINISHES

Use manufacturer's standard textures, patterns and finishes as specified for acoustical units and suspension system members. Treat ceiling suspension system components to inhibit corrosion.

2.6 COLORS AND PATTERNS

Use colors and patterns for acoustical units and suspension system components as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers.

PART 3 EXECUTION

3.1 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

Examine surfaces to receive directly attached acoustical units for unevenness, irregularities, and dampness that would affect quality and execution of the work. Rid areas, where acoustical units will be cemented, of oils, form residue, or other materials that reduce bonding capabilities of the adhesive. Complete and dry interior finish work such as plastering, concrete, and terrazzo work before installation. Complete and approve mechanical, electrical, and other work above the ceiling line prior to the start of acoustical ceiling installation. Provide acoustical work complete with necessary fastenings, clips, and other accessories required for a complete installation. Do not expose mechanical fastenings in the finished work. Lay out hangers for each individual room or space. Provide hangers to support framing around beams, ducts, columns, grilles, and other penetrations through ceilings. Keep main runners and carrying channels clear of abutting walls and partitions. Provide at least two main runners for each ceiling span. Wherever required to bypass an object with the hanger wires, install a subsuspension system so that all hanger wires will be plumb.

3.1.1 Suspension System

Install suspension system in accordance with [ASTM C636/C636M](#) and [UNE 102043](#) and as specified herein. Do not suspend hanger wires or other loads from underside of steel decking.

3.1.1.1 Plumb Hangers

Install hangers plumb and not pressing against insulation covering ducts and pipes. Where lighting fixtures are supported from the suspended ceiling system, provide hangers at a minimum of four hangers per fixture and located not more than **152 mm** from each corner of each fixture.

3.1.2 Wall Molding

Provide wall molding where ceilings abut vertical surfaces. Miter corners where wall moldings intersect or install corner caps. Secure wall molding not more than **76 mm** from ends of each length and not more than **406 mm** on centers between end fastenings. Provide wall molding springs at each acoustical unit in semi-exposed or concealed systems.

3.2 CLEANING

Following installation, clean dirty or discolored surfaces of acoustical units and leave them free from defects. Remove units that are damaged or improperly installed and provide new units as directed.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 69 13

RIGID GRID ACCESS FLOORING

11/15, CHG 1: 08/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 SPARE PARTS
- 1.4 QUALITY CONTROL
 - 1.4.1 Qualification of Manufacturer
- 1.5 DELIVERY, STORAGE, AND HANDLING
 - 1.5.1 Delivery
 - 1.5.2 Storage
 - 1.5.3 Handling
- 1.6 WARRANTY

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Design Requirements
 - 2.1.2 Allowable Tolerances
 - 2.1.2.1 Floor Panel Flatness
 - 2.1.2.2 Floor Panel Length
 - 2.1.2.3 Floor Panel Squareness
 - 2.1.3 Stringers
 - 2.1.4 Pedestals
 - 2.1.5 Bonding Strength of Pedestal Adhesive
 - 2.1.6 Bond Strength of Factory Installed Covering
 - 2.1.7 Seismic Calculations
 - 2.1.7.1 Navy Project Specific Requirements
- 2.2 FLOOR PANELS
 - 2.2.1 Floor System Drawings And Planer Quality
 - 2.2.2 Detailed Installation Drawings
 - 2.2.3 Panel Construction
 - 2.2.3.1 Hollow Formed Steel
 - 2.2.4 Floor Covering
 - 2.2.4.1 Solid Vinyl Tile
 - 2.2.5 Accessories
 - 2.2.6 Resilient Base
 - 2.2.7 Adhesives
 - 2.2.8 Lifting Device
- 2.3 PANEL SUPPORT SYSTEM
 - 2.3.1 Pedestals
 - 2.3.2 Stringers
 - 2.3.3 Gaskets
- 2.4 FACTORY TESTS
 - 2.4.1 Load Tests
 - 2.4.2 Bond Strength of Covering
- 2.5 REGISTERS AND GRILLES

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.6 CUT OUTS
- 2.7 EDGE CLOSURE
- 2.8 COLOR

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Preparation for Installation
 - 3.1.2 Pedestals
 - 3.1.3 Stringers
 - 3.1.4 Auxiliary Framing
 - 3.1.5 Panels
 - 3.1.6 Resilient Base
 - 3.1.7 Repair of Zinc Coating
- 3.2 CLEANING AND PROTECTION
 - 3.2.1 Cleaning
 - 3.2.2 Protection
 - 3.2.3 Surplus Material Removal
- 3.3 OPERATION AND MAINTENANCE MANUALS

-- End of Section Table of Contents --

SECTION 09 69 13

RIGID GRID ACCESS FLOORING
11/15, CHG 1: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM A780/A780M (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings

ASTM F1700 (2020) Standard Specification for Solid Vinyl Floor Tile

ASTM F1861 (2021) Standard Specification for Resilient Wall Base

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350 (2010; Version 1.1) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers

CEILINGS AND INTERIOR SYSTEMS CONSTRUCTION ASSOCIATION (CISCA)

CISCA Access Floors (2007) Recommended Test Procedures for Access Floors

GREEN SEAL (GS)

GS-36 (2013) Adhesives for Commercial Use

ICC EVALUATION SERVICE, INC. (ICC-ES)

ICC-ES AC308 (2014) Acceptance Criteria for Access Floors

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2018) International Building Code

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2017) Adhesive and Sealant Applications

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2019, with Change 1, 2022) Structural
Engineering

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detailed Installation Drawings; G

Fabrication Drawings; G

SD-03 Product Data

Access Flooring System; G

Indoor Air Quality For Pedestal Adhesive; S

Indoor Air Quality For Adhesives; S

SD-04 Samples

Floor Panels

Floor Covering; G

Panel Support System

Accessories; G

Cut Outs; G

Seismic Calculations

SD-06 Test Reports

Factory Tests

Concentrated Load

Uniform Live Load

Rolling Load

Impact Load

Ultimate Load

Stringer Load

Pedestal Axial Load

Bonding Strength of Pedestal Adhesive

SD-07 Certificates

Compliance with ICC-ES AC308

Compliance with ICC IBC

Certificate of Compliance

Qualification of Manufacturer

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

SD-11 Closeout Submittals

Lifting Device

Warranty

1.3 SPARE PARTS

Provide four floor panels complete with specified floor covering for future use.

1.4 QUALITY CONTROL

1.4.1 Qualification of Manufacturer

Access flooring manufacturer must have at least 5 years experience in manufacturing access flooring systems. Certify that the manufacturer of the access flooring system meets requirements specified under paragraph entitled QUALIFICATION OF MANUFACTURER.

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 Delivery

Deliver materials to site in undamaged condition, in original containers or packages, complete with accessories and instructions. Label packages with manufacturer's name and brand designations. Package materials covered by specific references bearing specification number, type and class as applicable.

1.5.2 Storage

Store all materials in original protective packaging in a safe, dry, and clean location. Store panels at temperatures between 4 and 32 degrees C, and between 20 and 70 percent humidity. Replace defective or damaged materials.

1.5.3 Handling

Handle and protect materials in a manner to prevent damage during the

entire construction period.

1.6 WARRANTY

Minimum manufacturer warranty must have no dollar limit, cover full system, and must have a minimum duration of 1 years. Include an agreement to repair or replace floor panels, pedestals or stringers that fail within the warranty period in the standard performance guarantee or warranty. Failures include, but are not limited to, sagging and warping of panels; rusting and manufacturers defects of panels or support system. For solid vinyl tile provide manufacturer's standard performance guarantees or warranties that extend beyond a one-year period for finish materials. For conductive vinyl tile provide manufacturer's standard performance guarantees or warranties that extend beyond one year, standard warranty must not be less than a five year wear warranty and ten year conductivity warranty.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

- a. Provide for self-alignment of floor panels, adjustable pedestals and readily removable floor panels covered as specified.
- b. Lateral stability of floor support system must be independent of panels. Provide a finished assembly that is rigid and free of vibration, noises, and rocking panels.
- c. Submit [certificate of compliance](#) attesting that the installed access floor system meets specification requirements, including all special equipment loads and specific electrical and or cable requirements for the complete access flooring system including, but not limited to the following:
 - (1) [Compliance with ICC-ES AC308](#) and [Compliance with ICC IBC](#) Acceptance Criteria for Access Floors.
 - (2) Load-bearing capabilities of pedestals, floor panels, and pedestal adhesive resisting force.
 - (3) Supporting independent laboratory test reports. For panel, stringer and pedestal load test results include concentrated loads at center of panel, panel edge midpoint, ultimate loads and uniform loads.
 - (4) Floor electrical characteristics.
 - (5) Material requirements.
 - (6) An elevated floor system free of defects in materials, fabrication, finish, and installation, that will remain so for a period of not less than 1 years after completion.
- d. Submit manufacturer's product data for [access flooring system](#) consisting of descriptive data, catalog cuts, and installation instructions. Include in the data information about any design and production techniques, total system including all accessories and finish coatings of under-floor components, procedures and policies used to conserve energy, reduce material, improve waste management or

incorporate green building/recycled products into the manufacturer of their components or products. Include cleaning and maintenance instructions. Systems which contain zinc electroplated anti-corrosion coatings are prohibited.

2.1.1.1 Design Requirements

Conduct floor panel testing in accordance with **CISCA Access Floors**. When tested as specified, make all deflection and deformation measurements at the point of load application on the top surface of the panel. Floor panels must be capable of supporting the following loads:

- a. **Concentrated load** of **6670 N** on **645 square mm**, at any point on panel, without a top-surface deflection more than **2.54 mm**, and a permanent set not to exceed **0.25 mm** in any of the specified tests. Testing must be in accordance with **CISCA Access Floors**, Section 1 Concentrated Loads with test panels being supported by understructure to be used with installed system instead of steel support blocks.
- b. **Uniform live load** of **16.76 kPa**, without a top-surface deflection more than **1.5 mm**, and a permanent set not to exceed **0.25 mm** in any of the specified tests, when tested in accordance with **CISCA Access Floors**, Section 7 Uniform Load Test with test panels being supported by understructure to be used with installed system instead of steel support blocks.
- c. A **rolling load** of **5560 N** applied through a **75 mm** diameter by **30 mm** wide caster for 10 cycles over the same path, without developing a local overall surface deformation greater than **1 mm**. In accordance with **CISCA Access Floors**, Section 3 Rolling Loads, the permanent deformation limit under rolling load must be satisfied in all of the specified tests.
- d. An **impact load** of **670 N** anywhere on the panel dropped from a height of **914 mm** onto a **645 square mm** area without failure of the system, according to **CISCA Access Floors**, Section 8 Drop Impact Load Test.
- e. **Ultimate Load**. Panels must meet manufactures published Ultimate Load rating of **11120 N** when tested in accordance with **CISCA Access Floors**, Section 2 Ultimate Loading.
- f. **Safety Factor**. Panels must provide a minimum Safety Factor of 5 times the uniform load specified above in accordance with **ICC-ES AC300**.

2.1.1.2 Allowable Tolerances

2.1.1.2.1 Floor Panel Flatness

Plus or minus **0.89 mm** on diagonal on top of panel or underneath edge.

2.1.1.2.2 Floor Panel Length

Plus or minus **0.4 mm**.

2.1.1.2.3 Floor Panel Squareness

Plus or minus **0.5 mm** in panel length.

2.1.3 Stringers

Provide stringers capable of supporting a **N 2000 N** concentrated load at midspan without permanent deformation in excess of **0.25 mm**, when tested in accordance with **CISCA Access Floors**, Section 4 **Stringer Load** Testing.

2.1.4 Pedestals

Pedestals must be capable of supporting a **22.24 kN** axial load without permanent deformation, when tested in accordance with **CISCA Access Floors**, Section 5 **Pedestal Axial Load** Test.

2.1.5 Bonding Strength of Pedestal Adhesive

Adhesive for anchoring pedestal bases must have a bonding strength capable of resisting an overturning moment of **226 Nm** when a force is applied to the top of the pedestal in any direction, when tested in accordance with **CISCA Access Floors**, Section 6 Pedestal Overturning Moment Test. Pedestal adhesive must meet emissions requirement of **CDPH SECTION 01350** (use the office or classroom requirements, regardless of space type). Provide validation of **indoor air quality for pedestal adhesive**.

2.1.6 Bond Strength of Factory Installed Covering

Bond strength of floor covering must be sufficient to permit handling of the panels by use of the panel lifting device, and to withstand moving caster loads up to **3560 N**, without separation of the covering from the panel.

2.1.7 Seismic Calculations

2.1.7.1 Navy Project Specific Requirements

Submit seismic calculations for lateral bracing, sealed by a Professional Engineer. Document that access flooring system complies with seismic requirements of **ICC IBC** and **ASCE 7-16** for Occupancy Importance Factor (I_p) of 1.5, and seismic horizontal force (F_p) determined in accordance with **UFC 3-301-01** and Section 1615 of the **ICC IBC** and **ASCE 7-16**, Minimum Design Loads for buildings and other structures.

2.2 FLOOR PANELS

2.2.1 Floor System Drawings And Planer Quality

- a. Submit **Fabrication Drawings** for elevated floor systems consisting of fabrication and assembly details to be performed in the factory.
- b. Indicate on Location Drawings exact location of pedestals, ventilation openings, cable cutouts, and the panel installation pattern.
- c. Provide Detail Drawings showing details of the pedestals, pedestal-floor interlocks, floor panels, panel edging, floor openings, floor opening edging, floor registers, floor grilles, cable cutout treatment, perimeter base, expansion, and peripheral support facilities.
- d. Design and workmanship of the floor, as installed, must be completely planar within plus or minus **1.5 mm in 3050 mm**, **2.5 mm** for the entire floor, and **0.7 mm** across panel joints.

- e. Floor-panel joint-width tolerances must not exceed 0.43 mm as measured with a feeler gage at any point in any joint when the panels are installed and as long as the air leakage requirements specified in this section are met.
- f. Submit one complete sample of floor panels.

2.2.2 Detailed Installation Drawings

Submit Detailed Installation Drawings that as a minimum indicate the following:

- a. Location of panels
- b. Layout of supports, panels, and cutout locations
- c. Stair, handrail, and ramp framing
- d. Sizes and details of components
- e. Details at floor perimeter and height above structural floor
- f. Method of anchorage to structural subfloor
- g. Lateral bracing
- h. Typical cutout details
- i. Gasketing, return air grilles, supply air registers, and perforated panels. Include air transfer capacity of grilles, registers and panels
- j. Description of shop coating
- k. Floor finishes
- l. Location of connection to building grounding electrode

2.2.3 Panel Construction

- a. Base access floor system on a 600 by 600 mm square module providing minimum of 300 mm clearance between structural floor and underside of panel and stringer. Fabricate so accurate job cutting and fitting may be done using standard sizes for perimeters and around columns.
- b. Do not expose metal on finished top surface of panels. Provide cutouts and cutout closures to accommodate utility systems and equipment intercabling. Reinforce cutouts to meet design load requirements. Provide extra support pedestals at each corner of cutout for cutout panels that do not meet specified design load requirements.
- c. Panel design must provide for convenient panel removal for underfloor servicing and for openings for new equipment. Use panels of uniform dimensions within specified tolerances. Permanently mark panels to indicate load rating and model number.
- d. Machine square floor panels to within plus or minus 0.38 mm with edge straightness plus or minus 0.064 mm. If plastic edging is applied to

the panel, the tolerances apply to the panel before the plastic edging is applied.

2.2.3.1 Hollow Formed Steel

Steel panels must be of die-formed construction, consisting of a flat steel top sheet welded to one or more formed steel stiffener sheets or components. Panels must be chemically cleaned, bonderized, and painted with the manufacturer's standard finish.

2.2.4 Floor Covering

Surface floor panels with factory applied finish materials firmly bonded in place with waterproof adhesive. The electrical resistance must remain stable over the life expectancy of the floor covering. Any anti-static agent used in the manufacturing process must be an integral part of the material, not surface applied. Bolt heads or similar attachments must not rise above the traffic surface. Submit three separate samples of each specified floor covering finish and color.

2.2.4.1 Solid Vinyl Tile

Provide factory applied conductive vinyl tile that is a homogeneous vinyl product and conforms to [ASTM F1700](#), Class I monolithic (minimum wear layer thickness [3 mm](#) and minimum overall thickness [3 mm](#)), Type A smooth surface. Finish material must consist of one piece to cover the face of the panel. Provide edge detail that is integral to the finish material or is an applied trim piece that finishes the edges of the panel, is flush with floor finish, and is PVC.

2.2.5 Accessories

Provide the manufacturer's standard registers, grilles, perforated panels, and plenum dividers type where indicated. Provide registers, grilles, and perforated panels designed to support the same static loads as floor panels without structural failure, and capable of delivering the air volumes indicated. Registers and perforated panels must be 25 percent open area and equipped with adjustable dampers. Submit three samples and colors of each accessory.

2.2.6 Resilient Base

Conform to [ASTM F1861](#), Type TS (vulcanized thermoset rubber) or Type TP (thermoplastic rubber), Style B (coved - installed with resilient flooring). Provide [100 mm](#) high and a minimum [3.175 mm](#) thick wall base. Provide job formed corners in matching height, shape, and color.

2.2.7 Adhesives

Provide adhesives as recommended by the manufacturer. Provide non-aerosol adhesive products that meet either emissions requirements of [CDPH SECTION 01350](#) (use the requirements for either office or classroom, regardless of space type) or VOC content requirements of [SCAQMD Rule 1168](#). Provide aerosol adhesives that meet either emissions requirements of [CDPH SECTION 01350](#) (use the requirements for office or classroom, regardless of space type) or VOC content requirements of [GS-36](#). Provide validation of [indoor air quality for adhesives](#). Provide conductive adhesive as recommended by the manufacturer of the static-control flooring.

2.2.8 Lifting Device

At turn over provide one floor panel lifting device standard with the floor manufacturer, for each individual floor area (room or corridor). Furnish a minimum of two devices.

2.3 PANEL SUPPORT SYSTEM

Design support system to allow for 360 degree clearance in laying out cable and cutouts for service to machines and so that panel and stringer together take up maximum of 50 mm. Submit one sample of suspension system proposed for use.

2.3.1 Pedestals

Provide pedestals made of steel or aluminum or a combination thereof. Ferrous materials must have a factory-applied corrosion-resistant finish. Provide pedestal base plates with a minimum of 10,300 square mm of bearing surface and a minimum of 3 mm thickness. Pedestal shafts must be threaded to permit height adjustment within a range of approximately 50 mm, to permit overall floor adjustment within plus or minus 2.5 mm of the required elevation, and to permit leveling of the finished floor surface within 1.56 mm in 3000 mm in all directions. Provide locking devices to positively lock the final pedestal vertical adjustments in place. Pedestal caps must interlock with panels to preclude tilting or rocking of the panels.

2.3.2 Stringers

Provide stringers of rolled steel or extruded aluminum, to interlock with the pedestal heads to prevent lateral movement. Provide stringers that can be added or removed after floor is in place.

2.3.3 Gaskets

Provide continuous gasketing at contact surfaces between panel and stringers to deaden sound and seal off the underfloor cavity from above for air tightness, and to maintain panel alignment.

2.4 FACTORY TESTS

Factory test access flooring, using an independent laboratory, at the same position and maximum design elevation and in the same arrangement as shown on the drawings for installation so as to duplicate service conditions as much as possible.

2.4.1 Load Tests

Conduct floor panel, stringer, and pedestal testing in accordance with CISCA Access Floors to determine deformation and permanent set of panels and sytem due to concentrated, Uniform, rolling, impact and ultimate loading when panels are supported by actual understructure.

2.4.2 Bond Strength of Covering

Conduct test for bond strength of covering in accordance with CISCA Access Floors for rolling loads, except as specified. Panels must be tested with specified hard surface flooring and on the pedestals and stringers as specified for the installed floor. Brace the supports as

necessary to prevent sideways movement during the test. Impose a test load of 5560 N on the test assembly through a 75 mm in diameter and 25 mm wide hard plastic caster. Roll the caster completely across the center of the panel. The panel shall withstand 20 passes of the caster with no delamination or separation of the covering.

2.5 REGISTERS AND GRILLES

Registers and grilles must be 600 mm by 600 mm long with a minimum free area of 50 percent, made from extruded aluminum, in mill finish, to sustain point loads of 1100 newton per vane without failure or permanent deformation. No part of a grille may project more than 3 mm above the floor. Registers and grills are not permitted in a laminate floor tile system.

2.6 CUT OUTS

Provide cable cutouts finished with rigid polyvinylchloride or molded polypropylene edging to conform to the appearance level of the floor surface and to cover raw edges of the cutout panel. Extrusion must be of a configuration to permit its effective and convenient use when new cable openings are required. Provide at least 7300 mm of additional extrusion for future use. Submit three color samples for cut outs.

- a. Provide non-metallic adapter for openings less than 100 mm wide. Secure adapter adhesively in cutout to preclude removal from panel. Provide at least two adapters per 10 square meter for future use.
- b. Openings larger than 100 mm wide must use rigid polyvinylchloride or molded polypropylene edging. Perform cutting of panels, including cutouts, outside of the building.
- c. When size of cutout reduces the performance requirement of panel, provide intermediate stringers adjacent to cutouts.

2.7 EDGE CLOSURE

Provide 1.5 mm aluminum closure plate and extruded aluminum nosing at exposed edge of floor. Back up the closure plates with aluminum or steel framing braced diagonally, or anchor at bottom to continuous angle.

2.8 COLOR

Color must be as indicated. Color listed is not intended to limit the selection of equal colors from other manufacturers.

PART 3 EXECUTION

3.1 INSTALLATION

Install access flooring at the location and elevation and in the arrangement shown on the approved detailed installation drawings. The floor system must be of the rigid grid stringer type, complete with all supplemental items, and be the standard product of a manufacturer specializing in access flooring systems.

Install the floor system in accordance with the manufacturer's instructions. Open ends of the floor, where the floor system does not abut wall or other construction, must have positive anchorage and rigid

support. Maintain areas to receive access flooring between 16 and 32 degrees C, and between 20 and 70 percent humidity for 24 hours prior to and during installation.

3.1.1 Preparation for Installation

Clear out all debris in the area in which the floor system is to be installed. Thoroughly clean structural floor surfaces and remove all dust. Install floor coatings, required for dust or vapor control, prior to installation of pedestals, only if the pedestal adhesive will not damage the coating. If the coating and adhesive are not compatible, apply the coating after the pedestals have been installed and the adhesive has cured.

3.1.2 Pedestals

Pedestals must be accurately spaced, and set plumb and in true alignment. Set base plates in full and firm contact with the structural floor, and secured to the structural floor with adhesive or steel expansion anchors in accordance with manufacturer's instructions.

3.1.3 Stringers

Interlock stringers with the pedestal caps to preclude lateral movement, spaced uniformly in parallel lines at the indicated elevation.

3.1.4 Auxiliary Framing

Provide auxiliary framing or pedestals around columns and other permanent construction, at sides of ramps, at open ends of the floor, and beneath panels that are substantially cut to accommodate utility systems. Use special framing for additional lateral support as shown on the approved detailed installation drawings. Provide additional pedestals and stringers designed to specific heights and lengths to meet structural irregularities and design loads. Connect auxiliary framing to main framing.

3.1.5 Panels

Interlock panels with supports in a manner that will preclude lateral movement. Fasten perimeter panels, cutout panels, and panels adjoining columns, stairs, and ramps to the supporting components to form a rigid boundary for the interior panels. Level floors within the specified tolerances. Cut edges of steel and wood-core panels must be finished as recommended by the panel manufacturer. Secure extruded vinyl edging in place at all cut edges of all panel cut-outs to prevent abrasion of cables.

3.1.6 Resilient Base

Provide base at vertical wall intersections as indicated in the drawings. Apply the base after the floor system has been completely installed. Install wall base in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's printed directions. Tighten base joints and make even with adjacent flooring. Fill voids along the top edge of base at masonry walls with caulk. Roll entire vertical surface of base with hand roller, and press toe of base with a straight piece of wood to ensure proper alignment. Avoid excess adhesive in corners.

3.1.7 Repair of Zinc Coating

Repair zinc coating that has been damaged, and cut edges of zinc-coated components and accessories, by the application of a galvanizing repair paint conforming to [ASTM A780/A780M](#). Areas to be repaired must be thoroughly cleaned prior to application of the paint.

3.2 CLEANING AND PROTECTION

3.2.1 Cleaning

Keep the space below the completed floor free of all debris. Before any traffic or other work on the completed raised floor is started, clean the completed floor in accordance with the floor covering manufacturer's instructions.

3.2.2 Protection

Protect traffic areas of raised floor systems with a covering of building paper, fiberboard, or other suitable material to prevent damage to the surface. Cover cutouts with material of sufficient strength to support the loads to be encountered. Place plywood or similar material on the floor to serve as runways for installation of heavy equipment not in excess of design load capacity. Maintain protection until the raised floor system is accepted.

3.2.3 Surplus Material Removal

Clean surfaces of the work, and adjacent surfaces soiled as a result of the work. Remove all installation equipment, surplus materials, and rubbish from the work site.

3.3 OPERATION AND MAINTENANCE MANUALS

Submit maintenance instructions for proper care of the floor panel surface. When conductive flooring is specified, also submit maintenance instructions to identify special cleaning and maintenance requirements to maintain "conductivity" properties of the panel finish.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 90 00

PAINTS AND COATINGS

02/21

PART 1 GENERAL

- 1.1 RELATED REQUIREMENTS
 - 1.1.1 Painting Included
 - 1.1.1.1 Exterior Painting
 - 1.1.1.2 Interior Painting
 - 1.1.2 Painting Excluded
 - 1.1.3 Mechanical and Electrical Painting
 - 1.1.4 Miscellaneous Painting
 - 1.1.4.1 Lettering Building
- 1.2 REFERENCES
- 1.3 DEFINITIONS
 - 1.3.1 Qualification Testing
 - 1.3.2 Batch Quality Conformance Testing
 - 1.3.3 Coating
 - 1.3.4 DFT or dft
 - 1.3.5 DSD
 - 1.3.6 EXT
 - 1.3.7 INT
 - 1.3.8 Loose Paint
 - 1.3.9 micron / microns
 - 1.3.10 mm
 - 1.3.11 MPI Gloss Levels
 - 1.3.12 MPI System Number
 - 1.3.13 Paint
- 1.4 SCHEDULING
- 1.5 SUBMITTALS
- 1.6 QUALITY ASSURANCE
 - 1.6.1 Regulatory Requirements
 - 1.6.1.1 Environmental Protection
 - 1.6.1.2 Lead Content
 - 1.6.1.3 Chromate Content
 - 1.6.1.4 Asbestos Content
 - 1.6.1.5 Mercury Content
 - 1.6.1.6 Silica
 - 1.6.1.7 Human Carcinogens
 - 1.6.1.8 Carbon Based Fibers / Tubes
 - 1.6.2 Coating Contractor's Qualification
 - 1.6.3 Approved Products List
 - 1.6.4 Paints and Coatings Indoor Air Quality Certifications
- 1.7 PACKAGING, LABELING, AND STORAGE
- 1.8 SAFETY AND HEALTH
 - 1.8.1 Toxic Materials
- 1.9 ENVIRONMENTAL REQUIREMENTS
 - 1.9.1 Coatings
 - 1.9.2 Post-Application

PART 2 PRODUCTS

- 2.1 MATERIALS
- 2.2 COLOR SELECTION OF FINISH COATS

PART 3 EXECUTION

- 3.1 PROTECTION OF AREAS AND SPACES NOT TO BE PAINTED
- 3.2 SURFACE PREPARATION
- 3.3 PREPARATION OF METAL SURFACES
 - 3.3.1 Existing and New Ferrous Surfaces
 - 3.3.2 Final Ferrous Surface Condition:
 - 3.3.2.1 Tool Cleaned Surfaces
 - 3.3.2.2 Abrasive Blast Cleaned Surfaces
 - 3.3.2.3 Waterjet Cleaned Surfaces
 - 3.3.3 Galvanized Surfaces
 - 3.3.4 Non-Ferrous Metallic Surfaces
 - 3.3.5 Terne-Coated Metal Surfaces
 - 3.3.6 Existing Surfaces with a Bituminous or Mastic-Type Coating
- 3.4 PREPARATION OF CONCRETE AND CEMENTITIOUS SURFACE
 - 3.4.1 Concrete and Masonry
 - 3.4.2 Gypsum Board, Plaster, and Stucco
 - 3.4.2.1 Surface Cleaning
 - 3.4.2.2 Repair of Minor Defects
 - 3.4.2.3 Allowable Moisture Content
- 3.5 APPLICATION
 - 3.5.1 Coating Application
 - 3.5.2 Mixing and Thinning of Paints
 - 3.5.3 Two-Component Systems
 - 3.5.4 Coating Systems
- 3.6 COATING SYSTEMS FOR METAL
- 3.7 COATING SYSTEMS FOR CONCRETE AND CEMENTITIOUS SUBSTRATES
- 3.8 PIPING IDENTIFICATION
- 3.9 INSPECTION AND ACCEPTANCE
- 3.10 PAINT TABLES
 - 3.10.1 Exterior Paint Tables
 - 3.10.1.1 MPI Division 3: Exterior Concrete Paint Table
 - 3.10.1.2 MPI Division 5: Exterior Metal, Ferrous and Non-Ferrous Paint Table
 - 3.10.1.3 MPI Division 9: Exterior Stucco Paint Table
 - 3.10.2 Interior Paint Tables
 - 3.10.2.1 MPI Division 9: Interior Plaster, Gypsum Board, Textured Surfaces Paint Table

-- End of Section Table of Contents --

SECTION 09 90 00

PAINTS AND COATINGS
02/21

PART 1 GENERAL

1.1 RELATED REQUIREMENTS

1.1.1 Painting Included

Where a space or surface is indicated to be painted, include the following unless indicated otherwise.

- a. Surfaces behind portable objects and surface mounted articles readily detachable by removal of fasteners, such as screws and bolts.
- b. New factory finished surfaces that require identification or color coding and factory finished surfaces that are damaged during performance of the work.
- c. Existing coated surfaces that are damaged during performance of the work.

1.1.1.1 Exterior Painting

Includes new surfaces of the buildings and appurtenances. Also included are existing coated surfaces made bare by cleaning operations.

1.1.1.2 Interior Painting

Includes new surfaces of the buildings and appurtenances as indicated and existing coated surfaces made bare by cleaning operations. Where a space or surface is indicated to be painted, include the following items, unless indicated otherwise.

- a. Exposed columns, girders, beams, joists, and metal deck; and
- b. Other contiguous surfaces.

1.1.2 Painting Excluded

Do not paint the following unless indicated otherwise.

- a. Surfaces concealed and made inaccessible by panelboards, fixed ductwork, machinery, and equipment fixed in place.
- b. Surfaces in concealed spaces. Concealed spaces are defined as enclosed spaces above suspended ceilings, furred spaces, attic spaces, crawl spaces, elevator shafts and chases.
- c. Steel to be embedded in concrete.
- d. Copper, stainless steel, aluminum, anodized aluminum, brass, and lead except existing coated surfaces.
- e. Hardware, fittings, and other factory finished items.

1.1.3 Mechanical and Electrical Painting

Includes field coating of interior and exterior new surfaces.

- a. Where a space or surface is indicated to be painted, include the following items unless indicated otherwise.
 - (1) Exposed piping, conduit, and ductwork;
 - (2) Supports, hangers, air grilles, and registers;
 - (3) Miscellaneous metalwork and insulation coverings.
- b. Do not paint the following, unless indicated otherwise:
 - (1) New zinc-coated, aluminum, and copper surfaces under insulation
 - (2) New aluminum jacket on piping
 - (3) New interior ferrous piping under insulation.

1.1.4 Miscellaneous Painting

1.1.4.1 Lettering Building

Provide lettering as scheduled on the drawings type, . Samples must be approved before application.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

ACGIH 0100 (2017; Suppl 2020) Documentation of the Threshold Limit Values and Biological Exposure Indices

ASTM INTERNATIONAL (ASTM)

ASTM D235 (2002; R 2012) Mineral Spirits (Petroleum Spirits) (Hydrocarbon Dry Cleaning Solvent)

ASTM D523 (2014; R 2018) Standard Test Method for Specular Gloss

ASTM D4263 (1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method

ASTM D4444 (2013; R 2018) Standard Test Method for Laboratory Standardization and Calibration of Hand-Held Moisture Meters

ASTM D6386 (2016a) Standard Practice for Preparation of Zinc (Hot-Dip Galvanized) Coated Iron

and Steel Product and Hardware Surfaces
for Painting

ASTM F1869

(2016a) Standard Test Method for Measuring
Moisture Vapor Emission Rate of Concrete
Subfloor Using Anhydrous Calcium Chloride

CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC)

Intelligence Bulletin 65

(2013) Occupational Exposure to Carbon
Nanotubes and Nanofibers

MASTER PAINTERS INSTITUTE (MPI)

MPI 3

(2016) Primer, Alkali Resistant, Water
Based

MPI 9

(2016) Alkyd, Exterior Gloss (MPI Gloss
Level 6)

MPI 10

(2016) Latex, Exterior Flat (MPI Gloss
Level 1)

MPI 11

(2016) Latex, Exterior Semi-Gloss, MPI
Gloss Level 5

MPI 23

(2015) Primer, Metal, Surface Tolerant

MPI 79

(2016) Primer, Alkyd, Anti-Corrosive for
Metal

MPI 94

(2016) Alkyd, Exterior, Semi-Gloss (MPI
Gloss Level 5)

MPI 101

(2016) Primer, Epoxy, Anti-Corrosive, for
Metal

MPI 107

(2016) Primer, Rust-Inhibitive, Water Based

MPI 113

(2018) Elastomeric, Pigmented, Exterior,
Water Based, Flat

MPI 119

(2016) Latex, Exterior, Gloss (MPI Gloss
Level 6)

MPI 144

(2016) Latex, Interior, Institutional Low
Odor/VOC, (MPI Gloss Level 2)

MPI 145

(2016) Latex, Interior, Institutional Low
Odor/VOC, (MPI Gloss Level 3)

MPI 146

(2016) Latex, Interior, Institutional Low
Odor/VOC, (MPI Gloss Level 4)

MPI 147

(May 2016) Latex, Interior, Institutional
Low Odor/VOC, Semi-Gloss (MPI Gloss Level
5)

MPI 149

(2016) Primer Sealer, Interior,

Institutional Low Odor/VOC

MPI 214	(2016) Latex, Exterior (MPI Gloss Level 2)
MPI ASM	(2019) Architectural Painting Specification Manual
MPI GPS-1-14	(2014) Green Performance Standard GPS-1-14
MPI GPS-2-14	(2014) Green Performance Standard GPS-2-14
MPI MRM	(2015) Maintenance Repainting Manual

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC 7/NACE No.4	(2007) Brush-Off Blast Cleaning
SSPC Glossary	(2011) SSPC Protective Coatings Glossary
SSPC PA 1	(2016) Shop, Field, and Maintenance Coating of Metals
SSPC SP 1	(2015) Solvent Cleaning
SSPC SP 2	(2018) Hand Tool Cleaning
SSPC SP 3	(2018) Power Tool Cleaning
SSPC SP 6/NACE No.3	(2007) Commercial Blast Cleaning
SSPC SP 10/NACE No. 2	(2015) Near-White Blast Cleaning
SSPC VIS 1	(2002; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning
SSPC VIS 3	(2004) Guide and Reference Photographs for Steel Surfaces Prepared by Hand and Power Tool Cleaning
SSPC VIS 4/NACE VIS 7	(1998; E 2000; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Waterjetting
SSPC-SP WJ-1/NACE WJ-1	(2012) Clean to Bare Substrate, Waterjet Cleaning of Metals
SSPC-SP WJ-2/NACE WJ-2	(2012) Very Thorough Cleaning, Waterjet Cleaning of Metals
SSPC-SP WJ-3/NACE WJ-3	(2012) Thorough Cleaning, Waterjet Cleaning of Metals
SSPC-SP WJ-4/NACE WJ-4	(2012) Light Cleaning, Waterjet Cleaning of Metals

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety -- Safety and Health
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Requirements Manual

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-101

(2014; Rev C) Color Code for Pipelines and
for Compressed Gas Cylinders

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1000

Air Contaminants

1.3 DEFINITIONS

1.3.1 Qualification Testing

Qualification testing is the performance of all test requirements listed in the product specification. This testing is accomplished by MPI to qualify each product for the MPI Approved Product List, and may also be accomplished by Contractor's third-party testing lab if an alternative to Batch Quality Conformance Testing by MPI is desired.

1.3.2 Batch Quality Conformance Testing

Batch quality conformance testing determines that the product provided is the same as the product qualified to the appropriate product specification. This testing must be accomplished by an MPI testing lab.

1.3.3 Coating

SSPC Glossary; (1) A liquid, liquefiable, or mastic composition that is converted to a solid protective, decorative, or functional adherent film after application as a thin layer; (2) Generic term for paint, lacquer, enamel.

1.3.4 DFT or dft

Dry film thickness, the film thickness of the fully cured, dry paint or coating.

1.3.5 DSD

Degree of Surface Degradation, the MPI system of defining degree of surface degradation. Five levels are generically defined under the Assessment sections in the MPI MRM, MPI Maintenance Repainting Manual.

1.3.6 EXT

MPI short term designation for an exterior coating system.

1.3.7 INT

MPI short term designation for an interior coating system.

1.3.8 Loose Paint

Paint or coating that can be removed with a dull putty knife.

1.3.9 micron / microns

The metric measurement for 0.001 mm or one one-thousandth of a millimeter.

1.3.10 mm

The metric measurement for millimeter, 0.001 meter or one one-thousandth of a meter.

1.3.11 MPI Gloss Levels

MPI system of defining gloss. Seven gloss levels (G1 to G7) are generically defined under the Evaluation sections of the MPI Manuals. Traditionally, Flat refers to G1/G2, Eggshell refers to G3, Semigloss refers to G5, and Gloss refers to G6.

Gloss levels are defined by MPI as follows:

Gloss Level	Description	Units at 60 degree angle	Units at 80 degree angle
G1	Matte or Flat	0 to 5	10 max
G2	Velvet	0 to 10	10 to 35
G3	Eggshell	10 to 25	10 to 35
G4	Satin	20 to 35	35 min
G5	Semi-Gloss	35 to 70	
G6	Gloss	70 to 85	
G7	High Gloss		

Gloss is tested in accordance with [ASTM D523](#). Historically, the Government has used Flat (G1 / G2), Eggshell (G3), Semi-Gloss (G5), and Gloss (G6).

1.3.12 MPI System Number

The MPI coating system number in each MPI Division found in either the MPI Architectural Painting Specification Manual or the Maintenance Repainting Manual and defined as an exterior (EXT/REX) or interior system (INT/RIN).

1.3.13 Paint

[SSPC Glossary](#); (1) Any pigmented liquid, liquefiable, or mastic composition designed for application to a substrate in a thin layer that is converted to an opaque solid film after application. Used for protection, decoration, identification, or to serve some other functional purposes; (2) Application of a coating material.

1.4 SCHEDULING

Allow paint, polyurethane, varnish, and wood stain installations to cure prior to the installation of materials that adsorb VOCs, including unprimed gypsum wallboard, acoustical ceiling panels.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Samples of specified materials may be taken and tested for compliance with specification requirements.

SD-02 Shop Drawings

Piping Identification

SD-03 Product Data

Coating; G

Product Data Sheets

SD-04 Samples

Color; G

SD-08 Manufacturer's Instructions

Mixing

Manufacturer's Safety Data Sheets

SD-10 Operation and Maintenance Data

Coatings, Data Package 1; G

1.6 QUALITY ASSURANCE

1.6.1 Regulatory Requirements

1.6.1.1 Environmental Protection

In addition to requirements specified elsewhere for environmental protection, provide coating materials that conform to the restrictions of the local Air Pollution Control District and regional jurisdiction. Notify Contracting Officer of any paint specified herein which fails to conform.

1.6.1.2 Lead Content

Do not use coatings having a lead content over 0.06 percent by weight of nonvolatile content.

1.6.1.3 Chromate Content

Do not use coatings containing zinc-chromate or strontium-chromate.

1.6.1.4 Asbestos Content

Provide asbestos-free materials.

1.6.1.5 Mercury Content

Provide materials free of mercury or mercury compounds.

1.6.1.6 Silica

Provide abrasive blast media containing no free crystalline silica.

1.6.1.7 Human Carcinogens

Provide materials that do not contain ACGIH 0100 confirmed human carcinogens (A1) or suspected human carcinogens (A2).

1.6.1.8 Carbon Based Fibers / Tubes

Materials must not contain carbon based fibers such as carbon nanotubes or carbon nanofibers. Intelligence Bulletin 65 ranks toxicity of carbon nanotubes on a par with asbestos.

1.6.2 Coating Contractor's Qualification

Submit the name, address, telephone number, and e-mail address of the Contractor that will be performing all surface preparation and coating application. Submit evidence that key personnel have successfully performed surface preparation and application of coatings on a minimum of three similar projects within the past three years. List information by individual and include the following:

a. Name of individual and proposed position for this work.

b. Information about each previous assignment including:

Position or responsibility

Employer (if other than the Contractor)

Name of facility owner

Mailing address and telephone number of facility owner

Name of individual in facility owner's organization who can be contacted as a reference

Location, size and description of structure

Dates work was carried out

Description of work carried out on structure

1.6.3 Approved Products List

The current MPI, "Approved Product List" which lists paint by brand, label, product name and product code as of the date of Contract award, will be used to determine compliance with the submittal requirements of this specification. The Contractor may choose to use a subsequent MPI "Approved Product List", however, only one list may be used for the entire Contract and each coating system is to be from a single manufacturer. Provide all coats on a particular substrate from a single manufacturer. No variation from the MPI Approved Products List is acceptable.

1.6.4 Paints and Coatings Indoor Air Quality Certifications

Provide paint and coating products certified to meet indoor air quality requirements by MPI GPS-1-14, MPI GPS-2-14 or provide certification by other third-party programs. Provide current product certification documentation from certification body.

1.7 PACKAGING, LABELING, AND STORAGE

Provide paints in sealed containers that legibly show the Contract specification number, designation name, formula or specification number, batch number, color, quantity, date of manufacture, manufacturer's formulation number, manufacturer's directions including any warnings and special precautions, and name and address of manufacturer. Furnish pigmented paints in containers not larger than 20 liters. Store paints and thinners in accordance with the manufacturer's written directions, and as a minimum, stored off the ground, under cover, with sufficient ventilation to prevent the buildup of flammable vapors, and at temperatures between 4 to 35 degrees C.

1.8 SAFETY AND HEALTH

Comply with applicable Federal, State, and local laws and regulations, and with the ACCIDENT PREVENTION PLAN, including the Activity Hazard Analysis as specified in Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and in Appendix A of EM 385-1-1. Include in the Activity Hazard Analysis the potential impact of painting operations on painting personnel and on others involved in and adjacent to the work zone.

1.8.1 Toxic Materials

To protect personnel from overexposure to toxic materials, conform to the most stringent guidance of:

- a. The applicable manufacturer's Safety Data Sheets (SDS) or local regulation.
- b. 29 CFR 1910.1000.
- c. ACGIH 0100, threshold limit values.

1.9 ENVIRONMENTAL REQUIREMENTS

Comply, at minimum, with manufacturer recommendations for space ventilation during and after installation.

1.9.1 Coatings

Do not apply coating when air or substrate conditions are:

- a. Less than 3 degrees C above dew point;
- b. Below 10 degrees C or over 35 degrees C, unless specifically pre-approved by the Contracting Officer and the product manufacturer. Do not, under any circumstances, violate the manufacturer's application recommendations.

1.9.2 Post-Application

Vacate space for as long as possible after application. Wait a minimum of 48 hours before occupying freshly painted rooms. Maintain one of the following ventilation conditions during the curing period, or for 72 hours after application:

- a. Supply 100 percent outside air 24 hours a day.
- b. Supply airflow at a rate of 6 air changes per hour, when outside temperatures are between 13 degrees C and 29 degrees C and humidity is between 30 percent and 60 percent.
- c. Supply airflow at a rate of 1.5 air changes per hour, when outside air conditions are not within the range stipulated above.

PART 2 PRODUCTS

2.1 MATERIALS

Conform to the coating specifications and standards referenced in PART 3. Submit Product Data Sheets for specified coatings and solvents. Provide preprinted cleaning and maintenance instructions for all coating systems. Submit Manufacturer's Instructions on Mixing: Detailed mixing instructions, minimum and maximum application temperature and humidity, pot life, and curing and drying times between coats.

2.2 COLOR SELECTION OF FINISH COATS

Provide colors of finish coats as indicated or specified. Allow Contracting Officer to select colors not indicated or specified. Manufacturers' names and color identification are used for the purpose of color identification only. Named products are acceptable for use only if they conform to specified requirements. Products of other manufacturers are acceptable if the colors are approximately the colors indicated and the product conforms to specified requirements.

Provide color, texture, and pattern of wall coating systems as indicated. Submit manufacturer's samples of paint colors. Cross reference color samples to color scheme as indicated. Submit color stencil codes. Tint each coat progressively darker to enable confirmation of the number of coats.

PART 3 EXECUTION

3.1 PROTECTION OF AREAS AND SPACES NOT TO BE PAINTED

Prior to surface preparation and coating applications, remove, mask, or otherwise protect hardware, hardware accessories, machined surfaces, radiator covers, plates, lighting fixtures, public and private property, and other such items not to be coated that are in contact with surfaces to be coated. Following completion of painting, reinstall removed items by workmen skilled in the trades. Restore surfaces contaminated by coating materials, to original condition and repair damaged items.

3.2 SURFACE PREPARATION

Remove dirt, splinters, loose particles, grease, oil, and other foreign matter and substances deleterious to coating performance as specified for

each substrate before application of paint or surface treatments. Remove oil and grease prior to mechanical cleaning. Schedule cleaning so that dust and other contaminants will not fall on wet, newly painted surfaces. Spot-prime exposed ferrous metals such as nail heads on or in contact with surfaces to be painted with water-thinned paints, with a suitable corrosion-inhibitive primer capable of preventing flash rusting and compatible with the coating specified for the adjacent areas. Refer to MPI ASM and MPI MRM for additional more specific substrate preparation requirements.

3.3 PREPARATION OF METAL SURFACES

3.3.1 Existing and New Ferrous Surfaces

- a. Ferrous Surfaces including Shop-coated Surfaces and Small Areas That Contain Rust, Mill Scale and Other Foreign Substances: Solvent clean or detergent wash in accordance with SSPC SP 1 to remove oil and grease. Where shop coat is missing or damaged, clean according to SSPC SP 2. Brush-off blast remaining surface in accordance with SSPC 7/NACE No.4; Protect shop-coated ferrous surfaces from corrosion by treating and touching up corroded areas immediately upon detection.
- b. Surfaces With More Than 20 Percent Rust, Mill Scale, and Other Foreign Substances: Clean entire surface in accordance with SSPC SP 6/NACE No.3 / SSPC-SP WJ-3/NACE WJ-3.

3.3.2 Final Ferrous Surface Condition:

3.3.2.1 Tool Cleaned Surfaces

Comply with SSPC SP 2 and SSPC SP 3. Use as a visual reference, photographs in SSPC VIS 3 for the appearance of cleaned surfaces.

3.3.2.2 Abrasive Blast Cleaned Surfaces

Comply with SSPC 7/NACE No.4, SSPC SP 6/NACE No.3, and SSPC SP 10/NACE No. 2. Use as a visual reference, photographs in SSPC VIS 1 for the appearance of cleaned surfaces.

3.3.2.3 Waterjet Cleaned Surfaces

Comply with SSPC-SP WJ-1/NACE WJ-1, SSPC-SP WJ-2/NACE WJ-2, SSPC-SP WJ-3/NACE WJ-3 or SSPC-SP WJ-4/NACE WJ-4. Use as a visual reference, photographs in SSPC VIS 4/NACE VIS 7 for the appearance of cleaned surfaces.

3.3.3 Galvanized Surfaces

- a. New or Existing Galvanized Surfaces With Only Dirt and Zinc Oxidation Products: Clean with solvent, or non-alkaline detergent solution in accordance with SSPC SP 1. Completely remove coating by brush-off abrasive blast if the galvanized metal has been passivated or stabilized. Do not "passivate" or "stabilize" new galvanized steel to be coated. If the absence of hexavalent stain inhibitors is not documented, test as described in ASTM D6386, Appendix X2, and remove by one of the methods described therein.

3.3.4 Non-Ferrous Metallic Surfaces

Aluminum and aluminum-alloy, lead, copper, and other nonferrous metal surfaces.

Surface Cleaning: Solvent clean in accordance with **SSPC SP 1** and wash with mild non-alkaline detergent to remove dirt and water soluble contaminants.

3.3.5 Terne-Coated Metal Surfaces

Solvent clean surfaces with mineral spirits, **ASTM D235**. Wipe dry with clean, dry cloths.

3.3.6 Existing Surfaces with a Bituminous or Mastic-Type Coating

Remove chalk, mildew, and other loose material by washing with a solution of **0.20 liter** trisodium phosphate, **0.1 liter** household detergent, **1.6 liters** 5 percent sodium hypochlorite solution and **4.8 liters** of warm water.

3.4 PREPARATION OF CONCRETE AND CEMENTITIOUS SURFACE

3.4.1 Concrete and Masonry

- a. Curing: Allow concrete, stucco and masonry surfaces to cure at least 30 days before painting, and concrete slab on grade to cure at least 90 days before painting.
- b. Surface Cleaning: Remove the following deleterious substances.
 - (1) Dirt, Grease, and Oil: Wash new surfaces with a solution composed of **0.2 liter** trisodium phosphate, **0.1 liter** household detergent, and **6.4 liters** of warm water. Then rinse thoroughly with fresh water. For large areas, water blasting may be used.
 - (2) Fungus and Mold: Wash new surfaces with a solution composed of **0.2 liter** trisodium phosphate, **0.1 liter** household detergent, **1.6 liters** 5 percent sodium hypochlorite solution and **4.8 liters** of warm water. Rinse thoroughly with fresh water.
 - (3) Paint and Loose Particles: Remove by wire brushing.
 - (4) Efflorescence: Remove by scraping or wire brushing followed by washing with a 5 to 10 percent by weight aqueous solution of hydrochloric (muriatic) acid. Do not allow acid to remain on the surface for more than five minutes before rinsing with fresh water. Do not acid clean more than **0.4 square meter** of surface, per workman, at one time.
- c. Cosmetic Repair of Minor Defects: Repair or fill mortar joints and minor defects, including but not limited to spalls, in accordance with manufacturer's recommendations and prior to coating application.
- d. Allowable Moisture Content: Latex coatings may be applied to damp surfaces, but not to surfaces with droplets of water. Do not apply epoxies to damp vertical surfaces as determined by **ASTM D4263** or horizontal surfaces that exceed **169.0 micrograms moisture per second, square meter** as determined by **ASTM F1869**. In all cases follow manufacturer's recommendations. Allow surfaces to cure a minimum of

30 days before painting.

3.4.2 Gypsum Board, Plaster, and Stucco

3.4.2.1 Surface Cleaning

Verify that plaster and stucco surfaces are free from loose matter and that gypsum board is dry. Remove loose dirt and dust by brushing with a soft brush, rubbing with a dry cloth, or vacuum-cleaning prior to application of the first coat material. A damp cloth or sponge may be used if paint is water-based.

3.4.2.2 Repair of Minor Defects

Prior to painting, repair joints, cracks, holes, surface irregularities, and other minor defects with patching plaster or spackling compound and sand smooth.

3.4.2.3 Allowable Moisture Content

Latex coatings may be applied to damp surfaces, but not surfaces with droplets of water. Do not apply epoxies to damp surfaces as determined by [ASTM D4263](#). Verify that new plaster to be coated has a maximum moisture content of 8 percent, when measured in accordance with [ASTM D4444](#), Method A, unless otherwise authorized. In addition to moisture content requirements, allow new plaster to age a minimum of 30 days before preparation for painting.

3.5 APPLICATION

3.5.1 Coating Application

- a. Comply with applicable federal, state and local laws enacted to ensure compliance with Federal Clean Air Standards. Apply coating materials in accordance with [SSPC PA 1](#). [SSPC PA 1](#) methods are applicable to all substrates, except as modified herein.
- b. At the time of application, paint must show no signs of deterioration. Maintain uniform suspension of pigments during application.
- c. Unless otherwise specified or recommended by the paint manufacturer, paint may be applied by brush, roller, or spray. Use trigger operated spray nozzles for water hoses. Use rollers for applying paints and enamels of a type designed for the coating to be applied and the surface to be coated. Wear protective clothing and respirators when applying oil-based paints or using spray equipment with any paints.
- d. Only apply paints, except water-thinned types, to surfaces that are completely free of moisture as determined by sight or touch.
- e. Thoroughly work coating materials into joints, crevices, and open spaces. Pay special attention to ensure that all edges, corners, crevices, welds, and rivets receive a film thickness equal to that of adjacent painted surfaces.
- f. Apply each coat of paint so that dry film is of uniform thickness and free from runs, drops, ridges, waves, pinholes or other voids, laps, brush marks, and variations in color, texture, and finish. Completely

hide all blemishes.

- g. Touch up damaged coatings before applying subsequent coats. Broom clean and clear dust from interior areas before and during the application of coating material.
- h. Drying Time: Allow time between coats, as recommended by the coating manufacturer, to permit thorough drying, but not to present topcoat adhesion problems. Provide each coat in specified condition to receive next coat.
- i. Primers, and Intermediate Coats: Do not allow primers or intermediate coats to dry more than 30 days, or longer than recommended by manufacturer, before applying subsequent coats. Follow manufacturer's recommendations for surface preparation if primers or intermediate coats are allowed to dry longer than recommended by manufacturers of subsequent coatings. Cover each preceding coat or surface completely by ensuring visually perceptible difference in shades of successive coats.
- j. Finished Surfaces: Provide finished surfaces free from runs, drops, ridges, waves, laps, brush marks, and variations in colors.
- k. Thermosetting Paints: Apply topcoats over thermosetting paints (epoxies and urethanes) within the overcoat window recommended by the manufacturer.

3.5.2 Mixing and Thinning of Paints

Reduce paints to proper consistency by adding fresh paint, except when thinning is mandatory to suit surface, temperature, weather conditions, application methods, or for the type of paint being used. Obtain written permission from the Contracting Officer to use thinners. Verify that the written permission includes quantities and types of thinners to use.

When thinning is allowed, thin paints immediately prior to application with not more than 0.125 L of suitable thinner per liter. The use of thinner does not relieve the Contractor from obtaining complete hiding, full film thickness, or required gloss. Thinning cannot cause the paint to exceed limits on volatile organic compounds. Do not mix paints of different manufacturers.

3.5.3 Two-Component Systems

Mix two-component systems in accordance with manufacturer's instructions. Follow recommendation by the manufacturer for any thinning of the first coat to ensure proper penetration and sealing for each type of substrate.

3.5.4 Coating Systems

- a. Systems by Substrates: Apply coatings that conform to the respective specifications listed in the following Tables:

Table for Exterior Applications	
MPI Division	Substrate Application

Table for Exterior Applications	
MPI Division 3	Exterior Concrete Paint Table
MPI Division 4	Exterior Concrete Masonry Units Paint Table
MPI Division 5	Exterior Metal, Ferrous and Non-Ferrous Paint Table
MPI Division 6	Exterior Wood; Dressed Lumber, Paneling, Decking, Shingles Paint Table
MPI Division 9	Exterior Stucco Paint Table
MPI Division 10	Exterior Cloth Coverings and Bituminous Coated Surfaces Paint Table
Table for Interior Applications	
MPI Division	Substrate Application
MPI Division 3	Interior Concrete Paint Table
MPI Division 4	Interior Concrete Masonry Units Paint Table
MPI Division 5	Interior Metal, Ferrous and Non-Ferrous Paint Table
MPI Division 6	Interior Wood Paint Table
MPI Division 9	Interior Plaster, Gypsum Board, Textured Surfaces Paint Table

- b. Minimum Dry Film Thickness (DFT): Apply paints, primers, varnishes, enamels, undercoats, and other coatings to a minimum dry film thickness of **0.038 mm** each coat unless specified otherwise in the Tables. Coating thickness, where specified, refers to the minimum dry film thickness.
- c. Coatings for Surfaces Not Specified Otherwise: Coat unspecified surfaces the same as surfaces having similar conditions of exposure.
- d. Existing Surfaces Damaged During Performance of the Work, Including New Patches In Existing Surfaces: Coat surfaces with the following:
 - (1) One coat of primer.
 - (2) One coat of undercoat or intermediate coat.
 - (3) One topcoat to match adjacent surfaces.
- e. Existing Coated Surfaces To Be Painted: Apply coatings conforming to the respective specifications listed in the Tables herein, except that pretreatments, sealers and fillers need not be provided on surfaces where existing coatings are soundly adhered and in good condition. Do not omit undercoats or primers.

3.6 COATING SYSTEMS FOR METAL

Apply coatings of Tables in MPI Division 5 for Exterior and Interior.

- a. Apply specified ferrous metal primer to steel surfaces on the same day that surface is cleaned, to surfaces that meet all specified surface preparation requirements at time of application.
- b. Inaccessible Surfaces: Prior to erection, use one coat of specified primer on metal surfaces that will be inaccessible after erection.
- c. Shop-primed Surfaces: Touch up exposed substrates and damaged coatings to protect from rusting prior to applying field primer.
- d. Surface Previously Coated with Epoxy or Urethane: Apply MPI 101, 0.038 mm DFT immediately prior to application of epoxy or urethane coatings.
- e. Pipes and Tubing: The semitransparent film applied to some pipes and tubing at the mill is not to be considered a shop coat. Overcoat these items with the specified ferrous-metal primer prior to application of finish coats.
- f. Exposed Nails, Screws, Fasteners, and Miscellaneous Ferrous Surfaces. On surfaces to be coated with water thinned coatings, spot prime exposed nails and other ferrous metal with latex primer MPI 107.

3.7 COATING SYSTEMS FOR CONCRETE AND CEMENTITIOUS SUBSTRATES

Apply coatings of Tables in MPI Division 3, 4 and 9 for Exterior and Interior.

3.8 PIPING IDENTIFICATION

Piping Identification, Including Surfaces In Concealed Spaces: Provide in accordance with MIL-STD-101. Place stenciling in clearly visible locations. On piping not covered by MIL-STD-101, stencil approved names or code letters, in letters a minimum of 13 mm high for piping and a minimum of 50 mm high elsewhere. Stencil arrow-shaped markings on piping to indicate direction of flow using black stencil paint.

3.9 INSPECTION AND ACCEPTANCE

In addition to meeting previously specified requirements, demonstrate mobility of moving components, including swinging and sliding doors, cabinets, and windows with operable sash, for inspection by the Contracting Officer. Perform this demonstration after appropriate curing and drying times of coatings have elapsed and prior to invoicing for final payment.

3.10 PAINT TABLES

All DFT's are minimum values. Acceptable products are listed in the MPI Green Approved Products List, available at <http://www.specifygreen.com/APL/ProductIdxByMPInum.asp>.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

3.10.1 Exterior Paint Tables

3.10.1.1 MPI Division 3: Exterior Concrete Paint Table

A. Concrete; Vertical Surfaces, Undersides of Balconies and Soffits

(1) New and uncoated existing concrete; vertical surfaces, including undersides of balconies and soffits but excluding tops of slabs

Latex					
New and uncoated existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI EXT 3.1A-G1 (Flat)	MPI REX 3.1A-G1 (Flat)	MPI 3	MPI 10	MPI 10	88 microns
MPI EXT 3.1A-G2 (Velvet)	MPI REX 3.1A-G2 (Velvet)	MPI 3	MPI 214	MPI 214	88 microns
MPI EXT 3.1A-G5 (Semigloss)	MPI REX 3.1A-G5 (Semigloss)	MPI 3	MPI 11	MPI 11	88 microns
MPI EXT 3.1A-G6 (Gloss)	MPI REX 3.1A-G6 (Gloss)	MPI 3	MPI 119	MPI 119	88 microns
Primer as recommended by manufacturer. Topcoat: Coating to match adjacent surfaces.					

3.10.1.2 MPI Division 5: Exterior Metal, Ferrous and Non-Ferrous Paint Table

A. Steel / Ferrous Surfaces

(1) New Steel that has been hand or power tool cleaned to SSPC SP 2 or SSPC SP 3

Alkyd					
New	Existing, uncoated	Primer	Intermediate	Topcoat	System DFT
MPI EXT 5.1Q-G5 (Semigloss)	MPI REX 5.1D-G5 (Semigloss)	MPI 23	MPI 94	MPI 94	131 microns
MPI EXT 5.1Q-G6 (Gloss)	MPI REX 5.1D-G6 (Gloss)	MPI 23	MPI 9	MPI 9	131 microns
Topcoat: Coating to match adjacent surfaces.					

(2) New Steel that has been blast-cleaned to SSPC SP 6/NACE No.3

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Alkyd					
New	Existing, uncoated	Primer	Intermediate	Topcoat	System DFT
MPI EXT 5.1D-G5 (Semigloss)	MPI REX 5.1D-G5 (Semigloss)	MPI 79	MPI 94	MPI 94	131 microns
MPI EXT 5.1D-G6 (Gloss)	MPI REX 5.1D-G6 (Gloss)	MPI 79	MPI 9	MPI 9	131 microns
Topcoat: Coating to match adjacent surfaces.					

3.10.1.3 MPI Division 9: Exterior Stucco Paint Table

A. New stucco

Latex					
New	Existing	Primer	Intermediate	Topcoat	System DFT
MPI EXT 9.1A-G1 (Flat)	MPI REX 9.1A-G2 (Flat)	MPI 10	MPI 10	MPI 10	112 microns
MPI EXT 9.1A-G5 (Semigloss)	MPI REX 9.1A-G5 (Semigloss)	MPI 11	MPI 11	MPI 11	112 microns
MPI EXT 9.1A-G6 (Gloss)	MPI REX 9.1A-G6 (Gloss)	MPI 119	MPI 119	MPI 119	112 microns
Primer as recommended by manufacturer. Topcoat: Coating to match adjacent surfaces. On existing stucco, apply primer based on surface condition.					

B. New stucco, elastomeric system

Elastomeric Coating					
New	Existing	Primer	Intermediate	Topcoat	System DFT
MPI EXT 9.1C-G1 (Flat)	MPI REX 9.1C-G1 (Flat)	N/A	MPI 113	MPI 113	400 microns
Primer as recommended by manufacturer. Topcoat: Coating to match adjacent surfaces. Surface preparation and number of coats in accordance with manufacturer's instructions Apply sufficient coats of MPI 113 to achieve a minimum dry film thickness of 400 microns.					

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

3.10.2 Interior Paint Tables

3.10.2.1 MPI Division 9: Interior Plaster, Gypsum Board, Textured
Surfaces Paint Table

A. Interior New Plaster and Wallboard not otherwise specified

Institutional Low Odor / Low VOC Latex, New

Institutional Low Odor / Low VOC Latex				
New	Primer	Intermediate	Topcoat	System DFT
MPI INT 9.2M-G2 (Flat)	MPI 149	MPI 144	MPI 144	100 microns
MPI INT 9.2M-G3 (Eggshell)	MPI 149	MPI 145	MPI 145	100 microns
MPI INT 9.2M-G4 (Satin)	MPI 149	MPI 146	MPI 146	100 microns
MPI INT 9.2M-G5 (Semigloss)	MPI 149	MPI 147	MPI 147	100 microns
Topcoat: Coating to match adjacent surfaces.				

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 97 13.17

THREE COAT EPOXY INTERIOR COATING OF WELDED STEEL PETROLEUM FUEL TANKS

05/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Contract Errors, Omissions, and Other Discrepancies
 - 1.4.2 Corrective Action (CA)
 - 1.4.2.1 Corrective Action Procedures
 - 1.4.2.2 Corrective Action Request (CAR) Form
 - 1.4.2.3 Corrective Action Log
 - 1.4.3 Coatings Work Plan
 - 1.4.4 Design Data
 - 1.4.4.1 Environmental Control System
 - 1.4.5 Test Reports
 - 1.4.5.1 Coatings Qualification Test Reports
 - 1.4.5.2 Joint Sealant Qualification Test Reports
 - 1.4.5.3 Ferrous Metallic Abrasive Qualification Test Reports
 - 1.4.5.4 Non-Metallic Abrasive Qualification Test Reports
 - 1.4.5.5 Recycled Ferrous Metallic Abrasive Field Test Reports (Daily and Weekly)
 - 1.4.6 Qualifications
 - 1.4.6.1 Qualifications of Certified Industrial Hygienist (CIH)
 - 1.4.6.2 Qualifications of Certified Protective Coatings Specialist (PCS)
 - 1.4.6.3 Qualifications of Coatings Inspection Company
 - 1.4.6.4 Qualifications of Quality Assurance Coatings Inspector
 - 1.4.6.5 Qualifications of Coatings Contractors
 - 1.4.6.6 Qualifications of Individuals Performing Abrasive Blasting
 - 1.4.6.7 Qualifications of Individuals Applying Coatings
 - 1.4.6.8 Qualifications of Testing Laboratory for Coatings
 - 1.4.6.9 Qualifications of Testing Laboratory for Abrasive
 - 1.4.6.10 Coating Materials Certificate of Conformance
 - 1.4.6.11 Joint Sealant Materials Certificate of Conformance
 - 1.4.6.12 Joint Sealant Compatibility
 - 1.4.6.13 Ferrous Metallic Abrasive Certificate of Conformance
 - 1.4.6.14 Non-Metallic Abrasive Certificate of Conformance
 - 1.4.7 QA and QC Personnel
 - 1.4.7.1 QC Manager
 - 1.4.7.2 Protective Coatings Specialist (PCS)
 - 1.4.7.3 Quality Assurance Coatings Inspector
 - 1.4.7.4 Coatings Contractor QC Coatings Inspector
 - 1.4.8 Pre-Application Meeting
- 1.5 PRODUCT DATA
 - 1.5.1 Joint Sealant Instructions
 - 1.5.2 Coating System Instructions

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 1.6 DELIVERY AND STORAGE
- 1.7 COATING HAZARDS
- 1.8 WORK SEQUENCE
- 1.9 JOB SITE REFERENCES

PART 2 PRODUCTS

- 2.1 COATING SYSTEM
 - 2.1.1 Epoxy Primer, Intermediate, and Topcoats
 - 2.1.1.1 Epoxy Primer Coat
 - 2.1.1.2 Epoxy Intermediate Coat
 - 2.1.1.3 Epoxy Topcoat
- 2.2 JOINT SEALANT
- 2.3 COATING FIELD COLLECTION KIT
- 2.4 ABRASIVE FIELD COLLECTION KIT
- 2.5 INSPECTION TEST KITS
 - 2.5.1 Test Kit for Measuring Chloride, Sulfate, and Nitrate Ions on Steel and Coated Surfaces
 - 2.5.2 Test Kit for Measuring Chlorides in Abrasives
 - 2.5.3 Test Kit for Identifying Amine Blush on Epoxy Surfaces
- 2.6 ABRASIVE
 - 2.6.1 Ferrous Metallic Abrasive
 - 2.6.1.1 New and Remanufactured Steel Grit
 - 2.6.1.2 Recycled Steel Grit
 - 2.6.2 Non-Metallic Abrasive

PART 3 EXECUTION

- 3.1 FIELD SAMPLE COLLECTION AND TESTING
 - 3.1.1 Coating Field Sample Collection
 - 3.1.2 Abrasive Field Sample Collection
 - 3.1.3 Coating Field Test Reports
 - 3.1.4 Abrasive Field Test Reports
- 3.2 LIGHTING
- 3.3 ENVIRONMENTAL CONDITIONS
 - 3.3.1 Tank Containment
 - 3.3.2 Control System Requirements
 - 3.3.2.1 Automated Monitoring Requirements
 - 3.3.2.2 Humidity Control for Surface Preparation and Primer Application
 - 3.3.2.3 Humidity Control for Application of Intermediate and Topcoats and Initial Curing
- 3.4 EQUIPMENT USED IN TANK
- 3.5 SURFACES TO BE COATED
- 3.6 SURFACE PREPARATION
 - 3.6.1 Abrasive Blasting Equipment
 - 3.6.2 Operational Evaluation of Abrasive
 - 3.6.3 Surface Standard
 - 3.6.4 Pre-Preparation Testing for Surface Contamination
 - 3.6.4.1 Pre-Preparation Testing for Oil and Grease Contamination
 - 3.6.4.2 Pre-Preparation Testing for Soluble Salts Contamination
 - 3.6.5 Abrasive Blasting
 - 3.6.6 Disposal of Used Abrasive
 - 3.6.7 Pre-Application Testing for Surface Contamination
 - 3.6.7.1 Pre-Application Testing for Oil and Grease Contamination
 - 3.6.7.2 Pre-Application Testing for Soluble Salts Contamination
 - 3.6.7.3 Pre-Application Testing for Surface Cleanliness
- 3.7 MIXING AND APPLICATION OF COATING SYSTEM AND SEALANT
 - 3.7.1 Preparation of Sealant and Coating Materials for Application

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.7.1.1 Mixing
- 3.7.1.2 Pot Life
- 3.7.1.3 Application Conditions and Recoat Windows
- 3.7.2 Amine Blush Testing of Epoxy Coat Prior to Overcoating
- 3.7.3 Application of Coating System and Joint Sealant
 - 3.7.3.1 Application of STRIPE COAT
 - 3.7.3.2 Application of Primer
 - 3.7.3.3 Application of Intermediate Coat
 - 3.7.3.4 Application of Topcoat
 - 3.7.3.5 Application of Joint Sealant
- 3.7.4 Holiday Testing
- 3.7.5 Tank Occupancy After Coating Application
- 3.7.6 Procedure for Holiday and Spot Repairs of Newly Applied Coating
- 3.7.7 Extended Cure of Coating System Prior to Immersion Service
- 3.8 PROJECT IDENTIFICATION
- 3.9 FIELD QUALITY CONTROL
 - 3.9.1 Field Inspection
 - 3.9.1.1 Inspection and Documentation Requirements
 - 3.9.1.2 Inspection Report Form
 - 3.9.1.3 Daily Inspection Reports
 - 3.9.1.4 Inspection Logbook
 - 3.9.1.5 Inspection Equipment
 - 3.9.1.5.1 Black Light
 - 3.9.2 Coatings Contractor QC Coatings Inspector's Field Responsibilities
 - 3.9.3 Quality Assurance Coatings Inspector's Field Responsibilities
- 3.10 FINAL CLEANUP

-- End of Section Table of Contents --

SECTION 09 97 13.17

THREE COAT EPOXY INTERIOR COATING OF WELDED STEEL PETROLEUM FUEL TANKS
05/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API Std 653 (2014; Addendum 1 2018) Tank Inspection, Repair, Alteration, and Reconstruction

ASTM INTERNATIONAL (ASTM)

ASTM C920 (2018) Standard Specification for Elastomeric Joint Sealants

ASTM D3276 (2015; E 2016) Standard Guide for Painting Inspectors (Metal Substrates)

ASTM D3335 (1985a; R 2020) Low Concentrations of Lead, Cadmium, and Cobalt in Paint by Atomic Absorption Spectroscopy

ASTM D3718 (1985a; R 2015) Low Concentrations of Chromium in Paint by Atomic Absorption Spectroscopy

ASTM D3925 (2002; R 2015) Sampling Liquid Paints and Related Pigmented Coatings

ASTM D4285 (1983; R 2018) Indicating Oil or Water in Compressed Air

ASTM D4417 (2021) Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel

ASTM D4940 (2015) Standard Test Method for Conductimetric Analysis of Water Soluble Ionic Contamination of Blast Cleaning Abrasives

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 8502-3 (2017) Preparation of Steel Substrates Before Application of Paints and Related Products - Tests for the Assessment of Surface Cleanliness - Part 3: Assessment of Dust on Steel Surfaces Prepared for

Painting (Pressure-Sensitive Tape Method)

ISO 9001

(2015) Quality Management Systems-
Requirements

NACE INTERNATIONAL (NACE)

NACE SP0178

(2007) Design, Fabrication, and Surface
Finish Practices for Tanks and Vessels to
be Lined for Immersion Service

NACE SP0188

(1999; R 2006) Discontinuity (Holiday)
Testing of New Protective Coatings on
Conductive Substrates

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC AB 1

(2015; E 2017) Mineral and Slag Abrasives

SSPC AB 2

(2015; E 2016) Cleanliness of Recycled
Ferrous Metallic Abrasive

SSPC AB 3

(2003; E 2004) Ferrous Metallic Abrasive

SSPC Guide 12

(2023) Guide for Illumination of
Industrial Painting Projects

SSPC PA 1

(2016) Shop, Field, and Maintenance
Coating of Metals

SSPC PA 2

(2015; E 2018) Procedure for Determining
Conformance to Dry Coating Thickness
Requirements

SSPC PA Guide 11

(2020) Protecting Edges, Crevices, and
Irregular Steel Surfaces by Stripe Coating

SSPC QP 1

(2019) Standard Procedure for Evaluating
the Qualifications of Industrial/Marine
Painting Contractors (Field Application to
Complex Industrial Steel Structures and
Other Metal Components)

SSPC QP 2

(2019) Standard Procedure for Evaluating
the Qualifications of Industrial/Marine
Painting Contractors (Removal of Hazardous
Coatings from Structures)

SSPC QP 5

(2022) Standard Procedure for Evaluating
the Qualifications of Coating and Lining
Inspection Companies

SSPC QS 1

(2015) Standard Procedure for Evaluating a
Contractor's Advanced Quality Management
System

SSPC SP 1

(2015) Solvent Cleaning

SSPC SP 10/NACE No. 2

(2015) Near-White Blast Cleaning

SSPC SP 11 (2020) Surface Preparation Standard No. 11
- Power Tool Cleaning to Bare Metal

SSPC SP COM (2016; E 2017) Surface Preparation
Commentary for Steel and Concrete
Substrates

SSPC VIS 1 (2002; E 2004) Guide and Reference
Photographs for Steel Surfaces Prepared by
Dry Abrasive Blast Cleaning

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE AMS-STD-595A (2017) Colors used in Government
Procurement

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-DTL-24441 (2009; Rev D; Notice 1 2021) Paint,
Epoxy-Polyamide, General Specification for

MIL-DTL-24441/29 (2009; Rev B; Notice 1 2021) Paint,
Epoxy-Polyamide, Green Primer, Formula
150, Type IV

MIL-DTL-24441/31 (2009; Rev B; Notice 1 2021) Paint,
Epoxy-Polyamide, White, Formula 152, Type
IV

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.134 Respiratory Protection

29 CFR 1910.1000 Air Contaminants

29 CFR 1926.59 Hazard Communication

1.2 DEFINITIONS

Definitions are generally provided throughout this Section in the paragraphs where used and denoted by capital letters. The following definitions are used throughout this Section:

- a. ROOF - Interior tank surfaces that extend from the horizontal plane at the designated maximum fuel line upward, including the upper portion of the tank shell (walls), columns, structural steel, the underside of the roof plates, and other steel components in this area.
- b. SHELL - Interior tank surfaces that extend along the vertical tank walls between the horizontal planes approximately **1 meter** above the shell-to-bottom joint upward to the horizontal plane at the designated fuel line, including columns, wall plates, and other steel components in this area.
- c. BOTTOM - Interior tank surfaces below the horizontal plane approximately **1 meter** above the shell-to-bottom joint, including columns, wall plates, piping, pipe supports, bottom plates, and other steel components in this area.

- d. INDEPENDENT THIRD-PARTY - Impartial third-party not a part or affiliated with Contractor or subcontractor principal or subsidiary businesses, and not a materials supplier.
- e. STRIPE COAT - An additional corrosion protection measure on edges, outside corners, crevices, bolt heads, welds, and other irregular surfaces, including minor surface preparation on sharp edges.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contract Errors, Omissions, and Other Discrepancies

Corrective Action Procedures

Corrective Action Request (CAR) Form

Coatings Work Plan

Inspection Report Form

SD-05, Design Data

Environmental Control System

SD-06 Test Reports

Coatings Qualification Test Reports

Joint Sealant Qualification Test Reports

Non-Metallic Abrasive Qualification Test Reports; G

Ferrous Metallic Abrasive Qualification Test Reports

Coating Field Test Reports

Abrasive Field Test Reports

Recycled Ferrous Metallic Abrasive Field Test Reports (Daily and Weekly)

Daily Inspection Reports

SD-07 Certificates

Qualifications of Certified Industrial Hygienist (CIH)

Qualifications of Certified Protective Coatings Specialist (PCS)

Qualifications of Coatings Inspection Company

- Qualifications of Quality Assurance Coatings Inspector
- Qualifications of Coatings Contractors
- Qualifications of Individuals Performing Abrasive Blasting
- Qualifications of Individuals Applying Coatings
- Qualifications of Testing Laboratory for Coatings
- Qualifications of Testing Laboratory for Abrasive
- Coating Materials Certificate of Conformance
- Joint Sealant Materials Certificate of Conformance
- Joint Sealant Compatibility
- Non-Metallic Abrasive Certificate of Conformance
- Ferrous Metallic Abrasive Certificate of Conformance

SD-08 Manufacturer's Instructions

- Joint Sealant Instructions
- Coating System Instructions

SD-11 Closeout Submittals

- Disposal of Used Abrasive; G
- Inspection Logbook; G
- Corrective Action Log; G

1.4 QUALITY ASSURANCE

1.4.1 Contract Errors, Omissions, and Other Discrepancies

Submit all errors, omissions, and other discrepancies in contract documents the Contracting Officer within 30 days of contract award for all work covered in this Section, other than the work that will not be uncovered until a later date. All such discrepancies must be addressed and resolved, and the Coatings Work Plan modified, prior to beginning the Initial and Follow-Up phases of work. Discrepancies that become apparent only after work is uncovered must be identified at the earliest discoverable time and submitted for resolution. Schedule time (float) must be built into the project schedule at those points where old work is to be uncovered or where access is not available during the first 30 days after award, to allow for resolution of contract discrepancies.

1.4.2 Corrective Action (CA)

CA must be included in the Contractor Quality Control Plan as outlined in Section 01 45 00 QUALITY CONTROL.

1.4.2.1 Corrective Action Procedures

Develop procedures for determining the root cause of each non-compliance, developing a plan to eliminate the root cause so that the non-compliance does not recur, and following up to ensure that the root cause was eliminated.

1.4.2.2 Corrective Action Request (CAR) Form

Develop Corrective Action Request (CAR) forms for initiating CA and for tracking and documenting each step. The CAR should be included with the Corrective Action Procedures. A CAR must be initiated by either the Contractor or the Contracting Officer. The Protective Coatings Specialist (PCS) must approve each CAR at the root cause identification stage, the plan for elimination stage, and the close out stage after verification that the root cause has been eliminated.

1.4.2.3 Corrective Action Log

When a CAR is initiated, the Contractor must take action to identify and eliminate the root cause of each non-compliance so as to prevent recurrence. These actions must apply to non-compliance in the work, and to non-compliance in the Quality Control (QC) System. Corrective actions must be appropriate to the effects of the non-compliance encountered. The corrective action must be documented in a report that is serialized and tracked in the Corrective Action Log until project completion and acceptance by the Contracting Officer. All corrective action reports must be retained in project records. The Corrective Action Log, showing status of each CAR, must be submitted to the Contracting Officer monthly.

1.4.3 Coatings Work Plan

The Coatings Work Plan must be considered as part of the Contractor Quality Control Plan as outlined in Section 01 45 00 QUALITY CONTROL.

The Coatings Work Plan must be submitted and approved by the PCS prior to mobilization. The Coatings Work Plan must explain in detail all procedures including, but not limited to, all sequential processes, quality control for each process, quality assurance for each process, and safety considerations. Subsections must include at least the following:

- a. Purpose;
- b. Introduction;
- c. Safety, fire, and health information;
- d. Contractor and worker qualifications with certifications;
- e. Project management organization and documents;
- f. Timeline in a Gantt chart;
- g. Project document references;
- h. Reference to all applicable standards (e.g., AMPP, NACE, SSPC, ISO, and ASTM);
- i. Coatings manufacturer's supporting documentation;

- j. Descriptions and explanations of any exceptions from the coating manufacturer;
- k. Coating and blasting equipment, model names, and, if applicable, calibration dates;
- l. Containment design and details;
- m. Environmental testing;
- n. Material delivery, storage, and handling details;
- o. Surface preparation;
- p. Pre-application test panel validation for field-applied external coating as outlined in SURFACE STANDARD;
- q. Coating materials, mixing, application, recoat windows, and coating curing times, if applicable;
- r. Coating repairs and rework;
- s. Non-conformance;
- t. Spent material handling and effluent discharge containment and disposal;
- u. Inspection test plan (as outlined in FIELD INSPECTION, and including inspection hold points, both Quality Assurance and Coating Contractor QC Coatings Inspector's responsibilities, and daily documentation and delivery);
- v. Instruments and test kits;
- w. Soluble salt testing (include procedures that must be used if greater-than-zero soluble salt levels are not able to be removed from the steel surface);
- x. Warranty (in writing, signed by the Contractor and the coating manufacturer's representative);
- y. Demobilization;
- z. PCS and PM approval;

1.4.4 Design Data

1.4.4.1 Environmental Control System

Submit design details of the proposed environmental control system to include ventilation, humidity control, and temperature regulation. Provide calculations for humidity control during separate surface preparation and coating application procedures, ventilation requirements during coating application, and maximum allowable coating application rates to coincide with ventilation. Include basis of design data on local conditions. Provide equipment layout sketches and procedures showing function of each piece of equipment and fail-safe measures. A Certified Industrial Hygienist must approve calculations, work procedures, and

personal protective equipment.

1.4.5 Test Reports

1.4.5.1 Coatings Qualification Test Reports

Submit qualification test results from an INDEPENDENT THIRD-PARTY laboratory of representative samples of each coating material. U.S. Department of Defense laboratories are considered to be independent laboratories. Samples must have been tested within the last three years.

The purpose of qualification testing is to pre-qualify the coating materials to MIL-DTL-24441. Submit test results for materials in conformance to the requirements of MIL-DTL-24441. Note that this is the same testing that is required for listing on the Qualified Products List (QPL). The coating materials must remain on the QPL for the entire project.

1.4.5.2 Joint Sealant Qualification Test Reports

Submit qualification test results from an INDEPENDENT THIRD-PARTY laboratory of representative samples of joint sealant material that will be used on this project. Samples must have been tested within the last three years. Submit results of conformance to ASTM C920.

1.4.5.3 Ferrous Metallic Abrasive Qualification Test Reports

Submit results for abrasive as required in paragraph 4 REQUIREMENTS of SSPC AB 3. Submit test results from an INDEPENDENT THIRD-PARTY laboratory of representative samples of each abrasive to be used on the jobsite. Samples must have been tested within the last three years. Note that this testing is for the purpose of pre-qualifying the abrasive.

1.4.5.4 Non-Metallic Abrasive Qualification Test Reports

Submit results for abrasive as required in paragraph 4 REQUIREMENTS of SSPC AB 1. Submit test results from an INDEPENDENT THIRD-PARTY laboratory of representative samples of each abrasive to be used on the jobsite. Samples must have been tested within the last three years. Note that this testing is for the purpose of pre-qualifying the abrasive.

1.4.5.5 Recycled Ferrous Metallic Abrasive Field Test Reports (Daily and Weekly)

Submit test results from an INDEPENDENT THIRD-PARTY laboratory of daily and weekly Quality Control testing required by SSPC AB 2, as modified in paragraph ABRASIVE.

1.4.6 Qualifications

1.4.6.1 Qualifications of Certified Industrial Hygienist (CIH)

Submit name, address, telephone number, fax number, and e-mail address of the INDEPENDENT THIRD-PARTY CIH. Submit documentation that the hygienist is certified by the American Board of Industrial Hygiene in comprehensive practice, including certification number and date of certification/recertification. The CIH must remain certified during the entire project, and the Contracting Officer must be notified of any change in certification status within 10 days of the change. If a CIH's

certification expires, the hygienist will not be allowed to perform any hygienist functions, and all hygienist work must stop, until the certification is reissued or another CIH is approved. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Provide evidence of experience with hazards involved in industrial coating application work.

1.4.6.2 Qualifications of Certified Protective Coatings Specialist (PCS)

Submit name, address, telephone number, fax number, and e-mail address of the INDEPENDENT THIRD-PARTY PCS. Submit documentation that specialist is certified by the Association for Materials Protection and Performance (AMPP) (formerly SSPC: The Society for Protective Coatings (SSPC)) as a PCS, including certification number and date of certification/recertification. If the PCS is employed by the same coatings inspection company to which the Quality Assurance Coatings Inspector is employed, this does not violate the INDEPENDENT THIRD-PARTY requirements. The PCS must remain certified during the entire project, and the Contracting Officer shall be notified of any change in certification status within 10 days of the change. If a PCS's certification expires, the PCS will not be allowed to perform any PCS functions, and all coatings work must stop, until the certification is reissued or another PCS is approved. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. The PCS must not be the designated Quality Assurance Coatings Inspector. The PCS's responsibilities are outlined in PROTECTIVE COATINGS SPECIALIST (PCS).

1.4.6.3 Qualifications of Coatings Inspection Company

Submit documentation that the coatings inspection company that will be performing all quality assurance coatings inspection functions is certified by AMPP to the requirements of **SSPC QP 5** prior to contract award. The coatings inspection company that is submitted and approved, must remain and cannot be changed through completion of the contract. The coatings inspection company must remain **SSPC QP 5** certified for the duration of the coating work and the Contracting Officer must be notified of any change in certification status within 10 days of the change. If a coatings inspection company's certification expires, the firm will not be allowed to perform any inspection functions, and all surface preparation and coating application work must stop, until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of all scheduled and unannounced on-site audits from AMPP and furnish a copy of all audit reports. The coatings inspection company must not engage in any activities that may conflict with their independence of judgment and integrity in relation to their inspection activities. In particular, they must not be engaged in the manufacture, supply, application, surface preparation, purchase, or maintenance of the applied coating in this project.

1.4.6.4 Qualifications of Quality Assurance Coatings Inspector

Submit documentation that each Quality Assurance Coatings Inspector is employed by the **SSPC QP 5** company and is qualified to a minimum certification of NACE CIP Level II. Each inspector must remain employed

by the coatings inspection company while performing any coatings inspection functions. The Quality Assurance Coatings Inspector's responsibilities are outlined in QUALITY ASSURANCE COATINGS INSPECTOR'S FIELD RESPONSIBILITIES. The roles of the Quality Assurance Coatings Inspector are in addition to, and distinct from, the role of the QC Coatings Inspector employed by the coatings Contractor.

1.4.6.5 Qualifications of Coatings Contractors

All Contractors and Subcontractors that perform surface preparation or coating application must be certified to both **SSPC QP 1** and **SSPC QS 1** prior to contract award, and must remain certified while accomplishing any surface preparation or coating application. If a Contractor's or Subcontractor's certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in Contractor certification status. Notify the Contracting Officer of all scheduled and unannounced on-site audits from AMPP and furnish a copy of all audit reports.

For projects located outside the United States, Guam, and Puerto Rico, the certifications for the coatings Contractor (**SSPC QP 1** and **SSPC QS 1**) can be substituted if the coatings Contractor meets all of the below requirements:

- a. **ISO 9001** certified;
- b. Eight years of experience with industrial coatings;
- c. Evidence of recent work that has Contractor Performance Assessment Report System (CPARS) ratings, and other quality/performance ratings, that are equivalent to, or exceed, "Above Average";
- d. Evidence of an INDEPENDENT THIRD-PARTY audit from AMPP demonstrating equivalency to **SSPC QP 1** and **SSPC QS 1** within the last two years.
- e. Evidence of an INDEPENDENT THIRD-PARTY audit from AMPP demonstrating equivalency to **SSPC QP 2** within the last two years.

The coatings Contractors and coatings Subcontractors must be certified to **ISO 9001** prior to contract award and must remain so certified for the duration of the project. If a Contractor's or Subcontractor's certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in Contractor certification status. Notify the Contracting Officer of all scheduled and unannounced on-site inspections from the ISO certifying organization and furnish a copy of all inspection reports.

1.4.6.6 Qualifications of Individuals Performing Abrasive Blasting

Submit name, address, and telephone number of each person that will be performing abrasive blasting. Submit documentation that each blaster is qualified by AMPP to the SSPC C7 Abrasive Blaster Qualification Program or CAS Coating Application Specialist Level 2 Certification Program (Interim

Status). Each blaster must remain qualified during the entire period of abrasive blasting, and the Contracting Officer shall be notified of any change in qualification status within 10 days of the change. If a blaster's qualification expires, the blaster will not be allowed to perform any blasting functions until the qualification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive qualification will not be considered and liquidated damages will apply.

1.4.6.7 Qualifications of Individuals Applying Coatings

Submit name, address, and telephone number of each person that will be applying coatings. Submit documentation that each applicator is qualified by AMPP to the SSPC CAS Coating Application Specialist Level 2 Certification Program (Interim Status) or SSPC C12 Spray Application Certification. Each applicator must remain certified during the entire period of coating application, and the Contracting Officer must be notified of any change in qualification status within 10 days of the change. If an applicator's qualification expires, the applicator will not be allowed to perform any coatings application functions until the qualification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive qualification will not be considered and liquidated damages will apply.

1.4.6.8 Qualifications of Testing Laboratory for Coatings

Submit name, address, telephone number, fax number, and e-mail address of the INDEPENDENT THIRD-PARTY laboratory or laboratories selected to perform testing of coating samples for qualification testing and for field sample testing for compliance with this Section. Submit documentation that the laboratory is regularly engaged in testing of paint samples for conformance with specifications and that the employees performing testing are qualified.

1.4.6.9 Qualifications of Testing Laboratory for Abrasive

Submit name, address, telephone number, fax number, and e-mail address of the INDEPENDENT THIRD-PARTY laboratory or laboratories selected to perform testing of abrasive for compliance with this Section. Submit documentation that the laboratory has experience in testing samples of abrasive for conformance with specifications and that the employees performing testing are qualified.

1.4.6.10 Coating Materials Certificate of Conformance

Provide manufacturer's certification of conformance to MIL-DTL-24441.

1.4.6.11 Joint Sealant Materials Certificate of Conformance

Provide manufacturer's certification of conformance to ASTM C920 and as modified in this Section.

1.4.6.12 Joint Sealant Compatibility

Provide manufacturer's certification that the selected joint sealant is compatible with the epoxy topcoat.

1.4.6.13 Ferrous Metallic Abrasive Certificate of Conformance

Provide manufacturer's certification of conformance that the materials are currently in conformance with SSPC AB 3 and as modified in this Section, and have been tested within the last three years.

1.4.6.14 Non-Metallic Abrasive Certificate of Conformance

Provide manufacturer's certification of conformance that the materials are currently in conformance with SSPC AB 1 and as modified in this Section, and have been tested within the last three years.

1.4.7 QA and QC Personnel

1.4.7.1 QC Manager

The QC Manager is as defined in Section 01 45 00 QUALITY CONTROL.

1.4.7.2 Protective Coatings Specialist (PCS)

The PCS must be considered a QC Specialist and must report to the QC Manager, as specified in Section 01 45 00 QUALITY CONTROL. The PCS must approve all submittals prior to submission to the QC Manager for approval or submission to the government for approval.

The PCS's responsibilities include, but are not limited to, the following:

- a. Obtain, review, and understand all project documentation including, but not limited to, this Section, project program, Coatings Work Plan, inspection and testing plan (ITP), and all submittals before the project starts, during the project, and all coatings related re-work;
- b. Attend all pre-job coatings meetings (in-person, phone, or virtually);
- c. Attend pre-final coatings walk-through (mandatory) and attend final coatings walk-through (as required).

1.4.7.3 Quality Assurance Coatings Inspector

The Quality Assurance Coatings Inspector must be considered a QC Specialist and must report to the QC Manager, as specified in Section 01 45 00 QUALITY CONTROL. The Quality Assurance Coatings Inspector must be present during all pre-preparation testing, surface preparation, coating application, initial cure of the coating system, during all coating repair work, and during completion activities. The Quality Assurance Coatings Inspector must provide complete documentation of conditions and occurrences on the job site, and be aware of conditions and occurrences that are potentially detrimental to the coating system. The requirements for inspection listed in this Section are in addition to the QC inspection and reporting requirements specified in Section 01 45 00 QUALITY CONTROL. The responsibilities of the Quality Assurance Coatings Inspector are defined in QUALITY ASSURANCE COATING INSPECTOR'S FIELD RESPONSIBILITIES. These responsibilities are separate and distinct from the responsibilities of the Coatings Contractor QC Coatings Inspector.

1.4.7.4 Coatings Contractor QC Coatings Inspector

The Coatings Contractor QC Coatings Inspector must stop non-compliant work. The responsibilities of the Coatings Contractor QC Coatings Inspector are

defined in COATINGS CONTRACTOR QC COATINGS INSPECTOR'S FIELD RESPONSIBILITIES. These responsibilities are separate and distinct from the responsibilities of the Quality Assurance Coatings Inspector.

1.4.8 Pre-Application Meeting

After approval of submittals, but prior to the initiation of coatings work, Contractor representatives, including at a minimum, project superintendent, QC manager, paint foreman, Quality Assurance Coatings Inspector, and PCS, must have a pre-application coating preparatory meeting. This meeting must be in addition to the pre-construction conference. Specific items addressed must include: corrective action requirements and procedures, coatings work plan, safety plan, coordination with other Sections, inspection standards, inspection requirements and tools, test procedures, environmental control system, and test logs. Notify Contracting Officer at least ten days prior to meeting.

1.5 PRODUCT DATA

1.5.1 Joint Sealant Instructions

Submit manufacturer's printed instructions including detailed mixing and application procedures, minimum and maximum application temperatures, and curing procedures. Include Safety Data Sheets (SDS) for materials to be used at the job site in accordance with 29 CFR 1926.59.

1.5.2 Coating System Instructions

Submit manufacturer's printed instructions including detailed mixing and application procedures, number and types of coats required, minimum and maximum application temperatures, and curing procedures. Include Safety Data Sheets (SDS) for materials to be used at the job site in accordance with 29 CFR 1926.59.

1.6 DELIVERY AND STORAGE

Ship, store, and handle materials in accordance with SSPC PA 1, and as modified in this Section. Maintain temperature in storage spaces between 5 and 24 degrees C, and air temperature more than 3 degrees C above the dew-point at all times. Inspect materials for damage prior to use and return non-compliant materials to manufacturer. Remove materials with expired shelf life from government property immediately and notify the Contracting Officer.

If materials are approaching shelf life expiration and an extension is desired, samples must be sent to the manufacturer, along with complete records of storage conditions, with a request for shelf life extension. If the manufacturer finds the samples and storage data suitable for shelf life extension, the manufacturer must issue an extension, referencing the product evaluation and the review of storage records. Products must not be extended longer than allowed in the product specification.

1.7 COATING HAZARDS

Ensure that employees are trained in all aspects of the safety plan. Specified coatings may have potential health hazards if ingested or improperly handled. The coating manufacturer's written safety precautions must be followed throughout mixing, application, and curing of the coatings. During tank cleaning, cleanup, surface preparation, and paint

application phases, ensure that employees are protected from toxic and hazardous chemical agents which exceed concentrations in 29 CFR 1910.1000. Comply with respiratory protection requirements in 29 CFR 1910.134. The CIH must approve work procedures and personal protective equipment.

1.8 WORK SEQUENCE

Coat tank interior following tank tightness testing.

1.9 JOB SITE REFERENCES

Make available to the Contracting Officer at least one copy each of API Std 653, ASTM C920, ASTM D3276, ASTM D3925, ASTM D4285, ASTM D4417, ASTM D4940, NACE SP0178 and companion visual comparator, NACE SP0188, SSPC AB 1, SSPC AB 2, SSPC AB 3, SSPC SP COM, SSPC SP 1, SSPC SP 10/NACE No. 2, SSPC SP 11, SSPC PA 1, SSPC PA 2, SSPC Guide 12, SSPC VIS 1, SSPC QP 1, SSPC QS 1, and an SSPC Certified Contractor Evaluation Form at the job site.

PART 2 PRODUCTS

2.1 COATING SYSTEM

Coating systems must be as specified herein; alternate systems will not be considered. All primer, intermediate, and topcoat materials must be manufactured by one manufacturer and supplied by one supplier.

2.1.1 Epoxy Primer, Intermediate, and Topcoats

The epoxy coating materials must be approved by the Naval Sea Systems Command and listed on their current Qualified Products List (QPL) for the specified materials.

2.1.1.1 Epoxy Primer Coat

Epoxy polyamide, MIL-DTL-24441/29 (Formula 150, Type IV, Green).

2.1.1.2 Epoxy Intermediate Coat

Epoxy polyamide, MIL-DTL-24441/31 (Formula 152, Type IV, White (Tinted)). Tint to approximately SAE AMS-STD-595A color number 27778 parchment using pigment dispersions prepared for epoxy paint tinting. Manufacturer shall tint material and appropriately label. All other requirements of this Military Specification apply.

2.1.1.3 Epoxy Topcoat

Epoxy polyamide, MIL-DTL-24441/31 (Formula 152, Type IV, White).

2.2 JOINT SEALANT

Industrial grade, two component, minimum 95 percent solids by volume, polysulfide type caulking material that has a minimum history of 10 years acceptable service in fuel tanks. Sealant must be compatible with the coating and suitable for direct application to prepared steel surfaces. Sealant must contain no more than 0.06 percent by dry weight lead, no more than 0.06 percent by dry weight cadmium, and no chromium. Joint sealant must be qualified to ASTM C920, Type M, Grade NS or P.

2.3 COATING FIELD COLLECTION KIT

Provide a kit for each sample to be collected. Each kit must contain: a one **liter** can for the base of the coating material; one appropriately sized can for the activator of the coating material; dipping cups for each component to be sampled; a shipping box sized for the samples to be shipped; and packing materials. Mark cans for the appropriate coating material and component (base or activator), including manufacturer's name, address, batch numbers, batch size shipped to the project site, and date of manufacture. Provide shipping documents, including either pre-paid shipping labels or a shipping number that can be used by the QC Manager to arrange pickup, addressed to the INDEPENDENT THIRD-PARTY coating testing laboratory.

2.4 ABRASIVE FIELD COLLECTION KIT

Provide a kit for each sample to be collected. Each kit must contain one suitable plastic bag or container for each sample to be collected. Mark containers with manufacturer's name, address, batch number, batch size, and date of manufacture. Provide shipping documents, including either pre-paid shipping labels or a shipping number that can be used by the QC Manager to arrange pickup, addressed to the approved coating testing laboratory.

2.5 INSPECTION TEST KITS

2.5.1 Test Kit for Measuring Chloride, Sulfate, and Nitrate Ions on Steel and Coated Surfaces

Provide test kits that meet all of the following requirements:

- a. Contains all materials, supplies, tools, and instructions for field testing and on-site quantitative evaluation of chloride, sulfate, and nitrate ions;
- b. Extract solution is acidic, factory pre-measured, pre-packaged, and of uniform concentration;
- c. Components and solutions are mercury free and environmentally friendly;
- d. Contains new materials and solutions for each test extraction;
- e. Contains an extraction test container (vessel, sleeve, cell) creates a sealed, encapsulated environment during salt ion extraction;
- f. Contains a test extract container is suitable for testing the following steel surfaces: horizontal (up/down configuration), vertical, flat, curved, smooth, pitted, and rough;
- g. All salt ion concentrations are directly measured in micrograms per square centimeter.

2.5.2 Test Kit for Measuring Chlorides in Abrasives

Provide test kits that meet all of the following requirements:

- a. Is a completely self-contained test kit with all materials, supplies, tools, and instructions to take tests and identify results;

- b. Uses identifiable, consistent, factory pre-measured test extract solution;
- c. Provides for testing equal volumes of abrasive and test solution;
- d. Provides for taking direct measurements of the chloride ion in parts per million (PPM), without using conversion charts or tables;
- e. Provides all new components for extraction and titration for each test;
- f. Provides a factory sealed titration device for each test;
- g. Uses the extract sampling container as the titration container.

2.5.3 Test Kit for Identifying Amine Blush on Epoxy Surfaces

Provide test kits that meet all of the following requirements:

- a. Is a completely self-contained field test kit with all materials, supplies, tools, and instructions to perform tests and indicate the presence of unreacted amines;
- b. Uses an identifiable, consistent, uniform, pre-packaged, factory pre-measured indicating solution;
- c. Contains no mercury or lead and is environmentally friendly;
- d. Contains a solution of an unreacted amine for the purpose of "self checking" the indicator solution;

2.6 ABRASIVE

Use abrasive that is specifically selected to provide a sharp, angular profile to the specified depth. Abrasive must meet all requirements of this Section each time that it is placed in the blast pot. A maximum limit for soluble salt contamination (chloride) is specified herein; however, this maximum level of contamination does not guarantee that contamination will not be transferred to the steel surface during abrasive blasting. Other factors, such as on-site handling and recycling, can allow contamination of abrasive that can be transferred to the steel surface. Contractors are cautioned to verify that the chosen abrasive, along with work and storage processes, allow the final surface cleanliness requirements to be achieved. Successful testing of contamination in abrasive does not negate the final acceptance testing of steel surfaces.

2.6.1 Ferrous Metallic Abrasive

2.6.1.1 New and Remanufactured Steel Grit

New and remanufactured steel grit abrasive must conform to the chemical and physical properties of **SSPC AB 3** Class 1 (Steel) only; Class 2 (Iron) abrasive must not be used. Modify the requirements of **SSPC AB 3** to substitute the requirement in paragraph 4.2.2 CONDUCTIVITY for one chloride test as measured using the test kit described in this Section (paragraph TEST KIT FOR MEASURING CHLORIDES IN ABRASIVES). The maximum allowable chloride content is 25 parts per million (PPM).

To develop a suitable work mix from new steel abrasive, a minimum of 200 to 400 recycles is required; therefore, it may be advantageous for a

Contractor to use remanufactured steel grit or grit reclaimed from a previous project. Such grit must be traced to new grit conforming to [SSPC AB 3](#) Class 1 and it meets all cleanliness requirements of [SSPC AB 3](#) Class 1 when brought to the current jobsite. Submit one representative sample of this work mix to the INDEPENDENT THIRD-PARTY laboratory for testing, along with samples of new material. Acceptance and use of this work mix must not be used to justify any deviation from surface preparation requirements.

2.6.1.2 Recycled Steel Grit

Recycled steel grit abrasive media must conform to the chemical and physical properties of [SSPC AB 2](#) except that:

- a. The maximum allowable chromium and cadmium content of the work mix must be 0.1 percent by weight when tested in accordance with [ASTM D3718](#) for chromium and [ASTM D3335](#) for cadmium. Modify the requirements of [SSPC AB 2](#) to add requirement for one chromate test and one cadmium test for each "LEAD" test required.
- b. The maximum allowable chloride content is 25 parts per million (PPM) as measured with the test kit described in paragraph TEST KIT FOR MEASURING CHLORIDES IN ABRASIVES. Modify the requirements of [SSPC AB 2](#) to substitute requirement for one chloride test for each "WATER SOLUBLE CONTAMINANTS" test.

2.6.2 Non-Metallic Abrasive

Non-metallic abrasive must be graded to the appropriate surface profile range and must conform to the chemical and physical properties of [SSPC AB 1](#), Class A except that:

- a. The maximum allowable chromium and cadmium content of the work mix must be less than 0.1 percent by weight when tested in accordance with [ASTM D3718](#) for chromium and [ASTM D3335](#) for cadmium.
- b. Must contain less than 7 PPM chlorides when tested with the kit provided in paragraph TEST KIT FOR MEASURING CHLORIDES IN ABRASIVES.

PART 3 EXECUTION

Perform all work, rework, and repair in accordance with approved procedures in the Coatings Work Plan. The Coatings Work Plan must be submitted and approved by the PCS prior to mobilization, in accordance with the paragraph COATINGS WORK PLAN.

3.1 FIELD SAMPLE COLLECTION AND TESTING

Sample and test materials delivered to the jobsite as required in TEST REPORTS and subsequent subparagraphs. Notify the Contracting Officer three days in advance of sampling. The QC Manager, and either the PCS or Quality Assurance Coatings Inspector, shall witness all sampling.

3.1.1 Coating Field Sample Collection

Coatings that are on the [MIL-DTL-24441](#) QPL require one sample to be collected. This sample must be collected and set aside for the duration of the project, and must be tested if unforeseen coatings issues arise or if testing is requested by the Contracting Officer. Coatings that are not on

the MIL-DTL-24441 QPL require a random field sample from each lot of coating material used on-site in accordance with ASTM D3925. Each random sample must be tested.

For sampling, utilize sample collection kits as outlined in the paragraph COATING FIELD SAMPLE COLLECTION KIT. Each sample must consist of one liter sample of each batch of each base material, and a sample of the activator that is proportional to the mix ratio of the coating type. Prior to sampling, mix contents of each sealed container to ensure uniformity. As an alternative to collecting small samples from kits, entire kits may be randomly selected and shipped to the INDEPENDENT THIRD-PARTY laboratory, observing all requirements for witnessing and traceability. For purposes of quality conformance inspection, a lot is defined as that quantity of materials from a single, uniform batch produced and offered for delivery at one time. A batch is defined as that quantity of material processed by the manufacturer at one time and identified by number on the label. Identify samples by designated name, specification number, batch number, project contract number, sample date, intended use, and quantity involved. If testing is required, the QC Manager will take possession of the packaged samples, contact the shipping company to arrange for pickup, and ship one complete sample of each material in question (including base and activator) with all batch information to the INDEPENDENT THIRD-PARTY laboratory for testing as required in paragraph COATING FIELD TESTING REPORTS.

3.1.2 Abrasive Field Sample Collection

Utilize the sample collection kits as required in paragraph ABRASIVE FIELD SAMPLE COLLECTION KIT to obtain samples from each lot of abrasive delivered to site using the sampling techniques and schedule of one sample per every 50 bags for ferrous metallic abrasive, paragraph 4 REQUIREMENTS FOR RECYCLED WORK MIX ABRASIVES of SSPC AB 2 for recycled ferrous metallic abrasives, or paragraph 5.3 SAMPLING FOR QUALITY CONTROL TESTS of SSPC AB 1 for non-metallic abrasives.

For purposes of quality conformance inspection, a lot must consist of all abrasive materials of the same type from a single, uniform batch produced and offered for delivery at one time. The addition of any substance to a batch must constitute a new lot. Identify samples by designated name, specification number, lot number, project contract number, sample date, intended use, and quantity involved. The QC manager will take possession of the packaged samples, contact the shipping company to arrange for pickup, and relinquish the samples only to the shipping representative for shipment to the approved laboratory for testing required in paragraph ABRASIVE FIELD SAMPLE TEST REPORTS.

3.1.3 Coating Field Test Reports

Submit test results for each sample that requires testing in paragraph COATING FIELD SAMPLE COLLECTION. Test samples of primer, intermediate, and topcoat materials for compliance with requirements of MIL-DTL-24441. Reject entire lot represented by samples that fail one or more tests, select new lots, and test samples.

3.1.4 Abrasive Field Test Reports

Submit test results for each lot of abrasive delivered to the jobsite. Test samples of ferrous metallic abrasive to the requirements of paragraph 5.2 TEST PARAMETERS of SSPC AB 3, excluding paragraph 5.2.4 DURABILITY.

Test samples of recycled ferrous metallic abrasives to the requirements of paragraph 4 REQUIREMENTS FOR RECYCLED WORK MIX ABRASIVES of SSPC AB 2. Test samples of non-metallic abrasive to the requirements of paragraph 5.3 SAMPLING FOR QUALITY CONTROL TESTS of SSPC AB 1. Reject entire lot represented by samples that fail one or more tests, select new lots, and test samples.

3.2 LIGHTING

Provide lighting for all work areas as prescribed in SSPC Guide 12.

3.3 ENVIRONMENTAL CONDITIONS

3.3.1 Tank Containment

Maintain exterior tank containment in full working condition during interior surface preparation, coating application, and initial curing to aid in maintaining interior environmental conditions.

3.3.2 Control System Requirements

Provide and utilize dehumidification and ventilation equipment to control humidity, temperature, and vapor levels in tank from beginning of abrasive blasting through coating application and for four days after the last coating is applied. System must maintain vapor concentrations at or below 10 percent of Lower Explosive Limit (LEL). System may incorporate any combination of solid desiccant and direct expansion refrigeration equipment. No liquid, granular, calcium chloride, or lithium chloride drying systems will be accepted. Use only electric, indirect fired combustion, indirect friction, or steam coil auxiliary heaters. System must be compatible with removal of dust and solvent vapors, and must have fail-safe measures to ensure reliability during operations.

3.3.2.1 Automated Monitoring Requirements

Provide continuous monitoring of dehumidification equipment, temperature, relative humidity, and dew point data at pertinent points on the structure, during surface preparation, coating application, and initial cure. This data does not suffice for documentation of conformity to surface conditions during application and cure of coating. Locate sensors to provide pertinent data for the surface preparation and coat application being performed. Describe the location plan, including required moves, in the Coatings Work Plan. Provide monitoring equipment to perform as follows:

- a. Data is collected in the field unit in 15-minute increments and available for download (on-site) in a standard database format. Contractor must collect these data and make available to the Contracting Officer, Quality Assurance Coatings Inspector, and QC Manager;
- b. Monitoring equipment must have backup power such that data collection will be uninterrupted during the entire period of the dehumidification requirement;
- c. Monitoring equipment must have capability to measure surface temperatures at a minimum of four locations anywhere on a structure, regardless of the size of the structure;
- d. Monitoring equipment must have capability to measure interior and

exterior dry bulb temperature (DB), relative humidity (RH), and dewpoint temperature (DP);

There is no requirement for connectivity of the monitoring system to control the dehumidification equipment; therefore, any combination of equipment having the required functionality will be accepted.

3.3.2.2 Humidity Control for Surface Preparation and Primer Application

Provide and utilize dehumidification equipment to maintain relative humidity at appropriate level to prevent prepared steel surfaces from corroding at all times during abrasive blasting through primer application. Failure of humidity control system, or failure to maintain proper conditions, during surface preparation stage may allow surface rusting, which will be rejected and require rework. All surfaces to be coated must meet all requirements at time of primer application. Failure of humidity control system during primer application stage will be cause for removal and replacement of all materials applied and cured while conditions were not as prescribed above.

3.3.2.3 Humidity Control for Application of Intermediate and Topcoats and Initial Curing

Provide and utilize dehumidification equipment to maintain relative humidity at the coldest steel surface in tank below 55 percent at all times during coating application, and during the first four days of initial curing after application of topcoat. This measurement is not the same as measuring the relative humidity of ambient air in the tank, and will require either electronic equipment to monitor relative humidity at the steel surface, or complex calculations to convert relative humidity of air in tank to relative humidity at steel surface. An approved alternative method of monitoring dehumidification that requires less sophisticated equipment or calculations is to maintain a minimum dew point depression of 10 degrees C below coldest steel surface temperature. This is in lieu of specific relative humidity and dew point requirements in this Section. Failure to maintain specified humidity control during this phase will be cause for extension of humidity controlled cure time to ensure four consecutive days at specified relative humidity at steel surfaces. Formation of condensation in coating application stage prior to the indicated dry-hard time will be cause for removal and replacement of all materials contacted by condensation.

3.4 EQUIPMENT USED IN TANK

Equipment used in the tank after surface preparation begins must not leave any oily residue from exhaust or other sources. Internal combustion driven equipment, other than that powered by natural or bottled gas, must not be used.

3.5 SURFACES TO BE COATED

Prepare and coat interior tank surfaces, including BOTTOM, SHELL, and ROOF. Remove interior piping to ensure complete coverage of the bottom and underside of pipe supports.

3.6 SURFACE PREPARATION

Prepare steel surfaces in accordance with SSPC PA 1 and as specified herein.

3.6.1 Abrasive Blasting Equipment

Use abrasive blasting equipment of conventional air, force-feed, or pressure type. Maintain a minimum pressure of 650 kPa at nozzle. Confirm that air supply for abrasive blasting is free of oil and moisture when tested in accordance with ASTM D4285. Test air quality at each startup, but in no case, less often than every five operating hours.

3.6.2 Operational Evaluation of Abrasive

Test abrasive for salt contamination and oil contamination as required in SSPC AB 1 for non-metallic abrasives, SSPC AB 2 for recycled ferrous abrasives, and SSPC AB 3 for ferrous abrasives. Modify the schedule of testing to be daily, at startup, and every five operating hours thereafter.

3.6.3 Surface Standard

Inspect surfaces to be coated, and select plate with similar properties and surface characteristics for use as a surface standard. Blast clean one or more 300 mm square steel panels as specified in paragraph SURFACE PREPARATION. Record blast nozzle type and size, air pressure at nozzle and compressor, distance of nozzle from panel, and angle of blast to establish procedures for blast cleaning. Measure surface profile in accordance with ASTM D4417, Method C. When the surface standard complies with all specified requirements, seal with a clearcoat protectant. Use the surface standard for comparison to abrasive blasted surfaces throughout the course of work.

3.6.4 Pre-Preparation Testing for Surface Contamination

Perform testing, abrasive blasting, and testing in the prescribed order.

3.6.4.1 Pre-Preparation Testing for Oil and Grease Contamination

- a. Inspect all surfaces for oil or grease contamination using two or more of the following inspection techniques: 1) Visual inspection, 2) WATER BREAK TEST, 3) BLACK LIGHT TEST, and 4) CLOTH RUB TEST. Reject oil- or grease-contaminated surfaces, clean in accordance with SSPC SP 1, and recheck for contamination until surfaces are free of oil and grease.
- b. WATER BREAK TEST - Spray atomized mist of distilled water onto surface and observe for water beading. If water wets surface rather than beading up, surface can be considered free of oil or grease contamination. Beading of water (water forms droplets) is evidence of oil or grease contamination.
- c. BLACK LIGHT TEST - Inspect surfaces for oil and grease contamination using the light specified in the paragraph BLACK LIGHT. Use light no more than 381 mm from surface unless testing indicates that the specific oil or grease found in tank fluoresce at a greater distance. Use light in tank that is completely sealed from light infiltration, under a hood, or at night. Any fluorescing on steel surfaces is indication of petroleum oil/grease contamination. Use either WATER BREAK TEST or CLOTH RUB TEST to confirm both contaminated and non-contaminated areas detected by BLACK LIGHT TEST. The BLACK LIGHT TEST must not be used during inspection of prepared surfaces for oil and grease contamination unless proven to fluoresce the oil and grease

found in the specific tank and documented during testing prior to abrasive blasting. Generally, only petroleum oil/grease will fluoresce; however, some may not fluoresce sufficiently to be recognized and other methods, such as the WATER BREAK TEST or CLOTH RUB TEST, must be used to confirm findings of the BLACK LIGHT TEST.

- d. CLOTH RUB TEST - Rub a clean, white, lint-free, cotton cloth onto the surface and observe for discoloration. To confirm oil or grease contamination in lightly stained areas, a non-staining solvent must be used to aid in oil or grease extraction. Any visible discoloration is evidence of oil or grease contamination.

3.6.4.2 Pre-Preparation Testing for Soluble Salts Contamination

Test surfaces for soluble salts, and wash as required, prior to abrasive blasting. This phase is required since pre-preparation testing and washing are generally more advantageous than attempting to remove soluble salt contamination after abrasive blasting. The purpose of soluble salts testing prior to surface preparation is to establish a baseline reading. Test all surfaces at rate of three tests for the first 100 square meters, plus one test for each additional 200 square meters, or part thereof. Perform 30 percent of tests on bare steel at welds, divided equally between horizontal and vertical welds. One or more readings greater than non-detectable for chlorides, sulfates, or nitrates is evidence of soluble salt contamination. Reject contaminated surfaces, wash as described below, allow to dry, and re-test until all required tests show allowable results. Re-blast tested areas using vacuum equipped blast equipment. Label all test tubes and retain for test verification. Soluble salts testing is also required in paragraph PRE-APPLICATION TESTING FOR SOLUBLE SALTS CONTAMINATION as a final acceptance test of prepared surfaces after abrasive blasting, and successful completion of this phase does not negate that requirement. Effective removal of soluble salts will require removal of any barrier to the steel surface, including rust. This procedure may necessitate combinations of wet abrasive blasting, high pressure water rinsing, and cleaning using a solution of water and soluble salts remover. The soluble salts remover shall be acidic, biodegradable, non-toxic, non-corrosive, and after application, will not interfere with primer adhesion. Delays between testing and preparation, or testing and coating application, may allow for the formation of new contamination. Use potable water, or potable water modified with soluble salt remover, for all washing or wet abrasive blasting. Test methods and equipment used in this phase are as stated in the Coatings Work Plan.

3.6.5 Abrasive Blasting

Abrasive blast steel surfaces to near-white metal in accordance with SSPC SP 10/NACE No. 2. Prepared surfaces shall conform to SSPC VIS 1 and shall match the prepared test-panels as specified in paragraph SURFACE STANDARD. Provide a 50 to 100 micron surface profile. Reject profile greater than 100 microns, discontinue abrasive blasting, and modify processes and materials to provide the specified profile. Measure surface profile in accordance with ASTM D4417, Method A and Method C. The appearance of the surface after blasting must have the appearance of a Sand or Grit comparator. A rounded profile shape or peened surface is not acceptable. Record all measurements required in this standard. Measure profile at rate of three test areas for the first 100 square meters plus one test area for each additional 100 square meters or part thereof. When surfaces are re-blasted for any reason, retest profile as specified. Following abrasive blasting, remove dust and debris by vacuum cleaning.

Dust and debris tend to collect at welds, plate overlaps, and surface irregularities. Do not attempt to wipe surface clean.

3.6.6 Disposal of Used Abrasive

Dispose of used abrasive off Government property in accordance with Federal, State, and Local mandated regulations.

3.6.7 Pre-Application Testing for Surface Contamination

3.6.7.1 Pre-Application Testing for Oil and Grease Contamination

Ensure tank surfaces are free of contamination as described in paragraph PRE-PREPARATION TESTING FOR OIL AND GREASE CONTAMINATION.

3.6.7.2 Pre-Application Testing for Soluble Salts Contamination

Test surfaces for soluble salts contamination using the test kit described in paragraph TEST KIT FOR MEASURING CHLORIDE, SULFATE, AND NITRATE IONS ON STEEL AND COATED SURFACES. Test all surfaces at rate of three tests for the first 100 square meters; plus one test for each additional 200 square meters; or part thereof. Perform 30 percent of tests on bare steel at welds, divided equally between horizontal and vertical welds. Label all test tubes and retain for test verification. One or more readings greater than non-detectable for chlorides, sulfates, or nitrates is evidence of soluble salt contamination. Reject contaminated surfaces, wash as required in paragraph PRE-PREPARATION TESTING FOR SOLUBLE SALTS CONTAMINATION, allow to dry, and re-test until all required tests show acceptable results. Re-blast tested areas using vacuum equipped blast equipment. An atmospheric event, such as a coastal storm blowing onshore, can bring chloride contamination. Following an atmospheric event, spot testing must be accomplished to verify satisfactory conditions and to avoid intercoat contamination. Where visual examination or spot testing indicates contamination, perform sufficient testing to verify non-contamination, or to define extent of contamination for appropriate treatment.

3.6.7.3 Pre-Application Testing for Surface Cleanliness

Apply coatings to dust free surfaces. To test surfaces, use ISO 8502-3. Use a kit that is compliant with ISO 8502-3. If the test does not result in a rating of 2 or better, then reject contaminated surfaces, clean by vacuum cleaning, and retest. Test surfaces at rate of three tests for the first 100 square meters, plus one test for each additional 100 square meters, or part thereof. Provide two additional tests for each failed test or questionable test. Document test results in the Daily Inspection Report and attach tape to the Daily Inspection Log.

Ferrous abrasives may become magnetized and difficult to remove from the steel substrate. If ferrous abrasives are used, additional visual inspection must be performed to ensure no surface contamination by the abrasive is present.

3.7 MIXING AND APPLICATION OF COATING SYSTEM AND SEALANT

3.7.1 Preparation of Sealant and Coating Materials for Application

Each of the different products, primer, intermediate, topcoat, and sealant, is a two-component material supplied in separate containers.

3.7.1.1 Mixing

Mix in accordance with coating system instructions, which may differ for each product. Do not mix partial kits unless standardized measuring cups are utilized. Do not alter mix ratios. All mixing processes must be witnessed by the Quality Assurance Coatings Inspector. Mix materials in same temperature and humidity conditions specified in paragraph DELIVERY AND STORAGE. Allow mixed material to stand for the required induction time based on its temperature.

3.7.1.2 Pot Life

Apply mixed products within stated pot life for each product. Stop applying when material becomes difficult to apply in a smooth, uniform wet film. Do not add solvent to extend pot life. All required solvent at time of mixing. Pot life is based on standard conditions at 21 degrees C and 50 percent relative humidity. For every 10 degrees C rise in temperature, pot life is reduced by approximately half, and for every 10 degrees C drop, it is approximately doubled. Usable pot life depends on the temperature of the material at the time of mixing and the sustained temperature at the time of application. Other factors such as the shape of the container and volume of mixed material may also affect pot life. In hot climates, pre-cooling or exterior icing of components for at least 24 hours to a minimum of 10 degrees C will extend pot life. The approximate pot life time for the epoxy primer and intermediate coat materials is four hours. The approximate pot life time for the sealant materials is as specified by the manufacturer.

3.7.1.3 Application Conditions and Recoat Windows

The application condition requirements for the coating system are very time and temperature sensitive, and are intended to avoid the delamination problems frequently found on industrial structures.

- a. Plan coating application to ensure that specified temperature, humidity, and condensation conditions are met. If conditions do not allow for orderly application of primer, STRIPE COAT, intermediate coat, topcoat, and sealant, use appropriate means of controlling air and surface temperatures, as required. Partial or total enclosures, insulation, heating or cooling, or other appropriate measures may be required to control conditions to allow for orderly application of all required coats.
- b. Maintain air and steel surface temperature within the range allowable by the manufacturer's PDS during application and the first four hours of cure for each epoxy coat. Maintain steel surface temperature more than 3 degrees C above the dew-point of the ambient air for the same period. These conditions may require environmental controls through containment.
- c. If coating is not applied during recoat window specified by the coating manufacturer, or if surface temperature exceeds the temperature recommended in the manufacturer's PDS between applications, provide GLOSS REMOVAL. If next planned coat is topcoat, apply FILL COAT if required to fill sanding marks. Sanding marks from GLOSS REMOVAL of intermediate coat reflecting through topcoat will be considered as non-compliant. Apply FILL COAT within 24 hours of GLOSS REMOVAL, then apply topcoat within RECOAT WINDOW of FILL COAT. The

topcoat must be free of defects and be of uniform appearance in accordance with **SSPC PA 1**. Lack of hiding by the finish coat must require additional applications to obtain uniform appearance.

- d. FILL COAT - Where indicated, apply coat of intermediate coat epoxy, at **50 to 75 microns** DFT, then apply next specified full coat within recoat window of FILL COAT. A FILL COAT may be used to adjust coating thickness to comply with requirements or to fill sanding marks in intermediate coat.
- e. GLOSS REMOVAL - Where required, hand sand in a circular fashion to remove gloss using 120-200 grit wet/dry sandpaper, followed by solvent wiping with a clean rag soaked with denatured alcohol to remove all dust. GLOSS REMOVAL of primer or intermediate coats is to scarify surface. If steel is exposed during GLOSS REMOVAL, repair in accordance with PROCEDURE FOR HOLIDAY AND SPOT REPAIRS OF NEWLY APPLIED COATING. GLOSS REMOVAL of the topcoat may include removal of up to **75 microns** of coating to avoid excess thickness, prior to application of FILL COAT.

3.7.2 Amine Blush Testing of Epoxy Coat Prior to Overcoating

Test epoxy surfaces prior to application of each epoxy coat or sealant for amine blush contamination using the test kit described in paragraph TEST KIT FOR IDENTIFYING AMINE BLUSH ON EPOXY SURFACES. Test all surfaces at rate of three tests for the first **100 square meters**, plus one test for each additional **200 square meters**, or part thereof. If one or more tests show positive results for amine blush contamination, either treat all surfaces using the approved amine blush removal procedure or increase testing to ensure that all contamination is located, and then treat identified contamination using the approved procedure.

3.7.3 Application of Coating System and Joint Sealant

Apply coatings in accordance with **SSPC PA 1**, **SSPC PA Guide 11** and as specified herein. Apply coatings and sealant to surfaces that meet all stated surface preparation requirements.

- a. Apply each coat in a consistent wet film, at 90 degrees to previous coat. Ensure that primer and intermediate overlaps are no less than **150 mm** from welds. Apply STRIPE COAT by brush, working the material into corners, crevices, pitted areas, and welds, and onto outside corners and angles. For convenience, STRIPE COAT material may be delivered by spray if followed immediately with brush-out and approved procedures include appropriate controls on thickness. Apply all other coats by spray application. Use appropriate controls to prevent airborne coating fog from drifting beyond **3 meters** from the tank perimeter. The cleanliness, temperature, recoat windows, and airborne paint containment requirements may necessitate the use of portable shelters or other appropriate controls.
- b. After application of primer coat, and prior to application of each subsequent coat, perform testing prescribed in paragraph PRE-APPLICATION TESTING FOR SURFACE CONTAMINATION to ensure minimal intercoat contamination. If contamination is detected, wash per **SSPC SP 1** and re-inspect. This testing may be reduced to one half of the prescribed rate for bare steel if the testing indicates no contamination when sampling is evenly distributed over surfaces being tested. If contamination is found between coats, revert to the

specified testing rate. Generally, oil and grease contamination and soluble salts contamination are not encountered if subsequent coats are applied within specified recoat windows and the quality of air entering tank is controlled. Spot testing must be accomplished to verify satisfactory conditions and to avoid intercoat contamination. Where visual examination or spot testing indicates contamination, perform sufficient testing to verify non-contamination or to define extent of contamination for appropriate treatment.

Apply coatings at the following specified thickness and in the following order:

<u>Coat</u>	<u>Minimum DFT (Microns)</u>	<u>Maximum DFT (Microns)</u>
Primer	75	125
Intermediate	75	125
Top	75	125
Total system	225	375

Measure coating thickness in accordance with **SSPC PA 2** to confirm that coating application is within the specified range and within the tolerances of that standard. For non-compliant areas, increase number of test areas to identify all non-compliant application as required by **SSPC PA 2**. Add coating as required to correct low DFT areas, and remove coating with excess thickness to bare steel and reapply as specified in PROCEDURE FOR HOLIDAY AND SPOT REPAIRS OF NEWLY APPLIED COATING.

3.7.3.1 Application of STRIPE COAT

Apply STRIPE COAT of epoxy primer material prior to application of general primer coat on ROOF and SHELL. Apply STRIPE COAT of epoxy intermediate coat material after application of general primer coat on BOTTOM. A STRIPE COAT must be applied to areas where joint sealant will be applied. This application must be consistent with APPLICATION OF COATING SYSTEM AND JOINT SEALANT. The STRIPE COAT must be in a contrasting color to the preceding and subsequent coats and extend a width of no less than **38 mm** on each side of the feature being protected.

3.7.3.2 Application of Primer

Apply primer coat within recoat window of STRIPE COAT.

3.7.3.3 Application of Intermediate Coat

Apply intermediate coat within recoat window of primer coat.

3.7.3.4 Application of Topcoat

Make all required repairs to primer and intermediate coats as specified in PROCEDURE FOR HOLIDAY AND SPOT REPAIRS OF NEWLY APPLIED COATING prior to applying topcoat. Apply topcoat within recoat window of intermediate coat. Consult manufacturer for application procedures for anticipated

temperature and humidity conditions. Touch-up blemishes and defects within recoat window of epoxy topcoat.

3.7.3.5 Application of Joint Sealant

After a full coating system has been installed, holiday tested, and repaired as necessary, apply sealant to the roof-to-shell joint, to all roof plate lap joints, and to roof-to-rafter joints up to 25 mm gap to exclude moisture from these marginally prepared crevice areas.

3.7.4 Holiday Testing

When the coating is dry to handle, but before the joint sealant is applied, perform holiday testing in accordance with the low voltage wet sponge method of NACE SP0188. Dry to handle is defined as curing to the degree that the surface will not be marred or damaged by normal foot traffic. Repair holidays per PROCEDURE FOR HOLIDAY AND SPOT REPAIRS OF NEWLY APPLIED COATING.

3.7.5 Tank Occupancy After Coating Application

Verify the coating has reached a cured state that will allow foot traffic. Use clean canvas, or other approved, shoe covers when walking on coated surfaces, regardless of curing time allowed. Provide cushioned mats for all traffic areas.

3.7.6 Procedure for Holiday and Spot Repairs of Newly Applied Coating

Repair coating film defects at the earliest practicable time, and before application of the succeeding coat. Any holiday found must have a STRIPE COAT applied in the area prior to application of the finish coat. Observe all requirements for soluble salts contamination, cleanliness between coats, and application conditions. Prepare defective area in accordance with SSPC SP 10/NACE No. 2, and feather coating as required to leave 100 mm of each succeeding coat feathered and abraded. If spot repair locations are less than 0.5 percent of the surface area and no greater than 150 mm in diameter, prepare surface to SSPC SP 10/NACE No. 2 vacuum blasting or SSPC SP 11 using an impact tool to create an acceptable profile. Do not abrade the epoxy topcoat. Protect adjacent areas from damage and overspray. Remove dust and solvent wipe the prepared area plus an additional 100 mm beyond the prepared area with clean denatured alcohol. Apply each coat within recoat window of preceding coat. Within four hours of preparation, apply primer to prepared steel and feather onto prepared intermediate area. Apply intermediate coat to primed area and feather to prepared intermediate area. Apply topcoat to intermediate coat and feather to prepared topcoat. Apply each repair coat to approximate thickness of surrounding coating system. If one percent or more of the total surface area, or more than one spot per 200 square meters, of the BOTTOM area requires repair to any coat or coats, including feathered areas, the entire BOTTOM coating system must be removed and reapplied. The limit on BOTTOM repairs includes repairs made before and after floating pan installation. If 5 percent or more of the total surface area, or more than one spot per 100 square meters, of the ROOF area requires repair to any coat or coats, including feathered areas, the entire ROOF coating system must be removed and reapplied. Repairs on the SHELL are not limited.

3.7.7 Extended Cure of Coating System Prior to Immersion Service

Allow a cure time of at least 14 days after the final coating material has been applied before introducing water or fuel into tank.

3.8 PROJECT IDENTIFICATION

At the completion of the tank work, stencil the following information on the exterior of the tank adjacent to the main manway opening in 3/4- to one-inch Helvetica style letters of contrasting color using acrylic stencil paint:

Date Interior coated:

Project Number:

Contractor:

Address:

Coating System

Surface Prep: SSPC SP _____ Profile: _____

Primer: _____ Thickness: _____

Intermediate: _____ Thickness: _____

Topcoat: _____ Thickness: _____

Total Thickness: _____

3.9 FIELD QUALITY CONTROL

Project documentation, including inspection and testing records, must be used to determine the Contractor's compliance with contract requirements and approved procedures. The Contractor's certifications of completion, for both invoices and for project completion, shall be based on documented evidence of compliance with all requirements and approved Coatings Work Plan procedures. Track inspections and tests in the Test Plan & Log.

3.9.1 Field Inspection

3.9.1.1 Inspection and Documentation Requirements

- a. Perform field inspection in accordance with **ASTM D3276** and the approved Coatings Work Plan.
- b. Provide all tools and instruments required to perform the required testing, as well as any tools or instruments that the inspector considers necessary to perform the required inspections and tests. Document each inspection and test, including required hold points and other required inspections and tests, as well as those inspections and tests deemed prudent from on-site evaluation to document a particular process or condition, as follows:
 - (1) Location or area;
 - (2) Purpose (required or special);
 - (3) Method;
 - (4) Criteria for evaluation;
 - (5) Results;
 - (6) Determination of compliance;

- (7) List of required rework;
- (8) Observations.
- c. Collect and record environmental conditions as described in [ASTM D3276](#) on a 24 hour basis, as follows:
 - (1) During surface preparation, every hour, or when changes occur;
 - (2) During coating application and the first four days of initial cure, every hour, or when changes occur;
 - (3) Note location, time, and temperature of the highest and lowest surface temperatures each day;
 - (4) Use a non-contact thermometer to locate temperature extremes, then verify with contact thermometers.
- d. Data collected on environmental conditions in AUTOMATED MONITORING REQUIREMENTS may be used for overnight data; however, the data must be constantly verified as to location of sensors and validity of data with respect to the coating work being accomplished.
- e. Document all equipment used in inspections and testing, including manufacturer, model number, serial number, last calibration date and future calibration date, and results of on-site calibration performed. Work documented using data from equipment found to be out of calibration must be considered as non-compliant since last calibration or calibration check, as required.
- f. Document Contractors compliance with the Coatings Work Plan.

3.9.1.2 Inspection Report Form

Develop project-specific report forms, as required, to report measurement and test results and observations being complete and compliant with contract requirements. This includes all direct requirements of the contract documents and indirect requirements of referenced documents. Show acceptance criteria with each requirement and indication of compliance of each inspected item. Annotation of non-compliance must be conspicuous so as to facilitate identification and transfer to the Rework Log. Report forms must include requirements and acceptance and rejection criteria, and must be legible and presented so that entered data can be quickly compared to the appropriate requirement. The data may be in any format, but must be legible and presented so that entered data can be quickly compared to the appropriate requirement.

3.9.1.3 Daily Inspection Reports

Submit one copy of daily inspection report completed each day when performing work under this Section, to the Contracting Officer. Note all non-compliance issues, and all issues that were reported for rework, in accordance with QC procedures of Section [01 45 00](#) QUALITY CONTROL. Each report must be signed by the Quality Assurance Coatings Inspector and the QC Manager. Submit report within 24 hours of date recorded on the report.

3.9.1.4 Inspection Logbook

A continuous record of all activity related to this Section must use an

electronic reporting program as outlined in Table I and be maintained on a daily basis. The computer / software package must be used to record all information provided in the Daily Inspection Reports, as well as other pertinent observations and information including photo documentation where appropriate.

In areas where photography is not allowed, the computer must come with verification that the camera / photo capability has been removed. Alternatively, a continuous record of all activity related to this Section must be maintained in an Inspection Logbook on a daily basis. The logbook must be hard or spiral-bound book or digital program with consecutively numbered pages, and must be used to record all information provided in the Daily Inspection Reports, as well as other pertinent observations and information. Submit the original Inspection Logbook to the Contracting Officer upon completion of the of the project and prior to final payment.

3.9.1.5 Inspection Equipment

All equipment must be in good condition, operational within its design range, and calibrated as required by the specified standard for use of each device.

3.9.1.5.1 Black Light

Use a black light having a 365-nanometer intensity of 4,000 microwatts per square centimeter minimum at 380 mm.

3.9.2 Coatings Contractor QC Coatings Inspector's Field Responsibilities

The Coatings Contractor QC Coatings Inspector responsibilities include complete documentation of all daily inspection and production activities for the entire coatings project as outlined in the Coatings Work Plan, project program, and this Section. This includes, but is not limited to, the following:

- a. Attending and documenting the pre-job meeting and acquiring the project program, inspection and testing plan (ITP), schedule, and a list of who will receive the QC daily inspection reports;
- b. Performing a project site walk-through with the Quality Assurance Evaluator (QAE) or asset owner, Coatings Contractor QC, QC Manager, and Quality Assurance Coatings Inspector, inspecting at least the following:
 - (1) Asset(s) to be coated;
 - (2) Equipment and placement of equipment;
 - (3) Materials delivery and storage;
 - (4) Facility operational requirements during the project.
- c. Perform all daily and hold point inspections including, but not limited to, the following:
 - (1) Check equipment, including blotter test to verify compressed air cleanliness;
 - (2) Perform non-visible contaminants testing (in accordance with

PRE-PREPARATION TESTING FOR SOLUBLE SALT CONTAMINATION and
PRE-APPLICATION TESTING FOR SOLUBLE SALT CONTAMINATION);

- (3) Perform visible contaminants testing (in accordance with
PRE-PREPARATION TESTING FOR OIL AND GREASE CONTAMINATION and
PRE-APPLICATION TESTING FOR OIL AND GREASE CONTAMINATION);
 - (4) Obtain environmental readings;
 - (5) Perform abrasive field testing per **SSPC AB 1**, **SSPC AB 2**, or
SSPC AB 3;
 - (6) Perform surface preparation monitoring and testing;
 - (7) Perform surface cleanliness testing;
 - (8) Perform dust quantity testing;
 - (9) Record materials storage documentation (record all coating and
abrasive materials information, batch numbers, segregation, and
storage temperature);
 - (10) Witness all coatings materials mixing and record mix materials
temperatures, with verification of time of coatings pot life;
 - (11) Verify, witness, and record application method;
 - (12) Perform random wet film thickness (WFT) readings;
 - (13) Perform inspection of coatings application;
 - (14) Obtain dry film thickness (DFT) readings per **SSPC PA 2**;
 - (15) Perform holiday testing in accordance with HOLIDAY TESTING;
 - (16) Observe label asset identification (label stickers);
 - (17) Write Correction Action Reports (CAR), if needed;
 - (18) Write Non-Conformance Reports (NCR), if needed.
- d. Writing a daily detailed summary of the work shift inspections,
testing, and the day's events, including any meetings and prevalent
conversations. The final daily report must include a project summary
that must be part of the last daily coatings inspection report.
- e. The Coatings Contractor QC Coatings Inspector must stop all
non-compliant work.

3.9.3 Quality Assurance Coatings Inspector's Field Responsibilities

The Quality Assurance Coatings Inspector's field responsibilities include complete documentation of all on-site work associated with the coatings project. These responsibilities include, but are not limited to, the following:

- a. Attending and documenting the pre-job meeting and acquiring the
project program, ITP, schedule, and a list of who will receive the QC
daily inspection reports;

- b. Performing a project site walk-through with the QAE or asset owner, prime Contractor, and coatings Contractor (QC Coatings Inspector and QC Manager), inspecting at least the following:
 - (1) Asset(s) to be coated;
 - (2) Equipment and placement of equipment;
 - (3) Materials delivery and storage;
 - (4) Facility operational requirements during the project.
- c. Verifying all daily and hold point inspections performed by the Coatings Contractor QC Coatings Inspector or QC Manager by performing mirror inspections including, but not limited to, the following:
 - (1) Verify equipment check, including blotter test to verify compressed air cleanliness;
 - (2) Verify visible contaminants testing;
 - (3) Take environmental readings;
 - (4) Perform surface preparation monitoring and testing;
 - (5) Perform surface cleanliness testing;
 - (6) Perform dust quantity test;
 - (7) Record materials storage documentation (record all coating and abrasive materials information, batch numbers, segregation, and storage temperature);
 - (8) Witness all coatings materials mixing and record mix materials temperatures, with verification of time of coatings pot life;
 - (9) Verify, witness, and record application method;
 - (10) Inspect coatings application;
 - (11) Perform dry film thickness (DFT) readings per [SSPC PA 2](#);
 - (12) Inspect asset identification (label stickers);
 - (13) Write Correction Action Reports (CAR), if needed;
 - (14) Write Non-Conformance Reports (NCR), if needed.
- d. The following testing is witnessed by the Quality Assurance Coatings Inspector and performed by the Coatings Contractor QC Coatings Inspector or QC Manager:
 - (1) Wet film thickness (WFT) readings by coatings applicator(s);
 - (2) Non-visible contaminants testing for chlorides, sulfates, and nitrates (CSN);
 - (3) Abrasive field testing per [SSPC AB 1](#), [SSPC AB 2](#), or [SSPC AB 3](#);

(4) Holiday testing.

- e. Writing a daily detailed summary of the work shift inspections, testing, and the day's events, including any meetings and prevalent conversations. The final daily report must include a project summary that will be part of the last daily coatings inspection report.

3.10 FINAL CLEANUP

Following completion of the work, remove debris, equipment, and materials from the site. Remove all foreign matter such as blast media, dust, dirt, debris, grease, and oils. Wipe all dry to handle coated surfaces with damp lint-free cloth. Remove temporary connections to Government- or Contractor- furnished water and electrical services. Restore existing facilities in and around the work areas to their original condition.

Table I
QA/QC Reporting Program Requirements

Administrative Controls:

Administrators must be able to turn on and off unique access to specific jobs and contracts.

Administrators must be able to remotely enable and disable access for users.

Administrators must be able to associate contract specific documents and specification limits quickly and easily.

Administrators must be able to associate PDS, SDS, blueprints, scope of work, and contracts uniquely to each job.

Objectivity Controls:

Data entry fields must be by multi-selectable choices, numeric keypads, pickers and skip logic to ensure repeatable data entry in a way that makes running analytics and metrics easy and objective.

Retrievable storage must be provided for all job-related reports and documents for a minimum time of five years from completion of the job or project. Archiving of the documents after five years will be the responsibility of the Government.

Document Library:

All reports must be in searchable and annotatable Portable Document Format (PDF).

Annotations and modifications must be locked and associated with the document. Only the Administrator has rights to modify or delete annotations or allow modifications to the document library especially all related inspection reports.

Customization:

The program must be capable of being customized to specific jobs, contracts or specifications.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 97 13.27

HIGH PERFORMANCE COATING FOR STEEL STRUCTURES

02/21, CHG 1: 08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Contract Errors, Omissions, and Other Discrepancies
 - 1.4.2 Corrective Action (CA)
 - 1.4.2.1 Corrective Action Procedures
 - 1.4.2.2 Implement Corrective Action
 - 1.4.3 Coating Work Plan
 - 1.4.4 Design Data
 - 1.4.4.1 Containment System
 - 1.4.5 Test Reports
 - 1.4.5.1 Joint Sealant Qualification Test Reports
 - 1.4.5.2 Coatings Qualification Test Reports
 - 1.4.5.3 Metallic Abrasive Qualification Test Reports
 - 1.4.5.4 Recycled Metallic Abrasive Field Test Reports (Daily and Weekly)
 - 1.4.6 Qualifications
 - 1.4.6.1 Qualifications of Certified Industrial Hygienist (CIH)
 - 1.4.6.2 Qualifications of Certified Protective Coatings Specialist (PCS)
 - 1.4.6.3 Qualifications of Coating Inspection Company
 - 1.4.6.4 Qualifications of QC Specialist Coating Inspector
 - 1.4.6.5 Qualifications Of Individuals Performing Abrasive Blasting
 - 1.4.6.6 Qualifications of Testing Laboratory for Coatings
 - 1.4.6.7 Qualifications of Testing Laboratory for Abrasive
 - 1.4.6.8 Qualifications of Coating Contractors
 - 1.4.6.9 Joint Sealant Materials
 - 1.4.6.10 Coating Materials
 - 1.4.6.11 Coating System Component Compatibility
 - 1.4.6.12 Non-metallic Abrasive
 - 1.4.6.13 Metallic Abrasive
 - 1.4.7 Protective Coating Specialist (PCS)
 - 1.4.8 Pre-Application Meeting
- 1.5 PRODUCT DATA
 - 1.5.1 Joint Sealant Instructions
 - 1.5.2 Coating System Instructions
- 1.6 DELIVERY AND STORAGE
- 1.7 COATING HAZARDS
- 1.8 JOB SITE REFERENCES

PART 2 PRODUCTS

- 2.1 JOINT SEALANT

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.2 COATING SYSTEM
 - 2.2.1 Zinc-Rich Epoxy Primer Coat
 - 2.2.2 Epoxy Intermediate Coat
 - 2.2.3 Polyurethane Topcoat
- 2.3 COATING SAMPLE COLLECTION AND SHIPPING KIT
- 2.4 ABRASIVE SAMPLE COLLECTION AND SHIPPING KIT
- 2.5 TEST KITS
 - 2.5.1 Test Kit for Measuring Chloride, Sulfate and Nitrate Ions on Steel and Coated Surfaces
 - 2.5.2 Test Kit for Identifying Amine Blush on Epoxy Surfaces
- 2.6 ABRASIVE
 - 2.6.1 Non-metallic Abrasive
 - 2.6.2 Metallic Abrasive
 - 2.6.2.1 New and Remanufactured Steel Grit
 - 2.6.2.2 Recycled Steel Grit
- 2.7 WHITE ALUMINUM OXIDE NON-SKID GRIT

PART 3 EXECUTION

- 3.1 COATING AND ABRASIVE SAMPLE COLLECTION AND TESTING
 - 3.1.1 Coating Sample Collection
 - 3.1.2 Abrasive Sample Collection
 - 3.1.3 Coating Sample Test Reports
 - 3.1.4 Abrasive Sample Test Reports
- 3.2 SURFACES TO BE COATED
- 3.3 LIGHTING
- 3.4 ENVIRONMENTAL CONDITIONS
 - 3.4.1 Containment
 - 3.4.2 Automated Monitoring Requirements
- 3.5 SURFACE PREPARATION
 - 3.5.1 Abrasive Blasting Equipment
 - 3.5.2 Operational Evaluation of Abrasive
 - 3.5.3 Surface Standard
 - 3.5.4 Pre-Preparation Testing for Surface Contamination
 - 3.5.4.1 Pre-Preparation Testing for Oil and Grease Contamination
 - 3.5.4.2 Pre-Preparation Testing for Soluble Salts Contamination
 - 3.5.5 Abrasive Blasting
 - 3.5.6 Disposal of Used Abrasive
 - 3.5.7 Pre-Application Testing For Surface Contamination
 - 3.5.7.1 Pre-Application Testing for Oil and Grease Contamination
 - 3.5.7.2 Pre-Application Testing for Soluble Salts Contamination
 - 3.5.7.3 Pre-Application Testing for Surface Cleanliness
- 3.6 MIXING AND APPLICATION OF SEALANT AND COATING SYSTEM
 - 3.6.1 Preparation of Sealant and Coating Materials for Application
 - 3.6.1.1 Mixing Sealant, Primer and Intermediate Coat Materials
 - 3.6.1.2 Mixing Topcoat Material
 - 3.6.1.3 Pot Life
 - 3.6.1.4 Application Conditions and Recoat Windows
 - 3.6.2 Amine Blush Testing of Epoxy Coat Prior to Overcoating
 - 3.6.3 Application of Coating System and Joint Sealant
 - 3.6.3.1 Application of Primer
 - 3.6.3.2 Application of Stripe Coat
 - 3.6.3.3 Application of Intermediate Coat
 - 3.6.3.4 Non-skid for Stairs and Top
 - 3.6.3.5 Application of Topcoat
 - 3.6.3.6 Application of Joint Sealant
 - 3.6.3.7 Procedure for Holiday and Spot Repairs of Newly Applied Coating
 - 3.6.3.8 Structure Occupancy After Coating Application

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.7 PROJECT IDENTIFICATION
- 3.8 FIELD QUALITY CONTROL
 - 3.8.1 Coating Inspector
 - 3.8.2 Field Inspection
 - 3.8.2.1 Inspection Requirements
 - 3.8.2.2 Inspection Report Forms
 - 3.8.2.3 Daily Inspection Reports
 - 3.8.2.4 Inspection Logbook
 - 3.8.2.5 Inspection Equipment
- 3.9 FINAL CLEANUP

-- End of Section Table of Contents --

SECTION 09 97 13.27

HIGH PERFORMANCE COATING FOR STEEL STRUCTURES
02/21, CHG 1: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM D1200	(2010; R 2014) Viscosity by Ford Viscosity Cup
ASTM D1640/D1640M	(2014) Standard Test Methods for Drying, Curing, or Film Formation of Organic Coatings
ASTM D3276	(2015; E 2016) Standard Guide for Painting Inspectors (Metal Substrates)
ASTM D3925	(2002; R 2015) Sampling Liquid Paints and Related Pigmented Coatings
ASTM D4285	(1983; R 2018) Indicating Oil or Water in Compressed Air
ASTM D7127	(2017) Standard Test Method for Measurement of Surface Roughness of Abrasive Blast Cleaned Metal Surfaces using a Portable Stylus Instrument
ASTM E11	(2022) Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 9001	(2015) Quality Management Systems- Requirements
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SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC 7/NACE No.4	(2007) Brush-Off Blast Cleaning
SSPC AB 2	(2015; E 2016) Cleanliness of Recycled Ferrous Metallic Abrasive
SSPC AB 3	(2023; R 2023) Ferrous Metallic Abrasive
SSPC Guide 6	(2021) Guide for Containing Surface

Preparation Debris Generated During Paint
Removal Operations

SSPC Guide 12	(2023) Guide for Illumination of Industrial Painting Projects
SSPC PA 1	(2016) Shop, Field, and Maintenance Coating of Metals
SSPC PA 2	(2015; E 2018) Procedure for Determining Conformance to Dry Coating Thickness Requirements
SSPC QP 1	(2019) Standard Procedure for Evaluating the Qualifications of Industrial/Marine Painting Contractors (Field Application to Complex Industrial Steel Structures and Other Metal Components)
SSPC QP 5	(2022) Standard Procedure for Evaluating the Qualifications of Coating and Lining Inspection Companies
SSPC QS 1	(2015) Standard Procedure for Evaluating a Contractor's Advanced Quality Management System
SSPC SP 1	(2015) Solvent Cleaning
SSPC SP 10/NACE No. 2	(2015) Near-White Blast Cleaning
SSPC SP COM	(2016; E 2017) Surface Preparation Commentary for Steel and Concrete Substrates
SSPC VIS 1	(2002; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE AMS-STD-595A	(2017) Colors used in Government Procurement
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U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-A-22262	(1993; Rev B; Am 1 1994; Am 2 1996; Notice 1 2021) Abrasive Blasting Media Ship Hull Blast Cleaning
MIL-DTL-24441	(2009; Rev D; Notice 1 2021) Paint, Epoxy-Polyamide, General Specification for
MIL-DTL-24441/19	(2009; Rev C; Notice 1 2023) Paint, Epoxy-Polyamide, Zinc Primer, Formula 159, Type III
MIL-DTL-24441/31	(2009; Rev B; Notice 1 2021) Paint, Epoxy-Polyamide, White, Formula 152, Type

IV

MIL-PRF-85285

(2022; Rev F) Topcoat, Aircraft and
Support Equipment

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.134

Respiratory Protection

29 CFR 1910.1000

Air Contaminants

29 CFR 1926.59

Hazard Communication

1.2 DEFINITIONS

Definitions are provided throughout this Section, generally in the paragraph where used, and denoted by capital letters.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-05, Design Data

Containment System

SD-06 Test Reports

Joint Sealant Qualification Test Reports

Coatings Qualification Test Reports

Metallic Abrasive Qualification Test Reports

Coating Sample Test Reports

Abrasive Sample Test Reports

Inspection Report Forms

Daily Inspection Reports

Recycled Metallic Abrasive Field Test Reports (Daily and Weekly)

SD-07 Certificates

Contract Errors, Omissions, and Other Discrepancies

Corrective Action Procedures

Coating Work Plan

Qualifications of Certified Industrial Hygienist (CIH)

Qualifications Of Individuals Performing Abrasive Blasting

Qualifications of Certified Protective Coatings Specialist (PCS)

Qualifications of Coating Inspection Company

Qualifications of QC Specialist Coating Inspector

Qualifications of Testing Laboratory for Coatings

Qualifications of Testing Laboratory for Abrasive

Qualifications of Coating Contractors

Joint Sealant Materials

Coating Materials

Coating System Component Compatibility

Non-metallic Abrasive

Metallic Abrasive

SD-08 Manufacturer's Instructions

Joint Sealant Instructions

Coating System Instructions

SD-11 Closeout Submittals

Disposal of Used Abrasive

Inspection Logbook; G

1.4 QUALITY ASSURANCE

1.4.1 Contract Errors, Omissions, and Other Discrepancies

Submit all errors, omissions, and other discrepancies in contract documents the Contracting Officer within 30 days of contract award for all work covered in this Section, other than the work that will not be uncovered until a later date. All such discrepancies shall be addressed and resolved, and the Coating Work Plan modified, prior to beginning the Initial and Follow-Up phases of work. Discrepancies that become apparent only after work is uncovered shall be identified at the earliest discoverable time and submitted for resolution. Schedule time (Float) should be built into the project schedule at those points where old work is to be uncovered or where access is not available during the first 30 days after award, to allow for resolution of contract discrepancies.

1.4.2 Corrective Action (CA)

CA shall be included in the Quality Control Plan.

1.4.2.1 Corrective Action Procedures

Develop procedures for determining the root cause of each non-compliance, developing a plan to eliminate the root cause so that the non-compliance does not recur, and following up to ensure that the root cause was

eliminated. Develop Corrective Action Request (CAR) forms for initiating CA, and for tracking and documenting each step.

1.4.2.2 Implement Corrective Action

The Contractor shall take action to identify and eliminate the root cause of each non-compliance so as to prevent recurrence. These procedures shall apply to non-compliance in the work, and to non-compliance in the QC System. Corrective actions shall be appropriate to the effects of the non-compliance encountered. Each CAR shall be serialized, tracked in a Log to completion and acceptance by the Contracting Officer, and retained in project records. The Corrective Action Log, showing status of each CAR, shall be submitted to the Contracting Officer monthly. A CAR may be initiated by either the Contractor or the Contracting Officer. The Contracting Officer must approve each CAR at the root cause identification stage, the plan for elimination stage, and the close out stage after verification that the root cause has been eliminated.

1.4.3 Coating Work Plan

This work plan shall be considered as part of the Quality Control Plan.

Provide procedures for reviewing contract documents immediately after award to identify errors, omissions, and discrepancies so that any such issues can be resolved prior to project planning and development of detailed procedures.

Provide procedures for verification of key processes during Initial Phase to ensure that contract requirements can be met. Key processes shall include surface preparation, coating application and curing, inspection, and documentation, and any other process that might adversely impact orderly progression of work.

Provide procedures for all phases of coating operations, including planned work, rework, repair, inspection, and documentation. Address mobilization and setup, surface preparation, coating application, coating initial cure, tracking and correction of noncompliant work, and demobilization. Coordinate work processes with health and safety plans and confined space entry plans. For each process, provide procedures that include appropriate work instructions, material and equipment requirements, personnel qualifications, controls, and process verification procedures. Provide procedures for inspecting work to verify and document compliance with contract requirements, including inspection forms and checklists, and acceptance and rejection criteria.

Provide procedures for correcting noncompliant work. Detailed procedures are required in advance to avoid delays in meeting overcoat windows as well as to avoid delays in production. Provide procedures for repairing defects in the coating film, such as runs, drips, sags, holidays, overspray, as well as how to handle correct coating thickness noncompliance, any other areas of repair or rework that might be adversely affected by delays in preparing and approving new procedures.

If a procedure is based on a proposed or approved request for deviation, the deviation shall be referenced. Changes to procedures shall be noted by submittal number and date approved, clearly delineating old requirements and new requirements, so that the records provide a continuous log of requirements and procedures.

1.4.4 Design Data

1.4.4.1 Containment System

Submit complete design drawings and calculations for the scaffolding and containment system, including an analysis of the loads which will be added to the structure by the containment system and waste materials. A registered engineer shall approve calculations and scaffold system design.

1.4.5 Test Reports

1.4.5.1 Joint Sealant Qualification Test Reports

Submit test results from independent laboratory of representative samples of joint sealant material. Samples must have been tested within the last three years. Submit results as required in paragraph QUALITY ASSURANCE PROVISIONS of [ASTM C920](#). Note that testing in accordance with QUALITY ASSURANCE PROVISIONS is a pre-qualification requirement.

1.4.5.2 Coatings Qualification Test Reports

Submit test results from independent laboratory of representative samples of each coating material. U.S. Department of Defense laboratories are considered to be independent laboratories for purposes of compliance with "QUALIFICATION INSPECTION" requirements herein. Samples must have been tested within the last three years. Submit results for epoxy materials as required in paragraph QUALIFICATION INSPECTION of [MIL-DTL-24441](#), and as revised by paragraph COATING SYSTEM herein. Submit results for polyurethane materials as required in paragraph QUALIFICATION INSPECTION of [MIL-PRF-85285](#), and as revised by paragraph COATING SYSTEM herein. Note that requirement for QUALIFICATION INSPECTION is a pre-qualification requirement, and involves the same testing required for listing in the Qualified Products List of the respective material. See appropriate Military Specification for specific test requirements.

1.4.5.3 Metallic Abrasive Qualification Test Reports

Submit results for abrasive as required in paragraph 4 REQUIREMENTS of [SSPC AB 3](#). Submit test results from independent laboratory of representative samples of each abrasive to be used on the jobsite. Samples must have been tested within the last three years. Note that this testing is for the purpose of prequalifying the abrasive.

1.4.5.4 Recycled Metallic Abrasive Field Test Reports (Daily and Weekly)

Submit test results from independent laboratory of daily and weekly Quality Control testing required by [SSPC AB 2](#), as modified in paragraph ABRASIVE.

1.4.6 Qualifications

1.4.6.1 Qualifications of Certified Industrial Hygienist (CIH)

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party CIH. Submit documentation that hygienist is certified by the American Board of Industrial Hygiene in comprehensive practice, including certification number and date of certification/recertification. Provide evidence of experience with hazards involved in industrial coating application work.

1.4.6.2 Qualifications of Certified Protective Coatings Specialist (PCS)

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party PCS. Submit documentation that specialist is certified by SSPC: The Society for Protective Coatings (SSPC) as a PCS, including certification number and date of certification/recertification. If the PCS is employed by the same coating inspection company to which the coating inspector is employed, this does not violate the independent third-party requirements. The PCS shall remain certified during the entire project, and the Contracting Officer shall be notified of any change in certification status within 10 days of the change. The PCS shall not be the designated coating inspector.

1.4.6.3 Qualifications of Coating Inspection Company

Submit documentation that the coating inspection company that will be performing all coating inspection functions is certified by SSPC to the requirements of **SSPC QP 5** prior to contract award, and shall remain certified while accomplishing any coating inspection functions. The coating inspection company must remain so certified for the duration of the project. If a coating inspection company's certification expires, the firm will not be allowed to perform any inspection functions, and all surface preparation and coating application work must stop, until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in coating inspection company certification status.

1.4.6.4 Qualifications of QC Specialist Coating Inspector

Submit documentation that each coating inspector is employed, and qualified to **SSPC QP 5**, Level III, by the selected coating inspection company. Each inspector shall remain employed by the coating inspection company while performing any coating inspection functions.

1.4.6.5 Qualifications Of Individuals Performing Abrasive Blasting

Submit name, address, and telephone number of each person that will be performing abrasive blasting. Submit documentation that each blaster is qualified by SSPC to the SSPC C-7 Dry Abrasive Blaster Qualification Program. Each blaster shall remain qualified during the entire period of abrasive blasting, and the Contracting Officer shall be notified of any change in qualification status.

1.4.6.6 Qualifications of Testing Laboratory for Coatings

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party laboratory selected to perform testing of coating samples for compliance with specification requirements. Submit documentation that laboratory is regularly engaged in testing of paint samples for conformance with specifications, and that persons performing analyses are qualified.

1.4.6.7 Qualifications of Testing Laboratory for Abrasive

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party laboratory selected to perform testing of abrasive for compliance with specification requirements. Submit

documentation that laboratory has experience in testing samples of abrasive for conformance with specifications, and that persons performing analyses are qualified.

1.4.6.8 Qualifications of Coating Contractors

All Contractors and Subcontractors that perform surface preparation or coating application shall be certified to either ISO 9001 or SSPC QP 1 and SSPC QS 1 prior to contract award, and shall remain certified while accomplishing any surface preparation or coating application. The painting Contractors and painting Subcontractors must remain so certified for the duration of the project. If a Contractor's or Subcontractor's certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in Contractor certification status.

1.4.6.9 Joint Sealant Materials

Provide manufacturer's certification of conformance to contract requirements.

1.4.6.10 Coating Materials

Provide manufacturer's certification of conformance to contract requirements.

1.4.6.11 Coating System Component Compatibility

Provide certification from each manufacturer of components of the coating system, epoxy primer, epoxy intermediate, and polyurethane topcoat, that the supplied coating material is suitable for use in the specified coating system. Each manufacturer shall identify the specific products, including manufacturer's name, which their product may be used with. The certification shall provide the name of the manufacturer that will provide technical support for the entire system. When all coating materials are manufactured by one manufacturer, this certification is not required.

1.4.6.12 Non-metallic Abrasive

Provide manufacturer's certification that the materials are currently approved by the Naval Sea Systems Command and listed on the Qualified Products Lists (QPL) for the specified materials.

1.4.6.13 Metallic Abrasive

Provide manufacturer's certification of conformance to contract requirements and provide copies of test results.

1.4.7 Protective Coating Specialist (PCS)

The PCS shall be considered a QC Specialist and shall report to the QC Manager, as specified in Section 01 45 00 QUALITY CONTROL. The PCS shall approve all submittals prior to submission to the QC Manager for approval or submission to the government for approval.

1.4.8 Pre-Application Meeting

After approval of submittals but prior to the initiation of coating work, Contractor representatives, including at a minimum, project superintendent and QC manager, paint foreman, coating inspector, and PCS shall have a pre-application coating preparatory meeting. This meeting shall be in addition to the pre-construction conference. Specific items addressed shall include: corrective action requirements and procedures, coating work plan, safety plan, coordination with other Sections, inspection standards, inspection requirements and tools, test procedures, environmental control system, safety plan, and test logs. Notify Contracting Officer at least ten days prior to meeting.

1.5 PRODUCT DATA

1.5.1 Joint Sealant Instructions

Submit manufacturer's printed instructions including detailed application procedures, minimum and maximum application temperatures, and curing procedures. Include Safety Data Sheets (SDS) for materials to be used at the job site in accordance with 29 CFR 1926.59.

1.5.2 Coating System Instructions

Submit manufacturer's printed instructions including detailed mixing and application procedures, number and types of coats required, minimum and maximum application temperatures, and curing procedures. Include Safety Data Sheets (SDS) for materials to be used at the job site in accordance with 29 CFR 1926.59.

1.6 DELIVERY AND STORAGE

Ship, store, and handle materials in accordance with SSPC PA 1, and as modified in this Section. Maintain temperature in storage spaces between 5 and 29 degrees C, and air temperature more than 3 degrees C above the dew-point at all times. Inspect materials for damage prior to use and return non-compliant materials to manufacturer. Remove materials with expired shelf life from government property immediately and notify the Contracting Officer.

If materials are approaching shelf life expiration and an extension is desired, samples may be sent to the manufacturer, along with complete records of storage conditions, with a request for shelf life extension. If the manufacturer finds the samples and storage data suitable for shelf life extension, the manufacturer may issue an extension, referencing the product evaluation and the review of storage records. Products may not be extended longer than allowed in the product specification.

1.7 COATING HAZARDS

Ensure that employees are trained in all aspects of the safety plan. Specified coatings may have potential health hazards if ingested or improperly handled. The coating manufacturer's written safety precautions shall be followed throughout mixing, application, and curing of the coatings. During all cleaning, cleanup, surface preparation, and paint application phases, ensure that employees are protected from toxic and hazardous chemical agents which exceed concentrations in 29 CFR 1910.1000. Comply with respiratory protection requirements in 29 CFR 1910.134. The CIH shall approve work procedures and personal protective equipment.

1.8 JOB SITE REFERENCES

Make available to the Contracting Officer at least one copy each of [ASTM D3276](#), [ASTM D3925](#), [ASTM D4285](#), [ASTM D7127](#), [SSPC SP COM](#), [SSPC SP 1](#), [SSPC 7/NACE No.4](#), [SSPC SP 10/NACE No. 2](#), [SSPC PA 1](#), [SSPC PA 2](#), [SSPC Guide 6](#), [SSPC VIS 1](#), [SSPC QP 1](#), [SSPC QS 1](#), and an SSPC Certified Contractor Evaluation Form at the job site.

PART 2 PRODUCTS

2.1 JOINT SEALANT

TT-S-00230, Type II, Class B

2.2 COATING SYSTEM

Alternate systems or products will not be considered. Provide a complete system (primer, intermediate coat, top coat) material from one supplier.

The Military specification epoxy and polyurethane products specified in this Section do not require approval for listing on the QPL prior to contract award, as indicated in paragraph 3.2 of [MIL-DTL-24441](#) and paragraph 3.1 of [MIL-PRF-85285](#). Testing of products by an independent laboratory to the QUALIFICATION INSPECTION requirements of [MIL-DTL-24441](#) and [MIL-PRF-85285](#) prior to contract award is required. See specific submittal requirements in paragraph QUALITY ASSURANCE.

2.2.1 Zinc-Rich Epoxy Primer Coat

Epoxy polyamide, [MIL-DTL-24441/19](#) (Formula 159, Type III).

2.2.2 Epoxy Intermediate Coat

Epoxy polyamide, [MIL-DTL-24441/31](#) (Formula 152, Type IV, White (Tinted)). Tint to approximately [SAE AMS-STD-595A](#) color number 27778 parchment using pigment dispersions prepared for epoxy paint tinting. Manufacturer shall tint material and appropriately label. All other requirements of this Military Specification apply.

2.2.3 Polyurethane Topcoat

Polyurethane coating topcoat of [MIL-PRF-85285](#), Type II, White [SAE AMS-STD-595A](#) color number 17925.

Modify paragraph 3.6.4 of [MIL-PRF-85285](#), Viscosity and Pot Life, as follows:

The viscosity of the admixed coating, when tested in accordance with [ASTM D1200](#) through a No. 4 Ford cup, shall be as follows:

Time from mix (minimum)	Maximum time through a No. 4 Ford cup
Initially	30 seconds
2 hours	60 seconds

Time from mix (minimum)	Maximum time through a No. 4 Ford cup
4 hours	No gel

Modify paragraph 3.7.1 of MIL-PRF-85285, Drying Time, as follows:

When applied by spray techniques and when tested in accordance with ASTM D1640/D1640M, the coating shall be set-to-touch within four hours and dry-hard within eight hours (see 4.6 and table I).

2.3 COATING SAMPLE COLLECTION AND SHIPPING KIT

Provide a kit that contains one liter can for the base of each coating material, an appropriately sized can for each activator, dipping cups for each component to be sampled, a shipping box sized for the samples to be shipped, and packing material. Mark cans for the appropriate component. Provide shipping documents, including either pre-paid shipping or a shipper number that can be used by the QC Manager to arrange pickup, addressed to the approved coating testing laboratory.

2.4 ABRASIVE SAMPLE COLLECTION AND SHIPPING KIT

Provide a kit that contains one suitable plastic bag or container for each sample to be collected. Mark containers for the appropriate component. Provide shipping documents, including either pre-paid shipping or a shipper number that can be used by the QC Manager to arrange pickup, addressed to the approved coating testing laboratory.

2.5 TEST KITS

2.5.1 Test Kit for Measuring Chloride, Sulfate and Nitrate Ions on Steel and Coated Surfaces

Provide test kits that meet all of the following requirements:

- Contains all materials, supplies, tools, and instructions for field testing and on-site quantitative evaluation of chloride, sulfate, and nitrate ions;
- Extract solution is acidic, factory pre-measured, pre-packaged, and of uniform concentration;
- Components and solutions are mercury free and environmentally friendly;
- Contains new materials and solutions for each test extraction;
- Contains an extraction test container (vessel, sleeve, cell) that creates a sealed, encapsulated environment during salt ion extraction;
- Contains a test extract container suitable for testing the following steel surfaces: horizontal (up/down configuration), vertical, flat, curved, smooth, pitted, and rough;
- All salt ion concentrations are directly measured in micrograms per square centimeter.

2.5.2 Test Kit for Identifying Amine Blush on Epoxy Surfaces

Provide test kits that meet all of the following requirements:

- a. Is a completely self-contained field test kit with all materials, supplies, tools, and instructions to perform tests and indicate the presence of unreacted amines;
- b. Uses an identifiable, consistent, uniform, pre-packaged, factory pre-measured indicating solution;
- c. Contains no mercury or lead and is environmentally friendly;
- d. Contains a solution of an unreacted amine for the purpose of "self checking" the indicator solution;

2.6 ABRASIVE

The referenced abrasive specifications have maximum limits for soluble salts contamination, however, this maximum level of contamination does not guarantee that contamination will not be transferred to the steel surface during abrasive blasting. Other factors such as on-site handling and recycling can allow contamination of abrasive. Contractors are cautioned to verify that the chosen abrasive, along with work and storage processes, allow the final surface cleanliness requirements to be achieved. Successful testing of chlorides in abrasive does not negate the final acceptance testing of steel surfaces.

2.6.1 Non-metallic Abrasive

Conform to [MIL-A-22262](#), Type I (Inorganic materials). Abrasive shall be approved by the Naval Sea Systems Command and listed on the appropriate Qualified Products List (QPL) for the specified materials. Use sampling procedures and testing frequencies as prescribed in [MIL-A-22262](#). Use abrasive that is specifically selected and graded to provide a sharp, angular profile to the specified depth. Do not use ungraded abrasive. Make adjustments to processes or abrasive gradation to achieve specified surface profile. Recycled non-metallic abrasive shall meet all requirements of the specification each time that it is placed in the blast pot.

2.6.2 Metallic Abrasive

2.6.2.1 New and Remanufactured Steel Grit

Conform to the chemical and physical properties of [SSPC AB 3](#) Class 1 (Steel) only. Class 2 (Iron) abrasive shall not be used.

To develop a suitable work mix from new steel abrasive, a minimum of 200 - 400 recycles is required, therefore, it is advantageous for a Contractor to use remanufactured steel grit or grit reclaimed from a previous project. Such grit shall be considered to conform if it can be traced to new grit conforming to [SSPC AB 3](#) Class 1 and it meets all cleanliness requirements of [SSPC AB 3](#) Class 1 when brought to the current jobsite. Submit one representative sample of this work mix to the laboratory for testing, along with samples of new material. Acceptance and use of this work mix shall not be used to justify any deviation from surface preparation requirements.

2.6.2.2 Recycled Steel Grit

Conform to the chemical and physical properties of [SSPC AB 2](#)

2.7 WHITE ALUMINUM OXIDE NON-SKID GRIT

Size #60, dust free (washed and dry), minimum 99 percent pure, having the following sieve analysis when tested in accordance with [ASTM E11](#) using a [1000 gram](#) sample:

Sieve #	Percent Retained
40	0
50	15-40
60	60-85

PART 3 EXECUTION

Perform all work, rework, and repair in accordance with approved procedures in the Coating Work Plan.

3.1 COATING AND ABRASIVE SAMPLE COLLECTION AND TESTING

Sample and test materials delivered to the jobsite. Notify Contracting Officer three days in advance of sampling. The QC Manager and either the PCS or coating inspector shall witness all sampling.

3.1.1 Coating Sample Collection

Provide a sample collection kit as required in paragraph COATING SAMPLE COLLECTION AND SHIPPING KIT. From each lot, obtain a one [liter](#) sample of each base material, and proportional samples of each activator based on mix ratio, by random selection from sealed containers in accordance with [ASTM D3925](#). Prior to sampling, mix contents of each sealed container to ensure uniformity. As an alternative to collecting small samples from kits, entire kits may be randomly selected and shipped to laboratory, observing all requirements for witnessing and traceability. For purposes of quality conformance inspection, a lot is defined as that quantity of materials from a single, uniform batch produced and offered for delivery at one time. A batch is defined as that quantity of material processed by the manufacturer at one time and identified by number on the label. Identify samples by designated name, specification number, batch number, project contract number, sample date, intended use, and quantity involved. The QC manager will take possession of the packaged samples, contact the shipping company to arrange for pickup, and relinquish the samples only to the shipping representative for shipment to the approved laboratory for testing as required by the paragraph COATING SAMPLE TEST REPORTS.

3.1.2 Abrasive Sample Collection

Provide a sample collection kit as required in paragraph ABRASIVE SAMPLE COLLECTION AND SHIPPING KIT. For purposes of quality conformance inspection, a lot shall consist of all abrasive materials of the same type from a single, uniform batch produced and offered for delivery at one time. Obtain samples of each abrasive lot using the sampling techniques and schedule of [MIL-A-22262](#). The addition of any substance to a batch

shall constitute a new lot. Identify samples by designated name, specification number, lot number, project contract number, sample date, intended use, and quantity involved. The QC manager will take possession of the packaged samples, contact the shipping company to arrange for pickup, and relinquish the samples only to the shipping representative for shipment to the approved laboratory for testing as required by the paragraph ABRASIVE SAMPLE TEST REPORTS.

3.1.3 Coating Sample Test Reports

Submit test results for each lot of coating material delivered to the jobsite. Test samples of primer, intermediate, and topcoat materials for compliance with requirements of Table I. Reject entire lot represented by samples that fail one or more tests, select new lots, and test samples.

3.1.4 Abrasive Sample Test Reports

Submit test results for each lot of abrasive delivered to the jobsite. Test samples of metallic abrasive to the requirements of paragraph REQUIREMENTS of **SSPC AB 3**, except paragraph 4.1.5 DURABILITY. Test samples of non-metallic abrasive as required in paragraph QUALITY CONFORMANCE INSPECTION of **MIL-A-22262**. Reject entire lot represented by samples that fail one or more tests, select new lots, and test samples.

3.2 SURFACES TO BE COATED

Coat exterior surfaces of tanks including shell and other exterior appurtenances. Coat exterior of all piping that is aboveground carbon steel including valves, equipment, and appurtenances.

3.3 LIGHTING

Provide lighting for all work areas as prescribed in **SSPC Guide 12**.

3.4 ENVIRONMENTAL CONDITIONS

3.4.1 Containment

Design and provide a containment system for the capture, containment, collection, storage and disposal of the waste materials generated by the work under this Section, to meet the requirements of **SSPC Guide 6**, Class 2. Vapor concentrations shall be kept at or below 10 percent of Lower Explosive Limit (LEL) at all times. Containment may be designed as fixed containment for complete structure or portable containment for sections of structure, however, containment shall remain in any one place from beginning of abrasive blasting through initial cure of coating. Waste materials covered by this paragraph shall not include any material or residue from removal of coatings containing lead, chromium, cadmium, PCB, or any other hazardous material.

It is the Contractors responsibility to insure the feasibility and workability of the containment system. The Contractor shall perform his operations and work schedule in a manner as to minimize leakage of the containment system. The containment system shall be properly maintained and shall not deviate from the approved drawings. If the containment system fails to function satisfactorily, the Contractor shall suspend all operations, except those required to minimize adverse impact on the environment or government property. Operations shall not resume until modifications have been made to correct the cause of the failure.

3.4.2 Automated Monitoring Requirements

Provide continuous monitoring of temperature, relative humidity, and dew point data at pertinent points on the structure, during surface preparation, coating application, and initial cure. Locate sensors to provide pertinent data for the surface preparation and coat application being performed. Monitor any heating, cooling, or dehumidification equipment used. Make data available to the Contracting Officer through Internet access. Provide monitoring equipment to perform as follows:

- a. Data is collected in the field unit in one minute increments, and available for download (on-site) in a standard format. Contractor shall collect this data and make available to the Contracting Officer;
- b. Monitoring equipment shall have backup power such that data collection and transmission to web server will be uninterrupted during the entire period of the dehumidification requirement;
- c. Monitoring equipment shall have capability to measure surface temperatures at a minimum of four locations anywhere on a 150 foot diameter by 50 foot high tank;
- d. Monitoring equipment shall have capability to measure interior and exterior dry bulb temperature (DB), relative humidity (RH), and dewpoint temperature (DP);
- e. Data shall be available continuously through secure Internet connection, using widely available web browsers;
- f. Internet accessible data shall be collected and stored in maximum 15 minute increments, and lag time between data collection and online availability shall be no greater than 70 minutes;
- g. Internet accessible data shall be available for viewing online in tabular format, and graphical format using selected data;
- h. Internet accessible data shall be available for download in user-defined segments, or entire project to date, in a standard format usable by Microsoft Excel and other spreadsheet programs.
- i. Internet-based controls shall provide alerts to pre-designated parties through email messaging;
- j. Internet-based controls shall monitor data uploads from field unit and issue alert if data not initiated within 60 minutes of last upload;
- k. Internet-based controls shall monitor operation of DH equipment and issues alert when power remains off for more than 15 seconds, or if pre-determined temperature, RH, or DP conditions are exceeded;

The requirements listed here were developed around the Munters Exactaire Monitoring System, as this was the only monitoring system having Internet connectivity known to be commercially available. There is no requirement for connectivity of the monitoring system to control the DH equipment, therefore, any combination of equipment having the required functionality will be accepted.

3.5 SURFACE PREPARATION

3.5.1 Abrasive Blasting Equipment

Use abrasive blasting equipment of conventional air, force-feed, or pressure type. Maintain a minimum pressure of 650 kPa at nozzle. Confirm that air supply for abrasive blasting is free of oil and moisture when tested in accordance with ASTM D4285. Test air quality at each startup, but in no case less often than every five operating hours.

3.5.2 Operational Evaluation of Abrasive

Test abrasive for salt contamination and oil contamination as required by the appropriate abrasive specification daily at startup and every five operating hours thereafter.

3.5.3 Surface Standard

Inspect surfaces to be coated, and select plate with similar properties and surface characteristics for use as a surface standard. Blast clean one or more 300 mm square steel panels as specified in paragraph SURFACE PREPARATION. Record blast nozzle type and size, air pressure at nozzle and compressor, distance of nozzle from panel, and angle of blast to establish procedures for blast cleaning. Measure surface profile in accordance with ASTM D7127. When the surface standard complies with all specified requirements, seal with a clearcoat protectant. Use the surface standard for comparison to abrasive blasted surfaces throughout the course of work.

3.5.4 Pre-Preparation Testing for Surface Contamination

Perform testing, abrasive blasting, and testing in the prescribed order.

3.5.4.1 Pre-Preparation Testing for Oil and Grease Contamination

Inspect all surfaces for oil and/or grease contamination using two or more of the following inspection techniques: 1) Visual inspection, 2) WATER BREAK TEST, 3) CLOTH RUB TEST. Reject oil and/or grease contaminated surfaces, clean using a water based pH neutral degreaser in accordance with SSPC SP 1, and recheck for contamination until surfaces are free of oil and grease.

WATER BREAK TEST - Spray atomized mist of distilled water onto surface, and observe for water beading. If water "wets" surface rather than beading up, surface can be considered free of oil or grease contamination. Beading of water (water forms droplets) is evidence of oil or grease contamination.

CLOTH RUB TEST - Rub a clean, white, lint free, cotton cloth onto surface and observe for discoloration. To confirm oil or grease contamination in lightly stained areas, a non-staining solvent may be used to aid in oil or grease extraction. Any visible discoloration is evidence of oil or grease contamination.

3.5.4.2 Pre-Preparation Testing for Soluble Salts Contamination

Test surfaces for soluble salts, and wash as required, prior to abrasive blasting. Soluble salt testing is also required in paragraph PRE-APPLICATION TESTING FOR SOLUBLE SALTS CONTAMINATION as a final

acceptance test of prepared surfaces after abrasive blasting, and successful completion of this phase does not negate that requirement. This phase is recommended since pre-preparation testing and washing are generally more advantageous than attempting to remove soluble salt contamination after abrasive blasting. Effective removal of soluble salts will require removal of any barrier to the steel surface, including rust. This procedure may necessitate combinations of wet abrasive blasting, high pressure water rinsing, and cleaning using a solution of water washing and soluble salts remover. The soluble salts remover shall be acidic, biodegradable, nontoxic, noncorrosive, and after application, will not interfere with primer adhesion. Delays between testing and preparation, or testing and coating application, may allow for the formation of new contamination. Use potable water, or potable water modified with soluble salt remover, for all washing or wet abrasive blasting. Test methods and equipment used in this phase are selected at the Contractor's discretion.

3.5.5 Abrasive Blasting

Abrasive blast steel surfaces to near-white metal in accordance with SSPC SP 10/NACE No. 2. Prepared surfaces shall conform to SSPC VIS 1 and shall match the prepared test-panels. Provide a 50 to 75 micron surface profile. Reject profile greater than 75 microns, discontinue abrasive blasting, and modify processes and materials to provide the specified profile. Measure surface profile in accordance with ASTM D7127, using Rmax as the measure of profile height. Record all measurements required in this standard. Measure profile at rate of three test areas for the first 100 square meters plus one test area for each additional 100 square meters or part thereof. When surfaces are reblasted for any reason, retest profile as specified. Following abrasive blasting, remove dust and debris by vacuum cleaning. Do not attempt to wipe surface clean.

3.5.6 Disposal of Used Abrasive

Dispose of used abrasive off Government property in accordance with Federal, State, and Local mandated regulations.

3.5.7 Pre-Application Testing For Surface Contamination

3.5.7.1 Pre-Application Testing for Oil and Grease Contamination

Ensure surfaces are free of contamination as described in paragraph PRE-PREPARATION TESTING FOR OIL AND GREASE CONTAMINATION, except that only questionable areas need be checked for beading of water misted onto surface.

3.5.7.2 Pre-Application Testing for Soluble Salts Contamination

Test surfaces for chloride contamination using the Test Kit described in TEST KIT FOR MEASURING CHLORIDE, SULFATE AND NITRATE IONS ON STEEL AND COATED SURFACES. Test all surfaces at rate of three tests for the first 100 square meters plus one test for each additional 200 square meters or part thereof. One or more readings greater than 3 micrograms per square centimeter of chlorides or 10 micrograms per square centimeter of sulfates or 5 micrograms per square centimeter of nitrates is evidence of soluble salt contamination. Reject contaminated surfaces, wash as discussed in paragraph PRE-PREPARATION TESTING FOR SOLUBLE SALTS CONTAMINATION, allow to dry, and re-test until all required tests show allowable results. Reblast tested and cleaned areas as required. Label all test tubes and retain for test verification.

3.5.7.3 Pre-Application Testing for Surface Cleanliness

Apply coatings to dust free surfaces. To test surfaces, apply strip of clear adhesive tape to surface and rub onto surface with finger. When removed, the tape should show little or no dust, blast abrasive, or other contaminant. Reject contaminated surfaces and retest. Test surfaces at rate of three tests for the first 100 square meters plus one test for each additional 100 square meters or part thereof. Provide two additional tests for each failed test or questionable test. Attach test tapes to Daily Inspection Reports.

3.6 MIXING AND APPLICATION OF SEALANT AND COATING SYSTEM

3.6.1 Preparation of Sealant and Coating Materials for Application

Each of the sealant, primer, intermediate, and topcoat materials is a two-component material supplied in separate containers.

3.6.1.1 Mixing Sealant, Primer and Intermediate Coat Materials

Mix in accordance with manufacturer's instructions, which may differ for each product. Do not mix partial kits, or alter mix ratios. Mix materials in same temperature and humidity conditions specified in paragraph DELIVERY AND STORAGE. Allow mixed material to stand for the required induction time based on its temperature.

3.6.1.2 Mixing Topcoat Material

Do not mix partial kits, or alter mix ratios. Mix polyurethane coating materials in same temperature conditions specified in paragraph DELIVERY AND STORAGE. The polyurethane coating material is moisture sensitive and any introduction of moisture or water into the material during mixing or application will shorten usable pot life. Use a mixer that does not create a vortex. Do not add solvent without specific written recommendation from the manufacturer. No induction time is required, only thorough agitation of the mixed material.

3.6.1.3 Pot Life

Apply mixed products within stated pot life for each product. Stop applying when material becomes difficult to apply in a smooth, uniform wet film. Add all required solvent at time of mixing. Do not add solvent to extend pot life. Pot life is based on standard conditions at 21 degrees C and 50 percent relative humidity. For every 10 degrees C rise in temperature, pot life is reduced by approximately half, and for every 10 degrees C drop it is approximately doubled. Usable pot life depends on the temperature of the material at the time of mixing and the sustained temperature at the time of application. Other factors such as the shape of the container and volume of mixed material may also affect pot life. Precooling or exterior icing of components for at least 24 hours to a minimum of 10 degrees C in hot climates will extend pot life. High humidity at time of mixing and application shortens pot life of the Polyurethane topcoat material. Following are approximate pot life times:

Sealant	As specified by manufacturer
Epoxy primer and intermediate materials	4 hours
Polyurethane topcoat materials	2 hours.

3.6.1.4 Application Conditions and Recoat Windows

The application condition requirements for the coating system are very time and temperature sensitive, and are intended to avoid the delamination problems frequently found on industrial structures. Plan coating application to ensure that specified temperature, humidity, and condensation conditions are met. If conditions do not allow for orderly application of sealant, primer, stripe coat, intermediate coat and topcoat, use appropriate means of controlling air and surface temperatures, as required. Partial or total enclosures, insulation, heating or cooling, or other appropriate measures may be required to control conditions to allow for orderly application of all required coats.

Maintain air and steel surface temperature between **16 and 38 degrees C** during application and the first four hours of cure for epoxy coats and the first eight hours of cure for polyurethane coats. Maintain steel surface temperature more than **3 degrees C** above the dew-point of the ambient air for the same period.

Use Table entitled "RECOAT WINDOWS" to determine appropriate recoat windows for each coat after the initial coat. Apply each coat during appropriate RECOAT WINDOW of preceding coat. If a RECOAT WINDOW is missed, the minimum and maximum primer and intermediate coat thickness may be adjusted to accommodate a FILL COAT, however, requirements for total epoxy coating thickness and total coating thickness will not be modified. Missing more than one RECOAT WINDOW may require complete removal of coating if maximum total coating thickness requirements cannot be achieved.

If coating is not applied during RECOAT WINDOW, or if surface temperature exceeds **49 degrees C** between applications, provide GLOSS REMOVAL, apply next coat within 24 hours. If next planned coat is topcoat, apply FILL COAT if required to fill sanding marks. Sanding marks from GLOSS REMOVAL of intermediate coat reflecting through topcoat will be considered as noncompliant. Apply FILL COAT within 24 hours of GLOSS REMOVAL, then apply topcoat within RECOAT WINDOW of FILL COAT.

RECOAT WINDOWS						
EPOXY OVER EPOXY						
Temperature degrees C	16-21	22-27	28-32	33-38	39-43	44-49
RECOAT WINDOW (Hrs.)	24-72	18-60	16-48	12-36	8-18	4-6
POLYURETHANE OVER EPOXY						

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

RECOAT WINDOWS						
Temperature degrees C	16-21	22-27	28-32	33-38	39-43	44-49
RECOAT WINDOW (Hrs.)	24-96	24-72	16-48	12-36	10-24	8-16
<u>POLYURETHANE OVER POLYURETHANE</u>						
Temperature degrees C	16-21	22-27	28-32	33-38	39-43	44-49
RECOAT WINDOW (Hrs.)	8-48	6-48	4-36	3-24	2-12	1-2

The temperature ranges shown in the table above are for determining recoat windows. Choose recoat window based on the highest surface temperature that was sustained for one or more hours between coats. This applies to the entire time between coats. Measure and record air and surface temperatures on hourly basis to determine appropriate recoat windows. If surface temperature goes above **38 degrees C**, measure and record temperatures every half hour.

FILL COAT - Where indicated, apply coat of intermediate coat epoxy, at **50 to 75 microns** DFT, then apply next specified full coat within recoat window of FILL COAT. A FILL COAT may be used to adjust coating thickness to comply with requirements or to fill sanding marks in intermediate coat.

GLOSS REMOVAL - Where required, hand sand in a linear fashion to remove gloss using 120-200 grit wet/dry sandpaper, followed by solvent wiping with a clean rag soaked with denatured alcohol to remove all dust. GLOSS REMOVAL of primer coat is to scarify surface and shall consist of removal of approximately **25 microns** of coating. If steel is exposed during GLOSS REMOVAL, repair in accordance with paragraph PROCEDURE FOR HOLIDAY AND SPOT REPAIRS OF NEWLY APPLIED COATING. GLOSS REMOVAL of intermediate coat may include removal of up to **75 microns** of coating to avoid excess thickness, prior to application of FILL COAT.

3.6.2 Amine Blush Testing of Epoxy Coat Prior to Overcoating

Test epoxy surfaces prior to application of roof joint sealant, epoxy coat, or polyurethane topcoat for amine blush contamination using the Test Kit described in paragraph TEST KIT FOR IDENTIFYING AMINE BLUSH ON EPOXY SURFACES. Test all surfaces at rate of three tests for the first **100**

square meters plus one test for each additional 200 square meters or part thereof. Remove any identified contamination using an approved procedure.

3.6.3 Application of Coating System and Joint Sealant

Apply coatings in accordance with SSPC PA 1 and as specified herein. Apply coatings to surfaces that meet all stated surface preparation requirements.

After application of primer coat and prior to application of each subsequent coat, perform testing prescribed in paragraph PRE-APPLICATION TESTING FOR SURFACE CONTAMINATION, as necessary, to ensure minimal intercoat contamination. This testing may be reduced to one half of the prescribed rate for bare steel if the testing indicates no contamination when sampling is evenly distributed over surfaces being tested. If contamination is found between coats, revert to the specified testing rate. Generally, oil and grease contamination and soluble salts contamination are not encountered if subsequent coats are applied within specified recoat windows and unusual atmospheric events do not occur. Such atmospheric events as a coastal storm blowing onshore can bring unusual chloride contamination. Concern for intercoat contamination should be continually prevalent, and spot testing should be accomplished to verify satisfactory conditions. Where visual examination or spot testing indicates contamination, perform sufficient testing to verify non-contamination, or to define extent of contamination for appropriate treatment.

Apply each coat in a consistent wet film, at 90 degrees to previous coat. Ensure that primer and intermediate coat "cold joints" are no less than 150 mm from welds. Apply stripe coat by brush. For convenience, stripe coat material may be delivered by spray if followed immediately with brush-out and approved procedures include appropriate controls on thickness. Apply all other coats by spray application. Use appropriate controls to prevent airborne coating fog from drifting beyond three meters from the structure perimeter. Cover or protect all surfaces that will not be coated. The cleanliness, temperature, recoat windows, and airborne paint containment requirements may necessitate the use of enclosures, portable shelters, or other appropriate controls.

Apply coatings at the following specified thickness:

Coat	Minimum DFT (Microns)	Maximum DFT (Microns)
Primer	75	125
Intermediate	75	125
Top	50	75
Total system	200	325

3.6.3.1 Application of Primer

Apply primer coat, maintaining paint supply container height within 1 meter of the paint nozzle for applying zinc primer. Maintain constant agitation of paint pot to ensure that zinc does not settle in container.

3.6.3.2 Application of Stripe Coat

Apply a stripe coat of intermediate coat epoxy material within RECOAT WINDOW of primer, allowing sufficient dry time to allow application of intermediate coat within RECOAT WINDOW of primer. Apply by brush, working material into corners, crevices, angles, and welds, and onto outside corners and angles.

3.6.3.3 Application of Intermediate Coat

Apply intermediate coat within RECOAT WINDOW of primer coat.

3.6.3.4 Non-skid for Stairs and Top

Where non-skid is required, apply a second intermediate coat, and immediately follow with application of non-skid grit, broadcast at the rate of 2 pounds per 100 square feet, and backroll. Apply topcoat as specified.

3.6.3.5 Application of Topcoat

Make all required repairs to primer and intermediate coats as specified in paragraph entitled "Procedure for Holiday and Spot Repairs of Newly Applied Coating" prior to applying topcoat. Apply topcoat within RECOAT WINDOW of intermediate coat. The polyurethane topcoat may require multiple passes to achieve desired aesthetics and required thickness. Consult manufacturer for thinning and application procedures for anticipated temperature, humidity, and wind conditions. Touch-up blemishes and defects within recoat window of polyurethane topcoat. Retain sample of polyurethane topcoat, from the same batch used to coat structure, to make touch-ups that might be required later.

3.6.3.6 Application of Joint Sealant

Apply joint sealant to back-to-back steel joints that are less than 3/8 inches wide and are not seal welded. Apply sealant to top and bottom, or each side, of narrow joints. Apply sealant within 48 hours of application of the topcoat, and touch-up with topcoat after appropriate cure of the sealant.

3.6.3.7 Procedure for Holiday and Spot Repairs of Newly Applied Coating

Repair coating film defects at the earliest practicable time, preferably before application of the succeeding coat. Observe all requirements for soluble salts contamination, cleanliness between coats, and application conditions. Prepare defective area in accordance with SSPC SP 10/NACE No. 2, and feather coating as required to leave 100 mm of each succeeding coat feathered and abraded. Protect adjacent areas from damage and overspray. Remove dust and solvent wipe the prepared area plus an additional 100 mm beyond the prepared area with clean denatured alcohol. Apply each coat within RECOAT WINDOW of preceding coat. Within four hours of preparation, apply zinc-rich primer to prepared steel and feather onto prepared primer. Apply intermediate coat to primed area and feather to prepared intermediate area. Apply topcoat to intermediate coat and feather to prepared topcoat. Apply each repair coat to approximate thickness of surrounding coating system.

3.6.3.8 Structure Occupancy After Coating Application

Use clean canvas or other approved shoe covers when walking on coated surfaces, regardless of curing time allowed. For heavily trafficked areas, provide cushioned mats for additional protection.

3.7 PROJECT IDENTIFICATION

At the completion of the work, stencil the following information on the structure in 3/4 to one inch Helvetica style letters of contrasting color using acrylic stencil paint:

Date exterior coated:

Project Number:

Contractor:

Address:

Coating System

Surface Prep: SSPC SP _____ Profile: _____

Primer: _____ Thickness: _____

Intermediate: _____ Thickness: _____

Topcoat: _____ Thickness: _____

Total Thickness: _____

3.8 FIELD QUALITY CONTROL

For marking of tank surfaces, use chalk for marking bare steel, and water based markers for marking coated surfaces, and remove marks prior to coating. Do not use any wax or grease based markers, or any other markers that leave a residue or stain.

3.8.1 Coating Inspector

The coating inspector shall be considered a QC Specialist and shall report to the QC Manager, as specified in Section 01 45 00 QUALITY CONTROL. The Coating Inspector shall be present during all pre-preparation testing, surface preparation, coating application, initial cure of the coating system, during all coating repair work, and during completion activities as specified in Section 01 45 00 QUALITY CONTROL. The Coating Inspector shall provide complete documentation of conditions and occurrences on the job site, and be aware of conditions and occurrences that are potentially detrimental to the coating system. The requirements for inspection listed in this Section are in addition to the QC inspection and reporting requirements specified in Section 01 45 00 QUALITY CONTROL.

3.8.2 Field Inspection

3.8.2.1 Inspection Requirements

Perform field inspection in accordance with ASTM D3276 and the approved Coating Work Plan. Document Contractor's compliance with the approved Coating Work Plan.

Provide all tools and instruments required to perform the required testing, as well as any tools or instruments that the inspector considers necessary to perform the required inspections and tests. Document each inspection and test, including required hold points and other required inspections and tests, as well as those inspections and tests deemed prudent from on-site evaluation to document a particular process or condition, as follows:

- a. Location or area;
- b. Purpose (required or special);
- c. Method;
- d. Criteria for evaluation;
- e. Results;
- f. Determination of compliance;
- g. List of required rework;
- h. Observations.

Collect and record Environmental Conditions as described in [ASTM D3276](#) on a 24 hour basis, as follows:

- a. During surface preparation, every two hours or when changes occur;
- b. During coating application and the first four days of initial cure, every hour, or when changes occur;
- c. Note location, time, and temperature of the highest and lowest surface temperatures each day;
- d. Use a non-contact thermometer to locate temperature extremes, then verify with contact thermometers.

Document all equipment used in inspections and testing, including manufacturer, model number, serial number, last calibration date and future calibration date, and results of on-site calibration performed.

Document Contractors compliance with the approved Coating Work Plan.

3.8.2.2 [Inspection Report Forms](#)

Develop project-specific report forms as required to report measurements, test results, and observations being complete and conforming to contract requirements. This includes all direct requirements of the contract documents and indirect requirements of referenced documents. Show acceptance criteria with each requirement and indication of conformity of each inspected item. The data may be in any format, but must be legible and presented so that entered data can be quickly compared to the appropriate requirement.

3.8.2.3 [Daily Inspection Reports](#)

Submit one copy of daily inspection report completed each day when performing work under this Section, to the Contracting Officer. Note all non-compliance issues, and all issues that were reported for rework in accordance with QC procedures of Section [01 45 00](#) QUALITY CONTROL. Each report shall be signed by the coating inspector and the QC Manager. Submit report within 24 hours of date recorded on the report.

3.8.2.4 [Inspection Logbook](#)

A continuous record of all activity related to this Section shall be maintained in an Inspection Logbook on a daily basis. The logbook shall be hard or spiral bound with consecutively numbered pages, and shall be used to record all information provided in the Daily Inspection Reports, as well as other pertinent observations and information. The Coating Inspector's Logbook that is sold by NACE is satisfactory. Submit the original Inspection Logbook to the Contracting Officer upon completion of

the project and prior to final payment.

3.8.2.5 Inspection Equipment

All equipment shall be in good condition, operational within its design range, and calibrated as required by the specified standard for use of each device.

3.9 FINAL CLEANUP

Following completion of the work, remove debris, equipment, and materials from the site. Remove temporary connections to Government or Contractor furnished water and electrical services. Restore existing facilities in and around the work areas to their original condition.

TABLE 1						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ia - Zinc-rich Epoxy Primer Coat MIL-DTL-24441/19 Formula 159						
Test	Component A		Component B		Mixed	
	Min.	Max.	Min.	Max.	Min.	Max.
Pigment content, percent (zinc dust)	---	---	81.5	85.5	---	---
Volatiles, percent	42.8	44.3	8.0	8.4	---	---
Non-volatile vehicle percent	53.7	57.7	8.3	8.7	---	---
Weight, Kilograms/liter	0.87	1.01	3.30	3.40	2.80	2.91
Weight, Pounds/gallon	7.3	8.4	27.5	28.4	23.4	24.4
Flashpoint, Degrees C	35.6	---	37.8	---	---	---
Flashpoint, Degrees F	96	---	100	---	---	---
Consistency, grams	---	---	250	500	150	300
Set to touch time, hours at 23 degrees C, 73 degrees F	---	---	---	---	---	2
Dry hard time, hours at 23 degrees C, 73 degrees F	---	---	---	---	---	8

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

TABLE 1						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ia - Zinc-rich Epoxy Primer Coat MIL-DTL-24441/19 Formula 159						
Test	Component A		Component B		Mixed	
	Min.	Max.	Min.	Max.	Min.	Max.
Pot life, hours at 23 degrees C, 73 degrees F	---	---	---	---	4	---
Sag resistance, Micrometers	---	---	---	---	300	---
Sag resistance, Mils	---	---	---	---	12	---
VOC, Grams/liter	---	---	---	---	---	304
VOC, Pounds/gallon	---	---	---	---	---	2.5
NOTES: Test methods as specified in MIL-DTL-24441 .						

TABLE 1						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ib. - Epoxy Intermediate Coat MIL-DTL-24441/31 Formula 152 Type IV (White (Tinted))						
Test	Component A		Component B		Mixed	
	Min.	Max.	Min.	Max.	Min.	Max.
Pigment content, percent	44.0	49.0	33.0	38.0	---	---
Volatiles, percent	29.0	35.0	16.0	21.0	---	---
Non-volatile vehicle percent	17.5	23.5	44.0	49.0	---	---
Coarse particles, percent	---	0.3	---	0.3	---	---

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

TABLE 1						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ib. - Epoxy Intermediate Coat MIL-DTL-24441/31 Formula 152 Type IV (White (Tinted))						
Test	Component A		Component B		Mixed	
	Min.	Max.	Min.	Max.	Min.	Max.
Consistency, grams	180	320	300	470	180	245
Weight, Kilograms/liter	1.39	1.45	1.29	1.35	1.34	1.4
Weight, Pounds/gallon	11.6	12.1	10.8	11.3	11.2	11.7
Set to touch time, hours at 23 degrees C, 73 degrees F	---	---	---	---	---	3
Dry hard time, hours at 23 degrees C, 73 degrees F	---	---	---	---	---	8
Fineness of grind, Hegman	4	---	4	---	---	---
Flashpoint, Degrees C	35.5	---	37.8	---	---	---
Flashpoint, Degrees F	96	---	100	---	---	---
Titanium dioxide, percent of pigment	91	---	---	---	---	---
Pot life, hours at 23 degrees C, 73 degrees F	---	---	---	---	4	---
Sag resistance, Micrometers	---	---	---	---	300	---
Sag resistance, Mils	---	---	---	---	12	---
Color of dry film to approximate color of SAE AMS-STD-595A color 27778	---	---	---	---	---	Conform
Contrast ratio, at 75 micrometers, 3 mils DFT	---	---	---	---	.098	---
Gloss, 60 degree specular	---	---	---	---	35	---

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

TABLE 1						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ib. - Epoxy Intermediate Coat <u>MIL-DTL-24441/31</u> Formula 152 Type IV (White (Tinted))						
<u>Test</u>	<u>Component A</u>		<u>Component B</u>		<u>Mixed</u>	
	<u>Min.</u>	<u>Max.</u>	<u>Min.</u>	<u>Max.</u>	<u>Min.</u>	<u>Max.</u>
VOC, Grams/liter	---	---	---	---	---	340
VOC, Pounds/gallon	---	---	---	---	---	2.8
GENERAL NOTES: Test methods as specified in <u>MIL-DTL-24441</u> . Where "Conform" is indicated, refer to specific requirements of <u>MIL-DTL-24441/31</u> .						

TABLE I						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ic - Polyurethane TopcoatMIL-PRF-85285 Type II (White and Colors)						
Test	Component A		Component B		Mixed	
	Min.	Max.	Min.	Max.	Min.	Max.
Moisture content, percent	---	2	---	---	---	---
Course particles, percent	---	---	---	---	---	.5
Viscosity	---	---	---	---	---	See Note 1
Fineness of grind, Hegman	---	---	---	---	7	---
Drying to touch (See Note 2)	---	---	---	---	---	4
Dry-hard (See Note 2)	---	---	---	---	---	8
VOC, grams per liter	---	---	---	---	---	340
Color	---	---	---	---	delta E+-1.0	
Gloss 60 degree specular gloss						

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

TABLE I						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ic - Polyurethane Topcoat <u>MIL-PRF-85285</u> Type II (White and Colors)						
Test	Component A		Component B		Mixed	
	Min.	Max.	Min.	Max.	Min.	Max.
Gloss	---	---	---	---	---	90
Semi-gloss	---	---	---	---	15	45
Opacity	---	---	---	---	0.95	---
Flexibility	---	---	---	---	---	Conform
Fluid resistance	---	---	---	---	---	Conform
Heat resistance (cure)	---	---	---	---	---	Conform
Solvent resistance (cure)	---	---	---	---	---	Conform
Condition in container	---	---	---	---	---	Conform
Odor	---	---	---	---	---	Conform
Lead percent	---	---	---	---	---	0.06
Cadmium percent	---	---	---	---	---	0.06
Chromium percent	---	---	---	---	---	0.00
NOTES: (1) Modify paragraph 3.6.4 Viscosity and Pot Life, of <u>MIL-PRF-85285</u> as follows: The viscosity of the admixed coating, when tested in accordance with <u>ASTM D1200</u> through a No. 4 Ford cup, shall be as follows:						
Time from mix (minimum)			Maximum time through a No. 4 Ford Cup			
Initially			30 seconds			
2 hours			60 seconds			
4 hours			No gel			

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

TABLE I						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ic - Polyurethane Topcoat MIL-PRF-85285 Type II (White and Colors)						
<u>Test</u>	<u>Component A</u>		<u>Component B</u>		<u>Mixed</u>	
	<u>Min.</u>	<u>Max.</u>	<u>Min.</u>	<u>Max.</u>	<u>Min.</u>	<u>Max.</u>
(2) Modify paragraph 3.7.1 Drying Time, of MIL-PRF-85285 . When applied by spray techniques and when tested in accordance with ASTM D1640/D1640M , the coating shall be set-to-touch within four hours and dry-hard within eight hours (see 4.6 and table I).						
GENERAL NOTES: Test methods as specified in MIL-PRF-85285 , except those marked with "*". Where "Conform" is indicated, refer to specific requirements of MIL-PRF-85285 .						

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 03 00.00 20

BASIC MECHANICAL MATERIALS AND METHODS

08/10, CHG 3: 08/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 QUALITY ASSURANCE
 - 1.3.1 Material and Equipment Qualifications
 - 1.3.2 Alternative Qualifications
 - 1.3.3 Service Support
 - 1.3.4 Manufacturer's Nameplate
 - 1.3.5 Modification of References
 - 1.3.5.1 Definitions
 - 1.3.5.2 Administrative Interpretations
- 1.4 DELIVERY, STORAGE, AND HANDLING
- 1.5 ELECTRICAL REQUIREMENTS
- 1.6 INSTRUCTION TO GOVERNMENT PERSONNEL
- 1.7 ACCESSIBILITY

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 PAINTING OF NEW EQUIPMENT
 - 3.1.1 Factory Painting Systems
 - 3.1.2 Shop Painting Systems for Metal Surfaces

-- End of Section Table of Contents --

SECTION 23 03 00.00 20

BASIC MECHANICAL MATERIALS AND METHODS
08/10, CHG 3: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM B117 (2019) Standard Practice for Operating
Salt Spray (Fog) Apparatus

1.2 RELATED REQUIREMENTS

This section applies to all sections of Divisions: 21, FIRE SUPPRESSION; 22, PLUMBING; and 23, HEATING, VENTILATING, AND AIR CONDITIONING of this project specification, unless specified otherwise in the individual section.

1.3 QUALITY ASSURANCE

1.3.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design and workmanship. Standard products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

1.3.2 Alternative Qualifications

Products having less than a two-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

1.3.3 Service Support

The equipment items must be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. These service organizations must be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.3.4 Manufacturer's Nameplate

For each item of equipment, provide a nameplate bearing the manufacturer's

name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

1.3.5 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer.

1.3.5.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions must be considered mandatory, the word "should" is interpreted as "must." Reference to the "code official" must be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" must be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" must be interpreted to mean the "lessor." References to the "permit holder" must be interpreted to mean the "Contractor."

1.3.5.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, must be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

1.4 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

1.5 ELECTRICAL REQUIREMENTS

Furnish motors, controllers, disconnects and contactors with their respective pieces of equipment. Motors, controllers, disconnects and contactors must conform to and have electrical connections provided under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Furnish internal wiring for components of packaged equipment as an integral part of the equipment. Extended voltage range motors will not be permitted. Controllers and contactors shall have a maximum of 120 volt control circuits, and must have auxiliary contacts for use with the controls furnished. When motors and equipment furnished are larger than sizes indicated, the cost of additional electrical service and related work must be included under the section that specified that motor or equipment. Power wiring and conduit for field installed equipment must be provided under and conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

1.6 INSTRUCTION TO GOVERNMENT PERSONNEL

When specified in other sections, furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Instructors must be thoroughly familiar with all parts of the installation and must be trained in operating theory as well as practical operation and maintenance work.

Instruction must be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished must be as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

1.7 ACCESSIBILITY

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 PAINTING OF NEW EQUIPMENT

New equipment painting must be factory applied or shop applied, and must be as specified herein, and provided under each individual section.

3.1.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied will withstand 500 hours in a salt-spray fog. Salt-spray fog test must be in accordance with [ASTM B117](#), and for that test the acceptance criteria must be as follows: immediately after completion of the test, the paint must show no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen must show no signs of rust creepage beyond [3 mm](#) on either side of the scratch mark.

The film thickness of the factory painting system applied on the equipment must not be less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above [50 degrees C](#), the factory painting system must be designed for the temperature service.

3.1.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except metal surfaces subject to temperatures in excess of 50 degrees C must be cleaned to bare metal.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Color of finish coat must be aluminum or light gray.

- a. Temperatures Less Than 50 Degrees C: Immediately after cleaning, the metal surfaces subject to temperatures less than 50 degrees C must receive one coat of pretreatment primer applied to a minimum dry film thickness of 0.0076 mm, one coat of primer applied to a minimum dry film thickness of 0.0255 mm; and two coats of enamel applied to a minimum dry film thickness of 0.0255 mm per coat.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 05 93

TESTING, ADJUSTING, AND BALANCING FOR HVAC

11/15

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Similar Terms
- 1.3 WORK DESCRIPTION
 - 1.3.1 Air Distribution Systems
 - 1.3.2 TAB SCHEMATIC DRAWINGS
 - 1.3.3 Related Requirements
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Independent TAB Agency and Personnel Qualifications
 - 1.5.1.1 TAB Standard
 - 1.5.1.2 Qualifications
 - 1.5.1.3 TAB Related HVAC Submittals
 - 1.5.2 Responsibilities
 - 1.5.2.1 Contractor
 - 1.5.2.2 TAB Agency
 - 1.5.2.3 TAB Team Supervisor
 - 1.5.2.4 TAB Team Field Leader
 - 1.5.3 Sequencing and Scheduling
 - 1.5.3.1 Projects with Phased Construction
 - 1.5.3.2 DALT and TAB Submittal and Work Schedule
 - 1.5.3.3 TAB Pre-Field Engineering Report
 - 1.5.4 Subcontractor Special Requirements
 - 1.5.5 Instrument Calibration Certificates

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 WORK DESCRIPTIONS OF PARTICIPANTS
- 3.2 PRE-DALT/TAB MEETING
- 3.3 DALT PROCEDURES
 - 3.3.1 Instruments, Consumables and Personnel
 - 3.3.2 Advance Notice of Pre-Final DALT Field Work
 - 3.3.3 Ductwork To Be DALT'd
 - 3.3.4 DALT Testing
 - 3.3.5 Completed Pre-Final DALT Report
 - 3.3.6 Quality Assurance - COTR DALT Field Acceptance Testing
 - 3.3.7 Additional COTR Field Acceptance Testing
 - 3.3.8 Certified Final DALT Report
 - 3.3.9 Prerequisite for TAB Field Work
- 3.4 TAB PROCEDURES
 - 3.4.1 TAB Field Work

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.4.2 Preliminary Procedures
- 3.4.3 TAB Air Distribution Systems
 - 3.4.3.1 Units With Coils
 - 3.4.3.2 Air Handling Units
 - 3.4.3.3 Heating and Ventilating Units
 - 3.4.3.4 Exhaust Fans
- 3.4.4 TAB Work on Performance Tests Without Seasonal Limitations
 - 3.4.4.1 Performance Tests
 - 3.4.4.2 Ambient Temperatures
- 3.4.5 Workmanship
- 3.4.6 Deficiencies
- 3.4.7 TAB Reports
- 3.4.8 Quality Assurance - COTR TAB Field Acceptance Testing
 - 3.4.8.1 TAB Field Acceptance Testing
 - 3.4.8.2 Additional COTR TAB Field Acceptance Testing
 - 3.4.8.3 Prerequisite for Approval
- 3.5 MARKING OF SETTINGS
- 3.6 MARKING OF TEST PORTS
- 3.7 APPENDICES

-- End of Section Table of Contents --

SECTION 23 05 93

TESTING, ADJUSTING, AND BALANCING FOR HVAC
11/15

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 62.1 (2010) Ventilation for Acceptable Indoor Air Quality

ASSOCIATED AIR BALANCE COUNCIL (AABC)

AABC MN-1 (2002; 6th ed) National Standards for Total System Balance

AABC MN-4 (1996) Test and Balance Procedures

NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB)

NEBB MASV (2006) Procedural Standards for Measurements and Assessment of Sound and Vibration

NEBB PROCEDURAL STANDARDS (2015) Procedural Standards for TAB (Testing, Adjusting and Balancing) Environmental Systems

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1780 (2002) HVAC Systems - Testing, Adjusting and Balancing, 3rd Edition

SMACNA 1858 (2004) HVAC Sound And Vibration Manual - First Edition

SMACNA 1972 CD (2012) HVAC Air Duct Leakage Test Manual - 2nd Edition

1.2 DEFINITIONS

- a. AABC: Associated Air Balance Council
- b. COTR: Contracting Officer's Technical Representative
- c. DALT: Duct air leakage test
- d. DALT'd: Duct air leakage tested

- e. HVAC: Heating, ventilating, and air conditioning; or heating, ventilating, and cooling
- f. NEBB: National Environmental Balancing Bureau
- g. Out-of-tolerance data: Pertains only to field acceptance testing of Final DALT or TAB report. When applied to DALT work, this phase means "a leakage rate measured during DALT field acceptance testing which exceeds the leakage rate allowed by Appendix D REQUIREMENTS FOR DUCT AIR LEAK TESTING." When applied to TAB work this phase means "a measurement taken during TAB field acceptance testing which does not fall within the range of plus 5 to minus 5 percent of the original measurement reported on the TAB Report for a specific parameter."
- h. Season of maximum heating load: The time of year when the outdoor temperature at the project site remains within plus or minus 17.5 degrees Celsius of the project site's winter outdoor design temperature, throughout the period of TAB data recording.
- i. Season of maximum cooling load: The time of year when the outdoor temperature at the project site remains within plus or minus 3 degrees Celsius of the project site's summer outdoor design temperature, throughout the period of TAB data recording.
- j. Season 1, Season 2: Depending upon when the project HVAC is completed and ready for TAB, Season 1 is defined, thereby defining Season 2. Season 1 could be the season of maximum heating load, or the season of maximum cooling load.
- k. Sound measurements terminology: Defined in AABC MN-1, NEBB MASV, or SMACNA 1858 (TABB).
- l. TAB: Testing, adjusting, and balancing (of HVAC systems)
- m. TAB'd: HVAC Testing/Adjusting/Balancing procedures performed
- n. TAB Agency: TAB Firm
- o. TAB team field leader: TAB team field leader
- p. TAB team supervisor: TAB team engineer
- q. TAB team technicians: TAB team assistants
- r. TABB: Testing Adjusting and Balancing Bureau

1.2.1 Similar Terms

In some instances, terminology differs between the Contract and the TAB Standard primarily because the intent of this Section is to use the industry standards specified, along with additional requirements listed herein to produce optimal results.

The following table of similar terms is provided for clarification only. Contract requirements take precedent over the corresponding AABC, NEBB, or TABB requirements where differences exist.

SIMILAR TERMS			
Contract Term	AABC Term	NEBB Term	TABB Term
TAB Standard	National Standards for Testing and Balancing Heating, Ventilating, and Air Conditioning Systems	Procedural Standards for Testing, Adjusting and Balancing of Environmental Systems	International Standards for Environmental Systems Balance
TAB Specialist	TAB Engineer	TAB Supervisor	TAB Supervisor
Systems Readiness Check	Construction Phase Inspection	Field Readiness Check & Preliminary Field Procedures	Field Readiness Check & Prelim. Field Procedures

1.3 WORK DESCRIPTION

The work includes duct air leakage testing (DALT) and testing, adjusting, and balancing (TAB) of new heating, ventilating, and cooling (HVAC) air distribution systems including equipment and performance data, ducts, and piping which are located within, on, under, between, and adjacent to buildings.

Perform TAB in accordance with the requirements of the TAB procedural standard recommended by the TAB trade association that approved the TAB Firm's qualifications. Comply with requirements of AABC MN-1, NEBB PROCEDURAL STANDARDS, or SMACNA 1780 (TABB) as supplemented and modified by this specification section. All recommendations and suggested practices contained in the TAB procedural standards are considered mandatory.

1.3.1 Air Distribution Systems

Test, adjust, and balance systems (TAB) in compliance with this section. Obtain Contracting Officer's written approval before applying insulation to exterior of air distribution systems.

1.3.2 TAB SCHEMATIC DRAWINGS

Show the following information on TAB Schematic Drawings:

1. A unique number or mark for each piece of equipment or terminal.
2. Air quantities at air terminals.
3. Air quantities and temperatures in air handling unit schedules.
4. Ductwork Construction and Leakage Testing Table that defines the DALT test requirements, including each applicable HVAC duct system ID or mark, duct pressure class, duct seal class, and duct leakage test pressure. This table is included in the file for Graphics for Unified Facilities Guide Specifications:
<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-gra>

The Testing, Adjusting, and Balancing (TAB) Specialist must review the Contract Plans and Specifications and advise the Contracting Officer of any deficiencies that would prevent the effective and accurate TAB of the system, and systems readiness check. The TAB Specialist must provide a Design Review Report individually listing each deficiency and the corresponding proposed corrective action necessary for proper system operation.

Submit three copies of the [TAB Schematic Drawings and Report Forms](#) to the Contracting Officer, no later than 21 days prior to the start of TAB field measurements.

1.3.3 Related Requirements

Requirements for price breakdown of HVAC TAB work are specified in Section [01 20 00 PRICE AND PAYMENT PROCEDURES](#).

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section [01 33 00 SUBMITTAL PROCEDURES](#):

[SD-01 Preconstruction Submittals](#)

[Independent TAB Agency and Personnel Qualifications; G](#)

[TAB Design Review Report; G](#)

[SD-02 Shop Drawings](#)

[TAB Schematic Drawings and Report Forms; G](#)

[SD-03 Product Data](#)

[Equipment and Performance Data; G](#)

[TAB Related HVAC Submittals; G](#)

A list of the TAB Related HVAC Submittals, no later than 7 days after the approval of the TAB team engineer .

[SD-06 Test Reports](#)

[Completed Pre-Final DALT Report; G](#)

[Certified Final DALT Report; G](#)

[Prerequisite HVAC Work Checkout List; G](#)

[Certified Final TAB Report for ; G](#)

[SD-07 Certificates](#)

[Independent TAB Agency and Personnel Qualifications; G](#)

[DALT and TAB Submittal and Work Schedule; G](#)

TAB Pre-Field Engineering Report; G

Instrument Calibration Certificates; G

DALT and TAB Procedures Summary; G

Completed Pre-Final DALT Work Checklist; G

Advance Notice of Pre-Final DALT Field Work; G

Advance Notice of TAB Field Work; G

1.5 QUALITY ASSURANCE

1.5.1 Independent TAB Agency and Personnel Qualifications

To secure approval for the proposed agency, submit information certifying that the TAB agency is a first tier subcontractor who is not affiliated with any other company participating in work on this contract, including design, furnishing equipment, or construction. Further, submit the following, for the agency, to Contracting Officer for approval:

a. Independent AABC or NEBB or TABB TAB agency:

TAB agency: AABC registration number and expiration date of current certification; or NEBB certification number and expiration date of current certification; or TABB certification number and expiration date of current certification.

TAB team supervisor: Name and copy of AABC or NEBB or TABB TAB supervisor certificate and expiration date of current certification.

TAB team field leader: Name and documented evidence that the team field leader has satisfactorily performed full-time supervision of TAB work in the field for not less than 3 years immediately preceding this contract's bid opening date.

TAB team field technicians: Names and documented evidence that each field technician has satisfactorily assisted a TAB team field leader in performance of TAB work in the field for not less than one year immediately preceding this contract's bid opening date.

Current certificates: Registrations and certifications are current, and valid for the duration of this contract. Renew Certifications which expire prior to completion of the TAB work, in a timely manner so that there is no lapse in registration or certification. TAB agency or TAB team personnel without a current registration or current certification are not to perform TAB work on this contract.

b. TAB Team Members: TAB team approved to accomplish work on this contract are full-time employees of the TAB agency. No other personnel is allowed to do TAB work on this contract.

c. Replacement of TAB team members: Replacement of members may occur if each new member complies with the applicable personnel qualifications and each is approved by the Contracting Officer.

1.5.1.1 TAB Standard

Perform TAB in accordance with the requirements of the standard under which the TAB Firm's qualifications are approved, i.e., AABC MN-1, NEBB PROCEDURAL STANDARDS, or SMACNA 1780 unless otherwise specified herein. All recommendations and suggested practices contained in the TAB Standard are considered mandatory. Use the provisions of the TAB Standard, including checklists, report forms, etc., as nearly as practical, to satisfy the Contract requirements. Use the TAB Standard for all aspects of TAB, including qualifications for the TAB Firm and Specialist and calibration of TAB instruments. Where the instrument manufacturer calibration recommendations are more stringent than those listed in the TAB Standard, adhere to the manufacturer's recommendations.

All quality assurance provisions of the TAB Standard such as performance guarantees are part of this contract. For systems or system components not covered in the TAB Standard, TAB procedures must be developed by the TAB Specialist. Where new procedures, requirements, etc., applicable to the Contract requirements have been published or adopted by the body responsible for the TAB Standard used (AABC, NEBB, or TABB), the requirements and recommendations contained in these procedures and requirements are considered mandatory, including the latest requirements of ASHRAE 62.1.

1.5.1.2 Qualifications

a. TAB Firm

The TAB Firm must be either a member of AABC or certified by the NEBB or the TABB and certified in all categories and functions where measurements or performance are specified on the plans and specifications.

Certification must be maintained for the entire duration of duties specified herein. If, for any reason, the firm loses subject certification during this period, the Contractor must immediately notify the Contracting Officer and submit another TAB Firm for approval. Any firm that has been the subject of disciplinary action by either the AABC, the NEBB, or the TABB within the five years preceding Contract Award is not be eligible to perform any duties related to the HVAC systems, including TAB. All work specified in this Section and in other related Sections to be performed by the TAB Firm will be considered invalid if the TAB Firm loses its certification prior to Contract completion and must be performed by an approved successor.

These TAB services are to assist the prime Contractor in performing the quality oversight for which it is responsible. The TAB Firm must be a prime subcontractor of the Contractor and be financially and corporately independent of the mechanical subcontractor, reporting directly to and paid by the Contractor.

b. TAB Specialist

The TAB Specialist must be either a member of AABC, an experienced technician of the Firm certified by the NEBB, or a Supervisor certified by the TABB. The certification must be maintained for the entire duration of duties specified herein. If, for any reason, the

Specialist loses subject certification during this period, immediately notify the Contracting Officer and submit another TAB Specialist for approval. Any individual that has been the subject of disciplinary action by either the AABC, the NEBB, or the TABB within the five years preceding Contract Award is not eligible to perform any duties related to the HVAC systems, including TAB. All work specified in this Section and in other related Sections performed by the TAB Specialist will be considered invalid if the TAB Specialist loses its certification prior to Contract completion and must be performed by the approved successor.

c. TAB Specialist Responsibilities

TAB Specialist responsibilities include all TAB work specified herein and in related sections under his direct guidance. The TAB specialist is required to be onsite on a daily basis to direct TAB efforts. The TAB Specialist must participate in the commissioning process.

1.5.1.3 TAB Related HVAC Submittals

The TAB Specialist must prepare a list of the submittals from the Contract Submittal Register that relate to the successful accomplishment of all HVAC TAB. Accompany the submittals identified on this list with a letter of approval signed and dated by the TAB Specialist when submitted to the Government. Ensure that the location and details of ports, terminals, connections, etc., necessary to perform TAB are identified on the submittals.

1.5.2 Responsibilities

The Contractor is responsible for ensuring compliance with the requirements of this section. The following delineation of specific work responsibilities is specified to facilitate TAB execution of the various work efforts by personnel from separate organizations. This breakdown of specific duties is specified to facilitate adherence to the schedule listed in the paragraph TAB SUBMITTAL AND WORK SCHEDULE.

1.5.2.1 Contractor

- a. TAB personnel: Ensure that the DALT work and the TAB work is accomplished by a group meeting the requirements specified in the paragraph TAB PERSONNEL QUALIFICATION REQUIREMENTS.
- b. Pre-DALT/TAB meeting: Attend the meeting with the TAB Supervisor, and ensure that a representative is present for the sheetmetal contractor, mechanical contractor, electrical contractor, and automatic temperature controls contractor.
- c. HVAC documentation: Furnish one complete set of the following HVAC-related documentation to the TAB agency:
 - (1) Contract drawings and specifications
 - (2) Approved submittal data for equipment
 - (3) Construction work schedule
 - (4) Up-to-date revisions and change orders for the previously listed items

- d. Submittal and work schedules: Ensure that the schedule for submittals and work required by this section and specified in the paragraph TAB SUBMITTAL AND WORK SCHEDULE is met.

- e. Coordination of supporting personnel:

Provide the technical personnel, such as factory representatives or HVAC controls installer required by the TAB field team to support the DALT and the TAB field measurement work.

Provide equipment mechanics to operate HVAC equipment and ductwork mechanics to provide the field designated test ports to enable TAB field team to accomplish the DALT and the TAB field measurement work. Ensure these support personnel are present at the times required by the TAB team, and cause no delay in the DALT and the TAB field work.

Conversely, ensure that the HVAC controls installer has required support from the TAB team field leader to complete the controls check out.

- f. Deficiencies: Ensure that the TAB Agency supervisor submits all Design/Construction deficiency notifications directly to the Contracting officer within 3 days after the deficiency is encountered. Further, ensure that all such notification submittals are complete with explanation, including documentation, detailing deficiencies.
- g. Prerequisite HVAC work: Complete check out and debugging of HVAC equipment, ducts, and controls prior to the TAB engineer arriving at the project site to begin the TAB work. Debugging includes searching for and eliminating malfunctioning elements in the HVAC system installations, and verifying all adjustable devices are functioning as designed. Include as prerequisite work items, the deficiencies pointed out by the TAB team supervisor in the design review report.
- h. Prior to the TAB field team's arrival, ensure completion of the applicable inspections and work items listed in the TAB team supervisor's pre-field engineering report. Do not allow the TAB team to commence TAB field work until all of the following are completed.
 - (1) HVAC system installations are fully complete.
 - (2) HVAC prerequisite checkout work lists specified in the paragraph PRE-FIELD TAB ENGINEERING REPORT are completed, submitted, and approved. Ensure that the TAB Agency gets a copy of the approved prerequisite HVAC work checklist.
 - (3) DALT field checks for all systems are completed.
 - (4) HVAC system filters are clean.
- i. Advance notice: Furnish to the Contracting Officer with advance written notice for the commencement of the DALT field work and for the commencement of the TAB field work.
- j. Insulation work: For required DALT work, ensure that insulation is not installed on ducts to be DALT'd until DALT work on the subject ducts is complete. Later, ensure that openings in duct and machinery

insulation coverings for TAB test ports are marked, closed and sealed.

1.5.2.2 TAB Agency

Provide the services of a TAB team which complies with the requirements of the paragraph INDEPENDENT TAB AGENCY PERSONNEL QUALIFICATIONS. The work to be performed by the TAB agency is limited to testing, adjusting, and balancing of HVAC air systems to satisfy the requirements of this specification section.

1.5.2.3 TAB Team Supervisor

- a. Overall management: Supervise and manage the overall TAB team work effort, including preliminary and technical DALT and TAB procedures and TAB team field work.
- b. Pre-DALT/TAB meeting: Attend meeting with Contractor.
- c. Design review report: Review project specifications and accompanying drawings to verify that the air systems are designed in such a way that the TAB engineer can accomplish the work in compliance with the requirements of this section. Verify the presence and location of permanently installed test ports and other devices needed, including gauge cocks, thermometer wells, flow control devices, circuit setters, balancing valves, and manual volume dampers.
- d. Support required: Specify the technical support personnel required from the Contractor other than the TAB agency; such as factory representatives for temperature controls or for complex equipment. Inform the Contractor in writing of the support personnel needed and when they are needed. Furnish the notice as soon as the need is anticipated, either with the design review report, or the pre-field engineering report, the during the DALT or TAB field work.
- e. Pre-field DALT preliminary notification: Monitor the completion of the duct installation of each system and provide the necessary written notification to the Contracting Officer.
- f. Pre-field engineering report: Utilizing the following HVAC-related documentation; contract drawings and specifications, approved submittal data for equipment, up-to-date revisions and change orders; prepare this report.
- g. Prerequisite HVAC work checklist: Ensure the Contractor gets a copy of this checklist at the same time as the pre-field engineering report is submitted.
- h. Technical assistance for DALT work.
 - (1) Technical assistance: Provide immediate technical assistance to TAB field team.
 - (2) DALT field visit: Near the end of the DALT field work effort, visit the contract site to inspect the HVAC installation and the progress of the DALT field work. Conduct a site visit to the extent necessary to verify correct procedures are being implemented and to confirm the accuracy of the Pre-final DALT Report data which has been reported. Also, perform sufficient evaluation to allow the TAB supervisor to issue certification of

the final report. Conduct the site visit full-time for a minimum of one 8 hour workday duration.

- i. Final DALT report: Certify the DALT report. This certification includes the following work:
 - (1) Review: Review the Pre-final DALT report data. From these field reports, prepare the Certified Final DALT report.
 - (2) TAB Verification: Verify adherence, by the TAB field team, to the procedures specified in this section.
- j. Technical Assistance for TAB Work: Provide immediate technical assistance to the TAB field team for the TAB work.
 - (1) TAB field visit: Near the end of the TAB field work effort, visit the contract site to inspect the HVAC installation and the progress of the TAB field work. Conduct site visit full-time for a minimum of one 8 hour workday duration. Review the TAB final report data and certify the TAB final report.
- k. Certified TAB report: Certify the TAB report. This certification includes the following work:
 - (1) Review: Review the TAB field data report. From this field report, prepare the certified TAB report.
 - (2) Verification: Verify adherence, by the TAB field team, to the TAB plan prescribed by the pre-field engineering report and verify adherence to the procedures specified in this section.
- l. Design/Construction deficiencies: Within 3 working days after the TAB Agency has encountered any design or construction deficiencies, the TAB Supervisor must submit written notification directly to the Contracting Officer, with a separate copy to the Contractor, of all such deficiencies. Provide in this submittal a complete explanation, including supporting documentation, detailing deficiencies. Where deficiencies are encountered that are believed to adversely impact successful completion of TAB, the TAB Agency must issue notice and request direction in the notification submittal.
- m. TAB Field Check: The TAB team supervisor must attend and supervise TAB field check.

1.5.2.4 TAB Team Field Leader

- a. Field manager: Manage, in the field, the accomplishment of the work specified in Part 3, EXECUTION.
- b. Full time: Be present at the contract site when DALT field work or TAB field work is being performed by the TAB team; ensure day-to-day TAB team work accomplishments are in compliance with this section.
- c. Prerequisite HVAC work: Do not bring the TAB team to the contract site until a copy of the prerequisite HVAC Checklist, with all work items certified by the Contractor to be working as designed, reaches the office of the TAB Agency.

1.5.3 Sequencing and Scheduling

1.5.3.1 Projects with Phased Construction

This specification section is structured as though the HVAC construction, and thereby the TAB work, will be completed in a single phase. When the construction is completed in phases, the DALT work and TAB work must be planned, completed, and accepted for each construction phase.

a. Phasing of Work

This specification section is structured as though the HVAC construction, and thereby the TAB work, is going to be completed in a single phase. All elements of the TAB work are addressed on this premise. When a contract is to be completed in construction phases, including the TAB work, and the DALT work, the TAB work and DALT work must be planned for, completed and approved by the Contracting Officer with each phase. An example of this case would be one contract that requires the rehabilitation of the HVAC in each of several separated buildings. At the completion of the final phase, compile all approved reports and submit as one document.

1.5.3.2 DALT and TAB Submittal and Work Schedule

Comply with additional requirements specified in Appendix C: DALT AND TAB SUBMITTAL AND WORK SCHEDULE included at the end of this section.

Submit this schedule, and TAB Schematic Drawings, adapted for this particular contract, to the Contracting Officer (CO) for review and approval. Include with the submittal the planned calendar dates for each submittal or work item. Resubmit an updated version for CO approval every 90 calendar days. Compliance with the following schedule is the Contractor's responsibility.

Qualify TAB Personnel: Within 45 calendar days after date of contract award, submit TAB agency and personnel qualifications.

Pre-DALT/TAB Meeting: Within 30 calendar days after the date of approval of the TAB agency and personnel, meet with the COTR.

Design Review Report: Within 60 calendar days after the date of the TAB agency personnel qualifications approval, submit design review report.

Pre-Field DALT Preliminary Notification: On completion of the duct installation for each system, notify the Contracting Officer in writing within 5 days after completion.

Ductwork Selected for DALT: Within 7 calendar days of Pre-Field DALT Preliminary Notification, the COTR will select which of the project ductwork must be DALT'd.

DALT Field Work: Within 48 hours of COTR's selection, complete DALT field work on selected.

Submit Pre-final DALT Report: Within one working day after completion of DALT field work, submit Pre-final DALT Report. Separate Pre-final DALT reports may be submitted to allow phased testing from system to system.

DALT Work Field Check: Upon approval of the Pre-final DALT Report, schedule the COTR's DALT field check work with the Contracting Officer.

Submit Final DALT Report: Within 15 calendar days after completion of successful DALT Work Field Check, submit TAB report.

Pre-Field TAB Engineering Report: Within 30 calendar days after approval of the TAB agency Personnel Qualifications, submit the Pre-Field TAB Engineering Report.

Prerequisite HVAC Work Check Out List and Advanced Notice For TAB Field Work: At a minimum of 115 calendar days prior to CCD, submit prerequisite HVAC work check out list certified as complete, and submit advance notice of commencement of TAB field work.

TAB Field Work: At a minimum of 90 calendar days prior to CCD, accomplish TAB field work; submit TAB report; and conduct field check.

Complete TAB Work: Prior to CCD, complete all TAB work .

a. TAB Design Review Report

Submit typed report describing omissions and deficiencies in the HVAC system's design that would preclude the TAB team from accomplishing the duct leakage testing work and the TAB work requirements of this section. Provide a complete explanation including supporting documentation detailing the design deficiency. State that no deficiencies are evident if that is the case.

b. Pre-Field DALT Preliminary Notification

Notification: On completion of the installation of each duct system indicated to be DALT'd, notify the Contracting Officer in writing within 7 calendar days after completion.

1.5.3.3 TAB Pre-Field Engineering Report

Submit report containing the following information:

a. Step-by-step TAB procedure:

- (1) Strategy: Describe the method of approach to the TAB field work from start to finish. Include in this description a complete methodology for accomplishing each seasonal TAB field work session.
- (2) Air System Diagrams: Use the contract drawings and duct fabrication drawings if available to provide air system diagrams in the report showing the location of all terminal outlet supply, return, exhaust and transfer registers, grilles and diffusers. Use a key numbering system on the diagrams which identifies each outlet contained in the outlet airflow report sheets. Show intended locations of all traverses and static pressure readings.
- (3) Procedural steps: Delineate fully the intended procedural steps to be taken by the TAB field team to accomplish the required TAB work of each air distribution system and each water distribution system. Include intended procedural steps for TAB work for subsystems and system components.

- b. Pre-field data: Submit AABC or NEBB or **SMACNA 1780** data report forms with the following pre-field information filled in:
 - (1) Design data obtained from system drawings, specifications, and approved submittals.
 - (2) Notations detailing additional data to be obtained from the contract site by the TAB field team.
 - (3) Designate the actual data to be measured in the TAB field work.
 - (4) Provide a list of the types of instruments, and the measuring range of each, which are anticipated to be used for measuring in the TAB field work. By means of a keying scheme, specify on each TAB data report form submitted, which instruments will be used for measuring each item of TAB data. If the selection of which instrument to use, is to be made in the field, specify from which instruments the choice will be made. Place the instrument key number in the blank space where the measured data would be entered.
- c. **Prerequisite HVAC work checkout list**: Provide a list of inspections and work items which are to be completed by the Contractor. This list must be acted upon and completed by the Contractor and then submitted and approved by the Contracting Officer prior to the TAB team coming to the contract site.

At a minimum, a list of the applicable inspections and work items listed in the **NEBB PROCEDURAL STANDARDS**, Section III, "Preliminary TAB Procedures" under paragraph titled, "Air Distribution System Inspection" must be provided for each separate system to be TAB'd.

1.5.4 Subcontractor Special Requirements

Perform all work in this section in accordance with the paragraph SUBCONTRACTOR SPECIAL REQUIREMENTS in Section **01 30 00** ADMINISTRATIVE REQUIREMENTS, stating that all contract requirements of this section must be accomplished directly by a first tier subcontractor. No work may be performed by a second tier subcontractor.

1.5.5 Instrument Calibration Certificates

It is the responsibility of the TAB firm to provide instrumentation that meets the minimum requirements of the standard under which the TAB Firm's qualifications are approved for use on a project. Instrumentation must be in proper operating condition and must be applied in accordance with the instrumentation's manufacturer recommendations.

All instrumentation must bear a valid NIST traceable calibration certificate during field work and during government acceptance testing. All instrumentation must be calibrated within no later than one year of the date of TAB work or government acceptance testing field work.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 WORK DESCRIPTIONS OF PARTICIPANTS

Comply with requirements of this section as specified in Appendix A WORK DESCRIPTIONS OF PARTICIPANTS.

3.2 PRE-DALT/TAB MEETING

Meet with the Contracting Officer's technical representative (COTR) to develop a mutual understanding relative to the details of the DALT work and TAB work requirements. Ensure that the TAB supervisor is present at this meeting. Requirements to be discussed include required submittals, work schedule, and field quality control.

3.3 DALT PROCEDURES

3.3.1 Instruments, Consumables and Personnel

Provide instruments, consumables and personnel required to accomplish the DALT field work. Follow the same basic procedure specified below for TAB Field Work, including maintenance and calibration of instruments, accuracy of measurements, preliminary procedures, field work, workmanship and treatment of deficiencies. Calibrate and maintain instruments in accordance with manufacturer's written procedures.

3.3.2 Advance Notice of Pre-Final DALT Field Work

On completion of the installation of each duct system indicated to be DALT'd, notify the Contracting Officer in writing prior to the COTR's duct selection field visit.

3.3.3 Ductwork To Be DALT'd

From each duct system indicated as subject to DALT, the COTR will randomly select sections of each completed duct system for testing by the Contractor's TAB Firm. The sections selected will not exceed 20 percent of the total measured linear footage of duct systems indicated as subject to DALT. Sections of duct systems subject to DALT will include 20 percent of main ducts, branch main ducts, branch ducts and plenums for supply, return, exhaust, and plenum ductwork.

It is acceptable for an entire duct system to be DALT'd instead of disassembling that system in order to DALT only the 20 percent portion specified above.

3.3.4 DALT Testing

Perform DALT on the HVAC duct sections of each system as selected by the COTR. Use the duct class, seal class, leakage class and the leak test pressure data indicated on the drawings, to comply with the procedures specified in SMACNA 1972 CD.

In spite of specifications of SMACNA 1972 CD to the contrary, DALT ductwork of construction class of 746 Pa water gauge static pressure and below if indicated to be DALT'd. Complete DALT work on the COTR selected ductwork within 48 hours after the particular ductwork was selected for DALT. Separately conduct DALT work for large duct systems to enable the DALT work to be completed in 48 hours.

3.3.5 Completed Pre-Final DALT Report

After completion of the DALT work, prepare a Pre-final DALT Report meeting the additional requirements specified in Appendix B REPORTS - DALT and TAB. Data required by those data report forms shall be furnished by the TAB team. Prepare the report neatly and legibly; the Pre-final DALT report shall provide the basis for the Final DALT Report.

TAB supervisor shall review, approve and sign the Pre-Final DALT Report and submit this report within one day of completion of DALT field work. Verbally notify the COTR that the field check of the Pre-Final DALT Report data can commence.

3.3.6 Quality Assurance - COTR DALT Field Acceptance Testing

In the presence of the COTR and TAB team field leader, verify for accuracy Pre-final DALT Report data selected by the COTR. For each duct system, this acceptance testing shall be conducted on a maximum of 50 percent of the duct sections DALT'd.

Further, if any data on the Pre-final DALT report form for a given duct section is out-of-tolerance, then field acceptance testing shall be conducted on data for one additional duct section, preferably in the same duct system, in the presence of the COTR.

3.3.7 Additional COTR Field Acceptance Testing

If any of the duct sections checked for a given system are determined to have a leakage rate measured that exceeds the leakage rate allowed by SMACNA Leak Test Manual for an indicated duct construction class and sealant class, terminate data checking for that section. The associated Pre-final DALT Report data for the given duct system will be disapproved. Make the necessary corrections and prepare a revised Pre-final DALT Report. Reschedule a field check of the revised report data with the COTR.

3.3.8 Certified Final DALT Report

On successful completion of all field checks of the Pre-Final DALT Report data for all systems, the TAB Supervisor shall assemble, review, approve, sign and submit the Final DALT Report in compliance with Appendix B REPORTS - DALT and TAB to the Contracting Officer for approval.

3.3.9 Prerequisite for TAB Field Work

Do not commence TAB field work prior to the completion and approval, for all systems, of the Final DALT Report.

3.4 TAB PROCEDURES

3.4.1 TAB Field Work

Test, adjust, and balance the HVAC systems until measured flow rates (air) are within plus or minus 10 percent of the design flow rates as specified or indicated on the contract documents.

That is, comply with the the requirements of AABC MN-1 or SMACNA 1780 (TABB) and SMACNA 1858 (TABB), except as supplemented and modified by this section.

Provide instruments and consumables required to accomplish the TAB work. Calibrate and maintain instruments in accordance with manufacturer's written procedures.

Test, adjust, and balance the HVAC systems until measured flow rates (air) are within plus or minus 10 percent of the design flow rates as specified or indicated on the contract documents. Conduct TAB work, including measurement accuracy, and sound measurement work in conformance with the AABC MN-1 and AABC MN-4, or NEBB TABES and NEBB MASV, or SMACNA 1780 (used by TABB) and SMACNA 1858 sound measurement procedures, except as supplemented and modified by this section.

3.4.2 Preliminary Procedures

Use the approved pre-field engineering report as instructions and procedures for accomplishing TAB field work. TAB engineer is to locate, in the field, test ports required for testing. It is the responsibility of the sheet metal contractor to provide and install test ports as required by the TAB engineer.

3.4.3 TAB Air Distribution Systems

3.4.3.1 Units With Coils

Report heating and cooling performance capacity tests for hot water, chilled water, DX and steam coils for the purpose of verifying that the coils meet the indicated design capacity. Submit the following data and calculations with the coil test reports:

- a. For air handlers with capacities greater than 26,370 Watts cooling, such as factory manufactured units, central built-up units and rooftop units, conduct capacity tests in accordance with AABC MN-4, procedure 3.5, "Coil Capacity Testing."

Do not determine entering and leaving wet and dry bulb temperatures by single point measurement, but by the average of multiple readings in compliance with paragraph 3.5-5, "Procedures", (in subparagraph d.) of AABC MN-4, Procedure 3.5, "Coil Capacity Testing."

Submit part-load coil performance data from the coil manufacturer converting test conditions to design conditions; use the data for the purpose of verifying that the coils meet the indicated design capacity in compliance with AABC MN-4, Procedure 3.5, "Coil Capacity Testing," paragraph 3.5.7, "Actual Capacity Vs. Design Capacity" (in subparagraph c.).

- b. For units with capacities of 26370 Watts or less, such as fan coil units, duct mounted reheat coils associated with VAV terminal units, and unitary units, such as through-the-wall heat pumps:

Determine the apparent coil capacity by calculations using single point measurement of entering and leaving wet and dry bulb temperatures; submit the calculations with the coil reports.

3.4.3.2 Air Handling Units

Air handling unit systems including fans (air handling unit fans, exhaust fans and winter ventilation fans), coils, ducts, plenums, mixing boxes,

terminal units, variable air volume boxes, and air distribution devices for supply air, return air, outside air, mixed air relief air, and makeup air.

3.4.3.3 Heating and Ventilating Units

Heating and ventilating unit systems including fans, coils, ducts, plenums, roof vents, registers, diffusers, grilles, and louvers for supply air, return air, outside air, and mixed air.

3.4.3.4 Exhaust Fans

Exhaust fan systems including fans, ducts, plenums, grilles, and hoods for exhaust air.

3.4.4 TAB Work on Performance Tests Without Seasonal Limitations

3.4.4.1 Performance Tests

In addition to the TAB proportionate balancing work on the air distribution systems and the water distribution systems, accomplish TAB work on the HVAC systems which directly transfer thermal energy. TAB the operational performance of the heating systems and cooling systems.

3.4.4.2 Ambient Temperatures

On each tab report form used for recording data, record the outdoor and indoor ambient dry bulb temperature range and the outdoor and indoor ambient wet bulb temperature range within which the report form's data was recorded. Record these temperatures at beginning and at the end of data taking.

3.4.5 Workmanship

Conduct TAB work on the HVAC systems until measured flow rates are within plus or minus 10 percent of the design flow rates as specified or indicated on the contract documents. This TAB work includes adjustment of balancing valves, balancing dampers, and sheaves. Further, this TAB work includes changing out fan sheaves and pump impellers if required to obtain air flow rates specified or indicated. If, with these adjustments and equipment changes, the specified or indicated design flow rates cannot be attained, contact the Contracting Officer for direction.

3.4.6 Deficiencies

Strive to meet the intent of this section to maximize the performance of the equipment as designed and installed. However, if deficiencies in equipment design or installation prevent TAB work from being accomplished within the range of design values specified in the paragraph WORKMANSHIP, provide written notice as soon as possible to the Contractor and the Contracting Officer describing the deficiency and recommended correction.

Responsibility for correction of installation deficiencies is the Contractor's. If a deficiency is in equipment design, call the TAB team supervisor for technical assistance. Responsibility for reporting design deficiencies to Contractor is the TAB team supervisor's.

3.4.7 TAB Reports

Additional requirements for TAB Reports are specified in Appendix B
REPORTS - DALT and TAB

3.4.8 Quality Assurance - COTR TAB Field Acceptance Testing

3.4.8.1 TAB Field Acceptance Testing

During the field acceptance testing, verify, in the presence of the COTR, random selections of data (water, air quantities, air motion,) recorded in the TAB Report. Points and areas for field acceptance testing are to be selected by the COTR. Measurement and test procedures are the same as approved for TAB work for the TAB Report.

Field acceptance testing includes verification of TAB Report data recorded for the following equipment groups:

Group 1: All air handling units.

Group 2: 25 percent of the supply diffusers, registers, grilles associated with constant volume air handling units.

Group 3: 25 percent of the return grilles, return registers, exhaust grilles and exhaust registers.

Group 4: 25 percent of the supply fans, exhaust fans, and pumps.

Further, if any data on the TAB Report for Groups 2 through 5 is found not to fall within the range of plus 5 to minus 5 percent of the TAB Report data, additional group data verification is required in the presence of the COTR. Verify TAB Report data for one additional piece of equipment in that group. Continue this additional group data verification until out-of-tolerance data ceases to be found.

3.4.8.2 Additional COTR TAB Field Acceptance Testing

If any of the acceptance testing measurements for a given equipment group is found not to fall within the range of plus 10 to minus 10 percent of the TAB Report data, terminate data verification for all affected data for that group. The affected data for the given group will be disapproved. Make the necessary corrections and prepare a revised TAB Report. Reschedule acceptance testing of the revised report data with the COTR.

3.4.8.3 Prerequisite for Approval

Compliance with the field acceptance testing requirements of this section is a prerequisite for the final Contracting Officer approval of the TAB Report submitted.

3.5 MARKING OF SETTINGS

Upon the final TAB work approval, permanently mark the settings of HVAC adjustment devices including valves, gauges, splitters, and dampers so that adjustment can be restored if disturbed at any time. Provide permanent markings clearly indicating the settings on the adjustment devices which result in the data reported on the submitted TAB report.

3.6 MARKING OF TEST PORTS

The TAB team is to permanently and legibly mark and identify the location points of the duct test ports. If the ducts have exterior insulation, make these markings on the exterior side of the duct insulation. Show the location of test ports on the as-built mechanical drawings with dimensions given where the test port is covered by exterior insulation.

3.7 APPENDICES

Appendix A WORK DESCRIPTIONS OF PARTICIPANTS
Appendix B REPORTS - DALT and TAB
Appendix C DALT AND TAB SUBMITTAL AND WORK SCHEDULE
Appendix D REQUIREMENTS FOR DUCT AIR LEAK TESTING

Appendix A

WORK DESCRIPTIONS OF PARTICIPANTS

The Contractor is responsible for ensuring compliance with all requirements of this specification section. However, the following delineation of specific work items is provided to facilitate and co-ordinate execution of the various work efforts by personnel from separate organizations.

1. Contractor

- a. HVAC documentation: Provide pertinent contract documentation to the TAB Firm, to include the following: the contract drawings and specifications; copies of the approved submittal data for all HVAC equipment, air distribution devices, and air/water measuring/balancing devices; the construction work schedule; and other applicable documents requested by the TAB Firm. Provide the TAB Firm copies of contract revisions and modifications as they occur.
- b. Schedules: Ensure the requirements specified under the paragraph "DALT and TAB Schedule" are met.
- c. Pre-DALT and TAB meeting: Arrange and conduct the Pre-DALT and TAB meeting. Ensure that a representative is present for the sheet metal contractor, the mechanical contractor, the electrical contractor, and the automatic temperature controls contractor.
- d. Coordinate Support: Provide and coordinate support personnel required by the TAB Firm in order to accomplish the DALT and TAB field work. Support personnel may include factory representatives, HVAC controls installers, HVAC equipment mechanics, sheet metal workers, pipe fitters, and insulators. Ensure support personnel are present at the work site at the times required.
- e. Correct Deficiencies: Ensure the notifications of Construction Deficiencies are provided as specified herein. Refer to the paragraph CONSTRUCTION DEFICIENCIES. Correct each deficiency as soon as practical with the Contracting Officer, and submit revised schedules and other required documentation.
- f. Pre-TAB Work Checklists: Complete check out and debugging of HVAC equipment, ducts, and controls prior to the TAB engineer arriving at the project site to begin the TAB work. Debugging includes searching for and eliminating malfunctioning elements in the HVAC system installations, and verifying all adjustable devices are functioning as designed. Include as pre-TAB work checklist items, the deficiencies pointed out by the TAB team supervisor in the design review report.

Prior to the TAB field team's arrival, ensure completion of the applicable inspections and work items listed in the TAB team supervisor's DALT and TAB Work Procedures Summary. Do not allow the TAB team to commence TAB field work until all of the following are completed.

- g. Give Notice of Testing: Submit advance notice of proportional balancing TAB field work accompanied by completed prerequisite HVAC Work List

- h. Insulation work: Ensure that no insulation is shall not be installed on ducts to be DALT'd until DALT work on the subject ducts is complete.

Ensure the duct and piping systems are properly insulated and vapor sealed upon the successful completion and acceptance of the DALT and TAB work.

2. TAB Team Supervisor

- a. Overall management: Supervise and manage the overall TAB team work effort, including preliminary and technical DALT and TAB procedures and TAB team field work.
- b. Schedule: Ensure the requirements specified under the paragraph "DALT and TAB Schedule" are met.
- c. Submittals: Provide the submittals specified herein.
- d. Pre-DALT/TAB meeting: Attend meeting with Contractor. Ensure TAB personnel that will be involved in the TAB work under this contract attend the meeting.
- e. Design Review Report: Submit typed report describing omissions and deficiencies in the HVAC system's design that would preclude the TAB team from accomplishing the duct leakage testing work and the TAB work requirements of this section. Provide a complete explanation including supporting documentation detailing the design deficiency. State that no deficiencies are evident if that is the case.
- f. Support required: Specify the technical support personnel required from the Contractor other than the TAB agency; such as factory representatives for temperature controls or for complex equipment. Inform the Contractor in writing of the support personnel needed and when they are needed. Furnish the notice as soon as the need is anticipated, either with the design review report, or the DALT and TAB Procedures Summary, the during the DALT or TAB field work.

Ensure the Contractor is properly notified and aware of all support personnel needed to perform the TAB work. Maintain communication with the Contractor regarding support personnel throughout the duration of the TAB field work, including the TAB field acceptance testing checking.

Ensure all inspections and verifications for the Pre-Final DALT and Pre-TAB Checklists are completely and successfully conducted before DALT and TAB field work is performed.

- g. Advance Notice: Monitor the completion of the duct system installations and provide the Advance Notice for Pre-Final DALT field work as specified herein.
- h. Technical Assistance: Provide technical assistance to the DALT and TAB field work.
- i. Deficiencies Notification: Ensure the notifications of Construction Deficiencies are provided as specified herein. Comply with requirements of the paragraph CONSTRUCTION DEFICIENCIES. Resolve each deficiency as soon as practical and submit revised schedules and other

required documentation.

- j. Procedures: Develop the required TAB procedures for systems or system components not covered in the TAB Standard.

3. TAB Team Field Leader

- a. Field manager: Manage, in the field, the accomplishment of the work specified in Part 3, EXECUTION.
- b. Full time: Be present at the contract site when DALT field work or TAB field work is being performed by the TAB team; ensure day-to-day TAB team work accomplishments are in compliance with this section.
- c. Prerequisite HVAC work: Do not bring the TAB team to the contract site until a copy of the prerequisite HVAC work list, with all work items certified by the Contractor to be working as designed, reaches the office of the TAB Agency.

Appendix B

REPORTS - DALT and TAB

All submitted documentation must be typed, neat, and organized. All reports must have a waterproof front and back cover, a title page, a certification page, sequentially numbered pages throughout, and a table of contents. Tables, lists, and diagrams must be titled. Generate and submit for approval the following documentation:

1. DALT and TAB Work Execution Schedule

Submit a detailed schedule indicating the anticipated calendar date for each submittal and each portion of work required under this section. For each work entry, indicate the support personnel (such as controls provider, HVAC mechanic, etc.) that are needed to accomplish the work. Arrange schedule entries chronologically.

2. DALT and TAB Procedures Summary

Submit a detailed narrative describing all aspects of the DALT and TAB field work to be performed. Clearly distinguish between DALT information and TAB information. Include the following:

- a. A list of the intended procedural steps for the DALT and TAB field work from start to finish. Indicate how each type of data measurement will be obtained. Include what Contractor support personnel are required for each step, and the tasks they need to perform.
- b. A list of the project's submittals that are needed by the TAB Firm in order to meet this Contract's requirements.
- c. The schematic drawings to be used in the required reports, which may include building floor plans, mechanical room plans, duct system plans, and equipment elevations. Indicate intended TAB measurement locations, including where test ports need to be provided by the Contractor.
- d. The data presentation forms to be used in the report, with the preliminary information and initial design values filled in.
- e. A list of DALT and TAB instruments to be used, edited for this project, to include the instrument name and description, manufacturer, model number, scale range, published accuracy, most recent calibration date, and what the instrument will be used for on this project.
- f. A thorough checklist of the work items and inspections that need to be accomplished before DALT field work can be performed. The Contractor must complete, submit, and receive approval of the Completed Pre-Final DALT Work Checklist before DALT field work can be accomplished.
- g. A thorough checklist of the work items and inspections that need to be accomplished before the TAB field work can be performed. The Contractor must complete, submit, and receive approval of the Completed Pre-TAB Work Checklist before the TAB field work can be accomplished.
- i. The checklists specified above shall be individually developed and

tailored specifically for the work under this contract. Refer to **NEBB PROCEDURAL STANDARDS**, Section III, "Preliminary TAB Procedures" under the paragraphs titled, "Air Distribution System Inspection" for examples of items to include in the checklists.

3. Design Review Report

Submit report containing the following information:

- a. Review the contract specifications and drawings to verify that the TAB work can be successfully accomplished in compliance with the requirements of this section. Verify the presence and location of permanently installed test ports and other devices needed, including gauge cocks, thermometer wells, flow control devices, circuit setters, balancing valves, and manual volume dampers.
- b. Submit a typed report describing omissions and deficiencies in the HVAC system's design that would preclude the TAB team from accomplishing the DALT work and the TAB work requirements of this section. Provide a complete explanation including supporting documentation detailing the design deficiency. If no deficiencies are evident, state so in the report.

4. Completed Pre-Final DALT Work Checklist

Report the data for the Pre-Final DALT Report meeting the following requirements:

- a. Submit a copy of the approved DALT and TAB Procedures Summary: Provide notations describing how actual field procedures differed from the procedures listed.
- b. Report format: Submit a comprehensive report for the DALT field work data using data presentation forms equivalent to the "Air Duct Leakage Test Summary Report Forms" located in the **SMACNA 1972 CD**. In addition, submit in the report, a marked duct shop drawing which identifies each section of duct tested with assigned node numbers for each section. Node numbers shall be included in the completed report forms to identify each duct section.
- c. Calculations: Include a copy of all calculations prepared in determining the duct surface area of each duct test section. Include in the DALT reports copy(s) of the calibration curve for each of the DALT test orifices used for testing.
- d. Instruments: List the types of instruments actually used to measure the data. Include in the listing each instrument's unique identification number, calibration date, and calibration expiration date. Instruments are to be calibrated within one year of the date of use in the field; instrument calibration is to be traceable to the measuring standards of the National Institute of Standards and Technology.
- e. TAB Supervisor Approval: Include on the submitted report the typed name of the TAB supervisor and the dated signature of the TAB supervisor.

5. Final DALT Report

On successful completion of all COTR field checks of the Pre-final DALT Report data for all systems, the TABS Supervisor shall assemble, review, sign and submit the Final DALT Report to the Contracting Officer for approval.

6. TAB Reports: Submit TAB Report for Proportional Balancing, Season 1, and Season 2 in the following manner:

- a. Procedure Summary: Submit a copy of the approved DALT and TAB Procedures Summary. When applicable, provide notations describing how actual field procedures differed from the procedures listed.
- b. Report format: Submit the completed data forms approved in the pre-field TAB Engineering Report completed by TAB field team, reviewed, approved and signed by the TAB supervisor. Bind the report with a waterproof front and back cover. Include a table of contents identifying by page number the location of each report. Report forms and report data shall be typewritten. Handwritten report forms or report data are not acceptable.
- c. Temperatures: On each TAB report form reporting TAB work accomplished on HVAC thermal energy transfer equipment, include the indoor and outdoor dry bulb temperature range and indoor and outdoor wet bulb temperature range within which the TAB data was recorded.
 - (1) Data shall be measured and compiled on a continuous basis for the period in which TAB work affecting those rooms is being done.
 - (2) Data shall be measured/recorded only after the HVAC systems installations are complete, the systems fully balanced and the HVAC systems controls operating in fully automatic mode. Provide a detailed explanation wherever a final measurement did not achieve the required value.
 - (3) Data may be compiled using direct digital controls trend logging where available. Otherwise, the Contractor shall temporarily install calibrated time versus temperature/humidity recorders for this purpose. The HVAC systems and controls shall have been fully operational a minimum of 24 hours in advance of commencing data compilation.
- d. Air System Diagrams: Provided updated diagrams with final installed locations of all terminals and devices, any numbering changes, and actual test locations.
- e. Air Static Pressure Profiles: Report static pressure profiles for air duct systems. Report static pressure data for all supply, return, relief, exhaust and outside air ducts for the systems listed. The static pressure report data shall include, in addition to AABC or NEBB or TABB required data, the following:
 - (1) Report supply fan, return fan, relief fan, and exhaust fan inlet and discharge static pressures.
 - (2) Report static pressure drop across chilled water coils, DX coils, hot water coils, steam coils, electric resistance heating coils and heat reclaim devices installed in unit cabinetry or the system ductwork.

- (3) Report static pressure drop across outside air, return air, and supply air automatic control dampers, both proportional and two-position, installed in unit cabinetry.
- (4) Report static pressure drop across air filters, acoustic silencers, moisture eliminators, air flow straighteners, air flow measuring stations or other pressure drop producing specialty items installed in unit cabinetry, or in the system ductwork. Examples of these specialty items are smoke detectors, white sound generators, RF shielding, wave guides, security bars, blast valves, small pipes passing through ductwork, and duct mounted humidifiers.

Do not report static pressure drop across duct fittings provided for the sole purpose of conveying air, such as elbows, transitions, offsets, plenums, manual dampers, and branch takes-offs.

- (5) Report static pressure drop across outside air and relief/exhaust air louvers.
- (6) Report static pressure readings of supply air, return air, exhaust/relief air, and outside air in duct at the point where these ducts connect to each air moving unit.

- g. Instruments: List the types of instruments actually used to measure the tab data. Include in the listing each instrument's unique identification number, calibration date, and calibration expiration date.

Instrumentation, used for taking wet bulb temperature readings shall provide accuracy of plus or minus 5 percent at the measured face velocities. Submit instrument manufacturer's literature to document instrument accuracy performance is in compliance with that specified.

- h. Performance Curves: The TAB Supervisor shall include, in the TAB Reports, factory fan curves for fans TAB'd on the job.
- i. Data From TAB Field Work: After completion of the TAB field work, prepare the TAB field data for TAB supervisor's review and approval signature, using the reporting forms approved in the pre-field engineering report. Data required by those approved data report forms shall be furnished by the TAB team. Except as approved otherwise in writing by the Contracting Officer, the TAB work and thereby the TAB report shall be considered incomplete until the TAB work is accomplished to within the accuracy range specified in the paragraph WORKMANSHIP.

Appendix C

DALT AND TAB SUBMITTAL AND WORK SCHEDULE

Perform the following items of work in the order listed adhering to the dates schedule specified below.

Submit TAB Agency and TAB Personnel Qualifications: Within 42 calendar days after date of contract award.

Submit the DALT and TAB Work Execution Schedule: within 14 days after receipt of the TAB agency and TAB personnel qualifications approval. Revise and re-submit this schedule 28 days prior to commencement of DALT work and 28 days prior to the commencement of TAB Season 1 work and TAB Season 2 work.

Submit the DALT and TAB Work Procedures Summary: within 14 days after receipt of the initial approved DALT and TAB Work Execution Schedule.

Meet with the COTR at the Pre-DALT/TAB Meeting: Within 28 calendar days after receipt of the approved initial DALT/TAB Execution Schedule.

Submit Design Review Report: Within 56 calendar days after the receipt of the approved initial DALT and TAB Work Execution Schedule.

Advance Notice of Pre-Final DALT Field Work: After the completed installation of the HVAC duct system to be DALT'd, submit to the Contracting Officer an Advance Notice of Pre-Final DALT Field Work accompanied by the completed Pre-Final DALT Work Checklist for the subject duct system.

Ductwork Selected for DALT: Within 14 calendar days after receiving an acceptable completed Pre-Final DALT Work Checklist, the Contracting Officer's technical representative (COTR) will select the project ductwork sections to be DALT'd.

DALT Field Work: Within 48 hours of COTR's selection, complete DALT field work on selected project ductwork.

Submit Pre-Final DALT Report: Within two working days after completion of DALT field work, submit Pre-final DALT Report. Separate Pre-final DALT reports may be submitted to allow phased testing from system to system.

Quality Assurance - COTR DALT Field Checks: Upon approval of the Pre-final DALT Report, the COTR's DALT field check work shall be scheduled with the Contracting Officer.

Submit Final DALT Report: Within 14 calendar days after completion of successful DALT Work Field Check, submit TAB report.

Advance Notice of TAB Field Work: At a minimum of 14 calendar days prior to TAB Field Work, submit advance notice of TAB field work accompanied by completed Pre-TAB Work Checklist.

TAB Field Work: At a minimum of 84 calendar days prior to CCD, accomplish TAB field work.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Submit TAB Report: Within 14 calendar days after completion of TAB field work, submit initial TAB report.

Quality Assurance - COTR TAB Field Check: 30 calendar days after initial TAB report is approved by the Contracting Officer, conduct field check.

Complete TAB Work: Prior to CCD, complete all TAB work and submit final.

Receive the approved TAB report: Within 21 calendar days, receive the report from Contracting Officer approved TAB report.

Receive the approved TAB report: Within calendar 21 days, receive the report from Contracting Officer.

Appendix D					
REQUIREMENTS FOR DUCT AIR LEAK TESTING					
		SYSTEMS			
					Exhaust Systems Unit No. EF-1, EF-2, EF-3, EF-4
Duct System Static Pressure, in millimeters W.C.	for Supply				n/a
	for Return				n/a
	for Exhaust			n/a	25
	for Outside Air			n/a	n/a
System Oval/Round Duct and Rectangular Duct SMACNA Seal Class	for Supply	A	A	A	A
	for Return	A	A	A	A
	for Exhaust	A	A	A	A
	for Outside Air	A	A	A	A

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Appendix D					
REQUIREMENTS FOR DUCT AIR LEAK TESTING					
		SYSTEMS			
					Exhaust Systems Unit No. EF-1, EF-2, EF-3, EF-4
System Oval/Round Duct SMACNA Leak Class	for Supply			12	n/a
	for Return			12	n/a
	for Exhaust			n/a	12
	for Outside Air			n/a	n/a
System Rectangular Duct SMACNA Leak Class	for Supply			24	n/a
	for Return			24	n/a
	for Exhaust			n/a	24
	for Outside Air			n/a	n/a
Duct Test Pressure, in millimeters W.C.	for Supply				n/a
	for Return				n/a
	for Exhaust			n/a	25
	for Outside Air			n/a	n/a

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 31 13.00 40

METAL DUCTS

05/16

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY CONTROL

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
- 2.2 COMPONENTS
 - 2.2.1 Reinforcement
 - 2.2.2 Fittings
 - 2.2.3 Turning Vanes
 - 2.2.4 Dampers
 - 2.2.5 Flexible Connectors for Sheet Metal
 - 2.2.6 Duct Hangers
 - 2.2.7 Mill-Rolled Reinforcing and Supporting Materials
 - 2.2.8 Flexible Duct Materials
 - 2.2.9 Manual Volume Dampers
 - 2.2.10 Power Operated Dampers
- 2.3 MATERIALS
 - 2.3.1 Galvanized Steel Ductwork Materials
 - 2.3.2 Brazing Materials
 - 2.3.3 Mill-Rolled Reinforcing and Supporting Materials

PART 3 EXECUTION

- 3.1 PREPARATION
 - 3.1.1 Construction Standards
- 3.2 INSTALLATION
 - 3.2.1 Jointing
 - 3.2.2 Ducts
 - 3.2.2.1 Ductwork Cleaning Provisions
- 3.3 APPLICATION
 - 3.3.1 Low Pressure Sheet Metal Ducts
 - 3.3.1.1 Longitudinal Duct Seams
 - 3.3.1.2 Joints and Gaskets
 - 3.3.1.3 Flexible Duct Joints
 - 3.3.1.4 Square Elbows
 - 3.3.1.5 Radius Elbows
 - 3.3.1.6 Outlets, Inlets, and Duct Branches
 - 3.3.1.7 Duct Transitions
 - 3.3.1.8 Branch Connections
 - 3.3.1.9 Manual Volume Dampers
 - 3.3.1.10 Flexible Connectors for Sheet Metal
 - 3.3.2 Rectangular Sheet Metal Ducts

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.3.2.1 Medium-Pressure Gages, Joints, and Reinforcement
- 3.3.2.2 Medium- And High-Pressure Branches, Inlets, Outlets
- 3.3.2.3 Duct Branch Transition
- 3.3.2.4 High-Pressure Gages, Joints, and Reinforcement
- 3.3.3 Joint Gaskets
- 3.3.4 Radius Elbows
- 3.3.5 Duct Supports
 - 3.3.5.1 Hangars
 - 3.3.5.2 Installation
 - 3.3.5.3 Strap-type Hangars
 - 3.3.5.4 Trapeze Hangars
 - 3.3.5.5 Purlins
- 3.3.6 Flexible Connectors for Steel Metal
- 3.3.7 Insulation Protection Angles
- 3.3.8 Duct Probe Access
- 3.3.9 Openings In Roofs and Walls
- 3.4 FIELD QUALITY CONTROL
 - 3.4.1 Ductwork Leakage Tests
 - 3.4.2 Inspection
- 3.5 CLOSEOUT ACTIVITIES
 - 3.5.1 Operation and Maintenance
 - 3.5.2 Record Drawings

-- End of Section Table of Contents --

SECTION 23 31 13.00 40

METAL DUCTS
05/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325 (2017) Steel Construction Manual

AISC 360 (2016) Specification for Structural Steel Buildings

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE EQUIP SI HDBK (2012) Handbook, HVAC Systems and Equipment (SI Edition)

ASHRAE FUN SI (2021) Fundamentals Handbook, SI Edition

AMERICAN WELDING SOCIETY (AWS)

AWS A5.8/A5.8M (2019) Specification for Filler Metals for Brazing and Braze Welding

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A123/A123M (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A924/A924M (2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A (2021) Standard for the Installation of Air Conditioning and Ventilating Systems

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION
(SMACNA)

- SMACNA 1966 (2020) HVAC Duct Construction Standards
Metal and Flexible, 4th Edition
- SMACNA 1987 (2006) HVAC Duct Systems Inspection Guide,
3rd Edition

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

- SAE AMS 2480 (2009; Rev H) Phosphate Treatment, Paint,
Base

UNDERWRITERS LABORATORIES (UL)

- UL 181 (2013; Reprint Dec 2021) UL Standard for
Safety Factory-Made Air Ducts and Air
Connectors

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Material, Equipment, and Fixture Lists; G

SD-03 Product Data

Galvanized Steel Ductwork Materials; G

Brazing Materials

Mill-Rolled Reinforcing and Supporting Materials

Turning Vanes; G

Flexible Connectors; G

Flexible Duct Materials

Power Operated Dampers; G

Manual Volume Dampers; G

SD-06 Test Reports

Ductwork Leakage Tests; G

SD-07 Certificates Listing of Product Installations

Brazing Materials

Mill-Rolled Reinforcing and Supporting Materials Dampers

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

Power Operated Dampers; G

SD-11 Closeout Submittals

Record Drawings; G

1.3 QUALITY CONTROL

When furnishing the [listing of product installations](#) for medium and high pressure ductwork systems include identification of at least 5 units, similar to those proposed for use, that have been in successful service for a minimum period of 5 years. Include purchaser, address of installation, service organization, and date of installation.

PART 2 PRODUCTS

Include the manufacturer's style or catalog numbers, specification and drawing reference numbers, warranty information, and fabrication site information within [material, equipment, and fixture lists](#).

2.1 SYSTEM DESCRIPTION

Provide low-pressure systems ductwork and plenums where maximum air velocity is [10.1 meter per second](#) and maximum static pressure is [500 pascal](#) water gage (wg), positive or negative.

2.2 COMPONENTS

2.2.1 Reinforcement

Support inner liners of both duct and fittings by metal spacers welded in position to maintain spacing and concentricity.

2.2.2 Fittings

Make divided flow fittings as separate fittings, not tap collars into duct sections, with the following construction requirements:

- a. Sound, airtight, continuous welds at intersection of fitting body and tap
- b. Tap liner securely welded to inner liner, with weld spacing not to exceed [75 millimeter](#).
- c. Pack insulation around the branch tap area for complete cavity filling.
- d. Carefully fit branch connection to cutout openings in inner liner without spaces for air erosion of insulation and without sharp projections that cause noise and airflow disturbance.

Continuously braze seams in the pressure shell of fittings. Protect galvanized areas that have been damaged by welding with manufacturer's standard corrosion-resistant coating.

Construct two-piece type elbows for angles through 35 degrees, three-piece

type for angles 36 through 71 degrees, and five-piece type for angles 72 through 90 degrees.

2.2.3 Turning Vanes

Provide double-wall type turning vanes, commercially manufactured for high-velocity system service.

2.2.4 Dampers

Construct low pressure drop, high-velocity manual volume dampers, and high-velocity fire dampers in accordance with [ASHRAE EQUIP SI HDBK](#), Chapter 16, [ASHRAE FUN SI](#), Chapter 32 and [SMACNA 1966](#).

2.2.5 Flexible Connectors for Sheet Metal

Use UL listed connectors, [915 gram per square meter](#), waterproof, fire-retardant, airtight, woven fibrous-glass cloth, double coated with chloroprene. Clear width, not including clamping section, is [150 to 200 millimeter](#).

2.2.6 Duct Hangers

For duct hangers in contact with galvanized duct surfaces, provide galvanized steel painted with inorganic zinc.

2.2.7 Mill-Rolled Reinforcing and Supporting Materials

Provide mill-rolled structural steel conforming to [ASTM A36/A36M](#). Whenever in contact with sheet metal ducting, provide galvanized steel in accordance with [ASTM A123/A123M](#).

In lieu of mill-rolled structural steel, submit equivalent strength, proprietary-design, rolled-steel structural support systems for approval.

2.2.8 Flexible Duct Materials

Ensure flexible duct connectors comply with [NFPA 90A](#), and conform with [UL 181](#), Class 1 material.

2.2.9 Manual Volume Dampers

Conform to [SMACNA 1966](#) for volume damper construction.

Equip dampers with an indicating quadrant regulator with a locking feature externally located and easily accessible for adjustment and standoff brackets to allow mounting outside external insulation.

2.2.10 Power Operated Dampers

Where indicated provide Class I, Division 1, Group D rated explosion proof motor actuated damper control with bronze bearings. Actuator shall have field adjustable travel. Actuator shall be equipped with a spring return mechanism which will return the damper to its normal operating position on loss of power. Power source will be 120 single phase; provide transformer as required.

2.3 MATERIALS

2.3.1 Galvanized Steel Ductwork Materials

Provide hot-dip galvanized carbon steel ductwork sheet metal of lock-forming quality, with regular spangle-type zinc coating, conforming to [ASTM A924/A924M](#) and [ASTM A653/A653M](#), Designation G90. Treat duct surfaces to be painted by annealing.

Conform to [ASHRAE EQUIP SI HDBK](#), Chapter 16, [ASHRAE FUN SI](#), Chapter 32 and [SMACNA 1966](#) for sheet metal thickness and reinforcement thickness.

Low pressure ductwork minimum thicknesses are:

MINIMUM SHEET METAL THICKNESS	
<u>DUCT WIDTH MILLIMETER</u>	<u>THICKNESS MILLIMETER</u>
0-305	0.45
330-762	0.61
787-1524	0.76

2.3.2 Brazing Materials

Provide silicon bronze brazing materials conforming to [AWS A5.8/A5.8M](#).

2.3.3 Mill-Rolled Reinforcing and Supporting Materials

Conform to [ASTM A36/A36M](#) for mill-rolled structural steel. Wherever in contact with sheet metal ducting, galvanize to conforming with [ASTM A123/A123M](#).

In lieu of mill-rolled structural steel, submit for approval, equivalent strength, proprietary design, rolled-steel structural support systems.

PART 3 EXECUTION

3.1 PREPARATION

For sheet metal surfaces to be painted, and surfaces to which adhesives are to be applied, clean surface of oil, grease, and deleterious substances.

Ensure strength is adequate to prevent failure under service pressure or vacuum created by fast closure of duct devices. Provide leaktight, automatic relief devices.

3.1.1 Construction Standards

Provide sheet metal construction in accordance with the recommendations for best practices in [ASHRAE EQUIP SI HDBK](#), Chapter 16, [ASHRAE FUN SI](#), Chapter 32, [SMACNA 1966](#), and [NFPA 90A](#).

Design and fabricate supplementary steel in accordance with [AISC 360](#) and

AISC 325.

Where construction methods for certain items are not described in the referenced standards or herein, perform the work in accordance with recommendations for best practice defined in ASHRAE EQUIP SI HDBK.

3.2 INSTALLATION

Fabricate an airtight system. Include reinforcements, bracing, supports, framing, gasketing, sealing, and fastening to provide rigid construction and freedom from vibration, airflow-induced motion and noise, and excessive deflection at specified maximum system air pressure and velocity.

Provide offsets and transformations as required to avoid interference with the building construction, piping, or equipment.

Make plenum anchorage provisions, sheet metal joints, and other areas airtight and watertight by caulking, mating galvanized steel and concrete surfaces with a two-component elastomer.

3.2.1 Jointing

Enclose dampers located behind architectural intake or exhaust louvers by a rigid sheet metal collar and sealed to building construction with elastomers for complete air tightness.

Provide outside air-intake ducts and plenums made from sheet metal with soldered watertight joints.

3.2.2 Ducts

Wherever ducts pass through firewalls or through walls or floors dividing conditioned spaces from unconditioned spaces, provide a flanged segment in that surface during surface construction.

Where interiors of ducting may be viewed through air diffusion devices, construct the viewed interior with sheet metal and paint flat black.

3.2.2.1 Ductwork Cleaning Provisions

Protect open ducting from construction dust and debris in a manner approved by the Contracting Officer. Clean dirty assembled ducting by subjecting all main and branch interior surfaces to airstreams moving at velocities two times specified working velocities, at static pressures within maximum ratings. This may be accomplished by: filter-equipped portable blowers which remain the Contractor's property; wheel-mounted, compressed-air operated perimeter lances which direct the compressed air and which are pulled in the direction of normal airflow; or other means approved by the Contracting Officer. Use water- and oil- free compressed air for cleaning ducting. After construction is complete, and prior to acceptance of the work, remove construction dust and debris from exterior surfaces. Clean in conformance with SMACNA 1987.

3.3 APPLICATION

3.3.1 Low Pressure Sheet Metal Ducts

Weld angle iron frames at corners and ends, whenever possible. Rivet or weld angle iron reinforcements to ducts not more than 150 millimeters on

center, with not less than two points of attachment. Spot welding, where used, is 75 millimeters on center.

Seal standard seam joints with an elastomer compound to comply with SMACNA 1966 Seal Class A, B or C as applicable.

Limit crossbreaking to 1220 millimeters and provide on all ducts 200 millimeters wide and wider. Provide bead reinforcement in lieu of crossbreaking where panel popping may occur. Where rigid insulation is applied, crossbreaking is not required.

3.3.1.1 Longitudinal Duct Seams

Provide Pittsburgh lock corner seams.

3.3.1.2 Joints and Gaskets

Bolt companion angle flanges together with 6 millimeters diameter bolts and nuts spaced 150 millimeters on center. Gasket flanged joints with chloroprene full-face gaskets 3 millimeters thick, with Shore A 40 durometer hardness. Use one piece gaskets at joints.

3.3.1.3 Flexible Duct Joints

Between flexible duct without sheet metal collars and round metal ductwork connections make joints by trimming the ends, coating the inside of the flexible duct for a distance equal to depth of insertion with elastomer caulk, and by securing with sheet metal screws or binding with a strap clamp.

3.3.1.4 Square Elbows

Provide single-vane duct turns in accordance with SMACNA 1966, use on ducts 300 millimeters in width and narrower.

3.3.1.5 Radius Elbows

Conform to SMACNA 1966 for radius elbows. Provide an inside radius equal to the width of the duct. Where installation conditions preclude use of standard elbows, the inside radius may be reduced to a minimum of 0.25 times duct width. Install turning vanes in accordance with the following schedule.

	RADIUS OF TURNING VANES IN PERCENT OF DUCT WIDTH		
WIDTH OF ELBOWS MILLIMETER	<u>VANE NO. 1</u>	<u>VANE NO. 2</u>	<u>VANE NO. 3</u>
Up to 406	56	--	--
430 to 1220	43	73	--
1245 and over	37	55	83

Where two elbows are placed together in the same plane for ducts 760 millimeters wide and larger, continue the guide vanes through both elbows rather than spaced in accordance with above schedule.

3.3.1.6 Outlets, Inlets, and Duct Branches

Install branches, inlets, and outlets so that air turbulence is reduced to a minimum and air volume properly apportioned. Install adjustable splitter dampers at all supply junctions to permit adjustment of the amount of air entering the branch. Wherever an air-diffusion device is shown as being installed on the side, top, or bottom of a duct, and whenever a branch take-off is not of the splitter type; provide a commercially manufactured 45 degree side-take-off (STO) fitting with manual volume damper to allow adjustment of the air quantity and to provide an even flow of air across the device or duct it services.

Where a duct branch is to handle more than 25 percent of the air handled by the duct main, use a complete 90-degree increasing elbow with an inside radius of 0.75 times branch duct width. Size of the leading end of the increasing elbow within the main duct with the same ratio to the main duct size as the ratio of the related air quantities handled.

Where a duct branch is to handle 25 percent or less of the air handled by the duct main, construct the branch connection with a 45 degree side take-off entry in accordance with [SMACNA 1966](#).

3.3.1.7 Duct Transitions

Where the shape of a duct changes, ensure the angle of the side of the transition piece does not exceed 15 degrees from the straight run of duct connected thereto.

Where equipment is installed in ductwork, ensure the angle of the side of the transition piece from the straight run of duct connected thereto does not exceed 15 degrees on the upstream side of the equipment and 22-1/2 degrees on the downstream side of the equipment.

3.3.1.8 Branch Connections

Construct radius tap-ins in accordance with [SMACNA 1966](#).

3.3.1.9 Manual Volume Dampers

Provide balancing dampers of the splitter, butterfly, or multilouver type, to balance each respective main and branch duct.

For dampers regulated through ceilings provide a regulator concealed in a box mounted in the ceiling, with a cover finish aesthetically compatible with ceiling surface. Where ceiling is of removable construction, set regulators above the ceiling, and mark the location on ceiling in a manner acceptable to the Contracting Officer.

3.3.1.10 Flexible Connectors for Sheet Metal

Connect air handling equipment, ducts crossing building expansion joints, and fan inlets and outlets to upstream and downstream components by treated woven-cloth connectors.

Install connectors only after system fans are operative, and vibration isolation mountings have been adjusted. When system fans are operating, ensure connectors are free of wrinkles caused by misalignment or fan reaction. Width of surface is curvilinear.

3.3.2 Rectangular Sheet Metal Ducts

3.3.2.1 Medium-Pressure Gages, Joints, and Reinforcement

Ensure minimum sheet metal gages, joints, and reinforcements between joints are in accordance with [ASHRAE EQUIP SI HDBK](#), Chapter 16, [ASHRAE FUN SI](#), Chapter 32 and [SMACNA 1966](#).

Ensure sheet metal minimum thickness, transverse reinforcement between joints, and joints of ducts are in accordance with the following:

LONGEST SIDE (mm)	SHEET METAL THICKNESS ALL SIDES	COMPANION ANGLE (mm)	REINFORCEMENT ANGLES 600 (mm) ON CENTER MAXIMUM (BACK TO BACK)
2450 to 2750	1.6	50 by 50 by 3, two tie rods along angle	Two 50 by 50 by 3, two tie rods along angle
2451 to 3350	1.6	50 by 50 by 5, two tie rods along angle	Two 50 by 50 by 5, two tie rods along angle
3351 and longer	2.0	50 by 50 by 5, with tie rods every 1200 mm	Two 50 by 50 by 5, with tie rods every 1200 mm

3.3.2.2 Medium- And High-Pressure Branches, Inlets, Outlets

Install branches, inlets, and outlets to minimize air turbulence and to ensure proper airflow.

Install dampers so that the amount of air entering duct mains is adjustable.

Provide commercially manufactured air extractors to allow adjustment of the air quantity and to provide an even flow of air across the device or duct served.

3.3.2.3 Duct Branch Transition

Where a duct branch handles over 25 percent of the air transported by the duct main, use a complete 90-degree increasing elbow, with an inside radius of 0.75 times duct branch width. Ensure the size of the trailing end of the increasing elbow within the main duct has the same ratio to the main duct size as the ratio of the relative air quantities handled.

Where a duct branch is to handle 25 percent or less of the air handled by the duct main, provide a branch connection with an inside radius of 0.75 times branch duct width, a minimum arc length of 45 degrees, and an outside radius of 1.75 times duct branch width. Place arc tangent to duct main.

3.3.2.4 High-Pressure Gages, Joints, and Reinforcement

Ensure sheet metal minimum thickness, joints, and reinforcement between joints are in accordance with [ASHRAE EQUIP SI HDBK](#), Chapter 16, [ASHRAE FUN SI](#), Chapter 32 and [SMACNA 1966](#).

Use the following types of **ASHRAE EQUIP SI HDBK**, Chapter 16, **ASHRAE FUN SI**, Chapter 32 and **SMACNA 1966** joints and seams:

Transverse Joints:

- a. Welded flange joint with angle
- b. Companion angle flanged joint

Longitudinal Seams:

- a. Approved lock seams, back brazed, or continuously brazed seams for ducts with largest dimension up to **1800 millimeters**
- b. Continuously welded or brazed seams for ducts with largest dimension greater than **1800 millimeters**

Sheet metal minimum thickness, transverse reinforcement between joints, and companion angle joints of ducts with longest side greater than **2550 millimeters** are in accordance with the following:

LONGEST SIDE (mm)	SHEET METAL THICKNESS ALL SIDES	COMPANION ANGLE (mm)	REINFORCEMENT ANGLES 600 (mm) ON CENTER MAXIMUM (BACK TO BACK)
2450 to 2750	1.6	50 by 50 by 3, two tie rods along angle	Two 50 by 50 by 3, two tie rods along angle
2451 to 3350	1.6	50 by 50 by 5, two tie rods along angle	Two 50 by 50 by 5, two tie rods along angle
3351 and longer	2.0	65 by 65 by 5, with tie rods every 600 mm	Two 65 by 65 by 5, with tie rods every 600 mm

3.3.3 Joint Gaskets

For flanged joints, use chloroprene full-face gaskets **3.2 millimeters** thick, with Shore A 40 durometer hardness. Use one-piece gaskets, vulcanized at joints.

3.3.4 Radius Elbows

Fabricate elbow proportions and radius elbows in accordance with **ASHRAE EQUIP SI HDBK**, Chapter 16, **ASHRAE FUN SI**, Chapter 32 and **SMACNA 1966**.

3.3.5 Duct Supports

Install duct support in accordance with **ASHRAE EQUIP SI HDBK**, Chapter 16, **ASHRAE FUN SI**, Chapter 32 and **SMACNA 1966**. Meet the minimum size for duct hangers as specified in **ASHRAE EQUIP SI HDBK**, Chapter 16, **ASHRAE FUN SI**, Chapter 32 and **SMACNA 1966**. Provide two hangers where necessary to eliminate sway. Support attachment to duct surfaces by bolt or welding

100 millimeters on center.

Take the following into account in selection of a hanging system:

- a. Location and precedence of work under other sections
- b. Interferences of various piping and electrical conduit
- c. Equipment, and building configuration
- d. Structural and safety factor requirements
- e. Vibration, and imposed loads under normal and abnormal service conditions

Support sizes, configurations, and spacing are given to show the minimal type of supporting components required. If installed loads are excessive for the specified hanger spacing, hangers, and accessories provide heavier-duty components or reduce hanger spacing. After system startup, replace any duct support device which due to length, configuration, or size, vibrates or causes possible failure of a member. Do not use a ductwork support system that allows a cascade-type failure to occur.

Do not hang ductwork and equipment from roof deck, piping, or other ducts or equipment. Maximum span between any two points is 3 meters, with lesser spans as required by duct assemblies, interferences, and permitted loads imposed.

Where support from metal deck systems is involved, coordinate support requirements with installation of metal deck.

3.3.5.1 Hangars

Attach hanger rods, angles, and straps to beam clamps. Receive approval from the Contracting Officer for concrete inserts, masonry anchors, and fasteners for the application.

Hardened high-carbon spring-steel fasteners fitted onto beams and miscellaneous structural steel are acceptable upon prior approval of each proposed application and upon field demonstration of conformance to specification requirements. Make fasteners from steel conforming to AISI Type 1055, treated and finished in conformance with SAE AMS 2480, Type Z (zinc phosphate base), Class 2 (supplementary treatment). Verify a 72-hour load-carrying capacity by a certified independent laboratory.

Where ductwork system contains heavy equipment, excluding air-diffusion devices and single-leaf dampers, hang such equipment independently of the ductwork by means of rods or angles of sizes adequate to support the load.

Cross-brace hangers to preclude swaying both vertically and laterally.

3.3.5.2 Installation

Ensure hanger spacing gives a 20-to-1 safety factor for supported load.

Maximum load supported by any two fasteners is 45 kilograms.

Install hangers on both sides of all duct turns, branch fittings, and transitions.

Friction rod assemblies are not acceptable.

3.3.5.3 Strap-type Hangars

Support rectangular ducts up to **900 millimeters** by strap-type hangers attached at not less than three places to not less than two duct surfaces in different planes.

Perforated strap hangers are not acceptable.

3.3.5.4 Trapeze Hangars

Support rectangular ducting, **900 millimeters** and larger, by trapeze hangers. Support ducts situated in unconditioned areas and required to have insulation with a vapor-sealed facing on trapeze hangers. Space hangers far enough out from the side of the duct to permit the duct insulation to be placed on the duct inside the trapeze. Do not penetrate the vapor-sealed facing with duct hangers.

Where trapeze hangers are used, support the bottom of the duct on angles sized as follows:

<u>WIDTH OF DUCT, MILLIMETER</u>	<u>MINIMUM BOTTOM ANGLE SIZE, MILLIMETER</u>
760 and smaller	32 by 32 by 3.2
761 to 1200	38 by 38 by 3.2
1201 to 1830	38 by 38 by 4.8
1831 to 2440	50 by 50 by 6.4
2441 and wider	50 by 50 by 6.4

3.3.5.5 Purlins

Do not support ducting from roof purlins at points greater than one-sixth of the purlin span from the roof truss. Do not exceed **875 kilograms** load per hanger.

If the hanger load must exceed the above limit, provide reinforcing of purlin(s) or additional support beam(s). When an additional beam is used, have the beam bear on the top chord of the roof trusses, and also bear over the gusset plates of top chord. Stabilize the beam by connection to roof purlin along bottom flange.

Purlins used for supporting fire-protection sprinkler mains, electrical lighting fixtures, electrical power ducts, or cable trays are considered fully loaded. Provide supplemental reinforcing or auxiliary support steel for these purlins when used to support ductwork.

3.3.6 Flexible Connectors for Steel Metal

Connect air-handling equipment, ducts crossing building expansion joints,

and fan inlets and outlets to upstream and downstream components with treated woven-cloth connectors.

Install connectors only after system fans are operative and all vibration isolation mountings have been adjusted. When system fans are operating, ensure connectors are free of wrinkles caused by misalignment or fan reaction. Width of surface is curvilinear.

3.3.7 Insulation Protection Angles

Provide galvanized 1 millimeter thick sheet, formed into an angle with a 50 millimeters exposed long leg with a 10 millimeters stiffening break at outer edge, and with a variable concealed leg, depending upon insulation thickness.

Install angles over all insulation edges terminating by butting against a wall, floor foundation, frame, and similar construction. Fasten angles in place with blind rivets through the protection angle, insulation, and sheet metal duct or plenum. Install angles after final insulation covering has been applied.

3.3.8 Duct Probe Access

Provide holes with neat patches, threaded plugs, or threaded or twist-on caps for air-balancing pitot tube access. Provide extended-neck fittings where probe access area is insulated.

3.3.9 Openings In Roofs and Walls

Existing building openings are fixed in size and can not be resized without authorization. Provide equipment to suit existing opening size.

3.4 FIELD QUALITY CONTROL

3.4.1 Ductwork Leakage Tests

Conduct complete leakage test of new ductwork in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Perform tests prior to installing ductwork insulation.

3.4.2 Inspection

Inspect ductwork in accordance with SMACNA 1987.

3.5 CLOSEOUT ACTIVITIES

3.5.1 Operation and Maintenance

Submit 6 copies of the operation and maintenance manuals 30 calendar days prior to testing the medium and high pressure ductwork systems. Update data and resubmit for final approval no later than 30 calendar days prior to contract completion.

Ensure operation and maintenance manuals are consistent with manufacturer's standard brochures, schematics, printed instructions, general operating procedures and safety precautions.

3.5.2 Record Drawings

Provide [record drawings](#) with current factual information. Include deviations from, and amendments to, the drawings. Include concealed or visible changes in the work. Label drawings "As-Built".

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 34 23.00 40

HVAC POWER VENTILATORS

02/17

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY CONTROL

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Design Requirements
- 2.2 COMPONENTS
 - 2.2.1 Housing Style
 - 2.2.2 Fan Type
 - 2.2.2.1 Type C-PRV Centrifugal, Direct Drive
 - 2.2.3 Fan Motor
 - 2.2.4 Bases
 - 2.2.5 Back-Draft Dampers
 - 2.2.6 Screens
- 2.3 MATERIALS
 - 2.3.1 Aluminum Alloy
 - 2.3.2 Zinc-Coated Steel

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Lubrication
- 3.2 FIELD QUALITY CONTROL
 - 3.2.1 Tests
 - 3.2.2 Acceptance
 - 3.2.3 Final Test Reports
- 3.3 CLOSEOUT ACTIVITIES

-- End of Section Table of Contents --

SECTION 23 34 23.00 40

HVAC POWER VENTILATORS
02/17

PART 1 GENERAL

Provide power roof ventilators complete with all components and accessory equipment as specified in this section.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM B37 (2018) Standard Specification for Aluminum for Use in Iron and Steel Manufacture

ASTM B209M (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)

UNDERWRITERS LABORATORIES (UL)

UL 705 (2017; Reprint Aug 2021) UL Standard for Safety Power Ventilators

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G

Installation Drawings; G

SD-03 Product Data

Housing; G

Fan; G

Motor; G

Bases; G

Dampers; G

Screens; G

SD-06 Test Reports

Final Test Reports; G

SD-11 Closeout Submittals

Record Drawings; G

1.3 QUALITY CONTROL

Rate and label ventilators in accordance with the applicable standards of the Air Movement Control Association (AMCA), and indicate if the license bears the AMCA seal for both air and sound.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

2.1.1 Design Requirements

Submit the manufacturer's catalog data, including equipment and performance data, for power roof ventilator(s). As a minimum, include the following data:

- a. Fan Type
- b. Fan Specifications, including:
 - (1) Number of rotating fan blades/vanes
 - (2) Number of stationary fan blades/vanes
 - (3) Rotating speed(s)
 - (4) Number of belts (if belt-driven)
 - (5) Belt lengths- measured at the pitch line (if belt-driven)
 - (6) Diameter of the drive sheave at the drive pitch line (if belt-driven)
 - (7) Diameter of the driven sheave at the drive pitch line (if belt-driven)
- c. Location of Installation
- d. Identification Number
- e. Date of Installation (Required or Actual Acceptance Date)

f. Applicable reference drawing number(s)

Submit detailed [shop drawings](#) for power roof ventilator systems.

Provide roof ventilators that comply with [UL 705](#) and are furnished complete with bases, curbs, flashing flanges, noise baffles, dampers, damper controls, louvers, and screens, as indicated.

Provide ventilators that are designed for windloads in accordance with [ASCE 7-16](#) with the installed design not less than [210 kilometer per hour](#) windload. Ensure that the structural bracing is properly spaced to accommodate this loading and meets the design requirements of the covering material. Ensure that ventilators are adequately reinforced and well-braced with the joints properly formed. Ensure that the edges are wired or beaded where necessary to ensure rigidity. Prevent galvanic action between different metals in direct contact by providing nonconductive separators. Make all soldering even and smooth.

Provide corrosion-resistant steel bolts, rivets, and other fastenings used in connection with protected metal.

2.2 COMPONENTS

2.2.1 [Housing](#) Style

Provide round-mushroom style power ventilator as indicated.

2.2.2 [Fan](#) Type

Provide fans of the following types:

2.2.2.1 Type C-PRV Centrifugal, Direct Drive

For Type C-PRV ventilators, provide a centrifugal roof ventilator with direct drive, nonoverloading, backward-inclined wheel. Provide a vibration-isolated drive with an elastomer. Provide drive components that are mounted in a compartment isolated from the airstream.

2.2.3 [Fan Motor](#)

Provide single-phase, voltage as indicated, 60 Hz, split-phase, direct-driven motors, with permanently lubricated ball bearings.

Provide motors with local disconnects to allow for fan and motor maintenance. Provide all motors with thermal-overload protection. For motors located in airstreams, use a totally enclosed type.

Use energy efficient permanent split capacitor motors, single phase, 60 Hz direct-drive motors.

2.2.4 [Bases](#)

When bases are provided with the ventilators, use factory-formed bases of the type indicated, constructed of the same material as the hoods, and of the thickness necessary to meet the design requirement for connection to the roof. Provide bases that are suitable for raised-curb mounting where indicated. Form curb flanges of the base as cap flashing, extending at least [50 millimeter](#) over the roofing base. Where indicated or required,

extend the shafts of ventilators a sufficient distance through the supporting construction to permit attachment of vent ducts.

2.2.5 Back-Draft [Dampers](#)

Provide gravity-operated back-draft dampers with adjustable counterweights constructed of the same material as fan housing.

Interlock damper-actuating motor with the fan motor.

2.2.6 [Screens](#)

Provide bird screens with frames constructed of the same material as that used in the ventilators and ensure the screens are securely attached in a manner that permits easy removal for access and cleaning.

2.3 MATERIALS

Provide manufacturers' standard materials.

2.3.1 Aluminum Alloy

Provide aluminum alloy in accordance with [ASTM B209M](#) and [ASTM B37](#).

2.3.2 Zinc-Coated Steel

Provide zinc-coated steel in accordance with [ASTM A653/A653M](#).

PART 3 EXECUTION

3.1 INSTALLATION

Submit [installation drawings](#) for power ventilators.

Install power ventilators in accordance with the manufacturer's installation instructions. Coordinate installation of ventilators with other work. Coordinate anchors, attachments, and other items to be built for installation as the work progresses. Rigidly install ventilators in a weathertight and watertight manner that is free from vibration.

3.1.1 Lubrication

Ensure the movable parts of dampers and related operating hardware are lubricated in accordance with manufacturer's printed instructions and that they operate smoothly and quietly without binding.

3.2 FIELD QUALITY CONTROL

3.2.1 Tests

After installation, test each power roof ventilator to demonstrate proper operation at indicated and specified performance requirements, including the running, balance, noise, and proper direction of fan rotation.

3.2.2 Acceptance

Prior to final acceptance, use precision alignment devices to demonstrate that the fan and motor are aligned as specified by the manufacturer.

Prior to final acceptance, verify conformance to specifications with vibration analysis. Ensure vibration levels are not more than .075 in/sec at 1 times the run speed and at the fan/blade frequency, and not more than 0.04 in/sec at other multiples of the run speed.

3.2.3 Final Test Reports

Provide [final test reports](#) to the Contracting Officer. Provide reports with a cover letter/sheet clearly marked with the system name, date, and the words "Final Test Reports - Forward to the Systems Engineer/Condition Monitoring Office/Predictive Testing Group for inclusion in the Maintenance Database."

3.3 CLOSEOUT ACTIVITIES

Submit detailed [record drawings](#) upon completion of the installation.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 81 00

DECENTRALIZED UNITARY HVAC EQUIPMENT

05/18, CHG 1: 02/21

PART 1 GENERAL

- 1.1 RELATED REQUIREMENTS
- 1.2 REFERENCES
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
- 1.5 DELIVERY, STORAGE, AND HANDLING
- 1.6 ENVIRONMENTAL REQUIREMENTS
- 1.7 WARRANTY

PART 2 PRODUCTS

- 2.1 ENERGY EFFICIENCY REQUIREMENTS
- 2.2 MATERIALS
 - 2.2.1 Standard Products
 - 2.2.2 Product Sustainability Criteria
 - 2.2.2.1 Energy Efficient Equipment
 - 2.2.2.2 Electrical Equipment / Motors
 - 2.2.2.3 Ozone Depleting Substances
 - 2.2.3 Nameplates
 - 2.2.4 Safety Devices
- 2.3 EQUIPMENT
 - 2.3.1 Mini-Split-System Air Conditioners Heat Pumps
 - 2.3.1.1 Small-Capacity Split-System Heat Pump (Not Exceeding 19 kW)
 - 2.3.1.1.1 Energy Efficiency
 - 2.3.1.1.2 Air-to-Refrigerant Coil
 - 2.3.1.1.3 Compressor
 - 2.3.1.1.4 Refrigeration Circuit
 - 2.3.1.1.5 Unit Controls
 - 2.3.1.1.6 Condensing Coil
 - 2.3.1.1.7 Remote Condenser or Condensing Unit
 - 2.3.1.1.7.1 Air-Cooled Condenser
 - 2.3.1.1.8 Air Filters
 - 2.3.1.1.9 Fans
- 2.4 COMPONENTS
 - 2.4.1 Refrigerant and Oil
 - 2.4.2 Fans
 - 2.4.3 Coil Frost Protection
 - 2.4.4 Pressure Vessels
 - 2.4.5 Cabinet Construction
 - 2.4.5.1 Indoor Cabinet
 - 2.4.5.2 Outdoor Cabinet
- 2.5 FINISHES
 - 2.5.1 Equipment and Components Factory Coating
 - 2.5.1.1 Phenolic Coating
 - 2.5.1.2 Chemical Conversion Coating with Polyelastomer Finish Coat

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.5.1.3 Vinyl Coating
- 2.5.2 Factory Applied Insulation
- 2.6 TESTS, INSPECTIONS, AND VERIFICATIONS

PART 3 EXECUTION

- 3.1 EXAMINATION
- 3.2 INSTALLATION
 - 3.2.1 Equipment
 - 3.2.2 Field Applied Insulation
 - 3.2.3 Field Painting
- 3.3 CLEANING AND ADJUSTING
- 3.4 TRAINING
- 3.5 REFRIGERANT TESTS, CHARGING, AND START-UP
 - 3.5.1 Refrigerant Leakage
 - 3.5.2 Contractor's Responsibility
- 3.6 SYSTEM PERFORMANCE TESTS
- 3.7 MAINTENANCE
 - 3.7.1 EXTRA MATERIALS
 - 3.7.2 Maintenance Service

-- End of Section Table of Contents --

SECTION 23 81 00

DECENTRALIZED UNITARY HVAC EQUIPMENT
05/18, CHG 1: 02/21

PART 1 GENERAL

1.1 RELATED REQUIREMENTS

Section 23 03 00.00 20 BASIC MECHANICAL MATERIALS AND METHODS, applies to this section with the additions and modifications specified herein.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 340/360 I-P (2015) Performance Rating of Commercial and Industrial Unitary Air-Conditioning and Heat Pump Equipment

AHRI 700 (2016) Specifications for Fluorocarbon Refrigerants

ANSI/AHRI 460 (2005) Performance Rating of Remote Mechanical-Draft Air-Cooled Refrigerant Condensers

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ANSI/ASHRAE 15 & 34 (2013) ANSI/ASHRAE Standard 15-Safety Standard for Refrigeration Systems and ANSI/ASHRAE Standard 34-Designation and Safety Classification of Refrigerants

ASHRAE 15 & 34 (2013) ASHRAE Standard 34-2016 Safety Standard for Refrigeration Systems/ASHRAE Standard 34-2016 Designation and Safety Classification of Refrigerants-ASHRAE Standard 34-2016

ASHRAE 52.2 (2012) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

ASHRAE 55 (2010) Thermal Environmental Conditions for Human Occupancy

ASHRAE 62.1 (2010) Ventilation for Acceptable Indoor Air Quality

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

ASHRAE 90.1 - SI (2019; Errata 1-4 2020; Addenda BY-CP 2020; Addenda AF-DB 2020; Addenda A-G 2020; Addenda F-AB 2021; Errata 5-7 2021; Interpretation 1-4 2020; Interpretation 5-8 2021) Energy Standard for Buildings Except Low-Rise Residential Buildings

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC SEC IX (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

ASME BPVC SEC VIII D1 (2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1

AMERICAN WELDING SOCIETY (AWS)

AWS Z49.1 (2021) Safety in Welding and Cutting and Allied Processes

ASTM INTERNATIONAL (ASTM)

ASTM B117 (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM C1071 (2019) Standard Specification for Fibrous Glass Duct Lining Insulation (Thermal and Sound Absorbing Material)

ASTM D520 (2000; R 2011) Zinc Dust Pigment

ASTM E84 (2023) Standard Test Method for Surface Burning Characteristics of Building Materials

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2016) Motors and Generators - Revision 1: 2018; Includes 2021 Updates to Parts 0, 1, 7, 12, 30, and 31

NEMA MG 2 (2014) Safety Standard for Construction and Guide for Selection, Installation and Use of Electric Motors and Generators

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-DTL-5541 (2006; Rev F) Chemical Conversion Coatings on Aluminum and Aluminum Alloys

UNDERWRITERS LABORATORIES (UL)

UL 207 (2009; Reprint Jan 2020) Refrigerant-Containing Components and Accessories, Nonelectrical

UL 1995 (2015) UL Standard for Safety Heating and Cooling Equipment

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Spare Parts

Posted Instructions

System Performance Tests

Training; G

Inventory

Supplied Products

Manufacturer's Standard Catalog Data; G

SD-06 Test Reports

Refrigerant Tests, Charging, and Start-Up; G

System Performance Tests; G

SD-07 Certificates

Service Organizations

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

SD-11 Closeout Submittals

Ozone Depleting Substances

1.4 QUALITY ASSURANCE

Carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to be performed and arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions.

1.5 DELIVERY, STORAGE, AND HANDLING

Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Properly protect and care for all material both before and during installation. Submit an [inventory](#) of all the stored items. Replace any materials found to be damaged, at no additional cost to the Government. During installation, cap piping and similar openings capped to keep out dirt and other foreign matter.

1.6 ENVIRONMENTAL REQUIREMENTS

Ventilation must meet or exceed [ASHRAE 62.1](#) and all published addenda. Meet or exceed filter media efficiency as tested in accordance with [ASHRAE 52.2](#). Thermal comfort must meet or exceed [ASHRAE 55](#).

1.7 WARRANTY

Provide equipment with the Manufacturer's Standard Warranty.

PART 2 PRODUCTS

2.1 ENERGY EFFICIENCY REQUIREMENTS

42 USC 8259b requires the procurement of energy efficient products in product categories covered by the Energy Star program or the Federal Energy Management Program for designated products. A list of covered product categories is available from the Federal Energy Management Web site at <http://energy.gov/eere/femp/covered-product-categories>.

Submit Material, Equipment, and Fixtures List of all [supplied products](#) within a covered product category, including manufacturer's catalog numbers, specification and drawing reference number, warranty information, fabrication site, and energy performance data. For product categories covered by the Federal Energy Management Program, submit documentation that the product meets or exceeds FEMP-designated efficiency requirements.

2.2 MATERIALS

Provide [Manufacturer's standard catalog data](#), at least 5 weeks prior to the purchase or installation of a particular component, highlighted to show material, size, options, performance charts and curves, etc. in adequate detail to demonstrate compliance with contract requirements. Data includes manufacturer's recommended installation instructions and procedures. If vibration isolation is specified for a unit, include vibration isolator literature containing catalog cuts and certification that the isolation characteristics of the isolators provided meet the manufacturer's recommendations. Submit data for each specified component. Minimum efficiency requirements must be in accordance with [ASHRAE 90.1 - SI](#).

2.2.1 Standard Products

Provide materials and equipment that are standard products of a manufacturer regularly engaged in the manufacturing of such products, which are of a similar material, design and workmanship. The standard products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2 year use includes applications of equipment and materials under similar circumstances and of similar size. The 2 years' experience must be satisfactorily completed by a product which has been sold or is offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures. Products having less than a 2 year field service record will be acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory tests, can be shown. Products must be supported by a service organization. Ensure system components are environmentally suitable for the indicated geographic locations.

2.2.2 Product Sustainability Criteria

2.2.2.1 Energy Efficient Equipment

Provide equipment meeting the efficiency requirements as stated within this section.

2.2.2.2 Electrical Equipment / Motors

Provide electrical equipment, motors, motor efficiencies, and wiring which are in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Electrical motor driven equipment specified must be provided complete with motors, motor starters, and controls. Electrical characteristics must be as shown, and unless otherwise indicated, all motors of 746 W and above with open, dripproof, totally enclosed, or explosion proof fan cooled enclosures, must be the premium efficiency type in accordance with NEMA MG 1. Field wiring must be in accordance with manufacturer's instructions. Each motor must conform to NEMA MG 1 and NEMA MG 2 and be of sufficient size to drive the equipment at the specified capacity without exceeding the nameplate rating of the motor. Motors must be continuous duty with the enclosure specified. Motor starters must be provided complete with thermal overload protection and other appurtenances necessary for the motor control indicated. Motors must be furnished with a magnetic across-the-line or reduced voltage type starter as required by the manufacturer. Motor duty requirements must allow for maximum frequency start-stop operation and minimum encountered interval between start and stop. Motors must be sized for the applicable loads. Motor torque must be capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Motor bearings must be fitted with grease supply fittings and grease relief to outside of enclosure. Manual or automatic control and protective or signal devices required for the operation specified and any control wiring required for controls and devices specified, but not shown, must be provided.

2.2.2.3 Ozone Depleting Substances

Unitary air conditioning equipment must not use CFC-based refrigerants. Refrigerant may be an approved alternative refrigerant in accordance with EPA's Significant New Alternative Policy (SNAP) listing.

2.2.3 Nameplates

Major equipment including compressors, condensers, receivers, heat exchanges, fans, and motors must have the manufacturer's name, address, type or style, model or serial number, and catalog number on a plate secured to the item of equipment. Plates must be durable and legible throughout equipment life and made of anodized aluminum or stainless steel. Fix plates in prominent locations with nonferrous screws or bolts.

2.2.4 Safety Devices

Exposed moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Safety devices must be installed so that proper operation of equipment is not impaired. Welding and cutting safety requirements must be in accordance with AWS Z49.1.

2.3 EQUIPMENT

2.3.1 Mini-Split-System Air Conditioners Heat Pumps

2.3.1.1 Small-Capacity Split-System Heat Pump (Not Exceeding 19 kW)

Provide an air-cooled, split system which employs a remote condensing unit, a separate ceiling mounted indoor unit, and interconnecting refrigerant piping. Provide the heat pump type unit conforming to applicable Underwriters Laboratories (UL) standards including UL 1995. Unit must be rated in accordance with AHRI 340/360 I-P. Provide indoor unit with necessary fans, air filters, and galvanized steel cabinet construction. The remote unit must be as specified in paragraph CONDENSING UNIT. Provide double-width, double inlet, forward curved backward inclined, or airfoil blade, centrifugal scroll type evaporator or supply fans. Provide the manufacturer's standard condenser or outdoor fans for the unit specified and may be either propeller or centrifugal scroll type. Design unit to operate at outdoor ambient temperatures up to 46 degrees C.

2.3.1.1.1 Energy Efficiency

Provide unit with an Energy Star label. Air Conditioners must have a minimum seasonal energy efficiency ratio (SEER) of 20 and a minimum Heating Seasonal Performance Factor (HSPF) of 10,

2.3.1.1.2 Air-to-Refrigerant Coil

Provide condensing coils with copper or aluminum tubes of 10 mm minimum diameter with copper or aluminum fins that are mechanically bonded or soldered to the tubes. Casing must be galvanized steel or aluminum. Avoid contact of dissimilar metals. Test coils in accordance with ASHRAE 15 & 34 at the factory and ensure suitability for the working pressure of the installed system. Dehydrate and seal each coil testing and prior to evaluation and charging.

2.3.1.1.3 Compressor

Provide direct drive hermetically sealed, digitally controlled, inverter driven twin rotary type compressor. Provide compressor with internal over temperature and pressure protector; sump heater; oil pump; high pressure and low pressure controls; and liquid line dryer.

2.3.1.1.4 Refrigeration Circuit

Refrigerant-containing components must comply with ASHRAE 15 & 34 and be factory tested, cleaned, dehydrated, charged, and sealed. Provide each unit with a factory operating charge of refrigerant and oil or a holding charge. Field charge unit shipped with a holding charge. Provide refrigerant charging valves. Provide filter-drier in liquid line to prevent freeze-up in event of loss of water flow during heating cycle.

2.3.1.1.5 Unit Controls

Provide unit internally prewired with a 24 volt control circuit powered by an internal transformer. Provide terminal blocks for power wiring and external control wiring. Internally protect unit by fuses or a circuit breaker in accordance with UL 1995. Equip units with three-phase power with phase monitoring protection to protect against problems caused by

phase loss, phase imbalance and phase reversal. Provide unit with microprocessor controls to provide all 24V control functions. Control unit by a programmable electronic thermostat with heating setback and cooling setup with 7-day programming capability.

2.3.1.1.6 Condensing Coil

Provide coils with nonferrous tubes of 10 mm minimum diameter with copper or aluminum fins that are mechanically bonded or soldered to the tubes. Provide galvanized steel or aluminum casing. Avoid contact of dissimilar metals. Test coils in accordance with ANSI/ASHRAE 15 & 34 at the factory and ensure suitability for the working pressure of the installed system. Dehydrate and seal each coil after testing and prior to evaluation and charging. Provide separate expansion devices for each compressor circuit.

2.3.1.1.7 Remote Condenser or Condensing Unit

Fit each remote condenser coil fitted with a manual isolation valve and an access valve on the coil side. Saturated refrigerant condensing temperature must not exceed 49 degrees C at 40 degrees C ambient. Provide unit with low ambient condenser controls to ensure proper operation in an ambient temperature of 13 degrees C. Provide fan and cabinet construction as specified in paragraph UNITARY EQUIPMENT ACCESSORIES. Condensing unit must have controls to initiate a refrigerant pump down cycle at system shut down on each refrigerant circuit.

2.3.1.1.7.1 Air-Cooled Condenser

Provide Unit in accordance with ANSI/AHRI 460 and conform to the requirements of UL 1995. Provide factory fabricated, tested, packaged, and self-contained unit; complete with casing, propeller or centrifugal type fans, heat rejection coils, connecting piping and wiring, and all necessary accessories.

2.3.1.1.8 Air Filters

Provide filters of the cleanable type that are capable of filtering the entire air supply. Mount filter(s) integral within the unit and make accessible by hinged access panel(s).

2.3.1.1.9 Fans

Provide direct driven, statically and dynamically balanced, centrifugal or propeller type fans. Design the outdoor fan so that condensate will evaporate without drip, splash, or spray on building exterior. Provide indoor fan with a minimum two-speed motor with built-in overload protection. Fan motors must be the inherently protected, permanent split-capacitor type.

2.4 COMPONENTS

2.4.1 Refrigerant and Oil

Refrigerant must be one of the fluorocarbon gases. Refrigerants must have number designations and safety classifications in accordance with ASHRAE 15 & 34. Refrigerants must meet the requirements of AHRI 700 as a minimum. Provide a complete charge of refrigerant for the installed system as recommended by the manufacturer. Lubricating oil must be of a type and grade recommended by the manufacturer for each compressor. Where

color leak indicator dye is incorporated, charge must be in accordance with manufacturer's recommendation.

2.4.2 Fans

Fan wheel shafts must be supported by either maintenance-accessible lubricated antifriction block-type bearings, or permanently lubricated ball bearings. Unit fans must be selected to produce the **LPS** required at the fan total pressure. Thermal overload protection must be of the manual or automatic-reset type. Fan wheels or propellers must be constructed of aluminum or galvanized steel. Centrifugal fan wheel housings must be of galvanized steel, and both centrifugal and propeller fan casings must be constructed of aluminum or galvanized steel. Steel elements of fans, except fan shafts, must be hot-dipped galvanized after fabrication or fabricated of mill galvanized steel. Mill-galvanized steel surfaces and edges damaged or cut during fabrication by forming, punching, drilling, welding, or cutting must be recoated with an approved zinc-rich compound. Fan wheels or propellers must be statically and dynamically balanced. Direct-drive fan motors must be of the multiple-speed variety. Propeller fans must be direct-drive drive type with fixed pitch blades. Each drive will be independent of any other drive. Drive bearings must be protected with water slingers or shields.

2.4.3 Coil Frost Protection

Provide each circuit with a manufacturer's standard coil frost protection system. The coil frost protection system must use a temperature sensor in the suction line of the compressor to shut the compressor off when coil frosting occurs. Use timers to prevent the compressor from rapid cycling.

2.4.4 Pressure Vessels

Pressure vessels must conform to **ASME BPVC SEC VIII D1** or **UL 207**, as applicable for maximum and minimum pressure or temperature encountered. Where referenced publications do not apply, test pressure components at 1-1/2 times design working pressure. Refrigerant wetted carbon steel surfaces must be pickled or abrasive blasted free of mill scale, cleaned, dried, charged, and sealed.

2.4.5 Cabinet Construction

Casings for the specified unitary equipment must be constructed of galvanized steel or aluminum sheet metal and galvanized or aluminum structural members. Minimum thickness of single wall exterior surfaces must be **1.3 mm** galvanized steel or **1.8 mm** thick aluminum on units with a capacity above **70 kW** and **1.0 mm** galvanized steel or **1.6 mm** thick aluminum on units with a capacity less than **70 kW**. Casing must be fitted with lifting provisions, access panels or doors, fan vibration isolators, electrical control panel, corrosion-resistant components, structural support members, insulated condensate drip pan and drain, and internal insulation in the cold section of the casing. Where double-wall insulated construction is proposed, minimum exterior galvanized sheet metal thickness must be **1.0 mm**. Provisions to permit replacement of major unit components must be incorporated. Penetrations of cabinet surfaces, including the floor, must be sealed. Unit must be fitted with a drain pan which extends under all areas where water may accumulate. Drain pan must be fabricated from Type 300 stainless steel, galvanized steel with protective coating as required, or an approved plastic material. Pan insulation must be water impervious. Extent and effectiveness of the

insulation of unit air containment surfaces must prevent, within limits of the specified insulation, heat transfer between the unit exterior and ambient air, heat transfer between the two conditioned air streams, and condensation on surfaces. Insulation must conform to [ASTM C1071](#). Paint and finishes must comply with the requirements specified in paragraph FACTORY COATING.

2.4.5.1 Indoor Cabinet

Indoor cabinets must be suitable for the specified indoor service and enclose all unit components.

2.4.5.2 Outdoor Cabinet

Outdoor cabinets must be suitable for outdoor service with a weathertight, insulated and corrosion-protected structure. Cabinets constructed exclusively for indoor service which have been modified for outdoor service are not acceptable.

2.5 FINISHES

2.5.1 Equipment and Components Factory Coating

Unless otherwise specified, equipment and component items, when fabricated from ferrous metal, must be factory finished with the manufacturer's standard finish, except that items located outside of buildings must have weather resistant finishes that will withstand 500 hours exposure to the salt spray test specified in [ASTM B117](#) using a 5 percent sodium chloride solution. Immediately after completion of the test, the specimen must show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond [3 mm](#) on either side of the scratch mark. Cut edges of galvanized surfaces where hot-dip galvanized sheet steel is used must be coated with a zinc-rich coating conforming to [ASTM D520](#), Type I.

Where stipulated in equipment specifications of this section, coat finned tube coils of the affected equipment as specified below. Apply coating at the premises of a company specializing in such work. Degrease and prepare for coating in accordance with the coating applicator's procedures for the type of metals involved. Completed coating must show no evidence of softening, blistering, cracking, crazing, flaking, loss of adhesion, or "bridging" between the fins.

2.5.1.1 Phenolic Coating

Provide a resin base thermosetting phenolic coating. Apply coating by immersion dipping of the entire coil. Provide a minimum of two coats. Bake or heat dry coils following immersions. After final immersion and prior to final baking, spray entire coil with particular emphasis given to building up coating on sheared edges. Total dry film thickness must be [0.064 to 0.076 mm](#).

2.5.1.2 Chemical Conversion Coating with Polyelastomer Finish Coat

Dip coils in a chemical conversion solution to molecularly deposit a corrosion resistant coating by electrolysis action. Chemical conversion coatings must conform to [MIL-DTL-5541](#), Class 1A. Cure conversion coating at a temperature of [43 to 60 degrees C](#) for a minimum of 3 hours. Coat coil surfaces with a complex polymer primer with a dry film thickness of

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

0.025 mm. Cure primer coat for a minimum of 1 hour. Using dip tank method, provide three coats of a complex polyelastomer finish coat. After each of the first two finish coats, cure the coils for 1 hour. Following the third coat, spray a fog coat of an inert sealer on the coil surfaces. Total dry film thickness must be 0.064 to 0.076 mm. Cure finish coat for a minimum of 3 hours. Coating materials must have 300 percent flexibility, operate in temperatures of minus 46 to plus 104 degrees C, and protect against atmospheres of a pH range of 1 to 14.

2.5.1.3 Vinyl Coating

Apply coating using an airless fog nozzle. For each coat, make at least two passes with the nozzle. Materials to be applied are as follows:

- a. Total dry film thickness, 0.165 mm maximum
- b. Vinyl Primer, 24 percent solids by volume: One coat 0.051 mm thick
- c. Vinyl Copolymer, 30 percent solids by volume: One coat 0.114 mm thick

2.5.2 Factory Applied Insulation

Refrigeration equipment must be provided with factory installed insulation on surfaces subject to sweating including the suction line piping. Where motors are the gas-cooled type, factory installed insulation must be provided on the cold-gas inlet connection to the motor in accordance with manufacturer's standard practice. Factory insulated items installed outdoors are not required to be fire-rated. As a minimum, factory insulated items installed indoors must have a flame spread index no higher than 75 and a smoke developed index no higher than 150. Factory insulated items (no jacket) installed indoors and which are located in air plenums, in ceiling spaces, and in attic spaces must have a flame spread index no higher than 25 and a smoke developed index no higher than 50. Flame spread and smoke developed indexes must be determined by ASTM E84. Insulation must be tested in the same density and installed thickness as the material to be used in the actual construction. Material supplied by a manufacturer with a jacket must be tested as a composite material. Jackets, facings, and adhesives must have a flame spread index no higher than 25 and a smoke developed index no higher than 50 when tested in accordance with ASTM E84.

2.6 TESTS, INSPECTIONS, AND VERIFICATIONS

All manufactured units must be inspected and tested, and documentation provided to demonstrate that each unit is in compliance with ANSI/AHRI and UL requirements and that the minimum efficiency requirements of ASHRAE 90.1 - SI have been met.

PART 3 EXECUTION

3.1 EXAMINATION

After becoming familiar with all details of the work, perform Verification of Dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

3.2 INSTALLATION

Perform work in accordance with the manufacturer's published diagrams,

recommendations, and equipment warranty requirements. Where equipment is specified to conform to the requirements of ASME BPVC SEC VIII D1 and ASME BPVC SEC IX, the design, fabrication, and installation of the system must conform to ASME BPVC SEC VIII D1 and ASME BPVC SEC IX.

3.2.1 Equipment

Provide refrigeration equipment conforming to ASHRAE 15 & 34. Provide necessary supports for all equipment, appurtenances, and pipe as required. Equipment must be properly leveled, aligned, and secured in place in accordance with manufacturer's instructions.

3.2.2 Field Applied Insulation

Apply field applied insulation as specified, except as defined differently herein.

3.2.3 Field Painting

Painting required for surfaces not otherwise specified, and finish painting of items only primed at the factory are specified in Section 09 90 00 PAINTS AND COATINGS.

3.3 CLEANING AND ADJUSTING

Equipment must be wiped clean, with all traces of oil, dust, dirt, or paint spots removed. Temporary filters must be provided for all fans that are operated during construction, and new filters must be installed after all construction dirt has been removed from the building. System must be maintained in this clean condition until final acceptance. Bearings must be properly lubricated with oil or grease as recommended by the manufacturer. Belts must be tightened to proper tension. Control valves and other miscellaneous equipment requiring adjustment must be adjusted to setting indicated or directed. Fans must be adjusted to the speed indicated by the manufacturer to meet specified conditions. Testing, adjusting, and balancing must be as specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS.

3.4 TRAINING

Conduct a training course for the operating staff as designated by the Contracting Officer. The training period must consist of a total 8 hours of normal working time and start after the system is functionally completed but prior to final acceptance tests.

- a. Submit a schedule, at least 2 weeks prior to the date of the proposed training course, which identifies the date, time, and location for the training.
- b. Submit the field posted instructions, at least 2 weeks prior to construction completion, including equipment layout, wiring and control diagrams, piping, valves and control sequences, and typed condensed operation instructions. The condensed operation instructions must include preventative maintenance procedures, methods of checking the system for normal and safe operation, and procedures for safely starting and stopping the system. The posted instructions must be framed under glass or laminated plastic and be posted where indicated by the Contracting Officer.

- c. The posted instructions must cover all of the items contained in the approved [operation and maintenance manuals](#) as well as demonstrations of routine maintenance operations. Submit 6 complete copies of an operation manual in bound [216 by 279](#) booklets listing step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown at least 4 weeks prior to the first training course. The booklets must include the manufacturer's name, model number, and parts list. The manuals must include the manufacturer's name, model number, service manual, and a brief description of all equipment and their basic operating features.
- d. Submit 6 complete copies of maintenance manual in bound [216 by 279 mm](#) booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide. The manuals must include piping and equipment layouts and simplified wiring and control diagrams of the system as installed.

3.5 [REFRIGERANT TESTS, CHARGING, AND START-UP](#)

Split-system refrigerant piping systems must be tested and charged as specified. Packaged refrigerant systems which are factory charged must be checked for refrigerant and oil capacity to verify proper refrigerant levels in accordance with manufacturer's recommendations. Following charging, packaged systems must be tested for leaks with a halide torch or an electronic leak detector. Submit 6 copies of each test containing the information described below in bound [216 by 279 mm](#) booklets. Individual reports must be submitted for the refrigerant system tests.

- a. The date the tests were performed.
- b. A list of equipment used, with calibration certifications.
- c. Initial test summaries.
- d. Repairs/adjustments performed.
- e. Final test results.

3.5.1 Refrigerant Leakage

If a refrigerant leak is discovered after the system has been charged, the leaking portion of the system must immediately be isolated from the remainder of the system and the refrigerant pumped into the system receiver or other suitable container. Under no circumstances must the refrigerant be discharged into the atmosphere.

3.5.2 Contractor's Responsibility

Take steps, at all times during the installation and testing of the refrigeration system, to prevent the release of refrigerants into the atmosphere. The steps must include, but not be limited to, procedures which will minimize the release of refrigerants to the atmosphere and the use of refrigerant recovery devices to remove refrigerant from the system and store the refrigerant for reuse or reclaim. At no time must more than [85 g](#) of refrigerant be released to the atmosphere in any one occurrence. Any system leaks within the first year must be repaired in accordance with the requirements herein at no cost to the Government including material, labor, and refrigerant if the leak is the result of defective equipment, material, or installation.

3.6 SYSTEM PERFORMANCE TESTS

Before each refrigeration system is accepted, conduct tests to demonstrate the general operating characteristics of all equipment by a registered professional engineer or an approved manufacturer's start-up representative experienced in system start-up and testing, at such times as directed. Six copies of the report provided in bound 216 by 279 mm booklets. The report must document compliance with the specified performance criteria upon completion and testing of the system. The report must indicate the number of days covered by the tests and any conclusions as to the adequacy of the system.

For equipment providing heating and cooling the system performance tests must be performed during the heating and cooling seasons.

- a. Submit a schedule, at least 2 weeks prior to the start of related testing, for the system performance tests. The schedules must identify the proposed date, time, and location for each test. Tests must cover a period of not less than 48 hours for each system and must demonstrate that the entire system is functioning in accordance with the drawings and specifications.
- b. Make corrections and adjustments, as necessary, tests must be re-conducted to demonstrate that the entire system is functioning as specified. Prior to acceptance, install and tighten service valve seal caps and blanks over gauge points. Replace any refrigerant lost during the system startup.
- c. If tests do not demonstrate satisfactory system performance, correct deficiencies and retest the system. Conduct tests in the presence of the Contracting Officer. Water and electricity required for the tests will be furnished by the Government. Provide all material, equipment, instruments, and personnel required for the test.
- d. Coordinate field tests with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS. Submit 6 copies of the report provided in bound 216 by 279 mm booklets. The report must document compliance with the specified performance criteria upon completion and testing of the system. The report must indicate the number of days covered by the tests and any conclusions as to the adequacy of the system. Submit the report including the following information (where values are taken at least three different times at outside dry-bulb temperatures that are at least 3 degrees C apart):
 - (1) Date and outside weather conditions.
 - (2) The load on the system based on the following:
 - (a) The refrigerant used in the system.
 - (b) Condensing temperature and pressure.
 - (c) Suction temperature and pressure.
 - (d) Ambient, condensing and coolant temperatures.
 - (e) Running current, voltage and proper phase sequence for each phase of all motors.
 - (3) The actual on-site setting of operating and safety controls.
 - (4) Thermostatic expansion valve superheat - value as determined by

field test.

- (5) Subcooling.
- (6) High and low refrigerant temperature switch set-points
- (7) Low oil pressure switch set-point.
- (8) Defrost system timer and thermostat set-points.
- (9) Moisture content.
- (10) Capacity control set-points.
- (11) Field data and adjustments which affect unit performance and energy consumption.
- (12) Field adjustments and settings which were not permanently marked as an integral part of a device.

3.7 MAINTENANCE

3.7.1 EXTRA MATERIALS

Submit [spare parts](#) data for each different item of equipment specified, after approval of detail drawings and not later than 2 months prior to the date of beneficial occupancy. Include in the data a complete list of parts and supplies, with current unit prices and source of supply, a recommended spare parts list for 1 year of operation, and a list of the parts recommended by the manufacturer to be replaced on a routine basis.

3.7.2 Maintenance Service

Submit a certified list of qualified permanent [service organizations](#), which includes their addresses and qualifications, for support of the equipment. The service organizations must be reasonably convenient to the equipment installation and be able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 25 - INTEGRATED AUTOMATION

SECTION 25 05 11

CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS

05/21

PART 1 GENERAL

- 1.1 RELATED REQUIREMENTS
- 1.2 REFERENCES
- 1.3 DEFINITIONS
 - 1.3.1 Administrator Account
 - 1.3.2 Computer
 - 1.3.3 Controller
 - 1.3.4 Mission Space
 - 1.3.5 Network
 - 1.3.6 Network Connected
 - 1.3.6.1 Wireless Network Connected
 - 1.3.7 Network Media
 - 1.3.8 User Account Support Levels
 - 1.3.8.1 FULLY Supported
 - 1.3.8.2 MINIMALLY Supported
 - 1.3.8.3 NOT Supported
 - 1.3.9 Manual Local Input
 - 1.3.10 Card Reader
 - 1.3.11 User Interface
 - 1.3.11.1 Local User Interface
 - 1.3.11.2 Remote User Interface
 - 1.3.11.3 Types of User Interface (by capability)
 - 1.3.11.3.1 Read-Only User Interface
 - 1.3.11.3.2 Limited User Interface
 - 1.3.11.3.3 Full User Interface
 - 1.3.11.3.4 View-Only User Interface
 - 1.3.11.4 Other User Interface Terminology
 - 1.3.11.4.1 Writable User Interface
 - 1.3.11.4.2 Privileged User Interface
- 1.4 ADMINISTRATIVE REQUIREMENTS
 - 1.4.1 Points of Contact
 - 1.4.2 Coordination
- 1.5 SUBMITTALS
- 1.6 CYBERSECURITY DOCUMENTATION
 - 1.6.1 Proposed STIG and SRG Applicability Report
 - 1.6.2 Network Communication Report
 - 1.6.3 Control System Inventory Report
 - 1.6.4 Software and Configuration Backups
 - 1.6.5 Cybersecurity Riser Diagram
 - 1.6.6 STIG, SRG and Vendor Guide Compliance Result Report
 - 1.6.7 Control System Cybersecurity Documentation
 - 1.6.7.1 Software Applications
 - 1.6.7.2 All Other Control System Devices
 - 1.6.7.3 Default Requirements for Control System Devices
- 1.7 SOFTWARE LICENSING

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 1.8 CYBERSECURITY DURING CONSTRUCTION
 - 1.8.1 Contractor Computer Equipment
 - 1.8.1.1 Operating System
 - 1.8.1.2 Anti-Malware Software
 - 1.8.1.3 Passwords and Passphrases
 - 1.8.1.4 User-Based Authentication
 - 1.8.1.5 Demonstration of Compliance
 - 1.8.1.6 Contractor Computer Cybersecurity Compliance Statements
- 1.9 CYBERSECURITY DURING WARRANTY PERIOD

PART 2 PRODUCTS

- 2.1 ETHERNET SWITCH
 - 2.1.1 Required Functionality
 - 2.1.2 Configuration Requirements
- 2.2 DAISY CHAIN IP CONTROLLERS

PART 3 EXECUTION

- 3.1 CYBERSECURITY HARDENING AND CONFIGURATION GUIDES
- 3.2 NETWORK REQUIREMENTS
 - 3.2.1 Non-IP Control Networks
 - 3.2.2 IP Control Networks
 - 3.2.2.1 IP Network Routers
 - 3.2.2.2 IP Devices With Multiple Ethernet Connection
 - 3.2.3 Cryptographic Protection
 - 3.2.4 Device Identification and Authentication
 - 3.2.4.1 Default Requirements for Control System Devices
 - 3.2.5 Cryptographic Module Authentication
- 3.3 ACCESS CONTROL REQUIREMENTS
 - 3.3.1 User Accounts
 - 3.3.1.1 Computers
 - 3.3.1.2 Controllers
 - 3.3.1.2.1 Default Requirements for Other Control Systems
 - 3.3.1.3 Additional User Account Expiration Requirements In MODERATE Impact Systems:
 - 3.3.1.3.1 For Control System Applications Running on Computers
 - 3.3.1.3.2 For Other Control System Devices FULLY Supporting Accounts
 - 3.3.2 Unsuccessful Logon Attempts
 - 3.3.2.1 Devices MINIMALLY Supporting Accounts
 - 3.3.2.2 Devices FULLY Supporting Accounts
 - 3.3.2.3 High Availability Interfaces Exempt from Unsuccessful Logon Attempts Requirements
 - 3.3.3 System Use Notification
 - 3.3.3.1 System Use Notification for Remote User Interfaces
 - 3.3.3.2 System Use Notification for Local User Interfaces
 - 3.3.4 Permitted Actions Without Identification or Authentication
 - 3.3.5 Enclosures
- 3.4 USER IDENTIFICATION AND AUTHENTICATION
 - 3.4.1 User Identification and Authentication for All System Types
 - 3.4.2 Implementation of Identification and Authorization Requirements
 - 3.4.3 Password-Based Authentication Requirements
 - 3.4.3.1 Passwords for Software and Applications Running on Computers
 - 3.4.3.2 Passwords for Controllers FULLY Supporting Accounts
 - 3.4.3.3 Passwords for Remote Interfaces
 - 3.4.3.4 Passwords for Devices Minimally Supporting Accounts
 - 3.4.3.5 Password Configuration and Reporting

- 3.4.4 Authenticator Feedback
- 3.5 CYBERSECURITY AUDITING
 - 3.5.1 Audit Events, Content of Audit Records, and Audit Generation
 - 3.5.1.1 Computers
 - 3.5.1.1.1 Audited Events
 - 3.5.1.1.2 Audit Event Information To Record
 - 3.5.1.2 Default Requirements for Control System Controllers
 - 3.5.1.2.1 Controllers Which FULLY Support Accounts
 - 3.5.1.2.1.1 Audited Events
 - 3.5.1.2.1.2 Audit Event Information To Record
 - 3.5.1.2.2 Controllers Which Do Not FULLY Support Accounts
 - 3.5.2 Audit Time Stamps
 - 3.5.3 Auditing Front End Software
 - 3.5.3.1 Import and Upload Requirements
 - 3.5.3.2 Export Requirements
 - 3.5.4 Audit Storage Capacity and Audit Upload
 - 3.5.4.1 Device Audit Record Upload Software
 - 3.5.5 Response to Audit Processing Failures
- 3.6 REQUIREMENTS FOR LEAST FUNCTIONALITY
 - 3.6.1 Device Capabilities
 - 3.6.2 Software
- 3.7 SYSTEM AND COMMUNICATION PROTECTION
 - 3.7.1 Collaborative Computing
 - 3.7.2 Denial of Service Protection
- 3.8 SAFE MODE AND FAIL SAFE OPERATION
- 3.9 SYSTEM MAINTENANCE TOOL SOFTWARE
- 3.10 VULNERABILITY SCANNING
 - 3.10.1 Computers and Software Running on Computers
 - 3.10.2 Controllers
- 3.11 SYSTEM AND INTEGRATION INTEGRITY
 - 3.11.1 Malicious Code Protection
- 3.12 CONTROL SYSTEM CYBERSECURITY TESTING
 - 3.12.1 Control System Cybersecurity Testing Procedures
 - 3.12.2 Control System Cybersecurity Testing Execution
 - 3.12.3 Control System Cybersecurity Testing Report
- 3.13 FIELD QUALITY CONTROL, CYBERSECURITY VALIDATION SUPPORT
- 3.14 CYBERSECURITY TRAINING

-- End of Section Table of Contents --

SECTION 25 05 11

CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS
05/21

PART 1 GENERAL

Many subparts in this Section contain text in curly braces ("{" and "}") indicating which cybersecurity control and control correlation identifier (CCI) the requirements of the subpart relate to. The text inside these curly braces is for Government reference only and enables coordination of the requirements of this Section with the RMF process throughout the design and construction process. Text in curly braces are not contractor requirements.

This Section refers to Security Requirements Guide (SRGs) and Security Technical Implementation Guide (STIGs). STIGs and SRGs are available online at the Information Assurance Support Environment (IASE) website at <https://public.cyber.mil/stigs/downloads/> and an SRG/STIG Applicability Guide and Collection Tool is available at <https://public.cyber.mil/stigs/SCAP/>. Not all control system components have applicable STIGs or SRGs. The "Control Systems SRG" does not apply to work performed under this Section; all requirements within this section to apply applicable SRGs DO NOT include the "Control Systems SRG."

1.1 RELATED REQUIREMENTS

This section does not contain sufficient requirements to procure a control system and must be used in conjunction with other Sections which specify control systems. This Section adds cybersecurity requirements to the control systems specified in other Sections, and as these requirements are conditioned on the control system being provided, there may be requirements in this Section that will not apply to this project. All Sections containing facility-related control systems or control system components are related to the requirements of this Section. Review all specification sections to determine related requirements.

In cases where a requirement is specified in both this Section and in another Section, the more stringent requirement must be met. In cases where a requirement in this Section conflicts with the requirements of another Section such that both requirements cannot be met at the same time, request direction from the Contracting Officer Representative to determine which requirement applies to the project.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.1x

(2010) Local and Metropolitan Area

Networks - Port Based Network Access
Control

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST FIPS 140-2

(2001) Security Requirements for
Cryptographic Modules

U.S. DEPARTMENT OF DEFENSE (DOD)

DODI 8551.01

(2014) Ports, Protocols, and Services
Management (PPSM)

DTM 08-060

(2008) Policy on Use of Department of
Defense (DoD) Information Systems -
Standard Consent Banner and User Agreement

1.3 DEFINITIONS

1.3.1 Administrator Account

An administrator account is an account with full permissions to a device, application, or operating system, including the ability to create and modify other user accounts.

Note that the operating system Administrator Account may be different than Administrator Accounts for applications hosted on that operating system. Also, most controllers will not have any support for accounts and will therefore not have an 'Administrator Account.'

1.3.2 Computer

A computer is one of the following:

- a. a device running a non-embedded desktop or server version of Microsoft Windows
- b. a device running a non-embedded version of MacOS
- c. a device running a non-embedded version of Linux
- d. a device running a version or derivative of the Android Operating System, where Android is considered separate from Linux
- e. a device running a version of Apple iOS

Unless otherwise indicated or clear from context use of the word "device" in this Section includes computers.

1.3.3 Controller

A device other than a computer or Ethernet switch. For Fire Protection systems this includes fire alarm control panels, remote operating consoles, and remote annunciators.

1.3.4 Mission Space

A device or media is in mission space if physical access to the device or media is controlled by the organization served by the device. For

example, a VAV box controller in a suspended ceiling is in mission space if the VAV box serves that room; an electrical switchgear in an electrical room or an AHU in a mechanical room or on a rooftop may still be considered to be in mission space if the organization (mission) served by that switchgear or AHU controls access to the electrical room, mechanical room or rooftop.

1.3.5 Network

A network is a group of two or more devices that can communicate using a network protocol. Network protocols must provide a method for addressing devices on the network; a communication method that does not provide an addressing scheme is not a networked form of communication. Devices that communicate using a method of communication that does not support device addressing are not using a network.

1.3.6 Network Connected

A component is network connected (or "connected to a network") only when the device has a network transceiver which is directly connected to the network and implements the network protocol. A device lacking a network transceiver (and accompanying protocol implementation) can never be considered network connected. Note that (unlike many IT definitions of "Network Connected") a device connected to a non-IP network is still considered network connected (an IP connection or IP address is not required for a device to be network connected).

1.3.6.1 Wireless Network Connected

Any device that supports wireless network communication is network connected to a wireless network, regardless of whether the device is communicating using wireless. Unless physically disabled, devices with wireless transceivers support wireless, it is not sufficient to disable the wireless in software.

1.3.7 Network Media

The thing that provides the communication channel between the devices on a network. Typically wire, but might include wireless, fiber optic, or even power line (some network protocols allow sending network signals over power wiring).

1.3.8 User Account Support Levels

The support for user accounts is categorized in this Section as one of three levels:

1.3.8.1 FULLY Supported

Device supports configurable individual accounts. Accounts can be created, deleted, modified, etc. Privileges can be assigned to accounts. These devices support user-based (as opposed to role-based) authentication.

1.3.8.2 MINIMALLY Supported

Device supports a small, fixed number of accounts (perhaps only one). Accounts cannot be modified. A device with only a "User" and an "Administrator" account would fit this category. Similarly, a device with

two PINs for logon - one for restricted and one for unrestricted rights would fit here (in other words, the accounts do not have to be the traditional "username and password" structure). These devices typically only support role-based authentication.

Examples of devices which MINIMALLY support accounts are a) a variable frequency drive with a single account which requires a PIN for access to configuration; and b) a room lighting control touchpad interface that has a single account.

1.3.8.3 NOT Supported

Device does not support any Access Enforcement therefore the whole concept of "account" is meaningless.

1.3.9 Manual Local Input

Manual Local Inputs are system analog or binary inputs that are adjustable by a person but are, by intrinsic hardware design, very limited in potential capabilities. Manual Local Inputs do not have touch screens or full keyboards, but may have a few buttons or dials to allow input. Manual Local Inputs do not have full graphic screens or dot-matrix displays, but may have simple lights (LEDs) or 7-segment displays. Manual Local Inputs do not have any sort of menu structure, each button has a single well-defined function.

Examples of Manual Local Inputs are H-O-A switches, simple thermostats, and disconnect switches.

1.3.10 Card Reader

A card reader is an input/output device whose primary function is to assist in two-factor authentication. A card reader must have an interface to read data from a card and may be able to write data to a card. A card reader may have a means (such as buttons, keypad, touchscreen, etc.) for a user to input a PIN or password, as well as a limited display.

1.3.11 User Interface

A User Interface (UI) is something other than a Manual Local Input or Card Reader that allows a person to interact with the system or device. Note that while a Card Reader is not by itself a User Interface, a User Interface may contain a Card Reader in order for it to authenticate its user. Within control systems, there are a wide range of User Interfaces.

Two important distinctions are 1) whether the user interface is Local or Remote, and 2) the effective capabilities of the User Interface to alter data, which is the "privilege" of the user interface (where effective privilege available to a specific user at a specific user interface is the combination of the greatest privilege offered by the user interface and the specific account the user is logged into).

1.3.11.1 Local User Interface

A Local User Interface is a user interface where the physical hardware the user interacts with (keyboard, buttons, display, etc.) is physically part of the device being affected. All of the relevant characteristics of the user interface are embodied within a single device.

Note that a Local UI may be able to access data in a different device, Local versus Remote in this context refers to the user interface itself; the capability to access data in a different device is covered under "Full User Interface."

1.3.11.2 Remote User Interface

A Remote User Interface implements a Client/Server model where the physical hardware the user interacts with (Client) is physically distinct from the device being affected (Server). Most or all of the security and functionality characteristics of the user interface are defined by the Server, not the Client. The Client and Server communicate via a network connection. A common example of a remote user interface is a web-based interface where the browser (client) is generally on different hardware than the web server (server). A Remote UI remains a Remote UI even if the user happens to be at a Client on the same hardware as the Server. What is important is that a) the Client may be on different hardware than the Server and b) the majority of the security and functional characteristics of the interface are defined at the Server.

Note that this definition of "remote" is consistent with that generally used in the control industry but is not aligned with the NIST 800-53 definition of "Remote", which refers to "outside the system". The term "Remote" here better aligns with the NIST 800-53 definition of "Network" (remote from within the system) Access.

1.3.11.3 Types of User Interface (by capability)

User interfaces are also categorized by their capabilities as being Read Only, Limited, or Full.

1.3.11.3.1 Read-Only User Interface

A Read Only User Interface (also referred to as a View-Only User Interface) is a user interface that only allows for reading data, it does not allow (have the capability to) modify data. A Read Only User Interface may be either Local or Remote. A User Interface that is configured to be Read Only (by some other means than the interface itself, such as using configuration software on a laptop) is a Read-Only Interface. Note a Read Only User Interface may have buttons (or touch screen, etc.) allowing the user to navigate through the presentation of data.

Examples of a Read Only User Interfaces are a) a publicly viewable "energy dashboard" showing weather data and energy usage within a building and b) digital wayfinding signage.

1.3.11.3.2 Limited User Interface

A Limited User Interface is a user interface that - by design - can only alter information local to the user interface. Note that the determination of "alter" includes only direct interactions, it explicitly excludes interactions that might occur as secondary effects. For example, an interface changing the flow setpoint in a pump controller is a direct interaction, the subsequent change in flow (as well as any subsequent downstream changes in valve position) are not direct interactions.

Two examples of LIMITED UIs are: a) a variable speed drive has a Limited Local User Interface which allows the user to change properties within the

drive, but does not allow affecting things outside the drive; and b) a typical home WiFi Router has a Limited Remote User Interface which allows configuration of the Router, but does not allow direct interaction with other devices.

1.3.11.3.3 Full User Interface

A Full User Interface can alter information in devices outside the device with the user interface. For example, a typical Local Display Panel is a Full Local User Interface while a browser-based front end is a Full Remote User Interface.

1.3.11.3.4 View-Only User Interface

See Read-Only User Interface

1.3.11.4 Other User Interface Terminology

In addition to defining whether a user interface is a Hardware Limited, Read-Only, Limited or Full, and whether it is Local or Remote, user interfaces are classified by whether they are writable or privileged.

1.3.11.4.1 Writable User Interface

Any User Interface that is not Read-Only is Writable. (Limited User Interfaces and Full User Interfaces are both writable user interfaces (as they are capable of changing a value)).

1.3.11.4.2 Privileged User Interface

A Privileged UI is a UI that has sufficient capabilities or functionality that it requires specific cybersecurity measures to be put in place to limit its unauthorized use. Ultimately, whether a specific user interface is considered a Privileged User Interface must be determined by usage. Unless otherwise specified, user interfaces can be determined to be privileged or not using the following:

- a. Read-Only User Interfaces are not privileged user interfaces.
- b. Full User interfaces for Fire Alarm Systems are privileged user interfaces as indicated and shown, or when another requirement of this Section establishes they are privileged. For all other systems, Full User Interfaces are privileged user interfaces.
- c. User interfaces that allow for configuration of auditing or allows for modification or deletion of audit logs are privileged user interface.
- d. User interfaces that allow for reprogramming a network connected device is a privileged user interface.
- e. For Fire Protection Systems, User Interfaces that can inhibit or force the activation of a fire suppression system (e.g. such as for a pre-action or deluge system) are privileged user interfaces.
- f. Except as specified above, a Limited User Interface must be determined to be privileged or not based on the specific capabilities and use case of the user interface. In general however, user interfaces that do not offer significant capabilities above and beyond those available at that location via other means (e.g. such as a disconnect switch,

breaker, or hand-off-auto switch, or physical attack) are not privileged.

1.4 ADMINISTRATIVE REQUIREMENTS

1.4.1 Points of Contact

Coordinate with the following Points of Contact as indicated in this Section and as required. Not all projects will require coordination with all Points of Contact. When coordination is required and no Point of Contact is indicated, coordinate with The Contracting Office Representative (COR).

- a. Password Point of Contact: The Contracting Office Representative (COR)

1.4.2 Coordination

Coordinate the execution of this Section with the execution of all other Sections related to control systems as indicated in the paragraph RELATED REQUIREMENTS. Items that must be considered when coordinating project efforts include but are not limited to:

- a. The Cybersecurity Interconnection Schedule must be coordinated with other work that will be interconnected to, and interconnections must be approved by the Government before relying on them for system functionality.
- b. Cybersecurity testing support must be coordinated across control systems and with the Government cybersecurity testing schedule.
- c. Passwords must be coordinated with the indicated contact for the project site.
- d. Contractor Computer Cybersecurity Compliance Statements must be provided for each contractor using contractor owned computers.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Device Account Lock Exception Request; G

Multiple Ethernet Connection Device Request; G

Contractor Computer Cybersecurity Compliance Statements; G

Proposed STIG and SRG Applicability Report; G

SD-02 Shop Drawings

Network Communication Report; G

Cybersecurity Riser Diagram; G

SD-03 Product Data

Control System Cybersecurity Documentation; G

SD-06 Test Reports

Control System Cybersecurity Testing Procedures; G

Control System Cybersecurity Testing Report; G

SD-07 Certificates

Software Licenses; G

SD-11 Closeout Submittals

Password Change Summary Report; G Enclosure Keys; G

Software and Configuration Backups; G

Auditing Front End Software; G

System Maintenance Tool Software; G

Control System Scanning Tools; G

STIG, SRG and Vendor Guide Compliance Result Report; G

Control System Inventory Report; G

1.6 CYBERSECURITY DOCUMENTATION

{For Government Reference Only: This subpart (and its subparts) relates to PL-7; CCI-003071}

1.6.1 Proposed STIG and SRG Applicability Report

For each model of network connected or network infrastructure device, use the DISA SRG/STIG Applicability Guide and Collection Tool (available at <https://public.cyber.mil/stigs/SCAP/>) to identify applicable STIGs or SRGs and provide a report indicating applicable STIGs and SRGs for each model.

1.6.2 Network Communication Report

{For Government Reference Only: This subpart (and its subparts) relates to CA-9, PL-8; CCI-003075; CCI-002102, CCI-002103, CCI-002104, CCI-002105, CCI-003072, CCI-003073, CCI-003075 and also the submittal requirements associated with CM-6, CM-7, SC-8 and SC-41 including CM-7(3), CCI-000388.}

Provide a network communication report. For each networked device, document the communication characteristics of the device including communication protocols, services used, encryption employed, and a general description of what information is communicated over the network. For each device using IP, document all TCP and UDP ports used. For non-IP communications, document communication protocol and media used. If other control system Sections used on this project include submittals documenting this information, provide copies of those submittals to meet this requirement.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Network Communication Report as an editable Microsoft Excel file.

1.6.3 Control System Inventory Report

{For Government Reference Only: This subpart (and its subparts) relates to CM-8(a), SI-17, IA-3; CCI-000389, CCI-000392, CCI-000398, CCI-002773, CCI-002774, CCI-002775, CCI-000777, CCI-000778, CCI-001958}

Provide a Control System Inventory report using the Inventory Spreadsheet listed under this Section at

<https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-1> documenting all devices, including networked devices, network infrastructure devices, non-networked devices, input devices (e.g. sensors) and output devices (e.g. actuators). For each device provide all applicable information for which there is a field on the spreadsheet in accordance with the instructions on the spreadsheet.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Control System Inventory Report as an editable Microsoft Excel file.

1.6.4 Software and Configuration Backups

{For Government Reference Only: This subpart (and its subparts) relates to CP-10; CCI-000550, CCI-000551, CCI-000552}

For each computer on which software is installed under this project, provide a recovery image of the final as-built computer. This image must allow for bare-metal restore such that restoration of the image is sufficient to restore system operation to the imaged state without the need for re-installation of software. If additional user permissions are required to meet this requirement, coordinate the creation of the image with the identified Government Computer Access Point of Contact.

For all ethernet switches provide a backup of the switch configuration. For all controllers, provide a backup of the controller configuration and the source code for all loaded application programs (all software that is not common to every controller of the same manufacturer and model).

If any or all of these are provided under another Section, provide documentation indicating this and referencing those submittals.

1.6.5 Cybersecurity Riser Diagram

{For Government Reference Only: This subpart (and its subparts) relates to PL-2(a), PL-8; CCI-003051, CCI-003053, CCI-003072, CCI-003073, CCI-003075}

Provide a cybersecurity riser diagram of the complete control system including all network and device hardware. If the control system specifications require a riser diagram submittal, provide a copy of that submittal as the cybersecurity riser diagram. Otherwise, provide a riser diagram in one-line format.

1.6.6 STIG, SRG and Vendor Guide Compliance Result Report

For every component (device or software) with an applicable STIG or SRG in the Proposed STIG and SRG Applicability Report, provide a result report documenting compliance with the STIG or SRG requirements. For components which are scannable by the SCAP (security content automation protocol) tool (available online at <https://public.cyber.mil/stigs/scap>), provide the SCAP report and raw scan results.

For every component (device or software) with manufacturer provided cybersecurity documentation, procedure, or method for secure configuration or installation, provide a report documenting how the component was configured and any deviation from the manufacturer instructions.

1.6.7 Control System Cybersecurity Documentation

{For Government Reference Only: This subpart (and its subparts) relates to SA-5 (a),(b),(c); CCIs: CCI-003124, CCI-003125, CCI-003126, CCI-003127, CCI-003128, CCI-003129, CCI-003130, CCI-003131}

Provide a Control System Cybersecurity Documentation submittal containing the indicated information for each device and software application.

1.6.7.1 Software Applications

For all software applications running on computers provide:

- a. administrator documentation that describes secure configuration of the software {For Government Reference Only: relates to CCI-003124}
- b. administrator documentation that describes secure installation of the software {For Government Reference Only: relates to CCI-003125}
- c. administrator documentation that describes secure operation of the software {For Government Reference Only: relates to CCI-003124}
- d. administrator documentation that describes effective use and maintenance of security functions or mechanisms for the software {For Government Reference Only: relates to CCI-003127}
- e. administrator documentation that describes known vulnerabilities regarding configuration and use of administrative (i.e. privileged) functions for the software {For Government Reference Only: relates to CCI-003128}
- f. user documentation that describes user-accessible security functions or mechanisms in the software and how to effectively use those security functions or mechanisms {For Government Reference Only: relates to CCI-003129}
- g. user documentation that describes methods for user interaction which enables individuals to use the software in a more secure manner {For Government Reference Only: relates to CCI-003130}
- h. user documentation that describes user responsibilities in maintaining the security of the software {For Government Reference Only: relates to CCI-003131}

1.6.7.2 All Other Control System Devices

For all Control System Devices which do not FULLY support user accounts, provide:

- a. Documentation that describes secure configuration of the device; or a specific indication that there are no secure configuration steps that apply {For Government Reference Only: relates to CCI-003124}
- b. Documentation that describes effective use and maintenance of security functions or mechanisms for the device; or a specific indication that there are no security functions or mechanisms in the device {For Government Reference Only: relates to CCI-003127}
- c. For devices which include a user interface, documentation that describes methods for user interaction which enables individuals to use the device in a more secure manner {For Government Reference Only: relates to CCI-003130}

1.6.7.3 Default Requirements for Control System Devices

For control system devices where Control System Cybersecurity Documentation requirements are not otherwise indicated in this Section, provide:

- a. Documentation that describes secure configuration of the device {For Government Reference Only: relates to CCI-003124}
- b. Documentation that describes secure installation of the device {For Government Reference Only: relates to CCI-003125}
- c. Documentation that describes secure operation of the device {For Government Reference Only: relates to CCI-003124}
- d. Documentation that describes effective use and maintenance of security functions or mechanisms for the device {For Government Reference Only: relates to CCI-003127}
- e. Documentation that describes known vulnerabilities regarding configuration and use of administrative (i.e. privileged) functions for the device {For Government Reference Only: relates to CCI-003128}
- f. Documentation that describes user-accessible security functions or mechanisms in the device and how to effectively use those security functions or mechanisms {For Government Reference Only: relates to CCI-003129}
- g. Documentation that describes methods for user interaction which enables individuals to use the device in a more secure manner {For Government Reference Only: relates to CCI-003130}
- h. Documentation that describes user responsibilities in maintaining the security of the device {For Government Reference Only: relates to CCI-003131}

1.7 SOFTWARE LICENSING

{For Government Reference Only: This subpart (and its subparts) relates to SI-2(a), SI-2(c), SI-7(14); CCI-001227, CCI-002605, CCI-002737}

For all software provided that has not already been licensed to the government or project site, provide a license to the Government for a period of no less than 5 years, and the license must also include the following software updates:

- a. Security and bug-fix patches issued by the software manufacturer.
- b. Security patches to address any vulnerability identified in the National Vulnerability Database at <http://nvd.nist.gov> with a Common Vulnerability Scoring System (CVSS) severity rating of MEDIUM or higher.

Provide a single [Software Licenses](#) submittal with documentation of the software licenses for all software provided

1.8 CYBERSECURITY DURING CONSTRUCTION

{For Government Reference Only: This subpart (and its subparts) relates to AC-18, SA-3; CCI-000258}

In addition to the control system cybersecurity requirements indicated in this section, meet following requirement throughout the construction process.

1.8.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. Contractor computers connected to the control system, control system network, or a control system component at any point during construction must meet the following requirements:

1.8.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.8.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. Computers used on this project must be scanned using the installed software at least once per day.

1.8.1.3 Passwords and Passphrases

The passwords and passphrases for computers, applications, and web-based applications supporting passwords must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.8.1.4 User-Based Authentication

Each user must have a unique account; sharing of a single account between multiple users is prohibited.

1.8.1.5 Demonstration of Compliance

The Government has the right to require demonstration of computer compliance with these requirements at any time during the project.

1.8.1.6 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at

<https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-1>

Each Statement must be signed by a cybersecurity representative for the relevant company.

1.9 CYBERSECURITY DURING WARRANTY PERIOD

All work performed on the control system after acceptance must be performed using Government Furnished Equipment or equipment specifically and individually approved by the Government.

PART 2 PRODUCTS

All products used on this project must meet the indicated requirements, but not all products specified here will be required by every project.

2.1 ETHERNET SWITCH

Provide Open Systems Interconnection (OSI) Layer 2 Ethernet switches with the following capabilities, and with an interface to support switch configuration for these capabilities:

2.1.1 Required Functionality

Switches must:

- a. Copper Ethernet ports must auto negotiate for 10, 100 and 1000 megabits-per-second links.
- b. Be capable of implementing port level access control by MAC address and limit the number of MAC addresses to one MAC address per port.
- c. For LOW Impact Systems, be capable of implementing per-port access control lists (ACLs) where the list can be filtered by source and destination IP addresses, and by source and destination UDP or TCP ports.

2.1.2 Configuration Requirements

Switches must:

- a. Support configuration save and restore.
- b. Support both manual IP address assignment and acquisition of a dynamic IP address via Dynamic Host Configuration Protocol (DHCP).
- c. Be capable of limiting access for configuration to one or more of: a web interface using HTTPS, a command line interface using SSH, or an

SNMP connection using SNMP version 3 or later.

- d. Support the ability to lock configuration capability to a dedicated management port.

2.2 DAISY CHAIN IP CONTROLLERS

Controllers used as Daisy Chain IP Controllers must be IP controllers with exactly two Ethernet network connections and basic built-in switch capabilities to allow implementation of an Ethernet network in a daisy chain architecture. Switches incorporated by Daisy Chain IP Controllers are not required to meet the requirements for Ethernet Switches as defined in this Section.

PART 3 EXECUTION

3.1 CYBERSECURITY HARDENING AND CONFIGURATION GUIDES

Install, configure, and harden all hardware and software furnished on this project in accordance with manufacturer provided documentation, procedures, or methods for secure configuration or installation. Do not implement specific hardening actions if that action would conflict with required functionality or another requirement of this Section.

3.2 NETWORK REQUIREMENTS

3.2.1 Non-IP Control Networks

When control system specifications require particular communication protocols, use only those communication protocols and only as specified. Do not implement any other communication protocol.

When control system specifications do not indicate requirements for communication protocols, use only those protocols required for operation of the system as specified.

3.2.2 IP Control Networks

{For Government Reference Only: This subpart relates to CM-6(a), CM-7(a), CM-7(b), CM-7(1)(b), SC-41; CCI-001588, CCI-000381, CCI-000380, CCI-000381, CCI-000382, CCI-001761, CCI-001762, CCI-002544, CCI-002545, CCI-002546.}

IP Networks must be Ethernet networks and must use switches which are Ethernet Switches or Daisy Chain IP Controllers as defined in this Section. Do not use nonsecure functions, ports, protocols and services as defined in **DODI 8551.01** unless those ports, protocols and services are specifically required by the control system specifications or otherwise specifically authorized by the Government. Do not use ports, protocols and services that are not specified in the control system specifications or required for operation of the control system.

3.2.2.1 IP Network Routers

Do not install any device that performs IP routing.

3.2.2.2 IP Devices With Multiple Ethernet Connection

Except for Ethernet Switches and Daisy Chain IP Controllers, devices must

not have more than one Ethernet connection to IP networks unless doing so is required by the project specifications and the specific application is approved. If a device with Multiple Ethernet Connections to IP networks is required, provide a [Multiple Ethernet Connection Device Request](https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-1) using the Multiple Ethernet Connection Device Request Template at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-1> to request approval for each device. If a device with Multiple Ethernet Connections to IP networks is not required, instead provide a document stating that no approval is being requested.

3.2.3 Cryptographic Protection

{For Government Reference Only: This subpart relates to IA-2(9), IA-3(1), SC-8, SC-13, SC-23(1), SC-23(3); CCI-001942, CCI-001959, CCI-001967, CCI-002418, CCI-002449, CCI-002450, CCI-001185, CCI-001188, CCI-001664.}

All remote user interfaces must use HTTPS for all traffic between the user interface client and user interface server.

For devices that have STIG/SRGs related to cryptographic protection (CCI-002450), comply with the requirements of those STIG/SRGs. Ensure that all network traffic is encrypted using NSA-approved cryptography; provision of digital signatures and hashing, and FIPS-validated cryptography.

3.2.4 Device Identification and Authentication

{For Government Reference Only: This subpart (and its subparts) relates to IA-3; CCI-000777, CCI-000778, CCI-001958.}

All computers must support [IEEE 802.1x](#) for device authentication to the network.

3.2.4.1 Default Requirements for Control System Devices

For control system devices where Device Identification and Authentication requirements are not otherwise indicated in this Section: Devices using HTTP as a control protocol must use HTTPS instead.

3.2.5 Cryptographic Module Authentication

{For Government Reference Only: This subpart (and its subparts) relates to IA-7; CCI-000803}

For devices (including but not limited to [NIST FIPS 140-2](#) compliant radios) that have STIG/SRGs related to cryptographic module authentication (CCI-000803), comply with the requirements of those STIG/SRGs.

3.3 ACCESS CONTROL REQUIREMENTS

3.3.1 User Accounts

{For Government Reference Only: This subpart (and its subparts) relate to AC-2(a), AC-3, AC-6(1), AC-6(10), AC-6(2), AC-6(9), CM-11(2), and IA-2; CCI-002110, CCI-000213, CCI-002235, CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-002235, CCI-000039, CCI-001419, CCI-002234, CCI-001812, and CCI-000764.}

Any user interface supporting user accounts (either FULLY or MINIMALLY)

must limit access according to specified limitations for each account. Install and configure any device having a STIG or SRG in accordance with that STIG or SRG.

All user interfaces FULLY supporting accounts must implement user-based authentication where each account is uniquely assigned to a specific user. User interfaces FULLY supporting accounts must implement at least three (3) levels of user account privilege including: 1) an account with read-only permissions 2) an account with full permissions including account creation and modification and 3) an account with greater permissions than read-only but without account creation and modification.

3.3.1.1 Computers

All computer operating systems must FULLY support user accounts and implement accounts for access. Each control system software application not supporting accounts and running on a computer must be installed such that use of the software is restricted by the computer operating system to specific users.

Applications running on computers shall not require the user be logged in to a computer operating system administrator account for normal operation. It is permissible to require the computer operating system administrator account for initial application installation and configuration.

3.3.1.2 Controllers

For user interfaces provided by controllers, provide access control in accordance with the User Interface Requirements table for the applicable control system and user interface type.

- a. For table entries of "NA": NA means Not Applicable, there are no interfaces in this category.
- b. For table entries of "None Required": The user interface is not required to support user accounts.
- c. For table entries of "MINIMALLY": The user interface must at least MINIMALLY support user accounts.
- d. For table entries of "FULLY": The user interface must at FULLY support user accounts.
- e. For table entries of "KEY": The user interface must have physical security in the form of either a key lock on the interface itself or be furnished inside a locked enclosure. Where this is required for a read only interface, this lock must prevent viewing of data on the interface; for other interfaces, this lock must prevent using the interface to alter data.
- f. For table entries of "Physical Security": For Local FULL interfaces, the interface must be located inside mission space. For Local Limited (not FULL) interfaces, the user interface must either a) be located within mission space or b) be protected by physical security at least as good as the control devices (and equipment controlled by the control devices) affected by the interface. For purposes of this requirement, 'affected' includes controllers with data that can be directly altered by the interface, as well as mechanical and/or

electrical equipment directly controlled by those controllers, but does not include other interactions.

- g. Entries of the form "X and Y" must meet both the requirement indicated for X and the requirement indicated for Y. For example, an entry of "MINIMALLY and Physical Security" indicates the user interface must both MINIMALLY support accounts and have physical security.
- h. Entries of the form "X or Y" must meet either the requirement indicated for X or the requirement indicated for Y.

3.3.1.2.1 Default Requirements for Other Control Systems

For control system devices where User Interface Requirements are not otherwise indicated in this Section, use the Default User Interface Requirements tables.

Default User Interface Requirements for LOW Impact Control Systems	
<u>User Interface Type</u>	<u>Access Control Requirement</u>
Local Read Only (see note 1)	MINIMALLY
Local Limited, Non-privileged	MINIMALLY
Local Limited, Privileged	MINIMALLY
Local Full	FULLY
Remote Read Only	MINIMALLY
Remote Limited, Non-Privileged	MINIMALLY
Remote Limited, Privileged AND Remote Full (see note 2)	FULLY
Notes: 1)Local Read Only User Interfaces are always Non-Privileged 2)Remote Full User Interfaces are always Privileged	

3.3.1.3 Additional User Account Expiration Requirements In MODERATE Impact Systems:

In addition to other user account requirements, user account expiration and auditing must be configured as indicated.

3.3.1.3.1 For Control System Applications Running on Computers

If temporary accounts are supported, expire temporary accounts 72 hours after creation. Expire all other accounts after 35 days of inactivity.

3.3.1.3.2 For Other Control System Devices FULLY Supporting Accounts

If temporary accounts are supported, expire temporary accounts 72 hours after creation. Expire all other accounts after 365 days of inactivity.

3.3.2 Unsuccessful Logon Attempts

{For Government Reference Only: This subpart (and its subparts) relate to AC-7 (a), AC-7 (b); CCI-000043, CCI-000044, CCI-001423, CCI-002236, CCI-002237, CCI-002238}

Except for high availability user interfaces indicated as exempt, devices must meet the indicated requirements for handling unsuccessful logon attempts. If a device cannot meet these requirements, document device capabilities to protect from subsequent logon attempts and propose alternate protections in a [Device Account Lock Exception Request](#) submittal. Do not implement alternate protection measures in lieu of the indicated requirements without explicit permission from the Government. If no Device Account Lock Exceptions are requested, provide a document stating that no approval is being requested as the Device Account Lock Exception Request.

3.3.2.1 Devices MINIMALLY Supporting Accounts

For LOW Impact Systems: Devices which MINIMALLY (but not FULLY) support accounts are not required to lock based on unsuccessful logon attempts.

3.3.2.2 Devices FULLY Supporting Accounts

Devices which FULLY support accounts must meet the following requirements.

- a. It must lock the user account when three unsuccessful logon attempts occur within a 15 minute interval.
- b. Once an account is locked, the account must stay locked until unlocked by an administrator. If the account being locked is the sole administrator account on the device, the account must stay locked for 1 hour and then automatically unlock.
- c. Once the indicated number of unsuccessful logon attempts occurs, delay further logon prompts by 5 seconds.

3.3.2.3 High Availability Interfaces Exempt from Unsuccessful Logon Attempts Requirements

There are no high availability interfaces which are exempt from unsuccessful logon attempts requirements.

3.3.3 System Use Notification

{For Government Reference Only: This subpart (and its subparts) relates to AC-8; CCI-000048, CCI-002247, CCI-002243, CCI-002244, CCI-002245, CCI-002246, CCI-000050, CCI-002248}

3.3.3.1 System Use Notification for Remote User Interfaces

Remote user interfaces must display a warning banner meeting the requirements of [DTM 08-060](#) on screen.

3.3.3.2 System Use Notification for Local User Interfaces

Devices which are connected to a network and have a local user interface must display a warning banner meeting the requirements of [DTM 08-060](#) on

the user interface screen if capable of doing so and must have a permanently affixed label with an approved banner from DTM 08-060 if unable to display the warning banner on the screen. Where it is impractical (perhaps due to device size) to affix the label to the device, affix the label to the device enclosure.

Labels must be machine printed or engraved, plastic or metal, designed for permanent installation, must use a font no smaller than 14 point, and must provide a high contrast between font and background colors.

3.3.4 Permitted Actions Without Identification or Authentication

{For Government Reference Only: This subpart (and its subparts) relates to AC-14; CCI-000061, CCI-000232}

The control system must require identification and authentication before allowing any actions except read-only actions by a user acting from a user interface which MINIMALLY or FULLY supports accounts.

3.3.5 Enclosures

Prior to final acceptance of the system, lock all lockable enclosures. Submit an Enclosure Keys submittal with all copies of keys for all enclosures and a key inventory list documenting all keys. Label each key with the matching enclosure identifier.

3.4 USER IDENTIFICATION AND AUTHENTICATION

{For Government Reference Only: This subpart (and its subparts) relates to IA-2, IA-2(1), IA-2(12), IA-5 IA-5(b), IA-5(c), IA-5(e), IA-5(g), IA-5(1), IA-5(11); CCI-000764, CCI-000765, CCI-001953, CCI-001954, CCI-001544, CCI-001989, CCI-000182, CCI-001610, CCI-000192, CCI-000193, CCI-000194, CCI-000205, CCI-001619, CCI-001611, CCI-001612, CCI-001613, CCI-001614, CCI-000195, CCI-001615, CCI-000196, CCI-000197, CCI-000199, CCI-000198, CCI-001616, CCI-001617, CCI-000200, CCI-001618, CCI-002041, CCI-002002, CCI-002003.}

This subpart indicates requirements for specific methods of identification and authentication for users and user accounts. Where these requirements conflict apply the following order of precedence: 1) If present, Device Specific Requirements take precedence over any other requirements; and then 2) multifactor authentication requirements take precedence over password requirements.

3.4.1 User Identification and Authentication for All System Types

Unless otherwise indicated, all user interfaces supporting accounts (either FULLY or MINIMALLY) must implement Identification and Authorization via passwords.

For LOW Impact Systems: User interfaces provided by computer operating systems must implement multifactor authentication via PIV.

3.4.2 Implementation of Identification and Authorization Requirements

Identification and Authorization must be met by one of the following methods:

- a. Direct implementation in the user interface.

- b. For user interfaces on a computer: inheriting the Identification and Authorization from the computer operating system, either by the operating system limiting access to specific applications by user, or by the application itself having permissions based on the user logged into the computer.
- c. For remote interfaces: an implementation shared between the remote user interface server and the remote user interface client. For example, a requirement for PIV authentication may be met on a remote user interface by a PIV reader on a web browser client which sends the authentication information via HTTPS to the remote server.

3.4.3 Password-Based Authentication Requirements

3.4.3.1 Passwords for Software and Applications Running on Computers

All software and applications running on computers supporting password-based authentication must enforce the following requirements:

- a. Minimum password length of 12 characters
- b. Password must contain at least one uppercase character.
- c. Password must contain at least one lowercase character.
- d. Password must contain at least one numeric character.
- e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.
- f. Password must have a minimum lifetime of 24 hours.
- g. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.
- h. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters (where location is significant, a character may be reused if it is in a different position).
- i. Passwords must be cryptographically protected during storage and transmission.

3.4.3.2 Passwords for Controllers FULLY Supporting Accounts

All controllers FULLY supporting accounts and supporting password-based authentication must enforce the following requirements:

- a. Minimum password length of twelve (12) characters.
- b. Password must contain at least one uppercase character.
- c. Password must contain at least one lowercase character.
- d. Password must contain at least one numeric character.

- e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.
- f. Password must have a maximum lifetime of sixty (60) days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.
- g. Password must differ from previous five (5) passwords, where differ is defined as changing at least fifty percent of the characters.
- h. Passwords must be cryptographically protected during storage and transmission.

3.4.3.3 Passwords for Remote Interfaces

Passwords for connecting to a Remote User Interface supporting password-based authentication must enforce the following requirements:

- a. Minimum password length of twelve (12) characters.
- b. Password must contain at least one uppercase character.
- c. Password must contain at least one lowercase character.
- d. Password must contain at least one numeric character.
- e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.
- f. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.
- g. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters (where location is significant, a character may be reused if it is in a different position).
- h. Passwords must be cryptographically protected during storage and transmission.

3.4.3.4 Passwords for Devices Minimally Supporting Accounts

Devices MINIMALLY supporting accounts must support passwords with a minimum length of four characters.

3.4.3.5 Password Configuration and Reporting

For all devices with a password, coordinate the changing of passwords with the project site following testing of the system but prior to turnover to the Government. Coordinate with Password Point of Contact to determine appropriate project site personnel to complete password changes. Accompany identified personnel to each device with a password and instruct personnel on the process of changing password. Record the time, date and personnel present when each device's password is changed and submit a [Password Change Summary Report](#) documenting this information.

Provide the Password Summary Report electronically in both PDF and Microsoft Excel.

3.4.4 Authenticator Feedback

{For Government Reference Only: This subpart relates to IA-6; CCI-000206}

Devices must never show authentication information, including passwords, on a display. Devices that momentarily display a character as it is entered, and then obscure the character, are acceptable. For devices that have STIGs or SRGs related to obscuring of authenticator feedback (CCI-000206), comply with the requirements of those STIGs/SRGs.

3.5 CYBERSECURITY AUDITING

Where an auditing requirement exists for email notification, notify via email the application administrator and Information System Security Officer (ISSO) of the event. Coordinate with the Email Address Point of Contact for email addresses. If outgoing email is not available to the system, configure the system for these notifications for future support of outgoing email.

3.5.1 Audit Events, Content of Audit Records, and Audit Generation

{For Government Reference Only: This subpart (and its subparts) relates to AU-2(a), AU-2(c), AU-2(d), AU-3, AU-10, AU-12, AU-13(3), AU-14(b), AU-14(1), AU-14(2), AU-14(3), CM-5(1), SC-7 (9); CCI-000123, CCI-001571, CCI-000125, CCI-001485, CCI-000130, CCI-000131, CCI-000132, CCI-00133, CCI-000134, CCI-001487, CCI-000166, CCI-001899, CCI-000169, CCI-001459, CCI-000171, CCI-000172, CCI-001910, CCI-001914, CCI-001919, CCI-001464, CCI-001462, CCI-001920, CCI-001814, CCI-002400.

For devices that have STIG/SRGs related to audit events, content of audit records or audit generation, comply with the requirements of those STIG/SRGs.

If auditing requirements can be met using existing control system alarm or event capabilities, those existing capabilities may be used to meet these requirements.

3.5.1.1 Computers

For each computer, provide the capability to select audited events and the content of audit logs. Configure computers to audit the indicated events, and to record the indicated information for each auditable event.

3.5.1.1.1 Audited Events

Configure each computer to audit the following events:

- a. Successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels)
- b. Successful and unsuccessful logon attempts
- c. Successful logouts
- d. Privileged activities or other system level access

- e. Concurrent logons from different workstations
- f. Successful and unsuccessful accesses to objects
- g. All program initiations
- h. All direct access to the information system
- i. All account creations, modifications, disabling, and terminations.
- j. All kernel module load, unload, and restart

3.5.1.1.2 Audit Event Information To Record

Configure each computer to record, for each auditable event, the following information (where applicable to the event):

- a. What type of event occurred
- b. When the event occurred
- c. Where the event occurred
- d. The source of the event
- e. The outcome of the event
- f. The identity of any individuals or subjects associated with the event

3.5.1.2 Default Requirements for Control System Controllers

For control system controllers where Audit Events, Content of Audit Records, and Audit Generation are not otherwise indicated in this Section:

3.5.1.2.1 Controllers Which FULLY Support Accounts

For each controller which FULLY supports accounts, provide the capability to select audited events and the content of audit logs. Configure controllers to audit the indicated events, and to record the indicated information for each auditable event.

3.5.1.2.1.1 Audited Events

Configure each controller to audit the following events:

- a. Successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels).
- b. Successful and unsuccessful logon attempts.
- c. Successful logouts.
- d. Concurrent logons from different workstations.
- e. All account creations, modifications, disabling, and terminations. For MODERATE Impact Systems, also provide email notification when these audit events occur.

- f. All kernel module load, unload, and restart.
- g. For privileged user interfaces in MODERATE Impact Systems: All user commands.

3.5.1.2.1.2 Audit Event Information To Record

Configure each controller to record, for each auditable event, the following information (where applicable to the event):

- a. what type of event occurred
- b. when the event occurred
- c. where the event occurred
- d. the source of the event
- e. the outcome of the event
- f. the identity of any individuals or subjects associated with the event
- g. For privileged user interfaces in MODERATE Impact Systems: Full text recording of the executed command and the user executing the command.

For MODERATE Impact Systems: Audit records must provide sufficient detail to reconstruct events to determine cause of compromise and magnitude of damage, malfunction, or security violation.

3.5.1.2.2 Controllers Which Do Not FULLY Support Accounts

For each controller which does not FULLY support accounts configure the controller to audit all controller shutdown and startup events and to record for each event the type of event and when the event occurred.

3.5.2 Audit Time Stamps

{For Government Reference Only: This subpart (and its subparts) relates to AU-8; CCI-000159, CCI-001889, CCI-001890. For MODERATE Impact systems, this subpart (and its subparts) also relates to AU-8 (1); CCI-001891, CCI-001892, CCI-002046.}

Any device (computer or controller) generating audit records must have an internal clock capable of providing time with a resolution of one second. Clocks must not drift more than 10 seconds per day. Configure the system so that each device (computer or controller) generating audit records maintains accurate time to within 1 second. Note that if the control system specifications include requirement for clocks, the most stringent requirement applies.

3.5.3 Auditing Front End Software

The project site currently has the following software to support control system auditing: none. If there is no existing auditing front end software or the software is not compatible with the provided control systems, provide Auditing Front End Software with audit log import and upload, export, notification, and analysis functionality. The Auditing Front End Software may be provided as a component of the control system

front end or as a separate software package, and a single package may serve multiple control systems provided under the same projects if they are sharing a cybersecurity authorization.

When the Auditing Front End Software is neither existing nor installed under the requirements of another Section, furnish the Auditing Front End Software media and license and install the software on the control system front end computer in the electrical room of the additive injector building. Submit copies of Auditing Front End Software if this function is not part of the software provided with the control system to meet requirements of other Sections.

3.5.3.1 Import and Upload Requirements

Auditing Front End Software must be capable of importing audit logs from the Device Audit Record Upload Software and of uploading audit logs over the network from all control system devices supporting network upload of audit logs.

3.5.3.2 Export Requirements

Auditing Front End Software must be capable of exporting to a file format supported by Microsoft Excel.

3.5.4 Audit Storage Capacity and Audit Upload

{For Government Reference Only: This subpart (and its subparts) relates to AU-4; CCI-001848, CCI-001849}

The creation of audit records must never interfere with normal device operation. Devices must cease collection of auditing information if required to maintain normal operation.

- a. For devices that have STIG/SRGs related to audit storage capacity (CCI-001848 or CCI-001849) comply with the requirements of those STIG/SRGs.
- b. For controllers capable of generating audit records, provide 60 days worth of secure local storage, assuming 10 auditable events per day.

3.5.4.1 Device Audit Record Upload Software

For each device (computer or controller) required to audit events and for which audit logs cannot be uploaded over the network by the Auditing Front End Software, provide and license to the Government software implementing a secure mechanism of uploading audit records from the device and exporting them to the Auditing Front End Software. Where different devices use different software, provide software of each type required to upload audit logs from all devices.

When Device Audit Record Upload Software is capable of uploading audit logs over the network, install Device Audit Record Upload Software on the same computer as the Auditing Front End Software. Submit copies of device audit record upload software if this function is not part of the software provided with the control system to meet requirements of other Sections. If there are no devices requiring this software, provide a document stating this in lieu of this submittal.

3.5.5 Response to Audit Processing Failures

{For Government Reference Only: This subpart (and its subparts) relates to AU-5; CCI-000139, CCI-000140, CCI-001490.}

In the case of a failure in the auditing system, computers associated with auditing must provide email notification. In case of an audit failure, if possible, continue to collect audit records by overwriting existing audit records.

3.6 REQUIREMENTS FOR LEAST FUNCTIONALITY

{For Government Reference Only: This subpart (and its subparts), along with the network communication report submittal specified elsewhere in this section, relates to CM-6(a), CM-6(c), CM-7, CM-7(1)(b), SC-41; CCI-000363, CCI-000364, CCI-000365, CCI-001588, CCI-001755, CCI-000381, CCI-000380, CCI-00382, CCI-001761, CCI-001762, CCI-002544, CCI-002545, CCI-002546.}

For devices that have a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port and device I/O access for least functionality), install and configure the device in accordance with that STIG or SRGs.

3.6.1 Device Capabilities

Unless specifically required by the government, do not provide a capability to update device firmware over the network.

3.6.2 Software

For software that has a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port access for least functionality), install and configure the software in accordance with that STIG or SRG.

3.7 SYSTEM AND COMMUNICATION PROTECTION

3.7.1 Collaborative Computing

{For Government Reference Only: This subpart relates to SC-15(a), SC-15(b); CCI-001150, CCI-001152.}

Without explicit approval from the project site, control systems must not use collaborative computing technologies.

3.7.2 Denial of Service Protection

{For Government Reference Only: This subpart relates to SC-5, SC-39, SC-7(a); CCI-001093, CCI-002385, CCI-002386, CCI-002430, CCI-001097.}

To the greatest extent practical, implement control logic without reliance on the network. Except when required to meet the requirements of the control system Section (where the requirement can only be met using computer hardware), do not implement control logic in computers.

3.8 SAFE MODE AND FAIL SAFE OPERATION

{For Government Reference Only: This subpart (and its subparts) relates

to CP-12, SI-10(3), SI-17; CCI-002855, CCI-002856, CCI-002857, CCI-002754, CCI-002773, CCI-002774, CCI-002775}

For all control system components with an applicable STIG or SRG, configure the component in accordance with all applicable STIGs and SRGs.

3.9 SYSTEM MAINTENANCE TOOL SOFTWARE

{For Government Reference Only: This subpart (and its subparts) relates to MA-3; CCI-000865.}

Submit and license to the Government all software required to operate, maintain and modify the control system such the Government or their agents are able to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer. Submit hard copies of user manuals for each software with the software submittal.

For software provided and licensed to the Government under the requirements of another Section, submit a statement indicating the Section and Submittal under which the software was provided. For software provided to meet the requirements of this Section and not provided and licensed under another Section, submit software and software user manuals on DVD or CD as a Technical Data Package and submit one hard copy of the software user manual for each piece of software.

3.10 VULNERABILITY SCANNING

{For Government Reference Only: This subpart (and its subparts) relates to RA-5 RA-5(a), RA-5(b), RA-5(c), RA-5(d); CCI-001054, CCI-001055, CCI-000156, CCI-001641, CCI-001643, CCI-001057, CCI-001058, CCI-001059.

All IP devices must be scannable, such that the device can be scanned by industry standard IP network scanning utilities without harm to the device, application, or functionality.

3.10.1 Computers and Software Running on Computers

Computers and applications running on computers must meet relevant vulnerability scanning STIGs/SRGs and respond to approved DoD vulnerability scanning tools.

3.10.2 Controllers

Controllers shall be scannable by standard control system discovery tools or control system browsers and return meaningful status information including the network inputs and outputs for the controller. This information shall contain sufficient detail to detect vulnerabilities or exploits of the controller.

Provide all software needed to scan the control system as the [Control System Scanning Tools](#) submittal. If the software required to scan the system is already installed at the project site or is provided under a separate section instead provide a statement indicating this.

3.11 SYSTEM AND INTEGRATION INTEGRITY

3.11.1 Malicious Code Protection

{For Government Reference Only: This subpart (and its subparts) relates to SI-3(c); CCI-001241, CCI-002623}

For all computers installed under this project, provide malware protection software media, provide licenses, and install and configure malware protection software as indicated. Coordinate with the Government Computer Access Point of Contact as required.

- a. Provide malware protection software licenses.
- b. Provide malware protection software media.
- c. Install and configure malware protection software in accordance with the relevant STIGs.

3.12 CONTROL SYSTEM CYBERSECURITY TESTING

{For Government Reference Only: For MODERATE Impact systems, this subpart (and its subparts) relates to SA-11(a), SA-11(b), SA-11(c), SA-11(d), SA-11(e); CCI-003171, CCI-003172, CCI-003173, CCI-003174, CCI-003175, CCI-003176, CCI-003177, CCI-003178.}

3.12.1 Control System Cybersecurity Testing Procedures

Prepare Control System Cybersecurity Testing Procedures explaining step-by-step, the actions and expected results that will demonstrate that the control system meets the requirements of this Section.

Submit 4 copies of the Control System Cybersecurity Testing Procedures. The Control System Cybersecurity Testing Procedures may be submitted as a Technical Data Package.

3.12.2 Control System Cybersecurity Testing Execution

Using the Control System Cybersecurity Testing Procedures verify that the control system meets the requirements of this Section. UNLESS GOVERNMENT WITNESSING OF A TEST IS SPECIFICALLY WAIVED BY THE GOVERNMENT, PERFORM ALL TESTS WITH A GOVERNMENT WITNESS. If testing reveals deficiencies in the system, correct the deficiency and retest until successful.

3.12.3 Control System Cybersecurity Testing Report

Prepare and submit a Control System Cybersecurity Testing Report documenting all tests performed and their results. Include all tests in the Control System Cybersecurity Testing Procedures and any additional tests performed during testing. Document test failures and repairs conducted with the test results.

Submit four copies of the Control System Cybersecurity Testing Report. The Control System Cybersecurity Testing Report may be submitted as a Technical Data Package.

3.13 FIELD QUALITY CONTROL, CYBERSECURITY VALIDATION SUPPORT

In addition to testing and testing support required by other Sections,

provide a minimum of 8 hours of technical support for cybersecurity testing of control systems to support the DoD Risk Management Framework process Cybersecurity assessment of the control system. This support is independent of (and in addition to) the Control System Cybersecurity Testing specified in this section.

3.14 CYBERSECURITY TRAINING

Provide eight hours of classroom training for six Government personnel on the cybersecurity operation and maintenance of the control system provided. This training is in addition to and must be coordinated with control system training specified in other Sections.

The Government will provide the training location. Training must cover, at a minimum: (a) applying software and firmware updates, (b) user account creation, modification and deletion, (c) audit log upload procedures and (d) identification of privileged user interfaces and system impact of those interfaces. Training session must include a question and answer period during which government staff questions about cybersecurity aspects of the control system are answered.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 08 00

APPARATUS INSPECTION AND TESTING

11/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Qualifications
 - 1.4.2 Acceptance Tests and Inspections Reports
 - 1.4.3 Acceptance Test and Inspections Procedure

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 ACCEPTANCE TESTS AND INSPECTIONS
- 3.2 SYSTEM ACCEPTANCE
- 3.3 PLACING EQUIPMENT IN SERVICE

-- End of Section Table of Contents --

SECTION 26 08 00

APPARATUS INSPECTION AND TESTING
11/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2021) Standard for Acceptance Testing
Specifications for Electrical Power
Equipment and Systems

1.2 RELATED REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM applies to this section with additions and modifications specified herein.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Acceptance Tests and Inspections; G

SD-07 Certificates

Qualifications of Organization, and Lead Engineering Technician; G

Acceptance Test and Inspections Procedure; G

1.4 QUALITY ASSURANCE

1.4.1 Qualifications

Engage the services of a qualified testing organization to provide inspection, testing, calibration, and adjustment of the electrical distribution system and generation equipment listed in paragraph ACCEPTANCE TESTS AND INSPECTIONS herein. Organization must be independent of the supplier, manufacturer, and installer of the equipment. The organization must be a first tier subcontractor. No work required by this section of the specification may be performed by a second tier subcontractor.

- a. Submit name and qualifications of organization. Organization must have been regularly engaged in the testing of electrical materials, devices, installations, and systems for a minimum of 5 years. The

organization must have a calibration program, and test instruments used must be calibrated in accordance with **NETA ATS**.

- b. Submit name and qualifications of the lead engineering technician performing the required testing services. Include a list of three comparable jobs performed by the technician with specific names and telephone numbers for reference. Testing, inspection, calibration, and adjustments must be performed by an engineering technician, certified by NETA (Level III) or the National Institute for Certification in Engineering Technologies (NICET) with a minimum of 5 years' experience inspecting, testing, and calibrating electrical distribution and generation equipment, systems, and devices.

1.4.2 Acceptance Tests and Inspections Reports

Submit certified copies of inspection reports and test reports. Include certification of compliance with specified requirements, identify deficiencies, and recommend corrective action when appropriate. Type and neatly bind test reports to form a part of the final record. Submit test reports documenting the results of each test not more than 10 days after test is completed.

1.4.3 Acceptance Test and Inspections Procedure

Submit test procedure reports for each item of equipment to be field tested at least 45 days prior to planned testing date. Do not perform testing until after test procedure has been approved.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 ACCEPTANCE TESTS AND INSPECTIONS

Testing organization will perform acceptance tests and inspections. Test methods, procedures, and test values must be performed and evaluated in accordance with **NETA ATS**, the manufacturer's recommendations, and paragraph FIELD QUALITY CONTROL of each applicable specification section. Tests identified as optional in **NETA ATS** are not required unless otherwise specified. Place equipment in service only after completion of required tests and evaluation of the test results have been completed. Supply to the testing organization complete sets of shop drawings, settings of adjustable devices, and other information necessary for an accurate test and inspection of the system prior to the performance of any final testing. Notify Contracting Officer at least 14 days in advance of when tests will be conducted by the testing organization. Perform acceptance tests and inspections on applicable equipment and systems specified in the following sections:

- a. Section **26 32 15.00** ENGINE-GENERATOR SET STATIONARY 15-2500 KW, WITH AUXILIARIES. Functional engine shutdown tests, vibration base-line test, and load bank test will not be performed by the testing organization, but by the start-up engineer.
- b. Section **26 12 19.10** THREE-PHASE, LIQUID-FILLED PAD-MOUNTED TRANSFORMERS
- c. Section **33 71 02** UNDERGROUND ELECTRICAL DISTRIBUTION. Medium voltage

cables and grounding systems only.

- d. Section 26 13 00 SF6/HIGH-FIREPOINT FLUID INSULATED PAD-MOUNTED SWITCHGEAR

3.2 SYSTEM ACCEPTANCE

Final acceptance of the system is contingent upon satisfactory completion of acceptance tests and inspections.

3.3 PLACING EQUIPMENT IN SERVICE

A representative of the approved testing organization must be present when equipment tested by the organization is initially energized and placed in service.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 12 19.10

THREE-PHASE, LIQUID-FILLED PAD-MOUNTED TRANSFORMERS

05/19, CHG 1: 11/19

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 DEFINITIONS
- 1.4 SUBMITTALS
 - 1.4.1 Reduced Submittal Requirements
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Pad-Mounted Transformer Drawings
 - 1.5.2 Regulatory Requirements
 - 1.5.3 Standard Products
 - 1.5.3.1 Alternative Qualifications
 - 1.5.3.2 Material and Equipment Manufacturing Date
- 1.6 MAINTENANCE
 - 1.6.1 Additions to Operation and Maintenance Data

PART 2 PRODUCTS

- 2.1 PRODUCT COORDINATION
- 2.2 THREE-PHASE PAD-MOUNTED TRANSFORMERS
 - 2.2.1 Compartments
 - 2.2.1.1 High Voltage, Dead-Front
 - 2.2.1.2 Low Voltage
 - 2.2.2 Transformer
 - 2.2.2.1 Specified Transformer Efficiencies
 - 2.2.3 Insulating Liquid
 - 2.2.3.1 Liquid-Filled Transformer Nameplates
 - 2.2.4 Corrosion Protection
- 2.3 WARNING SIGNS AND LABELS
- 2.4 ARC FLASH WARNING LABEL
- 2.5 GROUNDING AND BONDING
- 2.6 CAST-IN-PLACE CONCRETE
- 2.7 SOURCE QUALITY CONTROL
 - 2.7.1 Transformer Test Schedule
 - 2.7.2 Design Tests
 - 2.7.3 Routine and Other Tests

PART 3 EXECUTION

- 3.1 INSTALLATION
- 3.2 GROUNDING
 - 3.2.1 Grounding Electrodes
 - 3.2.2 Pad-Mounted Transformer Grounding
 - 3.2.3 Connections
 - 3.2.4 Grounding and Bonding Equipment
- 3.3 INSTALLATION OF EQUIPMENT AND ASSEMBLIES

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.4 FIELD APPLIED PAINTING
- 3.5 FOUNDATION FOR EQUIPMENT AND ASSEMBLIES
 - 3.5.1 Cast-In-Place Concrete
 - 3.5.2 Sealing
- 3.6 FIELD QUALITY CONTROL
 - 3.6.1 Performance of Acceptance Checks and Tests
 - 3.6.1.1 Pad-Mounted Transformers
 - 3.6.1.2 Grounding System
 - 3.6.1.3 Surge Arresters, Medium- and High-Voltage
 - 3.6.2 Follow-Up Verification

-- End of Section Table of Contents --

SECTION 26 12 19.10

THREE-PHASE, LIQUID-FILLED PAD-MOUNTED TRANSFORMERS
05/19, CHG 1: 11/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A240/A240M	(2020a) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM D92	(2012a) Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester
ASTM D97	(2017b) Standard Test Method for Pour Point of Petroleum Products
ASTM D877/D877M	(2019) Standard Test Method for Dielectric Breakdown Voltage of Insulating Liquids Using Disk Electrodes
ASTM D1535	(2014; R 2018) Standard Practice for Specifying Color by the Munsell System

FM GLOBAL (FM)

FM APP GUIDE	(updated on-line) Approval Guide http://www.approvalguide.com/
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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 386	(2016) Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV
IEEE C2	(2017; Errata 1-2 2017; INT 1 2017) National Electrical Safety Code
IEEE C37.47	(2011) Standard for High Voltage Distribution Class Current-Limiting Type Fuses and Fuse Disconnecting Switches
IEEE C57.12.00	(2021) General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers
IEEE C57.12.28	(2014) Standard for Pad-Mounted Equipment - Enclosure Integrity

IEEE C57.12.34	(2015) Standard Requirements for Pad-Mounted, Compartmental-Type, Self-Cooled, Three-Phase Distribution Transformers, 10 MVA and Smaller; High Voltage, 34.5 kV Nominal System Voltage and Below; Low Voltage, 15 kV Nominal System Voltage and Below
IEEE C57.12.80	(2010) Standard Terminology for Power and Distribution Transformers
IEEE C57.12.90	(2021) Test Code for Liquid-Immersed Distribution, Power, and Regulating Transformers
IEEE C57.98	(2011) Guide for Transformer Impulse Tests
IEEE C62.11	(2020) Standard for Metal-Oxide Surge Arresters for Alternating Current Power Circuits (>1kV)
IEEE Stds Dictionary	(2009) IEEE Standards Dictionary: Glossary of Terms & Definitions

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS	(2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems
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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 260	(1996; R 2004) Safety Labels for Pad-Mounted Switchgear and Transformers Sited in Public Areas
NEMA Z535.4	(2011; R 2017) Product Safety Signs and Labels

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code
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ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT (OECD)

OECD Test 203	(1992) Fish Acute Toxicity Test
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U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 712-C-98-075	(1998) Fate, Transport and Transformation Test Guidelines - OPPTS 835.3100- "Aerobic Aquatic Biodegradation"
EPA 821-R-02-012	(2002) Methods for Measuring the Acute Toxicity of Effluents and Receiving Waters to Freshwater and Marine Organisms

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 431

Energy Efficiency Program for Certain
Commercial and Industrial Equipment

UNDERWRITERS LABORATORIES (UL)

UL 467

(2013; Reprint Jun 2017) UL Standard for
Safety Grounding and Bonding Equipment

1.2 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING applies to this section, with the additions and modifications specified herein.

1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE Stds Dictionary.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Pad-mounted Transformer Drawings; G

SD-03 Product Data

Pad-mounted Transformers; G

SD-06 Test Reports

Acceptance Checks and Tests; G

SD-07 Certificates

Transformer Efficiencies; G

SD-09 Manufacturer's Field Reports

Transformer Test Schedule; G

Pad-mounted Transformer Design Tests; G

Pad-mounted Transformer Routine and Other Tests; G

SD-10 Operation and Maintenance Data

Transformer(s), Data Package 5; G

1.4.1 Reduced Submittal Requirements

Transformers designed and manufactured by ABB in Jefferson City, MO; by Eaton's Cooper Power Series Transformers in Waukesha, WI; by ERMCO in Dyersburg, TN; or by Howard Industries in Laurel, MS need not submit the entire submittal package requirements of this contract. Instead, submit the following items:

- a. A certification, signed by the manufacturer, stating that the manufacturer will meet the technical requirements of this specification.
- b. An outline drawing of the transformer with devices identified (paragraph PAD-MOUNTED TRANSFORMER DRAWINGS, item a).
- c. ANSI nameplate data of the transformer (paragraph PAD-MOUNTED TRANSFORMER DRAWINGS, item b).
- d. Manufacturer's published time-current curves in PDF format and in electronic format suitable for import or updating into the SKM PowerTools for Windows computer program of the transformer high side fuses (paragraph PAD-MOUNTED TRANSFORMER DRAWINGS, item e).
- e. Provide transformer test schedule and routine and other tests required by submittal item "SD-09 Manufacturer's Field Reports."
- f. Provide acceptance test reports required by submittal item "SD-06 Test Reports."
- g. Provide operation and maintenance manuals required by submittal item "SD-10 Operation and Maintenance Data."

1.5 QUALITY ASSURANCE

1.5.1 Pad-Mounted Transformer Drawings

Include the following as a minimum:

- a. An outline drawing, including front, top, and side views.
- b. IEEE nameplate data.
- c. Elementary diagrams and wiring diagrams with terminals identified of watt-hour meter and current transformers.
- d. One-line diagram, including switch(es), current transformers and fuses.
- e. Manufacturer's published time-current curves in PDF format and in electronic format suitable for import or updating into the SKM PowerTools for Windows computer program of the transformer high side fuses.

1.5.2 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, except of NFPA 70 when more stringent requirements are specified or indicated, as though the word "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of

similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with NFPA 70 unless more stringent requirements are specified or indicated.

1.5.3 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship, and:

- a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.
- b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.
- c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.3.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.3.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable.

1.6 MAINTENANCE

1.6.1 Additions to Operation and Maintenance Data

Submit operation and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein. In addition to requirements of Data Package 5, include the following on the actual transformer(s) provided:

- a. An instruction manual with pertinent items and information highlighted.
- b. An outline drawing, front, top, and side views.
- c. Prices for spare parts and supply list.
- d. Routine and field acceptance test reports.
- e. Fuse curves for primary fuses.
- f. Actual nameplate diagram.
- g. Date of purchase.

PART 2 PRODUCTS

2.1 PRODUCT COORDINATION

Products and materials not considered to be pad-mounted transformers and related accessories are specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

2.2 THREE-PHASE PAD-MOUNTED TRANSFORMERS

IEEE C57.12.34, IEEE C57.12.28 and as specified herein. Submit manufacturer's information for each component, device, insulating fluid, and accessory provided with the transformer.

2.2.1 Compartments

Provide high- and low-voltage compartments separated by steel isolating barriers extending the full height and depth of the compartments. Compartment doors: hinged lift-off type with stop in open position and three-point latching.

2.2.1.1 High Voltage, Dead-Front

High-voltage compartment contains: the incoming line, insulated high-voltage load-break connectors, bushing well inserts, six high-voltage bushing wells configured for loop feed application, load-break switch handle(s), access to oil-immersed bayonet fuses, dead-front surge arresters, tap changer handle, connector parking stands, and ground pad.

Minimum high-voltage compartment dimensions: IEEE C57.12.34, Figures 16 and 17.

- a. Insulated high-voltage load-break connectors: IEEE 386, rated 15 kV, 95 kV BIL. Current rating: 200 amperes rms continuous. Short time rating: 10,000 amperes rms symmetrical for a time duration of 0.17 seconds. Connector must have a steel reinforced hook-stick eye, grounding eye, test point, and arc-quenching contact material.

- b. Bushing well inserts: IEEE 386, 200 amperes, 15 kV Class. Provide a bushing well insert for each bushing well unless indicated otherwise.

- c. Load-break switch

Loop feed sectionalizer switches: Provide three, two-position, oil-immersed type switches to permit closed transition loop feed and sectionalizing. Each switch must be rated at 15 kV, 95 kV BIL, with a continuous current rating and load-break rating of 200 amperes, and a make-and-latch rating of 12,000 rms amperes symmetrical. Locate the switch handles in the high-voltage compartment. Operation of switches must be as follows:

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

ARRANGEMENT NO.	DESCRIPTION OF SWITCH ARRANGEMENT	SWITCH POSITION					
		LINE A SW.		LINE B SW		XFMR. SW	
		OPEN	CLOSE	OPEN	CLOSE	OPEN	CLOSE
1	Line A connected to Line B and both lines connected to transformer		X		X		X
2	Transformer connected to Line A only		X	X			X
3	Transformer connected to Line B only	X			X		X
4	Transformer open and loop closed		X		X	X	
5	Transformer open and loop open	X		X		X	

- d. Provide bayonet oil-immersed, expulsion fuses in series with oil-immersed, partial-range, current-limiting fuses. The bayonet fuse links sense both high currents and high oil temperature in order to provide thermal protection to the transformer. Coordinate transformer protection with expulsion fuse clearing low-current faults and current-limiting fuse clearing high-current faults beyond the interrupting rating of the expulsion fuse. Include an oil retention valve inside the bayonet assembly housing, which closes when the fuse holder is removed, and an external drip shield to minimize oil spills. Display a warning label adjacent to the bayonet fuse(s) cautioning against removing or inserting fuses unless the transformer has been de-energized and the tank pressure has been released.

Bayonet fuse assembly: 150 kV BIL.

Oil-immersed current-limiting fuses: **IEEE C37.47**; 50,000 rms amperes symmetrical interrupting rating at the system voltage specified. Connect current-limiting fuses ahead of the radial-feed load-break switch.

- e. Surge arresters: **IEEE C62.11**, rated 15 kV, fully shielded, dead-front, metal-oxide-varistor, elbow type with resistance-graded gap. Provide three arresters for radial feed circuits. Provide three arresters for loop feed circuits.

- f. Parking stands: Provide a parking stand near each bushing.

2.2.1.2 Low Voltage

Low-voltage compartment contains: low-voltage bushings with NEMA spade terminals, accessories, metering, stainless steel or laser-etched anodized aluminum diagrammatic transformer nameplate, and ground pad.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- a. Include the following accessories: drain valve with sampler device, fill plug, pressure relief device, liquid level gage, pressure-vacuum gage, and dial type thermometer with maximum temperature indicator.
- b. Metering: Provide as specified in Section 26 27 14.00 20 ELECTRICITY METERING.

2.2.2 Transformer

- a. Less-flammable bio-based liquid-insulated, two winding, 60 hertz, 65 degrees C rise above a 30 degrees C average ambient, self-cooled type.
- b. Transformer rated 500 kVA.
- c. Transformer voltage ratings: 12.47 kV Delta 480 V - 277 V GrdY.
- d. Tap changer: externally operated, manual type for changing tap setting when the transformer is de-energized. Provide four 2.5 percent full capacity taps, two above and two below rated primary voltage. Indicate which tap setting is in use, clearly visible when the compartment is opened.
- e. Minimum tested percent impedance at 85 degrees C:
4.03 for 500kVA rated units
- f. Comply with the following audible sound level limits:

KVA	DECIBELS (MAX)
500	56

- g. Include:
 - (1) Lifting lugs and provisions for jacking under base, with base construction suitable for using rollers or skidding in any direction.
 - (2) An insulated low-voltage neutral bushing with NEMA spade terminal, and with removable ground strap.
 - (3) Provide transformer top with an access handhole.
 - (4) kVA rating conspicuously displayed using 75 mm high yellow letters on its enclosure.

2.2.2.1 Specified Transformer Efficiencies

Provide transformer efficiency calculations utilizing the actual no-load and load loss values obtained during the routine tests performed on the actual transformer(s) prepared for this project. Reference no-load losses (NLL) at 20 degrees C. Reference load losses (LL) at 55 degrees C and at 50 percent of the nameplate load. The transformer is not acceptable if the calculated transformer efficiency is less than the efficiency indicated in the "KVA / Efficiency" table below. The table is based on requirements contained within 10 CFR 431, Subpart K. Submit certification, including supporting calculations, from the manufacturer

indicating conformance.

kVA	EFFICIENCY (percent)
500	99.35

2.2.3 Insulating Liquid

- a. Less-flammable transformer liquids: NFPA 70 and FM APP GUIDE for less-flammable liquids having a fire point not less than 300 degrees C tested per ASTM D92 and a dielectric strength not less than 33 kV tested per ASTM D877/D877M. Provide identification of transformer as "non-PCB" and "manufacturer's name and type of fluid" on the nameplate.

Provide a fluid that is a biodegradable, electrical insulating, and cooling liquid classified by UL and approved by FM as "less flammable" with the following properties:

- (1) Pour point: ASTM D97, less than -15 degree C
- (2) Aquatic biodegradation: EPA 712-C-98-075, ultimately biodegradable as designated by EPA.
- (3) Trout toxicity: OECD Test 203, zero mortality of EPA 821-R-02-012, pass

2.2.3.1 Liquid-Filled Transformer Nameplates

Provide nameplate information in accordance with IEEE C57.12.00 and as modified or supplemented by this section.

2.2.4 Corrosion Protection

Provide corrosion resistant bases and cabinets of transformers, fabricated of stainless steel conforming to ASTM A240/A240M, Type 304 or 304L. Base includes any part of pad-mounted transformer that is within 75 mm of concrete pad.

Paint entire transformer assembly Munsell 7GY3.29/1.5 green, with paint coating system complying with IEEE C57.12.28 regardless of base, cabinet, and tank material. The Munsell color notation is specified in ASTM D1535.

2.3 WARNING SIGNS AND LABELS

Provide warning signs for the enclosures of pad-mounted transformers having a nominal rating exceeding 600 volts in accordance with NEMA Z535.4 and NEMA 260.

- a. When the enclosure integrity of such equipment is specified to be in accordance with IEEE C57.12.28, such as for pad-mounted transformers, provide self-adhesive warning labels on the outside of the high voltage compartment door(s) with nominal dimensions of 178 by 255 mm with the legend "WARNING HIGH VOLTAGE" printed in two lines of nominal 50 mm high letters. Include the word "WARNING" in white letters on an orange background and the words "HIGH VOLTAGE" in black letters on a white background.

2.4 ARC FLASH WARNING LABEL

Provide arc flash warning label for the enclosure of pad-mounted transformers. Locate this self-adhesive warning label on the outside of the high voltage compartment door warning of potential electrical arc flash hazards and appropriate PPE required. Provide label format as indicated.

2.5 GROUNDING AND BONDING

UL 467. Provide grounding and bonding as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

2.6 CAST-IN-PLACE CONCRETE

Provide concrete associated with electrical work for other than encasement of underground ducts rated for 30 MPa minimum 28-day compressive strength unless specified otherwise. Conform to the requirements of Section 03 30 00 CAST-IN-PLACE CONCRETE.

2.7 SOURCE QUALITY CONTROL

2.7.1 Transformer Test Schedule

The Government reserves the right to witness tests. Provide transformer test schedule for tests to be performed at the manufacturer's test facility. Submit required test schedule and location, and notify the Contracting Officer 30 calendar days before scheduled test date. Notify Contracting Officer 15 calendar days in advance of changes to scheduled date.

a. Test Instrument Calibration

- (1) Provide a calibration program which assures that all applicable test instruments are maintained within rated accuracy.
- (2) Accuracy: Traceable to the National Institute of Standards and Technology.
- (3) Instrument calibration frequency schedule: less than or equal to 12 months for both test floor instruments and leased specialty equipment.
- (4) Dated calibration labels: visible on all test equipment.
- (5) Calibrating standard: higher accuracy than that of the instrument tested.
- (6) Keep up-to-date records that indicate dates and test results of instruments calibrated or tested. For instruments calibrated by the manufacturer on a routine basis, in lieu of third party calibration, include the following:
 - (a) Maintain up-to-date instrument calibration instructions and procedures for each test instrument.
 - (b) Identify the third party/laboratory calibrated instrument to verify that calibrating standard is met.

2.7.2 Design Tests

IEEE C57.12.00, and IEEE C57.12.90. Section 5.1.2 in IEEE C57.12.80 states that "design tests are made only on representative apparatus of basically the same design." Submit design test reports (complete with test data, explanations, formulas, and results), in the same submittal package as the catalog data and drawings for the specified transformer(s), with design tests performed prior to the award of this contract.

- a. Tests: certified and signed by a registered professional engineer.
- b. Temperature rise: "Basically the same design" for the temperature rise test means a pad-mounted transformer with the same coil construction (such as wire wound primary and sheet wound secondary), the same kVA, the same cooling type (KNAN), the same temperature rise rating, and the same insulating liquid as the transformer specified.
- c. Lightning impulse: "Basically the same design" for the lightning impulse dielectric test means a pad-mounted transformer with the same BIL, the same coil construction (such as wire wound primary and sheet wound secondary), and a tap changer, if specified. Design lightning impulse tests includes the primary windings only of that transformer.
 - (1) IEEE C57.12.90, paragraph 10.3 entitled "Lightning Impulse Test Procedures," and IEEE C57.98.
 - (2) State test voltage levels.
 - (3) Provide photographs of oscilloscope display waveforms or plots of digitized waveforms with test report.
- d. Lifting and moving devices: "Basically the same design" requirement for the lifting and moving devices test means a test report confirming that the lifting device being used is capable of handling the weight of the specified transformer in accordance with IEEE C57.12.34.
- e. Pressure: "Basically the same design" for the pressure test means a pad-mounted transformer with a tank volume within 30 percent of the tank volume of the transformer specified.
- f. Short circuit: "Basically the same design" for the short circuit test means a pad-mounted transformer with the same kVA as the transformer specified.

2.7.3 Routine and Other Tests

IEEE C57.12.00. Routine and other tests: performed in accordance with IEEE C57.12.90 by the manufacturer on the actual transformer prepared for this project to ensure that the design performance is maintained in production. Submit test reports, by serial number and receive approval before delivery of equipment to the project site. Required tests and testing sequence as follows:

- a. Phase relation
- b. Ratio
- c. No-load losses (NLL) and excitation current

- d. Load losses (LL) and impedance voltage
- e. Dielectric
 - (1) Impulse
 - (2) Applied voltage
 - (3) Induced voltage
- f. Leak

PART 3 EXECUTION

3.1 INSTALLATION

Conform to **IEEE C2**, **NFPA 70**, and to the requirements specified herein. Provide new equipment and materials unless indicated or specified otherwise.

3.2 GROUNDING

NFPA 70 and **IEEE C2**, except provide grounding systems with a resistance to solid earth ground not exceeding 25 ohms.

3.2.1 Grounding Electrodes

Provide driven ground rods as specified in Section **33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION**. Connect ground conductors to the upper end of ground rods by exothermic weld or compression connector. Provide compression connectors at equipment end of ground conductors.

3.2.2 Pad-Mounted Transformer Grounding

Provide a ground ring around the transformer with 4/0 AWG bare copper. Provide four ground rods in the ground ring, one per corner. Install the ground rods at least **3000 mm** apart from each other. Provide separate copper grounding conductors and connect them to the ground loop as indicated. When work in addition to that indicated or specified is required to obtain the specified ground resistance, the provision of the contract covering "Changes" applies.

3.2.3 Connections

Make joints in grounding conductors and loops by exothermic weld or compression connector. Install exothermic welds and compression connectors as specified in Section **33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION**.

3.2.4 Grounding and Bonding Equipment

UL 467, except as indicated or specified otherwise.

3.3 INSTALLATION OF EQUIPMENT AND ASSEMBLIES

Install and connect pad-mounted transformers furnished under this section as indicated on project drawings, the approved shop drawings, and as specified herein.

3.4 FIELD APPLIED PAINTING

Where field painting of enclosures is required to correct damage to the manufacturer's factory applied coatings, provide manufacturer's recommended coatings and apply in accordance with manufacturer's instructions.

3.5 FOUNDATION FOR EQUIPMENT AND ASSEMBLIES

Mount transformer on concrete slab as follows:

- a. Unless otherwise indicated, provide the slab with dimensions at least **200 mm** thick, reinforced with a **152 by 152 mm MW19 by MW19** mesh placed uniformly **100 mm** from the top of the slab.
- b. Place slab on a **150 mm** thick, well-compacted gravel base.
- c. Install slab such that top of concrete slab is approximately **100 mm** above the finished grade with gradual slope for drainage.
- d. Provide edges above grade with **15 mm** chamfer.
- e. Provide slab of adequate size to project at least **200 mm** beyond the equipment.

Stub up conduits, with bushings, **50 mm** into cable wells in the concrete pad. Coordinate dimensions of cable wells with transformer cable training areas.

3.5.1 Cast-In-Place Concrete

Provide cast-in-place concrete work in accordance with the requirements of Section **03 30 00** CAST-IN-PLACE CONCRETE.

3.5.2 Sealing

When the installation is complete, seal all entries into the equipment enclosure with an approved sealing method. Provide seals of sufficient strength and durability to protect all energized live parts of the equipment from rodents, insects, or other foreign matter.

3.6 FIELD QUALITY CONTROL

3.6.1 Performance of [Acceptance Checks and Tests](#)

Perform in accordance with the manufacturer's recommendations and include the following visual and mechanical inspections and electrical tests, performed in accordance with **NETA ATS**. Submit reports, including acceptance criteria and limits for each test in accordance with **NETA ATS** "Test Values."

3.6.1.1 Pad-Mounted Transformers

- a. Visual and mechanical inspection
 - (1) Compare equipment nameplate data with specifications and approved shop drawings.
 - (2) Inspect physical and mechanical condition. Check for damaged or

cracked insulators and leaks.

- (3) Inspect anchorage, alignment, and grounding.
- (4) Verify the presence of PCB content labeling.
- (5) Verify the bushings and transformer interiors are clean.
- (6) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.
- (7) Verify correct liquid level in tanks and bushings.
- (8) Verify that positive pressure is maintained on gas-blanketed transformers.
- (9) Perform specific inspections and mechanical tests as recommended by manufacturer.
- (10) Verify de-energized tap changer position is left as specified.
- (11) Verify the presence of transformer surge arresters.

b. Electrical tests

- (1) Perform resistance measurements through all bolted connections with low-resistance ohmmeter.
- (2) Verify proper secondary voltage phase-to-phase and phase-to-neutral after energization and prior to loading.

3.6.1.2 Grounding System

a. Visual and mechanical inspection

- (1) Inspect ground system for compliance with contract plans and specifications.

b. Electrical tests

- (1) Perform ground-impedance measurements utilizing the fall-of-potential method. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. Use an instrument equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test.
- (2) Submit the measured ground resistance of each ground rod and grounding system, indicating the location of the rod and grounding system. Include the test method and test setup (i.e., pin location) used to determine ground resistance and soil conditions at the time the measurements were made.

3.6.1.3 Surge Arresters, Medium- and High-Voltage

a. Visual and mechanical inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Inspect anchorage, alignment, grounding, and clearances.
- (4) Verify the arresters are clean.
- (5) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.
- (6) Verify that the ground lead on each device is individually attached to a ground bus or ground electrode.

b. Electrical tests

- (1) Perform resistance measurements through all bolted connections with low-resistance ohmmeter, if applicable.
- (2) Perform an insulation-resistance test on each arrester, phase terminal-to-ground.
- (3) Test grounding connection.

3.6.2 Follow-Up Verification

Upon completion of acceptance checks and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. As an exception to requirements stated elsewhere in the contract, notify the Contracting Officer 5 working days in advance of the dates and times of checking and testing.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 13 00

SF6/HIGH-FIREPOINT FLUIDS INSULATED PAD-MOUNTED SWITCHGEAR

05/21, CHG 1: 05/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 DEFINITIONS
 - 1.3.1 Switched Way
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Switchgear Drawings
 - 1.5.2 Paint Coating System
 - 1.5.3 Electronic Overcurrent Control Curves
- 1.6 MAINTENANCE
 - 1.6.1 SF6/High-Firepoint Fluid Insulated Pad-mounted Switchgear Operation and Maintenance

PART 2 PRODUCTS

- 2.1 SF6/HIGH-FIREPOINT FLUID INSULATED PAD-MOUNTED SWITCHGEAR
 - 2.1.1 Ratings and Test Requirements
 - 2.1.2 Switchgear Construction
 - 2.1.2.1 Pad-mounting Provisions
 - 2.1.3 Load Interrupting Switched Ways
 - 2.1.3.1 Three-Pole Group Operated Switched Ways
 - 2.1.3.2 Fault Interrupting Switched Ways
 - 2.1.4 Automatic Switch Controls
 - 2.1.4.1 Manual Operation
 - 2.1.4.2 SCADA Operation
 - 2.1.4.3 Source-Transfer Operation
 - 2.1.5 Low Voltage Test Points
 - 2.1.6 Dead-Front High-Voltage Bushings
- 2.2 INSULATED HIGH-VOLTAGE CONNECTORS
- 2.3 SF6 REFILL CYLINDERS
- 2.4 SOURCE QUALITY CONTROL
 - 2.4.1 Switchgear Design and Production Tests

PART 3 EXECUTION

- 3.1 INSTALLATION
- 3.2 GROUNDING
 - 3.2.1 Grounding Electrodes
 - 3.2.2 Switchgear Grounding
 - 3.2.3 Connections
 - 3.2.4 Grounding and Bonding Equipment
- 3.3 FOUNDATION FOR EQUIPMENT AND ASSEMBLIES
- 3.4 FIELD QUALITY CONTROL
 - 3.4.1 Performance of Acceptance Checks and Tests

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.4.1.1 Switchgear
- 3.4.1.2 Grounding System
- 3.4.2 Follow-Up Verification

-- End of Section Table of Contents --

SECTION 26 13 00

SF6/HIGH-FIREPOINT FLUIDS INSULATED PAD-MOUNTED SWITCHGEAR
05/21, CHG 1: 05/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

- | | |
|------------|---|
| ASTM D1535 | (2014; R 2018) Standard Practice for Specifying Color by the Munsell System |
| ASTM D2472 | (2000; R 2014) Standard Specification for Sulphur Hexafluoride |
| ASTM D6871 | (2017) Standard Specification for Natural (Vegetable Oil) Ester Fluids Used in Electrical Apparatus |

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

- | | |
|----------------|---|
| IEEE 386 | (2016) Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV |
| IEEE C2 | (2017; Errata 1-2 2017; INT 1 2017) National Electrical Safety Code |
| IEEE C37.60 | (2019) High-Voltage Switchgear and Controlgear - Part 111: Automatic Circuit Reclosers for Alternating Current Systems Up to 38 kV |
| IEEE C37.74 | (2014) Standard Requirements for Subsurface, Vault, and Pad-Mounted Load-Interrupter Switchgear and Fused Load-Interrupter Switchgear for Alternating Current Systems Up to 38 kV |
| IEEE C57.12.28 | (2014) Standard for Pad-Mounted Equipment - Enclosure Integrity |
| IEEE C57.12.29 | (2014) Standard for Pad-Mounted Equipment - Enclosure Integrity for Coastal Environments |

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

- | | |
|----------|--|
| NETA ATS | (2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems |
|----------|--|

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 61099 (2010; ED 2.0) Insulating Liquids -
Specifications for Unused Synthetic
Organic Esters for Electrical Purposes

IEC 62271-111 (2019) High Voltage Switchgear And
Controlgear - Part 111: Automatic Circuit
Reclosers for Alternating Current Systems
up to and including 38 kV

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA
20-1; TIA 20-2; TIA 20-3; TIA 20-4)
National Electrical Code

NFPA 70B (2019) Recommended Practice for Electrical
Equipment Maintenance

UNDERWRITERS LABORATORIES (UL)

UL 467 (2013; Reprint Jun 2017) UL Standard for
Safety Grounding and Bonding Equipment

1.2 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING, applies to this
section, with the additions and modifications specified herein.

1.3 DEFINITIONS

1.3.1 Switched Way

A switched way is considered a three-phase circuit entrance to the bus
through a switch. For single-phase switches, it is a single-phase
entrance to the bus through a switch.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S"
classification. Submittals not having a "G" or "S" classification are for
Contractor Quality Control approval. Submit the following in accordance
with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Switchgear Drawings; G

SD-03 Product Data

Electronic Overcurrent Control Curves; G

SF6/High-Firepoint Fluid Insulated Pad-mounted Switchgear; G

Include data on switches and associated accessories with each
submittal. Include manufacturer's information for each component,
device and accessory provided with the equipment with each

submittal.

Insulated High-Voltage Connectors; G

SD-06 Test Reports

Acceptance Checks and Tests; G

SD-07 Certificates

Paint Coating System; G

SD-09 Manufacturer's Field Reports

Switchgear Design and Production Tests; G

SD-10 Operation and Maintenance Data

SF6/High-Firepoint Fluid Insulated Pad-mounted Switchgear, Data
Package 5; G

Submit in accordance with Section 01 78 23 OPERATION AND
MAINTENANCE DATA.

1.5 QUALITY ASSURANCE

1.5.1 Switchgear Drawings

Furnish drawings that include, but are not limited to, the following:

- a. Overall dimensions, weights, plan view, and front view
- b. Ratings
- c. Single-line diagram.

1.5.2 Paint Coating System

Submit IEEE C57.12.28 and IEEE C57.12.29 paint coating system performance
requirement tests.

1.5.3 Electronic Overcurrent Control Curves

Provide time-current characteristic curves in PDF format and in electronic
format suitable for import or updating into the SKM PowerTools for
Windows computer program.

1.6 MAINTENANCE

1.6.1 SF6/High-Firepoint Fluid Insulated Pad-mounted Switchgear Operation and Maintenance

Submit Operation and Maintenance Manuals in accordance with Section
01 78 23 OPERATION AND MAINTENANCE DATA.

PART 2 PRODUCTS

2.1 SF6/HIGH-FIREPOINT FLUID INSULATED PAD-MOUNTED SWITCHGEAR

IEEE C37.74

2.1.1 Ratings and Test Requirements

The voltage rating of the switchgear must be 15.5 kV. Provide the corresponding ratings associated with the required switchgear voltage rating as follows:

Rated Maximum Voltage, kV	15.5
Rated Withstand Impulse Voltage, kV BIL	95
Continuous and Load Interrupting Current, A	600
Short-Time Current, kA rms Sym	12.5

2.1.2 Switchgear Construction

Provide switchgear with switch contacts and cable entrance terminations contained in a sealed, dielectric-filled stainless steel tank. Ship switchgear from factory, filled with appropriate levels of SF6 gas conforming to ASTM D2472 or less-flammable, high-firepoint biodegradable fluid conforming to ASTM D6871 or IEC 61099. Configure switchgear with load interrupting and fault interrupting switched ways as indicated. Provide switchgear with front accessible terminations suitable for cables entering from below with manual operating provisions either mounted on the rear or capable of hookstick operation per IEEE C37.74. Switch contact positions for switched ways must be visible through viewing windows in the switchgear tank located adjacent to the manual operating provisions. Provide internal gas pressure gage or fluid level gage in viewable location from switch operating handle. Provide each switched way with three position switch; Open, Closed, Ground.

2.1.2.1 Pad-mounting Provisions

Provide enclosed switchgear suitable for installation on a concrete pad. Fabricate switchgear support frame and enclosure base with type 304 or 304L stainless steel. Enclosure base must include any part of the switchgear enclosure that is within 75 mm of concrete pad. Paint enclosure including base ASTM D1535 Munsell 7GY3.29/1.5 green. Comply with IEEE C57.12.28 and IEEE C57.12.29 as applicable for the paint coating system regardless of equipment material.

2.1.3 Load Interrupting Switched Ways

2.1.3.1 Three-Pole Group Operated Switched Ways

Provide three-pole group operated load interrupting switched ways as indicated.

2.1.3.2 Fault Interrupting Switched Ways

IEEE C37.60, IEC 62271-111. Provide non-fused, non-reclosing, manual

reset, vacuum interrupters consisting of vacuum interrupter and a spring assisted operating mechanism. Each fault interrupting switched way must utilize internally mounted current transformers and an electronic overcurrent control to provide three-pole ganged tripping for single-phase and three-phase faults. Provide electronic overcurrent control with provisions for a minimum of ten field changeable overcurrent trip settings. Provide remote tripping via an external dry contact device for fault interrupting switched ways. Provide 120 Vac from the switchgear itself.

2.1.4 Automatic Switch Controls

Provide an automatic switch control system to execute Manual, SCADA, and Automatic Source-Transfer operations. Power the switch control system and associated communication port provisions from an integral battery-charger DC supply system. Use motor operators and associated motor operator controllers for switch way operation. Use 120 Vac control power from switchgear itself for automatic switch control.

2.1.4.1 Manual Operation

Provide the motor operator controllers with "Close" and "Open" pushbuttons for manual operation.

2.1.4.2 SCADA Operation

The automatic switch control system must execute remote commands received from a SCADA master station and transmit switchgear operation information to a SCADA master station via a DNP 3.0 100Base-FX Ethernet communication port. Include transfer of switch ways to "Close" and "Open" positions and enabling of the Source-Transfer operation for execution of remote commands. Include switch way position status, voltage and current readings, and DC supply system status with communication of switchgear information.

2.1.4.3 Source-Transfer Operation

Provide an automatic switch control system that opens an incoming switch way when voltage is lost and closes the alternate incoming switch way if voltage is present. Include with the Source-Transfer controls an overcurrent-lockout feature that prevents automatic closing of a switch way into a system fault. Include provisions for returning the system to the normal configuration via manual or automatic operations when voltage is restored.

2.1.5 Low Voltage Test Points

Provide load interrupting switch ways with internal load side voltage sensors that allow for low voltage checks with relay interface at test point of elbow connectors to confirm energized and in-phase conditions using a standard high-impedance voltmeter.

2.1.6 Dead-Front High-Voltage Bushings

IEEE 386. 15 kV, 95 kV BIL. Provide 600 ampere one-piece deadbreak apparatus bushings for each switched way.

2.2 INSULATED HIGH-VOLTAGE CONNECTORS

IEEE 386. Provide corresponding connector for each switched way. Provide a grounding eye and test point on each connector.

- a. 600 Ampere deadbreak connector ratings: Voltage: 15 kV, 95 kV BIL. Short time rating: 25,000 rms symmetrical amperes. Provide connectors with 200 ampere bushing interface.

2.3 SF6 REFILL CYLINDERS

Provide two SF6 refill cylinders, minimum size of 6 pounds of SF6; include regulator, valves, and hose for connection to the fill valve of the switch.

2.4 SOURCE QUALITY CONTROL

2.4.1 Switchgear Design and Production Tests

Furnish reports which include results of design and production tests performed according to **IEEE C37.74** and **IEEE C37.60**. Perform manufacturer production tests on each switchgear assembly to ensure that design performance is maintained in production.

PART 3 EXECUTION

3.1 INSTALLATION

Conform to **IEEE C2**, **NFPA 70**, and to the requirements specified herein.

3.2 GROUNDING

NFPA 70 and **IEEE C2**, except provide grounds and grounding systems with a resistance to solid earth ground not exceeding 25 ohms. When work, in addition to that indicated or specified, is directed to obtain the specified ground resistance, the provision of the contract covering "Changes" applies.

3.2.1 Grounding Electrodes

Provide driven ground rods as specified in Section **33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION** at each corner of switchgear pad.

3.2.2 Switchgear Grounding

Connect #4/0 bare copper conductor ground ring, not less than **600 mm** below grade, to the upper end of the ground rods by exothermic welds or compression connectors. Provide #4/0 bare copper conductors connecting the switchgear grounding provisions to two different ground rods.

3.2.3 Connections

Make joints in grounding conductors and ground ring by exothermic weld or compression connector. Install exothermic welds and compression connectors as specified in Section **33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION**.

3.2.4 Grounding and Bonding Equipment

UL 467, except as indicated or specified otherwise.

3.3 FOUNDATION FOR EQUIPMENT AND ASSEMBLIES

Mount switch on concrete slab. Provide slab with dimensions at least 300 mm thick, reinforced with a 152 by 152 - MW19 by MW19 mesh, placed uniformly 100 mm from the top of the slab. Place slab on a 150 mm thick, well-compacted gravel base. Install top of concrete slab approximately 100 mm above finished grade. Provide edges above grade with 15 mm chamfer. Provide slab of adequate size to project at least 200 mm beyond equipment.

Stub up conduits, with bushings, 50 mm into cable wells in the concrete pad. Coordinate dimensions of cable wells with switch cable training areas. Provide concrete work as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.4 FIELD QUALITY CONTROL

3.4.1 Performance of Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, NFPA 70B, NETA ATS and referenced ANSI standards.

Include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

3.4.1.1 Switchgear

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate information with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Check for proper anchorage, alignment, required area clearances, and grounding.
- (4) Perform mechanical operator tests in accordance with manufacturer's instructions.
- (5) Verify that insulating SF6 gas pressure or dielectric fluid level is correct.
- (6) Inspect all indicating devices for proper operation.

b. Electrical Tests

- (1) Perform contact-resistance tests.
- (2) Trip fault interrupters by operation of overcurrent control.
- (3) Perform insulation-resistance tests.
- (4) Perform an over-potential test on each switched way pole with the switched way in the open position in accordance with the manufacturer's instructions.
- (5) Set fault interrupter overcurrent control in accordance with

government provided settings. Request settings from government, in writing, a minimum of 30 days prior to scheduling electrical tests.

3.4.1.2 Grounding System

a. Visual and Mechanical Inspection

Inspect ground system for compliance with contract plans and specifications.

b. Electrical Tests

- (1) Perform ground-impedance measurements utilizing the fall-of-potential method. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. Use an instrument equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test.
- (2) Submit the measured ground resistance of each ground rod and grounding system, indicating the location of the rod and grounding system. Include the test method and test setup (i.e., pin location) used to determine ground resistance and soil conditions at the time the measurements were made.

3.4.2 Follow-Up Verification

Upon completion of acceptance checks and tests, show by demonstration in service that devices are in good operating condition and properly performing the intended function. Perform each test function not less than three times. As an exception to requirements stated elsewhere in the contract, notify the Contracting Officer five working days in advance of the dates and times for checks and tests.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 20 00

INTERIOR DISTRIBUTION SYSTEM

08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
- 1.3 RELATED REQUIREMENTS
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Regulatory Requirements
 - 1.5.2 Standard Products
 - 1.5.2.1 Alternative Qualifications
 - 1.5.2.2 Material and Equipment Manufacturing Date
- 1.6 MAINTENANCE
 - 1.6.1 Electrical Systems
- 1.7 WARRANTY

PART 2 PRODUCTS

- 2.1 MATERIALS AND EQUIPMENT
- 2.2 CONDUIT AND FITTINGS
 - 2.2.1 Rigid Metallic Conduit
 - 2.2.1.1 Rigid, Threaded Zinc-Coated Steel Conduit
 - 2.2.2 Electrical, Zinc-Coated Steel Metallic Tubing (EMT)
 - 2.2.3 Flexible Metal Conduit
 - 2.2.3.1 Liquid-Tight Flexible Metal Conduit, Steel
 - 2.2.4 Fittings for Metal Conduit, EMT, and Flexible Metal Conduit
 - 2.2.4.1 Fittings for Rigid Metal Conduit
 - 2.2.4.2 Fittings for EMT
- 2.3 OUTLET BOXES AND COVERS
 - 2.3.1 Outlet Boxes for Telecommunications System
- 2.4 CABINETS, JUNCTION BOXES, AND PULL BOXES
- 2.5 WIRES AND CABLES
 - 2.5.1 Conductors
 - 2.5.1.1 Minimum Conductor Sizes
 - 2.5.2 Color Coding
 - 2.5.2.1 Ground and Neutral Conductors
 - 2.5.2.2 Ungrounded Conductors
 - 2.5.3 Insulation
 - 2.5.4 Bonding Conductors
 - 2.5.4.1 Bonding Conductor for Telecommunications
 - 2.5.5 Cable Tray Cable or Power Limited Tray Cable
- 2.6 SPLICES AND TERMINATION COMPONENTS
- 2.7 DEVICE PLATES
- 2.8 SWITCHES
 - 2.8.1 Toggle Switches
 - 2.8.2 Disconnect Switches
- 2.9 RECEPTACLES

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.9.1 Weatherproof Receptacles
- 2.9.2 Ground-Fault Circuit Interrupter Receptacles
- 2.9.3 Special Purpose Receptacles
- 2.10 PANELBOARDS
 - 2.10.1 Enclosure
 - 2.10.2 Panelboard Buses
 - 2.10.3 Circuit Breakers
 - 2.10.3.1 Multipole Breakers
 - 2.10.3.2 Circuit Breaker With Ground-Fault Circuit Interrupter
- 2.11 MOTOR SHORT-CIRCUIT PROTECTOR (MSCP)
- 2.12 TRANSFORMERS
 - 2.12.1 Specified Transformer Efficiency
- 2.13 MOTORS
 - 2.13.1 High Efficiency Single-Phase Motors
 - 2.13.2 Premium Efficiency Polyphase and Single-Phase Motors
 - 2.13.3 Motor Sizes
 - 2.13.4 Wiring and Conduit
- 2.14 MOTOR CONTROLLERS
 - 2.14.1 Control Wiring
 - 2.14.2 Control Circuit Terminal Blocks
 - 2.14.2.1 Types of Terminal Blocks
 - 2.14.3 Control Circuits
 - 2.14.4 Enclosures for Motor Controllers
 - 2.14.5 Pushbutton Stations
 - 2.14.6 Pilot and Indicating Lights
 - 2.14.7 Reduced-Voltage Controllers
- 2.15 MANUAL MOTOR STARTERS (MOTOR RATED SWITCHES)
 - 2.15.1 Pilot Lights
- 2.16 MOTOR CONTROL CENTERS
 - 2.16.1 Bus Systems
 - 2.16.1.1 Horizontal and Main Buses
 - 2.16.1.2 Vertical Bus
 - 2.16.1.3 Ground Bus
 - 2.16.1.4 Neutral Bus
 - 2.16.2 Combination Motor Controllers
 - 2.16.3 Circuit Breaker Key Interlock
 - 2.16.4 Protective Devices
 - 2.16.4.1 Insulated-Case Breaker
 - 2.16.4.2 Molded-Case Circuit Breaker
 - 2.16.4.3 Electronic Trip Units
- 2.17 LOCKOUT REQUIREMENTS
- 2.18 TELECOMMUNICATIONS SYSTEM
- 2.19 GROUNDING AND BONDING EQUIPMENT
 - 2.19.1 Ground Rods
 - 2.19.2 Ground Bus
 - 2.19.3 Primary Bonding Busbar
- 2.20 HAZARDOUS LOCATIONS
- 2.21 MANUFACTURER'S NAMEPLATE
- 2.22 FIELD FABRICATED NAMEPLATES
- 2.23 WARNING SIGNS
- 2.24 METERING
 - 2.24.1 Digital Meters
- 2.25 SURGE PROTECTIVE DEVICES
- 2.26 EMERGENCY FUEL SHUTOFF PUSHBUTTON
- 2.27 EMERGENCY FUEL SHUTOFF (EFSO) PANEL AND COMPONENTS
 - 2.27.1 Standard Indicator Lights
 - 2.27.2 Pushbuttons
 - 2.27.3 Relays
 - 2.27.4 Nameplates

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.27.5 Terminal Blocks
- 2.28 FACTORY APPLIED FINISH
- 2.29 SOURCE QUALITY CONTROL
 - 2.29.1 Transformer Factory Tests

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Underground Service
 - 3.1.2 Hazardous Locations
 - 3.1.3 Service Entrance Identification
 - 3.1.3.1 Labels
 - 3.1.4 Wiring Methods
 - 3.1.4.1 Pull Wire
 - 3.1.5 Conduit Installation
 - 3.1.5.1 Restrictions Applicable to EMT
 - 3.1.5.2 Restrictions Applicable to Flexible Conduit
 - 3.1.5.3 Underground Conduit
 - 3.1.5.4 Conduit Through Floor Slabs
 - 3.1.5.5 Stub-Ups
 - 3.1.5.6 Conduit Support
 - 3.1.5.7 Directional Changes in Conduit Runs
 - 3.1.5.8 Locknuts and Bushings
 - 3.1.5.9 Flexible Connections
 - 3.1.5.10 Telecommunications and Signal System Pathway
 - 3.1.6 Boxes, Outlets, and Supports
 - 3.1.6.1 Boxes
 - 3.1.6.1.1 Wall-Mounted Telecommunications Outlet Box
 - 3.1.6.2 Pull Boxes
 - 3.1.7 Mounting Heights
 - 3.1.8 Conductor Identification
 - 3.1.8.1 Marking Strips
 - 3.1.9 Splices
 - 3.1.10 Covers and Device Plates
 - 3.1.11 Grounding and Bonding
 - 3.1.11.1 Ground Rods
 - 3.1.11.2 Grounding Connections
 - 3.1.11.3 Ground Bus
 - 3.1.11.4 Resistance
 - 3.1.11.5 Telecommunications System
 - 3.1.12 Equipment Connections
 - 3.1.13 Repair of Existing Work
 - 3.1.13.1 Workmanship
 - 3.1.13.2 Removal of Existing Electrical Distribution System
 - 3.1.13.3 Continuation of Service
 - 3.1.14 Watthour Meters
 - 3.1.15 Surge Protective Devices
- 3.2 FIELD FABRICATED NAMEPLATE MOUNTING
- 3.3 WARNING SIGN MOUNTING
- 3.4 FIELD APPLIED PAINTING
- 3.5 FIELD QUALITY CONTROL
 - 3.5.1 Devices Subject to Manual Operation
 - 3.5.2 600-Volt Wiring Test
 - 3.5.3 Transformer Tests
 - 3.5.4 Ground-Fault Receptacle Test
 - 3.5.5 Grounding System Test
 - 3.5.6 Watthour Meter
 - 3.5.7 Phase Rotation Test

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

-- End of Section Table of Contents --

SECTION 26 20 00

INTERIOR DISTRIBUTION SYSTEM
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C12.1 (2014; Errata 2016) Electric Meters - Code for Electricity Metering

ASTM INTERNATIONAL (ASTM)

ASTM B1 (2013) Standard Specification for Hard-Drawn Copper Wire

ASTM B8 (2011; R 2017) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

ASTM D709 (2017) Standard Specification for Laminated Thermosetting Materials

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 81 (2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System

IEEE C2 (2023) National Electrical Safety Code

IEEE C37.90 (2005; R 2011) Standard for Relays and Relay Systems Associated With Electric Power Apparatus

IEEE Stds Dictionary (2009) IEEE Standards Dictionary: Glossary of Terms & Definitions

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)

NECA NEIS 1 (2015) Standard for Good Workmanship in Electrical Construction

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C80.1	(2020) American National Standard for Electrical Rigid Steel Conduit (ERSC)
ANSI C80.3	(2020) American National Standard for Electrical Metallic Tubing (EMT)
NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)
NEMA ICS 1	(2022) Standard for Industrial Control and Systems: General Requirements
NEMA ICS 2	(2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V
NEMA ICS 3	(2005; R 2010) Medium-Voltage Controllers Rated 2001 to 7200 V AC
NEMA ICS 4	(2015) Application Guideline for Terminal Blocks
NEMA ICS 6	(1993; R 2016) Industrial Control and Systems: Enclosures
NEMA KS 1	(2013) Enclosed and Miscellaneous Distribution Equipment Switches (600 V Maximum)
NEMA MG 1	(2021) Motors and Generators
NEMA MG 10	(2017) Energy Management Guide for Selection and Use of Fixed Frequency Medium AC Squirrel-Cage Polyphase Induction Motors
NEMA MG 11	(1977; R 2012) Energy Management Guide for Selection and Use of Single Phase Motors
NEMA ST 20	(2014) Dry-Type Transformers for General Applications
NEMA WD 1	(1999; R 2020) Standard for General Color Requirements for Wiring Devices
NEMA WD 6	(2021) Wiring Devices Dimensions Specifications
NEMA Z535.4	(2011; R 2017) Product Safety Signs and Labels

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023) National Electrical Code
NFPA 70E	(2024) Standard for Electrical Safety in the Workplace

NFPA 780 (2023) Standard for the Installation of
Lightning Protection Systems

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-568.1 (2020e) Commercial Building
Telecommunications Infrastructure Standard

TIA-569 (2019e; Add 1 2022) Telecommunications
Pathways and Spaces

TIA-607 (2019d) Generic Telecommunications Bonding
and Grounding (Earthing) for Customer
Premises

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 431 Energy Efficiency Program for Certain
Commercial and Industrial Equipment

29 CFR 1910.147 The Control of Hazardous Energy (Lock
Out/Tag Out)

29 CFR 1910.303 Electrical, General

UNDERWRITERS LABORATORIES (UL)

UL 1 (2005; Reprint Jan 2022) UL Standard for
Safety Flexible Metal Conduit

UL 6 (2022) UL Standard for Safety Electrical
Rigid Metal Conduit-Steel

UL 20 (2018; Reprint May 2023) UL Standard for
Safety General-Use Snap Switches

UL 44 (2018; Reprint May 2021) UL Standard for
Safety Thermoset-Insulated Wires and Cables

UL 50 (2015) UL Standard for Safety Enclosures
for Electrical Equipment,
Non-Environmental Considerations

UL 67 (2018; Reprint May 2023) UL Standard for
Safety Panelboards

UL 83 (2017; Reprint Mar 2020) UL Standard for
Safety Thermoplastic-Insulated Wires and
Cables

UL 360 (2013; Reprint Apr 2023) UL Standard for
Safety Liquid-Tight Flexible Metal Conduit

UL 467 (2022) UL Standard for Safety Grounding
and Bonding Equipment

UL 486A-486B (2018; Reprint May 2021) UL Standard for
Safety Wire Connectors

UL 486C	(2018; Reprint May 2021) UL Standard for Safety Splicing Wire Connectors
UL 489	(2016; Rev 2019) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures
UL 498	(2017; Reprint May 2023) UL Standard for Safety Attachment Plugs and Receptacles
UL 506	(2017; Reprint Jan 2022) UL Standard for Safety Specialty Transformers
UL 508	(2018; Reprint Jul 2021) UL Standard for Safety Industrial Control Equipment
UL 510	(2020; Dec 2022) UL Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape
UL 514A	(2013; Reprint Jun 2022) UL Standard for Safety Metallic Outlet Boxes
UL 514B	(2012; Reprint May 2020) Conduit, Tubing and Cable Fittings
UL 514C	(2014; Reprint Feb 2020) UL Standard for Safety Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers
UL 674	(2022) UL Standard for Safety Electric Motors and Generators for Use in Hazardous (Classified) Locations
UL 797	(2007; Reprint Apr 2023) UL Standard for Safety Electrical Metallic Tubing -- Steel
UL 845	(2021) UL Standard for Safety Motor Control Centers
UL 943	(2016; Reprint Feb 2018) UL Standard for Safety Ground-Fault Circuit-Interrupters
UL 984	(1996; Reprint Sep 2005) Hermetic Refrigerant Motor-Compressors
UL 1004-1	(2012; Reprint Nov 2020) UL Standard for Safety Rotating Electrical Machines - General Requirements
UL 1063	(2017; Reprint Jun 2022) UL Standard for Safety Machine-Tool Wires and Cables
UL 1203	(2013; Reprint Apr 2022) UL Standard for Safety Explosion-Proof and Dust-Ignition-Proof Electrical Equipment for Use in Hazardous (Classified) Locations

UL 1449

(2021; Reprint Dec 2022) UL Standard for
Safety Surge Protective Devices

1.2 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in [IEEE Stds Dictionary](#).

1.3 RELATED REQUIREMENTS

Section [25 05 11](#) CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS applies to this section, with the additions and modifications specified herein.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section [01 33 00](#) SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Panelboards; G

Transformers; G

Motor Control Centers; G

Marking Strips Drawings; G

SD-03 Product Data

Receptacles; G

Circuit Breakers; G

Switches; G

Transformers; G

Motor Controllers; G

Combination Motor Controllers; G

Manual Motor Starters; G

Metering; G

Primary Bonding Busbar; G

Surge Protective Devices; G

Emergency Fuel Shutoff Pushbutton; G

Emergency Fuel Shutoff (EFSO) Panel And Components; G

SD-06 Test Reports

600-volt Wiring Test; G

Grounding System Test; G

Transformer Tests; G

Ground-fault Receptacle Test; G

SD-09 Manufacturer's Field Reports

Transformer Factory Tests

SD-10 Operation and Maintenance Data

Electrical Systems, Data Package 5; G

Metering, Data Package 5; G

1.5 QUALITY ASSURANCE

1.5.1 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with NFPA 70 unless more stringent requirements are specified or indicated. NECA NEIS 1 shall be considered the minimum standard for workmanship.

1.5.2 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and:

- a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.
- b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.
- c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.2.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.2.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable.

1.6 MAINTENANCE

1.6.1 Electrical Systems

Submit operation and maintenance data in accordance with Section 01 78 23, OPERATION AND MAINTENANCE DATA and as specified herein. Submit operation and maintenance manuals for electrical systems that provide basic data relating to the design, operation, and maintenance of the electrical distribution system for the building. Include the following:

- a. Single line diagram of the "as-built" building electrical system.
- b. Schematic diagram of electrical control system (other than HVAC, covered elsewhere).
- c. Manufacturers' operating and maintenance manuals on active electrical equipment.

1.7 WARRANTY

Provide equipment items supported by service organizations that are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

As a minimum, meet requirements of UL, where UL standards are established for those items, and requirements of NFPA 70 for all materials, equipment, and devices.

2.2 CONDUIT AND FITTINGS

Conform to the following:

2.2.1 Rigid Metallic Conduit

2.2.1.1 Rigid, Threaded Zinc-Coated Steel Conduit

ANSI C80.1, UL 6.

2.2.2 Electrical, Zinc-Coated Steel Metallic Tubing (EMT)

UL 797, ANSI C80.3.

2.2.3 Flexible Metal Conduit

UL 1, limited to 1829 mm.

2.2.3.1 Liquid-Tight Flexible Metal Conduit, Steel

UL 360, limited to 1829 mm.

2.2.4 Fittings for Metal Conduit, EMT, and Flexible Metal Conduit

UL 514B. Ferrous fittings: cadmium- or zinc-coated in accordance with UL 514B.

2.2.4.1 Fittings for Rigid Metal Conduit

Threaded-type. Split couplings unacceptable.

2.2.4.2 Fittings for EMT

Steel compression type.

2.3 OUTLET BOXES AND COVERS

UL 514A, cadmium- or zinc-coated, if ferrous metal. UL 514C, if nonmetallic.

2.3.1 Outlet Boxes for Telecommunications System

Provide the following:

- a. Standard type 100 mm square by 54 mm deep.
- b. Outlet boxes for wall-mounted telecommunications outlets: 100 by 54 by 54 mm deep.
- c. Depth of boxes: large enough to allow manufacturers' recommended conductor bend radii.

2.4 CABINETS, JUNCTION BOXES, AND PULL BOXES

UL 50; volume greater than 1640 mL, NEMA Type 1 enclosure; sheet steel, hot-dip, zinc-coated. Where exposed to wet, damp, or corrosive environments, NEMA Type 4X.

2.5 WIRES AND CABLES

Provide wires and cables in accordance applicable requirements of NFPA 70 and UL for type of insulation, jacket, and conductor specified or indicated. Do not use wires and cables manufactured more than 24 months prior to date of delivery to site.

2.5.1 Conductors

Provide the following:

- a. Conductor sizes and capacities shown are based on copper, unless indicated otherwise.
- b. Conductors No. 8 AWG and larger diameter: stranded.
- c. Conductors No. 10 AWG and smaller diameter: solid.
- d. Conductors for remote control, alarm, and signal circuits, classes 1, 2, and 3: stranded unless specifically indicated otherwise.
- e. All conductors: copper.

2.5.1.1 Minimum Conductor Sizes

Provide minimum conductor size in accordance with the following:

- a. Branch circuits: No. 12 AWG.
- b. Class 1 remote-control and signal circuits: No. 14 AWG.
- c. Class 2 low-energy, remote-control and signal circuits: No. 16 AWG.
- d. Class 3 low-energy, remote-control, alarm and signal circuits: No. 22 AWG.

2.5.2 Color Coding

Provide color coding for service, feeder, branch, control, and signaling circuit conductors.

2.5.2.1 Ground and Neutral Conductors

Provide color coding of ground and neutral conductors as follows:

- a. Grounding conductors: Green.
- b. Neutral conductors: White.
- c. Exception, where neutrals of more than one system are installed in same raceway or box, other neutrals color coding: white with a different colored (not green) stripe for each.

2.5.2.2 Ungrounded Conductors

Provide color coding of ungrounded conductors in different voltage systems as follows:

- a. 208/120 volt, three-phase
 - (1) Phase A - black
 - (2) Phase B - red
 - (3) Phase C - blue
- b. 480/277 volt, three-phase
 - (1) Phase A - brown
 - (2) Phase B - orange
 - (3) Phase C - yellow
- c. 120/240 volt, single phase: Black and red

2.5.3 Insulation

Unless specified or indicated otherwise or required by NFPA 70, provide power and lighting wires rated for 600-volts, Type THWN/THHN conforming to UL 83 or Type XHHW conforming to UL 44, except that grounding wire may be

type TW conforming to [UL 83](#); remote-control and signal circuits: Type TW or TF, conforming to [UL 83](#). Where equipment or devices require 90-degree Centigrade (C) conductors, provide only conductors with 90-degree C insulation or better.

2.5.4 Bonding Conductors

[ASTM B1](#), solid bare copper wire for sizes No. 8 AWG and smaller diameter; [ASTM B8](#), Class B, stranded bare copper wire for sizes No. 6 AWG and larger diameter.

2.5.4.1 Bonding Conductor for Telecommunications

Provide a copper conductor Bonding Conductor for Telecommunications between the telecommunications main grounding busbar (PBB) and the electrical service ground in accordance with [TIA-607](#). Size the bonding conductor for telecommunications the same as the TBB.

2.5.5 Cable Tray Cable or Power Limited Tray Cable

UL listed; type TC-ER or PLTC-ER.

2.6 SPLICES AND TERMINATION COMPONENTS

[UL 486A-486B](#) for wire connectors and [UL 510](#) for insulating tapes. Connectors for No. 10 AWG and smaller diameter wires: insulated, pressure-type in accordance with [UL 486A-486B](#) or [UL 486C](#) (twist-on splicing connector). Provide solderless terminal lugs on stranded conductors.

2.7 DEVICE PLATES

Provide the following:

- a. UL listed, one-piece device plates for outlets to suit the devices installed.
- b. For metal outlet boxes, plates on unfinished walls: zinc-coated sheet steel or cast metal having round or beveled edges.
- c. Plates on finished walls: satin finish stainless steel or brushed-finish aluminum, minimum [0.792 mm](#) thick.
- d. Screws: machine-type with countersunk heads in color to match finish of plate.
- e. Sectional type device plates are not be permitted.
- f. Plates installed in wet locations: gasketed and UL listed for "wet locations."

2.8 SWITCHES

2.8.1 Toggle Switches

[NEMA WD 1](#), [UL 20](#), poles as indicated, totally enclosed with bodies of thermoplastic or thermoset plastic and mounting strap with grounding screw. Include the following:

- a. Handles: white thermoplastic.
- b. Wiring terminals: screw-type, side-wired.
- c. Contacts: silver-cadmium and contact arm - one-piece copper alloy.
- d. Switches: rated quiet-type ac only, 120/277 volts, with current rating and number of poles indicated.

2.8.2 Disconnect Switches

NEMA KS 1. Provide heavy duty-type switches where indicated, where switches are rated higher than 240 volts, and for double-throw switches. Utilize Class R fuseholders and fuses for fused switches, unless indicated otherwise. Provide horsepower rated for switches serving as the motor-disconnect means. Provide switches in NEMA 1 or 4X Type 304 stainless steel, enclosure as indicated per **NEMA ICS 6.**

2.9 RECEPTACLES

Provide the following:

- a. **UL 498**, general purpose specification grade, grounding-type. Residential grade receptacles are not acceptable.
- b. Ratings and configurations: as indicated.
- c. Bodies: white as per **NEMA WD 1.**
- d. Face and body: thermoplastic supported on a metal mounting strap.
- e. Dimensional requirements: per **NEMA WD 6.**
- f. Screw-type, side-wired wiring terminals or of the solderless pressure type having suitable conductor-release arrangement.
- g. Grounding pole connected to mounting strap.
- h. The receptacle: containing triple-wipe power contacts and double or triple-wipe ground contacts.

2.9.1 Weatherproof Receptacles

Provide receptacles, UL listed for use in "wet locations" with integral GFCI protection. Include cast metal box with gasketed, hinged, lockable and weatherproof while-in-use, die-cast metal/aluminum extra-duty rated hood.

2.9.2 Ground-Fault Circuit Interrupter Receptacles

UL 943, duplex type for mounting in standard outlet box. Provide device capable of detecting current leak when the current to ground is 6 milliamperes or higher, and tripping per requirements of **UL 943** for Class A ground-fault circuit interrupter devices. Provide screw-type, side-wired wiring terminals or pre-wired (pigtail) leads.

2.9.3 Special Purpose Receptacles

Receptacles serving hazardous areas are special purpose. Provide in

ratings indicated. Provide NEMA 7 enclosures for indoor installations and NEMA 8 enclosures for outdoor or wet locations.

2.10 PANELBOARDS

Provide panelboards in accordance with the following:

- a. **UL 67** and **UL 50** having a short-circuit current rating as indicated.
- b. Panelboards: circuit breaker-equipped.
- c. Designed such that individual breakers can be removed without disturbing adjacent units or without loosening or removing supplemental insulation supplied as means of obtaining clearances as required by UL.
- d. "Specific breaker placement" is required in panelboards to match the breaker placement indicated in the panelboard schedule on the design drawings. If it is not possible to match "specific breaker placement" during construction, obtain Government approval prior to device installation.
- e. Use of "Subfeed Breakers" is not acceptable.
- f. Main breaker: "separately" mounted "above" or "below" branch breakers.
- g. Where "space only" is indicated, make provisions for future installation of breakers.
- h. Directories: indicate load served by each circuit in panelboard.
- i. Directories: indicate source of service to panelboard (e.g., Panel PA served from Panel MDP).
- j. Type directories and mount in holder behind transparent protective covering.
- k. Panelboards: listed and labeled for their intended use.
- l. Panelboard nameplates: provided in accordance with paragraph FIELD FABRICATED NAMEPLATES.

2.10.1 Enclosure

Provide panelboard enclosure in accordance with the following:

- a. **UL 50**.
- b. Cabinets mounted outdoors or flush-mounted: hot-dipped galvanized after fabrication.
- c. Cabinets: painted in accordance with paragraph FIELD APPLIED PAINTING.
- d. Front edges of cabinets: form-flanged or fitted with structural shapes welded or riveted to the sheet steel, for supporting the panelboard front.
- e. All cabinets: fabricated such that no part of any surface on the finished cabinet deviates from a true plane by more than **3 mm**.

- f. Holes: provided in the back of indoor surface-mounted cabinets, with outside spacers and inside stiffeners, for mounting the cabinets with a 15 mm clear space between the back of the cabinet and the wall surface.
- g. Flush doors: mounted on hinges that expose only the hinge roll to view when the door is closed.
- h. Each door: fitted with a combined catch and lock latch.
- i. Keys: two provided with each lock, with all locks keyed alike.
- j. Finished-head cap screws: provided for mounting the panelboard fronts on the cabinets.

2.10.2 Panelboard Buses

Support bus bars on bases independent of circuit breakers. Design main buses and back pans so that breakers may be changed without machining, drilling, or tapping. Provide isolated neutral bus in each panel for connection of circuit neutral conductors. Provide separate ground bus identified as equipment grounding bus per UL 67 for connecting grounding conductors; bond to steel cabinet.

2.10.3 Circuit Breakers

UL 489, thermal magnetic-type having a minimum short-circuit current rating equal to the short-circuit current rating of the panelboard in which the circuit breaker will be mounted. Breaker terminals: UL listed as suitable for type of conductor provided. Series rated circuit breakers and plug-in circuit breakers are unacceptable.

2.10.3.1 Multipole Breakers

Provide common trip-type with single operating handle. Design breaker such that overload in one pole automatically causes all poles to open. Maintain phase sequence throughout each panel so that any three adjacent breaker poles are connected to Phases A, B, and C, respectively.

2.10.3.2 Circuit Breaker With Ground-Fault Circuit Interrupter

UL 943 and NFPA 70. Provide with auto-monitoring (self-test) and lockout features, "push-to-test" button, visible indication of tripped condition, and ability to detect and trip when current imbalance is 6 milliamperes or higher per requirements of UL 943 for Class A ground-fault circuit interrupter devices.

2.11 MOTOR SHORT-CIRCUIT PROTECTOR (MSCP)

Motor short-circuit protectors, also called motor circuit protectors (MCPs): UL 508 and UL 489, and provided as shown. Provide MSCPs that consist of an adjustable instantaneous trip circuit breaker used only in conjunction with a combination motor controller which provides coordinated motor branch-circuit overload and short-circuit protection. Rate MSCPs in accordance with the requirements of NFPA 70.

2.12 TRANSFORMERS

Provide transformers in accordance with the following:

- a. **NEMA ST 20**, general purpose, dry-type, self-cooled, ventilated.
- b. Provide transformers in NEMA 1 enclosure.
- c. Taps for transformers 15 kVA and larger: Two 2.5 percent taps Full Capacity Above Nominal (FCAN) and two 2.5 percent taps Full Capacity Below Nominal (FCBN).
- d. Transformer insulation system:
 - (1) 220 degrees C insulation system for transformers 15 kVA and greater, with temperature rise not exceeding 115 degrees C under full-rated load in maximum ambient of 40 degrees C.
 - (2) 180 degrees C insulation for transformers rated 10 kVA and less, with temperature rise not exceeding 80 degrees C under full-rated load in maximum ambient of 40 degrees C.
- e. Transformer of 150 degrees C temperature rise is not acceptable.
- f. Transformer of 115 degrees C temperature rise: capable of carrying continuously 115 percent of nameplate kVA without exceeding insulation rating.
- g. Transformer of 80 degrees C temperature rise: capable of carrying continuously 130 percent of nameplate kVA without exceeding insulation rating.

2.12.1 Specified Transformer Efficiency

Transformers, indicated and specified with: 480V primary, 80 degrees C or 115 degrees C temperature rise, kVA ratings of 37.5 to 100 for single phase or 30 to 500 for three phase, energy efficient type. The transformer is not acceptable if the calculated transformer efficiency is less than the efficiency indicated in **10 CFR 431**, Subpart K. Transformer shall meet US Department of Energy 2016 efficiency standards.

2.13 MOTORS

Provide motors in accordance with the following:

- a. **NEMA MG 1** FIRE PUMPS.
- b. Hermetic-type sealed motor compressors: Also comply with **UL 984**.
- c. Provide the size in terms of hp or or full-load current, or a combination of these characteristics, and other characteristics, of each motor as indicated or specified.
- d. Determine specific motor characteristics to ensure provision of correctly sized starters and overload heaters.
- e. Rate motors for operation on 208-volt, 3-phase circuits with a terminal voltage rating of 200 volts, and those for operation on 480-volt, 3-phase circuits with a terminal voltage rating of 460

volts.

- f. Use motors designed to operate at full capacity with voltage variation of plus or minus 10 percent of motor voltage rating.
- g. Unless otherwise indicated, use continuous duty type motors if rated **746 Watts** and above.
- h. Where fuse protection is specifically recommended by the equipment manufacturer, provide fused switches in lieu of non-fused switches indicated.

2.13.1 High Efficiency Single-Phase Motors

Single-phase fractional-horsepower alternating-current motors: high efficiency types are not acceptable. In exception, for special purpose motors and motor-driven equipment with a minimum seasonal or overall efficiency rating, such as a SEER rating, provide equipment with motor to meet the overall system rating indicated.

2.13.2 Premium Efficiency Polyphase and Single-Phase Motors

Select polyphase and continuous-duty single phase motors based on high efficiency characteristics relative to typical characteristics and applications as listed in **NEMA MG 10** and **NEMA MG 11**. In addition, continuous rated, polyphase squirrel-cage medium induction motors must meet the requirements for premium efficiency electric motors in accordance with **NEMA MG 1**, including the NEMA full load efficiency ratings. In exception, for motor-driven equipment with a minimum seasonal or overall efficiency rating, such as a SEER rating, provide equipment with motor to meet the overall system rating indicated. Electronically Commutated Motors (ECM) to have an integrated controller that senses rotor position and electronically commutates the stator windings to provide synchronous rotation. The rotor must be of a permanent magnet type supported by ball bearings to increase motor efficiency and longevity. The motor must maintain 70 percent or greater efficiency for its design application. ECM motors to comply with **UL 1004-1**.

2.13.3 Motor Sizes

Provide size for duty to be performed, not exceeding the full-load nameplate current rating when driven equipment is operated at specified capacity under most severe conditions likely to be encountered. When motor size provided differs from size indicated or specified, make adjustments to wiring, disconnect devices, and branch circuit protection to accommodate equipment actually provided. Provide controllers for motors rated 1-hp and above with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage. Provide protection for motors from immediate restart by a time adjustable restart relay.

2.13.4 Wiring and Conduit

Provide internal wiring for components of packaged equipment as an integral part of the equipment. Provide power wiring and conduit for field-installed equipment, and motor control equipment forming part of motor control centers or switchgear assemblies, the conduit and wiring connecting such centers, assemblies, or other power sources to equipment as specified herein. Power wiring and conduit: conform to the

requirements specified herein. Control wiring: provided under, and conform to, the requirements of the section specifying the associated equipment.

2.14 MOTOR CONTROLLERS

Provide motor controllers in accordance with the following:

- a. UL 508, NEMA ICS 1, and NEMA ICS 2.
- b. Provide controllers with thermal overload protection in each phase, and one spare normally open auxiliary contact, and one spare normally closed auxiliary contact.
- c. Provide controllers for motors rated 1-hp 746 watts and above with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage.
- d. Provide protection for motors from immediate restart by a time adjustable restart relay.
- e. When used with pressure, float, or similar automatic-type or maintained-contact switch, provide a hand/off/automatic selector switch with the controller.
- f. Connections to selector switch: wired such that only normal automatic regulatory control devices are bypassed when switch is in "hand" position.
- g. Safety control devices, such as low and high pressure cutouts, high temperature cutouts, and motor overload protective devices: connected in motor control circuit in "hand" and "automatic" positions.
- h. Control circuit connections to hand/off/automatic selector switch or to more than one automatic regulatory control device: made in accordance with indicated or manufacturer's approved wiring diagram.
- i. Provide selector switch with the means for locking in any position.
- j. Provide a disconnecting means, capable of being locked in the open position, for the motor that is located in sight from the motor location and the driven machinery location. As an alternative, provide a motor controller disconnect, capable of being locked in the open position, to serve as the disconnecting means for the motor if it is in sight from the motor location and the driven machinery location.
- k. Overload protective devices: provide adequate protection to motor windings; be thermal inverse-time-limit type; and include manual reset-type pushbutton on outside of motor controller case.
- l. Cover of combination motor controller and manual switch or circuit breaker: interlocked with operating handle of switch or circuit breaker so that cover cannot be opened unless handle of switch or circuit breaker is in "off" position.

2.14.1 Control Wiring

Provide control wiring in accordance with the following:

- a. All control wire: stranded tinned copper switchboard wire with 600-volt flame-retardant insulation Type SIS meeting [UL 44](#), or Type MTW meeting [UL 1063](#), and passing the VW-1 flame tests included in those standards.
- b. Hinge wire: Class K stranding.
- c. Current transformer secondary leads: not smaller than No. 10 AWG.
- d. Control wire minimum size: No. 14 AWG.
- e. Power wiring for 480-volt circuits and below: the same type as control wiring with No. 12 AWG minimum size.
- f. Provide wiring and terminal arrangement on the terminal blocks to permit the individual conductors of each external cable to be terminated on adjacent terminal points.

2.14.2 Control Circuit Terminal Blocks

Provide control circuit terminal blocks in accordance with the following:

- a. [NEMA ICS 4](#).
- b. Control circuit terminal blocks for control wiring: molded or fabricated type with barriers, rated not less than 600 volts.
- c. Provide terminals with removable binding, fillister or washer head screw type, or of the stud type with contact and locking nuts.
- d. Terminals: not less than No. 10 in size with sufficient length and space for connecting at least two indented terminals for 10 AWG conductors to each terminal.
- e. Terminal arrangement: subject to the approval of the Contracting Officer with not less than four spare terminals or 10 percent, whichever is greater, provided on each block or group of blocks.
- f. Modular, pull apart, terminal blocks are acceptable provided they are of the channel or rail-mounted type.
- g. Submit data showing that any proposed alternate will accommodate the specified number of wires, are of adequate current-carrying capacity, and are constructed to assure positive contact between current-carrying parts.

2.14.2.1 Types of Terminal Blocks

- a. Short-Circuiting Type: Short-circuiting type terminal blocks: furnished for all current transformer secondary leads with provision for shorting together all leads from each current transformer without first opening any circuit. Terminal blocks: comply with the requirements of paragraph CONTROL CIRCUIT TERMINAL BLOCKS above.
- b. Load Type: Load terminal blocks rated not less than 600 volts and of adequate capacity: provided for the conductors for NEMA Size 3 and smaller motor controllers and for other power circuits, except those for feeder tap units. Provide terminals of either the stud type with contact nuts and locking nuts or of the removable screw type, having

length and space for at least two indented terminals of the size required on the conductors to be terminated. For conductors rated more than 50 amperes, provide screws with hexagonal heads. Conducting parts between connected terminals must have adequate contact surface and cross-section to operate without overheating. Provide each connected terminal with the circuit designation or wire number placed on or near the terminal in permanent contrasting color.

2.14.3 Control Circuits

Control circuits: maximum voltage of 120 volts derived from control transformer in same enclosure. Transformers: conform to [UL 506](#), as applicable. Transformers, other than transformers in bridge circuits: provide primaries wound for voltage available and secondaries wound for correct control circuit voltage. Size transformers so that 80 percent of rated capacity equals connected load. Provide disconnect switch on primary side. Provide fuses in each ungrounded primary feeder. Provide one fused secondary lead with the other lead grounded.

2.14.4 Enclosures for Motor Controllers

[NEMA ICS 6](#).

2.14.5 Pushbutton Stations

Provide with "start/stop" momentary contacts having one normally open and one normally closed set of contacts, and red lights to indicate when motor is running. Stations: heavy duty, oil-tight design.

2.14.6 Pilot and Indicating Lights

Provide LED cluster lamps.

2.14.7 Reduced-Voltage Controllers

Provide for polyphase motors 20 horsepower and larger. Reduced-voltage starters: single-step, closed transition solid state-type, or as indicated, with an adjustable time interval between application of reduced and full voltages to motors. Provide with integral bypass contractor. Provide with 300 percent start-up current limitation.

2.15 [MANUAL MOTOR STARTERS](#) (MOTOR RATED SWITCHES)

Double pole designed for surface mounting with overload protection and pilot lights.

2.15.1 Pilot Lights

Provide yoke-mounted, seven element LED cluster light module. Color: green.

2.16 [MOTOR CONTROL CENTERS](#)

Submit wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Identify circuit terminals on wiring diagrams and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Indicate on the drawings

adequate clearance for operation, maintenance, and replacement of operating equipment devices.

Provide motor control centers in accordance with the following:

- a. **UL 845, NEMA ICS 2, NEMA ICS 3.**
- b. Wiring: Class II, Type B, in NEMA Type 1 enclosure.
- c. Provide control centers suitable for operation on 480-volt, 3-phase, 4-wire, 60 Hz system with minimum short-circuit withstand and interrupting rating of amperes rms symmetrical as indicated.
- d. Incoming power feeder: cable entering at the bottom of enclosure and terminating on main protective device.
- e. Main protective device: low-voltage power circuit breaker rated at amperes rms symmetrical interrupting capacity no less than the MCC short circuit interrupting rating.
- f. Interconnecting wires: copper.
- g. Terminal blocks: plug-in-type so that controllers may be removed without disconnecting individual control wiring.

2.16.1 Bus Systems

Provide the following bus systems. Power bus: be braced to withstand fault current of amperes rms symmetrical as indicated. Wiring troughs: isolated from horizontal and vertical bus bars.

2.16.1.1 Horizontal and Main Buses

Horizontal bus: continuous current rating of amperes as indicated. Main bus: copper, silver-plated enclosed in isolated compartment at top of each vertical section. Main bus: isolated from wire troughs, starters, and other areas.

2.16.1.2 Vertical Bus

Vertical bus: continuous current rating of 450 amperes, and aluminum, tin-plated. Vertical bus: enclosed in flame-retardant, polyester glass "sandwich."

2.16.1.3 Ground Bus

Copper ground bus: provided full width of motor control center and equipped with necessary lugs.

2.16.1.4 Neutral Bus

Insulated neutral bus: provided continuous through the motor control center; neutral full rated. Provide lugs of appropriate capacity, as required.

2.16.2 Combination Motor Controllers

UL 508 and other requirements in paragraph, MOTOR CONTROLLERS. Provide in controller a molded case circuit breaker. Minimum short circuit withstand

rating of combination motor controller: 25,000 rms symmetrical amperes.
Circuit breakers for combination controllers: thermal magnetic.

2.16.3 Circuit Breaker Key Interlock

Provide the main and generator circuit breakers with a trapped key interlocking system to prevent the two breakers from being closed at the same time. System shall be a commercially available standard product based on a mechanical bolt and shall allow for key removal only when a circuit breaker is in the "open" position. Keys shall be controlled with only a single key provided for the system. Duplicate keys shall be controlled by the system manufacturer.

2.16.4 Protective Devices

2.16.4.1 Insulated-Case Breaker

Provide the following:

- a. UL 489. UL listed and labeled, 100 percent rated main breaker, manually operated, low voltage, power circuit breaker, with a short-circuit current rating as indicated at 480 volts.
- b. Breaker frame size: as indicated.
- c. Series rated circuit breakers are unacceptable.

2.16.4.2 Molded-Case Circuit Breaker

Provide the following:

- a. UL 489. UL listed and labeled, standard rated branch breakers, manually operated, low voltage molded-case circuit breaker, with a short-circuit current rating of as indicated at 480 volts.
- b. Breaker frame size: as indicated.
- c. Series rated circuit breakers are unacceptable.

2.16.4.3 Electronic Trip Units

Equip main and distribution breakers as indicated with a solid-state tripping system consisting of three current sensors and a microprocessor-based trip unit that provides true rms sensing adjustable time-current circuit protection. Include the following:

- a. Current sensors ampere rating: the same as the breaker frame rating.
- b. Trip unit ampere rating: as indicated.
- c. Electronic trip units: provide additional features as indicated:
 - (1) Breakers: include long delay pick-up and time settings, and LED indication of cause of circuit breaker trip.
 - (2) Main breakers: include short delay pick-up and time settings and instantaneous settings.
 - (3) Distribution breakers: include short delay pick-up and time

settings and instantaneous settings.

- (4) Main Breakers: include a digital display for phase and ground current.
- (5) Main Breakers: include a digital display for watts, vars, VA, kWh, kvarh, and kVAh.
- (6) Main Breakers: include a digital display for phase voltage, and percent THD voltage and current.

2.17 LOCKOUT REQUIREMENTS

Provide circuit breakers, disconnecting means, and other devices that are electrical energy-isolating capable of being locked out for machines and other equipment to prevent unexpected startup or release of stored energy in accordance with 29 CFR 1910.147, NFPA 70E and 29 CFR 1910.303. Comply with requirements of Division 23, "Heating, Ventilating, and Air Conditioning (HVAC)" for mechanical isolation of machines and other equipment.

2.18 TELECOMMUNICATIONS SYSTEM

Provide system of telecommunications wire-supporting structures (pathway), including: outlet boxes, conduits with pull wires and other accessories for telecommunications outlets and pathway in accordance with TIA-569 and as specified herein. Additional telecommunications requirements are specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

2.19 GROUNDING AND BONDING EQUIPMENT

2.19.1 Ground Rods

UL 467. Ground rods: cone pointed copper-clad steel, with minimum diameter of 19 mm and minimum length of 3050 mm.

2.19.2 Ground Bus

Copper ground bus: provided in the electrical equipment rooms as indicated.

2.19.3 Primary Bonding Busbar

Provide corrosion-resistant grounding busbar suitable for indoor installation in accordance with TIA-607. Busbars: plated for reduced contact resistance. If not plated, clean the busbar prior to fastening the conductors to the busbar and apply an anti-oxidant to the contact area to control corrosion and reduce contact resistance. Provide a Primary bonding busbar (PBB) in the telecommunications entrance facility. The Primary bonding busbar (PBB): sized in accordance with the immediate application requirements and with consideration of future growth. Provide Primary bonding busbars with the following:

- a. Predrilled copper busbar provided with holes for use with standard sized lugs,
- b. Minimum dimensions of 6 mm thick by 100 mm wide for the PBB with length as indicated;

- c. Listed by a nationally recognized testing laboratory.

2.20 HAZARDOUS LOCATIONS

Electrical materials, equipment, and devices for installation in hazardous locations, as defined by NFPA 70: specifically approved by Underwriters' Laboratories, Inc., or Factory Mutual for particular "Class," "Division," and "Group" of hazardous locations involved. Boundaries and classifications of hazardous locations: as indicated. Equipment in hazardous locations: comply with UL 1203 for electrical equipment and industrial controls and UL 674 for motors. Provide NEMA 7 enclosures for indoor installations and NEMA 8 enclosures for outdoor or wet locations.

2.21 MANUFACTURER'S NAMEPLATE

Provide on each item of equipment a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.22 FIELD FABRICATED NAMEPLATES

Provide field fabricated nameplates in accordance with the following:

- a. ASTM D709.
- b. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified or as indicated on the drawings.
- c. Each nameplate inscription: identify the function and, when applicable, the position.
- d. Nameplates: melamine plastic, 3 mm thick, white with black center core.
- e. Provide red laminated plastic label with white center core where indicated.
- f. Surface: matte finish. Corners: square. Accurately align lettering and engrave into the core.
- g. Minimum size of nameplates: 25 by 65 mm.
- h. Lettering size and style: a minimum of 6.35 mm high normal block style.

2.23 WARNING SIGNS

Provide warning signs for flash protection in accordance with NFPA 70E and NEMA 2535.4 for switchboards, panelboards, industrial control panels, and motor control centers that are in other than dwelling occupancies and are likely to require examination, adjustment, servicing, or maintenance while energized. Provide field installed signs to warn qualified persons of potential electric arc flash hazards when warning signs are not provided by the manufacturer. Provide marking that is clearly visible to qualified persons before examination, adjustment, servicing, or maintenance of the equipment.

2.24 METERING

2.24.1 Digital Meters

IEEE C37.90.1 for surge withstand. Provide true rms, plus/minus one percent accuracy, programmable, microprocessor-based meter enclosed in a sealed case with the following features.

a. Display capability:

- (1) Multi-Function Meter: Display a selected phase to neutral voltage, phase to phase voltage, percent phase to neutral voltage THD, percent phase to phase voltage THD; a selected phase current, neutral current, percent phase current THD, percent neutral current; selected total PF, kW, KVA, kVAR, FREQ, kVAh, kWh. Detected alarm conditons include over/under current, over/under voltage, over/under KVA, over/under frequency, over/under selected PF/kVAR, voltage phase reversal, voltage imbalance, reverse power, over percent THD. Include a Form C KYZ pulse output relay on the meter.
- (2) Power Meter: Display Watts, VARs, and selected KVA/PF. Detected alarm conditions include over/under KVA, over/under PF, over/under VARs, over/under reverse power.
- (3) Volt Meter: Provide capability to be selectable between display of the three phases of phase to neutral voltages and display of the three phases of the phase to phase voltages. Detected alarm conditions include over/under voltage, over/under voltage imbalance, over percent THD.
- (4) Ammeter: Display phase A, B, and C currents. Detected alarm conditions include over/under current, over percent THD.
- (5) Digital Watthour Meter: Provide a single selectable display for watts, total kilowatt hours (kWh) and watt demand (Wd). Include a Form C KYZ pulse output relay on the meter.

- b. Design meters to accept input from standard 5A secondary instrument transformers and direct voltage monitoring range to 600 volts, phase to phase.
- c. Provide programming via a front panel display and a communication interface accessible by a computer.
- d. Provide password secured programming stored in non-volatile EEPROM memory.
- e. Provide digital communications in a Modbus protocol via a RS232C serial port.
- f. Provide communications port in Ethernet format with RJ-45 jack. Cybersecurity requirements are specified in Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS.
- g. Provide meter that calculates and stores average max/min demand values with time and date for all readings based on a user selectable sliding window averaging period.

- h. Provide meter with programmable hi/low set limits with two Form C dry contact relays when exceeding alarm conditions.
- i. Include historical trend logging capability with the ability to store up to 100,000 data points with intervals of 1 second to 180 minutes. Provide a unit that can store and time stamp up to 1000 programmable triggered conditions.
- j. Provide event waveform recording triggered by the rms of 2 cycles of voltage or current exceeding programmable set points. Store waveforms for all 6 channels of voltage and current for a minimum of 10 cycles prior to the event and 50 cycles past the event.

2.25 SURGE PROTECTIVE DEVICES

Provide parallel type surge protective devices (SPD) which comply with [UL 1449](#) at the service entrance, MCC and panelboards where indicated. Provide surge protectors in a NEMA 1 enclosure per [NEMA ICS 6](#). Do not install SPD inside a panelboard or switchboard enclosure. However, SPD may be installed in a separate compartment of a switchgear provided that it is supplied by a dedicated circuit breaker. SPD must have the same short-circuit current rating as the protected equipment and must not be installed at a point of system where the available fault current is in excess of that rating. Use Type 1 or Type 2 SPD and connect on the load side of a dedicated circuit breaker. Submit performance and characteristic curves.

Provide the following modes of protection:

FOR SINGLE PHASE AND THREE PHASE WYE CONNECTED SYSTEMS-
Phase to phase (L-L)
Each phase to neutral (L-N)
Neutral to ground (N-G)
Phase to ground (L-G)

SPDs at the service entrance: provide with a minimum surge current rating of 80,000 amperes for L-L mode minimum and 40,000 amperes for other modes (L-N, L-G, and N-G) and downstream SPDs rated 40,000 amperes for L-L mode minimum and 20,000 amperes for other modes (L-N, L-G, and N-G).

Provide SPDs per NFPA 780 for the lightning protection system.

Maximum L-N, and N-G Voltage Protection Rating:

700V for 120/240V, three phase system
700V for 208Y/120V, three phase system
1,200V for 480Y/277V, three phase system

Maximum L-G Protection Rating:

700V for 208Y/120V, three phase system
1,200V for 480Y/277V, three phase system

Maximum L-L Voltage Protection Rating:

1,200V for 208Y/120V, three phase system
1,800V for 480Y/277V, three phase system

The minimum MCOV (Maximum Continuous Operating Voltage) rating for L-N and

L-G modes of operation: 120 percent of nominal voltage for 240 volts and below; 115 percent of nominal voltage above 240 volts to 480 volts.

2.26 EMERGENCY FUEL SHUTOFF PUSHBUTTON

NEMA ICS 1, NEMA ICS 2, and UL 508. Non-illuminated pushbuttons shall be heavy duty, NEMA 4, round, utilize a 22 mm mounting hole, and have the number and type of contacts as indicated on the drawings or elsewhere in the specifications. The emergency shutoff pushbutton shall be red mushroom head, 38 mm diameter, momentary contact. Pushbuttons shall be rated 600 V, 10 amps continuous. Pushbutton enclosure shall be rated NEMA 1 when installed indoors, NEMA 4X when installed outdoors and NEMA 7/8 explosion proof when installed in a Class I, Division 1 location, and include a clearweather resistant cover when installed outdoors. Legend plates shall be provided with each switch with words as indicated on the drawings.

2.27 EMERGENCY FUEL SHUTOFF (EFSO) PANEL AND COMPONENTS

NEMA ICS 1, NEMA 250, AND UL 508. The EFSO system shall be mounted within a NEMA 1 enclosure center as configured as shown on the drawings.

2.27.1 Standard Indicator Lights

NEMA ICS 1, NEMA ICS 2, and UL 508. Lights shall be heavy duty, NEMA 13, 25 mm mounting hole, round indicating lights operating at 120 volts ac/dc. Long life bulbs shall be used. Indicator lights shall have a legend plate with words as shown on drawings. Lens color as indicated on the drawings. Lights shall be "push to test (lamp)" type. LED type lamps of comparable size and color may be substituted for standard indicator lights.

2.27.2 Pushbuttons

NEMA ICS 1, NEMA ICS 2, and UL 508. Non-illuminated pushbuttons shall be heavy duty, NEMA 13, round, utilize a 22 mm mounting hole, and have the number and type of contacts as indicated on the drawings or elsewhere in the specifications. The emergency stop switch shall be a red mushroom head, 38 mm diameter, momentary contact type. Pushbuttons shall be rated 600 volt, 10 amperes continuous. Legend plates shall be provided with each push button with words as indicated on the drawings.

2.27.3 Relays

IEEE C37.90, NEMA ICS 2, UL 508.

2.27.4 Nameplates

Nameplates shall be made of laminated plastic with black outer layers and a white core. Edges shall be chamfered. Nameplates shall be fastened with black-finished round-head drive screws or approved non-adhesive metal fasteners.

2.27.5 Terminal Blocks

NEMA ICS 4. Terminal blocks for conductors exiting the EFSO Panel shall be two-way type with double terminals, one for internal wiring connections and the other for external wiring connections. Terminal blocks shall be made of bakelite or other suitable insulating material with full deep

barrier between each pair of terminals. A terminal identification strip shall form part of the terminal block and each terminal shall be identified by a number in accordance with the numbering scheme on the approved wiring diagrams.

2.28 FACTORY APPLIED FINISH

Provide factory-applied finish on electrical equipment in accordance with the following:

- a. **NEMA 250** corrosion-resistance test and the additional requirements as specified herein.
- b. Interior and exterior steel surfaces of equipment enclosures: thoroughly cleaned followed by a rust-inhibitive phosphatizing or equivalent treatment prior to painting.
- c. Exterior surfaces: free from holes, seams, dents, weld marks, loose scale or other imperfections.
- d. Interior surfaces: receive not less than one coat of corrosion-resisting paint in accordance with the manufacturer's standard practice.
- e. Exterior surfaces: primed, filled where necessary, and given not less than two coats baked enamel with semigloss finish.
- f. Equipment located indoors: ANSI Light Gray, and equipment located outdoors: ANSI Dark Gray.
- g. Provide manufacturer's coatings for touch-up work and as specified in paragraph FIELD APPLIED PAINTING.

2.29 SOURCE QUALITY CONTROL

2.29.1 Transformer Factory Tests

Submittal: include routine **NEMA ST 20** transformer test results on each transformer and also provide the results of NEMA "design" and "prototype" tests that were made on transformers electrically and mechanically equal to those specified.

PART 3 EXECUTION

3.1 INSTALLATION

Electrical installations, including weatherproof and hazardous locations and ducts, plenums and other air-handling spaces: conform to requirements of **NFPA 70** and **IEEE C2** and to requirements specified herein.

3.1.1 Underground Service

Underground service conductors and associated conduit: continuous from service entrance equipment to outdoor power system connection.

3.1.2 Hazardous Locations

Perform work in hazardous locations, as defined by **NFPA 70**, in strict accordance with **NFPA 70** for particular "Class," "Division," and "Group" of

hazardous locations involved. Provide conduit and cable seals where required by NFPA 70. Provide conduit with tapered threads.

3.1.3 Service Entrance Identification

Service entrance disconnect devices, switches, and enclosures: labeled and identified as such.

3.1.3.1 Labels

Wherever work results in service entrance disconnect devices in more than one enclosure, as permitted by NFPA 70, label each enclosure, new and existing, as one of several enclosures containing service entrance disconnect devices. Label, at minimum: indicate number of service disconnect devices housed by enclosure and indicate total number of enclosures that contain service disconnect devices. Provide laminated plastic labels conforming to paragraph FIELD FABRICATED NAMEPLATES. Use lettering of at least 6.35 mm in height, and engrave on black-on-white matte finish. Service entrance disconnect devices in more than one enclosure: provided only as permitted by NFPA 70.

3.1.4 Wiring Methods

Provide insulated conductors installed in rigid steel conduit or EMT, except where specifically indicated or specified otherwise or required by NFPA 70 to be installed otherwise. Grounding conductor: separate from electrical system neutral conductor. Provide insulated green equipment grounding conductor for circuit(s) installed in conduit and raceways. Minimum conduit size: 20 mm in diameter for low voltage lighting and power circuits.

3.1.4.1 Pull Wire

Install pull wires in empty conduits. Pull wire: plastic having minimum 890-N force tensile strength. Leave minimum 915 mm of slack at each end of pull wire.

3.1.5 Conduit Installation

Unless indicated otherwise, conceal conduit under floor slabs and within finished walls, ceilings, and floors. Install conduit parallel with or at right angles to ceilings, walls, and structural members where located above accessible ceilings and where conduit will be visible after completion of project. Run conduits under floor slab as if exposed.

3.1.5.1 Restrictions Applicable to EMT

- a. Do not install underground.
- b. Do not encase in concrete, mortar, grout, or other cementitious materials.
- c. Do not use in areas subject to physical damage including but not limited to equipment rooms where moving or replacing equipment could physically damage the EMT.
- d. Do not use in hazardous areas.
- e. Do not use outdoors.

3.1.5.2 Restrictions Applicable to Flexible Conduit

Use only as specified in paragraph FLEXIBLE CONNECTIONS.

3.1.5.3 Underground Conduit

Refer to Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

3.1.5.4 Conduit Through Floor Slabs

Where conduits rise through floor slabs, do not allow curved portion of bends to be visible above finished slab. Where conduit rises through slab-on grade, seal all electrical penetrations to address radon mitigation and prevent infiltration of air, insects, and vermin.

3.1.5.5 Stub-Ups

Provide conduits stubbed up through concrete floor for connection to free-standing equipment with adjustable top or coupling threaded inside for plugs, set flush with finished floor. Extend conductors to equipment in rigid steel conduit, except that flexible metal conduit may be used 150 mm above floor. Where no equipment connections are made, install screwdriver-operated threaded flush plugs in conduit end.

3.1.5.6 Conduit Support

Support conduit by pipe straps, wall brackets, slotted metallic channel, threaded rod conduit hangers, or ceiling trapeze. Plastic cable ties are not acceptable. Fasten by wood screws to wood; by toggle bolts on hollow masonry units; by concrete inserts or expansion bolts on concrete or brick; and by machine screws, welded threaded studs, or spring-tension clamps on steel work. Threaded C-clamps may be used on rigid steel conduit only. Do not weld conduits or pipe straps to steel structures. Do not exceed one-fourth proof test load for load applied to fasteners. Provide vibration resistant and shock-resistant fasteners attached to concrete ceiling. Do not cut main reinforcing bars for any holes cut to depth of more than 40 mm in reinforced concrete beams or to depth of more than 20 mm in concrete joints. Fill unused holes. In partitions of light steel construction, use sheet metal screws. In suspended-ceiling construction, run conduit above ceiling. Do not support conduit by ceiling support system. Conduit and box systems: supported independently of both (a) tie wires supporting ceiling grid system, and (b) ceiling grid system into which ceiling panels are placed. Do not share supporting means between electrical raceways and mechanical piping or ducts. Identify independent conduit support in both fire and non-fire rated assemblies per NFPA 70. Coordinate installation with above-ceiling mechanical systems to assure maximum accessibility to all systems. Spring-steel fasteners may be used for lighting branch circuit conduit supports in suspended ceilings in dry locations. Where conduit crosses building expansion joints, provide suitable watertight expansion fitting that maintains conduit electrical continuity by bonding jumpers or other means. For conduits greater than 63 mm inside diameter, provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction.

3.1.5.7 Directional Changes in Conduit Runs

Make changes in direction of runs with symmetrical bends or cast-metal

fittings. Make field-made bends and offsets with hickey or conduit-bending machine. Do not install crushed or deformed conduits. Avoid trapped conduits. Prevent plaster, dirt, or trash from lodging in conduits, boxes, fittings, and equipment during construction. Free clogged conduits of obstructions.

3.1.5.8 Locknuts and Bushings

Fasten conduits to sheet metal boxes and cabinets with two locknuts where required by NFPA 70, where insulated bushings are used, and where bushings cannot be brought into firm contact with the box; otherwise, use at least minimum single locknut and bushing. Provide locknuts with sharp edges for digging into wall of metal enclosures. Install bushings on ends of conduits, and provide insulating type where required by NFPA 70.

3.1.5.9 Flexible Connections

Provide flexible steel conduit between 915 and 1830 mm in length for recessed and semirecessed lighting fixtures; for equipment subject to vibration, noise transmission, or movement; and for motors. Install flexible conduit to allow 20 percent slack. Minimum flexible steel conduit size: 16 mm diameter. Provide liquid tight flexible conduit in wet and damp locations for equipment subject to vibration, noise transmission, movement or motors. Provide separate ground conductor across flexible connections. Plastic cable ties are not acceptable as a support method.

3.1.5.10 Telecommunications and Signal System Pathway

Install telecommunications pathway in accordance with TIA-569.

- a. Horizontal Pathway: Telecommunications pathways from the work area to the telecommunications room: installed and cabling length requirements in accordance with TIA-568.1. Size conduits in accordance with TIA-569.

3.1.6 Boxes, Outlets, and Supports

Provide boxes in wiring and raceway systems wherever required for pulling of wires, making connections, and mounting of devices or fixtures. Boxes for metallic raceways: cast-metal, hub-type when located in wet locations, when surface mounted on outside of exterior surfaces, when surface mounted on interior walls exposed up to 2135 mm above floors and walkways, or when installed in hazardous areas and when specifically indicated. Boxes in other locations: sheet steel. Provide each box with volume required by NFPA 70 for number of conductors enclosed in box. Boxes for mounting lighting fixtures: minimum 100 mm square, or octagonal, except that smaller boxes may be installed as required by fixture configurations, as approved. Provide gaskets for cast-metal boxes installed in wet locations and boxes installed flush with outside of exterior surfaces. Provide separate boxes for flush or recessed fixtures when required by fixture terminal operating temperature; provide readily removable fixtures for access to boxes unless ceiling access panels are provided. Support boxes and pendants for surface-mounted fixtures on suspended ceilings independently of ceiling supports. Fasten boxes and supports with wood screws on wood, with bolts and expansion shields on concrete or brick, with toggle bolts on hollow masonry units, and with machine screws or welded studs on steel. In open overhead spaces, cast boxes threaded to raceways need not be separately supported except where

used for fixture support; support sheet metal boxes directly from building structure or by bar hangers. Where bar hangers are used, attach bar to raceways on opposite sides of box, and support raceway with approved-type fastener maximum 610 mm from box. When penetrating reinforced concrete members, avoid cutting reinforcing steel.

3.1.6.1 Boxes

Boxes for use with raceway systems: minimum 40 mm deep, except where shallower boxes required by structural conditions are approved. Boxes for other than lighting fixture outlets: minimum 100 mm square, except that 100 by 50 mm boxes may be used where only one raceway enters outlet.

3.1.6.1.1 Wall-Mounted Telecommunications Outlet Box

Provide double gang electrical boxes, minimum standard size 100 mm square and 54 mm deep with plaster ring for connection of single gang faceplate. Design outlet box for recess mounting with the faceplate flush with the wall surface, at the same height as the electrical outlets.

3.1.6.2 Pull Boxes

Construct of at least minimum size required by NFPA 70 of galvanized sheet steel, except where cast-metal boxes are required in locations specified herein. Provide boxes with screw-fastened covers. Where several feeders pass through common pull box, tag feeders to indicate clearly electrical characteristics, circuit number, and panel designation.

3.1.7 Mounting Heights

Mount panelboards, circuit breakers, motor controller and disconnecting switches so height of center of grip of the operating handle of the switch or circuit breaker at its highest position is maximum 2007 mm above floor or working platform or as allowed in Section 404.8 per NFPA 70. Mount lighting switches 1220 mm above finished floor. Mount receptacles and telecommunications outlets 460 mm above finished floor, unless otherwise indicated. Wall-mounted telecommunications outlets: mounted at height 1525 mm above finished floor. Mount other devices as indicated. Measure mounting heights of wiring devices and outlets in non-hazardous areas to center of device or outlet. Measure mounting heights of receptacle outlet boxes in the hazardous areas to the bottom of the outlet box.

3.1.8 Conductor Identification

Provide conductor identification within each enclosure where tap, splice, or termination is made. For conductors No. 6 AWG and smaller diameter, provide color coding by factory-applied, color-impregnated insulation. For conductors No. 4 AWG and larger diameter, provide color coding by plastic-coated, self-sticking markers; colored nylon cable ties and plates; or heat shrink-type sleeves. Identify control circuit terminations in accordance with manufacturer's recommendations. Provide telecommunications system conductor identification as specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEMS.

3.1.8.1 Marking Strips

Provide marking strips for identification of power distribution, control, data, and communications cables in accordance with the following:

- a. Provide white or other light-colored plastic marking strips, fastened by screws to each terminal block, for wire designations.
- b. Use permanent ink for the wire numbers
- c. Provide reversible marking strips to permit marking both sides, or provide two marking strips with each block.
- d. Size marking strips to accommodate the two sets of wire numbers.
- e. Assign a device designation in accordance with **NEMA ICS 1** to each device to which a connection is made. Mark each device terminal to which a connection is made with a distinct terminal marking corresponding to the wire designation used on the Contractor's schematic and connection diagrams.
- f. The wire (terminal point) designations used on the Contractor's wiring diagrams and printed on terminal block marking strips may be according to the Contractor's standard practice; however, provide additional wire and cable designations for identification of remote (external) circuits for the Government's wire designations.
- g. Prints of the **marking strips** drawings submitted for approval will be so marked and returned to the Contractor for addition of the designations to the terminal strips and tracings, along with any rearrangement of points required.

3.1.9 Splices

Make splices in accessible locations. Make splices in conductors No. 10 AWG and smaller diameter with insulated, pressure-type connector. Make splices in conductors No. 8 AWG and larger diameter with solderless connector, and cover with insulation material equivalent to conductor insulation.

3.1.10 Covers and Device Plates

Install with edges in continuous contact with finished wall surfaces without use of mats or similar devices. Plaster fillings are not permitted. Install plates with alignment tolerance of **0.58 mm**. Use of sectional-type device plates are not permitted. Provide gasket for plates installed in wet locations.

3.1.11 Grounding and Bonding

Provide in accordance with **NFPA 70** and **NFPA 780**. Ground exposed, non-current-carrying metallic parts of electrical equipment, access flooring support system, metallic raceway systems, grounding conductor in metallic and nonmetallic raceways, telecommunications system grounds, and neutral conductor of wiring systems. Make ground connection at main service equipment, and extend grounding conductor to point of entrance of metallic water service. Make connection to water pipe by suitable ground clamp or lug connection to plugged tee. If flanged pipes are encountered, make connection with lug bolted to street side of flanged connection. Supplement metallic water service grounding system with additional made electrode in compliance with **NFPA 70**. Make ground connection to driven ground rods on exterior of building. Bond additional driven rods together with a minimum of 4/0 AWG soft bare copper wire buried to a depth of at least **300 mm**. Interconnect all grounding media in or on the structure to

provide a common ground potential. This includes lightning protection, electrical service, telecommunications system grounds, as well as underground metallic piping systems. Use main size lightning conductors for interconnecting these grounding systems to the lightning protection system. In addition to the requirements specified herein, provide telecommunications grounding in accordance with TIA-607.

3.1.11.1 Ground Rods

Provide ground rods and measure the resistance to ground using the fall-of-potential method described in IEEE 81. Do not exceed 25 ohms under normally dry conditions for the maximum resistance of a driven ground. If this resistance cannot be obtained with a single rod, provide additional rods, spaced 3000 mm on center. If the resultant resistance exceeds 25 ohms measured not less than 48 hours after rainfall, notify the Contracting Officer who will decide on the number of ground rods to add.

3.1.11.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, excepting specifically those connections for which access for periodic testing is required, by exothermic weld or high compression connector.

- a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.
- b. Make high compression connections using a hydraulic or electric compression tool to provide the correct circumferential pressure. Provide tools and dies as recommended by the manufacturer. Use an embossing die code or other standard method to provide visible indication that a connector has been adequately compressed on the ground wire.

3.1.11.3 Ground Bus

Provide a copper ground bus as indicated. Noncurrent-carrying metal parts of transformer neutrals and other electrical equipment: effectively grounded by bonding to the ground bus. Bond the ground bus to both the entrance ground, and to a ground rod or rods as specified above having the upper ends terminating approximately 100 mm above the floor. Make connections and splices of the brazed, welded, bolted, or pressure-connector type, except use pressure connectors or bolted connections for connections to removable equipment. For raised floor equipment rooms in computer and data processing centers, provide a minimum of four, one at each corner, ground buses connected to the building grounding system. Use bolted connections in lieu of thermoweld, so they can be changed as required by additions and alterations.

3.1.11.4 Resistance

Maximum resistance-to-ground of grounding system: do not exceed 5 ohms under dry conditions. Where resistance obtained exceeds 5 ohms, contact Contracting Officer for further instructions.

3.1.11.5 Telecommunications System

Provide telecommunications grounding in accordance with the following:

- a. Telecommunications Grounding Busbars: Provide a Primary bonding busbar (PBB) in the telecommunications entrance facility. Install the PBB as close to the electrical service entrance grounding connection as practicable. Install bonding busbars to maintain clearances as required by NFPA 70 and insulated from its support. A minimum of 50 mm separation from the wall is recommended to allow access to the rear of the busbar and adjust the mounting height to accommodate overhead or underfloor cable routing.
- b. Telecommunications Bonding Conductors: Provide main telecommunications service equipment ground consisting of separate bonding conductor for telecommunications, between the PBB and readily accessible grounding connection of the electrical service. Grounding and bonding conductors should not be placed in ferrous metallic conduit. If it is necessary to place grounding and bonding conductors in ferrous metallic conduit that exceeds one m in length, bond the conductors to each end of the conduit using a grounding bushing or a No. 6 AWG conductor, minimum.
- c. Telecommunications Grounding Connections: Telecommunications grounding connections to the PBB: utilize listed compression two-hole lugs, exothermic welding, suitable and equivalent one hole non-twisting lugs, or other irreversible compression type connections. Bond all metallic pathways, cabinets, and racks for telecommunications cabling and interconnecting hardware located within the same room or space as the PBB to the PBB. In a metal frame (structural steel) building, where the steel framework is readily accessible within the room; bond each PBB to the vertical steel metal frame using a minimum No. 6 AWG conductor. Where the metal frame is external to the room and readily accessible, bond the metal frame to the PBB with a minimum No. 6 AWG conductor. When practicable because of shorter distances and, where horizontal steel members are permanently electrically bonded to vertical column members, the PBB may be bonded to these horizontal members in lieu of the vertical column members. All connectors used for bonding to the metal frame of a building must be listed for the intended purpose.

3.1.12 Equipment Connections

Provide power wiring for the connection of motors and control equipment under this section of the specification. Except as otherwise specifically noted or specified, automatic control wiring, control devices, and protective devices within the control circuitry are not included in this section of the specifications and are provided under the section specifying the associated equipment.

3.1.13 Repair of Existing Work

Perform repair of existing work, demolition, and modification of existing electrical distribution systems as follows:

3.1.13.1 Workmanship

Lay out work in advance. Exercise care where cutting, channeling, chasing, or drilling of floors, walls, partitions, ceilings, or other

surfaces is necessary for proper installation, support, or anchorage of conduit, raceways, or other electrical work. Repair damage to buildings, piping, and equipment using skilled craftsmen of trades involved.

3.1.13.2 Removal of Existing Electrical Distribution System

Removal of existing electrical distribution system equipment includes equipment's associated wiring, including conductors, cables, exposed conduit, surface metal raceways, boxes, and fittings, back to equipment's power source as indicated.

3.1.13.3 Continuation of Service

Maintain continuity of existing circuits of equipment to remain. Maintain existing circuits of equipment energized. Restore circuits wiring and power which are to remain but were disturbed during demolition back to original condition.

3.1.14 Watthour Meters

ANSI C12.1.

3.1.15 Surge Protective Devices

Connect the surge protective devices in parallel to the power source, keeping the conductors as short and straight as practically possible. Maximum allowed lead length is 900 mm avoiding 90 degree bends. Do not locate surge protective devices inside a panelboard or switchboard enclosure.

3.2 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets. Provide nameplate on all equipment in access controlled spaces and areas.

3.3 WARNING SIGN MOUNTING

Provide the number of signs required to be readable from each accessible side. Space the signs in accordance with NFPA 70E.

3.4 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Painting: as specified in Section 09 90 00 PAINTS AND COATINGS.

3.5 FIELD QUALITY CONTROL

Furnish test equipment and personnel and submit written copies of test results. Give Contracting Officer 5 working days notice prior to each test. Where applicable, test electrical equipment in accordance with NETA ATS.

3.5.1 Devices Subject to Manual Operation

Operate each device subject to manual operation at least five times, demonstrating satisfactory operation each time.

3.5.2 600-Volt Wiring Test

Test wiring rated 600 volt and less to verify that no short circuits or accidental grounds exist. Perform insulation resistance tests on wiring No. 6 AWG and larger diameter using instrument which applies voltage of 1,000 volts DC for 600 volt rated wiring and 500 volts DC for 300 volt rated wiring per [NETA ATS](#) to provide direct reading of resistance. All existing wiring to be reused must also be tested.

3.5.3 Transformer Tests

Perform the standard, not optional, tests in accordance with the Inspection and Test Procedures for transformers, dry type, air-cooled, 600 volt and below; as specified in [NETA ATS](#). Measure primary and secondary voltages for proper tap settings. Tests need not be performed by a recognized independent testing firm or independent electrical consulting firm.

3.5.4 Ground-Fault Receptacle Test

Test ground-fault receptacles with a "load" (such as a plug in light) to verify that the "line" and "load" leads are not reversed. Press the TEST button and then the RESET button to verify by LED status that the device is a self-test model as specified in [UL 943](#).

3.5.5 Grounding System Test

Test grounding system to ensure continuity, and that resistance to ground is not excessive. Test each ground rod for resistance to ground before making connections to rod; tie grounding system together and test for resistance to ground. Make resistance measurements in dry weather, not earlier than 48 hours after rainfall. Submit written results of each test to Contracting Officer, and indicate location of rods as well as resistance and soil conditions at time measurements were made.

3.5.6 Watthour Meter

a. Visual and mechanical inspection

- (1) Examine for broken parts, shipping damage, and tightness of connections.
- (2) Verify that meter type, scales, and connections are in accordance with approved shop drawings.

b. Electrical tests

- (1) Determine accuracy of meter.
- (2) Calibrate watthour meters to one-half percent.
- (3) Verify that correct multiplier has been placed on face of meter, where applicable.

3.5.7 Phase Rotation Test

Perform phase rotation test to ensure proper rotation of service power prior to operation of new or reinstalled equipment using a phase rotation meter. Follow the meter manual directions performing the test.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 27 14.00 20

ELECTRICITY METERING

02/21, CHG 1: 05/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY ASSURANCE
 - 1.3.1 Installation Drawings
 - 1.3.2 Standard Products
 - 1.3.3 Material and Equipment Manufacturing Data
- 1.4 MAINTENANCE
 - 1.4.1 Additions to Operation and Maintenance Data
- 1.5 WARRANTY
- 1.6 SYSTEM DESCRIPTION
 - 1.6.1 System Requirements
 - 1.6.2 Selection Criteria

PART 2 PRODUCTS

- 2.1 ELECTRICITY METERS AND ACCESSORIES
 - 2.1.1 Physical and Common Requirements
 - 2.1.2 Current Transformer Requirements
 - 2.1.3 Electricity Meter Requirements
 - 2.1.4 Disconnect Method
 - 2.1.5 Installation Methods
- 2.2 COMMUNICATIONS INTERFACES

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Configuration Software
- 3.2 FIELD QUALITY CONTROL
 - 3.2.1 Performance of Acceptance Checks and Tests
 - 3.2.2 System Functional Verification

-- End of Section Table of Contents --

SECTION 26 27 14.00 20

ELECTRICITY METERING
02/21, CHG 1: 05/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP (2019; Errata 1 2019; Errata 2-6 2020; Addenda BY-CP 2020; Addenda AF-DB 2020; Addenda A-G 2020; Addenda F-Y 2021; Errata 7-8 2021; Interpretation 1-4 2020; Interpretation 5-8 2021; Addenda AS-CB 2022) Energy Standard for Buildings Except Low-Rise Residential Buildings

ASHRAE 90.1 - SI (2019; Errata 1-4 2020; Addenda BY-CP 2020; Addenda AF-DB 2020; Addenda A-G 2020; Addenda F-AB 2021; Errata 5-7 2021; Interpretation 1-4 2020; Interpretation 5-8 2021) Energy Standard for Buildings Except Low-Rise Residential Buildings

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C37.90.1 (2013) Standard for Surge Withstand Capability (SWC) Tests for Relays and Relay Systems Associated with Electric Power Apparatus

IEEE C57.13 (2016) Standard Requirements for Instrument Transformers

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for

Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES, the CONTRACT CLAUSES and DD Form 1423:

Technical data packages consisting of technical data and computer software (meaning technical data which relates to computer software) which are specifically identified in this project and which may be defined/required in other specifications must be delivered strictly in accordance with the CONTRACT CLAUSES and in accordance with the Contract Data Requirements List, DD Form 1423. Data delivered must be identified by reference to the particular specification paragraph against which it is furnished. All submittals not specified as technical data packages are considered 'shop drawings' under the Federal Acquisition Regulation Supplement (FARS) and must contain no proprietary information and be delivered with unrestricted rights.

SD-02 Shop Drawings

Installation Drawings; G

SD-03 Product Data

Electricity Meters; G

The most recent meter product data must be submitted as a Technical Data Package and must be licensed to the project site. Any software must be submitted on CD-ROM and 2 hard copies of the software user manual must be submitted for each piece of software provided.

Current Transformer; G

SD-06 Test Reports

Acceptance Checks and Tests; G

System Functional Verification; G

Meter Configuration Report; G

The meter configuration report must be submitted as a Technical Data Package.

SD-10 Operation and Maintenance Data

Electricity Meters and Accessories, Data Package 5; G

Submit operation and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein.

SD-11 Closeout Submittals

System Functional Verification; G

1.3 QUALITY ASSURANCE

1.3.1 Installation Drawings

Drawings must be provided in hard-copy and electronic format, and must

include but not be limited to the following:

- a. Wiring diagrams with terminals identified of advanced meter, current transformers, and communications interfaces.
- b. One-line diagram, including meters, switch(es), current transformers, communications interfaces, and fuses. Provide one-line diagram to the local Public Works department.

1.3.2 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for one year prior to bid opening. The one-year period must include applications of equipment and materials under similar circumstances and of similar size. The product, or an earlier release of the product, must have been on sale on the commercial market through advertisements, manufacturers catalogs, or brochures during the prior one-year period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.3.3 Material and Equipment Manufacturing Data

Products manufactured more than 1 year prior to date of delivery to site must not be used, unless specified otherwise.

1.4 MAINTENANCE

1.4.1 Additions to Operation and Maintenance Data

In addition to requirements of Data Package 5, include the following on the actual [electricity meters and accessories](#) provided:

- a. A condensed description of how the system operates
- b. Block diagram indicating major assemblies
- c. Troubleshooting information
- d. Preventive maintenance
- e. Prices for spare parts and supply list

1.5 WARRANTY

The equipment items and software must be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment and software on a regular and emergency basis during the warranty period of the contract.

1.6 SYSTEM DESCRIPTION

1.6.1 System Requirements

Electricity metering, consisting of meters and associated equipment, will

be used to record the electricity consumption and other values as described in the requirements that follow and as shown on the drawings. Communication system requirements are contained in a separate specification section as identified in paragraph COMMUNICATIONS INTERFACES.

1.6.2 Selection Criteria

Metering components and software are part of a system that includes the physical meter, data recorder function and communications method. Every building site identified must include sufficient metering components to measure the electrical parameters identified and to store and communicate the values as required.

Contractor must verify that the electricity meter installed on any building site is compatible with the base-wide metering system with respect to the types of meters selected and the method used to program the meters for initial use. Software and meter programming tools are necessary to set up the meters described by this specification. New software tools different from the meter programming methods currently used by base personnel will require an Authority to Operate (ATO) by Command Information Office at the Enterprise level.

Contractor must verify that the metering system installed on any building site is compatible with the facility-wide or base-wide communication and meter reading protocol system.

PART 2 PRODUCTS

2.1 ELECTRICITY METERS AND ACCESSORIES

Provide meter(s) and connect the meter(s) to the existing AMI DAS. The contractor must use the existing government laptop computers to configure the meter using existing software loaded on the computer. The contractor will not be allowed to modify any software or add any additional software to the computer. Alternatively, the government will configure the meter(s), which must be compatible with the existing system, using existing software. Contractor must insure that the meter(s) will transmit the specified data to the DAS. The current meters being used by Naval Station Rota are: ION 8650C with 32 Mb of memory and PM 8000 and communications method of fiber optic links.

Once the devices are connected to the government network, only government-owned equipment can be used to connect to AMI components. Principle characteristics of the NAVSTA Rota Spain Advanced Metering System (AMI) are:

- a. Schneider PowerLogic ION 8650C (firmware version 8650V403) and PM 8000 electrical and mechanical meters connect to a designated Ethernet Point of Presence (POP) at each monitoring site.
- b. ION Setup version 3.0 software from Schneider Electric is installed on the government-owned laptop and local Data Acquisition Server (DAS) for configuration and verification of settings for ION series power and energy meters.
- c. 2911 Cisco Router that routes traffic between the POP, SPM Server, and meter.
- d. StruxureWare Power Monitoring 7.0 (SPM7) local Data Acquisition Server

(DAS) software stores all meter data, power quality information and graphics screens.

- e. Enterprise Energy Manager (EEM 4.0) regional Data Acquisition Server provides energy analysis tools for regional data utilizing the SPM databases.
- f. Both EEM and SPM utilize MS SQL Server 2008.
- g. Extract, Transform, Load (ETL) software provides metering data from SPM databases at each installation to the Centralized and Integrated Reporting for the Navy's Comprehensive Utilities Information Tracking System (CIRCUITS) Meter Data Management (MDM) module following format described in Call for Consistency version: 2.4, 2015-05-28.
- h. Transport is provided by the Navy's Public Safety Network (PSNet).

2.1.1 Physical and Common Requirements

- a. Meter must have NEMA 3R stainless steel enclosure for surface mounting with bottom or rear penetrations.
- b. Surge withstand capability must conform to [IEEE C37.90.1](#).
- c. Use #12 SIS (XHHW, or equivalent) wiring with ring lugs for all meter connections. Color code and mark the conductors as follows:
 - (1) Red - Phase A CT - C1
 - (2) Orange - Phase B CT - C2
 - (3) Brown - Phase C CT - C3
 - (4) Gray with white stripe - neutral current return - C0
 - (5) Black - Phase A voltage - V1
 - (6) Yellow - Phase B voltage - V2
 - (7) Blue - Phase C voltage - V3
 - (8) White - Neutral voltage

2.1.2 [Current Transformer](#) Requirements

- a. Current transformer must be installed with a rating as shown in the schedule.
- b. Current transformers must have an Accuracy Class of 0.15 (with a maximum error of plus/minus 0.3 percent at 5.0 amperes) when operating within the specified rating factor.
- c. Current transformers must be solid-core, bracket-mounted for new installations using ring-tongue lugs for electrical connections. Current transformers must be accessible and the associated wiring must be installed in an organized and neat workmanship arrangement. Current transformers that are retrofitted onto existing switchgear busbar can be a busbar split-core design.
- d. Current transformers must have:
 - (1) Insulation Class: All 600 volt and below current transformers must be rated 10 KV BIL.
 - (2) Frequency: Nominal 60 Hz.

- (3) Burden: Burden class must be selected for the load.
- (4) Phase Angle Range: 0 to 60 degrees.
- e. Meter must accept current input from standard instrument transformers (5A secondary current transformers).
- f. Current inputs must have a continuous rating in accordance with **IEEE C57.13**.
- g. Provide one single-ratio current transformer for each phase per power transformer with characteristics listed in the following table.

Single-Ratio Current Transformer Characteristics				
kVA	Sec. Volt	CT Ratio	RF	Meter Acc. Class
500	480Y:277	600:5	1.5	0.3 thru B-0.5

2.1.3 Electricity Meter Requirements

Notwithstanding any other provision of this contract, meters must be Schneider Electric Power Logic ION 8650; no other product will be acceptable. All meters must meet NAVFAC Cyber Security Requirements.

2.1.4 Disconnect Method

- a. Provide a 10-pole safety disconnect complete with isolation devices for the voltage and current transformer inputs, including a shorting means for the current transformers.
- b. Voltage monitoring circuits must be equipped with disconnect switches to isolate the meter base or socket from the voltage source.

2.1.5 Installation Methods

- a. Stand Mounted Adjacent to Transformer ("STAND" in Metering Systems Schedule). Meter base must be mounted on a structural steel pole approximately **1.2 meters** from the transformer pad. This can be used for multiple meters associated with a single transformers.

2.2 COMMUNICATIONS INTERFACES

Meter must have two-way communication with the existing data acquisition system (DAS). Provide a communications interface utilizing optical fiber link connected to existing meter.

Provide interfacing software if a meter is used that is different than the existing meters at the Activity to ensure compatibility within the metering system.

Connect to the AMI network utilizing a fiber optic link to the nearest existing AMI meter. Provide a communication interface integrated into the NAVFAC Advanced Metering Infrastructure Communications Network. Make the physical connections using a Government provided firewall. The DAS server facilitates communications to the meter for data collection, analysis, and reporting.

PART 3 EXECUTION

3.1 INSTALLATION

Electrical installations must conform to [ASHRAE 90.1 - IP](#), [ASHRAE 90.1 - SI](#), [IEEE C2](#), [NFPA 70](#) (National Electrical Code), and to the requirements specified herein. Provide new equipment and materials unless indicated or specified otherwise.

3.1.1 Configuration Software

The standard meter must include the latest available version of firmware and software. Meter must either be programmed at the factory or must be programmed in the field. Meters must have a password that must be provided to the contracting officer upon project completion. When field programming is performed, turn field programming device over to the Contracting Officer at completion of project. When interfacing software is used for a meter that is different than the existing meters in use at the Activity, turn the software over to the Contracting Officer at completion of the project.

3.2 FIELD QUALITY CONTROL

Perform the following acceptance checks and tests on all installed meters.

3.2.1 Performance of [Acceptance Checks and Tests](#)

Perform in accordance with the manufacturer's recommendations and include the following visual and mechanical inspections and electrical tests, performed in accordance with [NETA ATS](#).

a. Meter Assembly

(1) Visual and mechanical inspection.

(a) Compare equipment nameplate data with specifications and approved shop drawings.

(b) Inspect physical and mechanical condition. Confirm the meter is firmly seated in the socket, the socket is not abnormally heated, the display is visible, and the ring and seal on the cover are intact.

(c) Inspect all electrical connections to ensure they are tight. For Class 200 services, verify tightness of the service conductor terminations for high resistance using low-resistance ohmmeter, or by verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method.

(d) Record model number, serial number, firmware revision, software revision, and rated control voltage.

(e) Verify operation of display and indicating devices.

(f) Record password and user log-in for each meter.

(g) Verify grounding of metering enclosure.

(h) Set all required parameters including instrument transformer

ratios, system type, frequency, power demand methods/intervals, and communications requirements. Verify that the CT ratio is properly included in the meter multiplier or the programming of the meter. Confirm that the multiplier is provided on the meter face or on the meter.

(i) Provide building meter installation sheet, per building for each facility. See example Graphic E-S1.

(2) Electrical tests.

(a) Apply voltage or current as appropriate to each analog input and verify correct measurement and indication.

(b) Confirm correct operation and setting of each auxiliary input/output feature including mechanical relay, digital, and analog.

(c) After initial system energization, confirm measurements and indications are consistent with loads present.

(d) Make note of, and report, any "Error-Code" or "Caution-Code" on the meter's display.

(3) Provide [meter configuration report](#).

b. Current Transformers

(1) Visual and mechanical inspection.

(a) Compare equipment nameplate data with specification and approved shop drawings.

(b) Inspect physical and mechanical condition.

(c) Verify correct connection, including polarity.

(d) Inspect all electrical connections to ensure they are tight.

(e) Verify that required grounding and shorting connections provide good contact.

(2) Electrical Tests.

Verify proper operation by reviewing the meter configuration report.

3.2.2 [System Functional Verification](#)

Verify that the installed meters are working correctly in accordance with the meter configuration report:

a. The correct meter form is installed.

b. All voltage phases are present.

c. Phase rotation is correct.

- d. Phase angles are correct.
- e. The new meter accurately measures power magnitude and direction, and can communicate as required by paragraph COMMUNICATIONS INTERFACES.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 32 15.00

ENGINE-GENERATOR SET STATIONARY 15-2500 KW, WITH AUXILIARIES

05/20

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED MATERIALS
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Conformance to Codes and Standards
 - 1.4.2 Torsional Analysis
 - 1.4.3 Experience
 - 1.4.4 Field Engineer
 - 1.4.5 Detailed Drawings
 - 1.4.6 Auxiliary Systems Engine-Generator Set and Auxiliary Equipment Drawing Requirements
 - 1.4.7 Fuel System Certification
 - 1.4.8 IBC Seismic Certificate of Compliance Sub Title
- 1.5 DELIVERY, STORAGE, AND HANDLING
- 1.6 EXTRA MATERIALS
- 1.7 MAINTENANCE SERVICES
 - 1.7.1 Operation Manual
 - 1.7.2 Maintenance Manual
 - 1.7.3 Assembled Operation and Maintenance Manuals
- 1.8 SITE CONDITIONS

PART 2 PRODUCTS

- 2.1 SYSTEM REQUIREMENTS
 - 2.1.1 Engine-Generator Parameter Schedule
 - 2.1.2 Rated Output Capacity
 - 2.1.2.1 Engine Emission Limits
 - 2.1.2.2 Performance Class
 - 2.1.3 Power Ratings
 - 2.1.4 Transient Response
 - 2.1.5 Reliability and Durability
 - 2.1.6 Engine-Generator Set Enclosure
 - 2.1.7 Starting Time Requirements
- 2.2 NAMEPLATES
 - 2.2.1 Materials
 - 2.2.2 Control Devices and Operation Indicators
 - 2.2.3 Equipment
- 2.3 SAFETY DEVICES
- 2.4 MATERIALS AND EQUIPMENT
 - 2.4.1 Circuit Breakers, Low Voltage
 - 2.4.2 Filter Elements
 - 2.4.3 Instrument Transformers
 - 2.4.4 Pipe (Fuel/Lube-Oil, Compressed Air, Coolant, and Exhaust)
 - 2.4.4.1 Flanges and Flanged Fittings

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.4.4.2 Pipe Welding Fittings
- 2.4.4.3 Threaded Fittings
- 2.4.4.4 Valves
- 2.4.4.5 Gaskets
- 2.4.5 Pipe Hangers
- 2.4.6 Electrical Enclosures
 - 2.4.6.1 Panelboards
- 2.4.7 Electric Motors
- 2.4.8 Motor Controllers
- 2.5 ENGINE
 - 2.5.1 Sub-base Mounting
 - 2.5.2 Assembly
 - 2.5.3 Turbocharger
 - 2.5.4 Intercooler
 - 2.5.5 Crankcase Protection
 - 2.5.6 Miscellaneous Engine Accessories
 - 2.5.7 Intercooler
- 2.6 FUEL SYSTEM
 - 2.6.1 Pumps
 - 2.6.1.1 Main Pump
 - 2.6.2 Fuel Filter
 - 2.6.3 Relief/Bypass Valve
 - 2.6.4 Integral Main Fuel Storage Tank
 - 2.6.4.1 Fuel Transfer Pump
 - 2.6.4.2 Local Fuel Fill
 - 2.6.4.3 Arrangement
 - 2.6.5 Strainer
 - 2.6.6 Fuel Oil Cooler
- 2.7 LUBRICATION
 - 2.7.1 Lube-Oil Filter
 - 2.7.2 Lube-Oil Sensors
 - 2.7.3 Precirculation Pump
- 2.8 COOLING SYSTEM
 - 2.8.1 Coolant Pumps
 - 2.8.2 Heat Exchanger
 - 2.8.2.1 Fin-Tube-Type Heat Exchanger (Radiator)
 - 2.8.3 Thermostatic Control Valve
 - 2.8.4 Temperature Sensors
- 2.9 SOUND LIMITATIONS
- 2.10 AIR INTAKE EQUIPMENT
- 2.11 PYROMETER
- 2.12 EMISSIONS
- 2.13 STARTING SYSTEM
 - 2.13.1 Controls
 - 2.13.2 Capacity
 - 2.13.3 Electrical Starting
 - 2.13.3.1 Battery
 - 2.13.3.2 Battery Charger
 - 2.13.4 Starting Aids
 - 2.13.4.1 Jacket-Coolant Heaters
 - 2.13.4.1.1 Standby Rated Sets
 - 2.13.4.2 Lubricating-Oil Heaters
- 2.14 GOVERNOR
- 2.15 GENERATOR
 - 2.15.1 Current Balance
 - 2.15.2 Voltage Balance
 - 2.15.3 Waveform
- 2.16 EXCITER
- 2.17 VOLTAGE REGULATOR

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.18 GENERATOR ISOLATION AND PROTECTION
 - 2.18.1 Devices
- 2.19 SAFETY SYSTEM
 - 2.19.1 Audible Signal
 - 2.19.2 Visual Signal
 - 2.19.3 Alarms and Action Logic
 - 2.19.3.1 Shutdown
 - 2.19.3.2 Problem
 - 2.19.4 Safety Indications and Shutdowns
 - 2.19.5 Time-Delay on Alarms
- 2.20 PANELS
 - 2.20.1 Enclosures
 - 2.20.2 Analog
 - 2.20.3 Parameter Display
- 2.21 SURGE PROTECTION
- 2.22 MANUAL ENGINE-GENERATOR-SET SYSTEM OPERATION
- 2.23 BASE
- 2.24 PAINTING AND FINISHING
- 2.25 FACTORY INSPECTION AND TESTS
 - 2.25.1 Factory Inspection
 - 2.25.2 Factory Tests

PART 3 EXECUTION

- 3.1 GENERAL INSTALLATION
- 3.2 ELECTRICAL INSTALLATION
- 3.3 FIELD PAINTING
- 3.4 ONSITE INSPECTION AND TESTS
 - 3.4.1 Test Conditions
 - 3.4.1.1 Data
 - 3.4.1.2 Power Factor
 - 3.4.1.3 Contractor Supplied Items
 - 3.4.1.4 Instruments
 - 3.4.1.5 Sequence
 - 3.4.2 Construction Tests
 - 3.4.2.1 Piping Test
 - 3.4.2.2 Electrical Equipment Tests
 - 3.4.3 Inspections
 - 3.4.4 Engine Tests
 - 3.4.5 Generator Tests
 - 3.4.5.1 Routine Tests
 - 3.4.5.2 Design Tests
 - 3.4.6 Assembled Engine-Generator Set Tests
 - 3.4.6.1 Initial Stabilization Readings
 - 3.4.6.2 Regulator Range Test
 - 3.4.6.3 Frequency Range Test
 - 3.4.6.4 Transient Response Test
 - 3.4.7 Pre-operational Tests
 - 3.4.7.1 Insulation Test
 - 3.4.7.2 Engine-Generator Connection Coupling Test
 - 3.4.8 Safety Run Test
 - 3.4.9 Performance Tests
 - 3.4.9.1 Continuous Engine Load Run Test
 - 3.4.9.2 Voltage and Frequency Droop Test
- 3.5 GROUNDING
 - 3.5.1 Grounding Electrodes
 - 3.5.2 Engine-Generator Set Grounding
 - 3.5.3 Connections
 - 3.5.4 Grounding and Bonding Equipment

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.6 START-UP ENGINEER
- 3.7 PREREQUISITES FOR FUNCTIONAL ACCEPTANCE TESTING
 - 3.7.1 Performance of Acceptance Checks and Tests
 - 3.7.2 Generator Sets
 - 3.7.3 Preliminary Operations
 - 3.7.4 Preliminary Assembled Operation and Maintenance Manuals
 - 3.7.5 Functional Acceptance Test Procedure
 - 3.7.6 Test Equipment
- 3.8 FIELD QUALITY CONTROL
 - 3.8.1 Acceptance Checks and Tests
 - 3.8.1.1 Circuit Breakers - Low Voltage Insulated Case/Molded Case
 - 3.8.1.2 Current Transformers
 - 3.8.1.3 Metering and Instrumentation
 - 3.8.1.4 Battery Systems
 - 3.8.1.5 Engine-Generator Set
 - 3.8.1.6 Grounding System
 - 3.8.2 Functional Acceptance Tests
- 3.9 DEMONSTRATION
 - 3.9.1 Instructor's Qualification Resume
 - 3.9.2 Training Plan
 - 3.9.2.1 Operating Personnel Training
 - 3.9.2.2 Maintenance Personnel Training
- 3.10 ONSITE TRAINING
- 3.11 INSTALLATION
- 3.12 FINAL TESTING AND INSPECTION
- 3.13 MANUFACTURER'S FIELD SERVICE
- 3.14 ACCEPTANCE

-- End of Section Table of Contents --

SECTION 26 32 15.00

ENGINE-GENERATOR SET STATIONARY 15-2500 KW, WITH AUXILIARIES
05/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

- | | |
|-------------|--|
| ASME B16.3 | (2021) Malleable Iron Threaded Fittings, Classes 150 and 300 |
| ASME B16.5 | (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard |
| ASME B16.11 | (2016) Forged Fittings, Socket-Welding and Threaded |

ASTM INTERNATIONAL (ASTM)

- | | |
|-----------------|---|
| ASTM A53/A53M | (2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless |
| ASTM A106/A106M | (2019a) Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service |
| ASTM A181/A181M | (2014; R 2020) Standard Specification for Carbon Steel Forgings, for General-Purpose Piping |
| ASTM A234/A234M | (2019) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service |
| ASTM D975 | (2020) Standard Specification for Diesel Fuel Oils |

ELECTRICAL GENERATING SYSTEMS ASSOCIATION (EGSA)

- | | |
|-----------|--|
| EGSA 101P | (1995) Performance Standard for Engine Driven Generator Sets |
|-----------|--|

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

- | | |
|--------|--|
| IEEE 1 | (2000; R 2011) General Principles for Temperature Limits in the Rating of Electric Equipment and for the Evaluation of Electrical Insulation |
|--------|--|

IEEE 43	(2013) Recommended Practice for Testing Insulation Resistance of Rotating Machinery
IEEE 81	(2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System
IEEE 100	(2000; Archived) The Authoritative Dictionary of IEEE Standards Terms
IEEE 115	(2019) Guide for Test Procedures for Synchronous Machines: Part I Acceptance and Performance Testing; Part II Test Procedures and Parameter Determination for Dynamic Analysis
IEEE 120	(1989; R 2007) Master Test Guide for Electrical Measurements in Power Circuits
IEEE 519	(2014) Recommended Practices and Requirements for Harmonic Control in Electrical Power Systems
IEEE C2	(2017; Errata 1-2 2017; INT 1 2017) National Electrical Safety Code
IEEE C50.12	(2005; R 2010) Standard for Salient Pole 50 Hz and 60 Hz Synchronous Generators and Generation/Motors for Hydraulic Turbine Applications Rated 5 MVA and above

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2018) International Building Code
---------	------------------------------------

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS	(2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems
----------	--

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 3046	(2002, 2006, 2009, 2001) Reciprocating Internal Combustion Engines - Performance--Part 1, 3, 4, 5, 6
ISO 8528	(1993; R 2018) Reciprocating Internal Combustion Engine Driven Alternating Current Generator Sets--Part 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-58	(2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation
-----------	--

MSS SP-80 (2019) Bronze Gate, Globe, Angle and Check Valves

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 2 (2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V

NEMA ICS 6 (1993; R 2016) Industrial Control and Systems: Enclosures

NEMA MG 1 (2016) Motors and Generators - Revision 1: 2018; Includes 2021 Updates to Parts 0, 1, 7, 12, 30, and 31

NEMA PB 1 (2011) Panelboards

NEMA/ANSI C12.11 (2006; R 2019) Instrument Transformers for Revenue Metering, 10 kV BIL through 350 kV BIL (0.6 kV NSV through 69 kV NSV)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 30 (2021; TIA 20-1; TIA 20-2) Flammable and Combustible Liquids Code

NFPA 37 (2021) Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE ARP892 (1965; R 1994) DC Starter-Generator, Engine

SAE J537 (2016) Storage Batteries

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 60 Standards of Performance for New Stationary Sources

UNDERWRITERS LABORATORIES (UL)

UL 142 (2006; Reprint Jan 2021) UL Standard for Safety Steel Aboveground Tanks for Flammable and Combustible Liquids

UL 467 (2013; Reprint Jun 2017) UL Standard for Safety Grounding and Bonding Equipment

UL 489 (2016; Rev 2019) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures

UL 1236 (2015; Reprint Feb 2021) UL Standard for
Safety Battery Chargers for Charging
Engine-Starter Batteries

UL 1437 (2006) Electrical Analog Instruments -
Panel Board Types

1.2 RELATED MATERIALS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, and Section 26 08 00
APPARATUS INSPECTION AND TESTING apply to this section, except as modified
herein.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S"
classification. Submittals not having a "G" or "S" classification are for
Contractor Quality Control approval. Submit the following in accordance
with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Engine-Generator Set and Auxiliary Equipment; G

Auxiliary Systems; G

Detailed Drawings; G

Acceptance; G

SD-03 Product Data

Engine-Generator Set Efficiencies; G

Emissions; G

filters; G

special tools; G

Engine-Generator Parameter Schedule

Heat Exchanger

Generator

Spare Parts

Onsite Training

Experience

Field Engineer

General Installation

Exciter

SD-05 Design Data

Performance Criteria

Sound Limitations; G

Integral Main Fuel Storage Tank

Power Factor

Heat Exchanger

Time-Delay on Alarms

Cooling System

Battery Charger

Capacity Calculations for Engine-Generator Set; G

Brake Mean Effective Pressure (BMEP) Calculations; G

Torsional Vibration Stress Analysis Computations; G

Capacity Calculations for Batteries; G

Turbocharger Load Calculations; G

SD-06 Test Reports

Performance Tests

Factory Inspection and Tests

Factory Tests

Onsite Inspection and Tests; G

Acceptance Checks and Tests; G

Functional Acceptance Tests; G

Maintenance Procedures; G

Operation and Maintenance Manuals; G

Inspections; G

Functional Acceptance Test Procedure; G

SD-07 Certificates

Cooling System

Reliability and Durability

Fuel System Certification; G

Start-Up Engineer; G

Instructor's Qualification Resume; G

Engine Emission Limits; G

Sound Limitations

Current Balance

Materials and Equipment

IBC Seismic Certificate of Compliance; G

Factory Inspection and Tests

SD-09 Manufacturer's Field Reports

Engine Tests; G

Generator Tests; G

Assembled Engine-Generator Set Tests; G

SD-10 Operation and Maintenance Data

Preliminary Assembled Operation and Maintenance Manuals; G

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and the paragraph ASSEMBLED OPERATION AND MAINTENANCE MANUALS.

SD-11 Closeout Submittals

Training Plan; G

1.4 QUALITY ASSURANCE

1.4.1 Conformance to Codes and Standards

Where equipment is specified to conform to requirements of any code or standard such as UL, NEMA, etc., the design, fabrication and installation must also conform to the code.

1.4.2 Torsional Analysis

Submit torsional analysis including prototype testing or calculations which certify and demonstrate that no damaging or dangerous torsional vibrations will occur when the prime mover is connected to the generator, at synchronous speeds, plus/minus 10 percent.

1.4.3 Experience

Each component manufacturer must have a minimum of 3 years' experience in the manufacture, assembly and sale of components used with stationary engine-generator sets for commercial and industrial use. The engine-generator set manufacturer/assembler must have a minimum of 3 years' experience in the manufacture, assembly and sale of stationary engine-generator sets for commercial and industrial use. Submit a statement showing and verifying these requirements.

1.4.4 Field Engineer

The engine-generator set manufacturer or assembler must furnish a qualified field engineer to supervise the complete installation of the engine-generator set, assist in the performance of the onsite tests, and instruct personnel as to the operational and maintenance features of the equipment. The field engineer must have attended the engine generator manufacturer's training courses on installation and operation and maintenance of engine generator sets. Submit a letter listing the qualifications, schools, formal training, and experience of the field engineer.

1.4.5 Detailed Drawings

Submit detailed drawings showing the following:

- a. Base-mounted equipment, complete with base and attachments, including anchor bolt template and recommended clearances for maintenance and operation.
- b. Starting system.
- c. Fuel system.
- d. Cooling system.
- e. Exhaust system.
- f. Electric wiring of relays, breakers, programmable controllers, and switches including single line and wiring diagrams.
- g. Lubrication system, including piping, pumps, strainers, filters, heat exchangers for lube oil and turbocharger cooling, electric heater, controls and wiring.
- h. Location, type, and description of vibration isolation devices for all applications.
- i. The safety system, including wiring schematics.
- j. One-line schematic and wiring diagrams of the generator, exciter, regulator, governor, and instrumentation.
- k. Panel layouts.
- l. Mounting and support for each panel and major piece of electrical equipment.
- m. Engine-generator set rigging points and lifting instructions.

1.4.6 Auxiliary Systems Engine-Generator Set and Auxiliary Equipment Drawing Requirements

Submit drawings pertaining to the engine-generator set and auxiliary equipment, including but not limited to the following:

- a. Certified outline, general arrangement (setting plan), and anchor bolt details. Show total weight and center of gravity of assembled

equipment on the steel sub-base.

- b. Detailed elementary, schematic wiring, and interconnection diagrams of the engine starting system, jacket coolant heating system, engine protective devices, engine alarm devices, engine speed governor system, generator and excitation system, and other integral devices.
- c. Detailed elementary, schematic wiring; and interconnection diagrams of the fuel system, starting battery system, engine-generator control panel, generator circuit breaker.
- d. Dimensional drawings or catalog cuts of exhaust silencers, radiator, fuel day tanks, fuel oil cooler, valves and pumps, intake filters, vibration isolators, and other auxiliary equipment not integral with the engine-generator set.

1.4.7 Fuel System Certification

When the fuel system requires a fuel oil cooler as described in the paragraph FUEL OIL COOLER, submit certification from the engine manufacturer that the fuel system design is satisfactory.

1.4.8 IBC Seismic Certificate of Compliance Sub Title

Generator set shall have an up-to-date IBC seismic Certificate of Compliance meeting the testing, labeling, and certification requirements of the IBC.

1.5 DELIVERY, STORAGE, AND HANDLING

Properly protect materials and equipment, in accordance with the manufacturers recommended storage procedures, before, during, and after installation. Protect stored items from the weather and contamination. During installation, cap piping and similar openings to keep out dirt and other foreign matter.

Deliver equipment on pallets or blocking wrapped in heavy-duty plastic, sealed to protect parts and assemblies from moisture and dirt. Protect and prepare batteries for shipment as recommended by the battery manufacturer. Store auxiliary equipment at the site in covered enclosures, protected from atmospheric moisture, dirt, and ground water.

1.6 EXTRA MATERIALS

Provide two sets of special tools and two sets of filters required for maintenance. Special tools are those that only the manufacturer provides, for special purposes, or to reach otherwise inaccessible parts. One handset must be provided for each electronic governor when required to indicate and/or change governor response settings. Furnish 4 liters one gallon of identical paint used on engine-generator set in manufacturer's sealed container with each engine-generator set.

Wrenches and tools specifically designed and required to work on the new equipment, which are not commercially available as standard mechanic's tools, must be furnished to the Contracting Officer.

Provide proposed operating instructions for the engine-generator set and auxiliary equipment laminated between matte-surface thermoplastic sheets and suitable for placement adjacent to corresponding equipment. After

approval, install operating instructions where directed.

1.7 MAINTENANCE SERVICES

Submit the operation and maintenance manuals and have them approved prior to commencing onsite tests.

1.7.1 Operation Manual

Provide three copies of the manufacturers standard operation manual . Sections must be separated by heavy plastic dividers with tabs which identify the material in the section. Fold drawings with the title block visible, and placed in 216 by 279 mm plastic pockets with reinforced holes. The manual must include:

- a. Step-by-step procedures for system startup, operation, and shutdown;
- b. Drawings, diagrams, and single-line schematics to illustrate and define the electrical, mechanical, and hydraulic systems with their controls, alarms, and safety systems;

1.7.2 Maintenance Manual

Provide three copies of the manufacturers standard maintenance manual . Separate each section by a heavy plastic divider with tabs. Fold drawings with the title block visible, and placed in plastic pockets with reinforced holes. The manual must include:

- a. Procedures for each routine maintenance item.Procedures for troubleshooting.Factory-service, take-down overhaul, and repair service manuals, with parts lists.
- b. The manufacturer's recommended maintenance schedule.
- c. A component list which includes the manufacturer's name, address, type or style, model or serial number, rating, and catalog number for the major components.
- d. A list of spare parts for each piece of equipment and a complete list of materials and supplies needed for operation.

1.7.3 Assembled Operation and Maintenance Manuals

The contents of the assembled operation and maintenance manuals must include the manufacturer's O&M information required by the paragraph SD-10, OPERATION AND MAINTENANCE DATA.

- a. Manuals must be in separate books or volumes, assembled and bound securely in durable, hard covered, water resistant binder, and indexed by major assembly and components in sequential order.
- b. A table of contents (index) must be made part of the assembled O&M. The manual must be assembled in the order noted in table of contents.
- c. The cover sheet or binder on each volume of the manuals must be identified and marked with the words, "Operation and Maintenance Manual."

1.8 SITE CONDITIONS

Protect the components of the engine-generator set, including cooling system components, pumps, fans, and similar auxiliaries when not operating and provide components capable of the specified outputs in the following environment:

- a. Site Location: Naval Station Rota; Rota, Spain.
- b. Site Elevation: 30 meters above mean sea level.
- c. Ambient Temperatures:
 - (1) Maximum 33.9 degrees C dry bulb, 23.9 degrees C wet bulb.
 - (2) Minimum 2.2 degrees C dry bulb.
- d. Design Wind Velocity (3-second gust): 177 km/h.
- e. Prevailing Wind Direction: NNE
- f. Seismic Design Catagory: D as defined by ICC IBC.

PART 2 PRODUCTS

2.1 SYSTEM REQUIREMENTS

- a. Provide and install each engine-generator set complete and totally functional, with all necessary ancillary equipment to include: air filtration; starting system; generator controls, protection, and isolation; instrumentation; lubrication; fuel system; cooling system; and engine exhaust system. Each engine-generator set must satisfy the requirements specified in the Engine-Generator Parameter Schedule. Submit certification that the engine-generator set and cooling system function properly in the ambient temperatures specified.
- b. Provide each engine-generator set consisting of one engine, one generator, and one exciter mounted, assembled, and aligned on one base; and all other necessary ancillary equipment which may be mounted separately. Assemble sets having a capacity of 750 kW or smaller and attach to the base prior to shipping. Provide set components that are environmentally suitable for the locations shown and that are the manufacturer's standard product offered in catalogs for commercial or industrial use. Provide a generator strip heater for moisture control when the generator is not operating. Identify any nonstandard products or components and the reason for their use.

2.1.1 Engine-Generator Parameter Schedule

Engine-Generator Set and Auxiliary Equipment Capacity Calculations for Engine-Generator Set

ENGINE-GENERATOR PARAMETER SCHEDULE	
Identification	Make/Model
Electrical Characteristics	

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

ENGINE-GENERATOR PARAMETER SCHEDULE	
Power Rating	Standby Gross bhp rating / Net brake power rating 250 kW at 0.8 power factor
Governor Type	Type Make / Model Isochronous
Service Load	244 kVA (maximum) 267 kVA (continuous)
Motor Starting kVA (Max.)	300 kVA
Power Factor	0.8 lagging
Engine-Generator Applications	stand-alone
Voltage Regulation (No Load to Full Load)(Stand-alone applications)	plus or minus 2 percent (maximum)
Voltage Bandwidth (steady state)	plus or minus 2 percent
Frequency	60 Hz
Voltage	480Y/277 volts
Phases	3 Phase, Wye
Minimum Generator Sub-transient Reactance	0.12 percent Sub-transient
Nonlinear Loads	0 kVA
Max Step Load Increase	50 percent of Service Load at 0.8 PF
Transient Recovery Time with Step Load Increase (Voltage)	6 seconds
Transient Recovery	5 seconds
Maximum Voltage Deviation with Step Load Increase	20 percent of rated voltage
Maximum Frequency Deviation with Step Load Increase	3.0 percent of rated frequency
Max Step Load Decrease (without shutdown)	100 percent of Service Load at 0.8 PF

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

ENGINE-GENERATOR PARAMETER SCHEDULE					
Frequency Bandwidth (steady state)	plus or minus 0.5 percent				
Frequency Regulation (droop) (No Load to Full Load)	3 percent (maximum)				
Reactances	Synchronous reactance, Xd Transient reactance, X'd Sub-transient reactance, X''d Negative sequence reactance, X2 Zero sequence reactance, Xo				
Capacity Calculations					
Calculations must verify that the engine-generator set power rating is adequate for the following load conditions:					
Lighting	5.8 kW				
Motor Starting Sequence	Starting Order	Size (hp)	Locked Rotor NEMA Code	Starting Method	Maximum Voltage Dip
	1	5	H	Full Voltage	25 Percent
	2	5	H	Full Voltage	25 Percent
	3	100	G	Soft Start (300% Current Limit)	25 Percent
	4	100	G	Soft Start (300% Current Limit)	25 Percent
Other Load	10.0 kW at 0.8 power factor				
Capacity Calculations for Batteries					
Calculation must verify that the engine starting battery capacity exceeds dc power requirements.					
Mechanical Characteristics					

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

ENGINE-GENERATOR PARAMETER SCHEDULE	
Engine Description	6 Number of cylinders 117 Bore and 135 Stroke, mm
Engine Speed	1800 rpm
Heat Exchanger Type	fin-tube (radiator)
Engine Cooling Type	water/ethylene glycol
Intercooler Type	Jacket Water
Induction Method	Turbocharged
Max Time to Start and be Ready to Assume Load	10 seconds
Max Summer Outdoor Temp (Ambient)	34.6 degrees C
Min Winter Outdoor Temp (Ambient)	3.2 degrees C
Installation Elevation	30 m above sea level
Engine-Generator Set Efficiencies	
Fuel Consumption	At 0.8 power factor, liters / hour for: 1 / 2 load 3 / 4 load Full Load
Generator Efficiency	At 0.8 power factor, (per cent) for: 1 / 2 load 3 / 4 load Full Load
Radiator Capacity	Coolant Type L/s coolant L/s air through radiator kW per hr of heat exchange based on optimum coolant temperature to and from engine
Engine-Generator Set Emissions Data	
Exhaust Temperature	Degrees C at full load
Weight of Exhaust Gas	Kg per hr at full load
Weight of Intake Air	Kg per hr at full load
Total Heat Rejected	kW per hr, at full load to: Jacket Coolant System Fuel Oil Cooling System

ENGINE-GENERATOR PARAMETER SCHEDULE	
Emissions	Kg per hr , at full load Total Suspended Particulate - 0.14 kg/hr Nitrogen Oxides (as NO₂)plus non-methane hydrocarbons - 2.8 kg/hr Carbon Monoxide - 2.45 kg/hr
Visible Emissions	Percent opacity at full load
Brake Mean Effective Pressure (BMEP) Calculations	
Calculation must verify that the engine meets the specified maximum BMEP, as follows: $\text{BMEP kPa} = (120,000 \times \text{bkw}) \times (792,000 \times \text{bhp})$ $(\text{rpm} \times \text{liters})$ Where: bkw bhp = bkw' + bkw" bhp' + bhp" bkw" bhp" is the Brake kW horsepower required by engine driven fan for cooling radiator or motor driven fan for cooling radiator. bkw' bhp' = kW/GEN.EFF. kW/(GEN.EFF. times 0.746) GEN.EFF. = Generator efficiency liters = Total engine piston displacement in liters rpm = Engine revolutions per minute kW = Minimum power rating	
Torsional Vibration Stress Analysis Computations	

ENGINE-GENERATOR PARAMETER SCHEDULE
Torsional vibrational stresses in the crankshaft and generator shaft of assembled engine and driven generator must not exceed 34,450 kPa when engine is driving generator at rated speed while assembled unit is loaded to rated engine-generator set power. Computations must be based on a mathematical model of the assembled generator set provided or based on calculations using measured values from tests on a unit identical to the one provided. Calculations based on models of, or measured data from, the unassembled engine and generator will not be acceptable. Calculations must include: a. A description of the system relating information pertinent to analysis such as operating speed range and identification plate data. b. A mass elastic assembly drawing, showing the arrangement of the units in the generator set and dimensions of shafting, including minimum diameters (or section moduli) of shafting in the system. c. A labeled line diagram of the mass elastic system indicating values of masses, stiffness, equivalent lengths, and equivalent diameters including basic assumptions and definition of terms. d. Sample computations showing procedures used to obtain resulting stress values. e. Computations indicating assembled engine-generator speed of 1800 rpm with assembly loaded to rated generator power and the resulting computed critical torsional stress values in the assembled engine crankshaft and generator shaft.
Turbocharger Load Calculations
NOTE: When the engine-generator set installation includes field installed exhaust system (i.e., the engine-generator set is installed internal to a building in lieu of in a self-contained outdoor enclosure), include the following paragraph. When the proposed exhaust system layout is different from that shown on the contract drawings, submit calculations showing that the external loads from the exhaust system such as weight and thermal expansion do not exceed the engine manufacturer's maximum allowed forces and moments on the turbocharger.

2.1.2 Rated Output Capacity

Provide each engine-generator-set with power equal to the sum of service load plus the machine's efficiency loss and associated ancillary equipment loads. Rated output capacity must also consider engine and/or generator oversizing required to meet requirements in paragraph Engine-Generator Parameter Schedule.

The engine must meet the specified maximum BMEP requirements at rated speed as calculated in accordance with the calculations in the engine-generator parameter schedule. The engine capacity must be based on

the following:

- a. Engine burning diesel fuel conforming to **ASTM D975**, Grade 2-D, or at an ambient temperature of **29 degrees C**. For stationary engines operated in the United States, diesel fuel requirements are found in **40 CFR 60** Subpart IIII.
- b. Engine cooled by a radiator fan mechanically driven by the engine or remote with a motor driven fan.
- c. Engine cooled by coolant mixture of water and ethylene glycol, 50 percent by volume of each.

Maximum BMEP, kPa	
	Turbocharged and Intercooled
Four-cycle engines	2289
Engine speed, rpm:	1800

2.1.2.1 Engine Emission Limits

Engine must be certified by the manufacturer to meet applicable EPA emission standards found in **40 CFR 60** Subpart IIII. In addition, engine must meet any applicable state or local emission requirements (ex: California SCAQMD).

2.1.2.2 Performance Class

The voltage and frequency behavior of the generator set must be in accordance with **ISO 8528** operating limit values for performance Class G2.

2.1.3 Power Ratings

Power ratings must be in accordance with **EGSA 101P**.

2.1.4 Transient Response

The engine-generator set governor and voltage regulator must cause the engine-generator set to respond to the maximum step load changes such that output voltage and frequency recover to and stabilize within the operational bandwidth within the transient recovery time. The engine-generator set must respond to maximum step load changes such that the maximum voltage and frequency deviations from bandwidth are not exceeded.

2.1.5 Reliability and Durability

Each standby engine-generator set must have both an engine and a generator capable of delivering the specified power on a standby basis with an anticipated mean time between overhauls of no less than 5,000 hours operating with a load factor of 70 percent. Cite two like engines and two like generators that have performed satisfactorily in a stationary power plant, independent and separate from the physical location of the manufacturer's and assembler's facilities, for standby without any failure to start, including all periodic exercise. Provide like engines and generators that have had no failures resulting in downtime for repairs in

excess of 72 hours during two consecutive years of service. Provide engines that are the same model, speed, bore, stroke, number and configuration of cylinders, and rated output capacity. Provide generators that are the same model, speed, pitch, cooling, exciter, voltage regulator and rated output capacity.

Submit a reliability and durability certification letter from the manufacturer and assembler to prove that existing facilities are and have been successfully utilizing the same components proposed to meet this specification, in similar service. Certification may be based on components, i.e. engines used with different models of generators and generators used with different engines, and does not exclude annual technological improvements made by a manufacturer in the basic standard-model component on which experience was obtained, provided parts interchangeability has not been substantially affected and the current standard model meets the performance requirements specified. Provide a list with the name of the installations, completion dates, and name and telephone number of a point of contact.

2.1.6 Engine-Generator Set Enclosure

Provide engine-generator set enclosures that are corrosion resistant and fully weather resistant. The enclosure must contain all set components and provide ventilation to permit operation at Service Load under secured conditions. Provide access doors to controls and equipment requiring periodic maintenance or adjustment. Provide removable panels for access to components requiring periodic replacement. The enclosure must be capable of being removed without disassembly of the engine-generator set or removal of components other than the exhaust system. The enclosure must reduce the noise of the generator set to within the limits specified in the paragraph SOUND LIMITATIONS.

2.1.7 Starting Time Requirements

Upon receipt of a signal to start, each engine generator set will start, reach rated frequency and voltage and be ready to assume load within the time specified.

2.2 NAMEPLATES

Provide the manufacturer's name, type or style, model or serial number and rating on a plate secured to the equipment for each major component of this specification. Provide plates and tags sized so that inscription is readily legible to operating or maintenance personnel and securely mounted to or attached in proximity of their identified controls or equipment. Lettering must be normal block lettering, a minimum of 6.4 mm high. As a minimum, provide nameplates for:

Engines

Generators

Regulators

Pumps and pump motors

Day tanks

Governors

Generator Breaker

Air Starting System

2.2.1 Materials

Construct ID plates and tags of 16 gage minimum thickness bronze or stainless steel sheet metal engraved or stamped with inscription. Construct plates and tags not exposed to the weather or high operational temperature of the engine of laminated plastic, 3.2 mm thick, matte white finish with black center core, with lettering accurately aligned and engraved into the core.

2.2.2 Control Devices and Operation Indicators

Provide ID plates or tags for control devices and operation indicators, including valves, off-on switches, visual alarm annunciators, gages and thermometers, that are required for operation and maintenance of provided mechanical systems. Plates or tags must be minimum of 13 mm high and 50 mm long and must indicate component system and component function.

2.2.3 Equipment

Provide ID plates of a minimum size of 75 mm 3 inches high and 130 mm long on provided equipment indicating the following information:

- a. Manufacturer's name, address, type and model number, serial number, and certificate of compliance with applicable EPA mission standards;
- b. Contract number and accepted date;
- c. Capacity or size;
- d. System in which installed; and
- e. System which it controls.

2.3 SAFETY DEVICES

Exposed moving parts, parts that produce high operating temperatures, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Install safety devices such that proper operation of the equipment is not impaired.

2.4 MATERIALS AND EQUIPMENT

Submit certification stating that where materials or equipment are specified to comply with requirements of UL, written proof of such compliance has been obtained. The label or listing of the specified agency, or a written certificate from an approved, nationally recognized testing organization equipped to perform such services, stating that the items have been tested and conform to the requirements and testing methods of the specified agency are acceptable as proof.

2.4.1 Circuit Breakers, Low Voltage

UL 489.

2.4.2 Filter Elements

Provide the manufacturer's standard fuel-oil, lubricating-oil, and combustion-air filter elements.

2.4.3 Instrument Transformers

NEMA/ANSI C12.11.

2.4.4 Pipe (Fuel/Lube-Oil, Compressed Air, Coolant, and Exhaust)

ASTM A53/A53M, or ASTM A106/A106M steel pipe. Pipe smaller than 50 mm must be Schedule 80. Pipe 50 mm and larger must be Schedule 40.

2.4.4.1 Flanges and Flanged Fittings

ASTM A181/A181M, Class 60, or ASME B16.5, Grade 1, Class 150.

2.4.4.2 Pipe Welding Fittings

ASTM A234/A234M, Grade WPB or WPC, Class 150 or ASME B16.11, 1360.7 kg.

2.4.4.3 Threaded Fittings

ASME B16.3, Class 150.

2.4.4.4 Valves

MSS SP-80, Class 150.

2.4.4.5 Gaskets

Manufacturer's standard.

2.4.5 Pipe Hangers

MSS SP-58.

2.4.6 Electrical Enclosures

NEMA ICS 6.

2.4.6.1 Panelboards

NEMA PB 1.

2.4.7 Electric Motors

Provide electric motors that conform to the requirements of NEMA MG 1. Motors must have sealed ball bearings and a maximum speed of 1800 rpm. Motors used indoors must have drip-proof frames; enclose those that are used outside. Alternating current motors larger than 373 W must be of the squirrel-cage induction type for operation on 208 volts or higher, 60 Hz, and three-phase power. Alternating current motors 373 W or smaller, must be suitable for operation on 120 volts, 60 Hz, and single-phase power.

2.4.8 Motor Controllers

Provide motor controllers and starters that conform to the requirements of

NFPA 70 and NEMA ICS 2.

2.5 ENGINE

ISO 3046. Diesel engines must be four-cycle naturally aspirated, or turbocharged, or turbocharged and intercooled; vertical in-line or vertical Vee type; designed for stationary service. Engines must be capable of immediate acceleration from rest to normal speed without intermediate idle/warm up period or pre-lubrication to provide essential electrical power. Two-cycle engines are not acceptable.

2.5.1 Sub-base Mounting

Mount each engine-generator set on a structural steel sub-base sized to support the engine, generator, and necessary accessories, auxiliaries and control equipment to produce a complete self-contained unit as standard with the manufacturer. Design the structural sub-base to properly support the equipment and maintain proper alignment of the engine-generator set in the specified seismic zone. In addition, provide sub-base with both lifting rings and jacking pads properly located to facilitate shipping and installation of the unit. Factory align engine and generator on the sub-base and securely bolt into place in accordance with the manufacturer's standard practice. Crankshaft must have rigid coupling for connection to the generator.

2.5.2 Assembly

Completely shop assemble each engine-generator set on its structural steel sub-base. Paint entire unit with manufacturer's standard paints and colors. After factory tests and before shipping, thoroughly clean and retouch painting as necessary to provide complete protection.

2.5.3 Turbocharger

If required by the manufacturer to meet the engine-generator set rating, provide turbine type driven by exhaust gas from engine cylinders, and direct connected to the blower supplying air to the engine intake manifold.

2.5.4 Intercooler

Provide manufacturer's standard intercooler for engine size specified.

2.5.5 Crankcase Protection

Provide manufacturer's standard method of preventing crankcase explosions and standard method of crankcase ventilation.

2.5.6 Miscellaneous Engine Accessories

Provide the following engine accessories where the manufacturer's standard design permits:

- a. Piping on engine to inlet and outlet connections, including nonstandard companion flanges.
- b. Structural steel sub-base and vibration isolators, foundation bolts, nuts, and pipe sleeves.
- c. Level jack screws or shims, as required.

- d. Rails, chocks, and shims for installation of sub-base on the foundation.
- e. Removable guard, around fan. Support guard, on engine sub-base, to suit manufacturer's standard.

2.5.7 Intercooler

Provide manufacturer's standard intercooler for engine size specified.

2.6 FUEL SYSTEM

Provide fuel system conforming to the requirements of **NFPA 30** and **NFPA 37** and containing the following elements.

2.6.1 Pumps

Pump must be horizontal, positive displacement. Direct-connect pump to motor through a flexible coupling. Equip each pump with a bypass relief valve, if not provided with an internal relief valve. Provide motor and controller in accordance with the paragraphs ELECTRIC MOTORS and MOTOR CONTROLLERS, respectively.

2.6.1.1 Main Pump

Provide engines with an engine driven pump. The pump must supply fuel at a minimum rate sufficient to provide the amount of fuel required to meet the performance indicated within the parameter schedule. Base the fuel flow rate on meeting the load requirements and all necessary recirculation.

2.6.2 Fuel Filter

Provide a minimum of one full-flow fuel filter for each engine. The filter must be readily accessible and capable of being changed without disconnecting the piping or disturbing other components. Mark the inlet and outlet connections of the filter.

Provide intake filter assemblies for each engine of the oil bath or dry type, as standard with the manufacturer. Filters must be capable of removing a minimum of 92 percent of dirt and abrasive 3 microns and larger from intake air. Size filters to suit engine requirements at 100 percent of rated full load. Design unit for field access for maintenance purposes.

2.6.3 Relief/Bypass Valve

Provide a relief/bypass valve to regulate pressure in the fuel supply line, return excess fuel to a return line and prevent the build-up of excessive pressure in the fuel system.

2.6.4 Integral Main Fuel Storage Tank

Provide each engine with an integral, double-walled main fuel tank. Each tank must be factory installed and provided as an integral part of the generator manufacturer's product. Provide each tank with connections for fuel supply line, fuel return line, local fuel fill port, gauge, vent line, and float switch assembly. Provide a fuel return line cooler as recommended by the manufacturer and assembler. The temperature of the fuel returning to the tank must be below the flash point of the fuel.

Mount the tank within the enclosure for each engine-generator set provided with weatherproof enclosures. The fuel fill line must be accessible without opening the enclosure.

- a. All Tanks: **UL 142**. Provide integral in skid double wall (110 percent containment) fuel tanks with a minimum capacity of 72 hours of engine-generator set operation at full-rated load . Epoxy coat day tanks inside and prime and paint outside. Construct tanks of not less than **4.76 mm** steel plate with welded joints and necessary stiffeners on exterior of tank. Provide a braced structural steel framework support. Weld tank top tight. Provide **114 mm** square inspection port with a 2 inch NPT fill connection and spill box. Provide proper normal and emergency venting for the primary tank and emergency venting only for the secondary tank / containment basin in accordance with **UL 142** requirements.
- b. Float Switches for Day Tanks: Provide tank-top mounted or external float cage, single-pole, single-throw type designed for use on fuel oil tanks. Arrange high level float switches to close on rise of liquid level, and low level float switches to close on fall of liquid level. Mount float cage units with isolating and drain valves. Contacts must be suitable for the station battery voltage.
 - (1) Critical low level float switch which must activate at 5 percent of normal liquid level must shut engine off.
 - (2) Low-low level float switch which must activate "low-low level" alarm at 30 percent of normal liquid level.
 - (3) Critical high level float switch which must activate "overfill fuel level" alarm at 95 percent of normal liquid level.
- c. Leak Detector Switch for All Tanks: Actuates when fuel is detected in the integral tank interstitial space.
- d. Control Panel for All Tanks: Provide **NEMA ICS 6**, Type 1 , enclosed control panel for each tank. Control panel must include the following accessories.
 - (1) Power available LED (green).
 - (2) Critical low fuel alarm contacts for shut down of engine.
 - (3) Low-low level fuel alarm LED.
 - (4) Critical high level fuel alarm LED.
 - (5) Leak detecting alarm LED.
 - (6) Alarm horn.
- e. Tank Gages for All Tanks: Provide buoyant force type gages for fuel tanks with dial indicator not less than **100 mm** in size and arranged for top mounting. Calibrate each reading dial or scale for its specific tank to read from empty to full, with intermediate points of 1/4, 1/2, and 3/4.
- f. Integral Base Tanks Used as Primary Tank: Provide a 2 inch opening at the tank fill port, fitted an overfill prevention valve (OPV).

Additionally, the fill opening must be perpendicular to the tank in order to allow operation of the OPV. Integral base tank must be sized and configured such that the filling and venting nozzles are outside the generator cabinet for ease of accessibility, inspection, and maintenance. Level gage must be in the line of sight from the fill port.

2.6.4.1 Fuel Transfer Pump

Fuel transfer pumps may be mounted on the day tank. Pump must be horizontal, positive displacement. Direct-connect pump to motor through a flexible coupling. Equip each pump with a bypass relief valve, if not provided with an internal relief valve. Provide motor and controller in accordance with the paragraphs ELECTRIC MOTORS and MOTOR CONTROLLERS, respectively.

2.6.4.2 Local Fuel Fill

Each local fuel fill port on the tank must have a screw-on cap.

2.6.4.3 Arrangement

Integral tanks may allow gravity flow into the engine. Gravity flow tanks and any tank that allows a fuel level above the fuel injectors must have an internal or external factory installed valve located as near as possible to the shell of the tank. The valve must close when the engine is not operating. Provide integral tanks with any necessary pumps to supply fuel to the engine as recommended by the generator set manufacturer. The fuel supply line from the tank to the manufacturer's standard engine connection must be welded pipe.

2.6.5 Strainer

Provide simplex strainers with engine fuel system.

2.6.6 Fuel Oil Cooler

Provide an air cooled fuel oil cooler if the temperature of the fuel returned to the tank from the engine will cause overheating of the tank fuel above the maximum fuel temperature allowed by the engine manufacturer when operating at maximum rated generator power output and low fuel level in the tank. The fuel oil cooler must be furnished by the engine manufacturer for the application and the installation must be complete including piping and power requirements.

2.7 LUBRICATION

Provide engine with a separate lube-oil system conforming to NFPA 30 and NFPA 37. Pressurize each system by engine-driven pumps. Regulate system pressure as recommended by the engine manufacturer. Provide a pressure relief valve on the crankcase for closed systems. Vent the crankcase in accordance with the manufacturer's recommendation. Do not vent the crankcase to the engine exhaust system. Crankcase breathers, if provided on engines installed in buildings or enclosures, must be piped to vent to the outside. The system must be readily accessible for service such as draining, refilling, etc. Each system must permit addition of oil and have oil-level indication with the set operating. The system must utilize an oil cooler as recommended by the engine manufacturer.

2.7.1 Lube-Oil Filter

Provide one full-flow filter for each pump. The filter must be readily accessible and capable of being changed without disconnecting the piping or disturbing other components. Mark inlet and outlet connections.

2.7.2 Lube-Oil Sensors

Equip each engine with lube-oil pressure sensors located downstream of the filters and provide signals for required indication and alarms. Submit two complete sets of filters, required for maintenance, supplied in a suitable storage box. Provide these filters in addition to filters replaced after testing.

2.7.3 Precirculation Pump

Provide a motor-driven precirculation pump powered by the station battery, complete with motor starter, if recommended by the engine manufacturer.

2.8 COOLING SYSTEM

Provide each engine with its own cooling system to operate automatically while its engine is running. The cooling system coolant must use a combination of water and ethylene-glycol sufficient for freeze protection at the minimum winter outdoor temperature specified. The maximum temperature rise of the coolant across each engine must not exceed that recommended below. Submit a letter which certifies that the engine-generator set and cooling system function properly in the ambient temperature specified, stating the following values:

- a. The maximum allowable inlet temperature of the coolant fluid.
- b. The minimum allowable inlet temperature of the coolant fluid through the engine.
- c. The maximum allowable temperature rise in the coolant fluid through the engine.
- d. The minimum allowable inlet fuel temperature.

2.8.1 Coolant Pumps

Provide centrifugal coolant pumps. Each engine must have an engine-driven primary pump. Provide secondary pumps that are electric motor driven and have automatic controllers.

2.8.2 Heat Exchanger

Provide heat exchanger with the size and capacity to limit the maximum allowable temperature rise in the coolant across the engine to that recommended and submitted for the maximum summer outdoor design temperature and site elevation. Submit manufacturer's data to quantify heat rejected to the space with the engine generator set at rated capacity. Provide heat exchangers that are corrosion resistant, suitable for service in ambient conditions of application.

2.8.2.1 Fin-Tube-Type Heat Exchanger (Radiator)

Heat exchanger may be factory coated with corrosion resistant film,

provided that corrective measures are taken to restore the heat rejection capability of the radiator to the initial design requirement via oversizing, or other compensating methods. Provide internal surfaces that are compatible with liquid fluid coolant used. Materials and coolant are subject to approval by the Contracting Officer. Provide heat exchangers that are pressure type incorporating a pressure valve, vacuum valve and a cap. Design caps for pressure relief prior to removal. Provide heat exchanger and cooling system that is capable of withstanding a minimum pressure of 48 kPa and protect with a strong grille or screen guard. Provide heat exchanger with at least two tapped holes; equip one tapped hole with a drain cock, and plug the rest.

Provide for each engine-generator set, as standard with the manufacturer.

- a. Design Conditions: Each radiator unit must have ample capacity to remove not less than the total kW per hour of heat rejected by its respective engine at 100 percent full-rated load to the jacket water, fuel oil, and lubricating oil system, and intercooler. Radiator capacity must be rated at optimum temperature of coolant leaving the engine and intercooler as recommended by the engine manufacturer with an ambient dry bulb air temperature outside the enclosure of 34 degrees C maximum, and 3 degrees C minimum at the site elevation specified in the paragraph SITE CONDITIONS, and with the coolant mixture specified in the paragraph ENGINE CAPACITY. Pressure drop through the radiator must not exceed 41.34 kPa when circulating the maximum required coolant flow. Radiator air velocity must be a maximum of 7.6 meters per second .
- b. Engine Mounted Radiator Construction: Radiator fan must direct airflow from the engine outward through the radiator. Fan must be V-belt driven directly from the engine crankshaft. Fan static capacity must be adjusted to suit the ductwork furnished. Cooling section must have a tube and fin-type core consisting of copper or copper base alloy tubes with nonferrous fins. Select engine-driven fans for quiet vibration-free operation. Make provision for coolant expansion either by self-contained expansion tanks or separately mounted expansion tanks, as standard with the manufacturer. Provide suitable guards for each fan and drive.
- c. Coolant solution must be a mixture of clean water and ethylene glycol, 50 percent by volume each. Provide an anti-freeze solution tester suitable for the mixture.

2.8.3 Thermostatic Control Valve

Provide a modulating type, thermostatic control valve in the coolant system to maintain the coolant temperature range submitted in paragraph SUBMITTALS.

2.8.4 Temperature Sensors

Equip each engine with coolant temperature sensors. Provide temperature sensors with signals for pre-high and high indication and alarms.

2.9 SOUND LIMITATIONS

Submit sound power level data for the packaged unit operating at 100 percent load in a free field environment. The data should demonstrate compliance with the sound limitation requirements of this specification.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Submit certification from the manufacturer stating that the sound emissions meet the specification. Do not exceed the following sound pressure levels in any of the indicated frequencies when measured in a free field at a radial distance of 23 meters at 45 degrees apart in all directions when operating at 100 percent load.

Frequency Band (Hz)	Maximum Acceptable Sound Level (Decibels)
31	87
63	77
125	70
250	70
500	64
1,000	61
2,000	60
4,000	60
8,000	62

The A-weighted sound pressure level (dBA) shall be less than 80.

2.10 AIR INTAKE EQUIPMENT

Locate filters and silencers in locations that are convenient for servicing. Provide high-frequency filter type silencers and locate in the air intake system as recommended by the engine manufacturer. Provide silencer to reduce the noise level at the air intake so that the indicated pressure levels specified in paragraph SOUND LIMITATIONS will not be exceeded. A combined filter-silencer unit meeting requirements for the separate filter and silencer items may be provided. Provide rubber expansion elements in air-intake lines.

Provide intake filter assemblies for each engine of the oil bath or dry type, as standard with the manufacturer. Filters must be capable of removing a minimum of 92 percent of dirt and abrasive 3 microns and larger from intake air. Size filters to suit engine requirements at 100 percent of rated full load. Design unit for field access for maintenance purposes.

2.11 PYROMETER

Provide a pyrometer, and thermocouple with calibrated leads to show the temperature of the combined exhaust. For a supercharged engine, provide additional points, thermocouples and leads to show the temperature in the turbocharger exhaust gas outlet and combustion air discharge passages. Graduated scale length less than 150 mm is not acceptable. Provide double pole selector switch with an "off" position, one set of points for each thermocouple, and suitable indicating dial. Calibrate the pyrometer, thermocouples, leads and compensating devices to show true exhaust temperature within plus or minus 1 percent above the highest temperature encountered at 110 percent load conditions.

2.12 EMISSIONS

The finished installation must comply with Federal, state, and local regulations and restrictions regarding the limits of emissions, as listed here: EPA Tier 3. Submit certification from the engine manufacturer stating that the engine exhaust emissions meet the federal, state, and local regulations and restrictions specified. At a minimum this certification must include emission factors for criteria pollutants including nitrogen oxides, carbon monoxide, particulate matter, sulfur dioxide, non-methane hydrocarbon, and for hazardous air pollutants (HAPs).

2.13 STARTING SYSTEM

Provide starting system for engine generator sets used in non-emergency applications as follows.

2.13.1 Controls

Provide an engine control switch with functions including: run/start (manual), off/reset, and, automatic mode. Provide start-stop logic for adjustable cycle cranking and cool-down operation. Arrange the logic for manual starting. Provide electrical starting systems with an adjustable cranking limit device to limit cranking periods from 1 second up to the maximum duration.

2.13.2 Capacity

Provide starting system with sufficient capacity, at the maximum outdoor summer temperature specified to crank the engine without damage or overheating. The system must provide a minimum of three cranking periods with 15 second intervals between cranks. Each cranking period must have a maximum duration of 15 seconds. Starting must be accomplished using an adequately sized dc starter system with a positive shift solenoid to engage the starter motor and to crank the engine continuously for 60 seconds without overheating.

2.13.3 Electrical Starting

Manufacturers recommended dc system, utilizing a negative circuit ground. Starting motors must be in accordance with [SAE ARP892](#).

2.13.3.1 Battery

Provide a starting battery system including the battery, battery rack, intercell connectors, spacers, automatic battery charger with overcurrent protection, metering and relaying. Provide battery in accordance with [SAE J537](#). Size critical system components (rack, protection, etc.) to withstand the seismic acceleration forces specified. Provide Absorbent Glass Mat (AGM) lead-acid battery with sufficient capacity, at the minimum outdoor and maximum outdoor temperature specified, to provide the specified cranking periods. Valve-regulated lead-acid batteries are not acceptable.

Provide maintenance free, sealed, lead-acid, SAE Type D engine starting batteries. Batteries must have sufficient capacity to provide 60 seconds of continuous cranking of the engine in an ambient temperature of 0 degrees C.

2.13.3.2 Battery Charger

Provide a current-limiting battery charger, conforming to **UL 1236**, that automatically recharges the batteries. Submit battery charger sizing calculations. The charger must be capable of an equalize charging rate for recharging fully depleted batteries within 24 hours and a floating charge rate for maintaining the batteries at fully charged condition. Provide an ammeter to indicate charging rate. Provide a voltmeter to indicate charging voltage. Provide a timer for the equalize charging-rate setting. A battery is considered to be fully depleted when the output voltage falls to a value which will not operate the engine generator set and its components.

Provide 120 volt ac, enclosed, automatic equalizing, dual-rate, solid-state, constant voltage type battery charger with automatic ac line compensation. DC output must be voltage regulated and current limited. Charger must have two ranges, float and equalize, and must provide continuous taper charging. The charger must have a continuous output rating of not less than 10 amperes and must be sized to recharge the engine starting batteries in a minimum of 8 hours while providing the control power needs of the engine-generator set. Enclosure must be **NEMA ICS 6**, Type 1 . The following accessories must be included:

- a. DC ammeter
- b. DC voltmeter
- c. Equalize light
- d. AC on light
- e. Low voltage light
- f. High voltage light
- g. Equalize test button/switch
- h. AC circuit breaker
- i. Low dc voltage alarm relay
- j. High dc voltage alarm relay
- k. Current failure relay
- l. AC power failure relay

2.13.4 Starting Aids

Provide one or more of other following methods to assist engine starting.

2.13.4.1 Jacket-Coolant Heaters

Mount a thermostatically controlled electric heater in the engine coolant jacketing to automatically maintain the coolant within plus or minus **1.7 degrees C** of the control temperature. The heater must operate independently of engine operation so that starting times are minimized. Power for the heaters must be 208 volts ac. Include necessary equipment, piping, controls, wiring, and accessories.

2.13.4.1.1 Standby Rated Sets

The control temperature must be the temperature recommended by the engine manufacturer to meet the starting time specified at the minimum winter outdoor temperature.

2.13.4.2 Lubricating-Oil Heaters

Mount a thermostatically controlled electric heater in the engine lubricating-oil system to automatically maintain the oil temperature within plus or minus 1.7 degrees C of the control temperature. The heater must operate independently of engine operation so that starting times are minimized. Power for the heaters must be 208 volts ac.

2.14 GOVERNOR

Provide a forward acting type engine speed governor system. Steady-state frequency band and frequency regulation (droop) must be in accordance with the operating limit values of the performance class specified in the paragraph PERFORMANCE CLASS.

Provide engine with a governor which maintains the frequency within a bandwidth of the rated frequency, over a steady-state load range of zero to 100 percent of rated output capacity. Configure the governor for safe manual adjustment of the speed/frequency during operation of the engine-generator set, without special tools, from 90 to 110 percent of the rated speed/frequency, over a steady state load range of 0 to 100 percent or rated capacity. Submit two complete sets of special tools required for maintenance (except for electronic governor handset). Special tools are those that only the manufacturer provides, for special purposes, or to reach otherwise inaccessible parts. Provide a suitable tool box for tools. Provide one handset for each electronic governor when required to indicate and/or change governor response settings. Maintain the midpoint of the frequency bandwidth at the same value for steady-state loads over the range of zero to 100 percent of rated output capacity for isochronous governors.

2.15 GENERATOR

Provide synchronous type, one or two bearing, generator conforming to the performance criteria in NEMA MG 1, equipped with winding terminal housings in accordance with NEMA MG 1, equipped with an amortisseur winding, and directly connected to the engine. Submit calculations of the engine and generator output power capability, including efficiency and parasitic load data. Provide Class F insulation.

- a. Provide salient-pole type, ac, brushless-excited, revolving field, air-cooled, self-ventilated, coupled type, synchronous generator conforming to NEMA MG 1, Part 16, and IEEE C50.12. Generator must be rated for standby duty at 100 percent of the power rating of the engine-generator set as specified in paragraph ENGINE-GENERATOR SET RATINGS AND PERFORMANCE. Temperature rise of each of the various parts of the generator must not exceed 130 degrees C as measured by resistance, based on a maximum ambient temperature of 40 degrees C. Winding insulation must be Class F.
- b. Rotor: The rotor must have connected amortisseur windings.

- c. Generator Space Heater: Provide 208 volt ac heaters. Heater capacity must be as recommended by the generator manufacturer to aid in keeping the generator insulation dry.
- d. Grounding: Provide non-corrosive steel grounding pads located at two opposite mounting legs.
- e. Filters: Provide manufacturer's standard generator cooling air filter assembly.
- f. Design generator to protect against mechanical, electrical and thermal damage due to vibration, 25 percent overspeeds, or voltages and temperatures at a rated output capacity of 110 percent for prime applications and 100 percent for standby applications.
- g. Provide generator ancillary equipment meeting the short circuit requirements of NEMA MG 1.
- h. Submit manufacturer's standard data for each generator (prototype data at the specified rating or above is acceptable), listing the following information:
 - (1) Direct-Axis sub-transient reactance (per unit).
 - (2) The generator kW rating and short circuit current capacity (both symmetric and asymmetric).

2.15.1 Current Balance

At 100 percent rated output capacity, and load impedance equal for each of the 3 phases, the permissible current difference between any 2 phases must not exceed 2 percent of the largest current on either of the 2 phases. Submit certification stating that the flywheel has been statically and dynamically balanced and is capable of being rotated at 125 percent of rated speed without vibration or damage.

2.15.2 Voltage Balance

At any balanced load between 75 and 100 percent of rated output capacity, the difference in line-to-neutral voltage among the 3 phases must not exceed 1 percent of the average line-to-neutral voltage. For a single phase load condition, consisting of 25 percent load at unity power factor placed between any phase and neutral with no load on the other 2 phases, the maximum simultaneous difference in line-to-neutral voltage between the phases must not exceed 3 percent of rated line to neutral voltage. The single-phase load requirement must be valid utilizing normal exciter and regulator control. The interpretation of the 25 percent load for single phase load conditions means 25 percent of rated current at rated phase voltage and unity power factor.

2.15.3 Waveform

The deviation factor of the line-to-line voltage at zero load and at balanced rated output capacity must not exceed 10 percent. The RMS of all harmonics must be less than 5.0 percent and that of any one harmonic less than 3.0 percent of the fundamental at rated output capacity. Design and configure engine-generator to meet the total harmonic distortion limits of IEEE 519.

2.16 EXCITER

Provide brushless generator exciter. Provide semiconductor rectifiers that have a minimum safety factor of 300 percent for peak inverse voltage and forward current ratings for all operating conditions, including 110 percent generator output at 40 degrees C ambient. The exciter and regulator in combination must maintain generator-output voltage within the limits specified.

Provide a brushless excitation system consisting of an exciter and rotating rectifier assembly integral with the generator and a voltage regulator. Insulation class for parts integral with the generator must be as specified in paragraph GENERATOR. System must provide a minimum short circuit of 300 percent rated engine-generator set current for at least 10 seconds. Steady state voltage regulation must be in accordance with the operating limit values of the performance class specified in the paragraph PERFORMANCE CLASS.

- a. Exciter and Rotating Rectifier Assembly: Rectifiers must be provided with surge voltage protection.
- b. Voltage Regulator: Voltage regulator must be solid state or digital, automatic, three-phase sensing, volts per hertz type regulator. Voltage variation for any 40 degree C change over the operating temperature range must be less than plus or minus 1.0 percent. Operating temperature must be minus 40 degree C to plus 70 degree C. Voltage adjust range must be plus to minus 5.0 percent of nominal. Inherent regulator features must include over excitation shutdown.

2.17 VOLTAGE REGULATOR

Provide a solid-state voltage regulator, separate from the exciter, for each generator. Maintain the voltage within a bandwidth of the rated voltage, over a steady-state load range of zero to 100 percent of rated output capacity. Configure regulator for safe manual adjustment of the engine-generator voltage output without special tools, during operation, from 90 to 110 percent of the rated voltage over the steady state load range of 0 to 100 percent of rated output capacity. Regulation drift exceeding plus or minus 0.5 percent for an ambient temperature change of 20 degrees C is not acceptable. Provide voltage regulator with a maximum droop of 2 percent of rated voltage over a load range from 0 to 100 percent of rated output capacity and automatically maintain the generator output voltage within the specified operational bandwidth.

2.18 GENERATOR ISOLATION AND PROTECTION

Provide necessary devices for electrical protection and isolation of each engine-generator set and its ancillary equipment. The generator circuit breaker (IEEE Device 52) ratings must be consistent with the generator rated voltage and frequency, with continuous, short circuit withstand, and interrupting current ratings to match the generator capacity. Provide manually operated generator circuit breaker. Provide monitoring and control devices as specified in paragraph GENERATOR PANEL.

The generator circuit breaker must comply with UL 489 requirements for molded case, adjustable thermal magnetic trip type circuit breaker. The circuit breaker continuous current rating must be adequate for the power rating of the engine-generator set and the circuit breaker must be rated to withstand the short circuit current provided by the generator set.

Provide circuit breaker mounted within generator enclosure.

2.18.1 Devices

Furnish with respective pieces of equipment. Motors, controllers, contactors, and disconnects must conform to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide electrical connections under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide controllers and contactors with maximum of 120-volt control circuits, and auxiliary contacts for use with controls furnished. When motors and equipment furnished are larger than size indicated, the cost of providing additional electrical service and related work must be included under this section.

2.19 SAFETY SYSTEM

Provide and install devices, wiring, remote panels, and local panels, etc., as a complete system to automatically activate the appropriate signals and initiate the appropriate actions. Provide a safety system with a self-test method to verify its operability. Provide alarm signals that have manual acknowledgment and reset devices. The alarm signal systems must reactivate for new signals after acknowledgment is given to any signal. Configure the systems so that loss of any monitoring device will be dealt with as an alarm on that system element.

2.19.1 Audible Signal

Provide audible alarm signal sound at a frequency of 70 Hz at a volume of 75 dB at 3.1 m. The sound must be continuously activated upon alarm and silenced upon acknowledgment. Locate signal devices as shown.

2.19.2 Visual Signal

The visual alarm signal must be a panel light. The light must be normally off, activated to be blinking upon alarm. The light must change to continuously lit upon acknowledgement. If automatic shutdown occurs, the display must maintain activated status to indicate the cause of failure and must not be reset until cause of alarm has been cleared and/or restored to normal condition. Shutdown alarms must be red; all other alarms must be amber.

2.19.3 Alarms and Action Logic

2.19.3.1 Shutdown

Accomplish simultaneous activation of the audible signal, activation of the visual signal, stopping the engine, and opening the generator main circuit breakers.

2.19.3.2 Problem

Accomplish activation of the visual signal.

2.19.4 Safety Indications and Shutdowns

Provide a local alarm panel with the following shutdown and alarm functions as indicated mounted either on or adjacent to the engine generator set.

A remote alarm panel is not required for audible alarms, e.g., in the control room.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Indicator Function (at battery voltage)	NFPA 110 Level 2 CV S
Overcrank	X X
Low water temperature	X NA
High engine temperature	X X
Low lube oil pressure	X X
Overspeed	X X
Low fuel main tank	X NA
Low coolant level	X O
High fuel main tank	X NA
Lamp test	X NA
Audible alarm silencing switch	NA NA
Symbology: CV: Control panel-mounted visual. S: Shutdown of EPS indication. X: Required. O: Optional. NA: Not applicable.	

2.19.5 Time-Delay on Alarms

For startup of the engine-generator set, install time-delay devices bypassing the low lubricating oil pressure alarm during cranking, and the coolant-fluid outlet temperature alarm. Submit the magnitude of monitored values which define alarm or action set points, and the tolerance (plus and/or minus) at which the devices activate the alarm or action for items contained within the alarm panels. The lube-oil time-delay device must return its alarm to normal status after the engine starts. The coolant time-delay device must return its alarm to normal status 5 minutes after the engine starts.

2.20 PANELS

Each panel must be of the type and kind necessary to provide specified functions. Mount panels on the engine-generator set base by vibration/shock absorbing type mountings. Mount instruments flush or semiflush. Provide convenient access to the back of panels to facilitate maintenance. Calibrate instruments using recognized industry calibration standards. Provide a panel identification plate identifying the panel function. Provide a plate identifying the device and its function for each instrument and device on the panel. Provide switch plates identifying the switch-position function.

2.20.1 Enclosures

Design enclosures for the application and environment, conforming to **NEMA ICS 6**. Locking mechanisms must be keyed alike.

Provide for each engine-generator set and fabricate from zinc coated or phosphatized and shop primed 16 gage minimum sheet steel in accordance with the manufacturer's standard design. Provide a complete, weatherproof enclosure for the engine, generator, and auxiliary systems and equipment. Support exhaust piping and silencer so that the turbocharger is not subjected to exhaust system weight or lateral forces generated in connecting piping that exceed the engine manufacturer's maximum allowed forces and moments. The housing must have sufficient louvered openings to allow entrance of outside air for engine and generator cooling at full load. Design louvered openings to exclude driving rain and snow. Provide properly arranged and sized, hinged panels in the enclosure to allow convenient access to the engine, generator, and control equipment for maintenance and operational procedures. Provide hinged panels with spring type latches which must hold the panels closed securely and will not allow them to vibrate. Brace the housing internally to prevent excessive vibration when the set is in operation

2.20.2 Analog

Provide analog electrical indicating instruments in accordance with **UL 1437** with semi-flush mounting. Switchboard, switchgear, and control-room panel-mounted instruments must have 250 degree scales with an accuracy of not less than 99 percent. Unit-mounted instruments must be the manufacturer's standard with an accuracy of not less than 98 percent. The instrument's operating temperature range must be **minus 20 to plus 65 degrees C**. Distorted generator output voltage waveform of a crest factor less than 5 must not affect metering accuracy for phase voltages, hertz and amps.

2.20.3 Parameter Display

Provide indication or readouts of the tachometer, lubricating-oil pressure, ac voltmeter, ac ammeter, frequency meter, and safety system parameters. Specify a momentary switch for other panels.

2.21 SURGE PROTECTION

Electrical and electronic components must be protected from, or designed to withstand the effects of surges from switching and lightning.

2.22 MANUAL ENGINE-GENERATOR-SET SYSTEM OPERATION

Provide complete facilities for manual starting and testing of each set without load, loading and unloading of each set, and synchronization of each set with an energized bus.

2.23 BASE

Provide a steel base. Design the base to rigidly support the engine-generator set, ensure permanent alignment of rotating parts, be arranged to provide easy access to allow changing of lube-oil, and ensure that alignment is maintained during shipping and normal operation. The base must permit skidding in any direction during installation and must

withstand and mitigate the affects of synchronous vibration of the engine and generator. Provide base with suitable holes for anchor bolts and jacking screws for leveling.

2.24 PAINTING AND FINISHING

Clean, prime and paintthe engine-generator set in accordance with the manufacturer's standard color and practice.

2.25 FACTORY INSPECTION AND TESTS

Submit six complete reproducible copies of the factory inspection result on the checklist format specified below. Perform the factory tests on each engine-generator set. The component manufacturer's production line test is acceptable as noted. Run each engine-generator set for at least 1 hour at rated output capacity prior to inspections. Complete inspections and make all necessary repairs prior to testing. Use engine generator controls and protective devices that are provided by the generator set manufacturer as part of the standard package for factory tests. When controls and switchgear are not provided as part of the generator set manufacturer's standard package, the actual controls and protective devices provided for the project are not required to be used during the factory test. The Contracting Officer may provide one or more representatives to witness inspections and tests.

2.25.1 Factory Inspection

Perform inspections prior to beginning and after completion of testing of the assembled engine-generator set. Look for leaks, looseness, defects in components, proper assembly, etc. and note any item found to be in need of correction as a necessary repair. Use the following checklist for the inspection:

INSPECTION ITEM	GOOD	BAD	NOTES
Drive belts			
Governor and adjustments			
Engine timing mark			
Starting motor			
Starting aids			
Coolant type and concentration			
Radiator drains			
Block coolant drains			
Coolant fill level			
All coolant line connections			

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

INSPECTION ITEM	GOOD	BAD	NOTES
All coolant hoses			
Combustion air filter			
Combustion air silencer			
Lube oil type			
Lube oil sump drain			
Lube-oil filter			
Lube-oil-level indicator			
Lube-oil-fill level			
All lube-oil line connections			
All lube-oil lines			
Fuel type and amount			
All fuel-line connections			
All fuel lines			
Fuel filter			
Coupling and shaft alignment			
Voltage regulators			
Battery-charger connections			
All wiring connections			
Instrumentation			
Hazards to personnel			
Base			
Nameplates			
Paint			
Exhaust-heat recovery unit			

2.25.2 Factory Tests

Submit a letter giving notice of the proposed dates of factory inspections

and tests at least 14 days prior to beginning tests, including:

- a. A detailed description of the manufacturer's procedures for factory tests at least 14 days prior to beginning tests.
- b. Six copies of the Factory Test data described below in 216 by 279 mm binders having a minimum of 3 rings from which material may readily be removed and replaced, including a separate section for each test. Separate sections by heavy plastic dividers with tabs. Provide full size (216 by 279 mm minimum) data plots showing grid lines, with full resolution.
 - (1) A detailed description of the procedures for factory tests.
 - (2) A list of equipment used, with calibration certifications.
 - (3) A copy of measurements taken, with required plots and graphs.
 - (4) The date of testing.
 - (5) A list of the parameters verified.
 - (6) The condition specified for the parameter.
 - (7) The test results, signed and dated.
 - (8) A description of adjustments made.

On engine-generator set tests where the engine and generator are required to be connected and operated together, the load power factor must be the power factor specified in the engine generator set parameter schedule . Perform electrical measurements in accordance with IEEE 120. Temperature limits in the rating of electrical equipment and for the evaluation of electrical insulation must be in accordance with IEEE 1. In the following tests where measurements are to be recorded after stabilization of an engine-generator set parameter (voltage, frequency, current, temperature, etc.), stabilization is considered to have occurred when measurements are maintained within the specified bandwidths or tolerances, for a minimum of four consecutive readings. Tests specifically for the generator may be performed utilizing any prime mover.

- a. Insulation Resistance for Stator and Exciter Test, IEEE 115 and IEEE 43, to the performance criteria in NEMA MG 1, Part 22. Generator manufacturer's production line test is acceptable.
- b. High Potential Test, in accordance with IEEE 115 and NEMA MG 1, test voltage in accordance with NEMA MG 1. Generator manufacturer's production line test is acceptable.
- c. Winding Resistance Test, Stator and Exciter, in accordance with IEEE 115. Generator manufacturer's production line test is acceptable.
- d. Phase Balance Voltage Test, to the performance criteria specified in paragraph GENERATOR. This test can be performed with any prime mover. Generator manufacturer's production line test results are acceptable.
 - (1) Start and operate the generator at no load.

- (2) Adjust a regulated phase voltage (line-to-neutral) to rated voltage.
 - (3) Read and record the generator frequency, line-to-neutral voltages, and the line-to-line voltages.
 - (4) Apply 75 percent rated load and record the generator frequency, line-to-neutral voltages, and the line-to-line voltages.
 - (5) Apply rated load and record the generator frequency, line-to-neutral voltages, and the line-to-line voltages.
 - (6) Calculate average line-neutral voltage and percent deviation of individual line-neutral voltages from average for each load condition.
- e. Current Balance on Stator Winding Test, by measuring the current on each phase of the winding with the generator operating at 100 percent of Rated Output Capacity, with the load impedance equal for each of the three phases: to the performance criteria specified in paragraph GENERATOR.
- f. Voltage Waveform Deviation and Distortion Test in accordance with **IEEE 115** to the performance criteria specified in paragraph GENERATOR. Use high-speed recording instruments capable of recording voltage waveform deviation and all distortion, including harmonic distortion. Include appropriate scales to provide a means to measure and interpret results.
- g. Voltage and Frequency Droop Test. Verify that the output voltage and frequency are within the specified parameters as follows:
- (1) With the generator operating at no load, adjust voltage and frequency to rated voltage and frequency. Record the generator output frequency and line-line and line-neutral voltages.
 - (2) Increase load to Rated Output Capacity. Record the generator output frequency and line-line and line-neutral voltages.
 - (3) Calculate the percent droop for voltage and frequency with the following equations:

$$\text{Voltage droop percent} = \frac{(\text{No-Load Volts}) - (\text{Rated Capacity Volts})}{(\text{Service-Load Volts})} \times 100$$

$$\text{Frequency droop percent} = \frac{(\text{No-Load Hertz}) - (\text{Rated Capacity Hertz})}{(\text{Service-Load Hertz})} \times 100$$

- (4) Repeat steps 1 through 3 two additional times without making any adjustments.
- h. Frequency and Voltage Stability and Transient Response. Verify that the engine-generator set responds to addition and dropping of blocks of load in accordance with the transient response requirements. Document maximum voltage and frequency variation from bandwidth and verify that voltage and frequency return to and stabilize within the

specified bandwidth, within the specified response time period. Document results in tabular form and with high resolution, high speed strip chart recorders or comparable digital recorders, as approved by the Contracting Officer. Include the following tabular data:

- (1) Ambient temperature (at 15 minute intervals).
- (2) Generator output current (before and after load changes).
- (3) Generator output voltage (before and after load changes).
- (4) Frequency (before and after load changes).
- (5) Generator output power (before and after load changes).
- (6) Graphic representations must include the actual instrument trace of voltage and frequency showing: charts marked at start of test; observed steady-state band; mean of observed band; momentary overshoot and undershoot (generator terminal voltage and frequency) and recovery time for each load change together with the voltage and frequency maximum and minimum trace excursions for each steady state load condition prior to and immediately following each load change. Generator terminal voltage and frequency transient recovery time for each step load increase and decrease.
 - (a) Perform and record engine manufacturer's recommended pre-starting checks and inspections.
 - (b) Start the engine, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period and no load. Verify stabilization of voltage and frequency within specified bandwidths.
 - (c) With the unit at no load, apply the Maximum Step Load Increase.
 - (d) Apply load in steps equal to the Maximum Step Load Increase until the addition of one more step increase will exceed the Service Load.
 - (e) Decrease load to the unit such that addition of the Maximum Step Load Increase will load the unit to 100 percent of Service Load.
 - (f) Apply the Maximum Step Load Increase.
 - (g) Decrease load to zero percent in steps equal to the Maximum Step Load Decrease.
 - (h) Repeat steps (c) through (g).

PART 3 EXECUTION

3.1 GENERAL INSTALLATION

Provide clear space for operation and maintenance in accordance with **NFPA 70** and **IEEE C2**. Submit a copy of the manufacturer's installation procedures and a detailed description of the manufacturer's recommended

break-in procedure. Install pipe, duct, conduit, and ancillary equipment to facilitate easy removal and replacement of major components and parts of the engine-generator set.

3.2 ELECTRICAL INSTALLATION

Perform electrical installation in compliance with NFPA 70, IEEE C2, and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. For vibration isolation, provide flexible fittings for conduit attached to engine-generator sets; provide flexible stranded conductor for metallic conductor cables installed on the engine generator set and from the engine generator set to equipment not mounted on the engine generator set; and provide crimp-type terminals or lugs for terminations of conductors on the engine generator set.

3.3 FIELD PAINTING

Perform field painting as specified in Section 09 90 00 PAINTS AND COATINGS.

3.4 ONSITE INSPECTION AND TESTS

Perform and report on factory tests and inspections prior to shipment. Provide certified copies of manufacturer's test data and results. Test procedures must conform to ASME, IEEE, and ANSI standards, and to ISO requirements on testing, as appropriate and applicable. The manufacturer performing the tests must provide equipment, labor, and consumables necessary for tests and measuring and indicating devices must be certified to be within calibration. Tests must indicate satisfactory operation and attainment of specified performance. If satisfactory, equipment tested will be given a tentative approval. Equipment must not be shipped before approval of the factory test reports for the following tests.

Submit a letter giving notice of the proposed dates of onsite inspections and tests at least 14 days prior to beginning tests.

- a. Submit a detailed description of the Contractor's procedures for onsite tests including the test plan and a listing of equipment necessary to perform the tests at least 14 days prior to beginning tests.
- b. Submit six copies of the onsite test data described below in 216 by 279 mm binders having a minimum of 3 rings from which material may readily be removed and replaced, including a separate section for each test. Separate sections by heavy plastic dividers with tabs. Provide full size (216 by 279 mm minimum) data plots showing grid lines, with full resolution.
 - (1) A detailed description of the procedures for onsite tests.
 - (2) A list of equipment used, with calibration certifications.
 - (3) A copy of measurements taken, with required plots and graphs.
 - (4) The date of testing.
 - (5) A list of the parameters verified.
 - (6) The condition specified for the parameter.

(7) The test results, signed and dated.

(8) A description of adjustments made.

3.4.1 Test Conditions

3.4.1.1 Data

Make and record measurements of all parameters necessary to verify that each set meets specified parameters. If the results of any test step are not satisfactory, make adjustments, replacements, or repairs and repeat the step until satisfactory results are obtained. Unless otherwise indicated, record data in 15 minute intervals during engine-generator set operation and include: readings of all engine-generator set meters and gauges for electrical and power parameters; oil pressure; ambient temperature; and engine temperatures available from meters and gauges supplied as permanent equipment on the engine-generator set. Perform electrical measurements in accordance with [IEEE 120](#). Definitions of terms are in accordance with [IEEE 100](#). Provide temperature limits in the rating of electrical equipment and for the evaluation of electrical insulations in accordance with [IEEE 1](#).

3.4.1.2 Power Factor

Submit the generator capability curve showing generator kVA output capability (kW vs. kvar) for both leading and lagging power factors ranging from 0 to 1.0. For all engine-generator set operating tests the load power factor must be the power factor specified in the engine-generator set parameter schedule.

3.4.1.3 Contractor Supplied Items

Provide equipment and supplies required for inspections and tests including fuel, test instruments, and loadbanks at the specified power factors.

3.4.1.4 Instruments

Verify readings of panel gauges, meters, displays, and instruments provided as permanent equipment during test runs, using test instruments of greater precision and accuracy. Test instrument accuracy must be within the following: current plus or minus 1.5 percent, voltage plus or minus 1.5 percent, real power plus or minus 1.5 percent, reactive power plus or minus 1.5 percent, power factor plus or minus 3 percent, frequency plus or minus 0.5 percent. Calibrate test instruments by a recognized standards laboratory within 30 days prior to testing.

3.4.1.5 Sequence

Provide the sequence of testing as specified in the approved testing plan unless variance is authorized by the Contracting Officer. Perform field testing in the presence of the Contracting Officer. Schedule and sequence tests in order to optimize run-time periods; however, follow the general order of testing: Construction Tests; Inspections; Pre-operational Tests; Safety Run Tests; Performance Tests; and Final Inspection.

3.4.2 Construction Tests

Perform individual component and equipment functional tests for fuel piping, coolant piping, and lubricating-oil piping, electrical circuit continuity, insulation resistance, circuit protective devices, and equipment not provided by the engine-generator set manufacturer prior to connection to the engine-generator set.

3.4.2.1 Piping Test

- a. Flush lube-oil and fuel-oil piping with the same type of fluid intended to flow through the piping, until the outflowing fluid has no obvious sediment or emulsion.

3.4.2.2 Electrical Equipment Tests

- a. Perform low-voltage cable insulation integrity tests for cables connecting the generator breaker to the MCC. Test low-voltage cable, complete with splices, for insulation resistance after the cables are installed, in their final configuration, ready for connection to the equipment, and prior to energization. Apply a test voltage of 500 volts dc for one minute between each conductor and ground and between all possible combinations conductors in the same trench, duct, or cable, with all other conductors in the same trench, duct, or conduit. Provide the minimum value of insulation as follows:
 - (1) $R \text{ in meg-ohms} = (\text{rated voltage in kV plus } 1) \times 304.8 / (\text{length of cable in meters})$
 - (2) $R \text{ in meg-ohms} = (\text{rated voltage in kV plus } 1) \times 1000 / (\text{length of cable in feet})$
 - (3) Each cable failing this test must be repaired or replaced. The repair cable must be retested until failures have been eliminated.
- b. Ground-Resistance Tests. Measure the resistance of each grounding electrode and the ground ring using the fall-of-potential method defined in [IEEE 81](#). On systems consisting of interconnected ground rods, perform tests after interconnections are complete. Take measurements in normally dry weather, not less than 48 hours after rainfall. Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance. The combined resistance of separate systems may be used to meet the requirements resistance, but the specified number of electrodes must still be provided as follows:
 - (1) Single rod electrode - 25 ohms.
 - (2) Ground ring - 10 ohms.
- d. Examine and test circuit breakers in accordance with the manufacturer's published instructions for functional testing.

3.4.3 Inspections

Perform the following inspections jointly by the Contracting Officer and the Contractor, after complete installation of each engine-generator set and its associated equipment, and prior to startup of the engine-generator set. Submit a letter certifying that all facilities are complete and

functional; that each system is fully functional; and that each item of equipment is complete, free from damage, adjusted, and ready for beneficial use. Perform checks applicable to the installation. Document and submit the results of those which are physical inspections (I) in accordance with paragraph SUBMITTALS. Present manufacturer's data for the inspections designated (D) at the time of inspection. Verify that equipment type, features, accessibility, installation and condition are in accordance with the contract specification. Provide manufacturer's statements to certify provision of features which cannot be verified visually.

3.4.4 Engine Tests

Perform customary commercial factory tests in accordance with [ISO 3046](#) on each engine and associated engine protective device, including, but not limited to the following:

- a. Perform dynamometer test at rated power. Record horsepower at rated speed and nominal characteristics such as lubricating oil pressure, jacket water temperature, and ambient temperature.
- b. Test and record the values that the low oil pressure alarm and protective shutdown devices actuate prior to assembly on the engine.
- c. Test and record values that the high jacket water temperature alarm and protective shutdown devices actuate prior to assembly on the engine.

3.4.5 Generator Tests

Tests must be performed on the complete factory assembled generator prior to shipment. Conduct tests in accordance with [IEEE 115](#), and [NEMA MG 1](#).

3.4.5.1 Routine Tests

Perform the following routine tests on the generators and their exciters:

- a. Resistance of armature and field windings.
- b. Mechanical balance.
- c. Phases sequence.
- d. Open circuit saturation curve and phase (voltage) balance test.
- e. Insulation resistance of armature and field windings.
- f. High potential test.

3.4.5.2 Design Tests

Submit the following design tests made on prototype machines that are physically and electrically identical to the generators specified.

- a. Temperature rise test
- b. Short circuit saturation curve and current balance test

3.4.6 Assembled Engine-Generator Set Tests

Submit the following tests made on prototype machines that are physically and electrically identical to the engine-generator set specified.

3.4.6.1 Initial Stabilization Readings

Operate the engine-generator set and allow the set to stabilize at rated kW at rated power factor, rated voltage, and rated frequency. During this period record instrument readings for output power (kW), terminal voltage, line current, power factor, frequency (rpm) generator (exciter) field voltage and current, lubricating oil pressure, jacket water temperature, and ambient temperature at minimum intervals of 15 minutes. Adjust the load, voltage, and frequency to maintain rated load at rated voltage and frequency. Adjustments to load, voltage, or frequency controls must be recorded on the data sheet at the time of adjustment. Stabilization must be considered to have occurred when four consecutive voltage and current recorded readings of the generator (or exciter) field either remain unchanged or have only minor variations about an equilibrium condition with no evident continued increase or decrease in value after the last adjustment to the load, voltage, or frequency has been made.

3.4.6.2 Regulator Range Test

Remove load and record instrument readings (after transients have subsided). Adjust voltage to the maximum attainable value or to a value just prior to actuation of the overvoltage protection device. Apply rated load and adjust voltage to the minimum attainable value or a value just prior to activation of the under-voltage protection device. The data sheets must indicate the voltage regulation as a percent of rated voltage and the maximum and minimum voltages attainable. Voltage regulation must be defined as follows:

$$\text{Percent Regulation} = \frac{((\text{No-Load Voltage}) - (\text{Rated-Load Voltage})) \times 100}{(\text{Rated-Load Voltage})}$$

3.4.6.3 Frequency Range Test

Adjust the engine-generator set frequency for the maximum attainable frequency at rated load. Record instrument readings. Adjust the engine-generator set frequency for the specified minimum attainable frequency at rated load. Record instrument readings. Reduce the load to zero and adjust the engine-generator set frequency for the maximum attainable frequency. Record instrument readings. Adjust the engine-generator set frequency for the minimum attainable frequency. Record instrument readings. The data sheet must show the maximum and minimum frequencies attained at rated load, and at no load.

3.4.6.4 Transient Response Test

Drop the load to no load and re-apply rated load three times to ensure that the no load and rated load voltage and frequency values are repeatable and that the frequency and voltage regulation is within the limits specified. Record generator terminal voltage and frequency using a high speed strip chart recorder. The data sheet must show the following results:

- a. Frequency

- (1) Stability bandwidth or deviation in percent of rated frequency.
- (2) Recovery time.
- (3) Overshoot and undershoot.

b. Voltage

- (1) Stability bandwidth or deviation in percent of rated voltage.
- (2) Recovery time.
- (3) Overshoot and undershoot.

3.4.7 Pre-operational Tests

3.4.7.1 Insulation Test

Test generator and exciter circuits insulation resistance in accordance with **IEEE 43**. Take stator readings including generator leads to the circuit breaker. Record results of insulation resistance tests. Readings must be within limits specified by the manufacturer. Verify mechanical operation, insulation resistance, and wiring continuity of cabling assembly. Do not damage generator components during test.

3.4.7.2 Engine-Generator Connection Coupling Test

When the generator provided is a two-bearing machine, inspect and check the engine-generator connection coupling by dial indicator to prove that no misalignment has occurred. Use the dial indicator to measure variation in radial positioning and axial clearance between the coupling halves. Take readings at four points, spaced 90 degrees apart. Align solid couplings and pin-type flexible couplings within a total indicator reading of **0.012 to 0.025 mm** for both parallel and angular misalignment. For gear-type or grid-type couplings, **0.05 mm** will be acceptable.

3.4.8 Safety Run Test

For the following tests, repeat the associated safety tests if any parts are changed, or adjustments made to the generator set, its controls, or auxiliaries.

- a. Perform and record engine manufacturer's recommended prestarting checks and inspections.
- b. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period.
- c. Activate the manual emergency stop switch and verify that the engine stops.
- d. Remove the high and pre-high lubricating oil temperature sensing elements from the engine and temporarily install a temperature gauge in their normal locations on the engine (required for safety, not for recorded data). Where necessary provide temporary wiring harness to connect the sensing elements to their permanent electrical leads.

- e. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period. Operate the engine-generator set at no load until the output voltage and frequency stabilize. Monitor the temporarily installed temperature gauges. If either temperature reading exceeds the value required for an alarm condition, activate the manual emergency stop switch.
- f. Immerse the elements in a vessel containing controlled-temperature hot oil and record the temperature at which the pre-high alarm activates and the temperature at which the engine shuts down. Remove the temporary temperature gauges and reinstall the temperature sensors on the engine.
- g. Remove the high and pre-high coolant temperature sensing elements from the engine and temporarily install a temperature gauge in their normal locations on the engine (required for safety, not for recorded data). Where necessary provide temporary wiring harness to connect the sensing elements to their permanent electrical leads.
- h. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period. Operate the engine generator-set at no load until the output voltage and frequency stabilize.
- i. Immerse the elements in a vessel containing controlled-temperature hot oil and record the temperature at which the pre-high alarm activates and the temperature at which the engine shuts down. Remove the temporary temperature gauges and reinstall the temperature sensors on the engine.
- j. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period.
- k. Operate the engine generator-set for at least 2 hours at 75 percent of Service Load.
- l. Verify proper operation and set-points of gauges and instruments.
- m. Verify proper operation of ancillary equipment.
- n. Manually adjust the governor to increase engine speed past the over-speed limit. Record the RPM at which the engine shuts down.
- o. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections and operate the engine generator-set for at least 15 minutes at 75 percent of Service Load.
- p. Manually adjust the governor to increase engine speed to within 2 percent of the over-speed trip speed previously determined and operate at that point for 5 minutes. Manually adjust the governor to the rated frequency.
- q. Manually fill the day tank to a level above the overflow limit. Record the level at which the overflow alarm sounds. Verify shutdown of the fuel transfer pump. Drain the day tank down below the overflow limit.

- r. Shut down the engine. Remove the time-delay low lube oil pressure alarm bypass and try to start the engine.
- s. Attach a manifold to the engine oil system (at the oil pressure sensor port) that contains a shutoff valve in series with a connection for the engine's oil pressure sensor followed by an oil pressure gauge ending with a bleed valve. Move the engine's oil pressure sensor from the engine to the manifold. Open the manifold shutoff valve and close the bleed valve.
- t. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections and operate the engine generator-set for at least 15 minutes at 75 percent of Service Load.
- u. Close the manifold shutoff valve. Slowly allow the pressure in the manifold to bleed off through the bleed valve while watching the pressure gauge. Record the pressure at which the engine shuts down. Catch oil spillage from the bleed valve in a container. Add the oil from the container back to the engine, remove the manifold, and reinstall the engine's oil pressure sensor on the engine.
- v. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections and operate the engine generator-set for at least 15 minutes at 100 percent of Service Load. Record the maximum sound level in each frequency band at a distance of 23 m from the end of the exhaust and air intake piping directly along the path of intake and discharge for horizontal piping; or at a radius of 10.7 m from the engine at 45 degrees apart in all directions for vertical piping. If a sound limiting enclosure is provided, modify or replace the enclosure, the muffler, and intake silencer must be modified or replaced as required to meet the sound requirements contained within this specification
- w. Manually drain off fuel slowly from the day tank to empty it to below the low fuel level limit and record the level at which the audible alarm sounds. Add fuel back to the day tank to fill it above low level alarm limits.

3.4.9 Performance Tests

In the following tests, where measurements are to be recorded after stabilization of an engine-generator set parameter (voltage, frequency, current, temperature, etc.), stabilization is considered to have occurred when measurements are maintained within the specified bandwidths or tolerances, for a minimum of four consecutive readings. For the following tests, repeat the associated tests if any parts are changed, or adjustments made to the generator set, its controls, or auxiliaries.

3.4.9.1 Continuous Engine Load Run Test

Test the engine-generator set and ancillary systems at service load to demonstrate durability; verify that heat of extended operation does not adversely affect or cause failure in any part of the system; and check all parts of the system. If the engine load run test is interrupted for any reason, repeat the entire test. Accomplish the engine load run test during daylight hours, with an average ambient temperature of 30 degrees C, during the month of August. After each change in load in the following test, measure the vibration at the end bearings (front and back of engine,

outboard end of generator) in the horizontal, vertical, and axial directions. Verify that the vibration is within the allowable range. Take data taken at 15 minute intervals and include the following:

Electrical: Output amperes, voltage, real and reactive power, power factor, frequency.

Pressure: Lube-oil.

Temperature: Coolant, Lube-oil, Exhaust, Ambient.

- a. Perform and record engine manufacturer's recommended prestarting checks and inspections. Include as a minimum checking of coolant fluid, fuel, and lube-oil levels.
- b. Start the engine, make and record engine manufacturer's after-starting checks and inspections during a reasonable warmup period.
- c. Operate the engine generator-set for 2 hours at 75 percent of Service Load.
- d. Increase load to 100 percent of Service Load and operate the engine generator-set for 4 hours.
- e. Decrease load to 100 percent of Service Load and operate the engine generator-set for 2 hours or until all temperatures have stabilized.
- f. Remove load from the engine-generator set.

3.4.9.2 Voltage and Frequency Droop Test

For the following steps, verify that the output voltage and frequency return to and stabilize within the specified bandwidth values following each load change. Record the generator output frequency and line-line and line-neutral voltages following each load change.

- a. With the generator operating at no load, adjust voltage and frequency to rated voltage and frequency.
- b. Increase load to 100 percent of Rated Output Capacity. Record the generator output frequency and line-line and line-neutral voltages.
- c. Calculate the percent droop for voltage and frequency with the following equations.

$$\text{Voltage droop percent} = \frac{\text{No-load volts} - \text{rated output capacity volts}}{\text{Rated output capacity volts}} \times 100$$

$$\text{Frequency droop percent} = \frac{\text{No load hertz} - \text{rated output capacity hertz}}{\text{Rated output capacity volts}} \times 100$$

- d. Repeat steps a. through c. two additional times without making any adjustments.

3.5 GROUNDING

NFPA 70 and IEEE C2, except that grounding systems must have a resistance

to solid earth ground not exceeding 5 ohms.

3.5.1 Grounding Electrodes

Provide driven ground rods as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION . Connect ground conductors to the upper end of ground rods by exothermic weld or compression connector. Provide compression connectors at equipment end of ground conductors.

3.5.2 Engine-Generator Set Grounding

Provide separate copper grounding conductors and connect them to the ground system as indicated. When work in addition to that indicated or specified is required to obtain the specified ground resistance, the provision of the contract covering "Changes" must apply.

3.5.3 Connections

Make joints in grounding conductors by exothermic weld or compression connector. Exothermic welds and compression connectors must be installed as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION paragraph regarding GROUNDING.

3.5.4 Grounding and Bonding Equipment

UL 467, except as indicated or specified otherwise.

3.6 START-UP ENGINEER

Provide the services of a qualified factory trained start-up engineer, regularly employed by the engine-generator set manufacturer. The start-up services must include conducting preliminary operations and functional acceptance tests. The start-up engineer must be present at the engine generator set installation-site, full-time, while preliminary operations and functional acceptance tests are being conducted.

3.7 PREREQUISITES FOR FUNCTIONAL ACCEPTANCE TESTING

Completion of the following requirements is mandatory prior to scheduling functional acceptance tests for the engine-generator set and auxiliary equipment.

3.7.1 Performance of Acceptance Checks and Tests

The acceptance checks and tests must be accomplished by the testing organization as described in Section 26 08 00 APPARATUS INSPECTION AND TESTING.

3.7.2 Generator Sets

Complete as specified in the paragraph ACCEPTANCE CHECKS AND TESTS.

3.7.3 Preliminary Operations

The start-up engineer must conduct manufacturer recommended start-up procedures and tests to verify that the engine-generator set and auxiliary equipment are ready for functional acceptance tests. Give the Contracting Officer 15 days' advance notice that preliminary operations will be conducted. After preliminary operation has been successfully conducted,

the start-up engineer will notify the Contracting Officer in writing stating the engine-generator set and auxiliary equipment are ready for functional acceptance tests.

3.7.4 Preliminary Assembled Operation and Maintenance Manuals

Preliminary assembled operation and maintenance manuals must have been submitted to and approved by the Contracting Officer. Manuals must be prepared as specified in the paragraph ASSEMBLED OPERATION AND MAINTENANCE MANUALS.

3.7.5 Functional Acceptance Test Procedure

Test procedure must be prepared by the start-up engineer specifically for the engine-generator set and auxiliary equipment. The test agenda must cover the requirements specified in the paragraph FUNCTIONAL ACCEPTANCE TESTS. The test procedure must indicate in detail how tests are to be conducted. A statement of the tests that are to be performed without indicating how the tests are to be performed is not acceptable. Indicate what work is planned on each workday and identify the calendar dates of the planned workdays. Specify what additional technical support personnel is needed such as factory representatives for major equipment. Specify on which testing workday each technical support personnel is needed. Data recording forms to be used to document test results are to be submitted with the proposed test procedure. A list of test equipment and instruments must also be included in the test procedure.

3.7.6 Test Equipment

Test equipment and instruments must be on hand prior to scheduling field tests or, subject to Contracting Officer approval, evidence must be provided to show that arrangements have been made to have the necessary equipment and instruments on-site prior to field testing.

3.8 FIELD QUALITY CONTROL

Give Contracting Officer 30 days' notice of dates and times scheduled for tests which require the presence of the Contracting Officer. The Contracting Officer will coordinate with the using activity and schedule a time that will eliminate or minimize interruptions and interference with the activity operations. The Contractor must be responsible for costs associated with conducting tests outside of normal working hours and with incorporating special arrangements and procedures, including temporary power conditions. The Contractor must provide labor, equipment, fuel, test load, and consumables required for the specified tests. The test load must be a cataloged product. Calibration of measuring devices and indicating devices must be certified. Refer to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, for requirements for a cataloged product. Perform the following field tests.

3.8.1 Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

3.8.1.1 Circuit Breakers - Low Voltage Insulated Case/Molded Case

a. Visual and Mechanical Inspection

- (1) Compare nameplate data with specifications and approved shop drawings.
- (2) Inspect circuit breaker for correct mounting.
- (3) Operate circuit breaker to ensure smooth operation.
- (4) Inspect case for cracks or other defects.
- (5) Verify tightness of accessible bolted connections and cable connections by calibrated torque-wrench method. Thermo-graphic survey is not required.
- (6) Inspect mechanism contacts and arc chutes in unsealed units.

b. Electrical Tests

- (1) Perform contact-resistance tests.
- (2) Perform insulation-resistance tests.
- (3) Adjust breaker(s) for final settings in accordance with engine-generator set manufacturer's requirements.

3.8.1.2 Current Transformers

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify correct connection.
- (4) Verify that adequate clearances exist between primary and secondary circuit.
- (5) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method. Thermo-graphic survey is not required.
- (6) Verify that all required grounding and shorting connections provide good contact.

b. Electrical Tests

- (1) Perform insulation-resistance tests.
- (2) Perform polarity tests.
- (3) Perform ratio-verification tests.

3.8.1.3 Metering and Instrumentation

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved

shop drawings.

- (2) Inspect physical and mechanical condition.
- (3) Verify tightness of electrical connections.

b. Electrical Tests

- (1) Determine accuracy of meters at 25, 50, 75, and 100 percent of full scale.
- (2) Calibrate watt-hour meters according to manufacturer's published data.
- (3) Verify all instrument multipliers.
- (4) Electrically confirm that current transformer secondary circuits are intact.

3.8.1.4 Battery Systems

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method. Thermo-graphic survey is not required.
- (4) Measure electrolyte specific gravity and temperature and visually check fill level.
- (5) Verify adequacy of battery support racks, mounting, anchorage, and clearances.

b. Electrical Tests

- (1) Set charger float and equalizing voltage levels.
- (2) Verify all charger functions and alarms.
- (3) Measure each cell voltage and total battery voltage with charger energized and in float mode of operation.
- (4) Perform a capacity load test.

3.8.1.5 Engine-Generator Set

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Inspect for correct anchorage and grounding.

b. Electrical and Mechanical Tests

- (1) Perform an insulation-resistance test on generator winding with respect to ground. Calculate polarization index.
- (2) Perform phase rotation test to determine compatibility with load requirements.

3.8.1.6 Grounding System

a. Visual and Mechanical Inspection

- (1) Inspect ground system for compliance with contract plans and specifications.

b. Electrical Tests

- (1) Perform ground-impedance measurements utilizing the fall-of-potential method defined in IEEE 81. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. Take measurements in normally dry weather, not less than 48 hours after rainfall. Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance.

3.8.2 Functional Acceptance Tests

The tests must be performed by the start-up engineer. Upon successful test completion, the start-up engineer must provide the Contracting Officer with a written test report within 15 calendar days showing the tests performed and the results of each test. The report must include the completed approved test data forms and certification from the start-up engineer that the test results fall within the manufacturer's recommended limits and meet the specified requirements performance. The report must be dated and signed by the start-up engineer, and submitted for approval by the Contracting Officer. The Contracting Officer will witness final acceptance tests. Testing must include, but not be limited to, the following:

- a. Verify proper functioning of each engine protective shutdown device and pre-shutdown alarm device. Testing of the devices must be accomplished by simulating device actuation and observing proper alarm and engine shutdown operation.
- b. Verify proper functioning of the engine over-speed trip device. Testing of the over-speed trip device must be accomplished by raising the speed of the engine-generator set until an over-speed trip is experienced.
- c. Verify proper functioning of the crank cycle/terminate relay. Testing of the relay must be accomplished by engaging the starter motor with the engine being prevented from running. Observe the complete crank/rest cycle as described in the paragraph STARTING SYSTEM.
- d. Verify proper functioning of the following automatic and manual operations. Testing must include, but not be limited to, the following:

(1) Manual starting.

(2) Emergency stop.

- e. Operate the engine-generator set at rated current (amperes) until the jacket water temperature stabilizes. Stabilization will be considered to have occurred when three consecutive temperature readings remain unchanged. Continue to operate the generator set for an additional 2 hours. Record instrument readings for terminal voltage, line current, frequency (Hz), engine speed rpm, lubricating oil pressure, jacket water temperature, and ambient temperature at 5 minute intervals for first 15 minutes and at 15 minute intervals thereafter.

3.9 DEMONSTRATION

Upon completion of the work and at a time approved by the Contracting Officer, the Contractor must provide instructions by a qualified instructor to the Government personnel in the proper operation and maintenance of the equipment. Government personnel must receive training comparable to the equipment manufacturer's factory training. The duration of instruction must be for not less than one 8 hour working day for instruction of operating personnel and not less than one 8 hour working day for instruction of maintenance personnel.

3.9.1 [Instructor's Qualification Resume](#)

Instructors must be regular employees of the engine-generator set manufacturer. The instruction personnel provided to satisfy the requirements above must be factory certified by the related equipment manufacturer to provide instruction services. Submit the name and qualification resume of instructor to the Contracting Officer for approval.

3.9.2 [Training Plan](#)

Submit training plan 30 calendar days prior to training sessions. Training plan must include scheduling, content, outline, and training material (handouts). Content must include, but not be limited to, the following:

3.9.2.1 Operating Personnel Training

This instruction includes operating the engine-generator set, auxiliary equipment including automatic transfer switches in all modes, and the use of all functions and features specified.

3.9.2.2 Maintenance Personnel Training

Training must include mechanical, hydraulic, electrical, and electronic instructions for the engine-generator set and auxiliary equipment including automatic transfer switches.

- a. Mechanical Training: Must include at least the following:

- (1) A review of mechanical diagrams and drawings.
- (2) Component location and functions.
- (3) Troubleshooting procedures and techniques.

- (4) Repair procedures.
 - (5) Assembly/disassembly procedures.
 - (6) Adjustments (how, when, and where).
 - (7) Preventive maintenance procedures.
 - (8) Review of flow diagram.
 - (9) Valve locations and function.
 - (10) Valve and hydraulic equipment adjustment and maintenance procedures.
 - (11) Hydraulic system maintenance and servicing.
 - (12) Lubrication points, type, and recommended procedures and frequency.
- b. Electrical and Electronic Maintenance Training: Must include at least the following:
- (1) A review of electrical and electronic systems including wiring diagrams and drawings.
 - (2) Troubleshooting procedures for the machine and control systems.
 - (3) Electrical and electronic equipment servicing and care.
 - (4) Use of diagnostics to locate the causes of malfunction.
 - (5) Procedures for adjustments (locating components, adjustments to be made, values to be measured, and equipment required for making adjustments).
 - (6) Maintenance and troubleshooting procedures for microprocessor or minicomputer where applicable.
 - (7) Circuit board repair procedures where applicable (with schematics provided).
 - (8) Use of diagnostic tapes.
 - (9) Recommended maintenance servicing and repair for motors, switches, relays, solenoids, and other auxiliary equipment and devices.

3.10 **ONSITE TRAINING**

Conduct a training course for the operating staff as designated by the Contracting Officer. The training period must consist of a total 8 hours of normal working time and must start after the system is functionally completed but prior to final acceptance.

- a. Submit a letter giving the date proposed for conducting the onsite training course, the agenda of instruction, a description of the digital video recording to be provided. The course instructions must cover pertinent points involved in operating, starting, stopping, servicing the equipment, as well as major elements of the operation

and maintenance manuals. Additionally, the course instructions must demonstrate routine [maintenance procedures](#) as described in the [operation and maintenance manuals](#).

- b. Submit a digital video recording of the entire training session or the manufacturers operating and maintenance training course if available.
- c. One full size reproducible Mylar ach drawing must accompany the booklets. Mylars must be rolled and placed in a heavy cardboard tube with threaded caps on each end. The manual must include step-by-step procedures for system startup, operation, and shutdown; drawings, diagrams, and single-line schematics to illustrate and define the electrical, mechanical, and hydraulic systems together with their controls, alarms, and safety systems; the manufacturer's name, model number, and a description of equipment in the system. The instructions must include procedures for interface and interaction with related systems to include manual transfer switch operation. Each booklet must include a CD containing an ASCII file of the procedures.
- d. Provide approved operation and maintenance manuals for the training course. Post approved instructions prior to the beginning date of the training course. Coordinate the training course schedule with the using service's work schedule, and submit for approval 14 days prior to beginning date of proposed beginning date of training.

3.11 INSTALLATION

Installation must conform to the applicable requirements of [IEEE C2](#), [NFPA 30](#), [NFPA 37](#), and [NFPA 70](#).

3.12 FINAL TESTING AND INSPECTION

- a. Start the engine, record the starting time, make and record all engine manufacturer's after-starting checks and inspections during a reasonable warm-up period.
- b. Increase the load in steps no greater than the Maximum Step Load Increase to 100 percent of Service Load, and operate the engine-generator set for at least 30 minutes. Measure the vibration at the end bearings (front and back of engine, outboard end of generator) in the horizontal, vertical, and axial directions. Verify that the vibration is within the same range as previous measurements and is within the required range.
- c. Remove load and shut down the engine-generator set after the recommended cool down period.
- d. Remove the lube oil filter and have the oil and filter examined by the engine manufacturer for excessive metal, abrasive foreign particles, etc. Verify any corrective action for effectiveness by running the engine for 8 hours at Service Load, then re-examine the oil and filter.
- e. Remove the fuel filter and examine the filter for trash, abrasive foreign particles, etc.
- f. Visually inspect and check engine and generator mounting bolts for tightness and visible damage.

- g. Replace air, oil, and fuel filters with new filters.

3.13 MANUFACTURER'S FIELD SERVICE

The engine generator-set manufacturer must furnish a qualified representative to supervise the installation of the engine generator-set, assist in the performance of the onsite tests, and instruct personnel as to the operational and maintenance features of the equipment.

3.14 ACCEPTANCE

Submit drawings which accurately depict the as-built configuration of the installation, upon acceptance of the engine-generator set installation. Revise layout drawings to reflect the as-built conditions and submit them with the as-built drawings. Final acceptance of the engine-generator set will not be given until the Contractor has successfully completed all tests and all defects in installation material or operation have been corrected.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 41 00

LIGHTNING PROTECTION SYSTEM

08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
 - 1.2.1 Verification of Dimensions
 - 1.2.2 System Requirements
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Component UL Listed and Labeled
 - 1.4.2 Lightning Protection and Grounding System Test Plan
 - 1.4.3 Lightning Protection System Inspection Certificate of Qualifications
- 1.5 SITE CONDITIONS

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Main and Bonding Conductors
 - 2.1.2 Conductors
- 2.2 COMPONENTS
 - 2.2.1 Air Terminals on Integral Systems
 - 2.2.2 Ground Rods
 - 2.2.3 Connections and Terminations
 - 2.2.4 Connector Fittings

PART 3 EXECUTION

- 3.1 LIGHTNING PROTECTION SYSTEMS
 - 3.1.1 Integral Lightning Protection System
 - 3.1.1.1 Roof-Mounted Components
 - 3.1.1.2 Air Terminals
 - 3.1.1.3 Roof Conductors
 - 3.1.2 Down Conductors
 - 3.1.3 Ground Connections
 - 3.1.4 Installation of Grounding Electrodes (Ground Rods)
- 3.2 FIELD QUALITY CONTROL
 - 3.2.1 Lightning Protection Systems Testing
 - 3.2.2 Grounding Systems Testing

-- End of Section Table of Contents --

SECTION 26 41 00

LIGHTNING PROTECTION SYSTEM
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 81 (2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

NFPA 780 (2023) Standard for the Installation of Lightning Protection Systems

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-575-01 (2012; with Change 1, 2021) Lightning and Static Electricity Protection Systems

UNDERWRITERS LABORATORIES (UL)

UL 96 (2016; May 2023) UL Standard for Safety Lightning Protection Components

UL 96A (2016; Reprint Oct 2022) UL Standard for Safety Installation Requirements for Lightning Protection Systems

UL Electrical Construction (2012) Electrical Construction Equipment Directory

1.2 RELATED REQUIREMENTS

1.2.1 Verification of Dimensions

Confirm all details of work, verify all dimensions in field, and advise Contracting Officer of any discrepancy before performing work. Obtain prior approval of Contracting Officer after engineering approval and before making any departures from the design.

1.2.2 System Requirements

Provide a system furnished under this specification consisting of the latest products of a manufacturer regularly engaged in production of lightning protection system components. Products must be UL listed for

use on lightning protection systems unless this rating does not exist for items in question.

Comply with NFPA 70, NFPA 780, UL 96, and UFC 3-575-01.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Lightning Protection and Grounding System Test Plan; G

Grounding Systems Testing; G

SD-07 Certificates

Component UL Listed and Labeled; G

Lightning Protection System Inspection Certificate; G

Roof Manufacturer's Warranty; G

1.4 QUALITY ASSURANCE

In each standard referred to herein, consider the advisory provisions to be mandatory, as though the word "shall" or "must" has been substituted for "should," wherever it appears. Interpret references that require LPS expertise in these standards to mean Base Civil Engineer (BCE) or BCE-designated representative with LPS training certification.

1.4.1 Component UL Listed and Labeled

Submit proof of compliance that components are UL Listed and Labeled for use on lightning protection systems. Listing alone in UL Electrical Construction, which is the UL Electrical Construction Directory, is not acceptable evidence. In lieu of Listed and Labeled, submit written certificate from an approved, nationally-recognized testing organization equipped to perform such services, stating that items have been tested and conform to requirements and testing methods of Underwriters Laboratories (UL).

1.4.2 Lightning Protection and Grounding System Test Plan

Provide lightning protection system and grounding system test plans in compliance with NFPA 780. Identify (by number or letter, at the preference of the base - 1.2.3. or a.b.c.) the system test points/locations along with a listing or description of the item to be tested and the type of test to be conducted. As a minimum, include a sketch of the facility and surrounding lightning protection system as part of the specific test plan for each structure. Include the requirements as "Testing of Integral Lightning Protection System" in the test plan.

1.4.3 Lightning Protection System Inspection Certificate of Qualifications

Provide an approved third-party inspector who meets training certification

requirements for final acceptance of the LPS. Inspection and certification of the LPS must be completed and certified in writing, before occupancy of the building. Compliance requirements are in **UFC 3-575-01**. Signed Certificate of Qualifications must be placed in the LPS records.

Note that the Contracting Officer may not accept the LPS system without a recommendation from a qualified person identified in the project documents.

1.5 SITE CONDITIONS

Confirm all details of work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before starting any work, if known at that time, or when it becomes known while performing work. Obtain prior approval of the BCE or BCE-designated representative with LPS training certification before design changes are made.

PART 2 PRODUCTS

2.1 MATERIALS

See **NFPA 70**, **NFPA 780**, **UFC 3-575-01**. Do not use a combination of materials that forms an electrolytic couple of such nature that corrosion is accelerated in the presence of moisture unless moisture is permanently excluded from the junction of such metals. Where unusual conditions exist which would cause corrosion of conductors, provide conductors with protective coatings, such as tin or lead, or oversize conductors. Where a mechanical hazard is involved, increase conductor size to compensate for the hazard or protect conductors. When metallic conduit or tubing is provided, electrically bond conductor to conduit or tubing at the upper and lower ends by clamp type connectors or welds (including exothermic). All lightning protection components, such as bonding plates, air terminals, air terminal supports and braces, chimney bands, clips, connector fittings, and fasteners are to comply with the requirements of **UL 96** classes, as applicable.

2.1.1 Main and Bonding Conductors

Main and Bonding Conductors should be **NFPA 780** and **UL 96** Class I or Class II materials, as applicable.

2.1.2 Conductors

Provide copper or aluminum conductors, as applicable. See **NFPA 780**, **UFC 3-575-01**.

2.2 COMPONENTS

2.2.1 Air Terminals on Integral Systems

Provide solid air terminals. Tubular air terminals are not permitted. Support air terminals more than **600 mm** in length by suitable brace, supported at not less than one-half the height of the terminal.

2.2.2 Ground Rods

Provide ground rods conforming to **NFPA 780**. Provide ground rods that are not less than **19 mm** in diameter and **3000 mm** in length. Do not mix ground rods of copper-clad steel and solid copper on the job.

2.2.3 Connections and Terminations

Provide connectors for splicing conductors that conform to [UL 96](#), class as applicable. Conductor connections can be made by compression clamps or welds (including exothermic). Provide style and size connectors required for the installation.

2.2.4 Connector Fittings

Provide connector fittings for "end-to-end", "Tee", or "Y" splices that conform to [NFPA 780](#) and [UL 96](#).

PART 3 EXECUTION

3.1 LIGHTNING PROTECTION SYSTEMS

Provide a lightning protection system that meets the requirements of [NFPA 780](#) and [UFC 3-575-01](#).

3.1.1 Integral Lightning Protection System

The integral type lightning protection system consists of air terminals, roof conductors, down conductors, ground connections, grounding electrodes and ground ring electrode conductor for the purpose of carrying lightning current from a direct strike to ground in a manner that will protect assets and personnel. Expose all conductors on the structures except where exterior down conductors are required to be in protective sleeves for prevention of mechanical damage ([1830 mm](#) above grade level). Do not run down conductors inside columns or other methods which will prevent visual access for required inspections. Visual access is required by [NFPA 780](#) so that the annual visual inspection may be performed on all LPS components.

Make interconnections within side-flash distances between down conductors and metallic equipment mounted on the exterior or interior of a facility, at or below the level of the grounded metallic parts. Calculate side-flash distances in accordance with [NFPA 780](#). Be aware of side flashes that may occur through exterior walls.

3.1.1.1 Roof-Mounted Components

Coordinate with the roofing manufacturer and provide certification that the [roof manufacturer's warranty](#) is not violated by the installation methods for air terminals and roof conductors. Adhesive fasteners may be used on the roof if the installation is observed by base-qualified personnel to be compliant with manufacturer's instructions. Installation must be observed until the observer is satisfied that the proper methods for preparing the surface are being performed. These fasteners will be added and considered a test point in Base records and must be part of the annual inspection. Observer must date and sign the test record.

No connection or physical attachment is allowed to any coping system on the roof as this violates the integrity and warranty of the coping system.

3.1.1.2 Air Terminals

Use of adhesive fasteners, when installing air terminals on "rubber" (EPDM) type roofs, should be in accordance with adhesive and roof

manufactures' recommendations. See paragraph ROOF-MOUNTED COMPONENTS.

3.1.1.3 Roof Conductors

Roof conductors should comply with NFPA 780.

3.1.2 Down Conductors

A minimum of two paths to ground shall be provided on any system. This may be by two down conductors or with a down conductor in combination with other metallic connections to ground. Protect exposed down conductors from physical damage from ground level up to 1830 mm. If this protection is a metal conduit, both ends must be bonded to the down conductor passing through it. For Schedule 80 conduit providing this protection, no bonding at both ends is necessary. Schedule 80 conduit or metallic conduit may be painted to match the surrounding surface. NO CONNECTIONS OR FASTENERS OF ANY LIGHTNING PROTECTION SYSTEM MAY BE PAINTED because this affects the ease with which lightning current can pass to ground.

3.1.3 Ground Connections

Attach each down conductor and ground ring electrode to a ground rod below grade by exothermic weld for all buried connections and exothermic weld or compression connectors for connections inside test wells. Terminate all down conductors to a grounding electrode inside a test well. Test connections and record resistances and continuity readings prior to covering.

Accessible connections above ground level and in test wells can be grounded with mechanical clamping, meeting installation requirements in NFPA 780.

3.1.4 Installation of Grounding Electrodes (Ground Rods)

Extend driven ground rods vertically into the existing undisturbed earth for a distance of not-less-than 3000 mm if exothermic welds are used for bonds. Inside test wells, the rod may be driven to a point above the base of the test well, that will provide working/testing access to the mechanically-fastened bond. Set ground rods not less than 915 mm nor more than 1830 mm (see NFPA 780) from the structural foundation, and at least 915 mm beyond the drip line for the facility. After the completed installation, measure the total resistance to ground using the fall-of-potential method described in IEEE 81. Maximum allowed resistance of a single driven ground rod is 25 ohms (NFPA 70). If resistance-to-ground of a single ground rod for the lightning protection system exceeds 25 ohms and another ground rod is driven in accordance with NFPA 780, 25 ohms does not apply to this test point. See NFPA 70 exception to Article 250.53.

3.2 FIELD QUALITY CONTROL

3.2.1 Lightning Protection Systems Testing

A 100 percent test point test and inspection is required by all services. Identify test points for lightning protection system connections and bonds. Provide a sketch (NTS) with identified test points as part of the contract acceptance. Number points in a manner consistent with the installation (Base or Post) nomenclature for existing lightning protection systems; for example, use letters or numbers to track the test results.

Test each lightning protection system connection to ensure continuity across each connection or bond is 1 ohm or less, and record the value in the testing report.

- a. Comply with [UL 96A](#). A third-party inspector (not the designer and not the installer) must be present for the 100 percent inspection and data documentation by the contractor. This is to take place prior to project acceptance. Note that in many cases Air Force personnel have been trained to inspect and accept projects as the third-party inspector. Verify this before contract award.

3.2.2 [Grounding Systems Testing](#)

A 100 percent test point resistance test and inspection is required. Identify all test points for grounding system connections and bonds. Provide a sketch (NTS) with identified test points as part of the contract acceptance. Number points in a manner consistent with the installation (Base or Post) nomenclature for existing grounding systems; for example, use letters or numbers to track the test results. Test each grounding system connection or bond to ensure resistance-to-ground is 25 ohms or less or as excepted by [NFPA 70](#) or [NFPA 780](#). Record the resistance measurement in the testing report. Test the ground rod for resistance to ground before making connections to the rod. Tie the grounding system together and test for resistance to ground. Make resistance measurements in dry weather, and not earlier than 48 hours after rainfall. Include in the written report: locations of test points, measured values for continuity and ground resistances, and soil conditions at the time that measurements were made. Submit results of each test to the Contracting Officer.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 42 17

IMPRESSED CURRENT CATHODIC PROTECTION (ICCP) SYSTEM

05/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Cathodic Protection
 - 1.2.2 Corrosion
 - 1.2.3 Alternating Current (AC) Corrosion
 - 1.2.4 AC Interference
 - 1.2.5 Uniform Attack
 - 1.2.6 Galvanic or Two-Metal Corrosion
 - 1.2.7 Crevice Corrosion
 - 1.2.8 Pitting Corrosion
 - 1.2.9 Intergranular Corrosion
 - 1.2.10 Selective Leaching
 - 1.2.11 Erosion Corrosion
 - 1.2.12 Stress-Corrosion Cracking
 - 1.2.13 Exothermic Welding
 - 1.2.14 Error-Free
- 1.3 ADMINISTRATIVE REQUIREMENTS
- 1.4 SUBMITTALS
 - 1.4.1 Material and Equipment Manufacturer Data
- 1.5 MAINTENANCE MATERIAL SUBMITTALS
 - 1.5.1 Spare Parts
 - 1.5.2 Extra Materials
- 1.6 QUALITY CONTROL
 - 1.6.1 Regulatory Requirements
 - 1.6.2 Qualifications
 - 1.6.3 Services of Corrosion Expert
- 1.7 DELIVERY, STORAGE AND HANDLING
- 1.8 PROJECT/SITE CONDITIONS
 - 1.8.1 Environmental Requirements
 - 1.8.2 Existing Conditions
- 1.9 WARRANTY

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Corrosion Control System Description
 - 2.1.2 Design Requirements
 - 2.1.2.1 Electrical Isolators
 - 2.1.2.2 Anode and Bond Wires
 - 2.1.2.3 Surge Protection
 - 2.1.2.4 Non-metallic Pipe System
 - 2.1.2.5 Coatings
 - 2.1.2.6 Drawings
 - 2.1.2.7 Summary of Services Required

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.1.2.8 Contractor's Modifications
- 2.1.2.9 Tests of Components
- 2.1.2.10 Electrical Potential Measurements
- 2.1.2.11 Typical Metallic Components on Non-metallic Systems
 - 2.1.2.11.1 Pipe Under Concrete Slab
- 2.1.2.12 Metallic Component Coating
- 2.1.2.13 Location of Test Stations
- 2.1.2.14 Electrical Isolation of Structures
 - 2.1.2.14.1 Isolation Joint Testing
 - 2.1.2.14.2 Underground Structure Coating
 - 2.1.2.14.3 Field Joints
 - 2.1.2.14.4 Inspection of Pipe Coatings
 - 2.1.2.14.5 Protective Coating for Aboveground Piping System
 - 2.1.2.14.6 Ferrous Surfaces
- 2.1.3 Performance Requirements
 - 2.1.3.1 Criteria of Cathodic Protection
 - 2.1.3.1.1 Steel
- 2.2 EQUIPMENT
 - 2.2.1 Remote Monitoring
 - 2.2.2 Rectifiers
 - 2.2.2.1 Air Cooled Enclosure
 - 2.2.2.2 Cabinet Construction
 - 2.2.2.3 Cabinet Paint System
 - 2.2.2.4 Electrical Ratings
 - 2.2.2.5 Rectifier Stacks
 - 2.2.2.6 Transformer
 - 2.2.2.7 Circuit Breaker
 - 2.2.2.8 Fuses
 - 2.2.2.9 Overload and Short Circuit Protection
 - 2.2.2.10 Surge Protection
 - 2.2.2.11 Wiring
 - 2.2.2.12 Shunt Resistors
 - 2.2.2.13 DC Output Control
 - 2.2.2.14 Output Voltage and Current Metering
 - 2.2.2.15 Efficiency
 - 2.2.2.16 Filter Chokes and Capacitors
 - 2.2.2.17 Grounding Provisions
 - 2.2.2.18 Wiring Diagram
 - 2.2.2.19 Optional Rectifier Special Required Features
 - 2.2.3 Polarization Cell Replacement (PCR) and (PCRH) for Hazardous Locations
 - 2.2.4 Solid State Decoupler (SSD)
- 2.3 COMPONENTS
 - 2.3.1 Test Stations
 - 2.3.1.1 Flush Mounted
 - 2.3.1.2 Post Top Mounted
 - 2.3.2 Shunts for Test Stations and Junction Boxes
 - 2.3.3 Junction Box Enclosures
 - 2.3.3.1 Nameplates
 - 2.3.4 Terminal Boards
 - 2.3.5 Anode Junction Boxes
 - 2.3.5.1 Enclosure
 - 2.3.5.2 Terminal Boards
- 2.4 MATERIALS
 - 2.4.1 Impressed Current Anodes
 - 2.4.1.1 Dimensions and Weights
 - 2.4.1.2 High-Silicon Chromium Bearing Cast Iron
 - 2.4.1.2.1 Chemical Composition (Nominal)
 - 2.4.2 Wire and Cable

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.4.2.1 Anode Lead Wire
- 2.4.2.2 Deep Anode Lead Wire
- 2.4.2.3 Anode Header Cable
- 2.4.2.4 Structure (Negative) Cable
- 2.4.2.5 Test Wires
- 2.4.2.6 Joint and Continuity Bond Cables
- 2.4.2.7 AC Power Supply Wiring
- 2.4.2.8 Polyethylene Insulation
 - 2.4.2.8.1 HMWPE
 - 2.4.2.8.2 High Density Polyethylene (HDPE)
- 2.4.2.9 Rectifier DC Negative (Structure) Cable(s)
- 2.4.3 Cable and Wire Identification Tags
- 2.4.4 Anode Connection
 - 2.4.4.1 End-Connected Anode
- 2.4.5 Coke Breeze
 - 2.4.5.1 Calcined Petroleum Coke Breeze (Dry)
 - 2.4.5.1.1 Electrical Resistivity
 - 2.4.5.1.2 General Backfill Specifications
 - 2.4.5.2 Metallurgical Coke Breeze (Processed)
 - 2.4.5.2.1 Electrical Resistivity (Nominal)
 - 2.4.5.2.2 General Backfill Specifications
- 2.4.6 Permanent Reference Electrodes
- 2.5 ACCESSORIES
 - 2.5.1 Wire Connectors
 - 2.5.2 Electrical Tape
 - 2.5.3 Cable Marker Tape
 - 2.5.4 Insulating Tape
 - 2.5.5 Underground Splices
 - 2.5.5.1 Cast-Type Splice
 - 2.5.6 Buried Cable Warning and Identification Tape
 - 2.5.7 Conduit
 - 2.5.8 Protective Coating
 - 2.5.8.1 Pipeline Metallic Components
 - 2.5.8.2 Field Joints
 - 2.5.9 Isolation Flange Sets/Kits
 - 2.5.10 Electrical Isolation of Structures
 - 2.5.10.1 Gaskets
 - 2.5.10.2 Isolation Washers and Sleeves
 - 2.5.11 Electrically Isolating Pipe Joints
 - 2.5.12 Stray Current Measurements
 - 2.5.13 Cast-In-Place Concrete
 - 2.5.14 Steel Flanges and Bolting
 - 2.5.14.1 Steel Flanges
 - 2.5.14.2 Bolting

PART 3 EXECUTION

- 3.1 SAFETY PRECAUTIONS AND HAZARDOUS LOCATIONS
- 3.2 INSTALLATION
 - 3.2.1 Anode Bed Installation
 - 3.2.2 Deep Anode Lead Wire
 - 3.2.2.1 Anode Centering
 - 3.2.2.2 Casing
 - 3.2.3 Backfill
 - 3.2.4 Cable Marker Tape
- 3.3 MISCELLANEOUS INSTALLATION
 - 3.3.1 Rectifier Installation
 - 3.3.2 Test Stations
 - 3.3.3 Permanent Reference Electrodes

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.3.3.1 Permanent Reference Electrode Verification
- 3.3.4 Bonding Boxes
- 3.3.5 Joint Bonds
- 3.3.6 Wire Connections
 - 3.3.6.1 Wire-To-Structure Connections
 - 3.3.6.2 Cable Protection
 - 3.3.6.3 Wire Splicing
- 3.3.7 Steel Surfaces
- 3.3.8 Electrical Continuity
- 3.3.9 Electrical Isolation of Structures
 - 3.3.9.1 Locations of Electrical Isolation
- 3.3.10 Ferrous Valves
- 3.3.11 Brass or Bronze Valves
- 3.4 INSTALLATION DETAILS
 - 3.4.1 Anode Deep Well Installation
 - 3.4.2 Underground Pipe Joint Bonds
 - 3.4.3 Anode Junction Boxes
 - 3.4.4 Bonding Cables
 - 3.4.5 Test Stations and Permanent Reference Electrodes
 - 3.4.6 Permanent Reference Electrode Calibration and Installation
 - 3.4.6.1 Field Drawings
- 3.5 ELECTRICAL ISOLATION OF STRUCTURES
 - 3.5.1 Isolation Fittings
 - 3.5.2 Joint Bonds
- 3.6 FIELD QUALITY CONTROL
 - 3.6.1 Tests and Measurements
 - 3.6.1.1 Native Potentials
 - 3.6.1.2 Protected Potentials
 - 3.6.1.3 Isolation Testing
 - 3.6.1.4 Isolation Tester
 - 3.6.1.5 Anode Output
 - 3.6.1.6 Reference Electrode Potential Measurements
 - 3.6.1.7 Holiday Test
 - 3.6.1.8 Rectifier Testing
 - 3.6.1.9 Stray Current Measurements
 - 3.6.1.10 Induced AC Testing
 - 3.6.1.11 Interference Tests
 - 3.6.1.12 Initial Cathodic Protection System Field Testing
 - 3.6.1.13 Government Field Testing
 - 3.6.1.14 One-Year-Warranty-Period-Testing
 - 3.6.1.15 Final Acceptance Field Testing
- 3.7 CLOSEOUT ACTIVITIES
 - 3.7.1 Reconditioning of Surfaces
 - 3.7.1.1 Concrete
 - 3.7.1.2 Restoration of Sod
 - 3.7.1.3 Restoration of Pavement
 - 3.7.1.4 Cleanup
 - 3.7.2 Training
 - 3.7.2.1 Instruction to Government Personnel

-- End of Section Table of Contents --

SECTION 26 42 17

IMPRESSED CURRENT CATHODIC PROTECTION (ICCP) SYSTEM
05/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1	(2003; R 2018) Unified Inch Screw Threads (UN and UNR Thread Form)
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.21	(2021) Nonmetallic Flat Gaskets for Pipe Flanges
ASME B18.2.1	(2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)
ASME B18.2.2	(2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

ASTM INTERNATIONAL (ASTM)

ASTM A194/A194M	(2023) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both
ASTM A307	(2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A518/A518M	(1999; R 2022) Standard Specification for Corrosion-Resistant High-Silicon Iron Castings
ASTM B3	(2013) Standard Specification for Soft or Annealed Copper Wire
ASTM B8	(2011; R 2017) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft
ASTM C94/C94M	(2022a) Standard Specification for Ready-Mixed Concrete
ASTM D709	(2017) Standard Specification for

Laminated Thermosetting Materials

- ASTM D1248 (2016) Standard Specification for Polyethylene Plastics Extrusion Materials for Wire and Cable
- ASTM D2028/D2028M (2015) Cutback Asphalt (Rapid-Curing Type)
- ASTM D3381/D3381M (2018) Standard Specification for Viscosity-Graded Asphalt Binder for Use in Pavement Construction

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

- IEEE C2 (2023) National Electrical Safety Code

NACE INTERNATIONAL (NACE)

- NACE SP0106 (2018) Control of Internal Corrosion in Steel Pipelines and Piping Systems
- NACE SP0176 (2007) Corrosion Control of Submerged Areas of Permanently Installed Steel Offshore Structures Associated with Petroleum Production
- NACE SP0177 (2019) Mitigation of Alternating Current and Lightning Effects on Metallic Structures and Corrosion Control Systems
- NACE SP0186 (1986; R 2007) Application of Cathodic Protection for External Surfaces of Steel Well Casings
- NACE SP0188 (1999; R 2006) Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates
- NACE SP0193 (2016) Application of Cathodic Protection to Control External Corrosion of Carbon Steel On-Grade Storage Tank Bottoms
- NACE SP0196 (2020) Galvanic Anode Cathodic Protection of Internal Submerged Surfaces of Steel Water Storage Tanks
- NACE SP0200 (2014) Standard Practice Steel-Cased Pipeline Practices
- NACE SP0285 (2021) External Corrosion Control of Underground Storage Tank Systems by Cathodic Protection
- NACE SP0286 (1997; R 2007) Standard Practice Electrical Isolation of Cathodically Protected Pipelines
- NACE SP0388 (2018) Impressed Current Cathodic Protection of Internal Submerged Surfaces

of Carbon Steel Water Storage Tanks - Item
No. 21040

NACE SP21424

(2018) Alternating Current Corrosion on
Cathodically Protected Pipelines: Risk
Assessment, Mitigation and Monitoring

NACE TPC 11

(2008) A Guide to the Organization of
Underground Corrosion-Control Coordinating
Committees

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C119.1

(2016) Electric Connectors - Sealed
Insulated Underground Connector Systems
Rated 600 Volts

NEMA ICS 6

(1993; R 2016) Industrial Control and
Systems: Enclosures

NEMA RN 1

(2005; R 2013) Polyvinyl-Chloride (PVC)
Externally Coated Galvanized Rigid Steel
Conduit and Intermediate Metal Conduit

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70

(2023) National Electrical Code

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-I-1361

(1985; Rev C; Notice 1 1991; Notice 2
2021) Instrument Auxiliaries, Electrical
Measuring: Shunts, Resistors and
Transformers

UNDERWRITERS LABORATORIES (UL)

UL 6

(2022) UL Standard for Safety Electrical
Rigid Metal Conduit-Steel

UL 83

(2017; Reprint Mar 2020) UL Standard for
Safety Thermoplastic-Insulated Wires and
Cables

UL 467

(2022) UL Standard for Safety Grounding
and Bonding Equipment

UL 486A-486B

(2018; Reprint May 2021) UL Standard for
Safety Wire Connectors

UL 489

(2016; Rev 2019) UL Standard for Safety
Molded-Case Circuit Breakers, Molded-Case
Switches and Circuit-Breaker Enclosures

UL 506

(2017; Reprint Jan 2022) UL Standard for
Safety Specialty Transformers

UL 510

(2020; Dec 2022) UL Standard for Safety
Polyvinyl Chloride, Polyethylene and

Rubber Insulating Tape

UL 514A

(2013; Reprint Jun 2022) UL Standard for
Safety Metallic Outlet Boxes

UL 514B

(2012; Reprint May 2020) Conduit, Tubing
and Cable Fittings

1.2 DEFINITIONS

It is convenient to classify corrosion by the forms in which it manifests itself, the basis for this classification being the appearance of the corroded metal. Each form can be identified by visual observation, although, in some cases, magnification is required. Valuable information for the solution of a corrosion problem can often be obtained through careful observation of the corroded test specimens or failed equipment. Examination before cleaning is particularly desirable. Cathodic Protection is a method used to control corrosion.

1.2.1 Cathodic Protection

Cathodic Protection (CP) is an electrochemical (half electrical and half chemical) method used to control corrosion of buried or submerged metallic structures. It prevents corrosion by making the protected structure a cathode by installing a more anodic metal (sacrificial or galvanic) anode or a metallic (Impressed Current) anode connected to a Direct Current (DC) power source. When the proper amount of current is applied to the structure, it becomes a cathode. Since all corrosion occurs at the anode, the structure no longer corrodes. The electrons move in the metallic path (electrical). Reduction (chemical) reactions occur at the surface of the cathode resulting in a hydrogen coating and more alkaline environment. Oxidation (chemical) reactions occur at the surface of the anode resulting in corrosion and a more acidic environment. After a CP system is installed and adjusted to provide adequate protection, the hydrogen coats the defects in the coating and polarizes in the negative direction (to a copper/copper sulfate reference electrode) over time the current and potentials remain relatively stable; changes in currents or potentials indicate a problem. An error-free measurement of negative 850 millivolts DC or more negative to the copper/copper-sulfate reference electrode proves the structure is a cathode and corrosion has been mitigated.

1.2.2 Corrosion

It is convenient to classify corrosion by the forms in which it manifests itself, the basis for this classification being the appearance of the corroded metal. Each form can be identified by visual observation, although, in some cases, magnification is required. Valuable information for the solution of a corrosion problem can often be obtained through careful observation of the corroded test specimens or failed equipment. Examination before cleaning is particularly desirable. Some of the eight forms of corrosion are unique, but all of them are more or less interrelated.

The eight forms of corrosion are: (1) Uniform Attack, (2) Galvanic or Two-Metal Corrosion, (3) Crevice Corrosion, (4) Pitting Corrosion, (5) Intergranular Corrosion, (6) Selective Leaching, (7) Erosion Corrosion, and (8) Stress Corrosion Cracking. This listing is arbitrary but covers practically all corrosion failures and problems. The forms are not listed in any particular order of importance. Below, the eight forms of

corrosion are discussed in terms of their characteristics, mechanisms, and preventive measures. Hydrogen damage, although not a form of corrosion, often occurs indirectly as a result of corrosive attack and is, therefore, included in this discussion.

1.2.3 Alternating Current (AC) Corrosion

AC corrosion occurs when there is a source of AC current, typically from a high-voltage overhead AC (OHAC) power-line, when there is a low soil resistivity (typically less than 5,000 ohm-cm) and there is a very small coating holiday. The AC corrosion pits typically have a tubercle of corrosion product at the pit. AC interference study modeling software can determine the mitigation solution to solve this problem. Typically, AC Corrosion mitigation is done in conjunction with high AC potentials and fault current mitigation.

1.2.4 AC Interference

AC interference occurs when a pipeline parallels a high-voltage overhead AC (OHAC) power-line. An interference study is required when this situation occurs as AC interference can cause high AC potentials along the pipeline (safety), can cause a fault condition between the pipeline and power-line and could cause AC corrosion to occur. The pipeline coating when exposed can have blisters/bubbles caused by the excessive AC. The interference study will use modeling software to determine what combination of interference may be occurring (if any) and provide the mitigation solution to solve the problem.

1.2.5 Uniform Attack

Uniform attack is the most common form of corrosion. It is normally characterized by a chemical or electrochemical reaction that proceeds uniformly over the entire exposed surface or over a large area. The metal becomes thinner and eventually fails. For example, a piece of steel or zinc immersed in dilute sulfuric acid normally dissolves at a uniform rate over its entire surface. A sheet iron roof shows essentially the same degree of rusting over its entire outside surface.

Uniform attack, or general overall corrosion, represents the greatest destruction of metal on a tonnage basis. This form of corrosion, however, is not of great concern from a technical standpoint, because the life of equipment can be accurately estimated on the basis of comparatively simple tests. Merely immersing specimens in the fluid involved is often sufficient. Uniform attack can be prevented or reduced by (1) materials, such as coatings, that reduce contact between metal and electrolytes, (2) inhibitors, or (3) cathodic protection.

1.2.6 Galvanic or Two-Metal Corrosion

A potential difference usually exists between two dissimilar-metals when they are immersed in a corrosive or conductive solution. If these metals are placed in contact (or otherwise electrically connected), this potential difference produces electron flow between them. Corrosion of the less corrosion-resistant metal is usually increased, and attack of the more resistant material is decreased, compared to the behavior of these metals when they are not in contact. The less resistant metal becomes anodic and the more resistant metal becomes cathodic. Usually the cathode or cathodic metal corrodes very little or not at all in this type of couple. Because of the electric currents and dissimilar-metals involved,

this form of corrosion is called galvanic, bi-metallic or two-metal, corrosion. Galvanic corrosion is restricted to electrochemical corrosion caused by dissimilar-metal effects. It is electrochemical corrosion, but this document must restrict the term galvanic to dissimilar-metal effects for purposes of clarity.

1.2.7 Crevice Corrosion

Intense localized corrosion frequently occurs within crevices and other shielded areas on metal surfaces exposed to corrosives. This type of attack is usually associated with small volumes of stagnant solution caused by holes, gasket surfaces, lap joints, surface deposits, and crevices under bolt and rivet heads. As a result, this form of corrosion is called crevice corrosion or, sometimes, deposit or gasket corrosion.

1.2.8 Pitting Corrosion

Pitting is a form of extremely localized attack that results in holes in the metal. These holes may be small or large in diameter, but in most cases they are relatively small. Pits are sometimes isolated or so close together that they look like a rough surface. Generally a pit may be described as a cavity or hole with the surface diameter about the same as or less than the depth. Pitting is one of the most destructive and insidious forms of corrosion. It causes equipment to fail because of perforation with only a small percent weight loss of the entire structure. It is often difficult to detect pits because of their small size and because the pits are often covered with corrosion products. In addition, it is difficult to measure quantitatively and compare the extent of pitting because of the varying depths and numbers of pits that may occur under identical conditions. Pitting is also difficult to predict by laboratory tests. Sometimes the pits require a long time (several months or a year) to show up in actual service. Pitting is particularly vicious because it is a localized and intense form of corrosion, and failures often occur with extreme suddenness.

1.2.9 Intergranular Corrosion

Grain boundary effects are of little or no consequence in most applications or uses of metals. If a metal corrodes, uniform attack results since grain boundaries are usually only slightly more reactive than the matrix. However, under certain conditions, grain interfaces are very reactive and intergranular corrosion results. Localized attack at and adjacent to grain boundaries, with relatively little corrosion of the grains, is intergranular corrosion. The alloy disintegrates (grains fall out) or loses its strength. Intergranular corrosion can be caused by impurities at the grain boundaries, enrichment of one of the alloying elements, or depletion of one of these elements in the grain-boundary areas. Small amounts of iron in aluminum, wherein the solubility of iron is low, have been shown to segregate in the grain boundaries and cause intergranular corrosion. It has been shown that, based on surface tension considerations, the zinc content of a brass is higher at the grain boundaries. Depletion of chromium in the grain-boundary regions results in intergranular corrosion of stainless steels.

1.2.10 Selective Leaching

Selective leaching is the removal of one element from a solid alloy by corrosion processes. The most common example is the selective removal of zinc in brass alloys (dezincification). Similar processes occur in other

alloy systems in which aluminum, iron, cobalt, chromium, and other elements are removed. Selective leaching is the general term to describe these processes, and its use precludes the creation of terms such as de-aluminumification, de-cobaltification. Parting is a metallurgical term that is sometimes applied, but selective leaching is preferred.

1.2.11 Erosion Corrosion

Erosion corrosion is the acceleration or increase in rate of deterioration or attack on a metal because of relative movement between a corrosive fluid and the metal surface. Generally, this movement is quite rapid, and mechanical wear effects or abrasion are involved. Metal is removed from the surface as dissolved ions, or it forms solid corrosion products, which are mechanically swept from the metal surface. Sometimes, movement of the environment decreases corrosion, particularly when localized attack occurs under stagnant conditions; this is not erosion corrosion because deterioration is not increased. Erosion corrosion is characterized in appearance by grooves, gullies, waves, rounded holes, and valleys and usually exhibits a directional pattern. In many cases, failures because of erosion corrosion occur in a relatively short time, and they are unexpected largely because evaluation corrosion tests were run under static conditions or because the erosion effects were not considered.

1.2.12 Stress-Corrosion Cracking

Stress-corrosion cracking refers to cracking caused by the simultaneous presence of tensile stress and a specific corrosive medium. Many investigators have classified all cracking failures occurring in corrosive media as stress-corrosion cracking, including failures due to hydrogen embrittlement. However, these two types of cracking failures respond differently to environmental variables. To illustrate, CP is an effective method for preventing stress-corrosion cracking; however, hydrogen-embrittlement may be caused when excessive current is applied, especially on stainless steel. Hence, the importance of considering stress-corrosion cracking and hydrogen embrittlement as separate phenomena is obvious. During stress-corrosion cracking, the metal or alloy is virtually unattacked over most of its surface, while fine cracks progress through it. This cracking phenomenon has serious consequences, since it can occur at stresses within the range of typical design stress.

1.2.13 Exothermic Welding

Exothermic welding is used in CP to connect a copper wire to a metallic structure, usually steel or cast-iron. It is a pyrotechnic composition of copper oxide, aluminum powder and magnesium powder. The magnesium powder is ignited with a spark gun or electronic ignition equipment. The aluminum powder serves as fuel, and melts the copper oxide, which bonds the wire to the structure. Although not explosive, it can create brief bursts of heat and high temperature in a small area.

1.2.14 Error-Free

Potential measurement error due to a voltage drop caused by current flowing through a resistor (the electrolyte) between the reference electrode and the protected structure.

1.3 ADMINISTRATIVE REQUIREMENTS

After award of the contract, but prior to commencement of any work at the

site, meet with the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Develop a mutual understanding relative to the administration of the value engineering, the safety program, preparation of the schedule of prices or earned value report. Review shop drawings, other submittals, scheduling programming, execution of the work, and clear expectations of the "Interim Department of Defense (DD) Form 1354" submittal. Major subcontractors who will engage in the work must also attend.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Preconstruction Survey

SD-02 Shop Drawings

Drawings; G, AE

Anode junction boxes, bonding boxes, and test stations

Joint bonds

Contractor's Modifications; G, AE

SD-03 Product Data

Qualifications

Equipment; G, AE

Anodes; G, AE

Anode junction boxes, bonding boxes, and test stations

Isolation flange kits

Wires

Cable and wire

Shunts

Permanent reference electrodes; G, AE

Spare Parts

SD-06 Test Reports

Tests and Measurements; G, AE

Contractor's Modifications; G, AE

SD-10 Operation and Maintenance Data

Cathodic Protection System; G, AE

Training Course; G, AE

Cathodic Protection System, Data Package 5; G, AE

SD-11, Closeout Submittals

Initial Cathodic Protection System Field Testing; G, AE

One Year Warranty Period Cathodic Protection System Field Test
Report; G, AE

Final Cathodic Protection System Field Test Report; G, AE

1.4.1 Material and Equipment Manufacturer Data

DATE	ISSUE NO.	REQUEST DATE	REQUESTED BY	REQUEST REF. NO.
MANUFACTURER NAME				
DESCRIPTION OF EQUIPMENT				

1.5 MAINTENANCE MATERIAL SUBMITTALS

1.5.1 Spare Parts

After approval of shop drawings, furnish spare parts data for each different item of material and equipment specified. The data must include a complete list of parts, special tools, and supplies, with current unit prices and source of supply.

After approval of shop drawings, furnish revised spare parts for any changes made from original submittal. One spare anode of each type must be furnished. In addition, supply information for material and equipment replacement for all other components of the complete system, including anodes, cables, splice kits and connectors, corrosion test stations, and any other components not listed above. Furnish one reference electrode on a hand reel with 800 meters of conductor, one digital voltmeter that can be used in the maintenance of this CP system. Demonstrate use of

furnished equipment in actual tests during the [training course](#). Provide a description of equipment of the pipe-to-soil protected structure and foreign structures at electrical isolation between the utility supplier and the facility piping.

1.5.2 Extra Materials

Furnish one reference electrode on a hand reel with **300 meters** of conductor, one high-input-impedance digital multimeter that can be used in the maintenance of this CP system. Demonstrate use of furnished equipment in actual tests during the training course. Provide a description of equipment of the pipe-to-soil protected structure and foreign structures at electrical isolation between the utility supplier and the facility piping. Include a description of the equipment and measurement of the pipe-to-soil potentials, anode voltage, anode current and soil condition.

1.6 QUALITY CONTROL

1.6.1 Regulatory Requirements

Obtain the services of a corrosion expert to supervise, inspect, and test the installation and performance of the CP system. The term "corrosion expert" refers to a person, who by thorough knowledge of the physical sciences and the principles of engineering and mathematics, acquired by professional education and related practical experience, is qualified to engage in the practice of corrosion control of buried or submerged metallic structures.

1.6.2 [Qualifications](#)

The corrosion expert must be accredited or certified by NACE International, as a CP-4 CP Specialist or be a NACE International certified Corrosion Specialist or a registered professional engineer who has certification or licensing that includes education and experience in CP of the type of CP system being installed. The corrosion expert must have not less than 10 years of experience in the type of CP for buried or submerged metallic structures under this contract. Submit evidence of qualifications of the corrosion expert including their name and qualifications certified in writing to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager prior to the start of construction. Certification must be submitted giving the name of the firm, the number of years of experience, and a list of not less than five of the firm's installations, three or more years old, that have been tested and found satisfactory.

1.6.3 Services of Corrosion Expert

The "corrosion expert" must make a minimum of three visits to the project site. The first of these visits will include obtaining soil resistivity data, acknowledging the type of pipeline coatings to be used and reporting to the contractor if Impressed Current Cathodic Protection (ICCP) or Galvanic Anode Cathodic Protection (GACP) is required. Once the submittals are approved and the materials delivered, the "corrosion expert" will revisit the site to verify the materials meet submittal requirements, ensure the contractor understands installation practices and that the contractor is capable and qualified to complete the installation.

The "corrosion expert" will be available (but not necessarily be onsite the entire time) during the installation of the CP system to answer

questions, approve any changes or additions required during construction, or to provide recommendations as required. The third visit is to complete the training and demonstrations to applicable personnel on proper testing and maintenance techniques and to complete testing the installed CP systems to ensure it has been installed properly and meets adequate CP criteria. An additional visit is required if the One-Year-Warranty-Period-Testing is required.

1.7 DELIVERY, STORAGE AND HANDLING

Storage area for corrosion material will be designated by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. If material are not stored in a building, tarps or similar protection must be used to protect material from inclement weather. Resack and add backfill to packaged anodes that are damaged as a result of improper handling or exposure to rain.

1.8 PROJECT/SITE CONDITIONS

1.8.1 Environmental Requirements

1.8.2 Existing Conditions

Prior to start of any onsite construction activities, perform a [Preconstruction Survey](#) of the project site with the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager, and take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record. Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the contractor's assigned storage area and access route(s), as applicable. The contractor and the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager will sign this survey report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract.

1.9 WARRANTY

Provide equipment items that are supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

2.1.1 Corrosion Control System Description

A corrosion control system consists of several systems which work together to mitigate corrosion on buried or submerged metallic structures. Failure to comply with the requirements of any one of these systems may result in inadequate corrosion control and premature failure of the structure being protected. Each system's guide specifications must be included in the

design and installation of a complete corrosion control system and must be adhered to in the design and execution of the corrosion control of a structure being protected. Determination of Need for CP must be made by government requirements and policy directives.

- a. Construction Design Requirements (CDR) for protected structures are found in the UFGS for the structure being protected. For water storage tanks refer to [NACE SP0388](#), and [NACE SP0196](#), underground storage tanks [NACE SP0285](#), aboveground storage [NACE SP0193](#), fuel storage piping Section [33 52 40 FUEL SYSTEMS PIPING \(NON-HYDRANT\)](#), offshore pipelines, offshore structures and [NACE SP0176](#), pipeline casings structures (railroad, highway and water crossing) [NACE SP0200](#) and for well casings [NACE SP0186](#).
- b. Coating Systems (CS) are a critical factor in performance of an ICCP system. All coatings, including coatings in structure guide specifications and Green Seal (GS) coatings, must be compatible with the structure and the CP system, and have high disbondment capabilities. A high resistance to cathodic disbondment is critical for long term service life of coatings on buried or submerged metallic structures under CP. For paints and coatings refer to Section [09 90 00 PAINTS AND COATINGS](#), and for discontinuity (Holiday) testing of new protective coatings on conductive substrates refer to [NACE SP0188](#).
- c. Mechanical Damage Systems (MDS) such as bedding and rock control barriers normally included in Structure GS may be required by design for some locations. Electrical Isolation is required for all galvanic anode CP systems. For an Electrical Isolation System (EIS) refer to [NACE SP0286](#).
- d. An Electrical Continuity System (ECS) of the protected structure is critical to the operation of the CP system. The types of joints such as bonding and couplings are normally included in Structure GS, this is particularly important to nonwelded pipelines to allow sufficient CP current to conduct to the entire structure.
- e. Stray Current (Interference) Systems must be considered in design, monitored during construction, and interference testing must be completed during the final testing. Design must consider all other cathodic protection systems which may affect other systems or systems which may affect the project, including foreign systems. All foreign systems must be contacted for information and notification and any joint testing which may be required. Corrosion Coordinating Committees may exist. Reference [NACE TPC 11](#).
- f. Pipelines that parallel overhead high-voltage AC transmission power systems are subject to induced AC. Induced AC has several potential adverse impacts on the safety of personnel and pipeline integrity. Assuming that these conditions exist, there are several measures that can be taken to mitigate the induced AC present on a pipeline. These induced AC mitigation strategies are detailed in various international standards including [NACE SP0177](#) and [NACE SP21424](#).
- g. An Impressed Current CP Monitoring System is a solution for remote monitoring (and optionally controlling) different kinds of CP installations. Normally provides for measuring AC voltage input, DC voltage and current output. Optionally may be used to include potentials with permanent reference cells. Remote monitoring systems may be set to provide notifications of high and low alarms on any of

the monitored inputs/outputs. Automated recording of all measurements may be scheduled and provide a historical database. Remote monitoring is widely used for underground pipelines used in oil and gas distribution systems, remote facilities such as missile launch facilities, for resistance bonds and remote test stations. These Monitoring Systems are detailed in various NACE international technical papers.

- h. When a project is connecting to an existing infrastructure with CP the design must be compatible with the existing structure(s) CP system. Existing structures may have Impressed Current CP (ICCP) Anode Systems using Remote Anode Systems, Deep Anode Systems, Linear Anode Systems, or Distributed Anode Systems. Existing structures may also have Galvanic Cathodic Protection (GACP) Systems which may be distributed or remote. Existing structures might not have CP. They may use alternative methods of corrosion mitigation instead of CP such as Inhibitor System/Internal Corrosion Control. For control of internal corrosion in steel pipelines and piping systems refer to [NACE SP0106](#). Due to the limited voltage and current of galvanic anodes the protected structure must be coated and isolated from other structures.
- i. A highly dielectric bonded coating is required to attain adequate CP. Unbonded coatings block the protective current from the pipeline or structure and must not be used with CP. Failure to isolate other metallic structures will result in loss of protection. Isolation from other metallic structures must be maintained.
- j. Continuity of the structure with low resistance is crucial to proper operation of a galvanic anode system. All joints must be continuous or be bonded to both sides of the joint.
- k. A conductive electrolyte is required to allow current flow from the galvanic anodes. Use of galvanic anode systems are normally restricted to electrolytes with resistivities below 30,000 ohm-cm. Small well-coated structures such as coated valves, tees and elbows have very high resistance to earth in high resistance soils. Galvanic anodes in electrolytes over 30,000 ohms also have very high resistance to earth. High circuit resistance with the low voltage of galvanic anodes will not allow sufficient current to meet instant off or depolarization criteria. Additional anodes under these conditions will not noticeably increase the current applied to the structure. Reference SP0169 for criteria in high resistance electrolytes.

2.1.2 Design Requirements

2.1.2.1 Electrical Isolators

Isolators are required to isolate the pipes from any other structure. Provide isolators at all locations where the indicated pipes contact any other metallic structure. Provide locations and detailed drawings of required installations. Include any requirements for lightning protection, test stations, surge protection, or other requirements and include locations and details in design drawings.

2.1.2.2 Anode and Bond Wires

For each CP system, bond metallic components and structures that are not electrically continuous by installing bond wires between the various structures. Bonding of existing buried structures may also be required to

preclude detrimental stray current effects and safety hazards. Return stray current to its source without damaging structures intercepting the stray current. Provide electrical isolation of underground facilities in accordance with acceptable industry practice. All tests must be witnessed by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager.

2.1.2.3 Surge Protection

Install sealed weatherproof lightning and surge arrestor devices across isolation flanges or fittings installed in underground piping as indicated on the drawings. Provide gapless, self-healing, solid state type arrestor. Provide number 4 AWG copper lead wires with High Molecular Weight Polyethylene (HMWPE) insulation. Lightning arrestors are not required for isolation flanges on metallic components used on non-metallic piping systems.

2.1.2.4 Non-metallic Pipe System

In the event pipe other than metallic pipe is approved and used in lieu of metallic pipe, protect all metallic components of this pipe system with CP. Submit detailed drawings of CP for each component to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager for approval within 45 days after date of receipt of notice to proceed, and before commencement of any work.

2.1.2.5 Coatings

Provide coatings for metallic components as required for metallic fittings. Complete and test protective coating on each metallic component (such as valves, hydrants and fittings). Provide coating as required for underground metallic pipe. Each test must be witnessed by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Select, apply, and inspect coatings as specified in these specifications. The use of non-metallic pipe does not change other requirements of the specifications. Submit any deviations due to the use of non-metallic pipe for approval.

2.1.2.6 Drawings

Submit an electronic copy of detail drawings consisting of a complete list of equipment and material including manufacturer's descriptive and technical literature, catalog cuts, results of system design calculations including soil-resistivity, installation instructions and certified test data showing location of anodes and stating the maximum recommended anode current output density. Include in the detail drawings complete wiring and schematic diagrams, isolation fittings, test stations, permanent reference electrodes and bonding and any other details required to demonstrate that the system has been coordinated and will function properly as a unit. Reference locations to two permanent facilities or mark points. Provide one electronic PDF copy and digital photos of the completed installation.

2.1.2.7 Summary of Services Required

Include the following scope of services:

- a. Close-interval potential surveys,

- b. CP Installation System,
- c. System testing,
- d. Interference testing,
- e. Training,
- f. Operating and maintenance manual,
- g. Isolation testing and bonding testing,
- h. Coating and holiday testing.

2.1.2.8 Contractor's Modifications

The specified system is based on a complete system with Impressed Current in ICCP anodes. The contractor may modify the CP system after review of the project, site verification, and analysis, if the proposed modifications include the anodes specified and will provide better overall system performance.

- a. Submit an electronic copy of detail drawings showing proposed changes in location, scope of performance indicating any variations from, additions to, or clarifications of contract drawings. Show proposed changes in anode arrangement, anode size and number, anode materials and layout details, conduit size, wire size, mounting details, wiring diagram, method for electrically-isolating the structure, and any other pertinent information for proper installation and performance of the system. Reference locations to two permanent facilities or mark points. Include in the detail drawings complete wiring and schematic diagrams, permanent reference electrodes and bonding and any other details required to demonstrate that the system has been coordinated and will function properly as a unit. The modifications must be fully described and must be approved by the Contracting Officer. Provide one electronic PDF copy and digital photos of the completed installation.
- b. The proposed system must achieve performance requirements found in section 2.1.3 Performance Requirements, meet the criteria of CP and be verified by testing requirements in section 3.6.1 Tests and Measurements.

2.1.2.9 Tests of Components

Perform a minimum of four tests at each metallic component in the piping system. Two measurements must be made directly over the anodes and the other two tests must be over the outer edge of the component, but at the farthest point from the anodes. Provide a field drawing showing the component, the structure, all components of the CP system and their relationship to each other. Also provide a narrative describing how the CP system will work and the testing at each component. Components requiring CP must include but not be limited to the following:

- a. Pipes under the floor slab or foundations.
- b. Post Indicator Valve (PIV).
- c. Shutoff valves.

- d. Metallic pipes extended from aboveground locations.
- e. Connectors or change-of-direction devices.
- f. Metallic pipe components or sections.
- g. Backflow preventers.

2.1.2.10 Electrical Potential Measurements

Make all potential tests at a minimum of 3 m intervals witnessed by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Provide submittals identifying test locations on separate drawings, showing all metal to be protected and all CP equipment. Distinguish and identify test points equipment and protected metal.

2.1.2.11 Typical Metallic Components on Non-metallic Systems

2.1.2.11.1 Pipe Under Concrete Slab

Pipe under concrete slab must have 2 permanent reference electrodes located under the slab. Locate one permanent reference electrode where the pipe enters the concrete slab. Route all conductors to a test station.

2.1.2.12 Metallic Component Coating

Coatings for metallic components will be required for metallic fittings as indicated. These metallic fittings will include fire hydrants, tees, elbows, and valves. Coatings must be selected, applied, and inspected as specified in the coating specifications referenced and be compatible with the structure being protected. All coatings must be in accordance with all applicable Federal, State, and local regulations. Unbonded coatings must not be used with CP.

2.1.2.13 Location of Test Stations

Provide test stations of the type and location shown and curb box or post mount. Provide buried isolation joints with test wire connections brought to a test station. Reference all test stations with GPS coordinates. Unless otherwise shown, locate other test stations as follows:

- a. At 300 m intervals or less.
- b. Where the pipe or conduit crosses any other metal pipe.

2.1.2.14 Electrical Isolation of Structures

As a minimum, provide isolating flanges or unions at the following locations:

Provide isolation between fuel line; and foreign pipes that cross the new lines within 3.05 m.

2.1.2.14.1 Isolation Joint Testing

An isolator checker or insulation tester will be used for isolation or insulating joint (flange or dielectric) electrical testing.

2.1.2.14.2 Underground Structure Coating

This coating specification takes precedence over any other project specification and drawing notes, whether stated or implied, and also applies to the pipeline or tank supplier. Variance in coating quality is not allowed by the contractor or Base Construction Representative without the written consent of the designer. All underground metallic pipelines and tanks to be cathodically protected must have a high quality factory-applied coating. This includes all carbon steel, cast-iron and ductile-iron pipelines or vessels. Select, apply, and inspect coatings as specified. If non-metallic pipelines are installed, coat all metallic fittings on pipe sections in accordance with this specification section.

2.1.2.14.3 Field Joints

Coat all field joints with materials compatible with the pipeline coating compound. Apply the field joint coating material to an equal thickness as the pipeline coating. Do not use unbonded coatings for these buried metallic components. This includes the elimination of all unbonded polymer wraps or tubes. Once the pipeline or vessel is set in the trench, conduct an inspection of the coating. This inspection must include electrical holiday detection. Repair any damaged areas of the coating. The Contracting Officer or the Contracting Officer's Representative, Technical Expert or Project Manager must be asked to witness inspection of the coating and testing using a holiday detector.

2.1.2.14.4 Inspection of Pipe Coatings

Any damage to the protective coating during transit and handling must be repaired before installation. After field coating has been applied, inspect the entire pipe using an electric holiday detector in accordance with NACE SP0188 using a full-ring, spring-type coil electrode. The holiday detector must be equipped with a bell, buzzer, or other type of audible signal which sounds when a holiday is detected. Upon detection, immediately repair all holidays in the protective coating. Occasional checks of holiday detector, operation will be made by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager to determine suitability of the detector. Provide all labor, materials, and equipment necessary for conducting the inspection.

2.1.2.14.5 Protective Coating for Aboveground Piping System

Provide finish painting as follows:

2.1.2.14.6 Ferrous Surfaces

Touch-up shop-primed surfaces with ferrous metal primer. Solvent-clean surfaces that have not been shop-primed. Surfaces that contain loose rust, loose mil scale, and other foreign substances must be mechanically-cleaned by power wire-brushing and primed with ferrous metal primer. Finish primed surface with two coats of exterior oil paint and vinyl paint.

2.1.3 Performance Requirements

The design must allow for synchronized interruption of all applied current.

2.1.3.1 Criteria of Cathodic Protection

Criteria for adequate CP must be identified by the designer or the contractor's corrosion engineer and approved by the government corrosion engineer. The method of voltage drop consideration must also be identified by the contractor's corrosion engineer and approved by the government. The 100 mV shift criteria are not applicable to bi-metallic structures.

- a. Determination of the on and polarized (instant off) potentials must be made with the protective current applied to the structure for a minimum of 4 days. Polarized potentials must be determined by interrupting all the current being applied to the structure or coupon.
- b. The potential measurements for the native measurement and the polarized potential must be made with the reference electrode at the same exact location. The polarization decay measurements must also be made with the reference electrode at the same exact location as the polarization potential.
- c. The polarization decay measurements will be the difference between the polarized potential and a voltage measurement made 24 hours after the interruption of protective current.

2.1.3.1.1 Steel

A negative polarized potential of 0.85 volts (850 millivolts) or more negative. The voltage must not be more negative than a negative polarized potential of 1.200 volts (1200 millivolts).

2.2 EQUIPMENT

2.2.1 Remote Monitoring

Remote monitoring equipment must be designed, manufactured and procured specifically for CP use and should be provided as per design and drawings to monitor potential (requires permanent reference electrode) test station(s) shunts and must match or be compatible with previously installed remote monitoring equipment in use at the installation. Rectifier monitoring must include DC Current output(using rectifier shunt), DC Voltage Output, AC voltage input.

2.2.2 Rectifiers

Rectifier will be Air Cooled.

2.2.2.1 Air Cooled Enclosure

NEMA ICS 6 Type 3R Air Cooled enclosure suitable for pad mounting. Enclosures must be of 3.1 mm steel or heavier. Enclosure must include front hinged door with padlock hasp provide three keys. Side doors fit with screened openings to provide for cooling by natural convection. Provide holes, conduit knockouts and threaded hubs of sufficient size and location. The cabinet and mounting support must be hot-dipped galvanized steel according to the manufacturer's standards.

2.2.2.2 Cabinet Construction

Cabinets are constructed of hot dipped galvanized steel, and provided with

a full door. The enclosure must have oil-resistant gasket. The door must be hinged and have a hasp that will permit the use of a padlock. The cabinet must be fitted with screened openings of the proper size to provide for adequate cooling. Holes, conduit knockouts, or threaded hubs of sufficient size and number conveniently located.

2.2.2.3 Cabinet Paint System

The cabinet and mounting support must be hot dipped galvanized with the manufacturer's standard painting system.

2.2.2.4 Electrical Ratings

Electrical ratings as follows: Input voltage at 60 Hz: 460 volts three phase.

- a. Output voltage, dc: 24 volts as indicated.
- b. Output current, dc: 50 amperes as indicated.

The rectifier must be capable of supplying continuous full rated output at an ambient temperature of 44 degrees C in full sunlight with expected life of 10 years minimum.

2.2.2.5 Rectifier Stacks

Rectifying elements must be silicon diodes connected to provide full-wave rectification. Silicon diodes must be protected by selenium surge cells or varistors against over-voltage surges, and by current-limiting devices against over-current surges.

2.2.2.6 Transformer

Transformer must conform to UL 506.

2.2.2.7 Circuit Breaker

A three-pole, flush-mounted, fully magnetic, properly rated non-terminal type circuit breaker must be installed in the primary circuit of the rectifier supply transformer. Properly rated secondary magnetic circuit breaker between rectifier supply transformer and stacks, two on three phase rectifiers.

2.2.2.8 Fuses

Cartridge-type fuses with suitable fuse holders must be provided in each leg of the DC circuit. Properly rated secondary fuse between rectifier supply transformer and stacks, two on three phase rectifiers.

2.2.2.9 Overload and Short Circuit Protection

Provide UL 489, three-pole, flush-mounted molded case circuit breaker, thermal-magnetic type, in the primary circuit of the rectifier supply transformer.

2.2.2.10 Surge Protection

Install approved sealed weatherproof lightning arrestor devices for the AC input and the DC output. The arrestor must be gapless, self-healing,

solid state type.

2.2.2.11 Wiring

Install wiring in accordance with **NFPA 70** utilizing type THWN insulation. Fittings for conduit and cable work must conform to **UL 514A**. Outlets and conduit must be of the threaded hub type with gasketed covers. Conduit must be securely fastened at **2.4 m** intervals or less. Splices made in outlet fittings only. Conductors must be color coded for identification. Wiring for anode header and distribution must be No. 2 AWG stranded copper wire with type Dual Extrusion HALAR/HMWPE Direct Burial Cable insulation.

2.2.2.12 Shunt Resistors

MIL-I-1361. Shunt must be located on the rectifier front panel and clearly marked with current and voltage for verification of panel ammeter. Install shunts calibrated in millivolts per amp with amp capacity at or slightly under rectifier output in series in the anode or the structure circuit between the stacks and the rectifier output.

2.2.2.13 DC Output Control

Provide adjustable DC output voltage by transformer taps.

2.2.2.14 Output Voltage and Current Metering

Provide separate panel voltmeter and ammeter, not less than **63.5 mm** rectangular 2 percent full scale accuracy at **30 degrees C**, temperature stability above and below **30 degrees C** of at least 1 percent per **5 degrees C**. Provide toggle switch for each meter.

2.2.2.15 Efficiency

Overall efficiency of 80 percent minimum when operated at full output.

2.2.2.16 Filter Chokes and Capacitors

An efficiency filter (choke) may be required as an option to improve the rectifiers efficiency. A capacitor may also be required to be used in conjunction with the filter to further improve the efficiency and minimize noise.

2.2.2.17 Grounding Provisions

Grounding provisions must comply with **NFPA 70** and **UL 467** including a ground terminal in the cabinet. The grounding conductor from the terminal to the earth grounding system must be solid or stranded copper not smaller than No. 6 AWG. The earth grounding system must consist of one or more ground rods. Ground rods must be of copper-clad steel conforming to **UL 467** not less than **16 mm** in diameter by **2.4 m** in length. Rods are driven full length into the earth. Sectional type rods may be used.

2.2.2.18 Wiring Diagram

A complete wiring diagram of the power unit showing both the AC supply and the DC connections to anodes located on the inside of the cabinet door. All components must be visible and labeled.

2.2.2.19 Optional Rectifier Special Required Features

1. An efficiency filter (choke) (may be required to improve the rectifiers efficiency). (A capacitor may also be required to be used in conjunction with the filter to further improve the efficiency and minimize noise.)
2. Convenience Outlet mounted on Faceplate
3. Safety shield panel covering Taps or all energized conductors on faceplate
4. Stainless Steel Perforated screens on Air Cooled Rectifiers
5. Heavy duty Draw-pull Stainless steel cabinet latch
6. Separate Slide-out equipment racks for Transformer and Stack
7. Additional 2 coarse voltage control link bar taps
8. Quick-change, heavy-duty knobs for changing tap link bars Minimum 8 mm diameter
9. Soldered tap changing studs 5 mm
10. Phenolic front panel
11. Nickel Plated and double-nutted or soldered connections
12. Terminal block for AC input wires
13. Terminal block for Remote Monitoring Connections

2.2.3 Polarization Cell Replacement (PCR) and (PCRH) for Hazardous Locations

PCRs and PCRHs must be designed, manufactured, and procured specifically for the application and must exceed the modeled AC steady-state current and fault conditions. For hazardous locations, the PCRH model must be used.

Characteristic	PCR	PCRH
AC steady-state current, rms	45A, 80A	45A
AC fault current, rms. at 0.5s	3.7 kA to 15 kA	3.7 kA to 15 kA
Lightning current, 8x20 micros, peak	100 kA	100 kA
Hazardous location certification	Division 2, Zone 2	Division 1, Zone 1

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Characteristic	PCR	PCRH
Rain Proof, IP66	Yes	Yes
Submersible, IP68 or NEMA 6P	Optional	No

PCRs must be installed with a protective ground-based enclosure to secure the cable connections and prevent electrical hazards. The PCRH must be installed with an explosion-proof enclosure and must be flange mounted. Structure and Grounding conductors must be properly sized for the application.

AC Fault Current Rating	Minimum Wire Size (AWG)	Minimum Wire Size (Metric)
1.2 kA, 2kA, 3.7 kA	#6	16mm ²
5kA 9kA 10kA	#2	35mm ²
14kA 15kA	#2/0	70mm ²

2.2.4 Solid State Decoupler (SSD)

SSDs must be designed, manufactured, and procured specifically for the application and must exceed the modeled AC steady-state current and fault conditions. For hazardous locations, the PCRH model must be used. SSDs must be installed with a protective ground-based enclosure to secure the cable connections and prevent electrical hazards.

Characteristic	SSD
AC steady-state current, rms	45A
AC fault current, rms. at 0.5s	1.2 to 5 kA
Lightning current, 8x20micros, peak	75-100 kA
Hazardous location certification	Division 2, Zone 2
Rain Proof, IP66	Yes

Characteristic	SSD

2.3 COMPONENTS

2.3.1 Test Stations

2.3.1.1 Flush Mounted

NEMA ICS 6. Metallic or non-metallic with terminal board, 5 terminal posts and lockable lid. A non-metallic enclosure must be molded of glass filled polycarbonate and urethane coated or Acrylonitrile Butadiene Styrene (ABS) plastic. The unit must be of standard design, manufactured for use as a CP test station, complete with cover, terminal board, shunts, and brass or Type 316 stainless steel hardware. The terminal board must be removable for easy access to wires. Provide traffic valve box capable of withstanding H-20 traffic loads. The cover must have a cast in legend "CP TEST."

2.3.1.2 Post Top Mounted

NEMA ICS 6. Metallic or non-metallic with terminal board, 5 terminal posts and lockable lid. A non-metallic enclosure must be high impact strength molded plastic. The unit must be of standard design, manufactured for use as a CP test station, complete with cover, terminal board, shunts, and brass or Type 316 stainless steel hardware. The terminal board must be removable for easy access to wires. The test station must be mounted atop **1830 mm** long polyethylene conduit with anchor. Terminal connections will be permanently tagged to identify each termination of conductors (e.g. identify the conductors connected to the protected structure, anodes, and reference electrodes).

2.3.2 Shunts for Test Stations and Junction Boxes

MIL-I-1361. 0.01 ohm, 8 ampere, accuracy plus or minus one percent, polycarbonate circuit board type, color coded for value recognition yellow for 0.01 ohm shunt with nickel-plated brass posts and standard **6.35 mm** holes on 2.54 cm centers to fit test stations and terminal boards, accuracy plus or minus one percent, manganin wire type.

2.3.3 Junction Box Enclosures

NEMA ICS 6, Type 3R enclosure with clamped cover and padlocked hasp. Enclosure must be of galvanized steel construction with terminal board. Knockout for conduit must be the size and location as per design drawings.

2.3.3.1 Nameplates

Provide nameplate in accordance with Section **26 20 00** INTERIOR DISTRIBUTION SYSTEM and **ASTM D709.** Provide laminated plastic nameplates for each enclosure as specified or as indicated on the drawings. Each nameplate inscription must identify the function. Nameplates will be melamine plastic, **3 mm** thick, white with black center core. Surface will

be matte finish. Corners will be square. Accurately align lettering and engrave into the core. Minimum size of nameplates must be 635 mm by 65 mm es. Lettering must be a minimum of 6.35 mm high normal block style.

2.3.4 Terminal Boards

Provide terminal boards for anode junction boxes, bonding boxes, and test stations made of phenolic plastic 6 mm thick with dimensions as indicated. Insulated terminal boards must have the required number of terminals (one terminal required for each conductor). Install solderless copper lugs and copper buss bars, shunts, and variable resistors on the terminal board as indicated. Test station terminal connections will be permanently tagged to identify each termination of conductors (e.g. identify the conductors connected to the protected structure, anodes, reference electrodes and coupons).

2.3.5 Anode Junction Boxes

2.3.5.1 Enclosure

NEMA ICS 6, Type 3R enclosure with clamped cover and padlocked hasp. Enclosure must be of galvanized steel construction with terminal board and labeled with nameplate. Provide nameplate in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.3.5.2 Terminal Boards

Provide terminal boards for anode junction boxes, bonding boxes, and test stations made of phenolic plastic 6 mm thick with dimensions as indicated. Insulated terminal boards must have the required number of terminals (one terminal required for each conductor). Install solderless copper lugs and copper buss bars, shunts, and variable resistors on the terminal board as indicated. Test station terminal connections will be permanently tagged to identify each termination of conductors (e.g. identify the conductors connected to the protected structure, anodes, and reference electrodes).

2.4 MATERIALS

2.4.1 Impressed Current Anodes

2.4.1.1 Dimensions and Weights

Bare anode weight 28 kg not including core.

2.4.1.2 High-Silicon Chromium Bearing Cast Iron

Cast-iron anodes must be of the size indicated and conform to the following requirements:

2.4.1.2.1 Chemical Composition (Nominal)

ASTM A518/A518M, Grade 3. Chemical composition as follows:

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Carbon	0.70 to 1.10 percent
Manganese	1.5 percent maximum
Silicon	14.20-14.75 percent
Chromium	3.25 to 5.00 percent
Copper	0.50 percent maximum
Molybdenum	0.20 percent maximum
Iron	Remainder

Anode dimensions: 75 mm by 1,575 mm. Centrifugally cast tubular anodes with uniform wall thickness.

2.4.2 Wire and Cable

2.4.2.1 Anode Lead Wire

Anodes must have lead wires installed at the factory. Anode connecting wire must be No. 8 AWG stranded copper wire with type CP HMWP insulation, 2.8 mm thick, 600 volt rating. Cable-to-anode contact resistance must be 0.003 ohms maximum. In the toughest of environments, use wire specified for deep anode lead wire.

2.4.2.2 Deep Anode Lead Wire

For deep anodes, each anode must have a separate, continuous wire extending from the anode to the junction box with individual shunts. No spliced connections will be permitted in deep anode cables. No spliced connections will be permitted in deep anode cables. Chlorine gas resistant cable and shield connecting wire must be No. 8 AWG, stranded copper wire with HMWPE protective jacketed cable with a fluorocopolymer inner or primary insulation Direct Burial Cable insulation with 600-volt rating. Cable-to-anode contact resistance must be 0.02 ohms maximum. For ceramic coated anodes, anode connecting wires must have molded multi-seal solder connections.

2.4.2.3 Anode Header Cable

Anode header cable from rectifier to anode junction box must be No. 2 AWG stranded copper wire with type CP HMWP insulation, 2.8 mm thick 600 volt rating. Anode Header Cable aboveground in conduit must be No. 2 AWG stranded copper wire with type CP high molecular weight insulation, 2.8 mm thick insulation, 600 volt rating.

2.4.2.4 Structure (Negative) Cable

Structure connecting wire must be No. 2 AWG stranded copper wire with type CP high molecular weight insulation, 2.8 mm thick insulation, 600 volt rating. Copper conductors conforming to ASTM B3 and ASTM B8.

2.4.2.5 Test Wires

Test wires must be No. 8 AWG stranded copper wire with NFPA 70 Type HMWPE TW insulation. Copper conductors conforming to ASTM B3 and ASTM B8.

2.4.2.6 Joint and Continuity Bond Cables

Provide bonds across joints or any electrically discontinuous connections in the piping, and other pipes and structures with other than welded or threaded joints included in this CP system. Unless otherwise specified, bonds between structures and across joints in pipe with other than welded or threaded joints must be with No. 4 AWG stranded copper cable with HMWPE insulation. Bonds between structures must contain sufficient slack for any anticipated movement between structures. Bonds across pipe joints must contain a minimum of 100 mm of slack to allow for pipe movement and soil stress. Bonds must be attached by exothermic welding. Exothermic weld areas must be insulated with coating compound and approved by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Continuity bonds must be installed as necessary to reduce stray current interference. Additional joint bonding must be done where determined during construction or testing or as directed. Joint bonding must include excavation and backfilling. There must be a minimum of 2 continuity bonds between each structure and other than welded or threaded joints. Electrical continuity must be tested across joints with other than welded or threaded joints and across metallic portions of sewage lift stations and water booster stations. Copper conductors conforming to ASTM B3 and ASTM B8.

2.4.2.7 AC Power Supply Wiring

UL 83, Type THWN, stranded copper conductors, gauge (AWG) and color coded as indicated.

2.4.2.8 Polyethylene Insulation

Polyethylene insulation must comply with the requirements of ASTM D1248 and of the following types, classes, and grades:

2.4.2.8.1 HMWPE

HMWP must be Type I, Class C, Grade E5.

2.4.2.8.2 High Density Polyethylene (HDPE)

HDPE must be Type III, Class C, Grade E3.

2.4.2.9 Rectifier DC Negative (Structure) Cable(s)

ASTM D1248 HMWPE insulation, stranded copper conductors, gauge (AWG) as indicated.

2.4.3 Cable and Wire Identification Tags

Laminated plastic material with black letters on a yellow background. Print letters and numbers a minimum of 5 mm in height. Provide identifier legend in accordance with the cable schedule.

2.4.4 Anode Connection

2.4.4.1 End-Connected Anode

Cast a recess 150 mm deep in one end of the anode. Attach the lead wire to the anode with an anchor device. Not more than 10 mm of bare wire must protrude from the anchor device. Attachment must withstand a 1446 Newton pull without loosening the wire or anchor device. Fill the recess with an epoxy sealing compound. Provide non-metallic plug flush with the anode end surface. Cable-to-anode contact resistance must not exceed 0.02 ohms.

2.4.5 Coke Breeze

2.4.5.1 Calcined Petroleum Coke Breeze (Dry)

Coke Breeze must conform to the following requirements:

2.4.5.1.1 Electrical Resistivity

Resistivity is not to exceed 1 milliohm-meter (0.1 ohm-cm) Great Lake Carbon C 12 A Test Method.

2.4.5.1.2 General Backfill Specifications

Bulk Density - 1044 to 1204 kg/cubic meter
Fixed Carbon - 99.0 percent or greater
Volatiles - 0.2 percent or less
Sizing - 100 percent less than 13 mm

2.4.5.2 Metallurgical Coke Breeze (Processed)

Coke Breeze must conform to the following requirements:

2.4.5.2.1 Electrical Resistivity (Nominal)

Nominal electrical resistivity must be:

- a. 100 milliohm-meter (10 ohm-centimeter) Max., tightly compacted.
- b. 100 milliohm-meter to 150 milliohm-meter, (10 to 15 ohm-centimeter,) lightly compacted.
- c. 150 to 200 milliohm-meter, (15 to 20 ohm-centimeter,) loose.

2.4.5.2.2 General Backfill Specifications

Bulk density - 608 to 672 kg per cubic meter
Fixed Carbon - 80 percent or greater
Sizing - 100 percent less than .10 mm

2.4.6 Permanent Reference Electrodes

Permanent reference electrodes must be copper/copper-sulfate with dual CP coupons specifically manufactured for underground use, 32 mm diameter, by 203 mm long, plastic tube with an ion trap to minimize contamination of the electrode. Must never need recharging, maintenance, or recalibration. Must have impregnated membrane which keeps electrode electrolytes from drying out or getting the reference electrode electrolyte contaminated. Must have ion trap to prevent reference electrode damage from hydrogen

sulfide or excess chloride ions. The electrode will be prepackaged by the manufacturer with a backfill material as recommended by the manufacturer. Provide electrodes with No. 12 AWG, RHW cable of sufficient length to extend to the test station without splicing. Reference electrodes will have a minimum 20-year life, stability of plus or minus 5 millivolts under 3 microamp load. The manufacturer must calibrate the PRE to 316 mV plus or minus 10mV referenced to a standard hydrogen electrode (SHE) and provide a calibration certificate detailing the results of the calibration. Procedures for evaluating the accuracy annually must be included in the Operation and Maintenance Manual.

2.5 ACCESSORIES

2.5.1 Wire Connectors

Safety Standard for Wire Connectors must conform to [UL 486A-486B](#).

2.5.2 Electrical Tape

Pressure-sensitive vinyl plastic electrical tape must conform to [UL 510](#).

2.5.3 Cable Marker Tape

Traceable marker tape must be manufactured for the purpose and clearly labeled "Cathodic Protection Cable Buried Below".

2.5.4 Insulating Tape

Pressure-sensitive vinyl plastic electrical tape and rubber insulated tape must conform to [UL 510](#).

2.5.5 Underground Splices

Provide splices with a compression connector on the conductors, and insulation and waterproofing using one of the following methods which are suitable for continuous submersion in water and comply with [ANSI C119.1](#).

2.5.5.1 Cast-Type Splice

Provide cast-type splice insulation by means of molded casting process employing a thermosetting epoxy resin insulating material applied by a gravity poured method or pressure injected method. Provide component materials of the resin insulation in a packaged form ready for convenient mixing without removing from the package.

2.5.6 Buried Cable Warning and Identification Tape

Polyethylene tape, manufactured for warning and identification of buried cable and conduit. Tape must be **75 mm** wide, Yellow in color and read "CAUTION CATHODIC PROTECTION CABLE BURIED BELOW". Color and lettering must be permanent and unaffected by moisture or other substances in backfill materials.

2.5.7 Conduit

[UL 6](#), rigid galvanized steel, Outlet boxes: [UL 514A](#) and fittings [UL 514B](#), threaded hubs. Metallic conduit and fittings to be PVC coated in accordance with [NEMA RN 1](#), Type A40.

2.5.8 Protective Coating

Except as otherwise specified, protective coating for underground metallic components including pipe and fittings must be applied mechanically in a factory or field plant specially equipped for the purpose. Valves and fittings that cannot be coated mechanically must have the protective coating applied by hand, preferably at the plant. Joints must be coated by hand. Hand coating must produce a coating equal in thickness to the coating applied mechanically. Piping and components installed in valve boxes or manholes must also receive the specified protective coating.

2.5.8.1 Pipeline Metallic Components

Underground metallic pipelines and structures must have a good quality factory-applied coating. This includes carbon steel, cast iron and ductile-iron pipelines or vessels. If non-metallic pipelines are installed, metallic fittings or pipe sections must be coated as follows.

- a. The nominal thickness of the metallic pipe joint or other component coating must be 0.6 mm, plus or minus 5 percent.
- b. Pipe and joint coating for factory-applied or field-repair material must be applied as recommended by the manufacturer and must be one of the following:
 - (1) Fusion bonded epoxy.
 - (2) Continuously-extruded polyethylene and adhesive coating system.
 - (3) Polyvinyl chloride pressure-sensitive adhesive tape.
 - (4) High-density polyethylene/bituminous rubber compound tape.
 - (5) Butyl rubber tape.
 - (6) Coal tar epoxy.

2.5.8.2 Field Joints

Coat field joints with material compatible with the existing pipeline coating compound. Apply the joint coating material to an equal thickness as the pipeline coating. Unbonded coatings must not be used on buried metallic piping. This prohibition includes unbonded wraps or tubes.

2.5.9 Isolation Flange Sets/Kits

Provide full-faced gaskets, isolating sleeves and washers, and steel washers. Provide [isolation flange kits](#) rated for operation at the rated pressure and temperature. Bolts must be replaced with similar type longer bolts if less than two threads extend beyond the end of both steel washers.

2.5.10 Electrical Isolation of Structures

Isolating fittings, including isolating flanges and couplings, must be installed above ground or in a concrete hand hole. As a minimum, isolating flanges or unions must be provided at the following locations:

- a. Connection of new piping to existing pipes.

- b. Pressure piping under floor slab to a building.
- c. Additionally, isolation must be provided between new pipe lines and foreign pipes that cross the new lines within 3 m.

2.5.10.1 Gaskets

ASME B16.21. Laminated phenolic material for operation at 2,000 KPa, 65 degrees C. Asbestos materials must not be used.

2.5.10.2 Isolation Washers and Sleeves

Two sets 3 mm laminated phenolic for operation at 65 degrees C. Isolating washers must fit within the bolt facing on the flange over the outside of the fabric reinforced phenolic sleeve.

2.5.11 Electrically Isolating Pipe Joints

Electrically isolating pipe joints for above- or below-ground use must be nonflexible flanged type electrically isolating pipe joints to be field assembled.

2.5.12 Stray Current Measurements

Perform stray current measurements as indicated. Alternate methods may be used when approved. The stray current test report must indicate location of test, type of pipes tested, method of testing.

2.5.13 Cast-In-Place Concrete

Concrete must be 20 MPa minimum ultimate 28-day compressive strength with 25 mm minimum aggregate conforming to ASTM C94/C94M Section 03 30 00 CAST-IN-PLACE CONCRETE.

2.5.14 Steel Flanges and Bolting

2.5.14.1 Steel Flanges

ASME B16.5, 1335 N.

2.5.14.2 Bolting

ASTM A307, Grade B for bolts; ASTM A194/A194M, Grade 2 for nuts. Dimensions: ASME B18.2.1 for bolts, ASME B18.2.2 for nuts. Threads: ASME B1.1, Class 2A fit for bolts, Class 2B fit for nuts. Bolts must extend completely through the nuts and may have reduced shanks of a diameter not less than the diameter at the roof of threads.

PART 3 EXECUTION

3.1 SAFETY PRECAUTIONS AND HAZARDOUS LOCATIONS

Any personnel performing operations that will generate heat, sparks, or flame in hazardous locations must first perform adequate safety precautions. A trained responsible person must ensure the area is safe to perform the operation. Required actions include ensuring adequate ventilation before work starts, air monitoring, and a fire watch must be provided and remain for 30 minutes after the operation is completed. A minimum of a 20 pound ABC type fire extinguisher must be available and

must be inspected before each use. Equipment being used must be inspected and used in accordance with manufacturer recommendations. Combustibles that are in the work area(s) must be moved or if they cannot be moved, be covered with fire retardant welding blankets. When performing exothermic welding, properly sized charges and inspection of the structure condition must be accomplished to ensure a safe operation.

3.2 INSTALLATION

3.2.1 Anode Bed Installation

IEEE C2NFPA 70 Anode configuration and size must be as indicated in design drawings. A minimum of 20 anodes must be installed in a vertical deep well configuration with locations as specified in design drawings. Materials must meet these specifications. The corrosion expert must verify that the materials are in accordance with the approved submittals. Anode beds must be installed to avoid interference with the structure being protected.

3.2.2 Deep Anode Lead Wire

Each anode must have a separate, continuous lead wire, without splices, from the anode and terminating in an aboveground junction box equipped with individual anode current shunts.

3.2.2.1 Anode Centering

Anodes must be centered in the well by means of centering devices.

3.2.2.2 Casing

The casing must be to a depth and elevation indicated in design drawings.

3.2.3 Backfill

Backfill the well with specified coke breeze or metallurgical coke breeze surrounding the anodes by a method that does not leave voids or bridging. The recommended method is to pump the backfill from the bottom upward. The well must be over-filled with coke breeze allowing for settlement so that the settled level after a number of days is as high as the level shown. The number of days allowed for settling of the coke breeze will be determined by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. If the top level of coke breeze is below the level shown after settlement, put additional coke breeze in the well. The backfill used must not require tamping. The top portion of the well must be sealed for 8 m to prevent surface water run-off. All vents must be vented above the high water mark and at a safe height.

3.2.4 Cable Marker Tape

Locate traceable marker tape in the same trench above CP cables including structure leads, anode leads, anode header cables, test station leads, bonding cables, and rectifier electrical power cables.

3.3 MISCELLANEOUS INSTALLATION

3.3.1 Rectifier Installation

Mounting must be as shown. Pole, concrete pad or wall mounting must be equipped with support channels, channel bracket, lifting eyes, and a keyhole at the top. Cross-arm brackets must accommodate a 102 by 102 mm cross-arm.

3.3.2 Test Stations

Provide test stations complete with an insulated terminal block having the indicated number of terminals; provided with a lockable cover and have a cast-in legend, "C.P. Test" and complete with an insulated terminal block having the required number of terminals. (One terminal required for each conductor). Provide sufficient test stations to monitor underground isolation points. Test-bond stations (potential measurement and stray current control) must be provided to monitor pipe-to-soil potential of proposed underground pipes or existing underground metallic structures which may conduct stray current from the new CP system. The location of the test-bond stations must ensure that the pipe-to-soil potential of metallic pipe not designated to be protected is not made less negative by the energization of the CP system. Test station terminal connections and the terminal conductor must be permanently tagged to identify each termination of the conductors (e.g. identify the conductors connected to the protected structures). Conductors must be permanently identified in the station by means of plastic or metal tags, or plastic sleeves to indicate termination. Each conductor must be color coded in accordance with the drawings. The station test facility, including permanent copper/copper-sulfate reference electrodes and test returns must be installed as indicated. Pavement inserts must be non-metallic and must allow copper/copper-sulfate reference electrode to contact the electrolyte beneath the pavement surface. Abbreviations must not be used. Welding of electrical connections must be as follows: Exothermic welds must be as specified. Use and selection of these materials and welding equipment must be in accordance with the manufacturer's recommendations.

3.3.3 Permanent Reference Electrodes

3.3.3.1 Permanent Reference Electrode Verification

Verify permanent reference electrodes against a calibrated portable electrode in the presence of the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager before installation. Verify in a non-metallic container of water. Permanent electrode must measure a reference potential agreeing with that measured by the portable electrode within plus or minus 0.010 volts when the sensing windows of the two electrodes being compared are not more than 2 mm apart but not touching. Remove permanent reference electrodes not within this potential range from the construction site by the end of the day and replace at the contractor's expense. The testing provision applies to replacement permanent reference electrodes as well.

3.3.4 Bonding Boxes

Provide structure bonding boxes in locations as indicated.

3.3.5 Joint Bonds

Provide joint bonds on metallic pipe to and across buried flexible couplings, mechanical joints, flanged joints except at places where isolation joints are specified and joints not welded or threaded to provide electrical continuity. Connect bond wire(s) to the structure(s) by use of exothermic weld kit(s). Clean the structure surface by scraping, filing or wire brushing to produce a clean, bright surface. Check and verify adherence of the bond to the substrate for mechanical integrity by striking the weld with a 908 gram hammer. Cover connections with an electrically insulating coating which is compatible with the existing coating on the structure.

3.3.6 Wire Connections

3.3.6.1 Wire-To-Structure Connections

Lead wire to structure connections must be by exothermic welding process. Weld charges made specifically for use on cast iron must be used on cast iron pipe. A backfill shield filled with a pipeline mastic sealant or material compatible with the coating must be placed over the weld connection and must cover the exposed metal adequately. If other methods of connection are required, they must be approved by the corrosion expert and the Contract Officer or the Contracting Officer's Representative.

3.3.6.2 Cable Protection

Positive cable to the ground bed and negative cable to the pipe to be protected must be buried a minimum depth of 750 mm except where above ground construction utilizing conduit is used.

3.3.6.3 Wire Splicing

Connecting wire splicing is not allowed.

3.3.7 Steel Surfaces

Connections to metallic structure must be made by exothermic weld methods as manufactured by an approved manufacturer for the type of structure. If other methods of connection are required, such as brazing, mechanical, electric-arc welding and other types of welded connections to ferrous pipe and structures, they must be approved by the corrosion expert and the Contracting Officer or the Contracting Officer's Representative before use.

3.3.8 Electrical Continuity

Underground pipe must be electrically continuous except at places where electrically isolating joints are specified. Pipe joined by means other than welding must meet the following electrical continuity requirements:

- a. Mechanical joints that are not factory designed to provide electrical continuity must be bonded by installing a metallic bond across the joint. The bonding connections must be made by the exothermic welding process.
- b. Mechanical joints designed to provide electrical continuity may be used.

3.3.9 Electrical Isolation of Structures

Perform electrical isolation of structures as follows:

3.3.9.1 Locations of Electrical Isolation

- a. Structural reinforcing steel
- b. Metallic piping
- c. Metallic conduits
- d. Fuel-storage tanks
- e. Gasoline-storage tanks

Provide electrical isolation at each building entrance, and at other locations as indicated.

3.3.10 Ferrous Valves

Dissimilar ferrous valves in a buried ferrous pipeline, including the pipe, must be coated with an underground type dielectric coating for a minimum distance of 10 diameters on each side of the valve.

3.3.11 Brass or Bronze Valves

Brass or bronze valves must not be used in a buried ferrous pipeline.

3.4 INSTALLATION DETAILS

3.4.1 Anode Deep Well Installation

Unless otherwise authorized, installation must not proceed without the presence of the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Anodes of the size specified must be installed to the depth indicated and at the locations shown. Locations may be changed to clear obstructions with the approval of the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Anodes will be installed in sufficient number and of the required type, size, and spacing to obtain a uniform current distribution over the surface of the structure. The anode system will be designed for a life of 25 years of continuous operation. Anodes must be installed as indicated.

3.4.2 Underground Pipe Joint Bonds

Underground pipe having other than welded or threaded coupling joints must be made electrically continuous by means of a bonding connection installed across the joint.

3.4.3 Anode Junction Boxes

Provide junction boxes and mark each of the wires terminating in each box.

3.4.4 Bonding Cables

Provide structure bonding cables in locations as indicated.

3.4.5 Test Stations and Permanent Reference Electrodes

Test stations will be of the type and location shown and will be curb box or post mounted. Provide buried isolation joints with test wire connections brought to a test station. Reference all test stations with GPS coordinates. Unless otherwise shown, locate other test stations as follows:

- a. At 305 meters intervals.
- b. At all isolation joints.
- c. At both ends of casings.
- d. Where the pipe crosses any other metal pipes.
- e. Where the pipe connects to an existing piping system.
- f. Where the pipe connects to a dissimilar metal pipe.

Do not fill the bottom of the test station with concrete unless otherwise specified. Do not place rubbish, scrap or other debris into the test station.

3.4.6 Permanent Reference Electrode Calibration and Installation

Calibrate permanent reference electrodes against a portable electrode in the presence of the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager before installation. Calibrate in a test tank containing water with the same composition as the tank to be protected. Permanent electrode must measure a reference potential agreeing with that measured by the portable electrode within plus or minus 0.005 volt when the sensing windows of the two electrodes being compared are not more than 2 mm apart but not touching. Remove permanent reference electrodes not within this potential range from the construction site by the end of the day and replace at the contractor's expense. The testing provision applies to replacement reference electrodes as well.

3.4.6.1 Field Drawings

Complete a field drawing of each anode installation showing location of anode, depth of anode, color and size of anode lead wire and any other pertinent details, test station location and diagram, Underground Pipe Joint bond locations and details, Anode Junction Box Location and details, Permanent Reference Electrode locations and Location of all Electrical Isolations. Submit copy with daily report to the government.

3.5 ELECTRICAL ISOLATION OF STRUCTURES

3.5.1 Isolation Fittings

Isolating fittings, including isolating flange kits, dielectric unions and couplings, must be installed aboveground, or within manholes or vaults, wherever possible. Where isolating joints must be covered with soil, they must be fitted with a proper joint cover specifically manufactured for covering the particular joint, and the space within the cover filled with hot coal-tar enamel. Isolating fittings in lines entering buildings must be located at least 305 mm above grade of floor level, when possible.

Isolating joints must protect against over-voltage surges or approved surge protection devices. The cells must provide a low resistance across isolating joint without excessive loss of cathodic current.

3.5.2 Joint Bonds

Provide joint bonds on metallic pipe to and across buried flexible couplings, mechanical joints, flanged joints except at places where isolation joints are specified and joints not welded or threaded to provide electrical continuity. Connect bond wire(s) to the structure(s) by use of exothermic weld kit(s). Clean the structure surface by scraping, filing or wire brushing to produce a clean, bright surface. Check and verify adherence of the bond to the substrate for mechanical integrity by striking the weld with a 908 gram hammer. Cover connections with an electrically insulating coating which is compatible with the existing coating on the structure.

3.6 FIELD QUALITY CONTROL

Field tests must be witnessed by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager or their designated representative. Advise the Contracting Officer or Contracting Officer's Representative 14 days prior to performing each field test. Quality control for the [cathodic protection system](#) must consist of the following:

- a. Initial field testing by the contractor upon construction.
- b. Government Field Testing after contractor initial field test report submission.
- c. Warranty period field testing by the contractor.
- d. Final field testing by the contractor after one year of service.

3.6.1 Tests and Measurements

3.6.1.1 Native Potentials

Notify the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager a minimum of 14 working days prior to each test. Base potential tests: At least 24 hours after installation of structure to be protected. Perform measurements at anode junction boxes, test stations and other locations suitable for test purposes, at intervals not exceeding 800 meters with readings at each end point and the midpoints as a minimum. Use the same measuring equipment that is specified for measuring protected potential measurements.

3.6.1.2 Protected Potentials

Systems must be tested and inspected by the contractor's corrosion engineer in the presence of the Contracting Officer or the Contracting Officer's Representative, Technical Project Manager corrosion expert or an approved representative. Notify the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager a minimum of five working days prior to each test. At least one week after native potential testing and energization of rectifier and anodes, measure protected structure-to-electrolyte potentials. The locations of

these measurements must be identical to the locations specified for native potential measurements. Use the same measuring equipment that is specified for measuring protected potential measurements. Record test data, including date, time, and locations of testing and submit report to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Contractor must correct and retest, at the contractor's expense, deficiencies in the materials and installation observed by these tests and inspections.

3.6.1.3 Isolation Testing

Before the anode system is connected to the pipe, an isolation test must be made at each isolating joint or fitting. This test will demonstrate that no metallic contact, or short circuit exists between the two isolated sections of the pipe. Any isolating fittings installed and found to be defective must be reported to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager.

3.6.1.4 Isolation Tester

An isolation tester designed and manufactured for use in CP, using the continuity check circuit, must be used for all isolating joint (flange) electrical testing. Testing must conform to the manufacturer's operating instructions. Test must be witnessed by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. An isolating joint that is good will read full scale on the meter. If an isolating joint is shorted, the meter pointer will be deflected or near zero on the meter scale. Location of the fault will be determined from the instructions, and the joint must be repaired.

3.6.1.5 Anode Output

As the cable bonds are connected to the pipe, current output will be measured with an approved clamp-on milliammeter, calibrated shunt with a suitable millivoltmeter or multimeter, or a low resistance ammeter. (Of the three methods, the low-resistance ammeter is the least desirable and most inaccurate. The clamp-on milliammeter is the most accurate.) The values obtained and the date, time, and location must be recorded.

3.6.1.6 Reference Electrode Potential Measurements

Upon completion of the installation and with the entire CP system in operation, electrode potential measurements must be made using a copper/copper sulfate reference electrode and a potentiometer-voltmeter, or a direct-current voltmeter having an internal resistance (sensitivity) of not less than 10 megohms per volt and a full scale of 10 volts. The locations of these measurements must be identical to the locations used for baseline potentials. The values obtained and the date, time, and locations of measurements must be recorded. No less than eight (8) measurements will be made over any length of line or component. Additional measurements will be made at the direction of the cathodic protection design engineer.

3.6.1.7 Holiday Test

Any damage to the protective coating during transit and handling must be repaired before installation. After field-coating has been applied, the entire pipe must be inspected by an electric holiday detector with Impressed Current in accordance with [NACE SP0188](#) using a full-ring,

spring-type coil electrode or wet sponge test. The holiday detector will be equipped with a bell, buzzer, or other type of audible signal which sounds when a holiday is detected. Holidays in the protective covering must be repaired upon detection. Occasional checks of holiday detector potential will be made by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager to determine suitability of the detector. Labor, materials, and equipment necessary for conducting the inspection must be furnished by the contractor. The coating system must be inspected for holes, voids, cracks, and other damage during installation.

3.6.1.8 Rectifier Testing

The purpose of the rectifier operational inspection is to determine the serviceability of all components required to impress current to the anodes of the Impressed Current system. The inspection must be thorough to ensure dependable current until the next inspection.

- a. Before energizing the rectifier, visually check all rectifier components, shunt box components, safety switches, circuit breakers, and other system power components.
- b. Tighten all accessible connections and check temperature of all the components.
- c. Ensure all permanent reference electrode caps are removed and placed inside the rectifier cabinet for verification of removal and for use during testing.
- d. Startup testing of the rectifier must include voltage and current testing at all tap settings up to the level of protection or maximum of the rectifier rated current, whichever is the lowest. Do not apply excessive current to the structure. For automatic rectifiers, record each tap setting (if available) before switching to automatic potential control.
- e. Using a dependable hand-held meter, measure the output voltage and current, and calibrate the rectifier meters, if present. For rectifiers with more than one circuit, measure the output voltage and current for each circuit using a dependable hand-held meter, and calibrate the rectifier meters, if present.
- f. For systems with permanent reference electrodes, using a calibrated reference electrode, measure the potential difference to each installed permanent reference electrode by placing both electrodes together in the electrolyte with CP current off (may be tested before installation). If the difference is more than 10 mV, replace the permanent reference electrode. For rectifiers with potential voltmeters, using a dependable hand-held meter, measure the potentials for each voltmeter, and calibrate that rectifier meter.
- g. Calculate the CP system circuit resistance of each circuit by dividing the rectifier DC voltage output of each circuit by the rectifier DC ampere output for that circuit.
- h. Calculate the rectifier efficiency.

3.6.1.9 Stray Current Measurements

Before final acceptance of the installation, stray current tests must be performed on any foreign pipes and other metallic structures in close proximity to the installed anodes. A full report of the tests giving all details must be made.

3.6.1.10 Induced AC Testing

Before final acceptance of the installation, induced AC voltage tests must be performed on the pipes near high AC voltage infrastructure and where crossing above ground and underground AC transmission systems. A full report of these tests must be included in the final testing reports with all details and data taken. The touch potential of any testing over 5 volts must be reported to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Any touch potential over 10 Volts must be mitigated by effective mitigation techniques.

3.6.1.11 Interference Tests

Before final acceptance of the installation, interference tests will be made with respect to any foreign structures in cooperation with the owner of the foreign structures. A full report of the tests giving all details must be made. Stray current measurements must be performed at all isolating locations and at locations where the new pipeline crosses foreign metallic pipes; results of stray current measurements must also be submitted for approval. The method of measurements and locations of measurements must be submitted for approval. As a minimum, stray current measurements must be performed at the following locations:

- a. Connection points of new structures to existing structures.
- b. Crossing points of new pipeline with other existing metallic pipelines.

3.6.1.12 Initial Cathodic Protection System Field Testing

Initial field testing must be completed by the contractor upon completion of construction. Field testing must be witnessed by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager or their designated representative. Advise the Contracting Officer or Contracting Officer's Representative 14 days prior to performing each field test. Field testing must include native and protected potentials, anode current and rectifier testing.

The contractor must submit an initial field test report of the CP system. All structure-to-electrolyte measurements, including native potentials, protected potentials, anode current testing, rectifier testing, and other required testing must be recorded on applicable forms. Identification of test locations, test station and anode test stations will coordinate with the as-built drawings and be provided on system drawings included in the report. The contractor must locate, correct, and report to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager any short circuits encountered during the checkout of the installed CP system.

3.6.1.13 Government Field Testing

The government corrosion engineer must review the contractor's initial

field testing report. Approximately four weeks after receipt of the contractor's initial test report, the system will be tested and inspected in the contractor's presence by the government corrosion engineer. The contractor must correct, at the contractor's expense, materials and installations observed by these tests and inspections not to be in conformance with the plans and specifications. The contractor will pay for all retesting done by the government engineer made necessary by the correction of deficiencies.

3.6.1.14 One-Year-Warranty-Period-Testing

The contractor must inspect, test, and adjust the CP system monthly for one year, 12 interim inspections total, to ensure its continued conformance with the criteria outlined below. The performance period for these tests will commence upon the completion of all CP work, including changes required to correct deficiencies identified during initial testing, and preliminary acceptance of the CP system by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Copies of the [One Year Warranty Period Cathodic Protection System Field Test Report](#), including field data, and certified by the contractor's corrosion engineer must be submitted to the Contracting Officer or Contracting Officer's Representative, the activity, and the geographic EFD corrosion engineer.

3.6.1.15 Final Acceptance Field Testing

Conduct final field testing of the CP system utilizing the same procedures specified under, "Initial Field Testing of the Cathodic Protection Systems". The contractor will inspect, test, and adjust the CP system after one year of operation to ensure its continued conformance with the criteria outlined below. The performance period for these tests will commence upon preliminary acceptance for the CP system by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. One electronic copy of the [Final Cathodic Protection System Field Test Report](#), certified by the contractor's corrosion engineer must be submitted to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager and the geographic EFD corrosion engineer for approval, and as an attachment to the Operation and Maintenance Manual in accordance with Section [01 78 23 OPERATION AND MAINTENANCE DATA](#). The government corrosion engineer must review the contractor's final field testing report.

3.7 CLOSEOUT ACTIVITIES

3.7.1 Reconditioning of Surfaces

3.7.1.1 Concrete

Concrete must be [20 MPa](#) minimum ultimate 28-day compressive strength with [25 mm](#) minimum aggregate conforming to [ASTM C94/C94M](#) Section [03 30 00](#) CAST-IN-PLACE CONCRETE.

3.7.1.2 Restoration of Sod

Restore unpaved surfaces disturbed during the installation of anodes and wires to their original elevation and condition. In areas where grass cover exists, it is possible that sod can be carefully removed, watered, and stored during construction operations, and replaced after the operations are completed since it is estimated that no section of pipeline

must remain uncovered for more than two (2) days. Where the surface is disturbed in a newly seeded area, re-seed the area with the same quality and formula of seed as that used in the original seeding. Seeding must be done as directed, in all unsurfaced locations where sod and topsoil could not be preserved and replaced. The use of sod in lieu of seeding will require approval by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager.

3.7.1.3 Restoration of Pavement

Repair pavement, sidewalks, curbs, and gutters where existing surfaces are removed or disturbed for construction. Saw cut pavement edges. Graded aggregate base course must have a maximum aggregate size of 40 millimeters. Prime base course with liquid asphalt, ASTM D2028/D2028M, Grade RC-70 prior to paving. Match base course thickness to existing but must not be less than 150 millimeters. Asphalt aggregate size must be 15 mm, asphalt cement must conform to ASTM D3381/D3381M, Grade AR-2000. Match asphalt concrete thickness to existing but must not be less than 50 millimeters. Repair Portland cement concrete pavement, sidewalks, curbs, and gutters using 20.67 MPa concrete conforming to Section 03 30 00 CAST-IN-PLACE CONCRETE. Match existing pavement, sidewalk, curb, and gutter thicknesses.

3.7.1.4 Cleanup

The contractor is responsible for cleanup of the construction site. All paper bags, wire clippings, must be disposed of as directed. Paper bags, wire clippings and other waste will not be put in bell holes or anodes excavation.

3.7.2 Training

3.7.2.1 Instruction to Government Personnel

During the warranty testing or at a time designated by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager, make available the services of a technician regularly employed or authorized by the manufacturer of the CP system for instructing government personnel in the proper operation, maintenance, safety, and emergency procedures of the CP system. The period of instruction must be not less than two hours and not more than two 8-hour working days. Conduct the training at the jobsite or at another location mutually satisfactory to the government and the contractor. The field instructions will cover all of the items contained in the Operation and Maintenance Manual.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 51 00

INTERIOR LIGHTING

05/20, CHG 2: 11/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 DEFINITIONS
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Luminaire Drawings
 - 1.5.2 Luminaire Design Data
 - 1.5.3 ANSI/IES LM-79 Test Report
 - 1.5.4 ANSI/IES LM-80 Test Report
 - 1.5.5 ANSI/IES TM-21 Test Report
 - 1.5.6 ANSI/IES TM-30 Test Report
 - 1.5.7 Test Laboratories
 - 1.5.8 Regulatory Requirements
 - 1.5.9 Standard Products
 - 1.5.9.1 Alternative Qualifications
 - 1.5.9.2 Material and Equipment Manufacturing Date
- 1.6 WARRANTY
 - 1.6.1 Luminaire Warranty
 - 1.6.2 Lighting Controls Warranty
- 1.7 OPERATION AND MAINTENANCE MANUALS
 - 1.7.1 Lighting System
 - 1.7.2 Lighting Control System

PART 2 PRODUCTS

- 2.1 PRODUCT COORDINATION
- 2.2 LUMINAIRES
 - 2.2.1 Luminaires
 - 2.2.2 Luminaires for Hazardous Locations
- 2.3 LIGHT SOURCES
 - 2.3.1 LED Light Sources
- 2.4 LED DRIVERS
- 2.5 LIGHTING CONTROLS
 - 2.5.1 System
 - 2.5.2 Lighting Contactor
 - 2.5.3 Devices
 - 2.5.3.1 Switches
 - 2.5.3.2 Occupancy/Vacancy Sensors
 - 2.5.3.2.1 Passive Infrared Sensors
 - 2.5.3.2.2 Ultrasonic Sensors
 - 2.5.3.2.3 Dual Technology Sensors
 - 2.5.3.2.4 Power Packs
 - 2.5.3.3 Photosensors
- 2.6 EXIT AND EMERGENCY LIGHTING EQUIPMENT

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.6.1 Exit Signs
 - 2.6.1.1 Exit Signs with Battery Backup
- 2.6.2 LED Emergency Drivers
- 2.6.3 Self-Diagnostic Circuitry for LED Drivers
- 2.7 LUMINAIRE MOUNTING ACCESSORIES
 - 2.7.1 Recess and Surface Mounted Luminaires
 - 2.7.2 Luminaire Support Hardware
 - 2.7.2.1 Wire
 - 2.7.2.2 Straps
- 2.8 EQUIPMENT IDENTIFICATION
 - 2.8.1 Manufacturer's Nameplate
 - 2.8.2 Labels
- 2.9 FACTORY APPLIED FINISH

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Light Sources
 - 3.1.2 Luminaires
 - 3.1.2.1 Recessed Luminaires
 - 3.1.3 LED Drivers
 - 3.1.4 Exit Signs
 - 3.1.5 Lighting Controls
 - 3.1.5.1 Occupancy/Vacancy Sensors
- 3.2 FIELD QUALITY CONTROL
 - 3.2.1 Tests
 - 3.2.1.1 Lighting Control Verification Tests
 - 3.2.1.2 Emergency Lighting Test
- 3.3 CLOSEOUT ACTIVITIES
 - 3.3.1 Commissioning

-- End of Section Table of Contents --

SECTION 26 51 00

INTERIOR LIGHTING
05/20, CHG 2: 11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A641/A641M (2019) Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A1008/A1008M (2021a) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable

ASTM B633 (2019) Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

ASTM D4674 REV A (2002; R 2010) Standard Practice for Accelerated Testing for Color Stability of Plastics Exposed to Indoor Office Environments

EUROPEAN UNION (EU)

Directive 2011/65/EU (2011) Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment

ILLUMINATING ENGINEERING SOCIETY (IES)

ANSI/IES LM-79 (2019) Approved Method: Electrical and Photometric Measurements of Solid State Lighting Products

ANSI/IES LM-80 (2020) Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays and Modules

ANSI/IES LS-1 (2020) Lighting Science: Nomenclature and Definitions for Illuminating Engineering

ANSI/IES TM-15	(2020) Technical Memorandum: Luminaire Classification System for Outdoor Luminaires
ANSI/IES TM-21	(2019) Technical Memorandum: Projecting Long-Term Lumen, Photon, and Radiant Flux Maintenance of LED Light Sources
ANSI/IES TM-30	(2020) Technical Memorandum: IES Method for Evaluating Light Source Color Rendition
IES Lighting Library	IES Lighting Library

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 100	(2000; Archived) The Authoritative Dictionary of IEEE Standards Terms
IEEE C2	(2017; Errata 1-2 2017; INT 1 2017) National Electrical Safety Code
IEEE C62.41	(1991; R 1995) Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)
NEMA ANSLG C78.377	(2017) Electric Lamps— Specifications for the Chromaticity of Solid State Lighting Products
NEMA C82.77-10	(2020) Harmonic Emission Limits - Related Power Quality Requirements
NEMA ICS 2	(2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V
NEMA ICS 6	(1993; R 2016) Industrial Control and Systems: Enclosures
NEMA SSL 1	(2016) Electronic Drivers for LED Devices, Arrays, or Systems
NEMA SSL 3	(2011) High-Power White LED Binning for General Illumination
NEMA WD 1	(1999; R 2020) Standard for General Color Requirements for Wiring Devices
NEMA WD 7	(2011; R 2016; R 2021) Occupancy Motion Sensors Standard

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA
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20-1; TIA 20-2; TIA 20-3; TIA 20-4)
National Electrical Code

NFPA 101 (2021) Life Safety Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

47 CFR 15 Radio Frequency Devices

UNDERWRITERS LABORATORIES (UL)

UL 94 (2013; Reprint Mar 2022) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances

UL 508 (2018; Reprint Jul 2021) UL Standard for Safety Industrial Control Equipment

UL 844 (2012; Reprint Oct 2021) UL Standard for Safety Luminaires for Use in Hazardous (Classified) Locations

UL 916 (2015; Reprint Oct 2021) UL Standard for Safety Energy Management Equipment

UL 924 (2016; Reprint May 2020) UL Standard for Safety Emergency Lighting and Power Equipment

UL 1598 (2021; Reprint Jun 2021) Luminaires

UL 2043 (2013) Fire Test for Heat and Visible Smoke Release for Discrete Products and Their Accessories Installed in Air-Handling Spaces

UL 8750 (2015; Reprint Sep 2021) UL Standard for Safety Light Emitting Diode (LED) Equipment for Use in Lighting Products

1.2 RELATED REQUIREMENTS

Materials not considered to be luminaires, luminaire accessories, or lighting equipment are specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Luminaires and accessories that are mounted in exterior environments and not attached to the exterior of the building are specified in Section 26 56 00 EXTERIOR LIGHTING.

1.3 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications and on the drawings, must be as defined in IEEE 100 and ANSI/IES LS-1.
- b. For LED luminaire light sources, "Useful Life" is the operating hours before reaching 70 percent of the initial rated lumen output (L70) with no catastrophic failures under normal operating conditions. This is also known as 70 percent "Rated Lumen Maintenance Life" as defined

in ANSI/IES LM-80.

- c. For LED luminaires, "Luminaire Efficacy" (LE) is the appropriate measure of energy efficiency, measured in lumens/watt. This is gathered from LM-79 data for the luminaire, in which absolute photometry is used to measure the lumen output of the luminaire as one entity, not the source separately and then the source and housing together.
- d. Total harmonic distortion (THD) is the root mean square (RMS) of all the harmonic components divided by the total fundamental current.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Luminaire Drawings; G

Lighting Control System One-Line Diagram; G

Sequence of Operation for Lighting Control System; G

SD-03 Product Data

Luminaires; G

Light Sources; G

LED Drivers; G

Luminaire Warranty; G

Lighting Controls Warranty; G

Lighting Contactor; G

Switches; G

Occupancy/Vacancy Sensors; G

Photosensors; G

Power Packs; G

Exit Signs; G

Emergency Drivers; G

SD-05 Design Data

Luminaire Design Data; G

SD-06 Test Reports

ANSI/IES LM-79 Test Report; G

ANSI/IES LM-80 Test Report; G

ANSI/IES TM-21 Test Report; G

ANSI/IES TM-30 Test Report; G

Occupancy/Vacancy Sensor Verification Test; G

Photosensor Verification Test; G

SD-10 Operation and Maintenance Data

Lighting System, Data Package 5; G

Lighting Control System, Data Package 5; G

1.5 QUALITY ASSURANCE

Data, drawings, and reports must employ the terminology, classifications and methods prescribed by the IES Lighting Library as applicable, for the lighting system specified.

1.5.1 Luminaire Drawings

Include dimensions, accessories installation details, and construction details. Photometric data, including CRI, CCT, LED driver type, zonal lumen data, and candlepower distribution data must accompany shop drawings.

1.5.2 Luminaire Design Data

- a. Provide safety certification and file number for the luminaire family that must be listed, labeled, or identified in accordance with the NFPA 70. Applicable testing bodies are determined by the US Occupational Safety Health Administration (OSHA) as Nationally Recognized Testing Laboratories (NRTL) and include: CSA (Canadian Standards Association), ETL (Edison Testing Laboratory), and UL (Underwriters Laboratories).
- b. Provide long term lumen maintenance projections for each LED luminaire in accordance with ANSI/IES TM-21. Data used for projections must be obtained from testing in accordance with ANSI/IES LM-80.

1.5.3 ANSI/IES LM-79 Test Report

Submit test report on manufacturer's standard production model of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data in IES format as outlined under "14.0 Test Report" in ANSI/IES LM-79.

1.5.4 ANSI/IES LM-80 Test Report

Submit report on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data as outlined under "8.0 Test Report" in ANSI/IES LM-80.

1.5.5 ANSI/IES TM-21 Test Report

Submit test report on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data, as well as required interpolation information as outlined under "7.0 Report" in ANSI/IES TM-21.

1.5.6 ANSI/IES TM-30 Test Report

Submit color vector graphic in accordance with ANSI/IES TM-30 on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Include spectral distribution of test LED light source.

1.5.7 Test Laboratories

Test laboratories for the ANSI/IES LM-79 and ANSI/IES LM-80 test reports must be one of the following:

- a. National Voluntary Laboratory Accreditation Program (NVLAP) accredited for solid-state lighting testing as part of the Energy-Efficient Lighting Products laboratory accreditation program for both LM-79 and LM-80 testing.
- b. One of the qualified labs listed on the Department of Energy - LED Lighting Facts Approved Testing Laboratories List for LM-79 testing.
- c. One of the EPA-Recognized Laboratories listed for LM-80 testing.

1.5.8 Regulatory Requirements

Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of NFPA 70, unless more stringent requirements are specified or indicated. Provide luminaires and assembled components that are approved by and bear the label of UL for the applicable location and conditions unless otherwise specified.

1.5.9 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design, and workmanship. Products must have been in satisfactory commercial or industrial use for six months prior to bid opening. The six-month period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the six-month period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.9.1 Alternative Qualifications

Products having less than a six-month field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.9.2 Material and Equipment Manufacturing Date

Do not use products manufactured more than six months prior to date of delivery to site, unless specified otherwise.

1.6 WARRANTY

Support all equipment items by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.6.1 Luminaire Warranty

Provide and transfer to the government the original LED luminaire manufacturers standard commercial warranty for each different luminaire manufacturer used in the project.

- a. Provide a written five year minimum replacement warranty for material, luminaire finish, and workmanship. Provide written warranty document that contains all warranty processing information needed, including customer service point of contact, whether or not a return authorization number is required, return shipping information, and closest return location to the luminaire location.

- (1) Finish warranty must include failure and substantial deterioration such as blistering, cracking, peeling, chalking, or fading.

- (2) Material warranty must include:

- (a) All LED drivers and integral control equipment.

- (b) Replacement when more than 15 percent of LED sources in any lightbar or subassembly(s) are defective, non-starting, or operating below 70 percent of specified lumen output.

- b. Warranty period must begin in accordance with the manufacturer's standard warranty starting date.

- c. Provide replacements that are promptly shipped, without charge, to the using Government facility point of contact and that are identical to or an improvement upon the original equipment. All replacements must include testing of new components and assembly.

1.6.2 Lighting Controls Warranty

Provide and transfer to the government the original lighting controls manufacturers standard commercial warranty for each different lighting controls manufacturer used in the project. Warranty coverage must begin from date of final system commissioning or three months from date of delivery, whichever is the earliest. Warranty service must be performed by a factory-trained engineer or technician.

- a. Unless otherwise noted, provide a written five year minimum warranty on the complete system for all systems with factory commissioning. Provide warranty that covers 100 percent of the cost of any replacement parts and services required over the five years which are directly attributable to the product failure. Failures include, but

are not limited to, the following:

- (1) Damage of electronic components due to transient voltage surges.
- (2) Failure of control devices, including but not limited to occupancy sensors, photosensors, and manual wall station control devices.

- b. Provide a written five year minimum warranty on all control components attached to luminaires against defect in workmanship or materials.

1.7 OPERATION AND MAINTENANCE MANUALS

1.7.1 Lighting System

Provide operation and maintenance manuals for the lighting system in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that provide basic data relating to the design, operation, and maintenance of the lighting system for the building. Include the following:

- a. Manufacturers' operating and maintenance manuals.
- b. Luminaire shop drawings for modified and custom luminaires.
- c. Luminaire Manufacturers' standard commercial warranty information as specified in paragraph LUMINAIRE WARRANTY.

1.7.2 Lighting Control System

Provide operation and maintenance manuals for the lighting control system in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that provide basic data relating to the design, operation, and maintenance of the lighting control system for the building. Include the following:

- a. Lighting control system layout and wiring plan.
- b. Lighting control system one-line diagram.
- c. Product data for all devices, including installation and programming instructions.
- d. Occupancy/vacancy sensor coverage layout.
- e. Training materials, such as videos or in-depth manuals, that cover basic operation of the lighting control system and instructions on modifying the lighting control system. Training materials must include calibration, adjustment, troubleshooting, maintenance, repair, and replacement.
- f. Sequence of operation descriptions for each typical room type, including final programming, schedules, and calibration settings.
- g. "As-built" lighting control panel schedules.

PART 2 PRODUCTS

2.1 PRODUCT COORDINATION

2.2 LUMINAIRES

UL 1598, NEMA C82.77-10. Provide luminaires as indicated in the luminaire schedule and details on project plans, complete with light source, wattage, and lumen output indicated. All luminaires of the same type must be provided by the same manufacturer. Luminaires must be specifically designed for use with the driver and light source provided.

2.2.1 Luminaires

UL 8750, ANSI/IES LM-79, ANSI/IES LM-80. For all luminaires, provide:

- a. Complete system with LED drivers and light sources.
- b. Housings constructed of non-corrosive materials. All new aluminum housings must be anodized or powder-coated. All new steel housings must be treated to be corrosion resistant.
- c. ANSI/IES TM-21, ANSI/IES LM-80. Minimum L70 lumen maintenance value of 50,000 hours unless otherwise indicated in the luminaire schedule. Luminaire drive current value must be identical to that provided by test data for luminaire in question.
- d. Minimum efficacy as specified in the luminaire schedule. Theoretical models of initial lamp lumens per watt are not acceptable. If efficacy values are not listed in the luminaire schedule, provide luminaires that meet the following minimum values:

Luminaire Style	Minimum Luminaire Efficacy
Recessed 1 by 4, 2 by 4, and 2 by 2	100 LPW
Food Service and Hazardous Locations	60 LPW
Exterior Wall Sconce	50 LPW

- e. UL listed for dry or damp location typical of interior installations. Any luminaire mounted on the exterior of the building must be UL listed for wet location typical of exterior installations.
- f. LED driver and light source package, array, or module are accessible for service or replacement without removal or destruction of luminaire.
- g. Lenses constructed of heat tempered borosilicate glass, UV-resistant acrylic, or silicone. Provide polycarbonate vandal-resistant lenses as indicated. Sandblasting, etching and polishing must be performed as indicated in the luminaire description.
- h. ANSI/IES TM-15. Provide exterior building-mounted luminaires that do not exceed the BUG ratings as listed in the luminaire schedule. If BUG ratings are not listed in the luminaire schedule, provide luminaires that meet the following minimum values for each application and mounting conditions:

Lighting Application	Mounting Conditions	BUG Rating
Exterior Wall Sconce	Above 1.2 meters AFF	B1-U0-G2
Exterior Wall Sconce	Below or at 1.2 meters AFF	B4-U0-G4

2.2.2 Luminaires for Hazardous Locations

In addition to requirements stated herein, provide LED luminaires for hazardous locations which conform to UL 844 or which have Factory Mutual certification for the class and division indicated.

2.3 LIGHT SOURCES

NEMA ANSLG C78.377, NEMA SSL 3. Provide type, delivered lumen output, and wattage as indicated in the luminaire schedule on project plans.

2.3.1 LED Light Sources

Provide LED light sources that meet the following requirements:

- NEMA ANSLG C78.377. Emit white light and have a nominal CCT of 4000 Kelvin.
- Minimum Color Rendering Index (CRI) of 80.
- Directive 2011/65/EU. Restriction of Hazardous Substances (RoHS) compliant.
- Light source color consistency by utilizing a binning tolerance within a 3-step McAdam ellipse.

2.4 LED DRIVERS

NEMA SSL 1, UL 8750. Provide LED drivers that are electronic, UL Class 1 or Class 2, constant-current type and that comply with the following requirements:

- The combined driver and LED light source system does not exceed the minimum luminaire efficacy values as listed in the luminaire schedule provided.
- Operates at a voltage of 120-277 volts at 50/60 hertz, with input voltage fluctuations of plus/minus 10 percent.
- Power Factor (PF) greater than or equal to 0.90 at full input power and across specified dimming range.
- Maximum Total Harmonic Distortion (THD) less than 20 percent at full input power and across specified dimming range.
- Operates for at least 50,000 hours at maximum case temperature and 90 percent non-condensing relative humidity.
- Withstands Category A surges of 2 kV without impairment of performance. Provide surge protection that is integral to the driver.
- Integral thermal protection that reduces the output power to protect

the driver and light source from damage if the case temperature approaches or exceeds the driver's maximum operating temperature.

- h. **47 CFR 15**. Complies with the requirements of the Federal Communications Commission (FCC) rules and regulations, Non-Consumer (Class A) for EMI/RFI (conducted and radiated).
- i. Class A sound rating.
- j. **Directive 2011/65/EU**. Restriction of Hazardous Substances (RoHS) compliant.

2.5 LIGHTING CONTROLS

2.5.1 System

Provide lighting control system that operates the lighting system as described in the lighting control strategies in the project plans. Submit **Sequence of Operation for Lighting Control System** describing the operation of the proposed lighting control system and devices. Sequence of Operation must provide the strategies identified in the lighting control strategies.

2.5.2 Lighting Contactor

NEMA ICS 2, NEMA ICS 6. Provide a mechanically-held lighting contactor housed in a NEMA 1 enclosure. Provide contactor with one normally-open(NO), double pole contacts, rated 600 volts, 30 amps. Provide coil operating voltage of 120 volts.

2.5.3 Devices

2.5.3.1 Switches

Provide line-voltage toggle switches as specified in Section **26 20 00** INTERIOR DISTRIBUTION SYSTEM. When used for non-digital loads, devices must be rated at 20 Amps inductive load, and be compatible with the lighting control systems.

2.5.3.2 Occupancy/Vacancy Sensors

IEEE C62.41, NEMA WD 1, UL 94, UL 916, UL 508, ASTM D4674 REV A, NEMA WD 7. Provide occupancy/vacancy sensors with coverage patterns as indicated on project plans. Provide no less quantity of sensors as shown on plans, but add additional sensors when required to fulfill coverage requirement for the specific model of sensor provided. Provide occupancy sensor operation that requires movement to activate luminaires controlled and turns luminaires off after a set time of inactivity. Provide ceiling mounted occupancy/vacancy sensors that meet the following requirements:

- a. Operating voltage of 120-277 volts.
- b. Time delay of 30 seconds to 30 minutes with at least four intermediate time delay settings.
- c. Sensors are ceiling mounted.
- d. Shielded or controlled by internal logic to adjust sensitivity to avoid false triggering due to ambient temperature, air temperature

variations or HVAC air movement.

- e. Sensor is equipped to automatically energize the connected load upon loss of normal power when located in a means of egress.
- f. Occupancy and vacancy operation is field-adjustable and programmable with push-button or dip switch on the sensor device.
- g. No leakage current to load when in the off mode.
- h. Utilize zero-crossing circuitry to prevent damage from high inrush current and to promote long life operation.

2.5.3.2.1 Passive Infrared Sensors

Provide Passive Infrared Sensors (PIR) sensors that detect occupancy by sensing heat and movement in the area of coverage. Provide sensors are constructed of a housing of high-impact, injection-molded thermoplastic. Provide PIR sensors that are temperature compensated, with a dual element sensor and a multi-element fresnel lens of POLY IR4 material.

2.5.3.2.2 Ultrasonic Sensors

Provide ultrasonic sensors that detect occupancy by sensing a change in pattern of reflected ultrasonic waves in the area of coverage. Provide sensors that are constructed of a housing of high-impact, injection-molded thermoplastic. Provide ultrasonic sensors that operate at 40 kHz.

2.5.3.2.3 Dual Technology Sensors

Provide dual technology sensors that meet the requirements for PIR sensors and ultrasonic sensors indicated above. If either the PIR or ultrasonic sensing registers occupancy, the luminaires must remain on.

2.5.3.2.4 Power Packs

UL 2043. Provide power packs to provide power to lighting control sensors as required in accordance with the manufacturer's specifications. Provide power packs that meet the following requirements:

- a. Operate at an input voltage of 277 VAC, with an output voltage 12-24 VDC at 225 mA.
- b. Constructed of plenum-rated, high-impact thermoplastic enclosure.
- c. Utilizes zero-crossing circuitry to prevent damage from inrush current.
- d. Maximum load rating of 16 amps for electronic lighting loads.
- e. **Directive 2011/65/EU.** Restriction of Hazardous Substances (RoHS) compliant.

2.5.3.3 Photosensors

Provide photosensors as described in Section **26 56 00** EXTERIOR LIGHTING.

2.6 EXIT AND EMERGENCY LIGHTING EQUIPMENT

2.6.1 Exit Signs

UL 924, NFPA 101. Provide wattage as indicated in the luminaire schedule on project plans. Provide LED Exit Signs that meet the following criteria:

- a. Housing constructed of UV-stable, thermo-plastic.
- b. UL listed for damp location.
- c. Configured for universal mounting.
- d. Provide with internationally recognized 'running man' symbol on green background. Provide directional arrows as required to indicated direction.
- e. Single or double face as indicated in project plans and luminaire schedule.

2.6.1.1 Exit Signs with Battery Backup

Equip with automatic power failure device, test switch, and pilot light, and fully automatic high/low trickle charger in a self-contained power pack. Battery must be sealed, maintenance free nickel-cadmium type, and must operate unattended for a period of not less than five years. Emergency run time must be a minimum of 1-1/2 hours. LEDs must have a minimum rated life of 10 years. Provide self-diagnostic circuitry integral to emergency LED driver.

2.6.2 LED Emergency Drivers

UL 924, NFPA 101. Provide LED emergency driver with automatic power failure detection, test switch and LED indicator (or combination switch/indicator) located on luminaire exterior, and fully-automatic solid-state charger, battery and inverter integral to a self-contained housing. Provide self-diagnostic function integral to emergency driver. Integral nickel-cadmium battery is required to supply a minimum of 90 minutes of emergency power at 10 watts, 10-50 VDC compatible with LED forward voltage requirements, constant output. Driver must be RoHS compliant, rated for installation in plenum-rated spaces and damp locations, and be warranted for a minimum of five years.

2.6.3 Self-Diagnostic Circuitry for LED Drivers

UL 924, NFPA 101. Provide emergency lighting unit with fully-automatic, integral self-testing/diagnostic electronic circuitry. Circuitry must provide for a one minute diagnostic test every 28 days, and a 30 minute diagnostic test every six months, minimum. Any malfunction of the unit must be indicated by LED(s) visible from the exterior of the luminaire. A manual test switch must also be provided to perform a diagnostic test at any given time.

2.7 LUMINAIRE MOUNTING ACCESSORIES

2.7.1 Recess and Surface Mounted Luminaires

Provide access to light source and LED driver from bottom of luminaire. Provide trim and lenses for the exposed surface of flush-mounted

luminaires as indicated on project drawings and specifications. For surface mounted luminaires with brackets, provide flanged metal stem attached to outlet box, with threaded end suitable for supporting the luminaire rigidly in design position. Flanged part of luminaire stud must be of broad base type, secured to outlet box at not fewer than three points.

2.7.2 Luminaire Support Hardware

2.7.2.1 Wire

ASTM A641/A641M. Galvanized, soft tempered steel, minimum 2.7 mm in diameter, or galvanized, braided steel, minimum 2 mm in diameter.

2.7.2.2 Straps

Galvanized steel, 25 by 4.76 mm, conforming to **ASTM A653/A653M**, with a light commercial zinc coating or **ASTM A1008/A1008M** with an electrodeposited zinc coating conforming to **ASTM B633**, Type RS.

2.8 EQUIPMENT IDENTIFICATION

2.8.1 Manufacturer's Nameplate

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.8.2 Labels

UL 1598. All luminaires must be clearly marked for operation of specific light sources and LED drivers. The labels must be easy to read when standing next to the equipment, and durable to match the life of the equipment to which they are attached. Note the following light source characteristics in the format "Use Only _____":

- a. Correlated Color Temperature (CCT) and Color Rendering Index (CRI) for all luminaires.
- b. Driver and dimming protocol.

All markings related to light source type must be clear and located to be readily visible to service personnel, but unseen from normal viewing angles when light sources are in place. LED drivers must have clear markings indicating dimming type and indicate proper terminals for the various outputs.

2.9 FACTORY APPLIED FINISH

NEMA 250. Provide all luminaires and lighting equipment with factory-applied painting system that as a minimum, meets requirements of corrosion-resistance testing.

PART 3 EXECUTION

3.1 INSTALLATION

IEEE C2, NFPA 70.

3.1.1 Light Sources

When light sources are not provided as an integral part of the luminaire, deliver light sources of the type, wattage, lumen output, color temperature (CCT), color rendering index (CRI), and voltage rating indicated to the project site and install just prior to project completion, if not already installed in the luminaires from the factory.

3.1.2 Luminaires

Set luminaires plumb, square, and level with ceiling and walls, in alignment with adjacent luminaires and secure in accordance with manufacturers' directions and approved drawings. Provide accessories as required for ceiling construction type indicated on Finish Schedule. Luminaire catalog numbers do not necessarily denote specific mounting accessories for type of ceiling in which a luminaire may be installed. Provide wires, straps, or rods for luminaire support in this section. Install luminaires with vent holes free of air blocking obstacles.

3.1.2.1 Recessed Luminaires

- a. Support recessed luminaires independently from the building structure by a minimum of two wires, straps or rods per luminaire and located near opposite corners of the luminaire. Secure horizontal movement with clips provided by manufacturer. Ceiling grid clips are not allowed as an alternative to independently supported luminaires.
- b. Do not support luminaires by acoustical tile ceiling panels.
- c. Luminaires installed in suspended ceilings must also comply with the requirements of Section 09 51 00 ACOUSTICAL CEILINGS.

3.1.3 LED Drivers

Provide LED drivers integral to luminaire as constructed by the manufacturer.

3.1.4 Exit Signs

NFPA 101. Wire exit signs and emergency lighting units ahead of the local switch, to the normal lighting circuit located in the same room or area.

3.1.5 Lighting Controls

3.1.5.1 Occupancy/Vacancy Sensors

- a. Provide quantity of sensor units indicated as a minimum. Provide additional units to give full coverage over controlled area. Full coverage must provide hand and arm motion detection for office and administration type areas and walking motion for industrial areas, warehouses, storage rooms and hallways.
- b. Locate ceiling-mounted sensors no closer than **2 meters** from the nearest HVAC supply or return diffuser.
- c. Locate the sensor(s) as indicated and in accordance with the manufacturer's recommendations.

3.2 FIELD QUALITY CONTROL

3.2.1 Tests

3.2.1.1 Lighting Control Verification Tests

Verify lighting control system and devices operate according to approved sequence of operations. Verification tests are to be completed after commissioning.

- a. Verify occupancy/vacancy sensors operate as described in sequence of operations. Provide testing of sensor coverage, sensitivity, and time-out settings in all spaces where sensors are placed. This is to be completed only after all furnishings have been installed. Submit [occupancy/vacancy sensor verification test](#).
- b. Verify photosensors operate to turn on exterior building mounted lighting when ambient light level drops below 3 fc and turn off when ambient light level raises above 5 fc. Adjust photosensor shade away from other site lighting and roadways where vehicle lights could interfere with operation. Perform [Photosensor Verification test](#).

3.2.1.2 Emergency Lighting Test

Interrupt power supply to demonstrate proper operation of emergency lighting. If adjustments are made to the lighting system, re-test system to show compliance with standards.

3.3 CLOSEOUT ACTIVITIES

3.3.1 Commissioning

[NFPA 101](#). Commission all components of the lighting system and lighting control system. Factory Trained Field Service Technician is responsible for calibration and programming sequences for input devices and systems in accordance with the requirements described in the sequence of operation.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 56 00

EXTERIOR LIGHTING

08/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 DEFINITIONS
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Drawing Requirements
 - 1.5.1.1 Luminaire Drawings
 - 1.5.1.2 Pole Drawings
 - 1.5.2 Luminaire Design Data
 - 1.5.3 ANSI/IES LM-79 Test Report
 - 1.5.4 ANSI/IES LM-80 Test Report
 - 1.5.5 ANSI/IES TM-21 Test Report
 - 1.5.6 Test Laboratories
 - 1.5.7 Regulatory Requirements
 - 1.5.8 Standard Products
 - 1.5.8.1 Alternative Qualifications
 - 1.5.8.2 Material and Equipment Manufacturing Date
- 1.6 DELIVERY, STORAGE, AND HANDLING OF POLES
 - 1.6.1 Aluminum Poles
- 1.7 WARRANTY
 - 1.7.1 Luminaire Warranty
 - 1.7.2 Lighting Controls Warranty
 - 1.7.3 Pole Warranty
- 1.8 OPERATION AND MAINTENANCE MANUALS
 - 1.8.1 Lighting System

PART 2 PRODUCTS

- 2.1 PRODUCT COORDINATION
- 2.2 LUMINAIRES
 - 2.2.1 Luminaires
- 2.3 LIGHT SOURCES
 - 2.3.1 LED Light Sources
- 2.4 LED DRIVERS
- 2.5 LIGHTING CONTROLS
 - 2.5.1 System
 - 2.5.2 Devices
 - 2.5.2.1 Photosensors
 - 2.5.2.2 Lighting Contactor
- 2.6 POLES
 - 2.6.1 Aluminum Poles
 - 2.6.2 Brackets and Supports
 - 2.6.3 Pole Foundations
- 2.7 EQUIPMENT IDENTIFICATION

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.7.1 Manufacturer's Nameplate
- 2.7.2 Labels
- 2.8 FACTORY APPLIED FINISH

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Luminaires
 - 3.1.2 LED Drivers
 - 3.1.3 Aluminum Poles
 - 3.1.4 Pole Setting
 - 3.1.5 Lighting Controls
 - 3.1.5.1 Photosensors
 - 3.1.6 Grounding
- 3.2 FIELD QUALITY CONTROL
 - 3.2.1 Tests
 - 3.2.1.1 Lighting Control Verification Test

-- End of Section Table of Contents --

SECTION 26 56 00

EXTERIOR LIGHTING
08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO LTS (2013; Errata 2013) Standard
Specifications for Structural Supports for
Highway Signs, Luminaires and Traffic
Signals

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum
Design Loads and Associated Criteria for
Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM A153/A153M (2023) Standard Specification for Zinc
Coating (Hot-Dip) on Iron and Steel
Hardware

ASTM B108/B108M (2019) Standard Specification for
Aluminum-Alloy Permanent Mold Castings

ASTM B117 (2019) Standard Practice for Operating
Salt Spray (Fog) Apparatus

EUROPEAN UNION (EU)

Directive 2011/65/EU (2011) Restriction of the Use of Certain
Hazardous Substances in Electrical and
Electronic Equipment

ILLUMINATING ENGINEERING SOCIETY (IES)

ANSI/IES LM-79 (2019) Approved Method: Electrical and
Photometric Measurements of Solid State
Lighting Products

ANSI/IES LM-80 (2020) Approved Method: Measuring Luminous
Flux and Color Maintenance of LED
Packages, Arrays and Modules

ANSI/IES LS-1 (2020) Lighting Science: Nomenclature and
Definitions for Illuminating Engineering

ANSI/IES RP-8	(2018; Addenda 1 2020; Errata 1-2 2021) Recommended Practice for Design and Maintenance of Roadway and Parking Facility Lighting
ANSI/IES TM-15	(2020) Technical Memorandum: Luminaire Classification System for Outdoor Luminaires
ANSI/IES TM-21	(2019) Technical Memorandum: Projecting Long-Term Lumen, Photon, and Radiant Flux Maintenance of LED Light Sources
IES Lighting Library	IES Lighting Library
INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)	
IEEE 100	(2000; Archived) The Authoritative Dictionary of IEEE Standards Terms
IEEE C62.41.2	(2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits
NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)	
ANSI C136.3	(2020) Roadway and Area Lighting Equipment - Luminaire Attachments
ANSI C136.13	(2020) Roadway and Area Lighting Equipment - Metal Brackets for Wood Poles
ANSI C136.21	(2014) American National Standard for Roadway and Area Lighting Equipment - Vertical Tenons Used with Post-Top-Mounted Luminaires
NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)
NEMA ANSLG C78.377	(2017) Electric Lamps— Specifications for the Chromaticity of Solid State Lighting Products
NEMA C82.77-10	(2020) Harmonic Emission Limits - Related Power Quality Requirements
NEMA C136.31	(2018) Roadway and Area Lighting Equipment - Luminaire Vibration
NEMA ICS 2	(2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V
NEMA IEC 60529	(2004) Degrees of Protection Provided by Enclosures (IP Code)
NEMA SSL 1	(2016) Electronic Drivers for LED Devices,

Arrays, or Systems

NEMA SSL 3 (2011) High-Power White LED Binning for
General Illumination

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA
20-1; TIA 20-2; TIA 20-3; TIA 20-4)
National Electrical Code

UNDERWRITERS LABORATORIES (UL)

UL 773 (2016; Reprint Jul 2020) UL Standard for
Safety Plug-In, Locking Type Photocontrols
for Use with Area Lighting

UL 773A (2016; Reprint Jun 2020) UL Standard for
Safety Nonindustrial Photoelectric
Switches for Lighting Control

UL 1310 (2018) UL Standard for Safety Class 2
Power Units

UL 1598 (2021; Reprint Jun 2021) Luminaires

UL 8750 (2015; Reprint Sep 2021) UL Standard for
Safety Light Emitting Diode (LED)
Equipment for Use in Lighting Products

1.2 RELATED REQUIREMENTS

Materials not considered to be luminaires, luminaire accessories, or lighting equipment are specified in Section(s) 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION . Luminaires and accessories installed in interior of buildings or attached to the exterior of a building are specified in Section 26 51 00 INTERIOR LIGHTING.

1.3 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications and on the drawings must be as defined in IEEE 100 and ANSI/IES LS-1.
- b. For LED luminaire light sources, "Useful Life" is the operating hours before reaching 70 percent of the initial rated lumen output (L70) with no catastrophic failures under normal operating conditions. This is also known as 70 percent "Rated Lumen Maintenance Life" as defined in ANSI/IES LM-80.
- c. For LED luminaires, "Luminaire Efficacy" (LE) is the appropriate measure of energy efficiency, measured in lumens/watt. This is gathered from LM-79 data for the luminaire, in which absolute photometry is used to measure the lumen output of the luminaire as one entity, not the source separately and then the source and housing together.
- d. Total Harmonic Distortion (THD) is the Root Mean Square (RMS) of all the harmonic components divided by the total fundamental current.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Luminaire Drawings; G

Pole Drawings; G

Control System One-Line Diagram; G

SD-03 Product Data

Luminaires; G

Light Sources; G

LED Drivers; G

Luminaire Warranty; G

Lighting Controls Warranty; G

Pole Warranty; G

Photosensors; G

Lighting Contactor; G

Poles; G

Brackets

SD-05 Design Data

Luminaire Design Data; G

SD-06 Test Reports

ANSI/IES LM-79 Test Report; G

ANSI/IES LM-80 Test Report; G

ANSI/IES TM-21 Test Report; G

SD-08 Manufacturer's Instructions

Poles

SD-10 Operation and Maintenance Data

Lighting System, Data Package 5; G

1.5 QUALITY ASSURANCE

Data, drawings, and reports must employ the terminology, classifications and methods prescribed by the [IES Lighting Library](#) as applicable, for the lighting system specified.

1.5.1 Drawing Requirements

1.5.1.1 Luminaire Drawings

Include dimensions, effective projected area (EPA), weight, accessories, and installation and construction details. Photometric data, including CRI, CCT, TM-15-11 BUG rating, LED driver type, zonal lumen data, and candlepower distribution data per LM-79 must accompany shop drawings.

1.5.1.2 Pole Drawings

Include dimensions, wind load determined in accordance with [ASCE 7-16](#), pole deflection, pole class, and other applicable information.

1.5.2 Luminaire Design Data

- a. Provide distribution data according to IES classification type as defined in [IES Lighting Library](#) and [ANSI/IES RP-8](#).
- b. B.U.G. rating for the installed position as defined by [ANSI/IES TM-15](#) and shielding as defined by [ANSI/IES RP-8](#).
- c. Provide safety certification and file number for the luminaire family. Include listing, labeling and identification in accordance with [NFPA 70](#) (NEC). Applicable testing bodies are determined by the US Occupational Safety Health Administration (OSHA) as Nationally Recognized Testing Laboratories (NRTL) and include: CSA (Canadian Standards Association), ETL (Edison Testing Laboratory), and UL (Underwriters Laboratories).
- d. Provide long term lumen maintenance projections for each LED luminaire in accordance with [ANSI/IES TM-21](#). Data used for projections must be obtained from testing in accordance with [ANSI/IES LM-80](#).
- e. Provide wind loading calculations for luminaires mounted on poles. Weight and effective projected area (EPA) of luminaires and mounting brackets must not exceed maximum rating of pole as installed in particular wind zone area.

1.5.3 [ANSI/IES LM-79 Test Report](#)

Submit test report on manufacturer's standard production model of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data as outlined under "14.0 Test Report" in [ANSI/IES LM-79](#).

1.5.4 [ANSI/IES LM-80 Test Report](#)

Submit report on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data as outlined under "8.0 Test Report" in [ANSI/IES LM-80](#).

1.5.5 ANSI/IES TM-21 Test Report

Submit test report on manufacturer's standard production LED light source (package, array or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data, as well as required interpolation information as outlined under "7.0 Report" in [ANSI/IES TM-21](#).

1.5.6 Test Laboratories

Test laboratories for the [ANSI/IES LM-79](#) and [ANSI/IES LM-80](#) test reports must be one of the following:

- a. National Voluntary Laboratory Accreditation Program (NVLAP) accredited for solid-state lighting testing as part of the Energy-Efficient Lighting Products laboratory accreditation program.
- b. One of the qualified labs listed on the Department of Energy - Energy Efficiency & Renewable Energy, Solid-State Lighting web site.
- c. One of the EPA-Recognized Laboratories listed at for LM-80 testing.

1.5.7 Regulatory Requirements

Equipment, materials, installation, and workmanship must be in accordance with the mandatory provisions of [NFPA 70](#) unless more stringent requirements are specified or indicated. Provide luminaires and assembled components that are approved by and bear the label of UL for the applicable location and conditions unless otherwise specified.

1.5.8 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for six months prior to bid opening. The six-month period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the six-month period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.8.1 Alternative Qualifications

Products having less than a six-month field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.8.2 Material and Equipment Manufacturing Date

Do not use products manufactured more than six months prior to date of delivery to site, unless specified otherwise.

1.6 DELIVERY, STORAGE, AND HANDLING OF POLES

1.6.1 Aluminum Poles

Do not store poles on ground. Support poles so they are at least 305 mm above ground level and growing vegetation. Do not remove factory-applied pole wrappings until just before installing pole.

1.7 WARRANTY

Support all equipment items by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.7.1 Luminaire Warranty

Provide and transfer to the government the original LED luminaire manufacturers standard commercial warranty for each different luminaire manufacturer used in the project.

- a. Provide a written five year minimum replacement warranty for material, luminaire finish, and workmanship. Provide written warranty document that contains all warranty processing information needed, including customer service point of contact, whether or not a return authorization number is required, return shipping information, and closest return location to the luminaire location.

- (1) Finish warranty must include failure and substantial deterioration such as blistering, cracking, peeling, chalking, or fading.

- (2) Material warranty must include:

- (a) All LED drivers and integral control equipment.

- (b) Replacement when more than 15 percent of LED sources in any lightbar or subassembly(s) are defective, non-starting, or operating below 70 percent of specified lumen output.

- b. Warranty period must begin in accordance with the manufacturer's standard warranty starting date.

- c. Provide replacements that are promptly shipped, without charge, to the using Government facility point of contact and that are identical to or an improvement upon the original equipment. All replacements must include testing of new components and installation.

1.7.2 Lighting Controls Warranty

Provide and transfer to the government the original lighting controls manufacturers standard commercial warranty for each different lighting controls manufacturer used in the project. Warranty coverage must begin from date of final system commissioning or three months from date of delivery, whichever is the earliest. Warranty service must be performed by a factory-trained engineer or technician.

- a. Unless otherwise noted, provide a written five year minimum warranty on the complete system for all systems with factory commissioning. Provide warranty that covers 100 percent of the cost of any

replacement parts and services required over the five years which are directly attributable to the product failure. Failures include, but are not limited to, the following:

- (1) Damage of electronic components due to transient voltage surges.
 - (2) Failure of control devices, including but not limited to photosensors and motion sensors.
- b. Provide a written five year minimum warranty on all input devices against defect in workmanship or materials provided by device manufacturer.
 - c. Provide a written five year minimum warranty on all control components attached to luminaires against defect in workmanship or materials.

1.7.3 Pole Warranty

Provide and transfer to the government the original pole manufacturers standard commercial warranty for each different pole manufacturer used in the project. Warranty coverage must begin from date of final system commissioning or three months from date of delivery, whichever is the earliest. Provide a written three year minimum replacement warranty for material, luminaire finish, and workmanship. Warranty service must be performed by a factory-trained engineer or technician.

1.8 OPERATION AND MAINTENANCE MANUALS

1.8.1 Lighting System

Provide operation and maintenance manuals for the lighting system in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that provide basic data relating to the design, operation, and maintenance of the lighting system. Additional requirements for the Navy are provided in Section 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI). Include the following:

- a. Control System One-Line Diagram.
- b. Manufacturers' operating and maintenance manuals.
- c. Luminaire shop drawings for modified and custom luminaires.
- d. Luminaire Manufacturers' standard commercial warranty information as specified in paragraph LUMINAIRE WARRANTY.

PART 2 PRODUCTS

2.1 PRODUCT COORDINATION

2.2 LUMINAIRES

UL 1598, NEMA C82.77-10. Provide luminaires as indicated in the luminaire schedule and XL plates or details on project plans, complete with light source, wattage, and lumen output indicated. All luminaires of the same type must be provided by the same manufacturer. Luminaires must be specifically designed for use with the LED driver and light source provided.

2.2.1 Luminaires

UL 8750, ANSI/IES LM-79, ANSI/IES LM-80. For all luminaires, provide:

- a. Complete system with LED drivers and light sources.
- b. Housing constructed of non-corrosive materials. All new aluminum housings must be anodized or powder-coated. All new steel housings must be treated to be corrosion resistant.
- c. ANSI/IES TM-21, ANSI/IES LM-80. Minimum L70 lumen maintenance value of 50,000 hours unless otherwise indicated in the luminaire schedule. Luminaire drive current value must be identical to that provided by test data for luminaire in question.
- d. Minimum efficacy as specified in the luminaire schedule. Theoretical models of initial lamp lumens per watt are not acceptable. If efficacy values are not listed in the luminaire schedule, provide luminaires that meet the following minimum values:

Luminaire Style	Minimum Luminaire Efficacy
Area and Roadway (pole mounted, arm mounted)	119 LPW

- e. Product rated for operation within an ambient temperature range of minus 30 degrees C to 40 degrees C.
- f. UL listed for wet locations. Optical compartment for LED luminaires must be sealed and rated a minimum of IP65 per NEMA IEC 60529.
- g. IES Lighting Library. Light distribution and NEMA field angle classifications as indicated in luminaire schedule on project plans.
- h. Housing finish that is baked-on enamel, anodized, or baked-on powder coat paint. Finish must be capable of surviving ASTM B117 salt fog environment testing for 2500 hours minimum without blistering or peeling.
- i. LED driver and light source package, array, or module are accessible for service or replacement without removal or destruction of luminaire.
- j. ANSI/IES TM-15. Does not exceed the BUG ratings as listed in the luminaire schedule. If BUG ratings are not listed in the luminaire schedule, provide luminaires that meet the following minimum values for each application and mounting conditions:

Lighting Application	Mounting Conditions	BUG Rating
Area and Roadway	All	B3-U0-G3

- k. Fully assembled and electrically tested prior to shipment from factory.
- l. Finish color is as indicated in the luminaire schedule or detail on the project plans.
- m. Lenses constructed of clear tempered glass or UV-resistant acrylic.

- n. All factory electrical connections are made using crimp, locking, or latching style connectors. Twist-style wire nuts are not acceptable.
- o. **NEMA C136.31**. Comply with 3G vibration testing.
- p. Luminaire arm bolts constructed from 304 stainless steel or zinc-plated steel.
- q. Wiring compartment on pole-mounted, street and area luminaires is accessible without the use of hand tools to manipulate small screws, bolts, or hardware.
- r. Incorporate modular electrical connections, and construct luminaires to allow replacement of all or any part of the optics, heat sinks, LED drivers, surge suppressors and other electrical components using only a simple tool, such as a manual or cordless electric screwdriver.
- s. **ANSI C136.3**. For all roadway and area luminaires, provide products with an integral tilt adjustment of plus or minus 5 degrees to allow the unit to be leveled.

2.3 LIGHT SOURCES

NEMA ANSLG C78.377, NEMA SSL 3. Provide type, lumen rating, and wattage as indicated in luminaire schedule on project plans.

2.3.1 LED Light Sources

Provide LED light sources that meet the following requirements:

- a. **NEMA ANSLG C78.377**. Emit white light and have a nominal Correlated Color Temperature (CCT) of 4000 Kelvin.
- b. Minimum Color Rendering Index (CRI) of 80.
- c. **Directive 2011/65/EU**. Restriction of Hazardous Substances (RoHS) compliant.
- d. Light source color consistency by utilizing a binning tolerance within a 4-step McAdam ellipse.

2.4 LED DRIVERS

NEMA SSL 1, UL 1310. Provide LED Drivers that are electronic, UL Class 1 or Class 2, constant-current type and meet the following requirements:

- a. The combined LED driver and LED light source system is greater than or equal to the minimum luminaire efficacy values as listed in the luminaire schedule provided.
- b. Operate at a voltage of 120-277 volts at 50/60 hertz, with input voltage fluctuations of plus or minus 10 percent.
- c. Power Factor (PF) greater than or equal to 0.90 at full input power and across specified dimming range.
- d. Maximum Total Harmonic Distortion (THD) less than or equal to 20 percent at full input power and across specified dimming range.

- e. Operates for at least 50,000 hours at maximum case temperature and 90 percent non-condensing relative humidity.
- f. Meets the "Elevated" (10kV/10kA) requirements per IEEE C62.41.2 -2002. Manufacturer must indicate whether failure of the electrical immunity system can possibly result in disconnect of power to luminaire. Provide surge protection that is integral to the LED driver.
- g. Contains integral thermal protection that reduces the output power to protect the driver and light source from damage if the case temperature approaches or exceeds the driver's maximum operating temperature.
- h. Complies with the requirements of the Federal Communications Commission (FCC) rules and regulations, Title 47 CFR part 15, Non-Consumer (Class A) for EMI/RFI (conducted and radiated).
- i. Class A sound rating for all drivers mounted under a covered structure, such as a canopy, or where otherwise appropriate.
- j. Directive 2011/65/EU. Restriction of Hazardous Substances (RoHS) compliant.
- k. UL listed for wet locations typical of exterior installations.
- l. Rated to operate between ambient temperatures of minus 30 degrees C and 40 degrees C.

2.5 LIGHTING CONTROLS

2.5.1 System

Provide exterior lighting control system that operates the exterior lighting system as described in the exterior lighting control strategies in the project plans.

2.5.2 Devices

2.5.2.1 Photosensors

UL 773, UL 773A. Provide Photosensors that meet the following requirements:

- a. Hermetically sealed, silicon diode light sensor type, rated at 1200 watts, 120 volts, 50/60 Hz with single-pole, double-throw contacts.
- b. Turns ON at 10 to 30 lux and turns OFF at 30 to 150 lux.
- c. Designed to fail to the ON position.
- d. Housing is constructed of UV stabilized polypropylene, rated to operate within a temperature range of minus 40 to 70 degrees C.
- e. Time delay that prevents accidental switching from transient light sources.
- f. Directional lens in front of the cell to prevent fixed light sources from creating a turnoff condition.

- g. Designed for 20-year service to match life expectancy of long-life LED fixtures and exceed 15,000 operations at full load. Provide photosensors with zero-cross technology to withstand severe in-rush current and extend relay life.
- h. Fixed base type housing with 12.7 mm threaded base for mounting to a junction box or conduit.
- i. Provide photosensors with metal oxide varistor (MOV) type surge protection.

2.5.2.2 Lighting Contactor

NEMA ICS 2. Provide a mechanically-held lighting contactor housed in a NEMA 1 enclosure conforming to NEMA ICS 6. Contactor must have 4 poles, configured as normally open (NO). Contacts must be rated 600 volts, 30 amperes for a resistive load. Coil operating voltage must be 120 volts. Contactor must have silver cadmium oxide double-break contacts and coil clearing contacts for mechanically held contactors and must require no arcing contacts. Provide contactor with hand-off-automatic selector switch. Provide contactor as specified above along with disconnect switch in integral NEMA 1 enclosure with flange-mounted handle to satisfy requirement for a "combination lighting contactor" when specified.

2.6 POLES

ASCE 7-16. Provide round tapered poles designed for wind loading of 177 km/hr while supporting luminaires and all other appurtenances indicated. The effective projected areas (EPA) of luminaires and appurtenances used in calculations must be specific for the actual products provided on each pole. Provide poles that are anchor-base type designed for use with underground supply conductors. Poles must have oval-shaped hand hole having a minimum clear opening of 80 by 130 mm. Secure hand hole covers by stainless steel captive screws. Provide metal poles with an internal grounding connection accessible from the hand hole near the bottom of each pole. Install a means of wire disconnection accessible from the hand hole. Do not install square poles. Provide poles from a Manufacturer with a standard provision for protecting the finish during shipment and installation. Do not install scratched, stained, chipped, or dented poles.

2.6.1 Aluminum Poles

Provide aluminum poles with anodized dark bronze finish unless otherwise noted in luminaire schedule on project plans. Do not paint aluminum poles. Provide poles that meet the following requirements:

- a. **AASHTO LTS.** Manufactured of corrosion resistant aluminum alloys for Alloy 6063-T6 or Alloy 6005-T5 for wrought alloys and Alloy 356-T4 (3,5) for cast alloys.
- b. Seamless extruded or spun seamless-type with minimum 4.8 mm wall thickness.
- c. Top of shaft is fitted with a round or tapered cover.
- d. **ASTM B108/B108M.** Pole base is mounted by anchor bolts, made of cast 356-T6 aluminum alloy. Base must be machined to receive the lower end of shaft.

- e. Joint between shaft and base is welded.
- f. **ASTM B108/B108M**. Base cover is cast 356-T6 aluminum alloy.
- g. All hardware other than anchor bolts are either 2024-T4 anodized aluminum alloy or stainless steel.
- h. Grounding connection is designed to prevent electrolysis when used with copper ground wire.

2.6.2 Brackets and Supports

ANSI C136.3, **ANSI C136.13**, and **ANSI C136.21**. Provide pole brackets that are not less than **31.75 mm** aluminum secured to pole. Slip-fitter or pipe-threaded brackets may be used, but brackets must be coordinated to luminaires provided, and brackets for use with one type of luminaire must be identical. Brackets for pole-mounted street lights must correctly position luminaire no lower than mounting height indicated. Mount brackets not less than **7320 mm** above street. Provide special mountings or brackets as indicated and of metal which will not promote galvanic reaction with luminaire head.

2.6.3 Pole Foundations

Provide anchor bolts consisting of a steel rod with a minimum yield strength of **344.5 MPa**; the top **305 mm** of the rod must be galvanized in accordance with **ASTM A153/A153M**. Concrete must be as specified in Section **03 30 00** CAST-IN-PLACE CONCRETE and Section **33 71 02** UNDERGROUND ELECTRICAL DISTRIBUTION.

2.7 EQUIPMENT IDENTIFICATION

2.7.1 Manufacturer's Nameplate

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.7.2 Labels

UL 1598. Luminaires must be clearly marked for operation of specific light sources and drivers according to proper light source type. Note the following luminaire characteristics in the format "Use Only ____":

- a. Correlated color temperature (CCT) and color rendering index (CRI) for all luminaires.
- b. Driver.

Markings related to light source type must be clear and located to be readily visible to service personnel, but unseen from normal viewing angles when light sources are in place. LED drivers must have clear markings indicating dimming type and indicate proper terminals for the various outputs.

2.8 FACTORY APPLIED FINISH

NEMA 250. Provide all luminaires and lighting equipment with factory-applied painting system that as a minimum meets requirements of corrosion-resistance testing.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Luminaires

Install all luminaires in accordance with the luminaire manufacturer's written instructions. Install all luminaires at locations and heights as indicated on the project plans. Level all luminaires in accordance to manufacturer's written instructions.

3.1.2 LED Drivers

Provide LED drivers integral to luminaire as constructed by the manufacturer.

3.1.3 Aluminum Poles

Provide pole foundations with galvanized steel anchor bolts, threaded at the top end and bent 90 degrees at the bottom end. Provide ornamental covers to match pole and galvanized nuts and washers for anchor bolts. Concrete for anchor bases, polyvinyl chloride (PVC) conduit bells, and ground rods must be as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE and Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Thoroughly compact backfill with compacting arranged to prevent pressure between conductor, jacket, or sheath and the end of conduit ell. Adjust poles as necessary to provide a permanent vertical position with the bracket arm in proper position for luminaire location.

3.1.4 Pole Setting

Set pole to depth as indicated. Dig holes large enough to permit the proper use of tampers to the full depth of the hole. Place backfill in the hole in 150 mm maximum layers and thoroughly tamp. Place surplus earth around the pole in a conical shape and pack tightly to drain water away.

3.1.5 Lighting Controls

3.1.5.1 Photosensors

Aim photosensor according to manufacturer's recommendations. Mount sensor where indicated. Set adjustable window slide for 30 lux photosensor turn-on.

3.1.6 Grounding

Ground noncurrent-carrying parts of equipment including metal poles, luminaires, mounting arms, brackets, and metallic enclosures as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Where copper grounding conductor is connected to a metal other than copper, provide specially treated or lined connectors suitable for this purpose.

3.2 FIELD QUALITY CONTROL

3.2.1 Tests

Upon completion of installation, verify that equipment is properly installed, connected, and adjusted. Perform initial operational test, consisting of the entire system energized for 72 consecutive hours without any failures of any kind occurring in the system. All circuits must test clear of faults, grounds, and open circuits.

3.2.1.1 Lighting Control Verification Test

Verify lighting control system and devices operate according to approved sequence of operations. Verification tests are to be completed after commissioning.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 27 - COMMUNICATIONS

SECTION 27 10 00

BUILDING TELECOMMUNICATIONS CABLING SYSTEM

08/11

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 DEFINITIONS
 - 1.3.1 Campus Distributor (CD)
 - 1.3.2 Telecommunications Room (TR)
 - 1.3.3 Entrance Facility (EF) (Telecommunications)
 - 1.3.4 Open Cable
 - 1.3.5 Pathway
- 1.4 SYSTEM DESCRIPTION
- 1.5 SUBMITTALS
- 1.6 QUALITY ASSURANCE
 - 1.6.1 Shop Drawings
 - 1.6.1.1 Telecommunications Drawings
 - 1.6.1.2 Telecommunications Space Drawings
 - 1.6.2 Telecommunications Qualifications
 - 1.6.2.1 Telecommunications Contractor
 - 1.6.2.2 Key Personnel
 - 1.6.2.3 Minimum Manufacturer Qualifications
 - 1.6.3 Test Plan
 - 1.6.4 Regulatory Requirements
 - 1.6.5 Standard Products
 - 1.6.5.1 Alternative Qualifications
 - 1.6.5.2 Material and Equipment Manufacturing Date
- 1.7 DELIVERY AND STORAGE
- 1.8 ENVIRONMENTAL REQUIREMENTS
- 1.9 WARRANTY
- 1.10 MAINTENANCE
 - 1.10.1 Operation and Maintenance Manuals
 - 1.10.2 Record Documentation

PART 2 PRODUCTS

- 2.1 COMPONENTS
- 2.2 TELECOMMUNICATIONS PATHWAY
- 2.3 TELECOMMUNICATIONS CABLING
 - 2.3.1 Horizontal Cabling
 - 2.3.1.1 Horizontal Copper
- 2.4 TELECOMMUNICATIONS SPACES
 - 2.4.1 Backboards
 - 2.4.2 Equipment Support Frame
 - 2.4.3 Connector Blocks
 - 2.4.4 Cable Guides
 - 2.4.5 Patch Panels
 - 2.4.5.1 Modular to 110 Block Patch Panel

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.5 TELECOMMUNICATIONS OUTLET/CONNECTOR ASSEMBLIES
 - 2.5.1 Outlet/Connector Copper
 - 2.5.2 Cover Plates
- 2.6 GROUNDING AND BONDING PRODUCTS
- 2.7 MANUFACTURER'S NAMEPLATE
- 2.8 FIELD FABRICATED NAMEPLATES
- 2.9 TESTS, INSPECTIONS, AND VERIFICATIONS
 - 2.9.1 Factory Reel Tests

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Cabling
 - 3.1.1.1 Horizontal Cabling
 - 3.1.2 Pathway Installations
 - 3.1.3 Service Entrance Conduit, Underground
 - 3.1.4 Work Area Outlets
 - 3.1.4.1 Terminations
 - 3.1.4.2 Cover Plates
 - 3.1.4.3 Cables
 - 3.1.4.4 Pull Cords
 - 3.1.5 Telecommunications Space Termination
 - 3.1.5.1 Connector Blocks
 - 3.1.5.2 Patch Panels
 - 3.1.5.3 Equipment Support Frames
 - 3.1.6 Grounding and Bonding
- 3.2 LABELING
 - 3.2.1 Labels
 - 3.2.2 Cable
 - 3.2.3 Termination Hardware
- 3.3 FIELD APPLIED PAINTING
 - 3.3.1 Painting Backboards
- 3.4 FIELD FABRICATED NAMEPLATE MOUNTING
- 3.5 TESTING
 - 3.5.1 Telecommunications Cabling Testing
 - 3.5.1.1 Inspection
 - 3.5.1.2 Verification Tests
 - 3.5.1.3 Performance Tests
 - 3.5.1.4 Final Verification Tests

-- End of Section Table of Contents --

SECTION 27 10 00

BUILDING TELECOMMUNICATIONS CABLING SYSTEM
08/11

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D709 (2017) Standard Specification for
Laminated Thermosetting Materials

ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

ECIA EIA/ECA 310-E (2005) Cabinets, Racks, Panels, and
Associated Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 100 (2000; Archived) The Authoritative
Dictionary of IEEE Standards Terms

INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)

ICEA S-90-661 (2021) Category 3 and 5E Individually
Unshielded Twisted Pairs, Indoor Cables
(With or Without an Overall Shield) for
Use in General Purpose and LAN
Communications Wiring Systems

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)

NECA/BICSI 568 (2006) Standard for Installing Building
Telecommunications Cabling

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA WC 66 (2019) Performance Standard for Category 6
and Category 7 100 Ohm Shielded and
Unshielded Twisted Pairs

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA
20-1; TIA 20-2; TIA 20-3; TIA 20-4)
National Electrical Code

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-568.0 (2020e) Generic Telecommunications Cabling
for Customer Premises

TIA-568.1	(2020e) Commercial Building Telecommunications Infrastructure Standard
TIA-568.2	(2018d) Balanced Twisted-Pair Telecommunications Cabling and Components Standards
TIA-568.3	(2016d; Add 1 2019) Optical Fiber Cabling Components Standard
TIA-569	(2019e) Telecommunications Pathways and Spaces
TIA-606	(2021d) Administration Standard for Telecommunications Infrastructure
TIA-607	(2019d) Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
TIA-1152	(2016; R 2021) Requirements for Field Test Instruments and Measurements for Balanced Twisted-Pair Cabling

U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

FCC Part 68	Connection of Terminal Equipment to the Telephone Network (47 CFR 68)
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UNDERWRITERS LABORATORIES (UL)

UL 50	(2015) UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations
UL 444	(2017; Reprint Jun 2021) UL Standard for Safety Communications Cables
UL 467	(2013; Reprint Jun 2017) UL Standard for Safety Grounding and Bonding Equipment
UL 514C	(2014; Reprint Feb 2020) UL Standard for Safety Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers
UL 723	(2018) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials
UL 969	(2017; Reprint Mar 2018) UL Standard for Safety Marking and Labeling Systems
UL 1863	(2004; Reprint Oct 2019) UL Standard for Safety Communication Circuit Accessories

1.2 RELATED REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and Section 33 82 00

TELECOMMUNICATIONS, OUTSIDE PLANT (OSP), apply to this section with additions and modifications specified herein.

1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in this specification shall be as defined in TIA-568.1, TIA-568.2, TIA-568.3, TIA-569, TIA-606 and IEEE 100 and herein.

1.3.1 Campus Distributor (CD)

A distributor from which the campus backbone cabling emanates. (International expression for main cross-connect (MC).)

1.3.2 Telecommunications Room (TR)

An enclosed space for housing telecommunications equipment, cable, terminations, and cross-connects. The room is the recognized cross-connect between the backbone cable and the horizontal cabling.

1.3.3 Entrance Facility (EF) (Telecommunications)

An entrance to the building for both private and public network service cables (including wireless) including the entrance point at the building wall and continuing to the equipment room.

1.3.4 Open Cable

Cabling that is not run in a raceway as defined by NFPA 70. This refers to cabling that is "open" to the space in which the cable has been installed and is therefore exposed to the environmental conditions associated with that space.

1.3.5 Pathway

A physical infrastructure utilized for the placement and routing of telecommunications cable.

1.4 SYSTEM DESCRIPTION

The building telecommunications cabling and pathway system shall include permanently installed backbone and horizontal cabling, horizontal and backbone pathways, service entrance facilities, work area pathways, telecommunications outlet assemblies, conduit, raceway, and hardware for splicing, terminating, and interconnecting cabling necessary to transport telephone and data (including LAN) between equipment items in a building. The horizontal system shall be wired in a star topology from the telecommunications work area to the campus distributor at the center or hub of the star. The backbone cabling and pathway system includes intrabuilding and interbuilding interconnecting cabling, pathway, and terminal hardware. The intrabuilding backbone provides connectivity from the floor distributors to the building distributors or to the campus distributor and from the building distributors to the campus distributor as required. The backbone system shall be wired in a star topology with the campus distributor at the center or hub of the star. The interbuilding backbone system provides connectivity between the campus distributors and is specified in Section 33 82 00 TELECOMMUNICATIONS OUTSIDE PLANT (OSP). Provide telecommunications pathway systems referenced herein as specified in Section 26 20 00 INTERIOR DISTRIBUTION

SYSTEM. In order for Telefonica to provide maintenance to this installation, it is recommended to contract Telefonica or a company designated by Telefonica to avoid future problems with any connection to the existing infrastructure. If another company performs the installation, Telefonica will not be responsible for the maintenance of the installation.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Telecommunications Drawings; G

Telecommunications Space Drawings; G

In addition to Section 01 33 00 SUBMITTAL PROCEDURES, provide shop drawings in accordance with paragraph SHOP DRAWINGS.

SD-03 Product Data

Telecommunications Cabling (backbone and horizontal); G

Patch Panels; G

Telecommunications Outlet/Connector Assemblies; G

Equipment Support Frame; G

Connector Blocks; G

Submittals shall include the manufacturer's name, trade name, place of manufacture, and catalog model or number. Include performance and characteristic curves. Submittals shall also include applicable federal, military, industry, and technical society publication references. Should manufacturer's data require supplemental information for clarification, the supplemental information shall be submitted as specified in paragraph REGULATORY REQUIREMENTS and as required in Section 01 33 00 SUBMITTAL PROCEDURES.

SD-06 Test Reports

Telecommunications Cabling Testing; G

SD-07 Certificates

Telecommunications Contractor Qualifications; G

Key Personnel Qualifications; G

Manufacturer Qualifications; G

Test Plan; G

SD-09 Manufacturer's Field Reports

Factory Reel Tests; G

SD-10 Operation and Maintenance Data

Telecommunications Cabling and Pathway System Data Package 5; G

SD-11 Closeout Submittals

Record Documentation; G

1.6 QUALITY ASSURANCE

1.6.1 Shop Drawings

In exception to Section 01 33 00 SUBMITTAL PROCEDURES, submitted plan drawings shall be a minimum of 279 by 432 mm in size using a minimum scale of 1 mm per 100 mm. Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Wiring diagrams shall identify circuit terminals and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Drawings shall indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices. Submittals shall include the nameplate data, size, and capacity. Submittals shall also include applicable federal, military, industry, and technical society publication references.

1.6.1.1 Telecommunications Drawings

Provide registered communications distribution designer (RCDD) approved, drawings in accordance with TIA-606. The identifier for each termination and cable shall appear on the drawings. Drawings shall depict final telecommunications installed wiring system infrastructure in accordance with TIA-606. The drawings should provide details required to prove that the distribution system shall properly support connectivity from the EF telecommunications and ER telecommunications, CD's to the telecommunications work area outlets. The following drawings shall be provided as a minimum:

- a. T1 - Layout of complete building per floor - Building Area/Serving Zone Boundaries, Backbone Systems, and Horizontal Pathways. Layout of complete building per floor. The drawing indicates location of building areas, serving zones, vertical backbone diagrams, telecommunications rooms, access points, pathways, grounding system, and other systems that need to be viewed from the complete building perspective.
- b. T2 - Serving Zones/Building Area Drawings - Drop Locations and Cable Identification (ID'S). Shows a building area or serving zone. These drawings show drop locations, telecommunications rooms, access points and detail call outs for common equipment rooms and other congested areas.
- c. T4 - Typical Detail Drawings - Faceplate Labeling, Firestopping, Americans with Disabilities Act (ADA), Safety, Department of Transportation (DOT). Detailed drawings of symbols and typicals such

as faceplate labeling, faceplate types, faceplate population installation procedures, detail racking, and raceways.

1.6.1.2 Telecommunications Space Drawings

Provide T3 drawings in accordance with TIA-606 that include telecommunications rooms plan views, pathway layout (cable tray, racks, ladder-racks, etc.), mechanical/electrical layout, and backboard and wall elevations. Drawings shall show layout of applicable equipment including incoming cable stub or connector blocks, building protector assembly, outgoing cable connector blocks, patch panels and equipment spaces and cabinet/racks. Drawings shall include a complete list of equipment and material, equipment rack details, proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearance for maintenance and operation. Drawings may also be an enlargement of a congested area of T1 or T2 drawings.

1.6.2 Telecommunications Qualifications

Work under this section shall be performed by and the equipment shall be provided by the approved telecommunications contractor and key personnel. Qualifications shall be provided for: the telecommunications system contractor, the telecommunications system installer, and the supervisor (if different from the installer). A minimum of 30 days prior to installation, submit documentation of the experience of the telecommunications contractor and of the key personnel.

1.6.2.1 Telecommunications Contractor

The telecommunications contractor shall be a firm which is regularly and professionally engaged in the business of the applications, installation, and testing of the specified telecommunications systems and equipment. The telecommunications contractor shall demonstrate experience in providing successful telecommunications systems within the past 3 years of similar scope and size. Submit documentation for a minimum of three and a maximum of five successful telecommunication system installations for the telecommunications contractor.

1.6.2.2 Key Personnel

Provide key personnel who are regularly and professionally engaged in the business of the application, installation and testing of the specified telecommunications systems and equipment. There may be one key person or more key persons proposed for this solicitation depending upon how many of the key roles each has successfully provided. Each of the key personnel shall demonstrate experience in providing successful telecommunications systems within the past 3 years.

Supervisors and installers assigned to the installation of this system or any of its components shall be Building Industry Consulting Services International (BICSI) Registered Cabling Installers, Technician Level. Submit documentation of current BICSI certification for each of the key personnel.

In lieu of BICSI certification, supervisors and installers assigned to the installation of this system or any of its components shall have a minimum of 3 years experience in the installation of the specified copper and fiber optic cable and components. They shall have factory or factory approved certification from each equipment manufacturer indicating that

they are qualified to install and test the provided products. Submit documentation for a minimum of three and a maximum of five successful telecommunication system installations for each of the key personnel. Documentation for each key person shall include at least two successful system installations provided that are equivalent in system size and in construction complexity to the telecommunications system proposed for this solicitation. Include specific experience in installing and testing telecommunications systems and provide the names and locations of at least two project installations successfully completed using copper telecommunications cabling systems. All of the existing telecommunications system installations offered by the key persons as successful experience shall have been in successful full-time service for at least 18 months prior to the issuance date for this solicitation. Provide the name and role of the key person, the title, location, and completed installation date of the referenced project, the referenced project owner point of contact information including name, organization, title, and telephone number, and generally, the referenced project description including system size and construction complexity.

Indicate that all key persons are currently employed by the telecommunications contractor, or have a commitment to the telecommunications contractor to work on this project. All key persons shall be employed by the telecommunications contractor at the date of issuance of this solicitation, or if not, have a commitment to the telecommunications contractor to work on this project by the date that the bid was due to the Contracting Officer.

Note that only the key personnel approved by the Contracting Officer in the successful proposal shall do work on this solicitation's telecommunications system. Key personnel shall function in the same roles in this contract, as they functioned in the offered successful experience. Any substitutions for the telecommunications contractor's key personnel requires approval from The Contracting Officer.

1.6.2.3 Minimum [Manufacturer Qualifications](#)

Cabling, equipment and hardware manufacturers shall have a minimum of 3 years experience in the manufacturing, assembly, and factory testing of components which comply with [TIA-568.1](#), [TIA-568.2](#) and [TIA-568.3](#).

1.6.3 [Test Plan](#)

Provide a complete and detailed test plan for the telecommunications cabling system including a complete list of test equipment for the components and accessories for each cable type specified, 60 days prior to the proposed test date. Include procedures for certification, validation, and testing.

1.6.4 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship shall be in accordance with the mandatory and advisory provisions of [NFPA 70](#) unless more stringent requirements are specified or indicated.

1.6.5 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products shall have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period shall include applications of equipment and materials under similar circumstances and of similar size. The product shall have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items shall be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.6.5.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.6.5.2 Material and Equipment Manufacturing Date

Products manufactured more than 1 year prior to date of delivery to site shall not be used, unless specified otherwise.

1.7 DELIVERY AND STORAGE

Provide protection from weather, moisture, extreme heat and cold, dirt, dust, and other contaminants for telecommunications cabling and equipment placed in storage.

1.8 ENVIRONMENTAL REQUIREMENTS

Connecting hardware shall be rated for operation under ambient conditions of 0 to 60 degrees C and in the range of 0 to 95 percent relative humidity, noncondensing.

1.9 WARRANTY

The equipment items shall be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.10 MAINTENANCE

1.10.1 Operation and Maintenance Manuals

Commercial off the shelf manuals shall be furnished for operation, installation, configuration, and maintenance of products provided as a part of the telecommunications cabling and pathway system, Data Package 5. Submit operations and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein not later than 2 months prior to the date of beneficial occupancy. In addition to requirements of Data Package 5, include the requirements of paragraphs TELECOMMUNICATIONS DRAWINGS, TELECOMMUNICATIONS SPACE DRAWINGS, and RECORD DOCUMENTATION. Ensure that these drawings and documents depict the

as-built configuration.

1.10.2 Record Documentation

Provide T5 drawings including documentation on cables and termination hardware in accordance with TIA-606. T5 drawings shall include schedules to show information for cut-overs and cable plant management, patch panel layouts and cover plate assignments, cross-connect information and connecting terminal layout as a minimum. T5 drawings shall be provided on electronic media using Windows based computer cable management software. Provide the following T5 drawing documentation as a minimum:

- a. Cables - A record of installed cable shall be provided in accordance with TIA-606. The cable records shall include the required data fields for each cable and complete end-to-end circuit report for each complete circuit from the assigned outlet to the entry facility in accordance with TIA-606. Include manufacture date of cable with submittal.
- b. Termination Hardware - A record of installed patch panels, cross-connect points, distribution frames, terminating block arrangements and type, and outlets shall be provided in accordance with TIA-606. Documentation shall include the required data fields as a minimum in accordance with TIA-606.

PART 2 PRODUCTS

2.1 COMPONENTS

Components shall be UL or third party certified. Where equipment or materials are specified to conform to industry and technical society reference standards of the organizations, submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. The certificate shall state that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard. Provide a complete system of telecommunications cabling and pathway components using star topology. Provide support structures and pathways, complete with outlets, cables, connecting hardware and telecommunications cabinets/racks. Cabling and interconnecting hardware and components for telecommunications systems shall be UL listed or third party independent testing laboratory certified, and shall comply with NFPA 70 and conform to the requirements specified herein.

2.2 TELECOMMUNICATIONS PATHWAY

Provide telecommunications pathways in accordance with TIA-569 and as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.3 TELECOMMUNICATIONS CABLING

Cabling shall be UL listed for the application and shall comply with TIA-568.0, TIA-568.1, TIA-568.2, TIA-568.3 and NFPA 70. Provide a labeling system for cabling as required by TIA-606 and UL 969. Ship cable on reels or in boxes bearing manufacture date for for unshielded twisted pair (UTP) in accordance with ICEA S-90-661 for all cable used on this

project. Cabling manufactured more than 12 months prior to date of installation shall not be used.

2.3.1 Horizontal Cabling

Provide horizontal cable in compliance with NFPA 70 and performance characteristics in accordance with TIA-568.1.

2.3.1.1 Horizontal Copper

Provide horizontal copper cable, UTP, 100 ohm in accordance with TIA-568.2, UL 444, ANSI/NEMA WC 66, ICEA S-90-661. Provide four each individually twisted pair, minimum size 24 AWG conductors, Category 6, with a blue thermoplastic jacket. Cable shall be imprinted with manufacturers name or identifier, flammability rating, gauge of conductor, transmission performance rating (category designation) and length marking at regular intervals in accordance with ICEA S-90-661. Provide plenum (CMP), riser (CMR), or general purpose (CM or CMG) communications rated cabling in accordance with NFPA 70. Substitution of a higher rated cable shall be permitted in accordance with NFPA 70. Cables installed in conduit within and under slabs shall be UL listed and labeled for wet locations in accordance with NFPA 70.

2.4 TELECOMMUNICATIONS SPACES

Provide connecting hardware and termination equipment in the telecommunications entrance facility to facilitate installation as shown on design drawings for terminating and cross-connecting permanent cabling. Provide telecommunications interconnecting hardware color coding in accordance with TIA-606.

2.4.1 Backboards

Provide void-free, interior grade A-C plywood 19 mm thick 1200 by 2400 mm . Backboards shall be fire rated by manufacturing process. Fire stamp shall be clearly visible. Paint applied over fire retardant backboard shall be UL 723 fire retardant paint. Provide label including paint manufacturer, date painted, UL listing and name of Installer. When painted, paint label and fire stamp shall be clearly visible.

2.4.2 Equipment Support Frame

Provide in accordance with ECIA EIA/ECA 310-E and UL 50.

- a. Bracket, wall mounted, 8 gauge aluminum. Provide hinged bracket compatible with 482.6 mm panel mounting.

2.4.3 Connector Blocks

Provide insulation displacement connector (IDC) Type 110 for Category 6 systems. Provide blocks for the number of horizontal and backbone cables terminated on the block plus 25 percent spare.

2.4.4 Cable Guides

Provide cable guides specifically manufactured for the purpose of routing cables, wires and patch cords horizontally and vertically on 482.6 mm equipment telecommunications backboards. Cable guides of ring or bracket

type devices mounted on panels backboard for horizontal cable management and individually mounted for vertical cable management. Mount cable guides with screws, or nuts and lockwashers.

2.4.5 Patch Panels

Provide ports for the number of horizontal and backbone cables terminated on the panel plus 25 percent spare. Provide pre-connectorized copper patch cords for patch panels. Provide patch cords, as complete assemblies, with matching connectors as specified.

2.4.5.1 Modular to 110 Block Patch Panel

Provide in accordance with TIA-568.1 and TIA-568.2. Panels shall be third party verified and shall comply with EIA/TIA Category 6 requirements. Panel shall be constructed of 2.2 mm minimum aluminum and shall be wall mounted and compatible with an ECIA EIA/ECA 310-E 482.6 mm equipment. Panel shall provide 48 non-keyed, 8-pin modular ports, wired to T568B. Patch panels shall terminate the building cabling on Type 110 IDCs and shall utilize a printed circuit board interface. The rear of each panel shall have incoming cable strain-relief and routing guides. Panels shall have each port factory numbered and be equipped with laminated plastic nameplates above each port.

2.5 TELECOMMUNICATIONS OUTLET/CONNECTOR ASSEMBLIES

2.5.1 Outlet/Connector Copper

Outlet/connectors shall comply with FCC Part 68, TIA-568.1, and TIA-568.2. UTP outlet/connectors shall be UL 1863 listed, non-keyed, 8-pin modular, constructed of high impact rated thermoplastic housing and shall be third party verified and shall comply with TIA-568.2 Category 6 requirements. Outlet/connectors provided for UTP cabling shall meet or exceed the requirements for the cable provided. Outlet/connectors shall be terminated using a Type 110 IDC PC board connector, color-coded for both T568A and T568B wiring. Each outlet/connector shall be wired T568B. UTP outlet/connectors shall comply with TIA-568.2 for 200 mating cycles.

2.5.2 Cover Plates

Telecommunications cover plates shall comply with UL 514C, and TIA-568.1; flush design constructed of 302 stainless material. Provide labeling in accordance with the paragraph LABELING in this section.

2.6 GROUNDING AND BONDING PRODUCTS

Provide in accordance with UL 467, TIA-607, and NFPA 70. Components shall be identified as required by TIA-606. Provide ground rods, bonding conductors, and grounding busbars as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.7 MANUFACTURER'S NAMEPLATE

Each item of equipment shall have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.8 FIELD FABRICATED NAMEPLATES

ASTM D709. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified or as indicated on the drawings. Each nameplate inscription shall identify the function and, when applicable, the position. Nameplates shall be melamine plastic, 3 mm thick, white with black center core. Surface shall be matte finish. Corners shall be square. Accurately align lettering and engrave into the core. Minimum size of nameplates shall be 25 by 65 mm. Lettering shall be a minimum of 6.35 mm high normal block style.

2.9 TESTS, INSPECTIONS, AND VERIFICATIONS

2.9.1 Factory Reel Tests

Provide documentation of the testing and verification actions taken by manufacturer to confirm compliance with TIA-568.1, TIA-568.2, TIA-568.3 cables.

PART 3 EXECUTION

3.1 INSTALLATION

Install telecommunications cabling and pathway systems, including the horizontal and backbone cable, pathway systems, telecommunications outlet/connector assemblies, and associated hardware in accordance with NECA/BICSI 568, TIA-568.1, TIA-568.2, TIA-569, NFPA 70, and UL standards as applicable. Provide cabling in a star topology network. Pathways and outlet boxes shall be installed as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Install telecommunications cabling with copper media in accordance with the following criteria to avoid potential electromagnetic interference between power and telecommunications equipment. The interference ceiling shall not exceed 3.0 volts per meter measured over the usable bandwidth of the telecommunications cabling. Cabling shall be run with horizontal and vertical cable guides in telecommunications spaces with terminating hardware and interconnection equipment.

3.1.1 Cabling

Install UTP, telecommunications cabling system as detailed in TIA-568.1. Screw terminals shall not be used except where specifically indicated on plans. Use an approved insulation displacement connection (IDC) tool kit for copper cable terminations. Do not exceed manufacturers' cable pull tensions for copper and optical fiber cables. Provide a device to monitor cable pull tensions. Do not exceed 110 N pull tension for four pair copper cables. Do not chafe or damage outer jacket materials. Use only lubricants approved by cable manufacturer. Do not over cinch cables, or crush cables with staples. For UTP cable, bend radii shall not be less than four times the cable diameter. Cables shall be terminated; no cable shall contain unterminated elements. Cables shall not be spliced. Label cabling in accordance with paragraph LABELING in this section.

3.1.1.1 Horizontal Cabling

Install horizontal cabling as indicated on drawings. Do not untwist Category 6 UTP cables more than 12 mm from the point of termination to maintain cable geometry. Provide slack cable in the form of a figure eight (not a service loop) on each end of the cable, 3 m in the

telecommunications room, and 304 mm in the work area outlet.

3.1.2 Pathway Installations

Provide in accordance with TIA-569 and NFPA 70. Provide building pathway as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

3.1.3 Service Entrance Conduit, Underground

Provide service entrance underground as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

3.1.4 Work Area Outlets

3.1.4.1 Terminations

Terminate UTP cable in accordance with TIA-568.1, TIA-568.2 and wiring configuration as specified.

3.1.4.2 Cover Plates

As a minimum, each outlet/connector shall be labeled as to its function and a unique number to identify cable link in accordance with the paragraph LABELING in this section.

3.1.4.3 Cables

Unshielded twisted pair and fiber optic cables shall have a minimum of 304 mm of slack cable loosely coiled into the telecommunications outlet boxes. Minimum manufacturer's bend radius for each type of cable shall not be exceeded.

3.1.4.4 Pull Cords

Pull cords shall be installed in conduit serving telecommunications outlets that do not have cable installed.

3.1.5 Telecommunications Space Termination

Install termination hardware required for Category 6 system. An insulation displacement tool shall be used for terminating copper cable to insulation displacement connectors.

3.1.5.1 Connector Blocks

Connector blocks shall be wall mounted in orderly rows and columns. Adequate vertical and horizontal wire routing areas shall be provided between groups of blocks. Install in accordance with industry standard wire routing guides in accordance with TIA-569.

3.1.5.2 Patch Panels

Patch panels shall be mounted on the plywood backboard with sufficient ports to accommodate the installed cable plant plus 25 percent spares.

- a. Copper Patch Panel. Copper cable entering a patch panel shall be secured to the panel with cable ties as recommended by the manufacturer to prevent movement of the cable.

3.1.5.3 Equipment Support Frames

Install in accordance with TIA-569:

- a. Bracket, wall mounted. Mount bracket to plywood backboard in accordance with manufacturer's recommendations. Mount rack so height of highest panel does not exceed 1980 mm above floor.

3.1.6 Grounding and Bonding

Provide in accordance with TIA-607, NFPA 70 and as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

3.2 LABELING

3.2.1 Labels

Provide labeling in accordance with TIA-606. Handwritten labeling is unacceptable. Stenciled lettering for voice and data circuits shall be provided using thermal ink transfer process.

3.2.2 Cable

Cables shall be labeled using color labels on both ends with identifiers in accordance with TIA-606.

3.2.3 Termination Hardware

Workstation outlets and patch panel connections shall be labeled using color coded labels with identifiers in accordance with TIA-606.

3.3 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Painting shall be as specified in Section 09 90 00 PAINTS AND COATINGS.

3.3.1 Painting Backboards

If backboards are required to be painted, then the manufactured fire retardant backboard must be painted with fire retardant paint, so as not to increase flame spread and smoke density and must be appropriately labeled. Label and fire rating stamp must be unpainted.

3.4 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.5 TESTING

3.5.1 Telecommunications Cabling Testing

Perform telecommunications cabling inspection, verification, and performance tests in accordance with TIA-568.1. Test equipment shall conform to TIA-1152. Perform optical fiber field inspection tests via attenuation measurements on factory reels and provide results along with manufacturer certification for factory reel tests. Remove failed cable

reels from project site upon attenuation test failure.

3.5.1.1 Inspection

Visually inspect UTP and optical fiber jacket materials for UL or third party certification markings. Inspect cabling terminations in telecommunications rooms and at workstations to confirm color code for T568A or T568B pin assignments, and inspect cabling connections to confirm compliance with [TIA-568.1](#) and [TIA-568.2](#). Visually confirm Category 6, marking of outlets, cover plates, outlet/connectors, and patch panels.

3.5.1.2 Verification Tests

UTP backbone copper cabling shall be tested for DC loop resistance, shorts, opens, intermittent faults, and polarity between conductors, and between conductors and shield, if cable has overall shield. Test operation of shorting bars in connection blocks. Test cables after termination but prior to being cross-connected.

3.5.1.3 Performance Tests

Perform testing for each outlet and MUTOA as follows:

- a. Perform Category 6 link tests in accordance with [TIA-568.1](#) and [TIA-568.2](#). Tests shall include wire map, length, insertion loss, NEXT, PSNEXT, ELFEXT, PSELFEXT, return loss, propagation delay, and delay skew.

3.5.1.4 Final Verification Tests

Perform verification tests for UTP systems after the complete telecommunications cabling and workstation outlet/connectors are installed.

- a. Voice Tests. These tests assume that dial tone service has been installed. Connect to the network interface device at the demarcation point. Go off-hook and listen and receive a dial tone. If a test number is available, make and receive a local, long distance, and DSN telephone call.
- b. Data Tests. These tests assume the Information Technology Staff has a network installed and are available to assist with testing. Connect to the network interface device at the demarcation point. Log onto the network to ensure proper connection to the network.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 28 - ELECTRONIC SAFETY AND SECURITY

SECTION 28 31 70

INTERIOR FIRE ALARM SYSTEM, ADDRESSABLE

08/20

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED SECTIONS
- 1.3 SUMMARY
 - 1.3.1 Scope
 - 1.3.2 Qualified Fire Protection Engineer (QFPE)
- 1.4 DEFINITIONS
 - 1.4.1 Interface Device
 - 1.4.2 Fire Alarm Control Unit (FACU)
 - 1.4.3 Terminal Cabinet
 - 1.4.4 Control Module and Relay Module
 - 1.4.5 Designated Fire Protection Engineer (DFPE)
 - 1.4.6 Qualified Fire Protection Engineer (QFPE)
- 1.5 SUBMITTALS
- 1.6 SYSTEM OPERATION
 - 1.6.1 Alarm Initiating Devices and Notification Appliances (Visual, Audible)
 - 1.6.2 Functions and Operating Features
- 1.7 TECHNICAL DATA AND SITE-SPECIFIC SOFTWARE
- 1.8 EXISTING EQUIPMENT
- 1.9 QUALITY ASSURANCE
 - 1.9.1 Submittal Documents
 - 1.9.1.1 Preconstruction Submittals
 - 1.9.1.2 Shop Drawings
 - 1.9.1.3 Nameplates
 - 1.9.1.4 Wiring Diagrams
 - 1.9.1.5 System Layout
 - 1.9.1.6 Notification Appliances
 - 1.9.1.7 Initiating Devices
 - 1.9.1.8 Battery Power
 - 1.9.1.9 Voltage Drop Calculations
 - 1.9.1.10 Product Data
 - 1.9.1.11 Operation and Maintenance (O&M) Instructions
 - 1.9.1.12 As-Built Drawings
 - 1.9.2 Qualifications
 - 1.9.2.1 Fire Alarm System Designer
 - 1.9.2.2 Supervisor
 - 1.9.2.3 Technician
 - 1.9.2.4 Installer
 - 1.9.2.5 Test Technician
 - 1.9.2.6 Manufacturer
 - 1.9.3 Regulatory Requirements
- 1.10 DELIVERY, STORAGE, AND HANDLING
- 1.11 MAINTENANCE
 - 1.11.1 Spare Parts

1.11.2 Special Tools

PART 2 PRODUCTS

- 2.1 GENERAL PRODUCT REQUIREMENT
- 2.2 MATERIALS AND EQUIPMENT
 - 2.2.1 Standard Products
 - 2.2.2 Nameplates
 - 2.2.3 Keys
 - 2.2.4 Instructions
- 2.3 FIRE ALARM CONTROL UNIT
 - 2.3.1 Cabinet
 - 2.3.2 Silencing Switches
 - 2.3.2.1 Alarm Silencing Switch
 - 2.3.2.2 Supervisory/Trouble Silencing Switch
 - 2.3.3 Non-Interfering
 - 2.3.4 Memory
 - 2.3.5 Field Programmability
 - 2.3.6 Input/Output Modifications
 - 2.3.7 Resetting
 - 2.3.8 Walk Test
 - 2.3.9 History Logging
 - 2.3.10 Manual Access
 - 2.3.11 Heat Detector Self-Test Routines
- 2.4 MANUAL STATIONS
- 2.5 SMOKE DETECTORS
 - 2.5.1 Spot Type Detectors
- 2.6 HEAT DETECTORS
 - 2.6.1 Heat Detectors
 - 2.6.1.1 Fixed Temperature Detectors
- 2.7 NOTIFICATION APPLIANCES
 - 2.7.1 Audible Notification Appliances
 - 2.7.1.1 Horns
 - 2.7.2 Visual Notification Appliances
- 2.8 ELECTRIC POWER
 - 2.8.1 Primary Power
- 2.9 SECONDARY POWER SUPPLY
 - 2.9.1 Batteries
 - 2.9.1.1 Capacity
 - 2.9.1.2 Battery Power Calculations
 - 2.9.2 Battery Chargers
- 2.10 SURGE PROTECTIVE DEVICES
- 2.11 WIRING
 - 2.11.1 Alarm Wiring
- 2.12 AUTOMATIC FIRE ALARM TRANSMITTERS
 - 2.12.1 Radio Transmitter and Interface Panels
 - 2.12.1.1 Operation
 - 2.12.1.2 Battery Power
 - 2.12.1.3 Transmitter Housing
 - 2.12.1.4 Antenna
 - 2.12.2 Signals to Be Transmitted to the Base Receiving Station
- 2.13 ENVIRONMENTAL ENCLOSURES OR GUARDS

PART 3 EXECUTION

- 3.1 VERIFYING ACTUAL FIELD CONDITIONS
- 3.2 INSTALLATION
 - 3.2.1 Fire Alarm Control Unit (FACU)
 - 3.2.2 Battery Cabinets

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.2.3 Manual Stations
- 3.2.4 Notification Appliances
- 3.2.5 Smoke and Heat Detectors
- 3.2.6 Ceiling Bridges
- 3.3 SYSTEM FIELD WIRING
 - 3.3.1 Wiring within Cabinets, Enclosures, and Boxes
 - 3.3.2 Terminal Cabinets
 - 3.3.3 Alarm Wiring
 - 3.3.4 Back Boxes and Conduit
 - 3.3.5 Conductor Terminations
- 3.4 CONNECTION OF NEW SYSTEM
- 3.5 FIRESTOPPING
- 3.6 PAINTING
- 3.7 FIELD QUALITY CONTROL
 - 3.7.1 Test Procedures
 - 3.7.2 Pre-Government Testing
 - 3.7.2.1 Verification of Compliant Installation
 - 3.7.2.2 Request for Government Final Test
 - 3.7.3 Correction of Deficiencies
 - 3.7.4 Government Final Tests
- 3.8 MINIMUM SYSTEM TESTS
 - 3.8.1 System Tests
 - 3.8.2 Audibility Tests
- 3.9 SYSTEM ACCEPTANCE
- 3.10 INSTRUCTION OF GOVERNMENT EMPLOYEES
 - 3.10.1 Instructor
 - 3.10.2 Required Instruction Time
- 3.11 EXTRA MATERIALS
 - 3.11.1 Repair Service/Replacement Parts
 - 3.11.2 Spare Parts
 - 3.11.3 Document Storage Cabinet

-- End of Section Table of Contents --

SECTION 28 31 70

INTERIOR FIRE ALARM SYSTEM, ADDRESSABLE
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide
<http://www.approvalguide.com/>

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41.1 (2002; R 2008) Guide on the Surges
Environment in Low-Voltage (1000 V and
Less) AC Power Circuits

IEEE C62.41.2 (2002) Recommended Practice on
Characterization of Surges in Low-Voltage
(1000 V and Less) AC Power Circuits

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 4 (2018) Standard for Integrated Fire
Protection and Life Safety System Testing

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA
20-1; TIA 20-2; TIA 20-3; TIA 20-4)
National Electrical Code

NFPA 72 (2022) National Fire Alarm and Signaling
Code

NFPA 170 (2021) Standard for Fire Safety and
Emergency Symbols

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-601-02 (2021) Fire Protection Systems Inspection,
Testing, and Maintenance

UFC 4-010-06 (2016; with Change 1, 2017) Cybersecurity
of Facility-Related Control Systems

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

47 CFR 15 Radio Frequency Devices

47 CFR 90 Private Land Mobile Radio Services

UNDERWRITERS LABORATORIES (UL)

UL 268	(2016; Reprint Nov 2021) UL Standard for Safety Smoke Detectors for Fire Alarm Systems
UL 464	(2016; Reprint Sep 2017) UL Standard for Safety Audible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories
UL 497A	(2001; Bul. 2019) UL Standard for Safety Secondary Protectors for Communications Circuits
UL 497B	(2004; Reprint Feb 2022) UL Standard for Safety Protectors for Data Communications and Fire Alarm Circuits
UL 521	(1999; Reprint Mar 2021) UL Standard for Safety Heat Detectors for Fire Protective Signaling Systems
UL 864	(2014; Reprint May 2020) UL Standard for Safety Control Units and Accessories for Fire Alarm Systems
UL 1283	(2017) UL Standard for Safety Electromagnetic Interference Filters
UL 1449	(2021) UL Standard for Safety Surge Protective Devices
UL 1638	(2016; Reprint Sep 2017) UL Standard for Safety Visible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories
UL 1971	(2002; Reprint Oct 2008) Signaling Devices for the Hearing Impaired
UL Fire Prot Dir	(2012) Fire Protection Equipment Directory

1.2 RELATED SECTIONS

Section 25 05 11 Cybersecurity for Facility-Related Control Systems, applies to this section, with the additions and modifications specified herein. In addition, refer to the following sections for related work and coordination:

Section 07 84 00 FIRESTOPPING for additional work related to firestopping.

1.3 SUMMARY

1.3.1 Scope

- a. This work includes designing and providing a new, complete, fire alarm system as described herein and on the contract drawings for the P-919

Bulk Tank Farm Improvements, Phase I. Include system wiring, raceways, pull boxes, terminal cabinets, outlet and mounting boxes, control equipment, initiating devices, notification appliances, supervising station fire alarm transmitters, and other accessories and miscellaneous items required for a complete operational system even though each item is not specifically mentioned or described. Provide system complete and ready for operation. Design and installation must comply with UFGS 25 05 11, UFC 4-010-06 and AFGM 2019-320-02.

- b. Provide equipment, materials, installation, workmanship, inspection, and testing in strict accordance with NFPA 72, except as modified herein. The system layout on the drawings show the intent of coverage and suggested locations. Final quantity, system layout, and coordination are the responsibility of the Contractor.
- c. The fire alarm system must be independent of the building security, building management, and energy/utility monitoring systems other than for control functions.

1.3.2 Qualified Fire Protection Engineer (QFPE)

Services of the QFPE must include:

- a. Reviewing SD-02, SD-03, and SD-05 submittal packages for completeness and compliance with the provisions of this specification. Construction (shop) drawings and calculations must be prepared by, or prepared under the immediate supervision of, the QFPE. The QFPE must affix their professional engineering stamp with signature to the shop drawings, calculations, and material data sheets, indicating approval prior to submitting the shop drawings to the DFPE.
- b. Providing a letter documenting that the SD-02, SD-03, and SD-05 submittal package has been reviewed and noting any outstanding comments.
- c. Performing in-progress construction surveillance prior to installation of ceilings (rough-in inspection).
- d. Witnessing pre-Government functional performance testing and performing a final installation review.
- e. Signing applicable certificates under SD-07.

1.4 DEFINITIONS

Wherever mentioned in this specification or on the drawings, the equipment, devices, and functions must be defined as follows:

1.4.1 Interface Device

An addressable device that interconnects hard wired systems or devices to an analog/addressable system.

1.4.2 Fire Alarm Control Unit (FACU)

A master control unit having the features of a fire alarm control unit (FACU).

1.4.3 Terminal Cabinet

A steel cabinet with locking, hinge-mounted door where terminal strips are securely mounted inside the cabinet.

1.4.4 Control Module and Relay Module

Terms utilized to describe emergency control function interface devices as defined by NFPA 72.

1.4.5 Designated Fire Protection Engineer (DFPE)

The DoD fire protection engineer that oversees that Area of Responsibility for that project. This is sometimes referred to as the "cognizant" fire protection engineer. Interpret reference to "authority having jurisdiction" and/or AHJ in referenced standards to mean the Designated Fire Protection Engineer (DFPE). The DFPE may be responsible for review of the contractor submittals having a "G" designation, and for witnessing final inspection and testing.

1.4.6 Qualified Fire Protection Engineer (QFPE)

A QFPE is an individual who is a licensed professional engineer (P.E.), who has passed the fire protection engineering written examination administered by the National Council of Examiners for Engineering and Surveying (NCEES) and has relevant fire protection engineering experience.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval.

Shop drawings (SD-02), product data (SD-03) and calculations (SD-05) must be prepared by the fire alarm designer and combined and submitted as one complete package. The QFPE must review the SD-02/SD-03/SD-05 submittal package for completeness and compliance with the Contract provisions prior to submission to the Government. The QFPE must provide a Letter of Confirmation that they have reviewed the submittal package for compliance with the contract provisions. This letter must include their registered professional engineer stamp and signature. Partial submittals and submittals not reviewed by the QFPE will be returned by the Government disapproved without review.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualified Fire Protection Engineer (QFPE); G

Fire alarm system designer; G

Supervisor; G

Technician; G

Installer; G

Test Technician; G

Fire Alarm System Site-Specific Software Acknowledgement; G

SD-02 Shop Drawings

Nameplates; G

Instructions; G

Wiring Diagrams; G

System Layout; G

Notification Appliances; G

Initiating devices; G

Battery Power; G

Voltage Drop Calculations; G

SD-03 Product Data

Fire Alarm Control Unit (FACU); G

Manual Stations; G

Smoke Detectors; G

Heat Detectors; G

Notification Appliances; G

Batteries; G

Battery Chargers; G

Supplemental Notification Appliance Circuit Panels; G

Auxiliary Power Supply Panels; G

Surge Protective Devices; G

Alarm Wiring; G

Back Boxes and Conduit; G

Ceiling Bridges for Ceiling-Mounted Appliances; G

Terminal Cabinets; G

Automatic Fire Alarm Transmitters (including housing); G

Radio Transmitter and Interface Panels; G

Environmental Enclosures or Guards; G

Document Storage Cabinet; G

SD-06 Test Reports

Test Procedures; G

SD-07 Certificates

Verification of Compliant Installation; G

Request for Government Final Test; G

SD-10 Operation and Maintenance Data

Operation and Maintenance (O&M) Instructions; G

Instruction of Government Employees; G

SD-11 Closeout Submittals

As-Built Drawings

Spare Parts

1.6 SYSTEM OPERATION

Fire alarm system components requiring power, except for the FACU(s) power supply, must operate on 24 volts DC unless noted otherwise in this section.

The interior fire alarm system must be a complete, supervised, noncoded, analog/addressable fire alarm system conforming to NFPA 72, and UL 864. Systems meeting UL 2017 only are not acceptable. The system must be activated into the alarm mode by actuation of an alarm initiating device. The system must remain in the alarm mode until the initiating device is reset and the control unit is reset and restored to normal.

1.6.1 Alarm Initiating Devices and Notification Appliances (Visual, Audible)

- a. Connect alarm initiating devices to initiating device circuits (IDC) Class "B", or to signaling line circuits (SLC) Class "B" and installed in accordance with NFPA 72.
- b. Connect notification appliances to notification appliance circuits (NAC) Class "B".

1.6.2 Functions and Operating Features

The system must provide the following functions and operating features:

- a. Power, annunciation, supervision, and control for the system. Addressable systems must be microcomputer (microprocessor or microcontroller) based with a minimum word size of eight bits with sufficient memory to perform as specified.
- b. Visual alarm notification appliances must be synchronized as required by NFPA 72.
- c. Electrical supervision of the primary power (AC) supply, presence of the battery, battery voltage, and placement of system modules within

the control unit.

- d. An audible and visual trouble signal to activate upon a single break or open condition, or ground fault. The trouble signal must also operate upon loss of primary power (AC) supply, absence of a battery supply, low battery voltage, or removal of alarm or supervisory control unit modules. After the system returns to normal operating conditions, the trouble signal must again sound until the trouble is acknowledged. A smoke detector in the process of being verified for the actual presence of smoke must not initiate a trouble condition.
- e. A trouble signal silence feature that must silence the audible trouble signal, without affecting the visual indicator.
- f. Alarm functions must override trouble or supervisory functions. Supervisory functions must override trouble functions.
- g. The system must be capable of being programmed from the control unit keyboard. Programmed information must be stored in non-volatile memory.
- h. The system must be capable of operating, supervising, and/or monitoring non-addressable alarm and supervisory devices.
- i. There must be no limit, other than maximum system capacity, as to the number of addressable devices that may be in alarm simultaneously.
- j. An alarm signal must automatically initiate the following functions:
 - (1) Transmission of an alarm signal to a remote supervising station.
 - (2) Visual indication of the device operated on the FACU.
 - (3) Actuation of alarm notification appliances.
 - (4) Recording of the event electronically in the history log of the FACU.
- k. A supervisory signal must automatically initiate the following functions:
 - (1) Visual indication of the device operated on the FACU.
 - (2) Transmission of a supervisory signal to a remote supervising station.
 - (3) Recording of the event electronically in the history log of the FACU.
- l. A trouble condition must automatically initiate the following functions:
 - (1) Visual indication of the device operated on the FACU.
 - (2) Transmission of a trouble signal to a remote supervising station.
 - (3) Recording of the event electronically in the history log of the FACU.

- m. System control equipment must be programmed to provide a 60-minute to 180-minute delay in transmission of trouble signals resulting from primary power failure.

1.7 TECHNICAL DATA AND SITE-SPECIFIC SOFTWARE

Technical data and site-specific software (meaning technical data that relates to computer software) that are specifically identified in this project, and may be required in other specifications, must be delivered, strictly in accordance with the CONTRACT CLAUSES. The fire alarm system manufacturer must submit written confirmation of this contract provision as "[Fire Alarm System Site-Specific Software Acknowledgement](#)". Identify data delivered by reference to the specification paragraph against which it is furnished. Data to be submitted must include complete system, equipment, and software descriptions. Descriptions must show how the equipment will operate as a system to meet the performance requirements of this contract. The site-specific software data package must also include the following:

- a. Items identified in NFPA 72, titled "Site-Specific Software".
- b. Identification of programmable portions of the system equipment and capabilities.
- c. Description of system revision and expansion capabilities and methods of implementation detailing both equipment and software requirements.
- d. Provision of operational software data on all modes of programmable portions for fire alarm .
- e. Description of Fire Alarm Control Unit equipment operation.
- f. Description of auxiliary and remote equipment operations.
- g. Library of application software.
- h. Operation and maintenance manuals.

1.8 EXISTING EQUIPMENT

- a. Equipment and devices must be compatible and operable with the existing fire alarm system and must not impair reliability or operational functions of existing supervising station fire alarm system. The supervising equipment is existing and consists of a Monaco radio system..

1.9 QUALITY ASSURANCE

1.9.1 Submittal Documents

1.9.1.1 Preconstruction Submittals

Within 36 days of contract award but not less than 14 days prior to commencing any work on site, the Contractor must submit the following for review and approval. SD-02, SD-03 and SD-05 submittals received prior to the review and approval of the qualifications of the fire alarm subcontractor and QFPE must be returned disapproved without review. All resultant delays must be the sole responsibility of the Contractor.

1.9.1.2 Shop Drawings

Shop drawings must not be smaller than ISO A1. Drawings must comply with the requirements of NFPA 72 and NFPA 170. Minimum scale for floor plans must be 1/8"=1'.

1.9.1.3 Nameplates

Nameplate illustrations and data to obtain approval by the Contracting Officer before installation.

1.9.1.4 Wiring Diagrams

Four copies of point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams must show connections from field devices to the FACU and remote FACU, initiating circuits, switches, relays and terminals, including pathway diagrams between the control unit and shared communications equipment within the protected premises. Point-to-point wiring diagrams must be job specific and must not indicate connections or circuits not being utilized. Provide complete riser diagrams indicating the wiring sequence of all devices and their connections to the control equipment. Include a color-code schedule for the wiring.

1.9.1.5 System Layout

Four copies of plan view drawing showing device locations, terminal cabinet locations, junction boxes, other related equipment, conduit routing, conduit sizes, wire counts, conduit fill calculations, wire color-coding, circuit identification in each conduit, and circuit layouts for all floors. Indicate candela rating of each visual notification appliance. Clearly identify the locations of isolation modules. Indicate the addresses of all devices, modules, relays, and similar. Indicate if the environment for the FACU is within its environmental listing (e.g. temperature/humidity).

Provide a complete description of the system operation in matrix format similar to the "Typical Input/Output Matrix" included in the Annex of NFPA 72.

1.9.1.6 Notification Appliances

Calculations and supporting data on each circuit to indicate that there is at least 25 percent spare capacity for notification appliances. Annotate data for each circuit on the drawings.

1.9.1.7 Initiating Devices

Calculations and supporting data on each circuit to indicate that there is at least 25 percent spare capacity for initiating devices. Annotate data for each circuit on the drawings.

1.9.1.8 Battery Power

Calculations and supporting data as required in paragraph Battery Power Calculations for alarm, alert, and supervisory power requirements. Calculations including ampere-hour requirements for each system component

and each control unit component, and the battery recharging period, must be included on the drawings.

1.9.1.9 Voltage Drop Calculations

Voltage drop calculations for each notification circuit indicating that sufficient voltage is available for proper operation of the system and all components, at a minimum rated voltage of the system operating on batteries. Include the calculations on the system layout drawings.

1.9.1.10 Product Data

Four copies of annotated descriptive data to show the specific model, type, and size of each item. Catalog cuts must also indicate the NRTL listing. The data must be highlighted to show model, size, and options that are intended for consideration. Data must be adequate to demonstrate compliance with all contract requirements. Product data for all equipment must be combined into a single submittal.

Provide an equipment list identifying the type, quantity, make, and model number of each piece of equipment to be provided under this submittal. The equipment list must include the type, quantity, make and model of spare equipment. Types and quantities of equipment submitted must coincide with the types and quantities of equipment used in the battery calculations and those shown on the shop drawings.

1.9.1.11 Operation and Maintenance (O&M) Instructions

Six copies of the Operation and Maintenance Instructions. The O&M Instructions must be prepared in a single volume or in multiple volumes, with each volume indexed, and may be submitted as a Technical Data Package. Manuals must be approved prior to training. The Interior Fire Alarm System Operation and Maintenance Instructions must include the following:

- a. "Manufacturer Data Package five" as specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA.
- b. Operating manual outlining step-by-step procedures required for system startup, operation, and shutdown. The manual must include the manufacturer's name, model number, service manual, parts list, and preliminary equipment list complete with description of equipment and their basic operating features.
- c. Maintenance manual listing routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guide. The manuals must include conduit layout, equipment layout and simplified wiring, and control diagrams of the system as installed.
- d. Complete procedures for system revision and expansion, detailing both equipment and software requirements.
- e. Software submitted for this project on CD/DVD media utilized.
- f. Printouts of configuration settings for all devices.
- g. Routine maintenance checklist. The routine maintenance checklist must be arranged in a columnar format. The first column must list all installed devices, the second column must state the maintenance

activity or state no maintenance required, the third column must state the frequency of the maintenance activity, and the fourth column provided for additional comments or reference. All data (devices, testing frequencies, and similar) must comply with [UFC 3-601-02](#).

- h. A final Equipment List must be submitted with the Operating and Maintenance (O&M) manual.

1.9.1.12 [As-Built Drawings](#)

The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final Government test of the system. At least one set of the as-built (marked-up) drawings must be provided at the time of, or prior to the final Government test.

1.9.2 [Qualifications](#)

1.9.2.1 [Fire Alarm System Designer](#)

The fire alarm system designer must be certified as a Level III (minimum) Technician by National Institute for Certification in Engineering Technologies (NICET) in the Fire Alarm Systems subfield of Fire Protection Engineering Technology or meet the qualifications for a QFPE.

1.9.2.2 [Supervisor](#)

The fire alarm technicians supervising the installation of equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.9.2.3 [Technician](#)

Fire alarm technicians with a minimum of four years of experience must be utilized to install and terminate fire alarm devices, cabinets and control units. The fire alarm technicians installing the equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings, and must be thoroughly experienced in the installation of air sampling detection systems.

1.9.2.4 [Installer](#)

Fire alarm installer with a minimum of two years of experience utilized to assist in the installation of fire alarm devices, cabinets and control units. A licensed electrician must be allowed to install wire, cable, conduit and backboxes for the fire alarm system system. The fire alarm installer must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.9.2.5 [Test Technician](#)

Fire alarm technicians with a minimum of eight years of experience and NICET Level III must be utilized in testing and certification of the installation of the fire alarm devices, cabinets and control units. The fire alarm technicians testing the equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment installed as part of this project.

1.9.2.6 Manufacturer

Components must be of current design and must be in regular and recurrent production at the time of installation. Provide design, materials, and devices for a protected premises fire alarm system, complete, conforming to NFPA 72, except as specified herein.

1.9.3 Regulatory Requirements

Equipment and material must be listed or approved. Listed or approved, as used in this Section, means listed, labeled or approved by a Nationally Recognized Testing Laboratory (NRTL) such as UL Fire Prot Dir or FM APP GUIDE. The omission of these terms under the description of any item of equipment described must not be construed as waiving this requirement. All listings or approvals by testing laboratories must be from an existing ANSI or UL published standard. The recommended practices stated in the manufacturer's literature or documentation must be considered as mandatory requirements.

1.10 DELIVERY, STORAGE, AND HANDLING

Protect equipment delivered and placed in storage from the weather, humidity, and temperature variation, dirt and dust, and other contaminants.

1.11 MAINTENANCE

1.11.1 Spare Parts

Furnish the following spare parts in the manufacturers original unopened containers:

- a. Five complete sets of system keys.
- b. Two of each type of fuse required by the system.
- c. One manual stations.
- d. Two of each type of detector installed.
- e. Two of each type of detector base and head installed.
- f. Two of each type of audible and visual alarm device installed.

1.11.2 Special Tools

Software, connecting cables and proprietary equipment, necessary for the maintenance, testing, and reprogramming of the equipment must be furnished to the Contracting Officer, prior to the instruction of Government employees.

PART 2 PRODUCTS

2.1 GENERAL PRODUCT REQUIREMENT

All fire alarm equipment must be listed for use under the applicable reference standards.

2.2 MATERIALS AND EQUIPMENT

2.2.1 Standard Products

Provide materials, equipment, and devices that have been tested by a nationally recognized testing laboratory and listed for fire protection service when so required by **NFPA 72** or this specification. Select material from one manufacturer, where possible, and not a combination of manufacturers, for any particular classification of materials. Material and equipment must be the standard products of a manufacturer regularly engaged in the manufacture of the products for at least 2 years prior to bid opening.

2.2.2 Nameplates

Major components of equipment must have the manufacturer's name, address, type or style, model or serial number, catalog number, date of installation, installing Contractor's name and address, and the contract number provided on a new name plate permanently affixed to the item or equipment. Major components include, but are not limited to, the following:

a. FACU

Nameplates must be etched metal or plastic, permanently attached by screws to control units or adjacent walls.

2.2.3 Keys

Keys and locks for equipment, control units and devices must be identical. Master all keys and locks to a single key as required by the Installation Fire Department.

2.2.4 Instructions

Provide a typeset printed or typewritten instruction card mounted behind a Lexan plastic or glass cover in a stainless steel or aluminum frame. Install the instructions on the interior of the FACU. The card must show those steps to be taken by an operator when a signal is received as well as the functional operation of the system under all conditions, normal, alarm, supervisory, and trouble. The instructions and their mounting location must be approved by the Contracting Officer before being posted.

2.3 FIRE ALARM CONTROL UNIT

Provide a complete **fire alarm control unit (FACU)** fully enclosed in a lockable steel cabinet as specified herein. Operations required for testing or for normal care, maintenance, and use of the system must be performed from the front of the enclosure. If more than a single unit is required at a location to form a complete control unit, the unit cabinets must match exactly. The system must be capable of defining any module as an alarm module and report alarm trouble, loss of polling, or as a supervisory module, and reporting supervisory short, supervisory open or loss of polling such as waterflow switches, valve supervisory switches, fire pump monitoring, independent smoke detection systems, relays for output function actuation.

a. Each control unit must provide power, supervision, control, and logic for the entire system, utilizing solid state, modular components,

internally mounted and arranged for easy access. Each control unit must be suitable for operation on a 220 volt, 50 hertz, normal building power supply. Provide each control unit with supervisory functions for power failure, internal component placement, and operation.

- b. Visual indication of alarm, supervisory, or trouble initiation on the FACU must be by liquid crystal display or similar means with a minimum of 80 characters.

2.3.1 Cabinet

Install control unit components in cabinets large enough to accommodate all components and also to allow ample gutter space for interconnection of control units as well as field wiring. The cabinet must be a sturdy steel housing, complete with back box, hinged steel door with cylinder lock, and semi-recessed mounting provisions. The enclosure must be identified by an engraved phenolic resin nameplate. Lettering on the nameplate must say "Fire Alarm control unit" and must not be less than 25 mm high. Provide prominent rigid plastic or metal identification plates for lamps, circuits, meters, fuses, and switches.

2.3.2 Silencing Switches

2.3.2.1 Alarm Silencing Switch

Provide an alarm silencing switch at the FACU that must silence the audible and visual notification appliances. Subsequent activation of initiating devices must cause the notification appliances to re-activate.

2.3.2.2 Supervisory/Trouble Silencing Switch

Provide supervisory and trouble silencing switch(es) that must silence the audible trouble and supervisory signal(s), but not extinguish the visual indicator. This switch must be overridden upon activation of a subsequent supervisory or trouble condition. Audible trouble indication must resound automatically every 24 hours after the silencing feature has been operated if the supervisory or trouble condition still exists.

2.3.3 Non-Interfering

Power and supervise each circuit such that a signal from one device does not prevent the receipt of signals from any other device. Initiating devices must be manually reset by switch from the FACU after the initiating device or devices have been restored to normal.

2.3.4 Memory

Provide each control unit with non-volatile memory and logic for all functions. The use of long life batteries, capacitors, or other age-dependent devices must not be considered as equal to non-volatile processors, PROMS, or EPROMS.

2.3.5 Field Programmability

Provide control units and control units that are fully field programmable for both input and output of control, initiation, notification, supervisory, and trouble functions. The system program configuration must be menu driven. System changes must be password protected. Any

proprietary equipment and proprietary software needed by qualified technicians to implement future changes to the fire alarm system must be provided as part of this contract.

2.3.6 Input/Output Modifications

The FACU must contain features that allow the bypassing of input devices from the system or the modification of system outputs. These control features must consist of a control unit mounted keypad. Any bypass or modification to the system must indicate a trouble condition on the FACU.

2.3.7 Resetting

Provide the necessary controls to prevent the resetting of any alarm, supervisory, or trouble signal while the alarm, supervisory or trouble condition on the system still exists.

2.3.8 Walk Test

The FACU must have a walk test feature. When using this feature, operation of initiating devices must result in limited system outputs, so that the notification appliances operate for only a few seconds and the event is indicated in the history log, but no other outputs occur.

2.3.9 History Logging

The control unit must have the ability to store a minimum of 400 events in a log. These events must be stored in a battery-protected memory and must remain in the memory until the memory is downloaded or cleared manually. Resetting of the control unit must not clear the memory.

2.3.10 Manual Access

An operator at the control unit, having a proper access level, must have the capability to manually access the following information for each initiating device.

- a. Primary status.
- b. Device type.
- c. Present average value.
- d. Present sensitivity selected.
- e. Detector range (normal, dirty).

2.3.11 Heat Detector Self-Test Routines

Automatic self-test routines must be performed on each detector that will functionally check detector sensitivity electronics and ensure the accuracy of the value being transmitted. Any detector that fails this test must indicate a trouble condition with the detector location at the control unit.

2.4 MANUAL STATIONS

Provide metal or plastic, semi-flush mounted, double-action, addressable manual stations, that are not subject to operation by jarring or

vibration. Stations must be equipped with screw terminals for each conductor. Stations that require the replacement of any portion of the device after activation are not permitted. Stations must be finished in red with molded raised lettering operating instructions of contrasting color. The use of a key must be required to reset the station.

2.5 SMOKE DETECTORS

2.5.1 Spot Type Detectors

Provide addressable photoelectric smoke detectors as follows:

- a. Provide analog/addressable photoelectric smoke detectors utilizing the photoelectric light scattering principle for operation in accordance with [UL 268](#). Smoke detectors must be listed for use with the FACU.
- b. Provide self-restoring type detectors that do not require any readjustment after actuation at the FACU to restore them to normal operation. The detector must have a visual indicator to show actuation.
- c. Vibration must have no effect on the detector's operation. Protect the detection chamber with a fine mesh metallic screen that prevents the entrance of insects or airborne materials. The screen must not inhibit the movement of smoke particles into the chamber.
- d. Provide twist lock bases with screw terminals for each conductor. The detectors must maintain contact with their bases without the use of springs.
- e. The detector address must identify the particular unit, its location within the system, and its sensitivity setting. Detectors must be of the low voltage type rated for use on a 24 VDC system.

2.6 HEAT DETECTORS

2.6.1 Heat Detectors

Heat detectors must be analog/addressable and designed for detection of fire by fixed temperature in accordance with [UL 521](#). The alarm condition must be determined by comparing detector value with the stored values. Detectors located in areas subject to moisture, exterior atmospheric conditions, or hazardous locations as defined by [NFPA 70](#), must be types approved for such locations.

2.6.1.1 Fixed Temperature Detectors

Detectors must be surface mounted in the horizontal orientation and supported independently of wiring connections. Detectors must be self-restoring. The detectors must have a specific temperature setting of [57.2 degrees C](#).

2.7 NOTIFICATION APPLIANCES

2.7.1 Audible Notification Appliances

Audible appliances must conform to the applicable requirements of [UL 464](#). Appliances must be connected into notification appliance circuits. Surface mounted audible appliances must be painted red. Recessed audible

appliances must be installed with a grill that is painted red.

2.7.1.1 Horns

Horns must be semi-flush mounted. Horns must produce a sound rating of at least 85 dBA at **3.1 meters**. Horns used in exterior locations must be specifically listed or approved for outdoor use and be provided with metal housing and protective grilles.

2.7.2 Visual Notification Appliances

Visual notification appliances must conform to the applicable requirements of **UL 1638**, **UL 1971** and conform to the Architectural Barriers Act (ABA). Visual Notification Appliances must have clear high intensity optic lens, xenon flash tubes, or light emitting diode (LED) and be marked "Alert" in letters of contrasting color. The light pattern must be disbursed so that it is visible above and below the strobe and from a 90 degree angle on both sides of the strobe. Strobe flash rate must be 1 flash per second and a minimum of 15 candela based on the **UL 1971** test. Strobe must be semi-flush mounted.

2.8 ELECTRIC POWER

2.8.1 Primary Power

Power must be 220 VAC 50 Hz service for the FACU from the AC service to the building in accordance with **NFPA 72**.

2.9 SECONDARY POWER SUPPLY

Provide for system operation in the event of primary power source failure. Transfer from normal to auxiliary (secondary) power or restoration from auxiliary to normal power must be automatic and must not cause transmission of a false alarm.

2.9.1 Batteries

Provide sealed, maintenance-free, sealed lead acid batteries as the source for emergency power to the FACU. Batteries must contain suspended electrolyte. The battery system must be maintained in a fully charged condition by means of a solid state battery charger. Provide an automatic transfer switch to transfer the load to the batteries in the event of the failure of primary power.

2.9.1.1 Capacity

Battery size must be the following capacity. This capacity applies to every control unit associated with this system, including **supplemental notification appliance circuit panels**, **auxiliary power supply panels**, and fire alarm transmitters.

- a. Sufficient capacity to operate the fire alarm system under supervisory and trouble conditions, including audible trouble signal devices for 48 hours and audible and visual signal devices under alarm conditions for an additional 15 minutes.

2.9.1.2 Battery Power Calculations

- a. Verify that battery capacity exceeds supervisory and alarm power

requirements for the criteria noted in the paragraph "Capacity" above.

- (1) Substantiate the battery calculations for alarm and supervisory power requirements. Include ampere-hour requirements for each system component and each control unit component, and compliance with [UL 864](#).
 - (2) Provide complete battery calculations for both the alarm and supervisory power requirements. Submit ampere-hour requirements for each system component with the calculations.
 - (3) Provide voltage drop calculations to indicate that sufficient voltage is available for proper operation of the system and all components. Calculations must be performed using the minimum rated voltage of each component.
- b. For battery calculations assume a starting voltage of 24 VDC for starting the calculations to size the batteries. Calculate the required Amp-Hours for the specified standby time, and then calculate the required Amp-Hours for the specified alarm time. Using 20.4 VDC as starting voltage, perform a voltage drop calculation for circuits containing device and/or appliances remote from the power sources.

2.9.2 Battery Chargers

Provide a solid state, fully automatic, variable charging rate battery charger. The charger must be capable of providing 120 percent of the connected system load and must maintain the batteries at full charge. In the event the batteries are fully discharged (20.4 Volts dc), the charger must recharge the batteries back to 95 percent of full charge within 48 hours after a single discharge cycle as described in paragraph CAPACITY above. Provide pilot light to indicate when batteries are manually placed on a high rate of charge as part of the unit assembly if a high rate switch is provided.

2.10 SURGE PROTECTIVE DEVICES

[Surge protective devices](#) must be provided to suppress all voltage transients which might damage fire alarm control unit components. Systems having circuits located outdoors, communications equipment must be protected against surges induced on any signaling line circuit. Cables and conductors, that serve as communications links, must have surge protection circuits installed at each end. The surge protective device must wire in series to the power supply of the protected equipment with screw terminations. Line voltage surge arrestor must be installed directly adjacent to the power panel where the FACU breaker is located.

- a. Surge protective devices for nominal 120 VAC must be [UL 1449](#) listed with a maximum 500 volt suppression level and have a maximum response time of 5 nanoseconds. The surge protective device must also meet [IEEE 62.41.1](#) and [IEEE 62.41.2](#) category B tests for surge capacity. The surge protective device must feature multi-stage construction and be provided with a long-life indicator lamp (either light emitting diode or neon) which extinguishes upon failure of protected components. Any unit fusing must be externally accessible.
- b. Surge protective devices for nominal 24 VAC, fire alarm telephone dialer, or ethernet connection must be [UL 497B](#) listed, meet [IEEE 62.41.1](#) and have a maximum response time of 1-nanosecond. The

surge protective device must feature multi-stage construction and be self-resetting. The surge protective device must be a base and plug style. The base assembly must have screw terminals for fire alarm wiring. The base assembly must accept "plug-in" surge protective module.

- c. All surge protective devices (SPD) must be the standard product of a single manufacturer and be equal or better than the following:
- (1) For 120 VAC nominal line voltage: UL 1449 and UL 1283 listed, series connected 120 VAC, 20A rated, surge protective device in a NEMA 4x enclosure. Minimum 50,000 amp surge current rating with EMI/RFI filtering and a dry contact circuit for remote monitoring of surge protection status.
 - (2) For 24-volt nominal line voltage: UL 497B listed, series connected low voltage, 24-volt, 5A rated, loop circuit protector, base and replaceable module.
 - (3) For alarm telephone dialers: UL 497A listed, series connected, 130-volt, 150 mA rated with self-resetting fuse, dialer circuit protector with modular plug and play.
 - (4) For IP-DACTS: UL 497B listed, series connected, 6.4-volt, 1.5A rated with 20 kA/pair surge current, data network protector with modular plug and play.

2.11 WIRING

Provide wiring materials under this section as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM with the additions and modifications specified herein.

2.11.1 Alarm Wiring

IDC and SLC wiring must be fiber optic or solid copper cable in accordance with the manufacturers requirements. Copper signaling line circuits and initiating device circuit field wiring must be No. 16 AWG size conductors at a minimum. Visual notification appliance circuit conductors, that contain audible alarm appliances, must be copper No. 14 AWG size conductors at a minimum. Wire size must be sufficient to prevent voltage drop problems. Circuits operating at 24 VDC must not operate at less than the listed voltages for the detectors and/or appliances. Power wiring, operating at 120 VAC minimum, must be a minimum No. 12 AWG solid copper having similar insulation. Acceptable power-limited cables are FPL, FPLR or FPLP as appropriate with red colored covering. Nonpower-limited cables must comply with NFPA 70.

2.12 AUTOMATIC FIRE ALARM TRANSMITTERS

2.12.1 Radio Transmitter and Interface Panels

Transmitters must be compatible with proprietary supervising station receiving equipment. Each radio alarm transmitter must be the manufacturer's recognized commercial product, completely assembled, wired, factory tested, and delivered ready for installation and operation. Transmitters must be provided in accordance with applicable portions of NFPA 72, Federal Communications Commission (FCC) 47 CFR 90 and Federal Communications Commission (FCC) 47 CFR 15. Transmitter electronics module

must be contained within the physical housing as an integral, removable assembly. The proprietary supervising station receiving equipment is Monaco and the transmitter must be fully compatible with this equipment. At the contractors option, and if listed, the transmitter may be housed in the same control unit as the FACU. The transmitter must be narrowband radio, with FCC certification for narrowband operation and meets the requirements of the NTIA (National Telecommunications and Information Administration) Manual of Regulations and Procedures for Federal Frequency Management.

2.12.1.1 Operation

Operate each transmitter from 220-volt ac power. In the event of 220-volt ac power loss, the transmitter must automatically switch to battery operation. Switchover must be accomplished with no interruption of protective service, and must automatically transmit a trouble message. Upon restoration of ac power, transfer back to normal ac power supply must also be automatic.

2.12.1.2 Battery Power

Transmitter standby battery capacity must provide sufficient power to operate the transmitter in a normal standby status for a minimum of 72 hours and be capable of transmitting alarms during that period.

2.12.1.3 Transmitter Housing

Use NEMA Type 1 for housing. The housing must contain a lock that is keyed identical to radio alarm transmitter housings on the Installation. Radio alarm transmitter housing must be factory painted with a suitable priming coat and not less than two coats of a hard, durable weatherproof enamel.

2.12.1.4 Antenna

Antenna must be omnidirectional, coaxial, halfwave dipole antennas for radio alarm transmitters with a driving point impedance to match transmitter output. The antenna and antenna mounts must be corrosion resistant and designed to withstand wind velocities of 161 km/hour. Do not mount antennas to any portion of the building roofing system. Protect the antenna from physical damage.

2.12.2 Signals to Be Transmitted to the Base Receiving Station

The following signals must be sent to the base receiving station:

- a. Manual pull stations
- b. Smoke detectors
- c. Heat detectors

2.13 ENVIRONMENTAL ENCLOSURES OR GUARDS

Environmental enclosures must be provided to permit fire alarm components to be used in areas that exceed the environmental limits of the listing. The enclosure must be listed for the device or appliance as either a manufactured part number or as a listed compatible accessory for the component is currently listed. Guards required to deter mechanical damage

must be either a listed manufactured part or a listed accessory for the category of the initiating device or notification appliance.

PART 3 EXECUTION

3.1 VERIFYING ACTUAL FIELD CONDITIONS

Before commencing work, examine all adjoining work on which the contractor's work is in any way dependent for perfect workmanship according to the intent of this specification section, and report to the Contracting Officer's Representative any condition which prevents performance of first class work. No "waiver of responsibility" for incomplete, inadequate or defective adjoining work will be considered unless notice has been filed before submittal of a proposal.

3.2 INSTALLATION

3.2.1 Fire Alarm Control Unit (FACU)

Locate the FACU where indicated on the drawings. Semi-recess the enclosure with the top of the cabinet **2 m** above the finished floor or center the cabinet at **1.5 m**, whichever is lower. Conductor terminations must be labeled and a drawing containing conductors, their labels, their circuits, and their interconnection must be permanently mounted in the FACU. Locate the document storage cabinet adjacent to the FACU unless the Contracting Officer directs otherwise.

3.2.2 Battery Cabinets

When batteries will not fit in the FACU, locate battery cabinets below or adjacent to the FACU. Battery cabinets must be installed at an accessible location when standing at floor level. Battery cabinets must not be installed lower than **300 mm** above finished floor, measured to the bottom of the cabinet, nor higher than **900 mm** above the floor, measured to the top of the cabinet. Installing batteries above drop ceilings or in inaccessible locations is prohibited. Battery cabinets must be large enough to accommodate batteries and also to allow ample gutter space for interconnection of control units as well as field wiring. The cabinet must be provided in a sturdy steel housing, complete with back box, hinged steel door with cylinder lock, and surface mounting provisions. The cabinet must be identified by an engraved phenolic resin nameplate. Lettering on the nameplate must indicate the control unit(s) the batteries power and must not be less than **25 mm** high.

3.2.3 Manual Stations

Locate manual stations as required by **NFPA 72** and as indicated on the drawings. Mount stations so they are located no farther than **1.5 m** from the exit door they serve, measured horizontally. Manual stations must be mounted at **1067 mm** measured to the operating handle.

3.2.4 Notification Appliances

- a. Locate notification appliance devices as required by **NFPA 72** and where indicated on the drawings. Where more than two visual notification appliances are located in the same room or corridor or field of view, provide synchronized operation. Devices must use screw terminals for all field wiring. Audible and visual notification appliances mounted in acoustical ceiling tiles must be centered in the tiles plus or

minus 50 mm.

- b. Audible and visual notification appliances mounted on the exterior of the building, within unconditioned spaces, or in the vicinity of showers must be listed weatherproof appliances installed on weatherproof backboxes.

3.2.5 Smoke and Heat Detectors

Locate detectors as required by NFPA 72 and their listing on a 100 mm mounting box. Install heat detectors not less than 100 mm from a side wall to the near edge. Heat detectors located on the wall must have the top of the detector at least 100 mm below the ceiling, but not more than 300 mm below the ceiling. Smoke detectors are permitted to be on the wall no lower than 300 mm from the ceiling with no minimum distance from the ceiling. Install smoke detectors no closer than 1 m from air handling supply diffusers. Detectors installed in acoustical ceiling tiles must be centered in the tiles plus or minus 50 mm .

3.2.6 Ceiling Bridges

Provide ceiling bridges for ceiling-mounted appliances. Ceiling bridges must be as recommended/required by the manufacturer of the ceiling-mounted notification appliance.

3.3 SYSTEM FIELD WIRING

3.3.1 Wiring within Cabinets, Enclosures, and Boxes

Provide wiring installed in a neat and workmanlike manner and installed parallel with or at right angles to the sides and back of any box, enclosure, or cabinet. Conductors that are terminated, spliced, or otherwise interrupted in any enclosure, cabinet, mounting, or junction box must be connected to screw-type terminal blocks. Mark each terminal in accordance with the wiring diagrams of the system. The use of wire nuts or similar devices is prohibited. Wiring to conform with NFPA 70.

Indicate the following in the wiring diagrams:

- a. Point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams must show connections from field devices to the FACU and remote fire alarm control units, initiating circuits, switches, relays and terminals.
- b. Complete riser diagrams indicating the wiring sequence of devices and their connections to the control equipment. Include a color code schedule for the wiring. Include floor plans showing the locations of devices and equipment.

3.3.2 Terminal Cabinets

Provide a terminal cabinet at the base of any circuit riser, on each floor at each riser, and where indicated on the drawings. Terminal size must be appropriate for the size of the wiring to be connected. Conductor terminations must be labeled and a drawing containing conductors, their labels, their circuits, and their interconnection must be permanently mounted in the terminal cabinet. Minimum size is 200 mm by 200 mm. Only

screw-type terminals are permitted. Provide an identification label, that displays "FIRE ALARM TERMINAL CABINET" with 50 mm lettering, on the front of the terminal cabinet.

3.3.3 Alarm Wiring

- a. Voltages must not be mixed in any junction box, housing or device, except those containing power supplies and control relays.
- b. Utilize shielded wiring where recommended by the manufacturer. For shielded wiring, ground the shield at only one point, in or adjacent to the FACU.
- c. T-tapping using screw terminal blocks is allowed for Class "B" signaling line circuits.
- d. Color coding is required for circuits and must be maintained throughout the circuit. Conductors used for the same functions must be similarly color coded. Conform wiring to NFPA 70.
- e. Pull all conductors splice free. The use of wire nuts, crimped connectors, or twisting of conductors is prohibited. Where splices are unavoidable, the location of the junction box or pull box where they occur must be identified on the as-built drawings. The number and location of splices must be subject to approval by the Designated Fire Protection Engineer (DFPE).

3.3.4 Back Boxes and Conduit

In addition to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, provide all wiring in rigid metal conduit or intermediate metal conduit unless specifically indicated otherwise. Minimum conduit size must be 19 mm in diameter. Do not use electrical non-metallic tubing (ENT) or flexible non-metallic tubing and associated fittings.

- a. Galvanized rigid steel (GRS) conduit must be utilized where exposed to weather, where subject to physical damage, and where exposed on exterior of buildings. Intermediate metal conduit (IMC) may be used in lieu of GRS as allowed by NFPA 70.
- b. Electrical metallic tubing (EMT) is permitted above suspended ceilings or exposed where not subject to physical damage. Do not use EMT underground, encased in concrete, mortar, or grout, in hazardous locations, where exposed to physical damage, outdoors or in fire pump rooms. Use die-cast compression connectors.
- c. For rigid metallic conduit (RMC), only threaded type fitting are permitted for wet or damp locations.
- d. Flexible metal conduit is permitted for initiating device circuits in length or less. Flexible metal conduit is prohibited for notification appliance circuits and signaling line circuits. Use liquid tight flexible metal conduit in damp and wet locations.
- e. Schedule 40 (minimum) polyvinyl chloride (PVC) is permitted where conduit is routed underground or underground below floor slabs. Convert non-metallic conduit, other than PVC Schedule 40 or 80, to plastic-coated rigid, or IMC, steel conduit before turning up through floor slab.

- f. Exterior wall penetrations must be weathertight. Conduit must be sealed to prevent the infiltration of moisture.

3.3.5 Conductor Terminations

Labeling of conductors at terminal blocks in terminal cabinets, FACU, and remote FACU must be provided at each conductor connection. Each conductor or cable must have a shrink-wrap label to provide a unique and specific designation. Each terminal cabinet, FACU, and remote FACU must contain a laminated drawing that indicates each conductor, its label, circuit, and terminal. The laminated drawing must be neat, using 12 point lettering minimum size, and mounted within each cabinet, control unit, or unit so that it does not interfere with the wiring or terminals. Maintain existing color code scheme where connecting to existing equipment.

3.4 CONNECTION OF NEW SYSTEM

The following new system connections must be made during the last phase of construction, at the beginning of the pre-Government tests. New system connections must include:

Connection of new system transmitter to existing installation fire reporting system.

Once these connections are made, system must be left energized. Report immediately to the Contracting Officer, coordination and field problems resulting from the connection of the above components.

3.5 FIRESTOPPING

Provide firestopping for holes at conduit penetrations through floor slabs, fire-rated walls, partitions with fire-rated doors, corridor walls, and vertical service shafts in accordance with Section 07 84 00 FIRESTOPPING.

3.6 PAINTING

- a. In unfinished areas (including areas above drop ceilings), paint all exposed electrical conduit (serving fire alarm equipment), fire alarm conduit, surface metal raceway, junction boxes and covers red. In lieu of painting conduit, the contractor may utilize red conduit with a factory applied finish.
- b. In finished areas, paint exposed electrical conduit (serving fire alarm equipment), fire alarm conduit, surface metal raceways, junction boxes, and electrical boxes to match adjacent finishes. The inside cover of the junction box must be identified as "Fire Alarm" and the conduit must have painted red bands 19 mm wide at 3-meter centers and at each side of a floor, wall, or ceiling penetration.
- c. Painting must comply with Section 09 90 00 PAINTS AND COATINGS.

3.7 FIELD QUALITY CONTROL

3.7.1 Test Procedures

Submit detailed test procedures, prepared and signed by the NICET Level III Fire Alarm Technician, and the representative of the installing

company, and reviewed by the QFPE 60 days prior to performing system tests. Detailed test procedures must list all components of the installed system such as initiating devices and circuits, notification appliances and circuits, signaling line devices and circuits, control devices/equipment, batteries, transmitting and receiving equipment, power sources/supply, annunciators, special hazard equipment, emergency communication equipment, interface equipment, and surge protective devices. Test procedures must include sequence of testing, time estimate for each test, and sample test data forms. The test data forms must be in a check-off format (pass/fail with space to add applicable test data; similar to the forma in NFPA 72 and NFPA 4.) The test procedures and accompanying test data forms must be used for the pre-Government testing and the Government testing. The test data forms must record the test results and must:

- a. Identify the NFPA Class of all Initiating Device Circuits (IDC), and Notification Appliance Circuits (NAC), and Signaling Line Circuits (SLC).
- b. Identify each test required by NFPA 72 Test Methods and required test herein to be performed on each component, and describe how these tests must be performed.
- c. Identify each component and circuit as to type, location within the facility, and unique identity within the installed system. Provide necessary floor plan sheets showing each component location, test location, and alphanumeric identity.
- d. Identify all test equipment and personnel required to perform each test (including equipment necessary for smoke detector testing. The use of magnets is not permitted.
- e. Provide space to identify the date and time of each test. Provide space to identify the names and signatures of the individuals conducting and witnessing each test.

3.7.2 Pre-Government Testing

3.7.2.1 Verification of Compliant Installation

Conduct inspections and tests to ensure that devices and circuits are functioning properly. Tests must meet the requirements of paragraph entitled "Minimum System Tests" as required by NFPA 72. The contractor and an authorized representative from each supplier of equipment must be in attendance at the pre-Government testing to make necessary adjustments. After inspection and testing is complete, provide a signed Verification of Compliant Installation letter by the QFPE that the installation is complete, compliant with the specification and fully operable. The letter must include the names and titles of the witnesses to the pre-Government tests. Provide all completion documentation as required by NFPA 72 including all referenced annex sections and the test reports noted below.

- a. NFPA 72 Record of Completion.
- b. NFPA 72 Record of Inspection and Testing.
- c. Fire Alarm and Emergency Communication System Inspection and Testing Form.

- d. Audibility test results with marked-up test floor plans.
- e. Documentation that all tests identified in the paragraph "Minimum System Tests" are complete.

3.7.2.2 Request for Government Final Test

When the verification of compliant installation has been completed, submit a formal request for Government final test to the Designated Fire Protection Engineer (DFPE). Government final testing will not be scheduled until the DFPE has received copies of the request for Government final testing and Verification of Compliant Installation letter with all required reports. Government final testing will not be performed until after the connections to the installation-wide fire reporting system has been completed and tested to confirm communications are fully functional. Submit request for test at least 15 calendar days prior to the requested test date.

3.7.3 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests must be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

3.7.4 Government Final Tests

The tests must be performed in accordance with the approved test procedures in the presence of the DFPE. Furnish instruments and personnel required for the tests. The following must be provided at the job site for Government Final Testing:

- a. The manufacturer's technical representative.
- b. Marked-up red line drawings of the system as actually installed.
- c. Loop resistance test results.
- d. Complete program printout including input/output addresses.
- e. Copy of pre-Government Test Certificate, test procedures and completed test data forms.

Government Final Tests will be witnessed by the Designated Fire Protection Engineer. At this time, any and all required tests noted in the paragraph "Minimum System Tests" must be repeated at their discretion.

3.8 MINIMUM SYSTEM TESTS

3.8.1 System Tests

Test the system in accordance with the procedures outlined in NFPA 72. The required tests are as follows:

- a. Loop Resistance Tests: Measure and record the resistance of each circuit with each pair of conductors in the circuit short-circuited at the farthest point from the circuit origin. The tests must be

witnessed by the Contracting Officer and test results recorded for use at the final Government test.

- b. Verify the absence of unwanted voltages between circuit conductors and ground. The tests must be accomplished at the pre-Government test with results available at the final system test.
- c. Verify that the control unit is in the normal condition as detailed in the manufacturer's O&M manual.
- d. Test each initiating device and notification appliance and circuit for proper operation and response at the control unit. Smoke detectors must be tested in accordance with manufacturer's recommended calibrated test method. Use of magnets is prohibited. Testing of duct smoke detectors must comply with the requirements of NFPA 72 except disconnect at least 20 percent of devices. If there is a failure at these devices, then supervision must be tested at each device.
- e. Test the system for specified functions in accordance with the contract drawings and specifications and the manufacturer's O&M manual.
- f. Test both primary power and secondary power. Verify, by test, the secondary power system is capable of operating the system for the time period and in the manner specified.
- g. Determine that the system is operable under trouble conditions as specified.
- h. Visually inspect wiring.
- i. Test the battery charger and batteries.
- j. Verify that software control and data files have been entered or programmed into the FACU. Hard copy records of the software must be provided to the Contracting Officer.
- k. Verify that red-line drawings are accurate.
- l. Measure the current in circuits to ensure there is the calculated spare capacity for the circuits.
- m. Measure voltage readings for circuits to ensure that voltage drop is not excessive.
- n. Disconnect the verification feature for smoke detectors during tests to minimize the amount of smoke needed to activate the sensor. Testing of smoke detectors must be conducted using real smoke or the use of canned smoke which is permitted.
- o. Measure the voltage drop at the most remote appliance (based on wire length) on each notification appliance circuit.
- p. Verify the documentation cabinet is installed and contains all as-built shop drawings, product data sheets, design calculations, site-specific software data package, and all documentation required by paragraph titled "Test Reports".

3.8.2 Audibility Tests

Sound pressure levels from audible notification appliances must be a minimum of 15 dBa over ambient with a maximum of 110 dBa in any occupiable area. The provisions for audible notification (audibility and intelligibility) must be met with doors, fire shutters, movable partitions, and similar devices closed.

3.9 SYSTEM ACCEPTANCE

Following acceptance of the system, as-built drawings and O&M manuals must be delivered to the Contracting Officer for review and acceptance. The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final Government test of the system. At least one set of as-built (marked-up) drawings must be provided at the time of, or prior to the Final Government Test.

- a. The drawings must be prepared electronically and sized no less than the contract drawings.
- b. Include complete wiring diagrams showing connections between devices and equipment, both factory and field wired.
- c. Include a riser diagram and drawings showing the as-built location of devices and equipment.
- d. Provide [Operation and Maintenance \(O&M\) Instructions](#).

3.10 [INSTRUCTION OF GOVERNMENT EMPLOYEES](#)

3.10.1 Instructor

Provide the services of an instructor, who has received specific training from the manufacturer for the training of other persons regarding the operation, inspection, testing, and maintenance of the system provided. The instructor must train the Government employees designated by the Contracting Officer, in the care, adjustment, maintenance, and operation of the fire alarm system. The instructor must be thoroughly familiar with all parts of this installation. The instructor must be trained in operating theory as well as in practical O&M work. Submit the instructors information and qualifications including the training history.

3.10.2 Required Instruction Time

Provide 8 hours of instruction after final acceptance of the system. The instruction must be given during regular working hours on such dates and times selected by the Contracting Officer. The instruction may be divided into two or more periods at the discretion of the Contracting Officer. The training must allow for rescheduling for unforeseen maintenance and/or fire department responses.

3.11 EXTRA MATERIALS

3.11.1 Repair Service/Replacement Parts

Repair services and replacement parts for the system must be available for a period of 10 years after the date of final acceptance of this work by the Contracting Officer. During the warranty period, the service

technician must be on-site within 24 hours after notification. All repairs must be completed within 24 hours of arrival on-site.

During the warranty period, the installing fire alarm contractor is responsible for conducting all required testing and maintenance in accordance with the requirements and recommended practices of NFPA 72 and the system manufacturer. Installing fire alarm contractor is NOT responsible for any damage resulting from abuse, misuse, or neglect of equipment by the end user.

3.11.2 Spare Parts

Spare parts furnished must be directly interchangeable with the corresponding components of the installed system. Spare parts must be suitably packaged and identified by nameplate, tagging, or stamping. Spare parts must be delivered to the Contracting Officer at the time of the Government testing and must be accompanied by an inventory list.

3.11.3 Document Storage Cabinet

Upon completion of the project, but prior to project close-out, place in the document storage cabinet copies of the following record documentation:

- a. As-built shop drawings
- b. Product data sheets
- c. Design calculations
- d. Site-specific software data package
- e. All documentation required by SD-06.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 31 - EARTHWORK

SECTION 31 00 00

EARTHWORK

08/23

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Structural Fill
 - 1.2.2 Embankment Fill
 - 1.2.3 Topsoil
 - 1.2.4 Utility Bedding Material
 - 1.2.5 Flowable Fill
 - 1.2.6 Satisfactory Materials
 - 1.2.7 Unsatisfactory Materials
 - 1.2.8 Cohesionless Materials (Fuel Line Bedding Material)
 - 1.2.9 Hard/Unyielding Materials
 - 1.2.10 Rock
 - 1.2.11 Degree of Compaction (Proctor)
 - 1.2.12 Degree of Compaction (Relative Density)
 - 1.2.13 Borrow
 - 1.2.14 Subgrade
- 1.3 SUBSURFACE DATA
- 1.4 CRITERIA FOR BIDDING
- 1.5 SUBMITTALS
- 1.6 QUALITY CONTROL
 - 1.6.1 Geotechnical Engineer
 - 1.6.2 Qualified Technician
 - 1.6.3 Lab Validation
 - 1.6.4 Preconstruction Meeting

PART 2 PRODUCTS

- 2.1 SOIL MATERIALS
 - 2.1.1 Structural Fill
 - 2.1.2 Embankment Fill
 - 2.1.3 Topsoil
 - 2.1.4 Utility Bedding Material
 - 2.1.4.1 Class I
 - 2.1.4.2 Class II
 - 2.1.4.3 Sand (Fuel Pipe Bedding Material)
 - 2.1.5 Gravel and Crushed Stone
- 2.2 FLOWABLE FILL
- 2.3 BURIED WARNING AND IDENTIFICATION MARKERS
 - 2.3.1 Warning Tape for Metallic Piping
 - 2.3.2 Detectable Warning Tape for Non-Metallic Piping
 - 2.3.3 Detection Wire for Non-Metallic Piping
- 2.4 BORROW

PART 3 EXECUTION

- 3.1 PROTECTION
 - 3.1.1 Underground Utilities
 - 3.1.2 Drainage and Dewatering
 - 3.1.2.1 Drainage
 - 3.1.2.2 Dewatering
 - 3.1.3 Shoring and Sheet piling
 - 3.1.4 Protection of Graded Surfaces
- 3.2 BORROW
 - 3.2.1 Contractor Furnished Borrow Area(s)
 - 3.2.2 Environmental Requirements for Off-Site Soil
- 3.3 SURFACE PREPARATION
 - 3.3.1 Clearing and Grubbing
 - 3.3.2 Stripping
 - 3.3.3 Proof Rolling
 - 3.3.4 Stockpiling Operations
- 3.4 EXCAVATION
 - 3.4.1 Ditches, Gutters, and Channel Changes
 - 3.4.2 Trench Excavation Requirements
 - 3.4.2.1 Bottom Preparation
 - 3.4.2.2 Removal of Unyielding Material
 - 3.4.2.3 Removal of Unstable Material
 - 3.4.2.4 Excavation for Appurtenances
 - 3.4.2.5 Water Lines
 - 3.4.3 Jacking, Boring, and Tunneling
 - 3.4.3.1 Pipeline Casing
 - 3.4.3.2 Bore Holes
 - 3.4.3.3 Cleaning
 - 3.4.3.4 End Seals
 - 3.4.4 Underground Utilities
 - 3.4.5 Structural Excavation
 - 3.4.6 Rock Excavation and Blasting
- 3.5 SUBGRADE PREPARATION
 - 3.5.1 General Requirements
 - 3.5.2 Subgrade for Structures, Spread Footings, and Concrete Slabs
 - 3.5.3 Subgrade for Pavements
 - 3.5.4 Subgrade for Shoulders
- 3.6 FILLING AND COMPACTION
 - 3.6.1 Trench Backfill
 - 3.6.1.1 Replacement of Unyielding Material
 - 3.6.1.2 Replacement of Unstable Material
 - 3.6.1.3 Bedding and Initial Backfill
 - 3.6.1.4 Final Backfill
 - 3.6.1.4.1 Buildings and Pavements
 - 3.6.1.4.2 Turfed or Seeded Areas and Miscellaneous Areas
 - 3.6.1.5 Electrical Distribution System
 - 3.6.1.6 Displacement of Features
 - 3.6.1.7 Buried Tape And Detection Wire
 - 3.6.1.7.1 Buried Warning and Identification Tape
 - 3.6.1.7.2 Buried Detection Wire
 - 3.6.2 Structural Fill Placement
 - 3.6.3 Backfill for Appurtenances
 - 3.6.4 Porous Fill Placement
 - 3.6.5 Flowable Fill
 - 3.6.6 Compaction
 - 3.6.6.1 General Site
 - 3.6.6.2 Adjacent Areas
- 3.7 EMBANKMENTS
 - 3.7.1 Earth Embankments

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.8 FINISHING/FINISH OPERATIONS
 - 3.8.1 Grading Around Structures
 - 3.8.2 Shoulder Construction
 - 3.8.3 Grading
 - 3.8.4 Topsoil and Seed
- 3.9 DISPOSITION OF SURPLUS MATERIAL
- 3.10 TESTING

-- End of Section Table of Contents --

SECTION 31 00 00

EARTHWORK
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 103500	(1994) Geotechnical. Compaction test. Standard Proctor.
UNE 103501	(1994) Geotechnic. Compactation Test. Modified Proctor.
UNE 103503	(1995) Determination "In Situ" Of Density Of Soil By The Sand Method.
UNE 103900	(2013) In situ determination of density and moisture content of soil and granular materials by nuclear methods: low depths.
UNE-EN 197-1	(2011) Cement - Part 1: Composition, specifications and conformity criteria for common cements.
UNE-EN 450-1	(2013) Fly ash for concrete - Part 1: Definition, specifications and conformity criteria.
UNE-EN 933-1	(2012) Tests for geometrical properties of aggregates - Part 1: Determination of particle size distribution - Sieving method
UNE-EN 934-2	(2010+A1:2012) Admixtures for concrete, mortar and grout. Part 2: Concrete admixtures. Definitions, requirements, conformity, marking and labeling.
UNE-EN 12620	(2003+A1:2009) Aggregates for concrete
UNE-EN 13139	(2003) Aggregates for mortar
UNE-EN 15167-1	(2008) Ground granulated blast furnace slag for use in concrete, mortar and grout - Part 1: Definitions, specifications and conformity criteria
UNE-EN ISO 3834-2	(2006) Quality requirements for fusion welding of metallic materials - Part 2: Comprehensive quality requirements (ISO

3834-2:2005)

UNE-EN ISO 14688-1 (2019) Geotechnical investigation and testing - Identification and classification of soil - Part 1: Identification and description (ISO 14688-1:2017)

UNE-EN ISO 17892-1 (2015) Geotechnical investigation and testing - Laboratory testing of soil - Part 1: Determination of water content (ISO 17892-1:2014)

UNE-EN ISO 17892-12 (2019) Geotechnical investigation and testing - Laboratory testing of soil - Part 12: Determination of liquid and plastic limits (ISO 17892-12:2018).

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO T 180 (2017) Standard Method of Test for Moisture-Density Relations of Soils Using a 4.54-kg (10-lb) Rammer and a 457-mm (18-in.) Drop

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C600 (2017) Installation of Ductile-Iron Mains and Their Appurtenances

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM C33/C33M (2018) Standard Specification for Concrete Aggregates

ASTM C136/C136M (2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates

ASTM C260/C260M (2010a; R 2016) Standard Specification for Air-Entraining Admixtures for Concrete

ASTM C618 (2023; E 2023) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete

ASTM C989/C989M (2022) Standard Specification for Slag Cement for Use in Concrete and Mortars

ASTM D1556/D1556M (2015; E 2016) Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method

ASTM D1557 (2012; E 2015) Standard Test Methods for

	Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft ³) (2700 kN-m/m ³)
ASTM D2216	(2019) Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
ASTM D2487	(2017; E 2020) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D2974	(2020; E 2020) Moisture, Ash, and Organic Matter of Peat and Other Organic Soils
ASTM D4253	(2016; E 2019) Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table
ASTM D4254	(2016) Standard Test Methods for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density
ASTM D4318	(2017; E 2018) Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
ASTM D6023	(2016) Standard Test Method for Density (Unit Weight), Yield, Cement Content, and Air Content (Gravimetric) of Controlled Low-Strength Material (CLSM)
ASTM D6103/D6103M	(2017; E 2021) Standard Test Method for Flow Consistency of Controlled Low Strength Material (CLSM)
ASTM D6938	(2017a) Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety -- Safety and Health Requirements Manual
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U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 600/4-79/020	(1983) Methods for Chemical Analysis of Water and Wastes
EPA SW-846.3-3	(1999, Third Edition, Update III-A) Test Methods for Evaluating Solid Waste: Physical/Chemical Methods

1.2 DEFINITIONS

1.2.1 Structural Fill

Soil material placed to support buildings, walls, pads, and other similar facilities.

1.2.2 Embankment Fill

Soil material placed to construct embankment.

1.2.3 Topsoil

Surface layer of primarily organic soil capable of supporting vegetation growth. See 32 92 19 SEEDING.

1.2.4 Utility Bedding Material

Fill placed to directly support pipes, conduits, cables, and appurtenant structures. Bedding may also be used to provide a cushion between utilities and bedrock, obstacles, obstructions and other unyielding materials.

1.2.5 Flowable Fill

Fill placed in a plastic or liquid form that flows to near its final placement location with limited assistance and subsequently cures or solidifies to provide a stable or impermeable barrier.

1.2.6 Satisfactory Materials

Satisfactory materials for fill, backfill, and/or any in-situ soils to remain in place comprise any materials classified by ASTM D2487 or UNE-EN ISO 14688-1 as GW, GP, GM, GP-GM, GW-GM, SW, SM, SW-SM, SC, SP-SM. Maximum particle size to be no greater than 75 mm in any dimension.

1.2.7 Unsatisfactory Materials

Materials which do not comply with the requirements for satisfactory materials are unsatisfactory. Unsatisfactory materials also include man-made fills; trash; refuse; backfills from previous construction; roots and other organic matter or frozen material. Notify the Contracting Officer when encountering any contaminated materials.

1.2.8 Cohesionless Materials (Fuel Line Bedding Material)

Cohesionless materials include materials classified in ASTM D2487 or UNE-EN ISO 14688-1 as SW and SP.

1.2.9 Hard/Unyielding Materials

Hard/Unyielding materials comprise weathered rock, dense consolidated deposits, or conglomerate materials which are not included in the definition of "rock" with stones greater than 75 mm in any dimension or as defined by the pipe manufacturer, whichever is smaller. These materials usually require the use of heavy excavation equipment, ripper teeth, or jack hammers for removal.

1.2.10 Rock

Solid homogeneous interlocking crystalline material with firmly cemented, laminated, or foliated masses or conglomerate deposits, neither of which can be removed without systematic drilling and blasting, drilling and the use of expansion jacks or feather wedges, or the use of backhoe-mounted pneumatic hole punchers or rock breakers; also large boulders, buried masonry, or concrete other than pavement exceeding 0.375 cubic meter in volume. Removal of hard material will not be considered rock excavation because of intermittent drilling and blasting that is performed merely to increase production.

1.2.11 Degree of Compaction (Proctor)

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum density obtained by the test procedure presented in ASTM D1557 or UNE 103501 abbreviated as a percent of laboratory maximum density. Express the degree of compaction for material having more than 30 percent by weight of their particles retained on the 19.0 mm sieve as a percentage of the maximum density in accordance with AASHTO T 180-21 paragraph 1.5, Note 1.

1.2.12 Degree of Compaction (Relative Density)

Degree of compaction required for soils with less than 5 percent passing the No. 200 sieve, is expressed as a relative percentage of the maximum index density/dry unit weight and minimum index density/dry unit weight, obtained by the test procedures in accordance with ASTM D4253 and ASTM D4254 or UNE 103500, respectively, abbreviated as a percent of laboratory relative density.

1.2.13 Borrow

Soil brought to the project site from an external location for the purposes of project construction.

1.2.14 Subgrade

Earth materials directly below foundations and directly below granular base materials in building slab and pavement areas including shoulders.

1.3 SUBSURFACE DATA

Subsurface soil boring logs are shown in project plans. These data represent available subsurface information; however, variations may exist between boring locations.

1.4 CRITERIA FOR BIDDING

Base bids on the following criteria:

- a. Surface elevations are as indicated.
- b. Pipes or other artificial obstructions, except those indicated, will not be encountered.
- c. Ground water elevations indicated by the boring log were those existing at the time subsurface investigations were made and do not necessarily represent ground water elevation at the time of

construction.

- d. Material character is indicated by the boring logs.
- e. Hard materials and rock will not be encountered at 6 meter below existing surface elevations.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Excavation and Trenching Plan; G
Borrow Plan; G
Dewatering Work Plan; G
Disposition of Surplus Materials; G
Preconstruction Meeting; G

SD-03 Product Data

Flowable Fill Mix Design; G

SD-06 Test Reports

Dewatering Performance Records; G
Material Test Report; G
Borrow Site Testing; G
Pipe Inspection Report; G
Geotechnical Evaluation Report; G

1.6 QUALITY CONTROL

1.6.1 Geotechnical Engineer

Provide a Professional Geotechnical Engineer to provide inspection of excavations and soil/groundwater conditions throughout construction. The Geotechnical Engineer is responsible for performing pre-construction and periodic site visits throughout construction to assess site conditions. The Geotechnical Engineer is responsible for preparing and updating the Excavation and Trenching Plan and Dewatering Work Plan as construction progresses to reflect changing conditions and submit an updated plan if necessary. Submit a monthly [Geotechnical Evaluation report](#), informing the Contractor and Contracting Officer of the status of the plan and an accounting of the Contractor's adherence to the plan addressing any present or potential problems. The Contractor is responsible for

arranging meetings with the Geotechnical Engineer and Contracting Officer throughout the contract duration.

1.6.2 Qualified Technician

Provide a Qualified Technician to inspect, monitor, sample, and performing field testing. The technician qualifications need to be one of the following: a current National Institute for Certification in Engineering Technologies (NICET) Level II minimum certification in Construction Materials Testing Soils; a Geologist-in-Training with minimum one-year experience; an Engineer-in-Training with minimum one-year experience; a Registered Geologist; or a Professional Engineer.

1.6.3 Lab Validation

Perform testing by a Corps validated commercial testing laboratory or Contractor established testing laboratory meeting the requirements of Section 01 45 00 (or similar number) entitled QUALITY CONTROL and approved by the Contracting Officer. Submit testing laboratory validation for the testing to be performed. Do not permit work requiring testing until testing facilities have been inspected, Corps validated and approved by the Contracting Officer.

1.6.4 Preconstruction Meeting

Conduct a preconstruction meeting at the jobsite at least five business days prior to the start of earthwork operations on the project. The [preconstruction meeting](#) is to be arranged by the Contractor and is to follow the written agenda submitted prior to the meeting. The purpose of this meeting is to review the requirements of this specification and the associated plans. The following individuals must be in attendance at this meeting: Contractor's Project Manager and Project Superintendent, earthwork subcontractor's Project Manager and Site Foreman, Contractor's Geotechnical Engineer and Testing Agency, Government Geotechnical Engineer and Civil Engineer, and Government Construction Manager and Engineering Technician.

The minutes of this meeting are to be recorded by the Contractor and published via email within 48 hours to all attendees. The minutes must be re-published within 48 hours via email pending any subsequent comments from the attendees.

PART 2 PRODUCTS

2.1 SOIL MATERIALS

2.1.1 Structural Fill

Materials classified as GW, GP, GM, GW-GM, GP-GM, SW, SM, SW-SM, or SP-SM in accordance with [ASTM D2487](#) or [UNE-EN ISO 14688-1](#). Select material type appropriate for the intended purpose.

2.1.2 Embankment Fill

Materials classified as GW, GP, GM, GC, GW-GM, GW-GC, GP-GM, GP-GC, GC-GM, SW, SP, SM, SW-SM, SC, SW-SC, or SP-SM in accordance with [ASTM D2487](#) or [UNE-EN ISO 14688-1](#). Select material type appropriate for the intended purpose.

2.1.3 Topsoil

Material suitable for topsoil obtained from offsite areas or excavations is defined as: Natural, friable soil representative of productive, well-drained soils in the area, free of subsoil, stumps, rocks larger than 25 mm diameter, brush, weeds, toxic substances, and other material detrimental to plant growth. Amend topsoil pH range to obtain a pH of 5.5 to 7.

2.1.4 Utility Bedding Material

Except as specified otherwise in the individual piping section, provide bedding for buried piping in accordance with AWWA C600. Install bedding for plastic piping to spring line of pipe. Utility bedding material may include the following:

2.1.4.1 Class I

Angular, 6 to 40 mm, graded stone, including a number of fill materials that have regional significance such as coral, slag, cinders, crushed stone, and crushed shells.

2.1.4.2 Class II

Coarse sands and gravels with maximum particle size of 40 mm, including various graded sands and gravels containing small percentages of fines, generally granular and noncohesive, either wet or dry. Soil Types GW, GP, SW, and SP are included in this class as specified in ASTM D2487 or UNE-EN ISO 14688-1.

2.1.4.3 Sand (Fuel Pipe Bedding Material)

Clean, coarse-grained sand classified as SW or SP by ASTM D2487 or UNE-EN ISO 14688-1 for bedding as indicated.

2.1.5 Gravel and Crushed Stone

Clean, coarsely graded natural gravel, crushed stone or a combination thereof identified as gravel, or having a classification of GW GP in accordance with ASTM D2487 or UNE-EN ISO 14688-1 for road surface as indicated. Do not exceed maximum particle size of 75 mm.

2.2 FLOWABLE FILL

Design and submit flowable fill mix design to consist of Portland cement, fly ash, and/or slag cement and fine aggregate. Include the dry weights of cementitious material(s); quality and gradation of aggregates in the saturated surface-dry weights along with gradation tests; quantities, types, and names of admixtures; and quantity of water per cubic yard. The maximum unconfined compressive strength to be 10 Mpa psi at 28 days. The aggregates in accordance with ASTM C33/C33M or UNE-EN 12620 or or UNE-EN 13139 Fine Aggregates. Air-entrain fill in accordance with ASTM C260/C260M or UNE-EN 934-2. The air content to be between 8 and 15 percent in accordance with ASTM D6023. The flow to be between 17 to 25 mm in accordance with ASTM D6103/D6103M. Portland cement to be CEM I or II in accordance with UNE-EN 197-1. Fly ash to be Class C in accordance with ASTM C618 or UNE-EN 450-1. Provide slag cement in Grade 100 or 120 in accordance with ASTM C989/C989M or UNE-EN 15167-1.

2.3 BURIED WARNING AND IDENTIFICATION MARKERS

Provide polyethylene plastic and metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic warning tape manufactured specifically for warning and identification of buried utility lines. Provide tape on rolls, 75 mm minimum width, color coded as specified below for the intended utility with warning and identification imprinted in bold black letters continuously over the entire tape length. Warning and identification to read, "CAUTION, BURIED (intended service) LINE BELOW" or similar wording. Provide permanent color and printing, unaffected by moisture or soil.

Warning Tape Color Codes	
Red	Electric
Yellow	Gas, Oil; Dangerous Materials
Orange	Telephone and Other Communications
Blue	Water Systems
Green	Sewer Systems
White	Steam Systems
Gray	Compressed Air

2.3.1 Warning Tape for Metallic Piping

Provide acid and alkali-resistant polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.08 mm and a minimum strength of 10.3 MPa lengthwise, and 8.6 MPa crosswise, with a maximum 350 percent elongation.

2.3.2 Detectable Warning Tape for Non-Metallic Piping

Provide polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.10 mm, and a minimum strength of 10.3 MPa lengthwise and 8.6 MPa crosswise. Manufacture tape with integral wires, foil backing, or other means of enabling detection by a metal detector when tape is buried up to 920 mm deep. Encase metallic element of the tape in a protective jacket or provide with other means of corrosion protection.

2.3.3 Detection Wire for Non-Metallic Piping

Insulate a single strand, solid copper detection wire with a minimum of 12 AWG.

2.4 BORROW

If borrow is not available on site, provide borrow materials from sources located outside of Government property meeting the requirements of paragraph STRUCTURAL FILL TOPSOIL.

PART 3 EXECUTION

3.1 PROTECTION

Perform all work specified in accordance with applicable requirements of the Corps of Engineers publication [EM 385-1-1](#) Safety and Health Requirements Manual. Provide a Geotechnical Engineer to monitor construction activities and to prepare necessary work plans and reports; see paragraph QUALITY CONTROL.

Use equipment of type and size appropriate for the site conditions (soil character and moisture content). Maintenance of exposed subgrades and fills is the responsibility of the Contractor. The Contractor is required to prevent damage by ineffective drainage, dewatering, and heavy loads and equipment by implementing precautionary measures. Repair or replace any defects or damage.

3.1.1 Underground Utilities

Location of the existing utilities indicated is approximate. Physically verify the location and elevation of the existing utilities indicated prior to starting construction. The Contractor is responsible for protecting utilities from damage during construction.

3.1.2 Drainage and Dewatering

Provide for the collection and disposal of surface and subsurface water encountered during construction.

3.1.2.1 Drainage

Provide for the collection and disposal of surface and subsurface water encountered during construction. Construct storm drainage features (ponds/basins) at the earliest stages of site development, and throughout construction grade the construction area to provide positive surface water runoff away from the construction activity and provide temporary ditches, swales, and other drainage features and equipment as required to keep soils from becoming unstable, prevent erosion, or undermining of foundations. Remove unstable material from working platforms for equipment operation and soil support for subsequent construction features and provide new material as specified herein. It is the responsibility of the Contractor to assess the site conditions to employ necessary measures to permit construction to proceed.

3.1.2.2 Dewatering

Control groundwater flowing toward or into excavations to prevent sloughing of excavation slopes and walls, boils, uplift and heave in the excavation and to eliminate interference with orderly progress of construction. French drains, sumps, ditches or trenches are not allowed within [one meter](#) of the foundation of any structure, except with specific written approval, and after specific contractual provisions for restoration of the foundation area have been made. Perform control measures by the time the excavation reaches the water level in order to maintain the integrity of the in-situ material. While the excavation is open, maintain the water level continuously, at least [0.3 meters](#) below the working level. Submit a [Dewatering Work Plan](#) outlining procedures for accomplishing dewatering work. Operate dewatering system continuously until construction work below existing water levels is complete. Measure

and record performance of dewatering system at same time each day by use of observation wells or piezometers installed in conjunction with the dewatering system. Submit [dewatering performance records](#) weekly. All dewatering affluent must be filtered before release to the environment. Any dewatering effluent found to contain contamination must be handled in accordance with [01 57 19.01 21 ENVIRONMENTAL CONTROLS](#) and [01 35 29.13 HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES](#).

3.1.3 Shoring and Sheet piling

Submit an [Excavation and Trenching Plan](#) to stabilize features, prevent undermining or unintended horizontal and vertical movement of adjacent structures, and prevent slippage or movement in banks or slopes adjacent to the excavation. Submit drawings and calculations, certified by a registered professional engineer, describing the methods for shoring and sheet piling of excavations. Drawings to include material sizes and types, arrangement of members, and the sequence and method of installation and removal. Calculations are to include data and references used.

3.1.4 Protection of Graded Surfaces

Protect newly backfilled, graded, and topsoiled areas from traffic, erosion, and settlements that may occur. Repair or reestablish damaged grades, elevations, or slopes.

3.2 BORROW

Select borrow material to meet the requirements and conditions of the fill or embankment for which it is to be used. Obtain borrow material from approved private sources. Submit a [Borrow Plan](#) that includes materials to be excavated, stockpile locations, proposed slopes, drainage, and closure. Unless otherwise provided in the contract, the Contractor is responsible for obtaining the right to procure material, pay royalties and other charges involved, and bear the expense of developing the sources, including rights-of-way for hauling from the owners. Unless specifically provided, do not obtain borrow within the limits of the project site without prior written approval.

3.2.1 Contractor Furnished Borrow Area(s)

Obtain approved borrow materials from approved offsite sources. If a borrow source is selected that is not a commercial entity from which soil material is directly purchased, submit a Borrow Plan that includes the borrow source location, geotechnical test results showing the fill material meets the Contract requirements, environmental test results in accordance with paragraph ENVIRONMENTAL REQUIREMENTS FOR OFF-SITE SOIL, and any Spanish national, regional and local permits required for excavation and reclamation of the borrow area.

3.2.2 Environmental Requirements for Off-Site Soil

Test offsite soils brought in for use as backfill for Total Petroleum Hydrocarbons (TPH), Benzene, Toluene, Ethyl Benzene, and Xylene (BTEX) and full Toxicity Characteristic Leaching Procedure (TCLP) including ignitability, corrosivity and reactivity. Backfill may not contain concentrations for hazardous waste characteristics above 10 parts per million (ppm) of total petroleum hydrocarbons (TPH) and 10 ppm of the sum of Benzene, Toluene, Ethyl Benzene, and Xylene (BTEX) and pass the TCLP test. Determine TPH concentrations by using [EPA 600/4-79/020](#) Method

418.1. Determine BTEX concentrations by using EPA SW-846.3-3 Method 5030/8020/8260B. Perform TCLP in accordance with EPA SW-846.3-3 Method 1311. Perform hazardous waste characteristic tests for ignitability, corrosivity, and reactivity in accordance with accepted standard methods. Perform PCB testing in accordance with accepted standard methods for sampling and analysis of bulk solid samples. Provide borrow site testing for petroleum hydrocarbons and BTEX from a grab sample of material from the area most likely to be contaminated at the borrow site (as indicated by visual or olfactory evidence), with at least one test from each borrow site. Provide borrow site testing for hazardous waste characteristics (TPH, BTEX and TCLP) from a composite sample of material, collected in accordance with standard soil sampling techniques. Do not bring borrow material to project site until Borrow Plan containing environmental test results has been received and approved by the Contracting Officer.

3.3 SURFACE PREPARATION

3.3.1 Clearing and Grubbing

Remove trees, stumps, logs, shrubs, brush and vegetation and other items that would interfere with construction operations. Remove stumps entirely. Grub out matted roots and roots over 75 mm in diameter to at least 500 mm below existing surface.

3.3.2 Stripping

Strip site where indicated on the plans. Strip existing surface materials to a depth of 400 mm below the existing ground surface in areas designated as Clear and Grub on the plans. Strip in all areas within the planned limits of disturbance. All stripped materials not suitable for reuse as topsoil will be wasted in specified disposal area. Screen all stripped soils to remove roots and organic materials prior disposal.

Strip suitable soil from the site where excavation or grading is indicated and stockpile separately from other excavated material. Protect topsoil and keep in segregated piles until needed.

3.3.3 Proof Rolling

Perform proof rolling on exposed subgrade that is unfrozen and free of surface water (wet conditions resulting from rainfall). Notify the Contracting Officer a minimum of three days prior to proof rolling.

After stripping, excavating, and rough grading to the planned elevation, proof roll the existing subgrade of all building, pavement and embankment locations with six passes of a 15 ton, pneumatic-tired or smooth drum roller. Operate the roller in a systematic manner to ensure the number of passes over all areas, and at speeds between 4 to 5.5 km per hour. Subgrade materials that exhibit excessive deflection and/or rutting during proof rolling need to be scarified, aerated, and re-compacted to specified density at plus or minus 2 percent of optimum moisture content prior to being considered for remedial action by the Contracting Officer. When proof rolling under buildings, the building subgrade is considered to extend 1.5 meters beyond the building lines, and make one-half of the passes with the roller in a direction perpendicular to the other passes.

3.3.4 Stockpiling Operations

Place and grade stockpiles of satisfactory and unsatisfactory and wasted

materials as specified. Keep stockpiles in a neat and well drained condition, giving due consideration to drainage at all times. Clear, grub, and seal by rubber-tired equipment, the ground surface at stockpile locations; separately stockpile excavated satisfactory and unsatisfactory materials. Protect stockpiles of satisfactory materials from contamination which may destroy the quality and fitness of the stockpiled material. Do not create stockpiles that could obstruct the flow of any stream, endanger a partly finished structure, impair the efficiency or appearance of any structure, or be detrimental to the completed work in any way. If the Contractor fails to protect the stockpiles, and any material becomes unsatisfactory, remove and replace such material with satisfactory material from approved sources.

3.4 EXCAVATION

Excavate to contours, elevation, and dimensions indicated. Excavate soil disturbed or weakened by Contractor's operations, and soils softened or made unstable for subsequent construction due to exposure to weather. Use material removed from excavations meeting the specified requirements in the construction of fills, embankments, subgrades, shoulders, bedding (as backfill), and for similar purposes to minimize surplus material and to minimize additional material brought on site. Do not excavate below indicated depths except to remove unstable material as determined by the Government and confirmed by the Contracting Officer. Remove and replace excavations below the grades shown with appropriate materials as directed by the Contracting Officer.

If at any time during excavation, including excavation from borrow areas, the Contractor encounters material that may be classified as rock or as hard/unyielding material, uncover such material, and notify the Contracting Officer. Do not proceed with the excavation of this material until the Contracting Officer has classified the materials as common excavation or rock excavation. Failure on the part of the Contractor to uncover such material, notify the Contracting Officer, and allow sufficient time for classification and delineation of the undisturbed surface of such material will cause the forfeiture of the Contractor's right of claim to any classification or volume of material to be paid for other than that allowed by the Contracting Officer for the areas of work in which such deposits occur.

3.4.1 Ditches, Gutters, and Channel Changes

Finish excavation of ditches, gutters, and channel changes by cutting accurately to the cross sections, grades, and elevations shown. Do not excavate below grades shown. Backfill excessive excavation as directed by the Contracting Officer, with satisfactory, compacted, material or with suitable stone or cobble to grades shown. Dispose excavated material as shown or as directed. Do not allow material to be deposited within **one meter** from edge of a ditch. Maintain excavations free from detrimental quantities of leaves, brush, sticks, trash, and other debris until final acceptance of the work.

3.4.2 Trench Excavation Requirements

Excavate the trench as recommended by the manufacturer of the pipe to be installed. Slope trench walls below the top of the pipe, or make vertical, and of such width as recommended by the manufacturer. Provide vertical trench walls where no manufacturer installation instructions are available. Do not exceed the trench width of **600 mm** below the top pipe

plus pipe outside diameter (O.D.) for pipes of less than 600 mm inside diameter, and do not exceed 900 mm plus pipe outside diameter for pipe sizes larger than 600 mm inside diameter. Where recommended trench widths are exceeded, provide redesign, stronger pipe, or special installation procedures. The Contractor is responsible for the cost of redesign, stronger pipe, or special installation procedures without any additional cost to the Government.

3.4.2.1 Bottom Preparation

Grade the bottoms of trenches accurately to provide uniform bearing and support for the bottom quadrant of each section of the pipe. Excavate bell holes to the necessary size at each joint or coupling to eliminate point bearing. Remove stones of 75 mm or greater in any dimension, or as recommended by the pipe manufacturer, whichever is smaller, to avoid point bearing.

3.4.2.2 Removal of Unyielding Material

Where unyielding material is encountered in the bottom of the trench, notify the Contracting Officer. Following approval, remove such material 100 mm below the required grade and replaced with suitable materials as provided in paragraph FILLING AND COMPACTION.

3.4.2.3 Removal of Unstable Material

Where unstable material is encountered in the bottom of the trench, remove such material to the depth directed and replace it to the proper grade with suitable material as provided in paragraph FILLING AND COMPACTION. When removal of unstable material is required due to the Contractor's fault or neglect in performing the work, the Contractor is responsible for excavating the resulting material and replacing it without additional cost to the Government.

3.4.2.4 Excavation for Appurtenances

Provide excavation for manholes, catch-basins, inlets, or similar structures sufficient to leave at least 300 mm clear between the outer structure surfaces and the face of the excavation or support members.

3.4.2.5 Water Lines

Excavate trenches to a depth that provides a minimum cover of 1.0 meters from the existing ground surface, or from the indicated finished grade, whichever is lower, to the top of the pipe.

3.4.3 Jacking, Boring, and Tunneling

Unless otherwise indicated, provide excavation by open cut except those sections requested and if, in the opinion of the Contracting Officer, can be safely and properly installed and backfill. In that event, provide a Jacking, Boring, and Tunneling Plan, signed and sealed by a licensed Professional Engineer that includes a work site layout, methods and procedures, equipment, alignment control, monitoring, and contingency for responding to unplanned movements.

3.4.3.1 Pipeline Casing

Provide new smooth wall steel pipeline casing under existing pavement by

the boring and jacking method of installation. Provide each new pipeline casing, where indicated and to the lengths and dimensions shown, complete and suitable for use with the new piped utility as indicated. Install pipeline casing by dry boring and jacking method as follows:

3.4.3.2 Bore Holes

Mechanically bore holes and case through the soil with a cutting head on a continuous auger mounted inside the casing pipe. Weld lengths of pipe together in accordance with AWS D1.1/D1.1M or UNE-EN ISO 3834-2. Do not use water or other fluids in connection with the boring operation.

3.4.3.3 Cleaning

Clean inside of the pipeline casing of dirt, weld splatters, and other foreign matter which would interfere with insertion of the piped utilities by attaching a pipe cleaning plug to the boring rig and passing it through the pipe.

3.4.3.4 End Seals

After installation of piped utilities in pipeline casing, provide watertight end seals at each end of pipeline casing between pipeline casing and piping utilities. Provide watertight segmented elastomeric end seals.

3.4.4 Underground Utilities

Perform work adjacent to utilities in accordance with procedures outlined by utility owner. Excavation made with power-driven equipment is not permitted within 600 mm of known utility or subsurface construction. For work immediately adjacent to or for excavations exposing a utility or other buried obstruction, excavate by hand. Start hand excavation on each side of the indicated obstruction and continue until the obstruction is uncovered or until clearance for the new grade is assured. Support uncovered lines or other existing work affected by the contract excavation until approval for backfill is granted by the Contracting Officer. Report damage to utility lines or subsurface construction immediately to the Contracting Officer.

3.4.5 Structural Excavation

Make excavations to the lines, grades, and elevations shown, or as directed. Provide trenches and foundation pits of sufficient size to permit the placement and removal of forms for the full length and width of structure footings and foundations as shown. Clean rock or other hard foundation material of loose debris and cut to a firm, level, stepped, or serrated surface. Remove loose disintegrated rock and thin strata.

Concrete placement is not allowed until footing subgrades have been inspected and approved by the Contracting Officer.

3.4.6 Rock Excavation and Blasting

Rock excavation and blasting is not expected.

3.5 SUBGRADE PREPARATION

3.5.1 General Requirements

Shape subgrade to line, grade, and cross section as indicated. Remove unsatisfactory and unstable material in surfaces to receive fill or in excavated areas, and replaced with structural fill. Do not place material on surfaces that are muddy, frozen, contain frost, or otherwise containing unstable material. Scarify the surface to a depth of 100 mm prior to placing fill. Step or bench sloped surfaces steeper than 1 vertical to 4 horizontal prior to scarifying. Place 100 mm of loose fill and blend with scarified material. When subgrade is part fill and part excavation or natural ground, scarify to a depth of 200 mm.

3.5.2 Subgrade for Structures, Spread Footings, and Concrete Slabs

Do not excavate below depth shown for structures, spread footings, and concrete slabs. If over excavation occurs, notify the Contracting Officer and remove, replace, and compact as directed. Compact disturbed material to 95 percent of ASTM D1557 or UNE 103501. After final rolling, the surface of the subgrade for buildings and pavements must not show deviations greater than 15 mm when tested with a 4 meter straightedge applied both parallel and at right angles to the centerline of the area.

3.5.3 Subgrade for Pavements

Compact top 300 mm of subgrade for pavements to at least 95 percent of ASTM D1557 or UNE 103501. After final rolling, the surface of the subgrade for buildings and pavements must not show deviations greater than 15 mm when tested with a 4 meter straightedge applied both parallel and at right angles to the centerline of the area.

3.5.4 Subgrade for Shoulders

Compact the upper 150 mm of subgrade for shoulders to at least 95 percent of ASTM D1557 or UNE 103501 for the depth below the surface of shoulder shown .

3.6 FILLING AND COMPACTION

Prepare ground surface on which backfill is to be placed and provide compaction requirements for backfill materials in conformance with the applicable portions of paragraphs for SUBGRADE PREPARATION. Do not place material on surfaces that are muddy, frozen, or contain frost. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, or other approved equipment well suited to the soil being compacted. Moisten material as necessary to provide the moisture content that will readily facilitate obtaining the specified compaction with the equipment used. Fill and backfill to contours, elevations, and dimensions indicated. Compact and test each lift before placing overlaying lift.

3.6.1 Trench Backfill

Backfill trenches to the grade shown. Backfill the trench to 0.3 meters above the top of pipe prior to performing the required pressure tests. Leave the joints and couplings uncovered during the pressure test.

3.6.1.1 Replacement of Unyielding Material

Replace unyielding material removed from the bottom of the trench with satisfactory material or initial backfill material.

3.6.1.2 Replacement of Unstable Material

Replace unstable material removed from the bottom of the trench or excavation with satisfactory material placed in layers not exceeding 150 mm loose thickness.

3.6.1.3 Bedding and Initial Backfill

Provide bedding of the type and thickness shown. Place initial backfill material and compact it with approved tampers to a height of at least 300 mm above the utility pipe or conduit. Bring up the backfill evenly on both sides of the pipe for the full length of the pipe. Take care to ensure thorough compaction of the fill under the haunches of the pipe. Except where shown or when specified otherwise in the individual piping section, provide bedding for buried piping in accordance with PART 2 paragraph UTILITY BEDDING MATERIAL. Compact backfill to top of pipe to 85 percent of ASTM D1557 or UNE 103501. Provide plastic piping with bedding to spring line of pipe.

3.6.1.4 Final Backfill

Do not begin backfill until construction below finish grade has been approved, underground utilities systems have been inspected, tested and approved, forms removed, and the excavation cleaned of trash and debris. Bring backfill to indicated finish grade. Where pipe is coated or wrapped for protection against corrosion, the backfill material up to an elevation 600 mm above sewer lines and 300 mm above other utility lines need to be free from stones larger than 25 mm in any dimension. Heavy equipment for spreading and compacting backfill are not to be operated closer to foundation or retaining walls than a distance equal to the height of backfill above the top of footing; compact remaining area in layers not more than 100 mm in compacted thickness with power-driven hand tampers suitable for the material being compacted. Place backfill carefully around pipes or tanks to avoid damage to coatings, wrappings, or tanks. Do not place backfill against foundation walls prior to 7 days after completion of the walls. As far as practicable, bring backfill up evenly on each side of the wall and sloped to drain away from the wall.

Fill the remainder of the trench, except for special materials for buildings and pavements with satisfactory material. Place backfill material and compact as follows:

3.6.1.4.1 Buildings and Pavements

Place backfill up to the required elevation as specified. Do not permit water flooding or jetting methods of compaction. Compact as specified for Structural Fill.

3.6.1.4.2 Turfed or Seeded Areas and Miscellaneous Areas

Deposit backfill in layers of a maximum of 300 mm loose thickness, and compact it to 85 percent maximum density for cohesive soils and 90 percent maximum density for cohesionless soils. Apply this requirement to all other areas not specifically designated above.

3.6.1.5 Electrical Distribution System

Provide a minimum cover of **600 mm** from the finished grade to direct burial cable and conduit or duct line, unless otherwise indicated.

3.6.1.6 Displacement of Features

After other required tests have been performed and the trench backfill compacted to the finished grade surface, inspect the pipe to determine whether unexpected or damaging displacement has occurred. Conduct walk-through inspection of pipe sizes larger than **1200 mm**. Inspect pipes smaller than **1200 mm** using remote methods using closed circuit television, sonar, or hybrid that can provide a 360-degree inspection of the pipe. Prepare and submit a **pipe inspection report** consisting of digital video or photos. If, in the judgment of the Contracting Officer, the interior of the pipe shows poor alignment or any other defects that would cause improper functioning of the system, replace or repair the defects as directed at no additional cost to the Government.

3.6.1.7 Buried Tape And Detection Wire

3.6.1.7.1 Buried Warning and Identification Tape

Provide buried utility lines with utility identification tape. Bury tape **300 mm** below finished grade; under pavements and slabs, bury tape **150 mm** below top of subgrade.

3.6.1.7.2 Buried Detection Wire

Bury detection wire directly above non-metallic piping at a distance not to exceed **300 mm** above the top of pipe. Extend the wire continuously and unbroken, from manhole to manhole. Terminate the ends of the wire inside the manholes at each end of the pipe, with a minimum of **0.9 meters** of wire, coiled, remaining accessible in each manhole. Furnish insulated wire over its entire length. Install wires at manholes between the top of the corbel and the frame, and extend up through the chimney seal between the frame and the chimney seal. For force mains, terminate the wire in the valve pit at the pump station end of the pipe.

3.6.2 Structural Fill Placement

Place fill and backfill beneath and adjacent to structures in successive horizontal layers of loose material not more than **200 mm** in depth, or in loose layers not more than **100 mm** in depth when using hand-operated compaction equipment. Do not place over wet or frozen materials. Compact to at least 90 percent of laboratory maximum density for cohesive materials or 95 percent of laboratory maximum density for cohesionless materials, except as otherwise specified. Perform compaction in such a manner as to prevent wedging action or eccentric loading upon or other damage to the structure. Moisture condition fill and backfill material to a moisture content that will readily facilitate obtaining the specified compaction.

3.6.3 Backfill for Appurtenances

After the manhole, catchbasin, inlet, or similar structure has been constructed and the concrete has been allowed to cure for 7 days, place backfill in such a manner that the structure is not be damaged by the

shock of falling earth. Deposit the backfill material, compact it as specified for final backfill, and bring up the backfill evenly on all sides of the structure to prevent eccentric loading and excessive stress.

3.6.4 Porous Fill Placement

Provide under floor and area-way slabs on a compacted subgrade. Place in a single lift and compact with a minimum of two passes of a hand-operated plate-type vibratory compactor.

3.6.5 Flowable Fill

Place fill in a manner to completely fill voids in the location indicated. Do not place when atmospheric temperatures are expected to be below **one degree C** at any time during the 3 day period following placement.

3.6.6 Compaction

3.6.6.1 General Site

Compact underneath areas designated for vegetation and areas outside the **1.5 meter** line of the paved area or structure to 85 percent of **ASTM D1557** or **UNE 103501**.

3.6.6.2 Adjacent Areas

Compact areas within **1.5 meters** of structures to 95 percent of **ASTM D1557** or **UNE 103501**.

3.7 EMBANKMENTS

3.7.1 Earth Embankments

Construct earth embankments from satisfactory materials free of organic or frozen material and rocks with any dimension greater than **75 mm**. Place the material in successive horizontal layers of loose material not more than **200 mm** in depth. Spread each layer uniformly on a soil surface that has been moistened or aerated as necessary and scarified or otherwise broken up so that the fill will bond with the surface on which it is placed. After spreading, plow, disk, or otherwise break up each layer; moisten or aerate as necessary; thoroughly mix; and compact to at least 90 percent laboratory maximum density for cohesive materials or 95 percent laboratory maximum density for cohesionless materials. Backfill and fill material are to be to a moisture content that will readily facilitate obtaining the specified compaction.

Compaction requirements for the upper portion of earth embankments forming subgrade for pavements are identical with those requirements specified in paragraph SUBGRADE PREPARATION. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment.

3.8 FINISHING/FINISH OPERATIONS

During construction, keep embankments and excavations shaped and drained. Maintain ditches and drains along subgrade to drain effectively at all times. Do not disturb the finished subgrade by traffic or other operation. Protect and maintain the finished subgrade in a satisfactory condition until ballast, subbase, base, or pavement is placed. Do not

permit the storage or stockpiling of materials on finished subgrade. Do not lay subbase, base course, ballast, or pavement until the subgrade has been checked and approved, and in no case place subbase, base, surfacing, pavement, or ballast on a muddy, spongy, frozen or otherwise unstable subgrade.

Finish the surface of excavations, embankments, and subgrades to a smooth and compact surface in accordance with the lines, grades, and cross sections or elevations shown. Provide the degree of finish for graded areas within 30 mm of the grades and elevations indicated except as indicated for subgrades specified in paragraph SUBGRADE PREPARATION. Finish gutters and ditches in a manner that will result in effective drainage. Finish the surface of areas to be turfed to a smoothness suitable for the application of turfing materials. Repair graded, topsoiled, or backfilled areas prior to acceptance of the work, and re-established grades to the required elevations and slopes.

3.8.1 Grading Around Structures

Construct areas within 1.5 meters outside of each building and structure line true-to-grade, shape to drain, and maintain free of trash and debris until final inspection has been completed and the work has been accepted.

3.8.2 Shoulder Construction

Construct shoulders of satisfactory material. Submit advanced notice on shoulder construction for rigid pavements. Construct shoulders immediately after adjacent paving is complete. In the case of rigid pavements, do not construct shoulders until permission of the Contracting Officer has been obtained. Compact the entire shoulder area to at least the percentage of maximum density as specified in paragraph SUBGRADE PREPARATION above, for specific ranges of depth below the surface of the shoulder. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment. Finish shoulder construction in proper sequence in such a manner that adjacent ditches will be drained effectively and that no damage of any kind is done to the adjacent completed pavement. Align the completed shoulders true to grade and shaped to drain in conformity with the cross section shown.

3.8.3 Grading

Finish grades as indicated within 30 mm. Grade areas to drain water away from structures. Maintain areas free of trash and debris. For existing grades that will remain but which were disturbed by Contractor's operations, grade as directed.

3.8.4 Topsoil and Seed

Provide as specified in Section 32 92 19 SEEDING .

3.9 DISPOSITION OF SURPLUS MATERIAL

Remove from Government property all surplus or other soil material not required or not suitable for filling or backfilling, along with brush, refuse, stumps, roots, and timber. Properly disposed of in accordance with all applicable laws and regulations. Prepare plan for Disposition of Surplus Materials to include permissions document to dispose of nonsalable products.

3.10 TESTING

Perform testing as indicated in Table 1. Submit [Material Test Reports](#) within 7 days of tests being completed.

Material Type	Location of Material	Test Method	Test Frequency
Tank Backfill Structural Subgrades		Density - ASTM D1556/D1556M or UNE 103503 ASTM D6938 or UNE 103900 . When ASTM D6938 is used, check the calibration curves and adjust using only the sand cone method.	One test per 2000 square meters , or fraction thereof, of each lift of fill or backfill areas compacted by other than hand-operated machines. Double testing frequency for areas compacted by hand-operated machines. If ASTM D6938 is used, check in-place densities by sound cone method as follows: One check test per lift for every 6 tests.
Tank Backfill Structural Subgrades		Moisture Content - ASTM D2216 or UNE-EN ISO 17892-1	Two tests per day for each type of fill and backfill. Sample taken immediately prior to compaction after moisture conditioning.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Material Type	Location of Material	Test Method	Test Frequency
Tank Backfill Structural Subgrades		Moisture Density Relationship - ASTM D1557 or UNE 103501	One representative test per 500 cubic meters of fill and backfill, or when any change in material occurs which may affect the optimum moisture content or laboratory maximum density. Sample to be taken from stockpile or location of placement.
Tank Backfill Structural Subgrades		Relative Density - ASTM D4253 and ASTM D4254 or UNE 103500	One test per 2000 square meters, or fraction thereof, of each lift of fill or backfill areas compacted by other than hand-operated machines. Double testing frequency for areas compacted by hand-operated machines.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Material Type	Location of Material	Test Method	Test Frequency
Tank Backfill Structural Subgrades		Gradation - ASTM C136/C136M or UNE-EN 933-1	One representative test per 500 cubic meters of fill and backfill, or when any change in material occurs which may affect the optimum moisture content or laboratory maximum density. Sample to be taken from stockpile or location of placement.
		Atterberg Limits - ASTM D4318 or UNE-EN ISO 17892-12	One representative test per 500 cubic meters of fill and backfill, or when any change in material occurs which may affect the optimum moisture content or laboratory maximum density. Sample to be taken from stockpile or location of placement.
		Organic Content Test - ASTM D2974, Method C	One representative test per 200 lineal meters of embankment.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 05 33

LANDSCAPE ESTABLISHMENT

08/17

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Pesticide
 - 1.2.2 Stand of Turf
 - 1.2.3 Planter Beds
- 1.3 RELATED REQUIREMENTS
- 1.4 SUBMITTALS
- 1.5 DELIVERY, STORAGE AND HANDLING
 - 1.5.1 Delivery
 - 1.5.2 Storage
 - 1.5.2.1 Fertilizer, Lime, Mulch Storage
 - 1.5.2.2 Antidesiccant's Storage
 - 1.5.3 Handling
- 1.6 MAINTENANCE

PART 2 PRODUCTS

- 2.1 POST-PLANT FERTILIZER
 - 2.1.1 Granular Fertilizer
- 2.2 WATER
- 2.3 MULCHES TOPDRESSING
 - 2.3.1 Inert Mulch Materials
 - 2.3.2 Organic Mulch Materials
 - 2.3.3 Recycled Organic Mulch
- 2.4 PESTICIDES

PART 3 EXECUTION

- 3.1 EXTENT OF WORK
 - 3.1.1 Policing
 - 3.1.2 Drainage System Maintenance
- 3.2 GROUNDCOVER ESTABLISHMENT PERIOD
 - 3.2.1 Frequency of Maintenance
 - 3.2.2 Promotion of Growth
 - 3.2.3 Mowing
 - 3.2.3.1 Turf
 - 3.2.3.2 Native Grasses
 - 3.2.3.3 Wildflowers
 - 3.2.4 Turf Edging and Trimming
 - 3.2.5 Post-Fertilizer Application
 - 3.2.6 Turf Watering
 - 3.2.7 Replanting
 - 3.2.8 Final Inspection and Acceptance
 - 3.2.9 Unsatisfactory Work

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.3 EXTERIOR PLANT ESTABLISHMENT PERIOD
 - 3.3.1 Frequency of Maintenance
 - 3.3.2 Promotion of Plant Growth and Vigor
 - 3.3.3 Planter Bed Maintenance
 - 3.3.3.1 Shrub Selective Maintenance
 - 3.3.3.2 Tree Maintenance
 - 3.3.4 Slope Erosion Control Maintenance
 - 3.3.5 Removal of Dying or Dead Plants
 - 3.3.6 Tracking of Unhealthy Plants
 - 3.3.7 Final Inspection
 - 3.3.7.1 Total Plants on Site
 - 3.3.7.2 Mulching and Weeding
 - 3.3.7.3 Tree Supports
 - 3.3.7.4 Remedial Work
 - 3.3.8 Unsatisfactory Work
- 3.4 FIELD QUALITY CONTROL
 - 3.4.1 Maintenance Inspection Report
 - 3.4.2 Plant Quantities
 - 3.4.3 Tree Staking and Guying Removal

-- End of Section Table of Contents --

SECTION 32 05 33

LANDSCAPE ESTABLISHMENT
08/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

TREE CARE INDUSTRY ASSOCIATION (TCIA)

TCIA Z133

(2017) American National Standard for
Arboricultural Operations - Pruning,
Repairing, Maintaining, and Removing
Trees, and Cutting Brush - Safety
Requirements

1.2 DEFINITIONS

1.2.1 Pesticide

Any substance or mixture of substances, including biological control agents, that may prevent, destroy, repel, or mitigate pests and are specifically labeled for use by the Spanish and Andalusian Environmental Protection Agencies. Also, any substance used as a plant regulator, defoliant, disinfectant, or biocide. Examples of pesticides include fumigants, herbicides, insecticides, fungicides, nematocides, molluscicides and rodenticides.

1.2.2 Stand of Turf

95 percent ground cover of the established species.

1.2.3 Planter Beds

A planter bed is defined as an area containing one or a combination of the following plant types: shrubs, vines, wildflowers, annuals, perennials, ground cover, and a mulch topdressing excluding turf. Trees may also be found in planter beds.

1.3 RELATED REQUIREMENTS

Section 32 92 19 SEEDING applies to this section for installation of seed requirements, with additions and modifications herein.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Integrated Pest Management Plan; G

SD-03 Product Data

Fertilizer; G

Mulches Topdressing

Organic Mulch Materials

SD-07 Certificates

Maintenance Inspection Report

Plant Quantities; G

SD-10 Operation and Maintenance Data

Maintenance

SD-11 Closeout Submittals

Tree Staking and Guying Removal

1.5 DELIVERY, STORAGE AND HANDLING

1.5.1 Delivery

Deliver fertilizer to the site in original containers bearing manufacturer's chemical analysis, name, trade name, or trademark, and indication of conformance to state and federal laws. Instead of containers, fertilizer may be furnished in bulk with a certificate indicating the above information.

1.5.2 Storage

1.5.2.1 Fertilizer, Lime, Mulch Storage

Store material in designated areas. Store lime and fertilizer in cool, dry locations away from contaminants.

1.5.2.2 Antidesiccant's Storage

Do not store with fertilizers or other landscape maintenance materials.

1.5.3 Handling

Do not drop or dump materials from vehicles.

1.6 MAINTENANCE

Submit Operation and Maintenance (O&M) Manuals for planting materials. Include instructions indicating procedures during one typical year including variations of maintenance for climatic conditions throughout the year. Provide instructions and procedures for watering; promotion of growth, including fertilizing, pruning, and mowing; and integrated pest

management. O&M Manuals must include pictures of planting materials cross referenced to botanical and common names, with a description of the normal appearance in each season.

Develop a water monitoring program for surface and ground water on the project site and consistent with the water management program utilized during construction operations.

PART 2 PRODUCTS

2.1 POST-PLANT FERTILIZER

Fertilizer for groundcover, wildflowers, and grasses is not permitted. Provide fertilizer for trees, plants, and shrubs as recommended by plant supplier, except synthetic chemical fertilizers are not permitted. Fertilizers containing petrochemical additives or that have been treated with pesticides or herbicides are not permitted.

2.1.1 Granular Fertilizer

Organic, granular controlled release fertilizer containing the following minimum percentages, by weight, of plant food nutrients:

- 10 percent available nitrogen
- 10 percent available phosphorus
- 10 percent available potassium

2.2 WATER

Source of water must be approved by the Contracting Officer, and be of suitable quality for irrigation. Use collected storm water or graywater when available.

2.3 MULCHES TOPDRESSING

Free from noxious weeds, mold, pesticides, or other deleterious materials.

2.3.1 Inert Mulch Materials

Provide riverbank stone ranging in size from 10 to 25 mm.

2.3.2 Organic Mulch Materials

Provide shredded hardwood from site when available. Wood cellulose fiber must be processed to contain no growth or germination-inhibiting factors, dyed with non-toxic, biodegradable dye to an appropriate color to facilitate visual metering of materials application. Paper-based hydraulic mulch must contain a minimum of 100 percent post-consumer recycled content. Wood-based hydraulic mulch must contain a minimum of 100 percent total recovered materials content.

2.3.3 Recycled Organic Mulch

Recycled mulch may include compost, tree trimmings, or pine needles with a gradation that passes through a 65 by 65 mm screen. Clean recycled mulch of all sticks a minimum 25 mm in diameter and plastic materials a minimum 75 mm length. The material must be treated to retard the growth of mold and fungi.

2.4 PESTICIDES

Pesticides and herbicides are not permitted. Submit an Integrated Pest Management Plan, including weed and pest management strategies, and proposed alternatives to herbicides and pesticides. Use biological pest controls as approved in the Plan.

PART 3 EXECUTION

3.1 EXTENT OF WORK

Provide landscape construction maintenance to include mowing, edging, overseeding, fertilizing, watering, weeding, pruning, and stake and guy adjusting for all newly installed landscape areas, unless indicated otherwise, and at all areas inside or outside the limits of the construction that are disturbed by the Contractor's operations.

3.1.1 Policing

Police all landscaped areas. Policing includes removal of leaves, branches and limbs regardless of length or diameter, dead vegetation, paper, trash, cigarette butts, garbage, rocks or other debris. Collected debris must be promptly removed and disposed of at an approved disposal site.

3.1.2 Drainage System Maintenance

Remove all obstructions from surface and subsurface drain lines to allow water to flow unrestricted in swales. Remove grates and clear debris in catch basins. Open drainage channels are to be maintained free of all debris and vegetation at all times. Edges of these channels must be clear of any encroachment by vegetation.

3.2 GROUNDCOVER ESTABLISHMENT PERIOD

Groundcover establishment period will commence on the date that inspection by the Contracting Officer shows that the new turf furnished under this contract has been satisfactorily installed to a 95 percent stand of coverage. The establishment period must continue for a period of 365 days.

3.2.1 Frequency of Maintenance

Begin maintenance immediately after turf has been installed. Inspect area once a week during the installation and establishment period and perform needed maintenance promptly.

3.2.2 Promotion of Growth

Maintain groundcover in a manner that promotes proper health, growth, natural color. Turf must have a neat uniform manicured appearance, free of bare areas, ruts, holes, weeds, pests, dead vegetation, debris, and unwanted vegetation that present an unsightly appearance. Mow, remove excess clippings, eradicate weeds, water, fertilize, overseed, and perform other operations necessary to promote growth, as approved by Contracting Officer and consistent with approved Integrated Pest Management Plan. Remove noxious weeds common to the area from planting areas by mechanical means.

3.2.3 Mowing

3.2.3.1 Turf

Mow turf at a uniform finished height. Mow turfed areas to a minimum average height of 102 mm when average height of grass becomes 125 mm for spring/summer maintenance and to a minimum average height of 102 mm when the average height of grass reaches 125 mm for fall maintenance. The height of turf is measured from the soil. Perform mowing of turf in a manner that prevents scalping, rutting, bruising, uneven and rough cutting. Prior to mowing, all rubbish, debris, trash, leaves, rocks, paper, and limbs or branches on a turf area must be picked up and disposed. Adjacent paved areas must be swept/vacuumed clean.

3.2.3.2 Native Grasses

Mow above height of native grass seedlings (approximately 89 to 102 mm). Mow during spring or early summer. Do not mow after early summer during the second growing season.

3.2.3.3 Wildflowers

Mow three times per season above height of the wildflowers (approximately 305 to 381 mm).

3.2.4 Turf Edging and Trimming

Perimeter of planter bed edges, sidewalks, driveways, curbs, and other paved surfaces must be edged. Uniformly edge these areas to prevent encroachment of vegetation onto paved surfaces and to provide a clear cut division line between planter beds, turf, and ground cover. Edging is to be accomplished in a manner that prevents scalping, rutting, bruising, uneven and rough cutting. Perform edging on the same day that turf is mowed. Use of string line trimmers is permitted in "soft" areas such as an edge between turfgrass and a planter bed. Exercise care to avoid damage to any plant materials, structures, and other landscape features.

Trimming around trees, fences, poles, walls, and other similar objects is to be accomplished to match the height and appearance of surrounding mowed turf growth. Trimming must be performed on the same day the turf's mowed. Care must be exercised to avoid "Girdling" trees located in turf areas. The use of protective tree collars on trees in turf areas may be utilized as a temporary means to avoid injury to tree trunks. At the end of the plant establishment period Contractor will be responsible for removing all protective tree collars.

3.2.5 Post-Fertilizer Application

Do not fertilize wildflowers, groundcover, and grasses. Apply turf fertilizer in a manner that promotes health, growth, vigor, color and appearance of cultivated turf areas. The method of application, fertilizer type and frequencies must be determined by the laboratory soil analysis results the requirements of the particular turf species. Apply fertilizer by approved methods in accordance with the manufacturer's recommendations.

3.2.6 Turf Watering

Perform irrigation in a manner that promotes the health, growth, color and

appearance of cultivated vegetation and that complies with all Federal, State, and local water agencies and authorities directives. The Contractor must be responsible to prevent over watering, water run-off, erosion, and ponding due to excessive quantities or rate of application. Abide by Spanish national, regional, local or other water conservation regulations or restrictions in force during the establishment period.

3.2.7 Replanting

Replant in accordance with Section 32 92 19 SEEDING and within specified planting dates areas which do not have a satisfactory stand of turf. Replant areas which do not have a satisfactory stand of other groundcover and grasses.

3.2.8 Final Inspection and Acceptance

Final inspection will be made upon written request from the Contractor at least 10 days prior to the last day of the turf establishment period. Final turf acceptance will be based upon a satisfactory stand of turf. Final acceptance of wildflower and grass areas will be based upon a stand of 95 percent groundcover of established species.

3.2.9 Unsatisfactory Work

When work is found to not meet design intent and specifications, maintenance period will be extended at no additional cost to the Government until work has been completed, inspected and accepted by Contracting Officer.

3.3 EXTERIOR PLANT ESTABLISHMENT PERIOD

The exterior plant establishment period will commence on the date that inspection by the Contracting Officer shows that the new plants furnished under this contract has been satisfactorily installed and must continue for a period of 365 days.

3.3.1 Frequency of Maintenance

Begin maintenance immediately after plants have been installed. Inspect exterior plants at least once a week during the installation and establishment period and perform needed maintenance promptly.

3.3.2 Promotion of Plant Growth and Vigor

Water, prune, fertilize, mulch, adjust stakes, guys and turnbuckles, eradicate weeds and perform other operations necessary to promote plant growth, and vigor.

3.3.3 Planter Bed Maintenance

Planter beds must be weeded, fertilized, irrigated, kept pest free, turf free, pruned, and mulch levels maintained. Planter beds will not be allowed to encroach into turf areas. A definite break must be maintained between turf areas and planter beds. Fertilize exterior planting materials to promote healthy plant growth without encouraging excessive top foliar growth. Remove noxious weeds common to the area from planting areas by mechanical means.

3.3.3.1 Shrub Selective Maintenance

In addition to the above requirements, shrubs must be selectively pruned, and shaped for health and safety when the following conditions exist: Remove growth in front of windows, over entrance ways or walks, and any growth which will obstruct vision at street intersections or of security personnel; Remove dead, damaged or diseased branches or limbs; where shrub growth obstructs pedestrian walkways; where shrub growth is found growing against or over structures; where shrub growth permits concealment of unauthorized persons. Dispose of all pruning debris in a proper manner.

3.3.3.2 Tree Maintenance

Tree maintenance must include adjustment of stakes, ties, guy supports and turnbuckles, watering, fertilizing, pest control, mulching, and pruning for health and safety. Fertilize exterior trees to promote healthy plant growth without encouraging excessive top foliar growth. Inspect and adjust stakes, ties, guy supports, and turnbuckles to avoid girdling and promote natural development. All trees within the project boundaries, regardless of caliper, must be selectively pruned for safety and health reasons. These include but are not limited to removal of dead and broken branches and correction of structural defects. Prune trees according to their natural growth characteristics leaving trees well shaped and balanced. Pruning of all trees including palm trees must be accomplished by or in the presence of a certified member of the International Society of Arboriculture and in accordance with TCIA Z133. All pruning debris generated must be disposed of in a proper manner.

3.3.4 Slope Erosion Control Maintenance

Provide slope erosion control maintenance to prevent undermining of all slopes in newly landscaped areas. Maintenance tasks include immediate repairs to weak spots in sloped areas, and maintaining clean, clear culverts, and graded berms, to intercept and direct water flow to prevent development of large gullies and slope erosion. Eroded areas must be filled with amended topsoil and replanted with the same plant species. Erosion control blankets damaged due to slope erosion must be reinstalled.

3.3.5 Removal of Dying or Dead Plants

Remove dead and dying plants and provide new plants immediately upon commencement of the specified planting season, and replace stakes, guys, mulch and eroded earth mound water basins. Provide an additional 90 day establishment period for replacement plants beyond the original warranty period. A tree must be considered dying or dead when the main leader has died back, or a minimum of 20 percent of the crown has died. A shrub or ground cover must be considered dying or dead when a minimum of 20 percent of the plant has died. This condition must be determined by scraping on a branch an area 2 mm square, maximum, to determine the cause for dying plant material and must provide recommendations for replacement. The Contractor must determine the cause for dying plant material and provide recommendations for replacement.

3.3.6 Tracking of Unhealthy Plants

Note plants not in healthy growing condition, as determined by the Contracting Officer, and as soon as seasonal conditions permit, remove and replace with plants of the same species and sizes as originally specified.

3.3.7 Final Inspection

Final inspection will be made upon written request from the Contractor at least 10 days prior to the last day of the establishment period. Final inspection will be based upon satisfactory health and growth of plants and on the following:

3.3.7.1 Total Plants on Site

Plants have been accepted and required number of replacements have been installed.

3.3.7.2 Mulching and Weeding

Planter beds and earth mound water basins are properly mulched and free of weeds.

3.3.7.3 Tree Supports

Stakes, guys, and turnbuckles are in good condition.

3.3.7.4 Remedial Work

Remedial measures directed by the Contracting Officer to ensure plant material survival and promote healthy growth have been completed.

3.3.8 Unsatisfactory Work

When work is found to not meet design intent and specifications, maintenance period will be extended at no additional cost to the Government until work has been completed, inspected and accepted by Contracting Officer.

3.4 FIELD QUALITY CONTROL

3.4.1 Maintenance Inspection Report

Provide maintenance inspection report to assure that landscape maintenance is being performed in accordance with the specifications and in the best interest of plant growth and survivability. Site observations must be documented at the start of the establishment period, then quarterly following the start, and at the end of establishment period. Submit results of site observation visits to the Contracting Officer within 7 calendar days of each site observation visit.

3.4.2 Plant Quantities

Provide Contracting Officer with the number of plant quantities. In addition, provide total exterior area of hardscape and landscaping such as turf and total number of shrubs.

3.4.3 Tree Staking and Guying Removal

Provide a certified letter that all stakes and guys are removed from all project trees at the end of the establishment period.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 11 23

AGGREGATE BASE COURSE

05/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Aggregate Base Course
 - 1.2.2 Degree of Compaction
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Sampling
 - 1.4.2 Tests
 - 1.4.2.1 Gradation Analysis
 - 1.4.2.2 Liquid Limit and Plasticity Index
 - 1.4.2.3 Moisture-Density Determinations
 - 1.4.2.4 Field Density Tests
 - 1.4.2.5 Wear Test
 - 1.4.2.6 Flat and Elongated Pieces
 - 1.4.2.7 Fractured Faces
- 1.5 ENVIRONMENTAL REQUIREMENTS
- 1.6 ACCEPTANCE
 - 1.6.1 Tolerances
 - 1.6.2 Test Section

PART 2 PRODUCTS

- 2.1 AGGREGATES
 - 2.1.1 Coarse Aggregate
 - 2.1.1.1 Aggregate Base Course
 - 2.1.2 Fine Aggregate
 - 2.1.2.1 Aggregate Base Course
 - 2.1.3 Gradation Requirements
- 2.2 LIQUID LIMIT AND PLASTICITY INDEX
- 2.3 TESTS, INSPECTIONS, AND VERIFICATIONS
 - 2.3.1 Initial Tests
 - 2.3.2 Approval of Material
- 2.4 EQUIPMENT, TOOLS, AND MACHINES

PART 3 EXECUTION

- 3.1 GENERAL REQUIREMENTS
- 3.2 OPERATION OF AGGREGATE SOURCES
- 3.3 STOCKPILING MATERIAL
- 3.4 PREPARATION OF UNDERLYING COURSE OR SUBGRADE
- 3.5 GRADE CONTROL
- 3.6 MIXING AND PLACING MATERIALS
 - 3.6.1 Mixing
 - 3.6.2 Placing

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.7 LAYER THICKNESS
- 3.8 COMPACTION
- 3.9 EDGES OF BASE COURSE
- 3.10 FINISHING
- 3.11 SMOOTHNESS TEST
- 3.12 FIELD QUALITY CONTROL
 - 3.12.1 In-Place Tests
 - 3.12.2 Approval of Material
- 3.13 TRAFFIC
- 3.14 MAINTENANCE
- 3.15 DISPOSAL OF UNSATISFACTORY MATERIALS

-- End of Section Table of Contents --

SECTION 32 11 23

AGGREGATE BASE COURSE
05/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 103501	(1994) Geotechnic. Compaction Test. Modified Proctor.
UNE 103503	(1995) Determination "In Situ" Of Density Of Soil By The Sand Method.
UNE 103900	(2013) In situ determination of density and moisture content of soil and granular materials by nuclear methods: low depths.
UNE-EN 932-1	(1997) Test For Geometrical Properties Of Aggregates. Part 1: Methods for Sampling.
UNE-EN 933-1	(2012) Tests for geometrical properties of aggregates - Part 1: Determination of particle size distribution - Sieving method
UNE-EN 933-2	(1996) Test For Geometrical Properties Of Aggregates. Part 2: Determination Of Particle Size Distribution. Test Sieves, Nominal Size Of Apertures.
UNE-EN 933-3	(2012) Tests for geometrical properties of aggregates - Part 3: Determination of particle shape - Flakiness index
UNE-EN 933-5	(1999) Test For Geometrical Properties Of Aggregates. Part 5: Determination Of Percentage Of Crushed And Broken Surfaces In Coarse Aggregate Particles.
UNE-EN 1097-2	(2010) Tests for mechanical and physical properties of aggregates - Part 2: Methods for the determination of resistance to fragmentation
UNE-EN ISO 17892-12	(2019) Geotechnical investigation and testing - Laboratory testing of soil - Part 12: Determination of liquid and plastic limits (ISO 17892-12:2018).

MINISTERIO DE FOMENTO (MF)

FOM/2523/2014 (2014) Orden de 12 de diciembre, por la que se actualizan determinados artículos del pliego de prescripciones técnicas generales para obras de carreteras y puentes, relativos a materiales básicos, a firmes y pavimentos, y a señalización, balizamiento y sistemas de contención de vehículos.

ASTM INTERNATIONAL (ASTM)

ASTM C117 (2017) Standard Test Method for Materials Finer than 75-um (No. 200) Sieve in Mineral Aggregates by Washing

ASTM C127 (2015) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Coarse Aggregate

ASTM C128 (2022) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Fine Aggregate

ASTM C131/C131M (2020) Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine

ASTM C136/C136M (2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates

ASTM D75/D75M (2019) Standard Practice for Sampling Aggregates

ASTM D1556/D1556M (2015; E 2016) Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method

ASTM D1557 (2012; E 2015) Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft³) (2700 kN-m/m³)

ASTM D2487 (2017; E 2020) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)

ASTM D4318 (2017; E 2018) Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils

ASTM D4718/D4718M (2015) Standard Practice for Correction of Unit Weight and Water Content for Soils Containing Oversize Particles

ASTM D4791 (2019) Flat Particles, Elongated

Particles, or Flat and Elongated Particles
in Coarse Aggregate

ASTM D5821

(2013; R 2017) Standard Test Method for
Determining the Percentage of Fractured
Particles in Coarse Aggregate

ASTM D6938

(2017a) Standard Test Method for In-Place
Density and Water Content of Soil and
Soil-Aggregate by Nuclear Methods (Shallow
Depth)

ASTM E11

(2022) Standard Specification for Woven
Wire Test Sieve Cloth and Test Sieves

1.2 DEFINITIONS

For the purposes of this specification, the following definitions apply.

1.2.1 Aggregate Base Course

Aggregate base course (ABC) is well graded, durable aggregate uniformly
moistened and mechanically stabilized by compaction.

1.2.2 Degree of Compaction

Degree of compaction required, except as noted in the second sentence, is
expressed as a percentage of the maximum laboratory dry density obtained
by the test procedure presented in ASTM D1557 or UNE 103501 abbreviated as
a percent of laboratory maximum dry density. Since ASTM D1557 applies
only to soils that have 30 percent or less by weight of their particles
retained on the 19.0 mm sieve, express the degree of compaction for
material having more than 30 percent by weight of their particles retained
on the 19.0 mm sieve as a percentage of the laboratory maximum dry density
in accordance with ASTM D1557 Method C and corrected with ASTM D4718/D4718M.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S"
classification. Submittals not having a "G" or "S" classification are for
Contractor Quality Control approval. Submit the following in accordance
with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Plant, Equipment, and Tools; G

SD-06 Test Reports

Initial Tests; G

In-Place Tests; G

1.4 QUALITY ASSURANCE

Perform sampling and testing using a laboratory approved in accordance
with Section 01 45 00 QUALITY CONTROL. Do not start work requiring
testing until the testing laboratory has been inspected and approved.
Test the materials to establish compliance with the specified requirements
and perform testing at the specified frequency. Furnish copies of test

results within 24 hours of completion of the tests.

1.4.1 Sampling

Take samples for laboratory testing in conformance with ASTM D75/D75M or with UNE-EN 932-1.

1.4.2 Tests

1.4.2.1 Gradation Analysis

Perform gradation analysis in conformance with ASTM C117 and ASTM C136/C136M using sieves conforming to ASTM E11 or with UNE-EN 933-1 using sieves conforming to UNE-EN 933-2.

1.4.2.2 Liquid Limit and Plasticity Index

Determine liquid limit and plasticity index in accordance with ASTM D4318 or UNE-EN ISO 17892-12.

1.4.2.3 Moisture-Density Determinations

Determine the laboratory maximum dry density and optimum moisture content in accordance with paragraph DEGREE OF COMPACTION.

1.4.2.4 Field Density Tests

Measure field density in accordance with ASTM D1556/D1556M, or ASTM D6938 or alternatively with UNE 103503 or UNE 103900. For the method presented in ASTM D6938 check the calibration curves and adjust them, if necessary, using only the sand cone method as described in Annex A2 of ASTM D6938 or UNE 103503. Use ASTM D6938 or UNE 103900 to determine the moisture content of the soil. Check the calibration curves furnished with the moisture gauges along with density calibration checks as described in ASTM D6938. Make the calibration checks of both the density and moisture gauges using the prepared containers of material method, as described in Annex A2 of ASTM D6938, on each different type of material being tested at the beginning of a job and at intervals as directed. Submit calibration curves and related test results prior to using the device or equipment being calibrated.

1.4.2.5 Wear Test

Perform wear tests on ABC course material in conformance with ASTM C131/C131M or UNE-EN 1097-2.

1.4.2.6 Flat and Elongated Pieces

Determine flat and elongated pieces on ABC course material in conformance with ASTM D4791, Method A, OR UNE-EN 933-3.

1.4.2.7 Fractured Faces

Perform fractured faces test on ABC coarse aggregate in conformance with ASTM D5821 or UNE-EN 933-5.

1.5 ENVIRONMENTAL REQUIREMENTS

Correct completed areas damaged by rainfall or other weather conditions to

meet specified requirements.

1.6 ACCEPTANCE

1.6.1 Tolerances

Acceptance of ABC is based on compliance with the tolerances presented in Table 1. Remove any materials found to be non-compliant and replace with compliant material or rework, as directed, to meet the requirements of this specification

TABLE 1	
Measurement	Tolerance
Grade	Plus 6 mm, Minus 13 mm
Smoothness	Plus/Minus 10 mm
Individual Test Total Thickness	Plus/Minus 13 mm 1/2 inch
Average Job Thickness	Plus/Minus 6 mm
Compaction	Minimum 100 percent

1.6.2 Test Section

A test section is not required.

PART 2 PRODUCTS

2.1 AGGREGATES

FOM/2523/2014, Art. 510 and the following. Provide ABC consisting of clean, sound, durable particles of crushed stone, crushed gravel, angular sand, or other approved material. Provide ABC that is free of lumps of clay, organic matter, and other objectionable materials or coatings. The portion retained on the 4.75 mm sieve is known as coarse aggregate; that portion passing the 4.75 mm sieve is known as fine aggregate. When the coarse and fine aggregate is supplied from more than one source, provide aggregate from each source that meets the specified requirements.

2.1.1 Coarse Aggregate

Provide coarse aggregates with angular particles of uniform density. Separately stockpile coarse aggregate supplied from more than one source.

- Crushed Gravel: Provide crushed gravel that has been manufactured by crushing gravels and that meets all the requirements specified below.
- Crushed Stone: Provide crushed stone consisting of freshly mined quarry rock, meeting all the requirements specified below.

2.1.1.1 Aggregate Base Course

Limit the percentage of loss of ABC coarse aggregate to a maximum of 50 percent when tested in accordance with ASTM C131/C131M or with

UNE-EN 1097-2. Provide aggregate that contains a maximum of 30 percent flat and elongated particles when tested in accordance with **ASTM D4791**, Method A or with **UNE-EN 933-3**. A flat particle is one having a ratio of width to thickness greater than 3; an elongated particle is one having a ratio of length to width greater than 3. In the portion retained on each sieve specified, provide crushed aggregates containing a minimum of 50 percent by weight of crushed pieces having two or more freshly fractured faces determined in accordance with **ASTM D5821** or with **UNE-EN 933-5**. When two fractures are contiguous, the angle between planes of the fractures is required to be a minimum of 30 degrees in order to count as two fractured faces. Manufacture crushed gravel from gravel particles 50 percent of which, by weight, are retained on the maximum size sieve listed in TABLE 2.

2.1.2 Fine Aggregate

Provide fine aggregates consisting of angular particles of uniform density.

2.1.2.1 Aggregate Base Course

Provide ABC fine aggregate that consists of screenings, angular sand, or other finely divided mineral matter processed or naturally combined with the coarse aggregate.

2.1.3 Gradation Requirements

Select a gradation from Table 2 and apply the specified gradation requirements to the completed base course. Provide aggregates that are continuously well graded within the limits specified in TABLE 2. Use sieves that conform to **ASTM E11** or equivalent in accordance with **UNE-EN 933-2**. The gradation indicated in Table 2 can be replaced by the following: No. 1 by ZA 0/32, No. 2 & 3 by ZA 0/20.

TABLE 2. GRADATION OF AGGREGATES			
Percentage By Weight Passing Square-Mesh Sieve			
Sieve Designation	No. 1	No. 2	No.3
50 mm	100	---	---
37.5 mm	70-100	100	---
25.0 mm	45-80	60-100	100
12.5 mm	30-60	30-65	40-70
4.75 mm	20-50	20-50	20-50
2.0 mm	15-40	15-40	15-40
0.425 mm	5-25	5-25	5-25

TABLE 2. GRADATION OF AGGREGATES			
0.075 mm	0-8	0-8	0-8

NOTE 1: The values are based on aggregates of uniform specific gravity. If materials from different sources are used for the coarse and fine aggregates, test the materials in accordance with [ASTM C127](#) and [ASTM C128](#) to determine their specific gravities. Correct the percentages passing the various sieves as directed if the specific gravities vary by more than 10 percent.

NOTE 2: Gradations containing more than 30 percent retained on the [37.5 mm](#) sieve can produce inconsistent compacted density values when tested in accordance with paragraph DEGREE OF COMPACTION.

2.2 LIQUID LIMIT AND PLASTICITY INDEX

Apply liquid limit and plasticity index requirements to the completed course and to any component that is blended to meet the required gradation. Limit the portion of any component or of the completed course passing the [0.425 mm](#) sieve to be either nonplastic or have a maximum liquid limit of 25 and a maximum plasticity index of 5.

2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

2.3.1 Initial Tests

Perform one of each of the following [initial tests](#) on the proposed material prior to commencing construction to demonstrate that the proposed material meets all specified requirements when furnished. Complete this testing for each source if materials from more than one source are proposed. Submit certified copies of test results for approval a minimum of 30 days before material is required for the work.

- Gradation Analysis.
- Liquid limit and plasticity index.
- Moisture-density relationship.
- Wear.
- Flat and Elongated Pieces.
- Fractured Faces.

2.3.2 Approval of Material

Tentative approval of material will be based on initial test results.

2.4 EQUIPMENT, TOOLS, AND MACHINES

All [plant, equipment, and tools](#) used in the performance of the work are subject to approval by the Government before the work is started. Maintain all plant, equipment, and tools in satisfactory working condition at all times. Submit a list of proposed equipment, including descriptive data. Use equipment capable of minimizing segregation, producing the required compaction, meeting grade controls, thickness control, and smoothness requirements as set forth herein.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

When the ABC is constructed in more than one lift, clean the previously constructed lift of loose and foreign matter by sweeping with power sweepers or power brooms. Use hand brooms in areas where power cleaning is not practicable. Provide adequate drainage during the entire period of construction to prevent water from collecting or standing on the working area.

3.2 OPERATION OF AGGREGATE SOURCES

Condition aggregate sources on private lands in accordance with local laws or authorities.

3.3 STOCKPILING MATERIAL

Clear and level storage sites prior to stockpiling of material. Stockpile all materials, including approved material available from excavation and grading, in the manner and at the locations designated. Stockpile aggregates on the cleared and leveled areas designated to prevent segregation. Stockpile materials obtained from different sources separately.

3.4 PREPARATION OF UNDERLYING COURSE OR SUBGRADE

Clean the underlying course or subgrade of all foreign substances prior to constructing the base course(s). Do not construct base course(s) on underlying course or subgrade that is muddy. Construct the surface of the underlying course or subgrade to meet specified compaction and surface tolerances. Correct ruts or soft yielding spots in the underlying courses, areas having inadequate compaction, and deviations of the surface from the specified requirements set forth herein by loosening and removing soft or unsatisfactory material and adding approved material, reshaping to line and grade, and recompact to specified density requirements. For cohesionless underlying courses or subgrades containing sands or gravels, as defined in [ASTM D2487](#), stabilize the surface prior to placement of the base course(s). Stabilize by mixing ABC into the underlying course and compacting by approved methods. Consider the stabilized material as part of the underlying course and meet all requirements of the underlying course. Do not allow traffic or other operations to disturb the finished underlying course and maintain in a compliant condition until the base course is placed.

3.5 GRADE CONTROL

Provide a finished and completed base course conforming to the lines, grades, and cross sections shown. Place line and grade stakes as necessary for control.

3.6 MIXING AND PLACING MATERIALS

3.6.1 Mixing

Mix the coarse and fine aggregates in a stationary plant. Make adjustments in mixing procedures or in equipment to obtain true grades, to minimize segregation or degradation, to obtain the required water content, and to produce a satisfactory base course meeting all requirements of this

specification.

3.6.2 Placing

Place the mixed material on the prepared subgrade or subbase in lifts of uniform thickness with an approved spreader. Place the lifts so that when compacted they are true to the grades or levels required with the least possible surface disturbance. Where the base course is placed in more than one lift, clean the previously constructed lift of loose and foreign matter by sweeping with power sweepers, power brooms, or hand brooms. Make adjustments in placing procedures or equipment to obtain true grades, to minimize segregation and degradation, to adjust the water content, and to produce an acceptable base course.

3.7 LAYER THICKNESS

Compact the completed base course to the thickness indicated. Limit individual compacted lifts to a maximum thickness of 150 mm and a minimum thickness of 75 mm. Compact the base course(s) to a total thickness that is within the tolerances of paragraph ACCEPTANCE of the thickness indicated. Where the measured thickness is more than 13 mm deficient, correct such areas by scarifying, adding new material of proper gradation, reblading, and recompacting as directed. Where the measured thickness is more than 13 mm thicker than indicated, the course will be considered as conforming to the specified thickness requirements. However, the requirements for wearing course thickness and plan grade are still applicable. The average job thickness will be the average of all thickness measurements taken for the job and within the tolerances of paragraph ACCEPTANCE of the thickness indicated.

3.8 COMPACTION

Compact each lift of the base course, as specified, with approved compaction equipment. For cohesive soils, maintain water content during the compaction procedure to within plus or minus 2 percent of the optimum water content determined from laboratory tests as specified and for cohesionless soils, maintain the water content to facilitate compaction without bulking. Begin rolling at the outside edge of the surface and proceed to the center, overlapping on successive trips at least one-half the width of the roller. Slightly vary the length of alternate trips of the roller. Adjust speed of the roller as needed so that displacement of the aggregate does not occur. Compact mixture with hand-operated power tampers in all places not accessible to the rollers. Continue compaction until each lift is compacted through the full depth to meet the compaction requirements of Table 1. Make such adjustments in compacting or finishing procedures to obtain true grades, to minimize segregation and degradation, to reduce or increase water content, and to produce a compliant base course. Remove any materials found to be non-compliant and replace with compliant material or rework, as directed, to meet the requirements of this specification.

3.9 EDGES OF BASE COURSE

Place the base course(s) so that the completed section is a minimum of 600 mm wider, on all sides, than the next lift that will be placed above it. Place approved material along the outer edges of the base course in sufficient quantity to compact to the thickness of the course being constructed. When the course is being constructed in two or more lifts, simultaneously roll and compact at least a 600 mm width of this shoulder

material with the rolling and compacting of each lift of the base course.

3.10 FINISHING

Finish the surface of the top lift of base course after final compaction by cutting any overbuild to grade and rolling with a steel-wheeled roller. Do not add thin lifts of material to the top lift of base course to meet grade. If the elevation of the top lift of base course exceeds the tolerances of paragraph ACCEPTANCE, scarify the top lift to a depth of at least 75 mm and blend new material in and compact to bring to grade. Make adjustments to rolling and finishing procedures to minimize segregation and degradation, obtain grades, maintain moisture content, and produce an acceptable base course. If the surface become rough, corrugated, uneven in texture, or traffic marked prior to completion, scarify the non-compliant portion and rework and recompact it or replace as directed.

3.11 SMOOTHNESS TEST

Construct the top lift so that the surface shows no deviations exceeding the tolerances of paragraph ACCEPTANCE when tested with a 3.66 meter straightedge. Test the entire area in both a longitudinal and a transverse direction on parallel lines. Perform the transverse lines at a maximum spacing of 4.5 m or less apart, as directed. Perform the longitudinal lines at the centerline of each placement lane, regardless of whether multiple lanes are allowed to be paved at the same time, and at the 1/8th point in from each side of the lane. Hold the straightedge in contact with the surface and moved ahead one-half the length of the straightedge for each successive measurement. Determine the amount of surface irregularity by placing the freestanding (unleveled) straightedge on the pavement surface and measuring the maximum gap between the straightedge and the pavement surface. Determine measurements along the entire length of the straight edge. Correct deviations exceeding this amount by removing material and replacing with new material, or by reworking existing material and compacting it to meet these specifications.

3.12 FIELD QUALITY CONTROL

3.12.1 In-Place Tests

Perform each of the following in-place tests on samples taken from the placed and compacted ABC. Determine sample locations using random sampling. Take samples and test at the rates indicated.

- a. Perform density tests on every lift of material placed and at a frequency of one set of tests for every 250 square meters, or portion thereof, of completed area. Gradations containing more than 30 percent retained on the 37.5 mm sieve can produce inconsistent compacted density values when tested in accordance with paragraph DEGREE OF COMPACTION.
- b. Perform gradation analysis on every lift of material placed and at a frequency of one sieve analysis for every 500 square meters, or portion thereof, of material placed.
- c. Perform liquid limit and plasticity index tests at the same frequency as the sieve analysis.
- d. Measure the thickness of the base course at intervals providing at

least one measurement for each 500 square meters of base course or part thereof. Measure the thickness using test holes, at least 75 mm in diameter through the base course.

3.12.2 Approval of Material

Final approval of the materials will be based on tests for gradation, liquid limit, and plasticity index performed on samples taken from the completed and fully compacted course(s).

3.13 TRAFFIC

Completed portions of the base course can be opened to limited traffic, provided there is no marring or distorting of the surface by the traffic. Do not allow heavy equipment on the completed base course except when necessary for construction. When it is necessary for heavy equipment to travel on the completed base course, protect the area against marring or damage to the completed work. Repair damage to meet these specifications.

3.14 MAINTENANCE

Maintain the base course in a satisfactory condition until the full pavement section is completed and accepted. Immediately repair any defects and repeat repairs as often as necessary to keep the area intact. Retest any base course that was not paved over prior to the onset of winter to verify that it still complies with the requirements of this specification. Rework or replace any area of base course that is damaged as necessary to comply with this specification.

3.15 DISPOSAL OF UNSATISFACTORY MATERIALS

Dispose of any unsuitable materials that have been removed outside the limits of Government-controlled land. No additional payments will be made for materials that have to be replaced.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 12 13

BITUMINOUS TACK AND PRIME COATS

05/17

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY ASSURANCE
- 1.4 DELIVERY, STORAGE, AND HANDLING
- 1.5 EQUIPMENT, TOOLS AND MACHINES
 - 1.5.1 General Requirements
 - 1.5.2 Bituminous Distributor
 - 1.5.3 Heating Equipment for Storage Tanks
 - 1.5.4 Power Brooms and Power Blowers
- 1.6 ENVIRONMENTAL REQUIREMENTS

PART 2 PRODUCTS

- 2.1 PRIME COAT
 - 2.1.1 Emulsified Asphalt
- 2.2 TACK COAT
 - 2.2.1 Emulsified Asphalt

PART 3 EXECUTION

- 3.1 PREPARATION OF SURFACE
- 3.2 APPLICATION RATE
 - 3.2.1 Tack Coat
 - 3.2.2 Prime Coat
- 3.3 APPLICATION TEMPERATURE
 - 3.3.1 Viscosity Relationship
 - 3.3.2 Temperature Ranges
- 3.4 APPLICATION
 - 3.4.1 General
 - 3.4.2 Prime Coat
 - 3.4.3 Tack Coat
- 3.5 CURING PERIOD
- 3.6 FIELD QUALITY CONTROL
- 3.7 SAMPLING AND TESTING
 - 3.7.1 Sampling
 - 3.7.2 Calibration Test
 - 3.7.3 Trial Applications
 - 3.7.3.1 Tack Coat Trial Application Rate
 - 3.7.3.2 Prime Coat Trial Application Rate
 - 3.7.4 Sampling and Testing During Construction
- 3.8 TRAFFIC CONTROLS

-- End of Section Table of Contents --

SECTION 32 12 13

BITUMINOUS TACK AND PRIME COATS
05/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE-EN 13808 (2013) Bitumen and bituminous binders - Framework for specifying cationic bituminous emulsions

CENTRO DE ESTUDIOS Y EXPERIMENTACION DE OBRAS PUBLICAS (CEDEX)

NLT-121 (1999) Toma de Muestras de los Materiales Bituminosos

NLT-138 (1999) Viscosidad Saybolt de las emulsiones bituminosas

NLT-348 (1990) Toma de Muestras de Mezclas Bituminosas para Pavimentacion.

MINISTERIO DE FOMENTO (MF)

FOM/891/2004 (2004) Orden Por La Que Se Actualizan Determinados Articulos Del Pliego De Prescripciones Tecnicas Generales Para Obras De Carreteras Y Puentes, Relativos A Firmes Y Pavimentos

FOM/2523/2014 (2014) Orden de 12 de diciembre, por la que se actualizan determinados artículos del pliego de prescripciones técnicas generales para obras de carreteras y puentes, relativos a materiales básicos, a firmes y pavimentos, y a señalización, balizamiento y sistemas de contención de vehículos.

MF PG3 (2014) Pliego de Prescripciones Tecnicas Generales para Obras de Carreteras y Puentes

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO T 102 (2009; R 2013) Standard Method of Test for Spot Test of Asphaltic Materials

ASTM INTERNATIONAL (ASTM)

ASTM D140/D140M (2016) Standard Practice for Sampling
Asphalt Materials

ASTM D2995 (1999; R 2009) Determining Application
Rate of Bituminous Distributors

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Sampling and Testing

1.3 QUALITY ASSURANCE

Certificates of compliance for asphalt materials delivered will be obtained and checked to ensure that specification requirements are met. Tack coat materials will not be diluted. Prime coat materials when emulsions are used can be diluted on site with potable water up to 1 part emulsion to 1 part water.

1.4 DELIVERY, STORAGE, AND HANDLING

Inspect the materials delivered to the site for contamination and damage. Unload and store the materials with a minimum of handling.

1.5 EQUIPMENT, TOOLS AND MACHINES

1.5.1 General Requirements

Equipment, tools and machines used in the work are subject to approval. Maintain in a satisfactory working condition at all times. Calibrate equipment such as asphalt distributors, scales, batching equipment, spreaders and similar equipment within 12 months of their use. If the calibration expires during project, recalibrate the equipment before work can continue.

1.5.2 Bituminous Distributor

Provide a self propelled distributor with pneumatic tires of such size and number to prevent rutting, shoving or otherwise damaging the surface being sprayed. Calibrate the distributor in accordance with ASTM D2995. Design and equip the distributor to spray the bituminous material in a uniform coverage at the specified temperature, at readily determined and controlled total liquid rates from 0.14 to 4.5 L/square meter, with a pressure range of 172.4 to 517.1 kPa and with an allowable variation from the specified rate of not more than plus or minus 5 percent, and at variable widths. Include with the distributor equipment a separate power unit for the bitumen pump, full-circulation spray bars, tachometer, pressure gauges, volume-measuring devices, adequate heaters for heating of materials to the proper application temperature, a thermometer for reading the temperature of tank contents, and a hand hose attachment suitable for applying bituminous material manually to areas inaccessible to the

distributor. The distributor will be capable of circulating and agitating the bituminous material during the heating process.

1.5.3 Heating Equipment for Storage Tanks

Use steam, electric, or hot oil heaters for heating the bituminous material. Provide steam heaters consisting of steam coils and equipment for producing steam, so designed that the steam cannot come in contact with the bituminous material. Fix an armored thermometer to the tank with a temperature range from 4.4 to 204.4 degrees C so that the temperature of the bituminous material may be determined at all times.

1.5.4 Power Brooms and Power Blowers

Use power brooms and power blowers suitable for cleaning the surfaces to which the bituminous coat is to be applied.

1.6 ENVIRONMENTAL REQUIREMENTS

Apply bituminous coat only when the surface to receive the bituminous coat is dry. A limited amount of moisture (approximately 0.14 liter/square meter) can be sprayed on the surface of unbound material when prime coat is used to improve coverage and penetration of asphalt material. Apply bituminous coat only when the atmospheric temperature in the shade is 10 degrees C or above and when the temperature has not been below 2 degrees C for the 12 hours prior to application, unless otherwise directed.

PART 2 PRODUCTS

2.1 PRIME COAT

Provide asphalt conforming to one of the following grades:

2.1.1 Emulsified Asphalt

Provide emulsified asphalt conforming to MF PG3 Art. 214 and UNE-EN 13808, type C60BF4. Asphalt emulsion can be diluted up to 1 part water to 1 part emulsion for prime coat use. Do not dilute asphalt emulsion for tack coat use.

2.2 TACK COAT

2.2.1 Emulsified Asphalt

Provide emulsified asphalt conforming to MF PG3 Art. 214 and UNE-EN 13808, types C60B3 or C60B3 TER. For prime coats the emulsified asphalt can be diluted with up to 1 part emulsion to 1 part water. No dilution is allowed for tack coat applications. The base asphalt used to manufacture the emulsion is required to show a negative spot when tested in accordance with AASHTO T 102 using standard naphtha.

PART 3 EXECUTION

3.1 PREPARATION OF SURFACE

Immediately before applying the bituminous coat, remove all loose material, dirt, clay, or other objectionable material from the surface to be treated by means of a power broom or blower supplemented with hand brooms. Apply treatment only when the surface is dry and clean.

3.2 APPLICATION RATE

The exact quantities within the range specified, which may be varied to suit field conditions, will be determined by the Contracting Officer.

3.2.1 Tack Coat

Apply bituminous material for the tack coat in quantities of not less than **0.14 L** nor more than **0.46 L/square meter** of residual asphalt onto the pavement surface as approved by the Contracting Officer. Do not dilute asphalt emulsion when used as a tack coat.

3.2.2 Prime Coat

Apply bituminous material for the prime coat in quantities of not less than **0.23 L** nor more than **0.54 L/square meter** of residual asphalt for asphalt emulsion.

3.3 APPLICATION TEMPERATURE

3.3.1 Viscosity Relationship

Apply asphalt at a temperature that will provide a viscosity between 10 and 60 seconds, Saybolt Furol, or between **20 and 120 square mm/sec**, kinematic. Furnish the temperature viscosity relation to the Contracting Officer.

3.3.2 Temperature Ranges

The viscosity requirements determine the application temperature to be used. Apply the emulsion so the viscosity is within 10 and 40 SSF per **NLT-138**. Comply with **FOM/891/2004** and **FOM/2523/2014**, Art. 531.5.2.

3.4 APPLICATION

3.4.1 General

Following preparation and subsequent inspection of the surface, apply the bituminous prime or tack coat with the bituminous distributor at the specified rate with uniform distribution over the surface to be treated. Properly treat all areas and spots, not capable of being sprayed with the distributor, with the hand spray. Until the succeeding layer of pavement is placed, maintain the surface by protecting the surface against damage and by repairing deficient areas at no additional cost to the Government. If required, spread clean dry sand to effectively blot up any excess bituminous material. No smoking, fires, or flames other than those from the heaters that are a part of the equipment are permitted within **8 meters** of heating, distributing, and transferring operations of cutback materials. Prevent all traffic, except for paving equipment used in constructing the surfacing, from using the underlying material, whether primed or not, until the surfacing is completed. The bituminous coat requirements are described herein.

3.4.2 Prime Coat

Apply a prime coat on aggregate base. Apply the bituminous material uniformly over the surface to be treated at a pressure range of **172.4 to 517.1 kPa**; the rate will be as specified above in paragraph APPLICATION

RATE. To obtain uniform application of the prime coat on the surface treated at the junction of previous and subsequent applications, spread building paper on the surface for a sufficient distance back from the ends of each application to start and stop the prime coat on the paper and to ensure that all sprayers will operate at full force on the surface to be treated. Immediately after application remove and destroy the building paper.

3.4.3 Tack Coat

Apply tack coat at the locations shown on the drawings. A tack coat should be applied to every bound surface (asphalt or concrete pavement) that is being overlaid with asphalt mixture and at transverse and longitudinal joints. Apply the tack coat when the surface to be treated is clean and dry. Immediately following the preparation of the surface for treatment, apply the bituminous material by means of the bituminous distributor, within the limits of temperature specified herein and at a rate as specified above in paragraph APPLICATION RATE. Apply the bituminous material so that uniform distribution is obtained over the entire surface to be treated. Treat lightly coated areas and spots missed by the distributor by spraying with a hand wand or using other approved method. Following the application of bituminous material, allow the surface to cure without being disturbed for period of time necessary to permit setting of the tack coat. Apply the bituminous tack coat only as far in advance of the placing of the overlying layer as required for that day's operation. Maintain and protect the treated surface from damage until the succeeding course of pavement is placed.

3.5 CURING PERIOD

Following application of the bituminous material and prior to application of the succeeding layer of asphalt mixture allow the bituminous coat to cure and water or volatiles to evaporate prior to overlaying. Maintain the tacked surface in good condition until the succeeding layer of pavement is placed, by protecting the surface against damage and by repairing and recoating deficient areas. Allow the prime coat to cure without being disturbed for a period of at least 48 hours or longer, as may be necessary to attain penetration into the treated course. Furnish and spread enough sand to effectively blot up excess bituminous material.

3.6 FIELD QUALITY CONTROL

Obtain certificates of compliance for all asphalt material delivered to the project. Obtain samples of the bituminous material under the supervision of the Contracting Officer. The sample may be retained and tested by the Government at no cost to the Contractor.

3.7 SAMPLING AND TESTING

Furnish certified copies of the manufacturer's test reports indicating temperature viscosity relationship for cutback asphalt or asphalt cement, compliance with applicable specified requirements, not less than 5 days before the material is required in the work.

3.7.1 Sampling

Unless otherwise specified, sample bituminous material in accordance with [ASTM D140/D140M](#) or [NLT-121](#) and [NLT-348](#).

3.7.2 Calibration Test

Furnish all equipment, materials, and labor necessary to calibrate the bituminous distributor. Calibrate using the approved job material and prior to applying the bituminous coat material to the prepared surface. Calibrate the bituminous distributor in accordance with ASTM D2995 or MF PG3, Art. 531.4.1 and 531.8.3.

3.7.3 Trial Applications

Before applying the spray application of tack or prime coat, apply three lengths of at least 30 meters for the full width of the distributor bar to evaluate the amount of bituminous material that can be satisfactorily applied.

3.7.3.1 Tack Coat Trial Application Rate

Unless otherwise authorized, apply the trial application rate of bituminous tack coat materials in the amount of 0.23 L/square meter. Make other trial applications using various amounts of material as may be deemed necessary.

3.7.3.2 Prime Coat Trial Application Rate

Unless otherwise authorized, apply the trial application rate of bituminous materials in the amount of 0.66 L/square meter. Make other trial applications using various amounts of material as may be deemed necessary.

3.7.4 Sampling and Testing During Construction

Perform quality control sampling and testing as required in paragraph FIELD QUALITY CONTROL.

3.8 TRAFFIC CONTROLS

Keep traffic off surfaces freshly treated with bituminous material. Provide sufficient warning signs and barricades so that traffic will not travel over freshly treated surfaces.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 12 16.16

ROAD-MIX ASPHALT PAVING

11/20

PART 1 GENERAL

- 1.1 PERCENT PAYMENT
 - 1.1.1 Method of Measurement
 - 1.1.2 Basis of Payment
 - 1.1.3 Lot Pay Factor
 - 1.1.4 Payment Adjustment for Laboratory Air Voids
 - 1.1.4.1 Pay Factor Example for Laboratory Air Voids
 - 1.1.5 Payment Adjustment for In-place Densities
 - 1.1.5.1 Pay Factor Example for In-place Density
 - 1.1.6 Payment Adjustment for Plan Grade
- 1.2 PAYMENT
 - 1.2.1 Method of Measurement
 - 1.2.2 Basis of Payment
- 1.3 REFERENCES
- 1.4 SUBMITTALS
- 1.5 ACCEPTANCE
 - 1.5.1 Acceptability of Work
 - 1.5.2 Acceptance Requirements
 - 1.5.3 Pavement Lots
 - 1.5.4 Sublot Sampling
 - 1.5.5 Additional Sampling and Testing
 - 1.5.6 Theoretical Maximum Density (TMD)
 - 1.5.7 Laboratory Air Voids
 - 1.5.7.1 Tolerance
 - 1.5.7.2 Calculating Laboratory Air Voids
 - 1.5.8 In-place Density
 - 1.5.8.1 Tolerance
 - 1.5.9 Surface Smoothness
 - 1.5.9.1 Smoothness Requirements
 - 1.5.9.1.1 Straightedge Testing
 - 1.5.9.2 Testing Method
 - 1.5.9.2.1 Straightedge Testing
 - 1.5.10 Plan Grade
 - 1.5.11 Laboratory Accreditation and Validation
- 1.6 ENVIRONMENTAL REQUIREMENTS

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Asphalt Mixing Plant
 - 2.1.1.1 Truck Scales
 - 2.1.1.2 Inspection of Plant
 - 2.1.1.3 Storage bins
 - 2.1.2 Hauling Equipment
 - 2.1.3 Asphalt Pavers

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.1.3.1 Receiving Hopper
- 2.1.4 Rollers
- 2.1.5 Diamond Grinding
- 2.2 AGGREGATES
 - 2.2.1 Coarse Aggregate
 - 2.2.2 Fine Aggregate
 - 2.2.3 Mineral Filler
 - 2.2.4 Aggregate Gradation
- 2.3 ASPHALT CEMENT BINDER
- 2.4 MIX DESIGN
 - 2.4.1 JMF Requirements
 - 2.4.2 Adjustments to JMF
- 2.5 RECYCLED HOT MIX ASPHALT
 - 2.5.1 RAP Aggregates and Asphalt Cement
 - 2.5.2 RAP Mix

PART 3 EXECUTION

- 3.1 CONTRACTOR QUALITY CONTROL
 - 3.1.1 General Quality Control Requirements
 - 3.1.2 Testing Laboratory
 - 3.1.3 Quality Control Testing
 - 3.1.3.1 Asphalt Content
 - 3.1.3.2 Aggregate Properties
 - 3.1.3.3 Moisture Content of Aggregate
 - 3.1.3.4 Moisture Content of Asphalt Mixture
 - 3.1.3.5 Temperatures
 - 3.1.3.6 VMA, Marshall Stability, and Flow
 - 3.1.3.7 In-Place Density
 - 3.1.3.8 Additional Testing
 - 3.1.3.9 QC Monitoring
 - 3.1.4 Sampling
- 3.2 PREPARATION OF ASPHALT BINDER MATERIAL
- 3.3 PREPARATION OF MINERAL AGGREGATE
- 3.4 PREPARATION OF ASPHALT MIXTURE
- 3.5 PREPARATION OF THE UNDERLYING SURFACE
- 3.6 TRANSPORTING AND PLACING
 - 3.6.1 Transporting
 - 3.6.2 Placing
- 3.7 COMPACTION OF MIXTURE
 - 3.7.1 General
 - 3.7.2 Segregation
- 3.8 JOINTS
 - 3.8.1 Transverse Joints
 - 3.8.2 Longitudinal Joints

-- End of Section Table of Contents --

SECTION 32 12 16.16

ROAD-MIX ASPHALT PAVING
11/20

PART 1 GENERAL

1.1 PERCENT PAYMENT

1.1.1 Method of Measurement

Measurement of the quantity of hot-mix asphalt pavement, per tonne placed and accepted, will be made for the purposes of assessing the pay factors stipulated below.

1.1.2 Basis of Payment

The measured quantity of hot-mixed asphalt pavement will be paid for and included in the lump sum contract price. If less than 100 percent payment is due based on the pay factors stipulated in paragraph PERCENT PAYMENT, a unit price of 400 US Dollars per metric ton (tonne) will be used for purposes of calculating the payment reduction.

1.1.3 Lot Pay Factor

The lot pay factor is determined by taking the lowest computed pay factor based on either laboratory air voids, in-place density, smoothness, or grade (each discussed below). Remove and replace lots when the lowest computed pay factor requires rejection. At the end of the project calculate the average pay factor for all lots. If this average lot pay factor exceeds 95.0 percent and no individual lot has a pay factor less than 75.0 percent, then the percent payment for the entire project will be 100 percent of the unit bid price. If the average lot pay factor is less than 95.0 percent, then each lot will be paid for at the unit price multiplied by the lot's pay factor. For any lots which are less than 2,000 tonnes, a weighted lot pay factor will be used to calculate the average lot pay factor. When work on a lot is required to be terminated before all four sublots are completed, the results from the completed sublots will be analyzed to determine the percent payment for the lot following the same procedures and requirements for full lots but with fewer or more test results as determined in paragraph PAVEMENT LOTS.

1.1.4 Payment Adjustment for Laboratory Air Voids

Laboratory air void calculations for each lot will use the average theoretical maximum density values obtained for the lot. Determine the average TMD in accordance with paragraph THEORETICAL MAXIMUM DENSITY (TMD). The mean absolute deviation of the laboratory air void contents (one from each subplot) from the JMF air void content will be evaluated as shown in the example below and a pay factor will be determined from Table 1. When 0 percent payment is determined, remove and replace the rejected lot at least 100 mm into the cold (existing) lane adjacent to the longitudinal joint.

Table 1. Pay Factor Based on Laboratory Air Voids	
Mean Absolute Deviation of Lab Air Voids from JMF	Pay Factor, percent
0.60 or less	100
0.61 - 0.80	98
0.81 - 1.00	95
1.01 - 1.20	90
Above 1.20	reject (0)

1.1.4.1 Pay Factor Example for Laboratory Air Voids

An example of the computation of mean absolute deviation for laboratory air voids is as follows: Assume that the laboratory air voids are determined from 4 sublots where one set of laboratory compacted specimens is from a single subplot. The laboratory air voids for the 4 sublots are determined to be 3.5, 3.0, 4.0, and 3.7. Assume that the target air voids from the JMF is 4.0. The mean absolute deviation is then:

$$\text{Mean Absolute Deviation} = (|3.5 - 4.0| + |3.0 - 4.0| + |4.0 - 4.0| + |3.7 - 4.0|)/4$$

$$\text{Mean Absolute Deviation} = (0.5 + 1.0 + 0.0 + 0.3)/4 = (1.8)/4 = 0.45$$

The mean absolute deviation for laboratory air voids is determined to be 0.45. It can be seen from Table 1 that the lot's pay factor based on laboratory air voids is 100 percent.

1.1.5 Payment Adjustment for In-place Densities

The average in-place mat and joint densities are expressed as a percentage of the average theoretical maximum density (TMD) for the lot. Determine the average TMD in accordance with paragraph THEORETICAL MAXIMUM DENSITY (TMD). The average in-place mat density and joint density for a lot are determined and compared with Table 2 to calculate a single pay factor per lot. Use the following process to determine the single pay factor for in-place density:

- a. Step 1: Determine the pay factors for mat density and joint density using Table 2.
- b. Step 2: Determine ratio of joint area to mat area. The area associated with the joint is considered to be 3 m wide times the length of completed longitudinal construction joint in the lot. This joint area will not exceed the total lot size. The length of joint to be considered will be that length where a new lane has been placed against an adjacent lane of asphalt pavement, either an adjacent freshly paved lane or one paved at any time previously.
- c. Step 3: Compute the weighted pay factor for the joint using the formula in the example below.

- d. Step 4: Compare weighted pay factor for joint density to pay factor for mat density and select the smaller. This selected pay factor is the pay factor based on density for the lot.

When 0 percent payment is determined for mat density, remove and replace the rejected lot at least **100 mm** into the cold (existing) lane adjacent to the longitudinal joint. When 0 percent payment is determined for joint density, remove and replace the rejected longitudinal joint with a **3 m** wide paving lane that is centered over the joint.

Table 2. Pay Factor Based on In-place Density		
Average Mat Density (4 Cores) (Percent of TMD)	Pay Factor, Percent	Average Joint Density (4 Cores) (Percent of TMD)
93.0 - 96.0	100.0	91.5 or above
92.9	100.0	91.4
92.8 or 96.1	99.9	91.3
92.7	99.8	91.2
92.6 or 96.2	99.6	91.1
92.5	99.4	91.0
92.4 or 96.3	99.1	90.9
92.3	98.7	90.8
92.2 or 96.4	98.3	90.7
92.1	97.8	90.6
92.0 or 96.5	97.3	90.5
91.9	96.3	90.4
91.8 or 96.6	94.1	90.3
91.7	92.2	90.2
91.6 or 96.7	90.3	90.1
91.5	87.9	90.0
91.4 or 96.8	85.7	89.9
91.3	83.3	89.8
91.2 or 96.9	80.6	89.7
91.1	78.0	89.6
91.0 or 97.0	75.0	89.5
below 91.0, above 97.0	0.0 (reject)	below 89.5

1.1.5.1 Pay Factor Example for In-place Density

An example of the computation of a pay factor based on in-place density, is as follows: Assume the following test results for field density made on the lot: (1) Average mat density = 92.2 percent (of lab TMD). (2) Average joint density = 90.5 percent (of lab TMD). (3) Total area of lot = **2,800 square meters**. (4) Length of completed longitudinal construction

joint = 610 meters.

- a. Step 1: Determine pay factor based on mat density and on joint density, using Table 2:

Mat density of 92.2 percent = 98.3 pay factor.

Joint density of 90.5 percent = 97.3 pay factor.

- b. Step 2: Determine ratio of joint area to mat area. Multiply the length of completed longitudinal construction joint by the specified 3 meter width and divide by the mat area (total paved area in the lot).

Ratio = Ratio of joint area to mat area

Ratio = 610 meters/2,800 square meters

Ratio = 0.6667

- c. Step 3: Weighted pay factor (wpf) for joint is determined as indicated below:

$wpf = \text{joint pay factor} + (100 - \text{joint pay factor}) \times (1 - \text{ratio})$

$wpf = 97.3 + (100 - 97.3) \times (1 - 0.6667) = 98.2 \text{ percent}$

- d. Step 4: Compare weighted pay factor for joint density to pay factor for mat density and select the smaller:

Pay factor for mat density: 98.3 percent.

Weighted pay factor for joint density: 98.2 percent

Selected pay factor: 98.2 percent

1.1.6 Payment Adjustment for Plan Grade

When more than 5 percent of all measurements made within a lot are outside the 15 mm tolerance, the pay factor based on grade for that lot will be 95 percent. For individual locations where the grade exceeds 22.5 mm tolerance, remove the surface lift full depth and replace the lift with asphalt pavement to meet specification requirements at no additional cost to the Government. High spots can be diamond ground as an alternative to remove and replace in order to meet grade requirements for the lot and at individual locations.

1.2 PAYMENT

1.2.1 Method of Measurement

Measurement of the quantity of hot-mix asphalt pavement per lot will be made for the purposes of assessing acceptance stipulated in paragraph ACCEPTANCE.

1.2.2 Basis of Payment

The measured quantity of hot-mixed asphalt pavement will be paid for and included in the lump sum contract price.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 146403	(2018) Determination of clay lumps and crumbly particles in aggregate for the making of mortars and concretes.
UNE-EN 58	(2012) Bitumen and bituminous binders - Sampling bituminous binders
UNE-EN 932-1	(1997) Test For Geometrical Properties Of Aggregates. Part 1: Methods for Sampling.
UNE-EN 933-1	(2012) Tests for geometrical properties of aggregates - Part 1: Determination of particle size distribution - Sieving method
UNE-EN 933-2	(1996) Test For Geometrical Properties Of Aggregates. Part 2: Determination Of Particle Size Distribution. Test Sieves, Nominal Size Of Apertures.
UNE-EN 933-4	(2008) Tests for geometrical properties of aggregates - Part 4: Determination of particle shape - Shape index
UNE-EN 933-5	(1999) Test For Geometrical Properties Of Aggregates. Part 5: Determination Of Percentage Of Crushed And Broken Surfaces In Coarse Aggregate Particles.
UNE-EN 933-8	(2012+A1:2015) Tests for geometrical properties of aggregates - Part 8: Assessment of fines - Sand equivalent test
UNE-EN 1097-2	(2010) Tests for mechanical and physical properties of aggregates - Part 2: Methods for the determination of resistance to fragmentation
UNE-EN 1097-5	(2009) Tests for mechanical and physical properties of aggregates - Part 5: Determination of the water content by drying in a ventilated oven
UNE-EN 1367-2	(2010) Tests for thermal and weathering properties of aggregates - Part 2: Magnesium sulfate test
UNE-EN 12697-5	(2020) Bituminous mixtures - Test methods - Part 5: Determination of the maximum density
UNE-EN 12697-8	(2003) Bituminous mixtures - Test methods

for hot mix asphalt - Part 8:
Determination of void characteristics of
bituminous specimens

- UNE-EN 12697-10 (2019) Bituminous mixtures - Test methods
- Part 10: Compactability
- UNE-EN 12697-12 (2019) Bituminous mixtures - Test methods
- Part 12: Determination of the water
sensitivity of bituminous specimens
- UNE-EN 12697-31 (2020) Bituminous mixtures - Test methods
- Part 31: Specimen preparation by
gyratory compactor
- UNE-EN 12697-34 (2013) Bituminous mixtures - Test methods
- Part 34: Marshall test
- UNE-EN 13108-20 (2007) Bituminous mixtures - Material
specifications - Part 20: Type Testing
- UNE-EN 13108-21 (2007) Bituminous mixtures - Material
specifications - Part 21: Factory
Production Control
- UNE-EN ISO/IEC 17025 (2017) General requirements for the
competence of testing and calibration
laboratories (ISO/IEC 17025:2017)

CENTRO DE ESTUDIOS Y EXPERIMENTACION DE OBRAS PUBLICAS (CEDEX)

- NLT-121 (1999) Toma de Muestras de los Materiales
Bituminosos
- NLT-156 (1994) Densidad aparente de los áridos.
- NLT-164 (1990) Contenido de ligante en mezclas
bituminosas.
- NLT-168 (1990) Densidad y Huecos en Mezclas
Bituminosas Compactadas
- NLT-176 (1992) Densidad aparente del polvo mineral
en tolueno.
- NLT-314 (1992) Toma de muestras testigos en
pavimento
- NLT-348 (1990) Toma de Muestras de Mezclas
Bituminosas para Pavimentación.
- NLT-384 (2000) Contenido de ligante en mezclas
bituminosas por el método de combustión.

MINISTERIO DE FOMENTO (MF)

- FOM/510/2018 WARNING: Text in tags exceeds the maximum
length of 300 characters

FOM/2523/2014	(2014) Orden de 12 de diciembre, por la que se actualizan determinados artículos del pliego de prescripciones técnicas generales para obras de carreteras y puentes, relativos a materiales básicos, a firmes y pavimentos, y a señalización, balizamiento y sistemas de contención de vehículos.
MF PG3	(2014) Pliego de Prescripciones Técnicas Generales para Obras de Carreteras y Puentes
OC 3/2019	(2019) Orden Circular Oc 3/2019 Sobre Mezclas Bituminosas Tipo SMA
RD 305/2011	(2011) Reglamento (UE) nº 305/2011 del Parlamento Europeo y del Consejo, de 9 de marzo de 2011, por el que se establecen condiciones armonizadas para la comercialización de productos de construcción y se deroga la Directiva 89/106/CEE del Consejo.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO T 304	(2011; R 2015) Standard Method of Test for Uncompacted Void Content of Fine Aggregate
AASHTO T 329	(2015) Standard Test Method for Moisture Content of Hot Mix Asphalt (HMA) by Oven Method

ASPHALT INSTITUTE (AI)

AI MS-2	(2015) Asphalt Mix Design Methods
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ASTM INTERNATIONAL (ASTM)

ASTM C117	(2017) Standard Test Method for Materials Finer than 75-um (No. 200) Sieve in Mineral Aggregates by Washing
ASTM C127	(2015) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Coarse Aggregate
ASTM C128	(2022) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Fine Aggregate
ASTM C136/C136M	(2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM D2041/D2041M	(2011) Theoretical Maximum Specific Gravity and Density of Bituminous Paving Mixtures

ASTM D2172/D2172M	(2017; E 2018) Standard Test Methods for Quantitative Extraction of Asphalt Binder from Asphalt Mixtures
ASTM D2726/D2726M	(2019) Standard Test Method for Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures
ASTM D3665	(2012; R 2017) Standard Practice for Random Sampling of Construction Materials
ASTM D3666	(2016) Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials
ASTM D5361/D5361M	(2016) Standard Practice for Sampling Compacted Asphalt Mixtures for Laboratory Testing
ASTM D6307	(2019) Standard Test Method for Asphalt Content of Asphalt Mixture by Ignition Method
ASTM D6926	(2020) Standard Practice for Preparation of Asphalt Mixture Specimens Using Marshall Apparatus

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Placement Plan; G

SD-03 Product Data

Diamond Grinding Plan; G
Mix Design; G
Contractor Quality Control; G

SD-04 Samples

Aggregates
Asphalt Cement Binder

SD-06 Test Reports

Aggregates; G
QC Monitoring

SD-07 Certificates

Asphalt Cement Binder; G
Laboratory Accreditation and Validation

1.5 ACCEPTANCE

1.5.1 Acceptability of Work

Acquire the services of an independent commercial laboratory to perform acceptance testing. Acceptance of the plant produced mix and in-place requirements will be on a lot to lot basis. The materials and the pavement itself will be accepted on the basis of production testing. The Government may make check tests from split samples to validate the results of the production testing. Testing performed by the Government does not reduce the required testing of the independent commercial laboratory. Split samples will be taken for Government testing to reduce the variability between the independent commercial laboratory and the Government's test results. When the difference between the independent commercial laboratory and the Government's test results for split samples exceed the acceptable range of two results for multilaboratory precision for the appropriate test method (i.e. ASTM) then at least one of the laboratories is determined to be in error. An evaluation of procedures and equipment in both laboratories will be made to determine the cause(s) for the differences. Develop steps to correct procedures and equipment to bring multilaboratory precision to within acceptable limits.

1.5.2 Acceptance Requirements

Provide all sampling and testing required for acceptance and payment adjustment. Where appropriate, adjustments in percent payment acceptance for individual lots of asphalt pavement will be made based on laboratory air voids, in-place density, smoothness, and grade in accordance with the following paragraphs. Surface smoothness and grade determinations will be made on the lot as a whole. Exceptions or adjustments to this will be made in situations where the mix within one lot is placed as part of both the intermediate and surface courses, thus smoothness and grade measurements for the entire lot cannot be made.

1.5.3 Pavement Lots

A standard lot for all requirements is equal to one day's production or 2,000 tonnes, whichever is smaller. Divide each lot into four equal sublots in order to evaluate laboratory air voids and in-place density. When operational conditions cause a lot to be terminated before the specified four sublots have been completed, use the following procedure to adjust the lot size and number of tests for the lot. Where three sublots have been completed, they constitute a lot. Where one or two sublots have been completed, incorporate them into the next lot and the total number of sublots (i.e. 5 or 6 sublots) is used for acceptance criteria. Include partial lots at the end of asphalt production into the previous lot. Complete and report all theoretical maximum density, laboratory air voids, and in-place density testing within 24 hours after construction of each lot.

1.5.4 Sublot Sampling

Take one mixture sample for each sublot in accordance with NLT-121, NLT-348 and UNE-EN 58 from a random truck or another location for determining theoretical maximum density, laboratory air voids, any additional testing the Government desires, and Contractor Quality Control. All samples will be selected randomly, using commonly recognized methods of assuring randomness conforming to ASTM D3665 and employing tables of random numbers

or computer programs.

1.5.5 Additional Sampling and Testing

The Government reserves the right to direct additional samples and tests for any area which appears to deviate from the specification requirements. The cost of any additional testing will be paid for by the Government. Testing in these areas will be treated as a separate lot. **Payment Acceptance** will be made for the quantity of asphalt pavement represented by these tests in accordance with the provisions of this section.

1.5.6 Theoretical Maximum Density (TMD)

Measure theoretical maximum density one time for each subplot in accordance with **ASTM D2041/D2041M** or **UNE-EN 12697-5** for purposes of calculating laboratory air voids and determining in-place density. The average TMD for each lot will be determined as the average TMD of the random subplot samples. When the TMD on both sides of a longitudinal joint is different, the average of these two TMD values will be used as the TMD needed to calculate the percent joint density.

1.5.7 Laboratory Air Voids

Prepare one set of laboratory compacted specimens for each subplot in accordance with **ASTM D6926** OR **UNE-EN 12697-34** using the hand-held hammer for the Marshall Method. Prepare one set of laboratory compacted specimens for each subplot using the Superpave gyratory compactor. Provide three test specimens prepared from the same sample for each set of laboratory compacted specimens. Compact the specimens within 2 hours of the time the mixture was loaded into trucks at the asphalt plant. Do not reheat samples prior to compaction. Provide insulated containers as necessary to maintain the sample temperature. Measure the bulk density of laboratory compacted specimens in accordance with **ASTM D2726/D2726M** or **NLT-168**. Determine laboratory air voids from one set (three laboratory compacted specimens) for each subplot sample.

1.5.7.1 Tolerance

Provide laboratory air voids with a mean absolute deviation of 1.00 percent or less from the JMF for each lot. Remove and replace lots that do not meet the laboratory air voids requirement at least **100 mm** into the cold (existing) lane adjacent to the longitudinal joint, at no additional cost to the Government. The mean absolute deviation of the laboratory air void contents from the JMF air void content will be evaluated as shown in the example below.

1.5.7.2 Calculating Laboratory Air Voids

Laboratory air void calculations for each lot will use the average theoretical maximum density values obtained for the lot. Determine the average TMD in accordance with paragraph THEORETICAL MAXIMUM DENSITY (TMD). The mean absolute deviation of the laboratory air void contents (one from each subplot) from the JMF air void content will be evaluated as in the following example:

Assume that the laboratory air voids are determined from 4 sublots where one set of laboratory compacted specimens is from a single subplot. The laboratory air voids for the 4 sublots are determined to be 3.5, 3.0, 4.0,

and 3.7. Assume that the target air voids from the JMF is 4.0. The mean absolute deviation is then:

$$\text{Mean Absolute Deviation} = (|3.5 - 4.0| + |3.0 - 4.0| + |4.0 - 4.0| + |3.7 - 4.0|)/4$$

$$\text{Mean Absolute Deviation} = (0.5 + 1.0 + 0.0 + 0.3)/4 = (1.8)/4 = 0.45$$

The mean absolute deviation for laboratory air voids is determined to be 0.45. It can be seen that 0.45 is less than 1.00 percent. The lot is acceptable for laboratory air voids.

1.5.8 In-place Density

Obtain one random 100 mm or 150 mm diameter core from the mat and joint of each subplot in accordance with ASTM D5361/D5361M or NLT-314. for determining in-place density. Cut samples neatly with a diamond core drill bit. Obtain random cores that are the full thickness of the layer being placed. Select core locations randomly. Locate cores for mat density no closer than 300 mm from a transverse or longitudinal joint including the pavement edge. Center all cores for joint density on the joint. Discard samples that are clearly defective as a result of sampling and take an additional random core. When the random core is less than 25 mm thick, it will not be included in the analysis. In this case, obtain another random core sample. Clean and tack coat dry core holes before filling with asphalt mixture. Fill all core holes with asphalt mixture and compact using a standard Marshall hammer to the density specified. Provide all tools, labor, and materials for cutting samples, cleaning, and filling the cored pavement. Measure in-place density in accordance with ASTM D2726/D2726M or NLT-168. using each core obtained from the mat and joint.

1.5.8.1 Tolerance

Provide a minimum in-place mat density of 93.0 percent and a minimum in-place joint density of 90.0 percent for each lot. The average in-place mat and joint densities are expressed as a percentage of the average theoretical maximum density (TMD) for the lot. Determine the average TMD in accordance with paragraph THEORETICAL MAXIMUM DENSITY (TMD). Remove and replace lots that do not meet the in-place mat density requirement at least 100 mm into the cold (existing) lane adjacent to the longitudinal joint, at no additional cost to the Government. Remove and replace the longitudinal joint when the lot does not meet the in-place joint density, at no additional cost to the Government. Use a 3 m wide paving lane that is centered over the joint.

1.5.9 Surface Smoothness

Use a straightedge and profilograph for measuring surface smoothness. Use the profilograph method for all longitudinal testing, except for paving lanes less than 400 meters in length. Use the straightedge method for transverse testing, for longitudinal testing where the length of each pavement lane is less than 400 meters, and at the ends of the paving limits for the project. Smoothness requirements do not apply over crowns or grade breaks. Maintain detailed notes of the testing results and provide a copy to the Government immediately after each day's testing.

1.5.9.1 Smoothness Requirements

1.5.9.1.1 Straightedge Testing

Provide finished surfaces of the pavements with no abrupt change of 6 mm or more when checked with an approved 4 m straightedge. Remove and replace surface lift lots when the surface smoothness exceeds 9 mm, at no additional cost to the Government. High spots can be diamond ground as an alternative to remove and replace in order to meet surface smoothness requirements at individual locations.

1.5.9.2 Testing Method

After the final rolling, but not later than 24 hours after placement, test the surface of the pavement in each entire lot in a manner to reveal surface irregularities exceeding the tolerances specified above. If any pavement areas are diamond ground, retest these areas immediately after diamond grinding. The maximum area allowed to be corrected by diamond grinding is 10 percent of the total area of the lot. Check a number of random locations along with any observed suspicious locations primarily at transverse and longitudinal joints with the straightedge.

1.5.9.2.1 Straightedge Testing

Use the straightedge to measure abrupt changes in surface smoothness. Hold the straightedge in contact with the pavement surface and measure the maximum distance between the straightedge and the pavement surface. Determine the amount of surface irregularity by placing the freestanding (unleveled) straightedge on the pavement surface and allowing it to rest upon the two highest spots covered by its length, and measuring the maximum gap between the straightedge and the pavement surface in the area between these two high points.

1.5.10 Plan Grade

Provide a final wearing surface of pavement conforming to the elevations and cross sections shown and not vary more than 15 mm from the plan grade established and approved at site of work. Within 5 working days after completion of a particular lot incorporating the final wearing course, test the final wearing surface of the pavement for conformance with specified plan grade requirements. Match finished surfaces at juncture with other pavements with finished surfaces of abutting pavements. Deviation from the plan elevation will not be permitted in areas of pavements where closer conformance with planned elevation is required for the proper functioning of drainage and other appurtenant structures involved. For roads, the grade will be determined by running lines of levels along the centerline at intervals of 7.6 m or less longitudinally to determine the elevation of the completed pavement surface. Measure transverse grades at appropriate intervals. For parking lots, the grade will be determined by running lines of levels at intervals of 7.6 m or less longitudinally and transversely to determine the elevation of the completed pavement surface. Diamond grinding can be used to remove high spots to meet grade requirements. Skin patching for correcting low areas or planing or milling for correcting high areas will not be permitted. Maintain detailed notes of the results of the testing and provide a copy to the Government immediately after each day's testing. Remove and replace surface lift lots when individual locations exceed the 15 mm tolerance, at no additional cost to the Government. High spots can be diamond ground as an alternative to remove and replace in order to meet

plan grade requirements at individual locations.

1.5.11 Laboratory Accreditation and Validation

Provide laboratories used to develop the Job Mix Formula (JMF), perform acceptance testing, and Contractor Quality Control testing that meet the requirements of [ASTM D3666](#) or [UNE-EN ISO/IEC 17025](#) and are ENAC accredited. Provide laboratories with a masonry saw having a diamond blade for trimming pavement cores and samples. Perform all required test methods by an accredited laboratory. Schedule and provide payment for laboratory inspections. Additional payment or a time extension due to failure to acquire the required laboratory accreditation is not allowed. Submit a certificate of compliance signed by the manager of the laboratory stating that it meets these requirements to the Government prior to the start of construction. At a minimum, include the following certifications:

- a. Qualifications of personnel; laboratory manager, supervising technician, and testing technicians.
- b. A listing of equipment to be used in developing the job mix.
- c. A copy of the laboratory's quality control system.

1.6 ENVIRONMENTAL REQUIREMENTS

Do not place the asphalt mixture upon a wet surface or when the surface temperature of the underlying course is less than specified in [Table 3](#). The temperature requirements may be waived by the Government, if requested; however, meet all other requirements including compaction.

Table 3 . Surface Temperature Limitations of Underlying Course	
Mat Thickness, mm	Degrees C
75 or greater	4
Less than 75	7

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Perform the work consisting of pavement courses composed of mineral aggregate and asphalt material heated and mixed in a central mixing plant and placed on a prepared course. Provide asphalt pavement designed and constructed in accordance with this section conforming to the lines, grades, thicknesses, and typical cross sections shown on the drawings. Construct each course to the depth, section, or elevation required by the drawings and rolled, finished, and approved before the placement of the next course. Submit proposed [Placement Plan](#) indicating lane widths and longitudinal joints for each course or lift.

2.1.1 Asphalt Mixing Plant

[RD 305/2011](#). Provide plants used for the preparation of asphalt mixture conforming to the requirements of [UNE-EN 13108-20](#) and [UNE-EN 13108-21](#) with the following changes:

2.1.1.1 Truck Scales

Weigh the asphalt mixture on approved scales, or on certified public scales at no additional expense to the Government. Inspect and seal scales at least annually by an approved calibration laboratory.

2.1.1.2 Inspection of Plant

Provide access to the Government at all times, to all areas of the plant for checking adequacy of equipment; inspecting operation of the plant; verifying weights, proportions, and material properties; checking the temperatures maintained in the preparation of the mixtures and for taking samples. Provide assistance as requested, for the Government to procure any desired samples.

2.1.1.3 Storage bins

The asphalt mixture can be stored in non-insulated storage bins for a period of time not exceeding 3 hours. The asphalt mixture can be stored in insulated storage bins for a period of time not exceeding 8 hours. Provide the mix drawn from bins that meets the same requirements as mix loaded directly into trucks.

2.1.2 Hauling Equipment

Provide trucks used for hauling asphalt mixture that have tight, clean, and smooth metal beds. To prevent the mixture from adhering to them, lightly coat the truck beds with a minimum amount of paraffin oil, lime solution, or other approved material. Do not use petroleum based products as a release agent. Provide each truck with a suitable cover to protect the mixture from adverse weather, contamination, and loss of material during hauling. When necessary due to long haul distance and cold weather, provide insulated truck beds with covers (tarps) that are securely fastened.

2.1.3 Asphalt Pavers

Provide mechanical spreading and finishing equipment consisting of a self-powered paver, capable of spreading and finishing the mixture to the specified line, grade, and cross section. Provide paver screed capable of laying a uniform mixture to meet the specified thickness, smoothness, and grade without physical or temperature segregation, the full width of the material being placed. Provide a paver with a vibrating screed to be used during all placement.

2.1.3.1 Receiving Hopper

Provide paver with a receiving hopper of sufficient capacity to permit a uniform spreading operation and a distribution system to place the mixture uniformly in front of the screed without segregation. Provide a screed that effectively produces a finished surface of the required evenness and texture without tearing, shoving, or gouging the mixture.

2.1.4 Rollers

Provide rollers in good condition and operate at slow speeds to avoid displacement of the asphalt mixture. Provide sufficient number, type, and weight of rollers to compact the mixture to the required density while it is still in a workable condition. Do not use equipment which causes

excessive crushing of the aggregate.

2.1.5 Diamond Grinding

Those performing diamond grinding are required to have a minimum of three years experience in diamond grinding. In areas not meeting the specified limits for surface smoothness and plan grade, reduce high areas to attain the required smoothness and grade, except as depth is limited below. Reduce high areas by diamond grinding the asphalt pavement with approved equipment. Perform diamond grinding by sawing with saw blades impregnated with an industrial diamond abrasive. Assemble the saw blades in a cutting head mounted on a machine designed specifically for diamond grinding that produces the required texture and smoothness level without damage to the asphalt pavement or joint faces. Provide diamond grinding equipment with saw blades that are 3 mm wide, a minimum of 60 blades per 300 mm of cutting head width, and capable of cutting a path a minimum of 0.9 m wide. Diamond grinding equipment that causes raveling, fracturing of aggregate, or disturbance to the underlying material will not be allowed. The maximum area corrected by diamond grinding the surface of the asphalt pavement is 10 percent of the total area of any lot. The maximum depth of diamond grinding is 12 mm. Provide diamond grinding machine equipped to flush and vacuum the pavement surface. Dispose of all debris from diamond grinding operations off Government property. Prior to diamond grinding, submit a [Diamond Grinding Plan](#) for review and approval. At a minimum, include the daily reports for the deficient areas, the location and extent of deficiencies, corrective actions, and equipment. Remove and replace all pavement areas requiring plan grade or surface smoothness corrections in excess of the limits specified.

Prior to production diamond grinding operations, perform a test section at the approved location, consisting of a minimum of two adjacent passes with a minimum length of 12 m to allow evaluation of the finish and transition between adjacent passes. Production diamond grinding operations cannot be performed prior to approval.

2.2 AGGREGATES

Notify the Government at least 7 days before sampling aggregates. Obtain samples in accordance with [UNE-EN 932-1](#) that are representative of the materials to be used for the project. Provide aggregates consisting of crushed stone, crushed gravel, crushed slag, screenings, natural sand, and mineral filler as required. The portion of material retained on the 4.75 mm sieve is coarse aggregate. The portion of material passing the 4.75 mm sieve and retained on the 0.075 mm sieve is fine aggregate. The portion passing the 0.075 mm sieve is defined as mineral filler. Submit sufficient materials to produce 90 kg of blended mixture for mix design verification. Submit all aggregate test results and samples to the Government at least 14 days prior to start of construction. Perform job aggregate testing no earlier than 6 months before contract award.

2.2.1 Coarse Aggregate

Provide coarse aggregate consisting of sound, tough, durable particles, free from films of material that would prevent thorough coating and bonding with the asphalt material and free from organic matter and other deleterious substances. Provide coarse aggregate particles meeting the following requirements:

- a. The percentage of loss not greater than 40 percent after 500

revolutions when tested in accordance with [UNE-EN 1097-2](#).

- b. The sodium sulfate soundness loss not exceeding 12 percent, or the magnesium sulfate soundness loss not exceeding 18 percent after five cycles when tested in accordance with [UNE-EN 1367-2](#).
- c. At least 75 percent by weight of coarse aggregate containing two or more fractured faces when tested in accordance with [UNE-EN 933-5](#) with fractured faces produced by crushing.
- d. The particle shape essentially cubical and the aggregate containing not more than 10 percent, by weight, of flat and elongated particles (5:1 ratio of length to thickness) when tested in accordance with [UNE-EN 933-4](#).
- e. Slag consisting of air-cooled, blast furnace slag with a compacted weight of not less than [1200 kg/cubic meter](#) when tested in accordance with [NLT-176](#).
- f. Clay lumps and friable particles not exceeding 0.3 percent, by weight, when tested in accordance with [UNE 146403](#).

2.2.2 Fine Aggregate

Provide fine aggregate consisting of clean, sound, tough, durable particles. Provide aggregate particles that are free from coatings of clay, silt, or any objectionable material, contain no clay balls, and meet the following requirements:

- a. Quantity of natural sand (noncrushed material) added to the aggregate blend not exceeding 15 percent by weight of total aggregate.
- b. Individual fine aggregate sources with a sand equivalent value greater than 45 when tested in accordance with [UNE-EN 933-8](#).
- c. Fine aggregate portion of the blended aggregate with an uncompacted void content greater than 45.0 percent when tested in accordance with [AASHTO T 304](#) Method A.
- d. Clay lumps and friable particles not exceeding 0.3 percent, by weight, when tested in accordance with [UNE 146403](#).

2.2.3 Mineral Filler

Provide mineral filler consisting of a nonplastic material meeting the requirements of [FOM/2523/2014](#), Table 542.7 for material passing [UNE-EN 933-2](#) sieve 0.063 and with density between 0.50 and 0.90.

2.2.4 Aggregate Gradation

Provide a combined aggregate gradation that conforms to gradations specified in [Table 4](#), when tested in accordance with [ASTM C136/C136M](#) and [ASTM C117](#), and does not vary from the low limit on one sieve to the high limit on the adjacent sieve or vice versa, but grades uniformly from coarse to fine. Provide a JMF within the specification limits; however, the gradation can exceed the limits when the allowable deviation from the JMF shown in [Tables 4 and 5](#) are applied.

Table 4. Aggregate Gradations			
Sieve Size, mm		Gradation 2 Percent Passing by Mass	
25.0		---	
19.0		100	
12.5		90-100	
9.5		69-89	
4.75		53-73	
2.36		38-60	
1.18		26-48	
0.60		18-38	
0.30		11-27	
0.15		6-18	
0.075		3-6	

2.3 ASPHALT CEMENT BINDER

Provide asphalt cement binder that conforms to BC 35/50 or PMB 25/55-65 in compliance with Art. 211 and 542 of MF PG3. Provide test data indicating grade certification by the supplier at the time of delivery of each load to the mix plant. Submit copies of these certifications to the Government. The supplier is defined as the last source of any modification to the binder. The Government may sample and test the binder at the mix plant at any time before or during mix production.

2.4 MIX DESIGN

Develop the mix design in accordance with FOM/2523/2014, Art. 542 and as modified by FOM/510/2018, Type AC-16 SURF 35/50D for wearing course and

AC-22 base 35/50 for the base course. Perform Job Mix formula (JMF) and aggregates testing no earlier than 6 months before contract award. Provide asphalt mixture composed of well-graded aggregate, mineral filler if required, and asphalt material. Provide aggregate fractions sized, handled in separate size groups, and combined in such proportions that the resulting mixture meets the grading requirements of AC-16 and AC-22. Do not produce asphalt pavement for [payment acceptance](#) until a JMF has been approved. [Prepare samples at various asphalt contents and compacted in accordance with UNE-EN 12697-10 and UNE-EN 12697-31.](#) Use laboratory compaction temperatures for Polymer Modified Asphalts as recommended by the asphalt binder manufacturer. Determine the Tensile Strength Ratio (TSR) of the composite mixture in accordance with [UNE-EN 12697-12](#). Compact the TSR specimens to an air void content of 7 percent plus or minus 1 percent. If the Tensile Strength Ratio (TSR) of the composite mixture is less than 75, reject the aggregates or treat the asphalt mixture with an anti-stripping agent. Add a sufficient amount of anti-stripping agent to produce a TSR of not less than 75. If an antistrip agent is required, provide it at no additional cost to the Government. Provide sufficient materials to produce **90 kg** of blended mixture to the Government for verification of mix design at least 14 days prior to construction of test section.

2.4.1 JMF Requirements

Submit the proposed JMF in writing, for approval, at least 14 days prior to the start of the test section including, as a minimum:

- a. Percent passing each sieve size.
- b. Percent of asphalt cement.
- c. Percent of each aggregate and mineral filler to be used.
- d. Asphalt performance grade or penetration grade.
- e. [Number of blows of hammer per side of molded specimen or number of Superpave gyratory compactor gyrations.](#)
- f. Laboratory mixing temperature.
- g. Laboratory compaction temperature.
- h. Temperature-viscosity relationship of the asphalt cement
- i. Plot of the combined gradation on the 0.45 power gradation chart, stating the nominal maximum size.
- j. Graphical plots and summary tabulation of [Marshall stability, flow,](#) air voids, voids in the mineral aggregate, and unit weight versus asphalt content as shown in [AI MS-2](#). Include summary tabulation that includes individual specimen data for each specimen tested.
- k. Specific gravity and absorption of each aggregate.
- l. Percent natural sand.
- m. Percent particles with two or more fractured faces (in coarse aggregate).

- n. Fine aggregate angularity.
- o. Percent flat or elongated particles in coarse aggregate.
- p. Tensile Strength Ratio and wet/dry specimen test results.
- q. Antistrip agent (if required).
- r. List of all modifiers.
- s. Percentage and properties (asphalt content, aggregate gradation, and aggregate properties) of RAP in accordance with paragraph RECYCLED ASPHALT PAVEMENT, if RAP is used.

Table 5. Mix Design Criteria			
Test Property	Marshall (50 Blows)	Marshall (75 Blows)	Superpave (50 or 75 gyrations)
Stability, N , minimum (NA for Superpave)	4450 ⁽¹⁾	8000 ⁽¹⁾	NA
Flow, 0.25 mm , (NA for Superpave)	8-18	8-16	NA
Air voids, percent	3-5	3-5	3-5
Minimum Percent Voids in Mineral Aggregate (VMA) ⁽²⁾			
Gradation 1	13.0	13.0	13.0
Gradation 2	14.0	14.0	14.0
Gradation 3	15.0	15.0	15.0
TSR, minimum percent	75	75	75
(1) This is a minimum requirement. Provide significantly higher average during construction to ensure compliance with the specifications.			
(2) Calculate VMA in accordance with AI MS-2 , based on ASTM C127 and ASTM C128 bulk specific gravity for the aggregate.			

2.4.2 Adjustments to JMF

The JMF for each mixture is in effect until a new formula is approved in writing by the Government. Should a change in sources of any materials be made, perform a new mix design and a new JMF approved before the new material is used. Make minor adjustments within the specification limits to the JMF to optimize mix volumetric properties. Adjustments to the original JMF are limited to plus or minus 4 percent on the **4.75 mm** and coarser sieves; plus or minus 3 percent on the **2.36 mm** to **0.30 mm** sieves; and plus or minus 1 percent on the **0.15 mm** sieve and **0.075 mm** sieve. Asphalt content adjustments are limited to plus or minus 0.40 from the original JMF. If adjustments are needed that exceed these limits, develop

a new mix design.

2.5 RECYCLED HOT MIX ASPHALT

Provide recycled asphalt mixture consisting of reclaimed asphalt pavement (RAP), coarse aggregate, fine aggregate, mineral filler, and asphalt cement. Provide RAP of a consistent gradation, asphalt content, and properties. Maintain RAP stockpiles free from contamination including coal-tar sealers. Limit the maximum RAP chunk size to 50 mm when feeding RAP into the plant. The individual aggregates in a RAP chunk are not to exceed the maximum size aggregate of the gradation specified in Table 4. Design the recycled asphalt mixture using procedures contained in AI MS-2. Provide RAP job mix that meets the requirements of paragraph MIX DESIGN. Limit the amount of RAP so the asphalt binder from the RAP does not exceed 30 percent of the total asphalt content.

2.5.1 RAP Aggregates and Asphalt Cement

Provide a blend of aggregates used in the recycled mix that meet the requirements of paragraph AGGREGATES. Establish the percentage of asphalt binder in the RAP for the mixture design according to FOM/2523/2014 using the appropriate dust correction procedure.

2.5.2 RAP Mix

Select the virgin asphalt binder as described below:

- a. For 0 to 20 percent recycled binder content - no change in virgin binder selection.
- b. For 20+ percent to 30 percent recycled binder content - select virgin binder one grade softer than normal.

PART 3 EXECUTION

MF PG3, FOM/2523/2014, Art. 542 and OC 3/2019.

3.1 CONTRACTOR QUALITY CONTROL

3.1.1 General Quality Control Requirements

Submit the Quality Control Plan. Do not produce hot-mix asphalt for payment acceptance until the quality control plan has been approved. In the quality control plan, address all elements which affect the quality of the pavement including, but not limited to:

- a. Mix Design and unique JMF identification code
- b. Aggregate Grading
- c. Quality of Materials
- d. Stockpile Management and procedures to prevent contamination
- e. Proportioning
- f. Mixing and Transportation
- g. Mixture Volumetrics
- h. Moisture Content of Mixtures
- i. Placing and Compaction
- j. Joints
- k. Surface Smoothness
- l. Truck bed release agent
- m. Correlation of mechanical hammer to hand hammer. Determine the number

of blows of the mechanical hammer required to provide the same density of the JMF as provided by the hand hammer. Use the average of three specimens per trial blow application.

3.1.2 Testing Laboratory

Provide a fully equipped asphalt laboratory located at the plant or job site that is equipped with heating and air conditioning units to maintain a temperature of 24 plus or minus 2.3 degrees C. Provide laboratory facilities that are kept clean and all equipment maintained in proper working condition. Provide the Government with unrestricted access to inspect the laboratory facility, to witness quality control activities, and to perform any check testing desired. The Government will advise in writing of any noted deficiencies concerning the laboratory facility, equipment, supplies, or testing personnel and procedures. When the deficiencies are serious enough to adversely affect test results, immediately suspend the incorporation of the materials into the work. Incorporation of the materials into the work will not be permitted to resume until the deficiencies are corrected.

3.1.3 Quality Control Testing

Perform all quality control tests applicable to these specifications and as set forth in the Quality Control Program. Use the independent commercial laboratory for acceptance testing in paragraph ACCEPTANCE. Use in-house capabilities or the independent commercial laboratory for quality control testing. Required elements of the testing program include, but are not limited to tests for the control of asphalt content, aggregate gradation, aggregate moisture, moisture in the asphalt mixture, temperatures, VMA, Marshall stability, flow, and in-place density. Develop a Quality Control Testing Plan as part of the Quality Control Program.

3.1.3.1 Asphalt Content

Comply with FOM/2523/2014, Table 542.10 and as modified by FOM/510/2018. Determine asphalt content a minimum of twice per lot (a lot is defined in paragraph PAVEMENT LOTS) using the ignition method in accordance with ASTM D6307 or NLT-384. Use the extraction method in accordance with ASTM D2172/D2172M or NLT-164. if the correction factor for the ignition method in ASTM D6307 is greater than 1.0. The asphalt content for the lot will be determined by averaging the test results.

3.1.3.2 Aggregate Properties

Determine aggregate gradations a minimum of twice per lot from mechanical analysis of extracted aggregate in accordance with UNE-EN 933-1 and UNE-EN 933-2. Determine the specific gravity of each aggregate size grouping for each 18,000 tonnes in accordance with NLT-156. Determine fractured faces for gravel sources for each 18,000 tonnes in accordance with UNE-EN 933-5. Determine the uncompacted void content of natural sand, manufactured sand, and blended aggregate for each 18,000 tonnes in accordance with UNE-EN 12697-8.

3.1.3.3 Moisture Content of Aggregate

Determine the moisture content of aggregate used for production a minimum of once per lot in accordance with UNE-EN 1097-5.

3.1.3.4 Moisture Content of Asphalt Mixture

Determine the moisture content of the asphalt mixture at least once per lot in accordance with [AASHTO T 329](#).

3.1.3.5 Temperatures

Check temperatures at least four times per lot, at necessary locations to determine the temperature at the dryer, the asphalt cement binder in the storage tank, the asphalt mixture at the plant, and the asphalt mixture at the job site.

3.1.3.6 VMA, Marshall Stability, and Flow

Obtain mixture samples at least four times per lot. Calculate the VMA of each specimen in accordance with [UNE-EN 12697-8](#). Provide VMA within the limits of [Table 542.12 of FOM/2523/2014](#).

3.1.3.7 In-Place Density

Conduct any necessary testing to ensure the specified density is achieved. A nuclear gauge or other non-destructive testing device can be used to monitor pavement density.

3.1.3.8 Additional Testing

Perform any additional testing deemed necessary to control the process.

3.1.3.9 QC Monitoring

Submit all QC test results to the Government on a daily basis as the tests are performed. The Government reserves the right to monitor any of the Contractor's quality control testing and to perform duplicate testing as a check to the Contractor's quality control testing.

3.1.4 Sampling

When directed by the Government, sample and test any material which appears to not meet specification requirements unless such material is voluntarily removed and replaced or deficiencies corrected. Perform all sampling in accordance with standard procedures specified.

3.2 PREPARATION OF ASPHALT BINDER MATERIAL

Heat the asphalt cement material while avoiding local overheating. Provide a continuous supply of the asphalt material to the mixer at a uniform temperature. Maintain the temperature of the asphalt delivered to the mixer to provide a suitable viscosity for adequate coating of the aggregate particles. [For hot-mix, do not heat unmodified asphalt to a temperature exceeding 160 degrees C when added to the aggregate. Do not heat modified asphalt to a temperature exceeding 175 degrees C when added to the aggregate.](#)

3.3 PREPARATION OF MINERAL AGGREGATE

Heat and dry the aggregate prior to mixing. Provide a rate of heating and a maximum temperature that does not damage the aggregates. Do not heat the aggregate to a temperature exceeding [175 degrees C](#) when the asphalt binder is added. Maintain the temperature no lower than is required to

obtain complete coating and uniform distribution on the aggregate particles and to provide a mixture of satisfactory workability.

3.4 PREPARATION OF ASPHALT MIXTURE

Weigh or meter the aggregates and the asphalt cement and introduce into the mixer the amount specified by the JMF. Mix the combined materials until the aggregate obtains a uniform coating of asphalt binder and is thoroughly distributed throughout the mixture. The moisture content of all asphalt mixture upon discharge from the plant is not to exceed 0.5 percent by total weight of mixture as measured by [AASHTO T 329](#).

3.5 PREPARATION OF THE UNDERLYING SURFACE

Immediately before placing the asphalt mixture, clean the underlying course of dust and debris. Apply a prime coat or tack coat in accordance with Section [32 12 13](#) BITUMINOUS TACK AND PRIME COATS.

3.6 TRANSPORTING AND PLACING

3.6.1 Transporting

[FOM/2523/2014](#), Art. 542.5.5 and the following. Transport asphalt mixture from the mixing plant to the site in clean, tight vehicles. Schedule deliveries so that placing and compacting of mixture is uniform with minimum stopping and starting of the paver. Provide adequate artificial lighting for night placements. Hauling over freshly placed material will not be permitted until the material has been compacted as specified, and allowed to cool to [60 degrees C](#).

3.6.2 Placing

[FOM/2523/2014](#), Art. 542.5.6 and the following. Place the mix in lifts of adequate thickness and compact at a temperature suitable for obtaining density, surface smoothness, and other specified requirements. Upon arrival, place the mixture to the full width by an asphalt paver; strike off in a uniform layer of such depth that, when the work is completed, the required thickness is obtained and the surface conforms to the grade and contour indicated. Do not broadcast waste mixture onto the mat or recycle into the paver hopper. Collect waste mixture and dispose off site. Regulate the speed of the paver to eliminate pulling and tearing of the asphalt mat. Begin placement of the mixture along the centerline of a crowned section or on the high side of areas with a one-way slope. Place the mixture in consecutive adjacent strips having a minimum width of [3 m](#). Offset the longitudinal joint in one course from the longitudinal joint in the course immediately below by at least [300 mm](#); however, locate the joint in the surface course at the centerline of the pavement. Offset transverse joints in one course by at least [3 m](#) from transverse joints in the previous course. Offset transverse joints in adjacent lanes a minimum of [3 m](#). On isolated areas where irregularities or unavoidable obstacles make the use of mechanical spreading and finishing equipment impractical, the mixture can be spread and luted by hand tools.

3.7 COMPACTION OF MIXTURE

3.7.1 General

- a. After placing, thoroughly and uniformly compact the mixture by rolling. Compact the surface as soon as possible without causing

displacement, cracking, or shoving. Determine the sequence of rolling operations and the type of rollers used with the exception that application of more than three passes with a vibratory roller in the vibrating mode is prohibited. Maintain the speed of the roller, at all times, sufficiently slow to avoid displacement of the asphalt mixture and to be effective in compaction. Correct at once any displacement occurring as a result of reversing the direction of the roller, or from any other cause.

- b. Furnish sufficient rollers to handle the output of the plant. Continue rolling until the surface is of uniform texture, true to grade and cross section, and the required field density is obtained. To prevent adhesion of the mixture to the roller, keep the wheels properly moistened, but excessive water is not permitted. In areas not accessible to the roller, thoroughly compact the mixture with hand tampers or small compactors. Remove the full depth of any mixture that becomes loose and broken, mixed with dirt, contains check-cracking, or is in any way defective. Replace with fresh asphalt mixture and immediately compact to conform to the surrounding area. Perform this work at no expense to the Government. Skin patching is not allowed.

3.7.2 Segregation

The Government can sample and test any material that looks deficient. When the in-place material appears to be segregated, the Government has the option to sample the material and have it tested and compared to the in-place density requirements in [Table 2 paragraph ACCEPTANCE](#). If the material fails to meet these specification requirements, remove and replace the extent of the segregated material the full depth of the layer of asphalt mixture at no additional cost to the Government. When segregation occurs in the mat, take appropriate action to correct the process so that additional segregation does not occur.

3.8 JOINTS

Construct joints to ensure a continuous bond between the courses and to obtain the required density. Provide all joints with the same texture as other sections of the course and meet the requirements for smoothness and grade.

3.8.1 Transverse Joints

Do not pass the roller over the unprotected end of the freshly laid mixture, except when necessary to form a transverse joint. When necessary to form a transverse joint, construct by means of placing a bulkhead or by tapering the course. Utilize a dry saw cut on the transverse joint full depth and width on a straight line to expose a vertical face prior to placing the adjacent lane. Remove the cutback material from the project. In both methods, provide a light tack coat of asphalt material to all contact surfaces before placing any fresh mixture against the joint.

3.8.2 Longitudinal Joints

Provide a joint that meets density and smoothness requirements for joints and has uniform texture. Cut back longitudinal joints which are irregular, damaged, uncompacted, cold (less than **80 degrees C** at the time of placing adjacent lanes), or otherwise defective, a maximum of **75 mm** from the top of the course with a cutting wheel to expose a clean, sound,

near vertical surface for the full depth of the course. Remove all cutback material from the project. Provide a light tack coat of asphalt material to all contact surfaces prior to placing any fresh mixture against the joint.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 15 00

AGGREGATE SURFACING

05/17

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEGREE OF COMPACTION
- 1.3 SUBMITTALS
- 1.4 EQUIPMENT, TOOLS, AND MACHINES
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Sampling
 - 1.5.2 Testing
 - 1.5.2.1 Sieve Analysis
 - 1.5.2.2 Liquid Limit and Plasticity Index
 - 1.5.2.3 Moisture-Density Determinations
 - 1.5.2.4 Field Density Tests
 - 1.5.2.5 Wear Test
- 1.6 ENVIRONMENTAL REQUIREMENTS

PART 2 PRODUCTS

- 2.1 AGGREGATES
 - 2.1.1 Coarse Aggregates
 - 2.1.2 Fine Aggregates
 - 2.1.3 Gradation Requirements
- 2.2 LIQUID LIMIT AND PLASTICITY INDEX
- 2.3 TESTS, INSPECTIONS, AND VERIFICATIONS
 - 2.3.1 Initial Tests
 - 2.3.2 Approval of Material

PART 3 EXECUTION

- 3.1 STOCKPILING MATERIAL
- 3.2 PREPARATION OF UNDERLYING SUBGRADE
- 3.3 GRADE CONTROL
- 3.4 MIXING AND PLACING MATERIALS
- 3.5 LAYER THICKNESS
- 3.6 COMPACTION
- 3.7 EDGES OF AGGREGATE SURFACE COURSE
- 3.8 SMOOTHNESS TEST
- 3.9 FIELD QUALITY CONTROL
 - 3.9.1 In-Place Tests
 - 3.9.2 Approval of Material
- 3.10 MAINTENANCE

-- End of Section Table of Contents --

SECTION 32 15 00

AGGREGATE SURFACING
05/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 103501	(1994) Geotechnic. Compaction Test. Modified Proctor.
UNE 103503	(1995) Determination "In Situ" Of Density Of Soil By The Sand Method.
UNE 103900	(2013) In situ determination of density and moisture content of soil and granular materials by nuclear methods: low depths.
UNE-EN 932-1	(1997) Test For Geometrical Properties Of Aggregates. Part 1: Methods for Sampling.
UNE-EN 933-1	(2012) Tests for geometrical properties of aggregates - Part 1: Determination of particle size distribution - Sieving method
UNE-EN 933-2	(1996) Test For Geometrical Properties Of Aggregates. Part 2: Determination Of Particle Size Distribution. Test Sieves, Nominal Size Of Apertures.
UNE-EN 1097-2	(2010) Tests for mechanical and physical properties of aggregates - Part 2: Methods for the determination of resistance to fragmentation
UNE-EN ISO 17892-12	(2019) Geotechnical investigation and testing - Laboratory testing of soil - Part 12: Determination of liquid and plastic limits (ISO 17892-12:2018).

CENTRO DE ESTUDIOS Y EXPERIMENTACION DE OBRAS PUBLICAS (CEDEX)

NLT-149	(1991) Resistencia al Desgaste de los Aridos por medio de la Maquina de Los Angeles
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AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO T 180	(2017) Standard Method of Test for
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Moisture-Density Relations of Soils Using
a 4.54-kg (10-lb) Rammer and a 457-mm
(18-in.) Drop

AASHTO T 224

(2010) Standard Method of Test for
Correction for Coarse Particles in the
Soil Compaction Test

ASTM INTERNATIONAL (ASTM)

ASTM D1557

(2012; E 2015) Standard Test Methods for
Laboratory Compaction Characteristics of
Soil Using Modified Effort (56,000
ft-lbf/ft³) (2700 kN-m/m³)

ASTM E11

(2022) Standard Specification for Woven
Wire Test Sieve Cloth and Test Sieves

1.2 DEGREE OF COMPACTION

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum laboratory dry density obtained by the test procedure presented in [UNE 103501](#) abbreviated as a percent of laboratory maximum dry density. The degree of compaction for material having more than 30 percent by weight of their particles retained on the [9.0 mm](#) sieve will be expressed as a percentage of the laboratory maximum dry density in accordance with [AASHTO T 180](#) Method D and corrected with [AASHTO T 224](#).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section [01 33 00](#) SUBMITTAL PROCEDURES:

[SD-03 Product Data](#)

[Plant, Equipment, and Tools; G](#)

[SD-06 Test Reports](#)

[Initial Tests; G](#)
[In-Place Tests; G](#)

1.4 EQUIPMENT, TOOLS, AND MACHINES

All [plant, equipment, and tools](#) used in the performance of the work will be subject to approval by the Contracting Officer before the work is started. Maintain all plant, equipment, and tools in satisfactory working condition at all times. Submit a list of proposed equipment, including descriptive data. Provide adequate equipment having the capability of minimizing segregation, producing the required compaction, meeting grade controls, thickness control, and smoothness requirements as set forth herein.

1.5 QUALITY ASSURANCE

Sampling and testing are the responsibility of the Contractor. Perform

sampling and testing using a laboratory approved in accordance with Section 01 45 00 QUALITY CONTROL. Work requiring testing will not be permitted until the testing laboratory has been inspected and approved. Test the materials to establish compliance with the specified requirements and perform testing at the specified frequency. The Contracting Officer may specify the time and location of the tests. Furnish copies of test results to the Contracting Officer within 24 hours of completion of the tests.

1.5.1 Sampling

Take samples for laboratory testing in conformance with UNE-EN 932-1. When deemed necessary, the sampling will be observed by the Contracting Officer.

1.5.2 Testing

1.5.2.1 Sieve Analysis

Perform sieve analysis in conformance with UNE-EN 933-1 using sieves conforming to UNE-EN 933-2.

1.5.2.2 Liquid Limit and Plasticity Index

Determine liquid limit and plasticity index in accordance with UNE-EN ISO 17892-12.

1.5.2.3 Moisture-Density Determinations

Determine the laboratory maximum dry density and optimum moisture content in accordance with paragraph DEGREE OF COMPACTION.

1.5.2.4 Field Density Tests

Measure field density in accordance with UNE 103503 or UNE 103900. For the method presented in UNE 103900 check the calibration curves and adjust them, if necessary, using only the sand cone method. Tests performed in accordance with UNE 103900 result in a wet unit weight of soil and UNE 103900 will be used to determine the moisture content of the soil. Also check the calibration curves furnished with the moisture gauges along with density calibration checks as described in UNE 103900. Make the calibration checks of both the density and moisture gauges using the prepared containers of material method, on each different type of material being tested at the beginning of a job and at intervals as directed. Submit calibration curves and related test results prior to using the device or equipment being calibrated.

1.5.2.5 Wear Test

Perform wear tests on aggregate surface course material in conformance with NLT-149 and UNE-EN 1097-2.

1.6 ENVIRONMENTAL REQUIREMENTS

Bring surfaces damaged by rainfall or other weather conditions to a satisfactory condition.

PART 2 PRODUCTS

2.1 AGGREGATES

Provide aggregates consisting of clean, sound, durable particles of natural gravel, crushed gravel, crushed stone, sand, slag, soil, or other approved materials processed and blended or naturally combined. Provide aggregates free from lumps and balls of clay, organic matter, objectionable coatings, and other foreign materials. The Contractor is responsible for obtaining materials that meet the specification and can be used to meet the grade and smoothness requirements specified herein after all compaction operations have been completed.

2.1.1 Coarse Aggregates

The material retained on the 5 mm sieve is known as coarse aggregate. Use only coarse aggregates that are reasonably uniform in density and quality. Use only coarse aggregate having a percentage of wear not exceeding 50 percent after 500 revolutions as determined by UNE-EN 1097-2. The amount of flat and/or elongated particles must not exceed 20 percent. A flat particle is one having a ratio of width to thickness greater than three; an elongated particle is one having a ratio of length to width greater than three. When the coarse aggregate is supplied from more than one source, aggregate from each source must meet the requirements set forth herein.

2.1.2 Fine Aggregates

The material passing the 5 mm sieve is known as fine aggregate. Fine aggregate consists of screenings, sand, soil, or other finely divided mineral matter that is processed or naturally combined with the coarse aggregate.

2.1.3 Gradation Requirements

Gradation requirements specified in TABLE I apply to the completed aggregate surface. It is the responsibility of the Contractor to obtain materials that will meet the gradation requirements after mixing, placing, compacting, and other operations. TABLE I shows permissible gradings for granular material used in aggregate surface roads and parking areas. Use sieves conforming to ASTM E11 or to UNE-EN 933-2.

TABLE I. GRADATION FOR AGGREGATE SURFACE COURSES Percentage by Weight Passing Square-Mesh Sieve		
Sieve Designation (mm)	No. 1	No. 2
25	100	100
9.5	50-85	60-100
4.7	35-65	50-85
2.00	25-50	40-70

TABLE I. GRADATION FOR AGGREGATE SURFACE COURSES Percentage by Weight Passing Square-Mesh Sieve		
Sieve Designation (mm)	No. 1	No. 2
0.425	15-30	24-45
0.075	8-15	8-15

2.2 LIQUID LIMIT AND PLASTICITY INDEX

The portion of the completed aggregate surface course passing the 0.425 mm sieve must have a maximum liquid limit of 35 and a plasticity index of 4 to 9.

2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

2.3.1 Initial Tests

Perform one of each of the following tests, on the proposed material prior to commencing construction, to demonstrate that the proposed material meets all specified requirements when furnished. Complete this testing for each source if materials from more than one source are proposed.

- Sieve Analysis.
- Liquid limit and plasticity index.
- Moisture-density relationship.
- Wear.

Submit certified copies of test results for approval not less than 30 days before material is required for the work.

2.3.2 Approval of Material

Tentative approval of material will be based on initial test results.

PART 3 EXECUTION

3.1 STOCKPILING MATERIAL

Prior to stockpiling the material, clear and level the storage sites. Stockpile all materials, including approved material available from excavation and grading, in the manner and at the locations designated. Stockpile aggregates in such a manner that will prevent segregation. Stockpile aggregates and binders obtained from different sources separately.

3.2 PREPARATION OF UNDERLYING SUBGRADE

Clean the subgrade and shoulders of all foreign substances. Correct ruts or soft yielding spots in the subgrade, areas having inadequate compaction and deviations of the surface from the requirements set forth herein by loosening and removing soft or unsatisfactory material and by adding approved material, reshaping to line and grade and recompacting to density

requirements specified in Section 31 00 00 EARTHWORK. Do not allow traffic or other operations to disturb the completed subgrade and maintain in a satisfactory condition until the surface course is placed.

3.3 GRADE CONTROL

During construction, maintain the lines and grades including crown and cross slope indicated for the aggregate surface course by means of line and grade stakes placed by the Contractor in accordance with the SPECIAL CONTRACT REQUIREMENTS.

3.4 MIXING AND PLACING MATERIALS

Mix and place the materials to obtain uniformity of the material and a uniform optimum water content for compaction. Make adjustments in mixing, placing procedures, or in equipment to obtain the true grades, to minimize segregation and degradation, to obtain the desired water content, and to ensure a satisfactory surface course.

3.5 LAYER THICKNESS

Place the aggregate material on the subgrade in layers of uniform thickness. Compact the completed aggregate surface course to the thickness indicated. No individual layer may be thicker than 150 mm nor be thinner than 75 mm in compacted thickness. Compact the aggregate surface course to a total thickness that is within 13 mm of the thickness indicated. Where the measured thickness is more than 13 mm deficient, correct such areas by scarifying, adding new material of proper gradation, reblading, and recompacting as directed. Where the measured thickness is more than 13 mm thicker than indicated, the course will be considered as conforming to the specified thickness requirements. The average job thickness will be the average of all thickness measurements taken for the job and must be within 6 mm of the thickness indicated. Measure the total thickness of the aggregate surface course at intervals of one measurement for each 500 square meters of surface course. Measure total thickness using 75 mm diameter test holes penetrating the aggregate surface course.

3.6 COMPACTION

Degree of compaction is a percentage of the maximum density obtained by the test procedure presented in ASTM D1557 abbreviated herein as percent laboratory maximum density. Compact each layer of the aggregate surface course with approved compaction equipment, as required in the following paragraphs. Maintain the water content during the compaction procedure at optimum or at the percentage specified by the Contracting Officer. Compact the mixture with mechanical tampers in locations not accessible to rollers. Continue compaction until each layer through the full depth is compacted to at least 100 percent of laboratory maximum density. Remove any materials that are found to be unsatisfactory and replace them with satisfactory material or rework them to produce a satisfactory material.

3.7 EDGES OF AGGREGATE SURFACE COURSE

Place approved material along the edges of the aggregate surface course in such quantity as to compact to the thickness of the course being constructed.

3.8 SMOOTHNESS TEST

Construct each layer so that the surface shows no deviations in excess of 10 mm when tested with a 3 m straightedge applied both parallel with and at right angles to the centerline of the area to be paved. Correct deviations exceeding this amount by removing material, replacing with new material, or reworking existing material and compacting, as directed.

3.9 FIELD QUALITY CONTROL

3.9.1 In-Place Tests

Perform each of the following tests on samples taken from the placed and compacted aggregate surface course. Take samples and test at the rates indicated.

- a. Perform density tests on every lift of material placed and at a frequency of one set of tests for every 250 square meters, or portion thereof, of completed area.
- b. Perform sieve analysis on every lift of material placed and at a frequency of one sieve analysis for every 500 square meters, or portion thereof, of material placed.
- c. Perform liquid limit and plasticity index tests at the same frequency as the sieve analysis.
- d. Measure the thickness of the aggregate surface course at intervals providing at least one measurement for each 500 square meters of base course or part thereof. Measure the thickness using test holes, at least 75 mm in diameter through the aggregate surface course.

3.9.2 Approval of Material

Final approval of the materials will be based on tests for gradation, liquid limit, and plasticity index performed on samples taken from the completed and full compacted aggregate surface course.

3.10 MAINTENANCE

Maintain the aggregate surface course in a condition that will meet all specification requirements until accepted.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 16 19

CONCRETE CURBS AND SIDEWALKS

05/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 EQUIPMENT, TOOLS, AND MACHINES
 - 1.3.1 General Requirements
- 1.4 ENVIRONMENTAL REQUIREMENTS
 - 1.4.1 Placing During Cold Weather
 - 1.4.2 Placing During Warm Weather

PART 2 PRODUCTS

- 2.1 CONCRETE
 - 2.1.1 Air Content
 - 2.1.2 Slump
 - 2.1.3 Precast Concrete Curb
- 2.2 CONCRETE CURING MATERIALS
 - 2.2.1 Impervious Sheet Materials
 - 2.2.2 Burlap
 - 2.2.3 White Pigmented Membrane-Forming Curing Compound
- 2.3 CONCRETE PROTECTION MATERIALS
- 2.4 JOINT FILLER STRIPS
 - 2.4.1 Contraction Joint Filler for Curb
 - 2.4.2 Expansion Joint Filler, Premolded
- 2.5 JOINT SEALANTS
- 2.6 FORM WORK
 - 2.6.1 Wood Forms
 - 2.6.2 Steel Forms
 - 2.6.3 Sidewalk Forms
- 2.7 DETECTABLE WARNING SYSTEM

PART 3 EXECUTION

- 3.1 SUBGRADE PREPARATION
 - 3.1.1 Sidewalk Subgrade
 - 3.1.2 Curb Subgrade
 - 3.1.3 Maintenance of Subgrade and Aggregate Base
- 3.2 FORM SETTING
 - 3.2.1 Sidewalks
- 3.3 SIDEWALK CONCRETE PLACEMENT AND FINISHING
 - 3.3.1 Formed Sidewalks
 - 3.3.2 Concrete Finishing
 - 3.3.3 Edge and Joint Finishing
 - 3.3.4 Surface and Thickness Tolerances
- 3.4 CURB CONCRETE PLACEMENT AND FINISHING
 - 3.4.1 Concrete Finishing

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.4.2 Joint Finishing
- 3.4.3 Surface and Thickness Tolerances
- 3.4.4 Installation of Precast Concrete Curbs
- 3.5 SIDEWALK JOINTS
 - 3.5.1 Sidewalk Contraction Joints
 - 3.5.2 Sidewalk Expansion Joints
- 3.6 CURING AND PROTECTION
 - 3.6.1 General Requirements
 - 3.6.1.1 Mat Method
 - 3.6.1.2 Impervious Sheeting Method
 - 3.6.1.3 Membrane Curing Method
 - 3.6.2 Backfilling
 - 3.6.3 Protection
 - 3.6.4 Protective Coating
 - 3.6.4.1 Application
 - 3.6.4.2 Precautions
- 3.7 FIELD QUALITY CONTROL
 - 3.7.1 General Requirements
 - 3.7.2 Concrete Testing
 - 3.7.2.1 Strength Testing
 - 3.7.2.2 Air Content
 - 3.7.2.3 Slump Test
 - 3.7.3 Thickness Evaluation
 - 3.7.4 Surface Evaluation
- 3.8 SURFACE DEFICIENCIES AND CORRECTIONS
 - 3.8.1 Thickness Deficiency
 - 3.8.2 High Areas
 - 3.8.3 Appearance
- 3.9 DETECTABLE WARNING SYSTEM

-- End of Section Table of Contents --

SECTION 32 16 19

CONCRETE CURBS AND SIDEWALKS
05/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 41107	(1961) Prefabricated Products. Elastics and of Low Transversal Dilation for the Filling of Expansion Joints in Concrete Paving
UNE-EN 12350-1	(2020) Testing fresh concrete - Part 1: Sampling and common apparatus
UNE-EN 12350-2	(2020) Testing fresh concrete - Part 2: Slump-test
UNE-EN 12350-6	(2020) Testing fresh concrete - Part 6: Density
UNE-EN 12350-7	(2020) Testing fresh concrete - Part 7: Air content - Pressure methods.
UNE-EN 12390-2	(2020) Testing hardened concrete. Part 2: Making and curing specimens for strength tests.
UNE-EN 13163	(2013+A2:2017) Thermal insulation products for buildings - Factory made expanded polystyrene (EPS) products - Specification

MINISTERIO DE FOMENTO (MF)

FOM/2523/2014	(2014) Orden de 12 de diciembre, por la que se actualizan determinados artículos del pliego de prescripciones técnicas generales para obras de carreteras y puentes, relativos a materiales básicos, a firmes y pavimentos, y a señalización, balizamiento y sistemas de contención de vehículos.
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AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO M 182	(2005; R 2017) Standard Specification for Burlap Cloth Made from Jute or Kenaf and Cotton Mats
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ASTM INTERNATIONAL (ASTM)

ASTM C173/C173M	(2016) Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method
ASTM C231/C231M	(2022) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM D5893/D5893M	(2016) Standard Specification for Cold Applied, Single Component, Chemically Curing Silicone Joint Sealant for Portland Cement Concrete Pavements

INTERNATIONAL CODE COUNCIL (ICC)

ICC A117.1	(2017) Standard And Commentary Accessible and Usable Buildings and Facilities
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Concrete

SD-04 Samples

Concrete Curbs

SD-06 Test Reports

Field Quality Control

1.3 EQUIPMENT, TOOLS, AND MACHINES

1.3.1 General Requirements

Plant, equipment, machines, and tools used in the work will be subject to approval and must be maintained in a satisfactory working condition at all times. Use equipment capable of producing the required product, meeting grade controls, thickness control and smoothness requirements as specified. Discontinue using equipment that produces unsatisfactory results. Allow the Contracting Officer access at all times to the plant and equipment to ensure proper operation and compliance with specifications.

1.4 ENVIRONMENTAL REQUIREMENTS

1.4.1 Placing During Cold Weather

Do not place concrete when the air temperature reaches 5 degrees C and is falling, or is already below that point. Placement may begin when the air temperature reaches 2 degrees C and is rising, or is already above 5 degrees C. Prepare and protect the underlying material so that it is entirely free of deleterious materials. Provide covering or other means as needed to maintain the concrete at a temperature of at least 10 degrees C for not less than 72 hours after placing, and at a temperature above freezing for the remainder of the curing period.

1.4.2 Placing During Warm Weather

The temperature of the concrete as placed must not exceed 30 degrees C except where an approved retarder is used. Cool the mixing water and aggregates as necessary to maintain a satisfactory placing temperature. The placing temperature must not exceed 35 degrees C at any time.

PART 2 PRODUCTS

2.1 CONCRETE

Provide concrete conforming to the applicable requirements of Section 03 30 00 CAST-IN-PLACE CONCRETE except as otherwise specified. Size of aggregate must not exceed 37.5 mm. Concrete for sidewalk and precast curb must have a minimum compressive strength of 25 MPA at 28 days. Concrete for the concrete foundation of the precast curb must have a minimum compressive strength of 20 MPa at 28 days. Submit copies of certified delivery tickets for all concrete used in the construction.

2.1.1 Air Content

Use concrete mixtures that have an air content by volume of concrete of 5 to 7 percent, based on measurements made immediately after discharge from the mixer.

2.1.2 Slump

Use concrete with a slump of 75 mm plus or minus 25 mm for hand placed concrete as determined in accordance with UNE-EN 12350-2.

2.1.3 Precast Concrete Curb

Provide precast concrete curbs as detailed in the drawings from an approved manufacturer. Hydraulically press precast sections in approved molds under conditions of controlled temperatures and humidity. Water or steam cure sections until the concrete attains 100 percent of the specified strength. Provide precast elements uniform in color, free from cracks, flaws or other defects, have a clean finish with smooth surfaces and have well defined arises. Provide precast elements free of segregation, honeycombing or broken corners will not be and without remedial measures. Sample and test precast concrete elements in accordance with paragraph CONCRETE TESTING. Provide concrete materials in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE except otherwise specified. Provide precast concrete curbs comprised of homogenous elements 60 cm or 90 cm long where laid in straight lines, but may be reduced to 20 cm long to be laid on curves.

2.2 CONCRETE CURING MATERIALS

2.2.1 Impervious Sheet Materials

Use impervious sheet materials, type optional, except that polyethylene film, if used, must be white opaque.

2.2.2 Burlap

Use burlap conforming to [AASHTO M 182](#).

2.2.3 White Pigmented Membrane-Forming Curing Compound

Use white pigmented membrane-forming curing compound conforming to [FOM/2523/2014](#), Art. 550.2.9.

2.3 CONCRETE PROTECTION MATERIALS

Use concrete protection materials consisting of a linseed oil mixture of equal parts, by volume, of linseed oil and either mineral spirits, naphtha, or turpentine. At the option of the Contractor, commercially prepared linseed oil mixtures, formulated specifically for application to concrete to provide protection against the action of deicing chemicals may be used, except that emulsified mixtures are not acceptable.

2.4 JOINT FILLER STRIPS

2.4.1 Contraction Joint Filler for Curb

Use hard-pressed fiberboard contraction joint filler for curb.

2.4.2 Expansion Joint Filler, Premolded

Unless otherwise indicated, use [13 mm](#) thick premolded expansion joint filler conforming to [UNE 41107](#) and [UNE-EN 13163](#).

2.5 JOINT SEALANTS

Use cold-applied joint sealant conforming to [ASTM C920](#) or [ASTM D5893/D5893M](#).

2.6 FORM WORK

Design and construct form work to ensure that the finished concrete will conform accurately to the indicated dimensions, lines, and elevations, and within the tolerances specified. Use wood or steel forms that are straight and of sufficient strength to resist springing during depositing and consolidating concrete.

2.6.1 Wood Forms

Use forms that are surfaced plank, [50 mm](#) nominal thickness, straight and free from warp, twist, loose knots, splits or other defects. Use forms with a nominal length of [3 m](#). Radius bends may be formed with [19 mm](#) boards, laminated to the required thickness.

2.6.2 Steel Forms

Use channel-formed sections with a flat top surface and welded braces at

each end and at not less than two intermediate points. Use forms with interlocking and self-aligning ends. Provide flexible forms for radius forming, corner forms, form spreaders, and fillers as needed. Use forms with a nominal length of 3 m and that have a minimum of 3 welded stake pockets per form. Use stake pins consisting of solid steel rods with chamfered heads and pointed tips designed for use with steel forms.

2.6.3 Sidewalk Forms

Use sidewalk forms that are of a height equal to the full depth of the finished sidewalk.

2.7 DETECTABLE WARNING SYSTEM

Detectable Warning Systems shown on the Contract plans are to meet requirements of ICC A117.1 - Section 705.

PART 3 EXECUTION

3.1 SUBGRADE PREPARATION

Construct subgrade and aggregate base to the specified grade and cross section prior to concrete placement.

3.1.1 Sidewalk Subgrade

Place and compact the subgrade and aggregate base in accordance with Section 31 00 00 EARTHWORK and Section 32 11 23 AGGREGATE BASE COURSE. Test the subgrade and aggregate base for grade and cross section with a template extending the full width of the sidewalk and supported between side forms.

3.1.2 Curb Subgrade

Place and compact the subgrade and aggregate base in accordance with Section 32 11 23 AGGREGATE BASE COURSE and Section 31 00 00 EARTHWORK and Section 32 11 23 AGGREGATE BASE COURSE. Test the subgrade and aggregate base for grade and cross section by means of a template extending the full width of the curb. Use subgrade materials equal in bearing quality to the subgrade under the adjacent pavement.

3.1.3 Maintenance of Subgrade and Aggregate Base

Maintain subgrade and aggregate base in a smooth, compacted condition in conformity with the required section and established grade until the concrete is placed. Prepare and protect the aggregate base so that it is free from frost when the concrete is deposited.

3.2 FORM SETTING

Set forms to the indicated alignment, grade and dimensions. Hold forms rigidly in place by a minimum of 3 stakes per form placed at intervals not to exceed 1.2 m. Use additional stakes and braces at corners, deep sections, and radius bends, as required. Use clamps, spreaders, and braces where required to ensure rigidity in the forms. Remove forms in a manner that will not injure the concrete. Do not use bars or heavy tools against the concrete when removing the forms. Promptly and satisfactorily repair concrete found to be defective after form removal. Clean forms and coat with form oil each time before concrete is placed. Wood forms may,

instead, be thoroughly wetted with water before concrete is placed, except that with probable freezing temperatures, oiling is mandatory.

3.2.1 Sidewalks

Set forms for sidewalks with the upper edge true to line and grade with an allowable tolerance of 3 mm in any 3 m long section. After forms are set, grade and alignment must be checked with a 3 m straightedge. Sidewalks must have a transverse slope of maximum 20 mm per meter. Unless otherwise indicated, construct sidewalks that are located adjacent to curbs with the low side adjacent to the curb. Do not remove side forms less than 12 hours after finishing has been completed.

3.3 SIDEWALK CONCRETE PLACEMENT AND FINISHING

3.3.1 Formed Sidewalks

Place concrete in the forms in one layer. When consolidated and finished, the sidewalks must be of the thickness indicated. Use a strike-off guided by side forms after concrete has been placed in the forms to bring the surface to proper section to be compacted. Consolidate concrete by tamping and spading or with an approved vibrator. Finish the surface to grade with a strike off.

3.3.2 Concrete Finishing

After straightedging, when most of the water sheen has disappeared, and just before the concrete hardens, finish the surface with a wood or magnesium float or darby to a smooth and uniformly fine granular or sandy texture free of waves, irregularities, or tool marks. Produce a scored surface by brooming with a fiber-bristle brush in a direction transverse to that of the traffic, followed by edging.

3.3.3 Edge and Joint Finishing

Finish all slab edges, including those at formed joints, with an edger having a radius of 3 mm. Edge transverse joints before brooming. Eliminate the flat surface left by the surface face of the edger with brooming. Clean and solidly fill corners and edges which have crumbled and areas which lack sufficient mortar for proper finishing with a properly proportioned mortar mixture and then finish.

3.3.4 Surface and Thickness Tolerances

Finished surfaces must not vary more than 8 mm from the testing edge of a 3 m straightedge. Permissible deficiency in section thickness will be up to 6 mm.

3.4 CURB CONCRETE PLACEMENT AND FINISHING

3.4.1 Concrete Finishing

Float and finish exposed surfaces with a smooth wood float until true to grade and section and uniform in texture. Brush floated surfaces with a fine-hair brush using longitudinal strokes. Round the edges of the top of the curb with an edging tool to a radius of 13 mm. Immediately after removing the front curb form, rub the face of the curb with a wood or concrete rubbing block and water until blemishes, form marks, and tool marks have been removed. Brush the front curb surface, while still wet,

in the same manner as the curb top.

3.4.2 Joint Finishing

Finish curb edges at formed joints as indicated.

3.4.3 Surface and Thickness Tolerances

Finished surfaces must not vary more than 6 mm from the testing edge of a 3 m straightedge. Permissible deficiency in section thickness will be up to 6 mm.

3.4.4 Installation of Precast Concrete Curbs

Place precast concrete curbs where indicated on the drawings, or as directed by the Contracting Officer, and prior to constructing pavement on roads or parking areas. Do not place precast curbs until after the subbase course of the pavement has been placed, compacted and approved by the Contracting Officer's Representation unless otherwise specifically directed in writing. Submit samples of the various types of concrete curbs for approval prior to commencing the work. Place precast concrete curb sections on a setting bed of mortar consisting of one part cement to two parts sand with constant width joints not exceeding 0.5 cm. Place the just damp mortar by ramming it in place with a hammer and cold chisel or piece of wood when required. Do not use mortar after it has partially set. Do not re-temper or use of any material from any previous mortar batch to mix a new batch of mortar. Prepared mortar only in a sufficient amount to be used within a short period of time and of such a quantity that a partial set does not occur before all the batch is used in the work.

3.5 SIDEWALK JOINTS

Construct sidewalk joints to divide the surface into rectangular areas. Space transverse contraction joints at a distance equal to the sidewalk width or 1.5 m on centers, whichever is less, and continuous across the slab. Construct longitudinal contraction joints along the centerline of all sidewalks 3 m or more in width. Construct transverse expansion joints at sidewalk returns and opposite expansion joints in adjoining curbs. Where the sidewalk is not in contact with the curb, install transverse expansion joints as indicated. Form expansion joints around structures and features which project through or into the sidewalk pavement, using joint filler of the type, thickness, and width indicated. Expansion joints are not required between sidewalks and curb that abut the sidewalk longitudinally.

3.5.1 Sidewalk Contraction Joints

Form contraction joints in the fresh concrete by cutting a groove in the top portion of the slab to a depth of at least one-fourth of the sidewalk slab thickness. Unless otherwise approved or indicated, either use a jointer to cut the groove or saw a groove in the hardened concrete with a power-driven saw. Construct sawed joints by sawing a groove in the concrete with a 3 mm blade. Provide an ample supply of saw blades on the jobsite before concrete placement is started. Provide at least one standby sawing unit in good working order at the jobsite at all times during the sawing operations.

3.5.2 Sidewalk Expansion Joints

Form expansion joints using 13 mm joint filler strips. Joint filler in expansion joints surrounding structures and features within the sidewalk may consist of preformed filler material conforming to UNE 41107 and UNE-EN 13163, or building paper. Hold joint filler in place with steel pins or other devices to prevent warping of the filler during floating and finishing. Immediately after finishing operations are completed, round joint edges using an edging tool having a radius of 3 mm. Remove any concrete over the joint filler. At the end of the curing period, clean the top of expansion joints and fill with cold-applied joint sealant. Use joint sealant that is gray or stone in color. Thoroughly clean the joint opening before the sealing material is placed. Do not spill sealing material on exposed surfaces of the concrete. Apply joint sealing material only when the concrete at the joint is surface dry and atmospheric and concrete temperatures are above 10 degrees C. Immediately remove any excess material on exposed surfaces of the concrete and clean the concrete surfaces.

3.6 CURING AND PROTECTION

3.6.1 General Requirements

Protect concrete against loss of moisture and rapid temperature changes for at least 7 days from the beginning of the curing operation. Protect unhardened concrete from rain and flowing water. All equipment needed for adequate curing and protection of the concrete must be on hand and ready for use before actual concrete placement begins. Protect concrete as necessary to prevent cracking of the pavement due to temperature changes during the curing period.

3.6.1.1 Mat Method

Cover the entire exposed surface with two or more layers of burlap. Overlap mats at least 150 mm. Thoroughly wet the mat with water prior to placing on concrete surface and keep the mat continuously in a saturated condition and in intimate contact with concrete for not less than 7 days.

3.6.1.2 Impervious Sheeting Method

Wet the entire exposed surface with a fine spray of water and then cover with impervious sheeting material. Lay sheets directly on the concrete surface with the light-colored side up and overlapped 300 mm when a continuous sheet is not used. Use sheeting that is not less than 450 mm wider than the concrete surface to be cured. Secure sheeting using heavy wood planks or a bank of moist earth placed along edges and laps in the sheets. Satisfactorily repair or replace sheets that are torn or otherwise damaged during curing. Sheeting must remain on the concrete surface to be cured for not less than 7 days.

3.6.1.3 Membrane Curing Method

Apply a uniform coating of white-pigmented membrane-curing compound to the entire exposed surface of the concrete as soon after finishing as the free water has disappeared from the finished surface. Coat formed surfaces immediately after the forms are removed and in no case longer than 1 hour after the removal of forms. Do not allow concrete surface to dry before application of the membrane. If drying has occurred, moisten the surface of the concrete with a fine spray of water and apply the curing compound

as soon as the free water disappears. Apply curing compound in two coats by hand-operated pressure sprayers at a coverage of approximately 5 square meters/L for the total of both coats. Apply the second coat in a direction approximately at right angles to the direction of application of the first coat. The compound must form a uniform, continuous, coherent film that will not check, crack, or peel and must be free from pinholes or other imperfections. If pinholes, abrasion, or other discontinuities exist, apply an additional coat to the affected areas within 30 minutes. Respray concrete surfaces that are subjected to heavy rainfall within 3 hours after the curing compound has been applied by the method and at the coverage specified above. Respray areas where the curing compound is damaged by subsequent construction operations within the curing period. Take precautions necessary to ensure that the concrete is properly cured at sawed joints, and that no curing compound enters the joints. Tightly seal the top of the joint opening and the joint groove at exposed edges before the concrete in the region of the joint is resprayed with curing compound. Use a method used for sealing the joint groove that prevents loss of moisture from the joint during the entire specified curing period. Provide approved standby facilities for curing concrete pavement at a location accessible to the jobsite for use in the event of mechanical failure of the spraying equipment or other conditions that might prevent correct application of the membrane-curing compound at the proper time. Adequately protect concrete surfaces to which membrane-curing compounds have been applied during the entire curing period from pedestrian and vehicular traffic, except as required for joint-sawing operations and surface tests, and from other possible damage to the continuity of the membrane.

3.6.2 Backfilling

After curing, remove debris and backfill, grade, and compact the area adjoining the concrete to conform to the surrounding area in accordance with lines and grades indicated.

3.6.3 Protection

Protect completed concrete from damage until accepted. Repair damaged concrete and clean concrete discolored during construction. Remove and reconstruct concrete that is damaged for the entire length between regularly scheduled joints. Refinishing the damaged portion will not be acceptable. Dispose of removed material as directed.

3.6.4 Protective Coating

Apply a protective coating of linseed oil mixture to the exposed-to-view concrete surface after the curing period, if concrete will be exposed to de-icing chemicals within 6 weeks after placement. Moist cure concrete to receive a protective coating.

3.6.4.1 Application

Complete curing and backfilling operation prior to applying two coats of protective coating. Concrete must be surface dry and clean before each application. Spray apply at a rate of not more than 11 square meters/L for first application and not more than 15.5 square meters/L for second application, except that the number of applications and coverage for each application for commercially prepared mixture must be in accordance with the manufacturer's instructions. Protect coated surfaces from vehicular and pedestrian traffic until dry.

3.6.4.2 Precautions

Do not heat protective coating by direct application of flame or electrical heaters and protect the coating from exposure to open flame, sparks, and fire adjacent to open containers or applicators. Do not apply material at ambient or material temperatures lower than 10 degrees C.

3.7 FIELD QUALITY CONTROL

Submit copies of all test reports within 24 hours of completion of the test.

3.7.1 General Requirements

Perform the inspection and tests described and meet the specified requirements for inspection details and frequency of testing. Based upon the results of these inspections and tests, take the action and submit reports as required below, and additional tests to ensure that the requirements of these specifications are met.

3.7.2 Concrete Testing

3.7.2.1 Strength Testing

Take concrete samples in accordance with UNE-EN 12350-1 not less than once a day nor less than once for every 190 cubic meters of concrete placed. Mold cylinders in accordance with UNE-EN 12390-2 for strength testing by an approved laboratory. Each strength test result must be the average of 2 test cylinders from the same concrete sample tested at 28 days, unless otherwise specified or approved. Concrete specified on the basis of compressive strength will be considered satisfactory if the averages of all sets of three consecutive strength test results equal or exceed the specified strength, and no individual strength test result falls below the specified strength by more than 4 MPa.

3.7.2.2 Air Content

Determine air content in accordance with ASTM C173/C173M or ASTM C231/C231M. Use ASTM C231/C231M with concretes and mortars made with relatively dense natural aggregates. Alternatively, determine air content in accordance with UNE-EN 12350-6 and UNE-EN 12350-7. Make two tests for air content on randomly selected batches of each class of concrete placed during each shift. Make additional tests when excessive variation in concrete workability is reported by the placing foreman or the Government inspector. Notify the placing foreman if results are out of tolerance. The placing foreman must take appropriate action to have the air content corrected at the plant. Additional tests for air content will be performed on each truckload of material until such time as the air content is within the tolerance specified.

3.7.2.3 Slump Test

Perform two slump tests on randomly selected batches of each class of concrete for every 190 cubic meters, or fraction thereof, of concrete placed during each shift. Perform additional tests when excessive variation in the workability of the concrete is noted or when excessive crumbling or slumping is noted along the edges of slip-formed concrete.

3.7.3 Thickness Evaluation

Determine the anticipated thickness of the concrete prior to placement by passing a template through the formed section or by measuring the depth of opening of the extrusion template of the curb forming machine. The thickness will be determined by measuring each edge of the completed slab.

3.7.4 Surface Evaluation

Provide finished surfaces for each category of the completed work that are uniform in color and free of blemishes and form or tool marks.

3.8 SURFACE DEFICIENCIES AND CORRECTIONS

3.8.1 Thickness Deficiency

When measurements indicate that the completed concrete section is deficient in thickness by more than 6 mm the deficient section will be removed, between regularly scheduled joints, and replaced.

3.8.2 High Areas

In areas not meeting surface smoothness and plan grade requirements, reduce high areas either by rubbing the freshly finished concrete with carborundum brick and water when the concrete is less than 36 hours old or by grinding the hardened concrete with an approved surface grinding machine after the concrete is 36 hours old or more. The area corrected by grinding the surface of the hardened concrete must not exceed 5 percent of the area of any integral slab, and the depth of grinding must not exceed 6 mm. Remove and replace pavement areas requiring grade or surface smoothness corrections in excess of the limits specified.

3.8.3 Appearance

Exposed surfaces of the finished work will be inspected by the Contracting Officer and deficiencies in appearance will be identified. Remove and replace areas which exhibit excessive cracking, discoloration, form marks, or tool marks or which are otherwise inconsistent with the overall appearances of the work.

3.9 DETECTABLE WARNING SYSTEM

Install Detectable Warning Systems required by Contract plans in accordance with ICC A117.1, Section 705, and by manufacturers' installation instructions.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 17 23

PAVEMENT MARKINGS

08/16, CHG 5: 11/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY ASSURANCE
 - 1.3.1 Regulatory Requirements
 - 1.3.2 Qualifications
- 1.4 DELIVERY AND STORAGE
- 1.5 PROJECT/SITE CONDITIONS
 - 1.5.1 Environmental Requirements
 - 1.5.1.1 Weather Limitations for Application
 - 1.5.1.2 Weather Limitations for Removal of Pavement Markings on Roads and Automotive Parking Areas
 - 1.5.2 Traffic Controls
 - 1.5.3 Lighting

PART 2 PRODUCTS

- 2.1 EQUIPMENT
 - 2.1.1 Surface Preparation and Paint Removal
 - 2.1.1.1 Surface Preparation Equipment for Roads and Automotive Parking Areas
 - 2.1.1.1.1 Waterblasting Equipment
 - 2.1.1.1.2 Grinding or Scarifying Equipment
 - 2.1.2 Application Equipment
 - 2.1.2.1 Paint Application Equipment
 - 2.1.2.1.1 Hand-Operated, Push-Type Machines
 - 2.1.2.1.2 Self-Propelled or Mobile-Drawn Spraying Machines
 - 2.1.2.1.2.1 Road Marking
 - 2.1.2.1.2.2 Hand Application
 - 2.1.2.2 Reflective Media Dispenser
- 2.2 MATERIALS
 - 2.2.1 Waterborne Paint
 - 2.2.2 Epoxy Paint
 - 2.2.2.1 Formulation
 - 2.2.2.2 Composition
 - 2.2.2.3 Epoxide Value
 - 2.2.2.4 Total Amine Value
 - 2.2.2.5 Toxicity
 - 2.2.2.6 Daylight Directional Reflectance
 - 2.2.2.7 Laboratory Drying Time
 - 2.2.2.8 Curing
 - 2.2.2.9 Adhesion to Concrete
 - 2.2.2.10 Hardness
 - 2.2.2.11 Abrasion Resistance
 - 2.2.2.12 Tensile Strength

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.2.2.13 Compressive Strength
- 2.2.3 Reflective Media
- 2.2.3.1 Reflective Media for Roads

PART 3 EXECUTION

- 3.1 EXAMINATION
 - 3.1.1 Testing for Moisture
 - 3.1.2 Surface Preparation Demonstration
 - 3.1.3 Test Stripe Demonstration
 - 3.1.4 Application Rate Demonstration
 - 3.1.5 Level of Performance Demonstration
- 3.2 EXTERIOR SURFACE PREPARATION
 - 3.2.1 Early Painting of Asphalt Pavements
- 3.3 APPLICATION
 - 3.3.1 Paint
 - 3.3.1.1 Waterborne Paint
 - 3.3.1.1.1 Roads
 - 3.3.1.2 Epoxy Paint
 - 3.3.2 Preformed Tape
 - 3.3.3 Cleanup and Waste Disposal
- 3.4 FIELD QUALITY CONTROL
 - 3.4.1 Sampling and Testing
 - 3.4.2 Material Inspection
 - 3.4.3 Dimensional Tolerances
 - 3.4.4 Bond Failure Verification
 - 3.4.5 Reflective Media and Coating Application Verification

-- End of Section Table of Contents --

SECTION 32 17 23

PAVEMENT MARKINGS
08/16, CHG 5: 11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 48076	(1992) Paints and varnishes. Determination of apparent viscosity or consistency. Method krebs-stormer.
UNE 48083	(1992) Paints And Varnishes. Conservation and Stability In Containers
UNE 135202	(2010) Road signalization equipment. Horizontal signalisation. Paints and cold plastic. Drying time "no pick-up".
UNE 135213	(2018) Road signalization equipment. Horizontal signalization. Paints. Contrast ratio of the dried film.
UNE-EN 1423	(2013) Road marking materials - Drop on materials - Glass beads, antiskid aggregates and mixtures of the two
UNE-EN 1436	(2018) Road marking materials - Road marking performance for road users and test methods
UNE-EN 1871	(2000) Road Marking Materials. Physical Properties.
UNE-EN 12802	(2012) Road marking materials - Laboratory methods for identification.
UNE-EN ISO 11664-2	(2011) Colorimetry - Part 2: CIE standard illuminants (ISO 11664-2:2007)

CENTRO DE ESTUDIOS Y EXPERIMENTACION DE OBRAS PUBLICAS.
LABORATORIO CENTRAL DE ESTRUCTURAS Y MATERIALES (MELC)

MELC 12.30	(1990) Determination of percentage of default glass microspheres in road signaling.
MELC 12.31	(1990) Refraction measurement of glass microspheres for road signaling.

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

MELC 12.32 (1990) Sample taking of glass microspheres
for road signaling.

MINISTERIO DE FOMENTO (MF)

MF PG3 (2014) Pliego de Prescripciones Tecnicas
Generales para Obras de Carreteras y
Puentes

Norma 8.2-IC (1987) Marcas Viales.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO M 247 (2013) Standard Specification for Glass
Beads Used in Pavement Markings

ASTM INTERNATIONAL (ASTM)

ASTM D476 (2015) Dry Pigmentary Titanium Dioxide
Pigments

ASTM D638 (2014) Standard Test Method for Tensile
Properties of Plastics

ASTM D695 (2010) Standard Test Method for
Compressive Properties of Rigid Plastics

ASTM D711 (2010; R 2015) No-Pick-Up Time of Traffic
Paint

ASTM D1652 (2011; E 2012) Standard Test Method for
Epoxy Content of Epoxy Resins

ASTM D2074 (2007; R2013) Standard Test Methods for
Total, Primary, Secondary, and Tertiary
Amine Values of Fatty Amines by
Alternative Indicator Method

ASTM D2240 (2015; E 2017) Standard Test Method for
Rubber Property - Durometer Hardness

ASTM D4060 (2019) Abrasion Resistance of Organic
Coatings by the Taber Abraser

ASTM D7234 (2012) Standard Test Method for Pull-Off
Adhesion Strength of Coatings on Concrete
Using Portable Pull-Off Adhesion Testers

ASTM E1347 (2006; R 2020) Standard Test Method for
Color and Color Difference Measurement by
Tristimulus (Filter) Colorimetry

MASTER PAINTERS INSTITUTE (MPI)

MPI 97 (2012) Traffic Marking Paint, Latex

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Surface Preparation Equipment List; G
Application Equipment List; G
Exterior Surface Preparation
Safety Data Sheets; G
Reflective media for roads; G
Waterborne Paint; G

SD-06 Test Reports

Reflective Media for Roads; G
Waterborne Paint; G

SD-07 Certificates

Qualifications; G
Reflective Media for Roads
Waterborne Paint
Volatile Organic Compound, (VOC); G

SD-08 Manufacturer's Instructions

Waterborne Paint; G

1.3 QUALITY ASSURANCE

1.3.1 Regulatory Requirements

Submit certificate stating that the proposed pavement marking paint meets the [Volatile Organic Compound](#), (VOC) regulations of the local Air Pollution Control District having jurisdiction over the geographical area in which the project is located. Submit [Safety Data Sheets](#) for each product.

1.3.2 Qualifications

Submit documentation certifying that pertinent personnel are qualified for equipment operation and handling of applicable chemicals. The documentation should include experience on five projects of similar size and scope with references for all personnel.

1.4 DELIVERY AND STORAGE

Deliver paint materials, thermoplastic compound materials, and reflective media in original sealed containers that plainly show the designated name, specification number, batch number, color, date of manufacture, manufacturer's directions, and name of manufacturer.

Provide storage facilities at the job site, only in areas approved by the Contracting Officer, for maintaining materials at temperatures recommended by the manufacturer. Make available paint stored at the project site or

segregated at the source for sampling not less than 30 days prior to date of required approval for use to allow sufficient time for testing. Notify the Contracting Officer when paint is available for sampling.

1.5 PROJECT/SITE CONDITIONS

1.5.1 Environmental Requirements

1.5.1.1 Weather Limitations for Application

Apply pavement markings to clean, dry surfaces, and unless otherwise approved, only when the air and pavement surface temperature is at least 3 degrees C above the dew point and the air and pavement temperatures are within the limits recommended by the pavement marking manufacturer. Allow pavement surfaces to dry after water has been used for cleaning or rainfall has occurred prior to striping or marking. Test the pavement surface for moisture before beginning work each day and after cleaning. Do not commence marking until the pavement is sufficiently dry and the pavement condition has been approved by the Contracting Officer. Employ the "plastic wrap method" to test the pavement for moisture as specified in paragraph TESTING FOR MOISTURE.

1.5.1.2 Weather Limitations for Removal of Pavement Markings on Roads and Automotive Parking Areas

Cease operation during thunderstorms, or during rainfall, except for waterblasting and removal of previously applied chemicals. Cease waterblasting where surface water accumulation alters the effectiveness of material removal.

1.5.2 Traffic Controls

Place warning signs near the beginning of the worksite and well ahead of the worksite for alerting approaching traffic from both directions. Place small markers along newly painted lines or freshly placed raised markers to control traffic and prevent damage to newly painted surfaces or displacement of raised pavement markers. Mark painting equipment with large warning signs indicating slow-moving painting equipment in operation.

When traffic must be rerouted or controlled to accomplish the work, provide necessary warning signs, flag persons, and related equipment for the safe passage of vehicles.

1.5.3 Lighting

When night operations are necessary, provide all necessary lighting and equipment. The Government reserves the right to accept or reject night work on the day following night activities by the Contractor.

PART 2 PRODUCTS

2.1 EQUIPMENT

2.1.1 Surface Preparation and Paint Removal

2.1.1.1 Surface Preparation Equipment for Roads and Automotive Parking Areas

Submit a [surface preparation equipment list](#) by serial number, type, model,

and manufacturer. Include descriptive data indicating area of coverage per pass, pressure adjustment range, tank and flow capacities, and safety precautions required for the equipment operation. Mobile equipment must allow for removal of markings without damaging the pavement surface or joint sealant. Maintain machines, tools, and equipment used in the performance of the work in satisfactory operating condition.

2.1.1.1.1 Waterblasting Equipment

Use mobile waterblasting equipment capable of producing a pressurized stream of water that effectively removes paint from the pavement surface without significantly damaging the pavement. Provide equipment, tools, and machinery which are safe and in good working order at all times.

2.1.1.1.2 Grinding or Scarifying Equipment

Use equipment capable of removing surface contaminants, paint build-up, or extraneous markings from the pavement surface without leaving any residue. Clean the surface by hydro blast to remove surface contaminants and ash after a weed torch is used to remove paint.

2.1.2 Application Equipment

Submit [application equipment list](#) appropriate for the material(s) to be used. Include manufacturer's descriptive data and certification for the planned use that indicates area of coverage per pass, pressure adjustment range, tank and flow capacities, and all safety precautions required for operating and maintaining the equipment. Provide and maintain machines, tools, and equipment used in the performance of the work in satisfactory operating condition, or remove them from the work site. Provide mobile and maneuverable application equipment to the extent that straight lines can be followed and normal curves can be made in a true arc.

2.1.2.1 Paint Application Equipment

2.1.2.1.1 Hand-Operated, Push-Type Machines

Provide hand-operated push-type applicator machine of a type commonly used for application of water based paint or two-component, chemically curing paint, thermoplastic, or preformed tape, to pavement surfaces for small marking projects, such as legends and cross-walks, automotive parking areas, or surface painted signs. Provide applicator machine equipped with the necessary tanks and spraying nozzles capable of applying paint uniformly at coverage specified. Hand operated spray guns may be used in areas where push-type machines cannot be used.

2.1.2.1.2 Self-Propelled or Mobile-Drawn Spraying Machines

Provide self-propelled or mobile-drawn spraying machine with suitable arrangements of atomizing nozzles and controls to obtain the specified results. Provide machine having a speed during application capable of applying the stripe widths indicated at the paint coverage rate specified herein and of even uniform thickness with clear-cut edges.

2.1.2.1.2.1 Road Marking

Provide equipment used for marking roads capable of placing the prescribed number of lines at a single pass as solid lines, intermittent lines, or a combination of solid and intermittent lines using a maximum of three

different colors of paint as specified.

2.1.2.1.2.2 Hand Application

Provide spray guns for hand application of paint in areas where the mobile paint applicator cannot be used.

2.1.2.2 Reflective Media Dispenser

Attach the dispenser for applying the reflective media to the paint or thermoplastic dispenser and designed to operate automatically and simultaneously with the applicator through the same control mechanism. The bead applicator must be capable of adjustment and designed to provide uniform flow of reflective media over the full length and width of the stripe at the rate of coverage specified in paragraph APPLICATION.

2.2 MATERIALS

Use waterborne paint or epoxy paint for roads. Use non-reflectorized waterborne paint for automotive parking areas. The maximum allowable VOC content of pavement markings is 150 grams per liter. Color of markings are indicated on the drawings and must conform to Norma 8.2-IC and Art. 700 of MF PG3 for roads and automotive parking areas. Provide materials conforming to the requirements specified herein.

2.2.1 Waterborne Paint

MPI 97 and the following. UNE-EN 1436 and UNE-EN 1871, 100% acrylic waterborne paint, color as indicated. Paints with content of mercury, lead, hexavalent chromium, toluene, chlorinated solvents, chlorine derivatives or ethylene-based glycol shall not be authorized. The paints shall have the following characteristics:

- Consistency: Krebs Units. UNE 48076. Min. 84 UK +/- 10 UK.
- Daylight diffuse illuminance factor (Beta). UNE-EN ISO 11664-2. Beta ≥ 0.85 minimum for white paint, drying time UNE 135202 ≤ 30 min.
- Contrast value with adjacent pavement: UNE 135213, $rc \geq 0.90$.
- Accelerated package stability: UNE 48083. Variation in consistency: less than 5 Krebs Units.
- Minimum content of solids: UNE-EN 12802. Minimum value: 75%. Maximum deviation versus declared value: +/- 2.5%.
- Accelerated aging: UNE-EN 1871. Maximum deviation of illuminance: Beta ≤ 0.05 , Class BR2.
- Bleeding of paint for markings on bituminous pavement. UNE-EN 1871. Min. value 0.95.
- Alkali resistance for marking on concrete pavement. UNE-EN 1871. Without superficial damages.

2.2.2 Epoxy Paint

2.2.2.1 Formulation

Epoxy pavement marking material will be a two component, 100 percent solids, material formulated to provide simple volumetric mixing ratio of two volumes of component A and one volume of component B unless otherwise recommended by the manufacturer.

2.2.2.2 Composition

The component A of both white and yellow must be within the following limits:

TABLE I		
	White	Yellow
Pigments	Minimum 18 percent by weight Titanium Dioxide (ASTM D476, Type II)	21-27 percent by weight
Epoxy Resin	75-82 percent	73-79 percent

The epoxy resin must be free of lead, cadmium, mercury, hexavalent chromium and other toxic heavy metals as defined by the Environmental Protection Agency. Submit a manufacturer's certification of compliance with this requirement.

2.2.2.3 Epoxide Value

Determine epoxide epoxy number of the epoxy resin in accordance with ASTM D1652 for white and yellow component A on pigment free basis. The epoxide number must be within plus or minus 50 of the published manufacturer's standard.

2.2.2.4 Total Amine Value

Determine the amine number on the curing agent (component B) in accordance with ASTM D2074. The amine number must be within plus or minus 50 of the published manufacturer's standard.

2.2.2.5 Toxicity

Upon heating to application temperature, the material must not produce fumes which are toxic or injurious to persons or property.

2.2.2.6 Daylight Directional Reflectance

Directional reflectance of white and yellow paint (without glass beads) in accordance with ASTM E1347: White 84 percent Yellow 55 percent.

2.2.2.7 Laboratory Drying Time

The epoxy pavement marking material must have a maximum no-pick-up time of 30 minutes when tested in accordance with ASTM D711.

2.2.2.8 Curing

The epoxy material must be capable of fully curing under a constant surface temperature of 7 Degrees C or above.

2.2.2.9 Adhesion to Concrete

The catalyzed epoxy pavement marking material must have a high degree of adhesion to the specified concrete surface (100 percent concrete failure) when tested according to ASTM D7234. The concrete substrate must have a

minimum compressive strength of 27.5 MPa. Condition prepared specimens at a temperature of 23.9 plus or minus 1.1 Degrees C for a minimum of 24 hours and a maximum of 72 hours prior to performance of the test.

2.2.2.10 Hardness

Epoxy pavement marking materials must have a Shore D Hardness between 75 and 100 when tested in accordance with ASTM D2240. Cure the samples at 23.9 plus or minus 1.1 Degrees C for a minimum of 72 hours and a maximum of 96 hours prior to performing the tests.

2.2.2.11 Abrasion Resistance

The wear index for a catalyzed sample must not exceed 82 when tested in accordance with ASTM D4060 using a 1000 gram load, CS-17 wheels and a test duration of 1000 cycles. Run the test on cured samples of material which have been applied at a film thickness of 15 plus or minus 0.5 mils to code S-16 stainless steel plates. Cure the samples at 23.9 plus or minus 1.1 Degrees C for a minimum of 48 hours prior to performing the tests.

2.2.2.12 Tensile Strength

Epoxy pavement marking materials must have a tensile strength of at least 41,370 kPa when tested in accordance with ASTM D638. Cast the Type IV specimens in a suitable mold and pull at the rate of 6 mm per minute using a suitable dynamic testing machine. Cure the samples at 23.9 plus or minus 1.1 Degrees C for a minimum of 12 hours and a maximum of 48 hours prior to performing the tests.

2.2.2.13 Compressive Strength

Catalyzed epoxy pavement marking materials must have a compressive strength of at least 82,700 kPa when tested in accordance with ASTM D695. Condition the cast sample at 23.9 plus or minus 1.1 Degrees C for a minimum of 12 hours and a maximum of 48 hours prior to performing the tests. The rate of compression of these samples must not exceed 6 mm per minute.

2.2.3 Reflective Media

2.2.3.1 Reflective Media for Roads

Post mixed microspheres per UNE-EN 1423 and MELC 12.30, MELC 12.31 and MELC 12.32. The size of beads will be between 125 and 600 microns. The spheres will have a crash resistance of 193 N minimum. The beads will have 2% maximum of spheres with a ratio between maximum diameter and minimum diameter less than 0.9. Additionally, the beads will comply with the following requirements:

- Roundness: MELC 12.30, 80% minimum of true sphere.
- Index of refraction: MELC 12.31, Class A, 1.5 minimum.
- Specific weight: between 2.30 and 2.50.
- Gradation: MELC 12.32.
Sieve UNE:
 - #800: pass 100%
 - #630: pass 95% to 100%
 - #500: pass 90% to 100%
 - #320: pass 30% to 70%
 - #125: pass 0% to 5%

- Chemical resistance: the beads shall not develop haze or dulling when immersed in sulfuric acid for 5 minutes and rinsed with distilled water; when immersed in calcium chloride for 3 hours and rinsed with distilled water; when immersed in distilled water and 0.1 N hydrochloric acid for 2 hours and rinsed with distilled water. The beads will not darken when immersed in a dilution of 50% sodium sulfide for 1 hour and rinsed with distilled water.

PART 3 EXECUTION

3.1 EXAMINATION

3.1.1 Testing for Moisture

Test the pavement surface for moisture before beginning pavement marking after each period of rainfall, fog, high humidity, or cleaning, or when the ambient temperature has fallen below the dew point. Do not commence marking until the pavement is sufficiently dry and the pavement condition has been approved by the Contracting Officer or authorized representative.

Employ the "plastic wrap method" to test the pavement for moisture as follows: Cover the pavement with a 300 mm by 300 mm section of clear plastic wrap and seal the edges with tape. After 15 minutes, examine the plastic wrap for any visible moisture accumulation inside the plastic. Do not begin marking operations until the test can be performed with no visible moisture accumulation inside the plastic wrap. Re-test surfaces when work has been stopped due to rain.

3.1.2 Surface Preparation Demonstration

Prior to surface preparation, demonstrate the proposed procedures and equipment. Prepare areas large enough to determine cleanliness, adhesion of remaining coating and rate of cleaning. Approved demonstration area establishes the standard for the remainder of the work.

3.1.3 Test Stripe Demonstration

Prior to paint application, demonstrate test stripe application within the work area using the proposed materials and equipment. Apply separate test stripes in each of the line widths and configurations required herein using the proposed equipment. Make the test stripes long enough to determine the proper speed and operating pressures for the vehicle(s) and machinery, but not less than 15 m long.

3.1.4 Application Rate Demonstration

During the Test Stripe Demonstration, demonstrate compliance with the application rates specified herein. Document the equipment speed and operating pressures required to meet the specified rates in each configuration of the equipment and provide a copy of the documentation to the Contracting Officer prior to proceeding with the work.

3.1.5 Level of Performance Demonstration

The Contracting Officer will be present at the application demonstrations to observe the results obtained and to validate the operating parameters of the vehicle(s) and equipment. If accepted by the Contracting Officer, the test stripe is the measure of performance required for this project. Do not proceed with the work until the demonstration results are

satisfactory to the Contracting Officer.

3.2 EXTERIOR SURFACE PREPARATION

Allow new pavement surfaces to cure for a period of not less than 30 days before application of marking materials. Thoroughly clean surfaces to be marked before application of the paint. Remove dust, dirt, and other granular surface deposits by sweeping, blowing with compressed air, rinsing with water, or a combination of these methods as required. Remove rubber deposits, existing paint markings, residual curing compounds, and other coatings adhering to the pavement by water blasting.

3.2.1 Early Painting of Asphalt Pavements

For asphalt pavement systems requiring painting application at less than 30 days, apply the paint and beads at half the normal application rate, followed by a second application at the normal rate after 30 days.

3.3 APPLICATION

Apply pavement markings to dry pavements only.

3.3.1 Paint

Apply paint with approved equipment at rate of coverage specified herein. Provide guidelines and templates as necessary to control paint application. Take special precautions in marking numbers, letters, and symbols. Manually paint numbers, letters, and symbols. Sharply outline all edges of markings. The maximum drying time requirements of the paint specifications will be strictly enforced, to prevent undue softening of bitumen, and pickup, displacement, or discoloration by tires of traffic. If there is a deficiency in drying of the markings, painting operations must cease until the cause of the slow drying is determined and corrected.

3.3.1.1 Waterborne Paint

3.3.1.1.1 Roads

Apply paint at a rate of 2.6 plus or minus 0.1 square meter per liter. Apply AASHTO M 247 Type 1 beads at a rate of 834 plus or minus 60 grams of glass spheres per liter.

3.3.1.2 Epoxy Paint

Apply paint evenly to the pavement surface at a wet film thickness of 0.508 mm plus or minus 0.025 mm to cover 2.0 plus or minus 0.1 square meters per liter. Apply glass spheres uniformly to the wet paint on road and street pavement at a rate of 834 plus or minus 60 grams of glass spheres per liter.

3.3.2 Preformed Tape

The pavement surface and ambient air temperature must be a minimum of 15 degrees C and rising. Place the preformed markings in accordance with the manufacturer's written instructions.

3.3.3 Cleanup and Waste Disposal

Keep the worksite clean and free of debris and waste from the removal and application operations. Dispose of debris at approved sites.

3.4 FIELD QUALITY CONTROL

3.4.1 Sampling and Testing

As soon as the paint and thermoplastic materials and reflective media are available for sampling, obtain by random selection from the sealed containers, two quart samples of each batch in the presence of the Contracting Officer. Accomplish adequate mixing prior to sampling to ensure a uniform, representative sample. A batch is defined as that quantity of material processed by the manufacturer at one time and identified by number on the label. Clearly identify samples by designated name, specification number, batch number, project contract number, intended use, and quantity involved.

At the discretion of the Contracting Officer, samples provided may be tested by the Government for verification.

3.4.2 Material Inspection

Examine material at the job site to determine that it is the material referenced in the report of test results or certificate of compliance. Provide test results substantiating conformance to the specified requirements with each certificate of compliance.

3.4.3 Dimensional Tolerances

Apply all markings in the standard dimensions provide in the drawings. New markings may deviate a maximum of 10 percent larger than the standard dimension. The maximum deviation allowed when painting over an old marking is up to 20 percent larger than the standard dimensions.

3.4.4 Bond Failure Verification

Inspect newly applied markings for signs of bond failure based on visual inspection and comparison to results from Test Stripe Demonstration paragraph.

3.4.5 Reflective Media and Coating Application Verification

Use a wet film thickness gauge to measure the application of wet paint. Use a microscope or magnifying glass to evaluate the embedment of glass beads in the paint. Verify the glass bead embedment with approximately 50 percent of the individual bead spheres embedded and 50 percent of the individual bead spheres exposed.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 31 13.53

HIGH-SECURITY FENCES (CHAIN LINK AND ORNAMENTAL) AND GATES

11/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

- 2.1 COMPONENTS
 - 2.1.1 Chain Link Fence Fabric
 - 2.1.1.1 General
 - 2.1.2 Posts
 - 2.1.2.1 Metal Posts for Chain Link Fence
 - 2.1.2.2 Accessories
 - 2.1.3 Chain Link Braces and Rails
 - 2.1.4 Chain Link Gates
 - 2.1.4.1 Gate Assembly
 - 2.1.4.2 Gate Leaves
 - 2.1.4.3 Gate Hardware and Accessories
 - 2.1.5 Padlocks
- 2.2 MATERIALS
 - 2.2.1 Wire
 - 2.2.1.1 Wire Ties
 - 2.2.1.2 Barbed Wire
 - 2.2.1.3 Tension Wire
 - 2.2.2 Concrete

PART 3 EXECUTION

- 3.1 PREPARATION
 - 3.1.1 Line and Grade
 - 3.1.2 Excavation
- 3.2 INSTALLATION
 - 3.2.1 Installation Drawings
 - 3.2.2 Security Fencing
 - 3.2.3 Posts
 - 3.2.3.1 Earth and Bedrock
 - 3.2.4 Fabric
 - 3.2.5 Supporting Arms
 - 3.2.6 Gate Installation
 - 3.2.7 Grounding
- 3.3 CLOSEOUT ACTIVITIES
 - 3.3.1 Cleanup

-- End of Section Table of Contents --

SECTION 32 31 13.53

HIGH-SECURITY FENCES (CHAIN LINK AND ORNAMENTAL) AND GATES
11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

- | | |
|-----------------|--|
| UNE-EN 10216-1 | (2014) Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties |
| UNE-EN 10244-3 | (2001) Steel wire and wire products -Non-ferrous metallic coatings on steel wire - Part 3: Aluminium coatings |
| UNE-EN 10297-1 | (2004) Seamless circular steel tubes for mechanical and general engineering purposes - Technical delivery conditions - Part 1: Non-alloy and alloy steel tubes |
| UNE-EN ISO 1461 | (2010) Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods (ISO1461:1999) |

ASTM INTERNATIONAL (ASTM)

- | | |
|-----------------|--|
| ASTM A121 | (2019) Standard Specification for Metallic-Coated Carbon Steel Barbed Wire |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware |
| ASTM A491 | (2011; R 2017) Standard Specification for Aluminum-Coated Steel Chain-Link Fence Fabric |
| ASTM A780/A780M | (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings |
| ASTM F567 | (2014a; R 2019) Standard Practice for Installation of Chain Link Fence |
| ASTM F626 | (2014; R 2019) Standard Specification for Fence Fittings |

ASTM F883	(2013; R 2022) Standard Performance Specification for Padlocks
ASTM F900	(2011; R 2017) Standard Specification for Industrial and Commercial Swing Gates
ASTM F1043	(2018) Standard Specification for Strength and Protective Coatings on Steel Industrial Fence Framework
ASTM F1083	(2018) Standard Specification for Pipe, Steel, Hot-Dipped Zinc Coated (Galvanized) Welded, for Fence Structures
ASTM F1184	(2016) Standard Specification for Industrial and Commercial Horizontal Slide Gates

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fence Installation Drawings; G

SD-03 Product Data

Fabric

Posts

Post Caps

Chain Link Braces

Line Posts

Rails

Tension Wire

Barbed Wire

Barbed Wire Supporting Arms

Latches

Hinges

Stops

Keepers

Rollers

Padlocks

Wire Ties

SD-07 Certificates

Chain Link Fence

Fabric

Barbed Wire

Gate Hardware and Accessories

Concrete

SD-10 Operation and Maintenance Data

Operating and maintenance instructions

1.3 DELIVERY, STORAGE, AND HANDLING

Deliver materials to site in an undamaged condition. Store materials elevated off of the ground to protect against oxidation caused by ground contact.

PART 2 PRODUCTS

2.1 COMPONENTS

2.1.1 Chain Link Fence Fabric

2.1.1.1 General

Provide [ASTM A491](#) or [UNE-EN 10244-3](#), aluminum-coated steel wire. Fabricate fence fabric of 9 gauge or 3.7 mm diameter wire woven in 50 mm diamond mesh.

2.1.2 Posts

2.1.2.1 Metal Posts for Chain Link Fence

Provide posts conforming to [ASTM F1083](#) or to [UNE-EN 10297-1](#) (seamless pipe) or [UNE-EN 10216-1](#), zinc-coated. Provide sizes as shown on the drawings. Use line posts and terminal (corner, gate, and pull) posts of the same designation throughout the fence. Provide gate post for the gate type specified subject to the limitation specified in [ASTM F900](#) or [ASTM F1184](#). Fabric height shall be 2.1 m for fuel farm fencing.

2.1.2.2 Accessories

- a. Provide accessories with zinc coating conforming to [ASTM F626](#).
- b. Provide truss rods (with turnbuckles or other means of adjustment) for each terminal post.
- c. Provide [barbed wire supporting arms](#) of the single 45 degree outward angle 3-strand arm type and of the design required for the post furnished. Secure arms by top tension wire.

- d. Furnish [post caps](#) in accordance with manufacturer's standard accessories with coating matching that of fence posts.
- e. Provide 9 gauge or 3.7 mm diameter tie wire for attaching fabric to rails, braces, and posts and match the material and coating of the fence fabric. Provide miscellaneous hardware coatings which conform to [ASTM A153/A153M](#) or [UNE-EN ISO 1461](#) unless modified.

2.1.3 Chain Link Braces and Rails

[ASTM F1083](#) or [UNE-EN 10297-1](#) or [UNE-EN 10216-1](#), zinc-coated, Group IA Regular Strength, steel pipe, size NPS 1-1/4. Use braces and rails that are Group IA Regular Strength, steel pipe, size NPS 1-1/4 or Group II, formed steel sections, size [42 mm](#) and be zinc coated and polyvinyl chloride-coated, minimum thickness, [0.25 mm](#) conforming to the requirements of [ASTM F1043](#). Group II, formed steel sections, size [42 mm](#), conforming to [ASTM F1043](#), may be used as braces and rails if Group II [line posts](#) are furnished. Provide rails and braces with polyvinyl chloride coating, minimum thickness, [0.25 mm](#) conforming to [ASTM F1043](#); color of PVC coating to match that of fabric. The fencing at the fuel farm must have top tension wire to match existing.

2.1.4 Chain Link Gates

2.1.4.1 Gate Assembly

Provide gate assembly conforming to [ASTM F900](#) and/or [ASTM F1184](#) of the type and swing shown. Provide gate frames conforming to strength and coating requirements of [ASTM F1083](#) for Group IA Regular Strength, steel pipe, nominal pipe size (NPS) 1-1/2. Use gate fabric that matches the specified chain link fabric.

2.1.4.2 Gate Leaves

For gate leaves, more than [2.44 m](#) wide, provide either intermediate members and diagonal truss rods or tubular members as necessary to provide rigid construction, free from sag or twist. For gate leaves less than [2.44 m](#) wide, provide truss rods or intermediate braces. Provide intermediate braces on all gate frames with an electro-mechanical lock. Attach fabric to the gate frame by method standard with the manufacturer. Welding is not an acceptable method for attaching fabric to gate frames.

2.1.4.3 Gate Hardware and Accessories

Submit manufacturer's catalog data. Furnish and install [latches](#), [hinges](#), [stops](#), [keepers](#), [rollers](#), and other hardware items as required for the operation of the gate. All items are required to match the material characteristics of the fence system being installed. Provide hinge with stainless steel pin. Arrange latches for padlocking so that the padlock will be accessible from both sides of the gate. Provide stops for holding the gates in the open position. For high security applications, extend each end member of gate frames sufficiently above the top member to carry three strands of barbed wire in horizontal alignment with barbed wire strands on the fence. Coating for steel latches, stops, hinges, keepers, and accessories, must be galvanized.

2.1.5 Padlocks

Provide padlocks conforming to ASTM F883, Type P01, Option A. Size 44 mm. Key all padlocks alike.

2.2 MATERIALS

2.2.1 Wire

2.2.1.1 Wire Ties

Submit samples as specified. Provide wire ties constructed of the same material and finish as the fencing fabric.

2.2.1.2 Barbed Wire

Provide barbed wire conforming to ASTM A121 aluminum-coated, Type A, with 12.5 gauge or 2.5 mm diameter wire with 14 gauge, round, 4-point barbs spaced no more than 125 mm apart.

2.2.1.3 Tension Wire

Provide metallic coated steel marcelled tension wire 3.7 mm diameter. Provide aluminum-coated (aluminized) steel wire with coating that weighs not less than 122 gram per square meter.

2.2.2 Concrete

Section 03 30 00 CAST-IN-PLACE CONCRETE, using 19 mm maximum size aggregate, and having minimum compressive strength of 25 MPa at 28 days. Use grout consisting of one part portland cement to three parts clean, well-graded sand and the minimum amount of water to produce a workable mix.

PART 3 EXECUTION

3.1 PREPARATION

Perform complete installation conforming to ASTM F567.

3.1.1 Line and Grade

Install fence to the lines and grades indicated. Clear the area on either side of the fence line to the extent indicated. Space line posts equidistant at intervals not exceeding 3 m. Set terminal (corner, gate, and pull) posts whenever abrupt changes in vertical and horizontal alignment are encountered. Provide continuous fabric between terminal posts; however, ensure runs between terminal posts do not exceed 152.4 m. Repair any damage to galvanized surfaces, including welding, with paint containing zinc dust in accordance with ASTM A780/A780M or in accordance with UNE-EN ISO 1461.

3.1.2 Excavation

Excavate holes to depths indicated. Clear all post holes of loose material and spread waste material where directed. Eliminate ground surface irregularities along the fence line to the extent necessary to maintain a 50 mm clearance between the bottom of the fabric and finish grade.

3.2 INSTALLATION

3.2.1 Installation Drawings

Submit complete [Fence Installation Drawings](#) for review and approval by the Contracting Officer prior to shipment. Submit drawing details that include, but are not limited to the following information: Fence Installation Drawings, Location of gate, corner, end, and pull posts, Gate Assembly, Turnstiles, and [Gate Hardware and Accessories](#). Install fence system per approved drawings.

3.2.2 Security Fencing

Install new security fencing, remove existing security fencing, and perform related work to provide continuous security for facility. Schedule and fully coordinate work with Contracting Officer.

3.2.3 Posts

3.2.3.1 Earth and Bedrock

- a. Set posts plumb and in alignment. Set posts in concrete to the depth indicated on the drawings.
- b. Set posts in holes not less than the diameter shown on the drawings. Thoroughly consolidate concrete around each post, free of voids and finished to form a dome. Allow concrete to cure for 72 hours prior to attachment of any item to the posts.
- c. Test fence post rigidity by applying a [222.4 newtons](#) force on the post, perpendicular to the fabric, at [1.52 m](#) above ground. Ensure post movement measured at the point where the force is applied is less than or equal to [19 mm](#) from the relaxed position. Test every tenth post for rigidity. When a post fails this test, make further tests on the next four posts on either side of the failed post. Remove, replace, and retest all failed parts at the Contractor's expense.

3.2.4 Fabric

- a. Set fabric height at [2.1 m](#) or as shown.
- b. Install chain link fabric on the side of the post indicated. Attach fabric to terminal posts with stretcher bars and tension bands. Space bands at approximately [381 mm](#) intervals. Install fabric and pull taut to provide a smooth and uniform appearance free from sag, without permanently distorting the fabric diamond or reducing the fabric height. Fasten fabric to line posts at approximately [381 mm](#) intervals and fastened to all rails and tension wires at approximately [305 mm](#) intervals.
- c. Cut fabric by untwisting and removing pickets. Accomplish splicing by weaving a single picket into the ends of the rolls to be joined. Install the bottom of the fabric [50 mm plus or minus 13 mm](#) above the ground.
- d. After the fabric installation is complete, exercise the fabric by applying a [222 newton](#) push-pull force at the center of the fabric between posts; use a [133 newton](#) pull at the center of the panel to ensure fabric deflection of not more than [63.5 mm](#) when pulling fabric

from the post side of the fence. Every second fence panel is required to meet this requirement. Resecure and retest all failed panels at the Contractor's expense.

3.2.5 Supporting Arms

Install barbed wire supporting arms and barbed wire as indicated on the drawings and as recommended by the manufacturer. Anchor supporting arms to the posts in a manner to prevent easy removal with hand tools. Pull barbed wire taut and attach to the arms with clips or other means that will prevent easy removal.

3.2.6 Gate Installation

- a. Install gates at the locations shown. Mount gates to swing as indicated. Install latches, stops, and keepers as required.
- b. Attach padlocks to gates or gate posts with chains. Weld or otherwise secure hinge pins, and hardware assembly to prevent removal.
- c. Submit 6 copies of [operating and maintenance instructions](#). Outline the step-by-step procedures required for system startup, operation, and shutdown. Include the manufacturer's name, model number, service manual, parts list, and brief description of all equipment and their basic operating features. Include in the maintenance instructions routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guide. Also include the general gate layout, equipment layout and simplified wiring and control diagrams of the system as installed.

3.2.7 Grounding

- a. Ground fencing as indicated on drawings and specified.
- b. Ground fences on each side of all gates, at each corner, at the closest approach to each building located within **15 m** of the fence, and where the fence alignment changes more than 15 degrees. Ensure grounding locations are located no more than **198 m** apart. Bond each gate panel with a flexible bond strap to its gate post. Ground fences crossed by powerlines of 600 volts or more at or near the point of crossing and at distances not exceeding **45 m** on each side of crossing.
- c. Provide ground conductor consisting of **3.3 mm** solid copper wire. Use grounding electrodes that measures **19 mm** by **3.05 m** long and are a copper-clad steel rod. Drive electrodes into the earth so that the top of the electrode is at least **152 mm** below the grade. Where driving is impracticable, bury electrodes a minimum of **305 mm** deep and radially from the fence. Install the top of the electrode to be less than **610 mm** or more than **2.4 m** from the fence. Clamp ground conductor to the fence and electrodes with bronze grounding clamps to create electrical continuity between fence posts, fence fabric, and ground rods. Measure total resistance of the fence to ground and ensure it is not greater than 25 ohms.

3.3 CLOSEOUT ACTIVITIES

3.3.1 Cleanup

Remove waste fencing materials and other debris from the work site each

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

workday.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 92 19

SEEDING

08/17, CHG 1: 08/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Stand of Turf
- 1.3 RELATED REQUIREMENTS
- 1.4 SUBMITTALS
- 1.5 DELIVERY, STORAGE, AND HANDLING
 - 1.5.1 Delivery
 - 1.5.1.1 Seed Protection
 - 1.5.1.2 Fertilizer Delivery
 - 1.5.2 Storage
 - 1.5.2.1 Seed and Fertilizer Storage
 - 1.5.2.2 Topsoil
 - 1.5.2.3 Handling
- 1.6 TIME RESTRICTIONS AND PLANTING CONDITIONS
 - 1.6.1 Restrictions
- 1.7 TIME LIMITATIONS
 - 1.7.1 Seed

PART 2 PRODUCTS

- 2.1 SEED
 - 2.1.1 Classification
 - 2.1.2 Seed Mixture
- 2.2 TOPSOIL
 - 2.2.1 On-Site Topsoil
 - 2.2.2 Off-Site Topsoil
 - 2.2.3 Composition
- 2.3 SOIL CONDITIONERS
 - 2.3.1 Aluminum Sulfate
 - 2.3.2 Sulfur
 - 2.3.3 Iron
 - 2.3.4 Sand
 - 2.3.5 Perlite
 - 2.3.6 Composted Derivatives
 - 2.3.6.1 Particle Size
 - 2.3.6.2 Nitrogen Content
 - 2.3.7 Gypsum
- 2.4 FERTILIZER
 - 2.4.1 Granular Fertilizer
- 2.5 WATER
- 2.6 EROSION CONTROL MATERIALS
 - 2.6.1 Erosion Control Fabric
 - 2.6.2 Erosion Control Material Anchors

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 EXTENT OF WORK

3.1.1.1 Topsoil

3.1.1.2 Soil Conditioner Application Rates

3.1.1.3 Fertilizer Application Rates

3.2 SEEDING

3.2.1 Seed Application Seasons and Conditions

3.2.2 Seed Application Method

3.2.2.1 Broadcast and Drop Seeding

3.2.3 Rolling

3.2.4 Erosion Control Material

3.2.5 Watering

3.3 PROTECTION OF TURF AREAS

3.4 RESTORATION

-- End of Section Table of Contents --

SECTION 32 92 19

SEEDING

08/17, CHG 1: 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE-ISO 10390 (2012) Soil quality. Determination of pH.

ASTM INTERNATIONAL (ASTM)

ASTM D4972 (2018) Standard Test Methods for pH of Soils

U.S. DEPARTMENT OF AGRICULTURE (USDA)

DOA SSIR 42 (1996) Soil Survey Investigation Report No. 42, Soil Survey Laboratory Methods Manual, Version 3.0

1.2 DEFINITIONS

1.2.1 Stand of Turf

95 percent ground cover of the established species.

1.3 RELATED REQUIREMENTS

Section 31 00 00 EARTHWORK, and Section 32 05 33 LANDSCAPE ESTABLISHMENT apply to this section for pesticide use and plant establishment requirements, with additions and modifications herein.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Fertilizer

Include physical characteristics, and recommendations.

SD-06 Test Reports

Topsoil Composition Tests (reports and recommendations).

SD-07 Certificates

Certification and Approval for Seed

SD-08 Manufacturer's Instructions

Erosion Control Materials

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 Delivery

1.5.1.1 Seed Protection

Protect from drying out and from contamination during delivery, on-site storage, and handling.

1.5.1.2 Fertilizer Delivery

Deliver to the site in original, unopened containers bearing manufacturer's chemical analysis, name, trade name, trademark, and indication of conformance to state and federal laws.

1.5.2 Storage

1.5.2.1 Seed and Fertilizer Storage

Store in cool, dry locations away from contaminants.

1.5.2.2 Topsoil

Prior to stockpiling topsoil, treat growing vegetation with application of appropriate specified non-selective herbicide. Clear and grub existing vegetation three to four weeks prior to stockpiling topsoil.

1.5.2.3 Handling

Do not drop or dump materials from vehicles.

1.6 TIME RESTRICTIONS AND PLANTING CONDITIONS

1.6.1 Restrictions

Do not plant when the ground is muddy, or when air temperature exceeds 32 degrees Celsius.

1.7 TIME LIMITATIONS

1.7.1 Seed

Apply seed within twenty four hours after seed bed preparation.

PART 2 PRODUCTS

2.1 SEED

2.1.1 Classification

Provide certified seed of the latest season's crop delivered in original sealed packages, bearing producer's guaranteed analysis for percentages of

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

mixtures, purity, germination, weedseed content, and inert material. Label in conformance with applicable state seed laws. Wet, moldy, or otherwise damaged seed will be rejected. Field mixes will be acceptable when field mix is performed on site in the presence of the Contracting Officer.

2.1.2 Seed Mixture

Botanical Name	Minimum	Maximum
Cynodon Dactylon	40%	60%
Festuca Rubra	20%	30%
Festuca Arundinacea	15%	20%
Medicago Sativa	5%	10%
Onobrychis viciifolia	5%	10%

Proportion seed mixtures by weight.

2.2 TOPSOIL

2.2.1 On-Site Topsoil

Surface soil stripped and stockpiled on site and modified as necessary to meet the requirements specified for topsoil in paragraph COMPOSITION. When available topsoil must be existing surface soil stripped and stockpiled on-site in accordance with Section 31 00 00 EARTHWORK.

2.2.2 Off-Site Topsoil

Conform to requirements specified in paragraph COMPOSITION. Additional topsoil must be furnished by the Contractor.

2.2.3 Composition

Containing from 5 to 10 percent organic matter as determined by the [topsoil composition tests](#) of the Organic Carbon, 6A, Chemical Analysis Method described in [DOA SSIR 42](#). Maximum particle size, [19 mm](#), with maximum 3 percent retained on [6 mm](#) screen. The pH must be tested in accordance with [ASTM D4972](#) or [UNE-ISO 10390](#). Topsoil must be free of sticks, stones, roots, and other debris and objectionable materials. Other components must conform to the following limits:

Silt	25-50 percent
Clay	10-30 percent
Sand	20-35 percent
pH	5.5 to 7.0

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Soluble Salts	600 ppm maximum
---------------	-----------------

2.3 SOIL CONDITIONERS

Add conditioners to topsoil as required to bring into compliance with "composition" standard for topsoil as specified herein.

2.3.1 Aluminum Sulfate

Commercial grade.

2.3.2 Sulfur

100 percent elemental

2.3.3 Iron

100 percent elemental

2.3.4 Sand

Clean and free of materials harmful to plants.

2.3.5 Perlite

Horticultural grade.

2.3.6 Composted Derivatives

Ground bark, nitrolized sawdust, humus or other green wood waste material free of stones, sticks, and soil stabilized with nitrogen and having the following properties:

2.3.6.1 Particle Size

Minimum percent by weight passing:

4.75 mm screen	95
2.36 mm screen	80

2.3.6.2 Nitrogen Content

Minimum percent based on dry weight:

Fir Sawdust	0.7
Fir or Pine Bark	1.0

2.3.7 Gypsum

Coarsely ground gypsum comprised of calcium sulfate dihydrate 80 percent, calcium 18 percent, sulfur 14 percent; minimum 96 percent passing through 850 micrometers, 100 percent passing thru 970 micrometers screen.

2.4 FERTILIZER

2.4.1 Granular Fertilizer

Organic, granular controlled release fertilizer containing the following

minimum percentages, by weight, of plant food nutrients:

- 10 percent available nitrogen
- 10 percent available phosphorus
- 10 percent available potassium

2.5 WATER

Source of water must be approved by Contracting Officer and of suitable quality for irrigation, containing no elements toxic to plant life.

2.6 EROSION CONTROL MATERIALS

Erosion control material must conform to the following:

2.6.1 Erosion Control Fabric

Fabric must be knitted construction of polypropylene yarn with uniform mesh openings 19 to 25 mm square with strips of biodegradable paper. Filler paper strips must have a minimum life of 6 months.

2.6.2 Erosion Control Material Anchors

Erosion control anchors must be as recommended by the manufacturer.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 EXTENT OF WORK

Provide soil preparation prior to planting (including soil conditioners as required), fertilizing, seeding, and surface topdressing of all newly graded finished earth surfaces, unless indicated otherwise, and at all areas inside or outside the limits of construction that are disturbed by the Contractor's operations.

3.1.1.1 Topsoil

Provide 102 mm of off-site topsoil to meet indicated finish grade. After areas have been brought to indicated finish grade, incorporate fertilizer, pH adjusters and soil conditioners into soil a minimum depth of 100 mm by disking, harrowing, tilling or other method approved by the Contracting Officer. Remove debris and stones larger than 19 mm in any dimension remaining on the surface after finish grading. Correct irregularities in finish surfaces to eliminate depressions. Protect finished topsoil areas from damage by vehicular or pedestrian traffic.

3.1.1.2 Soil Conditioner Application Rates

Apply soil conditioners at rates as determined by laboratory soil analysis of the soils at the job site.

3.1.1.3 Fertilizer Application Rates

Apply fertilizer at rates as determined by laboratory soil analysis of the soils at the job site.

3.2 SEEDING

3.2.1 Seed Application Seasons and Conditions

Immediately before seeding, restore soil to proper grade. Do not seed when ground is muddy or in an unsatisfactory condition for seeding. If special conditions exist that may warrant a variance in the above seeding dates or conditions, submit a written request to the Contracting Officer stating the special conditions and proposed variance. Apply seed within twenty four hours after seedbed preparation. Sow seed by approved sowing equipment. Sow one-half the seed in one direction, and sow remainder at right angles to the first sowing.

3.2.2 Seed Application Method

Seeding method must be broadcasted and drop seeding.

3.2.2.1 Broadcast and Drop Seeding

Seed must be uniformly broadcast at the rate of 15 kg per 100 square meter. Use broadcast or drop seeders. Sow one-half the seed in one direction, and sow remainder at right angles to the first sowing. Cover seed uniformly to a maximum depth of 6 mm in clay soils and 13 mm in sandy soils by means of spike-tooth harrow, cultipacker, raking or other approved devices.

3.2.3 Rolling

Immediately after seeding, firm entire area except for slopes in excess of 3 to 1 with a roller not exceeding 134 kg per m for each foot of roller width. If seeding is performed with cultipacker-type seeder or by hydroseeding, rolling may be eliminated.

3.2.4 Erosion Control Material

Install in accordance with manufacturer's instructions, where indicated or as directed by the Contracting Officer.

3.2.5 Watering

Start watering areas seeded as required by temperature and wind conditions. Apply water at a rate sufficient to insure thorough wetting of soil to a depth of 50 mm without run off. During the germination process, seed is to be kept actively growing and not allowed to dry out.

3.3 PROTECTION OF TURF AREAS

Immediately after turfing, protect area against traffic and other use.

3.4 RESTORATION

Restore to original condition existing turf areas which have been damaged during turf installation operations at the Contractor's expense. Keep clean at all times at least one paved pedestrian access route and one paved vehicular access route to each building. Clean other paving when work in adjacent areas is complete.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 01 50.55

CLEANING OF PETROLEUM STORAGE TANKS

02/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DEFINITIONS
 - 1.3.1 Certified Industrial Hygienist (CIH)
 - 1.3.2 Marine Chemist
 - 1.3.3 Hazardous Areas
 - 1.3.4 Personal Monitoring
 - 1.3.5 Reproductive Hazard
 - 1.3.6 Flammable Liquid
 - 1.3.7 Combustible Liquid
- 1.4 QUALIFICATIONS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Modification of References
 - 1.5.2 Copies of Standards
 - 1.5.3 Safety Permits and Equipment
 - 1.5.4 Regulatory Requirements
 - 1.5.5 Medical Examinations
 - 1.5.6 Medical Records
 - 1.5.7 CIH Responsibilities
 - 1.5.8 Training
 - 1.5.9 Respiratory Protection Program
 - 1.5.10 Pre-Construction Conference
 - 1.5.11 Certificates
 - 1.5.11.1 Qualifications of Marine Chemist
 - 1.5.11.2 Qualifications of Certified Industrial Hygienist (CIH)
 - 1.5.11.3 Testing Laboratory
 - 1.5.11.4 Safety Plan
 - 1.5.11.5 Work Plan
 - 1.5.11.6 Hazardous Waste Disposal Plan
 - 1.5.11.7 Tank Certification of Safety
 - 1.5.11.8 Training Certification
 - 1.5.11.9 Hazardous Waste Permits
 - 1.5.11.10 Non-Hazardous Waste Permits
 - 1.5.12 Test Results
 - 1.5.12.1 Required Test Reports
 - 1.5.12.2 Air Monitoring
- 1.6 DELIVERY AND STORAGE
- 1.7 JOB CONDITIONS
 - 1.7.1 Ventilation
- 1.8 SCHEDULING AND SEQUENCING
 - 1.8.1 Sequence of Primary Phases of the Cleaning Procedure
 - 1.8.2 General Scheduling

PART 2 PRODUCTS

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.1 MATERIALS
 - 2.1.1 Cleaning Agents
 - 2.1.2 Abrasive
 - 2.1.2.1 Abrasive for Blasting
 - 2.1.2.2 Recycled Abrasive
- 2.2 EQUIPMENT

PART 3 EXECUTION

- 3.1 PREPARATION FOR ENTRY
 - 3.1.1 Isolation From Piping
 - 3.1.1.1 Thermal Relief
 - 3.1.2 Lockout Tagout
 - 3.1.3 Removal of Ignition Sources
 - 3.1.4 Survey of Hazardous Areas
- 3.2 PROJECT CONDITIONS
 - 3.2.1 Permission for Each Entry Into a Tank
 - 3.2.2 Traffic Control
 - 3.2.3 Lavatory Facilities
 - 3.2.4 Miscellaneous
 - 3.2.4.1 Grounding and Bonding for Equipment
 - 3.2.4.2 Fire Extinguishers
 - 3.2.4.3 Disconnection of Pipelines
 - 3.2.4.4 Removal of Ignition Sources
 - 3.2.4.5 Survey of Hazardous Areas
 - 3.2.4.6 Exit from a Tank During Emergencies
- 3.3 INSPECTION
 - 3.3.1 Inspection of Equipment
 - 3.3.1.1 Respirators
 - 3.3.1.2 Air Hose from Breathing-Air Supply
 - 3.3.1.3 Safety Harness and Life Line
 - 3.3.1.4 Breathing-Air Supply Source
 - 3.3.1.5 Monitoring Equipment
 - 3.3.1.6 Other Equipment
 - 3.3.2 Personnel Inspection
 - 3.3.2.1 Clothing
 - 3.3.2.2 Breathing-Air Supply
 - 3.3.2.3 Harness and Lifeline
 - 3.3.2.4 Gum or Tobacco Chewing
 - 3.3.2.5 Physical Defects or Injuries
 - 3.3.2.6 Alcoholic Beverages and Drugs
 - 3.3.2.7 Counseling on Reproductive Hazards
 - 3.3.2.8 Hazardous Areas
- 3.4 TABLE OF TANK HISTORY
- 3.5 FUEL REMOVAL
- 3.6 TANK CLEANING
 - 3.6.1 Monitoring
 - 3.6.2 Water Removal and Disposal
 - 3.6.3 Sludge and Sediment Removal and Disposal
 - 3.6.3.1 Sludge Disposal Using Landfill
 - 3.6.4 Washing
 - 3.6.5 Wash Water, Detergent Solution, and Sediment Removal
 - 3.6.6 Disposal of Used Blasting Abrasive
 - 3.6.7 Special Precautions
- 3.7 FINAL CLEAN-UP
 - 3.7.1 Stenciling Tank

-- End of Section Table of Contents --

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

SECTION 33 01 50.55

CLEANING OF PETROLEUM STORAGE TANKS
02/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

- | | |
|--------------|---|
| API RP 500 | (2012; Errata 2014) Classification of Locations for Electrical Installations at Petroleum Facilities Classified as Class I, Division 1 and Division 2 |
| API RP 2003 | (2015; 8th Ed) Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents |
| API Std 521 | (2014; 6th Ed) Pressure-relieving and Depressuring Systems |
| API Std 2015 | (2018) Requirements for Safe Entry and Cleaning of Petroleum Storage Tanks |

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

- | | |
|-------------|--|
| ASME B16.5 | (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard |
| ASME B16.48 | (2015) Line Blanks |

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

- | | |
|----------|---|
| NFPA 70 | (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code |
| NFPA 306 | (2019) Standard for the Control of Gas Hazards on Vessels |
| NFPA 326 | (2015) Standard for Safeguarding of Tanks and Containers for Entry, Cleaning, or Repair |

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

- | | |
|--------------|-----------------------------------|
| NIOSH 99-109 | (Latest) Certified Equipment List |
|--------------|-----------------------------------|

U.S. ARMY CORPS OF ENGINEERS (USACE)

- | | |
|------------|------------------------------------|
| EM 385-1-1 | (2014) Safety -- Safety and Health |
|------------|------------------------------------|

Requirements Manual

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-A-22262 (1993; Rev B; Am 1 1994; Am 2 1996; Notice 1 2021) Abrasive Blasting Media Ship Hull Blast Cleaning

MIL-PRF-680 (2010; Rev C; Notice 1 2015) Degreasing Solvent

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS O-D-1276 (Rev B; Notice 1) Disinfectant-Detergent, General Purpose (Pine Oil)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.120 Hazardous Waste Operations and Emergency Response

29 CFR 1910.134 Respiratory Protection

29 CFR 1910.146 Permit-required Confined Spaces

29 CFR 1910.1025 Lead

29 CFR 1910.1028 Benzene

29 CFR 1910.1200 Hazard Communication

29 CFR 1926.55 Gases, Vapors, Fumes, Dusts, and Mists

40 CFR 260 Hazardous Waste Management System: General

40 CFR 261 Identification and Listing of Hazardous Waste

40 CFR 262 Standards Applicable to Generators of Hazardous Waste

40 CFR 263 Standards Applicable to Transporters of Hazardous Waste

40 CFR 264 Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities

40 CFR 265 Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities

40 CFR 266 Standards for the Management of Specific Hazardous Wastes and Specific Types of Hazardous Waste Management Facilities

UNDERWRITERS LABORATORIES (UL)

UL 844 (2012; Reprint Oct 2021) UL Standard for

Safety Luminaires for Use in Hazardous
(Classified) Locations

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Cleaning Agents

Abrasive for Blasting

Gasoline-Oil-Resisting Rubber Gloves and Boots

Cotton Coveralls and Hard Hat

Respiratory Protective Equipment

Disinfectant

SD-06 Test Reports

Blasting Abrasive Test

Tank Contents Tests

Cleaning Test Panel Results

Monitoring Results; G

SD-07 Certificates

Qualifications of Marine Chemist

Qualifications of Certified Industrial Hygienist (CIH)

Testing Laboratory

Safety Plan

Training Certification

Work Plan

Hazardous Waste Disposal Plan

Tank Certification of Safety

Tank Exhaust Blower

Respiratory Protective Equipment

Breathing-Air Supply Source

Combustible Gas Indicator

Oxygen Meter

Velometers

Lighting

First Aid Kit

Tank Exhaust Blower

SD-08 Manufacturer's Instructions

Tank Cleaning Agents

SD-11 Closeout Submittals

Safety Permits

1.3 DEFINITIONS

1.3.1 Certified Industrial Hygienist (CIH)

As used in this section, refers to an Industrial Hygienist employed by the Contractor and is certified by the American Board of Industrial Hygiene in comprehensive practice.

1.3.2 Marine Chemist

The holder of a valid Certificate issued by the National Fire Protection Association in accordance with the "Rules for Certification of Marine Chemists," establishing him as a person qualified to determine whether construction, alteration, repair, or shipbreaking of vessels, which may involve hazards covered by NFPA 306 can be undertaken with safety.

1.3.3 Hazardous Areas

Hazardous areas must be defined as any area within 30 meters of active aboveground storage tanks, areas within 30 meters of leaking sections of fuel pipelines or other vapor sources, areas within 60 meters of the downwind side of potential vapor emission sources (i.e., pressure-vacuum vents or open vents on active tanks, leaking sections of pipelines), areas within existing tanks, and areas within a dike.

1.3.4 Personal Monitoring

Sampling of lead concentrations within the breathing zone of an employee to determine the 8-hour time weighted average concentration in accordance with 29 CFR 1910.1025. Samples must be representative of the employee's work tasks. Breathing zone must be considered an area within a hemisphere, forward of the shoulders, with a radius of 150 to 225 mm and the center at the nose or mouth of an employee.

1.3.5 Reproductive Hazard

A reproductive hazard is defined as any occupational stressor (biological, chemical, or physical) that has the potential to adversely affect the human reproductive process. For example, it is well known that central nervous system problems often occur in the offspring of mothers exposed to

organic mercury during pregnancy. Therefore, based on the example cited, organic mercury can be classified as a reproductive stressor. Many reproductive hazards also cause other adverse health effects; for example, ethylene oxide is also known to be a carcinogen (i.e., produces cancer). Certain reproductive stressors can also have adverse effects on the male reproductive system. (If requested by the Contractor, the Contracting Officer will make available the Navy's standard on reproductive hazards.)

1.3.6 Flammable Liquid

Any liquid having a flash point below 38 degrees C and a vapor pressure not exceeding 275 kPa at 38 degrees C.

1.3.7 Combustible Liquid

Any liquid having a flash point at or above 38 degrees C.

1.4 QUALIFICATIONS

- a. To Be Considered Qualified: Show proof of having completed work on three previous projects.
- b. See certificate requirements for personnel as specified in paragraph CERTIFICATES.

1.5 QUALITY ASSURANCE

1.5.1 Modification of References

Except as modified herein, the work must conform with the recommendations of NFPA 326, API RP 500, API RP 2003, and API Std 2015. Where the word "should" appears in these publications, substitute "must."

1.5.2 Copies of Standards

Furnish four copies of NFPA 326, API RP 500, API RP 2003, and API Std 2015.

1.5.3 Safety Permits and Equipment

Acquire safety permits (specified by the facility safety authorities) and necessary safety equipment.

1.5.4 Regulatory Requirements

- a. Obtain permits required to comply with local, EU, and host nation regulations.
- b. Submit copies of permits required to comply with local, EU, and host nation regulations.
- c. Hazardous wastes, such as water, sediment, and sludge, must be packaged, labeled, stored, transported, treated and disposed of in accordance with 40 CFR 260 through 40 CFR 266 and State and local regulations. Transporters, sorters, treaters and disposers must be certified and have EPA ID numbers. Payment for disposal of hazardous waste will not be made until a completed hazardous waste manifest from the treatment or disposal facility is returned, and a copy furnished to the Government.

1.5.5 Medical Examinations

Before exposure to lead-contaminated fuel tank and at the completion of the work, provide workers with a comprehensive medical examination as required by 29 CFR 1910.1025 and 29 CFR 1910.1200. The initial examination will not be required if adequate records show that employees have been examined as required by 29 CFR 1910.1025 within the last year and the blood lead levels did not exceed 30 micrograms per 100 grams of whole blood.

1.5.6 Medical Records

Maintain complete and accurate medical records of employees for a period of at least 40-years or for the duration of employment plus 20-years, whichever is longer.

1.5.7 CIH Responsibilities

- a. Certify training.
- b. Review and approve safety plans and work plan for conformance to the applicable referenced standards.
- c. Inspect tank cleaning work for conformance with the approved safety and work plans.
- d. Direct monitoring.
- e. Ensure work is performed in strict accordance with specifications at all times.
- f. Ensure hazardous exposure to personnel and to the environment are adequately controlled at all times.

1.5.8 Training

Train each employee performing tank cleaning, waste disposal, and air sampling operations prior to the time of initial job assignment, in accordance with API Std 2015, 29 CFR 1910.120, 29 CFR 1910.134, and 29 CFR 1910.1200. The training must also include counseling of each employee on reproductive hazards involved in the work.

1.5.9 Respiratory Protection Program

- a. Furnish each employee required to wear a negative pressure respirator or other appropriate type with a respirator fit test at the time of initial fitting and at least every 6 months thereafter as required by 29 CFR 1910.1025 where lead exposure is involved. Fit testing is not required for positive pressure respirators.
- b. Establish and implement a respiratory protection program as required by 29 CFR 1910.134, and 29 CFR 1926.55. Also comply with 29 CFR 1910.1025 when exposure to lead is involved.

1.5.10 Pre-Construction Conference

Along with the CIH, marine chemist, or gas-free engineer, meet with the Contracting Officer to discuss in detail the tank cleaning work plan, including work procedures and precautions for the work plan.

1.5.11 Certificates

Submit certificates for the items listed. Where equipment or materials are specified to conform with the standards of organizations, such as National Institute for Occupational Safety and Health (NIOSH), Underwriters Laboratories (UL), and American Petroleum Institute (API), include a label or listing indicating compliance. In lieu of the label or listing, the Contractor may submit a test report from an approved testing organization stating that the item has been tested in accordance with the specified organization's test methods and that the item conforms with the organization's standard or code.

1.5.11.1 Qualifications of Marine Chemist

Submit name, address, and telephone number of the marine chemist selected to perform the required duties. Submit documentation that the marine chemist is certified by the National Fire Protection Association, including the certificate number and date of certification or recertification. The NFPA certification will be acceptable for non-ship work on this contract. Refer to [NFPA 306](#) to determine when a marine chemist is required, how a marine certificate is issued and maintained, and what to expect during an inspection.

1.5.11.2 Qualifications of Certified Industrial Hygienist (CIH)

Submit name, address, and telephone number of the CIH selected to perform responsibilities in paragraph CIH RESPONSIBILITIES. Provide previous experience of the CIH. Submit proper documentation that the Industrial Hygienist is certified by the American Board of Industrial Hygiene in comprehensive practice, including certification number and date of certification/recertification. The CIH must be familiar with the hazards involved in fuel systems work.

1.5.11.3 Testing Laboratory

Submit the name, address, and telephone number of the testing laboratory selected to perform the monitoring, testing, and reporting of airborne concentrations of lead and other contaminants. Provide proper documentation that persons performing the analysis have been judged proficient by successful participation within the last year in the National Institute for Occupational Safety and Health (NIOSH) Proficiency Analytical Testing (PAT) Program. The laboratory must be accredited by the American Industrial Hygiene Association (AIHA). Provide AIHA documentation along with date of accreditation/reaccreditation.

1.5.11.4 Safety Plan

Submit a safety plan within 45 calendar days after contract award and 30-days prior to commencing work. The safety plan must meet requirements of [EM 385-1-1](#), OSHA, and address the following:

- a. Identification and evaluation of the hazards and risks associated with each site being studied, including reproductive hazards and precautionary measures to be followed by workers for all hazards.
- b. Names and qualifications of each Contractor's representative in charge of the work and present at the job site when tank cleaning and repair work will be performed.

- c. Identification of supervisory personnel and alternates responsible for site safety/response operations.
- d. Determination of levels of personal protection to be worn for various site operations.
- e. List of equipment with adequate nomenclature by item, that will be used at the job site and the date and location where this equipment can be inspected by the Contracting Officer.
- f. Establishment of work zones (exclusion area, contamination area, and support area).
- g. Establishment of a tank entry and work permit program in accordance with 29 CFR 1910.146, EM 385-1-1, and NFPA 326.
- h. Establishment of decontamination methods and procedures.
- i. Determination of the number of people required to enter the contamination zones during the initial entries and subsequent operations.
- j. Establishment of emergency procedures, such as: escape routes, fire protection, signals for withdrawing work parties from site, emergency communications, wind indicators, including Navy notification.
- k. Identification and arrangements with nearest medical facility for emergency medical care for both routine-type injuries and toxicological problems. Submit name, location, and telephone number of this medical facility.
- l. Establishment of continual air and personnel monitoring procedures.
- m. Establishment of procedures for obtaining and handling potentially contaminated samples.
- n. Identification of medical monitoring program, including respirator medical qualification examination for each individual at the work site.
- o. Identification of training plan to be instituted, including contents of 29 CFR 1910.1200 and 29 CFR 1910.134; its training contents; and instructor with appropriate training certification. Training plan must also include counseling to each employee on reproductive hazards.
- p. Establishment of a respiratory protection program conforming to 29 CFR 1910.134.
- q. Establishment of a hazard communication program (29 CFR 1910.1200).

1.5.11.5 Work Plan

The shut down or interruption to normal operations or traffic must be listed on the progress schedule and submitted to the Contracting Officer.

1.5.11.6 Hazardous Waste Disposal Plan

Prepare a Hazardous Waste Disposal Plan and submit within 45 calendar days after contract award for approval by the Contracting Officer, or if there

are no hazardous wastes indicated by Government tests, submit the plan 21 days after the Contractor's tests indicate hazardous wastes. The Hazardous Waste Disposal Plan must comply with applicable requirements of local, EU, and host nation hazardous waste regulations and must address the following:

- a. Identification of hazardous wastes associated with the work, including a sampling and testing plan for each waste stream, the purpose of each test, and the rationale for evaluating the test results. Indicate the representative sampling and specific testing methods, number of samples, and the name and qualifications of the testing laboratory.
- b. Estimated quantities of wastes to be disposed in the cleaning of each tank and a description of arrangements made for storage and disposal.
- c. Names and qualifications of each Contractor that will be transporting, storing, treating, and disposing of the wastes. Include the facility location and a 24-hour point of contact. Furnish two copies of applicable hazardous waste documentation.
- d. Names and qualifications (experience and training) of personnel who will be working on-site with hazardous wastes.
- e. List of waste handling equipment to be used in performing the work, to include cleaning, treatment, volume reduction, and transport equipment.
- f. Spill prevention, containment, and cleanup contingency measures to be implemented.
- g. Work plan and schedule for waste removal and disposal.
- h. Cost for hazardous waste disposal according to this plan.

1.5.11.7 Tank Certification of Safety

Submit certification, in accordance with NFPA 326, from an NFPA certified "Marine Chemist" stating that tank is safe for hot work and that special precautionary measures have been taken for workers to enter the tank to perform the work.

1.5.11.8 Training Certification

Submit certifications signed and dated by the CIH specified in the testing plan and by each employee stating that the employee has received training on work practices and received counseling on and fully understands the reproductive hazards involved with lead and toluene exposure and the work.

1.5.11.9 Hazardous Waste Permits

Submit copies of all applicable hazardous waste documentation of the transporter, treatment, storage and disposal facility that will be accepting hazardous waste. Include the facility location and a 24-hour point of contact.

1.5.11.10 Non-Hazardous Waste Permits

Submit all applicable documentation for disposal site for non-hazardous residues and wastes.

1.5.12 Test Results

1.5.12.1 Required Test Reports

Submit test results required by MIL-A-22262, for blasting abrasive. Submit contractor's independent tests of tank contents (water, sediment, and sludge). Submit tank cleaning test panel results, including water pressure and temperature and nozzle distances used during tank washing procedure.

1.5.12.2 Air Monitoring

Submit monitoring results to the Contracting Officer within 2 working days after the samples are taken, signed by the testing laboratory employee performing the air monitoring, the employee that analyzed the sample, and the CIH.

1.6 DELIVERY AND STORAGE

Deliver equipment and materials to the site in an undamaged condition bearing the manufacturer's name and brand designation. Store equipment and materials off the ground to provide proper ventilation, drainage, and protection against dampness. Replace defective and damaged equipment and materials.

1.7 JOB CONDITIONS

1.7.1 Ventilation

Maintain a vapor-free condition throughout the course of the work inside the tank. The air movers must be non-sparking, explosion-proof, electrically operated or air-driven exhaust type. A rate of one air change per hour must be the lowest acceptable rate, for tanks under 3600 kL. For tanks greater than 3600 kL, use 4700 L/s. Air movers must be kept in operation whenever workers are in the tanks; except the air movers must be shut down 15 minutes before taking tests.

1.8 SCHEDULING AND SEQUENCING

1.8.1 Sequence of Primary Phases of the Cleaning Procedure

- a. Planning the operations
- b. Preparation for cleaning
- c. Vapor-freeing of the tank
- d. Cleaning the tank
- e. Clean-up, residue disposal, inspection, and acceptance.

1.8.2 General Scheduling

Complete the work specified in this section before any other work in the tank is started. The work includes the complete interior cleaning of the storage tanks.

PART 2 PRODUCTS

2.1 MATERIALS

Submit identification for the items by designated name, specification number, project contracting number, and intended use. Submit Safety Data Sheets for materials to be used at the job site, in accordance with 29 CFR 1910.1200.

2.1.1 Cleaning Agents

- a. Detergent: FS O-D-1276.
- b. Solvent: MIL-PRF-680, Type II, minimum flashpoint of 60 degrees C.
- c. Approved commercial cleaning agent.

2.1.2 Abrasive

2.1.2.1 Abrasive for Blasting

Provide sharp, washed, salt-free abrasive material, free from feldspar and other constituents that tend to break down and remain on the surface. Abrasive must not contain magnetic materials and must conform to MIL-A-22262.

2.1.2.2 Recycled Abrasive

Screen and air wash abrasive that is recycled at the job site, to remove dirt and fines. Add new abrasive so that the combined new and recycled abrasive mixture meets specified abrasive requirements for chemical composition, moisture, friability, silica, anchor pattern and oil content. Do not recycle abrasive which has picked up toxic or hazardous material. Do not recycle nickel slag.

2.2 EQUIPMENT

Furnish necessary clothing and equipment for the work and protection of people entering the tank. Electrical equipment and wiring must be in accordance with NFPA 70, Class 1, Group D, Division 1. Provide any item or items for the protection of these people including but not limited to the following:

- a. Gasoline-Oil-Resisting Rubber Gloves and Boots: Gauntlet type and conductive type respectively (acid-proof rubber is an acceptable material); furnished for each person entering or working inside the tank or handling sludge materials on the exterior of the tank, plus one extra pair each for emergency use.
- b. Cotton Coveralls and Hard Hat: Light colored; one change per person per day, and an adequate supply of chemical-resistant disposable coveralls to be worn over cotton coveralls.
- c. Respiratory Protection: Provide one of the following types of NIOSH-approved respiratory protective equipment for each person working inside the tank, plus one extra for emergency use. NIOSH 99-109 listing constitutes NIOSH approval.

- (1) Self-contained breathing apparatus with a full facepiece operated

in a positive pressure mode.

- (2) A combination respirator which includes a Type C supplied-air respirator with a full facepiece operated in a positive pressure mode and an auxiliary positive pressure self-contained breathing apparatus. Provide and use two-way communication equipment when cleaning underground tanks.
- (3) The CIH may specify airline (Type C) respirator in place of those specified above; however, the decision must be based on the results of personal monitoring.
- (4) Use Type CE respirator for abrasive blasting inside the tank.
- (5) CIH must specify respiratory protection if required for personnel handling sludge material outside of the tank.
- d. Safety Harness: For each person working inside tank, plus one extra for outside the tank.
- e. 13 mm Diameter Life Rope of Required Length: For each person working inside the tank.
- f. Breathing-Air Supply Source: 29 CFR 1910.134.
- g. Combustible Gas Indicator and Oxygen Meter.
- h. Shovels, Buckets, Brooms, Wrenches, Scrapers, Squeegees, Wire Brushes, Scrub-Brushes, Ladders, Staging, and Other Tools: Do not use brooms or brushes that have plastic or synthetic bristles.
- i. Lighting: UL 844, explosion-proof, minimum 540 lx, floodlight type, or Mining Enforcement and Safety Administration (MESA) approved, explosion-proof, portable battery-powered light.
- j. Air Movers for Tank Ventilation: Explosion proof electrically operated or air driven. Nonferrous fan blades. Use velometers for measuring velocity.
- k. Disinfectant for Cleaning Face Masks: Cleaner-sanitizer for cleaning and disinfecting respirator facepieces as specified in 29 CFR 1910.134, Appendix B-2.
- l. Soap for Personnel Washing: Non-phosphate type.
- m. A.B.C. Fire Extinguishers: UL listed 2A: 40B: C, 2A: 20B: C, or 4A: 30B: C; minimum 7 kg capacity.
- n. First Aid Kit: One 16-unit kit for each 25 persons.

PART 3 EXECUTION

3.1 PREPARATION FOR ENTRY

Prepare the tank for entry in accordance with NFPA 326. Isolate from sources of energy. Ensure vapors have been controlled or removed. Identify potential hazards and apply control measures to mitigate the hazards. Test and monitor atmospheric conditions to ensure conditions of the Marine Chemist Certificate safety designations have been met.

3.1.1 Isolation From Piping

For tank cleaning prior to out-of-service inspection and minor cold repairs, close the double block and bleed isolation valves and remove the body cavity plugs from the valves. Monitor the body cavity for fuel and maintain the valves in this condition for the duration of the cleaning, inspection, or minor cold repairs, not to exceed one month. Secure the valves in the closed position with mechanical means. Perform lockout tagout procedures on the valves.

For ordinary cleaning prior to out-of-service inspection and hot work or when nozzles are not equipped with double block and bleed valves, or for outages lasting longer than one month, disconnect piping connected to the tank. Provide a solid-plate line blank between two flanges near the tank in accordance with ASME B16.48, or remove a valve or piece of pipe and provide a blind flange compliant with ASME B16.5 to isolate tank. Isolation means must be of sufficient strength to withstand pressure which might be exerted by the product being blanked off, and must be gasketed on both sides if blind flange is inserted between two flanges. Do not disconnect piping or valves until it is certain the line has been defueled.

Isolate all piping connected to the tank. Perform lockout tagout procedures on any valve remaining connected to the tank.

3.1.1.1 Thermal Relief

Evaluate thermal relief capability on active facility piping isolated from the tank. Should isolation means result in a segment of active piping unprotected from thermal overpressure, provide temporary means consistent with API Std 521 to relieve the overpressure. Consult with operators to determine existing relief pressures and set temporary relief means to ensure overpressure does not occur.

3.1.2 Lockout Tagout

Perform lockout tagout on all electrical circuits and sources of energy to the tank in accordance with EM 385-1-1 and NFPA 326.

3.1.3 Removal of Ignition Sources

Remove sources of ignition from the cleaning area. Do not permit ignition producing devices, including matches, lighters or cigarettes, within 30 m upwind and 60 m downwind of a tank, or inside the tank farm, or within the tank firewall, whichever is farther.

3.1.4 Survey of Hazardous Areas

Carefully survey the entire area around the tank to be cleaned to ensure that there are no vapors present in the pit, low places, or hazardous areas and that all unauthorized personnel are cleared from the area. Ensure that there is no possibility of anyone smoking in the immediate vicinity. Hazardous areas are defined as follows:

- a. Interior of tanks.
- b. Areas within 30 meters from points having flammable vapor emissions which, for example, are from the exhaust manholes of tanks under repair, open vents or pressure vacuum vents (breather valves) of

active tanks in the vicinity of tanks under repairs or cleaning.
CAUTION: Allowance must be made for 4 or more miles per hour winds by increasing the size of the hazardous area to a minimum of 60 meters on the downward side.

3.2 PROJECT CONDITIONS

3.2.1 Permission for Each Entry Into a Tank

Obtain written permission from the Contracting Officer prior to each entry into a tank. Permission will be granted only under the following conditions:

- a. The Contractor's qualified supervisor is present.
- b. The Contractor's personnel have been briefed by the supervisor on the procedure and role of each employee in the event of an emergency.
- c. Required equipment is approved and properly located.
- d. Personnel are properly equipped with properly fitted protective equipment and have received adequate training from a qualified instructor.
- e. The entire area adjacent to the tank is secured.
- f. A minimum of two persons outside and two or more persons inside of each tank are provided at all times during cleaning operations.
- g. Tank air is monitored and corrective action is taken to ensure that the vapor concentration is less than 10 percent of the lower flammable limit (LFL) and oxygen content is a minimum of 19.5 percent.
- h. An NFPA certified "Marine Chemist" or CIH has certified that the tank is safe for hot work, and that the required special precautionary measures have been taken due to the potential health hazard to the worker that still exists, even when the vapor concentration is well below the LFL. The Contractor must be responsible for reviewing the record drawing(s) of the tank to be cleaned.
- i. People entering the area leave smoking materials such as cigarettes and flame-producing devices at a previously determined location.
- j. When work involves handling and disposal of hazardous waste, the Contractor has a copy of 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, and 40 CFR 266 in his possession.
- k. Permit only personnel authorized in the safety plan within 100 feet of the tank perimeter.

3.2.2 Traffic Control

Direct traffic minimum 60 meters away from the tank cleaning area. Set up road blocks and warning signs. Do not operate vehicles in hazardous areas.

3.2.3 Lavatory Facilities

Arrange for lavatory and toilet facilities.

3.2.4 Miscellaneous

Ensure that the manufacturers have labeled containers holding products involving hazards in use or storage, in accordance with 29 CFR 1910.1200. Label containers used to store, transport, or dispose of hazardous waste in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, and 40 CFR 266 and Host Nation Regulations. Remove small objects of ferrous metal within the working areas to prevent the accidental striking of a spark. Place equipment upwind of tank openings at highest elevation possible; do not place in a spot lower than the surrounding terrain. Review drawings of the tank to be cleaned and brief workers on the location of pits, sumps, piping, or other tank appurtenances which could be hazardous to personnel. Provide floodlights to illuminate the work area without the need for battery operated handlights. Provide scaffolding, platforms, and ladders for secure, safe accessibility to tank surfaces. Install electrical equipment in accordance with API RP 500. Provide floodlights to illuminate the work area without the need for battery operated handlights. Do not use artificial lights inside tank until the tank is vapor-free.

3.2.4.1 Grounding and Bonding for Equipment

Provide grounding and bonding for equipment which may generate static electricity, including air hose to sandblast nozzle. Do not pass the air hose through an area where flammable vapors may exist.

3.2.4.2 Fire Extinguishers

Furnish two carbon-dioxide fire extinguishers of minimum 7 kg capacity each, in the immediate vicinity of the work. Provide a continuous fire watch. CAUTION: Do not discharge high pressure carbon dioxide extinguishers where explosive vapors exist since the discharge can cause a spark which will ignite the vapors.

3.2.4.3 Disconnection of Pipelines

For normal tank cleaning prior to out-of-service (internal) inspection and minor cold repairs, close the double block and bleed valves on the tank nozzles connected to piping and remove the body cavity plugs from the bottoms of the valves and bleed the valves for the duration of the cleaning, inspection or minor cold repairs. Perform lock-out/tag-out procedures on the valves.

For cleaning prior to hot work or when nozzles are not equipped with double block and bleed valves, or for extended outages lasting longer than one month, disconnect pipelines connected to the tank. Provide a solid-plate blind flange between two flanges near the tank, or remove a valve or piece of pipe and provide a blind flange to prevent flammable material from entering the tank. Blind flanges must be of sufficient strength to withstand pressure which might be exerted by the material being blanked off, and must be gasketed on both sides if blind flange is inserted between two flanges. CAUTION: Do not disconnect piping or valves until it is certain the line has been emptied of fuel.

3.2.4.4 Removal of Ignition Sources

Remove sources of ignition from the cleaning area. Do not permit ignition producing devices, including matches, lighters or cigarettes, within 30 meters upwind and 60 meters downwind of a tank, or inside the tank farm,

or within the tank firewall, whichever is farther.

3.2.4.5 Survey of Hazardous Areas

Carefully survey the entire area around the tank to be cleaned to ensure that there are no vapors present in the pit, low places, or hazardous areas and that all unauthorized personnel are cleared from the area. Ensure that there is no possibility of anyone smoking in the immediate vicinity. Hazardous areas are defined as follows:

- a. Interior of tanks.
- b. Areas within **30 meters** from points having flammable vapor emissions which, for example, are from the exhaust manholes of tanks under repair, open vents or pressure vacuum vents (breather valves) of active tanks in the vicinity of tanks under repairs or cleaning. CAUTION: Allowance must be made for **6 or more km/h** winds by increasing the size of the hazardous area to a minimum of **60 meters** on the downwind side.

3.2.4.6 Exit from a Tank During Emergencies

To permit quick, free exit from a tank during emergencies, keep the area around the tank openings and emergency routes clear of obstructions.

3.3 INSPECTION

3.3.1 Inspection of Equipment

3.3.1.1 Respirators

Respirator users must inspect their respirators in strict accordance with the instructions provided by the manufacturer.

3.3.1.2 Air Hose from Breathing-Air Supply

If air line respirators are used, ensure that:

- a. There are no breaks in outside covering;
- b. Condition of gaskets is good;
- c. Connections are tight; and
- d. There are no restrictions in the hose.

3.3.1.3 Safety Harness and Life Line

Ensure that:

- a. There is no frayed or weak material; and
- b. Condition of harness is good.

3.3.1.4 Breathing-Air Supply Source

Ensure:

- a. Good working condition;

- b. Location in vapor-free area;
- c. Compliance with 29 CFR 1910.134 for breathing air quality, frequency of air analysis, and presence of safety devices; and
- d. Backup air supply source.

3.3.1.5 Monitoring Equipment

Calibrate each day before use:

- a. Combustible gas indicator
- b. Oxygen meter

3.3.1.6 Other Equipment

Ensure:

- a. Proper grounding and bonding;
- b. Explosion-proof motors; and
- c. Explosion-proof lighting.

3.3.2 Personnel Inspection

3.3.2.1 Clothing

Personnel for Proper Attire Commensurate with Hazards Involved: Check for:

- a. Clean clothing in good condition (wear freshly laundered clothing at the beginning of the job and at the start of each workday thereafter).
- b. Boots and gloves of approved type and in good condition.

3.3.2.2 Breathing-Air Supply

If air line respirators are used, ensure that air is supplied to the facepiece at a rate of 2 to 7 L/s. If self-contained breathing apparatus are used, ensure sufficient number of full replacement cylinders are available to last the duration of the job.

3.3.2.3 Harness and Lifeline

Harness and lifeline must be in good condition and properly attached.

3.3.2.4 Gum or Tobacco Chewing

Ensure that gum or tobacco chewing is prohibited.

3.3.2.5 Physical Defects or Injuries

Ensure that people have no physical defects or injuries which may prevent their wearing respirators or which may cause rescue to be difficult. No beards, sideburns, or large mustaches will be allowed on people who must wear respirators.

3.3.2.6 Alcoholic Beverages and Drugs

Ensure that people entering the tank are not under the influence of alcoholic beverages and drugs.

3.3.2.7 Counseling on Reproductive Hazards

Ensure that all employees have been counseled on and fully understand the reproductive hazards related to work in contaminated areas or in leaded gasoline or chemically contaminated tanks since they may be seriously affected by organic lead compounds or other chemical contaminants.

3.3.2.8 Hazardous Areas

Check hazardous areas as defined in paragraph SURVEY OF HAZARDOUS AREAS.

3.4 TABLE OF TANK HISTORY

Tank Number	Tank Location	Tank Capacity	Date Constructed	Type of Lining (If Applicable)	Type of Fuel	Remarks from the Last Inspection

3.5 FUEL REMOVAL

All possible fuel will be pumped or otherwise removed from the tank by the Government. Consider remaining fuel contaminated or waste fuel; pump into 55 gallon drums or other suitable containers for disposal in accordance with approved procedures meeting host nation regulations. Dispose of remaining fuel emulsions in accordance with applicable host nation regulations. Drums or tanks used for containerizing waste fuel will be furnished by the Contractor.

3.6 TANK CLEANING

For the interior of tanks, the shell, bottom, columns, roof, rafters, and interior accessory system components such as pumps, piping, and ladders must be cleaned, free of rust, dirt, scale, loose material, fuel, oil, grease, sludge, and other deleterious substances. Do not damage sound existing lining material. Remove unsound or loose lining or coating, and clean surfaces which became exposed to bare substrate. Immediately notify the Contracting Officer if lining or coating is deteriorated or loose.

3.6.1 Monitoring

Monitoring of airborne concentrations of lead must be in accordance with 29 CFR 1910.1025 of benzene in accordance with 29 CFR 1910.1028, and as specified herein. Air monitoring, testing, and reporting must be performed by a CIH or an Industrial Hygiene (IH) Technician who is under the direction of the CIH.

- The CIH or the IH Technician under the direction of the CIH must be on the jobsite directing the monitoring, and inspecting the work to ensure that the requirements of the Contract have been satisfied during the entire operation.
- Take personal air monitoring samples on employees who are anticipated to have the greatest risk of exposure as determined by the CIH. In

addition, take air monitoring samples on at least 25 percent of the work crew or a minimum of two employees, whichever is greater, during each work shift.

- c. Submit results of air monitoring samples, signed by the CIH, within 2 working days after the air samples are taken. Notify the Contracting Officer immediately of exposure to lead at or in excess of the action level of 30 micrograms per cubic meter of air outside of the lead control area, and 0.5 ppm for benzene.

3.6.2 Water Removal and Disposal

Pump or otherwise remove water from the tank. Ensure that the sludge and sediment are not pumped out or mixed with the water. There are 500 gallons of nonhazardous water that must be packaged, labeled, stored, transported, treated, and disposed of in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, and 40 CFR 266.

3.6.3 Sludge and Sediment Removal and Disposal

Squeegee or brush any sludge, sediment, or other loose material into piles, shovel into buckets or other suitable containers, and remove from the tank.

3.6.3.1 Sludge Disposal Using Landfill

There are 10 barrels of nonhazardous sediment and sludge in the tank that can be disposed of in sanitary landfill.

3.6.4 Washing

After water, fuel, and sludge have been removed, thoroughly wash the tank interior. Contractor must limit the water pressure during washing so that it does not cause damage to the existing coating. Contractor must clean a representative test panel with their planned washing procedure. Test panel access must be provided to and examined by the Contracting Officer to determine whether coating damage is occurring as a result of the Contractor's washing procedure. Contractor must modify water temperature and pressures based on cleaning test panel results. Maximum allowable pressure on coated surfaces must be 1380 Kpa and maximum allowable temperature of wash water must be 57 degree C. Prior to cleaning the tank, tests must be conducted to determine the minimum distance of the nozzle to the steel to prevent damage to the tank coating. Perform Quality Control to inspect the cleaning during the process to ensure the coating is not being removed. Minimize the use of water; substitute brush blasting when practical. Start washing at the top of the walls and columns and work down to the floor. Wash the floor last starting from the sides and working towards the sump. Wash to remove oil, sludge, wax, tar, and other fuel residue adhering to the surface. Wash by any one or a combination of the following methods:

- a. Use only fresh water under pressure.
- b. Apply a detergent conforming to FS 0-D-1276 by spray or brush and soak approximately 30 minutes.
- c. Hand-scrub the surfaces vigorously with long-handled stiff-bristle brushes. Wet the brushes intermittently during scrubbing process. For heavily oil-soaked areas which still appear to retain some residue

after first scrubbing, scrub until clean.

- d. Rinse the surfaces thoroughly with fresh water.
- e. Brush-off blast clean.

3.6.5 Wash Water, Detergent Solution, and Sediment Removal

During the washing process, operate a portable pump continuously with suction hose extended to the tank bottom to remove water, dirt, oil, or other loose materials washed off. Following the final rinse, pump, squeegee, and mop the tank dry.

- a. Prior to discharge or disposal, test the wash water, sediment, and sludge in accordance with paragraph WATER, SEDIMENT, AND SLUDGE ANALYSIS. The Contractor must furnish temporary tanks to hold water and detergent solution until testing is completed.
- b. The wash water is nonhazardous and can be disposed of in accordance with paragraph WATER, SEDIMENT, AND SLUDGE ANALYSIS.
- c. For bidding purposes, assume that the sediment is nonhazardous and can be disposed of in a sanitary landfill.

3.6.6 Disposal of Used Blasting Abrasive

Test used abrasive in accordance with 40 CFR 261 to determine if it is a hazardous waste using the EP toxicity test for metals. Handle and dispose of abrasive determined to be hazardous waste in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, and 40 CFR 266. Dispose of abrasive which is not hazardous waste at a landfill off Government property in accordance with applicable regulations. The contract price will be adjusted if the used abrasive is determined to be hazardous waste. However, payment for disposal of hazardous waste will not be made until a completed manifest from the treatment or disposal facility is returned, and a copy furnished to the Government.

3.6.7 Special Precautions

Special Precautions for Tanks with Pipe Columns and Braces:

- a. Pipes used for columns and braces are a potential source of explosive vapors even after the tank is cleaned. The tank may be determined to be vapor free below 4 percent of lower explosive limit; but after one or two hours, explosive readings must again be obtained from these sources. The Contractor must take readings at least every half hour when working in tanks after they have been cleaned and each floating roof or pan pontoon must be checked individually with a combustible gas indicator.

3.7 FINAL CLEAN-UP

After the Contracting Officer has inspected and accepted the tank cleaning and before final inspection, accomplish the following work:

3.7.1 Stenciling Tank

Stencil on the tank in 20 mm letters adjacent to the manhole openings the following data:

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Date Cleaned -

Contractor Name -

Address -

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 05 23.13

UTILITY HORIZONTAL DIRECTIONAL DRILLING

11/19

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY CONTROL
 - 1.3.1 Qualifications
 - 1.3.2 Safety
 - 1.3.3 Horizontal Directional Drilling Plan
 - 1.3.3.1 Layout Plan
 - 1.3.3.2 Utility Profile
 - 1.3.3.3 Equipment List
 - 1.3.3.4 Drilling Fluid Management Plan
 - 1.3.3.5 Pedestrian Access
 - 1.3.3.6 Method and Procedures
 - 1.3.3.7 Safety Data Sheets
 - 1.3.3.8 Revisions
 - 1.3.4 Steel Pipe Welding
 - 1.3.5 Pipe Welder Qualifications
- 1.4 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

- 2.1 EQUIPMENT
 - 2.1.1 Drill Rod
- 2.2 MATERIALS
 - 2.2.1 Pipe
 - 2.2.1.1 Carbon Steel
 - 2.2.2 Drilling Fluids
 - 2.2.3 Additives

PART 3 EXECUTION

- 3.1 EXAMINATION
- 3.2 INSTALLATION
 - 3.2.1 Drill Set-Up
 - 3.2.1.1 Drilling Fluids
 - 3.2.2 Drill Entrance and Exit Pits
 - 3.2.3 Drill Entrance and Exit Angle
 - 3.2.4 Pilot Hole
 - 3.2.5 Guidance Systems
 - 3.2.6 Reaming
 - 3.2.7 Pull Back
 - 3.2.8 Drilling Fluids Disposal
 - 3.2.9 Connection of Product Pipe to Pipeline
- 3.3 FIELD QUALITY CONTROL
 - 3.3.1 Daily Work Log

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

3.3.2 Drill Logs
3.3.3 Field Tests
3.4 CLOSEOUT ACTIVITIES

-- End of Section Table of Contents --

SECTION 33 05 23.13

UTILITY HORIZONTAL DIRECTIONAL DRILLING
11/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO T 180 (2017) Standard Method of Test for
Moisture-Density Relations of Soils Using
a 4.54-kg (10-lb) Rammer and a 457-mm
(18-in.) Drop

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926.652 Safety and Health Regulations for
Construction; Subpart P, Excavations;
Requirements for Protective Systems

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section **01 33 00** SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Statement of **Qualifications** and Records; **G**
Horizontal Directional Drilling Plan; G

SD-03 Product Data

Pipe; G
Drilling Fluids; G
Additives; G

SD-05 Design Data

Secondary Containment Plan; G

SD-06 Test Reports

Soil Test Data

SD-07 Certificates

Drill Rod
Pipe Welder Qualifications

SD-11 Closeout Submittals

Record Drawings

Complete Work Logs of Guided Directional Drill Operations

1.3 QUALITY CONTROL

1.3.1 Qualifications

Ensure that the field supervisor and workers assigned to this project are experienced in work of this nature and have successfully completed similar projects of similar length, pipe type, pipe size, and soil type using directional drilling in the last three (3) years. As part of the bid submission, submit project descriptions which include, at a minimum, a listing of the location(s), date of project(s), owner, pipe type and material, size installed, length of installation, manufacturer of equipment used, and other information relevant to the successful completion of the project.

1.3.2 Safety

Include in directional drilling equipment machine safety requirements a common grounding system to prevent electrical shock in the event of underground electrical cable strike. Ensure the grounding system connects all pieces of interconnecting machinery; the drill, mud mixing system, drill power unit, drill rod trailer, operator's booth, worker grounding mats, and any other interconnected equipment to a common ground. Equip the drill with an "electrical strike" audible and visual warning system that notifies the system operators of an electrical strike.

1.3.3 Horizontal Directional Drilling Plan

Provide a plan prepared, signed, and sealed by a licensed Professional Engineer. Submit supporting calculations, certifications, and material product data demonstrating the strength of the product pipes for acceptance before the beginning of the installation. Demonstrate that the proposed material satisfies the purpose of the utility and withstands the design and construction stresses and pressures. The HDD Plan will include the following:

1.3.3.1 Layout Plan

Provide a plan location of the operation, including entry and exit points, discussing the relationship of the equipment, pipe assembly, and staging areas.

1.3.3.2 Utility Profile

Provide a profile of the utility plotted at a scale appropriate for the work.

1.3.3.3 Equipment List

Provide a directional drilling equipment list including: drilling rig, drill bit, back-reamer, mud mixing and pumping systems, down-hole tools, guidance system, and rig safety system. Provide calibration records for guidance system.

1.3.3.4 Drilling Fluid Management Plan

Provide a drilling fluid management plan to include drilling fluid types and specifications, cleaning and recycling equipment, estimated flow rates, procedures for minimizing drilling fluid escape, and the method/location for final disposal of waste drilling fluids. Provide a frac out control plan, including frac control materials that will be onsite and contact information for emergency personnel.

1.3.3.5 Pedestrian Access

When and where installations disrupt pedestrian use of sidewalk for periods exceeding two consecutive days, provide an alternate route that meets current ADA requirements.

1.3.3.6 Method and Procedures

Provide an outline of the methods and procedures, describing the pilot hole drilling procedure, the reaming operation, and the pullback procedure, including drawings, schedule of operations, specifications, and method of operation. Include pipe storage and handling details and pipeline assembly and installation procedures.

1.3.3.7 Safety Data Sheets

Submit safety data sheets for fluids and additives.

1.3.3.8 Revisions

If site conditions change and require modification to the HDD Plan, submit revised drilling plan to achieve successful installation. Explain, in the revised submittal, the anticipated and encountered conditions that mandated the change in plans.

1.3.4 Steel Pipe Welding

Pipe welding operations, procedures, qualifications, and performance, test reporting procedures, and visual inspections must meet the requirements of Section 33 52 23.15 POL SERVICE PIPING WELDING paragraph QUALIFICATIONS. Pipe pressure and exterior coating testing procedures shall be per Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT) paragraph TESTING prior to being pulled into place per HDD procedures.

1.3.5 Pipe Welder Qualifications

Pipe welder shall be qualified to weld steel pipe of the materials, thicknesses, and diameters specified on this project per the requirements of Section 33 52 23.15 POL SERVICE PIPING WELDING paragraph QUALIFICATIONS.

1.4 DELIVERY, STORAGE, AND HANDLING

Inspect materials delivered to the site for damage. All materials found during inspection or during the progress of work to have cracks, flaws, surface abrasions, or other defects will be rejected. Remove defective materials from the job site.

Protect stored piping from moisture and dirt and place on level surface.

PART 2 PRODUCTS

2.1 EQUIPMENT

2.1.1 Drill Rod

Select the appropriate drill rod to be used. Submit certified statement that the **drill rod** has been inspected and is in satisfactory condition for its intended use.

2.2 MATERIALS

2.2.1 Pipe

2.2.1.1 Carbon Steel

Install **250 mm - 500 mm** carbon steel pipes as and where shown in the plans, with a factory applied coating, including abrasion resistant top coat, meeting the requirements of Section **33 52 40** FUEL SYSTEMS PIPING (NON-HYDRANT) paragraphs MATERIALS and CARBON STEEL PIPING. Piping shall meet or exceed the thickness and pressure class of the existing pipes of like diameter and use.

Pipes segments shall be joined by fluid tight, full circumferential welds per paragraph STEEL PIPE WELDING that are capable of withstanding horizontal directional drilling pullback forces, expected internal fluid pressures and external soil loadings without failure or leakage.

Where the factory applied exterior coating of the pipe is damaged due to welding or handling, the damaged areas shall be cleaned and prepared per coating manufacturer's instructions and field re-coated per the requirements of the original coating.

The interior of the lowest 100 m of HDD pipe shall be factory coated with material meeting the requirements of Section **33 52 40** FUEL SYSTEMS PIPING (NON-HYDRANT) paragraph CARBON STEEL PIPING. Where the interior coating of the pipe is damaged due to welding, the damaged areas shall be remotely cleaned and prepared per coating manufacturer's instructions and remotely re-coated per the requirements of the original coating.

All the interior and exterior pipe recoating work shall be done prior to installing the pipe.

2.2.2 Drilling Fluids

Use a high quality bentonite drilling fluid to ensure hole stability, cuttings transport, bit and electronics cooling, and hole lubrication to reduce drag on the drill pipe and the product pipe. Use only fluid with a composition which complies with all Spanish national, regional and local environmental regulations.

2.2.3 Additives

Use admixtures as required to address soil conditions and water conditions such as water hardness, acidity, and alkalinity.

PART 3 EXECUTION

3.1 EXAMINATION

a. Soil Test Data

Provide written documentation of conformance with [AASHTO T 180](#).

3.2 INSTALLATION

Ensure all utilities are located and clearly marked prior to start of excavation or drilling.

3.2.1 Drill Set-Up

Design and construct the drill entrance and exit pits.

3.2.1.1 Drilling Fluids

Mix the bentonite drilling fluid with potable water (of proper pH) to ensure no contamination is introduced into the soil during the drilling, reaming, or pipe installation process. Make any required additive adjustments.

3.2.2 Drill Entrance and Exit Pits

Drill entrance and exit pits are required. Maintain at minimum size to allow only the minimum amount of drilling fluid storage prior to transfer to mud recycling or processing system or removal from the site.

Do not allow drilling mud to flow freely on the site or around the entrance or exit pits. Remove spilled mud and restore ground to original condition. Provide shore pits in compliance with OSHA Standards, [29 CFR 1926.652](#).

Drilling near wetlands or water courses requires secondary containment to prevent drilling fluids from entering the wetlands. Secure written approval of a [secondary containment plan](#) from the Contracting Officer.

3.2.3 Drill Entrance and Exit Angle

Ensure entrance and exit angles and elevation profile maintains adequate cover to reduce risk of drilling fluid breakouts and ground exit occurs as specified herein. Ensure that entrance and exit angles generate pullback forces that do not exceed 7.5 percent strain on the pipe.

3.2.4 Pilot Hole

The type and size of the pilot string cutting head and the diameter of the drill pipe are at the Contractor's discretion.

Drill the pilot hole along the path shown on the plan and profile drawings. Pilot hole tolerances are as follows:

- a. Vertical Tolerance: Provide minimum cover below channel bottom as specified on the plans. Pilot hole may go deeper if necessary to prevent breakout.
- b. Horizontal Tolerance: Plus or minus - [152.4 cm](#) from the centerline of

the product pipe.

- c. Curve Radius: No curve is acceptable with a radius less than 1.2 m x the pipe diameter in mm.
- d. Entry Point Location: Make pilot hole entry point within plus or minus - 152.4 cm of the location shown on the drawings or as directed by the Contracting Officer in the field.
- e. Exit Point Location: Make the exit point location within plus/minus - 152.4 cm of the location shown on the drawings or as directed by the Contracting Officer in the field.
- f. Mandatory pipeline cover requirements are as shown on the drawings or as specified.

3.2.5 Guidance Systems

Walkover guidance systems are not acceptable for this project; use a gyroscopic guidance module and an at bit inclination assembly to track the initial pilot bore for each group of parallel pipes. Subsequent parallel bores can be guided by AC wire guidance run through the initial product pipe. Ensure proper calibration of all equipment before commencing directional drilling operation.

3.2.6 Reaming

Conduct reaming operations at the Contractor's discretion. Determine the type of back reamer to be utilized by the type of subsurface soil conditions that are encountered during the pilot hole drilling operation. The reamer type is at the Contractor's discretion.

3.2.7 Pull Back

Pre-assemble all of the longest segments of the entire pipeline to be installed via direction drill that can practicably be laid out in the available area prior to commencement of pull back operations minimizing welds that are needed during pullback.

Support the pipeline during pullback operations in a manner to enable it to move freely and prevent damage. Install the pipeline in one continuous pull when possible.

Minimize torsion stress by using a swivel to connect the pull section to the reaming assembly.

Maximum allowable tensile force imposed on each pull section is not to exceed 90 percent of the pipe manufacturer's safe pull (or tensile) strength for the specific pipe diameter.

Minimize external pressure during installation of the pullback section in the reamed hole. Replace damaged pipe resulting from external pressure at no cost to the Government. Buoyancy modification is at the discretion of the Contractor.

3.2.8 Drilling Fluids Disposal

Collect drilling fluid returns in the entrance pit, exit pit, or spoils recovery pit. Immediately clean up any drilling fluid spills or overflows

from these pits.

Dispose of fluids in a manner that is in compliance with all permits and applicable Spanish national, regional and local regulations. Disposal of the drilling fluids may occur on approved land owned by the Government subject to written approval from the Contracting Officer. Spread the drilling slurry over the Government-approved disposal area and plow into the soil.

Conduct disposal in compliance with all relative environmental regulations, right-of-way and work space agreements, and permit requirements.

3.2.9 Connection of Product Pipe to Pipeline

After the product pipe has been successfully installed, allow the product pipe to recover for 24 hours prior to connection of the pipeline. Ensure that a sufficient length of the product pipe has been pulled through the hole so that the pull-nose is not pulled back into bore hole due to stretch recovery of the product pipe.

3.3 FIELD QUALITY CONTROL

3.3.1 Daily Work Log

Maintain a work log of construction events and operations including, but not limited to, the following for each day's work:

- a. Hours worked.
- b. Log of each drill rod added or withdrawn during drilling, reaming, and pull back.
- c. Groundwater control operations.
- d. Description of soil conditions encountered.
- e. Tools and equipment in use, drilling fluid, fluid pumping rate, and drilling head location.
- f. Any unusual conditions or events.
- g. Reasons for operational shutdown in event work is halted.

3.3.2 Drill Logs

Maintain drilling logs that accurately provide drill bit location (both horizontally and vertically) at least every 5.1 cm along the drill path. In addition, keep logs that record, as a minimum the following, every 15 minutes throughout each drill pass, back ream pass, or pipe installation pass:

- a. Drilling Fluid Pressure
- b. Drilling Fluid Flow Rate
- c. Drill Thrust Pressure
- d. Drill Pullback Pressure

e. Drill Head Torque

Make all instrumentation, readings, and logs available to the Contracting Officer at all times during operation.

3.3.3 Field Tests

Perform field tests and provide labor, equipment, and incidentals required for testing. Submit test results, identifying any results that do not meet requirements, to the Contracting Officer within four days of test completion. Provide corrective action and retest pipe not meeting requirements. Provide corrective action as recommended by the pipe manufacturer and subject to approval by the Contracting Officer.

3.4 CLOSEOUT ACTIVITIES

Immediately upon completion of work, remove all rubbish and debris from the job site. Remove all construction equipment and implements of service leaving the entire area involved in a neat condition acceptable to the Contracting Officer.

Immediately clean "blow holes" or "breakouts" of drilling fluid to the surface and return the surface area to its original condition. Dispose of all drilling fluids, soils, and separated materials in compliance with Spanish national, regional, and local environmental regulations.

Provide a post-construction welding report conforming to Section 33 52 23.15 POL SERVICE PIPING WELDING paragraph QUALIFICATIONS.

Submit an electronic copy and three hard copies of the [record drawings](#) to the Contracting Officer within five days after completing the pull back. Include in the record drawings a plan, profile, and all information recorded during the progress of the work. Clearly tie the record drawings to the project's survey control. Maintain, and submit upon completion, signed [complete work logs of guided directional drill operations](#).

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 08 55.01

FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT)

08/19

PART 1 GENERAL

- 1.1 SUMMARY/APPLICABILITY
- 1.2 REFERENCES
- 1.3 ADMINISTRATIVE REQUIREMENTS
 - 1.3.1 Plans
 - 1.3.2 Copies of API Publication
 - 1.3.3 Existing System Cleanliness Confirmation Plan
 - 1.3.4 Fuel Testing Laboratory Qualifications
 - 1.3.5 System Start-up Plan
 - 1.3.6 Performance Testing Plan
 - 1.3.7 Fuel Provisioning Plan
 - 1.3.8 Certification (Ready for Start-Up and Performance Testing)
- 1.4 SUBMITTALS
- 1.5 CLOSEOUT SUBMITTALS
 - 1.5.1 Final Reports
- 1.6 QUALITY ASSURANCE
 - 1.6.1 Certification of Entire System
 - 1.6.2 Service Headquarters Definition
- 1.7 SYSTEM SUPPLIER INVOLVEMENT
- 1.8 DISPOSAL OF WASTE MATERIALS

PART 2 PRODUCTS

- 2.1 GOVERNMENT-FURNISHED MATERIAL AND EQUIPMENT
 - 2.1.1 Fuel
- 2.2 CONTRACTOR-FURNISHED MATERIAL AND EQUIPMENT
 - 2.2.1 Contractor-furnished
 - 2.2.2 Design Conditions

PART 3 EXECUTION

- 3.1 SEQUENCE OF EVENTS
- 3.2 PRELIMINARY REQUIREMENTS
 - 3.2.1 Safety
 - 3.2.2 Electrical Preparations
 - 3.2.3 Emergency Fuel Shutoff (EFSO) System Testing
 - 3.2.4 Storage Tanks
 - 3.2.5 Piping System
 - 3.2.6 Existing System Cleanliness Confirmation
- 3.3 PREPARATIONS FOR FLUSHING
 - 3.3.1 Protection of System Components
 - 3.3.2 Strainers
 - 3.3.3 Water Draw-off
- 3.4 INITIAL FUEL RECEIPT INTO STORAGE TANK
 - 3.4.1 General

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.4.2 Storage Tanks
- 3.4.3 Components
- 3.4.4 Fuel Quality
- 3.4.5 Fuel Receipt
 - 3.4.5.1 Fuel Receipt from Piping System
- 3.4.6 Initial Low Point Flush
- 3.4.7 Storage Tank Hydrostatic Tests
- 3.4.8 Storage Tank Tightness Tests
- 3.4.9 Piping Hydrostatic Tests
- 3.4.10 Piping Soak Tests
- 3.5 INITIAL FUEL RECEIPT DIRECTLY INTO PIPING SYSTEM
 - 3.5.1 General
 - 3.5.2 Fuel Quality
 - 3.5.3 Fuel Receipt
 - 3.5.3.1 Fuel Receipt from Existing Piping System
 - 3.5.4 Initial Low Point Flush
 - 3.5.5 Piping Hydrostatic Tests
 - 3.5.6 Piping Soak Tests
- 3.6 FLUSHING
 - 3.6.1 Flushing Requirements
 - 3.6.2 Fueling System Piping
 - 3.6.2.1 General Fuel Lines
 - 3.6.2.2 Receipt or Transfer Line
 - 3.6.2.3 Installation Pipeline
 - 3.6.2.4 Water Drain Tank Lines
 - 3.6.2.5 High Point Vent Lines, Low Point Drain Lines, Thermal Relief Piping, Instrumentation Piping and Tubing
 - 3.6.3 Piping Flushing Checklist
- 3.7 CLEANING
 - 3.7.1 Cleaning Requirements
 - 3.7.2 Storage Tank Piping
 - 3.7.3 High Point Vent Lines, Low Point Drain Lines, Thermal Relief Piping, Instrumentation Piping and Tubing
 - 3.7.4 Piping Cleaning Checklist
- 3.8 CONTROL VALVE ADJUSTMENT
 - 3.8.1 Rate of Flow Control Feature on Fueling Pump Non-Surge Check Valve
- 3.9 EQUIPMENT TESTS
 - 3.9.1 Control System and Control Valves
 - 3.9.2 Tank Level Alarms
 - 3.9.3 Fueling Pump Operation
 - 3.9.4 Emergency Shutdown
 - 3.9.5 Overfill Valve
- 3.10 PERFORMANCE TESTS
 - 3.10.1 Final Performance Test
 - 3.10.2 Satisfactory Performance
 - 3.10.3 Performance Testing Plan
 - 3.10.4 Equipment Tests
 - 3.10.5 Control Valve Tagging
 - 3.10.6 Final Acceptance
 - 3.10.6.1 Tank High Liquid Level Shut-Off Valve Test and Adjustments
 - 3.10.6.2 Tank Level Indicator Adjustments

-- End of Section Table of Contents --

SECTION 33 08 55.01

FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT)
08/19

PART 1 GENERAL

1.1 SUMMARY/APPLICABILITY

This project consists of two separate fueling components: an F-24 storage tank system and a JP-5 tanker pipeline. This specification covers the start-up and commissioning of the F-24 storage tank system. For start-up and commissioning of the JP-5 pipeline system, see Section 33 08 55.02. This specification defines the requirements and procedures for startup and performance testing of all equipment, components, control systems, devices, and associated appurtenances which are used for the receipt, storage, transfer and issue of petroleum fuel products for non-hydrant fuel facility systems. It covers requirements for safety, Government scheduling and coordination, device testing, existing and new system cleanliness, system flushing and cleaning, demonstration of indicated and specified system performance, and final acceptance and reporting. The types of fuel systems covered include:

- a. F-24 cut-and-cover Storage Tank, Additive Injector, and associated Piping System.

It does not cover private vehicle fueling stations such as exchange service stations, nor aircraft direct refueling hydrant systems.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API RP 1595 (2012; R 2019; 2nd Ed) Design, Construction, Operation, Maintenance, and Inspection of Aviation Pre-Airfield Storage Terminals

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1110-1-8167 (2016) Engineering and Design -- Petroleum, Oil, and Lubricants Mandatory Center Of Expertise

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-3004-1 (2018) Quality Assurance for Bulk Fuels, Lubricants and Related Products

1.3 ADMINISTRATIVE REQUIREMENTS

Contractor property management system shall comply with FAR 52.245-1 Government Property. Contractor shall confirm availability of required

Government-furnished material and equipment to support System Start-Up and incorporate into proposed schedule submitted with bid. Contractor shall notify Government no later than 180 days after notice-to-proceed if any Government-furnished material or equipment required for System Start-Up are not available. See Paragraph "System Start-up Plan" below for additional requirements. The contractor is required to notify the Government within the solicitation period and or the first 180 days after NTP that any element of the project specifications that are not present to have a complete system commissioning. Costs for equipment and materials required for System Start-Up that are not identified within specific period shall be borne by Contractor. Contractor is requested for acknowledgement of these requirements in their proposed schedule and/or bid documentation to be submitted.

1.3.1 Plans

The Contractor must submit detailed written plans (prepared by the System Supplier) for implementation of the System Start-Up Plan, the Existing System Cleanliness Confirmation Plan, the [Fuel Provisioning Plan](#), and the [Performance Testing Plan](#). The plans must specify a detailed plan incorporating in a sequential manner all work specified in PART 3 EXECUTION of this specification section. Plan elements must include:

- a. Personnel. List of Contractor's personnel by trade, list of key personnel, list of safety equipment, list of miscellaneous equipment such as two-way radios, and personnel transportation vehicles.
- b. Fluid Used for Hydrotesting. Identify what fluid will be used for each operation in each system section.
- c. Fuel. See the Fuel Provisioning Plan.
- d. Storage-Into. Identify where the contents of the piping will be pushed into including the capacity required, how the storage vessel will be prepared for receiving the piping content, how it will be emptied and how it will be cleaned after the contents are removed.
- e. Storage-From. Identify where the fluid that will be drained or pushed into the piping will come from including the capacity required, the number, size and type of the storage vessel, how the storage vessel is filled, how it will be emptied, and how it will be cleaned after the contents are removed.
- f. Piping. Notwithstanding the requirements described elsewhere, every pipe section and individual component in the system covered by this specification section must be filled, vented, hydrostatically tested, flushed, cleaned, and tested for fuel quality. It must also be able to be isolated and drained if found that it needs repair. The Plans must detail how this is to be done.
- g. Equipment Tests. Detailed procedures and schedules for each system component to perform all system tests under each operating scenario in accordance with paragraph EQUIPMENT TESTS.
- h. Performance Testing. Detailed procedures and schedules for each system component to perform all system tests under each operating scenario in accordance with paragraph PERFORMANCE TESTING.
- i. Hydrotesting Plans. See the Work Plan in Section [33 52 40 FUEL](#)

SYSTEMS PIPING (NON-HYDRANT).

- j. Schedule. Schedules must be generated with listing dates and durations of all system start-up activities as well as regular coordination and safety meetings and dates of key events for Government participation.
- k. Contingency plans. Information on spill and fire contingencies, along with the required Government Fire and Safety Office involvement and approvals.
- l. Coordination with Installation. Description of how Contractor and System Supplier must implement system start-up in coordination with ongoing operations at the Installation. Plan must incorporate all phasing and work restriction requirements of the Contract documents.

1.3.2 Copies of API Publication

Provide four copies of API RP 1595 to the Contracting Officer.

1.3.3 Existing System Cleanliness Confirmation Plan

Submit a detailed written plan prepared by the Contractor or System Supplier for the Contractor to verify and confirm the cleanliness of the existing system before connecting it to the new system or extension. The cleanliness confirmation test of the existing system (pre-cleanliness test) must be per MIL-STD-3004-1. Submit the plan 90 days prior to making the connection to the existing system. The Contractor and System Supplier are responsible for implementing the Existing System Cleanliness Confirmation Plan in coordination with ongoing base operations.

1.3.4 Fuel Testing Laboratory Qualifications

Contractor must submit the qualifications of the DoD Fuel Testing Laboratory which will be used to test the fuel samples.

1.3.5 System Start-up Plan

Submit a detailed written plan prepared by the Contractor and System Supplier for implementation of System Start-Up. A preliminary start-up plan must be submitted as a preconstruction item. Submit a finalized plan, including any changes needed due to project scope modification, a minimum of 180 (OCONUS) days prior to System Start-Up and no later than 180 days after notice to proceed. The Contractor and System Supplier are responsible for implementing System Start-Up in coordination with ongoing operations.

1.3.6 Performance Testing Plan

Submit a detailed written plan prepared by the Single System Supplier for implementation of Performance Testing. Submit the plan 180 days prior to System Start-Up. The Contractor and the System Supplier are responsible for implementing the Performance Testing Plan in coordination with ongoing operations.

1.3.7 Fuel Provisioning Plan

Submit a detailed written Fuel Provisioning Plan prepared by the Contractor. Submit the plan 90 days prior to receipt of fuel. The Fuel

Provisioning Plan must include a timeline for the required receipt date(s), grade of fuel, and quantity of fuel on each receipt date required to execute the Start-up and the Performance Testing plans.

1.3.8 Certification (Ready for Start-Up and Performance Testing)

As a prerequisite to fuel system start-up, the Contractor must submit a certificate that certifies all work provided on the fuel system, except for touch-up field painting, has been inspected and approved by the specified approving authorities. Further, the Contractor must certify on this certificate that all specified checks and inspections have been successfully completed prior to start-up. Submit the plan 45 notice prior to commencement of fuel system start-up. The Contractor must submit the Certificate of Completion to the Contracting Officer at least 7 calendar days prior to commencement of system start-up and performance testing. The Contracting Officer must then be responsible for scheduling the Government representatives and appropriate military command authority and designers for participation in the inspection, performance testing, and final approval activities. Any contractual deficiencies observed must be corrected by the Contractor without cost to the Government.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Copies of API RP 1595; G

Existing System Cleanliness Confirmation Plan; G

System Start-Up Plan; G

Performance Testing Plan; G

Fuel Provisioning Plan; G

SD-06 Test Reports

Final Reports; G

Cathodic Protection Reports; G

Equipment Tests; G

Piping Flushing Checklist; G

Piping Cleaning Checklist; G

SD-07 Certificates

Fuel Testing Laboratory Qualifications; G

Certification (Ready for Start-Up and Performance Testing); G

Certificate of Calibration; G

Disposal Of Waste Materials

SD-11 Closeout Submittals

Certification of Entire System

1.5 CLOSEOUT SUBMITTALS

1.5.1 Final Reports

Contractor must prepare a final report that documents the execution of the approved start-up and performance testing plan. All items of work specified in the Start-up and Performance Testing plans must be carried out and reported in this report unless otherwise approved by the Contracting Officer. Include as a part of this report:

- a. Verification letters of approved fuel storage tank hydrostatic tests and the approved piping hydrostatic tests, as generated under other specification sections.
- b. Final settings of the valves and switches.
- c. A copy of the flow and pressure output graphs and excel data and charts on CDR media with an explanation of what the graph indicates and what the system is doing.
- d. A schematic flow diagram drawing of the resultant system showing major valves with the "normal" position of each manual valve noted, equipment, and the type, location, and setpoints of each safety relief valve and thermal relief valve.
- e. A certified pipeline inventory - a detailed list with sizes, lengths, quantity, and volumes must be provided for the systems in this project. Such systems include, but are not limited to: marine receipt, pipeline receipt, truck off-loading receipt, pumphouse, pump pad, truck loading, marine loading, transfer pipeline, and other miscellaneous piping systems. Fuel system volume must be calculated using as constructed pipe lengths, internal diameters, fittings, and components. Totals must be provided for all items containing fuel with the exception of tanks which is covered by other specifications.

1.6 QUALITY ASSURANCE

1.6.1 Certification of Entire System

Prior to the acceptance of the newly constructed system by the Government, all installed mechanical and electrical system components must be inspected and approved by the Contracting Officer. Provide the Contracting Officer 60 days notice in order to schedule the Installation, DLA and Service Headquarters and/or their designated appointee for participation in the inspection, Performance Testing, and approval. Any deficiencies observed must be corrected by the Contractor without cost to the Government.

1.6.2 Service Headquarters Definition

SME is defined as Service Headquarters Subject Matter Experts: Air Force - The Air Force Fuels Facilities Subject Matter Expert (AFCEC/COS); Army -

Headquarters, U.S. Army Corps of Engineers; POL-MCX Facilities Proponent (CECW-EC) through the Army Petroleum Center (APC); Navy/Marine Corps - NAVFAC POL Facility Subject Matter Expert (NAVFAC EXWC, CI11).
ER 1110-1-8167 Engineering and Design Petroleum, Oil, and Lubricants Mandatory Center of Expertise must be followed.

The Service Control Point (SCP) is defined as follows: Army - Army Petroleum Center; Air Force - Air Force Petroleum Office; Navy/Marine Corps - Naval Supply Systems Command - Energy.

1.7 SYSTEM SUPPLIER INVOLVEMENT

The System Supplier is defined in Specification **33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT**. The Contractor and the System Supplier must work together to prepare the Start-Up Plan, the Existing System Cleanliness Confirmation Plan, Equipment Testing Plan, Performance Testing Plan, and the Final Reports. They must both be present during all activities described in this section. The System Supplier must be responsible to the Contractor for scheduling all Contractor, sub-Contractor, and manufacturer's service personnel during activities described in this specification section.

1.8 DISPOSAL OF WASTE MATERIALS

The Contractor must be responsible for properly disposing of any sludge, debris, filtration elements, waste water, contact water, and waste fuel resulting from piping and tank flushing and cleaning activities. Comply with all applicable Installation, local and **Host Nation** for hazardous waste disposal.

PART 2 PRODUCTS

2.1 GOVERNMENT-FURNISHED MATERIAL AND EQUIPMENT

The Government will furnish the following materials, equipment and services during the performance of the work under this section.

2.1.1 Fuel

Some fuel loss should be expected. Fuel that is lost during flushing and cleaning, equipment testing, and performance testing is viewed as a Start-up and Performance Testing expense and should be budgeted for in Contractor estimates. As Basis of Bid, the Contractor must allow for 200 gallons of fuel losses in the bid.

The Contractor is responsible for the disposal of any waste product.

The Contractor must establish a Department of Defense Activity Address Code and must reimburse the US Government for any US Government fuel lost resulting from poor workmanship or contamination.

The Government will provide fuel given that the requirements are submitted with sufficient lead time to allow the US Government to secure the grade and volume required and make the necessary transportation requirements. The Contractor must provide a copy of the Start-up, Fuel Provisioning, and Performance Testing plans a minimum of 90 days in advance of System Start-up. When Defense Working Capital Fund (DLA Owned) fuel is used, provide a copy of the Start-up, Fuel Provisioning, and Performance Testing plans to DLA Energy at the same time.

FLUSHING, CLEANING, EQUIPMENT TESTS and PERFORMANCE TESTING: The flushing, cleaning, and testing phases will be identified separately in the start-up and Performance Testing plans. Fuel used for flushing and cleaning, and testing will become contaminated, the fuel must be isolated and is considered off-specification until the quality can be verified. Fuel volumes required for flushing and cleaning and testing should be minimized to limit costs, so estimates should be made accordingly. The Contractor will be responsible for disposing of any waste product.

Upon satisfactory completion of the flushing and cleaning operations, the Government will supply the additional quantities of fuel required to complete the other work under this section. This larger volume of fuel will not be delivered to the system until the Contractor has satisfactorily completed all work and, in particular, the cleaning and coating of the interior surfaces of the storage tanks and the removal of preservatives and foreign matter from those portions coming in contact with the fuel valves, pumps, filter-separators and other such system components.

2.2 CONTRACTOR-FURNISHED MATERIAL AND EQUIPMENT

2.2.1 Contractor-furnished

Provide material, equipment and labor not specified to be Government-furnished and required for proper System Start-Up of the system. Equipment being provided must be calibrated and the [certificate of calibration](#) be submitted to the Contracting Officer prior to use. Equipment must include but not be limited to the following:

- a. Temporary strainers.
- b. Pipe spools.
- c. Flow meters.
- d. Pressure gauges.
- e. Electronic sensors and recorders for pressure and flow recording are included in the Pump Control Panel (PCP). This equipment must be used to monitor and record the system during the "Equipment Tests" and "Performance Testing" portions of this Specification Section. Recorded data must be used by the Contractor and equipment factory representatives to achieve final control valve and equipment adjustments. Recorded data must include:
 - (1) Fueling pumps discharge pressures.
 - (2) Storage Tank levels.
 - (3) [Drain](#) Tank levels.
 - (4) Fueling pumps suction pressures.
- f. Vacuum Trucks for removing water and fuel from pits, drains, and other systems.
- g. Fuel Bowser for removing water and fuel from pits, drains, and other systems.

- h. Temporary filtration/strainers.

2.2.2 Design Conditions

Use temporary flushing lines and equipment that are equal in strength, stability, and materials to the associated permanent components. However, spools may be carbon steel. Additional design conditions must be as specified in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

PART 3 EXECUTION

3.1 SEQUENCE OF EVENTS

The following events take place in the Start-Up and Performance Testing of a system and their definitions:

- a. System Start-Up Phase: The System Start-Up phase encompasses all of the events described below up to the Performance Tests Phase.
- b. Preliminary Requirements: Confirms completion of the system before fuel is received including verification of the cleanliness of the existing piping system being connected to.
- c. Preparations for Flushing: Activities that take place before initial receipt of fuel.
- d. Pneumatic Testing: Test for leaks in a piping system using air under pressure as the test medium. For requirements see Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- e. Hydrostatic Testing - Water: For requirements see Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- f. Initial Fuel Receipt: The point in time at which fuel is introduced into the system with all that implies for safety, spill prevention and control, and inventory control.
- g. Hydrostatic testing - Fuel: Use of fuel to test for leaks in a piping system using liquid under pressure as the test medium. For requirements see Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- h. Soak Testing: Perform "Soak Testing" of the piping system using fuel, as per Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- i. Flushing: In this context, refers to removing gross contamination (free water and solids) from a fuel system using fuel or water. Water is to be used for the water draw-off/water drain tank system only.
- j. Cleaning: In this context, refers to removing the remaining water and contamination from aviation fuel to the quality (allowable) levels suitable for issuing the final product using fuel as the cleaning medium.
- k. Control Valve Adjustment: Control valve adjustment so that equipment can be safely tested and adjusted.
- l. Equipment Tests: Tests performed by the Contractor and the System Supplier to confirm that individual components of the fuel system are

correctly installed and are operational.

- m. Performance Tests Phase: Government witnessed demonstration of the proper operation of the individual pieces of equipment and the system or systems as a whole.

3.2 PRELIMINARY REQUIREMENTS

All activities listed in paragraph PART 3 EXECUTION must be performed sequentially in the order they are presented. Prior to any on-site system start-up activities, the Contractor must ensure that all requirements of the paragraph SAFETY are satisfied. Project must be substantially complete, except for touch-up field painting, and Contractor's work area must be free of debris, trash and obstacles. Perform the following activities prior to receipt of fuel:

3.2.1 Safety

Prior to any on-site system start-up activities, the following safety procedures must be accomplished in all fueling areas to be started up and performance tested under this specification section: testing/operation of emergency showers and eyewash stations, placement of Contractor-provided portable eyewash units within 31 meters or 10 seconds from the fueling point, verification of proper grounding throughout system, coordination with Government Fire and Safety Office and Fuels personnel, placement of Contractor- or Government-provided spill pads and containment booms, placement of Contractor-provided fire extinguishers capable of extinguishing a fuel fire. Ensure that all radios/devices at all Class I, Division 1 areas are intrinsically safe.

3.2.2 Electrical Preparations

Prior to energizing the electrical system components, verify that short-circuit links have been removed from current transformer and that secondary circuits have been connected. Confirm that all tests required for fire detection and suppression systems have been performed and accepted. Verify all electrical transmitter connections and ensure proper calibration. Verify all electrical system components meet the classification and division as required by the design. Verify correct rotation of all motors prior to testing. Verify paddle type flow switches by physically actuating vanes and checking outputs. Conduit explosion-proof sealoffs must be poured after initial electrical checks but before fuel receipt.

3.2.3 Emergency Fuel Shutoff (EFSO) System Testing

Prior to initial fuel receipt, verify that each switch will trip the circuit breaker of the fuel pumps.

3.2.4 Storage Tanks

Ensure approved performance of storage tank integrity testing, hydrostatic tests, leak detection system, and coating application/inspection per the applicable specifications. Include verification letter of approved test results for information in the Final Reports. Ensure that tank interior is clean and free of any fuel-contaminating debris. Verify operation of tank level alarms. Ensure that certified strapping charts for all tanks are available for start-up personnel. API and STI tank inspections have been performed per applicable specifications and the reports are on hand prior

to commencement with the initial fill and Performance Testing. Verify correct orientation of internal tank inlet diffuser.

3.2.5 Piping System

Ensure that all piping weld integrity, leak detection systems, and coating inspections have been performed per the applicable specifications. Provide statement that required testing has been completed in the Final Reports. Evacuate all accumulated water from piping low point drains, valve cavities, and equipment drains. Verify all bolted connections are tightness tested to required torque using a calibrated torque wrench. Verify that all pressure gauges are properly calibrated, located and installed. Ensure that piping's cathodic protection system is tested, calibrated and operational. Ensure the cathodic protection inspections are performed per applicable specifications and the [cathodic protection reports](#) are on hand prior to initial fill and performance testing. Ensure that pipe marking and identification are provided as specified. Ensure that piping system thermal relief provisions are installed as designed. Verify the correct installation of piping, joints, and supports.

3.2.6 Existing System Cleanliness Confirmation

Contractor and System Supplier must conduct tests to confirm the cleanliness of the existing system before connecting it to the new system or an extension, upgrade or repair of the existing system. Confirm that the fuel in the existing system meets the intra-governmental receipt limits defined in [MIL-STD-3004-1](#). Sampling must be performed by the Government and testing must be done by the Service [or](#) a DoD regional fuels testing laboratory.

3.3 PREPARATIONS FOR FLUSHING

Upon completion of the system to the satisfaction of the Contracting Officer and the Service Headquarters and/or their designated appointee, make the following preparations for flushing the system.

3.3.1 Protection of System Components

The following items must be removed from the system prior to start of flushing operations and, where applicable, replaced with spools of pipe, diameter equal to the item removed.

- a. Control valves. This also includes control valve tubing exterior to the valve such as that which runs from tank level control valves up to tank float pilots.
- b. Sensors which are exposed to the fluid such as pressure gauges and thermometers.
- c. Meters.

After flushing, the above items must be reinstalled in the system and the spool sections turned over to the Contracting Officer.

3.3.2 Strainers

Ensure strainers are clean before flushing. Temporary 60 mesh strainers with minimum 300 percent open area must be provided in the suction line ahead of each fueling pump for the entire flushing operation. A temporary

strainer should be installed immediately upstream of the **drain tank** tank overflow valve. Any damaged permanent or temporary strainers must be replaced by the Contractor at no additional cost to the Government.

3.3.3 Water Draw-off

Remove any accumulated water from piping, equipment, and storage tanks' sumps and bottoms. Drain water and return fuel via filtration to storage tank. Repeat process until all water is removed. Dispose of petroleum contaminated water in accordance with Installation, local, and **Host Nation regulations**.

3.4 INITIAL FUEL RECEIPT INTO STORAGE TANK

3.4.1 General

Utilize one storage tank for initial fuel receipt to isolate contaminated fuel. Initial receipt of fuel must be done by gravity if possible. The Contractor must station personnel throughout piping system at high point vents to bleed air. All flanges and system components will be periodically inspected for leaks during filling procedures.

3.4.2 Storage Tanks

Receipt flow rate into an empty storage tank must not exceed **1 m/s** (FPS), as measured in the main receipt piping, until outlet of tank fill tube is submerged.

3.4.3 Components

Ensure that vessels are filled slowly by closing outlet valves and venting through air eliminators. Fill vessels using throttled downstream valves to maintain a packed condition in vessels throughout initial fill of piping system. Differential pressure across strainers must be continuously monitored. Any time a strainer differential pressure reaches **138 kPa**, it must be cleaned.

3.4.4 Fuel Quality

Fuel used during initial receipt must be considered contaminated and must be positively isolated, with blind flanges or closed, padlocked manual valves, from any active transfer, or aircraft fueling or tank truck loading operations. Fuel isolation must continue until all flushing and cleaning is completed. A method of checking fuel quality that is acceptable to the Contracting Officer must be provided at the beginning and end of any new piping section.

3.4.5 Fuel Receipt

3.4.5.1 Fuel Receipt from Piping System

Start-up personnel must meet with Government personnel in charge of existing fuel storage to discuss fuel transfer procedures. Topics must include: methods of communication to start/stop remote pumps; flow rate and head characteristics of pumps; methods of restricting initial receipt flow rate; methods of straining and filtering initial receipt fuel; accommodating multiple pump starts resulting from required strainer and filter cleaning operations; required quantity of fuel to be used. Contractor must provide a written summary of piping fill procedures to the

Contracting Officer.

3.4.6 Initial Low Point Flush

Perform an initial low point flush operation by flushing each low point drain through a portable basket strainer for 10 seconds at a system pressure of 207 kPa. Repeat flush until basket strainer collects no additional debris.

3.4.7 Storage Tank Hydrostatic Tests

Hydrostatically test the storage tanks per Specification Section 33 56 21.17 SINGLE WALL ABOVE GROUND FIXED ROOF STEEL POL STORAGE TANK. Upon completion of hydrostatic testing, perform "Soak Testing" of the tanks per API RP 1595. Duration of the test must be a minimum of 4 days and maximum of 7 days.

3.4.8 Storage Tank Tightness Tests

Perform tightness testing and manufacturer's tests of the storage tanks per Specification Section 33 56 10 FACTORY-FABRICATED FUEL STORAGE TANKS.

3.4.9 Piping Hydrostatic Tests

Hydrostatically test the piping system with fuel as required by Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).

3.4.10 Piping Soak Tests

Upon completion of hydrostatic testing with fuel, perform "Soak Testing" of the piping systems per API RP 1595. Duration of the test must be a minimum of 4 days and maximum of 7 days.

3.5 INITIAL FUEL RECEIPT DIRECTLY INTO PIPING SYSTEM

3.5.1 General

Initial receipt of fuel must be done by gravity if possible. The Contractor must station personnel throughout piping system at high point vents to bleed air. All flanges and system components will be periodically inspected for leaks during filling procedures.

3.5.2 Fuel Quality

Fuel used during initial receipt must be considered contaminated and must be positively isolated, with blind flanges or closed, padlocked manual valves, from any active transfer, or aircraft fueling or truck loading operations. Fuel isolation must continue until all flushing and cleaning is completed. A method of checking fuel quality that is acceptable to the Contracting Officer must be provided at the beginning and end of any new piping section.

3.5.3 Fuel Receipt

3.5.3.1 Fuel Receipt from Existing Piping System

Start-up personnel must meet with Government personnel in charge of existing fuel system to discuss fuel transfer procedures. Topics must include: methods of communication to start/stop remote pumps; flow rate

and head characteristics of pumps; methods of restricting initial receipt flow rate; methods of straining and filtering initial receipt fuel; accommodating multiple pump starts resulting from required strainer and filter cleaning operations; required quantity of fuel to be used. Contractor must provide a written summary of piping fill procedures to the Contracting Officer.

3.5.4 Initial Low Point Flush

Perform an initial low point flush operation by flushing each low point drain through a portable basket strainer for 10 seconds at a minimum system pressure of 200 kPa. Repeat flush until basket strainer collects no additional debris.

3.5.5 Piping Hydrostatic Tests

Hydrostatically test the piping system with fuel as required by Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).

3.5.6 Piping Soak Tests

Upon completion of hydrostatic testing with fuel, perform "Soak Testing" of the piping systems per API RP 1595. Duration of the test must be a minimum of 4 days and maximum of 7 days.

3.6 FLUSHING

Flushing procedures remove gross particulate and water contaminants and must precede cleaning procedures. All piping including but not limited to receipt piping, issue piping, transfer piping, supply and return lines to the storage tanks, water drain lines, thermal relief piping, high point vents, low point drains, control valve tubing on the control valves, control valve tubing exterior of the control valves, must be flushed with fuel.

In the event flushing identifies the presence of clay or soil like material in the system, Contractor must immediately notify Contracting Officer for direction.

3.6.1 Flushing Requirements

Begin flushing of fuel system piping at low flow rates using one pump in multiple pump systems at 25 percent of rated flow. Slowly increase flushing flow rate to full flow capacity of 75.7 lps. Flush for a minimum of 30 minutes. Sampling must be performed by the Government and testing must be done by the Service or a DoD regional fuels testing laboratory. Fuel must be free of gross contamination and visible free water to the satisfaction of the Contracting Officer.

In the event flushing identifies the presence of clay or soil like material in the system, Contractor must immediately notify Contracting Officer for direction.

3.6.2 Fueling System Piping

During flushing procedure periodically bleed air through high point vent and drain water through low point drains.

3.6.2.1 General Fuel Lines

The flushing of all of the system piping segments must be accomplished by pumping fuel through the fueling system piping and back to another tank in the fuel farm. After high-speed flush of main system piping, all piping laterals to fuel dispensing points must be flushed with at least three pipe volumes with fuel. Air must be bled from system high points. The procedure must be continued until the fuel being delivered into the tanks is acceptable to the Contracting Officer. After the piping system segments have been flushed to the satisfaction of the Contracting Officer (and periodically during the flushing operation), the Contractor must flush all high point vents and low point drains for a minimum of 10 seconds at a pressure of 207 kPa. Remove any accumulated water from tank sumps and bottoms by means of the Water Draw-off systems. Flush all plug valves of debris using the drain port at the bottom of the valve. Cone strainers must be kept clean in order to insure maximum flow rate. All accumulated material from the strainers must be reviewed and identified, including source if possible. Upon completion of the first flushing operations, the cone strainers must be cleaned, reinstalled, and remain in the system. In addition, baskets from all strainers must be removed and cleaned.

3.6.2.2 Receipt or Transfer Line

Flushing of the receipt pipeline and transfer line must occur during the initial receipt/filling operations. Samples of the incoming fuel must be taken at the point of connection with the upstream and downstream systems at the start and the end of the flushing and periodically between the two. These samples must be tested by the designated government agency and turned over to the Contracting Officer.

3.6.2.3 Installation Pipeline

Samples of the incoming fuel must be taken at the point of connection with the upstream and downstream systems at the start and the end of the flushing and periodically between the two. These samples must be tested by the designated government agency and turned over to the Contracting Officer.

3.6.2.4 Water Drain Tank Lines

During the flushing of pumphouse piping, operate the water draw-off pumps to issue product to the water drain tank. Fill the tank a minimum three times, each time utilizing the drain tank pump to drain it and issue product to a truck.

3.6.2.5 High Point Vent Lines, Low Point Drain Lines, Thermal Relief Piping, Instrumentation Piping and Tubing

During the flushing of pumphouse piping, operate valves and connections on all lines individually to flush their connection to a tank. Put at least 25 times the volume of the line through each line.

3.6.3 Piping Flushing Checklist

The Contractor must generate a comprehensive matrix of all new piping sections in the system. Matrix must serve as an Owner's piping inventory and a checklist for all Contractor-provided flushing operations. Column entries must include pipe section name, location, diameter, approximate

length, flushing fuel velocity and volume achieved and acceptable results of sampling.

3.7 CLEANING

After initial flushing is completed, clean each piping segment with product in accordance with the procedure specified hereafter. Isolate Tanks from the system and clean them as specified in Section 33 01 50.55 CLEANING OF PETROLEUM STORAGE TANKS.

In the event cleaning identifies the presence of clay or soil like material in the system, Contractor must immediately notify Contracting Officer.

3.7.1 Cleaning Requirements

Cleaning must continue until the Contracting Officer certifies that the fuel meets the intra-governmental receipt limits as defined in MIL-STD-3004-1. Sampling must be performed by the Contractor and testing must be done by an independent testing laboratory approved by the Contracting Officer.

3.7.2 Storage Tank Piping

Pipelines must be cleaned as follows:

- a. Position manual valves to circulate fuel from the issue tank, through the piping and into the receiving tank. Piping should be provided with temporary filters or conical strainers to assure removal of dirt and debris.
- b. Initially pump fuel at a flow rate of 38 lps, then increase flow rate up to the full capacity (all pumps running) starting manually one pump at a time.
- c. Monitor pressure drop through the filter during the cleaning operation.
- d. Periodically take samples from all sample connections. Continue cleaning until the fuel meets specified requirements of paragraph CLEANING REQUIREMENTS.
- e. During cleaning procedure periodically bleed air through high point vent and drain water through low point drains.

3.7.3 High Point Vent Lines, Low Point Drain Lines, Thermal Relief Piping, Instrumentation Piping and Tubing

Repeat the process described under initial flushing until samples taken at the end of the piping meet the requirements.

3.7.4 Piping Cleaning Checklist

The Contractor must generate a comprehensive matrix of all new piping sections in the system. Matrix must serve as an Owner's piping inventory and a checklist for all Contractor-provided cleaning operations. Column entries must include pipe section name, location, diameter, approximate length, cleaning fuel velocity and volume achieved and acceptable results of sampling.

3.8 CONTROL VALVE ADJUSTMENT

Check all control valve settings and field adjust from the factory settings at System Start-Up as necessary to provide a smooth operation, devoid of pressure surges, spikes, and hunting. Check the fueling pump non-surge check valve and adjust as follows:

3.8.1 Rate of Flow Control Feature on Fueling Pump Non-Surge Check Valve

Run one pump at a time and adjust rate of flow feature at nominal rated flow.

3.9 EQUIPMENT TESTS

After completion of flushing, cleaning, and control valve and electrical components adjusting operations, System Supplier must complete and submit the Equipment Tests (see Attachment 1 - Equipment Tests). Equipment Tests must be submitted prior to Performance Testing.

3.9.1 Control System and Control Valves

Field adjustment of automatic control valves and automatic pump controls while in operation must be made only by the valve manufacturer's authorized field test engineer. For final adjustment of installed electrical control equipment provide an experienced electrical engineer, factory representative of pump control panel manufacturer and factory representative of pressure indicating transmitter (PIT) manufacturer and . Both the mechanical and electrical components must be adjusted concurrently. Record required data necessary to prepare Equipment Tests Report.

3.9.2 Tank Level Alarms

Position valves to transfer fuel between tanks.

Start fueling pump and pump sufficient fuel out of the first tank to allow the low-low level alarm (LLLA) to stop the fueling pump. Repeat procedure for each fueling pump and each tank until the low-low level alarm LLLA stops the fueling pumps and sounds an alarm due to low liquid level in tank.

Repeat testing for high-high level alarm other alarms checking each one in turn. Repeat procedure for each fueling pump and each tank until the high-high level alarm (HHLA) stops the fueling pumps and sounds an alarm due to high liquid level in tank.

3.9.3 Fueling Pump Operation

Demonstrate operation of all pressure and flow devices and pump control start/stop station to start and stop the fueling pumps at the indicated pressure and flow rates in the presence of the Contracting Officer. Repeat the operating sequence with each of the pumps being selected as lead pump if the pumps are controlled by a Pump Control Panel or an automatic control system with a variable lead pump. For this test, measure the flow rates. Witness and record flow rates and test results.

3.9.4 Emergency Shutdown

With one fueling pump circulating fuel through the system, test each

"Emergency Stop" pushbutton station to verify that the pump stops . Repeat above procedure for each fueling pump and "Emergency Stop" pushbutton station. Conduct tests for both the automatic and manual modes. With all the fueling pumps circulating fuel through the system, push an "Emergency Stop" pushbutton station.

3.9.5 Overfill Valve

Use draw-off pumps to deliver quantity of product to the water drain tank and the FSII tank to demonstrate capability of valve to close. Empty the tank below the shut-off level to demonstrate capability of valve to open when fuel level drops below set point.

3.10 PERFORMANCE TESTS

Performance testing must occur after the Contractor has performed the Equipment Tests. Performance testing must demonstrate to the satisfaction of the Contracting Officer and Service Headquarters and/or their designated appointee these portions of the fueling system are working as specified. Performance testing must consist of repeating the Equipment Tests (indicated in previous paragraphs) and operating the fueling system during actual operations in the presence of Government Witnesses. The maximum rated capacity of the system must be demonstrated. The Contractor must notify the Contracting Officer 45 calendar days in advance of the test to permit arrangement for the use of Government-furnished items. During the time period of performance testing, no construction activities will be allowed on the project site. The project site must be considered an operational (fuel) zone (verses a construction zone) during this performance testing period. Personnel, dressed for fuel's operation, will be present to witness testing and participate in Contractor provided training.

3.10.1 Final Performance Test

A final performance test must consist of demonstrating the operation of the system for the purpose for which it is intended:

- a. Transferring fuel through the pipeline and piping system

3.10.2 Satisfactory Performance

In the event a portion of the system or any piece of equipment fails to meet the test, make the necessary repairs or adjustments and repeat the Performance Test until satisfactory performance is obtained. The determination of satisfactory performance must be made by the Contracting Officer and the Service Headquarters and/or their designated appointee.

3.10.3 Performance Testing Plan

System Supplier must edit the example Performance Testing plan and submit for approval. An example Performance Testing plan can be found at the end of this Section as Attachment 2 - Performance Testing Plan. Submit plan a minimum of 180 days prior to performance testing.

3.10.4 Equipment Tests

System Supplier must provide electronic copies of the completed Equipment Tests to the Service Headquarters and/or their designated appointee at the start of Performance Testing for validation during Performance Testing.

3.10.5 Control Valve Tagging

After the performance testing and system acceptance, tag the control valves with their final adjustments.

3.10.6 Final Acceptance

Fill the system with fuel and operate leak-free prior for acceptance. Anything wet with fuel is considered to be leaking.

3.10.6.1 Tank High Liquid Level Shut-Off Valve Test and Adjustments

During the final filling of tanks, check the tank automatic high liquid level shut-off valve for proper functioning at least three times by lowering the fuel level and refilling again. Adjust valve to achieve a safe fill level.

3.10.6.2 Tank Level Indicator Adjustments

Also, during the final filling of tanks, adjust and calibrate the tank level indicators including the final setting of the high high level (HHLA) and high level (HLA) alarms. Since the HHLA is at a point higher than the High Liquid Level Shut-Off Valve float set point, an artificial method of simulating HHL must be used.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 08 55.02

FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT)

08/19

PART 1 GENERAL

- 1.1 SUMMARY/APPLICABILITY
- 1.2 REFERENCES
- 1.3 ADMINISTRATIVE REQUIREMENTS
 - 1.3.1 Plans
 - 1.3.2 Existing System Cleanliness Confirmation Plan
 - 1.3.3 Fuel Testing Laboratory Qualifications
 - 1.3.4 System Start-up Plan
 - 1.3.5 Performance Testing Plan
 - 1.3.6 Fuel Provisioning Plan
 - 1.3.7 Certification (Ready for Start-Up and Performance Testing)
- 1.4 SUBMITTALS
- 1.5 CLOSEOUT SUBMITTALS
 - 1.5.1 Final Reports
- 1.6 QUALITY ASSURANCE
 - 1.6.1 Certification of Entire System
 - 1.6.2 Service Headquarters Definition
- 1.7 SYSTEM SUPPLIER INVOLVEMENT
- 1.8 DISPOSAL OF WASTE MATERIALS

PART 2 PRODUCTS

- 2.1 GOVERNMENT-FURNISHED MATERIAL AND EQUIPMENT
 - 2.1.1 Fuel
- 2.2 CONTRACTOR-FURNISHED MATERIAL AND EQUIPMENT
 - 2.2.1 Contractor-furnished
 - 2.2.2 Design Conditions

PART 3 EXECUTION

- 3.1 SEQUENCE OF EVENTS
- 3.2 PRELIMINARY REQUIREMENTS
 - 3.2.1 Safety
 - 3.2.2 Electrical Preparations
 - 3.2.3 Piping System
 - 3.2.3.1 Pier Piping Systems
 - 3.2.3.2 Transfer/Installation Pipeline Systems
 - 3.2.4 Existing System Cleanliness Confirmation
- 3.3 PREPARATIONS FOR FLUSHING
 - 3.3.1 Protection of System Components
 - 3.3.2 Water Draw-off
- 3.4 INITIAL FUEL RECEIPT INTO STORAGE TANK
 - 3.4.1 General
 - 3.4.2 Storage Tanks
 - 3.4.3 Components

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.4.4 Fuel Quality
- 3.4.5 Fuel Receipt
 - 3.4.5.1 Fuel Receipt by Pipeline
 - 3.4.5.2 Fuel Receipt from Piping System
 - 3.4.5.3 Fuel Receipt from Pier
- 3.4.6 Initial Low Point Flush
- 3.4.7 Piping Hydrostatic Tests
- 3.4.8 Piping Soak Tests
- 3.5 FLUSHING
 - 3.5.1 Flushing Requirements
 - 3.5.2 Fueling System Piping
 - 3.5.2.1 General Fuel Lines
 - 3.5.2.2 Installation Pipeline
 - 3.5.2.3 High Point Vent Lines, Low Point Drain Lines, Thermal Relief Piping, Instrumentation Piping and Tubing
 - 3.5.2.4 Pier Piping
 - 3.5.3 Piping Flushing Checklist
- 3.6 CLEANING
 - 3.6.1 Installation Pipeline
 - 3.6.2 High Point Vent Lines, Low Point Drain Lines, Thermal Relief Piping, Instrumentation Piping and Tubing
 - 3.6.3 Piping Cleaning Checklist
- 3.7 PERFORMANCE TESTS
 - 3.7.1 Final Performance Test
 - 3.7.2 Satisfactory Performance
 - 3.7.3 Performance Testing Plan
 - 3.7.4 Final Acceptance

-- End of Section Table of Contents --

SECTION 33 08 55.02

FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT)
08/19

PART 1 GENERAL

1.1 SUMMARY/APPLICABILITY

This specification defines the requirements and procedures for startup and performance testing of all equipment, components, control systems, devices, and associated appurtenances which are used for the receipt, storage, transfer and issue of petroleum fuel products for non-hydrant fuel facility systems. It covers requirements for safety, Government scheduling and coordination, device testing, existing and new system cleanliness, system flushing and cleaning, demonstration of indicated and specified system performance, and final acceptance and reporting. The types of fuel systems covered include:

- a. installation pipelines from pier to bulk storage

It does not cover private vehicle fueling stations such as exchange service stations, nor aircraft direct refueling hydrant systems.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API RP 1595 (2012; R 2019; 2nd Ed) Design, Construction, Operation, Maintenance, and Inspection of Aviation Pre-Airfield Storage Terminals

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1110-1-8167 (2016) Engineering and Design -- Petroleum, Oil, and Lubricants Mandatory Center Of Expertise

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-3004-1 (2018) Quality Assurance for Bulk Fuels, Lubricants and Related Products

1.3 ADMINISTRATIVE REQUIREMENTS

Contractor property management system shall comply with FAR 52.245-1 Government Property. Contractor shall confirm availability of required Government-furnished material and equipment to support System Start-Up and incorporate into proposed schedule submitted with bid. Contractor shall notify Government no later than 180 days after notice-to-proceed if any Government-furnished material or equipment required for System Start-Up are not available. See paragraph SYSTEM START-UP-PLAN for additional

requirements. The Contractor is required to notify the Government within the solicitation period and or the first 180 days after NTP that any element of the project specifications that are not present to have a complete system commissioning. Costs for equipment and materials required for System Start-Up that are not identified within specific period shall be borne by Contractor. Contractor is requested for acknowledgement of these requirements in their proposed schedule and / or bid documentation to be submitted.

1.3.1 Plans

The Contractor must submit detailed written plans (prepared by the System Supplier) for implementation of the System Start-Up Plan, the Existing System Cleanliness Confirmation Plan, the [Fuel Provisioning Plan](#), and the [Performance Testing Plan](#). The plans must specify a detailed plan incorporating in a sequential manner all work specified in PART 3 EXECUTION of this specification section. Plan elements must include:

- a. Personnel. List of Contractor's personnel by trade, list of key personnel, list of safety equipment, list of miscellaneous equipment such as two-way radios, and personnel transportation vehicles.
- b. Fluid Used for Hydrotesting, Cleaning Pigging, Verification Pigging, and Flushing and Cleaning. Identify what fluid will be used for each operation in each system section.
- c. Fuel. See the Fuel Provisioning Plan.
- d. Storage-Into. Identify where the contents of the piping will be pushed into including the capacity required, how the storage vessel will be prepared for receiving the piping content, how it will be emptied and how it will be cleaned after the contents are removed.
- e. Storage-From. Identify where the fluid that will be drained or pushed into the piping will come from including the capacity required, the number, size and type of the storage vessel, how the storage vessel is filled, how it will be emptied, and how it will be cleaned after the contents are removed.
- f. Piping. Notwithstanding the requirements described elsewhere, every pipe section and individual component in the system covered by this specification section must be filled, vented, hydrostatically tested, flushed, cleaned, and tested for fuel quality. It must also be able to be isolated and drained if found that it needs repair. The Plans must detail how this is to be done.
- g. Performance Testing. Detailed procedures and schedules for each system component to perform all system tests under each operating scenario in accordance with paragraph PERFORMANCE TESTING.
- h. Pigging Plans. See Section [33 52 40 FUEL SYSTEMS PIPING \(NON-HYDRANT\)](#).
- i. Hydrotesting Plans. See the Work Plan in Section [33 52 40 FUEL SYSTEMS PIPING \(NON-HYDRANT\)](#).
- j. Schedule. Schedules must be generated with listing dates and durations of all system start-up activities as well as regular coordination and safety meetings and dates of key events for Government participation.

- k. Contingency plans. Information on spill and fire contingencies, along with the required Government Fire and Safety Office involvement and approvals.
- l. Coordination with Installation. Description of how Contractor and System Supplier must implement system start-up in coordination with ongoing operations at the Installation. Plan must incorporate all phasing and work restriction requirements of the Contract documents.

1.3.2 Existing System Cleanliness Confirmation Plan

Submit a detailed written plan prepared by the Contractor or System Supplier for the Contractor to verify and confirm the cleanliness of the existing system before connecting it to the new system or extension. The cleanliness confirmation test of the existing system (pre-cleanliness test) must be per MIL-STD-3004-1. Submit the plan 90 (OCONUS) days prior to making the connection to the existing system. The Contractor and System Supplier are responsible for implementing the Existing System Cleanliness Confirmation Plan in coordination with ongoing base operations.

1.3.3 Fuel Testing Laboratory Qualifications

Contractor must submit the qualifications of the DoD Fuel Testing Laboratory which will be used to test the fuel samples.

1.3.4 System Start-up Plan

Submit a detailed written plan prepared by the Contractor and System Supplier for implementation of System Start-Up. A preliminary start-up plan must be submitted as a preconstruction item. Submit a finalized plan, including any changes needed due to project scope modification, a minimum of 180 (OCONUS) days prior to System Start-Up and no later than 180 days after notice-to-proceed. The Contractor and System Supplier are responsible for implementing System Start-Up in coordination with ongoing operations.

1.3.5 Performance Testing Plan

Submit a detailed written plan prepared by the Contractor or System Supplier for implementation of Performance Testing. Submit the plan 180 (OCONUS) days prior to System Start-Up. The Contractor and the System Supplier are responsible for implementing the Performance Testing Plan in coordination with ongoing operations.

1.3.6 Fuel Provisioning Plan

Submit a detailed written Fuel Provisioning Plan prepared by the Contractor. Submit the plan 90 days prior to System Start-up. The Fuel Provisioning Plan must include a timeline for the required receipt date(s), grade of fuel, and quantity of fuel on each receipt date required to execute the Start-up and the Performance Testing plans.

1.3.7 Certification (Ready for Start-Up and Performance Testing)

As a prerequisite to fuel system start-up, the Contractor must submit a certificate that certifies all work provided on the fuel system, except for touch-up field painting, has been inspected and approved by the specified approving authorities. Further, the Contractor must certify on

this certificate that all specified checks and inspections have been successfully completed prior to start-up. Submit the plan 45 (OCONUS) days prior to commencement of fuel system start-up. The Contractor must submit the Certificate of Completion to the Contracting Officer at least 7 calendar days prior to commencement of system start-up and performance testing. The Contracting Officer must then be responsible for scheduling the Government representatives and appropriate military command authority and designers for participation in the inspection, performance testing, and final approval activities. Any contractual deficiencies observed must be corrected by the Contractor without cost to the Government.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Existing System Cleanliness Confirmation Plan; G

System Start-Up Plan; G

Performance Testing Plan; G

Fuel Provisioning Plan; G

SD-06 Test Reports

Final Reports; G

Cathodic Protection Reports; G

Piping Flushing Checklist; G

Piping Cleaning Checklist; G

SD-07 Certificates

Fuel Testing Laboratory Qualifications; G

Certification (Ready for Start-Up and Performance Testing); G

Certificate of Calibration; G

Disposal Of Waste Materials

SD-11 Closeout Submittals

Certification of Entire System

1.5 CLOSEOUT SUBMITTALS

1.5.1 Final Reports

Contractor must prepare a final report that documents the execution of the approved start-up and performance testing plan. All items of work specified in the Start-up and Performance Testing plans must be carried

out and reported in this report unless otherwise approved by the Contracting Officer. Include as a part of this report:

- a. Verification letters of approved fuel storage tank hydrostatic tests and the approved piping hydrostatic tests, as generated under other specification sections.

1.6 QUALITY ASSURANCE

1.6.1 Certification of Entire System

Prior to the acceptance of the newly constructed system by the Government, all installed mechanical and electrical system components must be inspected and approved by the Contracting Officer. Provide the Contracting Officer 60 days notice in order to schedule the Installation, DLA and Service Headquarters and/or their designated appointee for participation in the inspection, Performance Testing, and approval. Any deficiencies observed must be corrected by the Contractor without cost to the Government.

1.6.2 Service Headquarters Definition

SME is defined as Service Headquarters Subject Matter Experts: Air Force - The Air Force Fuels Facilities Subject Matter Expert (AFCEC/COS); Army - Headquarters, U.S. Army Corps of Engineers; POL-MCX Facilities Proponent (CECW-EC) through the Army Petroleum Center (APC); Navy/Marine Corps - NAVFAC POL Facility Subject Matter Expert (NAVFAC EXWC, CI11). [ER 1110-1-8167](#) Engineering and Design Petroleum, Oil, and Lubricants Mandatory Center of Expertise must be followed.

The Service Control Point (SCP) is defined as follows: Army - Army Petroleum Center; Air Force - Air Force Petroleum Office; Navy/Marine Corps - Naval Supply Systems Command - Energy.

1.7 SYSTEM SUPPLIER INVOLVEMENT

The System Supplier is defined in Specification [33 52 43.11](#) AVIATION FUEL MECHANICAL EQUIPMENT. The Contractor and the System Supplier must work together to prepare the Start-Up Plan, the Existing System Cleanliness Confirmation Plan, Performance Testing Plan, and the Final Reports. They must both be present during all activities described in this section. The System Supplier must be responsible to the Contractor for scheduling all Contractor, sub-Contractor, and manufacturer's service personnel during activities described in this specification section.

1.8 DISPOSAL OF WASTE MATERIALS

The Contractor must be responsible for properly disposing of any sludge, debris, filtration elements, waste water, contact water, and waste fuel resulting from piping and tank flushing and cleaning activities. Comply with all applicable Installation, local and Host Nation for hazardous waste disposal.

PART 2 PRODUCTS

2.1 GOVERNMENT-FURNISHED MATERIAL AND EQUIPMENT

The Government will furnish the following materials, equipment and services during the performance of the work under this section.

2.1.1 Fuel

Some fuel loss should be expected. Fuel that is lost during flushing and cleaning and performance testing is viewed as a Start-up and Performance Testing expense and should be budgeted for in Contractor estimates. As Basis of Bid, the Contractor must allow for 200 gallons of fuel losses in the bid.

The Contractor is responsible for the disposal of any waste product.

The Contractor must establish a Department of Defense Activity Address Code and must reimburse the US Government for any US Government fuel lost resulting from poor workmanship or contamination.

The Government will provide fuel given that the requirements are submitted with sufficient lead time to allow the US Government to secure the grade and volume required and make the necessary transportation requirements. The Contractor must provide a copy of the Start-up, Fuel Provisioning, and Performance Testing plans a minimum of 90 days in advance of System Start-up. When Defense Working Capital Fund (DLA Owned) fuel is used, provide a copy of the Start-up, Fuel Provisioning, and Performance Testing plans to DLA Energy at the same time.

FLUSHING, CLEANING, EQUIPMENT TESTS and PERFORMANCE TESTING: The flushing, cleaning, and testing phases will be identified separately in the start-up and Performance Testing plans. Fuel used for flushing and cleaning, and testing will become contaminated, the fuel must be isolated and is considered off-specification until the quality can be verified. Fuel volumes required for flushing and cleaning and testing should be minimized to limit costs, so estimates should be made accordingly. The Contractor will be responsible for disposing of any waste product.

Upon satisfactory completion of the flushing and cleaning operations, the Government will supply the additional quantities of fuel required to complete the other work under this section. This larger volume of fuel will not be delivered to the system until the Contractor has satisfactorily completed all work and, in particular, the cleaning of the interior surfaces of the storage tanks and the removal of preservatives and foreign matter from those portions coming in contact with the fuel valves and other such system components.

2.2 CONTRACTOR-FURNISHED MATERIAL AND EQUIPMENT

2.2.1 Contractor-furnished

Provide material, equipment and labor not specified to be Government-furnished and required for proper System Start-Up of the system. Equipment being provided must be calibrated and the [certificate of calibration](#) be submitted to the Contracting Officer prior to use. Equipment must include but not be limited to the following:

- a. Temporary strainers.
- b. Pipe spools.
- c. Flow meters.
- d. Pressure gauges.

- e. Vacuum Trucks for removing water and fuel from pits, drains, and other systems.
 - f. Fuel Bowser for removing water and fuel from pits, drains, and other systems.
 - g. Barge(s) or Ship(s) to provide the fuel required for system start-up activities
 - h. Temporary Tankage for flushing, cleaning and draining of the system and for starting up and performance testing.
 - i. Temporary filtration/strainers.
 - j. Pigging equipment and services per Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT) including pig launching and receiving barrels, kicker line, and receiver line.
- 2.2.2 Design Conditions

Use temporary flushing lines and equipment that are equal in strength, stability, and materials to the associated permanent components. However, spools may be carbon steel. Additional design conditions must be as specified in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

PART 3 EXECUTION

3.1 SEQUENCE OF EVENTS

The following events take place in the Start-Up and Performance Testing of a system and their definitions:

- a. System Start-Up Phase: The System Start-Up phase encompasses all of the events described below up to the Performance Tests Phase.
- b. Preliminary Requirements: Confirms completion of the system before fuel is received including verification of the cleanliness of the existing piping system being connected to.
- c. Preparations for Flushing: Activities that take place before initial receipt of fuel.
- d. Pneumatic Testing: Test for leaks in a piping system using air under pressure as the test medium. For requirements see Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- e. Initial Fuel Receipt: The point in time at which fuel is introduced into the system with all that implies for safety, spill prevention and control, and inventory control.
- f. Hydrostatic testing - Fuel: Use of fuel to test for leaks in a piping system using liquid under pressure as the test medium. For requirements see Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- g. Soak Testing: Perform "Soak Testing" of the piping system using fuel, as per Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- h. Pipe Pigging - Cleaning with Fuel: Use of fuel propelled pigs to clean the line of gross contamination. Perform this as part of the work in

Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).

- i. Pipe Pigging - Verification with Fuel: Use of fuel propelled pigs to inspect the condition of the pipeline. For requirements see Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- j. Flushing: In this context, refers to removing gross contamination (free water and solids) from a fuel system using fuel.
- k. Cleaning: In this context, refers to removing the remaining water and contamination from aviation fuel to the quality (allowable) levels suitable for issuing the final product using fuel as the cleaning medium.
- l. Performance Tests Phase: Government witnessed demonstration of the proper operation of the individual pieces of equipment and the system or systems as a whole.

3.2 PRELIMINARY REQUIREMENTS

All activities listed in paragraph PART 3 EXECUTION must be performed sequentially in the order they are presented. Prior to any on-site system start-up activities, the Contractor must ensure that all requirements of the paragraph SAFETY are satisfied. Project must be substantially complete, except for touch-up field painting, and Contractor's work area must be free of debris, trash and obstacles. Perform the following activities prior to receipt of fuel:

3.2.1 Safety

Prior to any on-site system start-up activities, the following safety procedures must be accomplished in all fueling areas to be started up and performance tested under this specification section: testing/operation of emergency showers and eyewash stations, placement of Contractor-provided portable eyewash units within 31 meters or 10 seconds from the fueling point, verification of proper grounding throughout system, coordination with Government Fire and Safety Office and Fuels personnel, placement of Contractor- or Government-provided spill pads and containment booms, placement of Contractor-provided fire extinguishers capable of extinguishing a fuel fire. Ensure that all radios/devices at all Class I, Division 1 areas are intrinsically safe.

3.2.2 Electrical Preparations

Prior to energizing the electrical system components, verify that short-circuit links have been removed from current transformer and that secondary circuits have been connected. Verify all electrical transmitter connections and ensure proper calibration. Verify all electrical system components meet the classification and division as required by the design. Verify correct rotation of all motors prior to testing. Conduit explosion-proof sealoffs must be poured after initial electrical checks but before fuel receipt.

3.2.3 Piping System

Ensure that all piping weld integrity, leak detection systems, and coating inspections have been performed per the applicable specifications. Provide statement that required testing has been completed in the Final Reports. Evacuate all accumulated water from piping low point drains,

valve cavities, and equipment drains. Verify all bolted connections are tightness tested to required torque using a calibrated torque wrench. Verify that all pressure gauges are properly calibrated, located and installed. Ensure that piping's cathodic protection system is tested, calibrated and operational. Ensure the cathodic protection inspections are performed per applicable specifications and the [cathodic protection reports](#) are on hand prior to initial fill and performance testing. Ensure that pipe marking and identification are provided as specified. Ensure that piping system thermal relief provisions are installed as designed. Verify the correct installation of piping, joints, and supports.

3.2.3.1 Pier Piping Systems

For pier delivery/receipt systems or other over-water piping installations, ensure compliance with the Contractor's previously approved spill control plans.

3.2.3.2 Transfer/Installation Pipeline Systems

For pipeline systems or any other piping running outside the fuel facility fence, ensure compliance with the Contractor's previously approved spill control plans.

3.2.4 Existing System Cleanliness Confirmation

Contractor and System Supplier must conduct tests to confirm the cleanliness of the existing system before connecting it to the new system or an extension, upgrade or repair of the existing system. Confirm that the fuel in the existing system meets the intra-governmental receipt limits defined in [MIL-STD-3004-1](#). Sampling must be performed by the Government and testing must be done by the Service or a DoD regional fuels testing laboratory.

3.3 PREPARATIONS FOR FLUSHING

Upon completion of the system to the satisfaction of the Contracting Officer and the Service Headquarters and/or their designated appointee, make the following preparations for flushing the system.

3.3.1 Protection of System Components

The following items must be removed from the system prior to start of flushing operations and, where applicable, replaced with spools of pipe, diameter equal to the item removed.

- a. Sensors which are exposed to the fluid such as pressure gauges and thermometers.
- b. Pressure Indicating Transmitters.
- c. Meters.

After flushing, the above items must be reinstalled in the system and the spool sections turned over to the Contracting Officer.

3.3.2 Water Draw-off

Remove any accumulated water from piping and storage tanks' sumps and bottoms. Drain water and return fuel via filtration to storage tank.

Repeat process until all water is removed. Dispose of petroleum contaminated water in accordance with Installation, local, and Host Nation regulations.

3.4 INITIAL FUEL RECEIPT INTO STORAGE TANK

3.4.1 General

Utilize temporary tankage for initial fuel receipt to isolate contaminated fuel. The Contractor must station personnel throughout piping system at high point vents to bleed air. All flanges and system components will be periodically inspected for leaks during filling procedures.

3.4.2 Storage Tanks

Receipt flow rate into an empty storage tank must not exceed **1 m/s**, as measured in the main receipt piping, until outlet of tank fill tube is submerged.

3.4.3 Components

Differential pressure across strainers must be continuously monitored. Any time a strainer differential pressure reaches **138 kPa**, it must be cleaned.

3.4.4 Fuel Quality

Fuel used during initial receipt must be considered contaminated and must be positively isolated, with blind flanges or closed, padlocked manual valves, from any active transfer, or aircraft fueling or tank truck loading operations. Fuel isolation must continue until all flushing and cleaning is completed. A method of checking fuel quality that is acceptable to the Contracting Officer must be provided at the beginning and end of any new piping section.

3.4.5 Fuel Receipt

3.4.5.1 Fuel Receipt by Pipeline

Start-up personnel must meet with Government personnel in charge of existing fuel storage to discuss fuel transfer procedures. Topics must include: methods of communication to start/stop remote transfer pumps; flow rate and head characteristics of transfer pumps; methods of restricting initial receipt flow rate; methods of straining and filtering initial receipt fuel; accommodating multiple pump starts resulting from required strainer and filter cleaning operations; required quantity of fuel to be transferred. Contractor must provide a written summary of pipeline receipt procedures to the Contracting Officer.

3.4.5.2 Fuel Receipt from Piping System

Start-up personnel must meet with Government personnel in charge of existing fuel storage to discuss fuel transfer procedures. Topics must include: methods of communication to start/stop remote pumps; flow rate and head characteristics of pumps; methods of restricting initial receipt flow rate; methods of straining and filtering initial receipt fuel; accommodating multiple pump starts resulting from required strainer and filter cleaning operations; required quantity of fuel to be used. Contractor must provide a written summary of piping fill procedures to the

Contracting Officer.

3.4.5.3 Fuel Receipt from Pier

Start-up personnel must meet with Government personnel in charge of existing fuel storage to discuss fuel transfer procedures. Topics must include: methods of communication to start/stop pumps; flow rate and head characteristics of transfer pumps; methods of restricting initial receipt flow rate; methods of straining and filtering initial receipt fuel; accommodating multiple pump starts resulting from required strainer and filter cleaning operations; required quantity of fuel to be transferred. Contractor must provide a written summary of pipeline receipt procedures to the Contracting Officer.

3.4.6 Initial Low Point Flush

Perform an initial low point flush operation by flushing each low point drain through a portable basket strainer for 10 seconds at a system pressure of 207 kPa. Repeat flush until basket strainer collects no additional debris.

3.4.7 Piping Hydrostatic Tests

Hydrostatically test the piping system with fuel as required by Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).

3.4.8 Piping Soak Tests

Upon completion of hydrostatic testing with fuel, perform "Soak Testing" of the piping systems per API RP 1595. Duration of the test must be a minimum of 4 days and maximum of 7 days.

3.5 FLUSHING

Flushing procedures remove gross particulate and water contaminants and must precede cleaning procedures. All piping including but not limited to receipt piping, issue piping, transfer piping, supply and return lines to the storage tanks, water drain lines, installation pipelines, thermal relief piping, high point vents, low point drains, must be flushed with fuel. Fuel used for flushing shall be the same type and grade as intended service of piping system.

In the event flushing identifies the presence of clay or soil like material in the system, Contractor must immediately notify Contracting Officer for direction.

3.5.1 Flushing Requirements

Begin flushing of fuel system piping at low flow rate. Slowly increase flushing flow rate to specified velocity. Flush for a minimum of 30 minutes total at 2.5 m/s flow rate. Sampling must be performed by the Government and testing must be done by the Service or a DoD regional fuels testing laboratory. Fuel must be free of gross contamination and visible free water to the satisfaction of the Contracting Officer.

In the event flushing identifies the presence of clay or soil like material in the system, Contractor must immediately notify Contracting Officer for direction.

3.5.2 Fueling System Piping

During flushing procedure periodically bleed air through high point vent and drain water through low point drains.

3.5.2.1 General Fuel Lines

The flushing of all of the system piping segments must be accomplished by pumping fuel through the fueling system piping. Air must be bled from system high points. The procedure must be continued until the fuel being delivered is acceptable to the Contracting Officer. After the piping system segments have been flushed to the satisfaction of the Contracting Officer (and periodically during the flushing operation), the Contractor must flush all high point vents and low point drains for a minimum of 10 seconds at a pressure of 207 kPa. Remove any accumulated water from tank sumps and bottoms by means of the Water Draw-off systems. Flush all plug valves of debris using the drain port at the bottom of the valve. Baskets from all strainers must be removed and cleaned.

3.5.2.2 Installation Pipeline

Samples of the incoming fuel must be taken at the point of connection with the upstream and downstream systems at the start and the end of the flushing and periodically between the two. These samples must be tested by the designated government agency and turned over to the Contracting Officer.

3.5.2.3 High Point Vent Lines, Low Point Drain Lines, Thermal Relief Piping, Instrumentation Piping and Tubing

During the flushing of piping, operate valves and connections on all lines individually to flush their connection to a tank. Put at least 25 times the volume of the line through each line.

3.5.2.4 Pier Piping

Pier piping should be hydrostatically tested per the requirements of Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT). The Contractor will be allowed to use Government-furnished hoses. Perform flushing with product at 2.5 m/s for a total of 30 minutes.

3.5.3 Piping Flushing Checklist

The Contractor must generate a comprehensive matrix of all new piping sections in the system. Matrix must serve as an Owner's piping inventory and a checklist for all Contractor-provided flushing operations. Column entries must include pipe section name, location, diameter, approximate length, flushing fuel velocity and volume achieved and acceptable results of sampling.

3.6 CLEANING

3.6.1 Installation Pipeline

Pipelines must be cleaned as follows:

- a. Position manual valves to circulate fuel through the piping and into the receiving tank at the discharge end.

- b. First clean the pipe using pigs as called out in paragraph PIPE PIGGING-VERIFICATION, Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT). During this, low point drains and high point vents must be blown clean.
- c. Inspect the pipe as called out in paragraph PIPE PIGGING - VERIFICATION, Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- d. Initially pump fuel at a flow rate of 38 lps, then increase flow rate up to the full capacity.

3.6.2 High Point Vent Lines, Low Point Drain Lines, Thermal Relief Piping, Instrumentation Piping and Tubing

Repeat the process described under initial flushing until samples taken at the end of the piping meet the requirements.

3.6.3 Piping Cleaning Checklist

The Contractor must generate a comprehensive matrix of all new piping sections in the system. Matrix must serve as an Owner's piping inventory and a checklist for all Contractor-provided cleaning operations. Column entries must include pipe section name, location, diameter, approximate length, cleaning fuel velocity and volume achieved and acceptable results of sampling.

3.7 PERFORMANCE TESTS

Performance testing must occur after the Contractor has performed the Equipment Tests. Performance testing must demonstrate to the satisfaction of the Contracting Officer and Service Headquarters and/or their designated appointee these portions of the fueling system are working as specified. Performance testing must consist of operating the fueling system during actual operations in the presence of Government Witnesses. The Contractor must notify the Contracting Officer 45 calendar days in advance of the test to permit arrangement for the use of Government-furnished items. During the time period of performance testing, no construction activities will be allowed on the project site. The project site must be considered an operational (fuel) zone (verses a construction zone) during this performance testing period. Personnel, dressed for fuel's operation, will be present to witness testing and participate in Contractor provided training.

3.7.1 Final Performance Test

A final performance test must consist of demonstrating the operation of the system for the purpose for which it is intended:

- a. Transferring fuel through the pipeline and piping system

3.7.2 Satisfactory Performance

In the event a portion of the system or any piece of equipment fails to meet the test, make the necessary repairs or adjustments and repeat the Performance Test until satisfactory performance is obtained. The determination of satisfactory performance must be made by the Contracting Officer and the Service Headquarters and/or their designated appointee.

3.7.3 Performance Testing Plan

System Supplier must edit the example Performance Testing plan and submit for approval. Submit plan a minimum of 180 days prior to performance testing.

3.7.4 Final Acceptance

Fill the system with fuel and operate leak-free prior for acceptance. Anything wet with fuel is considered to be leaking.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 09 52

FUEL PUMP CONTROL AND ANNUNCIATION SYSTEM (NON-HYDRANT)

11/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 ADMINISTRATIVE REQUIREMENTS
- 1.3 SUBMITTALS
- 1.4 TOOLS AND SPARE PARTS
- 1.5 EXPERIENCE AND QUALIFICATIONS
- 1.6 WARRANTY

PART 2 PRODUCTS

- 2.1 MATERIALS AND SYSTEM COMPONENTS
 - 2.1.1 Pump Control Panel (PCP) and Components
 - 2.1.2 Ventilation System
 - 2.1.3 Ground Bar
 - 2.1.4 Standard Indicator Lights
 - 2.1.5 Selector Switches
 - 2.1.6 Pushbuttons
 - 2.1.7 Relays
 - 2.1.8 Nameplates
 - 2.1.9 Transient Voltage Surge Suppression Devices
 - 2.1.10 Terminal Blocks
 - 2.1.11 Circuit Breakers
 - 2.1.12 Uninterruptible Power Supplies
 - 2.1.13 Miscellaneous Power Supplies
 - 2.1.14 Alarms
 - 2.1.15 Alarm Horns
 - 2.1.16 Laptop Computer
 - 2.1.16.1 Hardware
 - 2.1.16.2 Software
 - 2.1.17 Personal Computer (PC)
 - 2.1.17.1 Hardware
- 2.2 PROGRAMMABLE LOGICAL CONTROLLER (PLC) HARDWARE AND SOFTWARE
 - 2.2.1 General
 - 2.2.2 Central Processing Unit Module
 - 2.2.3 Power Supply Module
 - 2.2.4 Program Storage/Memory Requirements
 - 2.2.5 Input/Output (I/O) Modules
 - 2.2.6 Interfacing
 - 2.2.7 Program Requirements
 - 2.2.8 Diagnostics

PART 3 EXECUTION

- 3.1 PUMP CONTROL PANEL (PCP) AND COMPONENTS
 - 3.1.1 General

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.1.2 Wiring
- 3.1.3 Certified Pump Control Panel (PCP) Shop Test Report
- 3.1.4 Ventilation System
- 3.1.5 Grounding
- 3.1.6 Indicator Lights, Switches, and Pushbuttons
- 3.1.7 Surge Protective Devices
- 3.1.8 Terminal Blocks
- 3.1.9 Circuit Breakers
- 3.1.10 Uninterruptible Power supplies
- 3.1.11 Power Supplies
- 3.1.12 Alarms and Horns
 - 3.1.12.1 Alarm Sequence
- 3.1.13 Personal Computer
 - 3.1.13.1 Screen Number 1
 - 3.1.13.2 Screen Number 2
 - 3.1.13.3 Screen Number 3
 - 3.1.13.4 Screen Number 4
 - 3.1.13.5 Screen Number 5
- 3.1.14 Laptop Computer
- 3.2 PROGRAMMABLE LOGICAL CONTROLLER (PLC) HARDWARE AND SOFTWARE
 - 3.2.1 General
 - 3.2.2 Programs
- 3.3 GRAPHICS DISPLAY SCREEN
 - 3.3.1 General
 - 3.3.2 Display Presentation
 - 3.3.3 Process Schematic
 - 3.3.4 Digital Tank Level Indicators
- 3.4 INSTALLATION
 - 3.4.1 Shop Drawing
 - 3.4.2 System Start-Up and Testing
 - 3.4.3 Training Plan for Instructing Personnel
- 3.5 PLC CONTROL SYSTEM SEQUENCE OF OPERATION
 - 3.5.1 Abbreviations
 - 3.5.2 Operating Tanks
 - 3.5.2.1 Level Control
 - 3.5.2.1.1 Low-Low Level
 - 3.5.2.1.2 Low Level
 - 3.5.2.1.3 High Level
 - 3.5.2.1.4 High-High Level
 - 3.5.2.2 Level Control
 - 3.5.2.2.1 Low-Low Level
 - 3.5.2.2.2 Low Level
 - 3.5.2.2.3 High Level
 - 3.5.2.2.4 High-High Level
 - 3.5.3 Drain Tank
 - 3.5.3.1 Drain Tank Pump
 - 3.5.3.2 Overfill Valve (OV)
 - 3.5.3.3 High Level Alarm
 - 3.5.3.4 High-High Level Alarm
 - 3.5.3.5 Leak Detection
 - 3.5.4 Fueling Pumps (FP)
 - 3.5.5 Flow Switch, Fueling Pump
 - 3.5.6 Transmitters
 - 3.5.6.1 Differential Pressure Transmitter (DPT)
 - 3.5.7 Safety Circuit
 - 3.5.7.1 Emergency Stop Status
 - 3.5.7.2 Circuit Power Status
 - 3.5.8 Pump Control Panel
 - 3.5.8.1 CPU Faults

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.5.8.2 Mode Select Switch
- 3.5.8.3 Lead Pump Selector Switch
- 3.5.8.4 PCP Temperature Alarm
- 3.6 OPERATING PROGRAM REQUIREMENTS
- 3.7 AUTOMATIC MODE - ISSUE OR TRANSFER CONDITION
- 3.8 OFF MODE
- 3.9 MANUAL OPERATION OF FUELING PUMPS

-- End of Section Table of Contents --

SECTION 33 09 52

FUEL PUMP CONTROL AND ANNUNCIATION SYSTEM (NON-HYDRANT)
11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C37.90 (2005; R 2011) Standard for Relays and Relay Systems Associated With Electric Power Apparatus

IEEE C62.41 (1991; R 1995) Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits

INTERNATIONAL SOCIETY OF AUTOMATION (ISA)

ISA 18.1 (1979; R2004) Annunciator Sequences and Specifications

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NEMA IA 2 (2005) Programmable Controllers - Parts 1 thru 8

NEMA ICS 1 (2000; R 2015) Standard for Industrial Control and Systems: General Requirements

NEMA ICS 2 (2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V

NEMA ICS 4 (2015) Application Guideline for Terminal Blocks

NEMA ICS 6 (1993; R 2016) Industrial Control and Systems: Enclosures

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code

U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

FCC Part 15

Radio Frequency Devices (47 CFR 15)

UNDERWRITERS LABORATORIES (UL)

UL 508

(2018; Reprint Jul 2021) UL Standard for
Safety Industrial Control Equipment

UL 1012

(2010; Reprint Apr 2016; Rev Mar 2021) UL
Standard for Safety Power Units Other than
Class 2

UL 1449

(2021) UL Standard for Safety Surge
Protective Devices

1.2 ADMINISTRATIVE REQUIREMENTS

- a. Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM applies to this section, with the additions and modifications specified herein.
- b. Programmable Logic Controllers (PLCs) receive information from pressure transmitters and other devices to control the pumps and control valves.
- c. The control system must be furnished by a single supplier. See Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT for other required components of the control system. The control system supplier must be responsible for providing a fully functional control system, in accordance with the drawings and specifications, including the field devices. Installation must be in accordance with NFPA 70.
- d. Submit six copies of Operation and Maintenance Manuals, within 7 calendar days following the completion of factory tests. Installation, Operation, and Maintenance manuals for all system components supplied must be furnished following the completion of shop tests and must include:
 - (1) Pump Control Panel including interior and exterior system components layout. Complete guide outlining step-by-step procedures for system startup and operation.
 - (2) All documents previously submitted and approved with all comments and field changes annotated. Complete description of the sequence of operation including that described in PART 3 and any subsystems not controlled by the PLC (e.g. alarm annunciator safety/circuit).
 - (3) Complete listing of all programming of the PLCs, laptop computer, and Personal Computer.
 - (4) Complete relay ladder logic diagrams, PLC input/output diagrams and control power distribution diagrams for the complete control system.
 - (5) Complete troubleshooting guide, which lists possible operational problems and corrective action to be taken, including all as-built conditions.
- e. Submit documents demonstrating the accuracy and completeness of the

list of material and components, that items proposed comply fully with contract requirements, and are otherwise suitable for the application indicated. Documents must consist of all data or drawings published by the manufacturer of individual items listed including manufacturer's descriptive and technical literature, performance data, catalog cuts, and installation instructions. Submit additional material if the listed items are not adequate to identify intent or conformance to technical requirements. Provide typed and electronic copies of these lists for approval. Any delays associated with resubmittals of incomplete or ambiguous initial submittals will be the Contractor's responsibility.

- f. Documents must be bound in a suitable binder adequately marked or identified on the spine and front cover. A table of contents page must be included and marked with pertinent contract information and contents of the manual. Tabs must be provided to separate different types of documents, such as catalog ordering information, drawings, instructions, and spare parts data. Index sheets must be provided for each section of the manual when warranted by the quantity of documents included under separate tabs or dividers.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawing; G

SD-03 Product Data

Tools and Spare Parts

Pump Control Panel (PCP) and Components; G

Laptop Computer; G

Personal Computer (PC); G

Programmable Logical Controller (PLC) Hardware and Software; G

Control Wiring Data Lists; G

Graphics Display Screen; G

SD-06 Test Reports

Certified Pump Control Panel (PCP) Shop Test Report

Record of Test

SD-07 Certificates

Experience and Qualifications; G

Testing Plan; G

Training Plan for Instructing Personnel; G

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

1.4 TOOLS AND SPARE PARTS

Provide the following:

- a. Any special tools necessary for operation and maintenance of the system components providing supplier name, current cost, catalog order number, and a recommended list of spare parts to be stocked.
- b. One spare set of fuses of each type and size.
- c. Recommended manufacturer list of spare parts, including part number, current unit price, and source of supply.
- d. One spare power supply module.
- e. One spare I/O module for discrete devices and one for analog devices.
- f. Two PLC RAM back-up batteries.
- g. Two complete sets of spare ink cartridges for the laser printer.
- h. Minimum of 10 spare lamps for the Alarm Annunciator.
- i. Minimum of 10 spare lamps of each type of non-LED lamps used on the Pump Control Panel.

1.5 EXPERIENCE AND QUALIFICATIONS

Submit the following data for approval:

- a. Certification stating that the manufacturer has manufactured, installed, and successfully completed at least five PLC-based systems for automatic cycling of pumps based upon varying dispensing demands ranging from 0 to 182 L/s utilizing multiple pumps. At least two of the five PLC-based systems must be for dispensing jet fuel into a pressurized, constant pressure, flow demand aircraft hydrant system.
- b. Certification that the control systems have successfully operated over the last 2-years and are currently in service.
- c. Project names, locations, and system description of these installations. Include user point-of-contact and current telephone numbers.

1.6 WARRANTY

The Pump Control and Annunciation System including devices, hardware and software must be warranted for a period of one year from the date of acceptance of the system by the Government. This warranty service must include parts and labor service for system components supplied under this specification. Upon notification by the Government of system or component failure, respond at the site with necessary parts within 48-hours of

notification.

PART 2 PRODUCTS

2.1 MATERIALS AND SYSTEM COMPONENTS

2.1.1 Pump Control Panel (PCP) and Components

NEMA ICS 1, NEMA ICS 6, NEMA 250, and UL 508. The PCP enclosure must be a freestanding NEMA Type 12, smooth, gasketed enclosure constructed of 12 gauge steel. All seams must be continuously welded and there must be no drilled holes or knockout prior to delivery to the job site. The pump control panel dimensions must be a maximum of 2.2 m high, maximum 1.6 m wide, and a maximum of 610 mm deep and must have removable lifting eyes. The interior surfaces of the panel must be properly cleaned, primed, and spray painted with white high-gloss enamel. Exterior surfaces must have standard factory finish. Access for the PCP must be front only and must consist of hinged doors having 3-point latching mechanisms. The doors must open approximately 120 degrees. Rack mounting angles, swing-out panels and other component mounting hardware must be installed such that servicing of one component must not require removal or disconnection of other components. No clearance will be required between the back of the panel and the room walls. Terminal facilities must be arranged for entrance of external conductors from the top or bottom of the enclosure.

Pump Control Panel face must be fitted with overview diagrams as shown on the drawings with the following color scheme: RAL 1024 for the bulk fuel tanks, RAL 1001 for the drain tank, and RAL 5014 for the additive injector building and system.

2.1.2 Ventilation System

Provide one supply fans, single phase, 115 volt. Each fan must supply a minimum of 47 L/s. The supply and exhaust grill must contain a filter that is easily removed from the exterior of the enclosure. Also provide two thermostats with an adjustable set point range of 21 degrees C to 60 degrees C. Locate the thermostats near the top in the interior of the PCP.

2.1.3 Ground Bar

The control panel must have a tin-plated copper equipment ground bar. The bar must have a minimum of twenty grounding screws.

2.1.4 Standard Indicator Lights

NEMA ICS 1, NEMA ICS 2, and UL 508. Lights must be heavy duty, NEMA 13, 22.5 mm mounting hole, round indicating lights operating at 120 volts ac/dc or 24 volts ac/dc. Long life bulbs must be used. Indicator lights must have a legend plate with words as shown on drawings. Lens color as indicated on the drawings. Lights must be "push to test (lamp)" type. LED type lamps of comparable size and color may be substituted for standard indicator lights.

2.1.5 Selector Switches

NEMA ICS 1, NEMA ICS 2, and UL 508. Non-illuminated lever operated selector switches must be heavy duty, NEMA 13, round, and utilize a 22.5 mm mounting hole. They must have the number of positions as indicated on the drawings. Switches must be rated 600 volt, 10 amperes continuous.

Legend plates must be provided with each switch with words as indicated on the drawings.

2.1.6 Pushbuttons

NEMA ICS 1, NEMA ICS 2, and UL 508. Non-illuminated pushbuttons must be heavy duty, NEMA 13, round, utilize a 22.5 mm mounting hole, and have the number and type of contacts as indicated on the drawings or elsewhere in the specifications. The emergency stop switch must be a red mushroom head, 38 mm diameter, momentary contact type. Pushbuttons must be rated 600 volt, 10 amperes continuous. Legend plates must be provided with each switch with words as indicated on the drawings.

2.1.7 Relays

IEEE C37.90, NEMA ICS 2, UL 508.

2.1.8 Nameplates

Nameplates must be made of laminated plastic with black outer layers and a white core. Edges must be chamfered. Nameplates must be fastened with black-finished round-head drive screws or approved nonadhesive metal fasteners.

Labeling of devices and functions must be with permanent machine-printed ink on labeling strips inverted into transparent windows. Text shall be in English and Spanish.

2.1.9 Transient Voltage Surge Suppression Devices

IEEE C62.41 for Category "B" transients, UL 1449.

2.1.10 Terminal Blocks

NEMA ICS 4. Terminal blocks for conductors exiting the PCP must be two-way type with double terminals, one for internal wiring connections and the other for external wiring connections. Terminal blocks must be made of bakelite or other suitable insulating material with full deep barriers between each pair of terminals. A terminal identification strip must form part of the terminal block and each terminal must be identified by a number in accordance with the numbering scheme on the approved wiring diagrams.

2.1.11 Circuit Breakers

UL 508. Individual, appropriately sized, terminal block mounted, circuit breakers must be provided for all 120 volt PCP mounted system components and for the 120 volt terminal boards shown on the drawings.

2.1.12 Uninterruptible Power Supplies

UL 1012. Input voltage must be 120 volts (nominal), 1 phase, 60 Hertz. Output voltage regulation must be plus/minus 5.0 percent for the following conditions:

- a. 20 to 100 percent load on output.
- b. Input voltage variation of minus 15 to 10 percent.

- c. Constant load power factor between 80 and 100 percent.

Response time must be 1.5 cycles or less. Battery capacity must be such as to provide an orderly shut down of operating programs or as a minimum 10 minutes.

2.1.13 Miscellaneous Power Supplies

UL 1012. Certain field devices may require power other than 120 VAC (i.e. 24 VDC). The power supplies must be convection cooled, have fully isolated independent outputs, have constant voltage, have short circuit and overvoltage protection, and have automatic current limiting.

2.1.14 Alarms

UL 508 and **ISA 18.1.** Alarms must be provided as visual annunciation, local and remote monitoring, constant or flashing visual and audible alarms as specified herein.

2.1.15 Alarm Horns

UL 508. The alarm horns must consist of 3-vibrating horns and 2-resonating horns. One vibrating horn is to be mounted in the PCP, and two vibrating and two resonating horns must be mounted outside of the additive injector building as shown on the drawings. The exterior horns must each produce 100 db at 3 m and must be provided in a weather proof housing. The PCP horn must produce 70 db at 3 m.

2.1.16 Laptop Computer

2.1.16.1 Hardware

The following are the minimum hardware requirements for the laptop computer:

- a. Latest Core i3 CPU operating at 3.9 GHz or faster
- b. 8 GB RAM
- c. 1 TB hard drive
- d. 16X Read Write DVD drive
- e. 381 mm 1080p LED - Backlit Display
- f. Keyboard
- g. Pointing device (e.g. mouse, track ball)
- h. 120VAC and Battery power supply
- i. All cables and connectors for interfacing with PLC
- j. HDMI Port
- k. Two USB 3.0 communications ports
- l. Provide a carrying case for the Laptop Computer

2.1.16.2 Software

The following is the minimum software to be loaded on the laptop. The software must be the most current versions and compatible with each other to make a complete and usable system. All software needs to be fully site licensed (provide with software key) and come with all disks to allow a full restore or reload of software in the event of a hard drive crash.

- a. Operating system (e.g. the latest commercially available MS Operating system)
- b. Software for programming the PLCs

2.1.17 Personal Computer (PC)

2.1.17.1 Hardware

The following are the minimum hardware requirements for the personal computer:

- a. Latest Core i3 CPU operating at 2.4 GHZ or faster
- b. 8 GB RAM
- c. 1 TB hard drive
- d. 16X Read Write DVD drive
- e. Color 432 mm 1080p LED-Backlit flat screen monitor
- f. Keyboard
- g. Pointing device (e.g. mouse)
- h. Serial communication port compatible with PLC (e.g. RS-232-C, RS-485)
- i. 120 VAC operating power
- j. All cables and connectors interfacing with PLC and Laser Printer
- k. Provide a modem capable of remote troubleshooting of the system. The modem will not be permanently connected to the System
- l. Four USB 3.0 communications ports
- m. Two HDMI ports

2.2 PROGRAMMABLE LOGICAL CONTROLLER (PLC) HARDWARE AND SOFTWARE

2.2.1 General

- a. **NEMA IA 2.** PLC must be able to receive discrete and analog inputs and through its programming it must control discrete and analog output functions, perform data handling operations and communicate with external devices and remote I/O racks. PLC must be a modular, field expandable design allowing the system to be tailored to the process control application. The capability must exist to allow for expansion to the system by the addition of hardware and user software. At a minimum the PLC must include mounting backplanes, power supply

modules, CPU module, communication modules, and I/O modules.

- b. Design and test PLC provided for use in the high electrical noise environment of an industrial plant. PLC module must comply with the **FCC Part 15** Part A for radio noise emissions. The programmable controller processor must be able to withstand conducted susceptibility tests as outlined in **NEMA ICS 2**, **IEEE C37.90**.
- c. PLC must function properly at temperatures between **0 and 50 degrees C**, at 5 to 95 percent relative humidity non-condensing and have storage temperatures between **minus 40 to 60 degrees C** at 5 to 95 percent relative humidity non-condensing.
- d. PLC must have manufacturer's standard system status indicators (e.g. power supply status, system fault, run mode status, back-up battery status).

2.2.2 Central Processing Unit Module

The CPU must be a modular self-contained unit that will provide time of day, scanning, application (ladder rung logic) program execution, storage of the application program, storage of numerical values related to the application process and logic, I/O bus traffic control, peripheral and external device communications and self-diagnostics.

2.2.3 Power Supply Module

- a. The power supply module must be plugged into the backplane not separately mounted. The power supply must be wired to utilize 120 VAC, 60 Hz power, the system must function properly within the range of minus 10 to plus 15 percent of nominal voltage. The power supply must provide an output to the backplane at a wattage and voltage necessary to support the attached modules. A single main power supply module must have the capability of supplying power to the CPU module and local communication and I/O modules. Auxiliary power supplies must provide power to remote racks.
- b. Each power supply must have an integral on/off disconnect switch to the module. If the manufacturers standard power supply does not have an on/off disconnect switch a miniature toggle type switch must be installed near the PLC and clearly labeled as to its function.
- c. The power supply must monitor the incoming AC line voltage for proper levels and have provisions for both over current and over voltage protection. If the voltage level is detected as being out of range the system must have adequate time to complete a safe and orderly shutdown.

2.2.4 Program Storage/Memory Requirements

- a. The PLC must have the manufacturers standard nonvolatile executive memory for the operating system. The PLC must also have EEPROM (Electrically Erasable Programmable Read Only Memory) for storage of the user program and battery backup RAM for application memory. The EEPROM must be loaded by use of the laptop computer or the personal computer.
- b. Submit a calculation of the required amount of EEPROM and RAM (random access memory) needed for this application plus an extra 50 percent.

- c. The number of times a normally open (N.O.) or normally closed (N.C.) contact of an internal output can be programmed must be limited only by the memory capacity to store these instructions.

2.2.5 Input/Output (I/O) Modules

- a. Provide all required I/O modules (analog input, analog output, discrete input, discrete output, and isolated discrete output) to manipulate the types of inputs and outputs as shown on the drawings and to comply with the sequence of operations. Also provide a minimum of 20 percent (round up for calculation) spare input and output points of each type provided, but not less than 2 of each type.
- b. I/O modules must be a self-contained unit housed within an enclosure to facilitate easy replacement. All user wiring to I/O modules must be through a heavy-duty terminal strip. Pressure-type screw terminals must be used to provide fast, secure wire connections. The terminal block must be removable so it is possible to replace any input or output module without disturbing field wiring.
- c. During normal operation, a malfunction in any remote input/output channel must affect the operation of only that channel and not the operation of the CPU or any other channel.
- d. Isolation must be used between all internal logic and external power circuits. This isolation must meet the minimum specification of 1500 VRMS. Provide optically isolated I/O components which are compatible with field devices.
- e. Each I/O module must contain visual indicators to display ON/OFF status of individual input or output points.
- f. Discrete output modules must be provided with self-contained fuses for overload and short circuit protection of the module.
- g. All input/output modules must be color coded and titled with a distinctive label.

2.2.6 Interfacing

The PLC must have communication ports and communication modules using the manufacturers standard communication architecture for connections of the Personal computer, and Laptop Computer. The PLC must have the ability to communicate with future Automated Fuel Handling System Remote Terminal Unit. Provide spare Ethernet and Modbus communications links which are compatible with Aveva Wonderware Orchestra SCADA software.

2.2.7 Program Requirements

- a. The programming format must be ladder diagram type as defined by **NEMA IA 2**.
- b. There must be a means to indicate contact or output status of the contact or output on the monitor (of the personal computer) or LCD screen (of the laptop computer). Each element's status must be shown independently, regardless of circuit configuration.
- c. The program must be full featured in its editing capabilities (e.g.

change a contact from normally open to normally closed, add instructions, change addresses).

2.2.8 Diagnostics

The CPU must continuously perform self-diagnostic routines that will provide information on the configuration and status of the CPU, memory, communications and I/O. The diagnostic routines must be regularly performed during normal system operation. A portion of the scan time of the controller should be dedicated to perform these housekeeping functions. In addition, a more extensive diagnostic routine should be performed at power up and during normal system shutdown. The CPU must log I/O and system faults in fault tables, which must be accessible for display. When a fault affects I/O or communication modules the CPU must shut down only the hardware affected and continue operation by utilizing healthy system components. All faults must be annunciated on the alarm annunciator.

PART 3 EXECUTION

3.1 PUMP CONTROL PANEL (PCP) AND COMPONENTS

3.1.1 General

- a. Where two or more system components performing the same function are required, they must be exact duplicates produced by the same manufacturer. All display instruments of each type must represent the same outward appearance, having the same physical size and shape, and the same size and style of numbers, characters, pointers, and lamp lenses.
- b. The PCP must include all required resident software programs and hardware to provide the specified sequence of operation. All software optical discs, including programming manuals, must be turned over to the Government at the completion of start-up so modification can be done in the field with no outside assistance.
- c. It is intended that process controlling devices except field devices, and motor controllers be attached to or mounted within the PCP enclosure and all interconnecting wiring installed prior to shipment to the job site. This is to allow shop testing of the system and to decrease field labor requirements.
- d. The PCP must be shipped fully assembled in one piece after the completion of the shop tests and all defects corrected.

3.1.2 Wiring

Wiring methods and practices must be in conformance with NEMA ICS 1, NEMA ICS 2, NEMA ICS 4, and NEMA ICS 6 recommendations as applicable. All wiring to instruments and control devices must be made with stranded wire, and wiring must be permanently labeled with conductor/wire numbers within one-inch of termination points. Labels must be tubular heat-shrinkable wire markers that remain legible after exposure to industrial fluids and abrasion. Position markers so that wire numbers can be read without disturbing or disconnecting wiring. Use of individual character-markers placed side-by-side is not acceptable. Numbers must match approved shop drawings. All wiring must be neatly laced from point of entry into enclosures to termination points with nylon lacing cord or plastic lacing

ties. Lacing within wiring channels is not required. Where the PCP contains intrinsically safe wiring, barriers or other devices, the panel must be arranged so that all intrinsically safe wiring is separated from non-intrinsically safe wiring by approved methods. Wiring channels containing intrinsically safe wiring must be blue and be labeled as "Intrinsically Safe Wiring Only" on the cover. All conduit penetrations into the PCP containing intrinsically safe wiring must be separated from non-intrinsically safe conduit penetrations by a minimum of 50 mm. Provide typed [Control Wiring Data Lists](#) within each terminal cabinet and the PCP. The data lists must include: conductor identification number, wire gauge, wire insulation type, "FROM" terminal identification, "TO" terminal identification, and remarks. The preliminary lists generated by the Contractor will be submitted for approval to the Contracting Officer and will be updated to As-Built conditions by the Contractor. The As-Built data lists must be placed in a document holder within each enclosure.

3.1.3 [Certified Pump Control Panel \(PCP\) Shop Test Report](#)

The manufacturer must shop test the PCP, Personal computer, and laptop computer. The procedure must include simulation of field components and must provide for fully testing the pump control and annunciator system as a unit before delivery to the project site. The test must, reveal system defects, including, but not limited to, functional deficiencies, operating program deficiencies, algorithm errors, timing problems, wiring errors, loose connections, short circuits, failed components and misapplication of components. The test must be performed prior to shipment to the site and problems detected must be corrected. The final testing and correction sequence must be repeated until no problems are revealed and then two additional successful tests must be performed. Submit certified test report within 15-days after completion of the test. The report must include a statement that the Pump Control Panel performs as specified. Notify the Contracting Officer and the Service Headquarters 30-days prior to the final shop testing date. The Contracting Officer may require a Government witness at the final test before the PCP is shipped to the site.

3.1.4 Ventilation System

Thermostat T-1, must control fan F-1. T-1 must be set at 27 degrees C to maintain interior air temperature to minus 7 degrees C above ambient. Thermostat T-2 , set at 38 degrees C, must provide a non-critical PCP HIGH TEMPERATURE alarm to the alarm annunciator.

3.1.5 Grounding

The PCP ground bar must be connected to the building counterpoise via a #10 AWG conductor. Within the enclosure all I/O racks, processor racks, and power supplies must be grounded to meet the manufacturer's specifications.

3.1.6 Indicator Lights, Switches, and Pushbuttons

Indicator lights, switches, and pushbuttons must be mounted through the PCP enclosure and must be arranged to allow easy vision and operation of each device. Each device must have a nameplate or legend plate as indicated on the drawings. Nameplate wordings must be as indicated on the drawings.

3.1.7 Surge Protective Devices

Surge protective devices (SPD) must be installed in the PCP to minimize effects of nearby lightning strikes, switching on and off of motors and other inductive loads. SPD must be provided for each control circuit ladder. Each ladder may contain any combination of the following devices: PLCs, power supplies (e.g., 24 volt), fans, relays, lights, switches. SPD must also be provided for PLC I/O originating outside of the building.

3.1.8 Terminal Blocks

As a minimum, any PCP device that connects to a field device (devices not located in the PCP) must be connected to a terminal block. A connection diagram similar to the drawings must be provided to the field Contractor for field connections to the PCP.

3.1.9 Circuit Breakers

As a minimum, any 120 volt PCP device i.e. (fans, lights, power receptacles, 24 VDC power supplies, PLC CPUs, PLC I/O racks) must be provided with an individual circuit breaker. Additionally 120 volt terminal boards connecting to field devices (devices not located in the PCP) must be protected by a 120 volt circuit breaker.

3.1.10 Uninterruptible Power supplies

The PCP must contain two uninterruptible power supplies (UPS) (each connected to a dedicated circuit). As shown on the drawings one UPS must supply the PLC System, and the second UPS must supply the miscellaneous device power. The UPSs output capacity must be sufficient to drive all the system components connected plus 25 percent. The UPSs must be mounted on shelves near the bottom of the PCP but not rest on the floor of the PCP.

3.1.11 Power Supplies

Provide and install all 120 VAC and 24 VDC power supply. Interconnecting wiring between UPSs and PLC power supplies must be completely installed prior to shipment to the job site.

3.1.12 Alarms and Horns

Signals must be initiated by hardwired field contacts or by PCP outputs as required. The annunciator must energize alarm horns, both an integral panel mounted vibrating horn and remote horns, and flash the appropriate alarm lamp. The drawings indicate panel layout and the alarms to be annunciated.

3.1.12.1 Alarm Sequence

Alarm sequence for each alarm must be as follows (ISA 18.1 sequence 'A').

- a. For a normal condition, visual indicator and horns will be off.
- b. For an alarm condition, visual indicator will flash, and horns will sound (this condition will be locked in).
- c. Upon acknowledgment of the alarm condition, visual indicator will be steady on and the horns will be off.

- d. If, after acknowledgment of an alarm condition, another alarm condition is established, the new alarm will cause the appropriate window to flash and the horn to sound.
- e. When condition returns to normal after acknowledgment, the visual indicator and the horn will be off.

3.1.13 Personal Computer

The personal computer must be a stand alone, desk top mounted unit. The personal computer must download system parameters from the PLCs for display. The personal computer must also upload new set point values that the operator has changed using the personal computer keyboard, after a password has been entered.

3.1.13.1 Screen Number 1

This must be a general opening screen. As a minimum it must display the name and location of the installation (e.g. Selfridge Air National Guard Base, Michigan), name of the project (e.g., Aviation Fuel Farm) and screen navigation information.

3.1.13.2 Screen Number 2

Screen number two must list the systems parameters monitored by the PCP and their current value. The values must be continuously updated, a 2 second delay maximum between updates will be acceptable.

3.1.13.3 Screen Number 3

This screen must be a depiction of the Graphic Display Screen showing a schematic of the process flow. This screen must be referred to as the graphical display. Many operating parameters must be displayed here as required in later paragraphs of this specification. This screen must show major equipment and piping as indicated on the flow schematic and mechanical instrumentation sheets on the drawings.

3.1.13.4 Screen Number 4

This screen must list all alarms which are displayed on the PCP face and their conditions. This screen must superimpose on the PC monitor whenever an alarm is activated.

3.1.13.5 Screen Number 5

This screen must be an alarm history screen. This screen must be referred to as the Alarm History Display. This screen must be capable of storing and displaying all alarms that have occurred in the system for at least a period of 30-days.

3.1.14 Laptop Computer

The Laptop computer must be used to create, edit, and load the ladder logic program into the PLCs and the operator interface graphics control program into the personal computer. The Laptop must also be used to monitor the PLCs memory and ladder logic program. The computer must be stored in a lockable cabinet located within the Pump Control Panel.

3.2 PROGRAMMABLE LOGICAL CONTROLLER (PLC) HARDWARE AND SOFTWARE

3.2.1 General

The basic operation of the PLC system is (Reference "Control System Block Diagram" on the drawings):

- a. The CPU and its associated I/O rack (I/O-1) sends system outputs to appropriate devices and receive input signals from field devices (PITs-1, DPTs-1, DPTs-3, flow switches, valve limit switches).

3.2.2 Programs

- a. Provide two copies of all working programs (i.e. PLC logic, personal computer) on read only optical discs as well as a printed program listing.
- b. Provide rung comments (documentation) in the ladder logic program. Each device, on the ladder, must be identified as to the type of device, i.e. limit switch XX, flow indicator XX, motor starter XX. Rung comments must be provided for input and output rungs. The programmer must also provide a comment describing the function of each rung or group of rungs that accomplish a specific function.

3.3 GRAPHICS DISPLAY SCREEN

3.3.1 General

The graphic display screen shall be capable of being displayed on the personal computer monitor.

3.3.2 Display Presentation

The Graphic Display shall depict the process fuel flow schematically as indicated on the drawings. Red, green, and amber symbols shall be integrated with the process schematic to provide current equipment status graphically. The symbols shall be located immediately adjacent to related equipment symbol.

3.3.3 Process Schematic

The process schematic graphic representation shall utilize conventional symbols when possible. Symbols and flow lines shall be sized and spaced so as to provide a clear representation of the system process. The Graphic Display shall be suitable for supervised field modification when future items are added. Minor changes may be incorporated to allow proper line width and spacing. Component arrangement, piping routing, and location of valves shall match the flow diagram. The Graphic Display layout shall be approved by the Government.

3.3.4 Digital Tank Level Indicators

The graphics display screen shall have digital displays for the tank levels as indicated on the drawings.

3.4 INSTALLATION

Installation must conform to the manufacturer's drawings, written

recommendations and directions.

3.4.1 Shop Drawing

The shop drawing must be clear and readable and preferably drawn using a computer aided drafting package. At the conclusion of the project the diagram drawings must be redrafted to include all as-built conditions. These updated drawings must be included in the O&M Manuals and appropriate section of the drawings placed in a data pocket located in each of the enclosures. The shop drawing at a minimum must show:

- a. Overall dimensions, front, side and interior elevation views of the PCP showing size, location and labeling of each device.
- b. Overall dimensions, front elevation of the GDP showing graphical layout and size, location and labeling of each device.
- c. Power ladder diagram indicating power connections between SDP, power conditioners, PLCs, power supplies and field and panel devices. Any terminal block connection numbers used must be indicated.
- d. Control ladder diagram indicating control connections between field and devices and PLC I/O modules. Terminal block connection numbers and PLC terminal numbers must be indicated.
- e. Communication connections between PLCs and I/O racks. Communication channel numbers must be indicated.
- f. Bill of materials.
- g. Written control sequence covering all inputs, outputs, and control scheme.

3.4.2 System Start-Up and Testing

- a. At PCP start-up and testing provide personnel, onsite, to provide technical assistance, program fine tuning, and to start-up and test the system. Start-up and testing must be coordinated with the overall fueling system start-up test specified in Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). Prior to this test, all connections must have been made between the PCP, the personal computer, the motor control center, and all field devices. In addition, wiring must have been checked for continuity and short circuits. Adjust set point values, timing values, and program logic as required to provide a functional hydrant fuel control system. Once the system has been fine tuned and passed the system test, the new system default values, must be loaded into the PLC EEPROM and the personal computer screens adjusted to indicate the new values.
- b. A step-by-step Testing Plan of the PCP must be submitted. The test must be designed to show that every device (lights, switches, personal computer display screens, alarms) on the PCP and personal computer is in working order and that the PLC program controls the system per specifications. The test must be performed in conjunction with Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). The plan must include a place for the Contractor and government representative to initial each step of the plan after satisfactory completion and acceptance of each step. The complete initialed Record of Test must be certified by the Contractor and then submitted.

3.4.3 Training Plan for Instructing Personnel

- a. Upon completion of the system start-up a competent technician regularly employed by the PCP manufacturer must hold a training class for the instruction of Government personnel in the operation and maintenance of the system. Provide both classroom type theory instruction and hands-on instruction using operating system components provided. The period of instruction must be a minimum of three 8-hour working days. The training must be designed to accommodate 8 operators, 4 maintenance personnel, and 2 programmers. The Contracting Officer must receive written notice a minimum of 14-days prior to the date of the scheduled classes.
- b. Furnish a written lesson plan and training schedule for Government approval at least 60-days prior to instructing operating, maintenance and programming personnel. Concurrently submit above to the Service Headquarters or officially designated alternate for their input into the review process. Approval of lesson plan will be based on both Government and Service Headquarters' concurrence. This plan must be tailored to suit the requirements of the Government. The training must be divided into three separate classes. Each class must be tailored to a specific group of personnel. The groups are: 1) Operators, those that will use the control system on a day-to-day basis; 2) Maintenance personnel, those that will perform routine and non-routine maintenance and trouble shooting of the control system; 3) Programmers, those that will make changes to and trouble shoot the PLC and personal computer programs. The training program must provide:
 - (1) a detailed overview of the control system including the complete step-by-step procedures for start-up, operation and shut-down of the control system.
 - (2) a general overview of programmable logic controllers.
 - (3) the maintenance of system components installed.
 - (4) the programming of the PLC and Personal Computer.
 - (5) troubleshooting of the system.
- c. Complete approved Operation and Maintenance manuals for this specification and 26 2000 INTERIOR DISTRIBUTION SYSTEM (specifically pertaining to the motor control center and its relay ladder diagrams) must be used for instructing operating personnel. Training must include both classroom and hands-on field instruction. The class must be recorded in DVD format.
- d. Provide training courses in DVD format covering system overview, operation, maintenance, trouble shooting, and programming. These DVDs must be produced onsite by the Contractor using the supplied Pump Control Panel as the teaching aid, or commercially produced DVDs by the PLC manufacturer or third party who specializes in training on PLC systems. Along with the DVDs, provide workbooks, which follow along with the DVDs.

3.5 PLC CONTROL SYSTEM SEQUENCE OF OPERATION

The following describes general functions of the fueling system components.

3.5.1 Abbreviations

- a. SYS: components of System including UPS, power supply, CPU, I/O, and system input and outputs.
- b. CPU: SYS PLC CPU.
- c. I/O: SYS PLC input/output modules.
- d. PCP: Pump Control Panel.
- e. PC: Personal Computer.
- f. UPS: Uninterruptible Power Supply.

3.5.2 Operating Tanks

3.5.2.1 Level Control

Each operating tank has four level float switches to measure low-low, low, high and high-high levels.

3.5.2.1.1 Low-Low Level

When the low-low level float is activated the associated tank's graphic display low-low level light must light up. If the selected tank's level drops below the low-low sensor the alarm annunciator's low-low level alarm sequence activates, issue pumps running in automatic mode must be disabled. If all tanks are at low-low level, no issue pumps in automatic mode must be enabled.

3.5.2.1.2 Low Level

When the low level float is activated the associated tank's graphic display low level light must light up and the alarm annunciator's non-critical low level alarm sequence activates.

3.5.2.1.3 High Level

When the high level float is activated the associated tank's graphic display high level light must light up and the alarm annunciator's non-critical high level alarm sequence activates.

3.5.2.1.4 High-High Level

When the high-high level float is activated the associated tank's graphic display high-high level light must light up, and the alarm annunciator's critical high-high level alarm sequence activates.

3.5.2.2 Level Control

Each operating tank has level switches to monitor low-low, low, high, and high-high fuel levels.

3.5.2.2.1 Low-Low Level

When the low-low level elevation is attained the associated tank's GDP low-low level light must light up. If the selected tank's level drops

below the low-low sensor the alarm annunciator's low-low level alarm sequence activates, issue pumps running in automatic mode must be disabled. If all tanks are at low-low level, no issue pumps in automatic mode must be enabled.

3.5.2.2.2 Low Level

When the low level elevation is attained the associated tank's GDP low level light must light up and the alarm annunciator's non-critical low level alarm sequence activates.

3.5.2.2.3 High Level

When the high level elevation is attained the associated tank's GDP high level light must light up and the alarm annunciator's non-critical high level alarm sequence activates.

3.5.2.2.4 High-High Level

When the high-high level elevation is attained, the associated tank's GDP high-high level light must light up and the alarm annunciator's critical high-high level alarm sequence activates. The pump control panel must de-energize the solenoid on the tank's high level shutoff valve to force it closed.

3.5.3 Drain Tank

3.5.3.1 Drain Tank Pump

The pump's motor controller has a status relay to indicate the on/off status of the pump. When status relay is open the pump's graphic display off light must activate. When the status relay is closed the pump's graphic display on light must light up. The status relay state must also be used to start and stop the pumps elapsed run time timer.

3.5.3.2 Overfill Valve (OV)

The tank's overfill valve has a limit switch to indicate valve position. The switch is SPST. The switch must close when the valve is fully closed. When the limit switch is closed, the tank's graphic display valve closed light must light up and the alarm annunciator's non-critical alarm sequence activates. When the limit switch is open the tank's graphic display valve open light must light up.

3.5.3.3 High Level Alarm

The tank has a high level alarm float switch. The switch is SPST. When the high level alarm float is activated the tank's graphic display high level light must light up and the alarm annunciator's critical alarm sequence activates.

3.5.3.4 High-High Level Alarm

The tank has a high-high level alarm float switch. When the high-high level alarm float is activated the tank's graphic display high-high level light must light up and the alarm annunciator's critical alarm sequence activates.

3.5.3.5 Leak Detection

The tank has a leak detection system. When the leak alarm is activated the alarm annunciator's non-critical alarm sequence activates.

3.5.4 Fueling Pumps (FP)

There are four fueling pumps with a maximum of two pumps running at one time. The lead pump selector switch must select the pump starting sequence. Each pump's motor controller has a status relay to indicate the on/off status of the pump. When status relay is open the associated pump's graphic display off light must activate and screen number 2 must indicate on. When the status relay is closed the associated pump's graphic display on light must activate and screen number 2 must indicate off. The status relay state must also be used to start and stop the pumps elapsed run time timer and must be displayed on screen number 2.

3.5.5 Flow Switch, Fueling Pump

On the discharge side of each pump is a flow switch to indicate positive flow (fail safe feature). If the PLC has given a signal to start a pump and the flow switch has not closed before the set point timer expires or if the flow switch opens after the pump has been running then the pump must be in a failure state and it must be disabled (taken out of the starting sequence), the alarm annunciator's non-critical alarm sequence must also be activated, and the next pump in the start sequence started. After the PLC has stopped all of the pumps, any failed pump must be added back into the start sequence.

3.5.6 Transmitters

3.5.6.1 Differential Pressure Transmitter (DPT)

The DPT's measure flow in L/s. The net flow is sent to the personal computer display. The issue rate, return rate and net flow must be displayed on the personal computer.

3.5.7 Safety Circuit

3.5.7.1 Emergency Stop Status

The emergency stop circuit status relay (ER1) N.O. contact must be connected to I/O-1, I/O-2 and UPS#3 as indicated on the Terminal Block Connection drawing. When the circuit is activated the alarm annunciator's critical alarm sequence is activated and any calls to start fueling pumps must be canceled and no additional pump start signals must be sent until the circuit has been reset. The fueling pumps will actually be stopped by an emergency stop circuit status relay (ESR2, ESR3, or ESR4) N.O. contact in the fuel pump motor control circuit located in the motor control center.

3.5.7.2 Circuit Power Status

When the safety circuit power status relay is closed the PCP emergency circuit power on light must light up.

3.5.8 Pump Control Panel

3.5.8.1 CPU Faults

The PCP mounted CPU on light is connected to SYS. The associated CPU light must light when no system faults are detected. When a fault is detected by the CPU the faulted CPU's on light must be turned off and the alarm annunciator's non-critical alarm sequence must be activated.

3.5.8.2 Mode Select Switch

The 2-position switch selects what mode of fueling is active: issue, or off. The screen number 2 status must indicate the active mode.

3.5.8.3 Lead Pump Selector Switch

The 2-position switch selects which pump must be the lead pump for each tank. The switch position must fix the starting sequence for all pumps. The off sequence must be the reverse of the start sequence, therefore, first on will be last off. A maximum of two pumps will be allowed to run at one time. If a pump fails to start or fails during operation, that pump will be disabled and the next pump in the sequence started. The screen number 2 status display must indicate the lead pump.

3.5.8.4 PCP Temperature Alarm

The alarm thermostat when activated must activate the alarm annunciator's non-critical alarm sequence.

3.6 OPERATING PROGRAM REQUIREMENTS

The control system's logic program must be stored on a EEPROM chip. Default values of operator adjustable parameters must be permanently stored on the chip with the capability of resetting the values in RAM to the values within the range specified below. The default values can be changed through the use of the personal computer (after the correct password has been entered). After loss of power and battery failure the adjustable settings must revert back to the default values located on the chip. The default values shown here must be reset to the values determined during the system start up and test.

SET POINT DESCRIPTION	SET POINT RANGE	DEFAULT VALUE
Timer to establish fueling pump failure	5 to 30 seconds	15 seconds
Should the operator enter a value not within the range for that parameter, the personal computer must indicate "INVALID ENTRY" and revert back to the previous value.		

SET POINT DESCRIPTION	SET POINT RANGE	DEFAULT VALUE
A number inside braces, {x}, in the following paragraphs indicates that the number may be changed by the operator via the operator interface within the Set Point Range listed above.		

3.7 AUTOMATIC MODE - ISSUE OR TRANSFER CONDITION

- a. The lead pump will start when a Pump Control Station start pushbutton is depressed.
- b. Depressing an additional Pump Control Station pushbutton will bring on additional pumps.
- c. Depress the Pump Control Station stop pushbutton will disable the lead pump.
- d. Depressing an additional Pump Control Station stop pushbuttons will disable the pumps in the order they were brought on.

3.8 OFF MODE

- a. Automatic starting of fueling pumps must be disabled. All other functions (graphic display, alarm annunciator, personal computer, control valve solenoids) must be active to allow manual control of the fueling pumps using the Hand-Off-Auto or Hand-Auto switch.
- b. The second, third and fourth fueling pumps maybe started or stopped manually as needed by the operator.

3.9 MANUAL OPERATION OF FUELING PUMPS

- a. If the PLC system is still active see paragraph OFF MODE.
- b. If the PLC system has no power or the CPUs have faulted (CPU light on PCP off) the pumping system will be in a completely manual mode. The safety circuit will need power so that the ESO solenoids on the non-surge check valves will be open and fuel can flow. The solenoids on the other solenoid controlled valves will be de-energized so the valves will have to be manually opened or enabled for the system to run. Other valves may need to be opened or closed manually by the operators for the system to work properly.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 11 00

WATER UTILITY DISTRIBUTION PIPING

02/18, CHG 1: 02/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Water Transmission Mains
 - 1.2.2 Water Mains
 - 1.2.3 Water Service Lines
 - 1.2.4 Additional Definitions
- 1.3 SUBMITTALS
- 1.4 QUALITY CONTROL
 - 1.4.1 Regulatory Requirements
 - 1.4.2 Qualifications
 - 1.4.2.1 Backflow Preventers
 - 1.4.2.1.1 Backflow Preventer Certificate
 - 1.4.2.1.1.1 Backflow Prevention Training Certificate
- 1.5 DELIVERY, STORAGE, AND HANDLING
 - 1.5.1 Delivery and Storage
 - 1.5.2 Handling

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Pipe, Fittings, Joints And Couplings
 - 2.1.1.1 Ductile-Iron Piping
 - 2.1.1.1.1 Pipe and Fittings
 - 2.1.1.1.2 Joints and Jointing Material
 - 2.1.1.2 Plastic Piping
 - 2.1.1.2.1 PVC Piping
 - 2.1.1.2.1.1 PVC Piping
 - 2.1.1.2.1.2 Fittings for PVC Pipe
 - 2.1.1.2.1.3 Joints and Jointing Material for PVC
 - 2.1.1.2.1.4 Corrosion Protection
 - 2.1.1.2.2 PVC Piping for Service Lines
 - 2.1.1.2.2.1 Pipe and Fittings
 - 2.1.1.2.2.2 Joints and Connections
 - 2.1.1.2.2.3 Solvent Joining
 - 2.1.1.3 Copper Pipe For Service Lines
 - 2.1.1.3.1 Copper Tubing and Associated Fittings
 - 2.1.2 Valves
 - 2.1.2.1 Gate Valves 80 mm Size and Larger on Buried Piping
 - 2.1.2.2 Gate Valves 75 mm Size and Larger in Valve Pit(s)
 - 2.1.2.3 Water Service Valves
 - 2.1.2.3.1 Gate Valves Smaller than 75 mm in Size on Buried Piping
 - 2.1.2.3.2 Gate Valves Smaller Than 75 mm Size in Valve Pits
 - 2.1.2.3.3 Check Valves Smaller than 50 mm in Size
 - 2.1.2.4 Valve Boxes

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.1.3 Fire Hydrants and Hose Houses
 - 2.1.3.1 Fire Hydrants
 - 2.1.3.1.1 Dry-Barrel Type Fire Hydrants
- 2.1.4 Backflow Preventers
 - 2.1.4.1 Backflow Preventer Enclosure
- 2.1.5 Disinfection
- 2.2 ACCESSORIES
 - 2.2.1 Pipe Restraint
 - 2.2.1.1 Thrust Blocks
 - 2.2.1.2 Joint Restraint
 - 2.2.2 Protective Enclosures
 - 2.2.2.1 Housing
 - 2.2.3 Tapping Sleeves
 - 2.2.4 Sleeve-Type Mechanical Couplings
 - 2.2.5 Insulating Joints
 - 2.2.6 Dielectric Fittings
 - 2.2.7 Tracer Wire for Nonmetallic Piping
 - 2.2.8 Water Service Line Appurtenances
 - 2.2.8.1 Corporation Stops
 - 2.2.8.2 Curb or Service Stops
 - 2.2.8.3 Service Clamps
 - 2.2.8.4 Goosenecks
 - 2.2.8.5 Curb Boxes
 - 2.2.9 Safety Shower

PART 3 EXECUTION

- 3.1 PREPARATION
 - 3.1.1 Connections to Existing System
 - 3.1.2 Operation of Existing Valves
 - 3.1.3 Earthwork
- 3.2 INSTALLATION
 - 3.2.1 Piping
 - 3.2.1.1 General Requirements
 - 3.2.1.1.1 Termination of Water Lines
 - 3.2.1.1.2 Pipe Laying and Jointing
 - 3.2.1.1.3 Tracer Wire
 - 3.2.1.1.4 Connections to Existing Water Lines
 - 3.2.1.1.5 Sewer Manholes
 - 3.2.1.1.6 Water Piping Crossing Sewer Piping
 - 3.2.1.1.7 Penetrations
 - 3.2.1.1.8 Flanged Pipe
 - 3.2.1.2 Ductile-Iron Piping
 - 3.2.1.3 PVC Water Main Pipe
 - 3.2.1.4 Plastic Service Piping
 - 3.2.1.4.1 Jointing
 - 3.2.1.4.2 Plastic Pipe Connections to Appurtenances
 - 3.2.1.5 Water Service Piping
 - 3.2.1.5.1 Location
 - 3.2.1.5.2 Water Service Line Connections to Water Mains
 - 3.2.2 Backflow Preventers
 - 3.2.2.1 Backflow Preventer Enclosure
 - 3.2.3 Disinfection
 - 3.2.4 Flushing
 - 3.2.5 Pipe Restraint
 - 3.2.5.1 Concrete Thrust Blocks
 - 3.2.5.2 Restrained Joints
 - 3.2.6 Valves
 - 3.2.6.1 Gate Valves

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.2.7 Fire Hydrants
- 3.3 FIELD QUALITY CONTROL
 - 3.3.1 Tests
 - 3.3.1.1 Hydrostatic Test
 - 3.3.1.2 Leakage Test
 - 3.3.1.3 Bacteriological Testing
 - 3.3.1.4 Backflow Preventer Tests
 - 3.3.1.5 Tracer Wire Continuity Test
- 3.4 SYSTEM STARTUP
- 3.5 CLEANUP

-- End of Section Table of Contents --

SECTION 33 11 00

WATER UTILITY DISTRIBUTION PIPING
02/18, CHG 1: 02/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

- | | |
|-------------------|--|
| UNE-EN 197-1 | (2011) Cement - Part 1: Composition, specifications and conformity criteria for common cements. |
| UNE-EN 545 | (2011) Ductile iron pipes, fittings, accessories and their joints for water pipelines - Requirements and test methods |
| UNE-EN 681-1 | (1996/A3:2006) Elastomeric Seals. Materials Requirements For Pipe Joint Seals Used In Water And Drainage Applications. Part 1: Vulcanized Rubber. |
| UNE-EN 805 | (2000) Water Supply. Requirements for Systems and Components Outside Buildings |
| UNE-EN 1074-1 | (2001) Valves for Water Supply - Fitnessfor Purpose Requirements and Appropriate Verification Tests - Part 1: General Requirements |
| UNE-EN 17176-2 | (2019) Plastics piping systems for water supply and for buried and above ground drainage, sewerage and irrigation under pressure - Oriented unplasticized poly(vinyl chloride) (PVC-O) - Part 2: Pipes |
| UNE-EN ISO 1452-2 | (2010) Plastics piping systems for water supply and for buried and above-ground drainage and sewerage under pressure - Unplasticized poly(vinyl chloride) (PVC-U) - Part 2: Pipes (ISO 1452-2:2009) |
| UNE-ENV 1452-6 | (2002) plastics Piping Systems for Water Supply - Unplasticized Poly (Vinyl Chloride) (Pvc-U) - Part 6: Guidance for Installation. |

MINISTERIO DE LA PRESIDENCIA (MP)

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| RD 140/2003 | (2003) Real Decreto 140/2003, de 7 de |
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febrero, por el que se establecen los
criterios sanitarios de la calidad del
agua de consumo humano.

MINISTERIO DE VIVIENDA (MV)

CTE DB-HS4 (2006) Technical Building Code - Basic
Document - Public Health - Water Supply

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.20.1 (2013; R 2018) Pipe Threads, General
Purpose (Inch)

ASME B1.20.3 (1976; R 2013) Dryseal Pipe Threads (Inch)

ASME B16.18 (2021) Cast Copper Alloy Solder Joint
Pressure Fittings

ASME B16.26 (2018) Standard for Cast Copper Alloy
Fittings for Flared Copper Tubes

ASME B18.2.2 (2022) Nuts for General Applications:
Machine Screw Nuts, and Hex, Square, Hex
Flange, and Coupling Nuts (Inch Series)

ASME B18.5.2.1M (2006; R 2011) Metric Round Head Short
Square Neck Bolts

ASME B18.5.2.2M (1982; R 2010) Metric Round Head Square
Neck Bolts

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA B300 (2018) Hypochlorites

AWWA B301 (2018) Liquid Chlorine

AWWA C105/A21.5 (2018) Polyethylene Encasement for
Ductile-Iron Pipe Systems

AWWA C111/A21.11 (2017) Rubber-Gasket Joints for
Ductile-Iron Pressure Pipe and Fittings

AWWA C115/A21.15 (2020) Flanged Ductile-Iron Pipe With
Ductile-Iron or Gray-Iron Threaded Flanges

AWWA C219 (2017) Bolted Sleeve-Type Couplings for
Plain-End Pipe

AWWA C500 (2019) Metal-Seated Gate Valves for Water
Supply Service

AWWA C502 (2018) Dry-Barrel Fire Hydrants

AWWA C503 (2021) Wet-Barrel Fire Hydrants

AWWA C509 (2015) Resilient-Seated Gate Valves for
Water Supply Service

AWWA C511	(2017; R 2021) Reduced-Pressure Principle Backflow Prevention Assembly
AWWA C515	(2020) Reduced-Wall, Resilient-Seated Gate Valves for Water Supply Service
AWWA C600	(2017) Installation of Ductile-Iron Mains and Their Appurtenances
AWWA C605	(2021) Underground Installation of Polyvinyl Chloride (PVC) and Molecularly Oriented Polyvinyl Chloride (PVC0) Pressure Pipe and Fittings
AWWA C651	(2014) Standard for Disinfecting Water Mains
AWWA C655	(2009) Field Dechlorination
AWWA C800	(2021) Underground Service Line Valves and Fittings
AWWA C900	(2016) Polyvinyl Chloride (PVC) Pressure Pipe, and Fabricated Fittings, 4 In. Through 60 In. (100 mm Through 1,500 mm)
AWWA M9	(2008; Errata 2013) Manual: Concrete Pressure Pipe
AWWA M23	(2020) Manual: PVC Pipe - Design and Installation - Third Edition
AWWA M41	(2009; 3rd Ed) Ductile-Iron Pipe and Fittings
AWWA M55	(2020; 2nd Ed) PE Pipe - Design and Installation

ASTM INTERNATIONAL (ASTM)

ASTM A307	(2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A536	(1984; R 2019; E 2019) Standard Specification for Ductile Iron Castings
ASTM A563M	(2007; R 2013) Standard Specification for Carbon and Alloy Steel Nuts (Metric)
ASTM B61	(2015; R 2021) Standard Specification for Steam or Valve Bronze Castings
ASTM B62	(2017) Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM B88M	(2020) Standard Specification for Seamless Copper Water Tube (Metric)

ASTM D1784 (2020) Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds

ASTM D2774 (2021) Underground Installation of Thermoplastic Pressure Piping

ASTM D2855 (2015) Standard Practice for Making Solvent-Cemented Joints with Poly(Vinyl Chloride) (PVC) Pipe and Fittings

ASTM D3139 (2019) Joints for Plastic Pressure Pipes Using Flexible Elastomeric Seals

ASTM F402 (2005; R 2012) Safe Handling of Solvent Cements, Primers, and Cleaners Used for Joining Thermoplastic Pipe and Fittings

FOUNDATION FOR CROSS-CONNECTION CONTROL AND HYDRAULIC RESEARCH (FCCCHR)

FCCCHR List (continuously updated) List of Approved Backflow Prevention Assemblies

FCCCHR Manual (10th Edition) Manual of Cross-Connection Control

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z358.1 (2014; R 2020) American National Standard for Emergency Eyewash and Shower Equipment

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 24 (2022) Standard for the Installation of Private Fire Service Mains and Their Appurtenances

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-600-01 (2016; with Change 6, 2021) Fire Protection Engineering for Facilities

UNDERWRITERS LABORATORIES (UL)

UL 246 (2011; Reprint Jul 2020) UL Standard for Safety Hydrants for Fire-Protection Service

UL 262 (2004; Reprint Oct 2011) Gate Valves for Fire-Protection Service

UNI-BELL PVC PIPE ASSOCIATION (UBPPA)

UBPPA UNI-PUB-08 (2016) Tapping Guide for PVC Pressure Pipe

1.2 DEFINITIONS

1.2.1 Water Transmission Mains

Water transmission mains include water piping having diameters greater than 350 mm, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

1.2.2 Water Mains

Water mains include water piping having diameters 100 through 350 mm, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

1.2.3 Water Service Lines

Water service lines include water piping from a water main to a building service at a point approximately 1.5 m from building or the point indicated on the drawings, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

1.2.4 Additional Definitions

For additional definitions refer to the definitions in the applicable referenced standard.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Connections; G

SD-02 Shop Drawings

Safety Shower Shop Drawing; G

SD-03 Product Data

Pipe, Fittings, Joints and Couplings; G

Valves; G

Valve Boxes; G

Fire Hydrants; G

Pipe Restraint; G

Tapping Sleeves; G

Corporation Stops; G

Backflow Preventer; G

Disinfection Procedures; G

SD-06 Test Reports

Backflow Preventer Tests; G

Bacteriological Samples; G

Leakage Test

Hydrostatic Test

SD-07 Certificates

Pipe, Fittings, Joints and Couplings
Lining

Lining for Fittings

Fire Hydrants

Backflow Prevention Training Certificate

Backflow Certificate

SD-08 Manufacturer's Instructions

Ductile-Iron Piping

PVC Piping

PVC Piping For Service Lines

Copper Pipe For Service Lines

1.4 QUALITY CONTROL

1.4.1 Regulatory Requirements

CTE DB-HS4 and UNE-EN 805. Comply with lead free content requirements.

Comply with NFPA 24 for materials, installation, and testing of fire main piping and components.

1.4.2 Qualifications

1.4.2.1 Backflow Preventers

1.4.2.1.1 Backflow Preventer Certificate

Certificate of Full Approval from FCCCHR List, University of Southern California, attesting that the design, size and make of each backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. Certificate of Provisional Approval will not be acceptable.

1.4.2.1.1.1 Backflow Prevention Training Certificate

Submit a certificate recognized by the State or local authority that

states the Contractor has completed at least 10 hours of training in backflow preventer installations. The certificate must be current.

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 Delivery and Storage

Inspect materials delivered to site for required pipe markings and damage. Unload and store with minimum handling and in accordance with manufacturer's instructions to prevent cuts, scratches and other damage. Store materials on site in enclosures or under protective covering. Store plastic piping, jointing materials and rubber gaskets under cover out of direct sunlight. Do not store materials directly on the ground. Keep inside of pipes, fittings, valves, fire hydrants, and other accessories free of dirt and debris or other contaminants.

1.5.2 Handling

Handle pipe, fittings, valves, fire hydrants, and other accessories in accordance with manufacturer's instructions and in a manner to ensure delivery to the trench in sound undamaged condition. Avoid injury to coatings and linings on pipe and fittings; make repairs if coatings or linings are damaged. Do not place other material, hooks, or pipe inside a pipe or fitting after the coating has been applied. Inspect the pipe for defects before installation. Carry, do not drag pipe to the trench. Use of pinch bars and tongs for aligning or turning pipe will be permitted only on the bare ends of the pipe. Clean the interior of pipe and accessories of foreign matter before being lowered into the trench and keep them clean during laying operations by plugging. Replace defective material without additional expense to the Government. Store rubber gaskets, not immediately installed, under cover or out of direct sunlight.

Handle ductile iron pipe, fittings, and accessories in accordance with AWWA C600 and AWWA M41. Handle PVC pipe, fittings, and accessories in accordance with AWWA C605.

PART 2 PRODUCTS

2.1 MATERIALS

All materials are intended for potable water use unless otherwise indicated.

Provide all materials in accordance with CTE DB-HS4 and RD 140/2003 and as indicated herein. Provide valves and fittings with pressure ratings equivalent to the pressure ratings of the pipe.

2.1.1 Pipe, Fittings, Joints And Couplings

Submit manufacturer's standard drawings or catalog cuts, except submit both drawings and cuts for push-on and rubber-gasketed bell-and-spigot joints. Include information concerning gaskets with submittal for joints and couplings.

2.1.1.1 Ductile-Iron Piping

2.1.1.1.1 Pipe and Fittings

Pipe, Pressure Class 150. Fittings in accordance with UNE-EN 545 with

joints compliant with [UNE-EN 681-1](#). Provide fittings with pressure ratings equivalent to that of the pipe. Provide compatible pipe ends and fittings for the specified joints. Provide cement-mortar [lining](#), per [UNE-EN 197-1](#) and [UNE-EN 545](#), standard thickness on pipe and fittings and exterior epoxy coating.

2.1.1.1.2 Joints and Jointing Material

Provide push-on joints for pipe and fittings unless otherwise indicated.

- a. Push-On Joints: Shape of pipe ends and fitting ends, gaskets, and lubricant for joint assembly as recommended by the fitting manufacturer and [UNE-EN 681-1](#).
- b. Mechanical Joints: Dimensional and material requirements for pipe ends, glands, bolts and nuts, and gaskets as recommended in [AWWA C111/A21.11](#).
- c. Flanged Joints: Bolts, nuts, and gaskets for flanged connections as recommended in Appendix A of [AWWA C115/A21.15](#). Provide ductile iron flanges, Class 125.
- d. Insulating Joints: Designed to prevent metal-to-metal contact at the joint between adjacent sections of piping. Provide flanged type joint with insulating gasket, insulating bolt sleeves, and insulating washers. Provide full face dielectric type gaskets, as recommended in the Appendix to [AWWA C115/A21.15](#). Bolts and nuts, as recommended in the Appendix to [AWWA C115/A21.15](#).
- e. Sleeve-Type Mechanical Coupled Joints: As specified in the paragraph [SLEEVE-TYPE MECHANICAL COUPLINGS](#).

2.1.1.2 Plastic Piping

2.1.1.2.1 PVC Piping

2.1.1.2.1.1 PVC Piping

[UNE-EN ISO 1452-2](#) or [UNE-EN 17176-2](#) plain end or gasket bell end pipe with a minimum Pressure Class PN16 with ductile iron outside diameter (DIOD).

2.1.1.2.1.2 Fittings for PVC Pipe

Ductile iron fittings or compact fittings, with cement-mortar [lining for fittings](#), standard thickness. Fittings with push-on joint ends are to conform to the same requirements as fittings with mechanical-joint ends, except for the factory modified bell design compatible for use with PVC pipe as specified per [UNE-EN 545](#).

Fittings from material that meets or exceeds [ASTM D1784](#) cell class 12454 and is the same material as the pipe with elastomeric gaskets, in conformance with [AWWA C605](#) and [AWWA C900](#).

2.1.1.2.1.3 Joints and Jointing Material for PVC

- a. Push-on joints: Use jointing material in accordance with [UNE-EN 681-1](#) between pipes, pipes and metal fittings, valves, and other accessories. Provide each joint connection with an elastomeric gasket compatible for the bell or coupling used. Gaskets for push-on joints for pipe, [UNE-EN 681-1](#).

- b. Mechanical Joint: Use mechanically coupled joints having a sleeve-type mechanical coupling, as specified in the paragraph SLEEVE-TYPE MECHANICAL COUPLINGS, as an optional jointing method for plain-end PVC pipe, subject to the limitations specified for mechanically coupled joints using a sleeve-type mechanical coupling as specified for compression-type joints in ASTM D3139. Provide jointing material in accordance with AWWA C111/A21.11 between pipe and sleeve-type mechanical couplings.

2.1.1.2.1.4 Corrosion Protection

Exterior protection: Completely encase buried ductile iron fittings with polyethylene film in accordance with AWWA C105/A21.5.

2.1.1.2.2 PVC Piping for Service Lines

2.1.1.2.2.1 Pipe and Fittings

Provide cell class pipe and fittings of the same PVC material.

- a. Pressure class PN16.

2.1.1.2.2.2 Joints and Connections

Fittings may be joined by the solvent-cement method or threading.

2.1.1.2.2.3 Solvent Joining

Provide solvent joints in accordance with the pipe manufacturer's recommendations.

2.1.1.3 Copper Pipe For Service Lines

2.1.1.3.1 Copper Tubing and Associated Fittings

Provide ASTM B88M, Type K copper tubing. Provide AWWA C800 fittings. AWWA C800 includes ASME B1.20.3, ASME B1.20.1, ASME B16.18 solder-type joint fittings.

2.1.2 Valves

UNE-EN 1074-1.

2.1.2.1 Gate Valves 80 mm Size and Larger on Buried Piping

AWWA C500 or UL 262 and:

- a. AWWA C500: nonrising stem type with double-disc gate and mechanical-joint ends or push-on joint ends compatible for the adjoining pipe
- b. AWWA C509 or AWWA C515: nonrising stem type with mechanical-joint ends
- c. UL 262: inside-screw type with operating nut, double-disc or split-wedge type gate, designed for a hydraulic working pressure of 1200kPa, and have mechanical-joint ends or push-on joint ends as appropriate for the pipe to which it is joined.

Match materials for **UL 262** gate valves to the reference standards specified in **AWWA C500**. Gate valves open by counterclockwise rotation of the valve stem. Stuffing boxes have O-ring stem seals for those valves. Stuffing boxes are bolted and constructed so as to permit easy removal of parts for repair. Use gate valves with special ends for connection to sleeve-type mechanical coupling in lieu of mechanical-joint ends and push-on joint ends. Provide valve ends and gaskets for connection to sleeve-type mechanical couplings that conform to the requirements specified respectively for the joint or coupling. **Gate valves on 50 mm service lines have ends compatible with joining to the pipe used; push-on joint ends or mechanical-joint ends for joining to ductile-iron pipe; with AWWA C111/A21.11 gaskets and pipe ends.** Provide all valves from one manufacturer.

2.1.2.2 Gate Valves **75 mm** Size and Larger in Valve Pit(s)

AWWA C500, AWWA C509, AWWA C515 or UL 262 and:

- a. **AWWA C500**: nonrising stem type with double-disc gates and flanged ends
- b. **AWWA C509** or **AWWA C515**: nonrising stem type with flanged ends
- c. **UL 262**: inside-screw type, with double-disc or split-wedge type gate and flanged ends, and designed for a hydraulic working pressure of **1200 kPa**

Match materials for **UL 262** gate valves to the reference standards specified in **AWWA C500**. Gate valves are nonrising stem type or inside-screw type. Provide gate valves with handwheels that open by counterclockwise rotation of the valve stem. Bolt and construct stuffing boxes so as to permit easy removal of parts for repair. Provide all valves from one manufacturer.

2.1.2.3 Water Service Valves

2.1.2.3.1 Gate Valves Smaller than **75 mm** in Size on Buried Piping

Gate valves smaller than **75 mm** size on Buried Piping, Class 150, solid wedge, nonrising stem, with flanged or threaded end connections, a union on one side of the valve, and a handwheel operator.

2.1.2.3.2 Gate Valves Smaller Than **75 mm** Size in Valve Pits

Class 150, solid wedge, inside screw, rising stem. Provide valves with flanged or threaded end connections, a union on one side of the valve, and a handwheel operator.

2.1.2.3.3 Check Valves Smaller than **50 mm** in Size

Provide check valves with a minimum working pressure of **1000 kPa** or as indicated with a clear waterway equal to the full nominal diameter of the valve. Valves open to permit flow when inlet pressure is greater than the discharge pressure, and close tightly to prevent return flow when discharge pressure exceeds inlet pressure. Cast the size of the valve, working pressure, manufacturer's name, initials, or trademark on the body of each valve.

Provide valves for screwed fittings, made of lead-free bronze, Class 150, compatible for the application.

2.1.2.4 Valve Boxes

Provide a valve box for each gate valve on buried piping. Construct adjustable valve boxes manufactured from cast iron of a size compatible for the valve on which it is used. Provide cast iron valve boxes with a minimum cover and wall thickness of 5 mm. Coat the cast-iron box with a heavy coat of bituminous paint. Cast the word "WATER" on the lid. The minimum diameter of the shaft of the box is 135 mm.

2.1.3 Fire Hydrants and Hose Houses

2.1.3.1 Fire Hydrants

Provide fire hydrants where indicated. Paint fire hydrants with at least one coat of primer and two coats of enamel paint. Paint barrel and bonnet colors in accordance with UFC 3-600-01. Stencil fire hydrant number and main size on the fire hydrant barrel using black stencil paint.

2.1.3.1.1 Dry-Barrel Type Fire Hydrants

Provide Dry-barrel type fire hydrants, AWWA C502 or UL 246, "Base Valve" with 150 mm inlet, 135 mm valve opening, one 115 mm pumper connection, and two 65 mm hose connections.

Provide mechanical-joint end only inlet with end matching requirements as specified in AWWA C502 or AWWA C503 or UL 246 for size and shape of operating nut, cap nuts, and threads on hose and pumper connections. Provide fire hydrants with frangible sections as mentioned in AWWA C502. Provide fire hydrant with special couplings joining upper and lower sections of fire hydrant barrel and upper and lower sections of fire hydrant stem that break from a force imposed by a moving vehicle.

2.1.4 Backflow Preventers

Provide a bronze AWWA C511 reduced pressure principle type backflow preventer meeting the following requirements:

- a. Size: 25 mm.
- b. Flanged, bronze mounted gate valve
- c. Strainer of the same material as the backflow preventer
- d. Stainless steel alloys, Type 304

The particular make, model, and size of backflow preventers to be installed must be included in the latest edition of the List of Approved Backflow Prevention Assemblies issued by the FCCCHR List and be accompanied by a backflow certificate of full approval from FCCCHR List. Select materials for piping, strainers, and valves used in assembly installation that are galvanically compatible. Materials joined, connected, or otherwise in contact are to have no greater than 0.25 V difference on the Anodic Index, unless separated by a dielectric type union or fitting.

2.1.4.1 Backflow Preventer Enclosure

Provide an insulated enclosure.

2.1.5 Disinfection

Chlorinating materials are to conform to: Chlorine, Liquid: AWWA B301; Hypochlorite, Calcium and Sodium: AWWA B300.

2.2 ACCESSORIES

2.2.1 Pipe Restraint

2.2.1.1 Thrust Blocks

Use concrete having a minimum compressive strength of 20 MPa at 28 days.

2.2.1.2 Joint Restraint

Provide restrained joints in accordance with NFPA 24, Chapter 10. Provide mechanical joint restraint.

2.2.2 Protective Enclosures

Provide Freeze-Protection Enclosures that are insulated and designed to protect aboveground water piping, equipment, or specialties from freezing and damage, with heat source to maintain minimum internal temperature of 5 degrees C when external temperatures reach as low as 10 degree C.

2.2.2.1 Housing

Reinforced and insulated aluminum or fiberglass construction; with anchoring devices for attaching housing to concrete base, access doors with locking devices, sized to allow access and service of the protected unit, drain openings, and an electric heating cable or heater with self-limiting temperature control.

2.2.3 Tapping Sleeves

Provide cast gray, ductile, malleable iron or stainless steel, split-sleeve type tapping sleeves of the sizes indicated for connection to existing main with flanged or grooved outlet, and with bolts, follower rings and gaskets on each end of the sleeve. Utilize similar metals for bolts, nuts, and washers to minimize the possibility of galvanic corrosion. Provide dielectric gaskets where dissimilar metals adjoin. Provide a tapping sleeve assembly with a maximum working pressure of 1000 kPa. Provide bolts with square heads and hexagonal nuts. Longitudinal gaskets and mechanical joints with gaskets as recommended by the manufacturer of the sleeve. When using grooved mechanical tee, utilize an upper housing with full locating collar for rigid positioning which engages a machine-cut hole in pipe, encasing an elastomeric gasket which conforms to the pipe outside diameter around the hole and a lower housing with positioning lugs, secured together during assembly by nuts and bolts as specified, pre-torqued to 67.8 Newton meters.

2.2.4 Sleeve-Type Mechanical Couplings

Use AWWA C219 couplings to join plain-end piping by compression of a ring gasket at each end of the adjoining pipe sections. The coupling consists of one middle ring flared or beveled at each end to provide a gasket seat; two follower rings; two resilient tapered rubber gaskets; and bolts and nuts to draw the follower rings toward each other to compress the

gaskets. Provide true circular middle ring and the follower rings sections free from irregularities, flat spots, and surface defects; provide for confinement and compression of the gaskets. For ductile iron and PVC pipe, use ASTM A536 ductile iron. Use gaskets for resistance to set after installation and to meet the requirements specified for gaskets for mechanical joint in AWWA C111/A21.11. Provide track-head type bolts ASTM A307, Grade A, with ASTM A563M, Grade A nuts or round-head square-neck type ASME B18.5.2.2M or ASME B18.5.2.1M bolts with ASME B18.2.2 hex nuts. Provide 16 mm diameter bolts. Shape bolt holes in follower rings to hold fast to the necks of the bolts used. Do not use mechanically coupled joints using a sleeve-type mechanical coupling as an optional method of jointing except where pipeline is adequately anchored to resist tension pull across the joint. Provide a tight flexible joint with mechanical couplings under reasonable conditions, such as pipe movements caused by expansion, contraction, slight settling or shifting in the ground, minor variations in trench gradients, and traffic vibrations. Match coupling strength to that of the adjoining pipeline.

2.2.5 Insulating Joints

Provide a rubber-gasketed insulating joint or dielectric coupling between pipe of dissimilar metals which will effectively prevent metal-to-metal contact between adjacent sections of piping.

2.2.6 Dielectric Fittings

Install dielectric fittings between threaded ferrous and nonferrous metallic pipe, fittings and valves, except where corporation stops join mains to prevent metal-to-metal contact of dissimilar metallic piping elements and compatible with the indicated working pressure.

2.2.7 Tracer Wire for Nonmetallic Piping

Provide a continuous bare copper or aluminum wire not less than 2.5 mm in diameter in sufficient length over each separate run of nonmetallic pipe.

2.2.8 Water Service Line Appurtenances

2.2.8.1 Corporation Stops

Ground key type; lead-free bronze, ASTM B61 or ASTM B62; compatible with the working pressure of the system and solder-joint, or flared tube compression type joint. Threaded ends for inlet and outlet of corporation stops, AWWA C800; coupling nut for connection to flared copper tubing, ASME B16.26.

2.2.8.2 Curb or Service Stops

Ground key, round way, inverted key type; made of lead-free bronze, ASTM B61 or ASTM B62; and compatible with the working pressure of the system. Provide compatible ends for connection to the service piping. Cast an arrow into body of the curb or service stop indicating direction of flow.

2.2.8.3 Service Clamps

Provide single or double flattened strap type service clamps used for repairing damaged cast-iron, steel or PVC pipe with a pressure rating not less than that of the pipe being repaired. Provide clamps with a

galvanized malleable-iron body with cadmium plated straps and nuts and a rubber gasket cemented to the body.

2.2.8.4 Goosenecks

Manufacture goosenecks from Type K copper tubing; provide joint ends for goosenecks compatible with connecting to corporation stop and service line.

2.2.8.5 Curb Boxes

Provide a curb box for each curb or service stop manufactured from cast iron, size capable of containing the stop where it is used. Provide a round head. Cast the word "WATER" on the lid. Factory coat the box with a heavy coat of bituminous paint.

2.2.9 Safety Shower

ANSI/ISEA Z358.1, floor supported free standing unit. Provide a deluge shower head, stay-open ball valve operated by pull rod ring or triangular handle. Provide eyewash and stay-open ball valve operated by foot treadle or push handle. Provide package, UL listed alarm system; including an amber strobe lamp, horn with externally adjustable loudness and horn silencing switch, mounting hardware and water flow service within NEMA Type 3 or 4 enclosures. Emergency shower and eyewash station shall be self-draining and freeze resistant. Submit safety shower shop drawing for review.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Connections to Existing System

Perform all connections to the existing water system in the presence of the Contracting Officer.

3.1.2 Operation of Existing Valves

Do not operate valves within or directly connected to the existing water system unless expressly directed to do so by the Contracting Officer.

3.1.3 Earthwork

Perform earthwork operations in accordance with Section 31 00 00 EARTHWORK 31 00 00 EARTHWORK.

3.2 INSTALLATION

Install all materials in accordance with the applicable reference standard, manufacturers instructions and as indicated herein.

3.2.1 Piping

3.2.1.1 General Requirements

Install pipe, fittings, joints and couplings in accordance with the applicable referenced standard, the manufacturer's instructions and as specified herein.

3.2.1.1.1 Termination of Water Lines

Terminate the work covered by this section at a point approximately 1.5 m from the building, unless otherwise indicated.

Do not lay water lines in the same trench with gas lines, fuel lines, electric wiring, or any other utility. Do not install copper tubing in the same trench with ferrous piping materials. Where nonferrous metallic pipe (i.e., copper tubing) crosses any ferrous piping, provide a minimum vertical separation of 300 mm between pipes.

3.2.1.1.2 Pipe Laying and Jointing

Remove fins and burrs from pipe and fittings. Before placing in position, clean pipe, fittings, valves, and accessories, and maintain in a clean condition. Provide proper facilities for lowering sections of pipe into trenches. Under no circumstances is it permissible to drop or dump pipe, fittings, valves, or other water line material into trenches. Cut pipe cleanly, squarely, and accurately to the length established at the site and work into place without springing or forcing. Replace a pipe or fitting that does not allow sufficient space for installation of jointing material. Blocking or wedging between bells and spigots is not permitted. Lay bell-and-spigot pipe with the bell end pointing in the direction of laying. Grade the pipeline in straight lines; avoid the formation of dips and low points. Support pipe at the design elevation and grade. Secure firm, uniform support. Wood support blocking is not permitted. Lay pipe so that the full length of each section of pipe and each fitting rests solidly on the pipe bedding; excavate recesses to accommodate bells, joints, and couplings. Provide anchors and supports for fastening work into place. Make provision for expansion and contraction of pipelines. Keep trenches free of water until joints have been assembled. At the end of each work day, close open ends of pipe temporarily with wood blocks or bulkheads. Do not lay pipe when conditions of trench or weather prevent installation. Provide a minimum of 760 mm depth of cover over top of pipe.

3.2.1.1.3 Tracer Wire

Install a continuous length of tracer wire for the full length of each run of nonmetallic pipe. Attach wire to top of pipe in such manner that it will not be displaced during construction operations.

3.2.1.1.4 Connections to Existing Water Lines

Make connections to existing water lines after coordination with the facility and with a minimum interruption of service on the existing line. Make connections to existing lines under pressure in accordance with the recommended procedures of the manufacturer of the pipe being tapped and as indicated, except as otherwise specified, tap concrete pipe in accordance with AWWA M9 for tapping concrete pressure pipe.

3.2.1.1.5 Sewer Manholes

No water piping is to pass through or come in contact with any part of a sewer manhole.

3.2.1.1.6 Water Piping Crossing Sewer Piping

- a. Normal Conditions: Provide a separation of at least 450 mm between the

bottom of the water piping and the top of the sewer piping in cases where water piping crosses above sewer piping.

- b. Unusual Conditions: When local conditions prevent a vertical separation described above, construct sewer piping passing over or under water piping of ductile iron water piping and perform hydrostatic sewer test, without leakage, prior to backfilling. Construct sewer crossing with a minimum 6.1 m length of the ductile iron water piping, centered at the point of the crossing so that joints are equidistant and as far as possible from the water piping. Protect water piping passing under sewer piping by providing a vertical separation of at least 450 mm between the bottom of the sewer piping and the top of the water piping; adequate structural support for the sewer piping to prevent excessive deflection of the joints and the settling on or damage to the water piping.

3.2.1.1.7 Penetrations

Provide ductile-iron or Schedule 40 steel wall sleeves for pipe passing through walls of valve pits and structures. Fill annular space between walls and sleeves with rich cement mortar. Fill annular space between pipe and sleeves with mastic.

3.2.1.1.8 Flanged Pipe

Only install flanged pipe aboveground or with the flanges in valve pits.

3.2.1.2 Ductile-Iron Piping

Unless otherwise specified, install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS and with the requirements of AWWA C600 for pipe installation, joint assembly, valve-and-fitting installation, and thrust restraint.

- a. Jointing: Make push-on joints with the gaskets and lubricant specified for this type joint; assemble in accordance with the applicable requirements of AWWA C600 and AWWA M41 for joint assembly. Make mechanical joints with the gaskets, glands, bolts, and nuts specified for this type joint; assemble in accordance with the applicable requirements of AWWA C600 and AWWA M41 for joint assembly and the recommendations of Appendix A to AWWA C111/A21.11. Assemble joints made with sleeve-type mechanical couplings in accordance with the recommendations of the coupling manufacturer. Make insulating joints with the gaskets, sleeves, washers, bolts, and nuts previously specified for this type joint. Assemble insulating joints as specified for flanged joints, except that bolts with insulating sleeves are to be full size for the bolt holes. Ensure that there is no metal-to-metal contact between dissimilar metals after the joint has been assembled.
- b. Allowable Deflection: Follow AWWA C600 and AWWA M41 for the maximum allowable deflection. If the alignment requires deflection in excess of the above limitations, provide special bends or a sufficient number of shorter lengths of pipe to achieve angular deflections within the limit set forth.
- c. Exterior Protection: Completely encase buried ductile iron pipelines using Method A or B, with polyethylene film, in accordance with AWWA C105/A21.5.

3.2.1.3 PVC Water Main Pipe

Unless otherwise specified, install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS and with the requirements of [AWWA C605](#) and [UNE-ENV 1452-6](#) for laying of pipe, joining PVC pipe to fittings and accessories, [setting of fire hydrants](#), valves, and fittings; and with the recommendations for pipe joint assembly and appurtenance installation in [AWWA M23](#), Chapter 7, "Installation".

- a. Jointing: Make push-on joints with the elastomeric gaskets specified for this type joint, using either elastomeric-gasket bell-end pipe or elastomeric-gasket couplings. For pipe-to-pipe push-on joint connections, use only pipe with push-on joint ends having factory-made bevel; for push-on joint connections to metal fittings, valves, and other accessories, cut spigot end of pipe off square and re-bevel pipe end to a bevel approximately the same as that on ductile-iron pipe used for the same type of joint. Use a lubricant recommended by the pipe manufacturer for push-on joints. Assemble push-on joints for pipe-to-pipe joint connections in accordance with the requirements of [AWWA C605](#) for laying the pipe and the recommendations in [AWWA M23](#), Chapter 7, "Installation," for pipe joint assembly. Assemble push-on joints for connection to fittings, valves, and other accessories in accordance with the requirements of [AWWA C605](#) for joining PVC pipe to fittings and accessories and with the requirements of [AWWA C600](#) for joint assembly. Make compression-type joints/mechanical joints with the gaskets, glands, bolts, nuts, and internal stiffeners previously specified for this type joint; assemble in accordance with the requirements of [AWWA C605](#) for joining PVC pipe to fittings and accessories, with the requirements of [AWWA C600](#) for joint assembly, and with the recommendations of Appendix A to [AWWA C111/A21.11](#). Cut off spigot end of pipe for compression-type joint/mechanical-joint connections and do not re-bevel. Assemble joints made with sleeve-type mechanical couplings in accordance with the recommendations of the coupling manufacturer using internal stiffeners as previously specified for compression-type joints.
- b. Joint Offset: Construct joint offset in accordance [AWWA C605](#). Do not exceed the minimum longitudinal bending as indicated by [AWWA C605](#).
- c. Fittings: Install in accordance with [AWWA C605](#).

3.2.1.4 Plastic Service Piping

Install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS and with the applicable requirements of [ASTM D2774](#) and [ASTM D2855](#) and [UNE-ENV 1452-6](#), unless otherwise specified. Handle solvent cements used to join plastic piping in accordance with [ASTM F402](#).

3.2.1.4.1 Jointing

Make plastic pipe joints to other pipe materials in accordance with the recommendations of the plastic pipe manufacturer.

3.2.1.4.2 Plastic Pipe Connections to Appurtenances

Connect plastic service lines to corporation stops and gate valves in accordance with the recommendations of the plastic pipe manufacturer.

3.2.1.5 Water Service Piping

3.2.1.5.1 Location

Connect water service piping to the building service where the building service has been installed. Where building service has not been installed, terminate water service lines approximately **1.5 m** from the building line at the points indicated; close such water service lines with plugs or caps.

3.2.1.5.2 Water Service Line Connections to Water Mains

Connect water service lines **50 mm** size to the main with a rigid connection or a corporation stop and gooseneck and install a gate valve on service line below the frostline. Connect water service lines to ductile-iron water mains in accordance with **AWWA C600** for service taps. Connect water service lines to PVC water mains in accordance with **UBPPA UNI-PUB-08** and the recommendations of **AWWA M23**, Chapter 9, "Service Connections" and with **UNE-ENV 1452-6**

3.2.2 Backflow Preventers

Install backflow preventers of type, size, and capacity indicated a minimum of **300 mm** and a maximum of **900 mm** above concrete base. Include valves and test cocks. Install according to the manufacturers requirements and the requirements of plumbing and health department and authorities having jurisdiction. Support NPS **63 mm** and larger backflow preventers, valves, and piping near floor with **300 mm** minimum air gap, and on concrete piers or steel pipe supports. Do not install backflow preventers that have a relief drain in vault or in other spaces subject to flooding. Do not install by-pass piping around backflow preventers.

3.2.2.1 Backflow Preventer Enclosure

Install a level concrete base with top of concrete surface approximately **50 mm** above grade. Install protective enclosure over valve and equipment. Anchor protective enclosure to concrete base.

3.2.3 Disinfection

Prior to disinfection, provide **disinfection procedures**, proposed neutralization and disposal methods of waste water from disinfection as part of the disinfection submittal. Disinfect new water piping and existing water piping affected by Contractor's operations in accordance with **AWWA C651**.

3.2.4 Flushing

Perform bacteriological tests prior to flushing. Flush solution from the systems with domestic water until maximum residual chlorine content is within the range of **0.2 to 0.5 mg/L**, the residual chlorine content of the distribution system, or acceptable for domestic use. Use **AWWA C655** neutralizing chemicals.

3.2.5 Pipe Restraint

3.2.5.1 Concrete Thrust Blocks

Install concrete thrust blocks where indicated.

3.2.5.2 Restrained Joints

Install restrained joints in accordance with the manufacturer's instructions.

3.2.6 Valves

3.2.6.1 Gate Valves

Install gate valves [AWWA C500](#) and [UL 262](#), in accordance with the requirements of [AWWA C600](#) for valve-and-fitting installation and with the recommendations of the Appendix ("Installation, Operation, and Maintenance of Gate Valves") to [AWWA C500](#). Install gate valves, [AWWA C509](#) or [AWWA C515](#), in accordance with the requirements of [AWWA C600](#) for valve-and-fitting installation and with the recommendations of the Appendix ("Installation, Operation, and Maintenance of Gate Valves") to [AWWA C509](#) or [AWWA C515](#). Install gate valves on PVC water mains in accordance with the recommendations for appurtenance installation in [AWWA M23](#), Chapter 7, "Installation." Make and assemble joints to gate valves as specified for making and assembling the same type joints between pipe and fittings.

3.2.7 Fire Hydrants

Install fire hydrants in accordance with [AWWA C600](#) for fire hydrant installation and as indicated. Make and assemble joints as specified for making and assembling the same type joints between pipe and fittings. Install fire hydrants with the [115 mm](#) connections facing the adjacent paved surface. If there are two paved adjacent surfaces, install fire hydrants with the [115 mm](#) connection facing the paved surface where the connecting main is located.

3.3 FIELD QUALITY CONTROL

3.3.1 Tests

Notify the Contracting Officer a minimum of five days in advance of hydrostatic testing. Coordinate the proposed method for disposal of waste water from hydrostatic testing. Perform field tests, and provide labor, equipment, and incidentals required for testing. Provide documentation that all items of work have been constructed in accordance with the Contract documents.

3.3.1.1 Hydrostatic Test

Test the water system in accordance with the applicable AWWA standard specified below. Test ductile-iron water mains in accordance with the requirements of [AWWA C600](#) for hydrostatic testing. The amount of leakage on ductile-iron pipelines with mechanical-joints or push-on joints is not to exceed the amounts given in [AWWA C600](#); no leakage will be allowed at joints made by any other methods. Test PVC and PVC-O plastic water systems made with PVC pipe in accordance with the requirements of [AWWA C605](#) for pressure and leakage tests. The amount of leakage on pipelines made of PVC water main pipe is not to exceed the amounts given in [AWWA C605](#), except that at joints made with sleeve-type mechanical couplings, no leakage will be allowed. Test PE pipe in accordance with the requirements of [AWWA M55](#) for hydrostatic testing. Test concrete water mains in accordance with the recommendations in [AWWA M9](#), "Hydrostatic Testing and Disinfection of Mains." The amount of leakage on concrete pipelines is

not to exceed 1.8 liters per 24 hours per millimeter of pipe diameter per mile of pipeline. Test water service lines in accordance with requirements of AWWA C600 for hydrostatic testing. No leakage will be allowed at copper pipe joints, copper tubing joints (soldered, compression type, brazed), plastic service pipe joints and flanged joints. Do not backfill utility trench or begin testing on any section of a pipeline where concrete thrust blocks have been provided until at least 7 days after placing of the concrete.

3.3.1.2 Leakage Test

For leakage test, use a hydrostatic pressure not less than the maximum working pressure of the system. Leakage test may be performed at the same time and at the same test pressure as the pressure test.

3.3.1.3 Bacteriological Testing

Perform bacteriological tests in accordance with AWWA C651 Option A. For new water mains use Option A and obtain two sets of samples for coliform analysis, each sample being collected at least 16 hours apart. Take samples every 370 m plus one set from the end of the line and at least one from each branch greater than one pipe length. Analyze samples by a certified laboratory, and submit the results of the bacteriological samples.

3.3.1.4 Backflow Preventer Tests

After installation conduct Backflow Preventer Tests and provide test reports verifying that the installation meets the FCCCHR Manual Standards.

3.3.1.5 Tracer Wire Continuity Test

Test tracer wire for continuity after service connections have been completed and prior to final pavement or restoration. Verify that tracer wire is locatable with electronic utility locating equipment. Repair breaks or separations and re-test for continuity.

3.4 SYSTEM STARTUP

Water mains and appurtenances must be completely installed, disinfected, flushed, and satisfactory bacteriological sample results received prior to permanent connections being made to the active distribution system. Obtain approval by the Contracting Officer prior to the new water piping being placed into service.

3.5 CLEANUP

Upon completion of the installation of water lines and appurtenances, remove all debris and surplus materials resulting from the work.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 40 00

STORMWATER UTILITIES

11/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY, STORAGE, AND HANDLING
 - 1.3.1 Delivery and Storage
 - 1.3.2 Handling

PART 2 PRODUCTS

- 2.1 PIPE FOR CULVERTS AND STORM DRAINS
 - 2.1.1 Concrete Pipe
 - 2.1.1.1 Reinforced Culvert and Storm Drain Pipe
- 2.2 PIPE JOINTS
 - 2.2.1 Concrete Pipe
 - 2.2.1.1 Rubber Gasket Joints
 - 2.2.1.2 Preformed Flexible Sealant Joints
- 2.3 MISCELLANEOUS MATERIALS
 - 2.3.1 Concrete
 - 2.3.2 Mortar
 - 2.3.3 Precast Concrete Segmental Blocks
 - 2.3.4 Brick
 - 2.3.5 Precast Reinforced Concrete Manholes
 - 2.3.6 Frame and Cover or Gratings
 - 2.3.7 Steel Ladder
 - 2.3.8 Flared End Sections
 - 2.3.8.1 Concrete Flared End Sections
 - 2.3.9 Concrete Headwalls

PART 3 EXECUTION

- 3.1 EXCAVATION FOR PIPE STORM DRAINS, AND DRAINAGE STRUCTURES
 - 3.1.1 Trenching
 - 3.1.2 Removal of Rock
 - 3.1.3 Removal of Unstable Material
- 3.2 BEDDING AND INITIAL BACKFILL
 - 3.2.1 Concrete Pipe
 - 3.2.1.1 Trenches
 - 3.2.1.2 Fill Sections
- 3.3 PLACING PIPE AND BOX CULVERT
 - 3.3.1 Concrete Pipe
- 3.4 JOINTING
 - 3.4.1 Concrete Pipe
 - 3.4.1.1 Plastic Sealing Compound Joints for Tongue-and-Grooved Pipe and Box Culverts
 - 3.4.1.2 Flexible Watertight Joints

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.5 DRAINAGE STRUCTURES
 - 3.5.1 Manholes and Inlets
 - 3.5.2 Walls and Headwalls
- 3.6 INSTALLATION OF TRACER WIRE AND WARNING TAPE
- 3.7 FINAL BACKFILL
- 3.8 FIELD QUALITY CONTROL
 - 3.8.1 Tests
 - 3.8.1.1 Deflection Testing
 - 3.8.1.2 Tracer Wire Continuity
 - 3.8.2 Inspection
 - 3.8.2.1 Post-Installation Inspection
 - 3.8.2.1.1 Concrete Pipe
 - 3.8.2.1.2 Post-Installation Inspection Report
 - 3.8.2.2 Low Impact Development Inspection
 - 3.8.3 Repair of Defects
 - 3.8.3.1 Deflection Testing
 - 3.8.3.2 Inspection
 - 3.8.3.2.1 Concrete Pipe
- 3.9 PROTECTION
- 3.10 WARRANTY PERIOD

ATTACHMENTS:

LID Verification Report

-- End of Section Table of Contents --

SECTION 33 40 00
STORMWATER UTILITIES
11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 127916	(2020) Concrete pipes and fittings, unreinforced, steel fibre and reinforced. National complement to the standard UNE-EN 1916:2008.
UNE-CEN/TR 1046	(2013) Thermoplastics piping and ducting systems - Systems outside building structures for the conveyance of water or sewage - Practices for underground installation (Endorsed by AENOR in December of 2014.)
UNE-EN 681-3	(2001) Elastomeric seals - Materials requirements for pipe joints seals used in water and drainage applications - Part 3: Cellular materials of vulcanized rubber.
UNE-EN 1561	(2012) Grey cast irons
UNE-EN 1562	(2020) Founding - Malleable cast irons
UNE-EN 1916	(2008) Concrete pipes and fittings, unreinforced, steel fibre and reinforced
UNE-EN 1917	(2008) Concrete manholes and inspection chambers, unreinforced, steel fibre and reinforced
UNE-EN 1996-1-1	(2006) Eurocode 6 - Design of masonry structures - Part 1-1: General rules for reinforced and unreinforced masonry structures
UNE-EN 13101	(2003) Steps for underground man entry chambers. Requirements, marking, testing and evaluation of conformity.
UNE-EN ISO 1461	(2010) Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods (ISO1461:1999)

ASTM INTERNATIONAL (ASTM)

ASTM C990M	(2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants (Metric)
ASTM D2321	(2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Pipe For Culverts And Storm Drains; G

SD-08 Manufacturer's Instructions

Placing Pipe and Box Culvert; G

SD-11 Closeout Submittals

Post-Installation Inspection Report; G
LID Verification Report; G

1.3 DELIVERY, STORAGE, AND HANDLING

1.3.1 Delivery and Storage

Inspect materials delivered to site for damage and unload and store materials with minimal handling. Do not store materials directly on the ground. Keep the inside of pipes and fittings free of dirt and debris. Before, during, and after installation, protect plastic pipe and fittings from any environment that would result in damage or deterioration to the material. Keep a copy of the manufacturer's instructions available at the construction site at all times and follow these instructions unless directed otherwise by the Contracting Officer. Store solvents, solvent compounds, lubricants, elastomeric gaskets, and any similar materials required to install plastic pipe in accordance with the manufacturer's recommendations and discard if the storage period exceeds the recommended shelf life. Discard solvents in use when the recommended pot life is exceeded.

1.3.2 Handling

Handle materials in a manner that ensures delivery to the trench in sound, undamaged condition. Carry pipe to the trench.

PART 2 PRODUCTS

2.1 PIPE FOR CULVERTS AND STORM DRAINS

Pipe sizes for culverts and storm drains are indicated on the drawings.

2.1.1 Concrete Pipe

2.1.1.1 Reinforced Culvert and Storm Drain Pipe

Manufactured in accordance with and conforming to **UNE-EN 1916**, Class 90 (breaking strength 90 kN/m²) for pipes installed in turfed areas no deeper than 1.00 m cover over the crown of the pipe; Class 135 (breaking strength 135 kN/m²) for pipes installed in non traffic areas deeper than 1.00 m over the crown of the pipe; and Class 180 (breaking strength 180 kN/m²) for pipes installed within vehicular traffic areas in accordance with **UNE 127916**.

2.2 PIPE JOINTS

2.2.1 Concrete Pipe

2.2.1.1 Rubber Gasket Joints

Provide rubber gasket joints of a design and physical requirements conforming to **UNE-EN 681-3**. Provide rubber gaskets oil resistant.

2.2.1.2 Preformed Flexible Sealant Joints

Provide joints made with preformed flexible joint sealant conforming to **ASTM C990M**.

2.3 MISCELLANEOUS MATERIALS

2.3.1 Concrete

Unless otherwise specified, provide concrete and reinforced concrete conforming to the requirements for 25 MPa concrete under Section **03 30 00** CAST-IN-PLACE CONCRETE. Provide air content by volume of concrete mixture, based on measurements made immediately after discharge from the mixer, of 5 to 7 percent when maximum size of coarse aggregate exceeds **37.5 mm**. Provide a minimum concrete covering over steel reinforcing of not less than **25 mm** thick for covers and not less than **40 mm** thick for walls and flooring. For concrete deposited directly against the ground, provide a covering thickness of at least **75 mm** between steel and ground. Provide expansion-joint filler material or provide be resin-impregnated fiberboard.

2.3.2 Mortar

Mortar is not allowed for pipe joints. Provide mortar for pipe connections to drainage structures and brick or block construction, Type M, except that the maximum placement time will be 1 hour. Provide a sufficient quantity of water in the mixture to produce a stiff workable mortar but in no case may the quantity exceed **19 liters** of water per sack of cement. Use water that is clean and free of harmful acids, alkalis, and organic impurities. Use the mortar within 30 minutes after the ingredients are mixed with water.

2.3.3 Precast Concrete Segmental Blocks

Provide precast concrete segmental block not more than **200 mm** thick, not less than **200 mm** long, and of such shape that joints can be sealed effectively and bonded with cement mortar.

2.3.4 Brick

Provide brick conforming to [UNE-EN 1996-1-1](#), Type solid or perforated for structural use. Provide mortar for jointing and plastering consisting of one part portland cement and two parts fine sand. Lime may be added to the mortar in a quantity not more than 25 percent of the volume of cement. Provide joints that are completely filled and that are smooth and free from surplus mortar on the inside of the structure. Plaster brick structures with [13 mm](#) of mortar over the entire outside surface of the walls. Lay brick in stretcher courses with a header course every sixth course for square or rectangular structures. Lay brick radially with every sixth course a stretcher course for round structures.

2.3.5 Precast Reinforced Concrete Manholes

Provide precast reinforced concrete manholes conforming to [UNE-EN 1917](#) . Provide joints between precast concrete risers and tops that are full-bedded in cement mortar and smoothed to a uniform surface on both interior and exterior of the structure or made with flexible watertight, rubber-type gaskets meeting the requirements of paragraph PIPE JOINTS.

2.3.6 Frame and Cover or Gratings

Provide certification on the ability of frame and cover or gratings to carry the imposed live load indicated on the drawings. Provide frame and cover or gratings made of cast gray iron, [UNE-EN 1561](#) or cast ductile iron, [UNE-EN 1562](#), Grade 65-45-12. Stamp or cast the word "Storm Sewer" into covers so that it is plainly visible.

2.3.7 Steel Ladder

Provide a steel ladder or steel steps with steel core and copolymer polypropylene cover in accordance with [UNE-EN 13101](#) where the depth of the storm drainage structure exceeds [1.20 m](#). Provide ladders or steps not less than [406 mm](#) in width, with [19 mm](#) diameter rungs spaced [305 mm](#) apart. Provide two stringers that are a minimum [10 mm](#) thick and [63 mm](#) wide. Galvanize inserts after fabrication in conformance with [UNE-EN ISO 1461](#).

2.3.8 Flared End Sections

2.3.8.1 Concrete Flared End Sections

Provide sections of a standard design fabricated with reinforced concrete.

2.3.9 Concrete Headwalls

Provide concrete headwalls at locations shown on the drawings.

PART 3 EXECUTION

3.1 EXCAVATION FOR PIPE STORM DRAINS, AND DRAINAGE STRUCTURES

Excavate trenches, excavate for appurtenances and backfill for culverts and storm drains, in accordance with the applicable portions of Section [31 00 00](#) EARTHWORK and the requirements specified below.

3.1.1 Trenching

Excavate trenches to the width indicated on the drawings or as specified herein. Trench width should permit satisfactory jointing and thorough tamping of the bedding material under and around the pipe. Place sheeting and bracing, where required, within the trench width as specified, without any overexcavation.

3.1.2 Removal of Rock

Replace rock in either ledge or boulder formation with suitable materials to provide a compacted earth cushion. Provide a compacted earth cushion between unremoved rock and the pipe with a thickness of at least 200 mm or 13 mm for each meter of fill over the top of the pipe, whichever is greater, but not more than three-fourths the nominal diameter of the pipe. Maintain the cushion under the bell as well as under the straight portion of the pipe where bell-and-spigot pipe is used. Provide a compacted earth cushion between unremoved rock and the box culvert of at least 200 mm in thickness for concrete box culverts. Excavate rock as specified and defined in Section 31 00 00 EARTHWORK.

3.1.3 Removal of Unstable Material

Where wet or otherwise unstable soil incapable of properly supporting the pipe or box culvert, as determined by the Contracting Officer, is unexpectedly encountered in the bottom of a trench, remove such material to the depth required and replace with select granular material to the proper grade. Compact select granular material as specified in paragraph FINAL BACKFILL. When removal of unstable material is due to the fault or neglect of the Contractor while performing shoring and sheeting, water removal, or other specified requirements, perform such removal and replacement at no additional cost to the Government.

3.2 BEDDING AND INITIAL BACKFILL

Provide a firm bedding foundation of uniform density throughout the entire length of the pipe or box culvert.

3.2.1 Concrete Pipe

Use select granular material conforming to Section 31 00 00 EARTHWORK for haunch and bedding material. Compact haunch and outer bedding to at least 90 percent laboratory maximum density and place in layers not exceeding 150 mm loose thickness for compaction by hand-operated compactors and 200 mm 8 inches for other than hand-operated machines. Loosely place middle bedding and do not compact. After the pipe has been properly bedded, place haunch material, at a moisture content that will facilitate compaction, evenly along both sides of the pipe and thoroughly compact each layer with mechanical tampers or rammers to the springline of the pipe. Thoroughly compact the haunch material under the haunches of the pipe. For bell and spigot pipe, form a depression in bedding material for bells so entire barrel of pipe is uniformly supported. Minimize the length, depth, and width of bell depressions to that required for properly making the particular type of joint.

3.2.1.1 Trenches

After the pipe has been properly bedded and haunch material placed to the midpoint (springline) of the pipe, backfill and compact the remainder of

the trench by spreading and rolling or compacting by mechanical rammers or tampers in layers not exceeding 150 mm. Test for density as necessary to ensure conformance to the compaction requirements specified below. Where it is necessary, in the opinion of the Contracting Officer, that sheeting or portions of bracing used be left in place, the contract will be adjusted accordingly. Leave untreated sheeting in place beneath structures or pavements.

3.2.1.2 Fill Sections

For pipe placed in fill sections, uniformly spread fill material longitudinally on both sides of the pipe in layers not exceeding 150 mm in compacted depth, and compact by rolling parallel with pipe or by mechanical tamping or ramming. Prior to commencing normal filling operations, the crown width of the fill at a height of 300 mm above the top of the pipe must extend a distance of not less than twice the outside pipe diameter on each side of the pipe or 4 m, whichever is less. After the backfill has reached at least 300 mm above the top of the pipe, place and thoroughly compact the remainder of the fill in layers not exceeding 200 mm.

3.3 PLACING PIPE AND BOX CULVERT

Submit printed copies of the pipe or box culvert manufacturer's recommended pipe or box culvert installation procedures prior to installation. Thoroughly examine each section of pipe or box culvert before being laid; do not use defective or damaged pipe. Protect plastic pipe, from exposure to direct sunlight prior to laying, if necessary to maintain adequate pipe stiffness and meet installation deflection requirements. Lay pipelines to the grades and alignment indicated. Provide proper facilities for lowering sections of pipe into trenches. Do not lay pipe in water or when trench conditions or weather are unsuitable for such work. Divert drainage or dewater trenches during construction as necessary. Deflection of installed flexible pipe must not exceed the following limits:

TYPE OF PIPE	MAXIMUM ALLOWABLE DEFLECTION (percent)
Ductile Iron Culvert	3
Plastic (PVC and PP)	5
Corrugated Steel and Aluminum	5

3.3.1 Concrete Pipe

Lay pipe proceeding upgrade with spigot ends of bell-and-spigot pipe and tongue ends of tongue-and-groove pipe pointing in the direction of the flow.

3.4 JOINTING

3.4.1 Concrete Pipe

3.4.1.1 Plastic Sealing Compound Joints for Tongue-and-Grooved Pipe and Box Culverts

Follow the recommendation of the particular manufacturer in regard to

sealing compound special installation requirements. When lubricants, primers, or adhesives are used, only apply on surfaces that are dry and clean. Affix sealing compounds to the pipe or box culvert not more than 3 hours prior to installation of the pipe or box culvert. Protect sealing compounds from the sun, blowing dust, and other deleterious agents at all times. Inspect sealing compounds before installation of the pipe or box culvert, and remove and replace any loose or improperly affixed sealing compound. Align the pipe or box culvert with the previously installed pipe or box culvert, and pull the joint together.

3.4.1.2 Flexible Watertight Joints

Use lubricants, cements, adhesives, and other special installation requirements for gaskets and jointing materials as recommended by the manufacturer. When lubricants, cements, or adhesives are used, only apply on surfaces that are clean and dry. Affix gaskets and jointing materials to the pipe not more than 24 hours prior to the installation of the pipe, and protect from the sun, blowing dust, and other deleterious agents at all times. Inspect gaskets and jointing materials before installing the pipe; remove and replace any loose or improperly affixed gaskets and jointing materials. Align the pipe with the previously installed pipe, and push the joint home. If the gasket becomes visibly dislocated when joining sections of pipe, remove the pipe and remake the joint.

3.5 DRAINAGE STRUCTURES

3.5.1 Manholes and Inlets

Construct manholes of precast reinforced concrete. Construct inlets of precast reinforced concrete. Provide manholes and inlets complete with frames and covers or gratings; and with fixed galvanized steel ladders as indicated. The wall along the line where steel ladders are installed must be vertical for its entire length. Adequately anchor ladders to the wall by means of steel inserts spaced not more than 1.83 m vertically, and install to provide at least 150 mm of space between the wall and the rungs.

3.5.2 Walls and Headwalls

Construct headwalls as indicated.

3.6 INSTALLATION OF TRACER WIRE AND WARNING TAPE

Install a continuous length of tracer wire for the full length of each run of nonmetallic pipe in accordance with Section 31 00 00 EARTHWORK. Attach wire to top of pipe in such a manner that it will not be displaced during construction operations. Install warning tape above all storm drain pipe in accordance with Section 31 00 00 EARTHWORK.

3.7 FINAL BACKFILL

Backfill trenches with satisfactory material deposited in layers of a maximum of 200 mm loose thickness and compacted to 90 percent of maximum density for cohesive soils and 95 percent of maximum density for cohesionless soils in accordance with Section 31 00 00 EARTHWORK. Testing is the responsibility of the Contractor and will be performed at no additional cost to the Government. Unless otherwise specified, determine field in-place density of final backfill at a frequency of one test per 15 linear meters, or fraction thereof, of each lift of backfill. Submit test

results in accordance with Section 31 00 00 EARTHWORK. Do not displace or damage pipe or box when compacting final backfill by rolling or operating heavy equipment parallel with the pipe or box. Movement of construction machinery over a culvert or storm drain at any stage of construction will be at the Contractor's risk. Repair or replace any damaged pipe. Protect concrete pipes with a minimum of 1 meter of cover prior to permitting heavy construction equipment to pass over them during construction. Provide minimum cover for construction loads over plastic pipes as specified in ASTM D2321 or UNE-CEN/TR 1046.

3.8 FIELD QUALITY CONTROL

3.8.1 Tests

Testing is the responsibility of the Contractor. Perform all testing and retesting at no additional cost to the Government.

3.8.1.1 Deflection Testing

Conduct deflection test no sooner than 30 days after completion of final backfill and compaction testing. Clean or flush all lines prior to testing. Perform a deflection test on entire length of installed flexible pipeline upon completion of work adjacent to and over the pipeline, including backfilling, placement of fill, grading, paving, placement of concrete, and any other superimposed loads. Deflection of pipe in the installed pipeline under external loads must not exceed the limits in paragraph PLACING PIPE above as percent of the average inside diameter of pipe. Use a laser profiler or mandrel to determine if allowable deflection has been exceeded.

3.8.1.2 Tracer Wire Continuity

Test tracer wire for continuity after initial and final backfilling of pipes. Verify that tracer wire is locatable with electronic utility location equipment. Repair breaks or separations and re-test for continuity.

3.8.2 Inspection

3.8.2.1 Post-Installation Inspection

Inspect each segment of pipe for alignment, settlement, joint separations, soil migration through the joint, cracks, buckling, bulging and deflection. An engineer must evaluate all defects to determine if any remediation or repair is required.

3.8.2.1.1 Concrete Pipe

An engineer must evaluate all pipes with cracks with a width greater than 0.25 mm, but less than 2.5 mm to determine if any remediation or repair is required.

3.8.2.1.2 Post-Installation Inspection Report

The deflection results and final post installation inspection report must include: pipe location identification, equipment used for inspection, inspector name, deviation from design, grade, deviation from line, deflection and deformation of flexible pipe, inspector notes, condition of joints, condition of pipe wall (e.g. distress, cracking, wall damage

dents, bulges, creases, tears, holes, etc.).

3.8.2.2 Low Impact Development Inspection

Inspect Low Impact Development (LID) features indicated on the design portion of the [LID Verification Report](#). Certify LID features were constructed according to plans and specifications or by submitting as-built drawings in accordance with UFGS [01 78 00](#) Closeout Submittals. When as-built drawings show deviations to the LID features, document the deviations on the LID Verification Report.

3.8.3 Repair of Defects

3.8.3.1 Deflection Testing

When deflection readings are in excess of the allowable deflection of average inside diameter of pipe are obtained, remove pipe which has excessive deflection and replace with new pipe. Retest 30 days after completing backfill, leakage testing and compaction testing.

3.8.3.2 Inspection

Replace pipe or repair defects indicated in the Post-Installation Inspection Report.

3.8.3.2.1 Concrete Pipe

Replace pipes having cracks with a width greater than [2.5 mm](#).

3.9 PROTECTION

Protect storm drainage piping and adjacent areas from superimposed and external loads during construction.

3.10 WARRANTY PERIOD

Pipe segments found to have defects during the warranty period must be replaced with new pipe and retested.

-- End of Section --

DESIGN

eProjects WON: 1671552

Designer of Record: Indicate the LID features listed below that require Contractor post construction analysis.

LID Features: select all that apply

- ☐ Bioretention _____
- ☐ Cisterns _____
- ☐ Dry Wells _____
- ☐ Grass Buffers _____
- ☐ Grass Swales _____
- ☐ Tree Preservation _____
- ☐ Infiltration Trenches _____
- ☐ Permeable Pavers _____
- ☐ Porous Concrete _____
- ☐ Rain Barrels _____
- ☐ Rain Gardens _____
- ☐ Re-vegetation _____
- ☐ Roof Leader Disconnection _____
- ☐ Tree Box Filters _____
- ☐ Tree Preservation _____
- ☒ Other Level Spreader C-102T _____
- ☐ Other _____
- ☐ Other _____

CONSTRUCTION

☐ I certify that the LID features indicated above were constructed in accordance with the plans and specifications and that there have been no deviations.

☐ I certify that as-built drawings have been submitted in accordance with UFGS 01 78 00 Closeout Submittals for the following deviations.

(Explain deviations here):

Contractor (Print Name): _____

Contractor (Signature): _____

Date: _____

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 52 23.15

POL SERVICE PIPING WELDING

11/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
- 1.3 SUBMITTALS
- 1.4 GENERAL REQUIREMENTS
- 1.5 PERFORMANCE
- 1.6 QUALIFICATIONS
 - 1.6.1 Welding Operations
 - 1.6.2 Welding Procedure Specification and Procedure Qualification Records
 - 1.6.3 Welder and Welding Operator Performance
 - 1.6.3.1 Certification
 - 1.6.3.2 Identification
 - 1.6.3.3 Renewal of Qualification
 - 1.6.4 Test Reports
 - 1.6.5 Inspection and NDE Personnel
 - 1.6.5.1 Inspector Certification
 - 1.6.5.2 NDE Personnel
- 1.7 DELIVERY, STORAGE, AND HANDLING
 - 1.7.1 Material Control
 - 1.7.1.1 Damaged Containers
 - 1.7.1.2 Partial Issues
 - 1.7.2 Damaged Materials
- 1.8 SYMBOLS
- 1.9 SAFETY

PART 2 PRODUCTS

- 2.1 WELDING MATERIALS

PART 3 EXECUTION

- 3.1 WELDING OPERATIONS
 - 3.1.1 Base Metal Preparation
 - 3.1.2 Weld Joint Fit-Up
 - 3.1.3 Butt Weld Joint Spacing
 - 3.1.4 Preheat and Interpass Temperatures
 - 3.1.5 Production Welding Instructions
 - 3.1.6 Postweld Heat Treatment
- 3.2 EXAMINATIONS, INSPECTIONS AND TESTS
 - 3.2.1 Visual Inspection
 - 3.2.2 NDE Testing Frequency
 - 3.2.3 NDE Testing
 - 3.2.4 Inspection and Tests by the Government
- 3.3 ACCEPTANCE STANDARDS

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.3.1 Visual
- 3.3.2 Magnetic Particle Examination
- 3.3.3 Liquid Penetrant Examination
- 3.4 CORRECTIONS AND REPAIRS
 - 3.4.1 Defect Removal
 - 3.4.1.1 Methods of Defect Removal
 - 3.4.1.2 Rewelding
 - 3.4.1.3 Peening or Caulking
- 3.5 MAINTAINING CLEANLINESS OF PIPING
- 3.6 COMMISSIONING

-- End of Section Table of Contents --

SECTION 33 52 23.15

POL SERVICE PIPING WELDING
11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

ASNT SNT-TC-1A (2020) Recommended Practice for Personnel Qualification and Certification in Nondestructive Testing

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B31.3 (2020) Process Piping

ASME BPVC SEC V (2017) BPVC Section V-Nondestructive Examination

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 (2012) Standard Symbols for Welding, Brazing and Nondestructive Examination

AWS A3.0M/A3.0 (2020) Standard Welding Terms and Definitions

AWS A5.1/A5.1M (2012) Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding

AWS A5.3/A5.3M (1999; R 2007) Specification for Aluminum and Aluminum-Alloy Electrodes for Shielded Metal Arc Welding

AWS A5.4/A5.4M (2012) Specification for Stainless Steel Electrodes for Shielded Metal Arc Welding

AWS A5.9/A5.9M (2017) Welding Consumables-Wire Electrodes, Strip Electrodes, Wires, and Rods for Arc Welding of Stainless and Heat Resisting Steels- Classification

AWS A5.10/A5.10M (2021) Welding Consumables - Wire Electrodes, Wires and Rods for Welding of Aluminum and Aluminum-Alloys - Classification

AWS A5.18/A5.18M (2021) Specification for Carbon Steel Electrodes and Rods for Gas Shielded Arc Welding

AWS A5.22/A5.22M	(2012) Specification for Stainless Steel Flux Cored and Metal Cored Welding Electrodes and Rods
AWS A5.32/A5.32M	(2021) Welding Consumables-Gases and Gas Mixtures for Fusion Welding and Allied Processes
AWS C5.5/C5.5M	(2003) Recommended Practices for Gas Tungsten Arc Welding
AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel
AWS D10.4	(1986; R 2000) Recommended Practices for Welding Austenitic Chromium-Nickel Stainless Steel Piping and Tubing
AWS D10.7/D10.7M	(2008) Guide for the Gas Shielded Arc Welding of Aluminum and Aluminum Alloy Pipe
AWS D10.10/D10.10M	(2021) Recommended Practices for Local Heating of Welds in Piping and Tubing
AWS D10.11M/D10.11	(2007) Guide for Root Pass Welding of Pipe Without Backing
AWS D10.12M/D10.12	(2000) Guideline for Welding Mild Steel Pipe
AWS QC1	(2016) Specification for AWS Certification of Welding Inspectors
AWS WHB-4.9	(2010) Welding Handbook, Volume 4 - Materials and Applications Part 1
AWS Z49.1	(2021) Safety in Welding and Cutting and Allied Processes

1.2 DEFINITIONS

Definitions must be in accordance with AWS A3.0M/A3.0 except as follows:

- a. Weld slag is defined as the crystalline residue remaining on the weld surface following a weld procedure which uses flux as a shielding method.
- b. POL service piping consists of piping and components used for petroleum, oil and lubricants (POL) under pressure or gravity force including modifications to existing hydrant fueling systems.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Welding Procedure Specifications (WPS); G

Welder Performance Qualification (WPQ); G

Procedure Qualification Record (PQR); G

Welding Operations; G

SD-02 Shop Drawings

POL Service Piping; G

SD-06 Test Reports

Welding Reports

Examinations, Inspections and Tests

SD-07 Certificates

Qualifications; G

1.4 GENERAL REQUIREMENTS

This section covers the welding of Petroleum, Oil and Lubricant (POL) Service systems. Deviations from applicable codes, approved procedures, and approved detail drawings will not be permitted without prior written approval by the Contracting Officer. Materials or components with welds made offsite will not be accepted if the welding does not conform to the requirements of this specification, unless otherwise specified. Procedures must be developed by the Contractor for welding all metals included in the work. Welding must not be started until welding procedures, welders, and welding operators have been qualified. Qualification testing must be performed by an approved testing laboratory, or by the Contractor if approved by the Contracting Officer. Costs of such testing must be borne by the Contractor. The Contracting Officer must be notified at least one week in advance of the time and place of the tests. If the Contracting Officer elects to witness the tests, the qualification tests must be performed at or near the worksite. The Contractor must maintain current records of the test results obtained in the welding procedure, welding operator, welder performance qualifications, and nondestructive examination (NDE) procedures readily available at the site for examination by the Contracting Officer. The procedures for making transition welds between different materials or between plates or pipes of different wall thicknesses must be qualified. Unless otherwise specified, the choice of welding process must be the responsibility of the Contractor. It should be noted that [ASME B31.3](#) incorporates by reference, other requirements of ASME Section V, Section IX and specific AWS requirements.

1.5 PERFORMANCE

The Contractor will be responsible for the quality of all joint preparation, welding, and examination. All materials used in the welding operations must be clearly identified and recorded. The inspection and testing defined in this specification are minimum requirements. Additional inspection and testing must be the responsibility of the

Contractor when he deems it necessary to achieve the quality required.

1.6 QUALIFICATIONS

Welding procedures, welders, and welding operators previously qualified by test may be accepted for the work without requalification, provided that all of the following conditions are fulfilled:

- a. Copies of the welding procedure specifications (WPS), the procedure qualification record (PQR) record, and the welder performance qualification (WPQ) are submitted and approved in accordance with paragraph SUBMITTALS.
- b. Testing was performed by an approved testing laboratory or approved technical consultant or by the Contractor's approved quality assurance organization.
- c. The welding procedures, welders, and welding operators were qualified in accordance with ASME B31.3 and base materials, filler materials, electrodes, equipment, and processes conformed to the applicable requirements of this specification.
- d. The requirements of paragraph RENEWAL OF QUALIFICATION below are met and records showing name of employer and period of employment using the process for which qualified are submitted as evidence of conformance.

1.6.1 Welding Operations

The Contractor must provide a description of how the critical welding operations will be accomplished. Provide the welding procedures to be used for each operation, the sequence of welding to minimize heat distortion, sequence of welding piping sections both in the trench and outside, machine welding if used, and multiple welders on same pipe weld. Submit detailed procedures which define methods of compliance to contract drawings and specifications.

1.6.2 Welding Procedure Specification and Procedure Qualification Records

The Contractor must record in detail and must qualify the Welding Procedure Specifications for every proposed welding procedure. Qualification for each welding procedure must conform to the requirements of ASME B31.3 and to this specification. The welding procedures must specify back purge gas requirements, end preparation for butt welds including cleaning, alignment, and root openings. Preheat, interpass temperature control, and postheat treatment of welds must be as required by approved welding procedures, unless otherwise indicated or specified. Copies of the welding procedure specifications and weld procedure qualification record results for each type of welding required must be submitted in accordance with paragraph SUBMITTALS. Approval of any procedure does not relieve the Contractor of the sole responsibility for producing acceptable welds. Welding procedures must be identified individually and must be referenced on the POL service piping shop drawings. Submit detail drawings showing location, length, and type of welds; and indicating preweld and postweld heat treatment and NDE as required. The drawings must show the welding procedure specification (WPS) to be used at each weld location.

1.6.3 Welder and Welding Operator Performance

Each welder and welding operator assigned to work must be qualified in accordance with **ASME B31.3**.

1.6.3.1 Certification

Before assigning welders or welding operators to the work, the Contractor must provide the Contracting Officer with their names together with certification that each individual is performance-qualified as specified. The certification must state the type of welding and positions for which each is qualified, the code and welding procedure specification under which each is qualified, date qualified, and the firm and individual certifying the qualification tests. The Contractor must provide a summary table showing all welders and the WPS with which they are qualified to weld.

1.6.3.2 Identification

Each particular weld must be identified with the personal number, letter, or symbol assigned to each welder or welding operator. To identify welds, written records indicating the location of welds made by each welder or welding operator must be submitted, and each welder or welding operator must apply the personal mark adjacent to the welds using a rubber stamp or felt-tipped marker with permanent, weatherproof ink or other methods approved by the Contracting Officer that do not deform the metal. Identification by die stamps or electric etchers will not be allowed.

1.6.3.3 Renewal of Qualification

Requalification of a welder or welding operator must be required under any of the following conditions:

- a. When a welder or welding operator has not used the specific welding procedure for a period of 3 months; the period may be extended to 6 months if the welder or welding operator has been employed on another welding procedure.
- b. When a welder or welding operator has not welded with any procedure during a period of 3 months, all the personal qualifications must be considered expired, including any extension by virtue of a. above.
- c. There is specific reason to question the person's ability to make welds that will meet the requirements of the specifications.
- d. The welder or welding operator was qualified by an employer, other than those firms performing work under this contract, and a qualification test has not been taken within the preceding 12 months.
- e. Renewal of qualification for a specific welding procedure under conditions a., b., and d., above, needs to be made on only a single test joint or pipe of a thickness, position, or material required by the welding procedure specifications to reestablish the welder's or welding operator's qualification for the previous qualification.

1.6.4 Test Reports

Test reports must consist of the following.

- a. Records made by the AWS certified inspector for all duties performed per paragraph 4.2 of AWS QC1.
- b. All NDE (radiograph) reports with unique weld ID for each weld tested.
- c. "Weld Maps." These maps/drawings correlate the shop drawings submitted to the NDE reports. The NDE report that shows a weld number as acceptable is correlated with weld number on the drawings.
- d. Provide the location of each weld, what procedure was used, which welder did the weld, the results of the visual test, and the results of the NDE.

1.6.5 Inspection and NDE Personnel

Contractor must provide a commercially independent organization for all weld examinations. All inspection and NDE personnel must be qualified in accordance with the following requirements. The contractor must submit the qualifications of all the testing personnel that will perform all field tests for review by the Contracting Officer. The qualifications of all personnel on the job site that will perform welding inspections and NDE must be submitted for approval. All inspectors and NDE personnel must have a minimum of one year experience inspecting the piping material being used and five years in military or commercial aircraft hydrant fueling systems or truck fueling systems, petroleum refineries, power generating plants, or chemical process plants.

1.6.5.1 Inspector Certification

Visual welding inspectors must be qualified in accordance with AWS QC1.

1.6.5.2 NDE Personnel

NDE personnel must be certified Level II in accordance with ASME Section V, ASME B31.3 and ASNT SNT-TC-1A for each NDE procedure he is required to use, and a written procedure for the control and administration of NDE personnel training, examination, and certification must be established. The procedures must be based on appropriate specific and general guidelines of training and experience recommended by ASNT SNT-TC-1A. Should the NDE examiner also be a welder, that individual is disqualified from examining their own work.

1.7 DELIVERY, STORAGE, AND HANDLING

All filler metals, electrodes, and other welding materials must be delivered to the site in manufacturers' original packages and stored in a dry space until used. Packages must be properly labeled and designed to give maximum protection from moisture and to insure safe handling.

1.7.1 Material Control

Materials must be stored in a controlled access and clean, dry area that is weathertight and is maintained at a temperature recommended by the manufacturer. The materials must not be in contact with the floor and must be stored on wooden pallets or cribbing.

1.7.1.1 Damaged Containers

Low-hydrogen steel electrodes must be stored in their sealed shipping

container. If the seal is damaged during shipment or storage, and the damage is not immediately detected, the covered electrodes in that container must be rebaked in accordance with the manufacturer's instructions prior to issuance or must be discarded. If a container is damaged in storage and the damage is witnessed, the electrodes from that container must be immediately placed in a storage oven. The storage oven temperature must be as recommended by the manufacturer or the welding material specification.

1.7.1.2 Partial Issues

When a container of covered electrodes is opened and only a portion of the content is issued, the remaining portion must, within the limits established by AWS D1.1/D1.1M be placed in a storage oven.

1.7.2 Damaged Materials

Materials which are damaged must be discarded. Covered electrodes which are oil or water-soaked, dirty, or on which the flux has separated from the wire must be discarded.

1.8 SYMBOLS

Symbols must be in accordance with AWS A2.4.

1.9 SAFETY

Safety precautions must conform to AWS Z49.1.

PART 2 PRODUCTS

2.1 WELDING MATERIALS

Welding materials for carbon steel, stainless steel and aluminum must comply with AWS WHB-4.9. Welding equipment, electrodes, welding wire, shielding and backing gas, and fluxes must be capable of producing satisfactory welds when used by a qualified welder or welding operator using qualified welding procedures. All field girth root pass welds must be made with non-covered electrodes or welding wire. All root passes must be made with shielding and backing gas. External welds on the pipe such as attaching pipe supports may be made with covered electrodes or welding wire. Electrodes, welding wire and fluxes must be in accordance with Table 1. Welding materials for aluminum and aluminum alloy must comply with AWS D10.7/D10.7M.

TABLE I				
AWS	Process	Alloy	Consumable	Use
			(Examples) (1)	
AWS A5.1/A5.1M	SMAW	Low Carbon	E7018	Fill
AWS A5.4/A5.4M	SMAW	Stainless	E308L, E309L	Fill

TABLE I				
AWS	Process	Alloy	Consumable	Use
			(Examples) (1)	
AWS A5.3/A5.3M	SMAW	Aluminum		Fill
AWS A5.9/A5.9M	GTAW	Stainless	ER308L,ER309L	Root and Fill
AWS A5.10/A5.10M	GTAW	Aluminum		Root and Fill
AWS A5.18/A5.18M	GTAW	Low Carbon	E70S-3,E70S-6	Root and Fill
AWS A5.22/A5.22M	GTAW	Stainless	E308LT1-1	Root (Backing and Shielding Gas Required)
AWS A5.32/A5.32M	GTAW	All		Shielding Gas
Note(1): The consumable material designations shown are examples only and are not intended to limit the Contractor's selection of consumable materials.				

PART 3 EXECUTION

3.1 WELDING OPERATIONS

Welding must be performed in accordance with qualified procedures using qualified welders and welding operators. Welding must not be done when the quality of the completed weld could be impaired by the prevailing working or weather conditions. The Contracting Officer must determine when weather or working conditions are unsuitable for welding. Welding of hangers, supports, and plates to structural members must conform to Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS.

Welding must be performed in accordance with ASME B31.3 and the applicable portions of AWS D10.4, AWS D10.7/D10.7M, AWS D10.10/D10.10M, AWS D10.11M/D10.11, AWS D10.12M/D10.12, and AWS C5.5/C5.5M.

All joints unless indicated otherwise, in carbon steel, aluminum and stainless steel piping systems must be welded. Unless otherwise approved, all girth welds must be complete penetration groove welds made in accordance with qualified welding procedures. The root pass on stainless steel, aluminum, and carbon steel pipe must be by the GTAW process.

- a. Weld Preparation must comply with the requirements of ASME B31.3 and the qualified Welding Procedure Specification. The use of "rice paper" as purge blocks is not permitted. Contractor must submit alternate method for approval.
- b. Backing Rings. The use of backing rings for making or repairing welds will not be permitted.

3.1.1 Base Metal Preparation

Oxy-fuel cutting must not be used on austenitic stainless steel or nonferrous materials.

Mechanical grinding of thermal cut ends must be used to remove the heat affected area but should be limited to maximum 1/8 inch.

3.1.2 Weld Joint Fit-Up

Parts that are to be joined by welding must be fitted, aligned, and retained in position during the welding operation by the use of bars, jacks, clamps, or other mechanical fixtures. End welds must be properly aligned prior to welding in accordance with Chapter V of ASME B31.3. All socket-welded joints must be properly fitted with gaps between the pipe and the bottom of the socket in accordance with ASME B31.3. Contractor must implement a program to ensure gaps are properly measured and documented. Welded temporary attachments must not be used except when it is impractical to use mechanical fixtures. When temporary attachments are used, they must be the same material as the base metal, and must be completely removed by grinding or thermal cutting after the welding operation is completed. If thermal cutting is used, the attachment must be cut to not less than 6 mm from the member and the balance removed by grinding. After the temporary attachment has been removed, the area must be visually examined.

3.1.3 Butt Weld Joint Spacing

Butt weld joints must be spaced a minimum of 150 mm apart. Measurement must be taken from the toe of the first weld to the toe of the second weld. The measurement must be taken at the closest point between the welds when visually inspected. If spacing is not possible due to existing conditions, a reduction in spacing to not less than 50 mm may be made.

3.1.4 Preheat and Interpass Temperatures

Preheat temperatures must meet the requirements specified by ASME B31.3. However, in no case will the preheat be below 10 degrees C for ferritic steel or austenitic stainless steel, or 0 degrees C for nonferrous alloys. The maximum interpass temperatures must not exceed 149 degrees C for austenitic stainless steels, nickel alloys, and copper alloys; and 260 degrees C for carbon steels. Preheat techniques must be such as to ensure that the full thickness of the weld joint preparation and adjacent base material, at least 75 mm in all directions, is at the specified temperature. Preheating by induction or resistance methods is preferred. When flame heating is used, only a neutral flame must be employed. Oxy-fuel heating must not be used on austenitic stainless steel; however, air-fuel heating is acceptable if controlled to ensure that the surface temperature does not exceed 66 degrees C. Interpass temperatures must be checked on the surface of the component within 25 mm of the weld groove and at the starting location of the next weld pass, and for a distance of about 150 mm ahead of the weld, but not on the area to be welded.

3.1.5 Production Welding Instructions

- a. Welding must not be done when the ambient temperature is lower than minus 18 degrees C.
- b. Welding is not permitted on surfaces that are wet or covered with ice, when snow or rain is falling on the surfaces to be welded, or during periods of high winds, unless the welders and the work are properly protected.

- c. Gases for purging and shielding must be welding grade and must have a dew point of **minus 40 degrees C** or lower.
- d. Back purges are required for austenitic stainless steels and nonferrous alloys welded from one side and must be set up such that the flow of gas from the inlet to the outlet orifice passes across the area to be welded. The oxygen content of the gas exiting from the purge vent must be less than 2 percent prior to welding. The flow rate must be that required by the approved weld procedure specification.
- e. The purge on groove welds must be maintained for at least two passes or **5 mm** whichever is greater.
- f. Removable purge dam materials must be made of expandable or flexible plugs, such as Plexiglas, plywood (which must be dry when used). Wood dams must be kiln-dried quality. Purge dams must not be made of polyvinyl alcohol.
- g. Any welding process which requires the use of external gas shielding must not be done in a draft or wind unless the weld area is protected by a shelter. This shelter must be of material and shape appropriate to reduce wind velocity in the vicinity of the weld to a maximum of **8 km/hour**.
- h. Tack welds to be incorporated in the final welds must have their ends tapered by grinding or welding technique. Tack welds that are cracked or defective must be removed and the groove must be retacked prior to welding. Temporary tack welds must be removed, the surface ground smooth, and visually inspected. For low-alloy and hardenable high-alloy steels, the area must be magnetic particle examination inspected.
- i. Grinding of completed welds is to be performed only to the extent required for NDE, including any inservice examination, and to provide weld reinforcement within the requirements of **ASME B31.3**. If the surface of the weld requires grinding, reducing the weld or base material below the minimum required thickness must be avoided. Minimum weld external reinforcement must be flush between external surfaces.
- j. Each qualified welder must be assigned an identification symbol. All welds must be permanently marked with the symbol of the individual who made the weld.
- k. Direct welded connection of carbon steel and stainless steel must not be made. See Section **33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT)** for method of connection.

3.1.6 Postweld Heat Treatment

- a. When required postweld heat treatment must be performed in accordance with **ASME B31.3**. Temperatures for local postweld heat treatment must be measured continuously by thermocouples in contact with the weldment.
- b. Postweld heat treatment of low-alloy steels, when required, must be performed immediately upon completion of welding and prior to the temperature of the weld falling below the preheat temperature. However, postweld heat treatment may be postponed after the completion

of the weld, if, immediately after the weld is completed, it is maintained at a minimum temperature of 149 degrees C or the preheat temperature, whichever is greater, for 2-hours per 25 mm of weld thickness.

3.2 EXAMINATIONS, INSPECTIONS AND TESTS

Weld inspection and NDE must be performed by the Contractor to detect surface and internal discontinuities in completed welds. The services of a qualified commercial inspection or testing laboratory or technical consultant meeting the requirements of paragraph INSPECTION AND NDE PERSONNEL, approved by the Contracting Officer, must be employed by the Contractor. All completed welds must be visually inspected in accordance with the visual inspection requirements of ASME B31.3 and AWS D1.1/D1.1M. Radiographic, Liquid penetrant, or Magnetic particle examination must be required as indicated below. When in-process weld quality control is required for tie-in welds, it must be performed in accordance with ASME B31.3. When inspection and testing indicates disqualifying defects in a weld joint, the weld must be repaired by a qualified welder in accordance with paragraph CORRECTIONS AND REPAIRS. The Contractor must submit weld inspection and NDE field testing reports to the Contracting Officer.

The person performing the weld inspection must perform the following:

- a. Verify that the base materials and consumable welding materials conform to the specifications and that welding filler metals used are as specified for each base material.
- b. Verify that the welding equipment to be used for the work is appropriate for use with the welding procedure specification and has the capability to meet the applicable requirements of the welding procedure.
- c. Verify that only approved or qualified welding procedures are used for the work.
- d. Verify that the edge preparation or joint geometry meet the requirements of the welding procedure and drawings.
- e. Verify that the specified filler metals are used and that filler metals are maintained in proper condition, per requirements, or as recommended by the manufacturer.
- f. Verify that the technique and performance of each welder, welding operator, and tack welder are as specified.
- g. Verify that the work conforms to requirements of the applicable standards, drawings, or other documents.
- h. Verify that the work inspected is identified and documented in accordance with specified requirements.
- i. Prepare clear and concise reports and verify that records of the results of examinations are maintained.
- j. Verify the approved WPS pre-heat and post heat procedures are being used.

Welders found making defective welds must be removed from the work or must be required to be requalified in accordance with [ASME B31.3](#).

3.2.1 Visual Inspection

Weld joints must be inspected visually as follows:

- a. Before welding - for compliance with requirements for joint preparation, alignment and fit-up, and cleanliness in accordance with [ASME B31.3](#).
- b. During welding - for cracks and conformance to the approved welding procedure only when in-process weld quality control is required by [ASME B31.3](#).
- c. After welding - for cracks, contour and finish, bead reinforcement, undercutting, overlap, weld slag on the interior of the pipe and size of welds in accordance with [ASME B31.3](#) and [AWS D1.1/D1.1M](#). Visual examination of the interior of the pipe may be performed by any of the remote means allowed by [ASME BPVC SEC V](#), visual inspection. Visual examination of the weld must be performed prior to any other NDE examinations as required by this specification.

3.2.2 NDE Testing Frequency

All pipe field welds, including high point vent pipe tees, insert butt welded weld-o-lets, and low point drain pipe, must be examined by radiographic methods to determine conformance to the paragraph ACCEPTANCE STANDARDS. All socket welds and sock-o-lets or weld-o-lets to pipe welds must be examined with either magnetic particle or liquid penetrant methods, in addition to the visual examinations. The services of a qualified commercial or testing laboratory approved by the Contracting Officer must be employed by the Contractor for testing of piping welds. Costs of testing, including retesting of repaired welds, must be borne by the Contractor.

- a. Provide 100 percent radiographic testing for all underground piping.
- b. Provide select radiographic testing in accordance with [ASME B31.3](#) for all aboveground piping. The inspection must include an examination of welds made by each welding operator or welder. If the testing reveals that any welds fail to meet minimum quality requirements, an additional percent of the welds in that same group must be inspected in accordance with [ASME B31.3](#). If all of the additional welds inspected meet the quality requirements, the entire group of welds represented must be accepted and the defective welds must be repaired. If any of the additional welds inspected also fail to meet the quality requirements, that entire group of welds must be rejected. The rejected welds must be removed and rewelded, or the rejected welds must be 100 percent inspected and all defective weld areas removed and rewelded.

3.2.3 NDE Testing

NDE must be as required by [ASME B31.3](#) and in accordance with written procedures. Procedures for radiographic liquid penetrant magnetic particle ultrasonic tests and methods must conform to [ASME BPVC SEC V](#). Only Radiography are acceptable test methods for butt welded joints. The approved procedure must be demonstrated to the satisfaction of the

Contracting Officer. In addition to the information required in [ASME BPVC SEC V](#), the written procedures must include the timing of the NDE in relation to the welding operations and safety precautions.

The services of a commercially independent qualified testing agency approved by the Contracting Officer must be employed by the Contractor for testing of piping welds. Costs of testing, including retesting of repaired welds, must be borne by the Contractor. Weld ripples or surface irregularities that might mask or be confused with the radiographic image of any objectionable defect must be removed by grinding, or other suitable mechanical means. The weld surface must be merged smoothly with the base metal surface.

3.2.4 Inspection and Tests by the Government

The Government may perform inspection and supplemental nondestructive or destructive tests as deemed necessary. The cost of supplemental NDE will be borne by the Government. The correction and repair of defects and the reexamination of weld repairs must be performed by the Contractor at no additional cost to the Government. Inspection and tests will be performed as required for visual inspection and NDE, except that destructive tests may be required also. When destructive tests are ordered by the Contracting Officer and performed by the Contractor and the specimens or other supplemental examinations indicate that the materials and workmanship do not conform to the contract requirements, the cost of the tests, corrections, and repairs must be borne by the Contractor. When the specimens or other supplemental examinations of destructive tests indicate that materials or workmanship do conform to the specification requirements, the cost of the tests and repairs will be borne by the Government. When destructive tests are made, repairs must be made by qualified welders or welding operators using welding procedures which will develop the full strength of the members cut. Welding must be subject to inspection and tests in the mill, shop, and field. When materials or workmanship do not conform to the specification requirements, the work may be rejected at any time before final acceptance of the system containing the weldment.

3.3 ACCEPTANCE STANDARDS

Acceptance standards must be in accordance with [ASME B31.3](#) paragraph 341.3.2, Chapter VI in addition to the following specified items.

Interpretation of test results and limitations on imperfections in welds must comply with the requirements of 100 percent radiography, per [ASME B31.3](#), paragraph 341.3.2, Chapter VI.

3.3.1 Visual

The following indications are unacceptable:

Weld Slag on the interior of the pipe.

3.3.2 Magnetic Particle Examination

The following relevant indications are unacceptable:

- a. Any linear indications.
- b. Rounded indications with dimensions greater than 5 mm.

- c. Four or more rounded indications in a line separated by 2 mm or less edge-to-edge.
- d. Ten or more rounded indications in any 3870 square mm of surface with the major dimension of this area not to exceed 150 mm with the area taken in the most unfavorable location relative to the indications being evaluated.

3.3.3 Liquid Penetrant Examination

Indications with major dimensions greater than 2 mm must be considered relevant. The following relevant indications are unacceptable:

- a. Any cracks or linear indications.
- b. Rounded indications with dimensions greater than 5 mm.
- c. Four or more rounded indications in a line separated by 2 mm or less edge-to-edge.
- d. Ten or more rounded indications in any 3870 square mm of surface with the major dimension of this area not to exceed 150 mm with the area taken in the most unfavorable location relative to the indications being evaluated.

3.4 CORRECTIONS AND REPAIRS

Disqualifying defects must be removed and repaired as specified in ASME B31.3 unless otherwise specified. Disqualifying defects discovered between weld passes must be repaired before additional weld material is deposited. After defect removal is complete and before rewelding, the area must be examined by the same test method which first revealed the defect to ensure that the defect has been eliminated. After rewelding, the repaired area must be reexamined by the same test method originally used for that area. Any indication of a defect must be regarded as a defect unless reevaluation by NDE or by surface conditioning shows that no disqualifying defects are present.

3.4.1 Defect Removal

Defective or unsound weld joints must be corrected by removing and replacing the entire weld joint, or for the following defects corrections must be made as follows:

- a. Excessive Convexity and Overlap: Reduce by removal of excess metal.
- b. Excessive Concavity of Weld, Undersized Welds, Undercutting: Clean and deposit additional weld metal.
- c. Excessive Weld Porosity, Inclusions, Lack of Fusion, Incomplete Penetration: Remove defective portions and reweld.
- d. Cracks or liner indications in Weld or Base Metal: Remove crack throughout its length, including sound weld metal for a distance of twice the thickness of the base metal or two inches, whichever is less, beyond each end of the crack, followed by the required rewelding. Complete removal must be confirmed by magnetic particle inspection for carbon steel or liquid penetrant inspection for

stainless steel. Inspection procedures must comply with the requirements of ASME B31.3.

- e. Poor Fit-Up: Cut apart improperly fitted parts, and reweld.

3.4.1.1 Methods of Defect Removal

The removal of weld metal or portions of the base metal must be done preferably by chipping, grinding, sawing, machining, or other mechanical means. Defects also may be removed by thermal cutting techniques. If thermal cutting techniques are used, the cut surfaces must be cleaned and smoothed by mechanical means to remove the heat affected zone. In addition, a maximum of 1/8-inch of metal must be removed by mechanical means from the cut surfaces of stainless steel.

Wherever a defect is removed, and repair by welding is not required, the affected area must be blended into the surrounding surface eliminating sharp notches, crevices, or corners.

3.4.1.2 Rewelding

Repair welds must be made using an electrode or filler wire smaller than that used in making the original weld. Rewelding must be done using qualified welding procedures. The surface must be cleaned before rewelding. Repair welds must meet the requirements of this specification.

3.4.1.3 Peening or Caulking

The use of force (peening) or foreign materials to mask, fill in, seal, or disguise any welding defects must not be permitted.

3.5 MAINTAINING CLEANLINESS OF PIPING

The Contractor must keep the interior and ends of all new piping affected by the Contractor's operations thoroughly cleaned of foreign matter and water before and after being installed. Piping systems must be kept clean during installation by means of plugs or other approved methods. When work is not in progress, open ends of piping and fittings must be closed so that no water or other foreign substance will enter the pipes or fittings. Piping must be inspected before placing into position. The interior of each length of pipe must be cleaned after welding; A swab, with a leather or canvas belt disc to fit the inside diameter of pipe, must be pulled through each length of pipe after welding in place. It must be the Contractor's responsibility for insuring that the interior of the piping is free of foreign matter including weld slag when it is connected into the system.

3.6 COMMISSIONING

For commissioning of POL service piping systems see Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). For repairs or modifications to hydrant systems see Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT).

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 52 40

FUEL SYSTEMS PIPING (NON-HYDRANT)

11/18, CHG 2: 11/20

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 REFERENCES
- 1.3 ADMINISTRATIVE REQUIREMENTS
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Contractor Qualifications
 - 1.5.2 System Supplier
 - 1.5.3 Work Plan
 - 1.5.4 Pigging Plan (JP-5 Pipeline)
 - 1.5.5 Hydrotesting Plan
 - 1.5.6 Design Data
 - 1.5.6.1 Pipeline Inventory
 - 1.5.7 Material and System Components Qualifications
 - 1.5.8 Nameplates
 - 1.5.9 Delivery, Storage, and Handling

PART 2 PRODUCTS

- 2.1 ELECTRICAL WORK
 - 2.1.1 General
 - 2.1.2 Grounding and Bonding
- 2.2 MATERIALS
 - 2.2.1 Types of Fuel
 - 2.2.2 Carbon Steel Piping
 - 2.2.3 Stainless Steel Piping
 - 2.2.4 External Protective Coatings for Aboveground Piping
 - 2.2.5 External Protective Coatings for Buried Steel Piping
 - 2.2.5.1 Carbon Steel Piping
 - 2.2.6 Fittings
 - 2.2.6.1 General
 - 2.2.6.2 Carbon Steel Fittings
 - 2.2.6.3 Stainless Steel Fittings
 - 2.2.7 Insulating Flange Kits (Electrically Isolating)
 - 2.2.8 Bolts, Nuts and Washers
 - 2.2.9 Flange Gaskets, Non-Metallic, Non-Electrically Isolating
 - 2.2.9.1 Nitrile Butadiene (Buna-N)
 - 2.2.9.2 Acrylonitrile Butadiene Rubber (NBR)
 - 2.2.9.3 Polytetrafluoroethylene (PTFE)
 - 2.2.9.4 Fluoro Rubber FKM
 - 2.2.9.5 Fluoroelastomer FPM
- 2.3 MANUAL VALVES
 - 2.3.1 Ball Valves
 - 2.3.1.1 Materials
 - 2.3.1.2 Full Port Ball (DBBV) Valves for Piggable Lines (JP-5

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- Pipeline)
 - 2.3.1.3 Electric Valve Actuator
- 2.3.2 Plug (Double Block and Bleed) Valves
 - 2.3.2.1 General
 - 2.3.2.2 Valve Operation
 - 2.3.2.3 Relief Valves
 - 2.3.2.4 Bleed Valves
 - 2.3.2.5 Electric Valve Actuator
- 2.3.3 Swing Check Valves
- 2.4 THERMAL RELIEF VALVE
 - 2.4.1 Valve Material
 - 2.4.2 Thermal Relief Valve (ASME Type)
 - 2.4.3 Thermal Relief Valve (Balanced Type)
- 2.5 PIPING ACCESSORIES
 - 2.5.1 Mechanically Adjustable Segmented Elastomeric Seal
 - 2.5.2 Pipe Sleeves
 - 2.5.3 Strainers
 - 2.5.3.1 Basket Type
 - 2.5.3.2 Cone Type
 - 2.5.4 Pressure Gauge
 - 2.5.5 Pipe Supports
 - 2.5.5.1 General
 - 2.5.5.2 Adjustable Pipe Supports
 - 2.5.5.3 U-bolt Half Round Supports
 - 2.5.5.4 Concrete and Grout
 - 2.5.6 Sample Connections
 - 2.5.7 Sight Flow Indicators
- 2.6 PIGGING SYSTEM COMPONENTS (JP-5 Pipeline System Only)
 - 2.6.1 Smart Pig Launchers and Receivers
 - 2.6.2 Launcher and Receiver Closure Door
 - 2.6.3 Signaler
- 2.7 MISCELLANEOUS ACCESSORIES
 - 2.7.1 Concrete Anchor Bolts
 - 2.7.2 Coatings for Bolts, Studs, Nuts, and Washers
 - 2.7.3 Polytetrafluoroethylene (PTFE) Tape
 - 2.7.4 Pipe Sleeves
 - 2.7.5 Buried Utility Tape
 - 2.7.6 Pipeline Markers
- 2.8 FINISHES
 - 2.8.1 Factory Coating
 - 2.8.1.1 Valves
 - 2.8.1.2 Equipment and Components
 - 2.8.2 Field Painting

PART 3 EXECUTION

- 3.1 GENERAL
- 3.2 VERIFICATION OF DIMENSIONS
- 3.3 CLEANING OF PIPING
- 3.4 TRENCHING AND BACKFILLING
- 3.5 PIPING LAYOUT REQUIREMENTS
 - 3.5.1 Pipe Fabrication
 - 3.5.2 Interferences and Measurements
 - 3.5.3 Space and Access
 - 3.5.4 Location
 - 3.5.5 Pipe Supports
 - 3.5.6 Structural Support
 - 3.5.7 Grade
 - 3.5.8 Size Changes

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.5.9 Direction Changes
- 3.5.10 Threaded End Connections
- 3.5.11 Existing Pipe Systems
- 3.5.12 Bolted Connections
- 3.5.13 Flanges and Unions
- 3.5.14 Manual Valves
- 3.5.15 Air Vents
- 3.5.16 Drains
- 3.5.17 Pipe Sleeves
- 3.6 SEISMIC REQUIREMENTS
- 3.7 STRUCTURAL ATTACHMENTS
- 3.8 WELDING
 - 3.8.1 General
- 3.9 INSTALLATION
 - 3.9.1 Precautions
 - 3.9.2 Protective Coatings for Buried Piping Including Stainless Steel Piping
 - 3.9.2.1 Application of Coating System
 - 3.9.2.2 Inspection and Testing
 - 3.9.2.3 Damage Repair
- 3.10 INTERIOR EPOXY COATING
- 3.11 INSTALLATION OF UNDERGROUND PIPE
 - 3.11.1 Pipe Assembly
 - 3.11.2 Warning Tapes in Earth Trenches
 - 3.11.3 Clearances
 - 3.11.4 Protective Coating
 - 3.11.5 Pipeline Markers
- 3.12 SYSTEM COMMISSIONING
- 3.13 TESTING (Both F-24 and JP-5 Systems)
 - 3.13.1 Before Backfilling
 - 3.13.1.1 Exterior Coating Holiday Test
 - 3.13.2 Pneumatic Test
 - 3.13.2.1 Pneumatic Test Procedure
 - 3.13.3 Hydrostatic Test
 - 3.13.4 Soak Testing
 - 3.13.5 Performance Testing
- 3.14 PIPE PIGGING - CLEANING (JP-5 Pipeline)
 - 3.14.1 General
 - 3.14.2 Use of Fuel in Cleaning Pigging
 - 3.14.3 Cleaning Pig Run
 - 3.14.4 Wire Brush Pig Run
- 3.15 PIPE PIGGING VERIFICATION (JP-5 Pipeline)
 - 3.15.1 Geometry/Ultrasonic Tool Reports
 - 3.15.2 Pipeline Internal Inspection Operations
 - 3.15.2.1 General
 - 3.15.2.2 Preparatory Work
 - 3.15.2.3 Pig Load And Launch
 - 3.15.2.4 Pipeline Operation During Pigging
 - 3.15.2.5 Brush and Gauging Survey
 - 3.15.2.6 Geometry/Ultrasonic Survey
 - 3.15.2.7 Pipe Wall Thickness Survey
 - 3.15.2.8 Lost Pig
- 3.16 DEMONSTRATIONS
- 3.17 POSTED OPERATING INSTRUCTIONS

-- End of Section Table of Contents --

SECTION 33 52 40

FUEL SYSTEMS PIPING (NON-HYDRANT)
11/18, CHG 2: 11/20

PART 1 GENERAL

1.1 SUMMARY

This section defines the requirements for pipe, piping components, and valves as related to an non-hydrant, non-service station, fuel distribution system. Such systems include, but are not limited to: marine receipt, pipeline receipt, truck off-loading receipt, pump house, pump pad, truck loading, marine loading, transfer pipeline, and other miscellaneous piping systems. Provide the entire fuel distribution system as a complete and fully operational system. Size, select, construct, and install equipment and system components to operate together as a complete system. Substitutions of functions specified herein will not be acceptable. Coordinate the work of the system manufacturer's service personnel during construction, testing, calibration, and acceptance of the system. System components and piping associated with the F-24 storage tank system must be designed to handle a working pressure of 1965 kPa (285 psi) at 38 deg C with a corresponding Class 150 Flange Rating. The JP-5 Pipeline from Pier '3' to Pit 'ML-4' must be designed for a working pressure of 2520 kPa (365 psi) at 38 deg C with a corresponding Class 300 Flange Rating. Components specified herein must be compatible with the fuel to be handled. Components to be suitable for outside, unsheltered location, and to function normally in ambient temperatures.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API RP 540	(1999; R 2004) Electrical Installations in Petroleum Processing Plants
API RP 1110	(2013; R 2018) Recommended Practice for the Pressure Testing of Steel Pipelines for the Transportation of Gas, Petroleum Gas, Hazardous Liquids, Highly Volatile Liquids, or Carbon Dioxide
API RP 1595	(2012; R 2019; 2nd Ed) Design, Construction, Operation, Maintenance, and Inspection of Aviation Pre-Airfield Storage Terminals
API RP 2003	(2015; 8th Ed) Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents
API RP 2009	(2002; R 2007; 7th Ed) Safe Welding, Cutting, and Hot Work Practices in

Refineries, Gasoline Plants, and
Petrochemical Plants

API RP 2200	(1999; R 2004) Electrical Installations in Petroleum Processing Plants
API STD 600	(2015) Steel Gate Valves-Flanged and Butt-welding Ends, Bolted Bonnets
API STD 608	(2012) Metal Ball Valves - Flanged, Threaded, And Welding End
API Spec 5L	(2018; 46th Ed; ERTA 2018) Line Pipe
API Spec 6D	(June 2018, 4th Ed; Errata 1 July 2018; Errata 2 August 2018) Specification for Pipeline and Piping Valves
API Spec 6FA	(1999; R 2006; Errata 2006; Errata 2008; R 2011) Specification for Fire Test for Valves
API Std 607	(2016) Fire Test for Quarter-turn Valves and Valves Equipped with Non-metallic Seats

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1	(2003; R 2018) Unified Inch Screw Threads (UN and UNR Thread Form)
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2018) Factory-Made Wrought Buttwelding Fittings
ASME B16.11	(2016) Forged Fittings, Socket-Welding and Threaded
ASME B16.21	(2021) Nonmetallic Flat Gaskets for Pipe Flanges
ASME B18.2.1	(2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)
ASME B18.2.2	(2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)
ASME B31.3	(2020) Process Piping
ASME BPVC SEC VIII D1	(2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
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ASTM A53/A53M	(2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A105/A105M	(2021) Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A182/A182M	(2021) Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service
ASTM A193/A193M	(2020) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications
ASTM A194/A194M	(2020a) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both
ASTM A234/A234M	(2019) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
ASTM A269/A269M	(2015; R 2019) Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service
ASTM A312/A312M	(2021) Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
ASTM A358/A358M	(2019) Standard Specification for Electric-Fusion-Welded Austenitic Chromium-Nickel Stainless Steel Pipe for High-Temperature Service and General Applications
ASTM A403/A403M	(2020) Standard Specification for Wrought Austenitic Stainless Steel Piping Fittings
ASTM A961/A961M	(2021) Standard Specification for Common Requirements for Steel Flanges, Forged Fittings, Valves, and Parts for Piping Applications
ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM B696	(2000; R 2015) Standard Specification for Coatings of Cadmium Mechanically Deposited
ASTM B766	(1986; R 2015) Standard Specification for Electrodeposited Coatings of Cadmium
ASTM D229	(2019) Standard Test Methods for Rigid

Sheet and Plate Materials Used for
Electrical Insulation

- ASTM D1418 (2010; R 2016) Standard Practice for Rubber and Rubber Lattices - Nomenclature
- ASTM D1655 (2018a) Standard Specification for Aviation Turbine Fuels
- ASTM D3308 (2012; R 2017) Standard Specification for PTFE Resin Skived Tape
- ASTM F336 (2002; R 2016) Standard Practice for Design and Construction of Nonmetallic Enveloped Gaskets for Corrosive Service
- ASTM F436 (2011) Hardened Steel Washers

BRITISH STANDARDS INSTITUTE (BSI)

- BS EN ISO 10497 (2010) Testing of Valves Fire Type-Testing Requirements

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

- IEEE 142 (2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book
- IEEE 1100 (2005) Emerald Book IEEE Recommended Practice for Powering and Grounding Electronic Equipment
- IEEE C62.41.2 (2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

- ISO 1629 (2013) Rubber and Latexes - Nomenclature

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS
INDUSTRY (MSS)

- MSS SP-58 (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation

NACE INTERNATIONAL (NACE)

- NACE SP0188 (1999; R 2006) Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

- NFPA 30 (2021; TIA 20-1; TIA 20-2) Flammable and Combustible Liquids Code

NFPA 70	(2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code
NFPA 77	(2014) Recommended Practice on Static Electricity
NFPA 407	(2022) Standard for Aircraft Fuel Servicing
NFPA 780	(2020) Standard for the Installation of Lightning Protection Systems

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 5/NACE No. 1	(2007) White Metal Blast Cleaning
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SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE AMS3275	(2009; Rev C) Sheet, Acrylonitrile Butadiene (NBR) Rubber and Non-Asbestos Fiber Fuel and Oil Resistant
SAE J514	(2012) Hydraulic Tube Fittings

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-DTL-5624	(2016; Rev W; Notice 1 2020) Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-PRF-13789	(1999; Rev E; Notice 1 2008; Notice 2 1016; Notice 3 2021) Strainers, Sediment: Pipeline, Basket Type
MIL-STD-161	(2005; Rev G; Notice 1 2010) Identification Methods for Bulk Petroleum Products Systems Including Hydrocarbon Missile Fuels
MIL-STD-3004-1	(2018) Quality Assurance for Bulk Fuels, Lubricants and Related Products

1.3 ADMINISTRATIVE REQUIREMENTS

Design conditions must be as specified in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT. Refer to Section 01 78 23 OPERATION MAINTENANCE DATA.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Work Plan; G

Pigging Plan; G

Hydrotesting Plan; G

Quality Assurance Plan; G

SD-02 Shop Drawings

Grounding and Bonding

Pipe Supports; G

Pigging System Components; G

SD-03 Product Data

Carbon Steel Piping; G

Stainless Steel Piping; G

External Protective Coatings For Aboveground Piping; G

External Protective Coatings For Buried Steel Piping; G

Fittings; G

Carbon Steel Fittings; G

Insulating Flange Kits; G

Lightning Surge Arrester; G

Bolts, Nuts And Washers; G

Ball Valves; G

Full Port Ball (DBBV) Valves; G

Plug (Double Block and Bleed) Valves; G

Electric Valve Actuator

Swing Check Valves

Thermal Relief Valve; G

Mechanically Adjustable Segmented Elastomeric Seal; G

Pipe Sleeves; G

Strainers; G

Sample Connections; G

Sight Flow Indicators; G

Buried Utility Tape; G

Pipeline Markers; G

Flange Gaskets, Non-Metallic, Non-Electrically Isolating; G

SD-06 Test Reports

Exterior Coating Holiday Test

Pneumatic Test

Hydrostatic Test

Performance Testing

Pipe Pigging Verification

SD-07 Certificates

Contractor Qualifications; G

System Supplier; G

Pipeline Inventory; G

Qualifications of Owner's Inspector; G

Survey Final Elevations; G

Demonstrations

SD-10 Operation and Maintenance Data

Insulating Flange Kits; G

Lightning Surge Arrester; G

Ball Valves; G

Full Port Ball (DBBV) Valves; G

Plug (Double Block and Bleed) Valves; G

Electric Valve Actuator; G

Swing Check Valves; G

Thermal Relief Valve; G

Mechanically Adjustable Segmented Elastomeric Seal; G

Strainers; G

Sample Connections; G

Sight Flow Indicators; G

1.5 QUALITY ASSURANCE

1.5.1 Contractor Qualifications

Each installation Contractor must have successfully completed at least 3

projects of the similar scope and the same size or larger within the last 6 years. Each installation Contractor must demonstrate specific installation experience in regard to the specific system installation to be performed. Each installation Contractor must have taken, if applicable, manufacturer's training courses on the installation of piping and must meet the licensing requirements in the state. Experience must include the erection of piping systems in compliance with the requirements of ASME B31.3, and NFPA 30. Submit a letter listing prior projects, the date of construction, a point of contact for each prior project, the scope of work of each prior project, and a detailed list of work performed. Provide in the letter evidence of prior manufacturer's training and state licensing.

1.5.2 System Supplier

As per requirements in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

1.5.3 Work Plan

Submit a comprehensive work plan that provides sufficient detail to demonstrate a thorough understanding of the project. Document that all components to be provided will function together and produce the result expected by the Government. Include any proposed dates for piping system shutdowns as well as the Contractor's ability to complete the work within the allotted shutdown periods. Show proposed dates and nature of piping system operations required of the Government. Include a list of manpower, spare piping and system component that will be on hand for each phase of the work. Describe, in detail, the means of:

- a. Coordinating work with Government and third parties.
- b. Preparing for safe piping repair work.
- c. Pneumatic pressure testing new piping sections.
- d. Hydrostatic pressure testing new piping sections.
- e. Interrupting or isolating an existing fuel service or system.
- f. Purging piping.
- g. Vapor monitoring.
- h. Preparations for containing and disposing of residual fuel.
- i. Cutting, sealing, and welding into existing piping systems.
- j. Welding tie-ins in place.
- k. Examining repair section tie-in welds.
- l. Collecting, storing and disposing of waste fuel generated during work.

1.5.4 Pigging Plan (JP-5 Pipeline)

The Contractor must submit a detailed written plan covering all aspects of the pipeline pigging operation, including anticipated pig runs, types of pigs, sequence of work, and retrieval/repair procedures. The Contractor

must identify for each pigging evolution, the characteristics of the pig (type/purpose) and method/medium for propulsion.

1.5.5 Hydrotesting Plan

The Contractor must submit a detailed written plan covering all aspects of the pipeline hydrostatic testing operations, including procedures and sequencing of testing, segments of piping to be tested, fluid to be used during testing, equipment removal and isolation, hydrostatic testing pressures, safety protocols, etc.

1.5.6 Design Data

1.5.6.1 Pipeline Inventory

Fuel system volume must be calculated using as constructed pipe lengths, internal diameters, fittings, and components. Totals must be provided for all items containing fuel with the exception of tanks which is covered by other specifications. A certified pipeline inventory - a detailed list with sizes, lengths, quantity, and volumes must be provided for the systems in this project. Such systems include, but are not limited to: marine receipt, pipeline receipt, truck off-loading receipt, pump house, pump pad, truck loading, marine loading, transfer pipeline, and other miscellaneous piping systems.

1.5.7 Material and System Components Qualifications

As per requirements in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

1.5.8 Nameplates

As per requirements in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

1.5.9 Delivery, Storage, and Handling

As per requirements in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT. If fuel is used for hydrotesting or cleaning pigging, comply with all the requirements in Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT).

PART 2 PRODUCTS

2.1 ELECTRICAL WORK

2.1.1 General

Motors, manual or automatic motor control system components except where installed in motor control centers, and protective or signal devices required for the operation specified herein must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Any wiring required for the operation specified herein, but not shown on the electrical plans, must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.1.2 Grounding and Bonding

Ground and bond as indicated on the drawings and in accordance with NFPA 70,

NFPA 77, NFPA 407, NFPA 780, API RP 540, API RP 2003, IEEE 142, and IEEE 1100. Provide jumpers to overcome the insulating effects of gaskets, paints, or nonmetallic components.

2.2 MATERIALS

Pipe and fittings in contact with fuel must be stainless steel, interior epoxy coated carbon steel, or interior uncoated carbon steel as indicated on the drawings or as specified herein. No zinc coated metals, brass, bronze or other copper bearing alloys must be used in contact with the fuel. All carbon steel and stainless steel underground piping must have an exterior protective coating and must be cathodically protected in accordance with Section 26 42 17 IMPRESSED CURRENT CATHODIC PROTECTION (ICCP) SYSTEM. Cathodic protection for metal components that attach to a tank must be coordinated and compatible with the tank corrosion control system. Identification of piping must be in accordance with MIL-STD-161 unless specified otherwise. Material for manual valves must be as specified hereinafter. Do not use aluminum valves.

2.2.1 Types of Fuel

Components must be suitable for use with F-24 turbine fuel (Jet A with additives FSII, CI/LE, and SDA); specific gravity 0.81 at 16 degrees C; viscosity 1.62 CS at 16 degrees C; Reid vapor pressure less than 0.35 kPa, ASTM D1655 or JP-5 turbine fuel; specific gravity 0.82 at 16 degrees C; viscosity 1.62 CS at 16 degrees C; Reid Vapor pressure less than 0.35 kPa, MIL-DTL-5624.

2.2.2 Carbon Steel Piping

Subject each length of pipe to factory hydrostatic testing and ultrasonic testing in accordance with their respective pipe specification.

- a. Piping 300 mm and Larger: API Spec 5L Product Specification Level (PSL) 1, Grade B, seamless; or ASTM A53/A53M Grade B, seamless; "Standard" schedule wall thickness (0.375 in.; 9.525 mm).
- b. Piping 65 through 250 mm: Schedule 40, API Spec 5L Product Specification Level (PSL) 1, Grade B, seamless Grade B; or Schedule 40, Seamless, ASTM A53/A53M Grade B.
- c. Piping 50 mm and Smaller: Schedule 80, API Spec 5L Product Specification Level (PSL) 1, Grade B, seamless Grade B; or Seamless, Schedule 80 ASTM A53/A53M Grade B.
- d. For JP-5 pipeline system, HDD piping segments - The lowest 100 meter of each HDD pipe run must be internally factory coated with an epoxy coating in accordance with MIL-PRF-4556. The ends of the pipe must be masked or wiped back a minimum of 25 mm but not more than 40 mm.

2.2.3 Stainless Steel Piping

- a. Piping:
 - (1) ASTM A358/A358M, Grade 304L, Class 1 or Class 3 with supplementary requirements of S1, S2 and S3, or ASTM A312/A312M Type 304L, seamless (only). Any agreements between the purchaser and the manufacturer or supplier as referenced in the applicable ASTM must include the Contracting Officer as a party to the agreement. All

longitudinal piping welds will receive 100 percent radiographic inspection, 100 percent liquid penetrant inspection, 100 percent visual inspection and all tests as required by the applicable ASTM Standard. All other welds must be inspected per Section 33 52 23.15 POL SERVICE PIPING WELDING. ASTM A312/A312M seamless piping must be provided with a minimum schedule 10S wall thickness for pipe 200 mm and larger; minimum schedule 40S for pipe smaller than 200 mm (except for threaded pipe which must be minimum Schedule 80S).

- (2) Pipe Ends: All Piping must be provided with beveled ends per Chapter V, ASME B31.3, and must be shipped with the ends capped.
 - (3) Factory Testing and Inspection Records: Per Table K341.3.2 of Chapter IX of ASME B31.3, visual, radiographic and liquid penetrant tests must be performed for each section of piping provided as all sections are subjected to cyclic conditions. All testing and inspections records must be submitted to the Contracting Officer and must indicate the pipe mark and installed location of each piping section on the project site. Observation by the Contracting Officer of the manufacturers and the fields testing and inspection procedures must be allowed under this contract. Pipe certification along with pipe markings must be submitted before the pipe arrives on the job site.
 - (4) Qualifications of Owner's Inspector: Provide a qualified inspector in accordance with Chapter VI of ASME B31.3. to act as the owner's inspector (for the Government) at the pipe manufacturer's facility in addition to the manufacturer's inspector.
 - (5) Quality Assurance Plan: Submit Quality Assurance Plan for the welding, inspecting and testing of the welded seam pipe.
- b. Stainless Steel Control Tubing: Seamless, fully annealed tubing conforming to ASTM A269/A269M, Grade TP316, Rockwell hardness B80 or less. Wall thickness for 13 mm tubing to be a minimum of 1.2 mm.

2.2.4 External Protective Coatings for Aboveground Piping

Provide exterior coating of aboveground piping and fittings, piping in pits, pipe supports, filter separators, and miscellaneous metal and system components in accordance with Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES. Color of finish coat must be white. Do not coat aboveground stainless steel or aluminum surfaces.

2.2.5 External Protective Coatings for Buried Steel Piping

2.2.5.1 Carbon Steel Piping

- a. New pipe and fittings must be factory coated in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEM.
- b. Field joints and repairs must be in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.
- c. Field joints and repairs in tight spots (valve pits when heaters are too big) must be liquid epoxy in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.

- d. Existing systems must match existing coating system and must be in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.
- e. Abrasion-resistant topcoat. Following the initial FBE coating application, provide a 20 mil thick abrasion-resistant FBE topcoat. Abrasion-resistant topcoat must be specifically suited for directional boring piping installation.

2.2.6 Fittings

2.2.6.1 General

Welding ells, caps, tees, reducers, must be of materials compatible for welding to the pipeline in which they are installed, and wall thickness, pressure and temperature ratings of the fittings must be not less than the adjoining pipeline. Unless otherwise specified herein or required by the conditions of installation, all elbows must be the 1.5 diameter (D) type. Miter joints are not acceptable. Make odd angle offsets with pipe bends or elbows cut to the proper angle. Butt weld fittings must be factory-made wrought fittings manufactured by forging or shaping. Fabricated fittings will not be permitted. Welding branch fittings must be insert type suitable for radiographic inspections specified herein.

Make branch connections with butt-welded tees except where the branch is at least two pipe sizes smaller than the run, in which case the branch connection can be made with a forged or seamless branch outlet fitting. The branch outlet fitting must be designed in such a way that the connection can be radiographed. The branch outlet fittings may be a non-radiographicable if: the piping it is connected to is aboveground, the branch outlet size is 65 mm or less in diameter, and the branch outlet is located in contained pumphouses, contained truck offloads, contained truck fill stands, and other visibly contained areas equipped with containment curb.

2.2.6.2 Carbon Steel Fittings

- a. Fittings 65 mm and Larger: Butt weld, conforming to ASTM A234/A234M, grade WPB and ASME B16.9 of the same wall thickness as the adjoining pipe. All welds must on buried piping be radiographically examined throughout the entire length of each weld. Each fitting must be subjected to the Supplementary Requirements S3 and S4, Liquid Penetration examination and Magnetic-Particle Examination. Detectable flaws will not be accepted in the supplementary examinations. Fittings must be identified to relate them to their respective radiograph.
- b. Fittings 50 mm and Smaller: Forged, butt weld or socket welded (except flanges, see below). If specifically indicated on drawings, non-flange fittings may be threaded. Socket welded or threaded fittings must be Class 3000, conforming to ASTM A105/A105M, Grade 2 and ASME B16.11. Threaded fittings must only be used for above grade applications. Underground and in pits, low point drain pipe, and high point vent pipe must be butt welded.
- c. Flanges: Class 150 or Class 300 (as indicated), weld neck, butt weld, forged flanges conforming to ASTM A105/A105M, and ASME B16.5 except flanges that are to be connected to pumps must match the pump flanges rating. Threaded and slip-on flanges are not allowed. Flanges to be 2 mm raised face with modified spiral serrated gasket surface finish,

except where required otherwise to match system components furnished. Match flange face to valves or system components furnished. Flange face must be machined to match valves or system components furnished. Use of spacing rings or gaskets discs are not allowed. Flanges must be subjected to the Supplementary Requirements S56, Liquid Penetrant Examination as outlined in [ASTM A961/A961M](#). Detectable flaws will not be accepted. For flanges 50 mm and smaller located in contained pumphouses, contained truck offloads, contained truck fill stands, and other visibly contained areas the fitting may be forged (socket welded), Class 150 or Class 300, (conforming to [ASTM A105/A105M](#), Grade 2) and [ASME B16.5](#). All flanges must be radiographicable, butt welded, weld neck type.

d. Piggable System ([JP-5 Pipeline](#))

(1) Provide barred tees on all branch outlets 50 mm and larger when within 6.1 meters of pig launcher or receiver barrel, including the barrels. Provide barred tees on all size outlets greater than 50 mm in size with any part of the outlet on the bottom half of the pipe. Provide barred tees on all branch connections equal to or greater than 50 percent of piggable line size.

(2) Use 3 D sweeps between pig launchers and receivers.

2.2.6.3 Stainless Steel Fittings

- a. Fittings 65 mm and Larger: Butt weld stainless steel conforming to [ASTM A403/A403M](#), Class WP, Type 304L, seamless or welded, and [ASME B16.9](#) of the same minimum wall thickness as the adjoining pipe. Welded fittings must be tested and inspected the same as the welded seam pipe and meet the same requirements as for the pipe.
- b. Fittings 50 mm and Smaller: Forged Type 304 or 304L, butt weld, or socket welded (except flanges, see below). If specifically indicated on drawings, non-flange fittings may be threaded. Socket welded or threaded fittings must be Class 3000 conforming to [ASTM A182/A182M](#) and [ASME B16.11](#). Threaded fittings must only be used for above grade applications. Underground and in pits, low point drain pipe, and high point vent pipe must be butt welded.
- c. Unions: Conforming to [ASTM A182/A182M](#), Grade 304 or 316.
- d. Flanges. Class 150 weld neck, butt weld, forged Type 304 stainless steel flanges conforming to [ASTM A182/A182M](#) and [ASME B16.5](#), except flange that are to be connected to pumps must match the pump flanges rating. Threaded and slip-on flanges are not allowed. Flanges to be 2mm raised-face with modified spiral serrated gasket surface finish, except where required otherwise to match system components furnished. Flange face must be machined to match valves or system components furnished. Match flange face to valves or system components furnished. Flanges must be subjected to the Supplementary Requirements S56, Liquid Penetrant Examination as outlined in [ASTM A961/A961M](#). Detectable flaws will not be acceptable. For flanges 50 mm and smaller located in contained pumphouses, contained truck offloads, contained truck fill stands, and other visibly contained areas the fitting may be forged (socket welded), Class 150, conforming to [ASTM A182/A182M](#) and [ASME B16.5](#). In pits, vaults, on thermal relief valve piping for pipeline routes, and other uncontained locations the flanges must be radiographicable, butt welded, weld neck type.

e. Piggable System (JP-5 Pipeline):

(1) Provide barred tees on all branch outlets 50 mm and larger when within 6 m of pig launcher or receiver barrel, including the barrels. Provide barred tees on all size outlets greater than 50 mm in size with any part of the outlet on the bottom half of the pipe. Provide barred tees on all branch connections equal to or greater than 50 percent of piggable line size.

(2) 3 D sweeps between pig launchers and receivers.

f. Stainless Steel Tube Fittings. Flareless, 316 stainless steel fittings conforming to SAE J514.

2.2.7 Insulating Flange Kits (Electrically Isolating)

Provide ASTM D229 electrical insulating material of 1,000 ohms minimum resistance or 500 Volts per mil (VPM) minimum dielectric strength; material must be resistant to the effects of aviation and non-aviation hydrocarbon fuels. Provide full face insulating gaskets between flanges. Provide full surface 0.75 mm thick wall thickness, spiral-wound mylar insulating sleeves between the bolts and the holes in flanges; bolts may have reduced shanks of a diameter not less than the diameter at the root of threads. Provide 3 mm thick high-strength phenolic insulating washers next to flanges and provide flat circular stainless steel washers over insulating washers and under bolt heads and nuts. Provide bolts 12 mm longer than standard length to compensate for the thicker insulating gaskets and the washers under bolt heads and nuts. Above grade flanges separated by electrically insulating flange kits must be provided with weatherproof lightning surge arrester devices. The surge arrester must bolt across flanges separated by insulating gasket kits per detail on contract drawings. Provide with flange protector as described in this section. The arrester must have the following features:

- a. Weatherproof NEMA 4 enclosure.
- b. Bidirectional and bipolar protection.
- c. Constructed of solid state components, no lights, fuses or relays and used without required maintenance or replacement.
- d. Withstand unlimited number of surges at 50,000 Amperes.
- e. Maximum clamping voltage of 700 Volts based on a IEEE C62.41.2 8x20 microsecond wave form at 50,000 Amperes peak measured at the device terminals (zero lead length).
- f. A UL listed arrester for installation in Class 1, Division 1, Group D, hazardous areas.

Install the mounting bracket and leads on the flange side of the bolt insulating sleeve and washer, and size in accordance with this schedule:

Line Size	Bolt Size
50 mm	16 mm

Line Size	Bolt Size
65 mm	16 mm
80 mm	16 mm
100 mm	16 mm
150 mm	19 mm
200 mm	19 mm
250 mm	22 mm
300 mm	22 mm
350 mm	25 mm
400 mm	25 mm
Note: Make allowance for the 1 mm thickness of the insulating sleeve around the bolts when sizing the mounting lugs.	

2.2.8 Bolts, Nuts and Washers

- a. Bolts and nuts for pipe flanges, flanged fittings, valves and accessories must conform to ASME B18.2.1 and ASME B18.2.2, except as otherwise specified.
- b. Bolts must be of sufficient length to obtain full bearing on the nuts and must project no more than three full threads and no less than two full threads beyond the nuts with the bolts tightened to the required torque.
- c. Bolts must be regular hexagonal bolts conforming to ASME B18.2.1 with material conforming to ASTM A193/A193M, Class 2, Grade B8, stainless steel, when connections are made where a stainless steel flange is involved, and Grade B7, chromium molybdenum alloy, when only carbon steel flanges are involved. Bolts and nuts chosen must have sufficient strength to seat gasket types chosen. Bolts must be threaded in accordance with ASME B1.1, Class 2A fit, Coarse Thread Series, for sizes 25 mm and smaller and Eight-Pitch Thread Series for sizes larger than 25 mm.
- d. Nuts must conform to ASME B18.2.2, hexagonal, heavy series with material conforming to ASTM A194/A194M, Grade 8, stainless steel for stainless steel bolts, and Grade 7, chromium molybdenum alloy for chromium molybdenum alloy bolts. Nuts must be threaded in accordance with ASME B1.1, Class 2B fit, Coarse Thread Series for sizes 25 mm and smaller and Eight-Pitch Thread Series for sizes larger than 25 mm.
- e. Provide washers under bolt heads and nuts. Use chromium molybdenum alloy washers dimensioned to ASTM F436 flat circular for chromium molybdenum bolts. Stainless steel washer dimensioned similar to ASTM F436 flat circular, use material the same as the bolt.
- f. Use torque wrenches to tighten all flange bolts to the torque recommended by the gasket manufacturer. Tight in the pattern recommended by the gasket manufacturer. Use anti-seize compound on stainless steel bolts.

2.2.9 Flange Gaskets, Non-Metallic, Non-Electrically Isolating

ASME B16.21, composition ring, using a Nitrile Rubber such as Buna-N and NBR, polytetrafluoroethylene (PTFE), or a fluoro rubber such as FKM, FPM and Viton®. The gasket must be 3 mm thick. Gaskets must be resistant to the effects of aviation and non-aviation hydrocarbon fuels and manufactured of fire-resistant materials. Full-face gaskets must be used for flat-face flanged joints. Ring gaskets must be used for raised-face flanged joints. Gaskets must be of one piece factory cut. Select a gasket suitable for the working and test pressure of the fluid.

2.2.9.1 Nitrile Butadiene (Buna-N)

Provide Buna-N material that conforms to SAE AMS3275.

2.2.9.2 Acrylonitrile Butadiene Rubber (NBR)

Provide NBR material that conforms to SAE AMS3275.

2.2.9.3 Polytetrafluoroethylene (PTFE)

Provide PTFE material that conforms to ASTM F336.

2.2.9.4 Fluoro Rubber FKM

Provide FKM material that conforms to ASTM D1418.

2.2.9.5 Fluoroelastomer FPM

Provide FPM material that conforms to ISO 1629.

2.3 MANUAL VALVES

All portions of a valve coming in contact with fuel in stainless steel pipelines or epoxy lined carbon steel pipelines must be of noncorrosive material. Valves in stainless steel pipelines or epoxy lined carbon steel pipelines must be Type 304 or Type 316 stainless steel or carbon steel internally plated with chromium or nickel or internally electroless nickel plated. Valves in unlined carbon steel pipelines must have carbon steel body. Stem and trim must be stainless steel for all valves. Manually operated valves 150 mm and larger must be worm-gear operated and valves smaller than 150 mm must be lever operated or handwheel operated. Valves smaller than 50 mm must have lever-type handles. Valves installed more than 2.4 m above finished floor must have chain operators and a position indicators visible from ground level. Sprocket wheel for chain operator must be aluminum. Valves in the piggable line flow path between the pig launchers and the pig receivers, including the valves in the isolation valve pits must be full bore, piggable, double block and bleed type. The full bore piggable valves at the launcher and the receiver must be ball type. Valves must be true full bore with no projections extending into the flow path of the pig train.

2.3.1 Ball Valves

Ball valves must be fire tested and qualified in accordance with the requirements of API Std 607 and API STD 608. Seal material for the fire test must be graphite, seal material for the project must be as indicated below. Ball valves must be nonlubricated valves that operate from fully open to fully closed with 90-degree rotation of the ball.

Valves 50 mm and larger must conform to applicable construction and dimension requirements of API Spec 6D, ANSI Class 150 or Class 300 (as indicated) and must have flanged ends. The balls in valves 250 mm and larger full port and 300 mm and larger regular port and larger must have trunnion type support bearings. Except as otherwise specified or indicated, reduced port or full port valves may be provided at the Contractor's option. Balls must be solid, not hollow cavity.

2.3.1.1 Materials

Ball must be stainless steel. Ball valves must have tetrafluoroethylene (TFE) or fluoroelastomer (FKM), commonly referred to as Viton seats, body seals and stem seals. Valves 100 mm and smaller must have a locking mechanism.

2.3.1.2 Full Port Ball (DBBV) Valves for Piggable Lines (JP-5 Pipeline)

Required for piggable lines, and where indicated elsewhere. Ball valves must be designed, manufactured, and tested to API Spec 6D, fire-safe and tested to API Spec 6FA, and BS EN ISO 10497 (BS 6755, Part 2). Seal material for the fire test must be graphite, seal material for the project must be as indicated below. Valves must be trunnion-mounted with independent spring and hydraulically actuated, floating, single piston effect, self-relieving seat rings, with bi-directional sealing. Ball must be solid type with full through-conduit opening, suitable for passage of pipeline pigs. Stem must be anti-static, blow-out-proof design with o-ring seals and provided with an emergency sealant injection fitting. Valves must be 3-piece, bolted body design with raised-faced ANSI Class 150 or Class 300 (as indicated) flanged connections, equipped with body drain/bleed valve and vent fitting, and suitable for double block and bleed service in the closed and open positions. Valves must be all stainless steel construction, or carbon steel with stainless steel stem, and all wetted parts electroless nickel-plated. Valves must have nylon, devlon, or polytetrafluoroethylene (TFM) seat inserts, FKM B body, stem, and seat o-rings, with stainless steel and graphite body gaskets and graphite secondary stem seals. Valves located in vaults or pits must be equipped with actuator extensions.

2.3.1.3 Electric Valve Actuator

Electric valve actuator must be as indicated for Plug (Double Block and Bleed) Valves, electric valve actuator.

2.3.2 Plug (Double Block and Bleed) Valves

API Spec 6D, API Spec 6FA, ANSI Class 150 or Class 300 (as indicated), non-lubricated, resilient, double seated, trunnion mounted, tapered lift plug capable of two-way shutoff. Valve must have tapered plug of steel or ductile iron with chrome or nickel plating and plug supported on upper and lower trunnions. Sealing slips must be steel or ductile iron, with Viton seals which are held in place by dovetail connections. Valve design must permit sealing slips to be replaced from the bottom with the valve mounted in the piping. Valves must operate from fully open to fully closed by rotation of the handwheel to lift and turn the plug. Valves must have weatherproof operators with mechanical position indicators. Indicator shaft must be stainless steel. Minimum bore size must be not less than 65 percent of the internal cross sectional area of a pipe of the same nominal diameter unless bore height of plug equals the nominal pipe diameter and manufacturer can show equal or better flow characteristics of the reduced

bore size design. Valves in the piggable line flow path between the pig launchers and the pig receivers, including the valves in the isolation valve pits, in fuel piping between the pig launchers and the pig receivers must be full bore, piggable. Valves must be true full bore with no projections extending into the flow path of the pig train. Full port plug valves in distribution piping must be provided with a 15 mm threaded body drain.

2.3.2.1 General

Valves in the operating tank suction and fill lines and the valves at the four valve manifold in the pump room in the tank fill lines must be provided with a factory-installed limit switch that is actuated by the valve closure. Tank fill line valve and four valve manifold limit switches must be provided with one double pole double throw contacts or four single pole, double throw contracts, two for open, two for closed. Tank suction line valve limit switches must be provided with one double pole double throw contacts or four single pole, double throw contacts, for closed, and one single pole double throw contact or two single pole, double throw contacts for open. All components must be watertight and U.L. listed for Class I, Division 1, Group D hazardous areas.

2.3.2.2 Valve Operation

Rotation of the handwheel toward open must lift the plug without wiping the seals and retract the sealing slips so that during rotation of the plug clearance is maintained between the sealing slips and the valve body. Rotation of the handwheel toward closed must lower the plug after the sealing slips are aligned with the valve body and force the sealing slips against the valve body for positive closure. When valve is closed, the slips must form a secondary fire-safe metal-to-metal seat on both sides of the resilient seal. Plug valves located in Isolation Valve Pits or vaults must be provided with handwheel extensions.

2.3.2.3 Relief Valves

ANSI Class 150 or Class 300 (as indicated). Provide plug valves with automatic thermal relief valves to relieve the pressure build up in the internal body cavity when the plug valve is closed. Relief valves must open at 175 kPa differential pressure and must discharge to the throat of, and to the upstream side, of the plug valve.

2.3.2.4 Bleed Valves

ANSI Class 150 or Class 300 (as indicated), stainless steel body valve. Provide manually operated bleed valves that can be opened to verify that the plug valves are not leaking when in the closed position.

2.3.2.5 Electric Valve Actuator

The actuator, controls and accessories must be the responsibility of the valve-actuator supplier for sizing, assembly, certification, field-testing and any adjustments necessary to operate the valve as specified. The electric valve actuator must include as an integral unit the electric motor, actuator unit gearing, limit switch gearing, position limit switches, torque switches, drive bushing or stem nut, declutch lever, wiring terminals for power, remote control, indication connections and handwheel. The electrically actuated plug valve must be set to open and close completely in 30 to 60 seconds against a differential pressure of 2

MPa. The actuator settings of torque and limit contacts must be adjustable. The valve actuator must be suitable for mounting in a vertical or horizontal position and be rated for 30 starts per hour. The valve actuator must be capable of functioning in an ambient environment temperature.

- a. The electrical enclosure must be specifically approved by UL or Factory Mutual for installation in Class I, Division 1, Group D locations.
- b. The electric motor must be specifically designed for valve actuator service and must be totally enclosed, non-ventilated construction. The motor must be capable of complete operation at plus/minus 10 percent of specified voltage. Motor insulation must be a minimum NEMA Class F. The motor must be a removable subassembly to allow for motor or gear ratio changes as dictated by system operational requirements. The motor must be equipped with an embedded thermostat to protect against motor overload and also be equipped with space heaters. It must de-energize when encountering a jammed valve.
- c. The reversing starter, control transformer and local controls must be integral with the valve actuator and suitably housed to prevent breathing or condensation buildup. The electromechanical starter must be suitable for 30 starts per hour. The windings must have short circuit and overload protection. A transformer, if needed, must be provided to supply all internal circuits with 24 VDC or 110 VAC may be used for remote controls.
- d. The actuator gearing must be totally enclosed in an oil-filled or grease-filled gearcase. Standard gear oil or grease must be used to lubricate the gearcase.
- e. The actuator must integrally contain local controls for Open, Close and Stop and a local/remote three position selector switch: Local Control Only, Off, and Remote Control plus Local Stop Only. A metallic handwheel must be provided for emergency operation. The handwheel drive must be mechanically independent of the motor drive. The remote control capability must be to open and close. Rim pull to operate valve manually must not exceed 28 kg.
- f. Position limit switches must be functional regardless of main power failure or manual operation. Four contacts must be provided with each selectable as normally open or normally closed. The contacts must be rated at 5A, 120 VAC, 30 VDC.
- g. Each valve actuator must be connected to a PLC supplied by "others."
- h. The actuator must have a local display of position even when power has been lost.
- i. The actuator must be supplied with a start-up kit comprising installation instruction, electrical wiring diagram and spare cover screws and seals.
- j. The actuator must be performance tested and a test certificate must be supplied at no extra charge. The test should simulate a typical valve load with current, voltage, and speed measured.

2.3.3 Swing Check Valves

Swing check valves must conform to **API STD 600**, regular type, ANSI Class 150 with flanged end connections. Discs and seating rings must be renewable without removing the valve from the line. The disc must be guided and controlled to contact the entire seating surface.

2.4 THERMAL RELIEF VALVE

2.4.1 Valve Material

Valves must have carbon steel bodies (stainless steel on stainless steel pipelines) and bonnets with stainless steel springs and trim. Valves must be Class 150 or **Class 300 (as indicated)** flanged end connections.

2.4.2 Thermal Relief Valve (ASME Type)

Thermal relief valves must be the fully enclosed, spring loaded, angle pattern, single port, hydraulically operated type with plain caps, and must be labeled in accordance with **ASME BPVC SEC VIII D1**. Valve stems must be fully guided between the closed and fully opened positions. The valves must be factory-set to open at pressures indicated on the drawings. Operating pressure must be adjustable by means of an enclosed adjusting screw. The valves must have a minimum capacity of 20 GPM at 10 percent overpressure. Valves must have a replaceable seat. Relief valves that do not relieve to a zone of atmospheric pressure or tank must be a balanced type relief valve.

2.4.3 Thermal Relief Valve (Balanced Type)

Thermal relief valves that do not relieve to a zone of atmospheric pressure or atmospheric tank must be a balanced type relief or regulator valve.

Thermal relief valves must be the fully enclosed, spring loaded, angle pattern, single port, fully balanced type (back pressure must not affect relief pressure) back pressure regulator/relief valve. Set valve at pressure indicated on drawings. Valve body must have **25 mm** (minimum) raised face flange connections unless otherwise indicated. Orifice must have a minimum orifice size of **15 mm** in diameter. Valve must have bubble-tight piston and seat design with stainless steel piston and Viton seat. Valve must be selected for the nominal flow condition of: pass a minimum of **18 liters per minute**, at a differential pressure of **380 kilopascal**, with a nominal set pressure of **345 kilopascal**. Valve must be factory configured to open at required set pressure but must be field adjustable by means of an enclosed adjusting screw.

2.5 PIPING ACCESSORIES

2.5.1 Mechanically Adjustable Segmented Elastomeric Seal

Mechanically adjustable segmented elastomeric seals must be constructed of fuel resistant Buna-N elastomers and Type 316 stainless steel fasteners and hardware.

2.5.2 Pipe Sleeves

Pipe sleeves must be installed where indicated and at all points where the piping passes through concrete construction. Such sleeves must be of

sufficient inside diameter to provide a minimum clear distance between the pipe and the sleeve of 13 mm. Sleeves through concrete pits or slabs must be standard weight carbon steel pipe with a protective coating. Each sleeve must extend through the respective pit wall or slab and must be provided with a wrap around Buna-N end seal (boot). (Viton when exposed to sunlight) and secured to the pipe sleeve and piping with adjustable stainless steel hose clamps. Sleeves where piping passes under roads or piping indicated to be double walled must be standard weight carbon steel pipe with a protective coating as previously specified. Alignment of the sleeve and piping must be such that the pipe is accurately centered within the sleeve by a nonconductive centering element. The sleeve must be securely anchored to prevent dislocation. Closure of space between the pipe and the pipe sleeve must be by means of a mechanically adjustable segmented elastomeric seal. The seal must be installed so as to be flush.

2.5.3 Strainers

2.5.3.1 Basket Type

Strainer must be single basket type arranged in a simplex configuration as indicated in compliance with MIL-PRF-13789, except as specified otherwise. Strainer end connections must be designed in accordance with ASME B16.5, Class 150 or Class 300 (as indicated). Strainer body material must be the same as the material specified for manual valves. Strainers must have removable baskets of 40 mesh wire screen with larger wire mesh reinforcement; wire must be stainless steel, Type 316. Pressure drop for clean strainer must not exceed 21 kPa at maximum design flow rate. The ratio of net effective strainer area to the area of the connecting pipe must be not less than three to one. Each strainer must be provided with a suitable drain at the bottom, equipped with a ball valve. The strainer must be equipped with a direct-reading, piston type differential pressure gauge that measures the differential pressure across the basket as perin accordance with Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

2.5.3.2 Cone Type

Strainer must be stainless steel type 304 or 316, 100 100 mesh screen with the ratio of net open area of strainer to the area of the connecting pipe must be not less than three to one at the pump suction. Pump suction strainer must have a 100 mesh screen with not less than 300 percent open area (ratio of the strainer open area to the cross section of pipe).

2.5.4 Pressure Gauge

See Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

2.5.5 Pipe Supports

2.5.5.1 General

Pipe supports must conform to MSS SP-58. Design pipe supports to meet the applicable requirements of ANSI/ASME B31.3 or ANSI/ASME B31.4. Provide hot-dip galvanized finish on rods, nuts, bolts, washers, and supports. Provide miscellaneous metal that conforms to ASTM A36/A36M, standard mill finished structural steel shapes, hot-dipped galvanized.

2.5.5.2 Adjustable Pipe Supports

Adjustable pipe supports must consist of a cast iron saddle and a threaded

nipple connected to a carbon steel pipe by means of a special reducer conforming to MSS SP-58. The supports must be provided with PTFE insulation strips.

2.5.5.3 U-bolt Half Round Supports

Supports must have anti-friction bearing half-round in contact with the bottom of the pipe. Provide Polytetrafluoroethylene or like hydrophobic, anti-corrosive material half-round with a compressive strength of at least 69 MPa or greater as required. U-bolts must be installed in either a loose or limited guide configuration as indicated on the design drawings. Provide hot-dip galvanized u-bolts with seamless non-metallic low friction coating. U-bolt connection must be double nutted on the bottom and single bolted on top.

2.5.5.4 Concrete and Grout

Concrete and grout for anchors and supports must comply with Section 03 30 00 CAST-IN-PLACE CONCRETE.

2.5.6 Sample Connections

- a. Sample connections must be factory assembled units specifically designed for obtaining representative samples from fuel pipelines. Each connection must include a 6 mm sampling probe where the probe faces upstream, ball valve and 6 mm quick disconnect coupling with dust plug, all assembled into a unit that is suitable for installation in a pipe nipple. The sampling probe must extend not less than 25 mm into the fuel pipe. All materials in the sample connections must be stainless steel or aluminum.
- b. Furnish two sampling hose assemblies to the Contracting Officer at the project site. Each assembly must consist of a 1.8 m length of 6 mm clear plastic tubing with internal bonding/grounding wire. One end of the tubing will contain a male connector that actuates flow when inserted into the quick disconnect coupler. Each end of the bonding/grounding wire must be equipped with clips for attaching to the pipe and metal sample container.

2.5.7 Sight Flow Indicators

Sight flow indicators must be ANSI Class 150 and must have flanged end connections. Sight flow indicators must consist of a housing containing a rotating propeller that is visible through a glass observation port. The housing must be stainless steel when installed in stainless steel lines and carbon steel when installed in carbon steel lines. The glass in the indicator must also meet the Class 150 rating.

2.6 PIGGING SYSTEM COMPONENTS (JP-5 Pipeline System Only)

2.6.1 Smart Pig Launchers and Receivers

Construct of the same materials as the pipe, valves and fittings for a Class 300 system. The length of the straight barrel and the line size section must each be 4.5 meter. Provide associated launcher kicker piping and receipt bypass piping, not less than 250 mm for 400 mm and larger pipelines.

2.6.2 Launcher and Receiver Closure Door

The closure must be hinged, swing bolted closure of the same material as the pipe and for a Class 300 system. Gasket must be nitrile or viton. Eye bolts must be pinned to lugs on the hub.

2.6.3 Signaler

The pig signaler must be mechanical flag type with manual reset, and be located on the pig launcher and the pig receiver. Material in contact with the fuel must be stainless steel. Units must be suitable for removal and installation under line pressure of 3.0 MPa (440 psi). Signaler must be capable of withstanding line pressure of a Class 300 system.

2.7 MISCELLANEOUS ACCESSORIES

2.7.1 Concrete Anchor Bolts

Concrete anchors must conform to ASTM A307, Grade C, hot-dipped galvanized.

2.7.2 Coatings for Bolts, Studs, Nuts, and Washers

Carbon steel bolts, studs, nuts, and washers must be provided with a factory applied cadmium coating that conforms to ASTM B696 or ASTM B766.

2.7.3 Polytetrafluoroethylene (PTFE) Tape

Tape must conform to ASTM D3308.

2.7.4 Pipe Sleeves

Provided sleeves constructed of Coated Carbon steel piping, ductile iron, or cast-iron pipe conforming to ASTM A53/A53M, Standard weight.

2.7.5 Buried Utility Tape

Provide detectable aluminum foil plastic-backed tape or detectable magnetic plastic tape for warning and identification of buried piping. Tape must be detectable by an electronic detection instrument. Provide tape in minimum 75 mm width rolls, color coded for the utility involved, with warning identification imprinted in bold black letters continuously and repeatedly over entire tape length. Warning identification must be at least 25 mm high and must state as a minimum "BURIED JET FUEL PIPING BELOW". Provide permanent code and letter coloring that is unaffected by moisture and other substances contained in trench backfill material.

2.7.6 Pipeline Markers

Provide pipeline markers constructed of 150 mm diameter, one-half inch thick bronze disk with a 75 mm long bronze headed bolt welded to the back of the disk. Engrave the front of the disk with the words "UNDERGROUND FUEL LINE" in the case of one line and "UNDERGROUND FUEL LINES" in the case of multiple fuel lines.

2.8 FINISHES

2.8.1 Factory Coating

2.8.1.1 Valves

Valve surfaces must be blasted clean according to [SSPC SP 5/NACE No. 1](#). Valve surfaces must be primed and coated in accordance with Section [09 97 13.27](#) HIGH PERFORMANCE COATING FOR STEEL STRUCTURES.

2.8.1.2 Equipment and Components

Unless otherwise specified, provide equipment and components fabricated from ferrous metal with the manufacturer's standard factory finish. Each factory finish must withstand 500 hours exposure to the salt spray test specified in [ASTM B117](#). For test acceptance, the test specimen must show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond [3 mm](#) on either side of the scratch mark immediately after completion of the test. For equipment and component surfaces subject to temperatures above [50 degrees C](#), the factory coating must be appropriately designed for the temperature service.

2.8.2 Field Painting

Painting required for surfaces not otherwise specified must be field painted as specified in Section [09 97 13.27](#) HIGH PERFORMANCE COATING FOR STEEL STRUCTURES. Do not paint aboveground stainless steel and aluminum surfaces. Do not coat equipment or components provided with a complete factory coating. Hot dip galvanized pipe support steel must not receive the Zinc-Rich Epoxy Primer Coat only the Epoxy Intermediate Coat and Polyurethane Topcoat must be applied to these surfaces. Prior to any field painting, clean surfaces to remove dust, dirt, rust, oil, and grease.

PART 3 EXECUTION

3.1 GENERAL

Installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing must be in accordance with [ASME B31.3](#) and [NFPA 30](#), except as modified herein. Safety rules as specified in [NFPA 30](#) and [NFPA 407](#) must be strictly observed. Never direct bury threaded connections, socket welded connections, unions, flanges, valves, air vents, or drains. Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible.

3.2 VERIFICATION OF DIMENSIONS

After becoming familiar with details of the work, verify dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

3.3 CLEANING OF PIPING

Keep the interior and ends of all new piping, affected by construction operations, thoroughly cleaned of foreign matter and water before and after being installed. Piping systems must be kept clean during installation by means of plugs or other approved methods. When work is not in progress, open ends of piping and fittings must be closed so that no water or other foreign substance will enter the pipes or fittings.

Piping must be inspected before placing into position. The interior of each length of pipe must be cleaned after welding ensuring that the interior of the piping is free of foreign matter when it is connected into the system.

3.4 TRENCHING AND BACKFILLING

Trenching and backfilling must conform to Section 31 00 00 EARTHWORK, and the following bedding and backfill requirements. The pipe must be laid in a bed of sand 150 mm deep, compacted in accordance with Section 31 00 00 EARTHWORK, paragraph BACKFILLING AND COMPACTION. Sand must meet the requirements of Section 31 00 00 EARTHWORK, paragraph SELECT GRANULAR MATERIAL. The full length of each section of pipe without any protective covering must be excavated to permit installation of the protective covering. Pipe that has the grade or joint disturbed after laying, must be taken up and relaid. Pipe must not be laid in water or when the trench or weather conditions are unsuitable for such work. After testing and application of protective covering to joints, sand backfill must be placed and compacted around the pipe or protective coating to a depth of 300 mm above top of pipe. The remainder of the backfill must be the same as for other types of pipe.

3.5 PIPING LAYOUT REQUIREMENTS

3.5.1 Pipe Fabrication

Fabricate piping to measurements established on the project site and position into place without springing or forcing. Make provisions for absorbing expansion and contraction without undue stress in any part of the system. The use of flexible connectors in permanently mounted pump suction and discharge lines as a method of compensating for piping misalignment is not acceptable.

3.5.2 Interferences and Measurements

Provide offsets, fittings, and accessories required to eliminate interferences and to match actual system components connection locations and arrangements. Verify measurements before commencing work. Submit discrepancies for clarification before proceeding with the installations to the Contracting Officer.

3.5.3 Space and Access

Keep piping, control tubing, which is not detailed close to structures and columns so as to take up a minimum amount of space. Ensure that access is provided for maintenance of system components, valves and gauges.

3.5.4 Location

Do not place unions in locations that will be inaccessible after the completion of the work. Place unions on each side of equipment.

3.5.5 Pipe Supports

Where provided hot dip galvanized pipe support steel must be surface prepped to SSPC SP 16, Brush-Off Blast Cleaning of Coated and Uncoated Galvanized Steel, Stainless Steels, and Non-Ferrous Metals. Once the steel is prepped, apply only the Epoxy Intermediate Coat and Polyurethane Topcoat per manufacturer's instructions.

Provide pipe supports with the maximum spacing as defined in Table 1 below, unless otherwise indicated. At the indicated locations. Provide additional pipe supports at concentrated piping loads (valves).

Table 1. Maximum Support Spacing									
Nominal Pipe Size (mm)	25 mm	40 mm	50 mm	80 mm	100 mm	150 mm	200 mm	250 mm	300 mm
Maximum Support Spacing (m)	2 m	2.75 m	3 m	3.5 m	4.25 m	5 m	5.75 m	6.50 m	7 m

Provide anchors where required to absorb or transmit thrust or eliminate vibration or pulsation. Provide supports near each change of direction. Select support components which do not restrict the movement of the pipe due to thermal expansion. Space supports uniformly and arrange symmetrically.

3.5.6 Structural Support

Provide supplementary or intermediate steel or other structural members as required for transmission of loads to members forming part of the supporting structure. Piping must not be supported from other piping.

3.5.7 Grade

Where profiles of piping lines are shown on the drawings, grade the line uniformly between changes in slope or direction. Maintain gradient to within 6 mm over the entire length of pipe. When backfilling has been completed to the top of the pipe, the pipe must be surveyed at each joint, and logged by station number. Submit to the Contracting Officer for approval the survey final elevations before backfilling can continue.

3.5.8 Size Changes

Make changes in pipe size with reducing fittings. Do not use bushings. Make branch connections with butt-welded tees except where the branch is at least two pipe sizes smaller than the run, in which case the branch connection can be made with a forged or seamless branch outlet fitting. The branch outlet fitting must be designed in such a way that the connection can be radiographed. The branch outlet fittings may be a non-radiographicable if: the piping it is connected to is aboveground, the branch outlet size is 65 mm or less in diameter, and the branch outlet is located either in a pump house or on a system component pad equipped with containment curb.

3.5.9 Direction Changes

Make changes in direction of pipes with 1.5 D fittings. Where piping is to be piggable, make changes in direction with 3 D sweeps as indicated. Provide special fittings when required. Make odd-angle offsets with pipe bends or elbows cut to the proper angle. Make changes in direction with

fittings, except that bending of pipe 100 mm and smaller will be permitted, provided a pipe bender is used and wide sweep bends are formed. Mitering or notching pipe or other similar construction to form elbows or tees will not be permitted. The centerline radius of bends must not be less than 6 diameters of the pipe if a pipe bender is used.

3.5.10 Threaded End Connections

Provide threaded end connections only on piping 50 mm in nominal size or smaller and only where indicated on the drawings. Provide threaded connections with PTFE tape or equivalent thread-joint compound applied to the male threads only. Not more than three threads must show after the joint is tightened.

3.5.11 Existing Pipe Systems

No interruptions or isolation of an existing fuel handling service or system may be performed unless the actions are appropriately documented in the approved work plan. Perform initial cutting of existing fuel pipe with a multiwheel pipe cutter, using a nonflammable lubricant. After cut is made, seal interior of piping with a gas barrier plug. Refer to API RP 2009 and API RP 2200. Purge interior of piping with carbon dioxide or nitrogen prior to performing any welding process.

3.5.12 Bolted Connections

For each bolted connection of stainless steel components (e.g., pipes, piping components, valves, and system components) use stainless steel bolts or studs, nuts, and washers. For each bolted connection of carbon steel components, use carbon steel bolts or studs, nuts, and washers. Bolts project no more than three full threads and no less than two full threads beyond the nuts with the bolts tightened to the required torque. Prior to installing nuts, apply a compatible anti-seize compound to the male threads.

3.5.13 Flanges and Unions

Provide flanged joints in each line immediately preceding the connection to system components or material requiring maintenance such as pumps, general valves, control valves, strainers, and other similar items and as indicated. Assemble flanged joints square and tight with matched flanges, gaskets, and bolts.

3.5.14 Manual Valves

Install isolation plug or ball valves on each side of each system equipment, at the midpoint of looped mains, and at any other points indicated or required for draining, isolating, or sectionalizing purpose. Install valves with stems vertically up unless otherwise indicated. Provide individual supports and anchors for each valve.

3.5.15 Air Vents

Provide 40 mm air vents at all high points and where indicated to ensure adequate venting of the piping system.

3.5.16 Drains

Provide 50 mm drains at all low points and where indicated to ensure

complete drainage of the piping. Drains must be schedule 120.

3.5.17 Pipe Sleeves

Provide a pipe sleeve around any pipe that penetrates a wall, floor, or slab. Do not install sleeves in structural members except where indicated or approved. Install pipe sleeves in masonry structures at the time of the masonry construction. Sleeves must be of such size as to provide a minimum of 12 mm all-around clearance between bare pipe and the sleeve. Align sleeve and piping such that the pipe is accurately centered within the sleeve by a nonconductive centering element. Securely anchor the sleeve to prevent dislocation. Closure of the space between the pipe and the pipe sleeve must be by means of a mechanically adjustable segmented elastomeric seal. The seal must be installed so as to be flush. For wall or floor penetrations, extend each sleeve through its respective wall or floor and cut flush with each surface. Seal around sleeves that penetrate through valve or fuel related pits with a Buna-N casing seal. Seal around sleeves that penetrate through non-fire-rated walls and floors in accordance with Section 07 92 00 JOINT SEALANTS. Seal around sleeves that penetrate through fire-rated walls and floors as specified in Section 07 84 00 FIRESTOPPING.

3.6 SEISMIC REQUIREMENTS

Support and brace piping and attach valves to resist seismic loads as shown on the drawings. Structural steel required for reinforcement to properly support piping, headers, and system components but not shown must be provided under this section. Material used for support must be as specified under Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS.

3.7 STRUCTURAL ATTACHMENTS

Provide attachment to building structure concrete and masonry by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Apply inserts and anchors with a safety factor not less than 5. Do not attach supports to metal decking. Construct masonry anchors for overhead applications of ferrous materials only. Structural steel brackets required to support piping, headers, and system components, but not shown, must be provided under this section. Material used for support must be as specified under Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS.

3.8 WELDING

3.8.1 General

All joints, unless indicated otherwise, in carbon steel and stainless steel piping systems must be welded. Do not weld carbon steel to stainless steel. Welding of fuel pipe joints must comply with Section 33 52 23.15 POL SERVICE PIPING WELDING.

3.9 INSTALLATION

3.9.1 Precautions

Take special care to ensure that the protective coating on buried pipe is not damaged during installation and that the completed system is free of rocks, sand, dirt, water, weld slag, and foreign objects including construction debris. Take the following steps to ensure these conditions.

- a. Coated pipe must be handled only with canvas or nylon slings or padded clamps. Any coating damaged by improper handling or storage must be repaired as specified.
- b. Pipe brought to the site must be stored on blocks or horses at least 450 mm above the ground and adequately supported to prevent sagging. Padded blocks or horses must be used for coated pipe. The method and height of storing coated pipe must be in accordance with the coating manufacturer's instructions. Pipe ends must be protected and capped against weather at all times, except to accommodate immediate installation.
- c. Visual inspection must be made of the inside of each length of pipe to ensure that it is clear and clean prior to installation.
- d. The open ends of the pipe system must be closed at the end of each day's work or when work is not in progress by use of expansion plugs and must not be opened until the work is resumed.
- e. A swab, with a leather or canvas belt disc to fit the inside diameter of pipe, must be pulled through each length of pipe after welding in place.
- f. Obstruction remaining in the pipe after completion of the system must be removed at the expense of the Contractor.
- g. Plasma cutters and torches are not to be used to make penetrations in the pipe or to cut pipe.
- h. After installation and backfill of the piggable piping is complete and before fuel is put in the pipe, the pipe will be cleaned per the paragraphs CLEANING PIG RUN and WIRE BRUSH RUN in this Section.

3.9.2 Protective Coatings for Buried Piping Including Stainless Steel Piping

3.9.2.1 Application of Coating System

Application of coating system must be in accordance with 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.

3.9.2.2 Inspection and Testing

The condition of factory field coated piping must be the responsibility of the Contractor and all damage to the protective covering during transit and handling must be repaired at no additional cost to the Government. All field coating must be subject to approval by the Contracting Officer. The entire pipe must be inspected as specified in sub-paragraph TESTING OF PROTECTIVE COATINGS under paragraph PROTECTIVE COATINGS FOR BURIED STEEL PIPING. The inspection for holidays must be performed just prior to lowering the pipe into the ditch and every precaution must be taken during lowering and backfilling to prevent damage to the protective covering.

3.9.2.3 Damage Repair

Damaged areas of coating must be repaired as specified in the preceding paragraph for fittings.

3.10 INTERIOR EPOXY COATING

When internally epoxy lined pipe is cut, the lining must be ground back from the end a minimum of 25 mm but not more than 38 mm.

3.11 INSTALLATION OF UNDERGROUND PIPE

Underground fuel pipelines must be pitched as shown on the drawings. Where not indicated they must be pitched a minimum of 50 mm per 30.5 m. 50 mm pipe size valved drain connections must be provided at all low points and 38 mm pipe size valved outlet vent connections must be provided at all high points. Vent and drain lines must terminate in male cam-type locking end connectors with matching female dust covers and installed in pits. The pipe must have cover as shown on the drawings. Drain lines must be installed at the slopes indicated.

3.11.1 Pipe Assembly

Pipe must be strung parallel and adjacent to or above a trench. The pipe must be supported on padded skids during welding and inspection of joints. Protective coating must be inspected and repaired prior to lowering the pipe into the trench. The pipe must be lowered using only canvas or nylon slings. The sling must be dug from underneath the pipe after placements and must not be pulled from underneath the pipe while in contact with it. Care must be taken to prevent damage to the pipe, welded joints or coating and any such damage must be repaired as directed by the Contracting Officer. Pressure testing of the pipe must be done after it has been placed in final position in the trench.

3.11.2 Warning Tapes in Earth Trenches

For the purpose of early warning and identification of buried pipes outside of building walls during future trenching or other excavation, continuous identification tapes must be provided in the trench. Provide metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic warning tape manufactured for the purpose of early warning and identification of utilities buried below the tape. Tape must be at least 80 mm in width. Color of tape must be as standard with the manufacturer with respect to the type of utility buried below the tape. Tape must have lettering at least 25 mm high with warning and identification imprinted in bold black letters continuously over the entire tape length with not less than the following identification on the tape: BURIED JET FUEL PIPING BELOW. Tape must be installed in accordance with the printed recommendations of the tape manufacturer, as modified herein. Tapes must be buried at a depth of 150 mm from the top of the subgrade or 300 mm below the top surface of earth. Provide permanent color and printing, unaffected by moisture or soil.

3.11.3 Clearances

Install pipe to be clear of contact with other pipes, pipe sleeves, casings, reinforcing steel, conduits, cables, or other metallic structures. Where pipes cross other pipes or structures with a separation of less than 150 mm, install an insulating separator. Protect the pipe from contact with a 300 mm square by 25 mm thick bituminous-impregnated canefiber board.

3.11.4 Protective Coating

When the protective coating on pipe is damaged, the Contracting Officer must be notified and must inspect the pipe before the coating is patched. If the damage to the pipe is deeper than 1.2 mm, the damage must be repaired by welding in accordance with paragraph WELDING. If the pipe is dented, out of round or damaged to the point that welding will not make it good as new, the length of pipe must be rejected.

3.11.5 Pipeline Markers

Provide above underground fuel piping spaced every 90 meters, at tees, and at changes in direction. For sections of underground piping less than 90 meters long, place at midpoint. Provide directly above pipe for single lines and between pipes where pipes run in pairs. Provide additional marker over each mid-line fitting connections for steel reinforced flexible pipe. Cast marker into 450 mm diameter, 300 mm thick concrete plug unless it is set in an area with concrete paving in which case it must be cast into the concrete paving.

3.12 SYSTEM COMMISSIONING

For F-24 storage tank system, conform to Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). For JP-5 pipeline system,, conform to Section 33 08 55.02.

3.13 TESTING (Both F-24 and JP-5 Systems)

Piping must be tested by pneumatic and hydrostatic pressure. Testing must comply with applicable requirements of ASME B31.3, NFPA 30 and the requirements specified herein. Hydrostatic testing must be performed using fuel as the liquid. Pressure and hydrostatic testing must be performed only after welding inspection has been completed. Labor, materials, equipment, electricity, repairs, and retesting necessary for any of the tests required herein must be furnished by the Contractor. Provide provisions to prevent displacement of the piping during testing. Keep personnel clear of the piping during pneumatic testing. Only authorized personnel must be permitted in the area during pneumatic and hydrostatic testing. Isolate system components such as pumps, tanks, filter separators, and meters from the piping system during the testing. Do not exceed the pressure rating of any component in the piping system during the testing. Following satisfactory completion of each test, relieve the test pressure and seal the pipe immediately. Piping to be installed underground must not receive field applied exterior coatings at the joints or be covered by backfill until the piping has passed the final pneumatic tests described herein.

3.13.1 Before Backfilling

Fuel piping must not be backfilled without completing the coating, pneumatic, weld radiograph and other piping examinations and approvals. Fuel piping in trench must not have exposed welds for longer than 30 days.

3.13.1.1 Exterior Coating Holiday Test

Following installation, test the exterior coating of direct buried piping for holidays using high-voltage spark testing in accordance with NACE SP0188. Repair holidays and retest to confirm holiday-free coating. Text must include all existing underground piping exposed for this project.

3.13.2 Pneumatic Test

Piping to be installed underground must not receive field applied protective covering at the joints or be covered by backfill until the piping has passed the pneumatic test described herein. To facilitate the tests, isolate various sections of the piping system and test each one separately. Where such sections terminate at flanged valve points, the line must be closed by means of blind flanges in lieu of relying on the valve. Furnish tapped flanges that can be attached to the end of the section of line being tested, and that will permit a direct connection between the piping and the air compressor or pressurizing pump. No taps in the permanent line will be permitted. Furnish all necessary equipment for testing; all gauges must be subject to testing and approval of the Contracting Officer. The air used for pneumatic testing must have a dew point of no more than 5 degrees C. Provide dehumidifying equipment on the suction or discharge side of the air compressor used to provide air for testing. Pressurizing pump must not exceed 4.7 L/s.

3.13.2.1 Pneumatic Test Procedure

Special safety measures, including the wearing of face mask, must be taken during testing under pressure. Only authorized personnel must be permitted in the area during testing. The pneumatic test pressure must be applied in increments. A preliminary 100 kPa (15 psi) test must be applied. Maintain the pressure while soapsuds or equivalent materials are applied to the exterior of the piping. While applying the soapsuds, visually inspect the entire run of piping, including the bottom surfaces, for leaks (bubble formations). If leaks are discovered, repair the leaks accordingly and retest. Repeat process until system is leak-free at 100 kPa (15 psi). The full test pressure must then be applied. Unless otherwise directed by the Contracting Officer, all piping must be tested at a pressure of 175 kPa (25 psi) for not less than 2-hours, during which time there must be no drop in pressure, only pressure rises with temperature. The pressure source must be disconnected during the final test period. Any leaks revealed by the test must be repaired and the test repeated.

3.13.3 Hydrostatic Test

Upon completion of pneumatic testing and after backfilling, hydrostatically test each underground piping system with fuel at 2.93 MPa (425 psi) for F-24 piping system; 3.8 MPa (550 psi) for JP-5 pipeline in accordance with ASME B31.3 and API RP 1110, with no leakage or reduction in gauge pressure for four hours. Upon completion of pneumatic testing, hydrostatically test each aboveground piping system with fuel at 1.97 MPa (285 psi) for F-24 piping system; 3.8 MPa (550 psi) for JP-5 pipeline in accordance with ASME B31.3 and API RP 1110, with no leakage or reduction in gauge pressure for four hours. Furnish electricity, instruments, connecting devices, and personnel for test. If fuel is used for testing, comply with all the requirements in Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). Upon completion of hydrostatic testing, perform "Soak Testing" of the piping systems per API RP 1595. Duration of the test must be a minimum of 4 days and maximum of 7 days. Defects in work must be corrected at the Contractor's expense, and the test repeated until the work is proven to be in compliance with the Contract requirements.

3.13.4 Soak Testing

Upon completion of hydrostatic testing with fuel, perform "Soak Testing" of the piping systems per [API RP 1595](#). Duration of the test must be a minimum of 4 days and maximum of 7 days.

3.13.5 Performance Testing

After the fuel system testing is completed (including pneumatic and hydrostatic testing) the fuel system must be flushed, cleaned and performance tested as specified in Section [33 08 55.01](#) FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). All control valves, both manual and automatic, must be checked for leaks (any area wetted with fuel) and proper operation and adjusted, repaired or replaced to correct any defects.

3.14 PIPE PIGGING - CLEANING ([JP-5 Pipeline](#))

3.14.1 General

Track all pigs, using transmitter and receivers, at no less than [805 m](#) increments, but no less than at four locations. The Contractor must prepare a contingency plan for retrieving a stuck pig and repairing any piping deformations. After pigging, plug valves must be flushed of all debris using the drain port at the bottom of the valve.

3.14.2 Use of Fuel in Cleaning Pigging

Cleaning pigging with fuel will take place after the initial receipt of fuel as per Section [33 08 55.01](#) FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). Ensure that the fuel that is returned to the storage tanks during the pig runs is free of gross contamination and passes the color assessment method, and meets the requirements of [MIL-STD-3004-1](#). Provide temporary storage tanks for the high particulate and dark color fuel that accumulates in front of and behind each pig. The contractor is responsible for cleaning the off-spec fuel in order to meet the requirements of [MIL-STD-3004-1](#) or dispose of the off-spec fuel off-base.

3.14.3 Cleaning Pig Run

Initially, a proving pig run (foam density [32 kg/m³](#) should be performed to ensure the system is fully piggable. Upon completion of the successful proving pig run, the piping system must be cleaned with a standard cleaning pig. This will provide thorough cleaning of the interior of the piping system. Cleaning pig must be the bi-directional disk scraper style with steel body and replaceable polyurethane guiding and sealing disks, as well as gauge plates of 80 percent pipe diameter with [3 mm](#) segmented aluminum fins. The pig body should include bypass nozzles and transmitter cavity. Propellant must be pressurized fuel using portable pumps. The pig must be examined after the initial run for signs of possible pipe damage, interior slag or other adhered particles. Additional runs must be performed until the amount of collected sludge or debris is minimized, as determined by the System Supplier.

3.14.4 Wire Brush Pig Run

After the cleaning pig runs, the piping system must be cleaned with a wire brush style pig. This will remove weld slag and adhered particles from the system. Wire brush pig must be the bi-directional disk style or directional cup style with two circular steel wire brushes. The pig body

should include bypass nozzles and transmitter cavity. Perform wire brush pig runs until the amount of collected weld slag or debris is minimized, as determined by the System Supplier.

3.15 PIPE PIGGING VERIFICATION (JP-5 Pipeline)

3.15.1 Geometry/Ultrasonic Tool Reports

After the system is installed and prior to performance testing, a field/preliminary report must be issued and a debrief given to Government personnel onsite on the condition of the piping system that was pigged. This must be comprised of raw data in the form of a PC download or equivalent which shows a continuous scan of each data unit output. Results of a preliminary interpretation of the data must be reported. These must include as a minimum all critical anomalies. A final report must include a description of the principle of operation, explanation of raw data, presentation of raw data, data to be clearly marked with distance traveled scale with classified anomaly location and all identifiable pipeline features, and all anomalies to be classified with locations in summary tabular form, pipe wall thickness survey, as well as the software necessary to read the data. Submittal must be in the form of digital media copied to a CD or DVD (flash drives are unacceptable).

3.15.2 Pipeline Internal Inspection Operations

3.15.2.1 General

The following pigs will be propelled through the pipeline with fuel in order to inspect the pipeline: 1.7 kg density foam swab, combination poly scraper-magnetic, stainless steel wire brush (plastic brush for internally lined piping), aluminum plate gauge, and geometry/ultrasonic tool. Tracking devices must be used on all pigs. At a minimum, the sequence of pig runs must be as follows: 1) foam swab for proving and cleaning, 2) wire brush for cleaning, 3) scraper-magnetic for cleaning, 4) aluminum plate gauge for gauging internal anomalies, 5) scraper-magnetic for cleaning, 6) wire brush for cleaning, 7) scraper-magnetic for cleaning, 8) foam swab for cleaning, (Note: the number of pig flights of each type of cleaning pigs must be determined by the amount and type of debris removed. The conclusion of the cleaning process must be when debris recovered is only that from the pigs themselves. This determination will be determined by the project's System Supplier and the Contracting Officer), 9) geometry/ultrasonic tool. The pipe wall must be continuously monitored on a real-time basis during the geometry/ultrasonic pig run. Anomalies such as patches, couplings, or flanges must also be identified, and the wall thickness given. The geometry/ultrasonic pig's technician will determine if additional runs are necessary. A permanent data set of internal inspection survey findings must be generated.

3.15.2.2 Preparatory Work

The Government will bring to the attention of the Contractor all statutes, rules and regulations relevant to the performance of the work on the site (on Government property) and will also provide the Contractor with a copy of its own site regulations (if any). Provide the pigging vendors with all-available pipeline records and drawings.

3.15.2.3 Pig Load And Launch

The pig must be loaded into the pig launcher by the Contractor. The

method of loading and lodging the front pig cup into the launcher must not involve the use of uncontrolled mechanical force applied to the rear of the pig.

3.15.2.4 Pipeline Operation During Pigging

All pig runs must be made with the line packed with product. The system pumps will be used to propel the pig. The new pig traps will be used for pig launch and retrieval.

3.15.2.5 Brush and Gauging Survey

Run a brush pig at least as often as previously indicated. The brush pig must be designed and provided by the geometry/ultrasonic pig vendor. Additional runs may be required based upon the amount of debris found in the pipeline. The onsite geometry/ultrasonic pig vendor's personnel must determine if additional runs are required. Immediately following the brush pig run and immediately prior to the geometry/ultrasonic survey, run, as a minimum, a single batching pig fitted with a gauge plate equal to 90 percent of the pipeline normal inside diameter. The plate is to be a segmented aluminum disk of 3 mm thickness. The plate gauge pig must also include a tracker and tracking equipment. Track the pig assembly above ground during the operation.

3.15.2.6 Geometry/Ultrasonic Survey

After a satisfactory gauging pig run, the pipeline geometric defects must be determined by a geometry/ultrasonic tool. The geometry/ultrasonic tool must provide accuracy geometric anomaly detection, and bend radius measuring capability. The data obtained must be presented in a PC software format to allow user friendly analysis and presentation. The geometry/ultrasonic tool assembly must be capable of:

- a. Operating in hydrocarbon liquid environment, specifically jet fuel, at a pressure of up to ANSI 300 rating.
- b. Traversing the pipeline with nominal wall thickness and possible bore restrictions down to 90 percent of nominal pipe inside diameter.
- c. Traversing the pipeline length at a speed of between 60 and 100 m/min when propelled by pumped jet fuel. Pressure differential across pig not to exceed 34 kPa.
- d. Traversing through smooth pipe bends as small as 3D (3 pipe diameters) radius and single miter bends of up to 10 degrees change of direction.
- e. Include a tracker and tracking equipment. Track the pig assembly above ground during the operation. The battery life of the tracker must not be less than 72-hours.
- f. Manual loading into the new horizontal pig trap.

The geometry/ultrasonic tool assembly instrumentation performance must be capable of:

- a. Battery life to be minimum 18-hours at operating conditions.
- b. Principle of operation to be electronically stored geometry system.

- c. Geometry sensing to span full circumference and length of pipe, with associated distance measuring method.
- d. Geometry system must be capable of:
 - (1) positive location and identification of each geometric anomaly.
 - (2) positive location and identification of each bend.
 - (3) positive location and identification of distance marker reference points of either magnetic or electronic type placed on or above the pipe.
- e. Classification of geometric anomalies to be as minimum:
 - (1) discrimination between ovality and intrusion anomalies.
 - (2) mechanical damage such as mill defects, dents, internal gouges, and buckles.
 - (3) pipeline weld defects (such as excess weld penetration).
 - (4) geometric thickness anomalies. As a minimum, these must be reported in the following categories within the listed accuracy.
 - (a) magnitude of anomaly (plus/minus 25 mm)
 - (b) span of anomaly (plus/minus 25 mm)
 - (c) ovality (plus/minus 2.5 mm)
 - (d) span of ovality (plus/minus 25 mm)
 - (e) anomaly station (plus/minus 1:2,000)

3.15.2.7 Pipe Wall Thickness Survey

The geometry/ultrasonic tool must provide accuracy measurement of pipe wall thickness (plus/minus 0.25 mm). The data obtained must be presented in a PC software format to allow user friendly analysis and presentation.

3.15.2.8 Lost Pig

The Contractor is responsible for a lost pig, finding the pig, retrieval of the pig, and all repairs, radiographs to the pipeline system and the pig.

3.16 DEMONSTRATIONS

As per requirements in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

3.17 POSTED OPERATING INSTRUCTIONS

As per requirements in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 52 43.11

AVIATION FUEL MECHANICAL EQUIPMENT

08/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 ADMINISTRATIVE REQUIREMENTS
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Types of Fuel
 - 2.1.2 Composition of Materials
 - 2.1.3 Gaskets
 - 2.1.4 Bolts and Nuts
- 2.2 EQUIPMENT AND MATERIAL
 - 2.2.1 General
 - 2.2.2 Supplier
- 2.3 ELECTRICAL EQUIPMENT
- 2.4 PRESSURE GAUGES
- 2.5 AUTOMATIC PUMP CONTROLS
 - 2.5.1 Flow Switches
- 2.6 METERS
- 2.7 HIGH POINT VENT AND LOW POINT DRAIN PITS
 - 2.7.1 Pit Assembly
 - 2.7.2 Pit
 - 2.7.3 Pit Cover, General Requirements
 - 2.7.4 Pit Cover Materials, Design, and Testing
 - 2.7.5 Pipe Riser Seal
- 2.8 SAFETY DEVICE AGAINST DETONATION, LIQUID TYPE
- 2.9 SAFETY DEVICE AGAINST DETONATION, DRY TYPE
- 2.10 QUICK DISCONNECT COUPLINGS
- 2.11 BULK STORAGE TANK LEVEL INDICATOR
- 2.12 BULK STORAGE TANK LEVEL SWITCHES
- 2.13 ADDITIVE INJECTION SYSTEM AND STORAGE TANK
 - 2.13.1 Injector Assembly
 - 2.13.2 Injector Storage Vessels (T-402)
 - 2.13.3 Additive Tubing and Conduit
 - 2.13.4 Additive Ball Valves and Strainer
- 2.14 BULK STORAGE TANK VENT

PART 3 EXECUTION

- 3.1 GENERAL
 - 3.1.1 Installation
 - 3.1.2 Anchoring
 - 3.1.3 Grouting

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.1.4 Leveling and Aligning
- 3.1.5 Direct Drives
 - 3.1.5.1 Rotation Direction and Speed
 - 3.1.5.2 End Play
 - 3.1.5.3 Shaft Leveling and Radial Alignment
 - 3.1.5.4 Angular Alignment and End Clearance
 - 3.1.5.5 Final Recheck
- 3.1.6 Precautions
- 3.2 INSTALLATION OF UNDERGROUND TANKS
 - 3.2.1 Coating Testing
 - 3.2.2 Steel Tanks
- 3.3 INSTALLATION OF FIBERGLASS PITS
- 3.4 POSTED OPERATING INSTRUCTIONS
 - 3.4.1 Each System
 - 3.4.2 Each Tank
 - 3.4.3 Each Item
 - 3.4.4 Diagrams
 - 3.4.5 Volume of Fuel

-- End of Section Table of Contents --

SECTION 33 52 43.11

AVIATION FUEL MECHANICAL EQUIPMENT
08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API RP 1615 (2011) Installation of Underground
Petroleum Storage Systems

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 (2020) Pipe Flanges and Flanged Fittings
NPS 1/2 Through NPS 24 Metric/Inch Standard

ASME B40.100 (2013) Pressure Gauges and Gauge
Attachments

ASTM INTERNATIONAL (ASTM)

ASTM C827/C827M (2016) Standard Test Method for Change in
Height at Early Ages of Cylindrical
Specimens of Cementitious Mixtures

ASTM D1655 (2018a) Standard Specification for
Aviation Turbine Fuels

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 30 (2021; TIA 20-1; TIA 20-2) Flammable and
Combustible Liquids Code

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-DTL-5624 (2016; Rev W; Notice 1 2020) Turbine Fuel,
Aviation, Grades JP-4 and JP-5

MIL-STD-130 (2007; Rev N; Change 1 2012)
Identification Marking of U.S. Military
Property

MIL-STD-161 (2005; Rev G; Notice 1 2010)
Identification Methods for Bulk Petroleum
Products Systems Including Hydrocarbon
Missile Fuels

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 280 Technical Standards and Corrective Action

Requirements for Owners and Operators of
Underground Storage Tanks (UST)

UNDERWRITERS LABORATORIES (UL)

UL 142

(2006; Reprint Jan 2021) UL Standard for
Safety Steel Aboveground Tanks for
Flammable and Combustible Liquids

1.2 ADMINISTRATIVE REQUIREMENTS

Submit detail drawings consisting of illustrations, schedules, performance charts, instructions, brochures, diagrams, and other information to illustrate the requirements and operation of the equipment and systems. Provide the drawings as one package with the design analysis. Shop fabrication drawings must include type of material, configuration, thickness, and necessary details of construction of the steel tank and vault. Shop drawings must also show the steel grating and supports. Submit Manufacturer's Catalog Data and Certificates of Compliance. Operation and maintenance information must be submitted for the equipment items or systems listed in PART 2. Automatic pump controls must include step-by-step procedures required for system startup, operation, and shutdown. Refer to Section 01 78 23 OPERATION AND MAINTENANCE DATA for the information to be submitted for various types of equipment and systems.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Meters; G

Bulk Storage Tank Vent; G

High Point Vent and Low Point Drain Pits; G

SD-03 Product Data

Pressure Gauges; G

Automatic Pump Controls; G

Meters; G

Bulk Storage Tank Vent; G.

High Point Vent and Low Point Drain Pits; G

Bulk Storage Tank Level Indicator; G

Bulk Storage Tank Level Switches; G

Quick Disconnect Couplings; G

Safety Device Against Detonation, Liquid Type; G

Safety Device Against Detonation, Dry Type; G

SD-06 Test Reports

Coating Testing; G

SD-07 Certificates

System Supplier; G

Safety Device Against Detonation, Dry Type; G

Safety Device Against Detonation, Liquid Type; G

SD-10 Operation and Maintenance Data

Automatic Pump Controls; G

Bulk Storage Tank Vent; G.

Bulk Storage Tank Level Indicator; G

Meters; G

Safety Device Against Detonation, Dry Type; G

Safety Device Against Detonation, Liquid Type; G

1.4 QUALITY ASSURANCE

Submit the following data for approval:

- a. Certification stating that the **system supplier** has provided and installed at least five PLC-based pump control systems in the last five years, for automatic cycling of pumps based upon varying dispensing demands, utilizing multiple pumps. These systems must be for dispensing jet fuel.
- b. Certification that six systems have been successfully operated over the last three years and are currently in service.
- c. Project names, locations, system description, and items provided at these installations. Include user point-of-contact and current telephone numbers.

PART 2 PRODUCTS

2.1 MATERIALS

Materials of construction must be stainless steel, aluminum or nonferrous material except meter case may be steel with electrolyses nickel plated internals coated to **0.075 mm** thickness. No ferrous or zinc-coated material bronze, brass or other copper bearing alloys must be used in contact with the fuel.

2.1.1 Types of Fuel

Components must be suitable for use with F-24 turbine fuel (Jet A with

additives FSII, CI/LE, and SDA); specific gravity 0.81 at 16 degrees C; viscosity 1.62 CS at 16 degrees C; Reid vapor pressure less than 0.35 kPa, ASTM D1655 or JP-5 turbine fuel; specific gravity 0.82 at 16 degrees C; viscosity 1.62 CS at 16 degrees C; Reid Vapor pressure less than 0.35 kPa, MIL-DTL-5624. Components to be ANSI Class 150 (1920 kPa at 38 degrees C.) unless noted otherwise. Components to be suitable for outside, unsheltered location, and to function normally in ambient temperatures.

2.1.2 Composition of Materials

Materials in contact with the fuel must be noncorrosive. No zinc-coated metals, brass, bronze, iron, lead or lead alloys, copper or copper alloys, or other light metal alloys containing more than 4 percent copper must be used in contact with the fuel.

2.1.3 Gaskets

Gaskets must be in accordance with Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).

2.1.4 Bolts and Nuts

Bolts and nuts must be in accordance with Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).

2.2 EQUIPMENT AND MATERIAL

2.2.1 General

All items of equipment and material must be new and of the best quality used for the purpose in commercial practice and must be products of reputable manufacturers. Each major component of equipment must have the manufacturer's name, address and catalog number on a plate securely affixed in a conspicuous place. The nameplate of a distributing agent only will not be acceptable. The gears, couplings, projecting set screws, keys and other rotating parts located so that any person may come in close proximity thereto must be fully enclosed or properly guarded. Equipment, assemblies and parts must be marked for identification in accordance with MIL-STD-130 and MIL-STD-161. Pump and filter vessel numbers must be as indicated on the drawings. In addition, filter vessels must include stenciled or embossing tape letters 19 to 25 mm tall which indicate element numbers, date element changed, due date of the next element change, and maximum differential pressure. Identification (ID) tags made of brass, stainless steel, or engraved anodized aluminum, indicating valve number and normally open (NO) or normally closed (NC) must be installed on valves. Tags must be 35 mm minimum diameter, and marking must be stamped or engraved. Indentations must be black, for reading clarity. Tags must be attached to valves with No 12 AWG, copper wire, stainless or aluminum hanging wires, or chrome-plated beaded chain designed for that purpose.

2.2.2 Supplier

Since the pump control system, including but not limited to pump control panel, transmitters, flow switches, fueling system pumps, all field instrumentation, and control valves with all hardware and software, is an integrated system it must be furnished by a single systems supplier regularly engaged in the supplying of this equipment. System Supplier must be a company whose regular, normal, and primary business is representing manufacturers in the distribution and start-up of aviation

fueling facilities, and have no affiliation with the Contractor other than as a seller to the Contractor. Supplier must provide all equipment and appurtenances regardless of manufacture, be a factory authorized certified representative, and be responsible to the Contractor for satisfactory operation of the entire system, and must oversee the installation of the equipment. Substitutions of functions specified will not be acceptable. The Contractor and the System Supplier must be present at the system commissioning, and must coordinate and schedule the work during construction, testing, calibration, and acceptance of the system. The System Supplier must be on-site with their mechanical and control personnel to supervise and assist the contractor during pre-commissioning check-out of the mechanical systems and control systems, initial fuel receipt, initial filling, hydrostatic testing, pigging, flushing, cleaning, Equipment Tests, Performance Testing, and training all for the owner's representatives. The System Supplier must be responsible to the Contractor for scheduling all Contractor, Sub-contractor, and manufacturer's service personnel during system start-up, Equipment Tests, Performance Testing, and training.

2.3 ELECTRICAL EQUIPMENT

Motors, manual or automatic motor control equipment except where installed in motor control centers, and protective or signal devices required for the operation specified herein must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Any wiring required for the operation specified herein, but not shown on the electrical plans, must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.4 PRESSURE GAUGES

Pressure gauges must conform to ASME B40.100 with metal cases and 100 mm diameter white dials. Gauges must be bottom connected, without back flanges. A pulsation dampener, adjustable to the degree of dampening required, must be provided for each gauge. Range of gauges must be as indicated. A ball valve must be provided for each pressure gauge. Gauges must be labeled with the calibration date.

2.5 AUTOMATIC PUMP CONTROLS

The pressure and flow transmitters specified in this paragraph must be obtained from a single supplier of such products. The same supplier must also furnish the associated venturi tubes and GPM meter. The supplier must be responsible for furnishing components that are compatible and that operate as a system to perform the required pump control functions. Control tubing between controls/instruments and fuel lines must be installed to eliminate air entrapment. Control tubing must be as specified in Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT). Each item of equipment specified hereafter must have manufacturer's authorized service personnel present to assist in PERFORMANCE TESTING as specified in Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). Items specified under this paragraph must be submitted for approval concurrently with items specified in Section 33 09 52 FUEL PUMP CONTROL AND ANNUNCIATION SYSTEM (NON-HYDRANT).

2.5.1 Flow Switches

Switches must be actuating vane type flow switch with single adjustable set-point. Switches must mount on ASME B16.5 Class 150 raised face

flange. Flange material must match the piping material at their connection to the system. Provide snap action switch mechanism U.L. listed for Class I, Division 1, Group D hazardous locations. Switches to be double pole double throw (DPDT). Switch power must be 120 volts, single phase, 60 hertz, 10 amps minimum. Units installed on 50 mm piping and smaller may be threaded.

2.6 METERS

Meter must be a one-way flow, temperature compensating, positive displacement type meter designed for a continuous flow of 9 L/s at the truck off-loading stand. Meter must have ANSI Class 150 flanges and body working pressure of not less than 2 MPa and must be suitable for hydrostatic testing of 2 MPa. Meter must be factory calibrated for FSII Additivejet fuel and capable of being calibrated in the field. The register must have a non-setback total indicator and a setback type run indicator so that individual runs can be registered without affecting the total of all runs as shown on the indicator. The total indicator must have a minimum of eight figures and the setback run indicator must have a minimum of five figures. The register must read in gallons and the smallest unit of indicated delivery must be 4 L. Accuracy must be within +0.3 percent between ten percent and maximum rated flow. Meters must be provided with a suitable drain at the bottom, equipped with a ball valve. Pressure loss through the meter must not exceed 20 kPa at 9 L/s flow rate.

2.7 HIGH POINT VENT AND LOW POINT DRAIN PITS

2.7.1 Pit Assembly

Each pit must incorporate the following items built into a self-contained assembly.

2.7.2 Pit

The basic pit must consist of 6.25 mm wall fiberglass liner with a main body approximately 575 mm in diameter and a minimum of 925 mm deep. The pit must contain two integral concrete anchors. The fiberglass top flange must require no exposed corrosive material, weldments, or strongbacks within the pit to support the cast aluminum ring and cover assembly. The pits must be the standard products of a firm regularly engaged in the manufacture of such product and must essentially duplicate items that have been in satisfactory use for at least three (3) years prior to bid opening. Proof of experience will be submitted.

2.7.3 Pit Cover, General Requirements

The pit cover must include a removable outer ring frame and an interior 457 mm diameter (clear opening) hinged lid that opens 160 degrees. Each cover lid must move smoothly through its entire range of motion and must require a maximum opening force of 150 N to be applied at a single lifting handle. Each handle must provide a comfortable, secure grip for an average adult male's full gloved hand. Tools must not be required to engage the lifting handle. Projections of the lid's hinges or handles above the plane of the lid, whether temporary or permanent, must not be allowed. The pit service must be integrally cast in raised letters on the top surface of each lid. The lettering must be a minimum of 25 mm high and 1.6 mm deep. The weight bearing flanges of the fiberglass pit liner and the aluminum cover frame (and lid) must be machined to assure uniform weight distribution.

2.7.4 Pit Cover Materials, Design, and Testing

The cover frames and lids must be designed and manufactured by a qualified company having a minimum of five years successful experience in the production of similar airport apron slab fixtures. All cover lids and frames must be designed using an appropriate cast aluminum alloy or rolled aluminum plate. The aluminum alloy material selected for design must be ductile, corrosion-resistant, impact-resistant, and suitable for the intended use. All covers must be non-skid surface construction and free of injurious defects. Welding for the purpose of structural repair of casting defects must not be allowed. Minor cosmetic welding is acceptable.

2.7.5 Pipe Riser Seal

The riser pipe penetration through the pit floor must be sealed by means of a Buna-N boot. The boot must be secured to a metal collar welded to the pipe riser and to a flange at the floor opening by stainless steel clamps. Collar must be fabricated from the same material as the pipe.

2.8 SAFETY DEVICE AGAINST DETONATION, LIQUID TYPE

Liquid-type safety device against detonation, static flame arrestor, maintenance-free design, for indoor installation, to secure fill lines. The integral immersion pipes with upstream detonation arrestor, expansion space as well as dipping fluid tank. Casing shall be resistant to explosion, detonation and pressure shocks. Provide Operation and Maintenance manual. Submit manufacturer's data including specifications and performance test data. Provide Certificate of Conformity to 94/9/EC. Provide CE mark.

2.9 SAFETY DEVICE AGAINST DETONATION, DRY TYPE

Safety device against detonation with dry static flame arrestor. Casing with locked cover - resistant to explosion and detonation pressures. The easily replaceable flame trap, including flame filter - three each - as well as two-part cage and upstream detonation shock absorber. Marginal gap width of flame filter. Connection shall be angle, stepped or globe as indicated. Provide Operation and Maintenance manual. Submit manufacturer's data including specifications and performance test data. Provide Certificate of Conformity to 94/9/EC. Provide CE mark.

2.10 QUICK DISCONNECT COUPLINGS

Pumpouts, filter separators, and tank connection hose couplings shall be groove and shoulder type, complete with dust caps. Metal components shall be aluminum with jet fuel resistant seals. Hose coupling manufacturer, type and size shall be submitted to the contracting Officer for approval.

2.11 BULK STORAGE TANK LEVEL INDICATOR

The level indicating system must perform tank gauging and have local tank readout. The level indicating system must use a servo to measure all the various locations required for the primary measurement. The level indicating system must be able to measure and compute fuel level, fuel density, fuel actual volume, fuel and water corrected volume, and fuel ambient temperature. The reference point for all level measurements must be from the tank's datum plate. The servo system must attach to the tank's riser/ stilling well to minimize the effects of turbulence on the

measurements and still allow the government access to take quality control samples. The level indicating system must be able to measure in underground, aboveground and cut and cover tanks with all floor and roof types. The level indicating system must be able to measure multiple tanks with a single field interface unit. The level indicating system must be able to determine whether the tank is issuing or receiving fuel while in the transfer mode and also with the same unit be able to perform leak detection. The level indicating system must require no periodic calibration after installation is complete. The level indicating system must be approved for installation in a hazardous area and certified intrinsically safe by an approved agency and provide lightning protection. The level indicating system must be able to interface with government owned information systems. The level indicating system must provide five sets of alarm outputs; high intermediate high, low, intermediate low and static tank movement alarm.

Level accuracy + 1.25 mm

Corrected volume accuracy $\pm$ 0.1 percent

Density accuracy $\pm$ 1 percent

Temperature accuracy $\pm$ -0.6 degrees C

Detect water in the tank sump to a level equal to or slightly above the water draw-off pipe

2.12 BULK STORAGE TANK LEVEL SWITCHES

- a. System must be designed and installed in such a way that the system must be continuously and automatically self-checking. Switches must be mounted on top of the tank, in the pump house, as indicated. Electronic level sensors must be thermistors or optic type, and be intrinsically safe Class I, Division 1, Group D for hazardous environments, with recognized FM, CSA or UL approval. The sensor holder/junction box must be accessible.
- b. Level alarms must be mechanically and electrically independent and be totally isolated from the gauging system. The level switches must receive power and send their signal to the Pump Control Panel. Circuitry and cables from the PCP to the electronic level sensors in the tank must be intrinsically safe.

2.13 ADDITIVE INJECTION SYSTEM AND STORAGE TANK

The Injection System must incorporate and include all components necessary to mechanically inject the required additive products into the fuel stream at a nominal, adjustable rate of up to 0-1,000 ppm. The actual rates of injection for each of the required components will be as determined by the injector manufacturer.

2.13.1 Injector Assembly

The injector assembly must be a completely self contained 100 mm flanged unit rated for 150 ANSI service and installed within the Injector Building as indicated. The injector pump must be adjustable from 0 to 1,000 ppm minimum. The assembly must include a flow indicator/suction strainer, inlet valve, outlet valve and piping all manufactured from stainless steel materials.

2.13.2 Injector Storage Vessels (T-402)

The injector storage vessel must be a stainless steel tank with a 200 L

capacity. The vessel must meet all requirements of NFPA 30 and UL 142 for flammable/combustible liquids and must include a 30 year warranty against defects in material or workmanship. The vessel and all components must be suitable for Jet A1 Fuel in addition to the specific additive. All materials mounted inside the tank must be 304 SS minimum. The storage tanks must be mounted on a 1.5 m high steel structural stand. The injector storage tank gauge must be a vertical direct read with an 200 mm minimum dial face calibrated in L. The gauge assembly must mount on a 50 mm NPT fitting.

2.13.3 Additive Tubing and Conduit

The additive supply header feeding the fuel injection stations must be manufactured of one 25 mm nylon flexible tubing natural colored. The tubing must have an inner diameter of 21 mm, a working pressure of 1.4 MPa and a minimum bend radius of 175 mm. All tube fittings must be of stainless steel ferrule compression design. Support aboveground tubing as required to mount securely.

2.13.4 Additive Ball Valves and Strainer

The valves and strainer in the tanks supply line must be stainless steel and must meet the requirements of Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT) plus be equipped with gaskets, seals, seats and packing that are compatible with Jet A1 Fuel and the specific additive. PTFE and fluorocarbon (Viton) -litharge cured are recommended by the additive manufacturer. The strainer must include a stainless steel 50 mesh screen.

2.14 BULK STORAGE TANK VENT

- a. Tank vent outlet must be equipped with pressure-vacuum breather vent, aluminum construction with weather hood and with fluoroelastomer (FKM, Viton) pallet seat inserts, high density screens, stainless steel internals, with pressure relief setting at 215 Pa, and vacuum relief set at 215 Pa. Pressure venting capacity must be 190 L/s, vacuum capacity must be 190 L/s.

PART 3 EXECUTION

3.1 GENERAL

3.1.1 Installation

Install equipment and components in position, true to line, level and plumb, and measured from established benchmarks or reference points. Follow manufacturer's recommended practices for equipment installation. Provide required clearances between equipment components. Equipment, apparatus, and accessories requiring normal servicing or maintenance to be accessible.

3.1.2 Anchoring

Anchor equipment in place. Check alignment of anchor bolts before installing equipment and clean-out associated sleeves. Do not cut bolts because of misalignment. Notify Contracting Officer of errors and obtain the Contracting Officer's acceptance before proceeding with corrections. Cut anchor bolts of excess length to the appropriate length without damage to threads. Where anchor bolts or like devices have not been installed, provide appropriate self-drilling type anchors for construction condition.

3.1.3 Grouting

Equipment which is anchored to a pad is to be grouted in place. Before setting equipment in place and before placing grout, clean surfaces to be in contact with grout, including fasteners and sleeves. Remove standing water, debris, oil, rust, and coatings which impair bond. Clean contaminated concrete by grinding. Clean metal surfaces of mill scale and rust by hand or power tool methods. Provide necessary formwork for placing and retaining grout. Grout to be non-metallic, non-shrink, fluid precision grout of a hydraulic cementitious system with graded and processed silica aggregate, Portland cement, shrinkage compensating agents, plasticizing and water reducing agents; free of aluminum powder agents, oxidizing agents and inorganic accelerators, including chlorides; proportioned, pre-mixed and packaged at factory with only the addition of water required at the project site. Grouting must be in accordance with [ASTM C827/C827M](#). Perform all grouting in accordance with equipment manufacturer's and grout manufacturer's published specifications and recommendations.

3.1.4 Leveling and Aligning

Level and align equipment in accordance with respective manufacturer's published data. Do not use anchor bolt, jack-nuts or wedges to support, level or align equipment. Install only flat shims for leveling equipment. Place shims to fully support equipment. Wedging is not permitted. Shims to be fabricated flat carbon steel units of surface configuration and area not less than equipment bearing surface. Shims to provide for full equipment support. Shim to have smooth surfaces and edges, free from burrs and slivers. Flame or electrode cut edges not acceptable.

3.1.5 Direct Drives

Alignment procedure follows:

3.1.5.1 Rotation Direction and Speed

Check and correct drive shaft rotation direction and speed.

3.1.5.2 End Play

Run drive shafts at operational speed. Determine whether axial end play exists. Run drive shaft at operational speed and mark drive shaft axial position when end play exists. Block drive shaft in operating position when aligning drive shaft with driven shaft.

3.1.5.3 Shaft Leveling and Radial Alignment

Pump alignment must be accomplished by the factory technician or a millwright trained in pump alignment, and with the use of dial gauges or laser alignment equipment.

3.1.5.4 Angular Alignment and End Clearance

Check angular alignment and end clearance by inserting a feeler gage at 4 points, 90 degrees apart around outer edges of coupling halves.

3.1.5.5 Final Recheck

Check adjustments with dial indicator after completing recheck. Align shafts within **0.025 mm** tolerance, except as other-wise required by more stringent requirements of equipment manufacturer.

3.1.6 Precautions

Special care must be taken to ensure that equipment and materials are stored properly to prevent damage and maintain cleanliness, and that the completed system is free of rocks, sand, dirt, and foreign objects. Take the following steps to insure these conditions.

- a. Equipment brought to the site and not stored inside, must be stored on blocks or horses at least **450 mm** above ground.
- b. Visual inspection must be made of each piece of equipment to ensure that it is clean prior to installation.
- c. The open ends of equipment must be closed when work with that piece of equipment is not in progress.

3.2 INSTALLATION OF UNDERGROUND TANKS

Installation must be per tank manufacturer's recommendations, **API RP 1615**, **NFPA 30**, **40 CFR 280**, state and local codes and as specified herein. If recommendations require tank to be filled, only fuel will be allowed in tanks. Water filling is not acceptable. Before being placed in service, tank must be tightness tested in accordance with **NFPA 30**.

3.2.1 Coating Testing

The coating must be examined for flaws and tested for thickness. Provide the facilities, personnel, and equipment for testing for flaws and thickness. Thickness must be measured electronically. Coating must be tested directly before placement of the tank with an electric flaw detector, equipped with a bell, buzzer, or other type of audible signal that operates when a flaw is detected. The detector for the type of coating used must have an operating voltage of 10,000 to 35,000 volts. Check of the holiday detector potential may be made by the Contracting Officer at any time to determine the suitability of the detector. Damaged areas must be repaired with materials identical to those used originally, and after drying, must be retested electrically. Submit test results.

3.2.2 Steel Tanks

- a. Cover the concrete hold down slab with **150 mm** of tank bedding backfill evenly graded and thoroughly compacted, prior to tank placement.
- b. Each tank is to be unloaded and placed on the sand bed using cranes and the rigging procedures provided by the tank manufacturer. Use the tank lifting lugs for lifting the tank into place. The use of slings around the tank is not permitted, nor is the use of chock blocks of any sort. During handling, carefully inspect the tanks for coating damage and repair any damage whatsoever before proceeding. After placement, check each tank to ensure it is sloped as required. The elevation must be confirmed.
- c. Before proceeding with backfill, install the hold down straps and

tighten the turnbuckles securely and evenly throughout the length of the tanks. The bottom and sides of the tanks to be fully and evenly supported by hand shoveling and tamping. Use tank bedding backfill up to 303 mm above the top of tank. Hand-guided power equipment can be used to place fill in 150 mm layers, compacted to a minimum of 95 percent maximum density, after the bottom quadrant is filled. A minimum of four density tests per tank to be performed. Clean, noncorrosive, well tamped gravel to be used for backfill from a point 303 mm above the tanks to finished grade.

- d. Do not fill the tank, even partially, before the bottom quadrant is backfilled. The level of fuel product not to exceed the level of compacted backfill at any time.
- e. Coordinate tank installation with the installation of cathodic protection.

3.3 INSTALLATION OF FIBERGLASS PITS

Submit recommended installation procedures and setting tolerances from the pit manufacturer/supplier for the fiberglass pit and the aluminum cover. These procedures must indicate recommended methods of supporting the pit in its proper position in the open excavation prior to and during concrete placement operations. Also, required installation tolerances, especially for flatness/levelness of the fiberglass pit lip, must be provided. Follow these recommendations and apply other procedures as required to ensure the integrity of the pit liner and cover assemblies in their installed positions. All penetrations through the fiberglass pit liner must be tightly sealed by suitable means to preclude water infiltration, with consideration for potential relative movements between the penetrating objects and the pit liner. Reference the Contract drawings for additional installation requirements.

3.4 POSTED OPERATING INSTRUCTIONS

For each designated system or equipment item, provide instructions for guidance of operating and maintenance personnel. Following approval of content, prepare these instructions in a form and scale that will be readily legible when displayed in appropriate locations, to be designated by the Contracting Officer and meet the following requirements:

3.4.1 Each System

For each system, include diagrams of equipment, piping, wiring and control. Define control sequences.

3.4.2 Each Tank

For each tank provide a P.E. stamped certified tank calibration chart in 1/16-inch increments reading in gallons.

3.4.3 Each Item

For each equipment item, include starting, adjustment, operation, lubrication, safety precautions and shut-down procedures. Identify procedures to be performed in event of equipment failure. Provide other instructions recommended by the manufacturer.

3.4.4 Diagrams

Provide a professionally prepared isometric piping diagram of the fueling system apparatus. Diagram must be 914 by 1370 mm and must be color coded to match PCP color diagrams. Diagram must show the entire facility and must include all equipment and the operational sequences of all equipment with equipment numbers displayed. Diagram must show all valves along with the valve numbers shown on the drawings and listed as normally open/closed. It must be wall mounted under glass.

3.4.5 Volume of Fuel

Provide a certified system inventory of fuel in the pipe, tank, pumphouse, etc. The piping will show length of pipe, size of pipe, L/s, and total L. Verify during initial fill.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 52 43.14

AVIATION FUEL CONTROL VALVES

08/18, CHG 1: 02/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 ADMINISTRATIVE REQUIREMENTS
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Field Assistance
 - 1.4.2 Training
- 1.5 WARRANTY

PART 2 PRODUCTS

- 2.1 MATERIALS AND EQUIPMENT
- 2.2 CONTROL VALVES
 - 2.2.1 General
 - 2.2.1.1 Bodies, Bonnets, and Covers
 - 2.2.1.2 Valve Seats
 - 2.2.1.3 Valve Discs
 - 2.2.1.4 Diaphragm Assembly
 - 2.2.1.5 Bolts, Screws and Nuts
 - 2.2.1.6 Pilot Control System and Auxiliary Piping
 - 2.2.1.7 Pilot Valves
 - 2.2.2 Serviceability of Main Valve Internal Parts
 - 2.2.3 Total Lengths
 - 2.2.4 Flanges
 - 2.2.5 Identification
 - 2.2.5.1 Main Valve Body
 - 2.2.5.2 Main Valve Cover
 - 2.2.5.3 Brass Name Plates
 - 2.2.5.4 Inlet Name Plate
 - 2.2.5.5 Outlet Name Plate
 - 2.2.5.6 Pilot Valves
- 2.3 INDIVIDUAL CONTROL VALVE OPERATIONAL REQUIREMENTS
 - 2.3.1 High Liquid Level Shut-Off Valve (HLV-100 AND HLV-200)
 - 2.3.1.1 Size
 - 2.3.1.2 Flow
 - 2.3.1.3 Operation
 - 2.3.1.4 Manual Test Feature
 - 2.3.1.5 Strainer
 - 2.3.1.6 Pressure Sensitive Close Feature
 - 2.3.1.7 Opening and Closing Feature
 - 2.3.2 Non-Surge Check /Pump Control Valve (FCV-101, FCV-102, FCV-201, FCV-202)
 - 2.3.2.1 Size
 - 2.3.2.2 Flow
 - 2.3.2.3 Operation

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.3.2.4 Quick closure
- 2.3.2.5 Flow Control
- 2.3.2.6 Strainer
- 2.3.3 Overfill Valve for Drain Tank (OV-500)
 - 2.3.3.1 Size
 - 2.3.3.2 Capacity
 - 2.3.3.3 Operation
 - 2.3.3.4 Control Float
 - 2.3.3.5 Pressure Reservoir
 - 2.3.3.6 Thermal Relief
 - 2.3.3.7 Limit Switch
 - 2.3.3.8 Strainer

PART 3 EXECUTION

3.1 VALVE TESTING AND START-UP SUPPORT

-- End of Section Table of Contents --

SECTION 33 52 43.14

AVIATION FUEL CONTROL VALVES
08/18, CHG 1: 02/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 (2020) Pipe Flanges and Flanged Fittings
NPS 1/2 Through NPS 24 Metric/Inch Standard

ASME BPVC SEC VIII D1 (2019) BPVC Section VIII-Rules for
Construction of Pressure Vessels Division 1

ASTM INTERNATIONAL (ASTM)

ASTM A194/A194M (2020a) Standard Specification for Carbon
Steel, Alloy Steel, and Stainless Steel
Nuts for Bolts for High-Pressure or
High-Temperature Service, or Both

ASTM A216/A216M (2021) Standard Specification for Steel
Castings, Carbon, Suitable for Fusion
Welding, for High-Temperature Service

ASTM A269/A269M (2015; R 2019) Standard Specification for
Seamless and Welded Austenitic Stainless
Steel Tubing for General Service

ASTM A320/A320M (2021a) Standard Specification for
Alloy-Steel and Stainless Steel Bolting
for Low-Temperature Service

ASTM A352/A352M (2021) Standard Specification for Steel
Castings, Ferritic and Martensitic, for
Pressure-Containing Parts, Suitable for
Low-Temperature Service

ASTM A743/A743M (2021) Standard Specification for
Castings, Iron-Chromium,
Iron-Chromium-Nickel, Corrosion Resistant,
for General Application

ASTM B26/B26M (2018; E 2018) Standard Specification for
Aluminum-Alloy Sand Castings

ASTM D751 (2006; R 2011) Coated Fabrics

ASTM D2000 (2018) Standard Classification System for
Rubber Products in Automotive Applications

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE AMS 3216	(2005; Rev G) Fluorocarbon (FKM) Rubber High-Temperature - Fluid Resistant Low Compression Set 70 To 80
SAE J200	(2015) Classification System for Rubber Materials
SAE J429	(2014) Mechanical and Material Requirements for Externally Threaded Fasteners

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-A-8625	(1993; Rev F; Am 1 2003) Anodic Coatings, for Aluminum and Aluminum Alloys
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1.2 ADMINISTRATIVE REQUIREMENTS

Design conditions must be as specified in Section 33 52 43.11 AVIATION FUEL MECHANICAL EQUIPMENT. Components must be suitable for ANSI Class 150 (2 MPa at 38 degrees C).

- a. Control valves specified herein must be of one manufacturer. The valve manufacturer must also produce the hydraulically-operated pilots. For each type control valve required and specified, submit the following:
 - (1). Flow diagrams.
 - (2). Operational description of the control valve and pilot control system.
 - (3). Complete valve assembly list of materials, along with material Certificates of Conformance, used in the manufacture of the control valves and pilot systems.
 - (4). sectional drawings of main valve and control pilot systems.
- b. Before shipment, each individual control valve must be operationally tested and adjusted by manufacturer under actual flow conditions utilizing a hydrocarbon test fluid with a specific gravity comparable to Jet A (F-24) and/or JP-5 (F-44) . Manufacturer must submit certified records of test data.
- c. Operation and maintenance information must be submitted for each individual type control valve specified herein. Refer to Section 01 78 23 OPERATION AND MAINTENANCE DATA for the information to be submitted.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Control Valves; G.

SD-03 Product Data

Control Valves; G.

SD-06 Test Reports

Control Valves.

SD-07 Certificates

Previous Military Projects; G.

Qualified Engineers; G.

Field Assistance; G.

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G.

1.4 QUALITY ASSURANCE

1.4.1 Field Assistance

Provide the following:

- a. Proof of experience on [previous Military projects](#).
- b. Number of [qualified engineers](#) (factory trained) available to provide startup support.
- c. Written assurance as to ability to respond to specified time for field assistance.

1.4.2 Training

The manufacturer must conduct two eight hour training classes for Liquid Fuels Maintenance Technicians which include valve overhaul procedures, pilot overhaul procedures, valve adjustments, and valve diagnostics. The manufacturer must provide a **100 mm** valve mock-up with various trim components (i.e., rate of flow, solenoid control, and speed control features) to be used during training. Video recording of training must be allowed or provided at the time of the class, and an attendance roster maintained by the Contractor. The **100 mm** valve mock-up must become the property of the Government and must be turned over to the Contracting Officer. Submit copies of the [Operation and Maintenance Manuals](#) for approval.

1.5 WARRANTY

For this section, Subject Matter Expert (SME) is defined as Service Headquarters Subject Matter Experts. SME for this project is Air Force - The Air Force Fuels Facilities Subject Matter Expert (HQ AFCEC/COS) Navy Corps - NAVFAC POL Facility Subject Matter Expert (NAVFAC EXWC, CI11).

If a problem attributable to the valve's manufacturer or installation arises after the initial system start-up has been accomplished, and after system final acceptance date, 48 hours from the time of notification that

a problem exists is allowed to solve the problem. The problem must be solved to the satisfaction of the Contracting Officer. If the Contractor cannot effectuate a proper resolution to the problem as outlined above in the 48 hour period, provide a factory trained engineer from the manufacturer of the valve within 48 hours after the expiration of the Contractor's initial 48 hour period to effectuate a resolution of the problem above. All services provided by the valve manufacturer must be at no cost to the Government. When it has been determined by the Contractor, Contracting Officer, and the valve manufacturer's representative that the valve(s) cannot be repaired in its installed position in the fuel system, it must be replaced with a new valve and pilot assembly within 48 hours after the initial 96-hour period listed above expires and at no cost to the Government.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

The type of materials which come in contact with the fuel, if not specified herein before, must be noncorrosive.

2.2 CONTROL VALVES

2.2.1 General

Control valves must be single-seated globe type, diaphragm actuated, hydraulically operated valves. Valves must consist of 3 major components: the valve body, valve cover, and diaphragm assembly. The diaphragm assembly must be the only moving part. In the event of diaphragm failure, valve must fail closed against flow, unless otherwise indicated. The main valve must be drip-tight when closed. Each valve must have an external indicator to show the position of the valve disc at all times. Control valves must be shipped from the factory as a complete assembly with all pilot controls and pilot auxiliary piping properly installed on the main valve. Materials which come in contact with the fuel must be resistant to the effects of and not harmful to aircraft engine fuel and must be stainless steel, or electroless nickel plated cast steel unless noted otherwise. Materials for control valves, and items to be mounted on the valves must be as follows:

2.2.1.1 Bodies, Bonnets, and Covers

Bodies, bonnets, and covers must be constructed of one of the following materials:

- a. Cast steel conforming to [ASTM A216/A216M](#), Grade WCB internally plated with chromium, nickel or internally electroless nickel plated.
- b. Cast stainless steel conforming to [ASTM A743/A743M](#).
- c. Cast steel conforming to [ASTM A352/A352M](#) Grade LCB internally plated with chromium, nickel, or internally electroless nickel plated.
- d. Bodies must have flanged inlet and outlet connections. Valve must have a screwed bottom drain plug.

2.2.1.2 Valve Seats

Valve seats must be stainless steel in accordance with [ASTM A743/A743M](#).

It must be possible to remove the valve seat while the valve is connected in the line. Valve seat and upper stem bearing must be removable and screwed in the body and/or cover. The lower stem bearing must be concentrically contained in the valve seat and must be exposed to flow on all sides. The diameter of the valve seat must be the same size as the inlet and/or outlet flanges of the main valve.

2.2.1.3 Valve Discs

Valve discs must contain a resilient, fluoroelastomer (FKM), commonly referred to as Viton disc conforming to [SAE AMS 3216](#) having a rectangular cross section, contained on 3.5 sides by a disc retainer and a disc guide, forming a drip tight seal against the seat. The disc must be usable on either side. The disc guide must be the contoured type capable of holding disc firmly in place during high differential pressure conditions that may develop across the seating surface. The disc retainer must be capable of withstanding rapid closing shocks.

2.2.1.4 Diaphragm Assembly

Diaphragm Assembly must form a sealed chamber in the upper portion of the valve, separating the operating fluid from the line pressure. The diaphragm assembly must contain a valve stem which is fully guided at both ends by a bearing in the valve cover and an integral bearing in the valve seat. Valve body and cover must be sealed by the diaphragm. Valve stem must be stainless steel. The bearing material must be compatible with the fuel specified and must not contain zinc coated metals, brass, bronze, or other copper bearing alloys. The diaphragm must be of a nonwicking material or design, with a minimum of 2 layers of nylon fabric bonded with a minimum of 3 layers of synthetic rubber (valves [62 mm](#) and smaller one layer of nylon fabric). The edge area of the center hole for the valve stem must be sealed by vulcanization. Materials to be resistant to aromatics of up to 50 percent in accordance with [ASTM D2000 \(SAE J200\)](#). The diaphragm must have a MULLINS-burst rating according to [ASTM D751](#) of a minimum of [4.14 MPa](#) per layer of nylon fabric. All diaphragm sizes must be cycle tested to a minimum of 100,000 cycles, by alternately applying pressure under the diaphragm (main valve pressure) and above the diaphragm (cover chamber pressure). That test must be certified by the manufacturer. The diaphragm must not be used as a seating surface. The diaphragm must be fully supported by the body and cover in either the open or closed position.

2.2.1.5 Bolts, Screws and Nuts

a. For Ductile Iron, and Cast Steel Body Valves.

- (1) Bolts and Screws, cadmium plated steel in accordance with [SAE J429](#), Grade 5.
- (2) Nuts, cadmium plated steel in accordance with [ASTM A194/A194M](#), Grade 2 H.

b. For Stainless Steel Body Valves. Bolts, Screws and Nuts, [ASTM A320/A320M](#), Grade B8M C.1.1.

2.2.1.6 Pilot Control System and Auxiliary Piping

Pilot Control System and auxiliary piping must be stainless steel, seamless, fully annealed tubing conforming to [ASTM A269/A269M](#), Grade

TP316, Rockwell hardness B80 or less. Wall thickness for 13 mm tubing to be 0.9 mm. Threaded connections must be used in pilot system piping and unions must be o-ring type with FKM o-rings. Tubing connections must not be welded.

2.2.1.7 Pilot Valves

Pilot valves must have aluminum bodies conforming to ASTM B26/B26M Type 356-T6 anodized in accordance with MIL-A-8625 with stainless steel internal working parts. Disc and diaphragm assemblies must be as specified herein before. The setting of adjustable type pressure operated pilot valves must be easily adjusted by means of a single adjusting screw. The adjusting screw must be protected by a threaded cap drilled to accommodate a lead-seal wire and a lock nut must be provided on the adjusting screw to lock it in position at the desired setting. The lead seal wire must be installed after final acceptance of the system. Spare wire seals and the "embossing" tool will be turned over to the Contracting Officer for the installation.

2.2.2 Serviceability of Main Valve Internal Parts

Main valve movable parts including strainers, valve seat, stem bearings, and control system must be replaceable without removing the main valve from the line. All nonmetallic parts must be replaceable.

2.2.3 Total Lengths

The total valve length does not include the orifice plate flange (when used). If the control valve being supplied has the orifice plate built into its flange, the spacer provided must bring the valve face-to-face dimension equal to those listed below plus 2.2 mm. The lengths of the valves must be equal for the following materials: cast stainless steel, and cast steel.

SIZE mm	VALVE LENGTH mm
38	216
50	234
75	305
100	381
150	500
200	635
250	745
305	850
350	975
400	1034

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

SIZE mm	VALVE LENGTH mm
Note: Tolerance must be ± 0.75 mm for size 38 mm through 200 mm and ± 1.5 mm for size 250 thru 400 mm.	

Control valves not meeting these face to face dimensions must be supplied with spacers suitable for the proper installation of the valve.

2.2.4 Flanges

MATERIAL	SEALING SURFACE
Cast Steel, ASME B16.5 Class 150	Raised Face
Cast Stainless Steel, ASME B16.5	Raised Face Class 150
Note: The mating flange must be made the same as above.	

2.2.5 Identification

2.2.5.1 Main Valve Body

The following must be cast into the main valve body:

- Pressure Class
- Size
- Material
- Foundry Heat Number and Identification
- Manufacturer
- Flow Pattern

2.2.5.2 Main Valve Cover

The following must be cast into the main valve cover:

- Size
- Material
- Foundry Heat Number and Identification

2.2.5.3 Brass Name Plates

Brass name plates must be fastened to the valve. Body name plates must list the following:

- Size
- Model Number
- Stock Number
- Manufacturer/Supplier
- Manufacturer's Inspection Stamp

2.2.5.4 Inlet Name Plate

Inlet name plate must list the following:

- Size

- b. "Inlet" Marking
- c. Assembly Model Number
- d. Part Number

2.2.5.5 Outlet Name Plate

Outlet name plate must list the "Outlet" Marking.

2.2.5.6 Pilot Valves

Pilot valves must be tag identified. The valve must have the field adjusted start up setting engraved on a plastic tag, white with black lettering.

2.3 INDIVIDUAL CONTROL VALVE OPERATIONAL REQUIREMENTS

Operation, performance, and special features of the individual control valves must be as specified herein.

2.3.1 High Liquid Level Shut-Off Valve (HLV-100 AND HLV-200)

2.3.1.1 Size

250 mm

2.3.1.2 Flow

75 L/s

2.3.1.3 Operation

High liquid level shut-off valve must be hydraulically operated and must be provided with a tank exterior mounted float. Activation point of the float for opening and closing the high liquid level shut-off valve must be as shown on the drawings. Upon a rise in fluid level to the float activation point, the float control system must cause the main valve to close tightly. The main valve must remain closed until a drop in tank fluid level occurs. Upon a drop in fluid level beneath the float activation point, the float control must cause the main valve to open completely.

2.3.1.4 Manual Test Feature

Manual testing of high level shut-off valve and exterior mounted float's automatic opening and closing feature must be possible.

2.3.1.5 Strainer

A 40-mesh stainless steel wire, self-cleaning strainer must be provided in the pilot valve supply piping.

2.3.1.6 Pressure Sensitive Close Feature

If the upstream pressure rises to 1 MPa or above while closing, the valve will stop closing or open slightly until the pressure is less than 1 MPa.

2.3.1.7 Opening and Closing Feature

The valve must be equipped with an adjustable differential pressure pilot

and a quick cover exhaust system to allow the valve to open in 3-4 seconds when pressure is greater than 207 kPa.

2.3.2 Non-Surge Check /Pump Control Valve (FCV-101, FCV-102, FCV-201, FCV-202)

2.3.2.1 Size

150 mm

2.3.2.2 Flow

40 L/s; 13 L/s for FTP-1.

2.3.2.3 Operation

Non-surge check valve must open slowly. Opening speed must be adjustable from two (2) to 30 seconds without affecting closing of valve. Factory set for 15 seconds. The nonsurge check valves must fail closed against reverse flow in check condition.

2.3.2.4 Quick closure

Valve closure to be rapid, closing quickly when outlet pressure exceeds inlet pressure.

2.3.2.5 Flow Control

Valve to limit flow to 40 L/s. Sensing must be by orifice. Valve to modulate to limit flow without hunting. Rate of flow to be manually adjustable and utilize a downstream orifice plate holder.

2.3.2.6 Strainer

A 40-mesh, stainless steel wire, self-cleaning strainer must be provided in the pilot valve supply piping.

2.3.3 Overfill Valve for Drain Tank (OV-500)

2.3.3.1 Size

50 mm.

2.3.3.2 Capacity

3 L/s.

2.3.3.3 Operation

Hydraulically operated overfill valve must close automatically upon rising to Drain Tank 95 percent fill level. Valve must open automatically upon falling below level as indicated on the drawings.

2.3.3.4 Control Float

Automatic opening and closing of the valve must be initiated by a control float located within the Drain Tank. Control float must be provided with a manual tester, mounted external to the tank, for testing of overfill valve operation.

2.3.3.5 Pressure Reservoir

Valve must be provided with a pressure reservoir to supply required hydraulic pressure for operation. Reservoir pressure to be supplied by the withdrawal pump using 13 mm tubing connected upstream of the pump non-surge check valve. Valve must close upon loss of reservoir pressure. Reservoir must be a 4 L capacity bladder-type tank, carbon steel constructed, tested and stamped in accordance with ASME BPVC SEC VIII D1 for a working pressure of 800 kPa and precharged with air of 80-100 kPa. The tank will be epoxy lined. The tank will be fitted with an air charging valve and pressure gauge.

2.3.3.6 Thermal Relief

Overfill valve must be provided with a pressure sustaining control valve that must automatically, upon inlet pressure rising to 1.3 MPa, open allowing thermal relief around overfill valve. Pressure sustaining valve must automatically close upon inlet pressure dropping below 1.3 MPa.

2.3.3.7 Limit Switch

Limit switch must be single pole, single throw contract (SPST) and provided with valve for remote indication of valve open or closed position. Valve closed position will become an alarm condition at the pump control panel (PCP).

2.3.3.8 Strainer

Pressure reservoir inlet line must be provided with a shut-off valve, strainer and check valve.

PART 3 EXECUTION

3.1 VALVE TESTING AND START-UP SUPPORT

Provide the services of a factory trained and certified service engineer authorized/sanctioned/certified by the valve manufacturer to verify that each valve has been properly installed and to verify valves were factory operationally tested, adjusted and set per these specifications. The service engineer must assist the Contractor in the valve start-up adjustment process and will remain on site until all control valves function as required by the contract documents.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 52 43.23

AVIATION FUEL PUMPS

08/18, CHG 1: 02/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 ADMINISTRATIVE REQUIREMENTS
- 1.3 SUBMITTALS

PART 2 PRODUCTS

- 2.1 FUELING/WATER DRAW-OFF PUMP (VERTICAL TURBINE)(P-101, P-102, P-103, P-201, P-202, P-203)
 - 2.1.1 Capacity
 - 2.1.2 Assembly
 - 2.1.3 Materials
 - 2.1.3.1 Mechanical Seal
 - 2.1.4 Construction
 - 2.1.4.1 Couplings
 - 2.1.4.2 Impeller
 - 2.1.4.3 Wear Rings
 - 2.1.4.4 Shaft
 - 2.1.4.5 Finishing
 - 2.1.4.6 Bearings
 - 2.1.4.7 Drilling and Tapping
 - 2.1.4.8 Mounting Flange
 - 2.1.4.9 Pump Discharge
 - 2.1.4.10 Special Tools
 - 2.1.4.11 Service Nameplate
 - 2.1.4.12 Identification Nameplate
 - 2.1.4.13 Primer Coat
 - 2.1.4.14 Topcoat
 - 2.1.5 Motor
- 2.2 WITHDRAWAL PUMP (P-501)
 - 2.2.1 Capacity
 - 2.2.2 Assembly
 - 2.2.3 Materials
 - 2.2.3.1 Mechanical Seal
 - 2.2.4 Construction
 - 2.2.4.1 Couplings
 - 2.2.4.2 Impeller
 - 2.2.4.3 Wear Rings
 - 2.2.4.4 Shaft
 - 2.2.4.5 Finishing
 - 2.2.4.6 Bearings
 - 2.2.4.7 Drilling and Tapping
 - 2.2.4.8 Mounting Flange
 - 2.2.4.9 Special Tools
 - 2.2.4.10 Service Nameplate

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.2.4.11 Identification Nameplate
- 2.2.4.12 Exterior Primer Coat
- 2.2.4.13 Exterior Topcoat
- 2.2.5 Motor
- 2.3 OFFLOADING PUMP(P-601)
 - 2.3.1 Off-load Pump
 - 2.3.2 Service Nameplate
 - 2.3.3 Identification Nameplate
 - 2.3.4 Exterior Primer Coat
 - 2.3.5 Exterior Topcoat
- 2.4 LEAK DETECTION PUMP (P-104, P-204)
 - 2.4.1 Pump Construction
 - 2.4.1.1 Casing
 - 2.4.1.2 Impeller
 - 2.4.1.3 Bearings
 - 2.4.1.4 Lubrication
- 2.5 PUMP MOTOR

PART 3 EXECUTION

- 3.1 PREPARATION FOR SHIPMENT
 - 3.1.1 Rust Preventative
 - 3.1.2 Closure of Openings
 - 3.1.3 Assembly
 - 3.1.4 Bracing
 - 3.1.5 Vapor Inhibiting Wraps
 - 3.1.6 Shipping Identification
- 3.2 INSTALLATION
 - 3.2.1 Anchoring
 - 3.2.2 Grouting
 - 3.2.3 Leveling and Aligning
 - 3.2.4 Direct Drives
 - 3.2.4.1 Rotation Direction and Speed
 - 3.2.4.2 End Play
 - 3.2.4.3 Shaft Leveling and Radial Alignment
 - 3.2.4.4 Angular Alignment and End Clearance
 - 3.2.4.5 Final Recheck
 - 3.2.5 Start-up Representative

-- End of Section Table of Contents --

SECTION 33 52 43.23

AVIATION FUEL PUMPS
08/18, CHG 1: 02/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API STD 610 (2010; Errata 2011) Centrifugal Pumps for Petroleum, Petrochemical, and Natural Gas Industries

API STD 682 (2014) Pumps Shaft Sealing Systems For Centrifugal and Rotary Pumps

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC SEC IX (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

ASME BPVC SEC VIII D1 (2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1

ASTM INTERNATIONAL (ASTM)

ASTM C827/C827M (2016) Standard Test Method for Change in Height at Early Ages of Cylindrical Specimens of Cementitious Mixtures

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 112 (2017) Standard Test Procedure for Polyphase Induction Motors and Generators

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2016) Motors and Generators - Revision 1: 2018; Includes 2021 Updates to Parts 0, 1, 7, 12, 30, and 31

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC PA 1 (2016) Shop, Field, and Maintenance

Coating of Metals

SSPC SP 10/NACE No. 2

(2015) Near-White Blast Cleaning

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-DTL-24441

(2009; Rev D; Notice 1 2021) Paint,
Epoxy-Polyamide, General Specification for

MIL-PRF-4556

(1998; Rev F; Am 1 1999; CANC Notice 1
2011) Coating Kit, Epoxy, for Interior of
Steel Fuel Tanks

1.2 ADMINISTRATIVE REQUIREMENTS

Design conditions must be as specified in Section 33 52 43.11 AVIATION
FUEL MECHANICAL EQUIPMENT.

- a. Tests: Hydrostatic, performance, vibration, and NPSH tests must be conducted at the factory on each pump in accord with API 610. Test each pump with the actual motor which will drive the pump in the field, unless the water test media will cause overload of the motor. If so, provide vibration test report for motor separately. Vertical turbine pump vibration test must be run with field driver. All tests will be observed by the Contracting Officer or the designated representative. Provide the Contracting Officer 30 days notice prior to performance of factory tests in order to schedule observing such tests. Remote access via web cam must be made available. Performance testing must not occur prior to acceptance of shop drawing submittal.
- b. Test reports must bear the serial number of both pump and driver. Submit manufacturer's certified reports of hydrostatic, performance, and NPSH tests. Submit manufacturer's **certified test curves**.
- c. **Operation and Maintenance Manuals** must be submitted for the pumps and appurtenance specified herein. Refer to Section 01 78 23 **OPERATION AND MAINTENANCE DATA** for the information to be submitted.
- d. Motors, manual or automatic motor control equipment, except where installed in motor control centers, and protective or signal devices required for the operation specified herein must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Any wiring required for the operation specified herein, but not shown on the electrical plans, must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Motors must be high efficiency type and in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Withdrawal Pump; G.

Offloading Pump; G

Fueling/Water Draw-Off Pump (Vertical Turbine); G

Leak Detection Pump; G

SD-03 Product Data

Offloading Pump; G

Fueling/Water Draw-Off Pump (Vertical Turbine); G

Leak Detection Pump; G

SD-06 Test Reports

Certified Test Curves.

SD-07 Certificates

Offloading Pump; G

Fueling/Water Draw-Off Pump (Vertical Turbine); G

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals.

PART 2 PRODUCTS

2.1 FUELING/WATER DRAW-OFF PUMP (VERTICAL TURBINE)(P-101, P-102, P-103, P-201, P-202, P-203)

The following paragraphs also apply to the Water Draw-Off Pumps (P-103, P-203). Water draw-off pumps will be the same type and configuration as the fueling pumps, but a smaller size.

2.1.1 Capacity

Pump Capacity (flow rate and pressure) must be as indicated. Overall efficiency at design conditions of pump and driver, connected, must be minimum 70 percent. Pump head capacity must be continually rising and must be free of dips and valleys from design point to shut-off head. Pump must be capable of at least 5 percent head increase at rated conditions by installing a new impeller.

2.1.2 Assembly

The pump for this service must meet the requirements of API STD 610, latest edition, seventh edition for vibration. Wherever the information contained herein conflicts with said standard, the information herein must govern. The pump for this service must run at a nominal 1800 rpm and must be a multi-stage, vertical turbine pump. Pump and motor must be furnished as a complete unit as herein specified. Pump assembly must be statically and dynamically balanced for all flow rates from minimum flow to 120 percent of design flow. Flanged column, shaft, and bearing spacing must not exceed 1.5 m sections to facilitate pump disassembly within pump room.

2.1.3 Materials

The materials of construction for the pump shaft and the impeller must be stainless steel. All other materials must be material class S-1 with the wetted ferrous parts such as the bowl interiors enamel-lined, bowl exteriors, column interior and exterior, discharge head interior epoxy-coated per MIL-PRF-4556, and discharge head exterior epoxy-coated per MIL-DTL-24441.

2.1.3.1 Mechanical Seal

API STD 682, balanced type, API Class Code BSTHN.

2.1.4 Construction

Castings used for any part of pumps must be sound and free of shrink or blow holes, scale, blisters, and other similar casting defects. The surfaces of casting must be cleaned by sand or shot blasting, pickling, or other standard methods used by the manufacturer. All mold parting fins and remains of gates and risers must be either chipped, filed, or ground flush with the surface of the casting. The repair of casting leaks and defects by peening or by the use of cement compounds is prohibited. All welding to be per ASME BPVC SEC IX.

2.1.4.1 Couplings

Couplings must be flanged, rigid spacer type, CPAT or equal. The couplings must be of the spacer-type with a spacer of sufficient length to permit replacement of the mechanical seal assembly without removing the motor. The pump half coupling must be of such design that it can be removed without the use of heat. Coupling halves must fit tightly to the shafts of the pump and the driver so as not to become loose during operation. The coupling must be provided with an OSHA approved coupling guard.

2.1.4.2 Impeller

Impeller must be enclosed and double keyed to the shaft for radial loads and fixed in the axial position by shaft sleeve nuts, or other positive positioning device. Impellers must be held to the shaft so that the impeller will not become loose should the pump accidentally rotate in reverse direction. The impeller must be statically and dynamically balanced to 8 W/N.

2.1.4.3 Wear Rings

Renewable stainless steel wearing rings must be positively locked on the impeller. Wearing rings must fit with close tolerances so as to permit a minimum of recirculation. Wear ring hardened surfaces differential must be at least Brinell 50. Positive locking case wearing rings must be provided so that the case wearing rings will not rotate or change position in the case. Clearances must be established for hydrocarbon (Jet Fuel) service.

2.1.4.4 Shaft

Shaft must be designed with a high safety factor to easily withstand the torsional loads and other stresses to which it may be subjected. It must be so designed that there will be no detrimental vibration stresses.

Surfaces must be ground to accurate dimensions. Shaft deflection must be limited to **0.05 mm** maximum when measured at the face of the mechanical seal under the operating condition of zero flow at shut off head. Seal piping from the discharge to the mechanical seal must be provided. The pump shaft must be in maximum **1.5 m** sections, and couplings must be keyed and split ring type, not threaded.

2.1.4.5 Finishing

Passageways and impellers must be finished to permit maximum efficiency and provide noise reduction. Overall sound levels must not exceed OSHA limits.

2.1.4.6 Bearings

Bearings must be product-lubricated. Sleeve type, carbon graphite must be provided. Bearing spacing must be per **API STD 610**, eight edition, but must not exceed **1.5 m** in any case.

2.1.4.7 Drilling and Tapping

Casting must be drilled and tapped for drain and seal recirculation lines. All connections must be provided with plugs.

2.1.4.8 Mounting Flange

Mounting flange must be coordinated with the tank's mounting flange, and must be ANSI or API pattern, and contain a **25 mm** tapping for air eliminator discharge.

2.1.4.9 Pump Discharge

Pump discharge head must include a **25 mm** tapping at the highest point with valve, 100 mesh strainer, and air eliminator valve, as specified in Section **33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT)**, with check valve on outlet.

2.1.4.10 Special Tools

Pumps must be furnished with special tools necessary to dismantle and reassemble the unit.

2.1.4.11 Service Nameplate

A pump service nameplate, of type 18-8 stainless steel or monel, securely attached by stainless steel pins at an easily accessible point on the pump, must be furnished in addition to the identification nameplate. The pump service nameplate must be stamped with the following information:

Manufacturer's name
Serial number of pump
Capacity, **L/s**
Pumping head, **m**.
Maximum specific gravity of fluid to be pumped
Revolutions per minute
Horsepower of driver

2.1.4.12 Identification Nameplate

A pump identification nameplate of Type 18-8 stainless steel or monel must be provided and securely attached by stainless steel pins to a conspicuous place on the pump head. Tagging in letters 6 mm high must be the equipment number as shown on the drawings.

2.1.4.13 Primer Coat

Surfaces of the pump and baseplate must be primed by the manufacturer. Surface cleaning must meet requirements of SSPC SP 10/NACE No. 2. Metal primer must be zinc rich paint conforming to specification MIL-DTL-24441 Type 1, Class 3. Dry film thickness must be 0.05 to 0.2 mm.

2.1.4.14 Topcoat

Topcoat must be factory applied and must be white and conforming to specification MIL-DTL-24441.

2.1.5 Motor

- a. Motor must be furnished by the pump manufacturer and must be suitable for the environment and operating conditions to which it will be subjected and be provided with anti-reversing ratchet. The motors unity service factor may be used to conform to the non-overloading through-out the curve requirement for the 10% head increase condition only. Select the lowest horsepower for the motors that will meet the non-overloading requirement and co-ordinate the MCC if different than shown. Provide space heaters suitable for operation on 460 or 120 volts as indicated on the drawings within the motor enclosure to prevent moisture condensation after shut-down. Motor must be UL listed for use in Class 1, Division 1, Group D hazardous areas, and must have a maximum temperature rating of "T2D 216 degrees C" as defined by NFPA 70. The motor nameplate must include the temperature rating of the motor and locked-rotor indicating code letters in accordance with NFPA 70, Table 430-7(b).
- b. Voltage rating must be 460 volts, 3 phase, 60HZ. Motor nominal speed must match pump. Motors must be capable of delivering rated horsepower output successfully and continuously under conditions of voltage variations of 10 percent above or below rated voltage.
- c. Pump manufacturer must assure the specified output and proper operation of the pump without being overloaded at unity service factor when operating at any point on the pump performance curve based on the future potential of a 5 percent head increase. In addition to having sufficient horsepower-output rating at rated speed, motor must have performance characteristics which will allow, without injurious overheating of the motor, accelerating the load from standstill to rated speed under conditions of 10 starts per hour. Attention is specifically directed to the fact that thermal characteristics of motors with regard to capability for accelerating the load may vary greatly from motor manufacturer to motor manufacturer, notwithstanding that the horsepower rating may be the same. It is the pump manufacturer's responsibility to provide motors with adequate thermal starting characteristics as well as adequate rated-speed operating characteristics. Service factors must conform with NEMA standards; however, service factors are only applicable at rated nameplate voltage and frequency. Since all system voltages are subject to

variation, service factors above unity must not be applied in sizing motor.

- d. Motor must be squirrel-cage induction type, high thrust vertical P base, unless bearing frame pump is utilized. Motor must be NEMA Design B (normal-torque, low starting current).
- e. Motor insulation must be non-hydroscopic, NEMA Class F, 150 degrees C for motors. Motor windings must be supplied with extra dips and bakes.
- f. Winding temperature rise, (based on a maximum ambient temperature of 40 degrees C at 1006 m altitude) must not exceed 80 degrees C.
- g. Bearings must be ABMA minimum L10 life of 60,000 hours or L50 life of 300,000 hours suitable for the size, type, and application when the pump is operating at the specified flow and head.
- h. Motor enclosures must be totally enclosed, weather sealed, fan cooled, explosion-proof and must be listed and labeled for Class 1, Group D areas. Provide bronze ground bolt on motor enclosure. All motor external electrical connections must be terminated within a single terminal housing.
- i. The motors must be dynamically balanced and vibration measured per NEMA MG 1, vibration and balance under category "precision." Motor overspeed withstand capability and sound power levels of the motor must conform with NEMA standard requirements.
- j. The pump manufacturer must furnish the Contracting Officer with the recommended minimum run time for the motor.
- k. Pump motor must be provided with temperature limiting thermostats within the motor frame when required to meet Class 1, Group D requirements.
- l. Pump motor must be furnished with lifting lugs on the motor casing.
- m. Unless indicated otherwise, motors for conventional applications over 15 horsepower must be the premium efficient type. This requirement is not applicable to hermetically sealed motors, integrally mounted motors, motors specified as part of energy efficient equipment, wound rotor motors, or any application involving special construction or performance. Guaranteed minimum full load efficiencies must be (based on 1800 rpm, 4 pole, totally enclosed):

15 kW	92.0 percent	56 kW	95.5 percent
19 kW	92.0 percent	75 kW	93.5 percent
22 kW	92.0 percent	93 kW	94.5 percent
30 kW	92.0 percent	112 kW	94.5 percent
37 kW	92.5 percent	149 kW	94.5 percent

45 kw	92.5 percent	447 kw	94.5 percent
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- n. Other motors of different speed or housing classification must also be of the premium efficient type, as advertised by the motor manufacturer, with efficiency greater than the standard line. Motor efficiencies must have been verified in accordance with NEMA MG 1, 12.53.a., and determined using the dynamometer method as described in IEEE 112, Method B. All shop drawing submittals on motor driven equipment must include the motor efficiency.

2.2 WITHDRAWAL PUMP (P-501)

2.2.1 Capacity

Pump capacity (flow rate and pressure) must be as indicated. Overall efficiency at design conditions of pump and driver, connected, must be minimum 70 percent. Pump head capacity must be continually rising and must be free of dips and valleys from design point to shut-off head. Pump must be capable of at least 10 percent head increase at rated conditions by installing a new impeller.

2.2.2 Assembly

The pump for this service must meet the requirements of API STD 610, latest edition, seventh edition for vibration. Wherever the information contained herein conflicts with said standard, the information herein must govern. The pump for this service must run at a nominal 1800 rpm and must be a multi-stage, vertical turbine pump. Pump and motor must be furnished as a complete unit as herein specified. Pump assembly must be statically and dynamically balanced for all flow rates from minimum flow to 120 percent of design flow.

2.2.3 Materials

The materials of construction for the pump shaft and the impeller must be stainless steel. All other materials must be material class S-1 with the wetted ferrous parts such as the bowl interiors enamel-lined, bowl exteriors, column interior and exterior, discharge head interior epoxy-coated per MIL-PRF-4556, and discharge head exterior epoxy-coated per MIL-DTL-24441.

2.2.3.1 Mechanical Seal

API STD 682, balanced type, API Class Code BSTHN.

2.2.4 Construction

Castings used for any part of pumps must be sound and free of shrink or blow holes, scale, blisters, and other similar casting defects. The surfaces of casting must be cleaned by sand or shot blasting, pickling, or other standard methods used by the manufacturer. All mold parting fins and remains of gates and risers must be either chipped, filed, or ground flush with the surface of the casting. The repair of casting leaks and defects by peening or by the use of cement compounds is prohibited by ASME BPVC SEC VIII D1.

2.2.4.1 Couplings

Couplings must be flanged, rigid spacer type, CPAT or equal. The couplings must be of the spacer-type with a spacer of sufficient length to permit replacement of the mechanical seal assembly without removing the motor. The pump half coupling must be of such design that it can be removed without the use of heat. Coupling halves must fit tightly to the shafts of the pump and the driver so as not to become loose during operation. The coupling must be provided with an OSHA approved coupling guard.

2.2.4.2 Impeller

Impeller must be keyed to the shaft for radial loads and fixed in the axial position by shaft sleeve nuts, or other positive positioning device. Impellers must be held to the shaft so that the impeller will not become loose should the pump accidentally rotate in reverse direction. The impeller must be statically and dynamically balanced.

2.2.4.3 Wear Rings

Renewable wearing rings must be positively locked on the impeller. Wearing rings must fit with close tolerances so as to permit a minimum of recirculation. Positive locking case wearing rings must be provided so that the case wearing rings will not rotate or change position in the case.

2.2.4.4 Shaft

Shaft must be designed with a high safety factor to easily withstand the torsional loads and other stresses to which it may be subjected. It must be so designed that there will be no detrimental vibration stresses. Surfaces must be ground to accurate dimensions. Shaft deflection must be limited to 0.05 maximum when measured at the face of the mechanical seal under the operating condition of zero flow at shut off head. Shaft must be protected through the mechanical seal by means of a shaft sleeve. Seal piping from the discharge to the mechanical seal must be provided.

2.2.4.5 Finishing

Passageways and impellers must be finished to permit maximum efficiency and provide noise reduction. Overall sound levels must not exceed OSHA limits.

2.2.4.6 Bearings

Bearings must be product-lubricated. Sleeve type, carbon graphite must be provided. Bearing spacing must be per API STD 610.

2.2.4.7 Drilling and Tapping

Casting must be drilled and tapped for drain and seal recirculation lines. All connections must be provided with plugs.

2.2.4.8 Mounting Flange

Mounting flange must be coordinated with the tank's mounting flange, and must be ANSI or API pattern.

2.2.4.9 Special Tools

Pumps must be furnished with special tools necessary to dismantle and reassemble the unit.

2.2.4.10 Service Nameplate

A pump service nameplate, of type 18-8 stainless steel or monel, securely attached by stainless steel pins at an easily accessible point on the pump, must be furnished in addition to the identification nameplate. The pump service nameplate must be stamped with the following information:

Manufacturer's name
Serial number of pump
Capacity, **L/s**
Pumping head, **m**
Maximum specific gravity of fluid to be pumped
Revolutions per minute
Horsepower of driver

2.2.4.11 Identification Nameplate

A pump identification nameplate of Type 18-8 stainless steel or monel must be provided and securely attached by stainless steel pins to a conspicuous place on the pump head. Tagging in letters **6 mm** high must be the equipment number as shown on the drawings.

2.2.4.12 Exterior Primer Coat

Exterior surfaces of the pump and baseplate must be primed by the manufacturer. Surface cleaning must meet requirements of **SSPC SP 10/NACE No. 2**. Metal primer must be zinc rich paint conforming to specification **MIL-DTL-24441** Type 1, Class 3. Dry film thickness must be **0.05 to 0.1 mm**.

2.2.4.13 Exterior Topcoat

Manufacturer's standard exterior topcoat must be factory applied and must be white.

2.2.5 Motor

Refer to paragraph, Motors for the Fueling Pumps.

2.3 OFFLOADING PUMP(P-601)

2.3.1 Off-load Pump

- a. The pump must be a single-stage, horizontal, centrifugal type consisting of a centrifugal impeller combined with a vane-type rotary, positive evacuating, volumetric-displacement priming unit, mounted on a common shaft.
- b. The pump must be such that all rotating parts may be removed without disconnection of the suction or discharge piping.
- c. The pump must operate dry for not less than 1 hour without damage or permanent defamation of moving parts after the pump has been operated by the petroleum products.

- d. The priming unit must be a vane-type rotary, positive, volumetric-displacement unit mounted on the same shaft as the centrifugal impeller. The priming unit must evacuate air from the suction piping and thereby initially priming the centrifugal impeller and restoring lost prime during the operation of the pump. A stainless steel self-cleaning strainer must be provided in the priming-unit intake line.
- e. The motor must have sufficient power for the service required, be of a type approved by a manufacturer of the pump, be suitable for available electric service, be totally enclosed, fan cooled, TEFC, and conform to the requirements of 26 20 00 INTERIOR DISTRIBUTION SYSTEM.
- f. Capacity must be as indicated on the drawings. Pump head capacity must be continually rising and must be free of dips and valleys from design point to shut-off head. Pump shut-off head must have a 10 to 20 percent head rise to shut off. Pump must be capable of a least a 10 percent head increase at rated conditions by installing a new impeller. Pumps must not overheat or be damaged in any way while operating continuously at a minimum flow condition of 50 percent of rated flow continuously at a maximum flow condition of 125 percent required capacity. The net positive suction head required at the center of the impeller must not exceed 6 feet.
- g. Pump must consist of a centrifugal impeller combined with an integral variable-capacity vane type priming unit located within the main centrifugal housing. The vane type positive priming unit must be capable of initially priming the pump and of restoring prime during operation against back pressures to 8 psi. The air release must enable the pump to prime at any discharge head pressure from zero feet to maximum discharge head and must release drip-free to the atmosphere and drain any liquid back into the centrifugal pump.
- h. The vane pump must be positively driven by the main pump shaft. No gears or pumps must be required to operate the primary pump. All parts must be factory inspected so that parts are interchangeable. Pumps and motors must be furnished as complete units as herein specified. Pump assembly must be statically and dynamically balanced for all flow rates from minimum flow to 120 percent of design flow.
- i. The pump must be fitted with a mechanical-type pump shaft seal for closure between the stationary pump case and the pump shaft. The seal must be capable of effectively sealing a pressure equal to 1 1/2 times the pump shut off head. The mechanical-type shaft seal must be a standard product of proven material and design. The sealing surface must be self-aligning and must be readily removable for repair or replacement without removing the electric motor from the base. The sealing surfaces must be constructed of carbon/nitride, and the elastomers must be viton. The rotating member must be pin driven by the pump shaft, or must be firmly retained on the pump shaft by a snap ring or other suitable means. Materials used as sealing members or elastomers must be impervious to the deleterious action of the specified product. All parts of the seal must be resistant to corrosion and oxidation.
- j. The pump shaft must be fabricated from 410 stainless steel. The shaft must be turned, ground and polished, and hardened to resist wear where the shaft passes through the seal and bearings. The shaft must be

supported in the shaft housing by means of heavy duty, anti-fiction, sealed type ball bearings.

- k. The pump shaft must be coupled to the motor shaft by means of a flexible copling having sufficient torsional strength to accomodate the rated motor horsepower. The coupling must be capable of handling angular and non-parallel alignment.
- l. The pump case must be ductile iron. Flange ratings must be class 105 kg 125-pound per ANSI Standard, and flanges must be faced and drilled.
- m. The impeller must be of the closed typek and must be statically and dynamically blanced. The impeller must be constructed of aluminum 356-T6.

2.3.2 Service Nameplate

A pump service nameplate, of type 18-8 stainless steel or monel, attached by stainless steel pins at an accessible point on the pump, shall be furnished in addition to the identification nameplate. The pump service nameplate shall be stamped with the following information:

Manufacturer's name
Serial number of pump
Capacity, gpm
Pumping head, ft.
Maximum specific gravity of fluid to be pumped
Revolutions per minute
Horsepower of driver

2.3.3 Identification Nameplate

A pump identification nameplate of Type 18-8 stainless steel or monel shall be provided and securely attached by stainless steel pins to a conspicuous place on the pump head. Tagging in letter 1/4-inch high shall bear the equipment number as shown on the drawings.

2.3.4 Exterior Primer Coat

Exterior surfaces of the baseplate shall be primed by the manufacturer. Coating shall be applied meeting requirements of SSPC PA 1. Surface cleaning shall meet requirements of SSPC SP 10/NACE No. 2. Metal primer shall be zinc rich paint conforming to specification MIL-DTL-24441, Type 1, Class 3. Dry film thickness shall be 2 to 4 mils.

2.3.5 Exterior Topcoat

Manufacturer's standard exterior topcoat shall be applied at factory to the base plate.

2.4 LEAK DETECTION PUMP (P-104, P-204)

Provide submersible nonclog pumps as indicated. Provide UL listed pumps for explosion proof Class 1, Division 1, Groups C and D hazardous locations. Provide submersible, centrifugal sewage pumps of the nonclogging type. Provide pump capacity, number of pumps and motor characteristics as indicated on the drawings. Select pumps to continuously operate in a submerged or partially submerged condition.

2.4.1 Pump Construction

2.4.1.1 Casing

Provide hard, close-grained cast iron casing or steel that is free from blow holes, porosity, hard spots, shrinkage defects, cracks, and other unjurious defects. Provide casings permitting replacement of wearing parts. Ensure all joints are gasketed to prevent leakage. Ensure passageways permit smooth flow and are free of sharp turns and projections. Use free standing pump support legs of cast-iron providing clearance from vault floor.

2.4.1.2 Impeller

Provide a stainless steel impeller. Provide nonclogging type cast-iron impeller. Ensure the impeller has a smooth surface and allows free flowing. Fit and key, spline, or thread impeller on shaft, and lock in such manner that lateral movement is prevented and reverse rotation will not cause loosening.

2.4.1.3 Bearings

Provide heavy duty ball thrust bearing or roller type bearing sized to withstand imposed loads. Oil lubricate bearings.

2.4.1.4 Lubrication

Provide lubricated permanently sealed bearings.

2.5 PUMP MOTOR

Provide hermetically sealed electric motors. Motor horsepower must not be less than pump horsepower at any point on the pump performance curve. Fit motors with lifting "eyes" capable of supporting entire weight of pump and motor. Seal the power cable inside the motor end bell. Provide a waterproof power cable for its full length. Provide motors UL listed for explosion proof Class 1, Division 1, Groups C and D. Air filled motors are not acceptable. Ensure oil used may be disposed as non-hazardous waste.

PART 3 EXECUTION

3.1 PREPARATION FOR SHIPMENT

3.1.1 Rust Preventative

Exterior machine surfaces must be coated with a rust preventative. Pumps must be disassembled after the shop running tests and inspected, and internal parts must be coated with a rust preventative before reassembling.

3.1.2 Closure of Openings

Threaded openings must be provided with metallic plugs or caps. Flanges must be gasketed with rubber and closed with 4.8 mm thick plate of the same outside diameter as the match flange. A minimum of four full-diameter bolts must hold closure in place.

3.1.3 Assembly

Pumps must be shipped assembled or a field service engineer must be furnished to supervise the field assembly at no additional cost to the Government.

3.1.4 Bracing

Each unit must be suitably prepared for shipment, supported and braced, with auxiliary equipment secured to prevent damage during shipment.

3.1.5 Vapor Inhibiting Wraps

Exposed shafts and shaft couplings must be wrapped with waterproof moldable waxed cloth or vapor inhibitor paper. The seams must be sealed with adhesive tape.

3.1.6 Shipping Identification

Each pump must be identified with a metal tag showing the item number. Material shipped separately must be marked with a metal tag indicating the item number for which it is intended.

3.2 INSTALLATION

Install equipment and components true to line, level and plumb, and measured from established benchmarks or reference points. Follow manufacturer's recommended practices for equipment installation. Provide required clearances between equipment components. Equipment, apparatus, and accessories requiring normal servicing or maintenance must be easily accessible.

3.2.1 Anchoring

Anchor equipment in place as indicated on the drawings or per manufacturer's recommendations. Minimum anchor bolt size is 127 mm. Check alignment of anchor bolts and/or bolt holes before installing equipment and clean-out associated sleeves. Do not cut bolts due to misalignment. Notify the Contracting Officer of errors and obtain the Contracting Officer's acceptance before proceeding with corrections. Cut anchor bolts of excess length to the appropriate length without damage to threads.

3.2.2 Grouting

Equipment which is anchored to a pad must be grouted in place. Before setting equipment in place and before placing grout, clean surfaces to be in contact with grout, including fasteners and sleeves. Remove standing water, debris, oil, rust, coatings and other materials which impair bond. Clean contaminated concrete by grinding. Clean metal surfaces of mill scale and rust by hand or power tool methods. Provide formwork for placing and retaining grout. Grout to be non-metallic, non-shrink, fluid precision grout of a hydraulic cementitious system with graded and processed silica aggregate, portland cement, shrinkage compensating agents, plasticizing and water reducing agents; free of aluminum powder agents, oxidizing agents and inorganic accelerators, including chlorides; proportioned, pre-mixed and packaged at factory with only the addition of water required at the project site. Grouting to meet requirements of ASTM C827/C827M. Perform all grouting in accord with equipment

manufacturer's and grout manufacturer's published specifications and recommendations.

3.2.3 Leveling and Aligning

Level and align equipment in accord with respective manufacturer's published data. Do not use anchor bolt, jack-nuts or wedges to support, level or align equipment. Install only flat shims for leveling equipment. Place shims to fully support equipment. Wedging is not permitted. Shims to be fabricated flat carbon steel units of surface configuration and area not less than equipment bearing surface. Shims to provide for full equipment support. Shim to have smooth surfaces and edges, free from burrs and slivers. Flame or electrode cut edges not acceptable.

3.2.4 Direct Drives

Alignment procedure follows.

3.2.4.1 Rotation Direction and Speed

Check and correct drive shaft rotation direction and speed.

3.2.4.2 End Play

Run drive shafts at operational speed. Determine whether axial end play exists. Run drive shaft at operational speed and mark drive shaft axial position when end play exists. Block drive shaft in operating position when aligning drive shaft with driven shaft.

3.2.4.3 Shaft Leveling and Radial Alignment

Check shaft leveling by placing a straightedge across the two coupling half faces in both horizontal and vertical planes.

3.2.4.4 Angular Alignment and End Clearance

Pump alignment must be accomplished by the factory technician or a millwright trained in pump alignment, and with the use of dial gauges or laser alignment equipment.

3.2.4.5 Final Recheck

Check adjustments with dial indicator after completing recheck. Align shafts within 0.05 tolerance, except as otherwise required by more stringent requirements of equipment manufacturer.

3.2.5 Start-up Representative

A manufacturer's field service representative must be provided at no additional cost to the Government to check the pumps for proper operation prior to start-up and also to witness, as a minimum, the first two days of operation. Any additional time required due to delays or corrections must be provided at no additional cost to the Government. The manufacturer's field service representative must also instruct the required personnel in the proper operation and maintenance of the pumps.

-- End of Section --

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 52 80

LIQUID FUELS PIPELINE COATING SYSTEMS

08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SYSTEM DESCRIPTION
- 1.3 DEFINITIONS
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Contract Errors, Omissions, and Other Discrepancies
 - 1.5.2 Corrective Action (CA)
 - 1.5.2.1 Corrective Action Procedures
 - 1.5.2.2 Corrective Action Request (CAR) Form
 - 1.5.2.3 Corrective Action Log
 - 1.5.3 Coatings Work Plan
 - 1.5.4 Design Data
 - 1.5.4.1 Containment System for Field Coating
 - 1.5.5 Test Reports
 - 1.5.5.1 Ferrous Metallic Abrasive Qualification Test Reports
 - 1.5.5.2 Non-Metallic Abrasive Qualification Test Reports
 - 1.5.5.3 Recycled Ferrous Metallic Abrasive Field Test Reports (Daily and Weekly)
 - 1.5.6 Qualifications
 - 1.5.6.1 Qualifications of Certified Industrial Hygienist (CIH)
 - 1.5.6.2 Qualifications of Pipe Coating Shop
 - 1.5.6.3 Qualifications of Certified Protective Coatings Specialist (PCS)
 - 1.5.6.4 Qualifications of Coatings Inspection Company
 - 1.5.6.5 Qualifications of Quality Assurance Coatings Inspector
 - 1.5.6.6 Qualifications of Coatings Contractors for Field Coating
 - 1.5.6.7 Qualifications of Individuals Performing Abrasive Blasting for Field Coating
 - 1.5.6.8 Qualifications of Individuals Performing Coating Application for Field Coating
 - 1.5.6.9 Qualifications of Testing Laboratory for Coatings
 - 1.5.6.10 Qualifications of Testing Laboratory for Abrasive
 - 1.5.6.11 Ferrous Metallic Abrasive Certificate of Conformance
 - 1.5.6.12 Non-Metallic Abrasive Certificate of Conformance
 - 1.5.7 QA and QC Personnel
 - 1.5.7.1 QC Manager
 - 1.5.7.2 Protective Coatings Specialist (PCS)
 - 1.5.7.3 Quality Assurance Coatings Inspector
 - 1.5.7.4 Coatings Inspector for Shop Coating
 - 1.5.7.5 Coatings Contractor QC Coatings Inspector
 - 1.5.8 Pre-Application Meeting for Field Coating
- 1.6 PRODUCT DATA
 - 1.6.1 Coating System Instructions
- 1.7 DELIVERY AND STORAGE

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 1.8 COATING HAZARDS
- 1.9 JOB SITE REFERENCES - SHOP
- 1.10 JOB SITE REFERENCES - FIELD

PART 2 PRODUCTS

- 2.1 SHOP-APPLIED COATING FOR BURIED PIPING
 - 2.1.1 External Pipe Coating
 - 2.1.2 Internal Pipe Coating (Lining)
- 2.2 FIELD-APPLIED EXTERNAL PIPE COATING
- 2.3 COATING FIELD SAMPLE COLLECTION AND SHIPPING KIT
- 2.4 ABRASIVE FIELD SAMPLE COLLECTION AND SHIPPING KIT
- 2.5 INSPECTION TEST KITS
 - 2.5.1 Test Kit for Measuring Chloride, Sulfate, and Nitrate Ions on Steel and Coated Surfaces
 - 2.5.2 Test Kit for Measuring Chlorides in Abrasives
 - 2.5.3 Test Kit for Identifying Amine Blush on Epoxy Surfaces
- 2.6 ABRASIVE
 - 2.6.1 Ferrous Abrasive
 - 2.6.1.1 New and Remanufactured Steel Grit
 - 2.6.1.2 Recycled Steel Grit
 - 2.6.2 Non-Metallic Abrasive

PART 3 EXECUTION

- 3.1 FIELD SAMPLE COLLECTION AND TESTING
 - 3.1.1 Coating Field Sample Collection
 - 3.1.2 Abrasive Field Sample Collection
 - 3.1.3 Coating Field Test Reports
 - 3.1.4 Abrasive Field Test Reports
- 3.2 LIGHTING
- 3.3 ENVIRONMENTAL CONDITIONS
 - 3.3.1 Automated Monitoring Requirements for Shop Applied Coating
 - 3.3.2 Containment for Field Coating
 - 3.3.3 Automated Monitoring Requirements for Field Coating
- 3.4 SHOP APPLIED COATING OF BURIED PIPING
 - 3.4.1 Surface Preparation for Shop Applied Coating
 - 3.4.1.1 Abrasive Blasting Equipment for Shop Applied Coating of Buried Piping
 - 3.4.1.2 Contamination Evaluation of Abrasive for Shop Applied Coating of Buried Piping
 - 3.4.1.3 Pre-Preparation Testing for Surface Contamination for Shop Applied Coating of Buried Piping
 - 3.4.1.3.1 Pre-Preparation Testing for Oil and Grease Contamination for Shop Applied Coating of Buried Piping
 - 3.4.1.3.2 Pre-Preparation Testing for Soluble Salts Contamination for Shop Applied Coating of Buried Piping
 - 3.4.1.4 Abrasive Blasting External Surfaces for Shop Applied Coating of Buried Piping
 - 3.4.1.5 Disposal of Used Abrasive for Shop Applied Coating of Buried Piping
 - 3.4.1.6 Pre-Application Testing for Surface Contamination for Shop Applied Coating of Buried Piping
 - 3.4.1.6.1 Pre-Application Testing for Oil and Grease Contamination for Shop Applied Coating of Buried Piping
 - 3.4.1.6.2 Pre-Application Testing for Surface Cleanliness for Shop Applied Coating of Buried Piping
 - 3.4.2 Mixing and Application of Shop Applied Coating of Buried Piping
 - 3.4.2.1 Mixing Liquid Epoxy Coatings

- 3.4.2.2 External Pipe Shop Applied Coating Application
- 3.4.2.3 Internal Pipe Coating Application
- 3.4.2.4 Holiday Testing and Shop Repair
- 3.5 FIELD EXTERIOR COATING OF ABOVEGROUND PIPING
- 3.6 FIELD REPAIRS TO EXTERNAL COATING OF BURIED PIPING
 - 3.6.1 Surface Preparation for Field Coating of Buried Piping
 - 3.6.1.1 Abrasive Blasting Equipment for Field Coating of Buried Piping
 - 3.6.1.2 Contamination Evaluation of Abrasive for Field Coating of Buried Piping
 - 3.6.1.3 Surface Standard
 - 3.6.1.4 Pre-Preparation Testing for Surface Contamination for Field Coating of Buried Piping
 - 3.6.1.4.1 Pre-Preparation Testing for Oil and Grease Contamination for Field Coating of Buried Piping
 - 3.6.1.4.2 Pre-Preparation Testing for Soluble Salts Contamination for Field Coating of Buried Piping
 - 3.6.1.5 Abrasive Blasting for Field Coating of Buried Piping
 - 3.6.1.6 Disposal of Used Abrasive for Field Coating of Buried Piping
 - 3.6.1.7 Pre-Application Testing for Surface Contamination for Field Coating of Buried Piping
 - 3.6.1.7.1 Pre-Application Testing for Oil and Grease Contamination for Field Coating of Buried Piping
 - 3.6.1.7.2 Pre-Application Testing for Soluble Salts Contamination for Field Coating of Buried Piping
 - 3.6.1.7.3 Pre-Application Testing for Surface Cleanliness for Field Coating of Buried Piping
 - 3.6.2 Coating Mixing and Application for Field Coating of Buried Piping
 - 3.6.2.1 Mixing Coating Materials for Field Coating of Buried Piping
 - 3.6.2.2 Application Conditions and Recoat Windows for Field Coating of Buried Piping
 - 3.6.2.3 Coating Application for Field Coating of Buried Piping
 - 3.6.2.4 Holiday Testing and Field Repair
 - 3.6.3 Final Inspection of Pipeline Prior to Burial
 - 3.6.4 Final Cleanup
- 3.7 FIELD AND SHOP QUALITY CONTROL
 - 3.7.1 Inspection
 - 3.7.1.1 Inspection Requirements
 - 3.7.1.2 Inspection Report Forms
 - 3.7.1.3 Daily Inspection Reports
 - 3.7.1.4 Inspection Logbook
 - 3.7.1.5 Inspection Equipment
 - 3.7.2 Coatings Contractor QC Coatings Inspector's Responsibilities
 - 3.7.3 Quality Assurance Coatings Inspector's Responsibilities
- 3.8 PROJECT IDENTIFICATION
 - 3.8.1 Stencils

-- End of Section Table of Contents --

SECTION 33 52 80

LIQUID FUELS PIPELINE COATING SYSTEMS
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

- AWWA C210 (2015) Standard for Liquid Epoxy Coating Systems for the Interior and Exterior of Steel Water Pipelines
- AWWA C213 (2022) Fusion-Bonded Epoxy Coatings and Linings for Steel Water Pipe and Fittings

ASTM INTERNATIONAL (ASTM)

- ASTM D3276 (2015; E 2016) Standard Guide for Painting Inspectors (Metal Substrates)
- ASTM D3335 (1985a; R 2020) Low Concentrations of Lead, Cadmium, and Cobalt in Paint by Atomic Absorption Spectroscopy
- ASTM D3718 (1985a; R 2015) Low Concentrations of Chromium in Paint by Atomic Absorption Spectroscopy
- ASTM D3925 (2002; R 2015) Sampling Liquid Paints and Related Pigmented Coatings
- ASTM D4285 (1983; R 2018) Indicating Oil or Water in Compressed Air
- ASTM D4417 (2021) Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

- ISO 8502-3 (2017) Preparation of Steel Substrates Before Application of Paints and Related Products - Tests for the Assessment of Surface Cleanliness - Part 3: Assessment of Dust on Steel Surfaces Prepared for Painting (Pressure-Sensitive Tape Method)
- ISO 9001 (2015) Quality Management Systems- Requirements

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

NACE INTERNATIONAL (NACE)

NACE RP0402 (2002) Field-Applied Fusion-Bonded Epoxy (FBE) Pipe Coating Systems for Girth Weld Joints: Application, Performance, and Quality Control

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC AB 1 (2015; E 2017) Mineral and Slag Abrasives

SSPC AB 2 (2015; E 2016) Cleanliness of Recycled Ferrous Metallic Abrasive

SSPC AB 3 (2003; E 2004) Ferrous Metallic Abrasive

SSPC Guide 6 (2021) Guide for Containing Surface Preparation Debris Generated During Paint Removal Operations

SSPC Guide 12 (2023) Guide for Illumination of Industrial Painting Projects

SSPC PA 1 (2016) Shop, Field, and Maintenance Coating of Metals

SSPC PA 2 (2015; E 2018) Procedure for Determining Conformance to Dry Coating Thickness Requirements

SSPC QP 1 (2019) Standard Procedure for Evaluating the Qualifications of Industrial/Marine Painting Contractors (Field Application to Complex Industrial Steel Structures and Other Metal Components)

SSPC QP 3 (2010) Standard Procedure for Evaluating Qualifications of Shop Painting Applicators

SSPC QP 5 (2022) Standard Procedure for Evaluating the Qualifications of Coating and Lining Inspection Companies

SSPC QS 1 (2015) Standard Procedure for Evaluating a Contractor's Advanced Quality Management System

SSPC SP 1 (2015) Solvent Cleaning

SSPC SP 10/NACE No. 2 (2015) Near-White Blast Cleaning

SSPC VIS 1 (2002; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.134 Respiratory Protection

29 CFR 1910.1000

Air Contaminants

29 CFR 1926.59

Hazard Communication

1.2 SYSTEM DESCRIPTION

This section specifies the requirements for exterior coating of aboveground and buried carbon steel liquid fuel pipelines and exterior coating of aboveground and buried stainless steel liquid fuel pipelines. The exterior coating system for aboveground, carbon steel pipelines is specified with the same requirements as the exterior coating system for the exterior of fuel tanks, Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES. The exterior coating system for buried pipelines is an extruded polyolefin system, a fusion bonded epoxy (FBE) system, or a liquid epoxy system.

1.3 DEFINITIONS

Definitions are generally provided throughout this Section in the paragraphs where used and denoted by capital letters. The following definition is used throughout this Section:

- a. INDEPENDENT THIRD-PARTY - Impartial third-party not a part or affiliated with Contractor or subcontractor principal or subsidiary businesses, and not a materials supplier.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contract Errors, Omissions, and Other Discrepancies

Corrective Action Procedures

Corrective Action Request (CAR) Form

Coatings Work Plan

Inspection Report Forms

SD-05, Design Data

Containment System

SD-06 Test Reports

Daily Inspection Reports

Ferrous Metallic Abrasive Qualification Test Reports

Non-Metallic Abrasive Qualification Test Reports

Coating Field Test Reports

Abrasive Field Test Reports

Recycled Ferrous Metallic Abrasive Field Test Reports (Daily and Weekly)

SD-07 Certificates

Qualifications of Certified Industrial Hygienist (CIH)

Qualifications of Pipe Coating Shop

Qualifications of Certified Protective Coatings Specialist (PCS)

Qualifications of Coatings Inspection Company

Qualifications of Quality Assurance Coatings Inspector

Qualifications of Coatings Contractors for Field Coating

Qualifications of Individuals Performing Abrasive Blasting for Field Coating

Qualifications of Individuals Performing Coating Application for Field Coating

Qualifications of Testing Laboratory for Coatings

Qualifications of Testing Laboratory for Abrasive

Ferrous Metallic Abrasive Certificate of Conformance

Non-Metallic Abrasive Certificate of Conformance

SD-08 Manufacturer's Instructions

Coating System Instructions

SD-11 Closeout Submittals

Disposal of Used Abrasive Shop Applied Coating

Disposal of Used Abrasive Field Applied Coating

Inspection Logbook

Corrective Action Log; G

1.5 QUALITY ASSURANCE

1.5.1 Contract Errors, Omissions, and Other Discrepancies

Submit all errors, omissions, and other discrepancies in contract documents to the Contracting Officer within 30 days of contract award for all work covered in this Section, other than the work that will not be uncovered until a later date. All such discrepancies must be addressed and resolved, and the Coatings Work Plan modified, prior to beginning the Initial and Follow-Up phases of work. Discrepancies that become apparent only after work is uncovered must be identified at the earliest discoverable time and submitted for resolution. Schedule time (float) must be built into the project schedule at those points where old work is

to be uncovered or where access is not available during the first 30 days after award, to allow for resolution of contract discrepancies.

1.5.2 Corrective Action (CA)

CA must be included in the Contractor Quality Control Plan as outlined in Section 01 45 00 QUALITY CONTROL.

1.5.2.1 Corrective Action Procedures

Develop procedures for determining the root cause of each non-compliance, developing a plan to eliminate the root cause so that the non-compliance does not recur, and following up to ensure that the root cause was eliminated.

1.5.2.2 Corrective Action Request (CAR) Form

Develop Corrective Action Request (CAR) forms for initiating CA and for tracking and documenting each step. The CAR should be included with the Corrective Action Procedures. A CAR must be initiated by either the Contractor or the Contracting Officer. The Protective Coatings Specialist (PCS) must approve each CAR at the root cause identification stage, the plan for elimination stage, and the close out stage after verification that the root cause has been eliminated.

1.5.2.3 Corrective Action Log

When a CAR is initiated, Contractor must take action to identify and eliminate the root cause of each non-compliance so as to prevent recurrence. These actions must apply to non-compliance in the work, and to non-compliance in the Quality Control (QC) System. Corrective actions must be appropriate to the effects of the non-compliance encountered. The corrective action must be documented in a report that is serialized and tracked in the [Corrective Action Log](#) until project completion and its acceptance by the Contracting Officer. All corrective action reports must be retained in project records. The Corrective Action Log, showing status of each CAR, must be submitted to the Contracting Officer monthly.

1.5.3 Coatings Work Plan

This work plan must be considered as part of the Contractor Quality Control Plan as outlined in Section 01 45 00 QUALITY CONTROL.

The [Coatings Work Plan](#) must be submitted and approved by the PCS prior to mobilization. The Coatings Work Plan must explain in detail all procedures including, but not limited to, all sequential processes, quality control for each process, quality assurance for each process, and safety considerations. Subsections must include at least the following:

- a. Purpose;
- b. Introduction;
- c. Safety, fire, and health information;
- d. Contractor and worker qualifications with certifications;
- e. Project management organization and documents;

- f. Timeline in a Gantt chart;
- g. Project document references;
- h. Reference to all applicable standards (e.g., AMPP, NACE, SSPC, ISO, and ASTM);
- i. Coatings manufacturer's supporting documentation;
- j. Descriptions and explanations of any exceptions from the coating manufacturer;
- k. Coating and blasting equipment, model names, and, if applicable, calibration dates;
- l. Containment design and/or details;
- m. Environmental testing;
- n. Material delivery, storage, and handling details;
- o. Surface preparation;
- p. Pre-application test panel validation for field-applied external coating as outlined in SURFACE STANDARD;
- q. Coating materials, mixing, application, recoat windows, and coating curing times, if applicable;
- r. Coating repairs and rework;
- s. Non-conformance;
- t. Spent material handling and effluent discharge containment and disposal;
- u. Inspection test plan (as outlined in FIELD INSPECTION, and including inspection hold points, both Quality Assurance and Coating Contractor QC Coatings Inspector's responsibilities, and daily documentation and delivery);
- v. Instruments and test kits;
- w. Warranty (in writing, signed by the Contractor and the coating manufacturer's representative);
- x. Demobilization;
- y. PCS and PM approval;

1.5.4 Design Data

1.5.4.1 [Containment System](#) for Field Coating

Submit complete design drawings and calculations for the scaffolding and containment system, including an analysis of the loads which will be added to the structure by the containment system and waste materials. A registered engineer must approve calculations and scaffold system design.

1.5.5 Test Reports

1.5.5.1 Ferrous Metallic Abrasive Qualification Test Reports

Submit qualification testing results for abrasive as required in paragraph 4 REQUIREMENTS of **SSPC AB 3**. Submit test results from an INDEPENDENT THIRD-PARTY laboratory of representative samples of each abrasive to be used on the jobsite. Samples must have been tested within the last three years. Note that this testing is for the purpose of pre-qualifying the abrasive.

1.5.5.2 Non-Metallic Abrasive Qualification Test Reports

Submit results for abrasive as required in paragraph 4 REQUIREMENTS of **SSPC AB 1**. Submit test results from an INDEPENDENT THIRD-PARTY laboratory of representative samples of each abrasive to be used on the jobsite. Samples must have been tested within the last three years. Note that this testing is for the purpose of pre-qualifying the abrasive.

1.5.5.3 Recycled Ferrous Metallic Abrasive Field Test Reports (Daily and Weekly)

Submit test results from an INDEPENDENT THIRD-PARTY laboratory of daily and weekly Quality Control testing required by **SSPC AB 2**, as modified in paragraph ABRASIVE.

1.5.6 Qualifications

The qualifications specified in this Section must be met throughout the duration of this contract. Personnel and corporate entities must provide proof of active certification to all qualifications specified in this Section. Work must not be provided by personnel and corporate entities unless all specified qualifications are met.

1.5.6.1 Qualifications of Certified Industrial Hygienist (CIH)

Submit name, address, telephone number, fax number, and e-mail address of the INDEPENDENT THIRD-PARTY CIH. Submit documentation that the hygienist is certified by the American Board of Industrial Hygiene in comprehensive practice, including certification number and date of certification/recertification. The CIH must remain certified during the entire project, and the Contracting Officer must be notified of any change in certification status within 10 days of the change. If a CIH's certification expires, the hygienist will not be allowed to perform any hygienist functions, and all hygienist work must stop, until the certification is reissued or another CIH is approved. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Provide evidence of experience with hazards involved in industrial coating application work.

1.5.6.2 Qualifications of Pipe Coating Shop

Each shop that applies coatings to pipe must be certified to **SSPC QP 3** prior to contract award and must remain certified while accomplishing any surface preparation or coating application. If a shop's certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not

be considered and liquidated damages will apply. Notify the Contracting Officer of any change in the shop's certification status within 10 days of the change. If a shop's qualification expires, the shop will not be allowed to perform any coatings related functions until the qualification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive qualification will not be considered and liquidated damages will apply. Notify the Contracting Officer of all scheduled and unannounced on-site audits from AMPP and furnish a copy of all audit reports.

For shops located outside the United States, Guam, and Puerto Rico, the certifications for the shop (**SSPC QP 3**) can be substituted if the coatings Contractor meets all of the below requirements:

- a. **ISO 9001** certified;
- b. Eight years of experience with industrial coatings;
- c. Evidence of recent work that has Contractor Performance Assessment Report System (CPARS) ratings, and/or other quality/performance ratings, that are equivalent to, or exceed, "Above Average";
- d. Evidence of an INDEPENDENT THIRD-PARTY audit from AMPP demonstrating equivalency to **SSPC QP 3** within the last two years.

The shop must be certified to **ISO 9001** prior to contract award and must remain so certified for the duration of the project. Notify the Contracting Officer of any change in the shop's certification status within 10 days of the change. If a Contractor's qualification expires, the shop will not be allowed to perform any coatings related functions until the qualification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive qualification will not be considered and liquidated damages will apply. Notify the Contracting Officer of all scheduled and unannounced on-site inspections from the ISO certifying organization and furnish a copy of all inspection reports.

1.5.6.3 Qualifications of Certified Protective Coatings Specialist (PCS)

Submit name, address, telephone number, fax number, and e-mail address of the INDEPENDENT THIRD-PARTY PCS. Submit documentation that the PCS is certified by the Association for Materials Protection and Performance (AMPP) (formerly SSPC: The Society for Protective Coatings (SSPC)), including certification number and date of certification/recertification. If the PCS is employed by the same coatings inspection company to which the Quality Assurance Coatings Inspector is employed, this does not violate the INDEPENDENT THIRD-PARTY requirements. The PCS must be an INDEPENDENT THIRD-PARTY to the pipe coating shop. The PCS must remain certified during the entire project, and the Contracting Officer must be notified of any change in certification status within 10 days of the change. If a PCS's certification expires, the PCS will not be allowed to perform any PCS functions, and all coatings work must stop, until the certification is reissued or another PCS is approved. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. The PCS must not be the designated Quality Assurance Coatings Inspector. The PCS's responsibilities are outlined in PROTECTIVE COATINGS SPECIALIST (PCS).

1.5.6.4 Qualifications of Coatings Inspection Company

Submit documentation that the selected quality assurance coatings inspection company is accredited by AMPP to the requirements of **SSPC QP 5** prior to contract award. The coatings inspection company that is submitted and approved, must remain and cannot be changed through completion of the contract. The coatings inspection company must remain **SSPC QP 5** certified for the duration of the coatings work. If a coatings inspection company's certification expires, the firm will not be allowed to perform any inspection functions, and all surface preparation and coating application work must stop, until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in the coatings inspection company certification status within 10 days of the change. Notify the Contracting Officer of all scheduled and unannounced on-site audits from AMPP and furnish a copy of all audit reports. The coatings inspection company must not engage in any activities that may conflict with their independence of judgment and integrity in relation to their inspection activities. In particular, they must not be engaged in the manufacture, supply, application, surface preparation, purchase, or maintenance of the applied coating in this project.

1.5.6.5 Qualifications of Quality Assurance Coatings Inspector

Submit documentation that each Quality Assurance Coatings Inspector is employed by the **SSPC QP 5** company and is qualified to a minimum certification of NACE CIP Level II. Each inspector must remain employed by the coatings inspection company while performing any coatings inspection functions. The Quality Assurance Coatings Inspector's responsibilities are outlined in QUALITY ASSURANCE COATINGS INSPECTOR'S FIELD RESPONSIBILITIES. The roles of the Quality Assurance Coatings Inspector are in addition to, and distinct from, the role of the QC Coatings Inspector employed by the coatings Contractor and the Coatings Inspector for Shop Coating.

1.5.6.6 Qualifications of Coatings Contractors for Field Coating

All Contractors and Subcontractors that perform surface preparation or coatings application must be certified to **SSPC QP 1** and **SSPC QS 1** prior to contract award and must remain certified while accomplishing any surface preparation or coating application. If a Contractor's or Subcontractor's certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in Contractor certification status within 10 days of the change. If a Contractor's qualification expires, the Contractor will not be allowed to perform any coatings related functions until the qualification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive qualification will not be considered and liquidated damages will apply. Notify the Contracting Officer of all scheduled and unannounced on-site audits from AMPP and furnish a copy of all audit reports.

For projects located outside the United States, Guam, and Puerto Rico, the certifications for the coatings Contractor (**SSPC QP 1** and **SSPC QS 1**) can be substituted if the coatings Contractor meets all of the below

requirements:

- a. **ISO 9001** certified;
- b. Eight years of experience with industrial coatings;
- c. Evidence of recent work that has Contractor Performance Assessment Report System (CPARS) ratings, and/or other quality/performance ratings, that are equivalent to, or exceed, "Above Average";
- d. Evidence of an INDEPENDENT THIRD-PARTY audit from AMPP demonstrating equivalency to **SSPC QP 1** and **SSPC QS 1** within the last two years.

The coatings Contractors and coatings Subcontractors must be certified to **ISO 9001** prior to contract award and must remain so certified for the duration of the project. Notify the Contracting Officer of any change in Contractor certification status within 10 days of the change. If a Contractor's qualification expires, the Contractor will not be allowed to perform any coatings related functions until the qualification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive qualification will not be considered and liquidated damages will apply. Notify the Contracting Officer of all scheduled and unannounced on-site inspections from the ISO certifying organization and furnish a copy of all inspection reports.

1.5.6.7 Qualifications of Individuals Performing Abrasive Blasting for Field Coating

Submit name, address, and telephone number of each person that will be performing abrasive blasting. All individuals performing abrasive blasting must be certified by AMPP to the SSPC C7 Dry Abrasive Blaster Qualification Program or CAS Coating Application Specialist Level 2 Certification Program (Interim Status), and must remain certified during the entire period of abrasive blasting. The Contracting Officer must be notified of any change in qualification status within 10 days of the change. If a blaster's qualification expires, the blaster will not be allowed to perform any blasting functions until the qualification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive qualification will not be considered and liquidated damages will apply.

This requirement applies to all manual abrasive blasting performed in shop and field locations. This requirement does not apply to automated abrasive blasting performed in the shop.

1.5.6.8 Qualifications of Individuals Performing Coating Application for Field Coating

Submit name, address, telephone number, and evidence of certification of each person that will be performing coatings application by any method. All individuals performing coatings application must be certified by AMPP to the SSPC CAS Coating Application Specialist Level 2 Certification Program (Interim Status) or SSPC C12 Spray Application Program. Each applicator must remain certified during the entire period of coatings application, and the Contracting Officer must be notified of any change in qualification status within 10 days of the change. If an applicator's qualification expires, the applicator will not be allowed to perform any coatings application functions until the qualification is reissued. Requests for extension of time for any delay to the completion of the

project due to an inactive qualification will not be considered and liquidated damages will apply.

1.5.6.9 Qualifications of Testing Laboratory for Coatings

Submit name, address, telephone number, fax number, and e-mail address of the INDEPENDENT THIRD-PARTY laboratory or laboratories selected to perform testing of coating samples for qualification testing and for field sample testing for compliance with this Section. Submit documentation that the laboratory is regularly engaged in testing of paint samples for conformance with specifications and that the employees performing testing are qualified.

1.5.6.10 Qualifications of Testing Laboratory for Abrasive

Submit name, address, telephone number, fax number, and e-mail address of the INDEPENDENT THIRD-PARTY laboratory or laboratories selected to perform testing of abrasive for compliance with this section. Submit documentation that the laboratory has experience in testing samples of abrasive for conformance with specifications and that the employees performing testing are qualified.

1.5.6.11 Ferrous Metallic Abrasive Certificate of Conformance

Provide manufacturer's certification of conformance that the materials are currently in conformance with SSPC AB 3 and as modified in this Section, and have been tested within the last three years.

1.5.6.12 Non-Metallic Abrasive Certificate of Conformance

Provide manufacturer's certification that the materials are currently in conformance with SSPC AB 1 and as modified in this Section, and have been tested within the last three years.

1.5.7 QA and QC Personnel

1.5.7.1 QC Manager

The QC Manager is as defined in Section 01 45 00 QUALITY CONTROL.

1.5.7.2 Protective Coatings Specialist (PCS)

The PCS must be considered a QC Specialist and must report to the QC Manager, as specified in Section 01 45 00 QUALITY CONTROL. The PCS must approve all submittals prior to submission to the QC Manager for approval or submission to the government for approval.

The PCS's responsibilities include, but are not limited to, the following:

- a. Obtain, review, and understand all project documentation including, but not limited to, this Section, project program, Coatings Work Plan, inspection and testing plan (ITP), and all submittals before the project starts, during the project, and all coatings related re-work;
- b. Attend all pre-job coatings meetings (in-person, phone, or virtually);
- c. Attend pre-final coatings walk-through (mandatory) and attend final coatings walk-through (as required).

1.5.7.3 Quality Assurance Coatings Inspector

The Quality Assurance Coatings Inspector must be considered a QC Specialist and must report to the QC Manager, as specified in Section 01 45 00 QUALITY CONTROL. The Quality Assurance Coatings Inspector must be present during all pre-preparation testing, surface preparation, coating application, initial cure of the coating system, during all coating repair work, and during completion activities. The Quality Assurance Coatings Inspector must provide complete documentation of conditions and occurrences on the job site, and be aware of conditions and occurrences that are potentially detrimental to the coating system. The responsibilities of the Quality Assurance Coatings Inspector are defined in QUALITY ASSURANCE COATING INSPECTOR'S RESPONSIBILITIES. These responsibilities are separate and distinct from the responsibilities of the Coatings Contractor QC Coatings Inspector and Coatings Inspector for Shop Coating.

1.5.7.4 Coatings Inspector for Shop Coating

The coatings inspector must be the shop Quality Manager or appropriate designee. The coatings inspector must be considered a QC Specialist and must report to the prime Contractor's QC Manager, as specified in Section 01 45 00 QUALITY CONTROL. The coatings inspector must be present during all pre-preparation testing, surface preparation, coating application, initial cure of the coating system, and during all coating repair work required of the shop. The coatings inspector must provide complete documentation of conditions and occurrences on the job site, and be aware of conditions and occurrences that are potentially detrimental to the coating system.

1.5.7.5 Coatings Contractor QC Coatings Inspector

The Coatings Contractor QC Coatings Inspector must stop non-compliant work. The responsibilities of the Coatings Contractor QC Coatings Inspector are defined in COATINGS CONTRACTOR QC COATINGS INSPECTOR'S RESPONSIBILITIES. These responsibilities are separate and distinct from the responsibilities of the Quality Assurance Coatings Inspector.

1.5.8 Pre-Application Meeting for Field Coating

After approval of submittals, but prior to the initiation of coatings work, Contractor representatives, including at a minimum, project superintendent, QC manager, paint foreman, Quality Assurance Coatings Inspector, and PCS must have a pre-application coating preparatory meeting. This meeting must be in addition to the pre-construction conference. Specific items addressed must include: corrective action requirements and procedures, coatings work plan, safety plan, coordination with other Sections, inspection standards, inspection requirements and tools, test procedures, environmental control system, safety plan, and test logs. Notify Contracting Officer at least ten days prior to meeting.

1.6 PRODUCT DATA

1.6.1 Coating System Instructions

Submit manufacturer's printed instructions, including detailed mixing and application procedures, number and types of coats required, minimum and maximum application temperatures, and curing procedures of all coating materials. Include Safety Data Sheets (SDS) for materials to be used at

the job site in accordance with 29 CFR 1926.59.

1.7 DELIVERY AND STORAGE

Ship, store, and handle materials in accordance with SSPC PA 1, applicable standards, and as modified in this Section. Maintain temperature in storage spaces between 5 and 24 degrees C, and air temperature more than 3 degrees C above the dew-point at all times. Inspect materials for damage and return non-compliant materials to manufacturer. Remove materials with expired shelf life from government property immediately and notify the Contracting Officer.

If materials are approaching shelf life expiration and an extension is desired, samples must be sent to the manufacturer, along with complete records of storage conditions, with a request for shelf life extension. If the manufacturer finds the samples and storage data suitable for shelf life extension, the manufacturer must issue an extension, referencing the product evaluation and the review of storage records. Products must not be extended longer than allowed in the product specification.

1.8 COATING HAZARDS

Ensure that employees are trained in all aspects of the safety plan. Specified coatings may have potential health hazards if ingested or improperly handled. The coating manufacturer's written safety precautions must be followed throughout mixing, application, and curing of the coatings. During cleaning, cleanup, surface preparation, and paint application phases, ensure that employees are protected from toxic and hazardous chemical agents which exceed concentrations in 29 CFR 1910.1000. Comply with respiratory protection requirements in 29 CFR 1910.134. The CIH must approve work procedures and personal protective equipment.

1.9 JOB SITE REFERENCES - SHOP

Make available to the Contracting Officer a copy of each standard to which the shop will be applying coatings under this Section.

1.10 JOB SITE REFERENCES - FIELD

Make available to the Contracting Officer a copy of each standard to which field coatings will be applied to under this Section, and an SSPC Certified Contractor Evaluation Form at the job site.

PART 2 PRODUCTS

2.1 SHOP-APPLIED COATING FOR BURIED PIPING

2.1.1 External Pipe Coating

Fusion Bonded Epoxy Coating: AWWA C213, minimum 375 microns, maximum 500 microns.

2.1.2 Internal Pipe Coating (Lining)

For JP-5 pipeline system, HDD piping segments - The lowest 100 meter of each HDD pipe run must be internally factory coated with an epoxy coating in accordance with MIL-PRF-4556. The ends of the pipe must be masked or wiped back a minimum of 25 mm but not more than 40 mm.

2.2 FIELD-APPLIED EXTERNAL PIPE COATING

All coating materials used to coat external girth welds or repair existing coating must be from the same manufacture as the shop applied coating. Use one or more of the following repair methods, as modified herein, to repair shop applied coatings and coat external girth welds:

- a. Fusion-Bonded Epoxy (FBE) Coating: NACE RP0402, coating material to be same as applied to pipe in shop
- b. Liquid-Epoxy Coating System: AWWA C210

2.3 COATING FIELD SAMPLE COLLECTION AND SHIPPING KIT

Provide a kit for each sample to be collected. For FBE coatings, each kit must contain: a one liter can for the dry/powder; dipping cups for sampling; a shipping box sized for the samples to be shipped; and packing material. For liquid coatings, each kit must contain: a one liter can for the base of each liquid coatings material; one appropriately sized can for the activator of each liquid coatings material; dipping cups for each component to be sampled; a shipping box sized for the samples to be shipped; and packing material. Mark cans for the appropriate components (powder, base, or activator), including manufacturer's name, address, batch numbers, batch size shipped to the project site, and date of manufacture. Provide shipping documents, including either pre-paid shipping or a shipping number that can be used by the QC Manager to arrange pickup, addressed to the INDEPENDENT THIRD-PARTY coating testing laboratory.

2.4 ABRASIVE FIELD SAMPLE COLLECTION AND SHIPPING KIT

Provide a kit for each sample to be collected. Each kit must contain: one suitable plastic bag or container for each sample to be collected. Mark containers with manufacturer's name, address, batch number, batch size, and date of manufacture. Provide shipping documents, including either pre-paid shipping label or a shipping number that can be used by the QC Manager to arrange pickup, addressed to the approved coating testing laboratory.

2.5 INSPECTION TEST KITS

2.5.1 Test Kit for Measuring Chloride, Sulfate, and Nitrate Ions on Steel and Coated Surfaces

Provide test kits that meet the following requirements:

- a. Contains all materials, supplies, tools, and instructions for field testing and on-site quantitative evaluation of chloride, sulfate, and nitrate ions;
- b. Extract solution is acidic, factory pre-measured, pre-packaged, and of uniform concentration;
- c. Components and solutions are mercury free and environmentally friendly;
- d. Contains new materials and solutions for each test extraction;
- e. Contains an extraction test container (vessel, sleeve, cell) creates a sealed, encapsulated environment during salt ion extraction;

- f. Contains a test extract container is suitable for testing the following steel surfaces: horizontal (up/down configuration), vertical, flat, curved, smooth, pitted, and rough;
- g. All salt ion concentrations are directly measured in micrograms per square centimeter.

2.5.2 Test Kit for Measuring Chlorides in Abrasives

Provide test kits that meet all of the following requirements:

- a. Be a completely self-contained test kit with all materials, supplies, tools, and instructions to take tests and identify results;
- b. Use identifiable, consistent, factory pre-measured test extract solution;
- c. Provide for testing equal volumes of abrasive and test solution;
- d. Provide for taking direct measurements of the chloride ion in parts per million (PPM), without using conversion charts or tables;
- e. Provide all new components for extraction and titration for each test;
- f. Provide a factory sealed titration device for each test;
- g. Use the extract sampling container as the titration container.

2.5.3 Test Kit for Identifying Amine Blush on Epoxy Surfaces

Provide test kits that meet all of the following requirements:

- a. Be a completely self-contained field test kit with all materials, supplies, tools, and instructions to perform tests and indicate the presence of unreacted amines;
- b. Use an identifiable, consistent, uniform, pre-packaged, factory pre-measured indicating solution;
- c. Contains no mercury or lead and is environmentally friendly;
- d. Contains a solution of an unreacted amine for the purpose of "self checking" the indicator solution.

2.6 ABRASIVE

Use abrasive that is specifically selected to provide a sharp, angular profile to the specified depth. Abrasive must meet all requirements of this Section each time that it is placed in the blast pot. The referenced abrasive specifications have maximum limits for soluble salts contamination; however, this maximum level of contamination does not guarantee that contamination will not be transferred to the steel surface during abrasive blasting. Other factors, such as on-site handling and recycling, can allow contamination of abrasive that can be transferred to the steel surface. Contractors are cautioned to verify that the chosen abrasive, along with work and storage processes, allow the final surface cleanliness requirements to be achieved. Successful testing of contamination in abrasive does not negate the final acceptance testing of

steel surfaces.

Non-ferrous abrasives must be used for abrasive blasting stainless steel piping.

2.6.1 Ferrous Abrasive

2.6.1.1 New and Remanufactured Steel Grit

New and remanufactured steel grit abrasive must conform to the chemical and physical properties of [SSPC AB 3](#) Class 1 (Steel) only; Class 2 (Iron) abrasive must not be used. Modify the requirements of [SSPC AB 3](#) to substitute the requirement in paragraph 4.2.2 CONDUCTIVITY for one chloride test as measured using the test kit described in this Section (paragraph TEST KIT FOR MEASURING CHLORIDES IN ABRASIVES). The maximum allowable chloride content is 25 parts per million (PPM).

To develop a suitable work mix from new steel abrasive, a minimum of 200 to 400 recycles is required; therefore, it may be advantageous for a Contractor to use remanufactured steel grit or grit reclaimed from a previous project. Such grit must be traced to new grit conforming to [SSPC AB 3](#) Class 1 and it meets all cleanliness requirements of [SSPC AB 3](#) Class 1 when brought to the current jobsite. Submit one representative sample of this work mix to the INDEPENDENT THIRD-PARTY laboratory for testing, along with samples of new material. Acceptance and use of this work mix must not be used to justify any deviation from surface preparation requirements.

2.6.1.2 Recycled Steel Grit

Recycled steel grit abrasive media must conform to the chemical and physical properties of [SSPC AB 2](#) except that:

- a. The maximum allowable chromium and cadmium content of the work mix must be 0.1 percent by weight when tested in accordance with [ASTM D3718](#) for chromium and [ASTM D3335](#) for cadmium. Modify the requirements of [SSPC AB 2](#) to add requirement for one chromate test and one cadmium test for each "LEAD" test required.
- b. The maximum allowable chloride content is 25 parts per million (PPM) as measured with the test kit described in paragraph TEST KIT FOR MEASURING CHLORIDES IN ABRASIVES. Modify the requirements of [SSPC AB 2](#) to substitute requirement for one chloride test for each "WATER SOLUBLE CONTAMINANTS" test.

2.6.2 Non-Metallic Abrasive

Non-metallic abrasive must be graded to the appropriate surface profile range and must conform to the chemical and physical properties of [SSPC AB 1](#), Class A except that:

- a. The maximum allowable chromium and cadmium content of the work mix must be less than 0.1 percent by weight when tested in accordance with [ASTM D3718](#) for chromium and [ASTM D3335](#) for cadmium.
- b. Must contain less than 7 PPM chlorides when tested with the kit provided in paragraph TEST KIT FOR MEASURING CHLORIDES IN ABRASIVES. Modify the requirements of [SSPC AB 1](#) to substitute requirement for one chloride test for each "CONDUCTIVITY TEST" required in [SSPC AB 1](#) (one

random sample per 50 bags of abrasive or three random samples from each shipment, if abrasive is delivered in bulk).

PART 3 EXECUTION

Perform all work, rework, and repair in accordance with approved procedures in the Coatings Work Plan. The Coatings Work Plan must be submitted and approved by the PCS prior to mobilization in accordance with the paragraph entitled COATINGS WORK PLAN.

3.1 FIELD SAMPLE COLLECTION AND TESTING

Sample and test materials delivered to the jobsite as required in the subsequent subparagraphs. Notify Contracting Officer three days in advance of sampling. The QC Manager, and either the PCS or the Quality Assurance Coatings Inspector, must witness all sampling.

3.1.1 Coating Field Sample Collection

All coatings/resins applied in the shop and in the field require one sample to be collected from each coating type used. This sample must be collected and set aside for the duration of the project, and must be tested if unforeseen coatings issues arise or if testing is requested by the Contracting Officer.

Field applied coatings require a random field sample from each lot of coating material used on-site in accordance with [ASTM D3925](#). Each random sample must be tested. Utilize sample collection kits as outlined in the paragraph entitled COATING FIELD SAMPLE COLLECTION AND SHIPPING KIT. For powder based coatings, collect a one [liter](#) sample from each lot by random selection. For two component liquid coatings, obtain a one [liter](#) sample of each base material, and proportional samples of each activator based on mix ratio, by random selection from sealed containers from each lot. Prior to sampling, mix contents of each sealed container to ensure uniformity. As an alternative to collecting small samples from kits, entire kits may be randomly selected and shipped to the THIRD-PARTY INDEPENDENT laboratory, observing all requirements for witnessing and traceability. For purposes of quality conformance inspection, a lot is defined as that quantity of materials from a single, uniform batch produced and offered for delivery at one time. A batch is defined as that quantity of material processed by the manufacturer at one time and identified by number on the label. Identify samples by designated name, specification number, batch number, project contract number, sample date, intended use, and quantity involved. The QC manager will take possession of the packaged samples, contact the shipping company to arrange for pickup, and ship one complete sample of each material in question (including powder or base and activator) with all batch information to the INDEPENDENT THIRD-PARTY laboratory for testing as required in paragraph COATING FIELD TESTING REPORTS.

3.1.2 Abrasive Field Sample Collection

Utilize the sample collection kits as required in paragraph ABRASIVE FIELD SAMPLE COLLECTION AND SHIPPING KIT to obtain samples from each lot of abrasive delivered to site using the sampling techniques and schedule of one sample per every 50 bags for ferrous metallic abrasive, paragraph 4 REQUIREMENTS FOR RECYCLED WORK MIX ABRASIVES of [SSPC AB 2](#) for recycled ferrous metallic abrasives, or paragraph 5.3 SAMPLING FOR QUALITY CONTROL TESTS of [SSPC AB 1](#) for non-metallic abrasives.

For purposes of quality conformance inspection, a lot must consist of all abrasive materials of the same type from a single, uniform batch produced and offered for delivery at one time. The addition of any substance to a batch must constitute a new lot. Identify samples by designated name, specification number, lot number, project contract number, sample date, intended use, and quantity involved. The QC manager will take possession of the packaged samples, contact the shipping company to arrange for pickup, and relinquish the samples only to the shipping representative for shipment to the approved INDEPENDENT THIRD-PARTY laboratory for testing required in paragraph ABRASIVE SAMPLE TEST REPORTS.

3.1.3 Coating Field Test Reports

Submit test results for each sample that requires testing in the paragraph COATINGS FIELD SAMPLE COLLECTION. Test samples of coating material for compliance with requirements of AWWA C210. Reject entire batch represented by samples that fail one or more tests, select new lots, and test samples.

3.1.4 Abrasive Field Test Reports

Submit test results for each lot of abrasive delivered to the jobsite. Test samples of ferrous metallic abrasive to the requirements of paragraph 5.2 TEST PARAMETERS of SSPC AB 3, excluding paragraph 5.2.4 DURABILITY. Test samples of recycled ferrous metallic abrasives to the requirements of paragraph 4 REQUIREMENTS FOR RECYCLED WORK MIX ABRASIVES of SSPC AB 2. Test samples of non-metallic abrasive to the requirements of paragraph 5.3 SAMPLING FOR QUALITY CONTROL TESTS of SSPC AB 1. Reject entire lot represented by samples that fail one or more tests, select new lots, and test samples.

3.2 LIGHTING

Provide lighting for all work areas as prescribed in SSPC Guide 12.

3.3 ENVIRONMENTAL CONDITIONS

3.3.1 Automated Monitoring Requirements for Shop Applied Coating

Provide continuous monitoring of temperature, relative humidity, and dew point data at pertinent points in the shop, during surface preparation, coating application, and initial cure. This data does not suffice for documentation of conformity to surface conditions during application and cure of coating. Locate sensors to provide pertinent data for the surface preparation and coating application being performed, as well as the temperature extremes on the structure. Describe the location plan in the Coatings Work Plan. Monitor any heating, cooling, or dehumidification equipment used. Provide monitoring equipment to perform as follows:

- a. Data is collected in 15-minute increments, and available for download (on-site) in a standard database format. Contractor must collect these data and make available to the Contracting Officer and QC Manager;
- b. Monitoring equipment must have backup power such that data collection will be uninterrupted during the entire period of the dehumidification requirement;
- c. Monitoring equipment must have capability to measure interior and

exterior dry bulb temperature (DB), relative humidity (RH), and dewpoint temperature (DP).

There is no requirement for connectivity of the monitoring system to control the dehumidification equipment; therefore, any combination of equipment having the required functionality will be accepted.

3.3.2 Containment for Field Coating

Design and provide a containment system for the capture, containment, collection, storage, and disposal of the waste materials generated by the work under this Section, to meet the requirements of [SSPC Guide 6](#), Class 2. Vapor concentrations must be kept at or below 10 percent of Lower Explosive Limit (LEL) at all times. Containment may be designed as fixed containment for complete structure or portable containment for sections of structure; however, containment must remain in any one place from beginning of abrasive blasting through initial cure of coating. Waste materials covered by this paragraph must not include any material or residue from removal of coatings containing lead, chromium, cadmium, PCB, or any other hazardous material.

It is the Contractors responsibility to insure the feasibility and workability of the containment system. The Contractor must perform their operations and work schedule in a manner as to minimize leakage of the containment system. The containment system must be properly maintained and must not deviate from the approved drawings. If the containment system fails to function satisfactorily, the Contractor must suspend all operations, except those required to minimize adverse impact on the environment or government property. Operations must not resume until modifications have been made to correct the cause of the failure.

3.3.3 Automated Monitoring Requirements for Field Coating

Provide continuous monitoring of temperature, relative humidity, and dew point data at pertinent points on the structure, during surface preparation, coating application, and initial cure. This data does not suffice for documentation of conformity to surface conditions during application and cure of coating. Locate sensors to provide pertinent data for the surface preparation and coating application being performed, as well as the temperature extremes on the structure. Describe the location plan, including required moves, in the Coatings Work Plan. Monitor any heating, cooling, or dehumidification equipment used. Provide monitoring equipment to perform as follows:

- a. Data is collected in the field unit in 15-minute increments, and available for download (on-site) in a standard database format. Contractor must collect these data and make available to the Contracting Officer, Quality Assurance Coatings Inspector, and QC Manager;
- b. Monitoring equipment must have backup power such that data collection will be uninterrupted during the entire period of the dehumidification requirement;
- c. Monitoring equipment must have capability to measure surface temperatures at a minimum of four locations anywhere on a structure, regardless of the size of the structure;
- d. Monitoring equipment must have capability to measure interior and

exterior dry bulb temperature (DB), relative humidity (RH), and dewpoint temperature (DP).

There is no requirement for connectivity of the monitoring system to control the dehumidification equipment; therefore, any combination of equipment having the required functionality will be accepted.

3.4 SHOP APPLIED COATING OF BURIED PIPING

3.4.1 Surface Preparation for Shop Applied Coating

Prepare exterior steel surfaces in accordance with [AWWA C213](#).

3.4.1.1 Abrasive Blasting Equipment for Shop Applied Coating of Buried Piping

Confirm that air supply for abrasive blasting is free of oil and moisture when tested in accordance with [ASTM D4285](#). Test air quality at each startup, but in no case, less often than every five operating hours.

3.4.1.2 Contamination Evaluation of Abrasive for Shop Applied Coating of Buried Piping

Test abrasive for salt contamination and oil contamination as required [SSPC AB 1](#) for non-metallic abrasives, [SSPC AB 2](#) for recycled ferrous abrasives, and [SSPC AB 3](#) for ferrous metallic abrasives. Modify the schedule of testing to be daily, at startup, and every five operating hours thereafter.

3.4.1.3 Pre-Preparation Testing for Surface Contamination for Shop Applied Coating of Buried Piping

3.4.1.3.1 Pre-Preparation Testing for Oil and Grease Contamination for Shop Applied Coating of Buried Piping

Inspect all surfaces for oil or grease contamination using two or more of the following inspection techniques:

- 1) VISUAL INSPECTION - Observe surface for evidence of dirt or oil.
- 2) WATER BREAK TEST - Spray atomized mist of distilled water onto surface and observe for water beading. If water wets surface rather than beading up, surface can be considered free of oil or grease contamination. Beading of water (water forms droplets) is evidence of oil or grease contamination.
- 3) CLOTH RUB TEST - Rub a clean, white, lint-free, cotton cloth onto surface and observe for discoloration. To confirm oil or grease contamination in lightly stained areas, a non-staining solvent may be used to aid in oil or grease extraction. Any visible discoloration is evidence of oil or grease contamination.

Reject oil or grease contaminated surfaces, clean using a water based pH neutral degreaser in accordance with [SSPC SP 1](#), and recheck for contamination until surfaces are free of oil and grease.

3.4.1.3.2 Pre-Preparation Testing for Soluble Salts Contamination for Shop Applied Coating of Buried Piping

Test each lot of pipe for soluble salt contamination on the surfaces using the test kit described in paragraph TEST KIT FOR MEASURING CHLORIDE, SULFATE, AND NITRATE IONS ON STEEL AND COATED SURFACES. For pipes with an outer diameter (OD) of less than 2.54 centimeters, one test per 6.09 meters is required. For pipes with an OD of more than 2.54 centimeters, three tests per 9.29 square meters must be taken, with a minimum of one test per section of pipe. Label all test tubes and retain for test verification. Reject and wash surfaces of the entire lot if one or more readings greater than 3 micrograms per square centimeter of chlorides or 10 micrograms per square centimeter of sulfates or 5 micrograms per square centimeter of nitrates is measured.

Effective washing and removal of soluble salts will require removal of any barrier to the steel surface, including rust or mill scale. This procedure may necessitate combinations of wet abrasive blasting, high pressure water rinsing, and cleaning using a solution of water and soluble salt remover. The soluble salt remover must be acidic, biodegradable, non-toxic, non-corrosive, and after application, will not interfere with primer adhesion. Use potable water, or potable water modified with soluble salt remover, for all washing or wet abrasive blasting. Test methods and equipment used in this phase must be included in the Coatings Work Plan.

This phase is required because pre-preparation testing and washing are generally more advantageous than attempting to remove soluble salt contamination after abrasive blasting. Soluble salt testing is also required in paragraph PRE-APPLICATION TESTING FOR SOLUBLE SALTS CONTAMINATION as a final acceptance test of prepared surfaces after abrasive blasting. Successful completion of pre-preparation testing and washing does not negate the requirement for pre-application testing.

3.4.1.4 Abrasive Blasting External Surfaces for Shop Applied Coating of Buried Piping

For carbon steel, prepare surfaces in accordance with SSPC SP 10/NACE No. 2. The surface profile must be in accordance with AWWA C213 and measured in accordance with ASTM D4417, Method A and Method C. The appearance of the surface after blasting must have the appearance of a Sand or Grit comparator. A rounded profile shape or peened surface is not acceptable. Measure profile at a rate of three measurements per lot at the beginning of the day.

3.4.1.5 Disposal of Used Abrasive for Shop Applied Coating of Buried Piping

Provide Disposal of Used Abrasive Shop Applied Coating, in accordance with Local, EU, and Host Nation mandated regulations.

3.4.1.6 Pre-Application Testing for Surface Contamination for Shop Applied Coating of Buried Piping

3.4.1.6.1 Pre-Application Testing for Oil and Grease Contamination for Shop Applied Coating of Buried Piping

Ensure surfaces are free of contamination as described in paragraph PRE-PREPARATION TESTING FOR OIL AND GREASE CONTAMINATION FOR SHOP APPLIED COATING OF BURIED PIPING, except that only questionable areas need be

checked for beading of water misted onto surface.

3.4.1.6.2 Pre-Application Testing for Surface Cleanliness for Shop Applied Coating of Buried Piping

Apply coatings to dust free surfaces. To test surfaces, use [ISO 8502-3](#). Use a kit that is compliant with [ISO 8502-3](#). A rating of 2 or better must be achieved for acceptance. If a test does not result in a rating of 2 or better, reject contaminated surfaces, clean by vacuum cleaning, and retest. Test surfaces at rate of one test per joint or repair area where application will be performed. Provide two additional tests for each failed test or questionable test. Document test results in the Daily Inspection Report and attach tape to the Inspection Logbook.

Ferrous abrasives may become magnetized and difficult to remove from the steel substrate. If ferrous abrasives are used, additional visual inspection must be performed to ensure no surface contamination by the abrasive is present.

3.4.2 Mixing and Application of Shop Applied Coating of Buried Piping

3.4.2.1 Mixing Liquid Epoxy Coatings

Mix materials as stated in coating system instructions. Mix materials in same temperature and humidity conditions specified in paragraph DELIVERY AND STORAGE. Allow mixed material to stand for the required induction time.

Apply mixed products within stated pot life for each product. Stop applying when material becomes difficult to apply in a smooth, uniform wet film. Do not add solvent to extend pot life. Add all required solvent, if allowable by the coating manufacturer, at time of mixing.

3.4.2.2 External Pipe Shop Applied Coating Application

Apply coatings in accordance with [SSPC PA 1](#) and as specified in [AWWA C213](#). Apply coatings to surfaces that meet all stated surface preparation requirements.

For the application of [AWWA C213](#), the DFT must be a minimum of [300 microns](#). The coating thickness must not exceed the manufacturers recommendations.

3.4.2.3 Internal Pipe Coating Application

For [AWWA C213](#), apply coatings in accordance with [SSPC PA 1](#) and as specified in [AWWA C213](#). Apply coatings to surfaces that meet all stated surface preparation requirements.

For the application of [AWWA C213](#), the DFT must be a minimum of [300 microns](#). The coating thickness must not exceed the manufacturers recommendations.

3.4.2.4 Holiday Testing and Shop Repair

Perform holiday testing on all shop-applied, externally coated areas and repair as outlined in [AWWA C213](#).

3.5 FIELD EXTERIOR COATING OF ABOVEGROUND PIPING

Coat aboveground, carbon steel piping in accordance with Section

09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES.

3.6 FIELD REPAIRS TO EXTERNAL COATING OF BURIED PIPING

Field repairs must only be performed when repairing shop applied coatings in the field and coating external girth welds.

3.6.1 Surface Preparation for Field Coating of Buried Piping

Block or suspended pipeline at a height that will allow the blast nozzle to be perpendicular to the surface being blasted, and at the proper standoff distance, at all times.

3.6.1.1 Abrasive Blasting Equipment for Field Coating of Buried Piping

Use abrasive blasting equipment of conventional air, force-feed, or pressure type. Maintain a minimum pressure of 650 kPa at nozzle. Confirm that air supply for abrasive blasting is free of oil and moisture when tested in accordance with ASTM D4285. Test air quality at each startup, but in no case, less often than every five operating hours.

3.6.1.2 Contamination Evaluation of Abrasive for Field Coating of Buried Piping

Test abrasive for salt contamination and oil contamination as required SSPC AB 1 for non-ferrous abrasives, SSPC AB 2 for recycled ferrous abrasives, and SSPC AB 3 for ferrous abrasives. Modify the schedule of testing to be daily, at startup, and every five operating hours thereafter.

3.6.1.3 Surface Standard

Inspect surfaces to be coated, and select plate with similar properties and surface characteristics for use as a surface standard. Blast clean one or more 300 mm square steel panels in the manner specified in paragraph ABRASIVE BLASTING FOR FIELD COATING OF BURIED PIPING. Record blast nozzle type and size, air pressure at nozzle and compressor, distance of nozzle from panel, and angle of blast to establish procedures for blast cleaning. Measure surface profile in accordance with ASTM D4417, Method C. When the surface standard complies with all specified requirements, seal with a clearcoat protectant. Use the surface standard for comparison to abrasive blasted surfaces throughout the course of work.

3.6.1.4 Pre-Preparation Testing for Surface Contamination for Field Coating of Buried Piping

3.6.1.4.1 Pre-Preparation Testing for Oil and Grease Contamination for Field Coating of Buried Piping

Inspect all surfaces for oil or grease contamination using two or more of the following inspection techniques:

- 1) VISUAL INSPECTION - Observe surface for evidence of dirt or oil.
- 2) WATER BREAK TEST - Spray atomized mist of distilled water onto surface and observe for water beading. If water wets surface rather than beading up, surface can be considered free of oil or grease contamination. Beading of water (water forms droplets) is evidence of oil or grease contamination.

- 3) CLOTH RUB TEST - Rub a clean, white, lint-free, cotton cloth onto surface and observe for discoloration. To confirm oil or grease contamination in lightly stained areas, a non-staining solvent may be used to aid in oil or grease extraction. Any visible discoloration is evidence of oil or grease contamination.

Reject oil or grease contaminated surfaces, clean using a water based pH neutral degreaser in accordance with **SSPC SP 1**, and recheck for contamination until surfaces are free of oil and grease.

3.6.1.4.2 Pre-Preparation Testing for Soluble Salts Contamination for Field Coating of Buried Piping

Prior to abrasive blasting, test all exposed girth welds and areas to be recoated for soluble salt contamination using the test kit described in paragraph TEST KIT FOR MEASURING CHLORIDE, SULFATE, AND NITRATE IONS ON STEEL AND COATED SURFACES. Label all test tubes and retain for test verification. Reject and wash surfaces if one or more readings greater than 3 micrograms per square centimeter of chlorides or 10 micrograms per square centimeter of sulfates or 5 micrograms per square centimeter of nitrates is measured.

Effective washing and removal of soluble salts will require removal of any barrier to the steel surface, including rust. This procedure may necessitate combinations of wet abrasive blasting, high pressure water rinsing, and cleaning using a solution of water and soluble salts remover. The soluble salts remover must be acidic, biodegradable, non-toxic, non-corrosive, and after application, will not interfere with coating adhesion. Delays between testing and preparation, or testing and coating application, may allow for the formation of new contamination. Use potable water, or potable water modified with soluble salt remover, for all washing or wet abrasive blasting. Test methods and equipment used in this phase must be included in the Coatings Work Plan.

This phase is required because pre-preparation testing and washing are generally more advantageous than attempting to remove soluble salt contamination after abrasive blasting. Soluble salt testing is also required in paragraph PRE-APPLICATION TESTING FOR SOLUBLE SALTS CONTAMINATION FOR FIELD COATING OF BURIED PIPING as a final acceptance test of prepared surfaces after abrasive blasting. Successful completion of pre-preparation testing and washing does not negate the requirement for pre-application testing.

3.6.1.5 Abrasive Blasting for Field Coating of Buried Piping

For carbon steel, prepare girth welds and repairs to bare steel to **SSPC SP 10/NACE No. 2** immediately prior to coating application. Verify that prepared surfaces comply with **SSPC VIS 1** and match the prepared test-panels from the paragraph SURFACE STANDARD at the time of coating application. The surface profile must be in accordance with the coating manufacturer's recommended range. Measure the surface profile in accordance with **ASTM D4417**, Method A and Method C. The appearance of the surface after blasting must have the appearance of a Sand or Grit comparator. A rounded profile shape or peened surface is not acceptable. For pipes with an outer diameter (OD) of less than **2.54 centimeters**, one test per **6.09 meters** is required. For pipes with an OD of more than **2.54 centimeters**, three tests per **9.29 square meters** must be taken, with a minimum of one test per section of pipe. When surfaces are re-blasted for any reason, retest profile as specified. Following abrasive blasting,

remove dust and debris by vacuum cleaning. Dust and debris tend to collect at welds, plate overlaps, and surface irregularities. Do not attempt to wipe surface clean.

3.6.1.6 Disposal of Used Abrasive for Field Coating of Buried Piping

Provide [Disposal of Used Abrasive Field Applied Coating](#) off Government property in accordance with Local, [EU](#), and [Host Nation](#) mandated regulations.

3.6.1.7 Pre-Application Testing for Surface Contamination for Field Coating of Buried Piping

3.6.1.7.1 Pre-Application Testing for Oil and Grease Contamination for Field Coating of Buried Piping

Ensure surfaces are free of contamination as described in paragraph PRE-PREPARATION TESTING FOR OIL AND GREASE CONTAMINATION FOR FIELD COATING OF BURIED PIPING, except that only questionable areas need to be checked for beading of water misted onto surface.

3.6.1.7.2 Pre-Application Testing for Soluble Salts Contamination for Field Coating of Buried Piping

Test surfaces for soluble salts using the test kit described in paragraph TEST KIT FOR MEASURING CHLORIDE, SULFATE, AND NITRATE IONS ON STEEL AND COATED SURFACES, test girth welds and areas to be recoated for soluble salts. Label all test tubes and retain for test verification. One or more readings greater than 3 micrograms per square centimeter of chlorides or 10 micrograms per square centimeter of sulfates or 5 micrograms per square centimeter of nitrates is evidence of soluble salt contamination. Reject contaminated surfaces, wash as required in paragraph PRE-PREPARATION TESTING FOR SOLUBLE SALTS CONTAMINATION FOR FIELD COATING OF BURIED PIPING, allow to dry, and re-test, and re-wash until all required tests show allowable results. Re-blast tested areas using vacuum equipped blast equipment.

3.6.1.7.3 Pre-Application Testing for Surface Cleanliness for Field Coating of Buried Piping

Apply coatings to dust free surfaces. To test surfaces, use [ISO 8502-3](#). Use a kit that is compliant with [ISO 8502-3](#). A rating of 2 or better must be achieved for acceptance. If a test does not result in a rating of 2 or better, reject contaminated surfaces, clean by vacuum cleaning, and retest. Test surfaces at rate of one test per joint or repair area where application will be performed. Provide two additional tests for each failed test or questionable test. Document test results in the Daily Inspection Reports and attach tape to the Inspection Logbook.

Ferrous abrasives may become magnetized and difficult to remove from the steel substrate. If ferrous abrasives are used, additional visual inspection must be performed to ensure no surface contamination by the abrasive is present.

3.6.2 Coating Mixing and Application for Field Coating of Buried Piping

3.6.2.1 Mixing Coating Materials for Field Coating of Buried Piping

- a. Mix and apply coating materials in accordance with the approved

coating system instructions, which may differ for each product. Do not mix partial kits unless standardized measuring cups are utilized. Do not alter mix ratios. Mix materials in same temperature and humidity conditions specified in paragraph DELIVERY AND STORAGE. Allow mixed material to stand for the required induction time. All mixing processes must be witnessed by the Quality Assurance Coatings Inspector.

- b. Apply mixed products within stated pot life for each product. Stop applying when material becomes difficult to apply in a smooth, uniform wet film. Do not add solvent to extend pot life. Add all required solvent at time of mixing, as allowable per the coating system instructions.
- c. If coating materials are not liquid materials (i.e., powder or extrusion materials), this paragraph may not apply. Follow the recommended instructions per the approved coating system instructions.

3.6.2.2 Application Conditions and Recoat Windows for Field Coating of Buried Piping

The application condition requirements for coatings are time and temperature sensitive. Plan coating application to ensure that specified temperature, humidity, and condensation conditions are met. If conditions do not allow for proper application of the coating materials, use appropriate means of controlling air and surface temperatures, as required. Partial or total enclosures, insulation, heating, cooling, or other appropriate measures may be required to control conditions to allow for orderly application of all required coats.

Maintain air and steel surface temperature within the range allowable by the approved coating system instructions during application and initial cure. Maintain steel surface temperature more than 3 degrees C above the dew-point of the ambient air for the same period. These conditions may require environmental controls through containment.

3.6.2.3 Coating Application for Field Coating of Buried Piping

Apply coatings in accordance with SSPC PA 1 and as specified in AWWA C213. Apply coatings to surfaces that meet all stated surface preparation requirements.

3.6.2.4 Holiday Testing and Field Repair

Perform holiday testing on all field coated areas and repair as outlined in AWWA C213.

3.6.3 Final Inspection of Pipeline Prior to Burial

Verify that all surfaces of the pipeline are holiday-free at time of placement of backfill over pipe. Use holiday inspection requirements and acceptance criteria of the standards applicable to the coatings being tested.

3.6.4 Final Cleanup

Following completion of the work, remove debris, equipment, and materials from the site. Remove all foreign matter such as blast media, dust, dirt, debris, grease, and oils. Wipe all dry to handle coated surfaces with

damp lint-free cloth. Remove temporary connections to Government- or Contractor-furnished water and electrical services. Restore existing facilities in and around the work areas to their original condition.

3.7 FIELD AND SHOP QUALITY CONTROL

For marking of surfaces, use chalk for marking bare steel and water based markers for marking coated surfaces. Remove marks prior to coating. Do not use any wax or grease based markers, or any other markers that leave a residue or stain.

3.7.1 Inspection

3.7.1.1 Inspection Requirements

- a. Perform inspections in accordance with [ASTM D3276](#) and the approved Coatings Work Plan.
- b. Provide all tools and instruments required to perform the required testing, as well as any tools or instruments that the inspector considers necessary to perform the required inspections and tests. Document each inspection and test, including required hold points and other required inspections and tests, as well as those inspections and tests deemed prudent from on-site evaluation to document a particular process or condition, as follows:
 - (1) Location or area;
 - (2) Purpose (required or special);
 - (3) Method;
 - (4) Criteria for evaluation;
 - (5) Results;
 - (6) Determination of compliance;
 - (7) List of required rework;
 - (8) Observations.
- c. Collect and record environmental conditions as described in [ASTM D3276](#) on a 24 hour basis, as follows:
 - (1) During surface preparation, every hour, or when changes occur;
 - (2) During coating application and the first four days of initial cure, every hour, or when changes occur;
 - (3) Note location, time, and temperature of the highest and lowest surface temperatures each day;
 - (4) Use a non-contact thermometer to locate temperature extremes, then verify with contact thermometers.
- d. Data collected on environmental conditions in AUTOMATED MONITORING REQUIREMENTS may be used for overnight data; however, the data must be constantly verified as to location of sensors and validity of data

with respect to the coating work being accomplished.

- e. Document all equipment used in inspections and testing, including manufacturer, model number, serial number, last calibration date and future calibration date, and results of on-site calibration performed. Work documented using data from equipment found to be out of calibration must be considered as non-compliant since last calibration or calibration check, as required.
- f. Document Contractors compliance with the Coatings Work Plan.

3.7.1.2 Inspection Report Forms

Develop project-specific report forms as required to report measurements, test results, and observations being complete and conforming to contract requirements. This includes all direct requirements of the contract documents and indirect requirements of referenced documents. Show acceptance criteria with each requirement and indication of conformity of each inspected item. The data may be in any format, but must be legible and presented so that entered data can be quickly compared to the appropriate requirement.

3.7.1.3 Daily Inspection Reports

Submit one copy of daily inspection report completed each day when performing work under this Section, to the Contracting Officer. Note all non-compliance issues, and all issues that were reported for rework in accordance with QC procedures of Section 01 45 00 QUALITY CONTROL. Each report must be signed by the Quality Assurance Coatings Inspector and the QC Manager. Submit report within 24 hours of date recorded on the report.

3.7.1.4 Inspection Logbook

A continuous record of all activity related to this Section must be maintained in an Inspection Logbook on a daily basis. The logbook must be hard or spiral-bound book or digital program with consecutively numbered pages, and must be used to record all information provided in the Daily Inspection Reports, as well as other pertinent observations and information. Submit the original Inspection Logbook to the Contracting Officer upon completion of the project and prior to final payment.

3.7.1.5 Inspection Equipment

All equipment must be in good condition, operational within its design range, and calibrated as required by the specified standard for use of each device.

3.7.2 Coatings Contractor QC Coatings Inspector's Responsibilities

The Coatings Contractor QC Coatings Inspector responsibilities include complete documentation of all daily inspection and production activities for the entire coatings project as outlined in the Coatings Work Plan, project program, and this Section. This includes, but is not limited to, the following:

- a. Attending and documenting the pre-job meeting and acquiring the project program, inspection and testing plan (ITP), schedule, and a list of who will receive the QC daily inspection reports;

- b. Performing a project site walk-through with the Quality Assurance Evaluator (QAE) or asset owner, Coatings Contractor QC, QC Manager, and Quality Assurance Coatings Inspector, inspecting at least the following:
 - (1) Asset(s) to be coated;
 - (2) Equipment and placement of equipment;
 - (3) Materials delivery and storage;
 - (4) Facility operational requirements during the project.
- c. Perform all daily and hold point inspections including, but not limited to, the following:
 - (1) Check equipment, including blotter test to verify compressed air cleanliness;
 - (2) Perform non-visible contaminants testing (in accordance with PRE-PREPARATION TESTING FOR SOLUBLE SALT CONTAMINATION FOR SHOP APPLIED COATING OF BURIED PIPING, PRE-APPLICATION TESTING FOR SOLUBLE SALT CONTAMINATION FOR SHOP APPLIED COATING OF BURIED PIPING, PRE-PREPARATION TESTING FOR SOLUBLE SALT CONTAMINATION FOR FIELD COATING OF BURIED PIPING, and PRE-APPLICATION TESTING FOR SOLUBLE SALT CONTAMINATION FOR FIELD COATING OF BURIED PIPING);
 - (3) Perform visible contaminants testing (in accordance with PRE-PREPARATION TESTING FOR OIL AND GREASE CONTAMINATION FOR SHOP APPLIED COATING OF BURIED PIPING, PRE-APPLICATION TESTING FOR OIL AND GREASE CONTAMINATION FOR SHOP APPLIED COATING OF BURIED PIPING, PRE-PREPARATION TESTING FOR OIL AND GREASE CONTAMINATION FOR FIELD COATING OF BURIED PIPING, and PRE-APPLICATION TESTING FOR OIL AND GREASE CONTAMINATION FOR FIELD COATING OF BURIED PIPING);
 - (4) Obtain environmental readings;
 - (5) Perform abrasive testing per SSPC AB 1, SSPC AB 2, or SSPC AB 3;
 - (6) Perform surface preparation monitoring and testing;
 - (7) Perform surface cleanliness testing;
 - (8) Perform dust quantity testing;
 - (9) Record materials storage documentation (record all coatings and abrasive materials information, batch numbers, segregation, and storage temperature);
 - (10) Witness all coatings materials mixing and record mix materials temperatures, with verification of time of coatings pot life;
 - (11) Verify, witness, and record application method;
 - (12) Perform random wet film thickness (WFT) readings;
 - (13) Perform inspection of coatings application;

- (14) Obtain dry film thickness (DFT) readings per **SSPC PA 2**;
 - (15) Perform holiday testing in accordance with HOLIDAY TESTING AND SHOP REPAIR and HOLIDAY TESTING AND FIELD REPAIR;
 - (16) Label asset identification (label stickers);
 - (17) Write Correction Action Reports (CAR), if needed;
 - (18) Write Non-Conformance Reports (NCR), if needed.
- d. Writing a daily detailed summary of the work shift inspections, testing, and the day's events, including any meetings and prevalent conversations. The final daily report must include a project summary that must be part of the daily coatings inspection report.
- e. The Coatings Contractor QC Coatings Inspector must stop all non-compliant work.

3.7.3 Quality Assurance Coatings Inspector's Responsibilities

The Quality Assurance Coatings Inspector's responsibilities include complete documentation of all on-site work associated with the coatings project. These responsibilities include, but are not limited to, the following:

- a. Attending and documenting the pre-job meeting and acquiring the **project program**, inspection and testing plan (ITP), schedule, and a list of who will receive the QC daily inspection reports;
- b. Performing a project site walk-through with the Quality Assurance Evaluator (QAE) or asset owner, prime Contractor, and coatings Contractor (QC Coatings Inspector and QC Manager), inspecting at least the following:
 - (1) Asset(s) to be coated;
 - (2) Equipment and placement of equipment;
 - (3) Materials delivery and storage;
 - (4) Facility operational requirements during the project.
- c. Verifying all daily and hold point inspections performed by the Coatings Contractor QC Coatings Inspector or QC Manager by performing mirror inspections including, but not limited to, the following:
 - (1) Verify equipment check, including blotter test to verify compressed air cleanliness;
 - (2) Verify visible contaminants testing;
 - (3) Take environmental readings;
 - (4) Perform surface preparation monitoring and testing;
 - (5) Perform surface cleanliness testing;
 - (6) Perform dust quantity test;

- (7) Record materials storage documentation (record all coating and abrasive information, batch numbers, segregation, and storage temperature);
 - (8) Witness all coatings materials mixing and record mix materials temperatures, with verification of time of coatings pot life;
 - (9) Verify, witness, and record application method;
 - (10) Inspect coatings application;
 - (11) Perform dry film thickness (DFT) readings per **SSPC PA 2**;
 - (12) Inspect asset identification (label stickers);
 - (13) Write Correction Action Reports (CAR), if needed;
 - (14) Write Non-Conformance Reports (NCR), if needed.
- d. The following testing is witnessed by the Quality Assurance Coatings Inspector and performed by the Coatings Contractor QC Coatings Inspector or QC Manager:
- (1) Wet film thickness (WFT) readings by coatings applicator(s);
 - (2) Non-visible contaminants testing for chlorides, sulfates, and nitrates (CSN);
 - (3) Abrasive testing per **SSPC AB 1**, **SSPC AB 2**, or **SSPC AB 3**;
 - (4) Holiday testing.
- e. Writing a daily detailed summary of the work shift inspections, testing, and the day's events, including any meetings and prevalent conversations. The final daily report must include a project summary that will be part of the last daily coatings inspection report.

3.8 PROJECT IDENTIFICATION

At the completion of the work, affix pertinent coating data on structure at a location that is readily accessible and visible from the ground. Use either stencils or nameplates. The following list generally describes the pertinent coating data, but should be modified as required to describe the coating systems.

Date coated/accepted: _____/_____

Project Number: _____

Contractor: _____

Address: _____

Coating System

Manufacturer: _____

Surface Prep: SSPC SP ____ Profile: _____

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Primer: _____ Thickness: ____

Intermediate: _____ Thickness: ____

Topcoat: _____ Thickness: ____

Total Thickness: _____

3.8.1 Stencils

Use stencils on piping 203 mm or larger. Use stencils with 3/4- to one-inch Helvetica style letters and acrylic stencil paint of contrasting color.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 56 10

FACTORY-FABRICATED FUEL STORAGE TANKS

05/19, CHG 1: 11/20

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY ASSURANCE
 - 1.3.1 Contractor Qualifications
 - 1.3.2 Regulatory Requirements
 - 1.3.2.1 Permitting
 - 1.3.2.2 Registration
 - 1.3.2.3 Licensed Personnel
- 1.4 DELIVERY, STORAGE, AND HANDLING
- 1.5 PROJECT/SITE CONDITIONS
- 1.6 WARRANTY

PART 2 PRODUCTS

- 2.1 ELECTRICAL WORK
 - 2.1.1 Grounding and Bonding
- 2.2 MATERIALS AND SYSTEM COMPONENTS
- 2.3 NAMEPLATES
- 2.4 ABOVEGROUND STORAGE TANK (Tank 401)
 - 2.4.1 Aboveground Storage Tank (Single Wall Steel)
- 2.5 UNDERGROUND DRAIN TANK AND ACCESSORIES (Tank 500)
 - 2.5.1 Tank Construction
 - 2.5.1.1 Steel Tank with Vault
- 2.6 TANK PROTECTIVE COATINGS (T-401, T-500)
 - 2.6.1 Interior Surfaces
 - 2.6.1.1 Certifications of Coating Contractors
 - 2.6.2 Exterior Surfaces, Aboveground Tanks (Tank 401)
 - 2.6.2.1 Certifications of Coating Contractors
 - 2.6.3 Exterior Surfaces, Underground Tanks (Tank 500)
 - 2.6.3.1 FRP Coating System
 - 2.6.4 Tank Labeling (T-401)
- 2.7 TANK COMPONENTS
 - 2.7.1 Tank Manhole (Tank 401, Tank 500)
 - 2.7.2 Tank Piping Penetrations (Tank 401, Tank 500)
 - 2.7.3 Tank Striker/Impact Plates (Tank 401, Tank 500)
 - 2.7.4 Manual Gauging/Sampling Hatch (Tank 401, Tank 500)
 - 2.7.5 Tank Ladder (Tank 401, Tank 500)
 - 2.7.6 Tank Venting
 - 2.7.6.1 Atmospheric Vent (Tank 500)
 - 2.7.6.2 Pressure/Vacuum Vent (Tank 401)
 - 2.7.6.3 Desiccant Vent Dryer
 - 2.7.6.4 Emergency Vent (Tank 401)
- 2.8 INDEPENDENT LEVEL ALARM SYSTEM (Tank 401)
 - 2.8.1 Setpoints

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.8.2 Independent Level Alarm Control Panel
- 2.9 TANK GAUGES
 - 2.9.1 Stick Gauge (Tank 401, Tank 500)
 - 2.9.2 Tank Strapping Table (Tank 401, Tank 500)
 - 2.9.3 Mechanical Clock Gauge (Tank 401, Tank 500)
 - 2.9.4 Automatic Tank Gauging (ATG)
 - 2.9.4.1 ATG (Tank 401, Tank 500)
 - 2.9.4.2 Leak Detection (Tank 500)
 - 2.9.4.3 Control Panel (Tank 401, Tank 500)
- 2.10 ACCESSORIES
 - 2.10.1 Concrete Anchor Bolts
 - 2.10.2 Bolts and Studs
 - 2.10.3 Nuts
 - 2.10.4 Washers
 - 2.10.5 Polytetrafluoroethylene (PTFE) Tape

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Underground Storage Tank (Tank 500)
 - 3.1.1.1 Steel Underground Storage Tank Handling
 - 3.1.1.2 Steel Underground Storage Tank Installation Procedures
 - 3.1.2 Aboveground Storage Tank (Tank 401)
 - 3.1.2.1 Steel Aboveground Storage Tank Handling
 - 3.1.2.2 Steel Aboveground Tank Installation Procedures
 - 3.1.3 System Components
- 3.2 FIELD QUALITY CONTROL
 - 3.2.1 Aboveground Storage Tank Tightness Tests (Tank 401)
 - 3.2.2 Underground Storage Tank Tightness Tests (Tank 500)
 - 3.2.2.1 Repairs
 - 3.2.3 Tank Manufacturer's Tests (Tank 401, Tank 500)
 - 3.2.4 System Commissioning
 - 3.2.5 Tank Inspection Reports
- 3.3 DEMONSTRATIONS
- 3.4 Tank Fill Tests (Tank 401, Tank 500)
- 3.5 FIELD PAINTING

-- End of Section Table of Contents --

SECTION 33 56 10

FACTORY-FABRICATED FUEL STORAGE TANKS
05/19, CHG 1: 11/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API MPMS 2.2E	(2004; Errata 2009; R 2009) Petroleum and Liquid Petroleum Products - Calibration of Horizontal Cylindrical Tanks - Part 1: Manual Methods
API RP 540	(1999; R 2004) Electrical Installations in Petroleum Processing Plants
API RP 1615	(2011) Installation of Underground Petroleum Storage Systems
API RP 2003	(2015; 8th Ed) Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents

ASTM INTERNATIONAL (ASTM)

ASTM A193/A193M	(2020) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications
ASTM A194/A194M	(2020a) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both
ASTM A307	(2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A563	(2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts
ASTM D3308	(2012; R 2017) Standard Specification for PTFE Resin Skived Tape
ASTM F436	(2011) Hardened Steel Washers
ASTM F844	(2019) Standard Specification for Washers, Steel, Plain (Flat), Unhardened for General Use

ASTM F1554 (2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 142 (2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book

IEEE 1100 (2005) Emerald Book IEEE Recommended Practice for Powering and Grounding Electronic Equipment

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 30 (2021; TIA 20-1; TIA 20-2) Flammable and Combustible Liquids Code

NFPA 30A (2021; TIA 20-1) Code for Motor Fuel Dispensing Facilities and Repair Garages

NFPA 31 (2020) Standard for the Installation of Oil-Burning Equipment

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code

NFPA 77 (2014) Recommended Practice on Static Electricity

NFPA 407 (2022) Standard for Aircraft Fuel Servicing

NFPA 704 (2022) Standard System for the Identification of the Hazards of Materials for Emergency Response

NFPA 780 (2020) Standard for the Installation of Lightning Protection Systems

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC QP 3 (2010) Standard Procedure for Evaluating Qualifications of Shop Painting Applicators

SSPC QS 1 (2015) Standard Procedure for Evaluating a Contractor's Advanced Quality Management System

STEEL TANK INSTITUTE (STI)

STI 010-50-1000 (2011) Specification and Manual for External Corrosion Protection of Underground Steel Storage Tanks

STI 020-50-1000 (2010) ACT-100 Specification for External Corrosion Protection of FRP Composite

Steel USTs

STI 700-50-5007 (2010) Installation Instructions for Shop Fabricated Aboveground Tanks for Flammable, Combustible Liquids

STI SP001 (2018, 6th Ed) SP001 Standard for The Inspection of Aboveground Storage Tanks

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-161 (2005; Rev G; Notice 1 2010) Identification Methods for Bulk Petroleum Products Systems Including Hydrocarbon Missile Fuels

UNDERWRITERS LABORATORIES (UL)

UL 58 (2018) UL Standard for Safety Steel Underground Tanks for Flammable and Combustible Liquids

UL 142 (2006; Reprint Jan 2021) UL Standard for Safety Steel Aboveground Tanks for Flammable and Combustible Liquids

UL 1746 (2007; Reprint Dec 2014) External Corrosion Protection Systems for Steel Underground Storage Tanks

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Grounding and Bonding

Underground Drain Tank And Accessories; G

SD-03 Product Data

Aboveground Storage Tank (Single Wall Steel); G

Tank Protective Coatings; G

Atmospheric Vent; G

Pressure/Vacuum Vent; G

Emergency Vent; G

Independent Level Alarm System; G

Tank Gauges; G

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Automatic Tank Gauging (Atg)

Underground Drain Tank And Accessories; G

Desiccant Vent Dryer; G

SD-06 Test Reports

Aboveground Storage Tank Tightness Tests; G

Underground Storage Tank Tightness Tests; G

Tank Manufacturer's Tests

Tank Fill Tests

Tank Inspection Reports; G

SD-07 Certificates

Contractor Qualifications; G

Host Nation Certification; G

Pollution Liability Insurance

Permitting

Registration

Licensed Personnel

Demonstrations

STI SP001 Inspector's Certification; G

SD-08 Manufacturer's Instructions

Aboveground Storage Tank

Independent Level Alarm System

Tank Gauges

Automatic Tank Gauging (Atg)

SD-10 Operation and Maintenance Data

Aboveground Storage Tank; G

Independent Level Alarm System; G

Tank Gauges; G

Automatic Tank Gauging (Atg)

Underground Drain Tank And Accessories; G

1.3 QUALITY ASSURANCE

1.3.1 Contractor Qualifications

Each installation Contractor must have successfully completed at least 5 projects of the same scope, and the same size or larger within the last 3 years and demonstrated specific installation experience in regard to the specific system installation to be performed. Each installation Contractor must have taken, if applicable, manufacturer's training courses on the installation of storage tanks and must meet all applicable licensing requirements in the state. If applicable, state certified installers must be provided by the Contractor. Installers must also be trained and certified by the manufacturer to install the equipment and materials being installed and must be STI certified. Installers must submit certification from the **manufacturer and Host Nation (as applicable)**. If installing underground storage tanks and piping systems, installation Contractor must have **Pollution Liability Insurance**. Submit a letter listing prior projects, the date of construction, a point of contact for each prior project, the scope of work of each prior project, and a detailed list of work performed. The letter must also provide evidence of prior manufacturer's training, state licensing, and other related information.

1.3.2 Regulatory Requirements

1.3.2.1 Permitting

Obtain necessary permits in conjunction with the installation of storage tanks as required by **Host Nation** or local authority.

1.3.2.2 Registration

Obtain and complete all tank registration forms required by **host nation** and local authorities. Submit all completed tank registration forms within 30 days after contract award to the Contracting Officer. The Contracting Officer will ensure the Base Environmental staff for the DoD Installation submits the forms to the proper regulatory agencies.

1.3.2.3 Licensed Personnel

Tank installers must be licensed/certified by the **Host Nation** when the **Host Nation** requires licensed installers.

1.4 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect system components and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer, upon recommendation by Base Environmental for the DoD Installation. Replace damaged or defective items.

1.5 PROJECT/SITE CONDITIONS

Exposed moving parts, parts that produce high operating temperatures and pressures, parts that may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Install safety devices so that proper operation of equipment is not impaired.

1.6 WARRANTY

All factory-fabricated storage tanks must come with a manufacturer's warranty of a minimum period of 30 years. All warranty paperwork will be completed and submitted by Contractor to both the tank and system component manufacturers, the Contracting Officer, and the Base Environmental for the DoD Installation. This includes all applicable completed manufacturers' equipment installation checklists.

PART 2 PRODUCTS

2.1 ELECTRICAL WORK

Provide controllers, integral disconnects, contactors, controls, and control wiring with their respective pieces of system components. Provide electrical system components, including motors and wiring, as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Any wiring required for the operation specified herein, but not shown on the electrical plans, must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.1.1 Grounding and Bonding

Grounding and bonding must be in accordance with NFPA 70, NFPA 77, NFPA 407, NFPA 780, API RP 540, API RP 2003, IEEE 142, and IEEE 1100. Provide jumpers to overcome the insulating effects of gaskets, paints, or nonmetallic components.

2.2 MATERIALS AND SYSTEM COMPONENTS

Provide materials and system components that are standard products of a manufacturer regularly engaged in the manufacturing of such products, that are of a similar material, design and workmanship. Provide materials and system components that have been in satisfactory commercial or industrial use for a minimum 3 years prior to bid opening. The 3 year period must include applications of the system components and materials under similar circumstances and of similar size. Provide materials and system components that have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 3 year period.

Internal parts and components of system components, piping, piping components, and valves that could be exposed to fuel during system operation must not be constructed of zinc coated (galvanized) metal, brass, bronze, or other copper alloys. Do not install cast iron bodied valves in piping systems that could be exposed to fuel during system operation.

2.3 NAMEPLATES

Attach nameplates to all specified system components defined herein. List on each nameplate the manufacturer's name, address, contract number, component type or style, model or serial number, catalog number, capacity or size, and the system that is controlled. Construct plates of stainless steel. Install nameplates in prominent locations with nonferrous screws, nonferrous bolts, or permanent adhesive. Minimum size of nameplates must be 25 by 65 mm. Provide manufacturer's storage tank nameplates as required. Lettering must be the normal block style with a minimum 6 mm height. Accurately align all lettering on nameplates.

2.4 ABOVEGROUND STORAGE TANK (Tank 401)

2.4.1 Aboveground Storage Tank (Single Wall Steel)

Provide a factory-welded, single wall stainless steel tank manufactured to UL 142 and equipped to comply with NFPA 30 for use as a flammable or combustible liquid storage tank. Tank must be designed and manufactured for a horizontal cylindrical installation. Tank must be mounted on the tank manufacturer's standard UL listed tank saddles that elevates the tank above the underlying concrete slab and/or concrete support a maximum of 305 mm. Tank assembly must have lifting lugs that allow tank relocation. Provide tank assembly with the manufacturer's standard stairway and catwalk assembly, except as modified herein. The stairway and catwalk assembly must be constructed of structural steel and must allow personal access to the top of the entire length of the tank system. The catwalk must have protective railings on tanks higher than 1.3 m.

2.5 UNDERGROUND DRAIN TANK AND ACCESSORIES (Tank 500)

2.5.1 Tank Construction

Drain tank must be a U.L. labeled, double wall, steel tank, with interstitial monitor. Tank must be provided with calibrated gage stick, strapping chart, and a steel vault attached to tank. Vault must be provided with a rolling pit cover and removable access grating. Minimum inner and outer tank wall thickness must be 4 mm.

2.5.1.1 Steel Tank with Vault

- a. The design, fabrication, erection, testing, and inspection of the double wall tank must conform to the requirements of UL 58, Standard for Safety, Steel Underground Tanks for Flammable and Combustible Liquids, Type II. Tank will be provided with a welded steel vault, as indicated, to allow access to the manholes, equipment, and appurtenances. Steel vault will be welded to the tank and will be provided by the tank manufacturer.
- b. Material must be carbon steel plate.
- c. Lifting lugs must be located at the balance points.
- d. Provide anchor straps to attach tank to hold down slab. Straps must be separated from the tank by a pad made of inert insulating material. Number and location of straps must be as indicated on the drawings. Metal straps, turnbuckles, and anchors must be coated to resist corrosion.
- e. Tank capacity, connections and appurtenance must be as shown on the drawings and as described.
- f. The interior and exterior surfaces of tank and vault must be coated for corrosion protection. The interior surface must be coated in accordance with 09 97 13.17 THREE COAT EPOXY INTERIOR COATING OF WELDED STEEL PETROLEUM FUEL TANKS. The exterior surface must be coated in accordance with STI 010-50-1000 and the tank must bear the STI 010-50-1000 label.

2.6 TANK PROTECTIVE COATINGS (T-401, T-500)

2.6.1 Interior Surfaces

Coat 100 percent of a metal tank's interior surfaces including all metal piping and metal appurtenances as specified in Section 09 97 13.17 THREE COAT EPOXY INTERIOR COATING OF WELDED STEEL PETROLEUM FUEL TANKS as modified herein.

2.6.1.1 Certifications of Coating Contractors

All Contractors and Subcontractors that perform surface preparation or coating application must be certified to SSPC QP 3 and SSPC QS 1 prior to contract award, and must remain certified while accomplishing any surface preparation or coating application. Painting contractors and painting subcontractors must remain so certified for the duration of the project. If a contractor or subcontractor certification expires, the firm is not allowed to perform any work until the certification is reissued. Requests for extension of time due to delay as a result of an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in certification status. Notify the Contracting Officer of all scheduled or unannounced on site audits from SSPC and furnish a copy of all audit reports.

2.6.2 Exterior Surfaces, Aboveground Tanks (Tank 401)

Protect the exterior surfaces of each aboveground tank as specified in Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES as modified herein.

2.6.2.1 Certifications of Coating Contractors

All Contractors and Subcontractors that perform surface preparation or coating application must be certified to SSPC QP 3 and SSPC QS 1 prior to contract award, and must remain certified while accomplishing any surface preparation or coating application. Painting contractors and painting subcontractors must remain so certified for the duration of the project. If a contractor or subcontractor certification expires, the firm is not allowed to perform any work until the certification is reissued. Requests for extension of time due to delay as a result of an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in certification status. Notify the Contracting Officer of all scheduled or unannounced on site audits from SSPC and furnish a copy of all audit reports.

2.6.3 Exterior Surfaces, Underground Tanks (Tank 500)

Provide steel tanks with one of the following corrosion protection systems.

2.6.3.1 FRP Coating System

Coating system must be in accordance with UL 1746 Part IV and UL 58. The integrity of the coating must be certified by the manufacturer as meeting the thickness requirements and having no flaws prior to shipment. The UL label must be affixed and visible on the exterior surface of each coated tank.

2.6.4 Tank Labeling (T-401)

Tank must be labelled with the following information at a minimum:

- a. Product Stored and Tank Capacity (Per MIL-STD-161).
- b. Tank Number and Facility Number.
- c. NFPA 704 Diamond Hazmat Label.

2.7 TANK COMPONENTS

2.7.1 Tank Manhole (Tank 401, Tank 500)

Tank manholes must have an internal diameter of 915 mm. Provide each manhole with a matching flanged watertight manhole cover. Manhole covers must be UL listed, be constructed of pressed or mild steel, and include a UL listed gasket.

2.7.2 Tank Piping Penetrations (Tank 401, Tank 500)

Provide a welded-in-place flanged pipe nozzle for each tank piping connection, as indicated. All unused or spare tank piping penetrations must be sealed with as indicated.

2.7.3 Tank Striker/Impact Plates (Tank 401, Tank 500)

Provide an interior striker/impact plate under each tank manhole and pipe connection. Each plate must be a minimum of 6 mm in thickness, be larger in diameter than the tank penetration, fit the curvature of the tank bottom, and be completely coated in the same fashion as the interior tank bottom coating. Each plate must be welded to the tank bottom at the factory (full circumference connection). The welds must be non-destructive tested using the appropriate means.

2.7.4 Manual Gauging/Sampling Hatch (Tank 401, Tank 500)

Provide a combination gauging and sampling hatch assembly. The assembly must include a bronze top-seal type adapter with a corresponding locking type cap (adapter and cap both externally-mounted to the top of the tank) and a steel or aluminum stilling well pipe. The stilling well pipe must be a minimum 100 mm in size and extend downward through the top of the tank to within 75 mm of the tank bottom. Provide the entire length of pipe inside the tank with 13 mm wide by 300 mm long slots at alternate locations. Coat the pipe in the same fashion as the interior tank bottom coating. Aluminum stilling wells will not require coating.

2.7.5 Tank Ladder (Tank 401, Tank 500)

Provide interior tank ladders constructed of fiberglass. If steel, coat the ladder in the same fashion as the tank interior. The two stringers must be a minimum 10 mm thick and a minimum 50 mm wide. The rungs must be a minimum 20 mm rod on 300 mm centers. Members of the ladder must be securely affixed. Ladder must be of sufficient length to extend from the bottom of the tank to the top surface of the tank. Ladder must be rigidly connected to the tank bottom in accordance with the tank manufacturer's standard. Ladder must be connected to the top of the tank with pipe guides or slip bars to accommodate expansion of the two stringers.

2.7.6 Tank Venting

2.7.6.1 Atmospheric Vent (Tank 500)

Provide atmospheric, updraft type cap. Cap must be constructed of aluminum or carbon steel. Cap must have an internal brass or bronze insect screen, minimum 40-mesh. Cap must prevent rain, snow, or ice from entering the vent piping.

2.7.6.2 Pressure/Vacuum Vent (Tank 401)

Desiccant dryer outlet must be equipped with pressure-vacuum breather vent, aluminum construction with weather hood and with fluoroelastomer (FKM, Viton) pallet seat inserts, high density screens, stainless steel internals, with pressure relief setting, vacuum relief setting, and venting/vacuum capacity per tank manufacturer.

2.7.6.3 Desiccant Vent Dryer

Provide desiccant vent dryer with 90 kg capacity. Coordinate with tank manufacturer and desiccant manufacturer. Provide extra desiccant material for maintenance change out. Provide vessel with factory fabricated base. Provide sight glass for ease of maintenance. Ensure compatibility with all fuel/product types.

2.7.6.4 Emergency Vent (Tank 401)

Vent must be the normally-closed, UL listed type that vents outward and upward. Vent must conform with NFPA 30 and UL 142 and must be sized by the tank manufacturer. Provide vent with the Liters per second (L/s) rating permanently labeled on the vent's exterior.

2.8 INDEPENDENT LEVEL ALARM SYSTEM (Tank 401)

Provide an independent level alarm system that will monitor 4 programmable liquid level setpoints. The system must delineate between each individual setpoint. The system must produce an audible and visible alarm in the event of monitoring an alarm condition. Mechanically-actuated float assemblies must be field adjustable. The system must be totally independent of the tank gauging system.

2.8.1 Setpoints

Configure the alarm system's 4 setpoints in accordance with the following.

- a. High Level Setpoint. Produce an alarm condition when a tank's liquid level rises above 90 percent capacity.
- b. High-High Level Setpoint. Produce an alarm condition when a tank's liquid level rises above 95 percent capacity.
- c. Low Level Setpoint. Produce an alarm condition when a tank's liquid level drops below 15 percent capacity.
- d. Low-Low Level Setpoint. Produce an alarm condition when a tank's liquid level drops below the minimum pump submergence level at percent capacity.

2.8.2 Independent Level Alarm Control Panel

Level alarms shall be compatible with the ATE Control Panel.

2.9 TANK GAUGES

2.9.1 Stick Gauge (Tank 401, Tank 500)

For each tank, provide 2 wooden stick gauges. Gauge length must allow the measurement of the entire level of fuel in the corresponding tank. Gauges must be compatible with the fuel to be measured (no swelling or damage from fuel contact). Provide gauge with non-sparking caps on each end. Mark gauges in m and mm. The smallest unit of measure on the gauge must be 1 mm.

2.9.2 Tank Strapping Table (Tank 401, Tank 500)

Furnish 2 API MPMS 2.2E certified strapping tables (calibration charts) for each tank. One of the tables must indicate the liquid contents in L for each 1 mm of tank depth and the other in for of tank depth. Strapping table volumes for all tanks 19,000 L and larger must be determined using physical measurements and not calculated values. For each tank, provide an electronic media file of each strapping table.

2.9.3 Mechanical Clock Gauge (Tank 401, Tank 500)

Gauge must be the level sensing, mechanically actuated type that provides the tank level readout in a sealed glass cap contained in a gauge box. Gauge must be accurate to plus or minus 6 mm and must measure the liquid level over the full range of a tank's height. Gauge must have vapor tight seals to prevent condensation from fogging the viewing glass.

2.9.4 Automatic Tank Gauging (ATG)

2.9.4.1 ATG (Tank 401, Tank 500)

ATG must be the mechanically or electronically actuated type that can continuously monitor a tank's usable liquid level storage capacity. The system must provide a digital readout of a tank's liquid level in terms of mm and liters. The system must be accurate to plus or minus 2 mm. The system must measure water accumulation in mm from 19 to 125 mm off the bottom of a storage tank. Construct system components to be chemically compatible with the fuel to be handled.

2.9.4.2 Leak Detection (Tank 500)

- a. Provide an annular space between the primary and secondary shells to allow for the free flow and containment of all leaked product from the primary tank.
- b. Provide the tank with a leak monitoring system capable of sensing leaks in the secondary containment space and in the vault. The system must detect a leak of fuel through the inner shell to the area between the inner and outer shells or a leak of ground water through the outer shell into the area between the inner and outer shells. The detector and any equipment in the area of the fuel tanks and valve pits must be rated for the environment in which it is installed. The system must be a continuous surveillance type. The sensor must be electronic or hydraulic type and must be connected to a control panel. Totally

flooded containment space reservoir system must not be permitted. The alarm must be manually reset at the control panel. Use an inert gas that is heavier than air in containment space of the tanks to prevent the forming of condensation. The tank monitoring system must be compatible with the tank furnished and must be as recommended by the tank manufacturer. Provide instructions and equipment required for calibration of the monitoring system and calibration maintenance schedule. Access must be provided to the tank sensor for testing and maintenance.

2.9.4.3 Control Panel (Tank 401, Tank 500)

The control panel must be located where shown on the plans. Panel must be a standard industrial enclosure. Panel doors must swing left or right. The panel must display the digital readout of each monitored tank on an LCD mounted exterior to the panel. The panel must also have external controls to allow operators to toggle between information on the LCD without having to open the panel. The panel must provide an audible and visible alarm if a leak or preset high or high-high level is detected. Panel must discriminate and indicate if the leak is fuel or water. Leak detection, high level and high-high level alarms must be provided to the pump control panel(PCP), see Section 33 09 52 FUEL PUMP CONTROL AND ANNUNCIATION SYSTEM (NON-HYDRANT). Provide system operating instructions inside of the control panel. Unit must be Veeder-Root ATG TLS, or Government approved equal and compatible with the Base Fuels Automated System (FAS).

2.10 ACCESSORIES

2.10.1 Concrete Anchor Bolts

Concrete anchors must conform to [ASTM F1554](#), hot-dipped galvanized.

2.10.2 Bolts and Studs

Carbon steel bolts and studs must conform to [ASTM A307](#), Grade B, hot-dipped galvanized. Stainless steel bolts and studs that conform to [ASTM A193/A193M](#), Grade 8.

2.10.3 Nuts

Carbon steel nuts must conform to [ASTM A563](#), Grade A, hex style, hot-dipped galvanized. Stainless steel nuts must conform to [ASTM A194/A194M](#), Grade 8.

2.10.4 Washers

Provide flat circular washers under each bolt head and each nut. Washer materials must be the same as the connecting bolt and nut. For [ASTM F1554](#) concrete anchors, use [ASTM F436](#), Type 1, hot-dipped galvanized washers. For [ASTM A307](#) bolts and studs, carbon steel washers must conform to [ASTM F844](#), hot-dipped galvanized. Stainless steel washers must conform to [ASTM A194/A194M](#), Grade 8.

2.10.5 Polytetrafluoroethylene (PTFE) Tape

Tape must conform to [ASTM D3308](#).

PART 3 EXECUTION

3.1 INSTALLATION

Install work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Handle storage tanks with extreme care to prevent damage during placement and install in accordance with the manufacturer's installation instructions and NFPA 30, NFPA 30A, or NFPA 31 as applicable. Inspect the exterior surface of each tank for obvious visual damage prior to and during the placement of each storage tank. Repair surface damage to a storage tank according to manufacturer's requirements before proceeding with the system installation. Provide the termination of fill lines within a tank with an antisplash deflector. Provide nylon dielectric bushings on pipe connections to a steel tank.

3.1.1 Underground Storage Tank (Tank 500)

Install underground storage tanks in accordance with API RP 1615 except as modified herein. Place tank on a 3 mm per 30 mm slope with the fill point at the low end and the vent connection at the high end. Locate tank so that the fuel discharge pipes slope up uniformly toward the fuel outlet. Install containment sumps prior to any backfill being added above the storage tanks.

3.1.1.1 Steel Underground Storage Tank Handling

Store, handle, and place externally coated steel tanks with care and in a manner that will minimize damage to the coating and will not reduce its protective value. Place coated tanks in position carefully and with a minimum of handling. Prior to backfilling a tank, visually inspect the tank exterior protective coating for damage. Repair any damaged tank coating in accordance with the appropriate UL or STI standard (UL 1746, STI 020-50-1000, or UL 58).

3.1.1.2 Steel Underground Storage Tank Installation Procedures

Anchor tank to a reinforced concrete anchor pad as indicated using manufacturer's supplied holddown straps. Separate tank from an anchor pad by a minimum of 300 mm of backfill material. Coat metal straps, turnbuckles, anchors, and accessories to resist corrosion. Uniformly place backfill material around the entire tank and extend to grade level. Inspect tank cathodic protection anodes, if applicable, to ensure integrity during backfill operations.

3.1.2 Aboveground Storage Tank (Tank 401)

Install aboveground storage tanks in accordance with STI 700-50-5007 (STI R912) except as modified herein. Place tank that is equal to or greater than 18,900 L on a 3 mm per 30 mm slope with the fill point at the low end and the vent connection and issue pump at the high end.

3.1.2.1 Steel Aboveground Storage Tank Handling

Store, handle, and place externally coated steel tanks with care and in a manner that will minimize damage to the coating and will not reduce its protective value. Place coated tanks in position carefully and with a minimum amount of handling. Repair any damaged tank coating in accordance with the appropriate UL or STI standard (UL 1746, STI 020-50-1000, or UL 58). Do not move the tank unless it is empty.

3.1.2.2 Steel Aboveground Tank Installation Procedures

Tanks should be secured to the associated tank pad per tank manufacturer's recommendations using fasteners installed through the tank saddle base plate.

3.1.3 System Components

Properly level, align, and secure system components in place in accordance with manufacturer's instructions. Provide supports for system components, appurtenances, and pipe as required. Install anchors, bolts, nuts, washers, and screws where required for securing the work in place. Sizes, types, and spacings of anchors and bolts not indicated or specified must be as required for proper installation.

3.2 FIELD QUALITY CONTROL

3.2.1 Aboveground Storage Tank Tightness Tests (Tank 401)

Perform tightness tests on each aboveground storage tank prior to making piping connections. Perform testing in accordance with **STI 700-50-5007** (STI R912) except as modified herein. Gauges used to monitor the tests must have a scale with a maximum limit of **103 kPa**. Repair leaks discovered during the tightness tests in accordance with tank manufacturer's instructions. Following any repair, re-test the tank until the tank successfully passes the testing requirements of this paragraph.

3.2.2 Underground Storage Tank Tightness Tests (Tank 500)

Perform a tightness test on each underground storage tank on-site just prior to their placement into the ground. Pneumatically pressurize each storage tank's primary chamber to **35 kPa** and monitor for a drop in pressure over a 2-hour period during which there must be no drop in pressure in the tank greater than that allowed for thermal expansion and contraction. Following the successful completion of the primary chamber test, bleed the pressure from the primary chamber into the interstitial space. Maintain this pressure while applying soapsuds or equivalent material over the exterior of the tank. While applying the soapsuds, visually inspect the entire tank, including the bottom surfaces, for leaks (bubble formations). Inspection of the bottom surfaces of a tank may be performed by rotating the tank; however, a tank must only be rotated in strict accordance with the manufacturer's recommendations. Do not rotate a tank more than 90 degrees from the upright position. During testing, install a pressure relief device that relieves at the tank manufacturer's suggested pneumatic pressure limit. Gauges used in pneumatic tests must have a scale with a maximum limit of **103 kPa**.

3.2.2.1 Repairs

Repair leaks discovered in either the primary chamber or the interstitial space in accordance with the tank manufacturer's instructions. Following any tank repairs, re-test the tank until the tank successfully passes the testing requirements defined herein.

3.2.3 Tank Manufacturer's Tests (Tank 401, Tank 500)

In addition to the tests required herein, perform any additional tests (i.e., leak tests, cathodic protection verification tests, etc.) on each

storage test that is required by the tank manufacturer's written test procedures. Manufacturer's tests that are redundant to tests already required by this specification will only be performed once per tank. Repair all leaks discovered during the tests in accordance with manufacturer's instructions. Following tank repairs, re-test the tank until the tank successfully passes the manufacturer's testing requirements.

3.2.4 System Commissioning

System commissioning must conform to Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT).

3.2.5 Tank Inspection Reports

Prior to system commissioning, a STI SP001 certified inspector must inspect the completed aboveground and underground tank in accordance with STI SP001 or STI SP131 and deliver a full report to the Contracting Officer. The report must include a record of ultrasonic thickness measurements (UTMs), exclusive of the coating, of each single wall aboveground and underground tank shell. The report must include the tank dataplate information and photograph of the tank data plate. Provide electronic copies of the tank inspection reports to Service Headquarters, Service Control Points, and DLA-Energy. The paper and electronic copies of the report and UTMs must be provided to the Contracting Officer for filing with the tank's "as-built drawings." Refer to Section 01 45 00 QUALITY CONTROL for STI SP001 Inspector's Certification requirements.

3.3 DEMONSTRATIONS

Conduct a training session for designated Government personnel in the operation and maintenance procedures related to the system components and systems specified herein. Include pertinent safety operational procedures in the session as well as physical demonstrations of the routine maintenance operations. Furnish instructors who are familiar with the installation/system components and systems, both operational and practical theories, and associated routine maintenance procedures. The training session must consist of a total of 4 hours of normal working time and must start after the system is functionally completed, but prior to final system acceptance. Submit a letter, at least 14 working days prior to the proposed training date, scheduling a proposed date for conducting the onsite training.

3.4 Tank Fill Tests (Tank 401, Tank 500)

Tank fill tests must not be performed until after the flushing, cleaning, and adjusting requirements defined in Section 33 08 55.01 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). For the tank fill tests, initially fill each storage tank with product in order to verify the tank level alarm system operates properly and the tank overfill protection device functions as designed. Stop filling each tank immediately once the overfill devices operates. Do not overfill any storage tank more than the 98 percent level. Drain the system below the low liquid level setpoint to verify operation of the low level alarm. Correct and retest any problems with the level alarm system or the overfill device until each operate as specified herein. During the tests, verify that all tank gauges are calibrated and operating appropriately.

3.5 FIELD PAINTING

Do not paint stainless steel and aluminum surfaces. Do not coat system components or components provided with a complete factory coating. Prior to any field painting, clean surfaces to remove dust, dirt, rust, oil, and grease.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 56 19

FUEL IMPERMEABLE LINER SYSTEM

11/22

PART 1 GENERAL

- 1.1 MEASUREMENT AND PAYMENT
- 1.2 REFERENCES
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Material and Equipment Qualifications
 - 1.4.2 Field Engineer Qualifications
 - 1.4.3 Installer Qualifications
 - 1.4.4 Factory Seams
 - 1.4.5 Liner Manufacturer's Certification
- 1.5 DELIVERY, STORAGE, AND HANDLING
 - 1.5.1 Delivery
 - 1.5.2 Storage
 - 1.5.3 Handling
- 1.6 AMBIENT CONDITIONS
- 1.7 WARRANTY

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Fabric Reinforced Geomembrane Liner
 - 2.1.2 Non-Reinforced Smooth Geomembrane Liner
 - 2.1.3 Nonwoven Geotextile
 - 2.1.3.1 General Properties
 - 2.1.3.2 Nominal Unit Weight
 - 2.1.3.3 Grab Tensile Strength
 - 2.1.3.4 Grab Tensile Elongation
 - 2.1.3.5 Puncture Strength
 - 2.1.3.6 Trapezoid Tear Strength
 - 2.1.3.7 Ultraviolet (UV) Resistance
- 2.2 ACCESSORIES
 - 2.2.1 Liner Fittings
 - 2.2.2 Embedment Strip
 - 2.2.3 Batten / Mounting Strip System
 - 2.2.3.1 Gasket
 - 2.2.3.2 Concrete Anchor Bolts
 - 2.2.4 Sealant
 - 2.2.5 Temporary Ballast
- 2.3 EQUIPMENT

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Field Engineer
 - 3.1.2 Surface Preparation

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.1.3 Embedment Strip Installation
- 3.1.4 Liner Installation
 - 3.1.4.1 Liner Projections
- 3.2 FIELD SEAMING
 - 3.2.1 Trial Seams
 - 3.2.2 Field Seams
- 3.3 FIELD QUALITY CONTROL
 - 3.3.1 Visual Inspection of Field Seams
 - 3.3.2 Non-Destructive Field Seam Testing
 - 3.3.2.1 Liner Vacuum Box Test
 - 3.3.2.2 Liner Air Lance Test
 - 3.3.2.3 Liner Air Pressure Test
 - 3.3.2.4 Liner Point Stress Test
 - 3.3.3 Destructive Field Seam Testing
- 3.4 REPAIRS
- 3.5 PROTECTION AND BACKFILLING
- 3.6 PERMANENT BALLAST
 - 3.6.1 Cast in Place Concrete
- 3.7 PROJECT CLOSEOUT

-- End of Section Table of Contents --

SECTION 33 56 19

FUEL IMPERMEABLE LINER SYSTEM
11/22

PART 1 GENERAL

1.1 MEASUREMENT AND PAYMENT

Measure the total surface area in square **meters** covered by liner system. Final quantities will be based on as-built conditions. Allowance will be made for liner system in anchor and drainage trenches; however, no allowance will be made for waste, overlap, repairs, or materials used for the convenience of the Contractor. Liner system installed and accepted by the Contracting Officer will be paid for at the respective contract unit price in the bidding schedule.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM D413	(1998; R 2017) Standard Test Methods for Rubber Property - Adhesion to Flexible Substrate
ASTM D696	(2016) Standard Test Method for Coefficient of Linear Thermal Expansion of Plastics Between -30 degrees C and 30 degrees C With a Vitreous Silica Dilatometer
ASTM D751	(2006; R 2011) Coated Fabrics
ASTM D1004	(2013) Initial Tear Resistance of Plastic Film and Sheeting
ASTM D1505	(2018) Standard Test Method for Density of Plastics by the Density-Gradient Technique
ASTM D1603	(2020) Carbon Black Content in Olefin Plastics
ASTM D2136	(2002; R 2012) Coated Fabrics - Low-Temperature Bend Test
ASTM D3776/D3776M	(2009a; R 2017) Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3895	(2014) Oxidative-Induction Time of Polyolefins by Differential Scanning

Calorimetry

ASTM D4218	(2020) Determination of Carbon Black Content in Polyethylene Compounds by the Muffle-Furnace Technique
ASTM D4437/D4437M	(2016; R 2018) Standard Practice for Non-destructive Testing (NDT) for Determining the Integrity of Seams Used in Joining Flexible Polymeric Sheet Geomembranes
ASTM D4533	(2011) Trapezoid Tearing Strength of Geotextiles
ASTM D4632/D4632M	(2015a) Grab Breaking Load and Elongation of Geotextiles
ASTM D4833/D4833M	(2007; R 2020) Standard Test Method for Index Puncture Resistance of Geomembranes and Related Products
ASTM D5199	(2012) Measuring Nominal Thickness of Geosynthetics
ASTM D5261	(2010; R 2018) Standard Test Method for Measuring Mass Per Unit Area of Geotextiles
ASTM D5397	(2019a) Standard Test Method for Evaluation of Stress Crack Resistance of Polyolefin Geomembranes Using Notched Constant Tensile Load Test
ASTM D5596	(2003; R 2016) Standard Test Method For Microscopic Evaluation of the Dispersion of Carbon Black in Polyolefin Geosynthetics
ASTM D5641/D5641M	(2016) Standard Practice for Geomembrane Seam Evaluation by Vacuum Chamber
ASTM D5721	(2008; R 2013) Air-Oven Aging of Polyolefin Geomembranes
ASTM D5820	(1995; R 2018) Standard Practice for Pressurized Air Channel Evaluation of Dual Seamed Geomembranes
ASTM D5884/D5884M	(2004a; R 2015; E 2015) Standard Test Method for Determining Tearing Strength of Internally Reinforced Geomembranes
ASTM D5885/D5885M	(2017) Standard Test Method for Oxidative Induction Time of Polyolefin Geosynthetics by High-Pressure Differential Scanning Calorimetry
ASTM D5893/D5893M	(2016) Standard Specification for Cold Applied, Single Component, Chemically Curing Silicone Joint Sealant for Portland

Cement Concrete Pavements

ASTM D6241	(2014) Standard Test Method for the Static Puncture Strength of Geotextiles and Geotextile-Related Products Using a 50-mm Probe
ASTM D6392	(2012; R 2018) Standard Test Method for Determining the Integrity of Nonreinforced Geomembrane Seams Produced Using Thermo-Fusion Methods
ASTM D6693/D6693M	(2004; R 2015; E 2015) Standard Test Method for Determining Tensile Properties of Nonreinforced Polyethylene and Nonreinforced Flexible Polypropylene Geomembranes
ASTM D7238	(2006; R 2012) Standard Test Method for Effect of Exposure of Unreinforced Polyolefin Geomembrane Using Fluorescent UV Condensation Apparatus
ASTM E96/E96M	(2022a; E 2023) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials
ASTM E228	(2022) Standard Test Method for Linear Thermal Expansion of Solid Materials with a Push-Rod Dilatometer
ASTM G152	(2013; R 2021) Standard Practice for Operating Open Flame Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials
ASTM G153	(2013; R 2021) Standard Practice for Operating Enclosed Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Installation; G

Liner Projections; G

As-Built Drawings; G

SD-03 Product Data

Fabric Reinforced Geomembrane Liner; G

Non-Reinforced Smooth Geomembrane Liner; G

Nonwoven Geotextile; G

Liner Fittings; G

Embedment Strips; G

Mounting Strip System; G

Sealant; G

Manufacturer's Warranty

Installer's Warranty

SD-04 Samples

Destructive Field Seam Test Sample; G

SD-06 Test Reports

Trial Seam Logs

Non-Destructive Field Seam

Destructive Field Seam

SD-07 Certificates

Field Engineer Qualifications

Installer Qualifications

Fabricator Certification

Liner Manufacturer's Certification

Surface Preparation

SD-08 Manufacturer's Instructions

Liner Manufacturer's Installation Instructions

SD-10 Operation and Maintenance Data

Geomembrane Liner; G

1.4 QUALITY ASSURANCE

1.4.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of a manufacturer regularly engaged in the manufacturing of such products, that are of a similar material, design and workmanship. Materials and equipment must have been in satisfactory commercial or industrial use for a minimum 2 years prior to bid opening. The 2 year period must include applications of the equipment and materials under similar circumstances

and of similar size. Materials and equipment must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period. Products having less than a 2 year field service record will be acceptable if a certified record of satisfactory field operation, for not less than 6000 hours, exclusive of the manufacturer's factory tests, can be shown.

1.4.2 Field Engineer Qualifications

Provide a field engineer who has successfully completed manufacturer's training on handling and installing of the fuel impermeable liner to be installed. Demonstrate that the engineer has at least one million square feet of liner installation experience. Submit a letter providing evidence of the field engineer's experience, training, and licensing. In regard to the field engineer's experience, include in the submittal a point of contact, a phone number, the address, the type of installation, and the current status of each installation mentioned.

1.4.3 Installer Qualifications

The installer is responsible for field handling, deploying, seaming, anchoring, and field Quality Control (QC) testing of the geomembrane. Demonstrate that the installer has installed the proposed geomembrane material for at least 5 completed projects and a total minimum area of **93,000 square meters**. At least one seamer must have experience seaming a minimum of **46,500 square meters** of the proposed geomembrane using the same type of seaming equipment and same geomembrane specified for this project.

1.4.4 Factory Seams

Where possible, use geomembrane factory fabricated to project specific panels in order to minimize field seams. Fabricator must conduct visual inspections on completed seams, as well as non-destructive and destructive testing to verify compliance with the seam strength requirements stated in Table 2. Provide fabricator certification of factory seams, including documentation of and results from quality control testing conducted.

1.4.5 Liner Manufacturer's Certification

Following the successful installation and testing of the liner, an authorized representative from the liner manufacturer must submit a letter certifying that the liner installation and testing results are satisfactory and that each meets the company's quality expectations and warranty. The letter must also certify that the liner installed is compatible with and recommend for use with the fuel to be stored. Include in the letter the representative's name, address, phone number, and qualifications for being a manufacturer's representative.

1.5 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer.

1.5.1 Delivery

The QC inspector must be present during delivery and unloading of the geomembrane. Label each geomembrane roll/panel with the manufacturer's name, product identification number, roll/panel number, and roll/panel

dimensions.

1.5.2 Storage

Store geomembranes and geotextiles elevated off of the ground and covered to provide protection from precipitation, sunlight/ultraviolet light, puncture/abrasion, undesirable chemicals (as recommended by the manufacturer), and flames/welding sparks. Storage in temperatures between 0 degrees C and 71 degrees C, or as recommended by the manufacturer. Storage must not result in crushing the core of roll goods or flattening of the rolls. Do not store rolls more than two high. Store palletized materials on level surfaces and not stacked on top of one another. Remove damaged geomembrane from the site and replace with geomembrane that meets the specified requirements.

1.5.3 Handling

Do not drag, lift by one end, or drop rolls and panels. Use a pipe or solid bar, of sufficient strength to support the full weight of a roll without significant bending but small enough to be easily inserted through the core of the roll, for all handling activities. Link the ends of the pipe or bar to the ends of a spreader bar with chains. Use a spreader bar wide enough to prevent the chains from rubbing against the ends of the roll. Alternatively, a stinger bar protruding from the end of a forklift or other equipment may be used. Use a stinger bar at least three-fourths the length of the core and capable of supporting the full weight of the roll without significant bending. If recommended by the manufacturer, a sling handling method utilizing appropriate loading straps may be used.

1.6 AMBIENT CONDITIONS

Do not deploy or field-seam geomembrane in the presence of excess moisture (i.e., rain, fog, dew), in areas of ponded water, or in the presence of winds above 20km/hr. The relative humidity must be less than 80 percent, and temperature above the dew point. Unless authorized by the Contracting Officer, do not attempt placement or seaming at ambient temperatures below 5 degrees C or above 40 degrees C. Measure ambient temperature at a height no greater than 150 mm above the ground or geomembrane surface. Seaming is only allowed below 5 degrees C if recommended by the geomembrane manufacturer and if destructive tests of trial seams at the proposed temperature meet the seam property requirements listed in Table 1 and 2 for the respective material.

1.7 WARRANTY

State in the manufacturer's warranty that the installed geomembrane liner is warranted for 10 years against deterioration as installed for containment of the intended liquid. State in the installer's warranty that the geomembrane liner won't fail due to improper installation within 2 years.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Fabric Reinforced Geomembrane Liner

Provide a flexible, internally fabric reinforced geomembrane liner that is factory fabricated into widths that are designed to minimize field

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

fabricated seams. Make factory seams with a 50 mm overlap plus or minus 6 mm. Provide liner that, as a minimum, meets the physical properties in Table 1. Include liner's routine maintenance requirements as well as procedures for liner repair and troubleshooting.

TABLE 1 - FABRIC REINFORCED GEOMEMBRANE PROPERTIES (ENGLISH)		
PROPERTY	TEST VALUE	TEST METHOD
Overall Finished Thickness (minimum)	0.76 mm	ASTM D751 or ASTM D5199
Base Fabric Material	aramid fibre, polyester, or nylon	
Base Fabric Weight (minimum)	254 g/m2	ASTM D3776/D3776M
Fabric Coating Adhesion (minimum)	2.6 kN/m	ASTM D751 or ASTM D413
Tensile Strength, Grab (minimum)	2.67 kN in both warp and fill directions	ASTM D751, Grab Test Method
Bursting Strength (minimum)	3.55 kN	ASTM D751, Ball Tip Method
Hydrostatic Resistance (minimum)	5515 kN/m2	ASTM D751, Procedure A
Trapizoid Tearing Strength (minimum)	0.22 kN in both the warp and fill directions	ASTM D4533 or ASTM D751, Trapazoidal Tear Method
Puncture Strength (minimum)	1.11 kN	ASTM D4833/D4833M
Tearing Resistance (minimum)	0.53 kN	ASTM D5884/D5884M
Low Temperature Bend (minimum)	-34 degrees C	ASTM D2136
Vapor Transmission (maximum)	3.78 ml/m2 over 24 hours	ASTM E96/E96M, Procedure BW, Inverted Water Method, using kerosene instead of water
Coefficient of Thermal Expansion (maximum)	0.000038 cm/cm/degree C	ASTM E228 or ASTM D696
Weathering Resistance	No appreciable changes, stiffening or cracking of coating for a minimum of 8000 hours	ASTM G152 or ASTM G153
Bonded Seam Shear Strength(minimum)	2.33 kN	ASTM D751
Bonded Seam Peel Strength (minimum)	88.9 N	ASTM D413
Dead Load Seam Shear Strength (minimum)	1110 N at 21 degrees C; 555 N at 71 degrees C	ASTM D751

2.1.2 Non-Reinforced Smooth Geomembrane Liner

Provide non-reinforced smooth geomembrane liner of unreinforced high density polyethylene (HDPE), manufactured as wide as possible to minimize factory and field seams. Geomembrane sheets must be uniform in color, thickness, and surface texture. The sheets must be free of and resistant to fungal or bacterial attack and free of cuts, abrasions, holes, blisters, contaminants and other imperfections. Provide liner meeting the physical properties of Table 2.

Make resin used in manufacturing geomembrane sheets of virgin uncontaminated ingredients. Use no more than 10 percent regrind, reworked, or trim material in the form of chips or edge strips to manufacture the geomembrane sheets. All regrind, reworked, or trim materials must be from the same manufacturer and exactly the same formulation as the geomembrane sheet being produced. Do not use post consumer materials or water-soluble ingredients to produce the geomembrane. For geomembranes with plasticizers, use only primary plasticizers that are resistant to migration. Submit a copy of the test reports and QC certificates for materials used in the manufacturing of the geomembrane shipped to the site.

TABLE 2 - SMOOTH HDPE GEOMEMBRANE PROPERTIES (ENGLISH)		
PROPERTY	TEST VALUE	TEST METHOD
Thickness (min ave)	1.50 mm	ASTM D5199
Lowest individual of 10 values	-10 percent	ASTM D5199
Density (min)	0.940 g/cc	ASTM D1505
Tensile Properties(1)(min ave)		ASTM D6693/D6693M Type IV
Yield Strength	22 kN/m	
Break Strength	40 kN/m	
Yield Elongation	12 percent	
Break Elongation	700 percent	
Tear Resistance (min ave)	187 N	ASTM D1004
Puncture Resistance(min ave)	480 N	ASTM D4833/D4833M
Stress Crack Resistance (2)	300 hr	ASTM D5397 (Appendix)
Carbon Black Content	2.0-3.0 percent	ASTM D1603 (3)
Carbon Black Dispersion	Note (4)	ASTM D5596

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

TABLE 2 - SMOOTH HDPE GEOMEMBRANE PROPERTIES (ENGLISH)		
PROPERTY	TEST VALUE	TEST METHOD
Oxidative Induction Time (OIT)(min ave)(5)		
Std OIT	100 min	ASTM D3895
High Pres OIT	400 min	ASTM D5885/D5885M
Oven Aging at 85 deg C (min ave) (5), (6)		ASTM D5721
Std OIT	55 percent at 90 days	ASTM D3895
or High Pres OIT	80 percent at 90 days	ASTM D5885/D5885M
UV Resistance (min ave) (7)		ASTM D7238
High Pres OIT(8)(9)	50 percent at 1600 hours	ASTM D5885/D5885M
Seam Shear Strength (min) Hot Wedge, Hot Air, Ultrasonic, Extrusion Fillet (10)(12)	525 N/25mm	ASTM D6392
Seam Peel Strength (min) Hot Wedge, Hot Air, Ultrasonic (10)(11)(12)	398 N/25mm	ASTM D6392
Seam Peel Strength (min) Extrusion Fillet (10)(11)(12)	340 N/25mm	ASTM D6392
(1) Base minimum average machine direction and minimum average cross machine direction values on 5 test specimens in each direction. For HDPE geomembrane, yield elongation is calculated using a gauge length of 33 mm. For HDPE geomembrane, break elongation is calculated using a gauge length of 50 mm.		
(2) For HDPE geomembrane, use the manufacturer's mean value yield stress used to calculate the applied load for test method ASTM D5397 (Appendix).		
(3) Other methods such as ASTM D4218 or microwave methods are acceptable if an appropriate correlation to ASTM D1603 can be established.		
(4) Carbon black dispersion for 10 different views: - minimum 8 of 10 in Categories 1 or 2 - all 10 in Categories 1,2, or 3		
(5) The manufacturer has the option to select either one of the OIT methods to evaluate the antioxidant content.		
(6) Evaluate samples at 30 and 60 days and compare with the 90 day response.		

TABLE 2 - SMOOTH HDPE GEOMEMBRANE PROPERTIES (ENGLISH)		
PROPERTY	TEST VALUE	TEST METHOD
(7) The condition of the test must be a 20 hour UV cycle at 75 degrees C followed by a 4 hour condensation cycle at 60 degrees C.		
(8) Do not use the standard OIT test (ASTM D3895) in determining UV resistance.		
(9) UV resistance is based on percent retained value regardless of the original HP-OIT value.		
(10) Seam tests for peel and shear must fail in the Film Tear Bond mode. This is a failure in the ductile mode of one of the bonded sheets by tearing or breaking prior to complete separation of the bonded area.		
(11) Where applicable, test both tracks of a double hot wedge seam for peel adhesion.		
(12) For five samples tested, four must meet or exceed the given value, with the fifth within 80 percent of the given value.		

2.1.1.3 Nonwoven Geotextile

Provide a nonwoven, polypropylene, needle punched geotextile fabric. Provide geotextile having the following physical properties as a minimum. No substitute methods are allowed for verification of any property.

2.1.1.3.1 General Properties

Retard the growth of mildew and be compatible with the soil in contact.

2.1.1.3.2 Nominal Unit Weight

Provide a nominal unit weight of 407 g/m² as measured in accordance with ASTM D5261.

2.1.1.3.3 Grab Tensile Strength

Provide a minimum grab tensile strength of 1335 N when tested in accordance with ASTM D4632/D4632M.

2.1.1.3.4 Grab Tensile Elongation

Provide a minimum grab tensile elongation of 50 percent when tested in accordance with ASTM D4632/D4632M.

2.1.1.3.5 Puncture Strength

Provide a minimum puncture strength of 620 N when tested in accordance with the pin strength methods of ASTM D4833/D4833M, or 3.56 kN when tested in accordance with the CBR methods of ASTM D6241.

2.1.3.6 Trapezoid Tear Strength

Provide a minimum trapezoid tear strength of 512 N when tested in accordance with ASTM D4533.

2.1.3.7 Ultraviolet (UV) Resistance

Maintain 70 percent of its original strength after 500 hours of testing in accordance with ASTM D4533.

2.2 ACCESSORIES

2.2.1 Liner Fittings

Provide liner fittings (for example, boots and sleeves) that are factory prefabricated components produced from the same manufacturer that produces the fuel impermeable liner and having the same fabrication characteristics as the liner.

2.2.2 Embedment Strip

Provide embedment strips for attachment to new cast-in-place concrete structures by the same manufacturer as the geomembrane, and made of the same materials as the geomembrane to be used or otherwise be certified by the manufacturer for use with the geomembrane.

2.2.3 Batten / Mounting Strip System

Provide minimum 6 mm thick by 50 mm wide carbon steel mounting strips. Provide pre-punched bolt holes in the strip at maximum 300 mm on center to accommodate the concrete anchor bolts.

2.2.3.1 Gasket

Provide a minimum 6 mm thick by 50 mm wide nitrile gasket on one or both sides of the geomembrane as recommended by the geomembrane manufacturer at mounting strip locations to seal and protect the geomembrane from the concrete and the mounting strip.

2.2.3.2 Concrete Anchor Bolts

Use minimum 9 mm diameter mechanical wedge-type concrete anchors, constructed of type 316 stainless steel. Provide nuts, washers, and other accessories of the same materials.

2.2.4 Sealant

Use sealant conforming to ASTM C920, Type S, Grade NS, Class 25 or better, or ASTM D5893/D5893M, Type NS, or other sealant recommended by the geomembrane manufacturer. Sealant used must be compatible with the material being stored, as recommended by the sealant manufacturer.

2.2.5 Temporary Ballast

Temporary ballast used during geomembrane installation may include sandbags, tires, or other material as recommended by the geomembrane manufacturer. Use non-abrasive material, free of sharp edges or other features that may damage the geomembrane liner.

2.3 EQUIPMENT

Utilize equipment in performance of the work in accordance with the geomembrane manufacturer's recommendations and maintain in satisfactory working condition.

PART 3 EXECUTION

3.1 INSTALLATION

Make equipment/parts subject to degradation or requiring adjustment, inspection or repair accessible and capable of convenient removal. Prior to any fabrication or erection, submit detailed [installation](#) drawings that show the proposed panel layout of the liner over the entire containment area. As a minimum, indicate the direction and location of factory and field fabricated seams, the termination of the panels at the perimeter of lined areas, details and methods of sealing around penetrations, details and methods for anchoring, and any applicable site specific installation instructions.

3.1.1 Field Engineer

Provide a field engineer to supervise the complete installation of the liner and perform each liner inspection and test.

3.1.2 Surface Preparation

Prepare surfaces in accordance with liner manufacturer's recommendations. Surfaces to be covered must be free of vegetation, gravel, rocks, debris, etc., graded true, compacted, and be smooth with no abrupt projections. [Install the geotextile in direct contact with the existing subgrade to be covered.](#) [Install the geotextile in strict accordance with manufacturer's recommendations.](#) [Install geotextile to closely fit around projections \(for example, pipe penetrations, concrete foundations/pads, conduit penetrations, etc.\).](#) Submit a signed letter from the field engineer, prior to placing any geotextile or liner, that states the subgrade and [surface preparation](#) were adequate and in accordance with the liner manufacturer's recommendations.

3.1.3 Embedment Strip Installation

Install embedment strips in new cast-in-place concrete structures using methods and materials recommended by the embedment strip/geomembrane manufacturer. Extrusion weld all joints and intersections of embedment material together or otherwise join as recommended by the manufacturer to provided a continuous surface for attachment of the geomembrane. Fill any holes through the face of the embed material resulting from the concrete forming or placement process with material recommended by the manufacturer.

3.1.4 Liner Installation

See paragraph AMBIENT CONDITIONS. Place the liner over the prepared surface in accordance with the [liner manufacturer's installation instructions](#). The procedures and equipment used cannot elongate, wrinkle, scratch, or otherwise damage the geomembrane, [geotextile](#), or the underlying subgrade. Place the liner in such a manner as to assure minimum handling and field seams. Use sand bags (or other manufacturer approved means) to hold the lining down in position during installation. Do not drag or slide materials, equipment or other items across the

surface of the liner at anytime. Do not allow personnel walking or working on the lining to damage it. Replace or repair geomembrane damaged during installation. Allow adequate slack in the geomembrane to prevent the creation of tensile stress and avoid "trampolining" or "bridging". Do not exceed a wrinkle height to width ratio of 0.5 for installed geomembrane. In addition, geomembrane wrinkles must not exceed 75 mm in height. Cut and repair wrinkles that do not meet the above criteria in accordance with paragraph REPAIRS.

3.1.4.1 Liner Projections

Install lining sheets to closely fit around liner projections (for example, pipe penetrations, concrete foundations/pads, conduit penetrations, etc.). Install and center manufacturer supplied sleeves/boots around liner projections in strict accordance with manufacturer's recommendations. Compress the end of pipe sleeves to a pipe with a stainless steel band clamp assembly. For liner anchorage to precast or existing concrete, install a batten / mounting strip around the liner edge and mount with concrete anchor bolts. Apply sealant to the perimeter edge of exposed liner to include the edge of sleeves/boots. For liner anchorage to new cast-in-place concrete, install an embedment strip in the concrete using methods and materials recommended by the embedment strip/geomembrane manufacturer.

3.2 FIELD SEAMING

3.2.1 Trial Seams

Make trial seams under field conditions on strips of excess geomembrane. Make trial seams each day prior to production seaming, whenever there is a change in seaming personnel or seaming equipment settings, whenever environmental conditions change significantly (more than 10 degrees), and at least once every four hours, by each seamer and each piece of seaming equipment used that day. Collect and test trial seam samples in accordance with ASTM D6392 for non-reinforced liner or ASTM D751 and ASTM D413 for fabric-reinforced liner. Obtain one sample from each trial seam. This sample must be at least 450 mm long by 305 mm wide with the seam centered lengthwise. Cut ten random specimens from the sample, 25.4 mm wide. Field test five seam specimens for seam shear strength and field test 5 seam specimens for seam peel adhesion. To be acceptable, 4 out of 5 replicate test specimens must meet seam strength requirements specified in Table 2 and Table 1 respectively. If the field tests fail to meet these requirements, repeat the entire operation. If the additional trial seam fails, do not use the seaming apparatus or seamer until the deficiencies are corrected by the installer and 2 consecutive successful trial seams are achieved. Maintain logs of trial seams, which include date, time, weather, seaming personnel, seaming equipment, equipment settings, and trial seam field test results. Submit trial seam logs following the completion of installation.

3.2.2 Field Seams

Seam panels in accordance with the geomembrane manufacturer's recommendations. Comply with the geomembrane manufacturer's recommendation for the overlap between panels being seamed together; however, in no overlap less than 50 mm. In sumps, corners and odd-shaped geometric locations, minimize the number of field seams. Extend seaming to the outside edge of panels. Compact soft subgrades prior to seaming. Provide a clean and free of moisture, dust, dirt, debris, markings, and

foreign material area at the time of seaming. Completely unroll and layout adjacent liner panels/sheets before performing field seam welds. Repair fish mouths in seams.

3.3 FIELD QUALITY CONTROL

3.3.1 Visual Inspection of Field Seams

Visually inspect each field seam to confirm that the seams are tightly bonded. Perform the inspection of a seam within 30 hours after the manufacturer's suggested application, curing, and cooling time. Repair and re-inspect seams found to be defective in accordance with manufacturer's recommendations.

3.3.2 Non-Destructive Field Seam Testing

Perform non-destructive testing on all field seams over the full seam length and on any other areas showing damage or other distresses. Allowable methods are stated below. Alternate methods approved by the geomembrane manufacturer may be submitted upon approval by the Contracting Officer. Submit [non-destructive field seam](#) test reports following the completion of installation and prior to covering the geomembrane liner.

3.3.2.1 Liner Vacuum Box Test

Perform a vacuum box test in accordance with [ASTM D5641/D5641M](#) on each field seam, the area around the seams, and each liner surface showing injury due to scuffing, penetration by foreign objects, or distress from rough subgrade. If the vacuum box test indicates a continuous stream of bubbles on repeated testing at the same location, then the area being tested is considered damaged and must be repaired and retested. Perform repairs in accordance with manufacturer's recommendations.

3.3.2.2 Liner Air Lance Test

Perform an air lance test on seams to detect an unbonded area in accordance with [ASTM D4437/D4437M](#). Perform the test using a minimum [345 kPa](#) jet of air regulated and directed through a [5 mm](#) diameter nozzle. Apply the jet of air to the lip of a seam in a near perpendicular direction to the length of the seam. Hold the nozzle a maximum of [100 mm](#) from the seam and travel at a rate of not to exceed [12 mpm](#). Inflation of any section of the seam by the impinging air stream is indicative of an unbonded area. Repair unbonded areas in accordance with manufacturer's recommendations and retest.

3.3.2.3 Liner Air Pressure Test

Perform air pressure testing in accordance with [ASTM D5820](#). Upon completion of the test, relieve pressure from the opposite end of the seam being tested to verify continuity of the seam.

3.3.2.4 Liner Point Stress Test

Where other non-destructive test methods are not possible, perform point stress testing in accordance with [ASTM D4437/D4437M](#).

3.3.3 Destructive Field Seam Testing

Obtain a minimum of one destructive test sample per [230 m](#) of field seam

at locations specified by the Contracting Officer. Recommended locations include the ends of seams or anchor trenches, so as to minimize the impact of required repairs to the geomembrane; however, final sample locations will be determined by the Contracting Officer. Sample locations will not be identified prior to seaming. Collect samples a minimum of 305 mm wide by 910 mm long with the seam centered lengthwise. Cut each sample into two equal pieces, with one piece retained by the installer for field testing and the remaining destructive field seam test sample given to the Contracting Officer for QA testing and/or permanent record. Number each sample and cross reference to a field log which identifies: (1) panel number; (2) seam number; (3) date and time cut; (4) ambient temperature within 150 mm above the geomembrane; (5) seaming unit designation; (6) name of seamer; and (7) seaming apparatus temperature and pressures (where applicable). Field test in accordance with ASTM D6392 for non-reinforced liner or ASTM D751 and ASTM D413 for fabric-reinforced liner. Cut ten 25 mm wide replicate specimens from the installer's sample. Test five specimens for shear strength and 5 for peel adhesion. Jaw separation speed must be in accordance with the approved QC manual. To be acceptable, 4 out of 5 replicate test specimens must meet the seam strength requirements specified in Table 2 and Table 1 respectively. If the field tests fail, repair the seam in accordance with paragraph REPAIRS. Repair holes for destructive seam samples the same day they are cut. Submit destructive field seam test reports following completion of installation and prior to covering the geomembrane liner.

3.4 REPAIRS

Patch damaged areas, destructive testing areas, or other areas requiring repair with the geomembrane material. Round the corners of the patch material and extend a minimum of 150 mm in each direction from the damaged area. Small holes or snags less than 6 mm in diameter may be repaired by extrusion welding where such process is approved by the geomembrane manufacturer.

3.5 PROTECTION AND BACKFILLING

Prior to installation of permanent ballast material, the deployed geomembrane must be in intimate contact with the underlying surface, with no areas in sufficient tension to form "bridges" or "trampolines". Minimize waves/wrinkles in the geomembrane prior to placement of ballast materials; in no case allow wrinkles to fold over during placement of ballast materials. Install the cover geotextile or protection board in direct contact with the geomembrane liner to be covered, in strict accordance with manufacturer's recommendations. Install geotextile or protection board to closely fit around projections (for example, pipe penetrations, concrete foundations/pads, conduit penetrations, etc.). Cover the deployed and seamed geomembrane with the specified permanent material within 14 calendar days of acceptance. Do not drop material onto the geomembrane or overlying geotextile from a height greater than 1 m. Push the material out over the geomembrane in an outward/upward tumbling motion. Place ballast material from the bottom of the slope upward.

3.6 PERMANENT BALLAST

3.6.1 Cast in Place Concrete

Place concrete cover over containment geomembrane in accordance with contract documents

3.7 PROJECT CLOSEOUT

Provide [As-Built Drawings](#) which show the as constructed panel layout, factory and field seam locations, penetrations, destructive field seam testing location, and repair locations. Provide operation and maintenance data package for the [geomembrane liner](#) to include material cut sheets, liner's routine maintenance requirements as well as procedures for liner repair and troubleshooting.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 56 21.17

SINGLE WALL ABOVEGROUND FIXED ROOF STEEL POL STORAGE TANK

11/18, CHG 1: 11/20

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 COPIES OF API PUBLICATIONS
- 1.4 RELATED REQUIREMENTS
- 1.5 DESIGN REQUIREMENTS
 - 1.5.1 Seismic Design Requirements
 - 1.5.1.1 Columns
 - 1.5.2 Tank Nozzles
 - 1.5.3 Tank Roof
 - 1.5.4 Corrosion Allowance
 - 1.5.5 Design Metal Temperature
 - 1.5.6 Tank Bottom
 - 1.5.7 Concrete Encasement Structure
- 1.6 QUALIFICATIONS OF TANK ERECTOR
 - 1.6.1 Welding Qualifications
- 1.7 TANK CALIBRATION EXPERIENCE
- 1.8 QUALITY ASSURANCE
 - 1.8.1 Delivery and Storage Handling
 - 1.8.2 Steel Tank Drawing Requirements
 - 1.8.3 Data Requirements
 - 1.8.4 Weld Inspector Certification
 - 1.8.5 Test Reports
 - 1.8.6 Inspection and NDE Personnel
 - 1.8.6.1 NDE Personnel Certification
 - 1.8.6.2 Qualifications of Testing Agency
 - 1.8.7 Qualifications of API Std 653 Inspector

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Materials for System Components, Pipe, and Fittings
- 2.2 STRUCTURAL STEEL
- 2.3 CARBON STEEL, PIPE FITTINGS, FLANGES, GASKETS, AND BOLTING
- 2.4 STAINLESS STEEL PIPE, FITTINGS, FLANGES, GASKETS, AND BOLTING
- 2.5 ALUMINUM PIPING FOR STILLING WELLS
- 2.6 BOLTING AND ALUMINUM FLANGES FOR STILLING WELLS
- 2.7 WELDING FOR ALUMINUM PIPING
 - 2.7.1 Process for Aluminum
 - 2.7.2 Aluminum Welding Electrodes and Rods
- 2.8 BOLTING FOR SHELL MANHOLE COVERS
- 2.9 GASKETS FOR MANHOLE COVERS, STILLING WELL FLANGES, AND ROOF CENTER VENT
- 2.10 INTERIOR PROTECTIVE COATING SYSTEM
- 2.11 EXTERIOR PROTECTIVE COATING SYSTEM

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

2.12 APPURTENANCES

- 2.12.1 Sample Gauge Hatch
- 2.12.2 Mechanical Tape Level Gauge
- 2.12.3 Venting
- 2.12.4 Internal Ladders
- 2.12.5 Internal Ladder Access Hatch
- 2.12.6 Water Draw-Off System
- 2.12.7 Antiseize Compound

PART 3 EXECUTION

3.1 SAFETY PRECAUTIONS

3.2 API Std 653 INSPECTION REPORTS

3.3 CONSTRUCTION

- 3.3.1 Accessibility
- 3.3.2 Tank Erector Site Superintendent
- 3.3.3 Tank
- 3.3.4 Roof Plate Seams
 - 3.3.4.1 Prohibition of Protective Coatings on Surfaces to be Welded
- 3.3.5 Roof Supports
- 3.3.6 Surface Finishing
- 3.3.7 Attachments
- 3.3.8 Nozzles
- 3.3.9 Tank Bottom Sump

3.4 END CONNECTIONS FOR SYSTEM COMPONENTS, VALVES, PIPE, AND FITTINGS

3.5 FIELD QUALITY CONTROL

- 3.5.1 Tank Calibration Table
 - 3.5.1.1 Tank Calibration Method
- 3.5.2 Weld Inspection
- 3.5.3 Reports of Other Tests and Examinations
- 3.5.4 Tightness Tests
 - 3.5.4.1 Penetrating Oil Test
 - 3.5.4.2 Vacuum Box Test of Tank Bottom
 - 3.5.4.3 Hydrostatic Test and Settlement
- 3.5.5 Tank Bottom Puddle Test
- 3.5.6 Fill Test and Related Miscellaneous Tests
 - 3.5.6.1 Fill Test
- 3.5.7 Roof Puddle Test
 - 3.5.7.1 Stilling Well Plumbness Test
- 3.5.8 Retesting
- 3.5.9 Maintenance Instructions
- 3.5.10 Operator Instructions

-- End of Section Table of Contents --

SECTION 33 56 21.17

SINGLE WALL ABOVEGROUND FIXED ROOF STEEL POL STORAGE TANK
11/18, CHG 1: 11/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API MPMS 2.2A	(1995; R 2017) Manual of Petroleum Measurement Standards Chapter 2-Tank Calibration Section 2A-Measurement and Calibration of Upright Cylindrical Tanks by the Manual Tank Strapping Method
API MPMS 2.2B	(1989; R 2019) Manual of Petroleum Measurement Standards Chapter 2: Tank Calibration - Section 2B: Calibration of Upright Cylindrical Tanks Using the Optical Reference Line Method
API MPMS 2.2C	(2002; R 2013) Manual of Petroleum Measurement Standards Chapter 2: Tank Calibration - Section 2C - Calibration of Upright Cylindrical Tanks Using the Optical-Triangulation Method
API MPMS 2.2D	(2003; R 2014) Manual of Petroleum Measurement Standards Chapter 2: Tank Calibration - Section 2D Calibration of Upright Cylindrical Tanks Using the Internal Electro-Optical Distance-Ranging Method
API RP 2009	(2002; R 2007; 7th Ed) Safe Welding, Cutting, and Hot Work Practices in Refineries, Gasoline Plants, and Petrochemical Plants
API Std 650	(2013; Errata 1 2013; Addendum 1 2014; Errata 2 2014; Addendum 2 2016; Addendum 3 2018) Welded Tanks for Oil Storage
API Std 653	(2014; Addendum 1 2018) Tank Inspection, Repair, Alteration, and Reconstruction

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

ASNT SNT-TC-1A	(2020) Recommended Practice for Personnel Qualification and Certification in Nondestructive Testing
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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2018) Factory-Made Wrought Buttwelding Fittings
ASME B16.11	(2016) Forged Fittings, Socket-Welding and Threaded
ASME B31.3	(2020) Process Piping

AMERICAN WELDING SOCIETY (AWS)

AWS A5.10/A5.10M	(2021) Welding Consumables - Wire Electrodes, Wires and Rods for Welding of Aluminum and Aluminum-Alloys - Classification
AWS QC1	(2016) Specification for AWS Certification of Welding Inspectors

ASTM INTERNATIONAL (ASTM)

ASTM B241/B241M	(2016) Standard Specification for Aluminum and Aluminum-Alloy Seamless Pipe and Seamless Extruded Tube
ASTM B247M	(2020) Standard Specification for Aluminum and Aluminum-Alloy Die Forgings, Hand Forgings, and Rolled Ring Forgings (Metric)
ASTM E329	(2021) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection

NACE INTERNATIONAL (NACE)

NACE SP0178	(2007) Design, Fabrication, and Surface Finish Practices for Tanks and Vessels to be Lined for Immersion Service
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NORTH ATLANTIC TREATY ORGANIZATION (NATO)

AFLP-3747	(2013; Rev 9) Guide Specifications (Minimum Quality Standards) for Aviation Turbine Fuels (F-24, F-27, F-34, F-35, F-37, F-40 And F-44)
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U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.23	(Nov 2016) Ladders
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification.
Submittals not having a "G" classification are for Contractor Quality

Control approval. Submit the following in accordance with Section 01 33 00
SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Copies of API Publications; G

Acknowledgement of API Std 650; G

NACE Visual Comparator; G

Acknowledgement of Surface Finish Requirements; G

SD-02 Shop Drawings

Steel Tank; G

Water Draw-Off System; G

SD-03 Product Data

Structural Steel; G

Carbon Steel, Pipe Fittings, Flanges, Gaskets, and Bolting; G

Aluminum Piping; G

Aluminum Flanges; G

Gaskets for Manhole Covers, Stilling Well Flanges, and Roof Center Vent; G

Sample Gauge Hatch; G

Antiseize Compound; G

SD-05 Design Data

Steel Tank Design; G

SD-06 Test Reports

API Std 653 Inspection Reports; G

Visual Examination of Vertical Shell-Seam Tack Welds

Visual Examination of Initial Pass of Internal Shell-to-Bottom Weld

Vacuum Box Testing of Internal Shell-to-Bottom Initial Weld Pass

Visual Examination of Completed Internal and External Shell-to-Bottom Welds

Radiographic Examination of Shell Butt Weld

Visual Examination of Shell Butt Welds

Visual Examination of Fillet Welds

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

Visual Examination of Tank Bottom Plates

Vacuum Box Testing of Tank Bottom Fillet Weld

Pneumatic Tests of Reinforcing Plates

Hydrostatic Testing; G

Shell Settlement Measurements Taken Before, During, and After
Hydrostatic Testing; G

Internal Bottom Elevation Readings Taken Before and After
Hydrostatic Testing; G

Shell Plumbness; G

Shell Roundness; G

Maximum Local Deviations, Shell; G

Tightness Test Records; G

Tank Bottom Puddle Test; G

Roof Puddle Test; G

Stilling Well Plumbness Test; G

SD-07 Certificates

Qualifications of Tank Erector

Welding Procedure Specifications (WPS); G

Welding Procedure Qualification Records (PQRs); G

Welder Performance Qualification Records (WPQ); G

Tank Calibration Experience; G

Weld Inspector Certification; G

NDE Personnel Certification; G

Qualifications of Testing Agency; G

Qualifications of API Std 653 Inspector; G

SD-09 Manufacturer's Field Reports

Mill Test Reports; G

Impact Test Data; G

SD-10 Operation and Maintenance Data

API Std 653 Inspection Reports, Data Package 2; G

Tank Calibration Table, Data Package 2; G

Electronic Calibration Table, Data Package 2; G

Maintenance Instructions, Data Package 2; G

Operator Instructions, Data Package 2; G

Submit in accordance with Section 01 78 23 OPERATION AND
MAINTENANCE DATA.

1.3 COPIES OF API PUBLICATIONS

Provide four copies of API RP 2009, API Std 650, and API MPMS 2.2A, API MPMS 2.2B, API MPMS 2.2C and API MPMS 2.2D to the Contracting Officer.

1.4 RELATED REQUIREMENTS

Materials, design, fabrication, welding, erection, testing, and appurtenances must be in accordance with API Std 650 and as indicated and specified herein. Submit acknowledgement of API Std 650 as required standard. Products to be stored in the tank is APLP-3747 Jet A F-24. Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM apply to this section except as specified otherwise.

1.5 DESIGN REQUIREMENTS

Design tank to resist the loads and forces listed on the structural drawings and for the following:

- a. Design specify gravity of liquid is 0.84.
- b. Design shell and nozzles for a design liquid level equal to overflow condition.

1.5.1 Seismic Design Requirements

Seismic loads and forces must be in accordance with API Std 650 Appendix E.

1.5.1.1 Columns

Provide tank columns as indicated in structural drawings.

1.5.2 Tank Nozzles

Design tank nozzles to accommodate external piping loads in accordance with API Std 650.

1.5.3 Tank Roof

Provide tank roof plates, lap welded with inner plates on top, and at least 6 mm thick (includes 1.5 mm corrosion allowance).

1.5.4 Corrosion Allowance

Provide corrosion allowance of 1.6 mm in thickness of steel for the interior of the shell, roof, and interior structural members.

1.5.5 Design Metal Temperature

API Std 650.

1.5.6 Tank Bottom

Provide tank bottom plates, butt welded, and at least 8 mm thick.

1.5.7 Concrete Encasement Structure

Materials, form work (including shoring), fabrication and testing must be in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE.

1.6 QUALIFICATIONS OF TANK ERECTOR

The Contractor must be regularly engaged in the erection of API Std 650 tanks. The Contractor must certify successful completion of at least 12 field erected API Std 650 aboveground tanks in the past three years. The information provided in the Contractor's certification must include the date of the notice to proceed, date of completion, location of tank, Owner, Owner's point of contact, tank size, configuration (e.g. vertical AST, horizontal AST), product stored, and material of construction.

1.6.1 Welding Qualifications

Submit Welding Procedure Specifications (WPS), Welding Procedure Qualification Records (PQRs), and Welder Performance Qualification Records (WPQ). Qualify all welders on site. Complete all WPQs specifically for this project. Give the Contracting Officer notice and opportunity to witness each of the welder performance qualification tests 24 hours in advance of the performance of each of the tests.

1.7 TANK CALIBRATION EXPERIENCE

Perform calibration of the tank using a qualified organization that can certify to having performed successful and accurate calibration of at least eight tanks of comparable type and size within the last two years. Submit certified data on tank calibration experience.

1.8 QUALITY ASSURANCE

1.8.1 Delivery and Storage Handling

Handle, store, and protect system components and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

1.8.2 Steel Tank Drawing Requirements

Drawings for the steel tank and floating pan must be prepared, sealed, signed, and dated by a registered professional engineer. Include erection diagrams and detail drawings of the tank roof, shell plates, wind girders, openings, and connections for fittings and appurtenances. The steel tank drawings must include the following:

- a. Tank erection details showing dimensions, sizes, thickness, gages, materials, finishes, and erection procedures.

b. Tank component details to include as a minimum:

- (1) Internal pipe and fittings, including supports and bearing plates.
- (2) Location of alarm and control switches.
- (3) Location of nozzles including nozzles for gauges and alarms.
- (4) Roof support system details.
- (5) Roof manhole.
- (6) Circulation vents/inspection hatches.
- (7) Center roof vent.
- (8) Overflow port/circulation vent.
- (9) Shell manholes and davits.
- (10) Stairway, including replaceable stair tread installation and platforms.
- (11) Channel mounting pads.
- (12) Tank Data Plate Plan and Information.
- (13) Shell to bottom connection.
- (14) Stilling wells.
- (15) Grounding lugs.
- (16) Sump.

1.8.3 Data Requirements

Calculations for the [steel tank design](#) must be prepared by a State registered Professional Engineer. Steel tank design calculations must include calculations for the design of the shell, as well as calculations for the design of the roof, and roof support.

1.8.4 [Weld Inspector Certification](#)

Contractor must arrange for the services of an independent (impartial third party not a part or affiliated with Contractor or subcontractor principal or subsidiary businesses) weld inspector certified by the American Welding Society to oversee all weld tests and examinations required by [API Std 650](#).

1.8.5 Test Reports

Test Reports must consist of the following:

- a. Records made by the AWS certified inspector for all duties performed per paragraph 4.2 of [AWS QC1](#).
- b. All Nondestructive Examination (NDE) (e.g.; radiograph, ultrasound, etc) reports with unique weld ID for each weld tested.

- c. "Weld Map." These maps/drawings correlate the shop drawings submitted to the NDE reports. The NDE report that shows a weld number as acceptable is correlated with weld number on the drawings.

Provide the location of each weld, what procedure was used, which welder made the weld, the results of the visual test, and the results of the NDE.

1.8.6 Inspection and NDE Personnel

All inspection and NDE personnel must be qualified in accordance with the following requirements. The Contractor must submit the qualifications of all the testing personnel that will perform all field tests for review by the Contracting Officer. The qualifications of all personnel on the job site that will perform welding inspections and NDE must be submitted for approval. All inspectors and NDE examiners must have a minimum of one-year experience inspecting the piping or plate material being used and five-years in military or commercial fueling systems or petroleum refineries, power generating plants, paper mills, or chemical process plants.

1.8.6.1 NDE Personnel Certification

A written procedure/quality assurance program for the training, examination, certification, control and administration of NDE personnel must be established. The procedures must be based on appropriate specific and general guidelines of training and experience recommended by [ASNT SNT-TC-1A](#). Submit proof of compliance of nondestructive test examiners with [API Std 650](#) including, but not limited to, examiners performing radiographic (RT), visual (VT), penetrant (PT), ultrasonic (UT), and/or magnetic particle (MT) testing.

1.8.6.2 Qualifications of Testing Agency

The testing agency, testing laboratory, technical consultant or contractor's approved quality assurance organization must meet the requirements of [ASTM E329](#).

1.8.7 Qualifications of API Std 653 Inspector

Contractor must arrange for the services of an independent (impartial third party not a part or affiliated with Contractor or subcontractor principal or subsidiary businesses) [API Std 653](#) inspector. [API Std 653](#) Inspector must have a minimum of five years of experience. Submit copy of current certificate.

PART 2 PRODUCTS

2.1 MATERIALS

Conform to the following requirements except that materials not definitely specified must conform to [API Std 650](#). All materials must be carbon steel unless otherwise noted.

2.1.1 Materials for System Components, Pipe, and Fittings

- a. All piping and fittings outside the tank (except for the tank fill line, tank issue line, and tank low suction line) must be stainless steel or carbon steel as indicated on the drawings. The tank fill

line, tank issue line, and tank low suction line piping and fittings must be interior and exterior coated carbon steel. All valves (except DBB valves) and ball joints must be stainless steel. DBB valves must be as specified.

- b. All piping and fittings inside the tank must be exterior and interior epoxy coated carbon steel except for piping 50 mm and smaller which must have an uncoated interior. Stilling well and ladder material must be as indicated.
- c. Do not weld stainless steel to carbon steel, except where specifically indicated or specified.
- d. If materials for system components are not specified, they must be stainless steel.
- e. Provide stainless steel HLV float control chamber, pilot, level switch housings, and level switch probe holders.

2.2 STRUCTURAL STEEL

API Std 650. Provide mill test reports for shell plates, shell nozzle reinforcing plates, shell insert plates, and all steel plate used in construction of shell penetrations. Provide impact test data when required by API Std 650 for the material group and thickness provided.

2.3 CARBON STEEL, PIPE FITTINGS, FLANGES, GASKETS, AND BOLTING

Carbon steel, pipe fittings, flanges, gaskets, and bolting must be provided in accordance with Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT), except gaskets inside tank and on roof nozzles must be non-asbestos, fuel resistant composition, or preformed type. Flanges must be weld-neck type in accordance with ASME B16.5. Threaded fittings must conform to ASME B16.11 (Class 3000), and butt-welded fittings must conform to ASME B16.9.

2.4 STAINLESS STEEL PIPE, FITTINGS, FLANGES, GASKETS, AND BOLTING

Stainless steel pipe, pipe fittings, flanges, gaskets, and bolting must be provided in accordance with Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT) and API Std 650, except: flanges must be weld-neck type in accordance with ASME B16.5, threaded fittings must conform to ASME B16.11 (Class 3000), and butt-welded fittings must conform to ASME B16.9.

2.5 ALUMINUM PIPING FOR STILLING WELLS

Aluminum pipe must be ASTM B241/B241M, alloy 6061-T6, Schedule 40 for pipe sizes 50 mm through 300 mm; Schedule 80 for pipe sizes 50 mm and smaller.

2.6 BOLTING AND ALUMINUM FLANGES FOR STILLING WELLS

Aluminum flanges must be ASME B16.5, Class 150 or Class 300 where indicated, Flat Face Type, except aluminum must conform to ASTM B247M, alloy 6061-T6 or alloy 356-T6. Aluminum flanges may be welding neck or slip-on type. Provide bolting in accordance with Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT). Provide electrical isolation for separation of dissimilar metals.

2.7 WELDING FOR ALUMINUM PIPING

2.7.1 Process for Aluminum

ASME B31.3, Gas Tungsten Arc Welding (GTAW) Process or Gas Metal Arc Welding (GMAW) Process. Backing rings are not permitted.

2.7.2 Aluminum Welding Electrodes and Rods

AWS A5.10/A5.10M, ER5356 electrodes.

2.8 BOLTING FOR SHELL MANHOLE COVERS

Bolting for shell manholes must be in accordance with Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).

2.9 GASKETS FOR MANHOLE COVERS, STILLING WELL FLANGES, AND ROOF CENTER VENT

Provide composition asbestos-free, fuel and fire-resistant gaskets for shell manhole covers, stilling well flanges, and roof center vent.

2.10 INTERIOR PROTECTIVE COATING SYSTEM

Section 09 97 13.17 THREE COAT EPOXY INTERIOR COATING OF WELDED STEEL PETROLEUM FUEL TANKS.

2.11 EXTERIOR PROTECTIVE COATING SYSTEM

Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES.

2.12 APPURTENANCES

2.12.1 Sample Gauge Hatch

Provide sample gauge hatch on top of stilling well where indicated for manual gauging. The hatch must be non-sparking and must have a flanged connection. Provide gasket for dissimilar metal protection. The tank erector must measure the stilling well gauge hatch hold-off distance with the tank empty, half full, and full, and place the information on a permanently affixed marker attached to the sample gauge hatch. This must be performed during tank hydrostatic testing. The stilling well gauge hatch hold-off distance is defined as the distance from the tank bottom datum plate to the top of the gauge hatch.

2.12.2 Mechanical Tape Level Gauge

The mechanical tape level gauge must be complete with all necessary components to be flange mounted and extend down into the tank via a stilling well. The gauge must automatically provide the location of the fuel level within plus or minus 1.6 mm of the actual liquid level. The head must be made of aluminum and must be mounted in the pump house as shown on the drawings. The head must contain a glass covered window complete with an inside wiper. The seals must be made of Teflon. The shafts, graduated tape, and tape drum assembly must be made of stainless steel. The tape must be of sufficient length to measure the liquid level from the bottom to the top of the storage tank. Gauge measurements must be graduated in 1.6 mm increments.

2.12.3 Venting

Provide tank venting as indicated.

2.12.4 Internal Ladders

OSHA 29 CFR 1910.23. Provide an internal ladder extending from the internal ladder access hatch to the tank bottom as indicated. Provide with aluminum safety rail as indicated. Provide removable aluminum safety rail extension as indicated. Provide two 63 mm long. Weld one of the pipes (align vertically) to the top rail of the roof perimeter guardrail near the internal ladder access hatch. Weld the second pipe (align vertically) to the toeboard directly below the first for storing the removable safety rail extension.

2.12.5 Internal Ladder Access Hatch

Provide internal ladder access hatch and 6 mm stainless steel cover with rain lip as indicated for access to the interior of the tank through the roof. Provide with stainless steel hardware (flat bar, round bar, eyebolt).

2.12.6 Water Draw-Off System

A water draw-off system must be provided for each Tank. Provide a water draw-off system complete with all equipment and controls and connected to the Tank as indicated. System must remove fuel/water from its associated tank sump, pump the fuel/water to the water Drain Tank. When the water Drain Tank is full, the fuel/water contents are removed by truck for proper disposal.

2.12.7 Antiseize Compound

Provide marine grade antiseize compound for fasteners on tank exterior flanges and bolted connections and covers. On tank interior fasteners, use oil only.

PART 3 EXECUTION

3.1 SAFETY PRECAUTIONS

API RP 2009 for fire and explosion hazard areas.

3.2 API Std 653 INSPECTION REPORTS

The API Std 653 inspector must inspect the completed tank in accordance with API Std 653 and deliver a full report to the Contracting Officer. The report must include a record of ultrasonic thickness measurements (UTMs), exclusive of the coating, of each tank bottom plate, each bottom shell course plate at 5 random locations per plate, the shell along the circumferential stairway at 5 locations per shell course. The record of UTMs must include sketches of the tank bottom plate and shell plate layouts. The location on each plate, where each ultrasonic thickness measurement (UTM) is taken, must be recorded. Five UTMs must be recorded on each tank bottom plate and on each lowest shell course plate. Five UTMs must be recorded for each of the shell courses above the lowest shell course and must be taken along the circumferential stairway. The report must include the tank dataplate information and photograph of the tank data plate. Provide electronic copies of the tank inspection reports to

Service Headquarters or officially designated alternate, Service Control Points, and DLA-Energy. The paper and electronic copies of the report and UTMs must be provided to the Contracting Officer for filing with the tank's "as-built drawings." Refer to Section 01 45 00 QUALITY CONTROL for API Std 653 inspector certification requirements.

3.3 CONSTRUCTION

3.3.1 Accessibility

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and system components requiring access in locations freely accessible through access doors.

3.3.2 Tank Erector Site Superintendent

Tank erector site superintendent must be on site at all times during any work by that crew.

3.3.3 Tank

Provide tank of welded construction encased in concrete. Slope the tank bottom down to the sump as indicated. Butt weld bottom plates to provide smooth interior bottom surface. Reinforce openings larger than 50 mm in diameter through plating of the tank shell and roof.

3.3.4 Roof Plate Seams

Tank roof plate must be butt welded to provide smooth exterior top surface.

3.3.4.1 Prohibition of Protective Coatings on Surfaces to be Welded

Remove protective coatings on surfaces to be welded and on surfaces within 25 mm from weld preparation. "Weld-through" inorganic zinc coatings and similar coatings will not be permitted.

3.3.5 Roof Supports

Provide column base plates as indicated. Weld the columns to the column base plates. Center the column base plates on the bearing plates and weld the column base clip-guides to the bearing plates. Do not weld the column base plates to their bearing plates. Continuously seal weld the bearing plates to the tank bottom so as to provide a seal against the entry of water or other liquids into the space between the column bearing plates and the tank bottom. Bearing plates must be larger than the base plates by at least 150 mm in either direction. Provide seal-welded cap plates on all columns. Roof support columns must be of pipe or round structural tubing.

3.3.6 Surface Finishing

Provide Contracting Officer with NACE visual comparator as described in NACE SP0178 Section 5. Finish interior surfaces before hydrotesting, in accordance with Section 5 of NACE SP0178 and accompanying Visual Comparator, to the condition described and shown for NACE Weld Designation "C" welds. Finish exterior surfaces, in accordance with Section 5 of NACE SP0178 and accompanying Visual Comparator, to the condition described and shown for NACE Weld Designation "D" welds. Submit acknowledgement of

surface finish requirements. Remove all weld splatter, sharp corners, edges and points from all carbon steel surfaces before coating.

3.3.7 Attachments

All exterior shell and roof attachments must be connected to the tank using continuously welded mounting plates. Mounting plates must exceed the size of the attachment by a minimum of 25 mm. All mounting plate corners must have a 50 mm radius. Attachment must be seal welded to the mounting plate with structurally sound welds of sufficient size to support the intended loads.

3.3.8 Nozzles

All shell nozzles must be flanged type. All nozzles penetrating the tank must be flanged with a steel collar embedded in the concrete encasement. Nozzles for pipe connections inside the tank must be flanged inside near the shell and outside the tank. Reinforcing plates for shell nozzles must be rolled to the curvature of the shell.

3.3.9 Tank Bottom Sump

Weld sump to the underside of the tank bottom at the lowest point of the tank bottom as indicated.

3.4 END CONNECTIONS FOR SYSTEM COMPONENTS, VALVES, PIPE, AND FITTINGS

All valve, system components, pipe and fitting connections including, but not limited to, piping for the Water Draw-Off System, drains, thermal reliefs, HLV float pilot chamber, and level switches must be welded or flanged except as indicated. Piping and fittings 63 mm and larger must be butt welded. Piping and fittings 50 mm and smaller may be butt welded or socket welded. Threaded connections are not allowed except where welded or flanged connections to appurtenances are not available, e.g., pressure gauges, fuel sample connections, level switch probes, HLV float pilot chamber.

3.5 FIELD QUALITY CONTROL

The Contracting Officer will conduct field inspections and witness field tests and trial operations specified in this section. The Contractor must perform all trial operations and field tests and provide all labor, equipment and incidentals required for testing.

3.5.1 Tank Calibration Table

After installation of the tank is complete, provide two calibration tables stamped by a Professional Engineer, one in English units and one in metric units. Tables must be laminated. Both tables must show the volume of the fuel for all liquid levels in the tank starting at the bottom of the sump and going up to the top of the shell. The English unit calibration table must show the volume of fuel in gallons and in barrels of 42 gallons and the level of the fuel in 1/16-inch increments. The metric table must show the volume of the fuel in liters and in m3 and the level of the fuel in 2.0 mm increments. The table must include notes at the bottom indicating 42 gallons = 1 barrel; and one kiloliter = one cubic meter. Volume calculations must be in the smaller units. Larger units may be obtained by rounding off. The 0 mm level must be the level of the bottom of the shell. Level below the bottom of the shell must be shown in negative

units starting at the bottom of the shell. The level of the bottom of the shell, alarm set points, high level shut off valve actuation point, and the level of the [top of the shell](#) must be identified on the calibration table (strapping chart). The table must not include tank volume above the level of the [top of the shell](#). Also, provide [Electronic Calibration Table](#) compatible with the Electronic Automatic Tank Gauging System. Contact Contracting Officer for direction on required format.

3.5.1.1 Tank Calibration Method

The tank gauging systems must be calibrated in accordance with the API Manual of Petroleum Measurement Standards (API MPMS) for critical measurement using methods outlined in one of the following chapters.

- a. [API MPMS 2.2A](#), Measurement and Calibration of Upright Cylindrical Tanks by the Manual Strapping Method.
- b. [API MPMS 2.2B](#), Calibration of Upright Cylindrical Tanks Using the Optical Reference Line Method.
- c. [API MPMS 2.2C](#), Calibration of Upright Cylindrical Tanks Using the Optical Triangulation Method.
- d. [API MPMS 2.2D](#), Calibration of Upright Cylindrical Tanks Using the Internal Electro-Optical Distance Ranging Method.

3.5.2 Weld Inspection

Perform inspection of welds in accordance with [API Std 650](#). Inspect butt welds requiring complete penetration and complete fusion by the radiographic method. Inspect roof support column welds below design liquid level by visual and dye penetrant methods. Submit the following weld inspection reports to the Contracting Officer:

- a. [Visual examination of vertical shell-seam tack welds](#), if left in place, in butt welds.
- b. [Visual examination of initial pass of internal shell-to-bottom weld](#).
- c. [Vacuum box testing of internal shell-to-bottom initial weld pass](#).
- d. [Visual examination of completed internal and external shell-to-bottom welds](#).
- e. [Radiographic examination of shell butt weld](#).

Submit reports for inspection of welds and radiographs to the Contracting Officer.

- f. [Visual examination of shell butt welds](#).
- g. [Visual examination of fillet welds](#).
- h. [Visual examination of tank bottom plates](#) after welding.
- i. [Vacuum box testing of tank bottom fillet weld](#).
- j. [Pneumatic tests of reinforcing plates](#).

3.5.3 Reports of Other Tests and Examinations

Submit reports of the results of the following examinations and tests required by **API Std 650** to the contracting officer:

- a. **Hydrostatic testing.**
- b. **Shell settlement measurements taken before, during, and after hydrostatic testing.**
- c. **Internal bottom elevation readings taken before and after hydrostatic testing.**
- d. **Shell Plumbness.**
- e. **Shell Roundness.**
- f. **Maximum local deviations, shell.**

3.5.4 Tightness Tests

Perform tightness tests described under this paragraph in accordance with **API Std 650**, as modified herein. Perform the tests after finishing welds in accordance with the paragraph SURFACE FINISHING, but prior to blast cleaning and application of the protective coating. Submit **tightness test records** to the Contracting Officer.

3.5.4.1 Penetrating Oil Test

Inspect tank shell-to-bottom, inside corner welds using the penetrating oil test prior to any vacuum box testing. After the initial inside fillet weld is made, apply No. 2 Diesel to the outside of the inside corner weld (before the outside weld is made). After 4 hours, inspect the inside fillet weld for oil penetration through defects. The contractor must correct any defects. Remove oil completely prior to finishing weld joint. Then, complete the remainder of the shell-to-bottom weld joint.

3.5.4.2 Vacuum Box Test of Tank Bottom

Perform a vacuum box test of the tank bottom immediately after installation and after completion of the penetrating oil test and prior to installing any columns. Test seams in bottom of tank and shell-to-bottom joint by applying a commercial soap film and subjecting the seam to a vacuum. Use a glass top vacuum box with hypalon or neoprene sealing gasket. Apply a commercial bubble forming solution to the weld or area to be tested; position the vacuum box over the area and slowly pull a partial vacuum. Observe the solution film for bubble formation between **0-14 kPa** differential pressure. Continue to open the valve until a differential pressure of **34.5 kPa** or **3.50 meters** of water or **259 mm** of mercury is achieved and hold for at least 20 seconds while continuing to observe the solution for bubbles.

3.5.4.3 Hydrostatic Test and Settlement

Perform hydrostatic test with fresh water only. Prior to hydrostatic testing, check the capacity and condition of the tank venting to insure they are adequate to handle the potential rate of fill. This procedure must be accomplished prior to application of coatings and before connecting product/operating piping to the tank. Shell settlement must be

measured before, during, and after hydrostatic testing in accordance with **API Std 650**. Hydrostatic test the shell by filling tank with water and maintaining it full for a period of not less than 24 hours or until the settlement of the tank stabilizes, then inspect shell for leaks. The appearance of damp spots must be considered evidence of leakage. Minimize water retention time to limit rusting of tank interior. Repair leaks disclosed by the test; then, retest the tank to prove the tank is leak-free. water to hydrostatically test the tanks will not be provided free of charge by the Government . Water used on one tank must be recycled to the fullest extent possible for use in testing subsequent tanks. No water must be released to the sanitary or storm sewer systems without the expressed, written approval of the Contracting Officer.

3.5.5 Tank Bottom Puddle Test

Test slope of the tank bottom in the presence of the Contracting Officer by examining the plate immediately after hydrotesting. Puddling deeper than **5 mm** anywhere on the tank bottom plates must not be accepted.

3.5.6 Fill Test and Related Miscellaneous Tests

3.5.6.1 Fill Test

After other tightness testing is complete and after application and cure of the interior and exterior coatings, fill test the tank using fuel. Tank piping and appurtenances must be ready for service. The Government will provide the necessary fuel and labor to fill the tank with fuel. The Government will provide a fill test plan to fill the tanks. The Contractor must remain on standby during fill test to assist in witnessing leaks and performing necessary repair activities. Advise the Contracting Officer, in writing, at least 90 calendar days in advance of the need for this service and provide access to the interior of the tank for the Contracting Officer's inspection to ensure the tank is clean and dry to the Government's satisfaction prior to receiving fuel.

a. Fill Test Stages

The Contractor, with support from the Government, will fill tank at increments as indicated within fill test plan. The fill test plan is generally comprised of the various tank filling activities, fuel quantities needed, hold points at different tank levels and duration, and inspection activities to be performed during the test. Before the tank fill test, check to ensure drain valves are closed. Padlock drain valves closed for the duration of the test and provide one set of keys to the Contracting Officer. If there are no damp spots, discoloration, leaks or a measurable drop in the fuel level during the fill test period, the tank will be accepted. If leakage becomes apparent during the filling or the test period, immediately notify the Contracting Officer and Government personnel will pump the fuel from the tank. Free the tank of vapor, clean it, and then carefully inspect the tank for evidence of failures at the Contractor's expense. Repair defects found and repeat fill tests.

b. Tank High Level Shutoff Valve

Check the operation of the high level shutoff valve on the inlet to the tank to insure that the valve closes completely and as indicated, no later than the high-high level. Check closing by the float valve. Before the tank high level is reached, verify operation of the valve by the manual operation of the float. Check for proper operation when the tank is

filled using appropriate safety measures.

c. Water Draw-Off System

Check System Operation

3.5.7 Roof Puddle Test

After coating, test slope of the finished tank roof plate in the presence of the Contracting Officer by applying water for five minutes, evenly in all directions, at a rate of not more than 20 liters per minute, near the center of the roof, and examining the roof plate for puddling. Puddling deeper than 5 mm anywhere on the tank roof plates will not be accepted.

3.5.7.1 Stilling Well Plumbness Test

All stilling wells must be aligned vertically and tested with a plumb bob in the presence of the Contracting Officer to ensure that they are plumb and are directly centered over the datum plates or sump.

3.5.8 Retesting

Deficiencies found must be rectified and work effected by such deficiencies must be completely retested at the Contractor's expense.

3.5.9 Maintenance Instructions

Provide the following instructions in the Operation and Maintenance Data as follows: Schedule periodic recalibration of ATG at 15 year intervals in accordance with API Manual of Petroleum Measurement Standard (API MPMS) Chapter 2.0 for tanks in custody transfer service and at 15-20 year intervals for all others, or when operating variables of the storage tank change, or when internal dimensions and structural variables of the tank change.

3.5.10 Operator Instructions

Provide the following instructions in the Operation and Maintenance Data as follows:

- a. Test the low-low, low, high and high-high level alarm switches semiannually. Test level switches by simulating product levels either manually or by operating the water stripping system pump and level alarm/control test/drain header valves.
- b. Examine and clear the tank venting semi-annually to ensure the vents have not become plugged.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 71 02

UNDERGROUND ELECTRICAL DISTRIBUTION

08/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 DEFINITIONS
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Precast Underground Structures
 - 1.5.2 Certificate of Competency for Cable Splicer/Terminator
 - 1.5.3 Cable Installer Qualifications
 - 1.5.4 Regulatory Requirements
 - 1.5.5 Standard Products
 - 1.5.5.1 Alternative Qualifications
 - 1.5.5.2 Material and Equipment Manufacturing Date

PART 2 PRODUCTS

- 2.1 CONDUIT, DUCTS, AND FITTINGS
 - 2.1.1 Rigid Metal Conduit
 - 2.1.1.1 Rigid Metallic Conduit, PVC Coated
 - 2.1.2 Intermediate Metal Conduit
 - 2.1.3 Plastic Conduit for Direct Burial and Riser Applications
 - 2.1.4 Plastic Duct for Concrete Encasement
 - 2.1.5 Duct Sealant
 - 2.1.6 Fittings
 - 2.1.6.1 Metal Fittings
 - 2.1.6.2 PVC Conduit Fittings
 - 2.1.6.3 PVC Duct Fittings
- 2.2 LOW VOLTAGE INSULATED CONDUCTORS AND CABLES
 - 2.2.1 Conductor Types
 - 2.2.2 Conductor Material
 - 2.2.3 Jackets
 - 2.2.4 Cable Marking
- 2.3 LOW VOLTAGE WIRE CONNECTORS AND TERMINALS
- 2.4 LOW VOLTAGE SPLICES
 - 2.4.1 Heat Shrinkable Splice
 - 2.4.2 Cold Shrink Rubber Splice
- 2.5 MEDIUM VOLTAGE CABLE
 - 2.5.1 Cable Configuration
 - 2.5.2 Conductor Material
 - 2.5.3 Insulation
 - 2.5.4 Shielding
 - 2.5.5 Jackets
- 2.6 MEDIUM VOLTAGE CABLE TERMINATIONS
 - 2.6.1 Cold-Shrink Type
 - 2.6.2 Heat Shrinkable Type

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.6.3 Separable Insulated Connector Type
- 2.7 MEDIUM VOLTAGE CABLE JOINTS
 - 2.7.1 Heat-Shrinkable Joint
 - 2.7.2 Cold-Shrink Rubber-Type Joint
- 2.8 TELECOMMUNICATIONS CABLING
- 2.9 TAPE
 - 2.9.1 Insulating Tape
 - 2.9.2 Buried Warning and Identification Tape
 - 2.9.3 Fireproofing Tape
- 2.10 PULL ROPE
- 2.11 GROUNDING AND BONDING
 - 2.11.1 Driven Ground Rods
 - 2.11.2 Grounding Conductors
- 2.12 CAST-IN-PLACE CONCRETE
- 2.13 UNDERGROUND STRUCTURES
 - 2.13.1 Cast-In-Place Concrete Structures
 - 2.13.2 Precast Concrete Structures, Risers and Tops
 - 2.13.2.1 General
 - 2.13.2.2 Design for Precast Structures
 - 2.13.2.3 Construction
 - 2.13.2.4 Joints
 - 2.13.3 Manhole Frames and Covers
 - 2.13.4 Handhole Frames and Covers
 - 2.13.5 Composite/Fiberglass Handholes and Covers
- 2.14 CABLE SUPPORTS (RACKS, ARMS, AND INSULATORS)
 - 2.14.1 Cable Rack Stanchions
 - 2.14.2 Rack Arms
 - 2.14.3 Insulators
- 2.15 CABLE TAGS IN MANHOLES
 - 2.15.1 Polyethylene Cable Tags
- 2.16 SOURCE QUALITY CONTROL
 - 2.16.1 Arc-Proofing Test for Cable Fireproofing Tape
 - 2.16.2 Medium Voltage Cable Qualification and Production Tests

PART 3 EXECUTION

- 3.1 INSTALLATION
- 3.2 CABLE INSPECTION
- 3.3 UNDERGROUND STRUCTURE CONSTRUCTION
 - 3.3.1 Cast-In-Place Concrete Structures
 - 3.3.2 Precast Concrete Construction
 - 3.3.3 Pulling-In Irons
 - 3.3.4 Cable Racks, Arms and Insulators
 - 3.3.5 Field Painting
- 3.4 UNDERGROUND CONDUIT AND DUCT SYSTEMS
 - 3.4.1 Requirements
 - 3.4.2 Treatment
 - 3.4.3 Conduit Cleaning
 - 3.4.4 Galvanized Conduit Concrete Penetrations
 - 3.4.5 Multiple Conduits
 - 3.4.6 Conduit Plugs and Pull Rope
 - 3.4.7 Conduit and Duct Without Concrete Encasement
 - 3.4.7.1 Encasement Under Roads and Structures
 - 3.4.8 Duct Encased in Concrete
 - 3.4.8.1 Connections to Manholes
 - 3.4.8.2 Connections to Existing Underground Structures
 - 3.4.8.3 Connections to Existing Concrete Pads
 - 3.4.8.4 Removal of Ducts
 - 3.4.9 Duct Sealing

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 3.5 CABLE PULLING
 - 3.5.1 Cable Lubricants
- 3.6 CABLES IN UNDERGROUND STRUCTURES
 - 3.6.1 Cable Tag Installation
- 3.7 CONDUCTORS INSTALLED IN PARALLEL
- 3.8 LOW VOLTAGE CABLE SPLICING AND TERMINATING
- 3.9 MEDIUM VOLTAGE CABLE TERMINATIONS
- 3.10 MEDIUM VOLTAGE CABLE JOINTS
 - 3.10.1 Joints in Shielded Cables
- 3.11 CABLE END CAPS
- 3.12 FIREPROOFING OF CABLES IN UNDERGROUND STRUCTURES
 - 3.12.1 Fireproofing Tape
- 3.13 GROUNDING SYSTEMS
 - 3.13.1 Grounding Electrodes
 - 3.13.2 Grounding Connections
 - 3.13.3 Grounding Conductors
 - 3.13.4 Ground Cable Crossing Expansion Joints
 - 3.13.5 Manhole Grounding
- 3.14 EXCAVATING, BACKFILLING, AND COMPACTING
 - 3.14.1 Reconditioning of Surfaces
 - 3.14.1.1 Unpaved Surfaces
 - 3.14.1.2 Paving Repairs
- 3.15 CAST-IN-PLACE CONCRETE
 - 3.15.1 Concrete Slabs (Pads) for Equipment
 - 3.15.2 Sealing
- 3.16 FIELD QUALITY CONTROL
 - 3.16.1 Performance of Field Acceptance Checks and Tests
 - 3.16.1.1 Medium Voltage Cables
 - 3.16.1.2 Low Voltage Cables, 600-Volt
 - 3.16.1.3 Grounding System
 - 3.16.2 Follow-Up Verification

-- End of Section Table of Contents --

SECTION 33 71 02

UNDERGROUND ELECTRICAL DISTRIBUTION
08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO HB-17 (2002; Errata 2003; Errata 2005, 17th Edition) Standard Specifications for Highway Bridges

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318M (2014; ERTA 2015) Building Code Requirements for Structural Concrete & Commentary

ACI SP-66 (2004) ACI Detailing Manual

ASSOCIATION OF EDISON ILLUMINATING COMPANIES (AEIC)

AEIC CS8 (2013) Specification for Extruded Dielectric Shielded Power Cables Rated 5 Through 46 kV

ASTM INTERNATIONAL (ASTM)

ASTM B1 (2013) Standard Specification for Hard-Drawn Copper Wire

ASTM B3 (2013) Standard Specification for Soft or Annealed Copper Wire

ASTM B8 (2011; R 2017) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

ASTM B496 (2016; R 2021) Standard Specification for Compact Round Concentric-Lay-Stranded Copper Conductors

ASTM C309 (2019) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete

ASTM C478M (2018) Standard Specification for Precast Reinforced Concrete Manhole Sections

(Metric)

ASTM C857 (2016) Standard Practice for Minimum Structural Design Loading for Underground Precast Concrete Utility Structures

ASTM C990M (2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants (Metric)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 48 (2020) Test Procedures and Requirements for Alternating-Current Cable Terminations Used on Shielded Cables Having Laminated Insulation Rated 2.5 kV through 765 kV or Extruded Insulation Rated 2.5 kV through 500 kV

IEEE 81 (2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System

IEEE 386 (2016) Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV

IEEE 400.2 (2013) Guide for Field Testing of Shielded Power Cable Systems Using Very Low Frequency (VLF)

IEEE 404 (2012) Standard for Extruded and Laminated Dielectric Shielded Cable Joints Rated 2500 V to 500,000 V

IEEE 495 (2007) Guide for Testing Faulted Circuit Indicators

IEEE C2 (2017; Errata 1-2 2017; INT 1 2017) National Electrical Safety Code

IEEE Stds Dictionary (2009) IEEE Standards Dictionary: Glossary of Terms & Definitions

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C119.1 (2016) Electric Connectors - Sealed Insulated Underground Connector Systems Rated 600 Volts

NEMA RN 1 (2005; R 2013) Polyvinyl-Chloride (PVC) Externally Coated Galvanized Rigid Steel

	Conduit and Intermediate Metal Conduit
NEMA TC 2	(2020) Standard for Electrical Polyvinyl Chloride (PVC) Conduit
NEMA TC 9	(2020) Standard for Fittings for Polyvinyl Chloride (PVC) Plastic Utilities Duct for Underground Installation
NEMA WC 74/ICEA S-93-639	(2012) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy
NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)	
NFPA 70	(2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code
SOCIETY OF CABLE TELECOMMUNICATIONS ENGINEERS (SCTE)	
ANSI/SCTE 77	(2013) Specification for Underground Enclosure Integrity
TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)	
TIA-758	(2012b) Customer-Owned Outside Plant Telecommunications Infrastructure Standard
U.S. DEPARTMENT OF AGRICULTURE (USDA)	
RUS Bull 1751F-644	(2002) Underground Plant Construction
U.S. GENERAL SERVICES ADMINISTRATION (GSA)	
CID A-A-60005	(Basic; Notice 2) Frames, Covers, Gratings, Steps, Sump And Catch Basin, Manhole
UNDERWRITERS LABORATORIES (UL)	
UL 6	(2007; Reprint Sep 2019) UL Standard for Safety Electrical Rigid Metal Conduit-Steel
UL 44	(2018; Reprint May 2021) UL Standard for Safety Thermoset-Insulated Wires and Cables
UL 83	(2017; Reprint Mar 2020) UL Standard for Safety Thermoplastic-Insulated Wires and Cables
UL 94	(2013; Reprint Mar 2022) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances
UL 467	(2013; Reprint Jun 2017) UL Standard for Safety Grounding and Bonding Equipment

UL 486A-486B	(2018; Reprint May 2021) UL Standard for Safety Wire Connectors
UL 510	(2020) UL Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape
UL 514B	(2012; Reprint May 2020) Conduit, Tubing and Cable Fittings
UL 651	(2011; Reprint Mar 2020) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
UL 1072	(2006; Reprint Apr 2020) Medium-Voltage Power Cables
UL 1242	(2006; Reprint Aug 2020) Standard for Electrical Intermediate Metal Conduit -- Steel

1.2 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING applies to this section, with the additions and modifications specified herein.

1.3 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE Std Dictionary.
- b. In the text of this section, the words conduit and duct are used interchangeably and have the same meaning.
- c. In the text of this section, "medium voltage cable splices," and "medium voltage cable joints" are used interchangeably and have the same meaning.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Submit modified drawings and engineering calculations associated with design changes required for use of aluminum conductors.

Precast Underground Structures; G

SD-03 Product Data

Medium Voltage Cable; G

Medium Voltage Cable Joints; G

Medium Voltage Cable Terminations; G

Precast Concrete Structures; G

Sealing Material

Pulling-In Irons

Manhole Frames and Covers; G

Handhole Frames and Covers; G

Composite/Fiberglass Handholes; G

Cable Supports (racks, arms and insulators); G
SD-06 Test Reports

Medium Voltage Cable Qualification and Production Tests; G

Field Acceptance Checks and Tests; G

Arc-proofing Test for cable fireproofing tape; G

SD-07 Certificates

Cable splicer/terminator; G

Cable Installer Qualifications; G

1.5 QUALITY ASSURANCE

1.5.1 Precast Underground Structures

Submittal required for each type used. Provide calculations and drawings for precast manholes and handholes bearing the seal of a registered professional engineer including:

- a. Material description (i.e., f'_c and F_y)
- b. Manufacturer's printed assembly and installation instructions
- c. Design calculations
- d. Reinforcing shop drawings in accordance with ACI SP-66
- e. Plans and elevations showing opening and pulling-in iron locations and details

1.5.2 Certificate of Competency for Cable Splicer/Terminator

Submit certification of the qualification of the cable splicer/terminator for approval, 30 days before splices or terminations are to be made in medium voltage (5 kV to 35 kV) cables. Include the training, and experience of the individual on the specific type and classification of cable to be provided under this contract. Indicate that the individual has had three or more years recent experience splicing and terminating medium voltage cables. List a minimum of three splices/terminations that have been in operation for more than one year. In addition, the individual may be required to perform a dummy or practice

splice/termination in the presence of the Contracting Officer, before being approved as a qualified cable splicer. If that additional requirement is imposed, the Contractor must provide short sections of the approved types of cables along with the approved type of splice/termination kit, and detailed manufacturer's instructions for the cable to be spliced. The Contracting Officer reserves the right to require additional proof of competency or to reject the individual and call for certification of an alternate cable splicer.

1.5.3 Cable Installer Qualifications

Provide at least one onsite person in a supervisory position with a documentable level of competency and experience to supervise all cable pulling operations. Provide a resume showing the cable installers' experience in the last three years, including a list of references complete with points of contact, addresses and telephone numbers. Cable installer must demonstrate experience with a minimum of three medium voltage cable installations. The Contracting Officer reserves the right to require additional proof of competency or to reject the individual and call for an alternate qualified cable installer.

1.5.4 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of IEEE C2 and NFPA 70 unless more stringent requirements are specified or indicated.

1.5.5 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.5.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.5.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable, unless specified otherwise.

PART 2 PRODUCTS

2.1 CONDUIT, DUCTS, AND FITTINGS

2.1.1 Rigid Metal Conduit

UL 6.

2.1.1.1 Rigid Metallic Conduit, PVC Coated

NEMA RN 1, Type A40, except that hardness must be nominal 85 Shore A durometer, dielectric strength must be minimum 15.75 kV per mm at 60 Hz, and tensile strength must be minimum 25 MPa.

2.1.2 Intermediate Metal Conduit

UL 1242.

2.1.3 Plastic Conduit for Direct Burial and Riser Applications

UL 651 and NEMA TC 2, EPC-40.

2.1.4 Plastic Duct for Concrete Encasement

Provide Type EPC-40 per UL 651 and NEMA TC 2.

2.1.5 Duct Sealant

UL 94, Class HBF. Provide high-expansion urethane foam duct sealant that expands and hardens to form a closed, chemically and water resistant, rigid structure. Sealant must be compatible with common cable and wire jackets and capable of adhering to metals, plastics and concrete. Sealant must be capable of curing in temperature ranges of 2 degrees C to 35 degrees C. Cured sealant must withstand temperature ranges of -29 degrees C to 93 degrees C without loss of function.

2.1.6 Fittings

2.1.6.1 Metal Fittings

UL 514B.

2.1.6.2 PVC Conduit Fittings

UL 514B, UL 651.

2.1.6.3 PVC Duct Fittings

NEMA TC 9.

2.2 LOW VOLTAGE INSULATED CONDUCTORS AND CABLES

Insulated conductors must be rated 600 volts and conform to the requirements of NFPA 70, including listing requirements. Wires and cables manufactured more than 24 months prior to date of delivery to the site are not acceptable.

2.2.1 Conductor Types

Cable and duct sizes indicated are for copper conductors and THHN/THWN unless otherwise noted. Conductors No. 10 AWG and smaller must be solid. Conductors No. 8 AWG and larger must be stranded. All conductors must be copper.

2.2.2 Conductor Material

Unless specified or indicated otherwise or required by NFPA 70, wires in conduit, other than service entrance, must be 600-volt, Type THWN/THHN conforming to UL 83 or Type XHHW conforming to UL 44. Copper conductors must be annealed copper complying with ASTM B3 and ASTM B8.

2.2.3 Jackets

Provide multiconductor cables with an overall PVC outer jacket.

2.2.4 Cable Marking

Insulated conductors must have the date of manufacture and other identification imprinted on the outer surface of each cable at regular intervals throughout the cable length.

Identify each cable by means of a fiber, laminated plastic, or non-ferrous metal tags in each manhole, handhole, junction box, and each terminal. Each tag must contain the following information; cable type, conductor size, circuit number, circuit voltage, cable destination and phase identification.

Color code conductors. Provide conductor identification within each enclosure where a tap, splice, or termination is made. Conductor identification must be by color-coded insulated conductors, plastic-coated self-sticking printed markers, colored nylon cable ties and plates, heat shrink type sleeves, or colored electrical tape. Properly identify control circuit terminations. Color must be green for grounding conductors and white for neutrals; except where neutrals of more than one system are installed in same raceway or box, other neutrals may be white with a different colored (not green) stripe for each. Color of ungrounded conductors in different voltage systems are as follows:

- a. 208/120 volt, three-phase
 - (1) Phase A - black
 - (2) Phase B - red
 - (3) Phase C - blue
- b. 480/277 volt, three-phase
 - (1) Phase A - brown
 - (2) Phase B - orange
 - (3) Phase C - yellow

2.3 LOW VOLTAGE WIRE CONNECTORS AND TERMINALS

Provide a uniform compression over the entire conductor contact surface. Use solderless terminal lugs on stranded conductors.

- a. For use with copper conductors: [UL 486A-486B](#).

2.4 LOW VOLTAGE SPLICES

Provide splices in conductors with a compression connector on the conductor and by insulating and waterproofing using one of the following methods which are suitable for continuous submersion in water and comply with [ANSI C119.1](#).

2.4.1 Heat Shrinkable Splice

Provide heat shrinkable splice insulation by means of a thermoplastic adhesive sealant material applied in accordance with the manufacturer's written instructions.

2.4.2 Cold Shrink Rubber Splice

Provide a cold-shrink rubber splice which consists of EPDM rubber tube which has been factory stretched onto a spiraled core which is removed during splice installation. The installation must not require heat or flame, or any additional materials such as covering or adhesive. It must be designed for use with inline compression type connectors, or indoor, outdoor, direct-burial or submerged locations.

2.5 MEDIUM VOLTAGE CABLE

Cable (conductor) sizes are designated by American Wire Gauge (AWG) and Thousand Circular Mils (Kcmil). Conductor and conduit sizes indicated are for copper conductors unless otherwise noted. Insulated conductors must have the date of manufacture and other identification imprinted on the outer surface of each cable at regular intervals throughout cable length. Wires and cables manufactured more than 12 months prior to date of delivery to the site are not acceptable. Provide single conductor type cables unless otherwise indicated.

2.5.1 Cable Configuration

Provide Type MV cable, conforming to [NEMA WC 74/ICEA S-93-639](#) and [UL 1072](#). Provide cables manufactured for use in duct applications. Cable must be rated 15 kV with 133 percent insulation level.

2.5.2 Conductor Material

Provide concentric-lay-stranded, Class B conductors. Provide soft drawn copper cables complying with [ASTM B3](#) and [ASTM B8](#) for regular concentric and compressed stranding or [ASTM B496](#) for compact stranding.

2.5.3 Insulation

Provide ethylene-propylene-rubber (EPR) insulation conforming to the requirements of ANSI/NEMA WC 74/ICEA S-93-639 and [AIEIC CS8](#).

2.5.4 Shielding

Cables rated for 2 kV and above must have a semiconducting conductor shield, a semiconducting insulation shield, and an overall copper tape shield for each phase.

2.5.5 Jackets

Provide cables with a PVC jacket. Provide PVC jackets with a separator that prevents contact with underlying semiconducting insulating shield.

2.6 MEDIUM VOLTAGE CABLE TERMINATIONS

IEEE 48 Class 1; of the molded elastomer, prestretched elastomer, or heat-shrinkable elastomer. Acceptable elastomers are track-resistant silicone rubber or track-resistant ethylene propylene compounds, such as ethylene propylene rubber or ethylene propylene diene monomer. Separable insulated connectors may be used for apparatus terminations, when such apparatus is provided with suitable bushings. Provide terminations, where required, with mounting brackets suitable for the intended installation and with grounding provisions for the cable shielding, metallic sheath, or armor. Provide terminations in a kit, including: skirts, stress control terminator, ground clamp, connectors, lugs, and complete instructions for assembly and installation. Terminations must be the product of one manufacturer, suitable for the type, diameter, insulation class and level, and materials of the cable terminated. Do not use separate parts of copper or copper alloy in contact with aluminum alloy parts in the construction or installation of the terminator.

2.6.1 Cold-Shrink Type

Terminator must be a one-piece design, utilizing the manufacturer's latest technology, where high-dielectric constant (capacitive) stress control is integrated within a skirted insulator made of silicone rubber. Termination must not require heat or flame for installation. Termination kit must contain all necessary materials (except for the lugs). Design termination for installation in low or highly contaminated indoor and outdoor locations and must resist ultraviolet rays and oxidative decomposition.

2.6.2 Heat Shrinkable Type

Terminator must consist of a uniform cross section heat shrinkable polymeric construction stress relief tubing and environmentally sealed outer covering that is nontracking, resists heavy atmospheric contaminants, ultra violet rays and oxidative decomposition. Provide heat shrinkable sheds or skirts of the same material. Design termination for installation in low or highly contaminated indoor or outdoor locations.

2.6.3 Separable Insulated Connector Type

IEEE 386. Provide connector with steel reinforced hook-stick eye, grounding eye, test point, and arc-quenching contact material. Provide connectors of the loadbreak or deadbreak type as indicated, of suitable construction for the application and the type of cable connected, and that include cable shield adaptors. Provide external clamping points and test points. Do not use separable connectors in manholes/handholes.

- a. 200 Ampere loadbreak connector ratings: Voltage: 15 kV, 95 kV BIL.

Short time rating: 10,000 rms symmetrical amperes.

- b. Install one set of faulted circuit indicators, complying with IEEE 495, on the test points of each set of separable insulated connectors. Indicators must be self powered; with automatic trip with mechanical flag indication upon overcurrent followed by loss of system voltage, and automatic reset upon restoration of system voltage. Indicators must be compact, sealed corrosion resistant construction with provision for hotstick installation and operation.

2.7 MEDIUM VOLTAGE CABLE JOINTS

Provide joints (splices) in accordance with IEEE 404 suitable for the rated voltage, insulation level, insulation type, and construction of the cable. Joints must be certified by the manufacturer for waterproof, submersible applications. Upon request, supply manufacturer's design qualification test report in accordance with IEEE 404. Connectors for joint must be tin-plated electrolytic copper, having ends tapered and having center stops to equalize cable insertion.

2.7.1 Heat-Shrinkable Joint

Consists of a uniform cross-section heat-shrinkable polymeric construction with a linear stress relief system, a high dielectric strength insulating material, and an integrally bonded outer conductor layer for shielding. Replace original cable jacket with a heavy-wall heat-shrinkable sleeve with hot-melt adhesive coating.

2.7.2 Cold-Shrink Rubber-Type Joint

Joint must be of a cold shrink design that does not require any heat source for its installation. Splice insulation and jacket must be of a one-piece factory formed cold shrink sleeve made of black EPDM rubber. Splice should be packaged three splices per kit, including complete installation instructions.

2.8 TELECOMMUNICATIONS CABLING

Provide telecommunications cabling in accordance with Section 33 82 00 TELECOMMUNICATIONS OUTSIDE PLANT (OSP).

2.9 TAPE

2.9.1 Insulating Tape

UL 510, plastic insulating tape, capable of performing in a continuous temperature environment of 80 degrees C.

2.9.2 Buried Warning and Identification Tape

Provide detectable tape in accordance with Section 31 00 00 EARTHWORK.

2.9.3 Fireproofing Tape

Provide tape composed of a flexible, conformable, unsupported intumescent elastomer. Tape must be not less than 0.762 mm thick, noncorrosive to cable sheath, self-extinguishing, noncombustible, adhesive-free, and must not deteriorate when subjected to oil, water, gases, salt water, sewage, and fungus.

2.10 PULL ROPE

Plastic or flat pull line (bull line) having a minimum tensile strength of **890 N**.

2.11 GROUNDING AND BONDING

2.11.1 Driven Ground Rods

Provide copper-clad steel ground rods conforming to **UL 467** not less than **19 mm** in diameter by **3.1 m** in length. Sectional type rods may be used for rods **6.2 m** or longer.

2.11.2 Grounding Conductors

Stranded-bare copper conductors must conform to **ASTM B8**, Class B, soft-drawn unless otherwise indicated. Solid-bare copper conductors must conform to **ASTM B1** for sizes No. 8 and smaller. Insulated conductors must be of the same material as phase conductors and green color-coded, except that conductors must be rated no more than 600 volts. Aluminum is not acceptable.

2.12 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section **03 30 00** CAST-IN-PLACE CONCRETE. In addition, provide concrete for encasement of underground ducts with **20 MPa** minimum 28-day compressive strength. Concrete associated with electrical work for other than encasement of underground ducts must be **30 MPa** minimum 28-day compressive strength unless specified otherwise.

2.13 UNDERGROUND STRUCTURES

Provide precast concrete underground structures or standard type cast-in-place manhole types as indicated, conforming to **ASTM C857** and **ASTM C478M**. Top, walls, and bottom must consist of reinforced concrete. Walls and bottom must be of monolithic concrete construction. Locate duct entrances and windows near the corners of structures to facilitate cable racking. Covers must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Install a pulling-in iron in the wall opposite each duct line entrance. Cable racks, including rack arms and insulators, must be adequate to accommodate the cable.

2.13.1 Cast-In-Place Concrete Structures

Concrete must conform to Section **03 30 00** CAST-IN-PLACE CONCRETE. Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers. Concrete block is not allowed in areas subject to aircraft loading.

2.13.2 Precast Concrete Structures, Risers and Tops

Precast concrete underground structures may be provided in lieu of cast-in-place subject to the requirements specified below. Precast units

must be the product of a manufacturer regularly engaged in the manufacture of precast concrete products, including precast manholes.

2.13.2.1 General

Precast concrete structures must have the same accessories and facilities as required for cast-in-place structures. Likewise, precast structures must have plan area and clear heights not less than those of cast-in-place structures. Concrete materials and methods of construction must be the same as for cast-in-place concrete construction, as modified herein. Slope in floor may be omitted provided precast sections are poured in reinforced steel forms. Concrete for precast work must have a 28-day compressive strength of not less than 30 MPa. Structures may be precast to the design and details indicated for cast-in-place construction, precast monolithically and placed as a unit, or structures may be assembled sections, designed and produced by the manufacturer in accordance with the requirements specified. Structures must be identified with the manufacturer's name embedded in or otherwise permanently attached to an interior wall face.

2.13.2.2 Design for Precast Structures

ACI 318M. In the absence of detailed on-site soil information, design for the following soil parameters/site conditions:

- a. Angle of Internal Friction (ϕ) = 0.523 rad
- b. Unit Weight of Soil (Dry) = 1760 kg/m³, (Saturated)
= 2080 kg/m³
- c. Coefficient of Lateral Earth Pressure (K_a) = 0.33
- d. Ground Water Level = 915 mm below ground elevation
- e. Vertical design loads must include full dead, superimposed dead, and live loads including a 30 percent magnification factor for impact. Live loads must consider all types and magnitudes of vehicular (automotive, industrial, or aircraft) traffic to be encountered. The minimum design vertical load must be for H20 highway loading per AASHTO HB-17.
- f. Horizontal design loads must include full geostatic and hydrostatic pressures for the soil parameters, water table, and depth of installation to be encountered. Also, horizontal loads imposed by adjacent structure foundations, and horizontal load components of vertical design loads, including impact, must be considered, along with a pulling-in iron design load of 26,700 N.
- g. Each structural component must be designed for the load combination and positioning resulting in the maximum shear and moment for that particular component.
- h. Design must also consider the live loads induced in the handling, installation, and backfilling of the manholes. Provide lifting devices to ensure structural integrity during handling and installation.

2.13.2.3 Construction

Provide a uniform thickness for structure top, bottom, and wall not less than 150 mm. Thin-walled knock-out panels for designed or future duct bank entrances are not permitted. Provide quantity, size, and location of duct bank entrance windows as directed, and cast completely open by the precaster. Size of windows must exceed the nominal duct bank envelope dimensions by at least 305 mm vertically and horizontally to preclude in-field window modifications made necessary by duct bank misalignment. However, the sides of precast windows must be a minimum of 150 mm from the inside surface of adjacent walls, floors, or ceilings. Form the perimeter of precast window openings to have a keyed or inward flared surface to provide a positive interlock with the mating duct bank envelope. Provide welded wire fabric reinforcing through window openings for in-field cutting and flaring into duct bank envelopes. Provide additional reinforcing steel comprised of at least two No. 4 bars around window openings. Provide drain sumps a minimum of 305 mm in diameter and 100 mm deep for precast structures.

2.13.2.4 Joints

Provide tongue-and-groove joints on mating edges of precast components. Shiplap joints are not allowed. Design joints to firmly interlock adjoining components and to provide waterproof junctions and adequate shear transfer. Seal joints watertight using preformed plastic strip conforming to ASTM C990M. Install sealing material in strict accordance with the sealant manufacturer's printed instructions. Provide waterproofing at conduit/duct entrances into structures, and where access frame meets the top slab, provide continuous grout seal.

2.13.3 Manhole Frames and Covers

Provide cast iron frames and covers for manholes conforming to CID A-A-60005. Cast the words "ELECTRIC" or "TELECOMMUNICATIONS" in the top face of power and telecommunications manhole covers, respectively.

2.13.4 Handhole Frames and Covers

Frames and covers of steel must be welded by qualified welders in accordance with standard commercial practice. Provide rolled-steel floor plate covers having an approved antislip surface. Hinges must be of stainless steel with bronze hinge pin, 125 by 125 mm by approximately 4.75 mm thick, without screw holes, and must be for full surface application by fillet welding. Hinges must have nonremovable pins and five knuckles. The surfaces of plates under hinges must be true after the removal of raised antislip surface, by grinding or other approved method.

2.13.5 Composite/Fiberglass Handholes and Covers

ANSI/SCTE 77. Provide handholes and covers of polymer concrete, reinforced with heavy weave fiberglass with a design load (Tier rating) appropriate for or greater than the intended use. All covers are required to have the Tier level rating embossed on the surface which must not exceed the design load of the box.

2.14 CABLE SUPPORTS (RACKS, ARMS, AND INSULATORS)

Zinc coat the metal portion of racks and arms after fabrication.

2.14.1 Cable Rack Stanchions

The wall bracket or stanchion must be 100 mm by approximately 38 mm by 4.76 mm channel steel, or 100 mm by approximately 25 mm glass-reinforced nylon with recessed bolt mounting holes, 1220 mm long (minimum) in manholes. Space slots for mounting cable rack arms at 200 mm intervals.

2.14.2 Rack Arms

Cable rack arms must be steel or malleable iron or glass reinforced nylon and must be of the removable type. Rack arm length must be a minimum of 200 mm and a maximum of 305 mm.

2.14.3 Insulators

Insulators for metal rack arms must be dry-process glazed porcelain. Insulators are not required for nylon arms.

2.15 CABLE TAGS IN MANHOLES

Provide polyethylene tags for each power cable located in manholes. Do not provide handwritten letters. The first position on the power cable tag denotes the voltage. The second through sixth positions on the tag identifies the circuit. The next to last position denotes the phase of the circuit and include the Greek "phi" symbol. The last position denotes the cable size. As an example, a tag could have the following designation: "11.5 NAS 1-8(Phase A)500," denoting that the tagged cable is on the 11.5kV system circuit number NAS 1-8, underground, Phase A, sized at 500 kcmil.

2.15.1 Polyethylene Cable Tags

Provide tags of polyethylene having an average tensile strength of 22.4 MPa; and that are 2 millimeter thick (minimum), non-corrosive non-conductive; resistive to acids, alkalis, organic solvents, and salt water; and distortion resistant to 77 degrees C. Provide 1.3 mm (minimum) thick black polyethylene tag holder. Provide a one-piece nylon, self-locking tie at each end of the cable tag, having a minimum loop tensile strength of 778.75 N and black block letters, numbers, and symbols 25 mm high on a yellow background. Letters, numbers, and symbols must not fall off or change positions regardless of the cable tags' orientation.

2.16 SOURCE QUALITY CONTROL

2.16.1 Arc-Proofing Test for Cable Fireproofing Tape

Manufacturer must test one sample assembly consisting of a straight lead tube 305 mm long with a 65.5 mm outside diameter, and a 3.175 mm thick wall, and covered with one-half lap layer of arc and fireproofing tape per manufacturer's instructions. The arc and fireproofing tape must withstand extreme temperature of a high-current fault arc 13,000 degrees K for 70 cycles as determined by using an argon directed plasma jet capable of constantly producing and maintaining an arc temperature of 13,000 degrees K. Temperature (13,000 degrees K) of the ignited arc between the cathode and anode must be obtained from a dc power source of 305 (plus or minus 5) amperes and 20 (plus or minus 1) volts. Direct the arc toward the sample assembly accurately positioned 5 (plus or minus 1) millimeters downstream in the plasma from the anode orifice by fixed flow rate of argon gas (0.18 g per second). Test each sample assembly at three unrelated points.

Start time for tests must be taken from recorded peak current when the specimen is exposed to the full test temperature. Surface heat on the specimen prior to that time must be minimal. The end point is established when the plasma or conductive arc penetrates the protective tape and strikes the lead tube. Submittals for arc-proofing tape must indicate that the test has been performed and passed by the manufacturer.

2.16.2 Medium Voltage Cable Qualification and Production Tests

Results of **AEIC CS8** qualification and production tests as applicable for each type of medium voltage cable.

PART 3 EXECUTION

3.1 INSTALLATION

Install equipment and devices in accordance with the manufacturer's published instructions and with the requirements and recommendations of **NFPA 70** and **IEEE C2** as applicable. In addition to these requirements, install telecommunications in accordance with **TIA-758** and **RUS Bull 1751F-644**.

3.2 CABLE INSPECTION

Inspect each cable reel for correct storage positions, signs of physical damage, and broken end seals prior to installation. If end seal is broken, remove moisture from cable prior to installation in accordance with the cable manufacturer's recommendations.

3.3 UNDERGROUND STRUCTURE CONSTRUCTION

Provide standard type cast-in-place construction as specified herein and as indicated, or precast construction as specified herein. Horizontal concrete surfaces of floors must have a smooth trowel finish. Cure concrete by applying two coats of white pigmented membrane forming-curing compound in strict accordance with the manufacturer's printed instructions, except that precast concrete may be steam cured. Curing compound must conform to **ASTM C309**. Locate duct entrances and windows in the center of end walls (shorter) and near the corners of sidewalls (longer) to facilitate cable racking and splicing. Covers for underground structures must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Manhole locations, as indicated, are approximate. Coordinate exact manhole locations with other utilities and finished grading and paving.

3.3.1 Cast-In-Place Concrete Structures

Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers.

3.3.2 Precast Concrete Construction

Set commercial precast structures on **150 mm** of level, 90 percent compacted granular fill, **19 mm to 25 mm** size, extending **305 mm** beyond the structure on each side. Compact granular fill by a minimum of four passes with a

plate type vibrator. Installation must additionally conform to the manufacturer's instructions.

3.3.3 Pulling-In Irons

Provide steel bars bent as indicated, and cast in the walls and floors. Alternatively, pipe sleeves may be precast into the walls and floors where required to accept U-bolts or other types of pulling-in devices possessing the strengths and clearances stated herein. The final installation of pulling-in devices must be made permanent. Cover and seal exterior projections of thru-wall type pulling-in devices with an appropriate protective coating. In the floor, locate the irons a minimum of 150 mm from the edge of the sump, and in the walls, locate the irons within 150 mm of the projected center of the duct bank pattern or precast window in the opposite wall. However, the pulling-in iron must not be located within 150 mm of an adjacent interior surface, or duct or precast window located within the same wall as the iron. If a pulling-in iron cannot be located directly opposite the corresponding duct bank or precast window due to this clearance limitation, locate the iron directly above or below the projected center of the duct bank pattern or precast window the minimum distance required to preserve the 150 mm clearance previously stated. In the case of directly opposing precast windows, pulling-in irons consisting of a 915 mm length of No. 5 reinforcing bar, formed into a hairpin, may be cast-in-place within the precast windows simultaneously with the end of the corresponding duct bank envelope. Irons installed in this manner must be positioned directly in line with, or when not possible, directly above or below the projected center of the duct bank pattern entering the opposite wall, while maintaining a minimum clear distance of 75 mm from any edge of the cast-in-place duct bank envelope or any individual duct. Pulling-in irons must have a clear projection into the structure of approximately 100 mm and must be designed to withstand a minimum pulling-in load of 26,700 N. Hot-dip galvanize irons after fabrication.

3.3.4 Cable Racks, Arms and Insulators

Cable racks, arms and insulators must be sufficient to accommodate the cables. Space racks in power manholes not more than 915 mm apart, and provide each manhole wall with a minimum of two racks. Space racks in signal manholes not more than 420 mm apart with the end rack being no further than 305 mm from the adjacent wall. Methods of anchoring cable racks are as follows:

- a. Provide a 15 mm diameter by 125 mm long anchor bolt with 75 mm foot cast in structure wall with 50 mm protrusion of threaded portion of bolt into structure. Provide 15 mm steel square head nut on each anchor bolt. Coat threads of anchor bolts with suitable coating immediately prior to installing nuts.
- b. Provide concrete channel insert with a minimum load rating of 1192 kg per meter. Insert channel must be steel of the same length as "vertical rack channel;" and cast flush in structure wall. Provide 15 mm steel nuts in channel insert to receive 15 mm diameter by 75 mm long steel, square head anchor bolts.
- c. Provide concrete "spot insert" at each anchor bolt location, cast flush in structure wall. Each insert must have minimum 365 kg load rating. Provide 15 mm diameter by 75 mm long steel, square head anchor bolt at each anchor point. Coat threads of anchor bolts with

suitable coating immediately prior to installing bolts.

3.3.5 Field Painting

Clean cast-iron frames and covers not buried in concrete or masonry of mortar, rust, grease, dirt and other deleterious materials, and coat with bituminous paint.

3.4 UNDERGROUND CONDUIT AND DUCT SYSTEMS

3.4.1 Requirements

Run conduit in straight lines except where a change of direction is necessary. Provide numbers and sizes of ducts as indicated. Provide a 4/0 AWG bare copper grounding conductor above medium-voltage distribution duct banks. Bond bare copper grounding conductor to ground rings (loops) in all manholes and to ground rings (loops) at all equipment slabs (pads). Route grounding conductor into manholes with the duct bank (sleeving is not required). Ducts must have a continuous slope downward toward underground structures and away from buildings, laid with a minimum slope of **75 mm per 30 m**. Depending on the contour of the finished grade, the high-point may be at a terminal, a manhole, a handhole, or between manholes or handholes. Terminate all PVC conduit end points in utility holes, switching cabinets, transform handholes and buildings with end bells. The bell end of the conduits that enter manholes and handholes must be flush with the wall.

Perform changes in ductbank direction as follows:

- a. Short-radius manufactured 90-degree duct bends may be used only for pole or equipment risers, unless specifically indicated as acceptable.
- b. The minimum manufactured bend radius must be **450 mm** for ducts of less than **80 mm** diameter, and **900 mm** for ducts **80 mm** or greater in diameter.
- c. As an exception to the bend radius required above, provide field manufactured longsweep bends having a minimum radius of **7.6 m** for a change of direction of more than 5 degrees, either horizontally or vertically, using a combination of curved and straight sections. Maximum manufactured curved sections allowed for use in field manufactured longsweep bend: 30 degrees.

3.4.2 Treatment

Keep ducts clean of concrete, dirt, or foreign substances during construction. Make field cuts requiring tapers with proper tools and match factory tapers. Use a coupling recommended by the duct manufacturer whenever an existing duct is connected to a duct of different material or shape. Store ducts to avoid warping and deterioration with ends sufficiently plugged to prevent entry of any water or solid substances. Thoroughly clean ducts before being laid. Store plastic ducts on a flat surface and protected from the direct rays of the sun.

3.4.3 Conduit Cleaning

As each conduit run is completed, for conduit sizes **75 mm** and larger, draw a flexible testing mandrel approximately **305 mm** long with a diameter less than the inside diameter of the conduit through the conduit. After which,

draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs. For conduit sizes less than 75 mm, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs.

3.4.4 Galvanized Conduit Concrete Penetrations

Galvanized conduits which penetrate concrete (slabs, pavement, and walls) in wet locations must be PVC coated and extend from at least 50 mm within the concrete to the first coupling or fitting outside the concrete (minimum of 150 mm from penetration).

3.4.5 Multiple Conduits

Separate multiple conduits by a minimum distance of 75 mm, except that light and power conduits must be separated from control, signal, and telephone conduits by a minimum distance of 300 mm. Stagger the joints of the conduits by rows (horizontally) and layers (vertically) to strengthen the conduit assembly. Provide plastic duct spacers that interlock vertically and horizontally. Spacer assembly must consist of base spacers, intermediate spacers, ties, and locking device on top to provide a completely enclosed and locked-in conduit assembly. Install spacers per manufacturer's instructions, but provide a minimum of two spacer assemblies per 3050 mm of conduit assembly.

3.4.6 Conduit Plugs and Pull Rope

Provide new conduit indicated as being unused or empty with plugs on each end. Plugs must contain a weephole or screen to allow water drainage. Provide a plastic pull rope having 915 mm of slack at each end of unused or empty conduits.

3.4.7 Conduit and Duct Without Concrete Encasement

Depths to top of the conduit must be not less than 610 mm below finished grade. Provide not less than 75 mm clearance from the conduit to each side of the trench. Grade bottom of trench smooth; where rock, soft spots, or sharp-edged materials are encountered, excavate the bottom for an additional 75 mm, fill and tamp level with original bottom with sand or earth free from particles, that would be retained on a 6.25 mm sieve. The first 150 mm layer of backfill cover must be sand compacted as previously specified. The rest of the excavation must be backfilled and compacted in 75 to 150 mm layers. Provide color, type and depth of warning tape as specified in Section 31 00 00 EARTHWORK.

3.4.7.1 Encasement Under Roads and Structures

Under roads, paved areas, and railroad tracks, install conduits in concrete encasement of rectangular cross-section providing a minimum of 75 mm concrete cover around ducts. Extend concrete encasement at least 1525 mm beyond the edges of paved areas and roads, and 3660 mm beyond the rails on each side of railroad tracks. Depths to top of the concrete envelope must be not less than 610 mm below finished grade.

3.4.8 Duct Encased in Concrete

Construct underground duct lines of individual conduits encased in concrete. Depths to top of the concrete envelope must be not less than

450 mm below finished grade, except under roads and pavement, concrete envelope must be not less than 610 mm below finished grade. Do not mix different kinds of conduit in any one duct bank. Concrete encasement surrounding the bank must be rectangular in cross-section and provide at least 75 mm of concrete cover for ducts. Separate conduits by a minimum concrete thickness of 75 mm. Before pouring concrete, anchor duct bank assemblies, prevent floating during concrete pouring by driving reinforcing rods adjacent to duct spacer assemblies and attaching the rods to the spacer assembly. Provide steel reinforcing in the concrete envelope as indicated. Provide color, type and depth of warning tape as specified in Section 31 00 00 EARTHWORK.

3.4.8.1 Connections to Manholes

Duct bank envelopes connecting to underground structures must be flared to have enlarged cross-section at the manhole entrance to provide additional shear strength. Dimensions of the flared cross-section must be larger than the corresponding manhole opening dimensions by no less than 300 mm in each direction. Perimeter of the duct bank opening in the underground structure must be flared toward the inside or keyed to provide a positive interlock between the duct bank and the wall of the structure. Use vibrators when this portion of the encasement is poured to assure a seal between the envelope and the wall of the structure.

3.4.8.2 Connections to Existing Underground Structures

For duct bank connections to existing structures, break the structure wall out to the dimensions required and preserve steel in the structure wall. Cut steel and extend into the duct bank envelope. Chip the perimeter surface of the duct bank opening to form a key or flared surface, providing a positive connection with the duct bank envelope.

3.4.8.3 Connections to Existing Concrete Pads

For duct bank connections to concrete pads, break an opening in the pad out to the dimensions required and preserve steel in pad. Cut the steel and extend into the duct bank envelope. Chip out the opening in the pad to form a key for the duct bank envelope.

3.4.8.4 Removal of Ducts

Where duct lines are removed from existing underground structures, close the openings to waterproof the structure. Chip out the wall opening to provide a key for the new section of wall.

3.4.9 Duct Sealing

Seal all electrical penetrations for radon mitigation, maintaining integrity of the vapor barrier, and to prevent infiltration of air, insects, and vermin.

3.5 CABLE PULLING

Test existing duct lines with a mandrel and thoroughly swab out to remove foreign material before pulling cables. Pull cables down grade with the feed-in point at the manhole or buildings of the highest elevation. Use flexible cable feeds to convey cables through manhole opening and into duct runs. Do not exceed the specified cable bending radii when installing cable under any conditions, including turnups into switches,

transformers, switchgear, switchboards, and other enclosures. Cable with tape shield must have a bending radius not less than 12 times the overall diameter of the completed cable. If basket-grip type cable-pulling devices are used to pull cable in place, cut off the section of cable under the grip before splicing and terminating.

3.5.1 Cable Lubricants

Use lubricants that are specifically recommended by the cable manufacturer for assisting in pulling jacketed cables.

3.6 CABLES IN UNDERGROUND STRUCTURES

Do not install cables utilizing the shortest path between penetrations, but route along those walls providing the longest route and the maximum spare cable lengths. Form cables to closely parallel walls, not to interfere with duct entrances, and support on brackets and cable insulators. Support cable splices in underground structures by racks on each side of the splice. Locate splices to prevent cyclic bending in the spliced sheath. Install cables at middle and bottom of cable racks, leaving top space open for future cables, except as otherwise indicated for existing installations. Provide one spare three-insulator rack arm for each cable rack in each underground structure.

3.6.1 Cable Tag Installation

Install cable tags in each manhole as specified, including each splice. Tag wire and cable provided by this contract. Install cable tags over the fireproofing, if any, and locate the tags so that they are clearly visible without disturbing any cabling or wiring in the manholes.

3.7 CONDUCTORS INSTALLED IN PARALLEL

Group conductors such that each conduit of a parallel run contains one Phase A conductor, one Phase B conductor, one Phase C conductor, and one neutral conductor.

3.8 LOW VOLTAGE CABLE SPLICING AND TERMINATING

Make terminations and splices with materials and methods as indicated or specified herein and as designated by the written instructions of the manufacturer. Do not allow the cables to be moved until after the splicing material has completely set. Make splices in underground distribution systems only in accessible locations such as manholes, handholes, or aboveground termination pedestals.

3.9 MEDIUM VOLTAGE CABLE TERMINATIONS

Make terminations in accordance with the written instruction of the termination kit manufacturer.

3.10 MEDIUM VOLTAGE CABLE JOINTS

Provide power cable joints (splices) suitable for continuous immersion in water. Make joints only in accessible locations in manholes or handholes by using materials and methods in accordance with the written instructions of the joint kit manufacturer.

3.10.1 Joints in Shielded Cables

Cover the joined area with metallic tape, or material like the original cable shield and connect it to the cable shield on each side of the splice. Provide a bare copper ground connection brought out in a watertight manner and grounded to the manhole grounding loop as part of the splice installation. Ground conductors, connections, and rods must be as specified elsewhere in this section. Wire must be trained to the sides of the enclosure to prevent interference with the working area.

3.11 CABLE END CAPS

Cable ends must be sealed at all times with coated heat shrinkable end caps. Cables ends must be sealed when the cable is delivered to the job site, while the cable is stored and during installation of the cable. The caps must remain in place until the cable is spliced or terminated. Sealing compounds and tape are not acceptable substitutes for heat shrinkable end caps. Cable which is not sealed in the specified manner at all times will be rejected.

3.12 FIREPROOFING OF CABLES IN UNDERGROUND STRUCTURES

Fireproof (arc proof) wire and cables which will carry current at 2200 volts or more in underground structures.

3.12.1 Fireproofing Tape

Tightly wrap strips of fireproofing tape around each cable spirally in half-lapped wrapping. Install tape in accordance with manufacturer's instructions.

3.13 GROUNDING SYSTEMS

NFPA 70 and **IEEE C2**, except provide grounding systems with a resistance to solid earth ground not exceeding 25 ohms.

3.13.1 Grounding Electrodes

Provide cone pointed driven ground rods driven full depth plus **150 mm**, installed to provide an earth ground of the appropriate value for the particular equipment being grounded.

If the specified ground resistance is not met, provide an additional ground rod in accordance with the requirements of **NFPA 70** (placed not less than 6 feet from the first rod). Should the resultant (combined) resistance exceed the specified resistance, measured not less than 48 hours after rainfall, notify the Contracting Officer immediately.

3.13.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, by exothermic weld or compression connector.

- a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.

- b. Make compression connections using a hydraulic compression tool to provide the correct circumferential pressure. Tools and dies must be as recommended by the manufacturer. An embossing die code or other standard method must provide visible indication that a connector has been adequately compressed on the ground wire.

3.13.3 Grounding Conductors

Provide bare grounding conductors, except where installed in conduit with associated phase conductors. Ground cable sheaths, cable shields, conduit, and equipment with No. 6 AWG. Ground other noncurrent-carrying metal parts and equipment frames of metal-enclosed equipment. Ground metallic frames and covers of handholes and pull boxes with a braided, copper ground strap with equivalent ampacity of No. 6 AWG. Provide direct connections to the grounding conductor with 600 v insulated, full-size conductor for each grounded neutral of each feeder circuit, which is spliced within the manhole.

3.13.4 Ground Cable Crossing Expansion Joints

Protect ground cables crossing expansion joints or similar separations in structures and pavements by use of approved devices or methods of installation which provide the necessary slack in the cable across the joint to permit movement. Use stranded or other approved flexible copper cable across such separations.

3.13.5 Manhole Grounding

Loop a 4/0 AWG grounding conductor around the interior perimeter, approximately 305 mm above finished floor. Secure the conductor to the manhole walls at intervals not exceeding 914 mm. Connect the conductor to the manhole grounding electrode with 4/0 AWG conductor. Connect all incoming 4/0 grounding conductors to the ground loop adjacent to the point of entry into the manhole. Bond the ground loop to all cable shields, metal cable racks, and other metal equipment with a minimum 6 AWG conductor.

3.14 EXCAVATING, BACKFILLING, AND COMPACTING

Provide in accordance with NFPA 70 and Section 31 00 00 EARTHWORK.

3.14.1 Reconditioning of Surfaces

3.14.1.1 Unpaved Surfaces

Restore to their original elevation and condition unpaved surfaces disturbed during installation of duct. Preserve sod and topsoil removed during excavation and reinstall after backfilling is completed. Replace sod that is damaged by sod of quality equal to that removed. When the surface is disturbed in a newly seeded area, re-seed the restored surface with the same quantity and formula of seed as that used in the original seeding, and provide topsoiling, fertilizing, liming, seeding, sodding, sprigging, or mulching. Provide work in accordance with Section 32 92 19 SEEDING.

3.14.1.2 Paving Repairs

Where trenches, pits, or other excavations are made in existing roadways and other areas of pavement where surface treatment of any kind exists,

restore such surface treatment or pavement the same thickness and in the same kind as previously existed, except as otherwise specified, and to match and tie into the adjacent and surrounding existing surfaces.

3.15 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.15.1 Concrete Slabs (Pads) for Equipment

Unless otherwise indicated, the slab must be at least 200 mm thick, reinforced with a 152 mm by 152 mm - MW19 by MW19 (6 by 6 - W2.9 by W2.9) mesh, placed uniformly 100 mm from the top of the slab. Place slab on a 150 mm thick, well-compacted gravel base. Top of concrete slab must be approximately 100 mm above finished grade with gradual slope for drainage. Edges above grade must have 15 mm chamfer. Slab must be of adequate size to project at least 200 mm beyond the equipment.

Stub up conduits, with bushings, 50 mm into cable wells in the concrete pad. Coordinate dimensions of cable wells with transformer cable training areas.

3.15.2 Sealing

When the installation is complete, seal all conduit and other entries into the equipment enclosure with an approved sealing compound. Seals must be of sufficient strength and durability to protect all energized live parts of the equipment from rodents, insects, or other foreign matter.

3.16 FIELD QUALITY CONTROL

3.16.1 Performance of Field Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

3.16.1.1 Medium Voltage Cables

Perform tests after installation of cable, splices, and terminators and before terminating to equipment or splicing to existing circuits.

a. Visual and Mechanical Inspection

- (1) Inspect exposed cable sections for physical damage.
- (2) Verify that cable is supplied and connected in accordance with contract plans and specifications.
- (3) Inspect for proper shield grounding, cable support, and cable termination.
- (4) Verify that cable bends are not less than ICEA or manufacturer's minimum allowable bending radius.
- (5) Inspect for proper fireproofing.
- (6) Visually inspect jacket and insulation condition.

(7) Inspect for proper phase identification and arrangement.

b. Electrical Tests

- (1) Perform a shield continuity test on each power cable by ohmmeter method. Record ohmic value, resistance values in excess of 10 ohms per 1000 feet of cable must be investigated and justified.
- (2) Perform acceptance test on new cables before the new cables are connected to existing cables and placed into service, including terminations and joints. Perform maintenance test on complete cable system after the new cables are connected to existing cables and placed into service, including existing cable, terminations, and joints. Tests must be very low frequency (VLF) alternating voltage withstand tests in accordance with IEEE 400.2. VLF test frequency must be 0.05 Hz minimum for a duration of 60 minutes using a sinusoidal waveform. Test voltages must be as follows:

CABLE RATING AC TEST VOLTAGE for ACCEPTANCE TESTING	
15 kV	20kV rms(peak)

CABLE RATING AC TEST VOLTAGE for MAINTENANCE TESTING	
15 kV	16kV rms(peak)

3.16.1.2 Low Voltage Cables, 600-Volt

Perform tests after installation of cable, splices and terminations and before terminating to equipment or splicing to existing circuits.

a. Visual and Mechanical Inspection

- (1) Inspect exposed cable sections for physical damage.
- (2) Verify that cable is supplied and connected in accordance with contract plans and specifications.
- (3) Verify tightness of accessible bolted electrical connections.
- (4) Inspect compression-applied connectors for correct cable match and indentation.
- (5) Visually inspect jacket and insulation condition.
- (6) Inspect for proper phase identification and arrangement.

b. Electrical Tests

- (1) Perform insulation resistance tests on wiring No. 6 AWG and larger diameter using instrument which applies voltage of approximately 1000 volts dc for one minute.
- (2) Perform continuity tests to insure correct cable connection.

3.16.1.3 Grounding System

a. Visual and mechanical inspection

Inspect ground system for compliance with contract plans and specifications.

b. Electrical tests

Perform ground-impedance measurements utilizing the fall-of-potential method in accordance with [IEEE 81](#). On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. The instrument must be equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test. Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance.

3.16.2 Follow-Up Verification

Upon completion of acceptance checks and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. As an exception to requirements stated elsewhere in the contract, the Contracting Officer must be given 5 working days advance notice of the dates and times of checking and testing.

.... -- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 82 00

TELECOMMUNICATIONS OUTSIDE PLANT (OSP)

04/06

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 DEFINITIONS
 - 1.3.1 Campus Distributor (CD)
 - 1.3.2 Entrance Facility (EF) (Telecommunications)
 - 1.3.3 Pathway
- 1.4 SYSTEM DESCRIPTION
- 1.5 SUBMITTALS
- 1.6 QUALITY ASSURANCE
 - 1.6.1 Shop Drawings
 - 1.6.1.1 Telecommunications Outside Plant Shop Drawings
 - 1.6.1.2 Telecommunications Entrance Facility Drawings
 - 1.6.2 Telecommunications Qualifications
 - 1.6.2.1 Telecommunications Contractor Qualifications
 - 1.6.2.2 Key Personnel Qualifications
 - 1.6.2.3 Minimum Manufacturer's Qualifications
 - 1.6.3 Outside Plant Test Plan
 - 1.6.4 Standard Products
 - 1.6.4.1 Alternative Qualifications
 - 1.6.4.2 Material and Equipment Manufacturing Date
 - 1.6.5 Regulatory Requirements
 - 1.6.5.1 Independent Testing Organization Certificate
- 1.7 DELIVERY, STORAGE, AND HANDLING
- 1.8 MAINTENANCE
 - 1.8.1 Record Documentation
- 1.9 WARRANTY

PART 2 PRODUCTS

- 2.1 MATERIALS AND EQUIPMENT
- 2.2 TELECOMMUNICATIONS ENTRANCE FACILITY
 - 2.2.1 Building Protector Assemblies
 - 2.2.2 Protector Modules
 - 2.2.3 Fiber Optic Terminations
- 2.3 CLOSURES
 - 2.3.1 Copper Conductor Closures
 - 2.3.1.1 Underground Cable Closures
 - 2.3.2 Fiber Optic Closures
 - 2.3.2.1 In Vault or Manhole
- 2.4 CABLE SPLICES, AND CONNECTORS
 - 2.4.1 Copper Cable Splices
 - 2.4.2 Copper Cable Splice Connector
 - 2.4.3 Fiber Optic Cable Splices
 - 2.4.4 Fiber Optic Splice Organizer

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.4.5 Shield Connectors
- 2.5 CONDUIT
- 2.6 PLASTIC INSULATING TAPE
- 2.7 WIRE AND CABLE
 - 2.7.1 Copper Conductor Cable
 - 2.7.1.1 Underground
 - 2.7.1.2 Screen
 - 2.7.2 Fiber Optic Cable
 - 2.7.2.1 Strength Members
 - 2.7.2.2 Performance Requirements
 - 2.7.3 Grounding and Bonding Conductors
- 2.8 CABLE TAGS IN MANHOLES, HANDHOLES, AND VAULTS
 - 2.8.1 Stainless Steel
- 2.9 BURIED WARNING AND IDENTIFICATION TAPE
- 2.10 GROUNDING BRAID
- 2.11 MANUFACTURER'S NAMEPLATE
- 2.12 FIELD FABRICATED NAMEPLATES
- 2.13 TESTS, INSPECTIONS, AND VERIFICATIONS
 - 2.13.1 Factory Reel Test Data

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Contractor Damage
 - 3.1.2 Cable Inspection and Repair
 - 3.1.3 Underground Duct
 - 3.1.4 Reconditioning of Surfaces
 - 3.1.5 Cable Pulling
 - 3.1.5.1 Cable Tensions
 - 3.1.5.2 Pulling Eyes
 - 3.1.5.3 Installation of Cables in Manholes, Handholes, and Vaults
 - 3.1.6 Cable Splicing
 - 3.1.6.1 Copper Conductor Splices
 - 3.1.6.2 Fiber Optic Splices
 - 3.1.7 Surge Protection
 - 3.1.8 Grounding
 - 3.1.8.1 Telecommunications Master Ground Bar (TMGB)
 - 3.1.8.2 Incoming Cable Shields
 - 3.1.8.3 Campus Distributor Grounding
 - 3.1.9 Cut-Over
- 3.2 LABELING
 - 3.2.1 Labels
 - 3.2.2 Cable Tag Installation
 - 3.2.3 Termination Hardware
- 3.3 FIELD FABRICATED NAMEPLATE MOUNTING
- 3.4 FIELD QUALITY CONTROL
 - 3.4.1 Pre-Installation Tests
 - 3.4.1.1 Cable Capacitance
 - 3.4.1.2 Loop Resistance
 - 3.4.1.3 Pre-Installation Test Results
 - 3.4.2 Acceptance Tests
 - 3.4.2.1 Copper Conductor Cable
 - 3.4.2.2 Fiber Optic Cable
 - 3.4.3 Soil Density Tests

-- End of Section Table of Contents --

SECTION 33 82 00

TELECOMMUNICATIONS OUTSIDE PLANT (OSP)
04/06

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

- ASTM B1 (2013) Standard Specification for Hard-Drawn Copper Wire
- ASTM B8 (2011; R 2017) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft
- ASTM D709 (2017) Standard Specification for Laminated Thermosetting Materials

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

- IEEE 100 (2000; Archived) The Authoritative Dictionary of IEEE Standards Terms
- IEEE C2 (2017; Errata 1-2 2017; INT 1 2017) National Electrical Safety Code

INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)

- ICEA S-87-640 (2016) Optical Fiber Outside Plant Communications Cable; 4th Edition
- ICEA S-98-688 (2012) Broadband Twisted Pair Telecommunication Cable, Aircore, Polyolefin Insulated, Copper Conductors Technical Requirements
- ICEA S-99-689 (2012) Broadband Twisted Pair Telecommunication Cable Filled, Polyolefin Insulated, Copper Conductors Technical Requirements

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

- ANSI C62.61 (1993) American National Standard for Gas Tube Surge Arresters on Wire Line Telephone Circuits

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

- NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA

20-1; TIA 20-2; TIA 20-3; TIA 20-4)
National Electrical Code

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-455-21	(1988a; R 2012) FOTP-21 - Mating Durability of Fiber Optic Interconnecting Devices
TIA-455-78-B	(2020c) FOTP-78 Optical Fibres - Part 1-40: Measurement Methods and Test Procedures - Attenuation
TIA-455-107	(1999a) FOTP-107 Determination of Component Reflectance or Link/System Return Loss using a Loss Test Set
TIA-472D000	(2007b) Fiber Optic Communications Cable for Outside Plant Use
TIA-492CAAA	(1998; R 2002) Detail Specification for Class IVa Dispersion-Unshifted Single-Mode Optical Fibers
TIA-526-7	(2015a) OFSTP-7 Measurement of Optical Power Loss of Installed Single-Mode Fiber Cable Plant
TIA-526-14	(2015c) OFSTP-14A Optical Power Loss Measurements of Installed Multimode Fiber Cable Plant
TIA-568.1	(2020e) Commercial Building Telecommunications Infrastructure Standard
TIA-568.2	(2018d) Balanced Twisted-Pair Telecommunications Cabling and Components Standards
TIA-568.3	(2016d; Add 1 2019) Optical Fiber Cabling Components Standard
TIA-569	(2019e) Telecommunications Pathways and Spaces
TIA-590	(1997a) Standard for Physical Location and Protection of Below Ground Fiber Optic Cable Plant
TIA-606	(2021d) Administration Standard for Telecommunications Infrastructure
TIA-607	(2019d) Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
TIA-758	(2012b) Customer-Owned Outside Plant Telecommunications Infrastructure Standard

TIA/EIA-455	(1998b) Standard Test Procedure for Fiber Optic Fibers, Cables, Transducers, Sensors, Connecting and Terminating Devices, and Other Fiber Optic Components
TIA/EIA-598	(2014D; Add 2 2018) Optical Fiber Cable Color Coding
TIA/EIA-604-2	(2004b; R 2014) FOCIS 2 Fiber Optic Connector Intermateability Standard
TIA/EIA-604-3	(2004b; R 2014) Fiber Optic Connector Intermateability Standard (FOCIS), Type SC and SC-APC, FOCIS-3
TIA/EIA-604-10	(2002a) FOCIS 10 Fiber Optic Connector Intermateability Standard - Type LC

U.S. DEPARTMENT OF AGRICULTURE (USDA)

RUS 1755	Telecommunications Standards and Specifications for Materials, Equipment and Construction
RUS Bull 345-65	(1985) Shield Bonding Connectors (PE-65)
RUS Bull 345-72	(1985) Filled Splice Closures (PE-74)
RUS Bull 345-83	(1979; Rev Oct 1982) Gas Tube Surge Arrestors (PE-80)
RUS Bull 1751F-643	(2002) Underground Plant Design
RUS Bull 1751F-815	(1979) Electrical Protection of Outside Plant
RUS Bull 1753F-201	(1997) Acceptance Tests of Telecommunications Plant (PC-4)
RUS Bull 1753F-401	(1995) Splicing Copper and Fiber Optic Cables (PC-2)

UNDERWRITERS LABORATORIES (UL)

UL 83	(2017; Reprint Mar 2020) UL Standard for Safety Thermoplastic-Insulated Wires and Cables
UL 497	(2001; Reprint Jul 2013) Protectors for Paired Conductor Communication Circuits
UL 510	(2020) UL Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape

1.2 RELATED REQUIREMENTS

Section 27 10 00, BUILDING TELECOMMUNICATIONS CABLING SYSTEM, and Section 33 71 02, UNDERGROUND ELECTRICAL DISTRIBUTION apply to this section with

additions and modifications specified herein.

1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in this specification shall be as defined in TIA-568.1, TIA-568.2, TIA-568.3, TIA-569, TIA-606, and IEEE 100 and herein.

1.3.1 Campus Distributor (CD)

A distributor from which the campus backbone cabling emanates.
(International expression for main cross-connect - (MC).)

1.3.2 Entrance Facility (EF) (Telecommunications)

An entrance to the building for both private and public network service cables (including antennae) including the entrance point at the building wall and continuing to the entrance room or space.

1.3.3 Pathway

A physical infrastructure utilized for the placement and routing of telecommunications cable.

1.4 SYSTEM DESCRIPTION

The telecommunications outside plant consists of cable, conduit, manholes, poles, etc. required to provide signal paths from the closest point of presence to the new facility, including free standing frames or backboards, interconnecting hardware, terminating cables, lightning and surge protection modules at the entrance facility. The work consists of providing, testing and making operational cabling, interconnecting hardware and lightning and surge protection necessary to form a complete outside plant telecommunications system for continuous use. Based on the agreement (HC1021-15-H-1007) between USA Government and Telefonica, any new Telecommunications infrastructure constructed at Naval Station Rota, Spain that connects to the existing Telefonica infrastructure is owned and maintained by Telefonica. For this reason, any action affecting the existing Telecommunication Infrastructure (new construction, modifications, or demolition) must be performed in accordance with Telefonica's regulations and will be executed by Telefonica or a company designated and supervised by Telefonica and coordinated by the Base Communications Office, NCTAMS.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Telecommunications Outside Plant; G

Telecommunications Entrance Facility Drawings; G

In addition to Section 01 33 00 SUBMITTAL PROCEDURES, provide shop drawings in accordance with paragraph SHOP DRAWINGS.

SD-03 Product Data

Wire and Cable; G

Cable Splices, and Connectors; G

Closures; G

Building Protector Assemblies; G

Protector Modules; G

Submittals shall include the manufacturer's name, trade name, place of manufacture, and catalog model or number. Submittals shall also include applicable federal, military, industry, and technical society publication references. Should manufacturer's data require supplemental information for clarification, the supplemental information shall be submitted as specified in paragraph REGULATORY REQUIREMENTS and as required for certificates in Section 01 33 00 SUBMITTAL PROCEDURES.

SD-06 Test Reports

Pre-installation Tests; G

Acceptance Tests; G

Outside Plant Test Plan; G

SD-07 Certificates

Telecommunications Contractor Qualifications; G

Key Personnel Qualifications; G

Minimum Manufacturer's Qualifications; G

SD-08 Manufacturer's Instructions

Building Protector Assembly Installation; G

Cable Tensions; G

Fiber Optic Splices; G

Submit instructions prior to installation.

SD-09 Manufacturer's Field Reports

Factory Reel Test Data; G

SD-10 Operation and Maintenance Data

Telecommunications Outside Plant (OSP), Data Package 5; G

Commercial off-the-shelf manuals shall be provided for operation, installation, configuration, and maintenance of products provided as a part of the telecommunications outside plant (OSP). Submit

operations and maintenance data in accordance with Section 01 78 23, OPERATION AND MAINTENANCE DATA and as specified herein not later than 2 months prior to the date of beneficial occupancy. In addition to requirements of Data package 5, include the requirements of paragraphs TELECOMMUNICATIONS OUTSIDE PLANT SHOP DRAWINGS and TELECOMMUNICATIONS ENTRANCE FACILITY DRAWINGS.

SD-11 Closeout Submittals

Record Documentation; G

In addition to other requirements, provide in accordance with paragraph RECORD DOCUMENTATION.

1.6 QUALITY ASSURANCE

1.6.1 Shop Drawings

Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Wiring diagrams shall identify circuit terminals and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Drawings shall indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices. Submittals shall include the nameplate data, size, and capacity. Submittals shall also include applicable federal, military, industry, and technical society publication references.

1.6.1.1 Telecommunications Outside Plant Shop Drawings

Provide Outside Plant Design in accordance with TIA-758 and RUS Bull 1751F-643 for underground system design. Provide T0 shop drawings that show the physical and logical connections from the perspective of an entire campus, such as actual building locations, exterior pathways and campus backbone cabling on plan view drawings, major system nodes, and related connections on the logical system drawings in accordance with TIA-606. Drawings shall include wiring and schematic diagrams for fiber optic and copper cabling and splices, copper conductor gauge and pair count, fiber pair count and type, pathway duct and innerduct arrangement, associated construction materials, and any details required to demonstrate that cable system has been coordinated and will properly support the switching and transmission system identified in specification and drawings. Provide Registered Communications Distribution Designer (RCDD) approved drawings of the telecommunications outside plant. The telecommunications outside plant (OSP) shop drawings shall be included in the operation and maintenance manuals.

1.6.1.2 Telecommunications Entrance Facility Drawings

Provide T3 drawings for EF Telecommunications as specified in the paragraph TELECOMMUNICATIONS SPACE DRAWINGS of Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEMS. The telecommunications entrance facility shop drawings shall be included in the operation and maintenance manuals.

1.6.2 Telecommunications Qualifications

Work under this section shall be performed by and the equipment shall be

provided by the approved telecommunications contractor and key personnel. Qualifications shall be provided for: the telecommunications system contractor, the telecommunications system installer, the supervisor (if different from the installer), and the cable splicing and terminating personnel. A minimum of 30 days prior to installation, submit documentation of the experience of the telecommunications contractor and of the key personnel.

1.6.2.1 Telecommunications Contractor Qualifications

The telecommunications contractor shall be a firm which is regularly and professionally engaged in the business of the applications, installation, and testing of the specified telecommunications systems and equipment. The telecommunications contractor shall demonstrate experience in providing successful telecommunications systems that include outside plant and broadband cabling within the past 3 years. Submit documentation for a minimum of three and a maximum of five successful telecommunication system installations for the telecommunications contractor. Each of the key personnel shall demonstrate experience in providing successful telecommunications systems in accordance with TIA-758 within the past 3 years.

1.6.2.2 Key Personnel Qualifications

Provide key personnel who are regularly and professionally engaged in the business of the application, installation and testing of the specified telecommunications systems and equipment. There may be one key person or more key persons proposed for this solicitation depending upon how many of the key roles each has successfully provided. Each of the key personnel shall demonstrate experience in providing successful telecommunications systems within the past 3 years.

Cable splicing and terminating personnel assigned to the installation of this system or any of its components shall have training in the proper techniques and have a minimum of 3 years experience in splicing and terminating the specified cables. Modular splices shall be performed by factory certified personnel or under direct supervision of factory trained personnel for products used.

Supervisors and installers assigned to the installation of this system or any of its components shall have factory or factory approved certification from each equipment manufacturer indicating that they are qualified to install and test the provided products.

Submit documentation for a minimum of three and a maximum of five successful telecommunication system installations for each of the key personnel. Documentation for each key person shall include at least two successful system installations provided that are equivalent in system size and in construction complexity to the telecommunications system proposed for this solicitation. Include specific experience in installing and testing telecommunications outside plant systems, including broadband cabling, and provide the names and locations of at least two project installations successfully completed using optical fiber and copper telecommunications cabling systems. All of the existing telecommunications system installations offered by the key persons as successful experience shall have been in successful full-time service for at least 18 months prior to the issuance date for this solicitation. Provide the name and role of the key person, the title, location, and completed installation date of the referenced project, the referenced

project owner point of contact information including name, organization, title, and telephone number, and generally, the referenced project description including system size and construction complexity.

Indicate that all key persons are currently employed by the telecommunications contractor, or have a commitment to the telecommunications contractor to work on this project. All key persons shall be employed by the telecommunications contractor at the date of issuance of this solicitation, or if not, have a commitment to the telecommunications contractor to work on this project by the date that the bid was due to the Contracting Officer.

Note that only the key personnel approved by the Contracting Officer in the successful proposal shall do work on this solicitation's telecommunications system. Key personnel shall function in the same roles in this contract, as they functioned in the offered successful experience. Any substitutions for the telecommunications contractor's key personnel requires approval from The Contracting Officer.

1.6.2.3 Minimum Manufacturer's Qualifications

Cabling, equipment and hardware manufacturers shall have a minimum of 3 years experience in the manufacturing, assembly, and factory testing of components which comply with, [TIA-568.1](#), [TIA-568.2](#) and [TIA-568.3](#). In addition, cabling manufacturers shall have a minimum of 3 years experience in the manufacturing and factory testing of cabling which comply with [ICEA S-87-640](#), [ICEA S-98-688](#), and [ICEA S-99-689](#).

1.6.3 Outside Plant Test Plan

Prepare and provide a complete and detailed test plan for field tests of the outside plant including a complete list of test equipment for the copper conductor and optical fiber cables, components, and accessories for approval by the Contracting Officer. Include a cut-over plan with procedures and schedules for relocation of facility station numbers without interrupting service to any active location. Submit the plan at least 30 days prior to tests for Contracting Officer approval. Provide outside plant testing and performance measurement criteria in accordance with [TIA-568.1](#) and [RUS Bull 1753F-201](#). Include procedures for certification, validation, and testing that includes fiber optic link performance criteria.

1.6.4 Standard Products

Provide materials and equipment that are standard products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and shall be the manufacturer's latest standard design that has been in satisfactory commercial or industrial use for at least 2 years prior to bid opening. The 2-year period shall include applications of equipment and materials under similar circumstances and of similar size. The product shall have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Products supplied shall be specifically designed and manufactured for use with outside plant telecommunications systems. Where two or more items of the same class of equipment are required, these items shall be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.6.4.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is provided.

1.6.4.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site shall not be used, unless specified otherwise.

1.6.5 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship shall be in accordance with the mandatory and advisory provisions of NFPA 70 unless more stringent requirements are specified or indicated.

1.6.5.1 Independent Testing Organization Certificate

In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. The certificate shall state that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.7 DELIVERY, STORAGE, AND HANDLING

Ship cable on reels in 305 meter length with a minimum overage of 10 percent. Radius of the reel drum shall not be smaller than the minimum bend radius of the cable. Wind cable on the reel so that unwinding can be done without kinking the cable. Two meters of cable at both ends of the cable shall be accessible for testing. Attach permanent label on each reel showing length, cable identification number, cable size, cable type, and date of manufacture. Provide water resistant label and the indelible writing on the labels. Apply end seals to each end of the cables to prevent moisture from entering the cable. Reels with cable shall be suitable for outside storage conditions when temperature ranges from minus 40 degrees C to plus 65 degrees C, with relative humidity from 0 to 100 percent. Equipment, other than cable, delivered and placed in storage shall be stored with protection from weather, humidity and temperature variation, dirt and dust, or other contaminants in accordance with manufacturer's requirements.

1.8 MAINTENANCE

1.8.1 Record Documentation

Provide the activity responsible for telecommunications system maintenance and administration a single complete and accurate set of record documentation for the entire telecommunications system with respect to this project.

Provide record documentation as specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

1.9 WARRANTY

The equipment items shall be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

Products supplied shall be specifically designed and manufactured for use with outside plant telecommunications systems.

2.2 TELECOMMUNICATIONS ENTRANCE FACILITY

2.2.1 Building Protector Assemblies

Provide self-contained 5 pin unit supplied with a field cable stub factory connected to protector socket blocks to terminate and accept protector modules for 12 pairs of outside cable. Building protector assembly shall have interconnecting hardware for connection to interior cabling at full capacity. Provide manufacturers instructions for building protector assembly installation. Provide copper cable interconnecting hardware as specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

2.2.2 Protector Modules

Provide in accordance with UL 497 three-electrode gas tube or solid state type 5 pin rated for the application. Provide gas tube protection modules in accordance with RUS Bull 345-83 and shall be maximum duty, A>20kA, B>1000, C>200A where A is the maximum single impulse discharge current, B is the impulse life and C is the AC discharge current in accordance with ANSI C62.61. The gas modules shall shunt high voltage to ground, fail short, and be equipped with an external spark gap and heat coils in accordance with UL 497. Provide the number of surge protection modules equal to the number of pairs of exterior cable of the building protector assembly.

2.2.3 Fiber Optic Terminations

Provide in accordance with TIA-455-21. Optical fiber connectors shall be duplex LC in accordance with TIA/EIA-604-10 with zirconia ceramic alignment sleeves, duplex SC in accordance with TIA/EIA-604-3 with zirconia ceramic or ST in accordance with TIA/EIA-604-2 with metallic ferrule, epoxyless crimp style compatible with 8/125 single-mode fiber. The connectors shall provide a maximum attenuation of 0.3 dB at 1310 nm with less than a 0.2 dB change after 500 mating cycles. Coordinate requirements with contracting officer for connections to AMI Metering RTU and AFHE RTU Equipment.

2.3 CLOSURES

2.3.1 Copper Conductor Closures

2.3.1.1 Underground Cable Closures

- a. In vault or manhole: Provide underground closure suitable to house a straight, butt, and branch splice in a protective housing into which can be poured an encapsulating compound. Closure shall be of suitable thermoplastic, thermoset, or stainless steel material supplying structural strength necessary to pass the mechanical and electrical requirements in a vault or manhole environment. Encapsulating compound shall be reenterable and shall not alter the chemical stability of the closure. Provide filled splice cases in accordance with [RUS Bull 345-72](#).

2.3.2 Fiber Optic Closures

2.3.2.1 In Vault or Manhole

Provide underground closure suitable to house splice organizer in a protective housing into which can be poured an encapsulating compound. Closure shall be of thermoplastic, thermoset, or stainless steel material supplying structural strength necessary to pass the mechanical and electrical requirements in a vault or manhole environment. Encapsulating compound shall be reenterable and shall not alter the chemical stability of the closure.

2.4 CABLE SPLICES, AND CONNECTORS

2.4.1 Copper Cable Splices

Provide multipair, in-line splices of a moisture resistant, insulation displacement connector held rigidly in place to assure maximum continuity in accordance with [RUS Bull 1753F-401](#). Cables greater than 25 pairs shall be spliced using multipair splicing connectors, which accommodate 25 pairs of conductors at a time. Provide correct connector size to accommodate the cable gauge of the supplied cable.

2.4.2 Copper Cable Splice Connector

Provide splice connectors with a polycarbonate body and cap and a tin-plated brass contact element. Connector shall accommodate 22 to 26 AWG solid wire with a maximum insulation diameter of [1.65 mm](#). Fill connector with sealant grease to make a moisture resistant connection, in accordance with [RUS Bull 1753F-401](#).

2.4.3 Fiber Optic Cable Splices

Provide fiber optic cable splices and splicing materials for mechanical methods at locations shown on the construction drawings. The splice insertion loss shall be 0.3 dB maximum when measured in accordance with [TIA-455-78-B](#) using an Optical Time Domain Reflectometer (OTDR). Splices shall be designed for a return loss of 40.0 db max for single mode fiber when tested in accordance with [TIA-455-107](#). Physically protect each fiber optic splice by a splice kit specially designed for the splice.

2.4.4 Fiber Optic Splice Organizer

Provide splice organizer suitable for housing fiber optic splices in a neat and orderly fashion. Splice organizer shall allow for a minimum of 1 m of fiber for each fiber within the cable to be neatly stored without kinks or twists. Splice organizer shall accommodate individual strain relief for each splice and allow for future maintenance or modification, without damage to the cable or splices. Provide splice organizer hardware, such as splice trays, protective glass shelves, and shield bond connectors in a splice organizer kit.

2.4.5 Shield Connectors

Provide connectors with a stable, low-impedance electrical connection between the cable shield and the bonding conductor in accordance with [RUS Bull 345-65](#).

2.5 CONDUIT

Provide conduit as specified in Section [33 71 02](#) UNDERGROUND ELECTRICAL DISTRIBUTION.

2.6 PLASTIC INSULATING TAPE

[UL 510](#).

2.7 WIRE AND CABLE

2.7.1 Copper Conductor Cable

Solid copper conductors, covered with an extruded solid insulating compound. Insulated conductors shall be twisted into pairs which are then stranded or oscillated to form a cylindrical core. For special high frequency applications, the cable core shall be separated into compartments. Cable shall be completed by the application of a suitable core wrapping material, a corrugated copper or plastic coated aluminum shield, and an overall extruded jacket. Telecommunications contractor shall verify distances between splice points prior to ordering cable in specific cut lengths. Gauge of conductor shall determine the range of numbers of pairs specified; 19 gauge (6 to 400 pairs), 22 gauge (6 to 1200 pairs), 24 gauge (6 to 2100 pairs), and 26 gauge (6 to 3000 pairs). Copper conductor shall conform to the following:

2.7.1.1 Underground

Provide filled cable meeting the requirements of [ICEA S-99-689](#) and [RUS 1755.390](#).

2.7.1.2 Screen

Provide screen-compartmental core cable filled cable meeting the requirements of [ICEA S-99-689](#) and [RUS 1755.390](#).

2.7.2 Fiber Optic Cable

Provide single-mode, 8/125-um, 0.10 aperture 1310 nm fiber optic cable in accordance with [TIA-492CAAA](#), [TIA-472D000](#), and [ICEA S-87-640](#) including any special requirements made necessary by a specialized design. Provide optical fibers as indicated. Fiber optic cable shall be specifically designed for outside use with loose buffer construction. Provide fiber

optic color code in accordance with TIA/EIA-598

2.7.2.1 Strength Members

Provide central, non-metallic strength members with sufficient tensile strength for installation and residual rated loads to meet the applicable performance requirements in accordance with ICEA S-87-640. The strength member is included to serve as a cable core foundation to reduce strain on the fibers, and shall not serve as a pulling strength member.

2.7.2.2 Performance Requirements

Provide fiber optic cable with optical and mechanical performance requirements in accordance with ICEA S-87-640.

2.7.3 Grounding and Bonding Conductors

Provide grounding and bonding conductors in accordance with RUS 1755.200, TIA-607, IEEE C2, and NFPA 70. Solid bare copper wire meeting the requirements of ASTM B1 for sizes No. 8 AWG and smaller and stranded bare copper wire meeting the requirements of ASTM B8, for sizes No. 6 AWG and larger. Insulated conductors shall have 600-volt, Type TW insulation meeting the requirements of UL 83.

2.8 CABLE TAGS IN MANHOLES, HANDHOLES, AND VAULTS

Provide tags for each telecommunications cable or wire located in manholes, handholes, and vaults. Cable tags shall be stainless steel and labeled in accordance with TIA-606. Handwritten labeling is unacceptable.

2.8.1 Stainless Steel

Provide stainless steel, cable tags 41.25 mm in diameter 1.58 mm thick minimum, and circular in shape. Tags shall be die stamped with numbers, letters, and symbols not less than 6.35 mm high and approximately 0.38 mm deep in normal block style.

2.9 BURIED WARNING AND IDENTIFICATION TAPE

Provide fiber optic media marking and protection in accordance with TIA-590. Provide color, type and depth of tape as specified in paragraph BURIED WARNING AND IDENTIFICATION TAPE in Section 31 00 00, EARTHWORK.

2.10 GROUNDING BRAID

Provide grounding braid that provides low electrical impedance connections for dependable shield bonding in accordance with RUS 1755.200. Braid shall be made from flat tin-plated copper.

2.11 MANUFACTURER'S NAMEPLATE

Each item of equipment shall have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.12 FIELD FABRICATED NAMEPLATES

Provide laminated plastic nameplates in accordance with ASTM D709 for each

patch panel, protector assembly, rack, cabinet and other equipment or as indicated on the drawings. Each nameplate inscription shall identify the function and, when applicable, the position. Nameplates shall be melamine plastic, 3 mm thick, white with black center core. Surface shall be matte finish. Corners shall be square. Accurately align lettering and engrave into the core. Minimum size of nameplates shall be 25 by 65 mm. Lettering shall be a minimum of 6.35 mm high normal block style.

2.13 TESTS, INSPECTIONS, AND VERIFICATIONS

2.13.1 Factory Reel Test Data

Test 100 percent OTDR test of FO media at the factory in accordance with TIA-568.1 and TIA-568.3. Use TIA-526-7 for single mode fiber and TIA-526-14 Method B for multi mode fiber measurements. Calibrate OTDR to show anomalies of 0.2 dB minimum. Enhanced performance filled OSP copper cables, referred to as Broadband Outside Plant (BBOSP), shall meet the requirements of ICEA S-99-689. Enhanced performance air core OSP copper cables shall meet the requirements of ICEA S-98-688. Submit test reports, including manufacture date for each cable reel and receive approval before delivery of cable to the project site.

PART 3 EXECUTION

3.1 INSTALLATION

Install all system components and appurtenances in accordance with manufacturer's instructions IEEE C2, NFPA 70, and as indicated. Provide all necessary interconnections, services, and adjustments required for a complete and operable telecommunications system.

3.1.1 Contractor Damage

Promptly repair indicated utility lines or systems damaged during site preparation and construction. Damages to lines or systems not indicated, which are caused by Contractor operations, shall be treated as "Changes" under the terms of the Contract Clauses. When Contractor is advised in writing of the location of a nonindicated line or system, such notice shall provide that portion of the line or system with "indicated" status in determining liability for damages. In every event, immediately notify the Contracting Officer of damage.

3.1.2 Cable Inspection and Repair

Handle cable and wire provided in the construction of this project with care. Inspect cable reels for cuts, nicks or other damage. Damaged cable shall be replaced or repaired to the satisfaction of the Contracting Officer. Reel wraps shall remain intact on the reel until the cable is ready for placement.

3.1.3 Underground Duct

Provide underground duct and connections to existing manholes, handholes, concrete pads, and existing ducts as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION with any additional requirements as specified herein.

3.1.4 Reconditioning of Surfaces

Provide reconditioning of surfaces as specified in Section 33 71 02
UNDERGROUND ELECTRICAL DISTRIBUTION.

3.1.5 Cable Pulling

Test duct lines with a mandrel and swab out to remove foreign material before the pulling of cables. Avoid damage to cables in setting up pulling apparatus or in placing tools or hardware. Do not step on cables when entering or leaving the manhole. Do not place cables in ducts other than those shown without prior written approval of the Contracting Officer. Roll cable reels in the direction indicated by the arrows painted on the reel flanges. Set up cable reels on the same side of the manhole as the conduit section in which the cable is to be placed. Level the reel and bring into proper alignment with the conduit section so that the cable pays off from the top of the reel in a long smooth bend into the duct without twisting. Under no circumstances shall the cable be paid off from the bottom of a reel. Check the equipment set up prior to beginning the cable pulling to avoid an interruption once pulling has started. Use a cable feeder guide of suitable dimensions between cable reel and face of duct to protect cable and guide cable into the duct as it is paid off the reel. As cable is paid off the reel, lubricate and inspect cable for sheath defects. When defects are noticed, stop pulling operations and notify the Contracting Officer to determine required corrective action. Cable pulling shall also be stopped when reel binds or does not pay off freely. Rectify cause of binding before resuming pulling operations. Provide cable lubricants recommended by the cable manufacturer. Avoid bends in cables of small radii and twists that might cause damage. Do not bend cable and wire in a radius less than 10 times the outside diameter of the cable or wire.

3.1.5.1 Cable Tensions

Obtain from the cable manufacturer and provide to the Contracting Officer, the maximum allowable pulling tension. This tension shall not be exceeded.

3.1.5.2 Pulling Eyes

Equip cables 32 mm in diameter and larger with cable manufacturer's factory installed pulling-in eyes. Provide cables with diameter smaller than 32 mm with heat shrinkable type end caps or seals on cable ends when using cable pulling grips. Rings to prevent grip from slipping shall not be beaten into the cable sheath. Use a swivel of 19 mm links between pulling-in eyes or grips and pulling strand.

3.1.5.3 Installation of Cables in Manholes, Handholes, and Vaults

Do not install cables utilizing the shortest route, but route along those walls providing the longest route and the maximum spare cable lengths. Form cables to closely parallel walls, not to interfere with duct entrances, and support cables on brackets and cable insulators at a maximum of 1220 mm. In existing manholes, handholes, and vaults where new ducts are to be terminated, or where new cables are to be installed, modify the existing installation of cables, cable supports, and grounding as required with cables arranged and supported as specified for new cables. Identify each cable with corrosion-resistant embossed metal tags.

3.1.6 Cable Splicing

3.1.6.1 Copper Conductor Splices

Perform splicing in accordance with requirements of [RUS Bull 1753F-401](#) except that direct buried splices and twisted and soldered splices are not allowed. Exception does not apply for pairs assigned for carrier application.

3.1.6.2 Fiber Optic Splices

Fiber optic splicing shall be in accordance with manufacturer's recommendation and shall exhibit an insertion loss not greater than 0.4 db for mechanical splices.

3.1.7 Surge Protection

All cables and conductors, except fiber optic cable, which serve as communication lines through off-premise lines, shall have surge protection installed at each end which meet the requirements of [RUS Bull 1751F-815](#).

3.1.8 Grounding

Provide grounding and bonding in accordance with [RUS 1755.200](#), [TIA-607](#), [IEEE C2](#), and [NFPA 70](#). Ground exposed noncurrent carrying metallic parts of telephone equipment, cable sheaths, cable splices, and terminals.

3.1.8.1 Telecommunications Master Ground Bar (TMGB)

The TMGB is the hub of the basic telecommunications grounding system providing a common point of connection for ground from outside cable, CD, and equipment. Establish a TMGB for connection point for cable stub shields to connector blocks and CD protector assemblies as specified in Section [26 20 00](#) INTERIOR DISTRIBUTION SYSTEM.

3.1.8.2 Incoming Cable Shields

Shields shall not be bonded across the splice to the cable stubs. Ground shields of incoming cables in the EF Telecommunications to the TMGB.

3.1.8.3 Campus Distributor Grounding

- a. Protection assemblies: Mount CD protector assemblies directly on the telecommunications backboard. Connect assemblies mounted on each vertical frame with No. 6 AWG copper conductor to provide a low resistance path to TMGB.

3.1.9 Cut-Over

All necessary transfers and cut-overs, shall be accomplished by the telecommunications contractor.

3.2 LABELING

3.2.1 Labels

Provide labeling for new cabling and termination hardware located within the facility in accordance with [TIA-606](#). Handwritten labeling is unacceptable. Stenciled lettering for cable and termination hardware

shall be provided using thermal ink transfer process.

3.2.2 Cable Tag Installation

Install cable tags for each telecommunications cable or wire located in manholes, handholes, and vaults including each splice. Tag only new wire and cable provided by this contract. The labeling of telecommunications cable tag identifiers shall be in accordance with TIA-606. Do not provide handwritten letters. Install cable tags so that they are clearly visible without disturbing any cabling or wiring in the manholes, handholes, and vaults.

3.2.3 Termination Hardware

Label patch panels, distribution panels, connector blocks and protection modules using color coded labels with identifiers in accordance with TIA-606.

3.3 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.4 FIELD QUALITY CONTROL

Provide the Contracting Officer 10 working days notice prior to each test. Provide labor, equipment, and incidentals required for testing. Correct defective material and workmanship disclosed as the results of the tests. Furnish a signed copy of the test results to the Contracting Officer within 3 working days after the tests for each segment of construction are completed. Perform testing as construction progresses and do not wait until all construction is complete before starting field tests.

3.4.1 Pre-Installation Tests

Perform the following tests on cable at the job site before it is removed from the cable reel. For cables with factory installed pulling eyes, these tests shall be performed at the factory and certified test results shall accompany the cable.

3.4.1.1 Cable Capacitance

Perform capacitance tests on at least 10 percent of the pairs within a cable to determine if cable capacitance is within the limits specified.

3.4.1.2 Loop Resistance

Perform DC-loop resistance on at least 10 percent of the pairs within a cable to determine if DC-loop resistance is within the manufacturer's calculated resistance.

3.4.1.3 Pre-Installation Test Results

Provide results of pre-installation tests to the Contracting Officer at least 5 working days before installation is to start. Results shall indicate reel number of the cable, manufacturer, size of cable, pairs tested, and recorded readings. When pre-installation tests indicate that

cable does not meet specifications, remove cable from the job site.

3.4.2 Acceptance Tests

Perform acceptance testing in accordance with [RUS Bull 1753F-201](#) and as further specified in this section. Provide personnel, equipment, instrumentation, and supplies necessary to perform required testing. Notification of any planned testing shall be given to the Contracting Officer at least 14 days prior to any test unless specified otherwise. Testing shall not proceed until after the Contractor has received written Contracting Officer's approval of the test plans as specified. Test plans shall define the tests required to ensure that the system meets technical, operational, and performance specifications. The test plans shall define milestones for the tests, equipment, personnel, facilities, and supplies required. The test plans shall identify the capabilities and functions to be tested. Provide test reports in booklet form showing all field tests performed, upon completion and testing of the installed system. Measurements shall be tabulated on a pair by pair or strand by strand basis.

3.4.2.1 Copper Conductor Cable

Perform the following acceptance tests in accordance with [TIA-758](#):

- a. Wire map (pin to pin continuity)
- b. Continuity to remote end
- c. Crossed pairs
- d. Reversed pairs
- e. Split pairs
- f. Shorts between two or more conductors

3.4.2.2 Fiber Optic Cable

Test fiber optic cable in accordance with [TIA/EIA-455](#) and as further specified in this section. Two optical tests shall be performed on all optical fibers: Optical Time Domain Reflectometry (OTDR) Test, and Attenuation Test. In addition, a Bandwidth Test shall be performed on all multimode optical fibers. These tests shall be performed on the completed end-to-end spans which include the near-end pre-connectorized single fiber cable assembly, outside plant as specified, and the far-end pre-connectorized single fiber cable assembly.

- a. OTDR Test: The OTDR test shall be used to determine the adequacy of the cable installations by showing any irregularities, such as discontinuities, micro-bendings or improper splices for the cable span under test. Hard copy fiber signature records shall be obtained from the OTDR for each fiber in each span and shall be included in the test results. The OTDR test shall be measured in both directions. A reference length of fiber, [20 m](#) minimum, used as the delay line shall be placed before the new end connector and after the far end patch panel connectors for inspection of connector signature. Conduct OTDR test and provide calculation or interpretation of results in accordance with [TIA-526-7](#) for single-mode fiber. Splice losses shall not exceed 0.3 db.

- b. Attenuation Test: End-to-end attenuation measurements shall be made on all fibers, in both directions, using a 1310 nanometer light source at one end and the optical power meter on the other end to verify that the cable system attenuation requirements are met in accordance with TIA-526-7 for single-mode fiber optic cables. The measurement method shall be in accordance with TIA-455-78-B. Attenuation losses shall not exceed 0.5 db/km at 1310 nm and 1550 nm for single-mode fiber.

3.4.3 Soil Density Tests

- a. Determine soil-density relationships as specified for soil tests in Section 31 00 00 EARTHWORK.

-- End of Section --

SECTION TABLE OF CONTENTS
DIVISION 40 - PROCESS INTERCONNECTIONS
SECTION 40 60 00
PROCESS CONTROL
05/20

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 SITE ENVIRONMENTAL CONDITIONS
- 1.4 SEQUENCING
 - 1.4.1 Sequencing for Submittals
 - 1.4.2 Sequencing of Activities
 - 1.4.3 Abbreviations

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Operation
 - 2.1.2 Points
 - 2.1.3 Building Telecommunications Cabling (BTC)
 - 2.1.4 System Reliability
- 2.2 MATERIALS AND EQUIPMENT
 - 2.2.1 Product Certifications
 - 2.2.2 Standard Products
 - 2.2.3 Nameplates
- 2.3 GENERAL REQUIREMENTS
- 2.4 SENSORS
 - 2.4.1 Transmitter
 - 2.4.2 Flow Rate Sensors and Meters
 - 2.4.2.1 Magnetic Flowmeter
 - 2.4.3 Level Instrumentation
 - 2.4.3.1 Displacement Type Level Switch
 - 2.4.3.2 Mercury Float Switch
 - 2.4.3.3 Non-Contact Ultrasonic Level Sensor
 - 2.4.3.4 Leak Detection
 - 2.4.4 Pressure Instrumentation
 - 2.4.4.1 Differential Pressure
 - 2.4.5 Limit Switch
 - 2.4.6 Emergency Stop
- 2.5 PROGRAMMABLE LOGIC CONTROLLER (PLC)
 - 2.5.1 PLC General Requirements
 - 2.5.2 Modular PLC
 - 2.5.2.1 Central Processing Unit (CPU) Module
 - 2.5.2.2 Communications Module
 - 2.5.2.3 Power Supply Module
 - 2.5.2.4 Input/Output (I/O) Modules
 - 2.5.3 Loop PLC
 - 2.5.3.1 Central Processing Unit (CPU)
 - 2.5.3.2 Power Requirements
 - 2.5.3.3 On-Off Switch

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

- 2.5.3.4 Parameter Input and Display
- 2.5.3.5 Self Tuning
- 2.5.3.6 Manual Tuning
- 2.5.4 Program Storage/Memory Requirements
- 2.5.5 Input/Output Characteristics
 - 2.5.5.1 Analog Inputs
 - 2.5.5.2 Binary Inputs
 - 2.5.5.3 Analog Outputs
 - 2.5.5.4 Binary Outputs
- 2.5.6 Wiring Connections
- 2.5.7 On-Off Switch
- 2.5.8 Diagnostics
- 2.5.9 Accuracy
- 2.5.10 Primary/Secondary PLC
- 2.6 PLC SOFTWARE
 - 2.6.1 Operating System
 - 2.6.1.1 Startup
 - 2.6.1.2 Failure Mode
 - 2.6.2 Functions
 - 2.6.2.1 Analog Monitoring
 - 2.6.2.2 Logic (Virtual)
 - 2.6.2.3 State Variables
 - 2.6.2.4 Analog Totalization
 - 2.6.3 Alarm Processing
 - 2.6.3.1 Binary Alarms
 - 2.6.3.2 Analog Alarms
 - 2.6.3.3 Pulse Accumulator (PA) Alarms
 - 2.6.4 Constraints
 - 2.6.4.1 Equipment Constraints Definitions
 - 2.6.4.2 Constraints Checks
 - 2.6.5 Control Sequences and Control Loops
 - 2.6.6 Command Priorities
 - 2.6.7 Resident Application Software
 - 2.6.7.1 Program Inputs and Outputs
 - 2.6.7.2 Failure Mode
- 2.7 CONTROL PANELS
 - 2.7.1 Components
 - 2.7.1.1 Enclosures
 - 2.7.1.2 Controllers
 - 2.7.1.3 Standard Indicator Light
 - 2.7.1.4 Selector Switches
 - 2.7.1.5 Push Buttons
 - 2.7.1.6 Relays
 - 2.7.1.7 Terminal Blocks
 - 2.7.1.8 Alarm Horns
 - 2.7.2 Panel Assembly
 - 2.7.3 Electrical Requirements
 - 2.7.4 Power Line Conditioner
 - 2.7.4.1 85 Percent Load
 - 2.7.4.2 Load Changes
 - 2.7.5 Grounding
 - 2.7.6 Convenience Outlet
 - 2.7.7 Panel Interior Light
 - 2.7.8 Ventilation System
- 2.8 FACTORY TEST
 - 2.8.1 Factory Test Setup
 - 2.8.2 Factory Test Procedure
 - 2.8.3 Factory Test Report

PART 3 EXECUTION

- 3.1 FACTORY TEST
- 3.2 EQUIPMENT INSTALLATION REQUIREMENTS
 - 3.2.1 Installation
 - 3.2.1.1 Isolation, Penetrations and Clearance from Equipment
 - 3.2.1.2 Device Mounting
 - 3.2.1.3 Pneumatic Tubing
 - 3.2.1.4 Grooved Mechanical Joints
 - 3.2.2 Sequences of Operation
- 3.3 INSTALLATION OF EQUIPMENT
 - 3.3.1 Control Panels
 - 3.3.2 Flow Measuring Device
 - 3.3.2.1 Magnetic Flowmeter
 - 3.3.3 Level Instruments
 - 3.3.3.1 Displacement Type Liquid Level Switch
 - 3.3.3.2 Mercury Float Switches
 - 3.3.3.3 Ultrasonic Level Sensor
 - 3.3.4 Pressure Instruments
 - 3.3.5 Output Devices
 - 3.3.6 Enclosures
 - 3.3.7 Transformers
- 3.4 WIRE, CABLE AND CONNECTING HARDWARE
 - 3.4.1 LAN Cables and Connecting Hardware
 - 3.4.2 Metering and Sensor Wiring
 - 3.4.2.1 Power Line Surge Protection
 - 3.4.2.2 Sensor and Control Wiring Surge Protection
- 3.5 SOFTWARE INSTALLATION
- 3.6 CONTROL DRAWINGS
 - 3.6.1 Control
 - 3.6.2 Contractor Design Drawings
 - 3.6.2.1 Draft As-Built
 - 3.6.2.2 Final As-Built
 - 3.6.3 Points Schedule
 - 3.6.3.1 Point Name
 - 3.6.3.2 Description
 - 3.6.3.3 DDC Hardware Identifier
 - 3.6.3.4 Settings
 - 3.6.3.5 Range
 - 3.6.3.6 Input or Output (I/O) Type
 - 3.6.3.7 Network Data Exchange Information
 - 3.6.3.8 Override Information
 - 3.6.3.9 Trend Object Information
 - 3.6.3.10 Alarm Information
- 3.7 FIELD TESTING AND ADJUSTING EQUIPMENT
 - 3.7.1 Testing, Adjusting and Commissioning
 - 3.7.2 Performance Verification Test (PVT)
 - 3.7.3 Endurance Test
 - 3.7.3.1 Phase I (Testing)
 - 3.7.3.2 Phase II (Assessment)
 - 3.7.3.3 Exclusions
- 3.8 FIELD TRAINING
 - 3.8.1 Preliminary Operator Training
 - 3.8.2 Additional Operator Training
 - 3.8.3 Maintenance Training
 - 3.8.4 Specialized Training
 - 3.8.4.1 Flow Meter Training
- 3.9 OPERATION AND MAINTENANCE DATA REQUIREMENTS

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

-- End of Section Table of Contents --

SECTION 40 60 00

PROCESS CONTROL
05/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C12.1 (2014; Errata 2016) Electric Meters - Code for Electricity Metering

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 142 (2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book

IEEE C37.90.1 (2013) Standard for Surge Withstand Capability (SWC) Tests for Relays and Relay Systems Associated with Electric Power Apparatus

IEEE C62.41.1 (2002; R 2008) Guide on the Surges Environment in Low-Voltage (1000 V and Less) AC Power Circuits

IEEE C62.41.2 (2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 61131-3 (2013) Programmable Controllers - Part 3: Programming Languages

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NEMA ICS 1 (2022) Standard for Industrial Control and Systems: General Requirements

NEMA ICS 2 (2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V

NEMA ICS 3 (2005; R 2010) Medium-Voltage Controllers Rated 2001 to 7200 V AC

NEMA ICS 4 (2015) Application Guideline for Terminal

Blocks

NEMA ICS 5	(2017) Industrial Control and Systems: Control Circuit and Pilot Devices
NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)	
NFPA 70	(2023) National Electrical Code
NFPA 79	(2015) Electrical Standard for Industrial Machinery
NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)	
NIST SP 250	(1991) Calibration Services Users Guide
U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)	
47 CFR 15	Radio Frequency Devices
UNDERWRITERS LABORATORIES (UL)	
UL 508A	(2018; Reprint Jul 2022) UL Standard for Safety Industrial Control Panels
UL 1059	(2019; Reprint Jul 2022) UL Standard for Safety Terminal Blocks

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Contractor Design Drawings; G

Draft As-Built Drawings; G

SD-03 Product Data

Control Drawings

Performance Verification Test (PVT)

Factory Test Procedure

SD-06 Test Reports

Factory Test Report

Testing, Adjusting and Commissioning

Performance Verification Test(PVT)

Endurance Test

SD-07 Certificates

Control and Sensor Wiring

Ground Rods

Wiring

Installation

SD-10 Operation and Maintenance Data

Training Manual; G

Control System; G

SD-11 Closeout Submittals

Final As-Built Drawings; G

1.3 SITE ENVIRONMENTAL CONDITIONS

The expected site environmental conditions are a minimum of 4 degrees C and a maximum of 32 degrees C .

1.4 SEQUENCING

TABLE I: PROJECT SEQUENCING specifies the sequencing of submittals as specified in paragraph SUBMITTALS (denoted by an 'S' in the 'TYPE' column) and activities as specified in PART 3 EXECUTION (denoted by an 'E' in the 'TYPE' column).

1.4.1 Sequencing for Submittals

The sequencing specified for submittals is the deadline by which the submittal must be initially submitted to the Government. Following submission there will be a Government review period as specified in Section 01 33 00 SUBMITTAL PROCEDURES. If the submittal is not accepted by the Government, revise the submittal and resubmit it to the Government within 14 days of notification that the submittal has been rejected. Upon re-submittal there will be an additional Government review period. If the submittal is not accepted the process repeats until the submittal is accepted by the Government.

1.4.2 Sequencing of Activities

The sequencing specified for activities indicates the earliest the activity may begin.

1.4.3 Abbreviations

In TABLE I the abbreviation AA0 is used for 'after approval of' and 'ACO' is used for 'after completion of'.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

The process control system must be used to monitor and control the

operation of process equipment as specified and in accordance with the sequence of operation and control schematics shown on the drawings. The process control system must provide for operator interaction, overall process control system supervision, and process equipment control and monitoring. The system must adhere to Section 25 05 11 CYBERSECURITY OF FACILITY-RELATED CONTROL SYSTEMS. Provide hardware configured and sized to support expansion as specified and shown on the drawings.

The process control system must be complete including sensors, field preamplifiers, signal conditioners, offset and span adjustments, amplifiers, transducers, transmitters, control devices, engineering units conversions and algorithms for the applications; and must maintain the specified end-to-end process control loop accuracy from the sensor to display and final control element. Connecting conductors must be suitable for installed controls. Enclosures must be rated for NEMA 4.

2.1.1 Operation

The process control system provided under this specification must operate using a combination of sequential function charts, function block diagrams, structured text, instruction, and ladder logic type as defined in IEC 61131-3 and supervisory control to provide the required sequences of operation. Input data to the controller must be obtained by using instruments and controls interfaced to mechanical, electrical, utility systems and other systems as shown and specified. All required setpoints, settings, alarm limits, and sequences of operation must be as identified in the database/settings tables and sequences of operation indicated.

2.1.2 Points

Provide inputs to and outputs from the process control system in accordance with the Input/Output (I/O) Summary Table indicated. Each connected analog output (AO), analog input (AI), binary output (BO), binary input (BI), pulse accumulator (PA) input and other input or output device connected to the control system must represent a "point" where referred to in this specification.

2.1.3 Building Telecommunications Cabling (BTC)

Provide data transmission systems for communication between PLCs and between PLCs and the central station and server as specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM, Section 33 82 00 TELECOMMUNICATIONS OUTSIDE PLANT (OSP) and as indicated.

2.1.4 System Reliability

The system must be designed for maximum reliability, safety and integrity while maintaining an availability of 99.99 percent or better.

2.2 MATERIALS AND EQUIPMENT

2.2.1 Product Certifications

Computing devices, as defined in FCC Part 15, supplied as part of the process control system must be certified to comply with the requirements of Class B computing devices.

2.2.2 Standard Products

Materials and equipment must be standard unmodified products of a manufacturer regularly engaged in the manufacturing of such products. Units of the same type of equipment must be products of a single manufacturer. Items of the same type and purpose must be identical and supplied by the same manufacturer, unless replaced by a new version approved by the Government.

2.2.3 Nameplates

Each major component of equipment must have the manufacturer's name and address, and the model and serial number in a conspicuous place. Laminated plastic nameplates must be provided for equipment devices and panels furnished. Each nameplate must identify the device, such as pump "P-1" or valve "VLV-402". Labels must be coordinated with the schedules and the process and instrumentation drawings. Laminated plastic must be 3 mm thick, white with black center core. Nameplates must be a minimum of 25 by 75 mm with minimum 6 mm high engraved block lettering. Nameplates for devices smaller than 25 by 75 mm must be attached by a nonferrous metal chain. All other nameplates must be attached to the device.

2.3 GENERAL REQUIREMENTS

Equipment located outdoors, not provided with climate-controlled enclosure, must be capable of operating in the ambient temperature range. Electrical equipment will conform to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Equipment and wiring must be in accordance with NFPA 70, with proper consideration given to environmental conditions such as moisture, dirt, corrosive agents, and hazardous area classification.

2.4 SENSORS

2.4.1 Transmitter

Unless indicated otherwise, each sensor must be provided with a transmitter, selected to match the sensor. Except where specifically indicated otherwise on the drawings, the transmitter must be provided with a four digit or analog visual display of the measured parameter and with a 4 to 20 mAdc output signal proportional to the level of the measured parameter. Accuracy must be plus or minus 0.5 percent of full scale reading with output error not exceeding plus or minus 0.25 percent of full scale. Transmitter must be located where indicated, mounted integrally with the sensor, pipe mounted, wall mounted or installed in the control panel. The distance between the sensor and transmitter must not exceed the manufacturer's recommendation. Field preamplifiers and signal conditioners must be included when necessary to maintain the accuracy from sensor to the programmable logic controller or recorder.

2.4.2 Flow Rate Sensors and Meters

Liquid flow indication must be provided in L/s. Pressure taps must incorporate appropriate snubbers. Unless indicated otherwise, the flow transmitter must produce a signal that is proportional to the volumetric flow rate, compensated for fluid temperature, and must have an accuracy of plus or minus 1 percent of the actual flow. Flow transmitter must be located within 5 m of the flow element. The flow transmitter must include a digital readout of the volumetric flow rate to 2 significant figures. The controller must be provided with a minimum of three sets of dry

contacts rated in accordance with NEMA ICS 1. The first set of contacts must close when the lower (warning) detection level has been exceeded. The second set of contacts must close when the upper (alarm) detection level has been exceeded. The third set of contacts must close when a controller malfunction has occurred, including loss of power or loss of sensor input. The alarm levels must be individually adjustable. The controller must be provided with an audible warning horn that sounds when the upper detection level has been exceeded, and a warning horn silence button. The controller must provide a 4-20 mA_{dc} output signal to the programmable logic controller, proportional to the measured parameter. The controller must be provided with an internal battery to maintain operation for a minimum of 12 hours if power is lost. Flow rate must be controlled to within plus or minus 3 percent of the design flow.

2.4.2.1 Magnetic Flowmeter

Magnetic flowmeter must be non-intrusive, DC pulse type and must measure fluid flow through the use of a self generated magnetic field. The meter must have automatic zeroing circuitry. The magnetic flow element must be encapsulated in type 300 stainless steel. Flowmeter must be capable of measuring up to a maximum flow velocity of 3 m/s. The metering tube must be constructed of 316 stainless steel. The meter must be rated for a process temperature range of 0 to 100C and -18 to 65C ambient. The maximum pressure drop across the meter and appurtenances must be 34 kPa at the maximum flow rate.

2.4.3 Level Instrumentation

Pressure taps must incorporate appropriate snubbers. Relays and housing must be intrinsically safe or explosion proof as required by the NFPA hazard rating for compatibility with the contents of the tank or sump. The controller must be provided with a minimum of three sets of dry contacts rated in accordance with NEMA ICS 1. The first set of contacts must close when the lower (warning) detection level has been exceeded. The second set of contacts must close when the upper (alarm) detection level has been exceeded. The third set of contacts must close when a controller malfunction has occurred, including loss of power or loss of sensor input. The alarm levels must be individually adjustable. The controller must be provided with an audible warning horn that sounds when the upper detection level has been exceeded, and a warning horn silence button. The controller must provide a 4-20 mA_{dc} output signal to the programmable logic controller, proportional to the measured parameter. The controller must be provided with an internal battery to maintain operation for a minimum of 12 hours if power is lost.

2.4.3.1 Displacement Type Level Switch

Liquid level switch must be displacement type, having a minimum of two tandem floats with each float independently activating a set of Form C contacts at two different level settings. Each switch must have an adjustable differential band. The mounting connections must be threaded, flanged or surface mounted to suit the application. All surfaces in contact with the tank contents must be austenitic stainless steel. The switch enclosure must be explosion proof for use in a hazardous environment, complete with a sealed water tight junction box, terminal block, and mounting plate. Each set of contacts must be snap action, dry contact type with one normally open and one normally closed, contact rated in accordance with NEMA ICS 1. The switch must be actuated by a magnetically equipped stainless steel displacer. Repetitive accuracy must

be plus or minus 6 mm of actual displacer setting.

2.4.3.2 Mercury Float Switch

Float switch assemblies for use in liquid systems must consist of wall bracket or mounting plate, galvanized steel rods, stainless steel bolts, explosion proof and corrosion resistant housing, and intrinsically safe relays. Each switch must consist of two normally open mercury switches, encapsulated in epoxy resin. The float casing must be polypropylene. The switch cable must be oil resistant neoprene or PVC cable with 4 No. 18 gauge stranded copper conductors, rated for 600 Volt application.

2.4.3.3 Non-Contact Ultrasonic Level Sensor

The sensor must be microprocessor based and must provide continuous, non-contact level measurement of liquids and solids utilizing microwave pulsed time of flight measurement method. Sensor must have pushbutton calibration and LCD display. The sensor must operate in a frequency band approved for industrial use. The sensor must be capable of measuring in a range of 0 to 7 m with an accuracy of plus or minus .25 percent of full scale and repeatability $\pm 3\text{mm}$. The sensor must be capable of distinguishing between real echoes, reflections and background noise. The sensor must automatically compensate for temperature changes. The sensor must be capable of operating in a temperature range from minus 25 degrees C to 40 degrees C. Assembly must be flange mounted include surface mounting bracket of sufficient size to eliminate echoing and suitable for the installed environment indicated. Mounting assembly must be suitable for service without requiring entry or drainage of the vessel where level is being measured.

2.4.3.4 Leak Detection

Double walled containment system leak detectors must use electrodes mounted in the interstices of double walled containment systems with a minimum time delay of 0.5 seconds. Leak detectors for open systems must be mounted at slab or floor level with either a minimum time delay of 0.5 seconds or a minimum built-in-vertical adjustment of 3 mm to prevent activation due to high humidity. Detector must have a contact rating of 1.0 amps resistive or 200 mA inductive at 28 VDC. Leak detector panel must indicate the location and detector causing the alarmed state. The indicator must be manual reset type. A framed, non-fading half-size as-built location map in laminated plastic must be provided for the cable leak detection system in double containment piping systems indicating the as installed system configuration; sensing string layout must be furnished. Marks in meters feet along the length of pipeline interstitial cable must be provided as references to locate leaks.

2.4.4 Pressure Instrumentation

Pressure taps must incorporate appropriate snubbers.

2.4.4.1 Differential Pressure

The sensor/transmitter assembly accuracy must be plus or minus 1 percent of full scale. The over pressure rating must be a minimum of 150 percent of the operating pressure. Transmitter must be suitable for installation with the low pressure connection removed.

2.4.5 Limit Switch

Limit switch must be induction/proximity type, contacts must be heavy duty and rated A600 or N600. Enclosure must be metal. The actuating head must vary based upon the application.

2.4.6 Emergency Stop

Emergency stop pushbutton must have red mushroom actuator and yellow background. Actuator must be self-latching and manually reset. Provide contacts that are NEMA ICS A600 and directly opened. Pushbutton must meet NFPA 79 SECTION 10.7.

2.5 PROGRAMMABLE LOGIC CONTROLLER (PLC)

2.5.1 PLC General Requirements

PLCs must be micro-processor based, capable of receiving binary and analog inputs and, through programming, must be able to control binary and analog output functions, perform data handling operations and communicate with external devices. PLCs must meet the requirements of Class A computing devices, and must be labeled as set forth in 47 CFR 15 and must be able to withstand conducted susceptibility test as outlined in NEMA ICS 1, NEMA ICS 2, NEMA ICS 3, and IEEE C37.90.1. PLCs must function properly at temperatures between 0 and 50 degrees C at 5 to 95 percent relative humidity non-condensing and must tolerate storage temperatures between minus 40 and plus 60 degrees C at 5 to 95 percent relative humidity non-condensing.

2.5.2 Modular PLC

PLCs must be based on a modular, field expandable design allowing the system to be tailored to the process control application. The system must be expandable through the use of additional hardware and/or user software. As a minimum, the PLC must include a mounting backplane, power supply module, central processing unit (CPU) module, communications module, and input/output (I/O) module. The modules must be grouped together in a mounting rack or cabinet. The mounting rack backplane must provide the communications mechanism to fully integrate the individual modules located within the rack. Modules other than I/O modules must plug directly into the backplane. The use of wire connectors between modules will not be allowed except for expansion of the system to include multiple backplanes. The rack size must be as needed to hold the equipment necessary while performing the required control functions. The system configuration must allow for the removal and/or installation of modules under power.

2.5.2.1 Central Processing Unit (CPU) Module

The CPU module must be a self contained, microprocessor based unit that provides time of day, scanning, application program execution, storage of application programs, storage of numerical values related to the application process and logic, I/O bus traffic control, peripheral and external device communications and self diagnostics. The scan time must be 250 milliseconds or better including spare I/O channels.

2.5.2.2 Communications Module

The communications module must allow peer-to-peer communication with other

PLCs and must allow the PLC to communicate with the central station, or workstation. The communication module must utilize the manufacturer's standard communication architecture and protocol, ethernet architecture and protocol or a combination of these. The communication module must allow programming of the PLC to be done locally through the use of a laptop computer or from the central station or remote workstation.

2.5.2.3 Power Supply Module

One or more power supply modules must be provided as necessary to power other modules installed in the same cabinet. Power supply modules must plug directly into the backplane. Auxiliary power supplies may be used to supply power to remote cabinets or modules.

- a. Power supply modules must use AC power with a nominal voltage of 120 VAC plus or minus 5 percent. The power supply module must monitor the incoming line voltage level and must provide over current and over voltage protection. If the voltage level is detected as being out of range the power supply module must continue to provide power for an adequate amount of time to allow for a safe and orderly shutdown. Power supply modules must be capable of withstanding a power loss for a minimum of 20 milliseconds while still remaining in operation and providing adequate power to all connected modules.
- b. Each power supply module must be provided with an on-off switch integral to the module. If the manufacturer's standard power supply module is not provided with an on-off switch, a miniature toggle type switch must be installed near the PLC and must be clearly labeled as to its function.
- c. Provide power supply modules with an indicating light which must be lit when the module is operating properly.

2.5.2.4 Input/Output (I/O) Modules

Modules must be self contained, microprocessor based units that provide an interface to field devices. The modules must be located in the same cabinet as the other PLC components. Each module must contain visual indication to display the on-off status of individual inputs or outputs. Each I/O must be protected against reversal of polarity of the signal. Analog inputs and analog outputs must have 'open, short and out of range circuit' detection. It must be configurable per channel.

2.5.3 Loop PLC

PLCs must be single or multiple loop controllers depending on the process control system requirements. Controllers must be self contained and must include a central processing unit (CPU), program memory, power supply, input/output capability, network communications capability and display/keyboard. The controller must have a scaleable process variable for each loop. Controller must have proportional, integral and derivative (PID) control logic. Analog outputs must be configured as direct acting or reverse acting. The controller must have keyboard, display, auto/manual selection for control of each loop output, remote setpoint, adjustment/local setpoint adjustment selection with adjustable high-end and low-end limits, ratio and bias adjustment on remote setpoint input. Controller must power analog output loops to 20 mAdc when connected to a load of 600 ohms.

2.5.3.1 Central Processing Unit (CPU)

The central processing unit must be microprocessor based and must provide time of day, scanning, application program (ladder rung logic) execution, storage of application programs, storage of numerical values related to the applications process and logic, I/O bus traffic control, peripheral and external device communications and self diagnostics.

2.5.3.2 Power Requirements

Each controller must be powered by DC power with a nominal voltage of 24 VDC. Power consumption must not exceed 25 watts. Controller must provide electrical noise isolation between the AC power line and the process variable inputs, remote setpoint inputs and output signals of not less than 100 dB at 60 Hertz common mode rejection ration and not less than 60 dB at 60 hertz normal-mode rejection ration.

2.5.3.3 On-Off Switch

Each controller must be provided with an integral on-off switch. If the controller is not provided with a manufacturer's standard on-off switch, a miniature toggle type switch must be installed near the controller and must be clearly labeled as to its function.

2.5.3.4 Parameter Input and Display

Enter and display control parameters directly, in the correct engineering units, through a series of keystrokes on a front panel display with decimal point and polarity indication. Provide display in metric or English units as selected by the operator.

2.5.3.5 Self Tuning

Provide controllers with self-tuning operation which applies to proportional, integral and derivative modes of control and modify the mode constants as required. Self-tuning must only be in operation when selected from the front panel.

2.5.3.6 Manual Tuning

Controllers must be provided with manual tuning operation which must apply to proportional, integral and derivative modes of control, by means of individually adjustable mode constants. These adjustments must be set for the appropriate value if a particular control mode action is required or to zero if that particular mode is not desired. The proportional mode constant must be adjustable from 0 to 200 percent of the input signal range. The integral mode constant must be adjustable from 0 to 20 repeats per minute. The derivative mode constant must be adjustable from 0 to 1 minute.

2.5.4 Program Storage/Memory Requirements

The CPU must utilize the manufacturer's standard non-volatile memory for the operating system. The controller must have electronically readable and writeable non volatile memory (EPROM, EEPROM, or Flash PROM) for storage of user programs. The user programs must be loaded through the controller keypad, central station or through the use of a laptop computer. The CPU memory capacity must be based on the system's control requirements. The memory capacity must be sized such that, when the

system is completely programmed and functional, no more than 50 percent of the memory allocated for these purposes is used.

2.5.5 Input/Output Characteristics

Each controller must allow for analog input, analog output, binary input and binary output. The number and type of inputs and outputs for the system must be as shown on the drawings and must comply with the sequence of control. The system capacity must include a minimum of 20 percent spare input and output points (no less than two points) for each point type provided. During normal operation, a malfunction in any input/output channel must affect the operation of that channel only and must not affect the operation of the CPU or any other channel. All input circuits must have a minimum optical isolation of 1500 VRMS and must be filtered to guard against high voltage transients from the externally connected devices. All output circuits must have a minimum optical isolation of 1500 VRMS and must be filtered to guard against high voltage transients from the externally connected devices.

2.5.5.1 Analog Inputs

Analog input circuits must be available in 4-20 mA.

2.5.5.2 Binary Inputs

Binary input circuits must be available in 10-30 VDC.

2.5.5.3 Analog Outputs

Analog output circuits must be available in 4-20 mA.

2.5.5.4 Binary Outputs

Binary output circuits must be available in 18-26 VDC.

2.5.6 Wiring Connections

Wiring connections must be heavy duty, self lifting, pressure type screw terminals to provide easy wire insertion and secure connections. The terminals must accept two #14 AWG wires. A hinged protective cover must be provided over the wiring connections. The cover must have write-on areas for identification of the external circuits.

2.5.7 On-Off Switch

Each controller must be provided with an integral on-off power switch. If the controller is not provided with a manufacturer's standard on-off switch, a miniature toggle type switch must be installed in the control panel near the controller and must be clearly labeled as to its function.

2.5.8 Diagnostics

Each PLC must have diagnostic routines implemented in firmware. The CPU must continuously perform self-diagnostic routines that will provide information on the configuration and status of the CPU, memory, communications and input/output. The diagnostic routines must be regularly performed during normal system operation. A portion of the scan time of the controller must be dedicated to performing these housekeeping functions. In addition, a more extensive diagnostic routine must be

performed at power up and during normal system shutdown. The CPU must log input/output and system faults in fault tables which must be accessible for display. When a fault affects input/output or communications modules the CPU must shut down only the hardware affected and continue operation by utilizing the healthy system components. All faults must be annunciated at the central station. Diagnostic software must be useable in conjunction with the portable tester. The following diagnostics must be performed:

- a. Analog Inputs: Sensor out of range, open or shorted loop, analog-to-digital converter check
- b. Analog Outputs: Open or shorted loop
- c. Configuration: Check compatibility and availability of selected I/O hardware and software
- d. Memory: Checksum, parity check End-to End CPU memory

2.5.9 Accuracy

Provide controllers with an accuracy of plus or minus 0.25 percent of input span.

2.5.10 Primary/Secondary PLC

The Primary/Secondary PLCs must have redundancy built into the process control system by having two systems (power supply and CPU) either of which is capable of controlling the system. Data must be transferred from the primary processor to the secondary processor each logic cycle. The I/O scan must be transferred from the PLC currently in charge to the other at the end of each logic execution and the logic must be executed. The data must be transferred via asynchronous transfer where the primary processor has two separate microprocessors embedded in its circuitry and at the end of logic execution all data must be passed to the second microprocessor and the second microprocessor must handle all transfer tasks while the first executes the next program scan. The Primary must perform the execution of the outputs unless a fault is detected in which case execution of the outputs is performed by the Secondary. Switchover must be automatic and indication of the switchover must be displayed on the Central Operator Workstation.

2.6 PLC SOFTWARE

Furnish all PLC software described in this specification as part of the complete control system.

2.6.1 Operating System

Each PLC must be provided with the manufacturer's standard operating system software package. The PLC must maintain a point database in its memory that includes all parameters, constraints and the latest value or status of all points connected to the PLC. Execution of the PLC application programs must use the data in memory resident files. The operating system must support a full compliment of process control functions. It must be possible to define these functions using a mix of ladder logic diagrams, function blocks, sequential function charts and text programming. Programming methods and interactions must be based on IEC 61131-3. A combination of the programming methods must be possible

within a single controller. The operating system must allow loading of control logic locally and data files from the portable tester. It must also support data entry and diagnostics using an operator interface panel attached directly to the PLC. Each PLC must be capable of operating in stand alone mode.

2.6.1.1 Startup

The PLC must have startup software that causes automatic commencement of operation without human intervention, including startup of all connected I/O functions. A PLC restart program based on detection of power failure at the PLC must be included in the PLC software. The restart program must include start time delays between successive commands to prevent demand surges or overload trips.

2.6.1.2 Failure Mode

Upon failure for any reason, each PLC must perform an orderly shutdown. Systems which are not Primary/Secondary must force all PLC outputs to a predetermined (failure mode) state, consistent with the failure modes shown and the associated control device. Primary/Secondary systems must transfer I/O scan and control to the PLC not currently failed.

2.6.2 Functions

The controller operating system must be able to scan inputs, control outputs, and read and write to its internal memory in order to perform the required control as indicated in the sequence of control on the drawings. The controller must periodically perform self diagnostics to verify that it is functioning properly. If the system is set up as a Primary/Secondary system the system must attempt to switch to the other PLC upon sensing a fault in the currently controlling PLC.

2.6.2.1 Analog Monitoring

Measure and transmit all analog values including calculated analog points.

2.6.2.2 Logic (Virtual)

Logic (virtual) points must be software points entered in the point database which are not directly associated with a physical I/O function. Logic (virtual) points must be analog or binary points created by calculation from any combination of binary and analog points, or other data having all the properties of real points, including alarms, without the associated hardware. Logic (virtual) points must be defined or calculated and entered into the database. The calculated analog point must have point identification in the same format as any other analog point.

2.6.2.3 State Variables

If an analog point represents more than two (up to 8) specific states, each state must be nameable. For example, a level sensor must be displayed at its measured engineering units plus a state variable with named states usable in programs or for display such as low alarm/low/normal/high/high alarm.

2.6.2.4 Analog Totalization

Any analog point must be operator assignable to the totalization program. Up to eight analog values must be totalized within a selectable time period.

2.6.3 Alarm Processing

Each PLC must have alarm processing software for AI, DI, and PA alarms for all real and virtual points connected to that PLC.

2.6.3.1 Binary Alarms

Binary alarms are those abnormal conditions indicated by BIs as specified and shown. The system must automatically suppress analog alarm reporting associated with a binary point when that point is turned off.

2.6.3.2 Analog Alarms

Analog alarms are those conditions higher or lower than a defined value, as measured by an AI. Analog readings must be compared to predefined high and low limits, and alarmed each time a value enters or returns from a limit condition. Unique high and low limits must be assigned to each analog point in the system. In control point adjustment (CPA) applications, key the limit to a finite deviation traveling with the setpoint. The system must automatically suppress analog alarm reporting associated with an analog point when that analog point is turned off.

2.6.3.3 Pulse Accumulator (PA) Alarms

Pulse accumulator alarms are those conditions calculated from totalized values of accumulator inputs or PA input rates that are outside defined limits as specified and shown. PA totalized values must be compared to predefined limits and alarmed each time a value enters a limit condition. Unique limits must be assigned to each PA point in the system.

2.6.4 Constraints

2.6.4.1 Equipment Constraints Definitions

Each control point in the database must have PLC resident constraints defined and entered by the Contractor, including as applicable: maximum starts (cycles) per hour; minimum off time; minimum on time; high limit (value in engineering units); and low limit (value in engineering units).

2.6.4.2 Constraints Checks

All control devices connected to the system must have the PLC constraints checked and passed before each command is issued. Each command point must have unique constraints assigned. High and low "reasonableness" values or one differential "rate-of-change" value must be assigned to each AI. Each individual point must be capable of being selectively disabled by the operator from the central station.

2.6.5 Control Sequences and Control Loops

Specific functions to be implemented are defined in individual system control sequences and database tables shown on the drawings, and must include, as applicable, the following functions: PI control must provide

proportional control and proportional plus integral control; two position control must provide control for a two state device by comparing a set point against a process variable and an established dead band; floating point control must exercise control when an error signal exceeds a selected dead band, and must maintain control until the error is within the dead band limits; signal selection must allow the selection of the highest or lowest analog value from a group of analog values as the basis of control and must include the ability to cascade analog values so that large numbers of inputs can be reduced to one or two outputs; signal averaging must allow the mathematical calculation of the average analog value from a group of analog values as the basis of control and must include the ability to "weight" the individual analog values so that the function output can be biased as necessary to achieve proper control; reset function must develop an AO based on up to two AIs and one operator specified reset schedule.

2.6.6 Command Priorities

A scheme of priority levels must be provided to prevent interaction of a command of low priority with a command of higher priority. Override commands entered by the operator must have higher priority than those emanating from applications programs.

2.6.7 Resident Application Software

Provide resident applications programs developed in accordance with paragraph Graphical Object Oriented Programming to achieve the sequences of operation, parameters, constraints, and interlocks necessary to provide control of the process systems connected to the process control system. All application programs must be resident in the PLC and must execute in the PLC, and must coordinate with each other, to ensure that no conflicts or contentions remain unresolved.

2.6.7.1 Program Inputs and Outputs

Use program inputs listed for each application program to calculate the required program outputs. Where specific program inputs are not available, a "default" value or virtual point appropriate for the equipment being controlled and the proposed sequence of operation must be provided to replace the missing input, thus allowing the application program to operate.

2.6.7.2 Failure Mode

In the event of a PLC failure, the controlled equipment must continue to function in the failure mode shown on the drawings. Systems that are Primary/Secondary must transfer control to the non-failed system.

2.7 CONTROL PANELS

2.7.1 Components

2.7.1.1 Enclosures

The enclosure for each control panel must conform to the requirements of **NEMA 250** for the types specified. Finish color must be the manufacturer's standard, unless otherwise indicated. Enclosures for installation in mechanical equipment rooms must be Type 12; those for installation in clean, dry indoor occupied space may be Type 1; other locations must be as

otherwise specified or shown. Enclosures for equipment installed outdoors must be Type 4 or as shown. Enclosures for installation in a corrosive environment must be Type 4X and must be constructed of stainless steel. Painted steel must not be allowed for use in a corrosive environment. Enclosure must be provided with a single, continuously hinged exterior door with print pocket, 3-point latching mechanism and key lock and a single, continuously hinged interior door.

2.7.1.2 Controllers

Provide controllers in accordance with paragraph Programmable Logic Controller (PLC).

2.7.1.3 Standard Indicator Light

Indicator lights showing on, off, stand-by, automatic, manual depending on the application must comply with NEMA ICS 1, NEMA ICS 2 and UL 508A. Lights must be heavy duty, round and must mount in a 22.5 mm inch mounting hole. Indicator lights must be LED type and must operate at 120 VAC or 24 VDC. Long life bulbs must be used. Indicator light must be provided with a legend plate labeled as shown on the drawings. Lens color must be as indicated on the drawings. Lights must be push to test (lamp) type.

2.7.1.4 Selector Switches

Selector switches must comply with NEMA ICS 1, NEMA ICS 2 and UL 508A. Selector switches must be heavy duty, round and must mount in a 22.5 mm mounting hole. The number of positions must be as indicated on the drawings. Switches must be as indicated on the drawings. Switches must be rated for 600 volts, 10 amperes continuous. Selector switches must be provided with a legend plate labeled as shown on the drawings. Where indicated or required, dual auxiliary contacts must be provided for the automatic position to provide position sensing at the central station or workstation. Auxiliary contacts must be rated for 120 VAC, 1A as a minimum. Where indicated on the drawings, switches must be key operated. All keys must be identical.

2.7.1.5 Push Buttons

Push buttons must comply with NEMA ICS 1, NEMA ICS 2 and UL 508A. Push buttons must be heavy duty, round and must mount in a 22.5 mm mounting hole. The number and type of contacts must be as indicated on the drawings or required by the Sequence of Control. Push buttons must be rated for 600 volts, 10 amperes continuous. Push buttons must be provided with a legend plate labeled as shown on the drawings.

2.7.1.6 Relays

Relays must comply with NEMA ICS 5 and derated for altitude above 1,500 m. Relays must be double-pole, double-throw (DPDT) or as required by the Sequence of Control. Relay coil must be 24 VDC and must be provided with matching mounting socket. Power consumption must not be greater than 3 watts. Coils must have a minimum current rating of 1 amps and minimum voltage rating of 120 volts. Contacts must have a minimum current rating of 5 amps and minimum voltage rating of 600 volts.

2.7.1.7 Terminal Blocks

Terminal blocks must comply with NEMA ICS 4 and UL 1059. Terminal blocks for conductors exiting control panels must be two-way type with double terminals, one for internal wiring connections and the other for external wiring connections. Terminal blocks must be made of bakelite or other suitable insulating material with full deep barriers between each pair of terminals. A terminal identification strip must form part of the terminal block and each terminal must be identified by a number in accordance with the numbering scheme on the approved wiring diagrams.

2.7.1.8 Alarm Horns

Alarm horns must be provided where indicated on the drawings. Horns must be vibrating type and must comply with UL 508A. Horns must provide 100 dB at 10 feet. Exterior mounted horns must be weather proof by design or must be mounted in a weather proof enclosure that does not reduce the effectiveness of the horn.

2.7.2 Panel Assembly

Control panels must be factory assembled and shipped to the jobsite as a single unit. Panels must be fabricated as indicated and devices must be mounted as shown or required. Each panel must be fabricated as a bottom-entry connection point for process control system electrical power, process control system wiring, communications system wiring to other control panels.

2.7.3 Electrical Requirements

Each panel must be powered by a dedicated 120 volts ac circuit, with a fuse, sized as recommended by the equipment manufacturer, and a disconnect switch located inside the panel. Wiring must terminate inside the panel on terminal blocks. Electrical work must be as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and as shown on the drawings.

2.7.4 Power Line Conditioner

Each control panel must be provided with a power line conditioner to provide both voltage regulation and noise rejection. The power line conditioner must be of the ferro-resonant design, with no moving parts and no tap switching, while electrically isolating the secondary from the power line side. The power line conditioner must be sized for 125 percent of the actual connected kva load. Characteristics of the power line conditioner must be as follows:

2.7.4.1 85 Percent Load

At 85 percent load, the output voltage must not deviate by more than plus or minus 1 percent of nominal voltage when the input voltage fluctuates between minus 20 percent to plus 10 percent of nominal voltage.

2.7.4.2 Load Changes

During load changes of zero to full load, the output voltage must not deviate by more than plus or minus 3 percent of nominal voltage. Full correction of load switching disturbances must be accomplished within 5 cycles, and 95 percent correction must be accomplished within 2 cycles of the onset of the disturbance.

2.7.5 Grounding

Control panel enclosures must be equipped with a solid copper ground bus or equivalent. The ground bus must be securely anchored to the enclosure so as to effectively ground the entire structure. Clamp-type terminals sized large enough to carry the maximum expected current must be provided on the ground bus for grounding cables. Where a definite circuit ground is required, a single wire not less than #10 AWG must run independently to the panel ground bus and must be fastened to the ground bus with a bolted terminal lug. Cases of instruments, relays and other devices must be effectively grounded through the enclosures steel structure unless otherwise indicated. Insulated wiring having a continuous rated current of not less than the circuit fuse rating must be used for grounding. Grounding terminals of power receptacles must be solidly grounded to the panel enclosure.

2.7.6 Convenience Outlet

A 120 volt ac, 20 amp, ground fault interruption (GFI) type duplex convenience outlet must be provided inside the panel. The outlet circuit must be separate from the panel power circuit.

2.7.7 Panel Interior Light

Where indicated, Each control panel must be provided with a 15 watt LED light. The light must be operated by a manual on-off switch mounted on the interior door of the enclosure. The light must be powered by the same circuit as the convenience outlet.

2.7.8 Ventilation System

Where indicated, Each control panel must be provided with two single phase, 120 volt ac ventilation fans. Each fan must supply a minimum of 50 L/s of ventilation air through the enclosure. Each fan must be provided with a line voltage thermostat. Thermostat setpoints must be adjustable in a range of 21 to 60 degrees C as a minimum. Each supply and exhaust grille must contain a filter that is easily removed for cleaning or replacement.

2.8 FACTORY TEST

The process control system must be tested at the factory prior to shipment. Written notification of planned testing must be given to the Government at least 21 days prior to testing, and in no case must notice be given until after the Contractor has received written Government approval of the test procedures.

2.8.1 Factory Test Setup

Assemble and integrate the factory test setup as specified to prove that performance of the system satisfies all requirements of this project, including system communications requirements in accordance with the approved test procedures. The factory test must take place during regular daytime working hours on weekdays. Equipment used must be the same equipment that is to be delivered to the site. The factory test setup must include the following:

Factory Test	
central station equipment	one each of the components
workstation	one of each type
control panel	not less than two control panels: at least one of each type used in the system plus at least one per DTS type
test set	one of each type
portable tester	one of each type
communications circuits	one of each type and speed to be utilized in the proposed system including bridges, modems, encoder/decoders, transceivers and repeaters
surge protection equipment	for power, communications, I/O functions and networks
I/O functions	sufficient to demonstrate the I/O capability and system normal operation
software	software required for proper operation of the proposed system including application programs and sequences of operation

2.8.2 Factory Test Procedure

Test procedures must define the tests required to ensure that the system meets technical, operational, and performance requirements. The test procedures must define location of tests, milestones for the tests, and identify simulation programs, equipment, personnel, facilities, and supplies required. Provide for testing all process control system capabilities and functions specified and shown. Cover actual equipment and sequences to be used for the specified project and include detailed instructions for test setup, execution, and evaluation of test results. The test reports must document results of the tests. Surge testing need not be conducted acceptable documented proof can be provided that such testing has been satisfactorily demonstrated to the Government with identical surge protection applied. The procedures must include the following:

	Test Procedure
equipment	block diagram
hardware and software	descriptions

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
NAVSTA ROTA, SPAIN

	Test Procedure
commands	operator commands
I/O functions	test database points with failure modes
passwords	required for each operator access level
each type of digital and analog point in the test database	description
test equipment	list
surge protection	circuit diagrams
inputs required (I/O point values and status) and corresponding expected results of each set of input values	for each application program
default values	for the application program inputs not implemented or provided for in the contract documents for the application programs to be tested

2.8.3 Factory Test Report

Submit original copies of data produced during the factory test, including results of each demonstration procedure within 7 days after completion of each test. Arrange the report so that commands, responses, and data acquired are correlated to allow logical interpretation of the data.

PART 3 EXECUTION

3.1 FACTORY TEST

Perform factory testing of the System as specified. The Contractor is responsible for providing personnel, equipment, instrumentation, and supplies necessary to perform required testing. Provide written notification of planned testing to the Government at least 21 days prior to testing, and do not give this notice until after receiving written Government approval of the specific Factory Test Procedures. Provide Factory Test Procedures which define the tests required to ensure that the system meets technical, operational, and performance specifications. Within the Procedures define location of tests, milestones for the tests, and identify simulation programs, equipment, personnel, facilities, and supplies required. Provide procedures which test all capabilities and functions specified and indicated. Perform the Factory Test using equipment and software of the same manufacturer, model and revision as will be used for the specified project. Include detailed instructions for test setup, execution, and evaluation of test results in the Procedures. Upon completion of the test, prepare a Factory Test Report, documenting the results of the Test, and submit it as specified. This report must be approved before any equipment is shipped.

Perform the Factory Test and provide Factory Test Submittals as shown in TABLE II. FACTORY TEST SEQUENCING.

TABLE II FACTORY TEST SEQUENCING

ITEM #	DESCRIPTION	SEQUENCING (START of ACTIVITY or DEADLINE FOR SUBMITTAL)
1	Submit Factory Test Procedure	14 days after notice to Proceed
2	Perform Factory Test	After Approval of #1
3	Submit Factory Test Report	7 days After Completion of #2

3.2 EQUIPMENT INSTALLATION REQUIREMENTS

3.2.1 Installation

Install system components and appurtenances in accordance with the manufacturer's instructions and provide necessary interconnections, services, and adjustments required for a complete and operable system. Adjust or replace devices not conforming to the required accuracies. Replace factory sealed devices, rather than adjusting. Installation, adjustment, and operation of the equipment specified must be supervised by a manufacturer's representative experienced in the installing, adjusting, and testing of the equipment.

- a. Install instrumentation and communication equipment and cable grounding as necessary to preclude ground loops, noise, and surges from adversely affecting system operation.
- b. Install wiring in exposed areas, including low voltage wiring, in EMT conduit as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Wiring in air plenum areas installed without conduit must be plenum-rated in accordance with NFPA 70.
- c. Submit detail drawings containing complete piping, wiring, schematic, flow diagrams and any other details required to demonstrate that the system has been coordinated and will properly function as a unit. Piping and Instrumentation (P&ID) drawings (prepared using industry recognized device symbols, clearly defined and describing piping designations to define the service and materials of individual pipe segments and instrument tags employing Instrument Society of America suggested identifiers). Include in the Drawings, as appropriate: product specific catalog cuts; a drawing index; a list of symbols; a series of drawings for each process control system using abbreviations, symbols, nomenclature and identifiers as shown; valve schedules; compressed instrument air station schematics and ASME air storage tank certificates for each type and make of compressed instrument air station.

3.2.1.1 Isolation, Penetrations and Clearance from Equipment

Dielectric isolation must be provided where dissimilar metals are used for connection and support. Penetrations through and mounting holes in the building exteriors must be made watertight. Holes in concrete, brick, steel and wood walls must be drilled or core drilled with proper

equipment; conduits installed through openings must be sealed with materials which are compatible with existing materials. Openings must be sealed with materials which meet the requirements of NFPA 70 and Section 07 84 00 FIRESTOPPING. Installation must provide clearance for control-system maintenance. Process control system installation must not interfere with the clearance requirements for mechanical and electrical system maintenance.

3.2.1.2 Device Mounting

Devices must be installed in accordance with manufacturers' recommendations and as shown. Control devices to be installed in piping must be provided with required gaskets, flanges, thermal compounds, insulation, piping, fittings, and manual valves for shutoff, equalization, purging, and calibration. Any deviations must be documented and submitted to the Government for approval prior to mounting. Damaged insulation must be replaced or repaired after devices are installed to match existing work. Damaged galvanized surfaces must be repaired by touching up with zinc paint.

3.2.1.3 Pneumatic Tubing

Tubing must be concealed in finished areas. Tubing may be run exposed in unfinished areas, such as mechanical equipment rooms. For tubing to be enclosed in concrete, rigid metal or intermediate metal conduit must be provided. Tubing must be installed parallel or perpendicular to building walls throughout. Maximum spacing between tubing supports must be 1.5 m 5 feet. Each tubing system must be tested pneumatically at 1.5 times the working pressure for 24 hours, with a maximum pressure drop of 0.15 kPa with compressed air supply turned off. Joint leaks must be corrected by remaking the joint. Caulking of joints will not be permitted. Tubing and two insulated copper phone wires for installation checkout may be run in the same conduit. Tubing and electrical power conductors must not be run in the same conduit; however, control circuit conductors may be run in the same conduit as polyethylene tubing.

3.2.1.4 Grooved Mechanical Joints

Grooves must be prepared according to the coupling manufacturer's instructions. Grooved fittings, couplings, and grooving tools must be the products of the same manufacturer. Pipe and groove dimensions must comply with the tolerances specified by the coupling manufacturer. The diameter of grooves made in the field must be measured using a "go/no-go" gauge, vernier or dial caliper, narrow-land micrometer, or other method specifically approved by the coupling manufacturer for the intended application. Groove width and dimension of groove from end of pipe must be measured and recorded.

3.2.2 Sequences of Operation

Study the operation and sequence of local equipment controls, as a part of the conditions report, and note any deviations from the described sequences of operation on the contract drawings. Perform necessary adjustments to make the equipment operate in an optimum manner and must fully document changes made.

3.3 INSTALLATION OF EQUIPMENT

Install equipment as specified, as shown and as required in the

manufacturer's instructions for a complete and fully operational control system.

3.3.1 Control Panels

Control panels must be located as indicated on the drawings. Devices located in the control panels must be as shown on the drawings or as needed to provide the indicated control sequences.

3.3.2 Flow Measuring Device

Fluid flow instruments must be installed in accordance with manufacturer's recommendations, unless otherwise indicated in the specification. The minimum straight unobstructed piping for the flowmeter installation must be 10.0 pipe diameters upstream and 5.0 pipe diameters downstream. Meters for gases and vapors must be installed in vertical piping, and meters for liquids must be installed in horizontal piping, unless otherwise recommended by the manufacturer or indicated in the specifications.

3.3.2.1 Magnetic Flowmeter

Meter must be installed in vertical piping so that the flow tube remains full of the process fluid under all operating conditions. A minimum of ten pipe diameters straight run upstream of the flowmeter and five pipe diameters straight run downstream of the flowmeter must be provided. The flowmeter and piping system must be grounded to earth ground.

3.3.3 Level Instruments

3.3.3.1 Displacement Type Liquid Level Switch

Level switches must be installed in accordance with the manufacturer's instructions. Switches must be accessible for maintenance and calibration. In applications where switches cannot be directly mounted to a tank by the threaded or flanged connection, a mounting bracket must be provided for connection to the inside tank wall.

3.3.3.2 Mercury Float Switches

Switches must be mounted in accordance with manufacturer's published instructions. Procedures must be those used for equipment in hazardous locations.

3.3.3.3 Ultrasonic Level Sensor

Sensor must be installed vertically in the top of the tank and in accordance with the manufacturer's instructions. Switches must be accessible for maintenance and calibration. In applications where switches cannot be directly mounted to a tank by the threaded or flanged connection, a mounting bracket must be provided for connection to the inside tank wall. Sensor must be positioned to maximize the return echo signal and minimize vessel obstructions in the sensors line of sight. The minimum recommended distance from the tank fill opening and from the side of the tank must be maintained.

3.3.4 Pressure Instruments

Pressure sensors and pressure transducers must be verified by calibration. All pressure taps must incorporate appropriate snubbers.

Pressure sensors and pressure switches must have valves for isolation, venting, and taps for calibration. Pressure switches and pressure transducers installed on liquid or steam lines must have drains. Pressure transducers, differential pressure sensors and differential pressure switches must have nulling valves. Pressure switches must be adjusted to the proper setpoint and must be verified by calibration. Switch contact ratings and duty must be selected for the application.

3.3.5 Output Devices

Output devices (transducers, relays, contactors, or other devices) which are not an integral part of the control panel, must be mounted in an enclosure mounted adjacent to the control panel, unless otherwise shown. Where H-O-A and/or override switches on the drawings or required by the control sequence, the switches must be installed so that the process control system controls the function through the automatic position and other controls work through the hand position.

3.3.6 Enclosures

All enclosure penetrations must be from the bottom of the enclosure, and must be sealed to preclude entry of water using a silicone rubber sealant.

3.3.7 Transformers

Transformers for control voltages below 120 VAC must be fed from the nearest power panel or motor control center, using circuits provided for the purpose. Provide a disconnect switch on the primary side and a fuse on the secondary side. Transformers must be enclosed in a steel cabinet with conduit connections.

3.4 WIRE, CABLE AND CONNECTING HARDWARE

3.4.1 LAN Cables and Connecting Hardware

LAN cables and connecting hardware must be installed in accordance with Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM and Section 33 82 00 TELECOMMUNICATIONS OUTSIDE PLANT (OSP).

3.4.2 Metering and Sensor Wiring

Metering and sensor wiring must be installed in accordance with the requirements of ANSI C12.1, NFPA 70, Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

3.4.2.1 Power Line Surge Protection

Control panels must be protected from power line surges. Protection must meet the requirements of IEEE C62.41.1 and IEEE C62.41.2. Fuses must not be used for surge protection.

3.4.2.2 Sensor and Control Wiring Surge Protection

Digital and analog inputs must be protected against surges induced on control and sensor wiring. Protect binary and analog outputs against surges induced on control and sensor wiring installed outdoors and as shown. Fuses must not be used for surge protection. Test the inputs and outputs in both the normal and common mode using the following two waveforms: The first waveform must be 10 microseconds by 1000

microseconds with a peak voltage of 1500 volts and a peak current of 60 amperes. The second waveform must be 8 microseconds by 20 microseconds with a peak voltage of 1000 volts and a peak current of 500 amperes. Submit certified test results for surge protection.

3.5 SOFTWARE INSTALLATION

Load software required for an operational process control system, including databases (for points specified and shown), operational parameters, and system, command, and application programs. Adjust, tune, debug, and commission all software and parameters for controlled systems to assure proper operation in accordance with the sequences of operation and database tables.

3.6 CONTROL DRAWINGS

3.6.1 Control

Control drawings, reproducible, with corresponding CADD files, must be provided for equipment furnished and for interfaces to equipment at each respective equipment location. Condensed operating instructions explaining preventive maintenance procedures, methods of checking the system for normal safe operation and procedures for safely starting and stopping the system manually must be prepared in typed form, reproducible, with corresponding word processor files and posted beside the diagrams. Diagrams and instructions must be submitted prior to posting. The framed instructions must be posted before acceptance testing of the system.

3.6.2 Contractor Design Drawings

Contractor Design Drawings as a single complete package: 6 hard copies and 1 copy in electronic form. As a minimum they must include wiring, logic, and layout. Submit hardcopy drawings on ISO A1 841 by 594 mm 34 by 22 inches sheets, and electronic drawings in PDF and in AutoCAD format. In addition, submit electronic drawings in editable Excel format for all drawings that are tabular, including but not limited to the Point Schedule and Equipment Schedule. Contractor Design Drawings must be approved prior to any fabrication.

3.6.2.1 Draft As-Built

[Draft As-Built Drawings](#) as a single complete package: 6 hard copies and 1 copy in electronic form. Submit hardcopy drawings on ISO A1 841 by 594 mm 34 by 22 inches sheets, and electronic drawings in PDF and in AutoCAD format. In addition, submit electronic drawings in editable Excel format for all drawings that are tabular, including but not limited to the Point Schedule and Equipment Schedule.

3.6.2.2 Final As-Built

[Final As-Built Drawings](#) as a single complete package: 6 hard copies and 1 copy in electronic form. Submit hardcopy drawings on ISO A1 841 by 594 mm 34 by 22 inches sheets, and electronic drawings in PDF and in AutoCAD format. In addition, submit electronic drawings in editable Excel format for all drawings that are tabular, including but not limited to the Point Schedule and Equipment Schedule.

3.6.3 Points Schedule

Provide a Points Schedule in tabular form for each system, with the indicated columns and with each row representing a hardware point, network point or configuration point in the system.

- a. When a Points Schedule was included in the Contract Drawing package, use the same fields as the Contract Drawing with updated information in addition to the indicated fields.
- b. When Point Schedules are included in the contract package, items requiring contractor verification or input have been shown in angle brackets (" $<$ " and " $>$ "), such as $< ___ >$ for a required entry or $< \text{value} >$ for a value requiring confirmation. Complete all items in brackets as well as any blank cells. Do not modify values which are not in brackets without approval. Points Schedule Columns must include:

3.6.3.1 Point Name

The abbreviated name for the point using the indicated naming convention.

3.6.3.2 Description

A brief functional description of the point such as "Supply Air Temperature".

3.6.3.3 DDC Hardware Identifier

The Unique DDC Hardware Identifier shown on the DDC Hardware Schedule and used across all drawings for the DDC Hardware containing the point.

3.6.3.4 Settings

The value and units of any setpoints, configured setpoints, configuration parameters, and settings related to each point.

3.6.3.5 Range

The range of values, including units, associated with the point, including but not limited to setpoint adjustment range, a sensor measurement range, or the status of a safety.

3.6.3.6 Input or Output (I/O) Type

The type of input or output signal associated with the point. Use the following abbreviations for entries in this column:

- a. AI: The value comes from a hardware (physical) Analog Input
- b. AO: The value is output as a hardware (physical) Analog Output
- c. BI: The value comes from a hardware (physical) Binary Input
- d. BO: The value is output as a hardware (physical) Binary Output
- e. PULSE: The value comes from a hardware (physical) Pulse Accumulator Input
- f. NET-IN: The value is provided from the network (generally from

another device). Use this entry only when the value is received from another device as part of scheduling or as part of a sequence of operation, not when the value is received on the network for supervisory functions such as trending, alarming, override or display at a user interface.

g. NET-OUT: The value is provided to another controller over the network.

Use this entry only when the value is transmitted to another device as part of scheduling or as part of a sequence of operation, not when the value is transmitted on the network for supervisory functions such as trending, alarming, override or display at a user interface.

3.6.3.7 Network Data Exchange Information

(Gets Data From, Sends Data To) Provide the DDC Hardware Identifier of other DDC Hardware the point is shared with.

3.6.3.8 Override Information

For each point requiring an Override, indicate if the Object for the point is Commandable.

3.6.3.9 Trend Object Information

For each point requiring a trend, indicate if the trend is Local or Remote. For remote trends provide the DDC Hardware Identifier for the device performing the trend.

3.6.3.10 Alarm Information

Indicate the Alarm Generation Type.

3.7 FIELD TESTING AND ADJUSTING EQUIPMENT

Provide personnel, equipment, instrumentation, and supplies necessary to perform site testing. The Government will witness the PVT, and written permission must be obtained from the Government before proceeding with the testing. Original copies of data produced, including results of each test procedure, during PVT must be turned over to the Government at the conclusion of each phase of testing prior to Government approval of the test. The test procedures must cover actual equipment and functions specified for the project.

3.7.1 Testing, Adjusting and Commissioning

After successful completion of the factory test as specified, the Contractor will be authorized to proceed with the installation of the system equipment, hardware, and software. Once the installation has been completed, tested, adjusted, and commissioned each control loop and system in accordance with [NIST SP 250](#) and must verify proper operation of each item in the sequences of operation, including hardware and software. Calibrate field equipment, including control devices, adjust control parameters and logic (virtual) points including control loop setpoints, gain constants, constraints, and verify data communications before the system is placed online. Test installed [ground rods](#) as specified in [IEEE 142](#) and submit certification stating that the test was performed in accordance with [IEEE 142](#). Calibrate each instrumentation device connected to the process control system control network by making a comparison

between the reading at the device and the display at the workstation, using a standard at least twice as accurate as the device to be calibrated. Check each control point within the process control system control network by making a comparison between the control command at the central station and field-controlled device. Deliver trend logs/graphs of all points showing to the Government that stable control has been achieved. Points on common systems must be trended simultaneously. One log must be provided showing concurrent samples taken once a minute for a total of 4 hours. One log must be provided showing concurrent samples taken once every 30 minutes, for a total of 24 hours. Verify operation of systems in the specified failure modes upon Process control system network failure or loss of power, and verify that systems return to process control system control automatically upon a resumption of process control system network operation or return of power. Deliver a report describing results of functional tests, diagnostics, calibrations and commissioning procedures including written certification to the Government that the installed complete system has been calibrated, tested, adjusted and commissioned and is ready to begin the PVT. The report must also include a copy of the approved PVT procedure.

3.7.2 Performance Verification Test (PVT)

Submit test procedures for the PVT. The test procedure must describe all tests to be performed and other pertinent information such as specialized test equipment required and the length of the PVT. The test procedures must explain, in detail, step-by-step actions and the expected results, to demonstrate compliance with all the requirements of the drawings and this specification. The test procedure must be site specific and based on the inputs and outputs, required calculated points and the sequence of control. Refer to the actions and expected results to demonstrate that the process control system performs in accordance with the sequence of control. Include a list of the equipment to be used during the testing plus manufacturer's name, model number, equipment function, the date of the latest calibration and the results of the latest calibration.

Demonstrate that the completed Process control system complies with the contract requirements. All physical and functional requirements of the project including communication requirements must be demonstrated and shown. Demonstrate that each system operates as required in the sequence of operation. The PVT as specified must not be started until after receipt of written permission by the Government, based on the written report including certification of successful completion of testing, adjusting and commissioning as specified, and upon successful completion of training as specified. Upon successful completion of the PVT, furnish test reports and other documentation.

3.7.3 Endurance Test

Use the endurance test to demonstrate the overall system reliability of the completed system. The endurance test must be conducted in phases. The endurance test must not be started until the Government notifies the Contractor in writing that the PVT is satisfactorily completed, training as specified has been completed, outstanding deficiencies have been satisfactorily corrected, and that the Contractor has permission to start the endurance test. Provide an operator to man the system 8 hours per day during daytime operations, including weekends and holidays, during Phase I endurance testing, in addition to any Government personnel that may be made available. The Government may terminate testing at any time when the system fails to perform as specified. Upon termination of testing by

the Government or by the Contractor, commence an assessment period as described for Phase II. Upon successful completion of the endurance test, deliver test reports and other documentation, as specified, to the Government prior to acceptance of the system.

3.7.3.1 Phase I (Testing)

The test must be conducted 24 hours per day, 7 days per week, for 14 consecutive calendar days, including holidays, and the system must operate as specified. Make no repairs during this phase of testing unless authorized by the Government in writing.

3.7.3.2 Phase II (Assessment)

After the conclusion of Phase I, identify failures, determine causes of failures, repair failures, and deliver a written report to the Government. The report must explain in detail the nature of each failure, corrective action taken, results of tests performed, and must recommend the point at which testing should be resumed. After delivering the written report, convene a test review meeting at the job site to present the results and recommendations to the Government. The meeting must not be scheduled earlier than 5 business days after receipt of the report by the Government. As a part of this test review meeting, demonstrate that failures have been corrected by performing appropriate portions of the performance verification test. The Government reserves the right to cancel the test review meeting if no failures or deficiencies occur during the Phase I testing. If the Government chooses to do so, the Contractor will be notified in writing. Based on the Contractor's report and the test review meeting, the Government will determine if retesting is necessary and the restart point. The Government reserves the right to require that the Phase I test be totally or partially rerun. Do not commence any required retesting until after receipt of written notification by the Government. After the conclusion of any retesting which the Government may require, the Phase II assessment must be repeated as if Phase I had just been completed.

3.7.3.3 Exclusions

The Contractor will not be held responsible for failures resulting from the following: Outage of the main power supply in excess of the capability of any backup power source, provided that the automatic initiation of all backup sources was accomplished and that automatic shutdown and restart of the process control system performed as specified. Failure of a Government furnished communications link, provided that the PLC automatically and correctly operates in the stand-alone mode as specified, and that the failure was not due to Contractor furnished equipment, installation, or software. Failure of existing Government owned equipment, provided that the failure was not due to Contractor furnished equipment, installation, or software.

3.8 FIELD TRAINING

Field training oriented to the specific system must be provided for designated personnel. Furnish a copy of the [training manual](#) for each trainee plus two additional copies. Manuals must include an agenda, the defined objectives for each lesson, and a detailed description of the subject matter for each lesson. Furnish audiovisual equipment and other training supplies and materials. Copies of the audiovisuals must be delivered with the printed training manuals. The Government reserves the

right to videotape training sessions for later use. A training day is defined as 8 hours of classroom instruction, excluding lunchtime, Monday through Friday, during the daytime shift in effect at the training facility. Submit the training manual and schedule to receive approval from the Government at least 30 days before the training.

3.8.1 Preliminary Operator Training

Prior to the start of field testing, preliminary operator training must be taught at the project site for 5 consecutive training days. Upon completion of this course, each student, using appropriate documentation, should be able to perform elementary operations with guidance and describe the general hardware architecture and functionality of the system. This course must include: general system architecture; functional operation of the system, including workstations; operator commands; application programs, control sequences, and control loops; database entry and modification; reports generation; alarm reporting; diagnostics; and historical files.

3.8.2 Additional Operator Training

Following the field testing, additional classroom training for operators must be taught for 5 consecutive training days; individual instruction sessions of 4-hour periods in the morning (or afternoon) of the same weekday for 2 consecutive weeks and an additional 1 day classroom session for answering operator questions. Individual instruction must consist of "hands-on" training under the constant monitoring of the instructor. Classroom training must include instruction on the specific hardware configuration of the installed process control system and specific instructions for operating the installed system. Schedule activities during this period so that the specified amount of time on the equipment will be available for each student. The final session will address specific topics that the students need to discuss and to answer questions concerning the operation of the system. Upon completion of the course, the students should be fully proficient in system operation and have no unanswered questions regarding operation of the installed process control system. Each student should be able to start the system, operate the system, recover the system after a failure and describe the specific hardware architecture and operation of the system and be fully proficient in all system operations. Report the skill level of each student at the end of this course.

3.8.3 Maintenance Training

Following the endurance test a minimum period of one training days must be provided by a factory representative or a qualified Contractor trainer for five designated personnel on maintenance of the equipment. The training must include: physical layout of each piece of hardware, calibration procedures, preventive maintenance procedures, schedules, troubleshooting, diagnostic procedures and repair instructions.

3.8.4 Specialized Training

Following the maintenance training, a minimum period of five, total training day(s) must be provided by a factory representative or a qualified Contractor trainer for ten people on the input devices.

3.8.4.1 Flow Meter Training

Each type of flow meter, to include calibration, maintenance and testing of flow elements and transducers.

3.9 OPERATION AND MAINTENANCE DATA REQUIREMENTS

Outline the step-by-step procedures required for system startup, operation and shutdown. Include in the instructions layout, wiring and control diagrams of the system as installed, the manufacturer's name, model number, service manual, parts list and a brief description of all equipment and their basic operating features. List routine maintenance procedures, possible breakdowns and repairs and troubleshooting guides.

-- End of Section --